Karlsruher Institut für Technologie

Module Handbook
Industrial Engineering and Management (M.Sc.)

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KIT DEPARTMENT OF BUSINESS AND ECONOMICS

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1 The Module Handbook: Your Guide Through Your Studies

Welcome to the **Module Handbook** for your study program at the KIT Department of Business and Economics. We are delighted to have you join us and wish you a successful start to the new semester! This handbook provides an essential introduction to the key terms and regulations governing the selection of modules, courses, and examinations within your program.

Your study program is structured around several disciplines (e.g., Business Administration, Economics, Operations Research). Each discipline is further divided into **modules**, with each module comprising one or more interrelated components that often lead to an examination. The scope of each module is defined by **Credit Points (CP)**, which are awarded upon successful completion. While some modules are compulsory, the program's interdisciplinary nature offers a wide range of individual specialization and opportunities to deepen your knowledge within many modules. This flexibility allows you to tailor the content and timeline of your studies according to your personal interests, career aspirations, and needs.

This module handbook serves as a comprehensive guide to the modules within your program. Specifically, it details:

- The **structure** of the modules
- Their **scope** (in CP)
- Module **dependencies**
- Expected **learning outcomes**
- **Assessment methods** and examinations

This handbook is an essential orientation and a helpful guide throughout your studies. Please note that it **complements, but does not replace, the course catalog**, which provides semester-specific and variable course details such as timetables and locations.

1.1 Module Selection and Completion

Each module and each examination can only be selected once. If an examination is selectable within multiple modules, the student's decision on its assignment to a specific module is made at the moment of registration for that examination.

A module is considered completed and passed when its module examination is successfully passed (achieving a grade of 4.0 or better). For modules where the module examination is subdivided into several partial examinations, the module is completed only when all necessary partial examinations have been successfully passed. If a module offers alternative partial examinations, the module examination is considered concluded with the examination that leads to or exceeds the required total credit points. The module grade, reflecting the weight of its predefined credit points, is then included in the overall grade calculation for the study program.

1.2 Module Versions and Revisions

Modules are periodically revised, for instance, due to the introduction of new courses or the cancellation of existing examinations. As a rule, a new module version is created, which applies to all students who are newly starting the module. Students who have already commenced a module, however, are generally "grandfathered" and remain subject to the conditions of the module version valid at the time they began. These students can complete the module under the same conditions as at the beginning (exceptions are regulated by the Examination Board). The decisive date for this is the student's "binding declaration" regarding the choice of the module, as defined by §5(2) of the Study and Examination Regulations. This binding declaration is established by registering for the first examination within that module.

1.3 Accessing Module Handbook Versions

All modules in this handbook are presented in their current version, with the version number indicated in each module description. Older module versions can be accessed via previous module handbooks in the archive at: http://www.wiwi.kit.edu/Archiv_MHB.php.

1.4 Examination Formats and Registration

Module examinations can be taken either as a comprehensive general examination or as partial examinations. If offered as a general examination, the entire learning content of the module will be assessed in a single examination. If the module exam is divided into partial exams, these may be taken over the course of several semesters, for example as individual exams for the corresponding courses.

Registration for examinations can be done online via the campus management portal. The following functions are available at https://campus.studium.kit.edu/:

- Register/deregister for examinations
- Check examination results
- Create transcript of records

For further and more detailed information, please refer to: https://campus.studium.kit.edu/faq.php.

1.5 Grading, Repetition Rules, and Types of Assessment

Examinations are categorized into written examinations, oral examinations, and alternative assessment types ("Prüfungsleistungen anderer Art"). All examinations are graded. Ungraded coursework ("Studienleistungen"), on the other hand, can be repeated multiple times and do not receive a grade.

Generally, a failed written exam, oral exam, or alternative assessment can be repeated only once. Should this repeat examination (including any provided oral repeat examination) also be failed, the right to take the examination is forfeited. A request for a second repetition must be submitted in writing to the Examination Board two months after forfeiting the examination right. For more information, please see: http://www.wiwi.kit.edu/hinweiseZweitwdh.php.

1.6 Examiners

The Examination Board has appointed the KIT examiners and lecturers listed in this module handbook for the modules and their respective courses as official examiners for the courses they offer.

1.7 Additional Achievements

Additional achievements are voluntarily undertaken graded assessments that do not impact the student's overall grade. They can pertain to single courses or entire modules. It is mandatory to declare an additional achievement as such at the time of registration for the respective assessment. Students may earn up to 30 CP in additional achievements and these may be listed on the final transcript.

1.8 Important Resources and Legal Framework

For current information regarding your studies at the KIT Department of Business and Economics, please visit our website www.wiwi.kit.edu, as well as our channels on Instagram, LinkedIn, and YouTube. Please also consult the current notices and announcements for students at: https://www.wiwi.kit.edu/studium.php.

Comprehensive information regarding the legal and regulatory framework of your study program can be found in the respective study and examination regulations. These documents are available under the Official Announcements of KIT at: http://www.sle.kit.edu/amtlicheBekanntmachungen.php.

2 General Information

2.1 Study program details

KIT-Department	KIT Department of Business and Economics
Academic Degree	Master of Science (M.Sc.)
Examination Regulations Version	2026
Regular semesters	4 semesters
Maximum semesters	7 semesters
Credits	120
Language	English
Grade calculation	Weighted average by credits
Additional Information	Department https://www.wiwi.kit.edu/index.php Business unit Studium und Lehre https://www.sle.kit.edu/english/vorstudium/master-industrial-engineering-management.php

2.2 Content

In the interdisciplinary master's degree program in Industrial Engineering and Management, you will deepen and supplement the academic qualifications you acquired in your bachelor's degree in an individual way - in the subjects of business administration, economics, computer science and engineering. The standard period of study is 4 semesters and you will earn 120 credit points (CP) according to ECTS.

In the compulsory area, the program focuses on developing technical and methodological skills in the fields of business administration and engineering, while in the compulsory elective area you can hone your personal skills profile by choosing your own specializations in the aforementioned range of subjects and/or additionally in statistics, law or sociology, for example.

The seminar module with two subject-independent seminars and the Supervised Profile Definition course also contribute to the development of your individual profile. In the final semester, you will complete your studies with your master's thesis.

The study program does not include a compulsory internship. You can complete a voluntary internship during your studies. If it serves the objectives of your studies and covers at least half of the lecture period, you can apply for a semester off (leave of absence). Such an internship offers you valuable experience outside the university and can make it easier for you to start your career later on.

You have various options for integrating a stay abroad into your master's degree course. For example, you can study at a foreign university of your choice for one or two semesters via Erasmus+ or as a freemover. You can also do an internship abroad. In addition, selective mobility is possible via Eucor - The European Campus: KIT students can flexibly attend individual courses in addition to courses at KIT or study full-time for an entire semester or year at an Eucor partner university.

At the end of your studies, you will complete your master's thesis (30 CP), an academic paper that allows you to apply and deepen the knowledge and skills you have acquired. It deals with a specific topic from your field of study and requires independent research, analysis and the preparation of a written thesis. The master's thesis takes 6 months to complete. The master's thesis can also be completed in collaboration with a company.

Characteristic features of Industrial Engineering and Management M.Sc. at KIT:

- the program is anchored at the KIT Department of Business and Economics
- individual curriculum design
- new English-language degree program with a high degree of internationality and a multicultural campus
- partner network with companies for company contacts and internships during the degree program
  department-internal International Relations Office to support stays abroad
  KIT-Gründerschmiede

2.3 Qualification Goals

- Graduates of the interdisciplinary Master's program in Industrial Engineering have advanced and in-depth knowledge in business administration, economics, computer science, operations research and engineering. This mainly has its focus on business administration and engineering.
- The areas of specialization depend on individual interests. Additional knowledge in statistics, law or sociology is also offered depending on one's interests.
- They have generalized or specialized expertise in the different disciplines.
- The graduates are in a position to define, describe and interpret the specifics, limits, terminologies and doctrines in these subjects, reproduce the current state of research and selectively use this as a basis for further development.
- Their extensive know-how enables them to think across the various disciplines and approach issues from different angles.
- They are able to select and combine appropriate courses of action for research-related topics. They can then transfer and apply these to specific problems.
- They can separately analyze extensive problems such as information and current challenges and review, compare and evaluate these using appropriate methods and concepts.
- They evaluate the complexity and risks, identify improvement potentials and choose sustainable solution processes and improvement methods.
- This puts them in a position where they are able to make responsible and science-based decisions.
- They are able to come up with innovative ideas and apply them accordingly.
- They can oversee these approaches either independently or in teams. They are able to explain and discuss their decisions.
- They can independently interpret, validate and illustrate the obtained results.
- The interdisciplinary use of knowledge also takes account of social, scientific and ethical insights.
- The graduates can communicate with expert representatives on a scientific level and assume prominent responsibility in a team.
- Karlsruhe's industrial engineers are characterized by their interdisciplinary thinking as well as their innovation and management capability.
- They are particularly qualified for industrial occupations, service sector or in public administration as well as a downstream scientific career (PhD).

Key qualifications

The Industrial Engineering and Management program is characterized by an exceptional degree of interdisciplinary. With the combination of subjects from business administration, economics, computer science, operations research, mathematics as well as engineering and natural sciences, the integration of knowledge from different disciplines is an inherent part of the program.

Interdisciplinary approaches and thinking in contexts are promoted naturally. In addition, the seminars and practical courses of the Master's degree program, in which students practice the highly qualified scientific processing and presentation of special subject areas, contribute significantly to the promotion of important key qualifications.

The key qualifications taught throughout the entire degree program can be assigned to the following areas:

Basic skills (soft skills):

- Teamwork, social communication and creativity techniques
- Presentation preparation and presentation techniques
- Logical and systematic argumentation and writing
- Structured problem-solving and communication

Practical orientation (enabling skills)

- Competence to act in a professional context
- Project management skills
- Basic knowledge of business administration
- English as a technical language

Orientation knowledge

- Imparting interdisciplinary knowledge
- Institutional knowledge of economic and legal systems
- Knowledge of international organizations
- Media, technology and innovation

The integrative teaching of key qualifications takes place in particular as part of a series of compulsory courses within the Master's programs, namely

- the seminar module
- Accompanying Master's thesis
- modules Business administration, economics, informatics.

In addition to the integrative teaching of key qualifications, the additive acquisition of key qualifications amounting to at least three credit points is provided for in the seminar module. All SQ courses offered by the HOC, the ZAK and language courses offered by the Language Center can be taken.

The SQ courses offered by the institutions can be found in the KIT VVZ under

- House of Competence (HOC) - Courses for all students >[Specializations]
- Studium Generale and Key Qualifications and Additional Qualifications (ZAK) >[Key Qualifications at the ZAK]
- Courses at the Language Center >[Language courses]

Further information on the concept and content of the SQ courses can be found on the respective homepage- to the courses offered by the HOC: http://www.hoc.kit.edu/lehrangebot- Key qualifications at ZAK: http://www.zak.kit.edu/sq.

2.4 Employment Prospects

As a graduate of the master's degree program in Industrial Engineering and Management, you will be characterized by your interdisciplinary way of thinking as well as your strong innovative and management skills. This qualifies you in particular as a manager in industry, the service sector or IT companies, where you can work in consulting, controlling, marketing, logistics and many other areas where both business and technical knowledge are required. Due to the international nature of the program, you will be able to carry out these activities both in Germany and abroad. Or you can opt for an academic career and tackle a doctorate as your next stage.

2.5 Acceptance Criterias

There are 100 places available each year. If more applications are received than there are places available, a ranking list is drawn up among those applicants who meet the admission requirements and have applied in due time and form.

The requirements for admission to the English-language master's degree program in Industrial Engineering and Management are based on the current admission regulations:

- **bachelor's degree**
  a bachelor's degree (or at least an equivalent degree) from a university, university of applied sciences, university of cooperative education or a foreign university in a business administration, economics, engineering, computer science, mathematics or physics program.
  The course must have a standard period of study of at least three years and have been completed with at least 180 ECTS credits.
- **eligibility for examination**
  there must be no final failure of an examination in a master's degree program in Industrial Engineering and Management, Industrial Engineering or a comparable degree program. The examination entitlement must still exist.
- **aptitude test**
  a successfully passed aptitude test is required.
  GMAT (Focus Edition) with a minimum score of 550 points or
  GRE with a minimum score of 315 points
- **proof of English language skills**
  proof of sufficient English language skills must be provided in accordance with the requirements of the KIT admission and enrollment regulations (see below)

The admission and selection commission of the master's degree program in Industrial Engineering and Management, in consultation with the examination board of the bachelor's degree program in Industrial Engineering and Management, decides on the equivalence of the bachelor's degree and the determination of the degree programs with essentially the same content. In the case of foreign degrees, the recommendations of the Standing Conference of the Ministers of Education and Cultural Affairs and existing university partnerships are taken into account.

2.6 Studies and Examination Regulations

The legal basis for the degree program and the examinations in the degree program is the

Study and Examination Regulations of the Karlsruhe Institute of Technology (KIT) for the Master's degree program in Industrial Engineering and Management

2.7 Organizational issues

Dates and events:

Current information on the degree programs as well as dates for information events and examinations can be found on the KIT department website (https://www.wiwi.kit.edu).

Recognition of achievements according to § 19 SPO

1. Achievements within the university system

   According to § 19 of the Study and Examination Regulations, study and examination achievements that have been completed in study programs at state or state-recognized universities and vocational academies in the Federal Republic of Germany or at foreign state or state-recognized universities can be recognized upon application by the student.

2. Achievements outside the higher education system

   Knowledge acquired outside the higher education system can also be recognized. A common example is the recognition of one or more internships through proof of relevant vocational training.
   For detailed information on the recognition process and the link to the application forms, please refer to the website of the KIT department.

Frequently asked questions

Answers to frequently asked questions from A for "thesis" to Z for "second repetition" can be found in our Hints A-Z.

2.8 Contact

If you have any questions about modules or exams, please contact the examination office of the KIT Department of Business and Economics:

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3 Study plan

The Master’s degree program in Industrial Engineering and Management (M.Sc.) has 4 semesters and consists of 120 credits (CP) including the Master's thesis. The Master’s programme further deepens or complements the scientific qualifications acquired in the Bachelor’s programme. Students will be capable of independently applying scientific knowledge and methods and evaluating their implications and scope concerning solutions of complex scientific and social problems.

Term	Credits	Quantitative Core	Engineering and Informatics		Business and Economics		Electives				Master Thesis
1	28	Quantitative Core Module 18 CP	ENG/INFO Module 1 9 LP	ENG/INFO Module 2 9 LP	BUS/ECON Module 1 9 LP	BUS/ECON Module 2 9 LP				Seminar Module 9 LP	
2	31										
3	31						Elective Module 1 9 LP	Elective Module 2 9 LP	Elective Module 3 9 LP		
4	30										Master Thesis 30 CP
	120	18	18		18		36				30

Figure 2: Structure of the Master Programme SPO2026 (Recommendation)

Figure 2 shows the structure of the subjects and the credits allocated to the subjects. In addition to the Quantitative Core module, two modules from the disciplines "Engineering and Informatics" and "Business and Economics" must be selected. Furthermore, three modules must be chosen from the subject "Electives". One of the modules can come from the disciplines law or sociology.

Furthermore, students have to attend two seminars with a minimum of 8 CP within the seminar module.

Figure 3 and 4 illustrates the examination load per semester in the Master's degree programme in Industrial Engineering and Management using an exemplary selection of modules.

Subject	Module	Course	Type	1. Sem		2. Sem		3. Sem		4. Sem	
				Ex	CP	Ex	CP	Ex	CP	CP	
Quantitative Core (18 LP)	Quantitative Core Module (18 LP)	Advanced Game Theory	L/E	WE	4,5						
		Applied Econometrics	L/E	WE	4,5						
		Applied Informatics	L/E	WE	4,5						
		Global Optimization I	L/E			WE	4,5				
Engineering and Informatics (18 LP)	Cooperative Autonomous Vehicles	Human Factors in Autonomous Driving	L/E	WE	4,5						
		Collective Perception in Autonomous Driving	L/E			WE	4,5				
	The Circular Factory	The Circular Factory	L			WE	9				
Business and Economics (18 LP)	Marketing and Sales Management	Economic Decision Making	L/E	EAT	4,5						
		Digital Marketing	L/E			EAT	4,5				
	Microeconomic Theory	Auction Theory	L/E	WE	4,5						
		Social Choice Theory	L/E			WE	4,5				
Electives (36 LP)	Mathematical Programming	Nonlinear Optimization I and II	L/E					WE	9		
	Advanced Topics in AI: Graph Neural Networks and Language Models	Knowledge Discovery	L/E					EAT	4,5		
		Advanced Lab Informatics (Master)	AL					EAT	4,5		
	Consumer Research	Current Directions in Consumer Psychology	L					EAT	4,5		
		Computational Modelling of Judgments, Decisions and Cognition	L					EAT	4,5		
	Seminar Module	Supervised Profile Definition		C	1						
		Seminar in Statistics A (Master)	S			EAT	4				
		Seminar in Operations Research A (Master)						EAT	4		
Master Thesis	Master Thesis	Master Thesis								30	
			Number of examinations:	6		6		6		0	**18**
			Credit Points:	28		31		31		30	**120**

L = Lecture
E = Exercise
AL = Advanced Lab
S = Seminar
WE = Written Examination
EAT = Examination of Another Type
C = Non-Graded Coursework
Ex = Type of Examination
CP = Credit Points
Sem = Semester

Figure 3: Examination load per semester based on an exemplary module selection

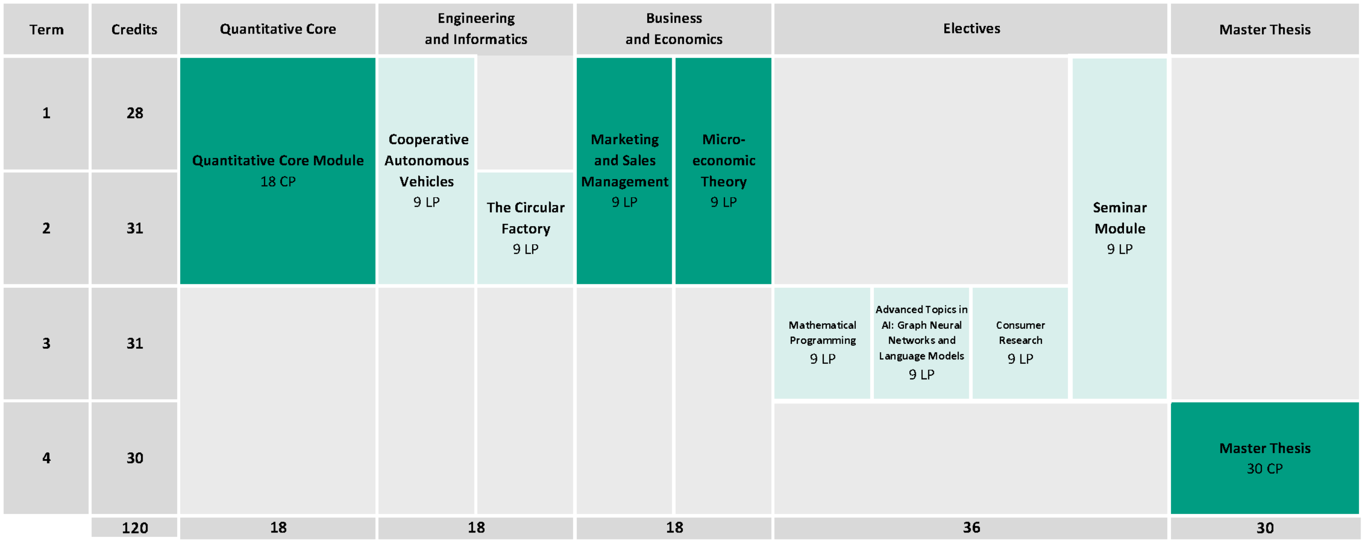

Figure 4: Structure of the Master Programme SPO2026 based on an exemplary module selection

It is left to the student's individual curriculum (taking into account the examination and module regulations), in which terms the chosen modules will be started and completed. However, it is highly recommended to complete all courses and seminars before beginning the Master's thesis.

4 Study Program Structure

Mandatory	
Master's Thesis	30 CP
Quantitative Core	18 CP
Engineering and Informatics	18 CP
Business and Economics	18 CP
Electives	36 CP
Voluntary	
Additional Examinations *This field will not influence the calculated grade of its parent.*	

4.1 Master's Thesis

Credits
30

ID	Name	Language	Term offered	Credits
Mandatory				
M-WIWI-107719	Master's Thesis	DE	WT+ST	30 CP

4.2 Quantitative Core

Credits
18

ID	Name	Language	Term offered	Credits
Mandatory				
M-WIWI-107187	Quantitative Core	EN	WT+ST	18 CP

4.3 Engineering and Informatics

Credits
18

ID	Name	Language	Term offered	Credits
Informatics (Elective:)				
M-WIWI-106804	Advanced Topics in AI: Graph Neural Networks and Language Models	DE/EN	WT+ST	9 CP
M-WIWI-106803	Advanced Topics in AI: Knowledge Graphs and the Web	DE/EN	WT+ST	9 CP
M-WIWI-105366	Artificial Intelligence	DE/EN	WT+ST	9 CP
M-WIWI-106631	Cooperative Autonomous Vehicles	EN	WT+ST	9 CP
Mechanical Engineering (Elective:)				
M-MACH-107636	Circular Factory	EN	ST	8 CP
M-MACH-107716	Data Driven Engineering	EN	WT+ST	9 CP
M-WIWI-101404	Extracurricular Module in Engineering		Once	9 CP
M-MACH-101282	Global Production and Logistics	EN	WT+ST	9 CP
M-MACH-105298	Logistics and Supply Chain Management	EN	ST	9 CP
M-WIWI-107718	Mechanical Engineering I	EN	WT+ST	9 CP
M-WIWI-107730	Mechanical Engineering II	EN	WT+ST	9 CP
Chemical and Process Engineering (Elective:)				
M-CIWVT-105407	Additive Manufacturing for Process Engineering	EN	ST	6 CP
M-CIWVT-103441	Biofilm Systems	EN	ST	4 CP
M-CIWVT-106837	Bioprocess Scale-up	EN	WT	6 CP
M-CIWVT-106838	Biosensors	EN	WT+ST	4 CP
M-CIWVT-106566	Chemical Hydrogen Storage	EN	WT	4 CP
M-CIWVT-106319	Data-Based Modeling and Control	EN	WT	6 CP
M-CIWVT-105206	Design of a Jet Engine Combustion Chamber	EN	WT	6 CP
M-CIWVT-107566	Electromagnetic Energy in Process Engineering	EN	WT+ST	6 CP
M-CIWVT-106320	Estimator and Observer Design	EN	WT	6 CP
M-WIWI-101404	Extracurricular Module in Engineering		Once	9 CP
M-CIWVT-106676	Introduction to Numerical Simulation of Reacting Flows	EN	WT	8 CP
M-CIWVT-106317	Optimal and Model Predictive Control	EN	ST	6 CP
M-CIWVT-103068	Physical Foundations of Cryogenics	EN	ST	6 CP
M-CIWVT-105891	Power-to-X – Key Technology for the Energy Transition	EN	WT+ST	6 CP
M-CIWVT-106313	Principles of Constrained Static Optimization	EN	WT	4 CP
M-CIWVT-106537	Reactor Modeling with CFD	EN	ST	6 CP
M-CIWVT-106564	Single-Cell Technologies	EN	WT	4 CP
Electrical Engineering and Information Technology (Elective:)				
M-ETIT-103802	Adaptive Optics	EN	WT	3 CP
M-ETIT-106815	Advanced Communications Engineering	EN	WT	6 CP
M-ETIT-106956	Antennas and Beamforming	EN	WT	4 CP
M-ETIT-107333	Applied Information Theory	EN	WT	6 CP
M-WIWI-101404	Extracurricular Module in Engineering		Once	9 CP
M-ETIT-107005	Batteries, Fuel Cells, and Electrolysis	EN	WT	6 CP
M-ETIT-100539	Communication Systems and Protocols	EN	ST	5 CP
M-ETIT-106953	Cyber-Physical Modeling	EN	ST	6 CP
M-ETIT-106690	Digital Real Time Simulations for Energy Technologies	EN	ST	3 CP
M-ETIT-100566	Field Propagation and Coherence	EN	WT	4 CP
M-ETIT-100449	Hardware Modeling and Simulation	EN	WT	4 CP
M-ETIT-107344	Integrated Photonics	EN	WT	6 CP
M-ETIT-105461	Introduction to Automotive and Industrial Lidar Technology	EN	WT	3 CP
M-ETIT-106789	IT/OT-Security Seminar	EN	WT	4 CP
M-ETIT-100434	Laser Metrology	EN	ST	3 CP
M-ETIT-106672	Medical Image Processing for Guidance and Navigation	EN	WT	9 CP
M-ETIT-105893	Mixed-Signal IC Design	EN	ST	3 CP
M-ETIT-105464	MMIC Design Laboratory	EN	WT+ST	6 CP

ID	Name	Language	Term offered	Credits
M-ETIT-105971	Mobile Communications	EN	WT	4 CP
M-ETIT-106244	Mobile Communications II	EN	ST	3 CP
M-ETIT-100430	Nonlinear Optics	EN	ST	6 CP
M-ETIT-106974	Optical Engineering and Machine Vision	EN	WT	6 CP
M-ETIT-100436	Optical Transmitters and Receivers	EN	WT	6 CP
M-ETIT-105608	Physics, Technology and Applications of Thin Films	EN	WT	4 CP
M-ETIT-107533	Power Electronic Systems	EN	WT	6 CP
M-ETIT-104567	Power Electronics	EN	ST	6 CP
M-ETIT-107156	Practical Amplifier Design	EN	ST	3 CP
M-ETIT-107343	Printed and Thin-Film Electronics	EN	WT	3 CP
M-ETIT-100420	Radar Systems Engineering	EN	WT	6 CP
M-ETIT-105123	Radio Frequency Integrated Circuits and Systems	EN	ST	6 CP
M-ETIT-106955	Radio-Frequency Electronics	EN	WT	6 CP
M-ETIT-107524	Robotics, Automation, and Systems Control Lab	EN	ST	6 CP
M-ETIT-106970	Seminar Industrial Process and Plant Engineering	EN	ST	4 CP
M-ETIT-107588	Seminar New Power Electronic Systems and Technologies	DE/EN	WT+ST	4 CP
M-ETIT-101971	Single-Photon Detectors	EN	WT	4 CP
M-ETIT-106684	Superconducting Magnet Technology	EN	ST	4 CP
M-ETIT-105521	Superconducting Materials	EN	WT	6 CP
M-ETIT-105609	Superconducting Nanowire Detectors	EN	ST	4 CP
M-ETIT-105611	Superconductivity for Engineers	EN	WT	6 CP
Civil Engineering, Geo and Environmental Sciences (Elective:)				
M-WIWI-101404	Extracurricular Module in Engineering		Once	9 CP
M-BGU-104448	Urban Water Technologies	EN	WT	9 CP

4.4 Business and Economics

Credits
18

ID	Name	Language	Term offered	Credits
Business and Economics (Elective: at least 18 credits)				
M-WIWI-105659	Advanced Machine Learning and Data Science	EN	WT+ST	9 CP
M-WIWI-101453	Applied Strategic Decisions	EN	WT+ST	9 CP
M-WIWI-101504	Collective Decision Making	EN	WT+ST	9 CP
M-WIWI-105714	Consumer Research	EN	WT+ST	9 CP
M-WIWI-101510	Cross-Functional Management Accounting	EN	WT+ST	9 CP
M-WIWI-105032	Data Science for Finance	EN	WT	9 CP
M-WIWI-103117	Data Science: Data-Driven Information Systems	EN	WT+ST	9 CP
M-WIWI-103118	Data Science: Data-Driven User Modeling	EN	WT+ST	9 CP
M-WIWI-101647	Data Science: Evidence-based Marketing	EN	WT+ST	9 CP
M-WIWI-104080	Designing Interactive Information Systems	EN	WT+ST	9 CP
M-WIWI-106258	Digital Marketing	EN	WT+ST	9 CP
M-WIWI-101502	Economic Theory and its Application in Finance	EN	WT+ST	9 CP
M-WIWI-107010	Economics in a Connected World	EN	WT+ST	9 CP
M-WIWI-107011	Economics of Innovation and Growth	EN	WT+ST	9 CP
M-WIWI-105894	Foundations for Advanced Financial -Quant and -Machine Learning Research	EN	see notes	9 CP
M-WIWI-105923	Incentives, Interactivity & Decisions in Organizations	EN	WT+ST	9 CP
M-WIWI-104068	Information Systems in Organizations	EN	WT+ST	9 CP
M-WIWI-101498	Management Accounting	EN	see notes	9 CP
M-WIWI-105312	Marketing and Sales Management	EN	ST	9 CP
M-WIWI-101500	Microeconomic Theory	EN	WT+ST	9 CP
M-WIWI-106660	Modeling the Dynamics of Financial Markets	EN	ST	9 CP
M-WIWI-101506	Service Analytics	EN	WT+ST	9 CP
M-WIWI-101503	Service Design Thinking	EN	WT	9 CP
M-WIWI-102754	Service Economics and Management	EN	WT+ST	9 CP
M-WIWI-102806	Service Innovation, Design & Engineering	EN	WT+ST	9 CP
M-WIWI-101448	Service Management	EN	WT+ST	9 CP
M-WIWI-105010	Student Innovation Lab (SIL) 1	EN	WT	9 CP
M-WIWI-107649	Sustainable Information Systems	EN	WT+ST	9 CP

4.5 Electives

Credits
36

Election notes
Please note that the elective modules listed include both English and German language offerings. Students are advised to verify the specific language of instruction in the individual module description before making their selection.

ID	Name	Language	Term offered	Credits
Mandatory				
M-WIWI-107204	Seminar Module	DE/EN	WT+ST	9 CP
Business Administration (Elective: at most 18 credits)				
M-WIWI-105659	Advanced Machine Learning and Data Science	EN	WT+ST	9 CP
M-WIWI-101410	Business & Service Engineering	DE/EN	WT+ST	9 CP
M-WIWI-105714	Consumer Research	EN	WT+ST	9 CP
M-WIWI-101498	Management Accounting	EN	see notes	9 CP
M-WIWI-101510	Cross-Functional Management Accounting	EN	WT+ST	9 CP
M-WIWI-105032	Data Science for Finance	EN	WT	9 CP
M-WIWI-103117	Data Science: Data-Driven Information Systems	EN	WT+ST	9 CP
M-WIWI-103118	Data Science: Data-Driven User Modeling	EN	WT+ST	9 CP
M-WIWI-101647	Data Science: Evidence-based Marketing	EN	WT+ST	9 CP
M-WIWI-105661	Data Science: Intelligent, Adaptive, and Learning Information Services	DE/EN	WT+ST	9 CP
M-WIWI-104080	Designing Interactive Information Systems	EN	WT+ST	9 CP
M-WIWI-106258	Digital Marketing	EN	WT+ST	9 CP
M-WIWI-102808	Digital Service Systems in Industry	DE	WT+ST	9 CP
M-WIWI-103720	eEnergy: Markets, Services and Systems	DE	WT+ST	9 CP
M-WIWI-101409	Electronic Markets	DE/EN	WT+ST	9 CP
M-WIWI-101451	Energy Economics and Energy Markets	DE/EN	WT+ST	9 CP
M-WIWI-101452	Energy Economics and Technology	DE/EN	WT+ST	9 CP
M-WIWI-101488	Entrepreneurship (EnTechnon)	DE/EN	WT+ST	9 CP
M-WIWI-101482	Finance 1	DE/EN	WT+ST	9 CP
M-WIWI-101483	Finance 2	DE/EN	WT+ST	9 CP
M-WIWI-101480	Finance 3	DE/EN	WT+ST	9 CP
M-WIWI-105894	Foundations for Advanced Financial -Quant and -Machine Learning Research	EN	see notes	9 CP
M-WIWI-105923	Incentives, Interactivity & Decisions in Organizations	EN	WT+ST	9 CP
M-WIWI-101471	Industrial Production II	DE/EN	WT	9 CP
M-WIWI-101412	Industrial Production III	DE/EN	ST	9 CP
M-WIWI-101411	Information Engineering	DE/EN	WT+ST	9 CP
M-WIWI-104068	Information Systems in Organizations	EN	WT+ST	9 CP
M-WIWI-101507	Innovation Management	DE/EN	WT+ST	9 CP
M-WIWI-101446	Market Engineering	DE/EN	WT+ST	9 CP
M-WIWI-105312	Marketing and Sales Management	EN	ST	9 CP
M-WIWI-106660	Modeling the Dynamics of Financial Markets	EN	ST	9 CP
M-WIWI-101506	Service Analytics	EN	WT+ST	9 CP
M-WIWI-101503	Service Design Thinking	EN	WT	9 CP
M-WIWI-102754	Service Economics and Management	EN	WT+ST	9 CP
M-WIWI-102806	Service Innovation, Design & Engineering	EN	WT+ST	9 CP
M-WIWI-101448	Service Management	EN	WT+ST	9 CP
M-WIWI-103119	Advanced Topics in Strategy and Management	DE	WT+ST	9 CP
M-WIWI-105010	Student Innovation Lab (SIL) 1	EN	WT	9 CP
M-WIWI-107649	Sustainable Information Systems	EN	WT+ST	9 CP
Economics (Elective: at most 18 credits)				
M-WIWI-101453	Applied Strategic Decisions	EN	WT+ST	9 CP
M-WIWI-101504	Collective Decision Making	EN	WT+ST	9 CP
M-WIWI-107010	Economics in a Connected World	EN	WT+ST	9 CP
M-WIWI-107011	Economics of Innovation and Growth	EN	WT+ST	9 CP
M-WIWI-101505	Experimental Economics	DE/EN	WT+ST	9 CP
M-WIWI-101500	Microeconomic Theory	EN	WT+ST	9 CP

ID	Name	Language	Term offered	Credits
M-WIWI-101406	Network Economics	DE/EN	WT+ST	9 CP
M-WIWI-101638	Econometrics and Statistics I	DE/EN	WT+ST	9 CP
M-WIWI-101502	Economic Theory and its Application in Finance	EN	WT+ST	9 CP
M-WIWI-101468	Environmental Economics	DE	WT+ST	9 CP
M-WIWI-101485	Transport Infrastructure Policy and Regional Development	DE/EN	WT+ST	9 CP
M-WIWI-101511	Advanced Topics in Public Finance	DE	WT+ST	9 CP
Informatics (Elective: at most 18 credits)				
M-WIWI-106803	Advanced Topics in AI: Knowledge Graphs and the Web	DE/EN	WT+ST	9 CP
M-WIWI-105366	Artificial Intelligence	DE/EN	WT+ST	9 CP
M-WIWI-106631	Cooperative Autonomous Vehicles	EN	WT+ST	9 CP
M-WIWI-101477	Development of Business Information Systems	DE	WT+ST	9 CP
M-WIWI-104520	Human Factors in Security and Privacy	DE	WT+ST	9 CP
M-WIWI-101472	Informatics	DE/EN	WT+ST	9 CP
M-WIWI-101456	Intelligent Systems and Services	DE/EN	WT+ST	9 CP
M-WIWI-103356	Machine Learning	DE/EN	WT+ST	9 CP
M-WIWI-101628	Emphasis in Informatics	DE/EN	WT+ST	9 CP
Operations Research (Elective: at most 18 credits)				
M-WIWI-101473	Mathematical Programming	DE/EN	WT+ST	9 CP
M-WIWI-102832	Operations Research in Supply Chain Management	DE/EN	WT+ST	9 CP
M-WIWI-102805	Service Operations	DE/EN	WT+ST	9 CP
M-WIWI-103289	Stochastic Optimization	DE/EN	WT+ST	9 CP
Engineering Sciences (Elective: at most 18 credits)				
M-ETIT-103802	Adaptive Optics	EN	WT	3 CP
M-CIWVT-105407	Additive Manufacturing for Process Engineering	EN	ST	6 CP
M-ETIT-106815	Advanced Communications Engineering	EN	WT	6 CP
M-ETIT-106956	Antennas and Beamforming	EN	WT	4 CP
M-ETIT-107333	Applied Information Theory	EN	WT	6 CP
M-WIWI-101404	Extracurricular Module in Engineering		Once	9 CP
M-ETIT-107673	Automation and Control	EN	WT	10 CP
M-MACH-101298	Automated Manufacturing Systems	DE	WT+ST	9 CP
M-MACH-101274	Rail System Technology	DE	WT+ST	9 CP
M-ETIT-107005	Batteries, Fuel Cells, and Electrolysis	EN	WT	6 CP
M-CIWVT-103441	Biofilm Systems	EN	ST	4 CP
M-MACH-101290	BioMEMS	DE	WT+ST	9 CP
M-CIWVT-106837	Bioprocess Scale-up	EN	WT	6 CP
M-CIWVT-106838	Biosensors	EN	WT+ST	4 CP
M-CIWVT-106566	Chemical Hydrogen Storage	EN	WT	4 CP
M-MACH-107636	Circular Factory	EN	ST	8 CP
M-ETIT-100539	Communication Systems and Protocols	EN	ST	5 CP
M-ETIT-106953	Cyber-Physical Modeling	EN	ST	6 CP
M-MACH-107716	Data Driven Engineering	EN	WT+ST	9 CP
M-CIWVT-106319	Data-Based Modeling and Control	EN	WT	6 CP
M-CIWVT-105206	Design of a Jet Engine Combustion Chamber	EN	WT	6 CP
M-ETIT-106690	Digital Real Time Simulations for Energy Technologies	EN	ST	3 CP
M-BGU-105592	Digitalization in Facility Management	DE	WT	9 CP
M-CIWVT-107566	Electromagnetic Energy in Process Engineering	EN	WT+ST	6 CP
M-MACH-101296	Energy and Process Technology I		WT	9 CP
M-MACH-101297	Energy and Process Technology II		ST	9 CP
M-BGU-100998	Design, Construction, Operation and Maintenance of Highways	DE	ST	9 CP
M-ETIT-101164	Generation and Transmission of Renewable Power	DE/EN	WT+ST	9 CP

ID	Name	Language	Term offered	Credits
M-CIWVT-106320	Estimator and Observer Design	EN	WT	6 CP
M-BGU-106453	Facility Management in Hospitals for Industrial Engineering	DE	WT	9 CP
M-MACH-101264	Handling Characteristics of Motor Vehicles	DE/EN	WT+ST	9 CP
M-MACH-101265	Vehicle Development	DE/EN	WT+ST	9 CP
M-MACH-101266	Automotive Engineering	DE/EN	WT+ST	9 CP
M-MACH-101276	Manufacturing Technology	DE	WT	9 CP
M-ETIT-100566	Field Propagation and Coherence	EN	WT	4 CP
M-MACH-101282	Global Production and Logistics	EN	WT+ST	9 CP
M-BGU-101064	Fundamentals of Transportation	DE	ST	9 CP
M-ETIT-100449	Hardware Modeling and Simulation	EN	WT	4 CP
M-ETIT-101163	High-Voltage Technology	DE	WT+ST	9 CP
M-ETIT-107344	Integrated Photonics	EN	WT	6 CP
M-MACH-101272	Integrated Production Planning	DE	ST	9 CP
M-ETIT-105461	Introduction to Automotive and Industrial Lidar Technology	EN	WT	3 CP
M-CIWVT-106676	Introduction to Numerical Simulation of Reacting Flows	EN	WT	8 CP
M-ETIT-106789	IT/OT-Security Seminar	EN	WT	4 CP
M-ETIT-100434	Laser Metrology	EN	ST	3 CP
M-BGU-101884	Lean Management in Construction	DE	WT	9 CP
M-MACH-105298	Logistics and Supply Chain Management	EN	ST	9 CP
M-MACH-101278	Material Flow in Networked Logistic Systems	DE	WT+ST	9 CP
M-WIWI-107718	Mechanical Engineering I	EN	WT+ST	9 CP
M-WIWI-107730	Mechanical Engineering II	EN	WT+ST	9 CP
M-ETIT-106672	Medical Image Processing for Guidance and Navigation	EN	WT	9 CP
M-MACH-101291	Microfabrication	DE	WT+ST	9 CP
M-MACH-101292	Microoptics	DE	WT+ST	9 CP
M-MACH-101287	Microsystem Technology	DE	WT+ST	9 CP
M-ETIT-105893	Mixed-Signal IC Design	EN	ST	3 CP
M-ETIT-105464	MMIC Design Laboratory	EN	WT+ST	6 CP
M-MACH-101267	Mobile Machines		WT	9 CP
M-ETIT-105971	Mobile Communications	EN	WT	4 CP
M-ETIT-106244	Mobile Communications II	EN	ST	3 CP
M-MACH-106496	Modern Mobility on Rails and Roads	DE	WT+ST	9 CP
M-MACH-101294	Nanotechnology	DE	WT+ST	9 CP
M-ETIT-100430	Nonlinear Optics	EN	ST	6 CP
M-ETIT-106974	Optical Engineering and Machine Vision	EN	WT	6 CP
M-ETIT-100436	Optical Transmitters and Receivers	EN	WT	6 CP
M-CIWVT-106317	Optimal and Model Predictive Control	EN	ST	6 CP
M-MACH-101295	Optoelectronics and Optical Communication	DE	WT+ST	9 CP
M-CIWVT-103068	Physical Foundations of Cryogenics	EN	ST	6 CP
M-ETIT-105608	Physics, Technology and Applications of Thin Films	EN	WT	4 CP
M-ETIT-107533	Power Electronic Systems	EN	WT	6 CP
M-ETIT-104567	Power Electronics	EN	ST	6 CP
M-CIWVT-105891	Power-to-X – Key Technology for the Energy Transition	EN	WT+ST	6 CP
M-ETIT-107156	Practical Amplifier Design	EN	ST	3 CP
M-CIWVT-106313	Principles of Constrained Static Optimization	EN	WT	4 CP
M-ETIT-107343	Printed and Thin-Film Electronics	EN	WT	3 CP
M-MACH-106590	Production Engineering	DE	WT+ST	9 CP
M-BGU-101888	Project Management in Construction	DE	WT	9 CP
M-ETIT-100420	Radar Systems Engineering	EN	WT	6 CP
M-ETIT-105123	Radio Frequency Integrated Circuits and Systems	EN	ST	6 CP

ID	Name	Language	Term offered	Credits
M-ETIT-106955	Radio-Frequency Electronics	EN	WT	6 CP
M-CIWVT-106537	Reactor Modeling with CFD	EN	ST	6 CP
M-ETIT-101157	Control Engineering II		WT+ST	9 CP
M-ETIT-107524	Robotics, Automation, and Systems Control Lab	EN	ST	6 CP
M-MACH-102626	Major Field: Integrated Product Development	DE	WT	16 CP
M-ETIT-106970	Seminar Industrial Process and Plant Engineering	EN	ST	4 CP
M-ETIT-107588	Seminar New Power Electronic Systems and Technologies	DE/EN	WT+ST	4 CP
M-ETIT-101158	Sensor Technology I	DE/EN	ST	9 CP
M-BGU-101066	Safety, Computing and Law in Highway Engineering	DE	WT	9 CP
M-CIWVT-106564	Single-Cell Technologies	EN	WT	4 CP
M-ETIT-101971	Single-Photon Detectors	EN	WT	4 CP
M-MACH-101268	Specific Topics in Materials Science	DE	WT+ST	9 CP
M-MACH-105455	Strategic Design of Modern Production Systems	DE	WT+ST	9 CP
M-ETIT-106684	Superconducting Magnet Technology	EN	ST	4 CP
M-ETIT-105521	Superconducting Materials	EN	WT	6 CP
M-ETIT-105609	Superconducting Nanowire Detectors	EN	ST	4 CP
M-ETIT-105611	Superconductivity for Engineers	EN	WT	6 CP
M-BGU-104448	Urban Water Technologies	EN	WT	9 CP
M-MACH-101275	Combustion Engines I		WT	8 CP
M-MACH-101303	Combustion Engines II	DE	WT+ST	9 CP
M-BGU-101110	Process Engineering in Construction	DE	WT	9 CP
M-BGU-101065	Transportation Modelling and Traffic Management	DE/EN	WT+ST	9 CP
M-MACH-104888	Advanced Module Logistics	DE	WT+ST	9 CP
M-CIWVT-101122	Water Chemistry and Water Technology II		WT+ST	12 CP
M-CIWVT-101121	Water Chemistry and Water Technology I	DE/EN	WT	9 CP
M-MACH-101286	Machine Tools and Industrial Handling	DE	WT	9 CP
Statistics (Elective: at most 18 credits)				
M-WIWI-101637	Analytics and Statistics	DE	WT+ST	9 CP
M-WIWI-101639	Econometrics and Statistics II	DE/EN	WT+ST	9 CP
Law or Sociology (Elective: at most 9 credits)				
M-INFO-101191	Commercial Law	DE	WT+ST	9 CP
M-INFO-101215	Intellectual Property Law	DE	WT+ST	9 CP
M-INFO-101216	Private Business Law	DE	WT+ST	9 CP
M-INFO-106754	Public Economic and Technology Law	DE/EN	WT+ST	9 CP
M-GEISTSOZ-101169	Sociology		WT+ST	9 CP

4.6 Additional Examinations

ID	Name	Language	Term offered	Credits
Additional Examinations (Elective: at most 30 credits)				
M-WIWI-104900	Business Administration	DE	WT+ST	9 CP
M-WIWI-104908	Economics	DE	WT+ST	9 CP
M-WIWI-104909	Informatics (Department of Informatics)	DE	WT+ST	9 CP
M-WIWI-104901	Informatics (KIT-Department of Economics and Management)	DE	WT+ST	9 CP
M-WIWI-104899	Operations Research	DE	WT+ST	9 CP
M-WIWI-104907	Engineering Sciences	DE	WT+ST	9 CP
M-WIWI-104902	Statistics	DE	WT+ST	9 CP
M-WIWI-104905	Mathematics	DE	WT+ST	9 CP
M-WIWI-104904	Natural Sciences	DE	WT+ST	9 CP
M-WIWI-104903	Law	DE	WT+ST	9 CP
M-WIWI-104906	Social Sciences	DE	WT+ST	9 CP
M-WIWI-104910	Interdisciplinary Qualifications	DE	WT+ST	9 CP
M-FORUM-106753	Supplementary Studies on Science, Technology and Society	DE	WT+ST	16 CP

5 Modules

5.1 Module: Adaptive Optics [M-ETIT-103802]

Coordinators: Ph.D. Szymon Gladysz
Prof. Dr. Ulrich Lemmer

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: Engineering and Informatics (Electrical Engineering and Information Technology)
Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each winter term	1 term	English	4	3

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-107644	Adaptive Optics	WT	3 CP	Gladysz, Lemmer

Assessment
Type of Examination: Oral examination

Duration of Examination: approx. 30 Minutes

Modality of Exam: The oral exam will be scheduled during the semester break.

Prerequisites
None.

Competence Goal
The students will:

- get familiar with Fourier description of imaging through aberrated optical systems and random media,
- understand the description of aberrations through Zernike modes,
- learn how to analytically compute the effects of turbulence on various optical observables such as image/beam motion, temporal power spectra, Zernike modes, scintillation, etc.,
- understand the effect of noise on various quantities and metrics pertinent to the design of adaptive optical systems,
- understand the advantages and disadvantages of various schemes for wavefront sensing and correction,
- learn how to simulate and design simple adaptive optics systems.

Content
Adaptive optics is a technology of correcting the effect of atmospheric turbulence on images of space objects and on laser beams propagating through random and highly aberrated media such as turbulence, tissue, and the inside of the human eye, to name just a few applications. The course will familiarize the students with theoretical basics of light propagation through random media, principles of wavefront sensing and reconstruction, as well as wavefront correction with deformable mirrors. The students will also receive solid introduction to statistical optics, the Kolmogorov theory of turbulence, practical aspects of turbulence simulation and modelling of adaptive optics.

1. Theory of turbulence (covariances, structure functions, power spectra, inertial range, dimensional argument of Kolmogorov)
2. Fourier optics (point-spread function, modulation transfer function)
3. Statistical optics (characteristic function, probability density function)
4. Sources and description of aberrations (Zernike polynomials, orthogonality, Marechal criterion)
5. Adaptive optics systems (open- and closed-loop systems, error budgets, tip-tilt correction)
6. Wavefront sensing (Shack-Hartmann wavefront sensor, wavefront reconstruction, wavefront-sensorless AO)
7. Wavefront correction (tip-tilt mirrors, deformable mirrors, piezoelectric effect, microelectromechanical systems, electrostatic actuation)
8. Simulation of adaptive optical systems (analytic vs. end-to-end modelling)
9. Propagation of laser beams through atmospheric turbulence (Gaussian beams, Rytov theory, scintillation index, beam wander)
10. Modelling of free-space optical communication systems (aperture averaging, mean signal-to-noise ratio, false-alarm rate and fade probability, bit error-rate)

Module Grade Calculation
The module grade is the grade of the oral exam.

Workload
total 90 h, hereof 30 h contact hours and 60 h homework and self-studies

Recommendations
Basic knowledge of statistics.
It is recommended that the students first complete the module Optical Engineering and Machine Vision [M-ETIT-106974] or Optical Engineering [M-ETIT-100456] or Fundamentals of Optics and Photonics [M-PHYS-101927] before they enroll for this course.

Literature
Robert K. Tyson, Principles of Adaptive Optics, CRC Press

Michael C. Roggemann, Byron M. Welsh, Imaging through Turbulence, CRC Press

M 5.2 Module: Additive Manufacturing for Process Engineering [M-CIWVT-105407]

Coordinators: TT-Prof. Dr. Christoph Klahn
Organisation: KIT Department of Chemical and Process Engineering
Part of: Engineering and Informatics (Chemical and Process Engineering)
Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each summer term	1 term	English	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-CIWVT-110902	Additive Manufacturing for Process Engineering - Examination	ST	5 CP	Klahn
T-CIWVT-110903	Practical in Additive Manufacturing for Process Engineering		1 CP	Klahn

Assessment
Learning control consists of:

- Practical (ungraded)
- Oral examination lastin approx. 30 minutes

Prerequisites
Practical in Additive Manufaturing for Process Engineeing is a prerequisite for the oral exam.

Competence Goal
Students are familiar with the concept of a fully digital fabrication chain using and linking together modeling and simulation, computer aided design and 3D printing. They know the most important 3D printing methods suitable for process engineering applications. Moreover, they are able to use standard tools for 3D data generation and they already own hands on practical experience with the use of a metal 3D printer for fabrication of highly precise parts with complex shape.

Content
The rationale for additive manufacturing and key aspects of this approach are explained. An overview of different methods and materials for 3D printing is given with a focus on the use of 3D printed parts or fully functional devices in chemical and process engineering. Tools for 3D data generation for additive manufacturing are introduced and design rules for selected 3D printing methods are explained. Illustrative examples for 3D printed components and functional devices in process engineering are presented and discussed based on literature and own research. In the practical, students will work together in small groups on a fully digital fabrication of functional parts by selective laser melting of metal powder going through a cycle of 3D data generation, 3D printing, and finishing of the printed parts.

Module Grade Calculation
Module grade is the grade of the oral examination.

Workload
Lectures: 30 h

Practical: 16 h (8 experiments)

Homework: 90 h

Exam Preparation: 44 h

Total: 180 h

Literature

- Ian Gibson, David Rosen, Brent Stucker, Mahyar Khorasani: Additive Manufacturing Technologies, Springer Nature Switzerland, 2021, DOI: 10.1007/978-3-030-56127-7
- Christoph Klahn, Mirko Meboldt, Filippo Fontana, Bastian Leutenecker-Twelsiek, Jasmin Jansen, Daniel Omidvarkarjan: Entwicklung und Konstruktion für die Additive Fertigung, Vogel Business Media, Würzburg, 2021, ISBN 978-3-8343-3469-5

5.3 Module: Advanced Communications Engineering [M-ETIT-106815]

Coordinators: Dr.-Ing. Holger Jäkel
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Engineering and Informatics (Electrical Engineering and Information Technology)
Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-113676	Advanced Communications Engineering	WT	6 CP	Jäkel

Assessment
The assessment takes place in the form of a written examination lasting 120 min,

Prerequisites
none

Competence Goal
The students are able to analyze and assess properties of communication systems and consider aspects of implementation. They can use mathematical methods in the context of communication systems for understanding involved derivations in the research literature; deriving and autonomously elaborating theoretical results, and checking their viability by simulations.

Content
The module is introducing and deriving results covering, but not being limited to, properties of linear modulation, channel description and diversity schemes, and processing of receiver signals, all based on detailed theoretical concepts. Topics already covered in previous modules are deduced thoroughly and mathematical derivations and reasoning are provided.

Module Grade Calculation
The module grade is the grade of the written exam.

Additional Information
Starting winter term 25/26

Workload

1.
    1. Attendance to the lecture: 20 * 1,5 h = 30 h
    2. Preparation and review: 20 * 3 h = 60 h
    3. Attendance to the tutorial: 6 * 1,5 h = 9 h
    4. Preparation and review: 6 * 3,5 h = 21 h
    5. Preparation for the exam: 60 h

    In total: 180 h = 6 LP

Recommendations
Basics knowledge of communication systems, as, e.g., provided in KIT's Bachelor courses “Grundlagen der Datenübertragung” and “Nachrichtensysteme”, is supposed. Furthermore, working knowledge in the areas of system theory and probability theory is assumed.

Teaching and Learning Methods
Lecture: 3 SWS, Exercise: 1 SWS

M 5.4 Module: Advanced Machine Learning and Data Science [M-WIWI-105659]

Coordinators: Prof. Dr. Maxim Ulrich
Organisation: KIT Department of Business and Economics
Part of: Business and Economics
Electives (Business Administration)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	1 term	English	4	3

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-WIWI-111305	Advanced Machine Learning and Data Science	WT+ST	9 CP	Ulrich

Assessment
The assessment is carried out in an alternative form. The grade is evaluated based on the intermediate presentations during the project, the quality of the implementation, the final written thesis, and a final presentation.

Prerequisites
None

Competence Goal
After a successful project, the students can:

- select and apply modern machine learning methods to solve a data science problem;
- organize themselves in a team in a goal-oriented manner and bring an extensive software project in the field of data science and machine learning to success;
- deepen their data science and machine learning skills
- solve a finance problem with the help of data science and machine learning algorithm.

Content
The course is targeted at students with a major in Data Science and/or Machine Learning and/or Quantitative Finance. It offers students the opportunity to develop hands-on knowledge on new developments in the intersection of quantitative financial markets, data science and machine learning. The result of the project should not only be a final thesis, but the implementation of methods or development of an algorithm in machine learning and data science. Typically, problems and data are taken from current research and innovations in the field of quantitative asset and risk management.

Workload
Total effort for 9 credit points: approx. 270 hours are divided into the following parts: Communication:Exchange during the project: 30 h, Final presentation: 10 h; Implementation and thesis: Preparation before development (Problem analysis and solution design): 70 h, Solution implementation: 110 h, Tests and quality assurance: 50 h.

Recommendations
None

Teaching and Learning Methods
Project

5.5 Module: Advanced Module Logistics [M-MACH-104888]

Coordinators: Prof. Dr.-Ing. Kai Furmans
Organisation: KIT Department of Mechanical Engineering

Part of: Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	1 term	German	4	10

ID	Name	Term offered	Credits	Lecturers
Specialization module logistics (Elective:)				
T-MACH-112213	Applied material flow simulation	WT	4,5 CP	Baumann
T-MACH-113950	Dimensioning of Material Flow Systems in Production and Logistics	WT	4 CP	Furmans
T-MACH-112113	Dynamic Systems of Technical Logistics	ST	4 CP	Mittwollen
T-MACH-112114	Dynamic Systems of Technical Logistics - Project	ST	2 CP	Mittwollen
T-MACH-105151	Energy Efficient Intralogistic Systems	WT	4 CP	Kramer, Schönung
T-WIWI-102718	Discrete-Event Simulation in Production and Logistics	ST	4,5 CP	Spieckermann
T-MACH-115017	Global Logistics	ST	4,5 CP	Furmans
T-MACH-113701	Industrial Mobile Robotics Lab	WT+ST	4,5 CP	Furmans
T-MACH-105187	IT-Fundamentals of Logistics	ST	4 CP	Thomas
T-MACH-105174	Warehousing and Distribution Systems	ST	4 CP	Furmans
T-MACH-105171	Safety Engineering	WT	4 CP	Kany

Assessment
The assessment is carried out as partial exams (according to Section 4 (2), 1-3 SPO) of the single courses of this module, whose sum of credits must meet the minimum requirement of 9 credits of this module. The assessment procedures are described for each course of the module separately. The overall grade of the module is the average of the grades for each course weighted by the credits and truncated after the first decimal.

Modeled Prerequisites
You have to fulfill one of 2 conditions:

1. The module M-MACH-105298 - Logistics and Supply Chain Management must have been passed.
2. The module M-MACH-106995 - Automation and Material Flow in Logistics must have been passed.

Competence Goal
The student aquires

- well-founded knowledge and method knowledge in the main topics of logistics,
- ability for modeling logistic systems with adequate accuracy by using simple models,
- ability to evaluate logistic systems and to identify cause-and-effects-chains within logistic systems.

Content
The Advanced Module Logistics provides a comprehensive and well-founded basics for the main topics of logistics. The module allows students to focus on various topics within the field of logistics.

Workload
270 hours

Teaching and Learning Methods
Lecture, tutorial.

5.6 Module: Advanced Topics in AI: Graph Neural Networks and Language Models [M-WIWI-106804]

Coordinators: Dr.-Ing. Tobias Käfer
Organisation: KIT Department of Business and Economics
Part of: Engineering and Informatics (Informatics)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	2 terms	German/English	4	1

ID	Name	Term offered	Credits	Lecturers
Compulsory Elective Courses (Elective: at least 2 items)				
T-WIWI-102666	Knowledge Discovery	WT	4,5 CP	Käfer
T-WIWI-110548	Advanced Lab Informatics (Master)	WT+ST	4,5 CP	Professorenschaft des Instituts AIFB

Assessment
The assessment mix of each course of this module is defined for each course separately. The final mark for the module is the average of the marks for each course weighted by the credits and truncated after the first decimal.

Prerequisites
None

Competence Goal
The student

- knows the basics of machine learning, data mining and knowledge discovery,
- can design, train and evaluate systems that are capable of learning,
- knows advanced concepts and methods, especially in the areas of Graph Neural Networks (GNNs) and Large Language Models (LLMs),
- can carry out knowledge discovery projects, taking into account algorithms, representations and applications,
- can apply interdisciplinary thinking to solve applied problems from different domains.

Content
The module focuses on machine learning and data mining methods for gaining knowledge from large data sets. In particular, advanced methods from the areas of Graph Neural Networks (GNNs) and Large Language Models (LLMs) are considered.

The lecture on Knowledge Discovery provides an overview of machine learning and data mining approaches for knowledge discovery from large data sets. These are examined in particular with regard to algorithms, applicability to different data representations and use in real application scenarios. Knowledge discovery is an established field of research with a large community investigating methods for discovering patterns and regularities in large amounts of data, including unstructured text.

A variety of methods exist to extract patterns and provide previously unknown insights. This information can be predictive or descriptive. The lecture will cover specific techniques and methods, challenges, and current and future research topics in this research area. The lecture covers the entire machine learning and data mining process, with topics on supervised and unsupervised learning methods and empirical evaluation.

The learning methods covered range from classical approaches such as decision trees, support vector machines and neural networks to selected approaches from current research. The learning problems considered include feature vector-based learning and text mining. The focus is particularly on advanced methods from the areas of Graph Neural Networks (GNNs) and Large Language Models (LLMs).In the practical course (Praktikum) on Knowledge Discovery, students apply the approaches and methods taught in the lecture to problems from different domains hands-on.

Workload
The total workload for this module is approximately 270 hours.

M 5.7 Module: Advanced Topics in AI: Knowledge Graphs and the Web [M-WIWI-106803]

Coordinators: Dr.-Ing. Tobias Käfer
Organisation: KIT Department of Business and Economics
Part of: Engineering and Informatics (Informatics)
Electives (Informatics)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	2 terms	German/English	4	1

ID	Name	Term offered	Credits	Lecturers
Compulsory Elective Courses (Elective: 2 items)				
T-WIWI-110848	Semantic Web Technologies	ST	4,5 CP	Käfer
T-WIWI-110548	Advanced Lab Informatics (Master)	WT+ST	4,5 CP	Professorenschaft des Instituts AIFB

Assessment
The assessment mix of each course of this module is defined for each course separately. The final mark for the module is the average of the marks for each course weighted by the credits and truncated after the first decimal.

Competence Goal
Students

- know the foundations and technologies used to build Knowledge Graphs.
- develop ontologies for semantic knowledge representation.
- are able to provide data and applications via a Web-based infrastructure.
- know foundations and advanced methods of symbolic reasoning on Knowledge Graphs.
- transfer the methods and technologies of semantic web technologies to various application domains.
- evaluate the potential of semantic web technologies for new application domains.
- are able to practically apply the aforementioned skills to solve problems from various application domains.

Content
This module covers a sub-area of artificial intelligence: (semantic) knowledge representation. The module presents the fundamentals, methods and applications for knowledge graph-based AI systems on the World Wide Web. We focus particularly on methods for semantic modelling and the decentralized provision of data and applications via the Web. Formal basics and practical aspects of semantic knowledge modeling are discussed in detail. Furthermore, technical details about the provision of data sets and their metadata on the Web based on Web standards are covered. In the practical course (Praktikum), students apply the skills acquired in the lecture to practical problems in different domains.

Workload
The total workload for this module is approximately 270 hours (9 credits). The allocation is based on the credits of the courses of the module. The workload for courses with 4.5 credits is about 135 hours.

The total number of hours per course results from the effort required to attend the lectures and exercises as well as the examination times and the time required to achieve the learning objectives of the module for an average student for an average performance.

M 5.8 Module: Advanced Topics in Public Finance (WW4VWL18) [M-WIWI-101511]

Coordinators: Prof. Dr. Berthold Wigger
Organisation: KIT Department of Business and Economics
Part of: Electives (Economics)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	2 terms	German	4	7

ID	Name	Term offered	Credits	Lecturers
Electives (Elective: between 1 and 2 items)				
T-WIWI-108711	Basics of German Company Tax Law and Tax Planning	WT	4,5 CP	Gutekunst, Wigger
T-WIWI-102740	Public Management	WT	4,5 CP	Wigger
Supplementary Courses (Elective: between 0 and 1 items)				
T-WIWI-111304	Fundamentals of National and International Group Taxation	ST	4,5 CP	Wigger
T-WIWI-102739	Public Revenues	ST	4,5 CP	Wigger

Assessment
The assessment is carried out as partial exams (according to Section 4 (2), 1-3 SPO) of the core course and further single courses of this module, whose sum of credits must meet the minimum requirement of credits of this module. The assessment procedures are described for each course of the module separately.

The overall grade of the module is the average of the grades for each course weighted by the credits and truncated after the first decimal.

Prerequisites
At least one of the courses "Public Management" or "Basics of German Company Tax Law and Tax Planning" is mandatory in the module and must be successfully examined.

Competence Goal
The student

- understands the theory and politics of taxation
- has knowledge in the area of public debt.
- understands efficiency problems of public organizations.
- is able to work on fiscal problems.

Content
As a branch of Economics, Public Finance is concerned with the theory and policy of the public sector and its interrelations with the private sector. It analyzes the economic role of the state from a normative as well as from a positive point of view. The normative view examines efficiency- and equity-oriented motives for government intervention and develops fiscal policy guidelines. The positive view explains the actual behavior of economic agents in public sector affairs.

In the course of the lectures within this module the students achieve knowledge in the areas of public revenues, national and international law of taxation and theory of public sector organizations.

Additional Information
The course T-WIWI-102790 "Specific Aspects in Taxation" will no longer be offered in the module as of winter semester 2018/2019.

Students who successfully passed the exam in „Public Management“ before the introduction of the module “Advanced Topics in Public Finance” in winter term 2014/15 are allowed to take both courses “Public Revenues” and “Specific Aspects in Taxation".

Workload
Total workload for 9 credit points: approx. 270 hours.

Attendance time: approx. 90 hours

Preparation and follow-up: approx. 135 hours

Exam and exam preparation: approx. 45 hours

The exact distribution is based on the credit points of the courses in the module.

Recommendations

Basic knowledge in the area of public finance and public management is required.

M 5.9 Module: Advanced Topics in Strategy and Management [M-WIWI-103119]

Coordinators: Prof. Dr. Hagen Lindstädt
Organisation: KIT Department of Business and Economics
Part of: Electives (Business Administration)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	2 terms	German	4	1

ID	Name	Term offered	Credits	Lecturers
Compulsory Elective Courses (Elective: 9 credits)				
T-WIWI-106188	Workshop Current Topics in Strategy and Management	Irreg.	3 CP	Lindstädt
T-WIWI-106189	Workshop Business Wargaming – Analyzing Strategic Interactions	Irreg.	3 CP	Lindstädt
T-WIWI-106190	Strategy and Management Theory: Developments and “Classics”	Irreg.	3 CP	Lindstädt

Assessment
The control of success takes place in the form of partial examinations (according to §4(2), 1-3 SPO) on the courses of the module, amounting to a total of 9 LP. The performance review is described for each course of this module. The overall grade of the module is formed from the LP-weighted grades of the partial examinations and truncated after the first decimal place.

Prerequisites
None

Competence Goal
After completing the module, students will be able to

- independently analyze strategic issues and derive recommendations in a structured manner using suitable models and reference frameworks of management theory.
- present their position convincingly in structured discussions using a well-thought-out line of argument.
- deal independently with a current, research-oriented question from strategic management.
- draw his/her own conclusions from the less structured information, drawing on his/her interdisciplinary knowledge and selectively developing current research findings.
- apply and discuss theoretical contents of management theory to real-life situations by intensively dealing with a large number of practice-relevant case studies.

Content
The content will focus on three areas. Firstly, strategic issues are discussed and analyzed on the basis of jointly selected case studies. Secondly, the students deal intensively with the topic of business wargaming in a workshop and analyze strategic interactions. Thirdly, topics of strategy and management theory are developed in a written paper.

Additional Information
The module is admission-restricted. Once you have been admitted to a course, you are guaranteed the opportunity to complete the module. The examinations are offered at least every second semester so that the entire module can be completed in two semesters.

Workload
Total effort for 9 credit points: approx. 270 hours. The exact distribution is done according to the credit points of the courses of the module. The workload for courses with 3 credits is approx. 90h.

M 5.10 Module: Analytics and Statistics [M-WIWI-101637]

Coordinators: Prof. Dr. Oliver Grothe
Organisation: KIT Department of Business and Economics
Part of: Electives (Statistics)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	2 terms	German	4	5

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-WIWI-103123	Advanced Statistics	WT	4,5 CP	Grothe
Supplementary Courses (Elective: between 4,5 and 5 credits)				
T-WIWI-114746	Selected Topics in Statistics and Analytics: Seminal Papers in Theory and Methods	Irreg.	4,5 CP	Grothe
T-WIWI-106341	Machine Learning 2 - Advanced Methods	ST	5 CP	Zöllner
T-WIWI-111247	Mathematics for High Dimensional Statistics	Irreg.	4,5 CP	Grothe
T-WIWI-103124	Multivariate Statistical Methods	Irreg.	4,5 CP	Grothe
T-WIWI-112109	Topics in Stochastic Optimization	WT	4,5 CP	Rebennack

Assessment
The assessment is carried out as partial written exams (according to Section 4(2), 1 of the examination regulation) of the single courses of this module, whose sum of credits must meet the minimum requirement of credits of this module. The examinations are offered every semester. Re-examinations are offered at every ordinary examination date. The assessment procedures are described for each course of the module seperately.

The overall grade of the module is the average of the grades for each course weighted by the credits and truncated after the first decimal.

Prerequisites
For all students, the course "Advanced Statistics" within this module is compulsory and must be taken.

Exception for Students of Economathematics: Students enrolled in the Economathematics degree program are exempt from the compulsory requirement to take the "Advanced Statistics" course. Instead, they are permitted to choose any two courses offered within this module. As the technical registration of this choice cannot be performed independently in the Campus system, students of Economathematics are kindly requested to contact the Examination Office of the Department of Economics and Management directly for correct module registration and exam enrollment.

Competence Goal
A Student

- Deepens the knowledge of descriptive and inferential statistics.
- Deals with simulation methods.
- Learns basic and advanced methods of statistical analysis of multivariate and high-dimensional data.

Content

- Deriving estimates and testing hypotheses
- Stochastic processes
- Multivariate statistics, copulas
- Dependence measures
- Dimension reduction
- High-dimensional methods
- Prediction

Additional Information
The planned lectures and courses for the next three years are announced online.

Workload
The total workload for this module is approximately 270 hours.

5.11 Module: Antennas and Beamforming [M-ETIT-106956]

Coordinators: Prof. Dr.-Ing. Marwan Younis
Prof. Dr.-Ing. Thomas Zwick

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: Engineering and Informatics (Electrical Engineering and Information Technology)
Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-113920	Antennas and Beamforming	WT	4 CP	Younis, Zwick

Assessment
The examination takes place in form of a written examination lasting 120 minutes.

Prerequisites
none

Competence Goal
After successfully participating in this course, students have in-depth knowledge of antennas, antenna systems and beamforming methods. This includes functionality, calculation methods as well as aspects of practical implementation. They are able to understand how typical electromagnetic radiators work and to develop and dimension them with specified properties. Students understand the principle and function of beamforming and the differences between digital, analog and hybrid beamforming. They know the theory, procedures and algorithms for beamforming. They can understand how beamforming is used for radio communication and radar.

Content
The lecture begins with a brief review of the basic knowledge of antennas and antenna arrays from the Bachelor's course. This is followed by a detailed discussion of all major antenna types (functionality, specifics). Furthermore, antenna measurement methods are presented. In the second part, the basic knowledge of noise, radio transmission and radar ambiguities is briefly refreshed, followed by a detailed presentation of the various beamforming algorithms, each with reference to radio communication and radar systems. Aspects such as digital and hybrid beamforming, as well as MIMO and equivalent virtual antenna configuration are explained.

The lecture will be accompanied by exercises. These are discussed in a room exercise and the corresponding solutions are presented in detail.

Module Grade Calculation
The module grade is the grade of the written exam.

Workload
The workload includes:

- Attendance study time lecture: 30 h
- Attendance study time exercise: 15 h
- Self-study time including exam preparation: 75 h

A total of 120 h

Recommendations
Knowledge of the basics of radio frequency technology and some basic knowledge on communication and radar systems is recommended.

M 5.12 Module: Applied Information Theory [M-ETIT-107333]

Coordinators: Dr.-Ing. Holger Jäkel
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Engineering and Informatics (Electrical Engineering and Information Technology)
Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-114412	Applied Information Theory	WT	6 CP	Jäkel

Assessment
The success control takes place in the form of an oral examination lasting approx. 25 minutes. Before the examination, there is a preparation phase of 15 minutes in which preparatory tasks are solved.

Prerequisites
none

Competence Goal
Students master the methods and concepts of information theory and are able to apply them to analyze communication technology issues.

Students acquire the ability to examine the information content of sources and the flow of information in systems and to evaluate their significance for the realization of telecommunications systems.

Content
The information theory founded by Shannon represents a central starting point for almost all questions of coding and encryption. In order to provide a solid foundation for later considerations, the concepts of information theory are developed at the beginning of the lecture. These are then applied to various areas of communications engineering and signal processing and used to analyze them.

Module Grade Calculation
The module grade is the grade of the oral examination.

Workload
1. attendance time lecture: 15 * 3 h = 45 h
2. preparation/follow-up lecture: 15 * 6 h = 90 h
3. attendance time exercise: 15 * 1 h = 15 h
4. preparation/post-exercise: 15 * 2 h = 30 h
5. exam preparation and attendance in the exam: included in preparation/follow-up work

Total: 180 h = 6 LP

Recommendations
Previous attendance of the lecture "Wahrscheinlichkeitstheorie" is recommended.

M 5.13 Module: Applied Strategic Decisions (WW4VWL2) [M-WIWI-101453]

Coordinators: Prof. Dr. Johannes Philipp Reiß
Organisation: KIT Department of Business and Economics
Part of: Business and Economics
Electives (Economics)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	1 term	English	4	7

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-WIWI-102861	Advanced Game Theory	WT	4,5 CP	Ehrhart, Puppe, Reiß
Supplementary Courses (Elective: between 4,5 and 5 credits)				
T-WIWI-113469	Advanced Corporate Finance	ST	4,5 CP	Ruckes
T-WIWI-102613	Auction Theory	WT	4,5 CP	Ehrhart
T-WIWI-115061	Experimental Research in Economics and Information Systems	WT	4,5 CP	Weinhardt
T-WIWI-102623	Financial Intermediation	WT	4,5 CP	Ruckes
T-WIWI-105781	Incentives in Organizations	ST	4,5 CP	Nieken
T-WIWI-112823	Platform & Market Engineering: Commerce, Media, and Digital Democracy	ST	4,5 CP	Weinhardt
T-WIWI-102862	Predictive Mechanism and Market Design	Irreg.	4,5 CP	Reiß

Assessment
The assessment is carried out as partial exams (according to Section 4 (2), 1-3 SPO) of the core course and further single courses of this module, whose sum of credits must meet the minimum requirement of credits of this module. The assessment procedures are described for each course of the module separately.

The overall grade of the module is the average of the grades for each course weighted by the credits and truncated after the first decimal.

Prerequisites
The course "Advanced Game Theory" is obligatory. Exception: The course "Introduction to Game Theory" was completed. Even those who have already successfully proven "Advanced Game Theory" in another master module can take the module. In this case you can choose freely from the rest of the offer. However, this choice can only be made by the examination office of the Department of Economics and Management.

Competence Goal
Students

- can model and analyze complex situations of strategic interaction using advanced game theoretic concepts;
- are provided with essential and advanced game theoretic solution concepts on a rigorous level and can apply them to understand real-life problems;
- learn about the experimental method, ranging from designing an economic experiment to data analysis.

Content
The module provides solid skills in game theory and offers a broad range of game theoretic applications. To improve the understanding of theoretical concepts, it pays attention to empirical evidence as well.

Additional Information
The course *Predictive Mechanism and Market Design* is not offered each year.

Workload
The total workload for this module is approximately 270 hours. The exact distribution is made according to the credit points of the courses of the module.

Recommendations
Basic knowledge in game theory is assumed.

5.14 Module: Artificial Intelligence [M-WIWI-105366]

Coordinators: Dr.-Ing. Tobias Käfer
Organisation: KIT Department of Business and Economics
Part of: Engineering and Informatics (Informatics)
Electives (Informatics)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	2 terms	German/English	4	2

ID	Name	Term offered	Credits	Lecturers
Compulsory Elective Courses (Elective: at least 2 items)				
T-WIWI-102666	Knowledge Discovery	WT	4,5 CP	Käfer
T-WIWI-110848	Semantic Web Technologies	ST	4,5 CP	Käfer
T-WIWI-110548	Advanced Lab Informatics (Master)	WT+ST	4,5 CP	Professorenschaft des Instituts AIFB

Assessment
The assessment mix of each course of this module is defined for each course separately. The final mark for the module is the average of the marks for each course weighted by the credits and truncated after the first decimal.

Prerequisites
None

Competence Goal
The student

- understands the koncepts behind Semantic Web and Linked Data technologies
- develops ontologies to be employed in semantic web-based applications and chooses suitable representation languages,
- is familiar with approaches in the area of knowledge representation and modelling,
- is able to transfer the methods and technologies of semantic web technologies to new application sectors,
- evaluates the potential of semantic web for new application sectors,
- understands the challenges in the areas of Data and system integration on the web is able to develop solutions.
- know the basics of machine learning, data mining and knowledge discovery
- can design, train and evaluate systems that are capable of learning
- carry out knowledge discovery projects, taking into account algorithms, representations and applications.

Content
The focus of the module is on Semantic Web Technologies as well as machine learning and data mining methods for knowledge acquisition from large databases.

The goal of the semantic web is the meaning (semantics) of data on the web for intelligent systems, e.g. in e-commerce and to make Internet portals usable. The representation of knowledge in the form of RDF and ontologies, the provision of data as Linked Data, as well as the request of data using SPARQL. In this lecture the basics of knowledge representation and processing for the corresponding technologies and application examples are presented.

The lecture "Knowledge Discovery" gives an overview of approaches of machine learning and data mining for knowledge extraction from large data sets. These are examined especially with regard to algorithms, applicability to different data representations and the use in real application scenarios.
Knowledge Discovery is an established research area with a large community that investigates methods for discovering patterns and regularities in large amounts of data, including unstructured text. A variety of methods exist to extract patterns and provide previously unknown insights. This information can be predictive or descriptive.
The lecture gives an overview of Knowledge Discovery. Specific techniques and methods, challenges and current and future research topics in this research area will be taught.
Contents of the lecture cover the entire machine learning and data mining process with topics on supervised and unsupervised learning and empirical evaluation. Covered learning methods range from classical approaches like decision trees, support vector machines and neural networks to selected approaches from current research. Learning problems considered include feature vector-based learning and text mining.

Workload
The total workload for this module is approximately 270 hours.

M 5.15 Module: Automated Manufacturing Systems (WW4INGMBWBK1) [M-MACH-101298]

Coordinators: Prof. Dr.-Ing. Jürgen Fleischer
Organisation: KIT Department of Mechanical Engineering

Part of: Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	1 term	German	4	3

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-MACH-114049	Basics of Production Automation	ST	5 CP	Fleischer
T-MACH-113999	Seminar Development of Automated Production Systems	WT	4 CP	Fleischer

Assessment
see individual courses

Prerequisites
none

Competence Goal
The students...

- are able to name and describe the automation tasks in production plants and the components required for implementation.
- can select components from the areas of ‘handling technique, ‘industrial robots’, ‘sensor technology’ and ‘control & drive technology’ for a given application.
- are able to compare different concepts of robotics and select the appropriate one for a given application.
- can assess the automation of production plants implemented on the basis of the examples given and apply them to new problems.
- know the process of planning automated production plants.
- can apply the theoretical knowledge they have acquired in practice.
- are able to develop a solution concept for an automation task, project it and implement the main features of it.

Content
The lecture provides an overview of the structure and functioning of automated production systems. The functioning and application of fundamental elements for the realisation of automated production systems are taught. These include:

- Drive and control technology
- Handling technology for handling workpieces and tools
- Industrial robots
- Quality assurance in automated production plants
- Structures and design of manufacturing and assembly systems
- Introduction to project planning of automated production plants

An interdisciplinary consideration of these sub-areas reveals interfaces to Industry 4.0 approaches. The theoretical foundations are supplemented by practical application examples and live demonstrations in the Karlsruhe Research Factory. Within exercises, the contents of the lecture are deepened and applied to specific problems and tasks.

The seminar Application of Production Automation focuses on the practical implementation of automation tasks. In small groups, solutions to automation problems are developed and basic functionalities are implemented in the Karlsruhe Research Factory. The results of this work are presented in a final presentation.

Workload
The work load is about 270 hours, corresponding to 9 credit points.

Teaching and Learning Methods
Lectures, exercise, excursion

M 5.16 Module: Automation and Control [M-ETIT-107673]

Coordinators: Prof. Dr.-Ing. Mike Barth
Prof. Dr.-Ing. Sören Hohmann
Dr.-Ing. Mathias Kluwe

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
10 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-114727	Multivariable Control Systems		6 CP	Hohmann
T-ETIT-112224	Digital Twin Engineering	WT	4 CP	Barth

Assessment
Multivariable Control Systems

The examination takes place in form of a written examination lasting 120 minutes.

Digital Twin Engineering

The examination takes place in form of other types of examination. It consists of a model library developed in the course of a semester-long project in the modeling language Modelica and a presentation of the library lasting 25 minutes. The quality of the model library is evaluated within the framework of the criteria: documentation, formal correctness, functionality, usability, HMI and modeling level of detail. The presentation is evaluated as an additional aspects. The overall impression is evaluated.

Prerequisites
none

Competence Goal
Multivariable Control Systems

- The students are able to describe linear multivariable systems in time and frequency domain with as well time continuous as time discrete models.
- In particular the students are familiar with the state space representation and its transformations in canonical forms.
- The students are able to analyze multivariable system with respect to stability, trajectory plots in state space, controllability, observability and zeros and poles.
- They master the control of linear multivariable systems in frequency domain by series decoupling with frequency control.
- The students are familiar with the basic control structures in time domain (pole placement with preliminary filter).
- The students are able to use different methods for direct pole placement in state space via state feedback like the Ackermann formula, modal synthesis and the complete modal synthesis.
- They are able to synthesize controllers for the decoupling control of multivariable systems in state space.
- The students are able to design time discrete controllers, in particular dead-beat control.
- They are familiar with the problem of determining non measurable state variables using complete and reduced state observers.
- As an extension, they are able to solve a generalized estimation problem in state space including noise by the design of Kalman filters.
- They can apply state feedback methods for the elimination of permanent deterministic disturbances like disturbance feedforward control, use of disturbance models or PI-State space controllers.
- The students can apply complete modal synthesis to design output feedback controllers in state space.
- They are able to extend the state feedback control with dynamical elements in order to create dynamical controllers.

Digital Twin Engineering

- The students will be able to analyze, structure and formally describe problems in the area of object-oriented physical system modeling.
- The students will be able to understand, apply and further develop the Modelica modeling language.
- The students are able to transfer bidirectionally acting systems into a model.
- The students are able to transfer physical equations into the modeling environment.
- The students are able to critically evaluate the different numerical integration methods for their applicability and to use them sensibly.
- The students are able to create system models and co-simulations using functional mockup units.
- The students will be able to implement a real system at the appropriate modeling depth for the task.
- The students will be able to abstract real system properties and, if necessary, decide whether they need to be modeled.
- The students know suitable simulation tools and their application.

Content
Multivariable Control Systems

The module is designed to provide students with basic knowledge and skills to analyze linear multivariable dynamic systems (described both in continuous and discrete time) and to design respective controllers. Furthermore the students are enabled to handle poor sensor information by state estimation methods or to eliminate disturbances and uncertainties, for example by dynamic state feedback.

Digital Twin Engineering

- This module is designed to provide students with the theoretical and practical aspects of object-theoretic equation-based modeling.
- This module also provides a definition of the digital twin and its aspects of the management shell.
    - In this context, a classification of simulation models in the I4.0 VWS takes place.
- Both system simulation in the Open Modelica Editor (OME) and co-simulation with Functional Mockup Units (FMU) will be covered.
- Students create a new model library of a mechatronic system in a semester-long project (teams of 3-4 students).
- The module provides an overview of modern system simulation methods based on bidirectional flow and potential modeling.
- Beyond theoretical and practical modeling, the module imparts the knowledge about practice-relevant modeling levels or depths.
- Furthermore, quality standards for simulation models with focus on the engineering of plants/systems are discussed.

Module Grade Calculation
The overall grade of the module is the average of the grades for each course weighted by the credits and truncated after the first decimal.

Workload
The workload includes:

1. attendance in lectures and exercises: 15*5 h = 75 h
2. preparation / follow-up: 15*6 h = 90 h
3. preparation of and attendance in examination: 60 h
4. implementation of the model library: 60 h
5. preparation of and attendance in the final presentation: 15 h

A total of 300 h = 10 CR

Recommendations
Multivariable Control Systems

Basic knowledge in the field of control theory is required (for example the contents of the bachelor module M-ETIT-106339 “Mess- und Regelungstechnik”).

M 5.17 Module: Automotive Engineering (WI4INGMB5) [M-MACH-101266]

Coordinators: Prof. Dr.-Ing. Marcus Geimer
Organisation: KIT Department of Mechanical Engineering

Part of: Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	1 term	German/English	4	8

ID	Name	Term offered	Credits	Lecturers
Automotive Engineering (Elective: at least 9 credits)				
T-MACH-102203	Automotive Engineering I	WT	6 CP	Gießler
T-MACH-112126	Data-Driven Algorithms in Vehicle Technology	WT	4,5 CP	Scheubner
T-MACH-100092	Automotive Engineering I (in German)	WT	8 CP	Gießler
T-MACH-102117	Automotive Engineering II (in German)	ST	4 CP	Gießler
T-MACH-102116	Fundamentals for Design of Motor-Vehicle Bodies I	WT	2 CP	Knoch
T-MACH-102119	Fundamentals for Design of Motor-Vehicle Bodies II	ST	2 CP	Knoch
T-MACH-110796	Python Algorithms for Vehicle Technology	ST	4 CP	Rhode
T-MACH-102156	Project Workshop: Automotive Engineering	WT+ST	6 CP	Frey, Gießler
T-MACH-111820	Control of Mobile Machines – Prerequisites	ST	0 CP	Becker, Geimer
T-MACH-111821	Control of Mobile Machines	ST	4 CP	Becker, Geimer
T-MACH-114435	Automotive Engineering II	ST	4,5 CP	Gießler
T-MACH-113912	Mathematical Methods in Hydraulics	WT+ST	4 CP	Geimer
T-MACH-113913	Tutorial Mathematical Methods in Hydraulics	WT	2 CP	Geimer

Assessment
The assessment is carried out as partial exams.

The partial exams consists of a written exam (duration 90 to 120 minutes) or an oral exam (duration 30 to 40 minutes).

Prerequisites
None

Competence Goal
The student

- knows the most important components of a vehicle,
- knows and understands the functioning and the interaction of the individual components,
- knows the basics of dimensioning the components.

Content
In the module Automotive Engineering the basics are taught, which are important for the development, the design, the production and the operation of vehicles. Particularly the primary important aggregates like engine, gear, drive train, chasis and auxiliary equipment are explained, but also all technical equipment, which make the operation safer and easier. Additionally the interior equipment is examined, which shall provide a preferably comfortable, optimum ambience to the user.

In the module Automotive Engineering the focus is on passenger cars and commercial vehicles, which are designed for road applications.

Workload
The total work load for this module is about 270 Hours (9 Credits). The partition of the work load is carried out according to the credit points of the courses of the module. The work load for courses with 6 credit points is about 180 hours, for courses with 4.5 credit points about 135 hours, for courses with 3 credit points about 90 hours, and for courses with 1.5 credit points about 45 hours. The total number of hours per course results from the time of visiting the lectures and exercises, as well as from the exam duration and the time that is required to achieve the objectives of the module as an average student with an average performance.

Recommendations
Knowledge of the content of the courses *Engineering Mechanics I* [2161238] and *Engineering Mechanics II* [1262276] is helpful.

Teaching and Learning Methods
The teaching and learning procedures (lecture, lab course, workshop) are described for each course of the module separately.

M 5.18 Module: Batteries, Fuel Cells, and Electrolysis [M-ETIT-107005]

Coordinators: Prof. Dr.-Ing. Ulrike Krewer
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Engineering and Informatics (Electrical Engineering and Information Technology)
Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each winter term	1 term	English	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-113986	Batteries, Fuel Cells, and Electrolysis	WT	5 CP	Krewer
T-ETIT-114957	Batteries, Fuel Cells, and Electrolysis - Group Project *This item will not influence the grade calculation of this parent.*	WT	1 CP	Krewer

Assessment
Success control takes place in the form of:

1. an ungraded written technical report (approx. 7-10 pages).
2. a graded written examination lasting 120 minutes.

Prerequisites
none

Competence Goal
Students gain an understanding of batteries, fuel cells and electrolysis including their application, design, and behavior. They acquire in-depth knowledge of the transport and charge transfer processes in them, their impact on performance and design, and the characteristics of the most frequent types of batteries, fuel and electrolysis cells. They understand how to analyze and characterize them using measurement methods and modeling. A practical insight into current areas of application and research topics of electrochemical energy storage and conversion allows them to relate the course work to demands of the society and for R&D. They are able to communicate with specialists from related disciplines in the field of (application of) batteries, fuel cells and electrolysis and can actively contribute to the opinion-forming process in society with regard to energy technology issues.

Content
The course introduces batteries, fuel cells and electrolysis and their use for sustainable mobile and stationary energy supply and storage. The course is divided into five sections. The first part covers the role of batteries, fuel cells and electrolysis for renewable energy storage and electrification of the energy system and the present applications. This is followed by a fundamentals part, where the processes in electrochemical cells at open circuit and during operation and their relation to cell performance and behavior are discussed. It contains thermodynamics, kinetics, transport and performance measures. The third part deals with the working principle, design and operation of fuel cells and electrolysis and the particularities of the different cell types. This is followed by a similar part for batteries. Finally, dynamic and stationary methods for characterizing the cells are covered.

Group project

As part of the coursework, student groups work on the design of a battery, fuel cell or electrolyser for a given application during the semester. This includes literature research on cell type, materials and material data as well as the dimensioning and energetic evaluation of the cell. The results are documented in a short technical report.

Module Grade Calculation
The module grade is the grade of the written examination.

Workload

1. Lecture attendance time: 15 * 2 h = 30 h
2. Preparation and follow-up time for lecture: 15 * 5 h = 75 h
3. Exercise attendance time: 7 * 2 h = 14 h
4. Preparation and follow-up time for exercise: 7 * 4 h = 28 h
5. Group work including writing of a report: 33 h
6. Exam preparation and attendance: included in preparation and follow-up time.

Total: 180 h = 6 CP

M 5.19 Module: Biofilm Systems [M-CIWVT-103441]

Coordinators: Dr. Andrea Hille-Reichel
Dr. Michael Wagner

Organisation: KIT Department of Chemical and Process Engineering

Part of: Engineering and Informatics (Chemical and Process Engineering)
Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each summer term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-CIWVT-106841	Biofilm Systems	ST	4 CP	Hille-Reichel, Wagner

Assessment
The learning control is an oral exam lasting approx. 20 minutes.

Prerequisites
None

Competence Goal
Students are able to describe the structure and function of biofilms in natural habitats and technical applications and explain the main influencing factors and processes for the formation of certain biofilms. They are familiar with methods for visualizing the structures.

Content
This lecture aims at providing an overview of biofilm systems, their development, functions, applications, and the techniques used to investigate them. Thus, topics involved will include basics of (biofilm) microbiology, natural (environmental) biofilm systems, their application in technical systems (reactors), and methods used to quantify biofilm development and performance (i.e., imaging techniques, digital image analysis).

Module Grade Calculation
Grande of the module is the grade of oral examination.

Workload
Attendance time: 30 h
Preparation/follow-up: 30 h
Examination + exam preparation: 60 h

5.20 Module: BioMEMS (WW4INGMBIMT1) [M-MACH-101290]

Coordinators: Prof. Dr. Jan Gerrit Korvink
Organisation: KIT Department of Mechanical Engineering

Part of: Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	1 term	German	4	3

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-MACH-100966	BioMEMS - Microsystems Technologies for Life-Sciences and Medicine I	WT	4 CP	Guber
BioMEMS (Elective: at least 6 credits)				
T-MACH-102164	Practical Training in Basics of Microsystem Technology	WT+ST	3 CP	Last
T-MACH-102165	Selected Topics on Optics and Microoptics for Mechanical Engineers	WT+ST	4 CP	Mappes
T-MACH-100967	BioMEMS - Microsystems Technologies for Life-Sciences and Medicine II	ST	4 CP	Guber
T-MACH-100968	BioMEMS - Microsystems Technologies for Life-Sciences and Medicine III	ST	4 CP	Guber
T-MACH-101910	Microactuators	ST	4 CP	Kohl
T-MACH-102176	Current Topics on BioMEMS	WT+ST	4 CP	Guber
T-MACH-111807	Introduction to Bionics	ST	4 CP	Hölscher
T-MACH-108312	Introduction to Microsystem Technology - Practical Course	WT+ST	3 CP	Last

Assessment
The assessment is carried out as partial exams

(according to Section 4 (2), 1-3 SPO) of the core course and further single courses of this module, whose sum of credits must meet the minimum requirement of credits of this module. The assessment procedures are described for each course of the module separately.

The overall grade of the module is the average of the grades for each course weighted by the credits and truncated after the first decimal.

Prerequisites
none

Competence Goal
The student

- has basic as well as extensive knowledge about different fields of applications of BioMEMS
- understands continuative aspects of the related subjects optics and microoptics, micro actuators, replications techniques and bionics

Content
Operations through small orifices, a pill which will take pictures on its way through your body or lab results right at the point of care - the need for easier and faster ways to help people is an important factor in research. The module BioMEMS (Bio(medical)-Micro-Electro-Mechanical-Systems) describes the application of microtechnology in the field of Life-Science, medical applications and Biotechnology and will teach you the necessary skills to understand and develop biological and medical devices.

The BioMEMS lectures will cover the fields of minimal invasive surgery, lab-on-chip systems, NOTES-Technology (Natural Orifice Transluminal Endoscopic Surgery), as well as endoscopic surgery and stent technology.

Additionally to the BioMEMS lectures you can specialize in various related fields like fabrication, actuation, optics and bionics. The course Replication processes will teach you some cost efficient and fast ways to produce parts for medical or biological devices. In the course Microactuation it is discussed how to receive movements in micrometer scale in a microsystem, this could be e.g. to drive micro pumps or micro valves. The necessary tools for optical measurement and methods of analysis to gain high resolution pictures are also part of this module. To deepen your knowledge and to get a hands-on experience this module contains a one week lab course. In the lecture bionics you can see how biological effects can be transferred into technical products.

Workload
270 hours

Teaching and Learning Methods
Lectures

5.21 Module: Bioprocess Scale-up [M-CIWVT-106837]

Coordinators: Prof. Dr.-Ing. Alexander Grünberger
Organisation: KIT Department of Chemical and Process Engineering
Part of: Engineering and Informatics (Chemical and Process Engineering)
Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each winter term	1 term	English	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-CIWVT-113712	Bioprocess Scale-up	WT	6 CP	Grünberger

Assessment
Learning control is an oral exam lasting approx. 30 minutes.

Competence Goal
Upon completion of the course, students will be able to:
Subject-Specific and Methodological Competencies

- Understand the fundamentals of scaling laws.
- Demonstrate knowledge of key scale-up strategies.
- Apply essential knowledge and toolsets required for the scale-up of bioprocesses.
- Recognize potential pitfalls and challenges during the scale-up process.
- Identify and implement best practices for the scale-up of bioprocesses.
- Bridge the gap between laboratory research and industrial production.

Social and Self-Competence

- Identify and summarize the key elements involved in bioprocess scale-up.
- Communicate effectively and collaborate with experts from various disciplines involved in bioprocess scale-up.
- Demonstrate critical thinking, creativity, and problem-solving skills necessary for scaling up novel bioprocesses.

Content
Biopharmaceuticals, enzymes, and biological materials used in food supplements are commonly produced through the cultivation of bacteria, yeast, fungi, plant, or animal cells in bioreactors. Regardless of the specific bioprocess, efficiency in terms of time, cost, and resource utilization is essential. Typically, these bioprocesses are developed initially at a small laboratory scale and then progressively transferred to larger volumes until reaching commercial industrial production. This critical transition is known as the scale-up of bioprocesses.

The objective of this course is to provide students with the fundamental knowledge and practical skills required to successfully scale-up biotechnological processes from laboratory to industrial scale. To achieve this, the course introduces key methods, concepts, and tools that form the foundation for effective scale-up of biochemical processes.

The course begins with an introduction to scaling laws, which are essential for understanding how process parameters change with scale. Examples from biology will be given. Following this, general scale-up methods are presented that enable transferring processes while maintaining performance and product quality. Industrial strategies and procedures are then discussed, supported by real-world examples and case studies. Finally, emerging trends and challenges in bioprocess scale-up are explored, highlighting innovative technologies and addressing future obstacles in the field. Through a combination of theoretical concepts, practical examples, and real-world case studies, this lecture aims to equip participants with the ability to develop and implement suitable scale-up strategies.

Module Grade Calculation
The module grade is the grade of the oral exam.

Workload

- Lectures and exercises: 45 hrs
- Homework: 95 hrs
- Exam preparation: 40 hrs

Recommendations
Fundamentals of Bioprocess Engineering.

Literature
No specific textbook is recommended.

M 5.22 Module: Biosensors [M-CIWVT-106838]

Coordinators: Dr. Gözde Kabay
Organisation: KIT Department of Chemical and Process Engineering
Part of: Engineering and Informatics (Chemical and Process Engineering)
Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-CIWVT-113714	Biosensors	WT+ST	4 CP	Kabay

M 5.23 Module: Business & Service Engineering (WW4BWLISM4) [M-WIWI-101410]

Coordinators: Prof. Dr. Christof Weinhardt
Organisation: KIT Department of Business and Economics
Part of: Electives (Business Administration)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	1 term	German/English	4	9

ID	Name	Term offered	Credits	Lecturers
Compulsory Elective Courses (Elective: 9 credits)				
T-WIWI-113160	Digital Democracy	WT	4,5 CP	Fegert
T-WIWI-112757	Digital Services: Innovation & Business Models	ST	4,5 CP	Satzger
T-WIWI-110887	Practical Seminar: Service Innovation	Irreg.	4,5 CP	Satzger
T-WIWI-102847	Recommender Systems	WT	4,5 CP	Geyer-Schulz
T-WIWI-113724	Special Topics in Information Systems	WT+ST	4,5 CP	Weinhardt

Assessment
The assessment is carried out as partial exams (according to Section 4 (2), 1-3 SPO) of the single courses of this module, whose sum of credits must meet the minimum requirement of credits of this module. The assessment procedures are described for each course of the module separately.

The overall grade of the module is the average of the grades for each course weighted by the credits and truncated after the first decimal.

Prerequisites
None

Competence Goal
The student should

- learn to develop and implement new markets with regards to the technological progresses of information and communication technology and the increasing economic networking
- learn to restructure and develop new business processes in markets under those conditions
- understand service competition as a sustainable competitive strategy and understand the effects of service competition on the design of markets, products, processes and services.
- improve his statistics skills and apply them to appropriate cases
- learn to elaborate solutions in a team

Content
This module addresses the challenges of creating new kinds of products, processes, services, and markets from a service perspective in the context of new developed information and communication technologies and the globalization process. The module describes service competition as a business strategy in the long term that leads to the design of business processes, business models, forms of organization, markets, and competition. This will be shown by actual examples from personalized services, recommender services and social networks.

Additional Information
All practical Seminars offered at the IM can be chosen for *Special Topics in Information Systems*. Please update yourself on www.iism.kit.edu/im/lehre .

From summer semester 2023, the course Service Innovation will be offered with a revised course concept and content. The focus will be on the closer integration of the topics of service innovation and digitalization. Current foundational content (e.g., on service innovation challenges or human-centered innovation methods) will remain. New content will cover topics such as digital platforms and ecosystems, IoT and smart service innovation, and business models.

Workload
The total workload for this module is approx. 270 hours (9 credits). The distribution is based on the credit points of the courses in the module. The workload for courses with 4.5 credits is approx. 135 hours, for courses with 5 credits approx. 150 hours.

The total number of hours per course is calculated from the time required to attend the lectures and exercises, as well as the examination times and the time required for an average student to achieve the learning objectives of the module for an average performance.

Recommendations
None

M 5.24 Module: Business Administration [M-WIWI-104900]

Coordinators: Professorenschaft des Fachbereichs Betriebswirtschaftslehre
Organisation: KIT Department of Business and Economics
Part of: Additional Examinations

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	pass/fail	Each term	1 term	German	4	15

ID	Name	Term offered	Credits	Lecturers
Elective Courses on Business Administration (Elective: at most 60 credits)				
T-WIWI-114749	(Gen)AI and Virtual Reality for Business	Irreg.	4,5 CP	Pfeiffer
T-WIWI-114210	(Gen)AI-based Automation in Organizations	ST	4,5 CP	Mädche
T-WIWI-113469	Advanced Corporate Finance	ST	4,5 CP	Ruckes
T-WIWI-109921	Advanced Machine Learning	ST	4,5 CP	Geyer-Schulz, Nazemi
T-WIWI-111305	Advanced Machine Learning and Data Science	WT+ST	9 CP	Ulrich
T-WIWI-102885	Advanced Management Accounting	WT	4,5 CP	Wouters
T-WIWI-114979	Advanced Quantitative Finance	ST	4,5 CP	Thimme
T-WIWI-111912	Advanced Topics in Digital Management	ST	3 CP	Nieken
T-WIWI-111913	Advanced Topics in Human Resource Management	Irreg.	3 CP	Nieken
T-WIWI-109819	Current Research Topics in Business Information Systems	see notes	1 CP	Mädche
T-WIWI-102596	Analytical CRM	ST	4,5 CP	Geyer-Schulz
T-WIWI-108715	Artificial Intelligence in Service Systems	WT	4,5 CP	Satzger
T-WIWI-114209	Artificial Intelligence in Service Systems II: Generative AI Applications & Adoption	ST	4,5 CP	Satzger
T-WIWI-102647	Asset Pricing	ST	4,5 CP	Ruckes, Uhrig-Homburg
T-WIWI-102613	Auction Theory	WT	4,5 CP	Ehrhart
T-WIWI-106495	Automated Financial Advisory	ST	3 CP	Ulrich
T-WIWI-111367	B2B Sales Management	WT	4,5 CP	Klarmann
T-WIWI-102742	Design, Construction and Sustainability Assessment of Buildings I	WT	4,5 CP	Lützkendorf
T-WIWI-102743	Design, Construction and Sustainability Assessment of Buildings II	ST	4,5 CP	Lützkendorf
T-WIWI-113471	Bayesian Statistics for Analyzing Data	Irreg.	4,5 CP	Scheibehenne
T-WIWI-114980	Behavioral Dimensions of Energy Transitions	ST	3,5 CP	Fichtner
T-WIWI-111393	Behavioral Experiments in Action	ST	4,5 CP	Scheibehenne
T-WIWI-113095	Behavioral Lab Exercise	Irreg.	4,5 CP	Nieken, Scheibehenne
T-WIWI-102819	Business Administration: Finance and Accounting	WT	4 CP	Ruckes, Uhrig-Homburg, Wouters
T-WIWI-102818	Business Administration: Production Economics and Marketing	ST	4 CP	Fichtner, Klarmann, Lützkendorf, Ruckes, Schultmann
T-WIWI-102817	Business Administration: Strategic Management and Information Engineering and Management	WT	3 CP	Nieken, Ruckes
T-WIWI-110995	Bond Markets	WT	4,5 CP	Uhrig-Homburg
T-WIWI-110997	Bond Markets - Models & Derivatives	WT	3 CP	Uhrig-Homburg
T-WIWI-110996	Bond Markets - Tools & Applications	WT	1,5 CP	Uhrig-Homburg
T-WIWI-112156	Brand Management	WT	4,5 CP	Kupfer
T-WIWI-106187	Business Data Strategy	WT	4,5 CP	Weinhardt
T-WIWI-102762	Business Dynamics	WT	4,5 CP	Geyer-Schulz, Glenn

ID	Name	Term offered	Credits	Lecturers
T-WIWI-105777	Business Intelligence Systems	WT	4,5 CP	Mädche
T-WIWI-114057	Circular Economy – Challenges and Potentials	ST	3,5 CP	Schultmann
T-WIWI-111392	Cognitive Modeling	Irreg.	4,5 CP	Scheibehenne
T-WIWI-114186	Collective Intelligence in Human Judgment and Decision Making	ST	4,5 CP	Scheibehenne
T-WIWI-106496	Computational FinTech with Python and C++	ST	1,5 CP	Ulrich
T-WIWI-114185	Computational Modelling of Judgments, Decisions and Cognition	ST	4,5 CP	Scheibehenne
T-WIWI-114292	Consumer Psychology	ST	4,5 CP	Scheibehenne
T-WIWI-109050	Corporate Risk Management	Irreg.	4,5 CP	Ruckes
T-WIWI-111100	Current Directions in Consumer Psychology	WT+ST	4,5 CP	Scheibehenne
T-WIWI-102595	Customer Relationship Management	WT	4,5 CP	Geyer-Schulz
T-WIWI-114089	Data Science for Business	ST	4,5 CP	Pfeiffer
T-WIWI-102643	Derivatives	ST	4,5 CP	Uhrig-Homburg
T-WIWI-114618	Design and Operation of Industrial Plants and Processes	WT	5,5 CP	Schultmann
T-WIWI-102866	Design Thinking	Irreg.	3 CP	Terzidis
T-WIWI-113465	Designing Interactive Systems: Human-AI Interaction	ST	4,5 CP	Mädche
T-WIWI-102806	Services Marketing and B2B Marketing	see notes	3 CP	Feurer
T-WIWI-113160	Digital Democracy	WT	4,5 CP	Fegert
T-WIWI-112693	Digital Marketing	ST	4,5 CP	Kupfer
T-WIWI-106981	Digital Marketing and Sales in B2B	ST	1,5 CP	Klarmann, Konhäuser
T-WIWI-112228	Digital Markets and Market Design	WT	4,5 CP	Hillenbrand
T-WIWI-111307	Digital Services: Foundations	ST	4,5 CP	Förster, Holtmann
T-WIWI-112757	Digital Services: Innovation & Business Models	ST	4,5 CP	Satzger
T-WIWI-114174	Economic Decision Making	WT	4,5 CP	Scheibehenne
T-WIWI-112822	Economics of Innovation	ST	4,5 CP	Ott
T-WIWI-102793	Efficient Energy Systems and Electric Mobility	ST	3,5 CP	Jochem
T-WIWI-110797	eFinance: Information Systems for Securities Trading	WT	4,5 CP	Weinhardt
T-WIWI-102746	Introduction to Energy Economics	ST	5,5 CP	Fichtner
T-WIWI-102634	Emissions into the Environment	WT	3,5 CP	Karl
T-WIWI-102650	Energy and Environment	ST	3,5 CP	Karl
T-WIWI-102607	Energy Policy	ST	3,5 CP	Wietschel
T-WIWI-107501	Energy Market Engineering	ST	4,5 CP	Weinhardt
T-WIWI-107503	Energy Networks and Regulation	WT	4,5 CP	Weinhardt
T-WIWI-112151	Energy Trading and Risk Management	ST	3,5 CP	N.N.
T-WIWI-106193	Engineering FinTech Solutions	WT+ST	9 CP	Ulrich
T-WIWI-113460	Engineering Interactive Systems: AI & Wearables	WT	4,5 CP	Mädche
T-WIWI-102608	Enterprise Risk Management	WT	4,5 CP	Werner
T-WIWI-113746	Enterprise Systems for Financial Accounting & Controlling	WT	4,5 CP	Fleig, Mädche
T-WIWI-102864	Entrepreneurship	WT+ST	3 CP	Terzidis
T-WIWI-114637	Entrepreneurship Basics	WT	4,5 CP	Terzidis
T-WIWI-113151	Entrepreneurship Seasonal School	Irreg.	3 CP	Terzidis
T-WIWI-102894	Entrepreneurship Research	ST	3 CP	Terzidis
T-WIWI-113663	Development of Sustainable, Digital Business Models	Irreg.	3 CP	Weissenberger-Eibl
T-WIWI-115061	Experimental Research in Economics and Information Systems	WT	4,5 CP	Weinhardt
T-WIWI-111395	Experimental Design	WT+ST	4,5 CP	Scheibehenne
T-WIWI-102852	Case Studies Seminar: Innovation Management	WT	3 CP	Weissenberger-Eibl
T-WIWI-107505	Financial Accounting for Global Firms	see notes	4,5 CP	Luedecke

ID	Name	Term offered	Credits	Lecturers
T-WIWI-102900	Financial Analysis	see notes	4,5 CP	Luedecke
T-WIWI-111238	Financial Data Science	ST	9 CP	Ulrich
T-WIWI-114633	Financial Data Science in Practice	WT+ST	9 CP	Ulrich
T-WIWI-102605	Financial Management	ST	4,5 CP	Ruckes
T-WIWI-102623	Financial Intermediation	WT	4,5 CP	Ruckes
T-WIWI-112694	FinTech	WT	4,5 CP	Thimme
T-WIWI-109816	Foundations of Interactive Systems	ST	4,5 CP	Mädche
T-WIWI-102865	Business Planning	Irreg.	3 CP	Terzidis
T-WIWI-112103	Global Manufacturing	WT	3,5 CP	Sasse
T-WIWI-102606	Fundamentals of Production Management	ST	5,5 CP	Schultmann
T-WIWI-108711	Basics of German Company Tax Law and Tax Planning	WT	4,5 CP	Gutekunst, Wigger
T-WIWI-112820	Introduction to Finance and Accounting	ST	5 CP	Luedecke, Ruckes, Wouters
T-WIWI-113745	HR-Management 1: HR Strategies in the Age of AI	WT	4,5 CP	Nieken
T-WIWI-114178	HR-Management 2: Organization, Fairness & Leadership	ST	4,5 CP	Nieken
T-WIWI-102838	Real Estate Economics and Sustainability Part 1: Basics and Valuation	WT	4,5 CP	Lorenz
T-WIWI-102839	Real Estate Economics and Sustainability Part 2: Reporting and Rating	ST	4,5 CP	Lorenz
T-WIWI-102893	Innovation Management: Concepts, Strategies and Methods	ST	3 CP	Weissenberger-Eibl
T-WIWI-102636	Insurance Risk Management	ST	2,5 CP	Maser
T-WIWI-111267	Intelligent Agent Architectures	WT	4,5 CP	Geyer-Schulz
T-WIWI-110915	Intelligent Agents and Decision Theory	ST	4,5 CP	Geyer-Schulz
T-WIWI-114893	International Entrepreneurship and Brand Development	see notes	6 CP	Kupfer, Terzidis
T-WIWI-102807	International Marketing	see notes	1,5 CP	Feurer
T-WIWI-102646	International Finance	see notes	3 CP	Uhrig-Homburg
T-WIWI-111028	Introduction to Machine Learning	WT	4,5 CP	Geyer-Schulz, Nazemi
T-WIWI-111029	Introduction to Neural Networks and Genetic Algorithms	ST	4,5 CP	Geyer-Schulz
T-WIWI-102604	Investments	ST	4,5 CP	Uhrig-Homburg
T-WIWI-109064	Joint Entrepreneurship Summer School	Irreg.	6 CP	Terzidis
T-WIWI-111109	KD²Lab Hands-On Research Course: New Ways and Tools in Experimental Economics	Irreg.	4,5 CP	Weinhardt
T-WIWI-107043	Liberalised Power Markets	WT	5,5 CP	Fichtner
T-WIWI-113107	Life Cycle Assessment – Basics and Application Possibilities in an Industrial Context	WT	3,5 CP	Schultmann
T-WIWI-102870	Logistics and Supply Chain Management	ST	3,5 CP	Schultmann
T-WIWI-113073	Machine Learning and Optimization in Energy Systems	WT	4 CP	Fichtner
T-WIWI-102800	Management Accounting 1	see notes	4,5 CP	Wouters
T-WIWI-102801	Management Accounting 2	see notes	4,5 CP	Wouters
T-WIWI-111594	Management and Marketing	WT	5 CP	Klarmann, Lindstädt, Nieken, Terzidis
T-WIWI-107720	Market Research	ST	4,5 CP	Klarmann
T-WIWI-103139	Marketing Analytics	WT	4,5 CP	Klarmann
T-WIWI-102805	Managing the Marketing Mix	ST	4,5 CP	Klarmann
T-WIWI-112711	Media Management	WT	4,5 CP	Kupfer
T-WIWI-113414	Modeling the Dynamics of Financial Markets	ST	9 CP	Ulrich
T-WIWI-111848	Online Concepts for Karlsruhe City Retailers	ST	3 CP	Kupfer

ID	Name	Term offered	Credits	Lecturers
T-WIWI-102597	Operative CRM	WT	4,5 CP	Geyer-Schulz
T-WIWI-102630	Managing Organizations	WT	4,5 CP	Lindstädt
T-WIWI-114750	Perceiving, Understanding and Communicating Risk and Uncertainty	WT	4,5 CP	Gradwohl, Scheibehenne
T-WIWI-114184	Pioneering Leadership in German SMEs	Irreg.	3 CP	Weissenberger-Eibl
T-WIWI-108016	Simulation Game in Energy Economics	ST	3,5 CP	Genoese
T-WIWI-112823	Platform & Market Engineering: Commerce, Media, and Digital Democracy	ST	4,5 CP	Weinhardt
T-WIWI-107506	Platform Economy	WT	4,5 CP	Weinhardt
T-WIWI-109935	Practical Seminar Interaction	WT+ST	4,5 CP	Mädche, Weinhardt
T-WIWI-109937	Practical Seminar Platforms	WT+ST	4,5 CP	Satzger, Weinhardt
T-WIWI-109939	Practical Seminar Servitization	WT+ST	4,5 CP	Mädche, Satzger
T-WIWI-114892	Practical Seminar: AI & Data Science	Irreg.	4,5 CP	Mädche
T-WIWI-112152	Practical Seminar: Artificial Intelligence in Service Systems	Irreg.	4,5 CP	Satzger
T-WIWI-110888	Practical Seminar: Digital Services	ST	4,5 CP	Satzger
T-WIWI-113459	Practical Seminar: Human-Centered Systems	WT+ST	4,5 CP	Mädche
T-WIWI-111914	Practical Seminar: Interactive Systems	WT+ST	4,5 CP	Mädche
T-WIWI-112154	Practical Seminar: Platform Economy	WT+ST	4,5 CP	Weinhardt
T-WIWI-110887	Practical Seminar: Service Innovation	Irreg.	4,5 CP	Satzger
T-WIWI-105946	Price Management	ST	4,5 CP	Geyer-Schulz, Glenn
T-WIWI-102883	Pricing	WT	4,5 CP	Klarmann
T-WIWI-102603	Principles of Insurance Management	ST	4,5 CP	Werner
T-WIWI-102871	Problem Solving, Communication and Leadership	ST	2 CP	Lindstädt
T-WIWI-111632	Production and Logistics	WT	3 CP	Fichtner, Nickel, Schultmann
T-WIWI-102820	Production Economics and Sustainability	WT	3,5 CP	Schultmann
T-WIWI-111602	Production, Logistics and Information Systems	WT	5 CP	Fichtner, Geyer-Schulz, Mädche, Nickel, Schultmann, Weinhardt
T-WIWI-102632	Production and Logistics Management	ST	5,5 CP	Schultmann
T-WIWI-103134	Project Management	WT	3,5 CP	Schultmann
T-WIWI-114340	Quantitative Finance	WT	4,5 CP	Thimme
T-WIWI-107446	Quantitative Methods in Energy Economics	WT	3,5 CP	Plötz
T-WIWI-102744	Real Estate Management I	WT	4,5 CP	Lützkendorf
T-WIWI-102745	Real Estate Management II	ST	4,5 CP	Lützkendorf
T-WIWI-102816	Financial Accounting and Cost Accounting	WT	4 CP	Strych
T-WIWI-102847	Recommender Systems	WT	4,5 CP	Geyer-Schulz
T-WIWI-100806	Renewable Energy-Resources, Technologies and Economics	WT	3,5 CP	Jochem
T-WIWI-111385	Responsible Artificial Intelligence	WT	4,5 CP	Pfeiffer
T-WIWI-102649	Risk Communication	WT	4,5 CP	Werner
T-WIWI-102826	Risk Management in Industrial Supply Networks	WT	3,5 CP	Schultmann
T-WIWI-103486	Seminar in Business Administration (Bachelor)	WT+ST	3 CP	Professorenschaft des Fachbereichs Betriebswirtschaftslehre
T-WIWI-103474	Seminar in Business Administration A (Master)	WT+ST	3 CP	Professorenschaft des Fachbereichs Betriebswirtschaftslehre
T-WIWI-103476	Seminar in Business Administration B (Master)	WT+ST	3 CP	Professorenschaft des Fachbereichs Betriebswirtschaftslehre
T-WIWI-110987	Seminar Methods along the Innovation process	see notes	3 CP	Beyer

ID	Name	Term offered	Credits	Lecturers
T-WIWI-106563	Practical Seminar Digital Service Systems	Irreg.	4,5 CP	Satzger
T-WIWI-108765	Practical Seminar: Advanced Analytics	WT+ST	4,5 CP	Weinhardt
T-WIWI-106207	Practical Seminar: Data-Driven Information Systems	Irreg.	4,5 CP	Satzger, Weinhardt
T-WIWI-102849	Service Design Thinking	Irreg.	9 CP	Satzger, Terzidis
T-WIWI-114635	SIL-LaunchLab	ST	9 CP	Terzidis
T-WIWI-107464	Smart Energy Infrastructure	WT	5,5 CP	Ardone, Pustisek
T-WIWI-107504	Smart Grid Applications	see notes	4,5 CP	Weinhardt
T-WIWI-109940	Special Topics in Information Systems	WT+ST	4,5 CP	Weinhardt
T-WIWI-113723	Special Topics in Information Systems	WT+ST	4,5 CP	Weinhardt
T-WIWI-113724	Special Topics in Information Systems	WT+ST	4,5 CP	Weinhardt
T-WIWI-113726	Special Topics in Information Systems	WT+ST	4,5 CP	Weinhardt
T-WIWI-113725	Special Topics in Information Systems	WT+ST	4,5 CP	Weinhardt
T-WIWI-111561	Startup Experience	WT+ST	6 CP	Terzidis
T-WIWI-106190	Strategy and Management Theory: Developments and “Classics”	Irreg.	3 CP	Lindstädt
T-WIWI-113090	Strategic Management	ST	4,5 CP	Lindstädt
T-WIWI-114636	Student2Startup	WT	4,5 CP	Terzidis
T-WIWI-102763	Supply Chain Management with Advanced Planning Systems	ST	3,5 CP	Bosch, Göbelt
T-WIWI-114995	Sustainable Information Systems	ST	4,5 CP	Staudt
T-WIWI-110968	Team Project Management and Technology	WT+ST	9 CP	Klarmann, Mädche
T-WIWI-110977	Team Project Management and Technology (BUS/ENG)	WT+ST	9 CP	Klarmann, Mädche
T-WIWI-102621	Valuation	WT	4,5 CP	Ruckes
T-WIWI-102695	Heat Economy	ST	3,5 CP	Fichtner
T-WIWI-114748	Science Communication in the Innovation Process	Irreg.	3 CP	Schwind
T-WIWI-106188	Workshop Current Topics in Strategy and Management	Irreg.	3 CP	Lindstädt
T-WIWI-106189	Workshop Business Wargaming – Analyzing Strategic Interactions	Irreg.	3 CP	Lindstädt

M 5.25 Module: Chemical Hydrogen Storage [M-CIWVT-106566]

Coordinators: TT-Prof. Dr. Moritz Wolf
Organisation: KIT Department of Chemical and Process Engineering
Part of: Engineering and Informatics (Chemical and Process Engineering)
Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-CIWVT-113234	Chemical Hydrogen Storage	WT	4 CP	Wolf

Assessment
The learning control is an oral exmaination lasting approx. 20 minutes.

Prerequisites
None

Competence Goal
The students are able to explain basic properties of hydrogen and hydrogen carriers, know the production methods of green hydrogen and can assess its role in the context of the energy transition, especially with regard to industrial use as feedstock. They understand sustainable and emerging technologies for chemical hydrogen storage, can describe the catalysts required for the various processes and know special associated challenges. The students can evaluate different chemical, but also physical storage technologies, assess the costs of individual process steps and describe the corresponding potential areas of application.

Content

- Introduction to various concepts of (chemical) hydrogen storage
    - Storage technologies
    - Carrier molecules
    - Storage cycles
- Processes and catalysts for chemical hydrogen storage technologies
    - Ammonia
    - Liquid organic hydrogen carriers (LOHCs)
    - Dimethylether
- Evaluation of storage processes in comparison with liquid hydrogen
    - Sustainability
    - Costs of production
    - Costs of transportation
    - Costs of hydrogen application

Module Grade Calculation
The module grade is the grade of the oral exam.

Workload

- Attendance time: 40 hrs
- Self-study: 40 hrs
- Exam preparation: 40 hrs

Literature
Announced in lectures/on slides.

Fundamentals:

- I. Chorkendorff, J. W. Niemantsverdriet, *Concepts of Modern Catalysis and Kinetics*, 2003, Wiley.
- R. Schlögl, *Chemical Energy Storage*, 2022, De Gruyter

M 5.26 Module: Circular Factory [M-MACH-107636]

Coordinators: Prof. Dr.-Ing. Gisela Lanza
Organisation: KIT Department of Mechanical Engineering

Part of: Engineering and Informatics (Mechanical Engineering)
Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
8 CP	graded	Each summer term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-MACH-114974	Circular Factory	ST	8 CP	Lanza

Assessment
written exam, duration 120 minutes

Prerequisites
none

Competence Goal
The students …

- can name dimensions of circularity and circular economy methods (e.g., repair, refurbish, recycle) and describe them in detail.
- can describe challenges in planning and control circular factories, including remanufacturing networks and disassembly systems.
- are able to apply guidelines for designing circular products.
- distinguish data acquisition techniques for metrologically assessing returned products and apply uncertainty-driven product modeling in circular production systems.
- have methodical knowledge on learning from human observation and disassembly automatization and apply this knowledge to new problem cases.
- can describe reprocessing methods, including reconditioning and material characterization.
- understand the challenges in intralogistics for circular products.

After completing this course, students are able to understand the challenges of establishing a circular economy. They are also able to evaluate possible solutions and assess them in relation to these challenges. In particular, students are ultimately able to understand circular production in a circular factory holistically and to relate it to existing concepts in industrial practice.

Content

The module provides a comprehensive overview of the principles, methods, and technologies of circular production and the circular economy within industrial engineering. It conveys a well-founded and practice-oriented understanding of how circularity can be systematically implemented across products, processes, and factories. At the beginning of the module, fundamental concepts of sustainability, circular economy, and circularity dimensions are introduced, including strategies such as repair, refurbishment, remanufacturing, and recycling.

Building on these fundamentals, the module addresses the planning and control of circular factories, with particular emphasis on remanufacturing networks, disassembly systems, and reprocessing operations. Key challenges related to uncertainty, product variety, and quality variability of returned products are discussed. Methods for data acquisition and metrological assessment of end-of-life and returned products are presented, alongside uncertainty-driven product modeling approaches for circular production systems.

Further content focuses on the design of circular products, including design guidelines that enable reuse, disassembly, and material recovery. In addition, the module covers learning from human observation and approaches to disassembly automation, enabling students to transfer methodical knowledge to new and unfamiliar problem cases. Reprocessing methods such as reconditioning and material characterization are systematically explained, and challenges in intralogistics for circular products are analyzed.

Based on examples from current research and industrial practice, the module highlights the holistic interaction between product design, production systems, logistics, and information flows in circular factories. By integrating these perspectives, students develop a comprehensive understanding of circular production and are able to evaluate solution approaches in relation to technical, economic, and organizational challenges encountered in industrial applications.

Module Grade Calculation

The module grade corresponds to the grade of the written examination.

Workload

regular attendance: 62 hours
self-study: 178 hours
In total: 240 h

Teaching and Learning Methods

Lecture and excursions

M 5.27 Module: Collective Decision Making (WW4VWL16) [M-WIWI-101504]

Coordinators: Prof. Dr. Clemens Puppe
Organisation: KIT Department of Business and Economics
Part of: Business and Economics
Electives (Economics)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	1 term	English	4	4

ID	Name	Term offered	Credits	Lecturers
Compulsory Elective Courses (Elective:)				
T-WIWI-102740	Public Management	WT	4,5 CP	Wigger
T-WIWI-102859	Social Choice Theory	ST	4,5 CP	Puppe

Assessment

The assessment is carried out as partial exams (according to Section 4(2), 1 or 2 of the examination regulation) of the single courses of this module, whose sum of credits must meet the minimum requirement of credits of this module. The assessment procedures are described for each course of the module seperately.

The overall grade of the the module is the average of the grades for each course weighted by the credits and truncated after the first decimal.

Prerequisites

None

Competence Goal

Students

- are able to model and assess problems in public economics and to analyze them with respect to positive and normative aspects,
- understand individual incentives and social outcomes of different institutional designs,
- are familiar with the functioning and design of democratic elections and can analyze them with respect to their individual incentives.

Content

The focus of the module is on mechanisms for public decision making including voting and the aggregation of preferences and judgements.

Workload

Total workload for 9 credit points: approx. 270 hours

The exact distribution is based on the credit points of the courses in the module.

M 5.28 Module: Combustion Engines I (WW4INGMB34) [M-MACH-101275]

Coordinators: Prof. Dr. Thomas Koch
Dr.-Ing. Heiko Kubach

Organisation: KIT Department of Mechanical Engineering

Part of: Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Level	Version
8 CP	graded	Each winter term	1 term	4	5

ID	Name	Term offered	Credits	Lecturers
Wahlpflicht (Elective: between 1 and 2 items)				
T-MACH-111550	CO2-Neutral Combustion Engines and their Fuels I	WT	4 CP	Koch
T-MACH-111585	Hydrogen and reFuels - Energy Conversion in Combustion Engines	WT	4 CP	Kubach

Assessment
The module examination contains of two oral examinations. The module score results from the two scores weighted according to the ECTS.

Prerequisites
None

Competence Goal
The student can name and explain the working princile of combustion engines. He is able to analyse and evaluate the combustion process. He is able to evaluate influences of gas exchange, mixture formation, fuels and exhaust gas aftertreatment on the combustion performance. He can solve basic research problems in the field of engine development.

The student can name all important influences on the combustion process. He can analyse and evaluate the engine process considering efficiency, emissions and potential.

Content
Working Principle og ICE

Characteristic Parameters

Characteristic parameters

Engine parts

Crank drive

Fuels

Gasolien engine operation modes

Diesel engine operation modes

Emissions

Fundamentals of ICE combustion

Thermodynamics of ICE

Flow field

Wall heat losses

Combsution in Gasoline and Diesel engines

Heat release calculation

Waste heat recovery

CO2-free engine technology

Workload
regular attendance: 62 hours
self-study: 208 hours

5.29 Module: Combustion Engines II (WW4INGMB35) [M-MACH-101303]

Coordinators: Dr.-Ing. Heiko Kubach
Julia Reichel

Organisation: KIT Department of Mechanical Engineering

Part of: Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	1 term	German	4	4

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-MACH-111560	CO2-Neutral Combustion Engines and their Fuels II	ST	4 CP	Koch
Verbrennungsmotoren II (Elective: at least 4 credits)				
T-MACH-105173	Analysis of Exhaust Gas and Lubricating Oil in Combustion Engines	ST	4 CP	Gohl
T-MACH-105649	Boosting of Combustion Engines	ST	4 CP	Kech, Lachenmaier
T-MACH-105184	Fuels and Lubricants for Combustion Engines	WT	4 CP	Kehrwald
T-MACH-110817	Development of Hybrid Drivetrains	ST	4 CP	Koch
T-MACH-110816	Large Diesel and Gas Engines for Ship Propulsions	ST	4 CP	Weisser
T-MACH-105044	Fundamentals of Catalytic Exhaust Gas Aftertreatment	ST	4 CP	Deutschmann, Grunwaldt, Kubach, Lox
T-MACH-105167	Analysis Tools for Combustion Diagnostics	ST	4 CP	Pfeil
T-MACH-105169	Engine Measurement Techniques	ST	4 CP	Bernhardt
T-MACH-111578	Sustainable Vehicle Drivetrains	WT	4 CP	Toedter
T-MACH-105985	Ignition Systems		4 CP	Toedter

Assessment
The assessment consists of an oral exam (60 min) taking place in the recess period (according to §4 (2), 2 of the examination regulation). The exam takes place in every semester. Reexaminations are offered at every ordinary examination date.

Prerequisites
None

Competence Goal
See courses.

Content
Compulsory:

Supercharging and air management

Engine mapsEmissions and Exhaust gas aftertreatment

Transient engine operationECU application

Electrification and alternative powertrains

Elective:

Fuels and lubricants for ICE

Fundamentals of catalytic EGA

Analysis tools for combustion diagnostics

Engine measurement techniques

Analysis of Exhaust Gas und Lubricating Oil in Combustion Engines

Workload
regular attendance: 62 h

self-study: 208 h

Teaching and Learning Methods
Lecture, Tutorial

M 5.30 Module: Commercial Law (IW1JURA2) [M-INFO-101191]

Coordinators: Dr. Yvonne Matz
Organisation: KIT Department of Informatics
Part of: Electives (Law or Sociology)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	3 terms	German	4	3

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-INFO-102013	Exercises in Civil Law	WT+ST	9 CP	Matz

Competence Goal
The student

- has in-depth knowledge of general and special law of obligations and property law,
- is able to understand the interaction of the legal regulations in the German Civil Code (BGB) (concerning the various types of contract and the associated liability issues, performance processing, performance disruptions, various types of transfer of ownership and security rights in rem) and in commercial and company law (in particular concerning the special features of commercial transactions, commercial representation and merchant law as well as the organizational forms provided by German company law for entrepreneurial activity),
- acquire the ability to solve legal problems methodically and accurately using legal means in the private law exercise.

Content
The module builds on the "Introduction to Private Law" module. Students will gain in-depth knowledge of special types of contracts under the German Civil Code (BGB) and more complex corporate law constructions. Furthermore, students are taught how to solve more complex legal issues in a methodically sound manner.

M 5.31 Module: Communication Systems and Protocols [M-ETIT-100539]

Coordinators: Dr.-Ing. Jens Becker
Prof. Dr.-Ing. Jürgen Becker
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Engineering and Informatics (Electrical Engineering and Information Technology)
Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
5 CP	graded	Each summer term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-101938	Communication Systems and Protocols	ST	5 CP	Becker, Becker

Assessment
The examination consists of a written examination of 120 min.

Prerequisites
none

Competence Goal
The students are able to:

- know basic communication systems and to name them
- categorize different communication systems in regards to possible constraints
- name basic mechanisms of communication systems
- carry out these mechanisms
- choose valid mechanisms suitable under given constraints
- design a communication system adhering to constraints, specifications and be able to choose suitable methods, components, and subsystems
- know current communication systems and know about their properties, mechanisms and application.

Content
The lecture will present the physical and technical basics for the design and construction of communication systems. Procedures and technical implementations for communication between electronic devices are presented. This includes, among other things, modulation methods, line model, arbitration, synchronization mechanisms, error correction mechanisms, multiplexing, communication systems, bus systems and on-chip communication. On the basis of selected practical examples, the application of the lecture contents in real systems is demonstrated.

- Information: Definition, Representation, Communication
- Physics: Media, Signals, Mathmatical Descriptions, Line Coupling & Termination, AD Conversion & Sampling, Line Codes, Modulation
- Data Transmission: Definition & Requirements, Transmission Channels, MultiUse of Channels, Multiplexing, Multiple Senders (Arbitration), Multiple Receivers (Addressing), Classification, Interfaces
- Bus Systems: Definitions, Protocols, Transmission of Dataframes, Classification
- Error Protection: Fundamentals, Errors, Error Detection/Correction: Error Handling
- Topologies: physical, logical, examples
- Networks: networks vs. busses, structure, Network specific topologies, routing, OSI Model, TCP/IP, Ethernet
- Classification of Com.Systems
- Real World Systems: Automotive Busses, PC Busses, Field Busses, Networks

Module Grade Calculation
The module grade is the grade of the written exam.

Workload
The workload includes:

1. Attendance in 15 lectures an 7 exercises: 33 h
2. Preparation / follow-up: 66 h (2 h per unit)
3. Preparation of and attendance in examination: 24 h + 2 h

A total of 125 h = 5 LP

Recommendations

Knowledge of the basics from the lecture "Digitaltechnik"is helpful.

M 5.32 Module: Consumer Research [M-WIWI-105714]

Coordinators: Prof. Dr. Benjamin Scheibehenne
Organisation: KIT Department of Business and Economics
Part of: Business and Economics
Electives (Business Administration)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	2 terms	English	4	6

ID	Name	Term offered	Credits	Lecturers
Compulsory Elective Courses (Elective:)				
T-WIWI-113471	Bayesian Statistics for Analyzing Data	Irreg.	4,5 CP	Scheibehenne
T-WIWI-113095	Behavioral Lab Exercise	Irreg.	4,5 CP	Nieken, Scheibehenne
T-WIWI-114186	Collective Intelligence in Human Judgment and Decision Making	ST	4,5 CP	Scheibehenne
T-WIWI-114185	Computational Modelling of Judgments, Decisions and Cognition	ST	4,5 CP	Scheibehenne
T-WIWI-111100	Current Directions in Consumer Psychology	WT+ST	4,5 CP	Scheibehenne
T-WIWI-114174	Economic Decision Making	WT	4,5 CP	Scheibehenne
T-WIWI-111109	KD²Lab Hands-On Research Course: New Ways and Tools in Experimental Economics	Irreg.	4,5 CP	Weinhardt
T-WIWI-107720	Market Research	ST	4,5 CP	Klarmann
T-WIWI-114750	Perceiving, Understanding and Communicating Risk and Uncertainty	WT	4,5 CP	Gradwohl, Scheibehenne

Assessment
The assessment is based on partial exams within the classes offered in this module. Please check the descriptions of the classes for details.

The overall grade of the module is the arithmetic mean of the grades for each course weighted by the number of credits and truncated after the first decimal.

Prerequisites
Willingness to actively engage with the topic.

Competence Goal

- Understand human judgment and decision making in an economic context
- Learn how to plan, program, conduct, statistically analyze, visualize, model, and report behavioral experiments
- Critically evaluate scientific findings in the aftermath of the replication crisis

Content
This module provides students with in-depth knowledge about consumer research at the intersection between Marketing, Psychology, and Cognitive Science. The module consists of classes that look into how individuals and groups make judgments and decisions and what factors influences their behavior (e.g. the lecture on judgment and decision making). Because most findings in this area of research rely on behavioral experiments, this module also focuses on methodological skills. This includes classes on how to plan and design behavioral experiments, conduct and report meaningful statistical analyses, and develop computational cognitive models. The module also includes classes about reproducibility and transparency in the behavioral sciences. The module is a pre-requisite for writing a Master thesis at the KIT Cognition and Consumer Behavior lab.

Workload
The total workload for this module is approximately 270 hours.

Recommendations
Interest in behavioral research.

5.33 Module: Control Engineering II (WI4INGETI2) [M-ETIT-101157]

Coordinators: Prof. Dr.-Ing. Sören Hohmann
Dr.-Ing. Mathias Kluwe
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Level	Version
9 CP	graded	Each term	2 terms	4	3

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-100666	Control of Linear Multivariable Systems	WT	6 CP	Kluwe
T-ETIT-100980	Nonlinear Control Systems	ST	3 CP	Kluwe

Assessment
The assessment is carried out as partial written exams of the single courses of this module (T-ETIT-100980 and T-ETIT-100666).

Prerequisites
none

Module Grade Calculation
The overall grade of the module is the average of the grades for each course weighted by the credits and truncated after the first decimal.

Recommendations
For this module a basic knowledge in system theory and control engineering is assumed. These subjects can be found in the course *System Dynamics and Control Engineering* (M-ETIT-102181) which is recommended to have been attended beforehand.

M 5.34 Module: Cooperative Autonomous Vehicles [M-WIWI-106631]

Coordinators: Prof. Dr. Alexey Vinel
Organisation: KIT Department of Business and Economics
Part of: Engineering and Informatics (Informatics)
Electives (Informatics)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	2 terms	English	4	2

ID	Name	Term offered	Credits	Lecturers
Compulsory Elective Courses (Elective:)				
T-WIWI-113363	Collective Perception in Autonomous Driving	ST	4,5 CP	Vinel
T-WIWI-112690	Cooperative Autonomous Vehicles	ST	4,5 CP	Vinel
T-WIWI-113059	Human Factors in Autonomous Driving	WT	4,5 CP	Vinel

Assessment
The assessment is carried out as partial exams of the single courses of this module, whose sum of credits must meet the minimum requirement of credits of this module. The assessment procedures are described for each course of the module separately.

Prerequisites
None.

Competence Goal
Students

- know the fundamentals of vehicular communications and networking,
- look critically into current research topics in the field of autonomous driving,
- explain basic concepts in cooperative vehicles,
- apply mathematical methods for the performance evaluation of cooperative driving systems,
- apply simulation tools for the modeling of cooperative autonomous vehicles.

Content
The module focuses on the aspects of communication, coordination, and cooperation of highly automated and autonomous vehicles. We explain the state-of-the-art of the vehicular communications (V2X) and respective cooperative driving applications from an interdisciplinary viewpoint. The module includes selected material from wireless networking, formal description methods, human-computer interaction, robotics, and machine learning. The students work with mathematical models, simulation environments and lab equipment.

Module Grade Calculation
The overall grade of the module is the average of the grades for each course, weighted by the credits and truncated after the first decimal.

Workload
The total workload for this module is approximately 270 hours. The exact distribution is made according to the credit points of the courses of the module.

5.35 Module: Cross-Functional Management Accounting (WW4BWLIBU2) [M-WIWI-101510]

Coordinators: Prof. Dr. Marcus Wouters
Organisation: KIT Department of Business and Economics
Part of: Business and Economics
Electives (Business Administration)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	1 term	English	4	14

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-WIWI-102885	Advanced Management Accounting	WT	4,5 CP	Wouters
Supplementary Courses (Elective: 4,5 credits)				
T-WIWI-105777	Business Intelligence Systems	WT	4,5 CP	Mädche
T-WIWI-105781	Incentives in Organizations	ST	4,5 CP	Nieken
T-WIWI-107720	Market Research	ST	4,5 CP	Klarmann
T-WIWI-111848	Online Concepts for Karlsruhe City Retailers	ST	3 CP	Kupfer
T-WIWI-102621	Valuation	WT	4,5 CP	Ruckes
T-WIWI-108651	Extraordinary Additional Course in the Module Cross-Functional Management Accounting	WT+ST	4,5 CP	Wouters

Assessment
The assessment is carried out as partial exams (according to Section 4 (2), 1-3 SPO) of the core course and further single courses of this module, whose sum of credits must meet the minimum requirement of credits of this module. The assessment procedures are described for each course of the module separately.
The overall grade of the module is the average of the grades for each course weighted by the credits and truncated after the first decimal.

Prerequisites
The course "Advanced Management Accounting" is compulsory.

The additional courses can only be chosen after the compulsory course has been completed successfully.

Competence Goal
Students will be able to apply advanced management accounting methods to managerial decision-making problems in marketing, finance, organization and strategy.

Content
The module includes a course on several advanced management accounting methods that can be used for various decisions in operations and innovation management. By selecting another course, each student looks in more detail at one interface between management accounting a particular field in management, namely marketing, finance, or organization and strategy.

Additional Information
The module "Cross-functional Management Accounting" always includes the compulsory course “Advanced Management Accounting.” Students look at the interface between management accounting and another field in management. Students build the module by adding a course from the specified list. Students can also suggest another suitable course for this module for evaluation by the coordinator.

Workload
Total workload for 9 credit points: approx. 270 hours

The exact distribution is based on the credit points of the courses in the module.

Recommendations
None

M 5.36 Module: Cyber-Physical Modeling [M-ETIT-106953]

Coordinators: Prof. Dr.-Ing. Mike Barth
Prof. Dr.-Ing. Sören Hohmann

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: Engineering and Informatics (Electrical Engineering and Information Technology)
Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each summer term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-113908	Cyber-Physical Modeling		6 CP	Barth, Hohmann

Assessment
The examination takes place in the form of a written examination lasting 90 min.

Prerequisites
none

Competence Goal

- The students are familiar with the concepts of Cyber-Physical System.
- Students understand the need for advanced methods and services in the field of automation.
- Students can validate different information models and ontologies for their applicability in CPS.
- Students will be able to model data, information and knowledge or extract them from existing systems.
- The students know suitable modeling tools and their application.
- The students understand the general model concept as well as the characteristics of physical and data-based modeling and can describe their differences.
- They can structure complex systems and systematically analyze dependencies of subsystems.
- They can explain the general procedure of physical and data-based modeling, apply it to technical systems, and analyze the results.
- They can apply causal and non-causal modeling approaches and distinguish between them.
- Students have gained an understanding of generalized, cross-domain, physical relationships and can develop models for electrical, mechanical, pneumatic and hydraulic systems.
- They can describe the relationship between generalized, cross-domain, physical models and basic procedures of physical-based control and explain their advantages / limitations based on basic knowledge of control engineering.
- The students can estimate and judge the effects of disturbances and real conditions on the identification results.

Content
This course aims at engineering students that focus on a system-based engineering curriculum, including architectures, modeling & simulation for Cyber Physical Systems. The module is designed to teach students the theoretical and practical aspects of Digital Twins and their interconnection with their physical counterpart. It encompasses fundamental topics along the complete process of modeling technical systems. For this purpose, it includes the conception and construction of digital twins including their model components. In terms of modeling and simulation of physical systems, two major areas will be covered: On the one hand, physical-based modeling techniques which derive formal model equations based on analyzing the physical first principles of technical systems. This includes generalized equivalent circuits and variational analysis (Euler-Lagrange of the first kind). On the other hand, data-based identification techniques will be covered which are used to identify concrete model parameters for a given technical system from experimental data sets. When combining the identification with an initial, non-physical, structural set up of model equations, the complete process is often referred to as data-based modeling or black-box modeling. Both modeling areas base on available information about the physical system which is structured in Meta- and Information-Models. Examples that are covered in this lecture are Metamodels, e.g. AutomationML or the asset administration shell principles. Also, semantic web principles and ontologies will be part of the lecture content.

Module Grade Calculation
The module grade is the grade of the written exam.

Workload

1. attendance in lectures an exercise: 3+1 SWS (60 h)
2. pre-/postprocessing of the lecture (90 h)
3. preparation of and attendance in the exam: (30 h)

   A total of 180 h = 6 CR

Recommendations
Interest in Modeling and Simulation of modern Cyber-Physical Systems in combination with concepts of digital twins, system architectures and Co-Simulation.
Sound understanding of engineering mechanics, electrical, mechatronic systems / physics / Software-Engineering should be fulfilled to successfully attend the lecture, exercise tasks / case studies, and exam.

Teaching and Learning Methods
lecture (3 SWS), excercise (1 SWS)

M 5.37 Module: Data Driven Engineering [M-MACH-107716]

Organisation: KIT Department of Mechanical Engineering

Part of: Engineering and Informatics (Mechanical Engineering)
Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	1 term	English	4	1

Election Notes
Hot Research Topics in AI for Engineering Applications (HotAI) only in the winter semester

ID	Name	Term offered	Credits	Lecturers
Data Driven Engineering (Election: 9 credits) (Elective:)				
T-MACH-115045	Project: Data-Driven Engineering	WT+ST	4,5 CP	Meyer
T-MACH-113669	Hot Research Topics in AI for Engineering Applications	WT	4,5 CP	Meyer
T-MACH-115011	Seminar: Machine Learning and Optimization in Engineering	WT+ST	4,5 CP	Meyer

Assessment
The module examination takes the form of partial examinations (according to §4(2), 1-3 SPO) on the courses of the module. The assessment of success is described for each course in this module.

The overall grade for the module is calculated from the LP-weighted grades of the partial examinations and cut off after the first decimal place.

Prerequisites

- Basic knowledge of artificial intelligence and machine learning
- Programming experience
- English proficiency

Competence Goal
Upon successful completion, students will be able to:

1. Identify, formulate, and analyze complex engineering problems as data-driven tasks.
2. Design, implement, and optimize solutions using advanced machine-learning, optimization, and simulation techniques.
3. Critically evaluate methodological choices and effectively communicate results in written, oral, and multimedia formats.

Content
The “Data-Driven Engineering” module is a flexible master-level offering that can be assembled from any two of three courses — "Data-Driven Engineering Project", "Hot Research Topics in AI for Engineering", and "Seminar: Machine Learning & Optimization" — providing theory, hands-on project work, and exposure to cutting-edge AI applications. It equips students with the skills to formulate, implement, and evaluate data-driven solutions in engineering contexts.

Core topics **Project: Data-Driven Engineering (DDE)**
Full-cycle project work: problem definition, method selection (ML, optimization, simulation), prototype implementation using tools such as PyTorch, scikit-learn, Isaac Sim, and evaluation

Core topics **Hot Research Topics in AI for Engineering (HotAI)**
State-of-the-art AI methods (LLMs, reinforcement learning, transformers), industrial application scenarios (automation, CAD generation, logistics), and hands-on prototype development

Core topics **Seminar: Machine Learning & Optimization in Engineering**
Hybrid ML-optimization approaches, algorithm design, computational performance, case studies in product design, process optimization, logistics

Students choose two of the three courses, allowing for a mix of theoretical principles and practical implementation.

Module Grade Calculation
Weighted average of the credit points from the completed partial achievements

Additional Information

The seminar 'Machine Learning and Optimization in Engineering' can be taken with a reduced credit load (3 CP) for the seminar module M-WIWI-101808. In this case, the course can no longer be selected for the 'Data-Driven Engineering' module.

Please note that in this scenario, credit is awarded via the assessment **'Seminar in Engineering Science Master (approval)' (T-WIWI-108763)**. Further information regarding this process can be found at https://www.wiwi.kit.edu/Anrechnung_und_Verbuchung_2015.php.

Workload

Total workload for 9 credit points: approx. 270 hours. The exact distribution is based on the credit points of the courses in the module.

Teaching and Learning Methods

Seminar and project work

M 5.38 Module: Data Science for Finance [M-WIWI-105032]

Coordinators: Prof. Dr. Maxim Ulrich
Organisation: KIT Department of Business and Economics
Part of: Business and Economics
Electives (Business Administration)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-WIWI-102878	Computational Risk and Asset Management	Irreg.	4,5 CP	Ulrich
T-WIWI-110213	Python for Computational Risk and Asset Management	WT	4,5 CP	Ulrich

Assessment
The module examination takes the form of an alternative exam assessment.
The alternative exam assessment consists of a Python-based "Takehome Exam". At the end of the third week of January, the student is given a "Takehome Exam" which he processes and sends back independently within 4 hours using Python. Precise instructions will be announced at the beginning of the course. The alternative exam assessment can be repeated a maximum of once. A timely repeat option takes place at the end of the third week in March of the same year. More detailed instructions will be given at the beginning of the course.

Competence Goal
The aim of the module is to use data science, machine learning and financial market theories to generate better investment, risk and asset management decisions. The student gets to know the characteristics of different asset classes in an application-oriented manner using real financial market data. We use Python and web scraping techniques to extract, visualize and examine patterns of publicly available financial market data. Interesting and non-public financial market data such as (option and futures data on shares and interest) are provided. Financial market theories are also discussed to improve data analysis through theoretical knowledge. Students get to know stock, interest rate, futures and options markets through the "data science glasses". Through "finance theory glasses" students understand how patterns can be communicated and interpreted using finance theory. Python is the link through which we bring data science and modern financial market modeling together.

Content
The course covers several topics, among them:

- Pattern detection in price and return data in equity, interest rate, futures and option markets
- Quantitative Portfolio Strategies
- Modeling Return Densities using tools from financial econometrics, data science and machine learning
- Valuation of equity, fixed-income, futures and options in a coherent framework to possibly exploit arbitrage opportunities
- Neural networks and Natural Language Processing

Workload
The total workload for this module is 270 hours (9 credit points).The total number of hours resulting from income from studying online video, answering quizzes, studying Ipython notebooks, active and interactive "Python Data Sessions" and reading literature you have heard.

Recommendations
Basic knowledge of capital markt theory.

Teaching and Learning Methods
Lectures and Exercises

M 5.39 Module: Data Science: Data-Driven Information Systems [M-WIWI-103117]

Coordinators: Prof. Dr. Alexander Mädche
Prof. Dr. Christof Weinhardt

Organisation: KIT Department of Business and Economics

Part of: Business and Economics
Electives (Business Administration)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	1 term	English	4	11

ID	Name	Term offered	Credits	Lecturers
Compulsory Elective Courses (Elective:)				
T-WIWI-114749	(Gen)AI and Virtual Reality for Business	Irreg.	4,5 CP	Pfeiffer
T-WIWI-114210	(Gen)AI-based Automation in Organizations	ST	4,5 CP	Mädche
T-WIWI-108715	Artificial Intelligence in Service Systems	WT	4,5 CP	Satzger
T-WIWI-114209	Artificial Intelligence in Service Systems II: Generative AI Applications & Adoption	ST	4,5 CP	Satzger
T-WIWI-113471	Bayesian Statistics for Analyzing Data	Irreg.	4,5 CP	Scheibehenne
T-WIWI-106187	Business Data Strategy	WT	4,5 CP	Weinhardt
T-WIWI-105777	Business Intelligence Systems	WT	4,5 CP	Mädche
T-WIWI-114089	Data Science for Business	ST	4,5 CP	Pfeiffer
T-WIWI-113160	Digital Democracy	WT	4,5 CP	Fegert
T-WIWI-113459	Practical Seminar: Human-Centered Systems	WT+ST	4,5 CP	Mädche
T-WIWI-111385	Responsible Artificial Intelligence	WT	4,5 CP	Pfeiffer
T-WIWI-106207	Practical Seminar: Data-Driven Information Systems	Irreg.	4,5 CP	Satzger, Weinhardt

Assessment
The assessment is carried out as partial exams (according to Section 4 (2), 1-3 SPO) of the core course and further single courses of this module, whose sum of credits must meet the minimum requirement of credits of this module. The assessment procedures are described for each course of the module separately.

Prerequisites
None.

Competence Goal
The student

- understands the strategic role of integrating, transforming, and analyzing large and complex enterprise data in modern business information systems and is capable of comparing and assessing strategic alternatives
- has the core skills to design, model, and control complex, inter-organisational analytical, processes, including various business functions as well as customers and markets
- understands the usage of performance indicators for a variety of controlling and management issues and is able to define models for generating the relevant performance indicators under considerations of data availability
- distinguishes different analytics methods and concepts and learn when to apply to better understand and anticipate business relationships and developments of industrial and in particular service companies to derive fact- and data-founded managerial actions and strategies.
- knows how to capture uncertainty in the data and how to appropriately consider and visualize uncertainty in decision support or business intelligence systems and analytical processes as a whole.

Content
The amount of business-related data availabe in modern enterprise information systems grows exponentially, and the various data sources are more and more integrated, transformed, and analyzed jointly to gain valuable business insights, pro-actively control and manage business processes, to leverage planning and decision making, and to provide appropriate, potentially novel services to customers based on relationships and developments observed in the data.

Also, data sources are more and more connected and single business unit that used to operate on separate data pools are now becoming highly integrated, providing tremendeous business opportunities but also challenges regarding how the data should be represented, integrated, preprocessed, transformed, and finally used in analytics planning and decision processes.

The courses of this module equip the students with core skills to understands the strategic role of integrating, transforming, and analyzing large and complex enterprise data in modern business information systems. Students will be capable to designing, comparing, and evaluating strategic alternatives. Also, students will learn how to design, model, and control complex analytical processes, including various business functions of industrial and service companies including customers and markets. Students learn core skills to understand fundamental strategies for integrating analytic models and operative controlling mechanisms while ensuring the technical feasibility of the resulting information systems..

Furthermore, the student can distinguish different methods and concepts in the realm of data science and learns when to apply. She/he will know the means of characterizing and analyzing hetergeneous, high-dimensional data available data in data warehouses and external data sources to gain additional insights valuable for enterprise planning and decision making. Also, the students know how to capture uncertainty in the data and how to appropriately consider and visualize uncertainty in business information and business intelligence systems.

The module offers the opportunity to apply and deepen this knowledge in a seminar and hands-on tutorials that are offered with all lectures.

Texteintrag

Workload
Attendance time: 90 h

Self-study: approx. 100 h

Exam preparation: approx. 80 h

Total workload: approx. 270 h

Recommendations
For successful participation in the module, basic knowledge of business informatics (in particular of tasks, systems and processes) is very helpful. In this context, prior attendance of the course Grundzüge der Wirtschaftsinformatik [2540450] is recommended. In addition, knowledge of operations research as well as descriptive and inferential statistics is advantageous in order to be able to follow the quantitative content of the module.

M 5.40 Module: Data Science: Data-Driven User Modeling [M-WIWI-103118]

Coordinators: Prof. Dr. Christof Weinhardt
Organisation: KIT Department of Business and Economics
Part of: Business and Economics
Electives (Business Administration)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	1 term	English	4	9

ID	Name	Term offered	Credits	Lecturers
Compulsory Elective Courses (Elective: at least 9 credits)				
T-WIWI-114089	Data Science for Business	ST	4,5 CP	Pfeiffer
T-WIWI-113160	Digital Democracy	WT	4,5 CP	Fegert
T-WIWI-115061	Experimental Research in Economics and Information Systems	WT	4,5 CP	Weinhardt
T-WIWI-111109	KD²Lab Hands-On Research Course: New Ways and Tools in Experimental Economics	Irreg.	4,5 CP	Weinhardt
T-WIWI-111385	Responsible Artificial Intelligence	WT	4,5 CP	Pfeiffer
T-WIWI-108765	Practical Seminar: Advanced Analytics	WT+ST	4,5 CP	Weinhardt

Assessment
The assessment is carried out as partial exams of the core course and further single courses of this module, whose sum of credits must meet the minimum requirement of credits of this module. The assessment procedures are described for each course of the module separately.

Prerequisites
None

Competence Goal
Students of this module

- learn methods for planning empirical studies, in particular laboratory experiments,
- acquire theoretical knowledge and practical skills in analysing empirical data,
- familiarize with different ways of modelling user behaviour, are able to critically discuss, and to evaluate them

Content
Understanding and supporting user interactions with applications better plays an increasingly large role in the design of business applications. This applies both to interfaces for customers and to internal information systems. The data that is generated during user interactions can be channelled straight into business processes, for instance by analysing and decomposing purchase decisions, and by feeding this data into product design processes.

The Crowd Analytics section considers the analysis of data from online platforms, particularly of those following crowd- or peer-to-peer based business models. This includes platforms like Airbnb, Kickstarter and Amazon Mechanical Turk.

Theoretical models of user (decision) behaviour help analyzing the empirically observed user behaviour in a systematic fashion. Testing these models and their predictions in controlled experiments (primarily in the lab) in turn helps refine theory and to generate practically relevant design recommendations. Analyses are carried out using advanced analytic methods.

Students learn fundamental theoretical models for user behaviour in systems and apply them to cases. Students are also taught methods and skills for conceptualizing and planning empirical studies and for analyzing the resulting data.

Recommendations
Basic knowledge of Information Management, Operations Research, Descriptive Statistics, and Inferential Statistics is assumed.

M 5.41 Module: Data Science: Evidence-based Marketing (WW4BWLMAR8) [M-WIWI-101647]

Coordinators: Prof. Dr. Martin Klarmann
Organisation: KIT Department of Business and Economics
Part of: Business and Economics
Electives (Business Administration)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	2 terms	English	4	5

ID	Name	Term offered	Credits	Lecturers
Compulsory Elective Courses (Elective: 9 credits)				
T-WIWI-103139	Marketing Analytics	WT	4,5 CP	Klarmann
T-WIWI-107720	Market Research	ST	4,5 CP	Klarmann

Assessment
The assessment is carried out as partial exams (according to Section 4 (2), 1-3 SPO) of the courses of this module, whose sum of credits must meet the minimum requirement of credits of this module. The assessment procedures are described for each course of the module separately.

The overall grade of the module is the average of the grades for each course weighted by the credits and truncated after the first decimal.

Prerequisites
Keine.

Competence Goal
Students

- possess advanced knowledge of relevant market research contents
- know many different qualitative and quantitative methods for measuring customer behavior, preparation of strategic decisions, making causal deductions, usage of social media data and sales forecasting
- possess the statistical skills required for working in marketing research

Content
This module provides in-depth knowledge of relevant quantitative and qualitative methods used in market research.Students can attend the following courses:

- The course “Market Research” provides contents of practical relevance for measuring customer attitudes and customer behavior. The participants learn using statistical methods for strategic decision-making in marketing. Students who are interested in writing their master thesis at the Marketing & Sales Research Group are required to take this course.
- The course "Marketing Analytics“ is based on "Market Research“ and teaches advanced statistical methods for analyzing relevant marketing and market research questions. Please note that a successful completion of "Market Research" is a prerequisite for the completion of "Marketing Analytics".

Workload
The total workload for this module is approximately 270 hours.

Recommendations
None

5.42 Module: Data Science: Intelligent, Adaptive, and Learning Information Services [M-WIWI-105661]

Coordinators: Prof. Dr. Andreas Geyer-Schulz
Organisation: KIT Department of Business and Economics
Part of: Electives (Business Administration)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	1 term	German/English	4	3

ID	Name	Term offered	Credits	Lecturers
Compulsory Elective Courses (Elective: 9 credits)				
T-WIWI-109921	Advanced Machine Learning	ST	4,5 CP	Geyer-Schulz, Nazemi
T-WIWI-114209	Artificial Intelligence in Service Systems II: Generative AI Applications & Adoption	ST	4,5 CP	Satzger
T-WIWI-102762	Business Dynamics	WT	4,5 CP	Geyer-Schulz, Glenn
T-WIWI-111267	Intelligent Agent Architectures	WT	4,5 CP	Geyer-Schulz
T-WIWI-110915	Intelligent Agents and Decision Theory	ST	4,5 CP	Geyer-Schulz
T-WIWI-102847	Recommender Systems	WT	4,5 CP	Geyer-Schulz

Assessment
The assessment is carried out as partial exams (according to Section 4(2), 1 or 2 of the examination regulation) of the single courses of this module, whose sum of credits must meet the minimum requirement of credits of this module. The assessment procedures are described for each course of the module seperately.

The overall grade of the module is the average of the grades for each course weighted by the credits and truncated after the first decimal.

Prerequisites
None

Competence Goal
The student

- models, analyzes and optimizes the structure and dynamics of complex economic changes.
- designs and develops intelligent, adaptive or learning agents as essential elements of information services.
- knows the essential learning methods for this and can apply them (also on modern architectures) in a targeted manner.
- develops and implements personalized services, especially in the area of recommender systems.
- develops solutions in teams.

Content
The Intelligent Architectures course addresses how to design modern agent-based systems. The focus here is on software architecture and design patterns relevant to learning systems. In addition, important machine learning methods that complete the intelligent system are discussed. Examples of systems presented include key-map architectures and genetic methods.
The impact of management decisions in complex systems is considered in Business Dynamics. Understanding, modeling, and simulating complex systems enables analysis, purposeful design, and optimization of markets, business processes, regulations, and entire enterprises.
Special problems of intelligent systems are covered in Personalization and Services and Recommendersystems. The content includes approaches and methods to design user-oriented services. The measurement and monitoring of service systems is discussed, the design of personalized offers is discussed and the generation of recommendations based on collected data from products and customers is shown. The importance of user modeling and recognition is addressed, as well as data security and privacy.

Additional Information
The module replaces from summer semester 2021 M-WIWI-101470 "Data Science: Advanced CRM".

Workload
The total workload for this module is approx. 270 hours (9 credits). The distribution is based on the credit points of the courses in the module. The workload for courses with 4.5 credits is approx. 135 hours.

The total number of hours per course is calculated from the time required to attend the lectures and exercises, as well as the examination times and the time required for an average student to achieve the learning objectives of the module for an average performance.

Recommendations
None

5.43 Module: Data-Based Modeling and Control [M-CIWVT-106319]

Coordinators: Prof. Dr.-Ing. Thomas Meurer
Organisation: KIT Department of Chemical and Process Engineering
Part of: Engineering and Informatics (Chemical and Process Engineering)
Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-CIWVT-112827	Data-Based Modeling and Control		6 CP	Meurer

Assessment
Learning control is an oral examination with a duration of about 45 minutes.

Prerequisites
none

Competence Goal
Students have an in-depth understanding of methods and concepts of data-based modeling and control of dynamic systems, including machine learning techniques and corresponding optimization methods. They understand the underlying mathematical concepts and can apply them to new problems. They are able to apply these methods independently to specific problems and familiarize themselves with further literature on their own.

Content
The module covers basic concepts and fundamentals of data-based approaches for modeling and control design for dynamical systems and processes. Data-based approaches for modeling, also called system identification, are used to identify a mathematical description of the considered system from the available input and output data. Data-based approaches for control design compute the controller without an a priori known model of the system. Extensions to learning-based control are addressed, where in principle machine learning techniques are used to learn a model or a controller for a given system.

Problem sets are considered in the exercises to apply the developed methods.

Module Grade Calculation
The grade of the module is the grade of the oral exam.

Workload
Attendance time: Lectures: 30 hrs. Exercises: 15 hrs.

Self-study: 75 hrs.

Exam preparation: 60 hrs.

Literature

- T. Meurer: Data-based Modeling and Control, Lecture Notes.
- S.L. Brunton, J.N. Kutz: Data-Driven Science and Engineering: Machine Learning, Dynamical Systems, and Control, Cambridge University Press, 2022.
- D. Bertsekas: Reinforcement Learning and Optimal Control, Athena Scientific, 2019.
- D.H. Owens: Iterative Learning Control, Springer, 2016.
- Various recent publications, which will be discussed in lecture.

M 5.44 Module: Design of a Jet Engine Combustion Chamber [M-CIWVT-105206]

Coordinators: Dr.-Ing. Stefan Raphael Harth
Organisation: KIT Department of Chemical and Process Engineering
Part of: Engineering and Informatics (Chemical and Process Engineering)
Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-CIWVT-110571	Design of a Jet Engine Combustion Chamber	WT	6 CP	Harth

Assessment
Learning control is an examination of another type.

The module grade consists of the grade of the oral examination (35 points maximum) and the cooperation / presentation during the project (65 points maximum).
The learning control ist passed when at least 45 points are achieved.

Prerequisites
None

Competence Goal

- The students are able to apply the relevant design parameters in order to design a jet engine combustor.
- The students are able to evaluate design modifications due to the performance of a jet engine combustor.
- The students are able to review literature studies and use them for their design aims.
- The students learn to work target oriented following a time schedule.
- The students learn to work in a team and to exchange information between the teams by definition of interfaces.
- The students learn to present clearly and in an acceptable time the work progress and the most important results.

Content
At the beginning the description and operating mode of a jet engine with emphasis on the combustor is explained in 4 lessons. Afterwards the design of the combustor based on geometrical boundary conditions (engine casing) and the performance conditions will start. The tasks to be solved for the design are the combustor aerodynamic (pressure loss, air split), thermal management (temperature distribution, wall cooling, material), calculation of emissions and the construction of the combustor. In order to solve the tasks the students have to be organized in groups which are responsible for the tasks mentioned. The work progress will be controlled by a time schedule and regular presentations. The complete design will be discussed in a final presentation.

Module Grade Calculation
The module grade is the grade of the examnation of another type.

Workload

- Attendance time (Lecture): 30 h
- Homework: 45 h
- Project: 80 h
- Exam Preparation: 45 h

Literature

- Lefebvre, Gas Turbine Combustion
- Rolls-Royce plc, the jet engine
- Müller, Luftstrahltriebwerke Grundlage, Charakteristiken, Arbeitsverhalten

M 5.45 Module: Design, Construction, Operation and Maintenance of Highways (bauiEX301-STREBBE) [M-BGU-100998]

Coordinators: Dr.-Ing. Matthias Zimmermann
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each summer term	1 term	German	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-BGU-106613	Design Basics in Highway Engineering	WT+ST	3 CP	Zimmermann
T-BGU-106300	Infrastructure Management	WT+ST	6 CP	Zimmermann

Assessment
- module component T-BGU-106613 with oral examination according to § 4 Par. 2 No. 2
- module component T-BGU-106300 with written examination according to § 4 Par. 2 No. 1

details about the learning controls, see module components

Prerequisites
The selection of this module excludes the selection of the module "Highway Engineering" (WI4INGBGU2)not offered anymore.

Competence Goal
The student:

- has in-depth knowledge of the design, construction, operation and maintenance of roads,
- is able to analyze and assess complex issues in road engineering.

Content
This module deals with road engineering, starting with the fundamentals relevant to dimensioning, the design of the traffic system as a three-dimensional spatial band, the construction of the road (earthworks and superstructure in various construction methods) through to the operation and maintenance of the entire infrastructure. In addition to the engineering-specific specialist knowledge, methods are taught that are required to analyze and assess complex issues in road engineering.

Module Grade Calculation
grade of the module is CP weighted average of grades of the partial exams

Additional Information
None

Workload
contact hours (1 HpW = 1 h x 15 weeks):

- Design Basics in Highway Engineering lecture: 30 h
- Design and Construction of Highways lecture: 30 h
- Operation and Maintenance of Highways lecture: 30 h

independent study:

- preparation and follow-up lecture Design Basics in Highway Engineering: 30 h
- examination preparation Design Basics in Highway Engineering: 30 h
- preparation and follow-up lecture Design and Construction of Highways: 30 h
- preparation and follow-up lecture Operation and Maintenance of Highways: 30 h
- examination preparation Infrastructure Management: 60 h

total: 270 h

Recommendations
None

5.46 Module: Designing Interactive Information Systems [M-WIWI-104080]

Coordinators: Prof. Dr. Alexander Mädche
Organisation: KIT Department of Business and Economics
Part of: Business and Economics
Electives (Business Administration)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	1 term	English	4	6

ID	Name	Term offered	Credits	Lecturers
Compulsory Elective Courses (Elective: at least 1 item)				
T-WIWI-113465	Designing Interactive Systems: Human-AI Interaction	ST	4,5 CP	Mädche
T-WIWI-113460	Engineering Interactive Systems: AI & Wearables	WT	4,5 CP	Mädche
Supplementary Courses (Elective: at most 1 item)				
T-WIWI-114749	(Gen)AI and Virtual Reality for Business	Irreg.	4,5 CP	Pfeiffer
T-WIWI-114210	(Gen)AI-based Automation in Organizations	ST	4,5 CP	Mädche
T-WIWI-111109	KD²Lab Hands-On Research Course: New Ways and Tools in Experimental Economics	Irreg.	4,5 CP	Weinhardt
T-WIWI-113459	Practical Seminar: Human-Centered Systems	WT+ST	4,5 CP	Mädche

Assessment
The assessment is carried out as partial exams (according to Section 4 (2), 1-3 SPO) of the core course and further single courses of this module, whose sum of credits must meet the minimum requirement of credits of this module. The assessment procedures are described for each course of the module separately.

Prerequisites
In this module, the courses "Designing Interactive Systems" or "Enineering Interactive Systems" must be compulsorily taken.

Competence Goal
The student

- has a comprehensive understanding of conceptual and theoretical foundations of interactive systems
- knows design processes for interactive systems
- is aware of the most important techniques and tools for designing interactive systems and knows how to apply them to real-world problems
- is able to apply design principles for the design of most important classes of interactive systems,
- creates new solutions of interactive systems teams

Content
Advanced information and communication technologies make interactive systems ever-present in the users' private and business life. They are an integral part of smartphones, devices in the smart home, mobility vehicles as well as at the working place in production and administration (e.g. in the form of dashboards).

With the continuous growing capabilities of computers, the design of the interaction between human and computer becomes even more important. This module focuses on design processes and principles for interactive systems. The contents of the module abstract from the technical implementation details and focus on foundational concepts, theories, practices and methods for the design of interactive systems. The students get the necessary knowledge to guide the successful implementation of interactive systems in business and private life.

Each lecture in the module is accompanied with a capstone project that is carried out with an industry partner.

Additional Information
See http://issd.iism.kit.edu/305.php for further information.

Workload
The total workload for this module is approximately 270 hours.

5.47 Module: Development of Business Information Systems (IW4INAIFB5) [M-WIWI-101477]

Coordinators: Prof. Dr. Andreas Oberweis
Organisation: KIT Department of Business and Economics
Part of: Electives (Informatics)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	1 term	German	4	9

ID	Name	Term offered	Credits	Lecturers
Compulsory Elective Courses (Elective: between 1 and 2 items)				
T-WIWI-113968	Management of IT-Projects	ST	4,5 CP	Alpers
T-WIWI-102895	Software Quality Management	ST	4,5 CP	Oberweis
Supplementary Courses (Elective: at most 1 item)				
T-WIWI-110346	Supplement Enterprise Information Systems	WT+ST	4,5 CP	Oberweis
T-WIWI-110548	Advanced Lab Informatics (Master)	WT+ST	4,5 CP	Professorenschaft des Instituts AIFB
T-WIWI-112914	Advanced Lab Realization of Innovative Services (Master)	WT+ST	4,5 CP	Oberweis

Assessment
The assessment mix of each course of this module is defined for each course separately. The final mark for the module is the average of the marks for each course weighted by the credits and truncated after the first decimal.

Prerequisites
The course *Datenbanksysteme und XML* or the course *Software Quality Management* must be examined.

Competence Goal
Students

- describe the structure and the components of enterprise information systems,
- explain functionality and architecture of the enterprise information system components ,
- choose and apply relevant components to solve given problems in a methodic approach,
- describe roles, activities and products in the field of software engineering management,
- compare process and quality models and choose an appropriate model in a concrete situation,
- write scientific theses in the areas of enterprise information system components and software engineering management and find own solutions for given problems and research questions.

Content
An enterprise information system contains the complete application software to store and process data and information in an organisation including design and management of databases, workflow management and strategic information planning.

Due to global networking and geographical distribution of enterprises as well as the increasing acceptation of eCommerce the application of distributed information systems becomes particular important.

This module teaches concepts and methods for design and application of information systems.

Additional Information
The course T-WIWI-102759 "Requirements Analysis and Requirements Management" will no longer be offered in the module as of winter semester 2018/2019.

Workload
See German version

5.48 Module: Digital Marketing [M-WIWI-106258]

Coordinators: Prof. Dr. Ann-Kristin Kupfer
Organisation: KIT Department of Business and Economics
Part of: Business and Economics
Electives (Business Administration)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	2 terms	English	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-WIWI-112693	Digital Marketing	ST	4,5 CP	Kupfer
Supplementary Courses (Elective: 4,5 credits)				
T-WIWI-106981	Digital Marketing and Sales in B2B	ST	1,5 CP	Klarmann, Konhäuser
T-WIWI-114174	Economic Decision Making	WT	4,5 CP	Scheibehenne
T-WIWI-114893	International Entrepreneurship and Brand Development	see notes	6 CP	Kupfer, Terzidis
T-WIWI-107720	Market Research	ST	4,5 CP	Klarmann
T-WIWI-112711	Media Management	WT	4,5 CP	Kupfer
T-WIWI-111848	Online Concepts for Karlsruhe City Retailers	ST	3 CP	Kupfer

Assessment
The assessment is carried out as partial exams of the core course and further single courses of this module, whose sum of credits must meet the minimum requirement of credits of this module. The assessment procedures are described for each course of the module separately.

The overall grade of the module is the average of the grades for each course, weighted by the credits and truncated after the first decimal.

Prerequisites
None

Competence Goal
Students

- have an advanced knowledge about central marketing contents
- have a fundamental understanding of the marketing instruments
- know current fundamental principles and latest trends in the field of digital marketing
- know and understand several strategic concepts and how to implement them
- are able to implement their extensive marketing knowledge in a practical context
- are able to critically discuss and question theoretical concepts and current practices in marketing
- have theoretical knowledge that is fundamental for writing a master thesis in the field of marketing
- have gained insight into scientific research that prepares them to independently write a master's thesis
- have the theoretical knowledge and skills necessary to work in or collaborate with the marketing department of a company

Content

The "Digital Marketing" module explores advanced marketing strategies within the context of digital transformation. Students develop a deep understanding of digital consumer behavior, the digital environment, and digital marketing strategies:

- **Principles of Digital Marketing:** In this mandatory course of the module, students learn about practices and strategies like search engine optimization, word-of-mouth, and influencer marketing.
- **Performance Analytics and Practical Applications:** Practical applications of SEO, SEA, and Content Management Systems. Students learn to monitor performance using Key Performance Indicators (KPIs), dashboards, and Return on Investment (RoI) calculations and apply their knowledge to real-world cases.
- **Contextualization:** Via course selections, students can learn about specific requirements for B2B digital strategies, media industries, and retailing.
- **Decision Psychology:** Students can decide to deepen their knowledge on consumer psychology, including heuristics, nudging, and risk perception.
- **Market Research & Methods:** Students can learn about advanced empirical techniques to interpret customer attitudes and prepare for scientific research.

By integrating these multidisciplinary perspectives, the module enables students to implement sophisticated marketing instruments and critically evaluate current practices in digital business environments.

Workload

Total effort for 9 credit points: approx. 270 hours.
The exact distribution is done according to the credit points of the courses of the module.

5.49 Module: Digital Real Time Simulations for Energy Technologies [M-ETIT-106690]

Coordinators: Prof. Dr.-Ing. Giovanni De Carne
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Engineering and Informatics (Electrical Engineering and Information Technology)
Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each summer term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-113449	Digital Real Time Simulations for Energy Technologies		3 CP	De Carne

Assessment
The examination takes place in form of other types of examination. It consists of an assessment from an exercise on HiL and an oral overall examination (approx. 15 minutes) explaining the exercise results. The overall impression is evaluated.

Prerequisites
none

Competence Goal
To give bachelor's and master's degree students an overview of the need, concept, implementation and execution of Hardware in the Loop (HiL) testing. At the end of the course:

- The students will be able to understand the setup of HiL systems (Device Under Tests, Real Time simulator, I/O, power amps for PHiL, interfacing principles and techniques)
- The students will be able to devise HiL test cases, creating models with an understanding of the trade-off between simulation fidelity and computational resources.
- The students can interface Device Under Tests, running Real Time simulations and executing tests.
- The students are able to perform an independent HiL project and deliver HiL experimental results.

Content
Lesson 1: Introduction

- Overview of control system development process (V-cycle and variants).
- Real-time simulation concept – What does it mean simulating in real time?
- Basic concepts of Real Control Prototyping and Control-Hardware in the Loop.
- Basic concept of Power-HiL

Lesson 2: Introduction to real time simulation (1/2)

- Initial definitions for real time simulations: Wall-clock time, simulation time, hard real-time and soft real-time.
- Off-line simulation software solvers
- Numerical integration, DAE solvers, numerical stability. Complexities induced by switches.
- Distinctions between state-space based and nodal approach-based solvers for CPU
- Difference between CPU and FPGA modeling
- Numerical examples & practical implementation of solvers

Lesson 3: Modelling in real time: grid modelling

- Modeling for HiL, level of model detail vs computational performance
- problems with parallelization due to latencies in transferring data, I/O latency considerations.
- Introducing transmission line decoupling, stublines, ITM and other techniques
- Introducing the State-Space Nodal approach

Lesson 4: Modelling in real time: power electronics

- Modelling components for fast transients (e.g, switches), average models, full switching models
- Interface FPGA-based with CPU-based modelling
- Practical examples and applications

Lesson 5: Modelling in real time: multi-time-scale networks

- Modeling for HiL: different time-scales phasor/RMS, EMT, fast EMT on FPGA.
- Stability issues associated with multi-time-scale hybrid simulation (e.g. RMS/EMT or EMT on CPU and FPGA.
- Multi-time-scale simulation with phasor/EMT/FPGA

Lesson 6: Rapid Control Prototyping (RCP)

- Main Concept and benefits from RCP, Applications
- Generating a code from Simulink and implementation in the real time simulator
- FPGA-based RCP, Datalogging
- Demonstration in classroom

Lesson 7: Controller Hardware In the Loop

- Definition of HIL and testing opportunities
- Analog and digital I/O in real time simulators
- Designing an HiL test – identifying controller functionality to be tested, creating appropriate model and test sequences, identifying potential failures and testing in failure/off-design conditions, use of test automation

Lesson 8: Power Hardware In the Loop

- Introduction to PHIL
- PHIL equipment: power amplifiers. 2Q/4Q applications, specifying power amplifiers
- Interface algorithms between Real Time Simulation and the Device Under Test
- Stability vs. interface accuracy, current state of the art
- Impedance-based stability analysis

Module Grade Calculation
The module grade results of the assessment of an exercise and the oral exam. Details will be given during the lecture.

Workload
The workload includes (2 SWS):

attendance in lectures and exercises: 15*2 h = 30 h
preparation / follow-up: 15*4 h = 60 h
final project, preparation of and attendance in examination: 30 h
A total of 120 h = 4 CR

Recommendations

- Good knowledge of power electronics, linear control theory, and power systems is required.
- Good knowledge of Matlab/Simulink simulation environment is required.

M 5.50 Module: Digital Service Systems in Industry (WW4BWLKSR6) [M-WIWI-102808]

Coordinators: Prof. Dr. Wolf Fichtner
Prof. Dr. Stefan Nickel

Organisation: KIT Department of Business and Economics

Part of: Electives (Business Administration)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	2 terms	German	4	9

ID	Name	Term offered	Credits	Lecturers
Compulsory Elective Courses (Elective: 9 credits)				
T-WIWI-112757	Digital Services: Innovation & Business Models	ST	4,5 CP	Satzger
T-WIWI-107043	Liberalised Power Markets	WT	5,5 CP	Fichtner
T-WIWI-106200	Modeling and OR-Software: Advanced Topics	WT	4,5 CP	Nickel
T-WIWI-106563	Practical Seminar Digital Service Systems	Irreg.	4,5 CP	Satzger
T-WIWI-114109	Service Operations and Cyber Security	ST	4,5 CP	Mohr

Assessment
The assessment is carried out as partial exams (according to Section 4 (2), 1-3 SPO), whose sum of credits must meet the minimum requirement of credits of this module. The assessment procedures are described for each course of the module separately.
The overall grade of the module is the average of the grades for each course weighted by the credits and truncated after the first decimal

Prerequisites
This module can only be assigned as an elective module.

Competence Goal
Students

- understand the basics of the management of digital services applied on an industrial context
- gain an industry-specific insight into the importance and most relevant characteristics of information systems as key components of the digitalization of business processes, products and services
- are able to transfer and apply the models and methods introduced on practical scenarios and simulations.
- understand the control and optimization methods in the sector of service management and are able to apply them properly.

Content
This module aims at deepening the fundamental knowledge of digital service management in the industrial context. Various mechanisms and methods to shape and control connected digital service systems in different industries are discussed and demonstrated with real life application cases.

Additional Information
From summer semester 2023, the course Service Innovation will be offered with a revised course concept and content. The focus will be on the closer integration of the topics of service innovation and digitalization. Current foundational content (e.g., on service innovation challenges or human-centered innovation methods) will remain. New content will cover topics such as digital platforms and ecosystems, IoT and smart service innovation, and business models.

Workload
Total workload for 9 credit points: approx. 270 hours. The allocation is based on the credit points of the courses in the module.

Recommendations
None

5.51 Module: Digitalization in Facility Management (bauiEX406-DIGIFM) [M-BGU-105592]

Coordinators: Prof. Dr.-Ing. Kunibert Lennerts
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each winter term	2 terms	German	4	2

Election Notes
Module components in extent of min. 3 CP has to be selected.

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-BGU-108941	Digitalization in Facility and Real Estate Management	WT+ST	6 CP	Lennerts
Compulsory Elective (Elective: at most 2 items as well as at least 3 credits)				
T-BGU-111211	Energetic Refurbishment	WT+ST	1,5 CP	Lennerts, Schneider
T-BGU-111212	Facility and Real Estate Management II	WT+ST	1,5 CP	Lennerts
T-BGU-111921	Turnkey Construction	WT+ST	3 CP	Haghsheno

Assessment
- module component T-BGU-108941 with examination of another type according to § 4 Par. 2 No. 3

according to selected course:

- module component T-BGU-111211 with oral examination according to § 4 Par. 2 No. 2
- module component T-BGU-111212 with oral examination according to § 4 Par. 2 No. 2
- module component T-BGU-111921 with written examination according to § 4 Par. 2 No. 1

details about the learning controls, see module components

Prerequisites
none

Competence Goal
Students are able to name and classify fundamental knowledge of the Internet of Things (IoT), sensor networks, and building automation. They can critically analyze and evaluate digital technologies (including cloud storage, data protection, artificial intelligence, image processing, and object detection) in the context of facility and real estate management. Furthermore, they are able to recognize and quantify the optimization potential of automation and digitalization solutions in buildings, as well as designing and practically implementing sensor networks of a certain complexity.

They can describe and explain the theoretical foundations of sensor networks, building automation, and industrial building production. In addition, they are familiar with the basics of building physics, energy-related improvements to the building envelope and building services engineering, as well as the calculation of energy requirements and energy performance assessment according to the German Building Energy Act (GEG). They master outsourcing, procurement regulations, data acquisition, transition management, CAFM, and performance metrics for SLAs and KPIs in facility management.

Students can convey and apply execution planning for shell and finishing, construction engineering fundamentals, construction work, technical building equipment, and processes in turnkey construction (planning, permitting, acceptance, warranty).

Content

- Digitalisation in Facility and Real Estate Management: fundamentals and key concepts of digitalisation in facility and real estate management; practical applications of the Internet of Things (IoT) in building automation; design, setup, and integration of more complex sensor networks into existing FM processes; integration of sensor signals into FM processes; data transmission, cloud-based storage, and requirements for data protection and data security; introduction to methods of artificial intelligence; preparation of a semester-long project with colloquium.
- Facility and Real Estate Management II: outsourcing and procurement regulations in facility management, data acquisition, transition, CAFM in facility management, presentation of metrics for SLAs and KPIs and their digitalisation.
- Energy Refurbishment: fundamentals of building physics, energy-related improvement of the building envelope, improvement of building services engineering and economic efficiency, energy sources and local energy conversion (e.g., photovoltaics, geothermal energy, heat pumps, etc.), calculation of the energy demands of buildings, energy performance assessment of buildings according to GEG.
- Turnkey Construction: execution planning for shell and finishing, construction engineering fundamentals, construction execution and technical building equipment; various processes in turnkey construction (planning, permitting, acceptance, warranty).

Module Grade Calculation
grade of the module is CP weighted average of grades of the partial exams

Additional Information
none

Workload
contact hours (1 HpW = 1 h x 15 weeks):

- Digitalization in Facility and Real Estate Management lecture/exercise: 60 h

according to selected courses or examinations respectively:

- Energetic Refurbishment II lecture: 15 h
- Facility and Real Estate Management II lecture: 15 h
- Turnkey Construction lecture/exercise: 30 h

independent study:

- preparation and follow-up lecture/exercises Digitalization in Facility and Real Estate Management: 40 h
- preparation of project Digitalization in Facility and Real Estate Management, incl. report and presentation (partial examination, compulsory): 80 h

according to selected courses or examinations respectively:

- preparation and follow-up lectures Energetic Refurbishment II: 15 h
- examination preparation Energetic Refurbishment II (selectable partial examination): 15 h
- preparation and follow-up lectures Facility and Real Estate Management II: 15 h
- examination preparation Facility and Real Estate Management II (selectable partial examination): 15 h
- preparation and follow-up lecture/exercises Turnkey Construction: 30 h
- examination preparation Turnkey Construction (selectable partial examination): 30 h

total: 270 h

Recommendations
none

M 5.52 Module: Econometrics and Statistics I [M-WIWI-101638]

Coordinators: Prof. Dr. Melanie Schienle
Organisation: KIT Department of Business and Economics
Part of: Electives (Economics)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	1 term	German/English	4	6

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-WIWI-111388	Applied Econometrics	WT	4,5 CP	Krüger
Supplementary Courses (Elective: between 4,5 and 5 credits)				
T-WIWI-103064	Financial Econometrics	WT	4,5 CP	Schienle
T-WIWI-103126	Non- and Semiparametrics	Irreg.	4,5 CP	Schienle
T-WIWI-103127	Panel Data	ST	4,5 CP	Heller
T-WIWI-110868	Predictive Modeling	Irreg.	4,5 CP	Krüger
T-WIWI-111387	Probabilistic Time Series Forecasting Challenge	Irreg.	4,5 CP	Krüger
T-WIWI-103065	Statistical Modeling of Generalized Regression Models	WT	4,5 CP	Heller
T-WIWI-110939	Financial Econometrics II	ST	4,5 CP	Schienle

Assessment
The assessment is carried out as partial exams (according to Section 4(2), 1-3 of the examination regulation) of the single courses of this module, whose sum of credits must meet the minimum requirement of credits of this module. The examinations are offered every semester. Re-examinations are offered at every ordinary examination date. The assessment procedures are described for each course of the module separately.
The overall grade of the module is the average of the grades for each course weighted by the credits and truncated after the first decimal.

Prerequisites
The course ‘Applied Econometrics’ [2520020] is compulsory and must be completed if it has not already been successfully completed in one of the modules ‘Economics of Innovation and Growth’ or ‘Economics in a Connected World’. If the course ‘Applied Econometrics’ has already been completed in another module, the module cannot be chosen by the student. In this case, please contact the Examinations Office of the WIWI-Department, which will adjust the elective requirements in the module.

Competence Goal
The student shows an in depth understanding of advanced Econometric techniques suitable for different types of data.He/She is able to apply his/her theoretical knowledge to real world problems with the help of statistical software and to evaluate performance of different approaches based on statistical criteria.

Content
The courses of this module offer students a broad range of advanced Econometric techniques for state-of-the art data analysis.

Workload
The total workload for this module is approximately 270 hours.

M 5.53 Module: Econometrics and Statistics II [M-WIWI-101639]

Coordinators: Prof. Dr. Melanie Schienle
Organisation: KIT Department of Business and Economics
Part of: Electives (Statistics)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	1 term	German/English	4	7

Election Notes
Special note on elective regulations: The partial achievement T-WIWI-111388 "Applied Econometrics" is generally a mandatory requirement for this module.

Exception: If this partial achievement has already been successfully completed as part of the module M-WIWI-101638 "Econometrics and Statistics I", the requirement for this module is waived. In this case, any other partial achievements from the module's offering may be selected instead to reach the required number of credits (LP).

Important note on registration: As the automated verification in the Campus System cannot account for this specific exception, students are unable to register for alternative partial achievements independently. In this case, please contact the **Examination Office of the KIT Department of Economics and Management** directly for manual registration.

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-WIWI-111388	Applied Econometrics	WT	4,5 CP	Krüger
Compulsory Elective Courses (Elective: at least 1 item)				
T-WIWI-103064	Financial Econometrics	WT	4,5 CP	Schienle
T-WIWI-110939	Financial Econometrics II	ST	4,5 CP	Schienle
T-WIWI-103124	Multivariate Statistical Methods	Irreg.	4,5 CP	Grothe
T-WIWI-103126	Non- and Semiparametrics	Irreg.	4,5 CP	Schienle
T-WIWI-103127	Panel Data	ST	4,5 CP	Heller
T-WIWI-103128	Portfolio and Asset Liability Management	ST	4,5 CP	Safarian
T-WIWI-110868	Predictive Modeling	Irreg.	4,5 CP	Krüger
T-WIWI-111387	Probabilistic Time Series Forecasting Challenge	Irreg.	4,5 CP	Krüger
T-WIWI-103123	Advanced Statistics	WT	4,5 CP	Grothe
T-WIWI-103065	Statistical Modeling of Generalized Regression Models	WT	4,5 CP	Heller
T-WIWI-103129	Stochastic Calculus and Finance	WT	4,5 CP	Safarian

Assessment
The assessment is carried out as partial exams (according to Section 4(2), 1-3 of the examination regulation) of the single courses of this module, whose sum of credits must meet the minimum requirement of credits of this module. The examinations are offered every semester. Re-examinations are offered at every ordinary examination date. The assessment procedures are described for each course of the module separately.
The overall grade of the module is the average of the grades for each course weighted by the credits and truncated after the first decimal.

Prerequisites
The partial achievement *T-WIWI-111388 "Applied Econometrics"* is generally a mandatory requirement for this module.

Exception: If this partial achievement has already been successfully completed as part of the module *M-WIWI-101638 "Econometrics and Statistics I"*, the requirement for this module is waived. In this case, any other partial achievements from the module's offering may be selected instead to reach the required number of credits (LP).

Important note on registration: As the automated verification in the Campus System cannot account for this specific exception, students are unable to register for alternative partial achievements independently. In this case, please contact the **Examination Office of the KIT Department of Economics and Management** directly for manual registration.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module M-WIWI-101638 - Econometrics and Statistics I must have been started.

Competence Goal
The student shows an in depth understanding of advanced Econometric techniques suitable for different types of data. He/She is able to apply his/her theoretical knowledge to real world problems with the help of statistical software and to evaluate performance of different approaches based on statistical criteria.

Content
This modula builds on prerequisites acquired in Module *"Econometrics and Statistics I"*. The courses of this module offer students a broad range of advanced Econometric techniques for state-of-the art data analysis.

Workload
Total workload for 9 credit points: approx. 270 hours. The allocation is based on the credit points of the courses in the module. The workload for courses with 4.5 credits is approx. 135 hours. The total number of hours per course is calculated from the time required to attend the lectures and exercises, as well as the examination times and the time required to achieve the learning objectives of the module for an average student for an average performance.

5.54 Module: Economic Theory and its Application in Finance (WW4VWL14) [M-WIWI-101502]

Coordinators: Prof. Dr. Kay Mitusch
Organisation: KIT Department of Business and Economics
Part of: Business and Economics
Electives (Economics)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	1 term	English	4	6

ID	Name	Term offered	Credits	Lecturers
Compulsory Elective Courses (Elective: 1 item)				
T-WIWI-102609	Advanced Topics in Economic Theory	Irreg.	4,5 CP	Brumm, Mitusch
T-WIWI-102861	Advanced Game Theory	WT	4,5 CP	Ehrhart, Puppe, Reiß
Supplementary Courses (Elective:)				
T-WIWI-113469	Advanced Corporate Finance	ST	4,5 CP	Ruckes
T-WIWI-102647	Asset Pricing	ST	4,5 CP	Ruckes, Uhrig-Homburg
T-WIWI-109050	Corporate Risk Management	Irreg.	4,5 CP	Ruckes
T-WIWI-102623	Financial Intermediation	WT	4,5 CP	Ruckes

Assessment
The assessment is carried out as partial exams (according to Section 4(2), 1 or 2 of the examination regulation) of the single courses of this module, whose sum of credits must meet the minimum requirement of credits of this module. The exams are offered at the beginning of the recess period about the subject matter of the latest held lecture. Re-examinations are offered at every ordinary examination date. The assessment procedures are described for each course of the module separately. The overall grade for the module is the average of the grades for each course weighted by the credits and truncated after the first decimal.

Prerequisites
One of the courses T-WIWI-102861 "Advanced Game Theory" and T-WIWI-102609 "Advanced Topics in Economic Theory" is compulsary.

Competence Goal
The students

- have learnt the methods of formal economic modeling, particularly of General Equilibrium Theory and contract theory
- will be able to apply these methods to the topics in Finance, specifically the areas of financial markets and institutions and corporate finance
- have gained many useful insights into the relationship between firms and investors and the functioning of financial markets

Content
The mandatory course "Advanced Topics in Economic Theory" is devoted in equal parts to General Equilibrium Theory and to contract theory. The course "Asset Pricing" will apply techniques of General Equilibrium Theory to valuation of financial assets. The courses "Corporate Financial Policy" and "Finanzintermediation" will apply the techniques of contract theory to issues of corporate finance and financial institutions.

Workload
Total workload for 9 credit points: approx. 270 hours

The exact distribution is based on the credit points of the courses in the module.

M 5.55 Module: Economics [M-WIWI-104908]

Coordinators: Professorenschaft des Fachbereichs Volkswirtschaftslehre
Organisation: KIT Department of Business and Economics
Part of: Additional Examinations

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	pass/fail	Each term	1 term	German	4	7

ID	Name	Term offered	Credits	Lecturers
Elective Courses on Economics (Elective: at most 45 credits)				
T-WIWI-102861	Advanced Game Theory	WT	4,5 CP	Ehrhart, Puppe, Reiß
T-WIWI-102609	Advanced Topics in Economic Theory	Irreg.	4,5 CP	Brumm, Mitusch
T-WIWI-114598	Digital Markets and Mechanisms	ST	4,5 CP	Rosar
T-WIWI-109194	Dynamic Macroeconomics	WT	4,5 CP	Brumm
T-WIWI-102892	Economics and Behavior	WT	4,5 CP	Mill
T-WIWI-102877	Introduction to Public Finance	WT	4,5 CP	Wigger
T-WIWI-114622	Introduction to Econometrics	ST	5 CP	Schienle
T-WIWI-102850	Introduction to Game Theory	ST	4,5 CP	Puppe, Reiß
T-WIWI-103213	Basic Principles of Economic Policy	see notes	4,5 CP	Ott
T-WIWI-112816	Growth and Development	WT	4,5 CP	Ott
T-WIWI-111304	Fundamentals of National and International Group Taxation	ST	4,5 CP	Wigger
T-WIWI-105781	Incentives in Organizations	ST	4,5 CP	Nieken
T-WIWI-102844	Industrial Organization	Irreg.	4,5 CP	Reiß
T-WIWI-112722	Introduction to Digital Economics	WT+ST	6 CP	Brumm, Reiß
T-WIWI-109121	Macroeconomic Theory	WT	4,5 CP	Brumm
T-WIWI-113264	Matching Theory	see notes	4,5 CP	Puppe
T-WIWI-114054	Methods in Economics	ST	1,5 CP	Ott
T-WIWI-102739	Public Revenues	ST	4,5 CP	Wigger
T-WIWI-102862	Predictive Mechanism and Market Design	Irreg.	4,5 CP	Reiß
T-WIWI-102740	Public Management	WT	4,5 CP	Wigger
T-WIWI-102712	Regulation Theory and Practice	see notes	4,5 CP	Mitusch
T-WIWI-102789	Seminar in Economic Policy	WT+ST	3 CP	Ott
T-WIWI-103487	Seminar in Economics (Bachelor)	WT+ST	3 CP	Professorenschaft des Fachbereichs Volkswirtschaftslehre
T-WIWI-103478	Seminar in Economics A (Master)	WT+ST	3 CP	Professorenschaft des Fachbereichs Volkswirtschaftslehre
T-WIWI-103477	Seminar in Economics B (Master)	WT+ST	3 CP	Professorenschaft des Fachbereichs Volkswirtschaftslehre
T-WIWI-102859	Social Choice Theory	ST	4,5 CP	Puppe
T-WIWI-103107	Spatial Economics	WT	4,5 CP	Ott
T-WIWI-113147	Telecommunications and Internet – Economics and Policy	WT	4,5 CP	Mitusch
T-WIWI-102863	Topics in Experimental Economics	Irreg.	4,5 CP	Reiß
T-WIWI-100007	Transport Economics	ST	4,5 CP	Mitusch, Szimba
T-WIWI-102616	Environmental and Resource Policy	ST	4 CP	Walz
T-WIWI-102615	Environmental Economics and Sustainability	WT	3 CP	Walz

ID	Name	Term offered	Credits	Lecturers
T-WIWI-102708	Economics I: Microeconomics	WT	5 CP	Puppe, Reiß
T-WIWI-102709	Economics II: Macroeconomics	ST	5 CP	Wigger
T-WIWI-100005	Competition in Networks	WT	4,5 CP	Mitusch
T-WIWI-102610	Welfare Economics	see notes	4,5 CP	Puppe

Prerequisites
None

5.56 Module: Economics in a Connected World [M-WIWI-107010]

Coordinators: Prof. Dr. Ingrid Ott
Organisation: KIT Department of Business and Economics
Part of: Business and Economics
Electives (Economics)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-WIWI-103107	Spatial Economics	WT	4,5 CP	Ott
Compulsory Elective Courses (Elective: at most 1 item)				
T-WIWI-102609	Advanced Topics in Economic Theory	Irreg.	4,5 CP	Brumm, Mitusch
T-WIWI-111388	Applied Econometrics	WT	4,5 CP	Krüger
T-WIWI-109194	Dynamic Macroeconomics	WT	4,5 CP	Brumm
T-WIWI-112822	Economics of Innovation	ST	4,5 CP	Ott
T-WIWI-112816	Growth and Development	WT	4,5 CP	Ott
T-WIWI-113147	Telecommunications and Internet – Economics and Policy	WT	4,5 CP	Mitusch

Assessment
The assessment is carried out as partial exams according to § 4 paragraph 2 Nr. 1 – Nr. 3 SPO of the examination regulation of the core course and further single courses of this module, whose sum of credits must meet the minimum requirement of credits of this module.

The overall grade of the module is the average of the grades for each course, weighted by the credits and truncated after the first decimal.

Prerequisites
None

Competence Goal

- Theoretical Understanding of Spatial and Network Economics:
  - develop a deep understanding of the economic forces shaping spatial distribution, locational choice, trade flows, and urban development in both physical and digital landscapes.
  - understand how processes of spatial concentration result from the interplay of agglomeration and dispersion forces.
- Equilibrium Analysis and Stability:
  - gain expertise in deriving economic equilibria and analyzing their stability properties
  - understand and identify bifurcation points, exploring their implications for dynamic changes in economic systems.
- Quantitative and Analytical Skills:
  - acquire advanced skills in applying mathematical and computational tools to model spatial and connected economics.
  - use simulations and calibrations to explore dynamic economic systems and assess the implications of policy interventions.
- Programming and Empirical Analysis:
  - build competence in programming tools (e.g., Python, MATLAB, or R) to implement economic models and analyze results.
  - use econometric methods to evaluate and interpret real-world data related to trade, locational choice, and infrastructure development.

Upon completion, these competence goals aim to prepare students for advanced research, policy-making, and strategic decision-making in economics, business engineering, and digital economics, enabling them to tackle real-world problems with a robust theoretical and quantitative foundation.

Content

This module explores the interplay of spatial, digital, and economic networks, focusing on trade, locational choice, and the role of physical and digital infrastructure in shaping economic outcomes. It combines theoretical and applied approaches, equipping students with advanced analytical tools and insights into interconnected economic systems. It investigates the economic forces shaping the spatial distribution of resources, firms, and individuals, with a focus on locational choice, trade flows, and urban development in physical landscapes.

A strong focus is laid on the derivation of economic equilibria, with particular attention to their stability properties. Stability analyses include the identification and exploration of bifurcation points to understand dynamic changes in economic systems. Furthermore, decentralized decisions are critically evaluated using the normative framework of social welfare, offering insights into trade-offs and efficiency in economic systems.

Workload

Total workload for 9 credit points: approx. 270 hours

The exact distribution is based on the credit points of the courses in the module.

Recommendations

Sound understanding of theoretical foundations in microeconomic theory, macroeconomic theory, and statistics according to international standards of a bachelor's degree in Economics, Business Administration or similar disciplines. Basic knowledge in programming (R, Python) is required.

M 5.57 Module: Economics of Innovation and Growth [M-WIWI-107011]

Coordinators: Prof. Dr. Ingrid Ott
Organisation: KIT Department of Business and Economics
Part of: Business and Economics
Electives (Economics)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Compulsory Elective Courses (Elective: at least 1 item)				
T-WIWI-112822	Economics of Innovation	ST	4,5 CP	Ott
T-WIWI-112816	Growth and Development	WT	4,5 CP	Ott
Supplementary Courses (Elective:)				
T-WIWI-102609	Advanced Topics in Economic Theory	Irreg.	4,5 CP	Brumm, Mitusch
T-WIWI-111388	Applied Econometrics	WT	4,5 CP	Krüger
T-WIWI-109194	Dynamic Macroeconomics	WT	4,5 CP	Brumm
T-WIWI-114054	Methods in Economics	ST	1,5 CP	Ott
T-WIWI-102789	Seminar in Economic Policy	WT+ST	3 CP	Ott

Assessment
The assessment is carried out as partial exams according to § 4 paragraph 2 Nr. 1 – Nr. 3 SPO of the examination regulation of the core course and further single courses of this module, whose sum of credits must meet the minimum requirement of credits of this module.

The overall grade of the module is the average of the grades for each course, weighted by the credits and truncated after the first decimal.

Prerequisites
None

Competence Goal

- Theoretical and Analytical Competences:
    - understand the key mechanisms underlying micro-based macro models.
      gain the ability to analyze formal theoretical models in economics, focusing on innovation, growth and development.
    - understand the identification of equilibria, assessing their stability characteristics, including their graphical derivations based on phase-diagrams.
    - understand the normative perspective of the models: get competence in evaluating decentralized decision-making processes based on the concept of social welfare and
    - understanding associated trade-offs between centralized and decentralized solutions.
- Quantitative and Computational Competences:
    - gain proficiency in calibrating models with real-world data and simulating outcomes to analyze dynamic economic systems.
    - gain experience in applying programming tools (e.g., Python, MATLAB, or R) to implement economic models, run simulations, and interpret computational results.
    - gain the ability to apply econometric techniques to evaluate the effects of physical and digital networks on trade, locational choices, and infrastructure investments.
- Regarding Content:
    - know the basic techniques for analyzing static and dynamic optimization models that are applied in the context of microeconomic and macroeconomic theories.
    - understand the important role of innovation to the overall economic growth and prosperity.
    - identify the importance of alternative incentive mechanisms for the emergence and dissemination of innovations.
    - explain, in which situations market interventions by the state, for example taxes and subsidies, can be legitimized, and evaluate them in the light of economic welfare.

Upon completion, students will have gained a strong theoretical foundation and practical skills to evaluate innovation and growth processes that are incentive-based, micro-founded while addressing the macroeconomic perspective. The module prepares students for advanced research or professional applications in academia, policy, and practice.

Content

This module provides a comprehensive introduction to the economic foundations, processes, and quantitative methods that drive innovation and growth and thus shape the development of nations over time. Based on empirical recurring patterns (so-called stylized facts), the overall goal is to gain a sound understanding of the determinants of long-run prosperity, structural change, and the role of government.

The module covers key theoretical concepts, discusses the role of policy frameworks and instruments, and applies quantitative and econometric techniques. Some programming techniques are also applied (R, Python). The courses enable students to understand and analyze economic dynamics from an aggregate perspective, although the theories are consequently micro-founded. Theoretical foundations are indispensable tools for theory building and serve as a normative compass for economic policy making, especially in addressing complex societal challenges.

Workload

Total workload for 9 credit points: approx. 270 hours.

The exact distribution is based on the credit points of the courses in the module.

Recommendations

Sound understanding of theoretical foundations in microeconomic theory, macroeconomic theory, and statistics according to international standards of a bachelor's degree in Economics, Business Administration or similar disciplines. Basic knowledge in programming (R, Python) is required.

M 5.58 Module: eEnergy: Markets, Services and Systems [M-WIWI-103720]

Coordinators: Prof. Dr. Christof Weinhardt
Organisation: KIT Department of Business and Economics
Part of: Electives (Business Administration)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	1 term	German	4	3

ID	Name	Term offered	Credits	Lecturers
Compulsory Elective Courses (Elective: at least 9 credits)				
T-WIWI-107501	Energy Market Engineering	ST	4,5 CP	Weinhardt
T-WIWI-107503	Energy Networks and Regulation	WT	4,5 CP	Weinhardt
T-WIWI-107504	Smart Grid Applications	see notes	4,5 CP	Weinhardt
T-WIWI-113726	Special Topics in Information Systems	WT+ST	4,5 CP	Weinhardt

Assessment
The assessment is carried out as partial exams (according to Section 4 (2), 1-3 SPO) of single courses of this module, whose sum of credits must meet the minimum requirement of credits of this module. The assessment procedures are described for each course of the module separately. The overall grade of the module is the average of the grades for each course weighted by the credits and truncated after the first decimal.

Prerequisites
None.

Competence Goal
The student

- is aware of design options for energy and especially electricity markets and can derive implications for the market results from the market design,
- knows about current trends regarding the Smart Grid and understands affiliated modelling aproaches,
- can evaluate business models of electricity grids according to the regulation regime
- is prepared for scientific contributions in the field of energy system analysis.

Content
The module conveys scientific and practical knowledge to analyse energy markets and according business models. To do so the scientific discussion on energy market designs is evaluated and analysed. Different energy market models are presented and their design implications are evaluated. Furthermore, the electricity system is analysed with regards to being a network industry and resulting regulation and business models are discussed. Besides these traditional areas of energy economics we will look at methods and models of digitalisation in the energy sector.

Additional Information
The lecture Smart Grid Applications will be available starting in the winter term 2018/19.

Workload
The total workload for this module is approx. 270 hours (9 CP). The allocation is based on the credit points of the courses in the module. The workload for courses with 4.5 CP is approx. 135 hours.

The total number of hours per course results from the time required to attend the lectures and exercises, as well as the examination times and the time required to achieve the qualification objectives of the module for an average student for an average performance.

M 5.59 Module: Electromagnetic Energy in Process Engineering [M-CIWVT-107566]

Coordinators: Prof. Dr.-Ing. Roland Dittmeyer
Dr. Alexander Navarrete Munoz

Organisation: KIT Department of Chemical and Process Engineering

Part of: Engineering and Informatics (Chemical and Process Engineering)
Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-CIWVT-114829	Electromagnetic Energy in Process Engineering - Oral Exam		5 CP	Navarrete Munoz
T-CIWVT-114830	Practical on Electromagnetic Energy in Process Engineering		1 CP	Navarrete Munoz

Assessment
The learning control constist of two module components:

- completed coursework: Practical
- oral exam lasting approx. 30 minutes

Prerequisites
None.

Competence Goal
This elective master course will prepare the students from Maxwell's equations to applications of electromagnetic heating in one semester. The studnets will be able to understand and grasp the fundamentals of operations with microwaves and electromagnetic activation as well as operate KIT's plasma rigs. They will also have the opportunity to test future process concepts with electromagnetic modelling and basic TEA.

Content
The lectures will combine fundamentals, research and real world applications.

1. Process Engineering towards Net-Zero. The need for renewable heat and green activation of processes.
2. Electromagnetic Fundamentals I. From Faraday to Maxwell equations in planar waves.
3. Electromagnetic Fundamentals II. Wave-matter interaction.
4. Induction heating fundamentals.
5. Dielectric and RF heating in food applications.
6. MW and RF heating of liquids and slurries.
7. Materials processing I. Applications involving metals
8. Materials processing II. Applications involving ceramics
9. Plasma fundamentals for chemical and process engineers
10. Plasma applications I. Gas based conversions (CO2 and N2 conversion).
11. Plasma applications II. Liquid based conversions (aqueous, fuel reforming and plasma assisted combustion)
12. Plasma applications III. Solid based conversions (catalytic surface activation, pyrolysis and sintering)
13. Process and energy integration, and Grid flexibility.
14. Review of techno-economic and sustainability analysis of electromagnetic based processes.

Exercises will take place every second week.

1. Skin-depth calculator and penetration-dept
2. Spreadsheet sizing of RF/MW heating device
3. The case of essential oils extraction. Heat and mass transfer aspects
4. Electromagnetic modeling in Comsol.
5. Conversion Vs power absorbed during Microwave Plasma splitting of CO2.
6. Energy-yield analysis for plasma based treatment of aqueous solutions.
7. Group TEA sheet for a plasma based process

Module Grade Calculation
The module grade is the grade of the oral exam.

Workload
Attendance time:

- Lecture: 30 hrs
- Exercise: 15 hrs
- Practical: 6 hrs

Selbststudium:

- Preparation and wrap-up of lectures and exercises: 65 hrs
- Preparation and wrap-up of the practical: 24 hrs
- Exam preparation: 40 hrs

Literature

1. Fridman, A. “Plasma Chemistry” (Cambridge, 2008)
2. Lieberman, M. & Lichtenberg, A. “Principles of Plasma Discharges and Materials Processing” (3 ed., Wiley, 2021)
3. Metaxas, A. C. & Meredith, R. J. “Industrial Microwave Heating” (IEE, 1983)
4. Metaxas, A. C. “Foundations of Electroheat” (Wiley, 1996)

M 5.60 Module: Electronic Markets (WW4BWLISM2) [M-WIWI-101409]

Coordinators: Prof. Dr. Andreas Geyer-Schulz
Organisation: KIT Department of Business and Economics
Part of: Electives (Business Administration)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	2 terms	German/English	4	9

ID	Name	Term offered	Credits	Lecturers
Compulsory Elective Courses (Elective: at least 9 credits)				
T-WIWI-102762	Business Dynamics	WT	4,5 CP	Geyer-Schulz, Glenn
T-WIWI-112823	Platform & Market Engineering: Commerce, Media, and Digital Democracy	ST	4,5 CP	Weinhardt
T-WIWI-105946	Price Management	ST	4,5 CP	Geyer-Schulz, Glenn
T-WIWI-113147	Telecommunications and Internet – Economics and Policy	WT	4,5 CP	Mitusch

Assessment
The assessment is carried out as partial exams (according to Section 4(2) of the examination regulation) of the single courses of this module, whose sum of credits must meet the minimum requirement of credits of this module. The assessment procedures are described for each course of the module seperately.

The overall grade of the module is the average of the grades for each course weighted by the credits and truncated after the first decimal.

Prerequisites
None

Competence Goal
The student

- knows coordination and motivation methods and analyzes them regarding their efficiency,
- classifies markets and describes the roles of the participants in a formal way,
- knows the conditions for market failure and knows and develops countermeasures,
- knows institutions and market mechanisms, their fundamental theories and empirical research results,
- knows the design criteria of market mechanisms and a systematical approach for creating new markets,
- models, analyzes and optimizes the structure and dynamics of complex business applications.

Content
What are the conditions that make electronic markets develop, and how can one analyze and optimize such markets?

In this module, the selection of the type of organization as an optimization of transaction costs is treated. Afterwards, the efficiency of electronic markets (price, information and allocation efficiency) as well as reasons for market failure are described. Finally, motivational issues like bounded rationality and information asymmetries (private information and moral hazard), as well as the development of incentive schemes, are presented. Regarding the market design, especially the interdependencies of market organization, market mechanisms, institutions and products are described, and theoretical foundations are lectured.

Electronic markets are dynamic systems that are characterized by feedback loops between many different variables. By means of the tools of business dynamics, such markets can be modelled. Simulations of complex systems allow the analysis and optimization of markets, business processes, policies, and organizations.

Topics include

- classification, analysis, and design of markets
- simulation of markets
- auction methods and auction theory
- automated negotiations
- nonlinear pricing
- continuous double auctions
- market-maker, regulation, control

Workload
The total workload for this module is approx. 270 hours (9 credits). The distribution is based on the credit points of the courses in the module. The workload for courses with 4.5 credits is approx. 135 hours.

The total number of hours per course is calculated from the time required to attend the lectures and exercises, as well as the examination times and the time required for an average student to achieve the learning objectives of the module for an average performance.

Recommendations
None

M 5.61 Module: Emphasis in Informatics (WW4INFO1) [M-WIWI-101628]

Coordinators: Dr.-Ing. Tobias Käfer
Prof. Dr.-Ing. Sanja Lazarova-Molnar
Prof. Dr. Andreas Oberweis
Prof. Dr. Harald Sack
Prof. Dr. Melanie Volkamer
Prof. Dr.-Ing. Johann Marius Zöllner

Organisation: KIT Department of Business and Economics

Part of: Electives (Informatics)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	1 term	German/English	4	23

ID	Name	Term offered	Credits	Lecturers
Compulsory Elective Area (Elective: between 1 and 3 items)				
T-WIWI-113363	Collective Perception in Autonomous Driving	ST	4,5 CP	Vinel
T-WIWI-102680	Computational Economics	WT	4,5 CP	Shukla
T-WIWI-112690	Cooperative Autonomous Vehicles	ST	4,5 CP	Vinel
T-WIWI-110346	Supplement Enterprise Information Systems	WT+ST	4,5 CP	Oberweis
T-WIWI-110372	Supplement Software- and Systemsengineering	WT+ST	4,5 CP	Oberweis
T-WIWI-114951	Advanced Topics in Digital Twins	Once	4,5 CP	Lazarova-Molnar
T-WIWI-113059	Human Factors in Autonomous Driving	WT	4,5 CP	Vinel
T-WIWI-109270	Human Factors in Security and Privacy	Irreg.	4,5 CP	Volkamer
T-WIWI-113968	Management of IT-Projects	ST	4,5 CP	Alpers
T-WIWI-102666	Knowledge Discovery	WT	4,5 CP	Käfer
T-WIWI-114863	Knowledge-Driven Artificial Intelligence	ST	4,5 CP	Sack
T-WIWI-106340	Machine Learning 1 - Basic Methods	WT	5 CP	Zöllner
T-WIWI-106341	Machine Learning 2 - Advanced Methods	ST	5 CP	Zöllner
T-WIWI-112685	Modeling and Simulation	ST	4,5 CP	Lazarova-Molnar
T-WIWI-102697	Business Process Modelling	WT	4,5 CP	Oberweis
T-WIWI-102679	Nature-Inspired Optimization Methods	ST	4,5 CP	Shukla
T-WIWI-109799	Process Mining	ST	4,5 CP	Schreiber
T-WIWI-110848	Semantic Web Technologies	ST	4,5 CP	Käfer
T-WIWI-102895	Software Quality Management	ST	4,5 CP	Oberweis
Seminars and Advanced Labs (Elective: at most 1 item)				
T-WIWI-110548	Advanced Lab Informatics (Master)	WT+ST	4,5 CP	Professorenschaft des Instituts AIFB
T-WIWI-112914	Advanced Lab Realization of Innovative Services (Master)	WT+ST	4,5 CP	Oberweis
T-WIWI-108439	Advanced Lab Security, Usability and Society	WT+ST	4,5 CP	Volkamer
T-WIWI-109985	Project Lab Cognitive Automobiles and Robots	WT	5 CP	Zöllner
T-WIWI-109983	Project Lab Machine Learning	ST	5 CP	Zöllner

Assessment
The assessment is carried out as partial exams (according to Section 4(2) of the examination regulation) of the single courses of this module, whose sum of credits must meet the minimum requirement of credits of this module. For passing the module exam in every singled partial exam the respective minimum requirements has to be achieved.

The examinations are offered every semester. Re-examinations are offered at every ordinary examination date. The assessment procedures are described for each course of the module separately.

When every singled examination is passed, the overall grade of the module is the average of the grades for each course weighted by the credits and truncated after the first decimal.

Prerequisites
None.

Competence Goal
The student

- has the ability to master methods and tools in a complex discipline and to demonstrate innovativeness regarding the methods used,
- knows the principles and methods in the context of their application in practice,
- is able to grasp and apply the rapid developments in the field of computer science, which are encountered in work life, quickly and correctly, based on a fundamental understanding of the concepts and methods of Informatics,
- is capable of finding and defending arguments for solving problems.

Content
The thematic focus will be based on the choice of courses in the areas of Applied Technical Cognitive Systems, Business Information Systems, Critical Information Infrastructures, Information Service Engineering, Security - Usability - Society or Web Science.

Workload
The total workload for this module is approximately 270 hours.

M 5.62 Module: Energy and Process Technology I (WW4INGMBITS1) [M-MACH-101296]

Coordinators: Prof. Dr. Dr. Michael Fischlschweiger
Organisation: KIT Department of Mechanical Engineering

Part of: Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Level	Version
9 CP	graded	Each winter term	1 term	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-MACH-102211	Energy and Process Technology I	WT	9 CP	Bauer, Fischlschweiger, Schwitzke, Velji

Assessment
The assessment is carried out as partial exams (according to Section 4 (2), 13 SPO) of the courses of this module, whose sum of credits must meet the requirement of credits of this module. The
assessment procedures are described for each course of the module separately.
The overall grade of the module is the average of the grades for each course weighted by the credits
and truncated after the first decimal.

Prerequisites
None

Competence Goal
In this modul students achieve a basic understanding of the technical properties of energy conversion processes and machines.

Content
Energy and Process Technology 1:

1. thermodynamic basics and cycle processes (ITT)
2. basics of piston engines (IFKM)
3. basics of turbomachines (FSM)
4. basics of thermal turbomachines (ITS)

Additional Information
All lectures and exams are hold in German only.

M 5.63 Module: Energy and Process Technology II (WW4INGMBITS2) [M-MACH-101297]

Coordinators: Prof. Dr. Dr. Michael Fischlschweiger
Organisation: KIT Department of Mechanical Engineering

Part of: Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Level	Version
9 CP	graded	Each summer term	1 term	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-MACH-102212	Energy and Process Technology II	ST	9 CP	Fischlschweiger, Schwitzke

Assessment
The assessment is carried out as partial exams (according to Section 4 (2), 13 SPO) of the courses of this module, whose sum of credits must meet the requirement of credits of this module. The
assessment procedures are described for each course of the module separately.
The overall grade of the module is the average of the grades for each course weighted by the credits
and truncated after the first decimal.

Prerequisites
None

Competence Goal
In this modul students achieve the ability to evaluate solitary and interconnected energy systems with respect to societal and economical aspects

Content
Energy and Process Technology 2:

1. basics in combustion and pollutant formation (ITT)
2. technical realisation and application of piston engines (IFKM) fluid flow engines (FSM) and thermal turbomachines (ITS)
3. technical aspects of energy supply systems and networks (ITS)

Additional Information
All lectures and exams are hold in German only.

M 5.64 Module: Energy Economics and Energy Markets (WW4BWLIIP4) [M-WIWI-101451]

Coordinators: Prof. Dr. Wolf Fichtner
Organisation: KIT Department of Business and Economics
Part of: Electives (Business Administration)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	1 term	German/English	4	9

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-WIWI-107043	Liberalised Power Markets	WT	5,5 CP	Fichtner
Supplementary Courses (Elective:)				
T-WIWI-114980	Behavioral Dimensions of Energy Transitions	ST	3,5 CP	Fichtner
T-WIWI-107501	Energy Market Engineering	ST	4,5 CP	Weinhardt
T-WIWI-112151	Energy Trading and Risk Management	ST	3,5 CP	N.N.
T-WIWI-108016	Simulation Game in Energy Economics	ST	3,5 CP	Genoese
T-WIWI-107446	Quantitative Methods in Energy Economics	WT	3,5 CP	Plötz
T-WIWI-102712	Regulation Theory and Practice	see notes	4,5 CP	Mitusch

Assessment
The assessment is carried out as partial written exams (according to Section 4(2), 1 of the examination regulation) of the single courses of this module, whose sum of credits must meet the minimum requirement of credits of this module. The examinations take place every semester. Re-examinations are offered at every ordinary examination date.The assessment procedures are described for each course of the module seperately.

The overall grade of the module is the average of the grades for each course weighted by the credits and truncated after the first decimal.

Prerequisites
The lecture Liberalised Power Markets has to be examined.

Competence Goal
The student

- gains detailed knowledge about the new requirements of liberalised energy markets,
- describes the planning tasks on the different energy markets,
- knows solution approaches to respective planning tasks.

Content

- Liberalised Power Markets: The European liberalisation process, energy markets, pricing, market failure, investment incentives, market power
- *Energy Trading and Risk Management:* trade centres, trade products, market mechanisms, position and risk management
- *Simulation Game in Energy Economics*: Simulation of the German electricity system
- *Behavioral Dimensions of Energy Transitions*: Explores how energy consumers make decisions regarding their energy use and how these decisions can be influenced by psychological and economic factors, such as financial incentives; examines the social acceptance of energy technologies and policies, and what determines different levels of acceptance.
- *Quantitative Methods in Energy Economics*: The aim of the course is to help the students understand real-world usage of quantitative methods of energy economics and apply these methods

Workload
The total workload for this module is approx. 270 hours (9 credits). The allocation is based on the credit points of the courses in the module. The workload for courses with 3.5 credits is approx. 105 hours, for courses with 5.5 credits approx. 165 hours.
The total number of hours per course is calculated from the time required to attend the lectures and exercises, as well as the examination times and the time required for an average student to achieve the learning objectives of the module for an average performance.

Recommendations

The courses are conceived in a way that they can be attended independently from each other. Therefore, it is possible to start the module in winter and summer term.

M 5.65 Module: Energy Economics and Technology (WW4BWLIIP5) [M-WIWI-101452]

Coordinators: Prof. Dr. Wolf Fichtner
Organisation: KIT Department of Business and Economics
Part of: Electives (Business Administration)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	1 term	German/English	4	5

ID	Name	Term offered	Credits	Lecturers
Compulsory Elective Courses (Elective: at least 9 credits)				
T-WIWI-102793	Efficient Energy Systems and Electric Mobility	ST	3,5 CP	Jochem
T-WIWI-102650	Energy and Environment	ST	3,5 CP	Karl
T-WIWI-113073	Machine Learning and Optimization in Energy Systems	WT	4 CP	Fichtner
T-WIWI-107464	Smart Energy Infrastructure	WT	5,5 CP	Ardone, Pustisek
T-WIWI-102695	Heat Economy	ST	3,5 CP	Fichtner

Assessment
The assessment is carried out as partial written exams (according to Section 4(2), 1 of the examination regulation) of the single courses of this module, whose sum of credits must meet the minimum requirement of credits of this module. The examinations take place every semester. Re-examinations are offered at every ordinary examination date.The assessment procedures are described for each course of the module seperately.

The overall grade of the module is the average of the grades for each course weighted by the credits and truncated after the first decimal.

Prerequisites
None

Competence Goal
The student

- gains detailed knowledge about present and future energy supply technologies (focus on final energy carriers electricity and heat),
- knows the techno-economic characteristics of plants for energy provision, for energy transport as well as for energy distribution and demand,
- is able to assess the environmental impact of these technologies.

Content

- **Heat Economy**: district heating, heating technologies, reduction of heat demand, statutory provisions
- **Energy and Environment***:* emission factors, emission reduction measures, environmental impact
- **Machine Learning and Optimization in Energy Systems**: Understanding and Applying the core principles of the most commonly used machine learning and optimization techniques within the energy sector

Workload
The total workload for this module is approx. 270 hours (9 credits). The allocation is based on the credit points of the courses in the module. The workload for courses with 3,5 credits is approx. 105 hours, and for courses with 5,5 credits approx. 165 hours.

The total number of hours per course is calculated from the time required to attend the lectures and exercises, as well as the examination times and the time required for an average student to achieve the learning objectives of the module for an average performance.

M 5.66 Module: Engineering Sciences [M-WIWI-104907]

Coordinators: Professorenschaft des Fachbereichs Ingenieurwissenschaften
Organisation: KIT Department of Business and Economics
Part of: Additional Examinations

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	pass/fail	Each term	1 term	German	4	13

ID	Name	Term offered	Credits	Lecturers
Elective Courses on Engineering Sciences (Elective: at most 60 credits)				
T-MACH-105173	Analysis of Exhaust Gas and Lubricating Oil in Combustion Engines	ST	4 CP	Gohl
T-MACH-102176	Current Topics on BioMEMS	WT+ST	4 CP	Guber
T-MACH-115042	Applied Data Science and Machine Learning in Engineering	ST	4 CP	Meyer
T-MACH-102141	Constitution and Properties of Wearresistant Materials	ST	4 CP	Ulrich
T-MACH-105150	Constitution and Properties of Protective Coatings	WT	4 CP	Ulrich
T-MACH-102160	Selected Applications of Technical Logistics	ST	4 CP	Mittwollen
T-MACH-108945	Selected Applications of Technical Logistics - Project	ST	2 CP	Mittwollen
T-MACH-102203	Automotive Engineering I	WT	6 CP	Gießler
T-MACH-114435	Automotive Engineering II	ST	4,5 CP	Gießler
T-MACH-102143	Rail System Technology	WT+ST	9 CP	Cichon
T-MACH-106424	Rail System Technology	WT+ST	4 CP	Cichon
T-BGU-101691	Construction Technology	WT+ST	6 CP	Haghsheno
T-BGU-103427	Site Management	ST	1,5 CP	Haghsheno
T-BGU-106613	Design Basics in Highway Engineering	WT+ST	3 CP	Zimmermann
T-BGU-101797	Methods and Models in Transportation Planning	WT+ST	3 CP	Vortisch
T-BGU-101860	Special Topics in Highway Engineering and Environmental Impact Assessment	WT+ST	3 CP	Zimmermann
T-MACH-105184	Fuels and Lubricants for Combustion Engines	WT	4 CP	Kehrwald
T-MACH-100966	BioMEMS - Microsystems Technologies for Life-Sciences and Medicine I	WT	4 CP	Guber
T-MACH-100967	BioMEMS - Microsystems Technologies for Life-Sciences and Medicine II	ST	4 CP	Guber
T-MACH-100968	BioMEMS - Microsystems Technologies for Life-Sciences and Medicine III	ST	4 CP	Guber
T-ETIT-104572	Business Innovation in Optics and Photonics	WT	4 CP	Nahm
T-MACH-114767	CAD Workshop	WT+ST	4 CP	Rönnau
T-MACH-114974	Circular Factory	ST	8 CP	Lanza
T-ETIT-101938	Communication Systems and Protocols	ST	5 CP	Becker, Becker
T-MACH-114078	Data Science Fundamentals in Engineering	WT	4 CP	Meyer
T-MACH-112126	Data-Driven Algorithms in Vehicle Technology	WT	4,5 CP	Scheubner
T-ETIT-112224	Digital Twin Engineering	WT	4 CP	Barth
T-MACH-113647	Digitalization from Product Concept to Production	WT	4,5 CP	Wawerla
T-BGU-101804	IT-Based Road Design	WT+ST	3 CP	Zimmermann
T-BGU-106609	Characteristics of Transportation Systems	WT+ST	3 CP	Vortisch
T-MACH-111807	Introduction to Bionics	ST	4 CP	Hölscher
T-BGU-101499	Introduction to Hydrogeology	WT	5 CP	Goldscheider
T-BGU-101500	Introduction to Engineering Geology	WT	5 CP	Blum
T-MACH-111814	Introduction to Nanotechnology	ST	4 CP	Hölscher

ID	Name	Term offered	Credits	Lecturers
T-MACH-102208	Introduction to Engineering Mechanics I: Statics and Strength of Materials	ST	5 CP	Fidlin
T-MACH-102210	Introduction to Engineering Mechanics II : Dynamics	WT	5 CP	Fidlin
T-BGU-101681	Introduction to GIS for Students of Natural, Engineering and Geo Sciences	WT	3 CP	Wursthorn
T-BGU-103541	Introduction to GIS for Students of Natural, Engineering and Geo Sciences, Prerequisite	WT	3 CP	Wursthorn
T-ETIT-110883	Electric Power Transmission & Grid Control	ST	6 CP	Leibfried
T-ETIT-100830	Power Network	WT	5 CP	Leibfried
T-ETIT-114244	Power Network	WT	6 CP	Leibfried
T-ETIT-112850	Electric Energy Systems	ST	6 CP	Hiller, Leibfried
T-ETIT-100533	Electrical Engineering for Business Engineers, Part I	WT	3 CP	Brodatzki
T-ETIT-100534	Electrical Engineering for Business Engineers, Part II	ST	5 CP	Brodatzki
T-MACH-102159	Elements and Systems of Technical Logistics	WT	4 CP	Mittwollen
T-MACH-108946	Elements and Systems of Technical Logistics - Project	WT	2 CP	Mittwollen
T-BGU-100010	Transportation Data Analysis	WT+ST	3 CP	Kagerbauer
T-MACH-102211	Energy and Process Technology I	WT	9 CP	Bauer, Fischlschweiger, Schwitzke, Velji
T-MACH-102212	Energy and Process Technology II	ST	9 CP	Fischlschweiger, Schwitzke
T-MACH-105151	Energy Efficient Intralogistic Systems	WT	4 CP	Kramer, Schönung
T-MACH-114015	Energy Efficient and Sustainable Tribological Systems (in German)	ST	4 CP	Dienwiebel
T-MACH-105564	Energy Conversion and Increased Efficiency in Internal Combustion Engines	WT	4 CP	Koch, Kubach
T-MACH-114016	Energy Efficient and Sustainable Tribological Systems	WT	4 CP	Dienwiebel
T-BGU-101801	Operation Methods for Earthmoving	WT+ST	1,5 CP	Schlick
T-ETIT-101924	Power Generation	ST	3 CP	Hoferer
T-CIWVT-113235	Exercises: Membrane Technologies	ST	1 CP	Horn, Saravia
T-MACH-102099	Experimental Lab Class in Welding Technology, in Groups	WT	4 CP	Dietrich
T-MACH-105152	Handling Characteristics of Motor Vehicles I	WT	4 CP	Weitz
T-MACH-105153	Handling Characteristics of Motor Vehicles II	ST	4 CP	Weitz
T-MACH-102207	Tires and Wheel Development for Passenger Cars	ST	4 CP	Leister
T-BGU-106301	Long-Distance and Air Traffic	WT+ST	3 CP	Vortisch
T-BGU-101636	Remote Sensing, Exam	WT+ST	4 CP	Cermak, Hinz, Weidner
T-BGU-101637	Systems of Remote Sensing, Prerequisite	ST	1 CP	Cermak, Hinz, Weidner
T-BGU-103542	Procedures of Remote Sensing		3 CP	Weidner
T-BGU-101638	Procedures of Remote Sensing, Prerequisite	ST	1 CP	Weidner
T-MACH-102166	Fabrication Processes in Microsystem Technology	WT+ST	4 CP	Brandner
T-CIWVT-106838	Fundamentals of Water Quality	WT	6 CP	Wagner
T-MACH-102197	Gas Engines	ST	4 CP	Golloch, Kubach
T-MACH-105157	Foundry Technology	ST	4 CP	Günther, Klan
T-MACH-114800	Global Production	WT	5 CP	Lanza
T-MACH-111003	Global Logistics	ST	4 CP	Furmans
T-MACH-100092	Automotive Engineering I (in German)	WT	8 CP	Gießler
T-MACH-102117	Automotive Engineering II (in German)	ST	4 CP	Gießler
T-MACH-105379	Global Logistics	ST	4 CP	Furmans
T-MACH-102111	Principles of Ceramic and Powder Metallurgy Processing	WT	4 CP	Schell
T-MACH-105044	Fundamentals of Catalytic Exhaust Gas Aftertreatment	ST	4 CP	Deutschmann, Grunwaldt, Kubach, Lox
T-MACH-114049	Basics of Production Automation	ST	5 CP	Fleischer
T-CIWVT-101874	Principles of Food Process Engineering		9 CP	Gaukel

ID	Name	Term offered	Credits	Lecturers
T-MACH-102116	Fundamentals for Design of Motor-Vehicle Bodies I	WT	2 CP	Knoch
T-MACH-102119	Fundamentals for Design of Motor-Vehicle Bodies II	ST	2 CP	Knoch
T-MACH-111389	Fundamentals in the Development of Commercial Vehicles	see notes	4 CP	Weber
T-MACH-114075	Principles of Whole Vehicle Engineering (in German)	WT+ST	4 CP	Harrer
T-BGU-106611	Freight Transport	WT+ST	3 CP	Szimba, Vortisch
T-ETIT-101915	High-Voltage Test Technique	WT	4 CP	Badent
T-ETIT-101913	High-Voltage Technology I	WT	4 CP	Badent
T-ETIT-101914	High-Voltage Technology II	ST	4 CP	Badent
T-MACH-113669	Hot Research Topics in AI for Engineering Applications	WT	4,5 CP	Meyer
T-BGU-101693	Hydrology	WT+ST	4 CP	Zehe
T-MACH-102209	Information Engineering	WT+ST	3 CP	Meyer, Ovtcharova
T-BGU-106608	Information Management for Public Mobility Services	WT	3 CP	Vortisch
T-MACH-102128	Information Systems and Supply Chain Management	ST	3 CP	Kilger
T-BGU-106300	Infrastructure Management	WT+ST	6 CP	Zimmermann
T-BGU-108943	Engineering Hydrology	WT+ST	3 CP	Loritz, Villinger
T-MACH-112882	Innovation2Business – Innovation Strategy in the Industrial Corporate Practice	WT	4,5 CP	Albers
T-ETIT-114418	Integrated Photonics		6 CP	Koos
T-MACH-105188	Integrative Strategies in Production and Development of High Performance Cars	ST	4 CP	Schlichtenmayer
T-MACH-105401	Integrated Product Development	WT	16 CP	Düser
T-MACH-109054	Integrated Production Planning in the Age of Industry 4.0	ST	9 CP	Lanza
T-MACH-114100	Introduction to Microsystem Technology I	WT	4 CP	Badilita, Korvink
T-MACH-114101	Introduction to Microsystem Technology II	ST	4 CP	Badilita, Korvink
T-MACH-106743	IoT Platform for Engineering	WT+ST	4 CP	Ovtcharova
T-MACH-105187	IT-Fundamentals of Logistics	ST	4 CP	Thomas
T-MACH-106457	I4.0 Systems Platform	WT+ST	4 CP	Maier, Ovtcharova
T-MACH-100287	Introduction to Ceramics	WT	6 CP	Pagnan Furlan
T-MACH-102182	Ceramic Processing Technology	ST	4 CP	Binder
T-MACH-108312	Introduction to Microsystem Technology - Practical Course	WT+ST	3 CP	Last
T-MACH-105174	Warehousing and Distribution Systems	ST	4 CP	Furmans
T-MACH-112763	Laser Material Processing	ST	4,5 CP	Schneider
T-ETIT-100741	Laser Physics	WT	4 CP	Eichhorn
T-BGU-108000	Lean Construction	WT+ST	4,5 CP	Haghsheno
T-MACH-105783	Learning Factory “Global Production“	WT	6 CP	Lanza
T-MACH-104739	Mechanical Design I and II - CIW	WT	6 CP	Matthiesen
T-MACH-102132	Mechanical Design I, Tutorial	WT	1 CP	Matthiesen
T-MACH-102133	Mechanical Design II, Tutorial	ST	1 CP	Matthiesen
T-BGU-101845	Construction Equipment	WT+ST	3 CP	Gentes
T-MACH-105189	Mathematical Models and Methods for Production Systems	ST	6 CP	Baumann, Furmans
T-MACH-112647	Mechatronical Systems and Products	WT	4 CP	Hohmann, Matthiesen
T-CIWVT-113236	Membrane Technologies in Water Treatment	ST	5 CP	Horn, Saravia
T-ETIT-112852	Measurement and Control Technology	ST	6 CP	Heizmann, Hohmann
T-MACH-105167	Analysis Tools for Combustion Diagnostics	ST	4 CP	Pfeil
T-MACH-105557	Microenergy Technologies	ST	4,5 CP	Kohl, Xu
T-MACH-101910	Microactuators	ST	4 CP	Kohl
T-BGU-103425	Mobility Services and New Forms of Mobility	WT+ST	3 CP	Kagerbauer

ID	Name	Term offered	Credits	Lecturers
T-MACH-102199	Model Based Application Methods	ST	4 CP	Kirschbaum
T-ETIT-100699	Modelling and Identification	WT	4 CP	Hohmann
T-BGU-101859	Morphodynamics	WT+ST	3 CP	Rodrigues Pereira da Franca
T-MACH-105169	Engine Measurement Techniques	ST	4 CP	Bernhardt
T-ETIT-114727	Multivariable Control Systems		6 CP	Hohmann
T-MACH-102152	Novel Actuators and Sensors	WT	4 CP	Kohl, Sommer
T-ETIT-100639	Optical Transmitters and Receivers	WT	6 CP	Freude
T-ETIT-101907	Optoelectronic Components	ST	4 CP	Huber-Loyola
T-MACH-105442	Intellectual Property Rights and Strategies in Industrial Companies	WT+ST	4 CP	Zacharias
T-ETIT-100724	Photovoltaic System Design	ST	3 CP	Grab
T-MACH-100530	Physics for Engineers	ST	5 CP	Nesterov-Müller
T-MACH-102102	Physical Basics of Laser Technology	WT	5 CP	Schneider
T-MACH-102153	PLM-CAD Workshop	WT+ST	4 CP	Ovtcharova
T-MACH-102137	Polymer Engineering I	WT	4 CP	Liebig
T-MACH-102138	Polymer Engineering II	ST	4 CP	Liebig
T-MACH-102192	Polymers in MEMS A: Chemistry, Synthesis and Applications	WT	4 CP	Worgull
T-MACH-102191	Polymers in MEMS B: Physics, Microstructuring and Applications	WT	4 CP	Worgull
T-MACH-102200	Polymers in MEMS C: Biopolymers and Bioplastics	ST	4 CP	Worgull
T-CIWVT-106840	Practical Course in Water Technology	WT	3 CP	Hille-Reichel, Horn
T-MACH-105556	Practical Course Polymers in MEMS	ST	2 CP	Worgull
T-MACH-105178	Practical Course Technical Ceramics	WT	4 CP	Ribas Gomes
T-MACH-102154	Laboratory Laser Materials Processing	WT+ST	4 CP	Schneider
T-MACH-108878	Laboratory Production Metrology	ST	4 CP	Benfer, Lanza
T-MACH-110318	Product- and Production-Concepts for Modern Automobiles	WT	4 CP	Kienzle, Steegmüller
T-MACH-110984	Production Technology for E-Mobility	ST	4 CP	Fleischer
T-MACH-109062	Seminar Production Technology	WT+ST	3 CP	Fleischer, Lanza, Schulze
T-MACH-102156	Project Workshop: Automotive Engineering	WT+ST	6 CP	Frey, Gießler
T-MACH-115045	Project: Data-Driven Engineering	WT+ST	4,5 CP	Meyer
T-MACH-115010	Project: Data-Driven Engineering Fundamentals	WT+ST	4 CP	Meyer
T-BGU-101007	Project Paper Lean Construction	WT	1,5 CP	Haghsheno
T-BGU-113720	Project Management	WT	3 CP	Haghsheno
T-BGU-103432	Project Management in Construction and Real Estate Industry I	WT	3 CP	Haghsheno
T-BGU-103433	Project Management in Construction and Real Estate Industry II	WT	3 CP	Haghsheno
T-BGU-101847	Project Studies	ST	3 CP	Gentes
T-BGU-101814	Project in Applied Remote Sensing		1 CP	Hinz, Weidner
T-BGU-113454	Examination Prerequisite Project Management	WT	0 CP	Haghsheno
T-MACH-102157	High Performance Powder Metallurgy Materials	ST	4 CP	Schell
T-MACH-110796	Python Algorithms for Vehicle Technology	ST	4 CP	Rhode
T-MACH-115104	Python-Basics Course	WT	1 CP	Meyer
T-MACH-102107	Quality Management	WT	4 CP	Lanza
T-ETIT-100740	Quantum Functional Devices and Semiconductor Technology	ST	3 CP	Koos
T-MACH-109122	X-ray Optics	WT+ST	4,5 CP	Last
T-MACH-105353	Rail Vehicle Technology	WT+ST	4 CP	Cichon
T-MACH-105170	Welding Technology	WT	4 CP	Farajian
T-MACH-108737	Seminar Data-Mining in Production	WT	3 CP	Lanza

ID	Name	Term offered	Credits	Lecturers
T-MACH-113999	Seminar Development of Automated Production Systems	WT	4 CP	Fleischer
T-WIWI-108763	Seminar in Engineering Science Master (approval)	WT+ST	3 CP	Fachvertreter ingenieurwissenschaftlicher Fakultäten
T-BGU-100014	Seminar in Transportation	WT+ST	3 CP	Kagerbauer, Vortisch
T-ETIT-100754	Seminar Creating a Patent Specification	ST	3 CP	Stork
T-MACH-115011	Seminar: Machine Learning and Optimization in Engineering	WT+ST	4,5 CP	Meyer
T-ETIT-101911	Sensors	ST	3 CP	Schröder
T-ETIT-100709	Sensor Systems	ST	3 CP	Menesklou
T-BGU-101674	Safety Management in Highway Engineering	WT+ST	3 CP	Zimmermann
T-MACH-105171	Safety Engineering	WT	4 CP	Kany
T-BGU-101800	Traffic Flow Simulation	WT+ST	3 CP	Vortisch
T-MACH-106493	Solar Thermal Energy Systems	WT	4,5 CP	Dagan
T-MACH-111821	Control of Mobile Machines	ST	4 CP	Becker, Geimer
T-MACH-111820	Control of Mobile Machines – Prerequisites	ST	0 CP	Becker, Geimer
T-MACH-105185	Control Technology	ST	4 CP	Gönnheimer
T-BGU-101798	Traffic Engineering	WT+ST	3 CP	Vortisch
T-MACH-114969	Strategic Decision-Making in Global Production Networks through Optimization and Simulation	ST	4,5 CP	Lanza, Martin
T-BGU-103426	Strategic Transport Planning	WT+ST	3 CP	Waßmuth
T-MACH-102170	Structural and Phase Analysis	WT	4 CP	Wagner
T-MACH-102103	Superhard Thin Film Materials	WT	4 CP	Ulrich
T-MACH-100531	Systematic Materials Selection	ST	4 CP	Dietrich, Schulze
T-BGU-101832	Operation Methods for Foundation and Marine Construction	WT+ST	1,5 CP	Schneider
T-BGU-101846	Tunnel Construction and Blasting Engineering	WT+ST	3 CP	Haghsheno
T-BGU-113671	Exercise Transportation Data Analysis	WT	0 CP	Kagerbauer
T-BGU-113971	Exercise Transportation Data Analysis	ST	0 CP	Vortisch
T-MACH-105177	Metal Forming	ST	4 CP	Herlan
T-MACH-102194	Combustion Engines I	WT	4 CP	Koch, Kubach
T-MACH-104609	Combustion Engines II	ST	5 CP	Koch, Kubach
T-BGU-101844	Process Engineering	WT+ST	3 CP	Schneider
T-BGU-101850	Disassembly Process Engineering	ST	3 CP	Gentes
T-CIWVT-106058	Process Fundamentals by the Example of Food Production		3 CP	Gaukel
T-BGU-106615	Laws concerning Traffic and Roads	WT+ST	3 CP	Hönig
T-BGU-101799	Traffic Management and Transport Telematics	WT+ST	3 CP	Vortisch
T-BGU-106610	Transportation Systems	WT+ST	3 CP	Vortisch
T-MACH-102139	Failure of Structural Materials: Fatigue and Creep	WT	4 CP	Gruber, Gumbsch
T-MACH-102140	Failure of Structural Materials: Deformation and Fracture	WT	4 CP	Gumbsch, Weygand
T-MACH-102148	Gear Cutting Technology	WT	4 CP	Hoffmeister, Klaiber
T-MACH-105292	Heat and Mass Transfer	WT+ST	4 CP	Fischlschweiger, Yu
T-BGU-101667	Hydraulic Engineering and Water Management	WT+ST	4 CP	Rodrigues Pereira da Franca
T-CIWVT-103351	Laboratory Work Water Chemistry		4 CP	Abbt-Braun, Horn
T-CIWVT-106802	Water Technology	WT	6 CP	Horn
T-MACH-102078	Materials Science I	WT	3 CP	Wagner
T-MACH-102079	Material Science II for Business Engineers	ST	5 CP	Wagner
T-MACH-110963	Machine Tools and High-Precision Manufacturing Systems	WT	9 CP	Fleischer

ID	Name	Term offered	Credits	Lecturers
T-BGU-101005	Tendering, Planning and Financing in Public Transport	WT+ST	3 CP	Vortisch
T-MACH-105234	Windpower	WT	4 CP	Lewald
T-MACH-112648	Workshop Mechatronical Systems and Products	WT	5 CP	Hohmann, Matthiesen

M 5.67 Module: Entrepreneurship (EnTechnon) (WW4BWLENT1) [M-WIWI-101488]

Coordinators: Prof. Dr. Orestis Terzidis
Organisation: KIT Department of Business and Economics
Part of: Electives (Business Administration)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	2 terms	German/English	4	16

ID	Name	Term offered	Credits	Lecturers
Mandatory part (Elective: 1 item)				
T-WIWI-102864	Entrepreneurship	WT+ST	3 CP	Terzidis
Compulsory Elective Courses (Elective: between 1 and 2 items)				
T-WIWI-115110	Business Planning for Founders	Irreg.	3 CP	Terzidis
T-WIWI-102866	Design Thinking	Irreg.	3 CP	Terzidis
T-WIWI-113151	Entrepreneurship Seasonal School	Irreg.	3 CP	Terzidis
T-WIWI-114893	International Entrepreneurship and Brand Development	see notes	6 CP	Kupfer, Terzidis
T-WIWI-109064	Joint Entrepreneurship Summer School	Irreg.	6 CP	Terzidis
T-WIWI-111561	Startup Experience	WT+ST	6 CP	Terzidis
Supplementary Courses (Elective: between 0 and 1 items)				
T-WIWI-102894	Entrepreneurship Research	ST	3 CP	Terzidis
T-MACH-112882	Innovation2Business – Innovation Strategy in the Industrial Corporate Practice	WT	4,5 CP	Albers
T-WIWI-102852	Case Studies Seminar: Innovation Management	WT	3 CP	Weissenberger-Eibl
T-WIWI-102893	Innovation Management: Concepts, Strategies and Methods	ST	3 CP	Weissenberger-Eibl

Assessment
The module examination takes place in the form of partial examinations (according to §4, 1-3 SPO) on

1. the Entrepreneurship lecture (3 CP),
2. one of the seminars of the Chair of Entrepreneurship and Technology Management (3 CP or 6 CP) and, if applicable
3. another course listed in the module.

The seminars of the chair are

- Startup Experience
- Design Thinking
- Business Planning for Founders
- Entrepreneurship research (this can be credited mainly in the seminar module, but also in the entrepreneurship module)
- Joint Entrepreneurship School
- Entrepreneurship Seasonal School
- International Entrepreneurship and Brand Development

The latter three seminars take place irregularly, as they are offered as part of projects.

The assessment of success is described for each course in the module. For courses with 3 CP in the compulsory elective and supplementary courses, 1/2 of the overall grade results from the entrepreneurship lecture, 1/4 from one of the chair's seminars with 3 CP and 1/4 from another course with 3 CP permitted in the module. If a course with 6 CP is selected in the compulsory elective or supplementary offer, this is included in the overall grade with a weighting of 1/2. The overall grade is cut off after the first decimal place.

Prerequisites
None

Competence Goal
Students are familiar with the basics and contents of entrepreneurship and ideally are able to start a company during or after their studies. The courses are therefore structured sequentially in modules, although in principle they can also be attended in parallel. In this way, the skills are taught to generate business ideas, to develop inventions into innovations, to write business plans for start-ups and to successfully establish a company. In the lecture, the basics of entrepreneurship will be developed, in the seminars, individual contents will be deepened. The overall learning objective is to enable students to develop and implement business ideas.

Content
The lectures form the basis of the module and give an overview of the overall topic. The seminars deepen the phases of the foundation processes, in particular the identification of opportunities, the development of a value proposition (especially based on inventions and technical innovations), the design of a business model, business planning, the management of a start-up, the implementation of a vision as well as the acquisition on resources and the handling of risks. The lecture Entrepreneurship provides an overarching and connecting framework for this.

Additional Information
Please note: Seminars offered by Prof. Terzidis (or the members of his research group) are not eligible for crediting in a seminar module of the WiWi degree programs. Exception: Seminar "Entrepreneurship Research".

Workload
Total workload for 9 credit points: approx. 270 hours

The exact distribution is based on the credit points of the courses in the module.

Recommendations
None

5.68 Module: Environmental Economics (WW4VWL5) [M-WIWI-101468]

Coordinators: Prof. Dr. Kay Mitusch
Organisation: KIT Department of Business and Economics
Part of: Electives (Economics)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	1 term	German	4	2

ID	Name	Term offered	Credits	Lecturers
Compulsory Elective Courses (Elective: at least 9 credits)				
T-WIWI-102650	Energy and Environment	ST	3,5 CP	Karl
T-WIWI-100007	Transport Economics	ST	4,5 CP	Mitusch, Szimba
T-WIWI-102615	Environmental Economics and Sustainability	WT	3 CP	Walz
T-WIWI-102616	Environmental and Resource Policy	ST	4 CP	Walz
T-BGU-111102	Environmental Law	WT	3 CP	Smeddinck

Assessment
The assessment is carried out as partial exams (according to Section 4(2), 1 or 2 of the examination regulation) of the single courses of this module, whose sum of credits must meet the minimum requirement of credits of this module. The exams are offered at the beginning of the recess period about the subject matter of the latest held lecture. Re-examinations are offerd at every ordinary examination date. The assessment procedures are described for each course of the module seperately.

The overall grade for the module is the average of the grades for each course weighted by the credits and truncated after the first decimal.

Prerequisites
None

Competence Goal
The students

- understand the treatment of non-market resources as well as future resource shortages
- are able to model markets of energy and environmental goods
- are able to assess the results of government intervention
- know legal basics and are able to evaluate conflicts with regard to legal situation

Content
Environmental degradation and increasing resource use are global challenges, which have to be tackled on a worldwide level. The module addresses these challenges from the perspective of economics, and imparts the fundamental knowledge of environmental and sustainability economics, and environmental and resource policy to the students. Additional courses address environmental law, environmental pressure, and applications to the transport sector.

Workload
Total workload for 9 credit points: approx. 270 hours

The exact distribution is based on the credit points of the courses in the module.

Recommendations
Knowledge of microeconomics is a prerequisite. For this reason, successful participation in the course *Economics I (Microeconomics)* [2600012] or a comparable course is strongly recommended.

M 5.69 Module: Estimator and Observer Design [M-CIWVT-106320]

Coordinators: Dr.-Ing. Pascal Jerono
Organisation: KIT Department of Chemical and Process Engineering
Part of: Engineering and Informatics (Chemical and Process Engineering)
Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-CIWVT-112828	Estimator and Observer Design		6 CP	Jerono

Assessment
Learning control is an oral examination with a duration of about 45 minutes.

Competence Goal
Students will gain an in-depth understanding of the concepts and methods used to estimate and identify the state of dynamic systems and will be familiar with their advantages and disadvantages. In addition, students will be able to analyze the observability and detectability properties of the underlying system dynamics and use this information to design suitable state observers for practical applications. Students are familiar with various numerical solution approaches, understand how they work, and can implement them for estimation and observer design tasks.

Content
State feedback control relies on the availability of the full state vector, which is in general not available from measurements. Moreover determining the states (or parameters) of a dynamical systems is of interest on its own as this allows to obtain insights into the system dynamics or to estimate quantities that are not or hardly measurable. The lecture addresses basic concepts of estimation and identification methods and the design of optimal state observers for linear and nonlinear dynamical systems both in a continuous and a discrete time setting. This includes:

- Introduction to fundamental concepts for system identification and state estimation
- State-space approaches for system identification
- Analysis of observability and detectability
- Design of linear and nonlinear observers as well as optimal state estimators (Kalman-Bucy and Kalman Filters)
- Numerical methods

Module Grade Calculation
The grade of the module is the grade of the oral exam.

Workload
Attendance time: Lectures: 30 hrs. Exercises: 15 hrs.

Self-study: 60 hrs.

Exam preparation: 75 hrs.

Literature

- P. Jerono: Estimator and Observer Design, Lecture Notes.
- L. Lennart: System identification. Birkhäuser, 1998.
- H. Nijmeijer, A. Van der Schaft: Nonlinear dynamical control systems, Springer-Verlag, 1990.
- Isidori: Nonlinear Control Systems, Springer-Verlag, 1995.
- Gelb: Applied optimal estimation. MIT Press, 1974.
- F.L. Lewis, X. Lihua, and D. Popa: Optimal and robust estimation: with an introduction to stochastic control theory, CRC Press, 2017.

M 5.70 Module: Experimental Economics (WW4VWL17) [M-WIWI-101505]

Coordinators: Prof. Dr. Johannes Philipp Reiß
Organisation: KIT Department of Business and Economics
Part of: Electives (Economics)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	2 terms	German/English	4	6

ID	Name	Term offered	Credits	Lecturers
Compulsory Elective Courses (Elective: 2 items)				
T-WIWI-115061	Experimental Research in Economics and Information Systems	WT	4,5 CP	Weinhardt
T-WIWI-105781	Incentives in Organizations	ST	4,5 CP	Nieken
T-WIWI-102862	Predictive Mechanism and Market Design	Irreg.	4,5 CP	Reiß
T-WIWI-102863	Topics in Experimental Economics	Irreg.	4,5 CP	Reiß

Assessment
The assessment is carried out as partial exams (according to Section 4 (2), 1-3 SPO) of the core course and further single courses of this module, whose sum of credits must meet the minimum requirement of credits of this module. The assessment procedures are described for each course of the module separately.
The overall grade of the module is the average of the grades for each course weighted by the credits and truncated after the first decimal.

Prerequisites
None.

Competence Goal
Students

- are acquainted with the methods of Experimental Economics along with its strengths and weaknesses;
- understand how theory-guided research in Experimental Economics interacts with the development of theory;
- are provided with foundations in data analysis;
- design an economic experiment and analyze its outcome.

Content
The module Experimental Economics offers an introduction into the methods and topics of Experimental Economics. It also fosters and extends knowledge in theory-guided experimental economics and its interaction with theory development. Throughout the module, readings of selected papers are required.

Additional Information
The course "Predictive Mechanism and Market Design" is offered every second winter semester, e.g. WS2013 / 14, WS2015 / 16, ...

Workload
Total workload for 9 credit points: approx. 270 hours

The exact distribution is based on the credit points of the courses in the module.

Recommendations
Basic knowledge in mathematics, statistics, and game theory is assumed.

5.71 Module: Extracurricular Module in Engineering (WW4INGAPL) [M-WIWI-101404]

Coordinators: Prüfungsausschuss der KIT-Fakultät für Wirtschaftswissenschaften
Organisation: KIT Department of Business and Economics
Part of: Engineering and Informatics (Mechanical Engineering)
Engineering and Informatics (Chemical and Process Engineering)
Engineering and Informatics (Electrical Engineering and Information Technology)
Engineering and Informatics (Civil Engineering, Geo and Environmental Sciences)
Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Level	Version
9 CP	graded	Once	1 term	4	5

ID	Name	Term offered	Credits	Lecturers
Compulsory Elective Courses (Elective: between 9 and 12 credits)				
T-WIWI-106291	PH APL-ING-TL01	Once	3 CP	
T-WIWI-106292	PH APL-ING-TL02	Once	3 CP	
T-WIWI-106293	PH APL-ING-TL03	Once	3 CP	
T-WIWI-106294	PH APL-ING-TL04 ub	Once	0 CP	
T-WIWI-106295	PH APL-ING-TL05 ub	Once	0 CP	
T-WIWI-106296	PH APL-ING-TL06 ub	Once	0 CP	
T-WIWI-108384	PH APL-ING-TL07	Once	3 CP	

Assessment
The assessment of the module is determined by the respective module corrdinator. It can either be in the form of a general exam or partial exams, and must contain at least 9 credit points (max. 12 credits) and at least 6 hours per week (max. 8 hours per week). The examination may contain presentations, experiments, laboratories, term papers, etc. At least 50 percent of the module examination has to be in the form of a written or an oral examination (according to Section 4 (2), 1 or 2 of the examination regulation).

The formation of the overall grade of the module will be determined by the respective module coordinator.

Prerequisites
The current regulations and guidance on the procedure for applying for an extracurricular module in engineering are explained in detail at https://www.wiwi.kit.edu/APIng-Modul.php.

Competence Goal
Through the extracurricular engineering module, the student is able to deal with technical topics and issues in depth.

The concrete learning objectives are coordinated with the respective module supervisor of the module.

Workload
The total workload for this module is about 270 hours (9 credits). The distribution is based on the credit points of the courses completed as part of the module.

M 5.72 Module: Facility Management in Hospitals for Industrial Engineering (bauiEX408-FMKH_WIWI) [M-BGU-106453]

Coordinators: Prof. Dr.-Ing. Kunibert Lennerts
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each winter term	1 term	German	4	2

Election Notes
Module components in extent of min. 3 CP has to be selected.

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-BGU-108004	Facility Management in Hospitals	WT	6 CP	Lennerts
Comnpulsory Elective (Elective: at most 2 items as well as at least 3 credits)				
T-BGU-111218	Upgrading of Existing Buildings	WT+ST	3 CP	Lennerts
T-BGU-111211	Energetic Refurbishment	WT+ST	1,5 CP	Lennerts, Schneider
T-BGU-111212	Facility and Real Estate Management II	WT+ST	1,5 CP	Lennerts

Assessment
- module components T-BGU-108004 with examination of another type according to § 4 Par. 2 No. 3

according to selected course:

- module components T-BGU-111218 with written examination according to § 4 Par. 2 No. 1
- module components T-BGU-111211 with oral examination according to § 4 Par. 2 No. 2
- module components T-BGU-111212 with oral examination according to § 4 Par. 2 No. 2

details about the learning controls, see module components

Prerequisites
none

Competence Goal
Students can explain the theoretical principles of facility management. They are able to analyze and optimize real estate-specific, facility and engineering processes of real estate.

Content

- Facility Management in Hospitals: facility management processes in hospitals, strategic planning and cost structure of selected facility management services, sustainable hospitals, master planning, room and function program and layout planning of hospitals, OR simulation and hygiene in hospitals
- Construction in Existing Buildings: maintenance strategies, service life and wear and tear of components, budgeting of maintenance costs, condition assessment and action planning, special procedures in existing buildings, monument protection and preservation
- Energy-efficient Refurbishment: fundamentals of building physics, improving the energy efficiency of the building envelope, improving building technology and cost-effectiveness, forms of energy and local energy conversion (e.g. photovoltaics, geothermal energy, heat pumps, etc.), calculating the energy requirements of buildings, energy assessment of buildings in accordance with the GEG
- Facility and Real Estate Management II: outsourcing and procurement regulations in facility management, data collection, transition, CAFM in facility management, presentation of measurement criteria for SLA and KPI and their digitalization

Module Grade Calculation
grade of the module is CP weighted average of grades of the partial exams

Additional Information
as from winter term 2025/26, the module component Project Development with Case Study (T-BGU-111217) will not be offered anymore as an option

Workload
contact hours (1 HpW = 1 h x 15 weeks):

- Facility Management in Hospitals lecture/exercise: 60 h

according to selected courses or examinations respectively:

- Upgrading of Existing Buildings lecture/exercise: 45 h
- Energetic Refurbishment II lecture: 15 h
- Facility and Real Estate Management II lecture: 15 h

independent study:

- preparation and follow-up lecture/exercises Facility Management in Hospitals: 45 h
- preparation of term paper Facility Management in Hospitals (partial examination, compulsory): 75 h

according to selected courses or examinations respectively:

- preparation and follow-up lecture/exercises Upgrading of Existing Buildings: 15 h
- examination preparation Upgrading of Existing Buildings (selectable partial examination): 30 h
- preparation and follow-up lectures Energetic Refurbishment II: 15 h
- examination preparation Energetic Refurbishment II (selectable partial examination): 15 h
- preparation and follow-up lectures Facility and Real Estate Management II: 15 h
- examination preparation Facility and Real Estate Management II (selectable partial examination): 15 h

total: 270 h

Recommendations
none

M 5.73 Module: Field Propagation and Coherence [M-ETIT-100566]

Coordinators: Prof.Dr.Dr.h.c. Wolfgang Freude
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Engineering and Informatics (Electrical Engineering and Information Technology)
Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-100976	Field Propagation and Coherence	WT	4 CP	Freude

Assessment
Type of Examination: oral exam

Duration of Examination: approx. 20 minutes

Modality of Exam: Oral examination, usually one examination day per month during the summer and winter terms. An extra questions-and-answers session will be held for preparation if students wish so.

Prerequisites
none

Competence Goal
Presenting in a unified approach the common background of various problems and questions arising in general optics and optical comunications

The students

- knwo the common properties of counting of modes, density of states and the sampling theorem
- comprehend the relationship between propagation in multimode waveguides, mode coupling, MMI and speckles
- can analyze propagation in homogeneous media with respect to system theory, antennas, and the resolution limit of optical instruments
- understand that coherence as a general concept comprises coherence in time, in space and in polarisation
- comprehend the implication of complete spatial incoherence, and what is the radiation efficiency of a source with a diameter smaller than a wavelength (the mathematical Hertzian dipole, for instance)
- can assess when can two incandescent bulbs form an interference pattern in time
- know under which conditions a heterodyne radio receiver, which is based on a non-stationary interference, actually works

Content
The following selection of topics will be presented:

- Light waves, modes and rays: Longitudinal and transverse modes, sampling theorem, counting and density of modes ("states")
- Propagation in multimode waveguides. Near-field and far-field. Impulse response and transfer function. Perurations and mode coupling. Multimode interference (MMI) coupler. Modal noise (speckle)
- Propagation in homogeneous media: Resolution limit. Non-paraxial and paraxial optics. Gaussian beam. ABCD matrix
- Coherence of optical fields: Coherence function and power spectrum. Polarisation, eigenstates and principal states. Measurement of coherence with interferometers (Mach-Zehnder, Michelson). Self-heterodyne and self-homodyne setups

Module Grade Calculation
The module grade is the grade of the oral exam.

Workload
total 120 h, hereof 45 h contact hours (30 h lecture, 15 h problem class), and 75 h homework and self-studies

Recommendations
Minimal background required: Calculus, differential equations and Fourier transform theory. Electrodynamics and field calculations or a similar course on electrodynamics or optics is recommended.
Solution of the problems on the exercise sheet, which can be downloaded as homework each week, is highly recommended. Also, active participation in the problem classes and studying in learning groups are strongly advised.

Literature

Detailed lecture notes as well as the presentation slides can be downloaded from the IPQ lecture pages. Additional reading:

Born, M.; Wolf, E.: Principles of optics, 6. Aufl. Oxford: Pergamon Press 1980

Ghatak, A.: Optics, 3. Ed. New Delhi: Tata McGraw Hill 2005

Hecht, E.: Optics, 2. Ed. Reading: Addison-Wesley 1974

Hecht, J.: Understanding fiber optics, 4. Ed. Upper Saddle River: Prentice Hall 2002

Iizuka, K.: Elements of photonics, Vol. I and II. New York: John Wiley & Sons 2002

Further textbooks in German (also in electronic form) can be named on request

M 5.74 Module: Finance 1 (WW4BWLFBV1) [M-WIWI-101482]

Coordinators: Prof. Dr. Martin Ruckes
Prof. Dr. Marliese Uhrig-Homburg
Organisation: KIT Department of Business and Economics
Part of: Electives (Business Administration)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	1 term	German/English	4	1

ID	Name	Term offered	Credits	Lecturers
Compulsory Elective Courses (Elective: 9 credits)				
T-WIWI-102643	Derivatives	ST	4,5 CP	Uhrig-Homburg
T-WIWI-102621	Valuation	WT	4,5 CP	Ruckes
T-WIWI-102647	Asset Pricing	ST	4,5 CP	Ruckes, Uhrig-Homburg

Assessment
The assessment is carried out as partial exams (according to Section 4(2), 1 or 2 of the examination regulation) of the single courses of this module, whose sum of credits must meet the minimum requirement of credits of this module. The assessment procedures are described for each course of the module seperately.

The overall grade of the module is the average of the grades for each course weighted by the credits and truncated after the first decimal.

Prerequisites
None

Competence Goal
The student

- has core skills in economics and methodology in the field of finance
- assesses corporate investment projects from a financial perspective
- is able to make appropriate investment decisions on financial markets

Content
The courses of this module equip the students with core skills in economics and methodology in the field of modern finance. Securities which are traded on financial and derivative markets are presented, and frequently applied trading strategies are discussed. A further focus of this module is on the assessment of both profits and risks in security portfolios and corporate investment projects from a financial perspective.

Workload
The total workload for this module is approx. 270 hours (9 credits). The distribution is based on the credit points of the courses in the module. The workload for courses with 4.5 credits is approx. 135 hours.

The total number of hours per course is calculated from the time required to attend the lectures and exercises, as well as the examination times and the time required for an average student to achieve the learning objectives of the module for an average performance.

M 5.75 Module: Finance 2 (WW4BWLFBV2) [M-WIWI-101483]

Coordinators: Prof. Dr. Martin Ruckes
Prof. Dr. Marliese Uhrig-Homburg

Organisation: KIT Department of Business and Economics

Part of: Electives (Business Administration)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	1 term	German/English	4	11

ID	Name	Term offered	Credits	Lecturers
Compulsory Elective Courses (Elective: at least 9 credits)				
T-WIWI-113469	Advanced Corporate Finance	ST	4,5 CP	Ruckes
T-WIWI-114979	Advanced Quantitative Finance	ST	4,5 CP	Thimme
T-WIWI-102647	Asset Pricing	ST	4,5 CP	Ruckes, Uhrig-Homburg
T-WIWI-110995	Bond Markets	WT	4,5 CP	Uhrig-Homburg
T-WIWI-110997	Bond Markets - Models & Derivatives	WT	3 CP	Uhrig-Homburg
T-WIWI-110996	Bond Markets - Tools & Applications	WT	1,5 CP	Uhrig-Homburg
T-WIWI-109050	Corporate Risk Management	Irreg.	4,5 CP	Ruckes
T-WIWI-102643	Derivatives	ST	4,5 CP	Uhrig-Homburg
T-WIWI-110797	eFinance: Information Systems for Securities Trading	WT	4,5 CP	Weinhardt
T-WIWI-102900	Financial Analysis	see notes	4,5 CP	Luedecke
T-WIWI-102623	Financial Intermediation	WT	4,5 CP	Ruckes
T-WIWI-102646	International Finance	see notes	3 CP	Uhrig-Homburg
T-WIWI-102621	Valuation	WT	4,5 CP	Ruckes

Assessment
The assessment is carried out as partial exams (according to Section 4(2), 1 or 2 of the examination regulation) of the single courses of this module, whose sum of credits must meet the minimum requirement of credits of this module. The assessment procedures are described for each course of the module seperately.

The overall grade of the module is the average of the grades for each course weighted by the credits and truncated after the first decimal.

Prerequisites
Students may only start the module *Finance 2* after they have successfully completed the module *Finance 1*. Exceptions are subject to approval by the examination committee; for this purpose, students must contact the examination office of the KIT Faculty of Economics and Management.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module M-WIWI-101482 - Finance 1 must have been passed.

Competence Goal
The student is in a position to discuss, analyze and provide answers to advanced economic and methodological issues in the field of modern finance.

Content
The module Finance 2 is based on the module Finance 1. The courses of this module equip the students with advanced skills in economics and methodology in the field of modern finance on a broad basis.

Workload
The total workload for this module is approximately 270 hours. For further information see German version.

5.76 Module: Finance 3 (WW4BWLFBV11) [M-WIWI-101480]

Coordinators: Prof. Dr. Martin Ruckes
Prof. Dr. Marliese Uhrig-Homburg

Organisation: KIT Department of Business and Economics

Part of: Electives (Business Administration)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	1 term	German/English	4	11

ID	Name	Term offered	Credits	Lecturers
Compulsory Elective Courses (Elective: at least 9 credits)				
T-WIWI-113469	Advanced Corporate Finance	ST	4,5 CP	Ruckes
T-WIWI-114979	Advanced Quantitative Finance	ST	4,5 CP	Thimme
T-WIWI-102647	Asset Pricing	ST	4,5 CP	Ruckes, Uhrig-Homburg
T-WIWI-110995	Bond Markets	WT	4,5 CP	Uhrig-Homburg
T-WIWI-110997	Bond Markets - Models & Derivatives	WT	3 CP	Uhrig-Homburg
T-WIWI-110996	Bond Markets - Tools & Applications	WT	1,5 CP	Uhrig-Homburg
T-WIWI-109050	Corporate Risk Management	Irreg.	4,5 CP	Ruckes
T-WIWI-102643	Derivatives	ST	4,5 CP	Uhrig-Homburg
T-WIWI-110797	eFinance: Information Systems for Securities Trading	WT	4,5 CP	Weinhardt
T-WIWI-102900	Financial Analysis	see notes	4,5 CP	Luedecke
T-WIWI-102623	Financial Intermediation	WT	4,5 CP	Ruckes
T-WIWI-102646	International Finance	see notes	3 CP	Uhrig-Homburg
T-WIWI-102621	Valuation	WT	4,5 CP	Ruckes
T-WIWI-110933	Web App Programming for Finance	Once	4,5 CP	Thimme

Assessment
The assessment is carried out as partial exams (according to Section 4(2), 1 or 2 of the examination regulation) of the single courses of this module, whose sum of credits must meet the minimum requirement of credits of this module. The assessment procedures are described for each course of the module seperately.

The overall grade of the module is the average of the grades for each course weighted by the credits and truncated after the first decimal.

Prerequisites
Students may only start the module *Finance 3* after they have successfully completed the modules *Finance 1* and *Finance 2*. Exceptions are subject to approval by the examination committee; for this purpose, students must contact the examination office of the KIT Faculty of Economics and Management.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module M-WIWI-101482 - Finance 1 must have been passed.
2. The module M-WIWI-101483 - Finance 2 must have been passed.

Competence Goal
The student is in a position to discuss, analyze and provide answers to advanced economic and methodological issues in the field of modern finance.

Content
The courses of this module equip the students with advanced skills in economics and methodology in the field of modern finance on a broad basis.

Workload
The total workload for this module is approximately 270 hours. For further information see German version.

5.77 Module: Foundations for Advanced Financial -Quant and -Machine Learning Research [M-WIWI-105894]

Coordinators: Prof. Dr. Maxim Ulrich
Organisation: KIT Department of Business and Economics
Part of: Business and Economics
Electives (Business Administration)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	see Annotations	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-WIWI-111846	Fundamentals for Financial -Quant and -Machine Learning Research	Irreg.	9 CP	Ulrich

Assessment
The module examination is an alternative exam assessment with a maximum score of 100 points to be achieved. These points are distributed over 4 worksheets to be submitted during the semester. The worksheets cover the respective material of the module and are handed out, worked on and assessed in lecture weeks 3 (10 points), 6 (20 points), 9 (30 points) and 12 (40 points).

The module-wide exam (all 4 worksheets) must be taken in the same semester.

The worksheets are a mixture of analytical tasks and programming tasks with financial data.

Competence Goal
This MSc module teaches students fundamental stats and analytics concepts, as well necessary financial economic intuition, necessary to identify, design and execute interesting research questions in quant finance and financial machine learning.
Topics include: Maximum Likelihood learning of arma-garch models, expectation maximization learning applied to stochastic volatility and valuation models, Kalman filter techniques to learn latent states, estimation of affine jump diffusion models with options and higher-order moments, stochastic calculus, dynamic modeling of asset markets (bond, equity, options), equilibrium determination of risk premiums, risk premiums for higher moment risk, risk decomposition (fundamental vs idiosyncratic), option-implied return distributions, mixture-density-networks and neural nets.

Content
Learning Objectives: Skills and understanding of how to successfully set-up, execute and interpret financial data driven research with the following methods: MLE, Kalman Filter, Expectation Maximization, Option Pricing, dynamic asset pricing theory, backward-looking historical return densities, forward-looking options-implied return densities, mixture-density-network, neural networks. Programming is not taught in this course, yet, some graded and non-graded exercises might make heavy use of software based data analysis. See the course's pre-requisites and comments in the modul handbook.

Additional Information

- Strongly recommended to have good knowledge in financial econometrics (MLE, OLS, GLS, ARMA-GARCH), mathematics (differential equations, difference equations and optimization), investments (CAPM, factor models), asset pricing (SDF, SDF pricing), derivatives (Black-Scholes, risk-neutral pricing), and programming of statistical concepts (Java or R or Python or Matlab or C or ...)
- Strongly recommended to have a strong interest for interdisciplinary research work in statistics, programming, applied math and financial economics.
- Students lacking the prior knowledge might find the resources of the Chair helpful: www.youtube.com/c/cram-kit.

Workload
The total workload for this course is approximately 270 hours. This is for a student with the appropriate prior knowledge in financial econometrics, finance, mathematics and programming. Students without programming experience of statistical concepts will need to invest extra time. Students who have struggled in math- or programming- or finance- oriented classes, will find this course very challenging. Please check the pre-requisites and comments in the module handbook.

Teaching and Learning Methods
Lecture and exercise.

M 5.78 Module: Fundamentals of Transportation (bauiEX305-VERGRDL) [M-BGU-101064]

Coordinators: Prof. Dr.-Ing. Peter Vortisch
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each summer term	2 terms	German	4	7

Election Notes
Characteristics of Transportation Systems [T-BGU-106609] is compulsory module component;
two further compulsory elective module components have to be selected

ID	Name	Term offered	Credits	Lecturers
Compulsory Examination (Elective: 1 item as well as 3 credits)				
T-BGU-106609	Characteristics of Transportation Systems	WT+ST	3 CP	Vortisch
Electives (Elective: 2 items as well as 6 credits)				
T-BGU-106611	Freight Transport	WT+ST	3 CP	Szimba, Vortisch
T-BGU-106301	Long-Distance and Air Traffic	WT+ST	3 CP	Vortisch
T-BGU-101005	Tendering, Planning and Financing in Public Transport	WT+ST	3 CP	Vortisch
T-BGU-100014	Seminar in Transportation	WT+ST	3 CP	Kagerbauer, Vortisch
T-BGU-112552	Seminar on Modeling and Simulation in Transportation	WT	3 CP	Kagerbauer, Vortisch
T-BGU-103425	Mobility Services and New Forms of Mobility	WT+ST	3 CP	Kagerbauer
T-BGU-103426	Strategic Transport Planning	WT+ST	3 CP	Waßmuth
T-BGU-106608	Information Management for Public Mobility Services	WT	3 CP	Vortisch
T-BGU-111057	Sustainability in Mobility Systems	WT+ST	3 CP	Kagerbauer
T-BGU-114772	Seminar on Transport Policy	WT	3 CP	Vortisch

Assessment
- module component T-BGU-106609 with written examination according to § 4 Par. 2 No. 1

according to selected course:

- module component T-BGU-106611 with written examination according to § 4 Par. 2 No. 1
- module component T-BGU-106301 with written examination according to § 4 Par. 2 No. 1
- module component T-BGU-101005 with oral examination according to § 4 Par. 2 No. 2
- module component T-BGU-100014 with examination of another type according to § 4 Par. 2 No. 3
- module component T-BGU-112552 with examination of another type according to § 4 Par. 2 No. 3
- module component T-BGU-103425 with oral examination according to § 4 Par. 2 No. 2
- module component T-BGU-103426 with oral examination according to § 4 Par. 2 No. 2
- module component T-BGU-106608 with examination of an other type according to § 4 Par. 2 No. 3
- module component T-BGU-111057 with written examination according to § 4 Par. 2 No. 1
- module component T-BGU-114772 with examination of another type according to § 4 Par. 2 No. 3

details about the learning controls, see module components

Prerequisites
None

Competence Goal
The student

- has basic knowledge in the field of transport studies from the perspective of professional practice,
- knows the decision-relevant aspects of transport studies from the perspective of management, politics and consulting,
- is able to analyze and evaluate transport projects from both perspectives.

Content
The subject of transport studies deals with issues relating to the transport sector, ranging from planning concepts based on society as a whole to technical problems of transport. Teaching is interdisciplinary and ranges from methodological principles (analytical approaches) to complex simulations. This module is aimed at those students who wish to focus initially on the field of transportation. This area can be further deepened with the module Transportation Modelling and Traffic Management [WI4INGBGU16]. An interest in transport planning and the transport sector is a prerequisite.

Module Grade Calculation
grade of the module is CP weighted average of grades of the partial exams

Additional Information
The module Transportation Modelling and Traffic Management [WI4INGBGU16] is also offered and recommended to deepen knowledge.

As from the winter term 2025/26, Transportation Systems [T-BGU-106610] will no longer be an elective option. Characteristics of Transportation Systems [T-BGU-106609] is therefore compulsory. Addtionally, the Seminar on Transportation Policy [T-BGU-114772] is available as an elective option in the compulsory elective area.

Workload
contact hours (1 HpW = 1 h x 15 weeks):

- Characteristics of Transportation Systems lecture: 30 h

according to selected courses or examinations respectively:

- Freight Transport lecture/exercise: 30 h
- Long-Distance and Air Traffic lecture: 30 h
- Tendering, Planning and Financing in Public Transport lecture: 30 h
- Seminar in Transportation: 30 h
- Seminar on Modeling and Simulation in Transportation: 30 h
- Mobility Services and New Forms of Mobility lecture/exercise: 30 h
- Strategic Transport Planning lecture: 30 h
- Information Management for Public Mobility Services lecture/exercise: 30 h
- Sustainability in Mobility Systems lecture: 30 h
- Seminar on Transport Policy: 30 h

independent study:

- preparation and follow-up lectures Characteristics of Transportation Systems: 30 h
- examination preparation Characteristics of Transportation Systems (partial examination, compulsory): 30 h

according to selected courses or examinations respectively:

- preparation and follow-up lecture/exercises Freight Transport: 30 h
- examination preparation Freight Transport (selectable partial examination): 30 h
- preparation and follow-up lectures Long-Distance and Air Traffic: 30 h
- examination preparation Long-Distance and Air Traffic (selectable partial examination): 30 h
- preparation and follow-up lectures Tendering, Planning and Financing in Public Transport: 30 h
- examination preparation Tendering, Planning and Financing in Public Transport (selectable partial examination): 30 h
- preparation of seminar paper in Transportation and presentation (selectable partial examination): 60 h
- work on a practical problem in the Seminar on Modeling and Simulation in Transportation (selectable partial examination): 60 h
- preparation and follow-up lecture/exercises Mobility Services and New Forms of Mobility: 30 h
- examination preparation Mobility Services and New Forms of Mobility (selectable partial examination): 30 h
- preparation and follow-up lectures Strategic Transport Planning: 30 h
- examination preparation Strategic Transport Planning (selectable partial examination): 30 h
- preparation and follow-up lecture/exercises Information Management for Public Mobility Services: 30 h
- preparation accompanying exercises Information Management for public Mobility Services (selectable partial examination): 30 h
- preparation and follow-up lectures Sustainability in Mobility Systems: 30 h
- examination preparation Sustainability in Mobility Systems (selectable partial examination): 30 h
- preparation and follow-up meetings Seminar on Transport Policy: 30 h
- preparation of presentation Seminar on Transport Policy (selectable partial examination): 30 h

total: 270 h

Recommendations
None

M 5.79 Module: Generation and Transmission of Renewable Power (WW4INGETIT7) [M-ETIT-101164]

Coordinators: Dr.-Ing. Bernd Hoferer
Prof. Dr.-Ing. Thomas Leibfried
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	2 terms	German/English	4	5

ID	Name	Term offered	Credits	Lecturers
compulsory optional subject (Elective: at least 9 credits)				
T-ETIT-110883	Electric Power Transmission & Grid Control	ST	6 CP	Leibfried
T-ETIT-101915	High-Voltage Test Technique	WT	4 CP	Badent

Assessment
Success controll takes place in form of examinations on the selected courses, which together fulfill the minimum requirement of credit points. The success controlls are described for each module component. The overall grade for the module is calculated from the LP-weighted grades of the two examinations and cut off after the first decimal place.

Prerequisites
The module "M-ETIT-101163 Hochspannungstechnik" must be passed.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module M-ETIT-101163 - High-Voltage Technology must have been passed.

Competence Goal
The student

- has wide knowledge of electrical power engineering,
- is capable to analyse and develop electrical power engineering systems.

Content
The module deals with wide knowledge about the electrical power engineering. This ranges from the electric power equipment networks in terms of function, structure and interpretation on the calculation of electrical power networks to special areas such as the FACTS elements or power transformers.

M 5.80 Module: Global Production and Logistics (WW4INGMB31) [M-MACH-101282]

Coordinators: Prof. Dr.-Ing. Gisela Lanza
Organisation: KIT Department of Mechanical Engineering

Part of: Engineering and Informatics (Mechanical Engineering)
Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	2 terms	English	4	9

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-MACH-114800	Global Production	WT	5 CP	Lanza
T-MACH-115017	Global Logistics	ST	4,5 CP	Furmans

Assessment
Oral exams: duration approx. 5 min per credit point

Written exams: duration approx. 20 - 25 min per credit point

Amount, type and scope of the success control can vary according to the individually choice.

Prerequisites
None

Competence Goal
The students

- are able to analyze the main topics of global production and logistics.
- can explain the main topics about planning and operations of global supply chains and are able to use simple models for planning.
- are capable to name the main topics about planning of global production networks.

Content
The module Global Production and Logistics provides comprehensive and well-founded basics for the main topics of global production and logistics. The lectures aim to show opportunities and market conditions for global enterprises. Part 1 focuses on economic backgrounds, opportunities and risks of global production. Part 2 focuses on the structure of international logistics, their modeling, design and analysis. The threats in international logistics are discussed in case studies.

Additional Information
The module is offered in English.

Workload
The work load is about 270 hours, corresponding to 9 credit points.

Teaching and Learning Methods
Lectures, seminars, workshops, excursions

M 5.81 Module: Handling Characteristics of Motor Vehicles (WI4INGMB6) [M-MACH-101264]

Coordinators: Prof. Dr.-Ing. Marcus Geimer
Organisation: KIT Department of Mechanical Engineering

Part of: Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	1 term	German/English	4	7

ID	Name	Term offered	Credits	Lecturers
Vehicle Properties (Elective: at least 9 credits)				
T-MACH-105152	Handling Characteristics of Motor Vehicles I	WT	4 CP	Weitz
T-MACH-105153	Handling Characteristics of Motor Vehicles II	ST	4 CP	Weitz
T-MACH-102156	Project Workshop: Automotive Engineering	WT+ST	6 CP	Frey, Gießler

Assessment
The assessment is carried out as partial exams (according to Section 4(2) of the examination regulation) of the single courses of this module, whose sum of credits must meet the minimum requirement of credits of this module. The assessment procedures are described for each course of the module seperately.

The overall grade of the module is the average of the grades for each course weighted by the credits and truncated after the first decimal.

Prerequisites
None

Competence Goal
The student

- knows and understands the characteristics of vehicles, owing to the construction and design tokens,
- knows and understands especially the factors being relevant for comfort and acoustics
- is capable of fundamentally evaluating and rating handling characteristics.

Content
See courses.

Workload
The total work load for this module is about 270 Hours (9 Credits). The partition of the work load is carried out according to the credit points of the courses of the module. The work load for courses with 4.5 credit points is about 135 hours, and for courses with 3 credit points about 90 hours. The total number of hours per course results from the time of visiting the lectures and exercises, as well as from the exam duration and the time that is required to achieve the objectives of the module as an average student with an average performance.

Recommendations
Knowledge of the content of the courses *Engineering Mechanics I* [2161238], *Engineering Mechanics II* [2162276] and *Basics of Automotive Engineering I* [2113805], *Basics of Automotive Engineering II* [2114835] is helpful.

M 5.82 Module: Hardware Modeling and Simulation [M-ETIT-100449]

Coordinators: Dr.-Ing. Jens Becker
Prof. Dr.-Ing. Jürgen Becker

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: Engineering and Informatics (Electrical Engineering and Information Technology)
Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each winter term	1 term	English	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-100672	Hardware Modeling and Simulation	WT	4 CP	Becker, Becker

Assessment
Achievement is examined in the form of a written examination lasting 120 minutes.

Prerequisites
none

Competence Goal
After completing this module, students will be familiar with different hardware description languages and their applications in various abstraction levels. They will gain knowledge of the SPICE Hardware Description Language and become proficient in building and deriving the analog matrix for spice simulation. In the realm of digital design, they will develop a comprehensive understanding of the hardware description language VHDL, encompassing the VHDL Standard and its extensions, such as VHDL 2008, the 9-valued logic, and the VHDL-AMS standard. Furthermore, students will achieve a profound comprehension of simulator principles, particularly the delta cycle model. They will also grasp the fundamentals of fault simulations for testing fabricated circuits and learn to derive test vectors. Additionally, students will acquire an understanding of higher-level hardware construction languages like Chisel and SystemC.

Content
In order to address the complexity of modern chips during development, it is essential to utilize modern hardware description languages. This course offers insights into the various levels of abstraction in these languages. It starts by covering the fundamentals of analog description using SPICE and then progresses through VHDL, VHDL-AMS, and Verilog. Additionally, the course introduces more abstract languages like Chisel and SystemC.

Topics covered in the course are:

- Design Process
- Basics of Modeling and Simulation
- Low Level Modeling
- VHDL
  - VHDL-AMS
  - 9-valued logic
  - Delta cycle simulation
  - Fault simulation
- Verilog
- Chisel
- SystemC

Module Grade Calculation
The module grade results from the grade of the written examination.

Workload
The workload is covered by:

1. Participating in lectures and tutorials: 33h
2. Preparing and wrap up of the above named units: 66h
3. Exam preparation and presence: 21h

Sum: 120h = 4 LP

5.83 Module: High-Voltage Technology (WW4INGETIT6) [M-ETIT-101163]

Coordinators: Dr.-Ing. Bernd Hoferer
Prof. Dr.-Ing. Thomas Leibfried

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	2 terms	German	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-110266	High-Voltage Technology	WT	6 CP	Leibfried, Suriyah
T-ETIT-100723	Electronics and EMC	ST	3 CP	Sack

Competence Goal
The student

- has wide knowledge of electrical power engineering,
- is capable to analyse and develop electrical power engineering systems.
- know coupling mechanisms and possible coupling paths for interference signals in electronic circuits and systems, as well as measures for interference suppression and for the functionally reliable construction of such systems.

M 5.84 Module: Human Factors in Security and Privacy [M-WIWI-104520]

Coordinators: Prof. Dr. Melanie Volkamer
Organisation: KIT Department of Business and Economics
Part of: Electives (Informatics)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	2 terms	German	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-WIWI-109270	Human Factors in Security and Privacy	Irreg.	4,5 CP	Volkamer
T-WIWI-108439	Advanced Lab Security, Usability and Society	WT+ST	4,5 CP	Volkamer

Assessment
The module examination is carried out in the form of partial examinations on the selected courses of the module, with which the minimum requirement at creditpoints is fulfilled. The learning control is described in each course. The overall score of the module is made up of the sub-scores weighted with creditpoints and is cut off after the first comma point.

Prerequisites
None

Competence Goal
Students …

- know why many existing security and privacy mechanisms are not usable and why many awareness/education/training approaches are not effective
- can explain for concrete examples why these are not usable / not effective including why people are likely to face problems with these
- can explain what mental models are, why they are important and how they can be identified
- know how to conduct a cognitive walkthrough to identify problems with existing mechanisms and approaches
- know how to conduct semi-structured interviews
- know how user studies in the security context differ from those conducted in other contexts
- can explain the process of human centered security / privacy by design
- know the advantages and disadvantages of various graphical password schemes
- know concepts such as just in time and place security interventions

Content

The history of information security and privacy has taught us that it takes more than technological innovation to develop effective security and privacy mechanisms: Many aspects of information security and privacy actually depend on both technical and human factors. As a result of focusing on the technical factors, we are seeing a persistent gap between theoretical security and actual security in real world which becomes an increasing problem in the age of digitalization. The gap is mainly caused by strong and actually unrealistic assumptions regarding the users' knowledge and behavior.

Human factors in security and privacy research addresses several types of security and privacy mechanisms, e.g., authentication mechanisms including text and graphical passwords, security and privacy indicators (such as the icons in the address bar of nowadays web browsers) and security and privacy interventions like warning messages, permission dialogs and security and privacy policies as well as corresponding configuration interfaces. Besides security and privacy mechanisms, human factors in security and privacy researchers deal with security and privacy awareness, education, and training approaches.

'Human factors in security & privacy' research areas are:

- identifying users' mental models using techniques such as (semi-)structured interviews or focus groups,
- evaluating existing approaches regarding their effectiveness in supporting their users in making secure decisions / informed decisions in the context of privacy using techniques such as cognitive walkthroughs, lab user studies or even field studies,
- proposing improved / new approaches and evaluating their effectiveness using the so called human-centered security / privacy by design approach.

This module discusses the various problems of existing security and privacy mechanisms and security and privacy awareness/ education/training approaches. The lecture addresses relevant psychological and sociological aspects which are important to know and to consider when developing more usable security/privacy mechanisms and more effective awareness/education/ training approaches. The human centered security and privacy by design approach is introduced. Furthermore, some of the methodologies used in this area are explained and a subset of them is applied. Finally, positive examples, such as graphical passwords, are introduced and discussed. Note,the main part of the exercise is replicating an interview based study. The main focus of the lab will be to replicate a quantitative based user study.

Additional Information

Please note:

The course Human Factors in Security and Privacy is offered only **irregularly**. Therefore, completion of this module within two semesters cannot be guaranteed.

Workload

The total workload for this module is approximately 270 hours.

M 5.85 Module: Incentives, Interactivity & Decisions in Organizations [M-WIWI-105923]

Coordinators: Prof. Dr. Petra Nieken
Organisation: KIT Department of Business and Economics
Part of: Business and Economics
Electives (Business Administration)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	2 terms	English	4	5

ID	Name	Term offered	Credits	Lecturers
Elective Offer (Elective:)				
T-WIWI-111912	Advanced Topics in Digital Management	ST	3 CP	Nieken
T-WIWI-111913	Advanced Topics in Human Resource Management	Irreg.	3 CP	Nieken
T-WIWI-113095	Behavioral Lab Exercise	Irreg.	4,5 CP	Nieken, Scheibehenne
T-WIWI-113465	Designing Interactive Systems: Human-AI Interaction	ST	4,5 CP	Mädche
T-WIWI-114174	Economic Decision Making	WT	4,5 CP	Scheibehenne
T-WIWI-115061	Experimental Research in Economics and Information Systems	WT	4,5 CP	Weinhardt
T-WIWI-105781	Incentives in Organizations	ST	4,5 CP	Nieken
T-WIWI-111109	KD²Lab Hands-On Research Course: New Ways and Tools in Experimental Economics	Irreg.	4,5 CP	Weinhardt
T-WIWI-111385	Responsible Artificial Intelligence	WT	4,5 CP	Pfeiffer

Assessment
The assessment is carried out as partial exams of the courses in this module. The assessment procedures are described for each course in the module separately.

The overall grade of the module is the average of grades for each course weighted by the credits and truncated after the first decimal.

Prerequisites
Please refer to the course descriptions for potential restrictions regarding an individual course.

Competence Goal
The student

- understands and analyses challenges and objectives within organizations
- applies economic models and empirical methods to analyze and solve challenges with a focus on the workplace and future of work
- understands the impact of digitalization and new information and communication technology on the work life and management decisions
- knows how to apply scientific research methods and understands the underlying problems

Content
The module „Incentives, Interactivity & Decisions in Organizations" offers an interdisciplinary approach to study incentive structures, the role of interactivity in information systems, and decision making in organizations. The module specifically focuses on topics related to the workplace and the future of work in organizations. The topics range from designing incentive systems and interactive systems to leadership, decision making, as well as understanding human behavior. All courses in the module foster active participation and allow students to learn state-of-the-art research methods and apply them to real-world challenges.

Workload
Total workload for 9 credits: approx. 270 hours.

Recommendations
Knowledge of Human Resource Management, microeconomics, game theory, and statistics is recommended.

M 5.86 Module: Industrial Production II (WW4BWLIIP2) [M-WIWI-101471]

Coordinators: Prof. Dr. Frank Schultmann
Organisation: KIT Department of Business and Economics
Part of: Electives (Business Administration)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each winter term	1 term	German/English	4	8

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-WIWI-114618	Design and Operation of Industrial Plants and Processes	WT	5,5 CP	Schultmann
Supplementary Courses (Elective: at most 1 item)				
T-WIWI-102763	Supply Chain Management with Advanced Planning Systems	ST	3,5 CP	Bosch, Göbelt
T-WIWI-102826	Risk Management in Industrial Supply Networks	WT	3,5 CP	Schultmann
T-WIWI-103134	Project Management	WT	3,5 CP	Schultmann
Supplementary Courses (Elective: at most 1 item)				
T-WIWI-114057	Circular Economy – Challenges and Potentials	ST	3,5 CP	Schultmann
T-WIWI-102634	Emissions into the Environment	WT	3,5 CP	Karl
T-WIWI-112103	Global Manufacturing	WT	3,5 CP	Sasse
T-WIWI-113107	Life Cycle Assessment – Basics and Application Possibilities in an Industrial Context	WT	3,5 CP	Schultmann

Assessment
The assessment is carried out as partial exams (according to section 4 (2), 1 SPO) of the core course "Design and Operation of Industrial Plants and Processes" and one further single course of this module, whose sum of credits must meet the minimum requirement of credits of this module. The assessment procedures are described for each course of the module separately.

The overall grade of the module is the average of the grades for each course weighted by the credits and truncated after the first decimal.

Prerequisites
The course "Design and Operation of Industrial Plants and Processes" and at least one additional activity are compulsory and must be examined.

Competence Goal

- Students shall be able to describe the tasks of tactical production management with special attention drawn upon industrial plants.
- Students shall understand the relevant tasks in plant management (projection, realisation and supervising tools for industrial plants).
- Students shall be able to describe the special need of a techno-economic approach to solve problems in the field of tactical production management.
- Students shall be proficient in using selected techno-economic methods like investment and cost estimates, plant layout, capacity planning, evaluation principles of production techniques, production systems as well as methods to design and optimize production systems.
- Students shall be able to evaluate techno-economical approaches in planning tactical production management with respect to their efficiency, accuracy and relevance for industrial use.

Content

- Planning and Management of Industrial Plants: Basics, circulation flow starting from projecting to techno-economic evaluation, construction and operating up to plant dismantling.

Additional Information
Apart from the core course the courses offered are recommendations and can be replaced by courses from the Module Industrial Production III.

Workload
Total effort will account to 270 hours (9 credit points) and can be allocated according to the credit point rating. Therefore, a course with 3.5 credits requires an effort of approximately 105h and a course with 5.5 credits 165h.

The total effort for each course consists of attending lectures and tutorials, examination times and the time an average student needs to prepare himself in order to pass the exam with an average grade.

5.87 Module: Industrial Production III (WW4BWLIIP6) [M-WIWI-101412]

Coordinators: Prof. Dr. Frank Schultmann
Organisation: KIT Department of Business and Economics
Part of: Electives (Business Administration)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each summer term	1 term	German/English	4	6

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-WIWI-102632	Production and Logistics Management	ST	5,5 CP	Schultmann
Supplementary Courses from Module Industrial Production II (Elective: at most 1 item)				
T-WIWI-102634	Emissions into the Environment	WT	3,5 CP	Karl
T-WIWI-112103	Global Manufacturing	WT	3,5 CP	Sasse
T-WIWI-113107	Life Cycle Assessment – Basics and Application Possibilities in an Industrial Context	WT	3,5 CP	Schultmann
Supplementary Courses (Elective: at most 1 item)				
T-WIWI-114057	Circular Economy – Challenges and Potentials	ST	3,5 CP	Schultmann
T-WIWI-103134	Project Management	WT	3,5 CP	Schultmann
T-WIWI-102826	Risk Management in Industrial Supply Networks	WT	3,5 CP	Schultmann
T-WIWI-102763	Supply Chain Management with Advanced Planning Systems	ST	3,5 CP	Bosch, Göbelt

Assessment
The assessment is carried out as partial exams (according to section 4 (2), 1 SPO) of the core course *Production and Logistics Management* [2581954] and further single courses of this module, whose sum of credits must meet the minimum requirement of credits of this module. The assessment procedures are described for each course of the module separately.

The overall grade of the module is the average of the grades for each course weighted by the credits and truncated after the first decimal.

Prerequisites
The course *Production and Logistics Management* [2581954] and at least one additional activity are compulsory and must be examined.

Competence Goal

- Students describe the tasks concerning general problems of an operative production and logistics management.
- Students describe the planning tasks of supply chain management.
- Students use proficiently approaches to solve general planning problems.
- Students explain the existing interdependencies between planning tasks and applied methods.
- Students describe the mail goals and set-up of software supporting tools in production and logistics management (i.e. APS, PPS-, ERP- and SCM Systems).
- Students discuss the scope of these software tools and their general disadvantages.

Content

- Planning tasks and exemplary methods of production planning and control in supply chain management.
- Supporting software tools in production and logistics management (APS, PPS- and ERP Systems).
- Project management in the field of production and supply chain management.

Additional Information
Apart from the core course the courses offered are recommendations and can be replaced by courses from the Module Industrial Production II.

Workload

The total amount of work for this module is approx. 270 hours (9 credits). The allocation is made according to the credit points of the courses of the module.

The total number of hours per course results from the effort required to attend the lectures and exercises, as well as the examination times and the time required to achieve the learning objectives of the module for an average student for an average performance.

M 5.88 Module: Informatics (WW4INFO1) [M-WIWI-101472]

Coordinators: Dr.-Ing. Tobias Käfer
Prof. Dr.-Ing. Sanja Lazarova-Molnar
Prof. Dr. Andreas Oberweis
Prof. Dr. Harald Sack
Prof. Dr. Alexey Vinel
Prof. Dr. Melanie Volkamer
Prof. Dr.-Ing. Johann Marius Zöllner

Organisation: KIT Department of Business and Economics

Part of: Electives (Informatics)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	1 term	German/English	4	22

ID	Name	Term offered	Credits	Lecturers
Compulsory Elective Area (Elective:)				
T-WIWI-113363	Collective Perception in Autonomous Driving	ST	4,5 CP	Vinel
T-WIWI-102680	Computational Economics	WT	4,5 CP	Shukla
T-WIWI-112690	Cooperative Autonomous Vehicles	ST	4,5 CP	Vinel
T-WIWI-110346	Supplement Enterprise Information Systems	WT+ST	4,5 CP	Oberweis
T-WIWI-110372	Supplement Software- and Systemsengineering	WT+ST	4,5 CP	Oberweis
T-WIWI-114951	Advanced Topics in Digital Twins	Once	4,5 CP	Lazarova-Molnar
T-WIWI-113059	Human Factors in Autonomous Driving	WT	4,5 CP	Vinel
T-WIWI-109270	Human Factors in Security and Privacy	Irreg.	4,5 CP	Volkamer
T-WIWI-113968	Management of IT-Projects	ST	4,5 CP	Alpers
T-WIWI-102666	Knowledge Discovery	WT	4,5 CP	Käfer
T-WIWI-114863	Knowledge-Driven Artificial Intelligence	ST	4,5 CP	Sack
T-WIWI-106340	Machine Learning 1 - Basic Methods	WT	5 CP	Zöllner
T-WIWI-106341	Machine Learning 2 - Advanced Methods	ST	5 CP	Zöllner
T-WIWI-112685	Modeling and Simulation	ST	4,5 CP	Lazarova-Molnar
T-WIWI-102697	Business Process Modelling	WT	4,5 CP	Oberweis
T-WIWI-102679	Nature-Inspired Optimization Methods	ST	4,5 CP	Shukla
T-WIWI-109799	Process Mining	ST	4,5 CP	Schreiber
T-WIWI-110848	Semantic Web Technologies	ST	4,5 CP	Käfer
T-WIWI-102895	Software Quality Management	ST	4,5 CP	Oberweis
Seminars and Advanced Labs (Elective: between 0 and 1 items)				
T-WIWI-110548	Advanced Lab Informatics (Master)	WT+ST	4,5 CP	Professorenschaft des Instituts AIFB
T-WIWI-112914	Advanced Lab Realization of Innovative Services (Master)	WT+ST	4,5 CP	Oberweis
T-WIWI-108439	Advanced Lab Security, Usability and Society	WT+ST	4,5 CP	Volkamer
T-WIWI-109985	Project Lab Cognitive Automobiles and Robots	WT	5 CP	Zöllner
T-WIWI-109983	Project Lab Machine Learning	ST	5 CP	Zöllner

Assessment

The assessment is carried out as partial exams of the single courses of this module, whose sum of credits must meet the minimum requirement of credits of this module. For passing the module exam in every singled partial exam the respective minimum requirements has to be achieved.

The examinations are offered every semester. Re-examinations are offered at every ordinary examination date. The assessment procedures are described for each course of the module separately.

When every singled examination is passed, the overall grade of the module is the average of the grades for each course weighted by the credits and truncated after the first decimal.

Prerequisites
It is only allowed to choose one lab.

Competence Goal
The student

- has the ability to master methods and tools in a complex discipline and to demonstrate innovativeness regarding the methods used,
- knows the principles and methods in the context of their application in practice,
- is able to grasp and apply the rapid developments in the field of computer science, which are encountered in work life, quickly and correctly, based on a fundamental understanding of the concepts and methods of computer science,
- is capable of finding and defending arguments for solving problems.

Content
The thematic focus will be based on the choice of courses in the areas of Applied Technical Cognitive Systems, Business Information Systems, Critical Information Infrastructures, Information Service Engineering, Security - Usability - Society or Web Science.

Workload
The total workload for this module is approximately 270 hours. The total number of hours per course is calculated from the time required to attend the lectures and exercises, as well as the examination times and the time required for an average student to achieve the learning objectives of the module.

M 5.89 Module: Informatics (Department of Informatics) [M-WIWI-104909]

Coordinators: Professorenschaft des Fachbereichs Informatik (KIT-Fakultät für Informatik)
Organisation: KIT Department of Business and Economics
Part of: Additional Examinations

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	pass/fail	Each term	1 term	German	4	5

ID	Name	Term offered	Credits	Lecturers
Elective Courses on Informatics (Elective: at most 30 credits)				
T-INFO-112775	Access Control Systems: Models and Technology	ST	5 CP	Hartenstein
T-INFO-100001	Algorithms I	ST	6 CP	Bläsius
T-INFO-102020	Algorithms II	WT	6 CP	Sanders
T-INFO-101305	Big Data Analytics	WT	5 CP	Böhm
T-INFO-113402	Data Science and Artificial Intelligence for Energy Systems	ST	6 CP	Schäfer
T-INFO-108405	Data Protection by Design	Irreg.	3 CP	Raabe
T-INFO-112344	Introduction to Quantum Computing (IQC)	WT	3 CP	Beckert, Schaefer
T-INFO-114192	Human-Computer-Interaction	ST	6 CP	Beigl
T-INFO-114193	Human-Computer-Interaction Pass	ST	0 CP	Beigl
T-INFO-101337	Internet of Everything	WT	4 CP	Zitterbart
T-INFO-101323	IT-Security Management for Networked Systems	WT	5 CP	Hartenstein
T-INFO-107499	Context Sensitive Systems	ST	5 CP	Beigl
T-INFO-101344	Low Power Design	ST	3 CP	Henkel
T-INFO-101319	Network Security: Architectures and Protocols	ST	4 CP	Zitterbart
T-INFO-101345	Parallel Computer Systems and Parallel Programming	ST	4 CP	Streit
T-INFO-101531	Programming	WT	5 CP	Koziolek, Reussner
T-INFO-101967	Programming Pass	WT+ST	0 CP	Koziolek, Reussner
T-INFO-103531	Computer Organization		6 CP	Karl
T-INFO-111255	Reinforcement Learning	WT	6 CP	Lioutikov, Neumann
T-INFO-101300	Requirements Engineering	ST	3 CP	Koziolek
T-INFO-114261	Software Architecture and Quality	WT	3 CP	Reussner
T-INFO-114188	Ubiquitous Computing	WT	5 CP	Beigl
T-INFO-102047	Seminar: Governance, Risk & Compliance		3 CP	Dreier
T-INFO-101271	Web Applications and Service-Oriented Architectures (II)	ST	4 CP	Abeck

5.90 Module: Informatics (KIT-Department of Economics and Management) [M-WIWI-104901]

Coordinators: Professorenschaft des Instituts AIFB
Organisation: KIT Department of Business and Economics
Part of: Additional Examinations

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	pass/fail	Each term	1 term	German	4	16

ID	Name	Term offered	Credits	Lecturers
Elective Courses on Informatics (Elective: at most 55 credits)				
T-WIWI-110340	Applied Informatics – Applications of Artificial Intelligence	WT	4,5 CP	Käfer
T-WIWI-114156	Applied Informatics – Cybersecurity	ST	4,5 CP	Volkamer
T-WIWI-110341	Applied Informatics – Database Systems	ST	4,5 CP	Oberweis
T-WIWI-113957	Applied Informatics – Mobile Computing	ST	4,5 CP	Schiefer
T-WIWI-110338	Applied Informatics – Modelling	WT	4,5 CP	Fritsch, Oberweis
T-WIWI-113363	Collective Perception in Autonomous Driving	ST	4,5 CP	Vinel
T-WIWI-102680	Computational Economics	WT	4,5 CP	Shukla
T-WIWI-112690	Cooperative Autonomous Vehicles	ST	4,5 CP	Vinel
T-WIWI-110346	Supplement Enterprise Information Systems	WT+ST	4,5 CP	Oberweis
T-WIWI-110372	Supplement Software- and Systemsengineering	WT+ST	4,5 CP	Oberweis
T-WIWI-102749	Foundations of Informatics I	ST	5 CP	Käfer
T-WIWI-102707	Foundations of Informatics II	WT	5 CP	Lazarova-Molnar
T-WIWI-113059	Human Factors in Autonomous Driving	WT	4,5 CP	Vinel
T-WIWI-109270	Human Factors in Security and Privacy	Irreg.	4,5 CP	Volkamer
T-WIWI-113968	Management of IT-Projects	ST	4,5 CP	Alpers
T-WIWI-102666	Knowledge Discovery	WT	4,5 CP	Käfer
T-WIWI-114863	Knowledge-Driven Artificial Intelligence	ST	4,5 CP	Sack
T-WIWI-106340	Machine Learning 1 - Basic Methods	WT	5 CP	Zöllner
T-WIWI-106341	Machine Learning 2 - Advanced Methods	ST	5 CP	Zöllner
T-WIWI-112685	Modeling and Simulation	ST	4,5 CP	Lazarova-Molnar
T-WIWI-102697	Business Process Modelling	WT	4,5 CP	Oberweis
T-WIWI-102679	Nature-Inspired Optimization Methods	ST	4,5 CP	Shukla
T-WIWI-110541	Advanced Lab Informatics (Bachelor)	WT+ST	4,5 CP	Professorenschaft des Instituts AIFB
T-WIWI-110548	Advanced Lab Informatics (Master)	WT+ST	4,5 CP	Professorenschaft des Instituts AIFB
T-WIWI-112915	Advanced Lab Realization of Innovative Services (Bachelor)	WT+ST	4,5 CP	Oberweis
T-WIWI-112914	Advanced Lab Realization of Innovative Services (Master)	WT+ST	4,5 CP	Oberweis
T-WIWI-108439	Advanced Lab Security, Usability and Society	WT+ST	4,5 CP	Volkamer
T-WIWI-109799	Process Mining	ST	4,5 CP	Schreiber
T-WIWI-102735	Introduction to Programming with Java	WT	5 CP	Zöllner
T-WIWI-102747	Advanced Programming - Java Network Programming	ST	4,5 CP	Ratz, Zöllner
T-WIWI-102748	Advanced Programming - Application of Business Software	WT	4,5 CP	Klink, Oberweis, Schreiber, Ullrich
T-WIWI-109985	Project Lab Cognitive Automobiles and Robots	WT	5 CP	Zöllner
T-WIWI-109983	Project Lab Machine Learning	ST	5 CP	Zöllner
T-WIWI-110848	Semantic Web Technologies	ST	4,5 CP	Käfer
T-WIWI-103485	Seminar in Informatics (Bachelor)	WT+ST	3 CP	Professorenschaft des Instituts AIFB

ID	Name	Term offered	Credits	Lecturers
T-WIWI-103479	Seminar in Informatics A (Master)	WT+ST	3 CP	Professorenschaft des Instituts AIFB
T-WIWI-103480	Seminar in Informatics B (Master)	WT+ST	3 CP	Professorenschaft des Instituts AIFB
T-WIWI-100809	Software Engineering	ST	4 CP	Oberweis
T-WIWI-102895	Software Quality Management	ST	4,5 CP	Oberweis
T-WIWI-110343	Applied Informatics – Software Engineering	see notes	4,5 CP	Oberweis

M 5.91 Module: Information Engineering (WW4BWLISM7) [M-WIWI-101411]

Coordinators: Prof. Dr. Christof Weinhardt
Organisation: KIT Department of Business and Economics
Part of: Electives (Business Administration)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	1 term	German/English	4	9

ID	Name	Term offered	Credits	Lecturers
Supplementary Courses (Elective:)				
T-WIWI-107501	Energy Market Engineering	ST	4,5 CP	Weinhardt
T-WIWI-112823	Platform & Market Engineering: Commerce, Media, and Digital Democracy	ST	4,5 CP	Weinhardt
T-WIWI-113727	Special Topics in Information Systems	WT+ST	4,5 CP	Weinhardt

Assessment
The assessment is carried out as partial exams (according to Section 4 (2), 1-3 SPO), whose sum of credits must meet the minimum requirement of credits of this module. The assessment procedures are described for each course of the module separately.

The overall grade of the module is the average of the grades for each course weighted by the credits and truncated after the first decimal.

Competence Goal
The student

- understands and analyzes the central role of information as an economic good, a production factor, and a competitive factor,
- identifies, evaluates, prices, and markets information goods,
- analyze and evaluate existing markets regarding the missing incentives and the optimal solution of a given market mechanism, respectively,
- develop solutions in teams.

Content
In the courses of the module the student can deepen his knowledge on the one hand on the design and operation of markets and on the other hand on the impact of digital goods in network industries regarding the pricing policies, business strategies and regulation issues. If choosen, the course Special Topics in Information Engineering & Management additionally provides an opportunity of practical research in the aforementioned range of subjects.

Additional Information
All practical Seminars offered at the IM can be chosen for *Special Topics in Information Systems*. Please update yourself on www.iism.kit.edu/im/lehre.

Workload
Total workload for 9 credit points: approx. 270 hours

The exact distribution is based on the credit points of the courses in the module.

5.92 Module: Information Systems in Organizations [M-WIWI-104068]

Coordinators: Prof. Dr. Alexander Mädche
Organisation: KIT Department of Business and Economics
Part of: Business and Economics
Electives (Business Administration)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	2 terms	English	4	5

ID	Name	Term offered	Credits	Lecturers
Compulsory Elective Courses (Elective: at least 9 credits)				
T-WIWI-105777	Business Intelligence Systems	WT	4,5 CP	Mädche
T-WIWI-113465	Designing Interactive Systems: Human-AI Interaction	ST	4,5 CP	Mädche
T-WIWI-114210	(Gen)AI-based Automation in Organizations	ST	4,5 CP	Mädche
T-WIWI-113459	Practical Seminar: Human-Centered Systems	WT+ST	4,5 CP	Mädche

Assessment
The assessment is carried out as partial exams (according to Section 4 (2), 1-3 SPO) of the core course and further single courses of this module, whose sum of credits must meet the minimum requirement of credits of this module. The assessment procedures are described for each course of the module separately.

The overall grade of the module is the average of the grades for each course weighted by the credits and truncated after the first decimal.

Prerequisites
None

Competence Goal
The student

- has a comprehensive understanding of conceptual and theoretical foundations of informations systems in organizations
- is aware of the most important classes of information systems used in organizations:. process-centric, information-centric and people-centric information systems.
- knows the most important activities required to execute in the pre-implementation, implementation and post-implementation phase of information systems in organizations in order to create business value
- has a deep understanding of key capabilities of business intelligence systems and/or interactive information systems used in organizations

Content
During the last decades we witnessed a growing importance of Information Technology (IT) in the business world along with faster and faster innovation cycles. IT has become core for businesses from an operational company-internal and external customer perspective. Today, companies have to rethink their way of doing business, from an internal as well as an external digitalization perspective.

This module focuses on the internal digitalization perspective. The contents of the module abstract from the technical implementation details and focus on foundational concepts, theories, practices and methods for information systems in organizations. The students get the necessary knowledge to guide the successful digitalization of organizations. Each lecture in the module is accompanied with a capstone project that is carried out in cooperation with an industry partner.

Additional Information
New module starting summer term 2018.

Workload
The total workload for this module is approximately 270 hours.

5.93 Module: Innovation Management (WW4BWLENT2) [M-WIWI-101507]

Coordinators: Prof. Dr. Marion Weissenberger-Eibl
Organisation: KIT Department of Business and Economics
Part of: Electives (Business Administration)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	1 term	German/English	4	14

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-WIWI-102893	Innovation Management: Concepts, Strategies and Methods	ST	3 CP	Weissenberger-Eibl
Compulsory Elective Courses (Elective: 1 item)				
T-WIWI-114977	Digital Transformation of Public Administration	Irreg.	3 CP	Bauer
T-WIWI-113663	Development of Sustainable, Digital Business Models	Irreg.	3 CP	Weissenberger-Eibl
T-WIWI-111823	Successful Transformation Through Innovation	ST	3 CP	Busch
T-WIWI-102852	Case Studies Seminar: Innovation Management	WT	3 CP	Weissenberger-Eibl
T-WIWI-110263	Methods in Innovation Management	Irreg.	3 CP	Weissenberger-Eibl
T-WIWI-114184	Pioneering Leadership in German SMEs	Irreg.	3 CP	Weissenberger-Eibl
T-WIWI-114748	Science Communication in the Innovation Process	Irreg.	3 CP	Schwind
Supplementary Courses (Elective: 1 item)				
T-WIWI-102866	Design Thinking	Irreg.	3 CP	Terzidis
T-WIWI-114977	Digital Transformation of Public Administration	Irreg.	3 CP	Bauer
T-WIWI-102864	Entrepreneurship	WT+ST	3 CP	Terzidis
T-WIWI-111823	Successful Transformation Through Innovation	ST	3 CP	Busch
T-WIWI-102852	Case Studies Seminar: Innovation Management	WT	3 CP	Weissenberger-Eibl
T-WIWI-110263	Methods in Innovation Management	Irreg.	3 CP	Weissenberger-Eibl
T-WIWI-114184	Pioneering Leadership in German SMEs	Irreg.	3 CP	Weissenberger-Eibl
T-WIWI-114748	Science Communication in the Innovation Process	Irreg.	3 CP	Schwind

Assessment
The module examination takes the form of partial examinations (according to §4(2), 1-3 SPO) on the core course and other courses of the module totaling at least 9 CP. The assessment of success is described for each course of the module.

The overall grade is based 50% on the lecture "Innovation Management: Concepts, Strategies and Methods", 25% on one of the seminars of the Chair of Innovation and Technology Management and 25% on another course permitted in the module. The overall grade is cut off after the first decimal place.

Prerequisites
The lecture “Innovation Management: Concepts, Strategies and Methods” and one of the seminars of the chair for Innovation and Technology Management are compulsory. The third course can be chosen from the courses of the module.

Competence Goal
Students develop a comprehensive understanding of the innovation process and its conditionality. There is an additional focus on the concepts and processes which are of particular relevance with regard to shaping the entire process. Various strategies and methods are then taught based on this.

After completing the module, students should have developed a systemic understanding of the innovation process and be able to shape this by developing and applying suitable methods.

Content
The Innovation Management: Concepts, Strategies and Methods lecture course teaches concepts, strategies and methods which help students to form a systemic understanding of the innovation process and how to shape it. Building on this holistic understanding, the seminar courses then go into the subjects in greater depth and address specific processes and methods which are central to innovation management.

Additional Information
Seminars offered by Prof. Terzidis (or the members of his research group) are not eligible for crediting in a seminar module of the WiWi degree programs. Exception: Seminar "Entrepreneurship Research".

Workload
Total workload for 9 credit points: approx. 270 hours

The exact distribution is based on the credit points of the courses in the module.

Recommendations
None

M 5.94 Module: Integrated Photonics [M-ETIT-107344]

Coordinators: Prof. Dr.-Ing. Christian Koos
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Engineering and Informatics (Electrical Engineering and Information Technology)
Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-114418	Integrated Photonics		6 CP	Koos

Assessment
The assessment takes place in the form of an oral examination (approx. 25 minutes); appointments individually on demand.

Prerequisites
none

Competence Goal
At the end of the course, students

- have a refreshed and deepened understanding of the basic principles of light-matter-interaction and wave propagation in dielectric media,
- know and understand the Lorentz model for frequency-dependent material properties and can use this knowledge to quantitatively analyze the dispersive properties of optical media using Sellmeier relations and scientific databases,
- understand and can quantitatively describe the formation and propagation of surface plasmon polaritons (SPP),
- know basic structures of integrated optical waveguides and understand the formation of guided modes in these structures,
- can quantitatively describe the propagation of signals in optical waveguides under the influence of dispersion,
- are familiar with state-of-the-art material systems and technology platforms of integrated photonics,
- understand basic analytical methods for modeling of photonic circuits such as eigenmode expansion (EME) methods, scattering-matrix formalisms, or coupled-mode theory and can apply these methods to specific use cases,
- understand the concepts and limitations of widely used computational techniques for integrated optical devices such as numerical mode solvers or finite-difference-time-domain (FDTD) simulators,
- know and can quantitatively analyze widely used building blocks and passive devices of integrated photonics, comprising, e.g., multi-mode interference (MMI) couplers, directional couplers, and waveguide gratings, as well as Mach-Zehnder interferometers (MZI), arrayed-waveguide gratings (AWG), lattice filters, and ring resonators.
- understand lasers and optical amplifiers and are familiar with the associated models, e.g., based on rate equations, and their predictions for dynamic behavior
- know material systems and device concepts used for electro-optic modulators
- understand photodetectors and the origin and quantitative description of noise in optoelectronic receiver systems.

Content

This course is geared towards engineering students that want to get a deeper insight into the vividly growing field of integrated photonics. The lecture and the associated tutorial provide an advanced understanding of the physical concepts and associated mathematical models of photonic integrated circuits, of the underlying material platforms and fabrication technologies, and of related application in optical communications, optical sensing, or microwave photonics. The concepts explained in the lecture are widely applicable also in fields outside integrated optics and can be a perfect complement to a variety of topics and adjacent fields such as communications engineering and high-speed data transmission, radio-frequency (RF) electronics, sensor system engineering and automation, photovoltaics, or quantum technologies.

The course covers the following aspects:

1. Review of fundamentals of wave propagation and light-matter interaction in photonics: Maxwell’s equations in optical media, wave equation and plane waves; material dispersion; Lorentz and Drude model of refractive index; Sellmeier equations
2. Plasmonics: Fundamentals of surface-plasmonic polariton (SPP) propagation
3. Integrated optical waveguides: Basic structures; formation of guided modes; mathematical description; orthogonality of modes
4. Propagation of optical signals in waveguides: Group velocity; dispersion; outlook to nonlinear optical effects
5. Optical fibers: Step-index fibers and associated modes; micro-structured fibers
6. Overview of material systems and integration platforms: Silicon photonics, silicon-nitride-based photonic circuits; III-V compound semiconductors; thin-film lithium niobate; polymer-based photonic circuits
7. Eigenmode expansion (EME) method: Concept and applications
8. Numerical methods in integrated photonics: Basics of numerical mode solvers and finite-difference-time-domain (FDTD) simulators
9. Scattering-matrices (S-matrices) for optical devices: Definition and properties of S-matrices
10. Coupled-mode theory and building blocks of integrated photonics: Multi-mode interference (MMI) devices; directional couplers; waveguide gratings
11. Selected passive devices: Mach-Zehnder interferometers (MZI); arrayed-waveguide gratings (AWG); lattice filters; ring resonators
12. Lasers and optical amplifiers: Material systems; quantitative model and rate equations; dynamic behaviour of lasers
13. Electro-optic modulators: Material systems, device concepts, and figures of merit
14. Photodetectors and noise in optoelectronic receiver systems: Photodetector concepts and implementations; direct and coherent detection; shot noise; thermal noise; noise figures

Module Grade Calculation

The module grade is the grade of the oral exam under consideration of any bonuses that may apply

The module grade is the grade of the oral exam under consideration of bonuses based on the problem sets that are solved during the term – details will be given during the lecture. A bonus of 0.3 or 0.4 grades will be granted on the final mark of the oral exam, except for grades worse than 4.0. Bonus points do not expire and are retained for any examinations taken at a later date.

Workload

The workload amounts to approximately 180 h (6 CP), comprising the following items:

Attendance of lectures and tutorials: 15 × (2 h +2 h) = 60 h
Preparation and follow-up of lectures: 15 × 3 h = 45 h
Preparation and follow-up of tutorials: 15 × 3 h = 45 h
Preparation of oral exam: 30 h

Recommendations

Basics understanding of the underlying topics such as electromagnetic fields and waves, semiconductors and solid-state electronics, and advanced calculus. These skills can, e.g., be acquired in the modules “Höhere Mathematik I-III”, “Elektromagnetische Felder und Wellen”, “Festkörperelektronik und Bauelemente“, and „Fundamentals of Photonics“ or in comparable lectures.

M 5.95 Module: Integrated Production Planning (WW4INGMB24) [M-MACH-101272]

Coordinators: Prof. Dr.-Ing. Gisela Lanza
Organisation: KIT Department of Mechanical Engineering

Part of: Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each summer term	1 term	German	4	3

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-MACH-109054	Integrated Production Planning in the Age of Industry 4.0	ST	9 CP	Lanza

Assessment
Written Exam (120 min)

Prerequisites
none

Competence Goal
The students

- can discuss basic questions of production technology.
- are able to apply the methods of integrated production planning they have learned about to new problems.
- are able to analyze and evaluate the suitability of the methods, procedures and techniques they have learned about for a specific problem.
- can apply the learned methods of integrated production planning to new problems.
- can use their knowledge targeted for efficient production technology.

Content
Within this engineering sciences-oriented module the students will get to learn principle aspects of organization and planning of production systems.

Workload
regular attendance: 63 hours
self-study: 207 hours

Teaching and Learning Methods
Lecture, exercise, excursion

M 5.96 Module: Intellectual Property Law (WW4JURA4) [M-INFO-101215]

Coordinators: N.N.
Organisation: KIT Department of Informatics
Part of: Electives (Law or Sociology)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	2 terms	German	4	5

ID	Name	Term offered	Credits	Lecturers
Intellectual Property Law (Elective: at least 1 item as well as at least 9 credits)				
T-INFO-101308	Copyright	WT	3 CP	N.N.
T-INFO-101313	Trademark and Unfair Competition Law	WT+ST	3 CP	Matz
T-INFO-101307	Internet Law	WT	3 CP	N.N.
T-INFO-108462	Selected Legal Issues of Internet Law	ST	3 CP	N.N.
T-INFO-101310	Patent Law	ST	3 CP	Straßburger

Assessment
see partial achievements

Prerequisites
None

Competence Goal
The student

- has detailed knowledge of the main intellectual property rights,
- analyzes and evaluates complex issues and leads them to a legal solution,
- translates the legal principles into contracts on the use of intellectual property and solves more complex infringement cases
- knows and understands the main features of registration procedures and has a broad overview of legal issues raised by the Internet.
- analyzes, assesses and evaluates relevant legal issues from a legal, information technology and legal policy perspective, economic and legal policy perspectives

Content
The module provides knowledge in the core areas of intellectual property law and core topics of internet law. It explains the requirements and the necessary procedure for protecting inventions and industrial marks nationally and internationally. In addition, the necessary know-how is taught to use intellectual property rights and to defend intellectual property rights against attacks by third parties.

Workload
The total workload for this module is approximately 270 hours (9 credits). The allocation is based on the credits of the courses of the module. The workload for courses with 3 credits is about 90 hours.The total number of hours per course results from the effort required to attend the lectures as well as the examination times and the time required to achieve the learning objectives of the module for an average student for an average performance.

M 5.97 Module: Intelligent Systems and Services (IW4INAIFB5) [M-WIWI-101456]

Coordinators: Dr.-Ing. Tobias Käfer
Organisation: KIT Department of Business and Economics
Part of: Electives (Informatics)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	1 term	German/English	4	9

ID	Name	Term offered	Credits	Lecturers
Compulsory Elective Courses (Elective: between 9 and 10 credits)				
T-WIWI-114951	Advanced Topics in Digital Twins	Once	4,5 CP	Lazarova-Molnar
T-WIWI-102666	Knowledge Discovery	WT	4,5 CP	Käfer
T-WIWI-114863	Knowledge-Driven Artificial Intelligence	ST	4,5 CP	Sack
T-WIWI-112685	Modeling and Simulation	ST	4,5 CP	Lazarova-Molnar
T-WIWI-110548	Advanced Lab Informatics (Master)	WT+ST	4,5 CP	Professorenschaft des Instituts AIFB
T-WIWI-110848	Semantic Web Technologies	ST	4,5 CP	Käfer

Assessment
The assessment mix of each course of this module is defined for each course separately. The final mark for the module is the average of the marks for each course weighted by the credits and truncated after the first decimal.

Algorithms for Internet Applications [T-WIWI-102658]: The examination will be offered latest until summer term 2017 (repeaters only).

Prerequisites
None

Competence Goal
Students

- know the different machine learning procedures for the supervised as well as the unsupervised learning,
- identify the pros and cons of the different learning methods,
- apply the discussed network learning methods in specific scenarios,
- compare the practicality of methods and algorithms with alternative approaches.

Content
In the broader sense learning systems are understood as biological organisms and artificial systems which are able to change their behavior by processing outside influences. Network leaning methods based on symbolic, statistic and neuronal approaches are the focus of Computer Sciences.

In this module the most important network learning methods are introduced and their applicability is discussed with regard to different information sources such as data texts and images considering especially procedures for knowledge acquirement via data and text mining, natural analogue procedures as well as the application of organic learning procedures within the finance sector.

Additional Information
Detailed information on the recognition of examinations in the field of Informatics can be found at http://www.aifb.kit.edu/web/Auslandsaufenthalt.

M 5.98 Module: Interdisciplinary Qualifications [M-WIWI-104910]

Coordinators: Studiendekan des KIT-Studienganges
Organisation: KIT Department of Business and Economics
Part of: Additional Examinations

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	pass/fail	Each term	1 term	German	4	5

ID	Name	Term offered	Credits	Lecturers
Interdisciplinary Qualifications (Elective: at most 40 credits)				
T-INFO-109862	Guide to Cruising through Informatics Program of Studies at KIT (eezi) *This item will not influence the grade calculation of this parent.*	WT	1 CP	Gheta, Glaubitz
T-WIWI-112967	Tutoring: Training and Practice *This item will not influence the grade calculation of this parent.*	WT+ST	2 CP	
T-WIWI-111438	Self-Booking-HOC-SPZ-FORUM-STK-Graded *This item will not influence the grade calculation of this parent.*		1 CP	
T-WIWI-111439	Self-Booking-HOC-SPZ-FORUM-STK-Graded *This item will not influence the grade calculation of this parent.*		2 CP	
T-WIWI-111440	Self-Booking-HOC-SPZ-FORUM-STK-Graded *This item will not influence the grade calculation of this parent.*		3 CP	
T-WIWI-113352	Self-Booking-HOC-SPZ-FORUM-STK-Graded *This item will not influence the grade calculation of this parent.*		1 CP	
T-WIWI-113353	Self-Booking-HOC-SPZ-FORUM-STK-Graded *This item will not influence the grade calculation of this parent.*		2 CP	
T-WIWI-113354	Self-Booking-HOC-SPZ-FORUM-STK-Graded *This item will not influence the grade calculation of this parent.*		3 CP	
T-WIWI-111441	Self-Booking-HOC-SPZ-FORUM-STK-Ungraded *This item will not influence the grade calculation of this parent.*		1 CP	
T-WIWI-111442	Self-Booking-HOC-SPZ-FORUM-STK-Ungraded *This item will not influence the grade calculation of this parent.*		2 CP	
T-WIWI-111443	Self-Booking-HOC-SPZ-FORUM-STK-Ungraded *This item will not influence the grade calculation of this parent.*		3 CP	
T-WIWI-113355	Self-Booking-HOC-SPZ-FORUM-STK-Ungraded *This item will not influence the grade calculation of this parent.*		1 CP	
T-WIWI-113356	Self-Booking-HOC-SPZ-FORUM-STK-Ungraded *This item will not influence the grade calculation of this parent.*		2 CP	
T-WIWI-113357	Self-Booking-HOC-SPZ-FORUM-STK-Ungraded *This item will not influence the grade calculation of this parent.*		3 CP	

M 5.99 Module: Introduction to Automotive and Industrial Lidar Technology [M-ETIT-105461]

Coordinators: Prof. Dr. Wilhelm Stork
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Engineering and Informatics (Electrical Engineering and Information Technology)
Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-111011	Introduction to Automotive and Industrial Lidar Technology	WT	3 CP	Stork

Assessment
The examination consists of an oral exam and a short oral presentation. The overall impression is rated.

Competence Goal

- The students are able to explain the basic principles of a lidar sensor
- The students can explain all relevant components of a lidar sensor and put them in context
- The students can explain different forms of execution and make a meaningful choice depending on the requirements
- The students can describe lidar sensors theoretically using the lidar equations and explain the interactions based on this theory
- The students are able to assess the eye safety of a system
- The students are able to suggest possible sensor concepts for different applications or to evaluate existing concepts

Content
In this course the functionality of a lidar sensor is explained and then put into context with relevant use cases. Typical criteria for the evaluation of the performance are then presented. In the following the concept of the sensor is presented in detail and all relevant components are introduced individually. Afterwards they are qualitatively related to each other and the whole system is quantitatively examined by means of the lidar equation. Finally, the interaction of the components is further considered to present meaningful combinations and design solutions. The eye safety of lidar sensors is always explicitly considered. The course concludes with a colloquium in which the students will give short presentations on what they have learned. This repetition is intended to repeat and deepen what has been learned and to lead to a discussion of open question

Module Grade Calculation
The module grade results of the assessment of the oral exam and the short oral presentation. Details will be given during the lecture.

Workload

1. participation in the lectures 12h - 8 dates á 1,5h
2. preparation and postprocessing 14 h (2h for VL dates 1-7)
3. preparation of the short lecture (16h)
4. preparation and participation in the oral exam : 48h

Recommendations
Basics of optics / optical technologies are helpful (e.g. optical engineering, optoelectronic, technical optics)

M 5.100 Module: Introduction to Numerical Simulation of Reacting Flows [M-CIWVT-106676]

Coordinators: Prof. Dr. Oliver Thomas Stein
Organisation: KIT Department of Chemical and Process Engineering
Part of: Engineering and Informatics (Chemical and Process Engineering)
Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
8 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-CIWVT-113435	Introduction to Numerical Simulation of Reacting Flows - Prerequisite		5 CP	Stein
T-CIWVT-113436	Introduction to Numerical Simulation of Reacting Flows		3 CP	Stein

Assessment
The learning control consists of two partial achievements:

1. Completed Coursework: As a prerequisite for the oral exam, reports on the tutorial have to be submitted. These document the processed task, the generated data and their analysis.
2. Oral examination lasting approx. 30 minutes.

Prerequisites
None

Competence Goal
Course participants know the fundamentals of both batch and flow reactors for the simulation of chemical kinetics and reacting flows. They are knowledgeable of numerical methods for temporal and spatial discretisation. In the related Python tutorials, they have obtained a first practical experience in setting up, running and post-processing chemical kinetics and reacting flow simulations, forming the basis for more advanced simulations.

Content

- Introduction to Python
- batch reactors for chemical kinetics simulations
- simple flow reactors
- Newton-Raphson method
- time and space discretisation

Module Grade Calculation
The module grade ist the grade of the oral exam.

Additional Information
The Python tutorials will be conducted on the students' laptops.

Workload

- Attendance time
  Lectures 2 SWS: 30 hrs
  Tutorials 2 SWS: 30 hrs
- Self-study
  Preparation and wrap-up lectures: 15 hrs
  Data analysis, preparation and submission of reports: 105 hrs
- Exam preparation:
  60 hrs

M 5.101 Module: IT/OT-Security Seminar [M-ETIT-106789]

Coordinators: Prof. Dr.-Ing. Mike Barth
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Engineering and Informatics (Electrical Engineering and Information Technology)
Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-113648	IT/OT-Security Seminar	WT	4 CP	Barth

Assessment
The examination takes place in the form of an oral examinationof approx. 25 min.

Prerequisites
none

Competence Goal
The students:
- know the definitions of terms and use-cases in the IT/OT-Security Domain
- know security requirements of both: the industrial information technology perspectiv as well as the production related operational technology domain
- can apply basic cryptographic mechanisms with focus on industrial IT networks
- know protection goals of IT/OT-security
- know various aspects of system security (buffer overflow, return-oriented programming, ...)
- can differentiate between classic information technology (IT) and operational technology (OT) in an industrial environment
- are familiar with attacks on industrial automation and control systems (Industrial Control Systems - ICS)
- are familiar with various concepts (defense-in-depth, security by design, ...) and specific security mechanisms (Public-Key-Infrastucture, network segmentation, ...) of OT security
- are familiar with current international security standards for ICS, in particular IEC 62443
- know the different roles involved and their challenges in the life cycle of ICS
- know and understand the concept of a risk analysis for security
- can evaluate the quality of security mechanisms and architectures for industrial systems
- know typical industrial communication protocols and can analyze and evaluate their security mechanisms

Content
- Industrial control and automation systems (ICS) are widely used in numerous domains and industries. They play a crucial role in areas such as industrial production, the process industry, critical infrastructures such as energy and water management, building automation and medical devices.

- In recent years, the frequency of vulnerabilities and attacks on these systems has increased, especially since the emergence of Stuxnet in 2014. As a result, the protection of ICS has become increasingly important.

- Compared to conventional IT systems, ICS have different boundary conditions and requirements. In particular, the focus is on availability and maintaining functional safety. Therefore, classic approaches to information security cannot be applied to industrial control systems without adaptation.

- This module first provides basic knowledge of security. Building on this, concepts, mechanisms and standards for the specific domain of ICS are introduced. This includes, for example:

o Defense-in-Depth concepts
o Risk-based approaches
o IEC 62443
o Structure and operation of cyber security management systems
o Security engineering
o Use of security information and event management systems in the industrial environment
o Secure use of Industry 4.0 technologies such as OPC UA

Module Grade Calculation
The module grade is the grade of the oral exam.

Workload
The workload includes:

1. attendance in seminar lectures and exercises: 12*2 h = 24 h
2. preparation / follow-up of seminar lectures: 12*3 h = 36 h
3. implementation of challenges and exercises: 12*3 h = 36 h
4. preparation of exam: 24 h.

A total of 120 h = 4 CR

Recommendations
Enjoy working with networked software systems in the production and industrial IT environment. Curiosity in the interplay between attackers and defenders as well as a general affinity to software related topics.

M 5.102 Module: Laser Metrology [M-ETIT-100434]

Coordinators: Prof. Dr. Marc Eichhorn
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Engineering and Informatics (Electrical Engineering and Information Technology)
Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each summer term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-100643	Laser Metrology	ST	3 CP	Eichhorn

Assessment
The exam will be taken as an oral examination (about 20 minutes). The individual appointments for examination are offered regularly at two previously determined dates.

Prerequisites
none

Competence Goal
The students understand the fundamental properties of laser light and possess the knowledge necessary to understand the metrologically obtainable information, understand the basics of various detectors as well as their limits and have the knowledge necessary to understand a multitude of laser metrological setups, mainly for interferometry, Moiré methods, distance and velocity measurements and absorption as well as scattering techniques.

Content
In the module several aspects of laser diagnostics will be discussed, beginning with the fundamental properties of laser light and the related metrologically useful information. In addition beam diagnostics and interferometric setups in general, as well as Moiré methods in particular, will be discussed. Further topics of the lecture will be commonly used setups, mainly for laser distance and velocity measurements along with widely used absorption and scattered light methods.

1. Laser diagnostics - theoretical considerations (laser beam properties, coherence, spectral emission of lasers, mode structure and selection, coherence length)
2. Metrological accessible information (propagation in homogeneous and isotropic, in inhomogeneous and in anisotropic media)
3. Beam diagnostics (photoelectric detectors, information theory, granulation properties of laser light)
4. Laser-Interferometer (fundamentals, two-beam Interferometer, interferometry applications in plasma physics, two- and multiwavelength-interferometry, laser gyroscopes)
5. Moiré technique (Moiré deflectometry, Fresnel- and Fraunhofer diffraction, applications and evaluation of the Moiré technique)
6. Laser range measurements (fundamentals, atmospheric influence on propagation, optical distance measurement techniques, accuracy, sensitivity, heterodyne detection, selected heterodyne detection schemes, tomoscopy)
7. Laser velocity measurement techniques (Doppler principle, measuring flow velocities using Doppler effect, the two-focus technique or laser anemometry; time-resolved imaging particle-trace anemometry)
8. Absorption and scattering techniques (absorption techniques, LIDARs, scattering processes in laser diagnostics, spontaneous scattering techniques, spectroscopic techniques, stimulated scattering, nonlinear optical laser light scattering techniques)

Module Grade Calculation
The module grade is the grade of the oral exam.

Workload
About 90 h in total, consisting of

30 h lectures

60 h recapitulation and self-studies

Recommendations
Solid mathematical background, basic knowledge in physics. Steady participation in the lecture as well as thorough preparation based on the scriptum is highly recommended.

Literature

M. Eichhorn, *Laser metrology* - Scriptum

A. E. Siegman, *Lasers* (university Science Books)

B. E. A. Saleh, M. C. Teich, *Fundamentals of Photonics* (Wiley-Interscience)

M 5.103 Module: Law [M-WIWI-104903]

Coordinators: Professorenschaft des Fachbereichs Recht
Organisation: KIT Department of Business and Economics
Part of: Additional Examinations

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	pass/fail	Each term	1 term	German	4	5

ID	Name	Term offered	Credits	Lecturers
Elective Courses on Law (Elective: at most 40 credits)				
T-INFO-111436	Employment Law	ST	3 CP	Hoff
T-INFO-108462	Selected Legal Issues of Internet Law	ST	3 CP	N.N.
T-INFO-103339	Civil Law for Beginners	WT	5 CP	Matz
T-INFO-108405	Data Protection by Design	Irreg.	3 CP	Raabe
T-INFO-101303	Data Protection Law	WT	3 CP	Eichenhofer
T-INFO-101312	European and International Law	ST	3 CP	Brühann
T-INFO-101307	Internet Law	WT	3 CP	N.N.
T-INFO-101313	Trademark and Unfair Competition Law	WT+ST	3 CP	Matz
T-INFO-101311	Public Media Law	WT	3 CP	
T-INFO-110300	Public Law I & II	ST	6 CP	Zufall
T-INFO-101310	Patent Law	ST	3 CP	Straßburger
T-INFO-102013	Exercises in Civil Law	WT+ST	9 CP	Matz
T-INFO-101288	Corporate Compliance	WT	3 CP	Herzig
T-INFO-115024	Seminar: Legal Studies Bachelor	WT+ST	3 CP	Professorenschaft des Fachbereichs Recht
T-INFO-115025	Seminar: Legal Studies Master I		3 CP	Professorenschaft des Fachbereichs Recht
T-INFO-115026	Seminar: Legal Studies Master II		3 CP	Professorenschaft des Fachbereichs Recht
T-INFO-111437	Tax Law	ST	3 CP	Dietrich
T-INFO-101309	Telecommunications Law	ST	3 CP	
T-BGU-111102	Environmental Law	WT	3 CP	Smeddinck
T-INFO-101308	Copyright	WT	3 CP	N.N.
T-INFO-102047	Seminar: Governance, Risk & Compliance		3 CP	Dreier
T-INFO-101316	Law of Contracts	ST	3 CP	Hoff
T-INFO-102036	Computer Contract Law	WT	3 CP	Menk

5.104 Module: Lean Management in Construction (bauiEX404-LEANMAN) [M-BGU-101884]

Coordinators: Prof. Dr.-Ing. Shervin Haghsheno
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each winter term	2 terms	German	4	4

Election Notes
The course Project Management in Construction and Real Estate Industry II is only allowed to be selected if the selectable course Project Management in Construction and Real Estate Industry I has been passed in the context of another module.

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-BGU-108000	Lean Construction	WT+ST	4,5 CP	Haghsheno
T-BGU-101007	Project Paper Lean Construction	WT	1,5 CP	Haghsheno
Electives (Elective: between 1 and 2 items as well as between 3 and 4,5 credits)				
T-BGU-111921	Turnkey Construction	WT+ST	3 CP	Haghsheno
T-BGU-111922	Civil Engineering Structures and Regenerative Energies	WT+ST	3 CP	Haghsheno
T-BGU-103427	Site Management	ST	1,5 CP	Haghsheno
T-BGU-111211	Energetic Refurbishment	WT+ST	1,5 CP	Lennerts, Schneider
T-BGU-103432	Project Management in Construction and Real Estate Industry I	WT	3 CP	Haghsheno
T-BGU-103433	Project Management in Construction and Real Estate Industry II	WT	3 CP	Haghsheno

Assessment
- module component T-BGU-108000 with written examination according to § 4 Par. 2 No. 1
- module component T-BGU-101007 with examination of another type according to § 4 Par. 2 No. 3

according to selected course:

- module component T-BGU-111921 with written examination according to § 4 Par. 2 No. 1
- module component T-BGU-111922 with written examination according to § 4 Par. 2 No. 1
- module component T-BGU-103427 with oral examination according to § 4 Par. 2 No. 2
- module component T-BGU-111211 with oral examination according to § 4 Par. 2 No. 2
- module component T-BGU-103432 with examination of another type according to § 4 Par. 2 No. 3
- module component T-BGU-103433 with examination of another type according to § 4 Par. 2 No. 3

details about the learning controls, see module components

Prerequisites
none

Competence Goal
Students can explain the theoretical principles of Lean Construction. They can also choose the right process management approach for a project, adapting and improving it as necessary. Furthermore, they are able to identify and analyze problems in construction projects from a process perspective. They can explain the different tools of Lean Construction and select, combine and apply them according to the problem.

Content

- Lean Construction (incl. Lean Construction project work): In this module, the theoretical basics of Lean Construction are presented at the beginning and deepened through learning simulations and exercises. Subsequently, the Last Planner System™, value stream mapping and cooperative contract forms, among others, are examined in depth. Aspects such as construction site logistics, cost and quality management and planning management from a lean perspective will also be discussed. In the exercise, students work in small groups on selected topics based on provided literature and analyze them in the context of the knowledge from the lecture. The results of the small group work are compiled in a written paper and presented at the end of the lecture. To consolidate and reflect on the learning objective, a joint follow-up of the small group work will take place, in which the individual works will be placed in an overall context.
- Turnkey Construction: In the subject area of turnkey construction besides the detailed design of shell construction, technical support and technical building services, there is also an explanation of the related basic knowledge in engineering. Also, basics of the technical support belong to the curriculum, e.g., heating installations, ventilation systems, A/C, electric installations. Most of all, there is a focus on regenerative energies. Furthermore, the explanation of the processes in turnkey construction, from design and construction permit to final acceptance of work, is part of the lecture.
- Civil Engineering Structures and Regenerative Energies: In the subject area of civil engineering structures and regenerative energies, besides basic knowledge in construction, there is also a focus on production methods for the construction and maintenance of the selected civil engineering structures. Adding to conventional construction methods there are topics like additive manufacturing in solid construction. This also includes the view on hydraulic constructions (e.g. water locks), waste disposal (e.g. waste disposal sites) and infrastructure constructions (e.g. steel composite bridge). Also, there is a focus on regenerative energies (e.g. wind power stations).
- Site Management: The course site management presents the work of foreman, site manager, and project manager and contains significant aspects of management processes of the construction site. In addition to performance reporting, work costing and site management, the technical, legal and economic tasks of the site manager as well as communication and correspondence on the construction site will be highlighted. In addition, accident prevention regulations, active and passive protection measures as well as the organization of the labor protection during operation and on site are discussed.
- Energy Refurbishment: fundamentals of building physics, energy-related improvement of the building envelope, improvement of building services engineering and economic efficiency, energy sources and local energy conversion (e.g., photovoltaics, geothermal energy, heat pumps, etc.), calculation of the energy demands of buildings, energy performance assessment of buildings according to GEG.
- Project Management in Construction and Real Estate Industry I: In Part I, selected topics in the field of project management in the construction and real estate industry are examined in depth (including project preparation, procurement models, awarding processes, project organization, contract models, quality/schedule/cost management, risk management, project culture, digitalization).
- Project Management in Construction and Real Estate Industry II: In part II, selected topics in the field of project management in the construction and real estate industry are examined in depth (including project preparation, procurement models, awarding processes, project organization, contract models, quality/deadline/cost management, risk management, project culture, digitization).

Module Grade Calculation
grade of the module is CP weighted average of grades of the partial exams

Additional Information
The number of participants in the course 'Lean Construction' is limited to 50. Registration modalities will be published in time on the institute's homepage or in the corresponding ILIAS course. Students of Profile 2 'Project Management and Lean Construction' in the degree program 'Technology and Management in Construction Management' always receive a confirmation of participation (basic module). Any necessary selection for the remaining places will be made with priority given to students from Civil Engineering (study focus module), Technology and Management in Construction Management (specialization module) and Industrial Engineering (elective module in Engineering Sciences), taking into account the degree program and the student's progress.

Workload
contact hours (1 HpW = 1 h x 15 weeks):

- Lean Construction lecture/exercise: 60 h

according to selected courses or examinations respectively:

- Turnkey Construction lecture/exercise: 30 h
- Civil Engineering Structures and Regenerative Energies lecture/exercise: 30 h
- Site Management lecture: 15 h
- Energetic Refurbishment lecture: 15 h
- Project Management in Construction and Real Estate Industry I lecture/exercise: 30 h
- Project Management in Construction and Real Estate Industry II lecture/xercise: 30 h

independent study:

- preparation and follow-up lectures, exercises Lean Construction: 30 h
- preparation of project report Lean Construction (partial examination, compulsory): 30 h
- examination preparation Lean Construction (partial examination, compulsory): 60 h

according to selected courses or examinations respectively:

- preparation and follow-up lecture/exercises Turnkey Construction: 30 h
- examination preparation Turnkey Construction (selectable partial examination): 30 h
- preparation and follow-up lecture/exercises Civil Engineering Structures and Regenerative Energies: 30 h
- examination preparation Civil Engineering Structures and Regenerative Energies (selectable partial examination): 30 h
- preparation and follow-up lectures Site Management: 15 h
- examination preparation Site Management (selectable partial examination): 15 h
- preparation and follow-up lectures Energetic Refurbishment: 15 h
- examination preparation Energetic Refurbishment (selectable partial examination): 15 h
- preparation and follow-up lecture/exercises Project Management in Construction and Real Estate Industry I: 30 h
- examination preparation Project Management in Construction and Real Estate Industry I (selectable partial examination): 30 h
- preparation and follow-up lecture/exercises Project Management in Construction and Real Estate Industry II: 30 h
- examination preparation Project Management in Construction and Real Estate Industry II (selectable partial examination): 30 h

total: 270 h

Recommendations
It is recommend to take the module Fundamentals of construction [WI3INGBGU3] from the Bachelor’s degree program.

Literature
Gehbauer, F. (2013) Lean Management Im Bauwesen. Skript des Instituts für Technologie und Management im Baubetrieb, Karlsruher Institut für Technologie (KIT).
Liker, J. & Meier, D. (2007) Praxisbuch, der Toyota Weg: für jedes Unternehmen. Finanzbuch Verlag.
Rother, M., Shook, J., & Wiegand, B. (2006). Sehen lernen: mit Wertstromdesign die Wertschöpfung erhöhen und Verschwendung beseitigen. Lean Management Institut.

5.105 Module: Logistics and Supply Chain Management [M-MACH-105298]

Coordinators: Prof. Dr.-Ing. Kai Furmans
Organisation: KIT Department of Mechanical Engineering

Part of: Engineering and Informatics (Mechanical Engineering)
Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each summer term	1 term	English	4	4

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-MACH-110771	Logistics and Supply Chain Management	ST	9 CP	Furmans

Assessment
The success control takes place in the form of an examination performance of a different kind. This is composed as follows:

- 50% assessment of a written examination (60 min) during the semester break
- 50% assessment of an oral examination (approx. 20 min) during the semester break

To pass the examination, both examination performances must be passed.

Prerequisites
None

Competence Goal
The student

- has comprehensive and well-founded knowledge of the central challenges in logistics and supply chain management, an overview of various practical issues and the decision-making requirements and models in supply chains,
- can model supply chains and logistics systems using simple models with sufficient accuracy,
- identifies cause-effect relationships in supply chains,
- is able to evaluate supply chains and logistics systems based on the methods they have mastered.

Content
Logistics and Supply Chain Management provides comprehensive and well-founded fundamentals for the crucial issues in logistics and supply chain management. Within the scope of the lectures, the interaction of different design elements of supply chains is emphasized. For this purpose, qualitative and quantitative description models are used. Methods for mapping and evaluating logistics systems and supply chains are also covered. The lecture contents are enriched by exercises and case studies and partially the comprehension of the contents is provided by case studies. The interacting of the elements will be shown, among other things, in the supply chain of the automotive industry.

Module Grade Calculation
grade of the module is grades of the exam

Workload
contact hours (1 HpW = 1 h x 15 weeks):

- lecture: 60 h

independent study:

- preparation and follow-up lectures: 90 h
- preparation of case studies: 60 h
- examination preparation: 60 h

total: 270 h

Recommendations
none

Teaching and Learning Methods
Lectures, tutorials, case studies.

Literature

Knut Alicke: Planung und Betrieb von Logistiknetzwerken: Unternehmensübergreifendes Supply Chain Management, 2003

Dieter Arnold et. al.: Handbuch Logistik, 2008

Marc Goetschalkx: Supply Chain Engineering, 2011

M 5.106 Module: Machine Learning [M-WIWI-103356]

Coordinators: Prof. Dr.-Ing. Johann Marius Zöllner
Organisation: KIT Department of Business and Economics
Part of: Electives (Informatics)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	1 term	German/English	4	3

ID	Name	Term offered	Credits	Lecturers
Compulsory Elective Courses (Elective:)				
T-WIWI-106340	Machine Learning 1 - Basic Methods	WT	5 CP	Zöllner
T-WIWI-106341	Machine Learning 2 - Advanced Methods	ST	5 CP	Zöllner
T-WIWI-109985	Project Lab Cognitive Automobiles and Robots	WT	5 CP	Zöllner
T-WIWI-109983	Project Lab Machine Learning	ST	5 CP	Zöllner

Assessment
The module examination is carried out in the form of partial examinations on the selected courses of the module, with which the minimum requirement at creditpoints is fulfilled. The learning control is described in each course. The overall score of the module is made up of the sub-scores weighted with creditpoints and is cut off after the first comma point.

Prerequisites
None

Competence Goal
Students

- Gain knowledge of basic methods in the field of machine learning.
- Understand advanced machine learning concepts and their possible applications.
- Can classify, formally describe and evaluate machine learning methods.
- Can apply their knowledge for the selection of suitable models and methods for selected problems in the field of machine learning.

Content
The topic of machine learning considering real-world challenges of complex application domains is a rapidly expanding field of knowledge and the subject of numerous research and development projects. Large parts of modern AI methods are based on machine-learned models.

The Machine Learning 1 course introduces students to the rapidly evolving field of machine learning by providing a solid foundation that covers the major concepts and techniques in the field. Students will explore various methods of supervised, unsupervised, and reinforcement learning, as well as associated model types ranging from simple linear classifiers to more complex models, such as Deep Neural Networks.

The lecture "Machine Learning 2" covers advanced and modern machine learning methods. Modern learning methods like Self-Supervised-Learning and Contrastive Learning as well as model architectures like Diffusion Models, Transformers, Graph Neural Networks, are introduced.

In the practical courses, scientific tasks in the field of autonomous driving or robotics are solved with modern machine learning methods. There, the techniques of machine learning are practically oriented.

Workload
The total workload for this module is approximately 270 hours.

5.107 Module: Machine Tools and Industrial Handling (WW4INGMB32) [M-MACH-101286]

Coordinators: Prof. Dr.-Ing. Jürgen Fleischer
Organisation: KIT Department of Mechanical Engineering

Part of: Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each winter term	1 term	German	4	5

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-MACH-110963	Machine Tools and High-Precision Manufacturing Systems	WT	9 CP	Fleischer

Assessment
Oral exam (approx. 45 minutes)

Prerequisites
None

Competence Goal
The students

- are able to assess the use and application of machine tools and high-precision manufacturing systems and to differentiate between them in terms of their characteristics and design.
- can describe and discuss the essential elements of machine tools and high-precision manufacturing systems (frame, main spindle, feed axes, peripheral equipment, control unit).
- are able to select and dimension the essential components of machine tools and high-precision manufacturing systems.
- are capable of selecting and evaluating machine tools and high-precision manufacturing systems according to technical and economic criteria.

Content
The module gives an overview of the construction, use and application of machine tools and high-precision manufacturing systems. In the course of the module a well-founded and practice-oriented knowledge for the selection, design and evaluation of machine tools and high-precision manufacturing systems is conveyed. First, the main components of the systems are systematically explained and their design principles as well as the integral system design are discussed. Subsequently, the use and application of machine tools and high-precision manufacturing systems will be demonstrated using typical machine examples. Based on examples from current research and industrial applications, the latest developments are discussed, especially concerning the implementation of Industry 4.0 and artificial intelligence.
Guest lectures from industry round off the module with insights into practice.

The individual topics are:

- Structural components of dynamic manufacturing Systems
- Feed axes: High-precision positioning
- Spindles of cutting machine Tools
- Peripheral Equipment
- Machine control unit
- Metrological Evaluation
- Maintenance strategies and condition Monitoring
- Process Monitoring
- Development process for machine tools and high-precision manufacturing Systems
- Machine examples

Workload
regular attendance: 63 hours
self-study: 207 hours

Teaching and Learning Methods
Lecture, exercise, excursions

M 5.108 Module: Major Field: Integrated Product Development [M-MACH-102626]

Coordinators: Prof. Dr.-Ing. Tobias Düser
Organisation: KIT Department of Mechanical Engineering

Part of: Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
16 CP	graded	Each winter term	1 term	German	4	5

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-MACH-105401	Integrated Product Development	WT	16 CP	Düser

Assessment
See course ("Teilleistung")

Prerequisites
None

Competence Goal
By working practically in experience-based learning arrangements with industrial development tasks, graduates are able to succeed in new and unknown situations when developing innovative products by using methodological and systematic approaches. They can apply and adapt strategies of development and innovation management, technical system analysis and team leadership to the situation. As a result, they are able to foster the development of innovative products in industrial development teams in prominent positions, taking into account social, economic and ethical aspects.

Content
Product development, innovation management, competitiveness

Product management: fundamentals, strategic project management, operational project management, lifecycle management

Process models and project management

Cross-generational system development

Problem solving

Leadership and organization

Development management: operational development management, methodological development management, digitization in development management; idea generation; evaluation methods, reaching decisions; conceptualization / modeling principle and form; validation and verification, prototypes

Cost, quality and risk management, target costing

Implementation of corporate strategies

Portfolio management

Invited Lectures

Additional Information
The participation in the course "Integrated Product Development" requires the simultaneous participation in the lecture(2145156), the workshop (2145157) and the product development project (2145300).

For organisational reasons, admission to the product development project is limited. Therefore, a selection process will take place. Registration for the selection process is done via a registration form, which is available annually from April to July on the IPEK homepage. The selection itself will then be made in personal interviews with Prof. Düser in accordance with §5 (4) SPO 2025 Master of Mechanical Engineering.

Workload
The work load is about 480 hours, corresponding to 16 credit points.

Recommendations
Attendance at the Advanced Systems Engineering lecture.

Teaching and Learning Methods

Lecture

Tutorial

Product development project

Literature

Klaus Ehrlenspiel - Integrierte Produktentwicklung. Denkabläufe, Methodeneinsatz, Zusammenarbeit, Hanser Verlag, 2009

M 5.109 Module: Management Accounting (WW3BWLIBU1) [M-WIWI-101498]

Coordinators: Prof. Dr. Marcus Wouters
Organisation: KIT Department of Business and Economics
Part of: Business and Economics
Electives (Business Administration)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	see Annotations	2 terms	English	4	3

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-WIWI-102800	Management Accounting 1	see notes	4,5 CP	Wouters
T-WIWI-102801	Management Accounting 2	see notes	4,5 CP	Wouters

Assessment
The course "Management Accounting 1" and the corresponding assessment for first-time examinees will be offered for the last time in the **summer semester 2026**.

The course "Management Accounting 2" and the corresponding assessment for first-time examinees will be offered for the last time in the **winter semester 2026/2027**.

The assessment is carried out as partial exams (according to Section 4 (2), 13 SPO) of the courses of this module. The assessment procedures are described for each course of the module separately.

The overall grade of the module is the average of the grades for each course weighted by the credits
and truncated after the first decimal.

Competence Goal
Students

• are familiar with various management accounting methods,

• can apply these methods for cost estimation, profitability analysis, and product costing,

• are able to analyze short-term and long-decisions with these methods,

• have the capacity to devise instruments for organizational control.

Content
The module consists of two courses "Management Accounting 1" and "Management Accounting 2". The emphasis is on structured learning of management accounting techniques.

Additional Information
Please note that the module can be taken **for the last time in the winter term of 2026/2027**.

The course "Management Accounting 1" and the corresponding assessment for first-time examinees will be offered for the last time in the **summer semester 2026**.

The course "Management Accounting 2" and the corresponding assessment for first-time examinees will be offered for the last time in the **winter semester 2026/2027**.

Workload
Total workload for 9 credit points: approx. 270 hours

The exact distribution is based on the credit points of the courses in the module.

M 5.110 Module: Manufacturing Technology (WW4INGMB23) [M-MACH-101276]

Coordinators: Prof. Dr.-Ing. Volker Schulze
Organisation: KIT Department of Mechanical Engineering

Part of: Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each winter term	1 term	German	4	7

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-MACH-113570	Additive Manufacturing Processes	ST	4 CP	Zanger
T-MACH-114058	Process Design for Machining of Metallic Components	WT	4 CP	Schulze
T-MACH-114686	Overview of Manufacturing Technologies für Industrial Engineers	WT	1 CP	Schulze

Assessment
Siehe einzelne Teilleistungen

Prerequisites
None

Competence Goal
The students

- can name different manufacturing processes, can describe their specific characteristics and are capable to depict the general function of manufacturing processes and are able to assign manufacturing processes to the specific main groups.
- are enabled to identify correlations between different processes and to select a process depending on possible applications.
- are capable to describe the theoretical basics for the manufacturing processes they got to know within the scope of the course and are able to compare the processes.
- are able to correlate based on their knowledge in materials science the processing parameters with the resulting material properties by taking into account the microstructural effects.
- are qualified to evaluate different processes on a material scientific basis.

Content
Within this engineering sciences-oriented module the students will get to learn principle aspects of manufacturing technology. Further information can be found at the description of the lectures.

Workload
regular attendance: 48 hours
self-study: 222 hours

M 5.111 Module: Market Engineering (WW4BWLISM3) [M-WIWI-101446]

Coordinators: Prof. Dr. Christof Weinhardt
Organisation: KIT Department of Business and Economics
Part of: Electives (Business Administration)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	1 term	German/English	4	10

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-WIWI-112823	Platform & Market Engineering: Commerce, Media, and Digital Democracy	ST	4,5 CP	Weinhardt
Supplementary Courses (Elective: 4,5 credits)				
T-WIWI-102613	Auction Theory	WT	4,5 CP	Ehrhart
T-WIWI-113160	Digital Democracy	WT	4,5 CP	Fegert
T-WIWI-110797	eFinance: Information Systems for Securities Trading	WT	4,5 CP	Weinhardt
T-WIWI-107501	Energy Market Engineering	ST	4,5 CP	Weinhardt
T-WIWI-107503	Energy Networks and Regulation	WT	4,5 CP	Weinhardt
T-WIWI-115061	Experimental Research in Economics and Information Systems	WT	4,5 CP	Weinhardt
T-WIWI-111109	KD²Lab Hands-On Research Course: New Ways and Tools in Experimental Economics	Irreg.	4,5 CP	Weinhardt
T-WIWI-107504	Smart Grid Applications	see notes	4,5 CP	Weinhardt

Assessment
The assessment is carried out as partial exams (according to Section 4 (2), 1-3 SPO) of the core course and further single courses of this module, whose sum of credits must meet the minimum requirement of credits of this module. The assessment procedures are described for each course of the module separately.

The overall grade of the module is the average of the grades for each course weighted by the credits and truncated after the first decimal.

Competence Goal
The students

- know the design criterias of market mechanisms and the systematic approach to create new markets,
- understand the basics of the mechanism design and auction theory,
- analyze and evaluate existing markets regarding the missing incentives and the optimal solution of a given market mechanism, respectively,
- develop solutions in teams.

Content
This module explains the dependencies between the design von markets and their success. Markets are complex interaction of different institution and participants in a market behave strategically according to the market rules. The development and the design of markets or market mechanisms has a strong influence on the behavior of the participants. A systematic approach and a thorough analysis of existing markets is inevitable to design, create and operate a market place successfully. the approaches for a systematic analysis are explained in the mandatory course *Market Engineering* [2540460] by discussing theories about mechanism design and institutional economics. The student can deepen his knowledge about markets in a second course.

Workload
The total workload for this module is approx. 270 hours (9 credits). The distribution is based on the credit points of the courses in the module. The workload for courses with 4.5 credits is approx. 135 hours for courses with 5 credits approx. 150 hours.

The total number of hours per course results from the time required to attend the lectures and exercises, as well as the examination times and the time required to achieve the learning objectives of the module for an average student for an average performance.

Recommendations
None

5.112 Module: Marketing and Sales Management [M-WIWI-105312]

Coordinators: Prof. Dr. Martin Klarmann
Organisation: KIT Department of Business and Economics
Part of: Business and Economics
Electives (Business Administration)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each summer term	1 term	English	4	10

ID	Name	Term offered	Credits	Lecturers
Compulsory Elective Courses (Elective:)				
T-WIWI-112693	Digital Marketing	ST	4,5 CP	Kupfer
T-WIWI-106981	Digital Marketing and Sales in B2B	ST	1,5 CP	Klarmann, Konhäuser
T-WIWI-114174	Economic Decision Making	WT	4,5 CP	Scheibehenne
T-WIWI-107720	Market Research	ST	4,5 CP	Klarmann
T-WIWI-111848	Online Concepts for Karlsruhe City Retailers	ST	3 CP	Kupfer
T-WIWI-102883	Pricing	WT	4,5 CP	Klarmann
T-WIWI-114893	International Entrepreneurship and Brand Development	see notes	6 CP	Kupfer, Terzidis

Assessment
The assessment is carried out as partial exams (according to Section 4(2) of the examination regulation) of the single courses of this module, whose sum of credits must meet the minimum requirement of credits of this module. For passing the module exam in every singled partial exam the respective minimum requirements has to be achieved.

When every singled examination is passed, the overall grade of the module is the average of the grades for each course weighted by the credits and truncated after the first decimal.

Prerequisites
None

Competence Goal
Students

- have an advanced knowledge about central marketing contents
- have a fundamental understanding of the marketing instruments
- know and understand several strategic concepts and how to implement them
- are able to implement their extensive marketing knowledge in a practical context
- know several qualitative and quantitative approaches to prepare decisions in Marketing
- have the theoretical knowledge to write a master thesis in Marketing
- have the theoretical knowledge to work in/together with the Marketing department

Content
This module provides students with a profound understanding of key principles and concepts of marketing management and enables them to independently analyze, develop, and critically reflect on marketing and sales strategies. The focus is on combining theoretical knowledge with its practical application in various contexts, from digital markets and B2B environments to international brand development.

Students work on realistic case studies and application-oriented projects, present their ideas in group settings, and receive continuous feedback. Complemented by collaboration with external partners and the use of qualitative and quantitative market research methods, students learn to make data-driven and strategic marketing decisions.

The module consists of the following courses: Digital Marketing, Digital Marketing and Sales in B2B, International Entrepreneurship and Brand Development, Market Research, Online Concepts for Local Retailers in Karlsruhe, and Pricing. The choice of options within the module allows students to set individual priorities.

Workload
The total workload for this module is approximately 270 hours.

M 5.113 Module: Master's Thesis [M-WIWI-107719]

Coordinators: Studiendekan des KIT-Studienganges
Organisation: KIT Department of Business and Economics
Part of: Master's Thesis

Credits	Grading	Recurrence	Duration	Language	Level	Version
30 CP	graded	Each term	1 term	German	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-WIWI-115109	Master's Thesis		30 CP	Studiendekan des KIT-Studienganges

Assessment
The Master's thesis is a written module component that demonstrates the student's ability to examine a subject-specific problem using scientific research methods. The detailed regulations are set forth in § 14 of the Study and Examination Regulations.

The thesis is supervised and assessed by at least two KIT examiners. At least one examiner must be a professor and should typically be a member of the KIT Department of Business and Economics.

The standard completion period is six months. Upon a justified request by the student, the Examination Board may extend this period by a maximum of three months. If the thesis is not submitted by the deadline, it will be graded as 'insufficient' (5.0), unless the student is not responsible for the delay (e.g., due to maternity leave). By agreement with the examiner, a presentation may be defined as an obligatory and graded part of the thesis. This presentation can take place either before or after the submission of the written work. The preparation time for the presentation is not included in the completion period for the written part, unless it is explicitly part of the overall workload.

The thesis must be written in English. Other languages require both the consent of the examiner and the approval of the Examination Board.

The student may withdraw from the assigned topic only once and only within the first month after registration.

If the Master's thesis is not passed, it may be repeated once. In this case, a new topic must be assigned. Returning to the original or a significantly similar topic is not permitted; the Examination Board shall decide in cases of doubt. The new topic may again be supervised by the examiners of the first attempt. This regulation also applies analogously after an official withdrawal from a registered topic.

The module grade corresponds to the grade of the Master's thesis.

Prerequisites
Admission to the Master's thesis requires the completion of at least 50 percent of the total credit points.

Furthermore, written confirmation from the examiner regarding the supervision of the thesis is mandatory. Please observe the specific internal regulations of the respective institute concerning thesis supervision.

In accordance with the Study and Examination Regulations, the Master's thesis must include the following declaration in both German and English:

- **German version**: 'Ich versichere wahrheitsgemäß, die Arbeit selbstständig verfasst, alle benutzten Hilfsmittel vollständig und genau angegeben und alles kenntlich gemacht zu haben, was aus Arbeiten anderer unverändert oder mit Abänderungen entnommen wurde sowie die Satzung des KIT zur Sicherung guter wissenschaftlicher Praxis in der jeweils gültigen Fassung beachtet zu haben.'
- **English version**: 'I herewith declare that the present thesis is original work written by me alone and that I have indicated completely and precisely all sources and aids used as well as all citations, whether changed or unchanged, of other theses and publications.'

The Master's thesis will not be accepted if these declarations are missing.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. You need to have earned at least 60 credits in the following fields:

Competence Goal
Students are able to independently address a complex and unfamiliar topic based on scientific criteria and the current state of research.

They are capable of critically analyzing and structuring researched information, as well as deriving underlying principles and patterns. Students demonstrate the ability to apply these results effectively to solve the assigned task. By integrating this expertise with their interdisciplinary knowledge, they can draw independent conclusions, identify potential for improvement, and propose as well as implement science-based decisions.

These processes are conducted while taking social and ethical aspects into account.

Furthermore, students can interpret, evaluate, and, if required, graphically represent their results. They are in a position to logically structure a research paper, document their findings, and communicate the results clearly in a scientific format.

Content
The Master’s thesis is a substantial piece of scientific research. The topic is selected by the student in consultation with the examiner. It must be relevant to the field of Industrial Engineering and Management and address either subject-specific or interdisciplinary problems.

Workload
The total workload for the preparation and presentation of the Master’s thesis is approximately 900 hours. In addition to the writing process, this figure encompasses all necessary activities, including literature research, initial familiarization with the topic and required tools, the conduct of studies or experiments, and regular supervision meetings.

M 5.114 Module: Material Flow in Networked Logistic Systems (WW4INGMB26) [M-MACH-101278]

Coordinators: Prof. Dr.-Ing. Kai Furmans
Organisation: KIT Department of Mechanical Engineering

Part of: Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	2 terms	German	4	9

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-MACH-113914	Mathematical Methods for Production Systems	ST	6 CP	Baumann, Furmans
Material flow in interconnected logistics systems (Elective:)				
T-MACH-112213	Applied material flow simulation	WT	4,5 CP	Baumann
T-MACH-113950	Dimensioning of Material Flow Systems in Production and Logistics	WT	4 CP	Furmans
T-MACH-105151	Energy Efficient Intralogistic Systems	WT	4 CP	Kramer, Schönung
T-MACH-115017	Global Logistics	ST	4,5 CP	Furmans
T-MACH-105187	IT-Fundamentals of Logistics	ST	4 CP	Thomas
T-MACH-105174	Warehousing and Distribution Systems	ST	4 CP	Furmans
T-MACH-105171	Safety Engineering	WT	4 CP	Kany

Assessment
The assessment is carried out as partial exams (according to Section 4 (2), 1-3 SPO) of the core course and further single courses of this module, whose sum of credits must meet the minimum requirement of credits of this module. The assessment procedures are described for each course of the module separately. The overall grade of the module is the average of the grades for each course weighted by the credits and truncated after the first decimal.

Prerequisites
none

Competence Goal
The student

- acquires in-depth knowledge on the main topics of logistics, gets an overview of different logistic questions in practice,
- is able to evaluate logistic systems by using the learnt methods,
- is able to analyze and explain the phenomena of industrial material and value streams.

Content
The module *Material Flow in networked Logistic Systems* provides in-depth basics for the main topics of logistics and industrial material and value streams. The obligatory lecture focuses on queuing methods to model production systems. To gain a deeper understanding, the course is accompanied by exercises.

Workload
Regular attendance: 270 hours (9 credits). Lectures with 180 hours attendance 6 credits. Lectures with 120 hours 4 credits.

Teaching and Learning Methods
Lecture, tutorial.

M 5.115 Module: Mathematical Programming (WW4OR9) [M-WIWI-101473]

Coordinators: Prof. Dr. Oliver Stein
Organisation: KIT Department of Business and Economics
Part of: Electives (Operations Research)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	1 term	German/English	4	8

ID	Name	Term offered	Credits	Lecturers
Compulsory Elective Courses (Elective: at most 2 items)				
T-WIWI-102719	Mixed Integer Programming I	Irreg.	4,5 CP	Stein
T-WIWI-102726	Global Optimization I	ST	4,5 CP	Stein
T-WIWI-103638	Global Optimization I and II	ST	9 CP	Stein
T-WIWI-102856	Convex Analysis	Irreg.	4,5 CP	Stein
T-WIWI-111587	Multicriteria Optimization	see notes	4,5 CP	Stein
T-WIWI-102724	Nonlinear Optimization I	WT	4,5 CP	Stein
T-WIWI-103637	Nonlinear Optimization I and II	WT	9 CP	Stein
T-WIWI-102855	Parametric Optimization	Irreg.	4,5 CP	Stein
Supplementary Courses (Elective: at most 2 items)				
T-WIWI-106548	Advanced Stochastic Optimization	Irreg.	4,5 CP	Rebennack
T-WIWI-102720	Mixed Integer Programming II	Irreg.	4,5 CP	Stein
T-WIWI-102727	Global Optimization II	ST	4,5 CP	Stein
T-WIWI-102723	Graph Theory and Advanced Location Models	Irreg.	4,5 CP	Nickel
T-WIWI-106549	Large-scale Optimization	ST	4,5 CP	Rebennack
T-WIWI-111247	Mathematics for High Dimensional Statistics	Irreg.	4,5 CP	Grothe
T-WIWI-103124	Multivariate Statistical Methods	Irreg.	4,5 CP	Grothe
T-WIWI-102725	Nonlinear Optimization II	WT	4,5 CP	Stein
T-WIWI-102715	Operations Research in Supply Chain Management	Irreg.	4,5 CP	Nickel
T-WIWI-112109	Topics in Stochastic Optimization	WT	4,5 CP	Rebennack

Assessment
The assessment is carried out as partial exams (according to Section 4(2), 1 or 2 of the examination regulation) of the single courses of this module, whose sum of credits must meet the minimum requirement of credits of this module. The assessment procedures are described for each course of the module seperately.

The overall grade of the module is the average of the grades for each course weighted by the credits and truncated after the first decimal.

Prerequisites
At least one of the courses "Mixed Integer Programming I", "Multicriteria Optimization", "Convex Analysis", "Parametric Optimization", "Nonlinear Optimization I" and "Global Optimization I" has to be taken.

Competence Goal
The student

- names and describes basic notions for advanced optimization methods, in particular from continuous and mixed integer programming,
- knows the indispensable methods and models for quantitative analysis,
- models and classifies optimization problems and chooses the appropriate solution methods to solve also challenging optimization problems independently and, if necessary, with the aid of a computer,
- validates, illustrates and interprets the obtained solutions,
- identifies drawbacks of the solution methods and, if necessary, is able to makes suggestions to adapt them to practical problems.

Content

This module provides an advanced education in mathematical foundations and sophisticated methods of optimization. It addresses complex decision-making situations where optimal solutions must be found under constraints, covering a wide range of specialized optimization classes.

The core content includes:

- **Global and Nonlinear Optimization**: Handling problems with non-convex structures where local optima are not necessarily global solutions. Teaching of methods for determining global optima.
- **Mixed-Integer Optimizatio**n: Modeling and solving problems where both continuous and integer (discrete) variables occur, which is essential for many applications from engineering and economics.
- **Multicriteria Optimization**: Methods for solving problems with multiple, often conflicting objectives (Pareto optimization).

Students learn both the mathematical modeling of complex systems and the application and implementation of modern solution algorithms to efficiently handle real-world optimization problems.

Additional Information

The lectures are partly offered irregularly. The curriculum of the next three years is available online (www.ior.kit.edu).

For the lectures of Prof. Stein a grade of 30 % of the exercise course has to be fulfilled. The description of the particular lectures is more detailed.

Workload

The total workload for this module is approximately 270 hours.

M 5.116 Module: Mathematics [M-WIWI-104905]

Coordinators: Professorenschaft des Fachbereichs Mathematik
Organisation: KIT Department of Business and Economics
Part of: Additional Examinations

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	pass/fail	Each term	1 term	German	4	2

ID	Name	Term offered	Credits	Lecturers
Elective Courses on Mathematics (Elective: at most 30 credits)				
T-MATH-102265	Seminar in Mathematics (Bachelor)		3 CP	Hug, Nestmann, Winter
T-MATH-102244	Fundamentals of Probability and Statistics for Students of Computer Science		4,5 CP	Bäuerle, Ebner, Fasen-Hartmann, Hug, Klar, Last, Trabs, Winter
T-MATH-111492	Mathematics I - Midterm Exam		5 CP	Hug, Nestmann, Winter
T-MATH-111493	Mathematics I - Final Exam		5 CP	Hug, Nestmann, Winter
T-MATH-111498	Mathematics III - Final Exam		4 CP	Hug, Nestmann, Winter
T-MATH-111495	Mathematics II - Midterm Exam		3,5 CP	Hug, Nestmann, Winter
T-MATH-111496	Mathematics II - Final Exam		3,5 CP	Hug, Nestmann, Winter
T-MATH-115204	Mathematics I - Midterm Exam for WiIng B.Sc. 2026		5 CP	Hug, Nestmann, Winter
T-MATH-115205	Mathematics I - Final Exam for WiIng B.Sc. 2026		5 CP	Hug, Nestmann, Winter
T-MATH-115206	Mathematics II - Midterm Exam for WiIng B.Sc. 2026		3,5 CP	Hug, Nestmann, Winter
T-MATH-115207	Mathematics II - Final Exam for WiIng B.Sc. 2026		3,5 CP	Hug, Nestmann, Winter
T-MATH-115208	Mathematics III - Final Exam for WiIng B.Sc. 2026		4 CP	Hug, Nestmann, Winter

M 5.117 Module: Mechanical Engineering I [M-WIWI-107718]

Coordinators: Studiendekan des KIT-Studienganges
Organisation: KIT Department of Business and Economics
Part of: Engineering and Informatics (Mechanical Engineering)
Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	2 terms	English	4	1

ID	Name	Term offered	Credits	Lecturers
Course Pool: Mechanical Engineering (Elective:)				
T-MACH-113929	Advanced Systems Engineering	ST	4,5 CP	Düser
T-MACH-110929	Applied Materials Simulation	ST	4,5 CP	Gumbsch, Schneider, Weygand
T-MACH-102203	Automotive Engineering I	WT	6 CP	Gießler
T-MACH-114435	Automotive Engineering II	ST	4,5 CP	Gießler
T-MACH-114149	Automotive Vision	ST	6 CP	Lauer, Stiller
T-MACH-115202	Autonomous Driving in Practice	WT	6 CP	Simon
T-MACH-114009	Beyond Conventional Materials - Metamaterials & Architected Structures	WT	4,5 CP	Bauer
T-MACH-113856	Biologically Inspired Robots	ST	3 CP	Rönnau
T-MACH-113857	CAD Engineering Project for Intelligent Systems	WT+ST	6 CP	Rönnau
T-MACH-113998	Chemically Reacting Flows	WT+ST	9 CP	Fischlschweiger
T-MACH-113926	Data and Artificial Intelligence for Numerical Simulations	ST	4,5 CP	August, Koeppe
T-MACH-112126	Data-Driven Algorithms in Vehicle Technology	WT	4,5 CP	Scheubner
T-MACH-113647	Digitalization from Product Concept to Production	WT	4,5 CP	Wawerla
T-MACH-110928	Exercises for Applied Materials Simulation	ST	2 CP	Gumbsch, Schneider, Weygand
T-MACH-113627	Fundamentals of Nuclear Energy and Radiation Protection	WT	4,5 CP	Dagan
T-MACH-115017	Global Logistics	ST	4,5 CP	Furmans
T-MACH-114800	Global Production	WT	5 CP	Lanza
T-MACH-114099	Heat and Mass Transfer	ST	4,5 CP	Fischlschweiger, Yu
T-MACH-112159	Hydrogen in Materials – Exercises and Lab Course	ST	4,5 CP	Wagner
T-MACH-110923	Hydrogen in Materials: from Energy Storage to Hydrogen Embrittlement	ST	4,5 CP	Pundt
T-MACH-113701	Industrial Mobile Robotics Lab	WT+ST	4,5 CP	Furmans
T-MACH-112882	Innovation2Business – Innovation Strategy in the Industrial Corporate Practice	WT	4,5 CP	Albers
T-MACH-110332	Nuclear Power and Reactor Technology		4,5 CP	Badea
T-MACH-112763	Laser Material Processing	ST	4,5 CP	Schneider
T-MACH-106739	Laser-Assisted Methods and Their Application for Energy Storage Materials	WT+ST	4,5 CP	Pfleging
T-MACH-105223	Machine Vision	WT	9 CP	Lauer, Stiller
T-MACH-105210	Machine Dynamics	ST	5 CP	Proppe
T-MACH-113914	Mathematical Methods for Production Systems	ST	6 CP	Baumann, Furmans
T-MACH-114062	Mathematical Models and Methods of the Theory of Thermochemical Processes	WT	4,5 CP	Bykov
T-MACH-105557	Microenergy Technologies	ST	4,5 CP	Kohl, Xu
T-MACH-114061	Model Reduction Methods for Modeling and Simulation of Reacting Flows	ST	4,5 CP	Bykov
T-MACH-114956	Non-ferrous Metals and Alloys	ST	4,5 CP	Heilmaier, White

ID	Name	Term offered	Credits	Lecturers
T-MACH-113977	Nuclear Power Plant and Fusion Technologies	WT+ST	9 CP	Badea, Cheng
T-MACH-114129	Particle Dynamics and Atomistic Simulation	ST	4,5 CP	Gumbsch, Schneider, Weygand
T-MACH-114095	Principles of Whole Vehicle Engineering	WT+ST	4,5 CP	Harrer
T-MACH-115056	Quality Management	WT	4 CP	Lanza
T-MACH-109122	X-ray Optics	WT+ST	4,5 CP	Last
T-MACH-106493	Solar Thermal Energy Systems	WT	4,5 CP	Dagan
T-MACH-114969	Strategic Decision-Making in Global Production Networks through Optimization and Simulation	ST	4,5 CP	Lanza, Martin
T-MACH-105422	Flows with Chemical Reactions	WT	4,5 CP	Class
T-MACH-113987	Sustainable Materials	WT	4,5 CP	Greiner
T-MACH-114608	Sustainable Materials Production	WT	4,5 CP	Dollmann, Greiner
T-MACH-114552	System Integration in Micro- and Nanotechnology I	ST	4,5 CP	Gengenbach
T-MACH-114553	System Integration in Micro- and Nanotechnology II	WT	4,5 CP	Gengenbach
T-MACH-113949	Topology Optimisation in Engineering	ST	4,5 CP	Deng
T-MACH-113982	Validation of Technical Systems	ST	4,5 CP	Düser

Assessment
The module assessment consists of several module components. These are conducted in the form of written or oral examinations, as well as alternative exam assessments, depending on the specific courses selected from the course pool.

The module is successfully completed once module components totaling at least 9 credit points have been passed.

The specific type of examination, the duration, and any prerequisites are specified in the respective description of each module component.

The overall grade of the module is calculated as the credit-point-weighted average of the individual grades of the module components.

Prerequisites
Any specific prerequisites for individual courses are defined and listed within the respective module component descriptions

Competence Goal
After completing the module, students possess a broad and in-depth understanding of fundamental and advanced topics in mechanical engineering. They are able to:

- Analyze complex technical systems and processes within various sub-disciplines of mechanical engineering (e.g., automotive engineering, materials science, or manufacturing technology).
- Apply engineering principles and specialized methods to identify, formalize, and solve technical problems at an advanced level.
- Evaluate different technological approaches and their impact on industrial applications and production environments.
- Integrate interdisciplinary knowledge to address challenges at the intersection of engineering and management.

Content
This module provides a comprehensive overview of diverse subject areas within the field of mechanical engineering. Due to the wide range of elective courses, the specific content depends on the individual student's choice. The curriculum covers key areas such as:

- **Vehicle and Propulsion Technology**: Design, simulation, and analysis of automotive systems and power units.
- **Materials and Micro-Engineering**: Simulation of materials, microsystem technology, and advanced structural mechanics.
- **Manufacturing and Automation**: Robotics, quality management, and the digitalization of production processes.
- **Systems and Process Engineering**: Methods for modeling and managing complex engineering systems.

The module allows students to customize their technical profile by selecting from a versatile pool of courses offered by the KIT Department of Mechanical Engineering.

Workload
The total workload for the module is at least 270 hours (calculated at 30 hours per credit point).

Due to the flexible selection of module components, the specific distribution of the workload varies depending on the chosen courses. In general, the workload for each course is divided into:

- Attendance time: Participation in lectures, tutorials, and seminars.
- Self-study: Preparation and follow-up of the course material, literature research, and completion of assignments or practical exercises.
- Exam preparation: Intensive study phase and participation in the respective examinations.

Detailed information on the specific workload for each course can be found in the individual module component descriptions.

Recommendations
As this module comprises a wide range of courses from the KIT Department of Mechanical Engineering, students are strongly advised to research specific course requirements in advance.

To ensure adequate prior knowledge for the selected module components, it is recommended to:

- Consult the official course catalog (Vorlesungsverzeichnis) at https://campus.studium.kit.edu/events/catalog.php for detailed course descriptions and logistics.
- Contact the respective lecturers at the KIT Department of Mechanical Engineering directly for guidance on necessary prerequisites or technical background.
- Review the specific syllabi or learning materials often provided on the institutes' websites or the ILIAS platform.

M 5.118 Module: Mechanical Engineering II [M-WIWI-107730]

Coordinators: Studiendekan des KIT-Studienganges
Organisation: KIT Department of Business and Economics
Part of: Engineering and Informatics (Mechanical Engineering)
Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	2 terms	English	4	1

ID	Name	Term offered	Credits	Lecturers
Course Pool: Mechanical Engineering (Elective:)				
T-MACH-113929	Advanced Systems Engineering	ST	4,5 CP	Düser
T-MACH-110929	Applied Materials Simulation	ST	4,5 CP	Gumbsch, Schneider, Weygand
T-MACH-102203	Automotive Engineering I	WT	6 CP	Gießler
T-MACH-114435	Automotive Engineering II	ST	4,5 CP	Gießler
T-MACH-114149	Automotive Vision	ST	6 CP	Lauer, Stiller
T-MACH-115202	Autonomous Driving in Practice	WT	6 CP	Simon
T-MACH-114009	Beyond Conventional Materials - Metamaterials & Architected Structures	WT	4,5 CP	Bauer
T-MACH-113856	Biologically Inspired Robots	ST	3 CP	Rönnau
T-MACH-113857	CAD Engineering Project for Intelligent Systems	WT+ST	6 CP	Rönnau
T-MACH-113998	Chemically Reacting Flows	WT+ST	9 CP	Fischlschweiger
T-MACH-113926	Data and Artificial Intelligence for Numerical Simulations	ST	4,5 CP	August, Koeppe
T-MACH-112126	Data-Driven Algorithms in Vehicle Technology	WT	4,5 CP	Scheubner
T-MACH-113647	Digitalization from Product Concept to Production	WT	4,5 CP	Wawerla
T-MACH-110928	Exercises for Applied Materials Simulation	ST	2 CP	Gumbsch, Schneider, Weygand
T-MACH-113627	Fundamentals of Nuclear Energy and Radiation Protection	WT	4,5 CP	Dagan
T-MACH-115017	Global Logistics	ST	4,5 CP	Furmans
T-MACH-114800	Global Production	WT	5 CP	Lanza
T-MACH-114099	Heat and Mass Transfer	ST	4,5 CP	Fischlschweiger, Yu
T-MACH-112159	Hydrogen in Materials – Exercises and Lab Course	ST	4,5 CP	Wagner
T-MACH-110923	Hydrogen in Materials: from Energy Storage to Hydrogen Embrittlement	ST	4,5 CP	Pundt
T-MACH-113701	Industrial Mobile Robotics Lab	WT+ST	4,5 CP	Furmans
T-MACH-112882	Innovation2Business – Innovation Strategy in the Industrial Corporate Practice	WT	4,5 CP	Albers
T-MACH-110332	Nuclear Power and Reactor Technology		4,5 CP	Badea
T-MACH-112763	Laser Material Processing	ST	4,5 CP	Schneider
T-MACH-106739	Laser-Assisted Methods and Their Application for Energy Storage Materials	WT+ST	4,5 CP	Pfleging
T-MACH-105223	Machine Vision	WT	9 CP	Lauer, Stiller
T-MACH-105210	Machine Dynamics	ST	5 CP	Proppe
T-MACH-113914	Mathematical Methods for Production Systems	ST	6 CP	Baumann, Furmans
T-MACH-114062	Mathematical Models and Methods of the Theory of Thermochemical Processes	WT	4,5 CP	Bykov
T-MACH-105557	Microenergy Technologies	ST	4,5 CP	Kohl, Xu
T-MACH-114061	Model Reduction Methods for Modeling and Simulation of Reacting Flows	ST	4,5 CP	Bykov
T-MACH-114956	Non-ferrous Metals and Alloys	ST	4,5 CP	Heilmaier, White

ID	Name	Term offered	Credits	Lecturers
T-MACH-113977	Nuclear Power Plant and Fusion Technologies	WT+ST	9 CP	Badea, Cheng
T-MACH-114129	Particle Dynamics and Atomistic Simulation	ST	4,5 CP	Gumbsch, Schneider, Weygand
T-MACH-114095	Principles of Whole Vehicle Engineering	WT+ST	4,5 CP	Harrer
T-MACH-115056	Quality Management	WT	4 CP	Lanza
T-MACH-109122	X-ray Optics	WT+ST	4,5 CP	Last
T-MACH-106493	Solar Thermal Energy Systems	WT	4,5 CP	Dagan
T-MACH-114969	Strategic Decision-Making in Global Production Networks through Optimization and Simulation	ST	4,5 CP	Lanza, Martin
T-MACH-105422	Flows with Chemical Reactions	WT	4,5 CP	Class
T-MACH-113987	Sustainable Materials	WT	4,5 CP	Greiner
T-MACH-114608	Sustainable Materials Production	WT	4,5 CP	Dollmann, Greiner
T-MACH-114552	System Integration in Micro- and Nanotechnology I	ST	4,5 CP	Gengenbach
T-MACH-114553	System Integration in Micro- and Nanotechnology II	WT	4,5 CP	Gengenbach
T-MACH-113949	Topology Optimisation in Engineering	ST	4,5 CP	Deng
T-MACH-113982	Validation of Technical Systems	ST	4,5 CP	Düser

Assessment
The module assessment consists of several module components. These are conducted in the form of written or oral examinations, as well as alternative exam assessments, depending on the specific courses selected from the course pool.

The module is successfully completed once module components totaling at least 9 credit points have been passed.

The specific type of examination, the duration, and any prerequisites are specified in the respective description of each module component.

The overall grade of the module is calculated as the credit-point-weighted average of the individual grades of the module components.

Prerequisites
Any specific prerequisites for individual courses are defined and listed within the respective module component descriptions

Competence Goal
After completing the module, students possess a broad and in-depth understanding of fundamental and advanced topics in mechanical engineering. They are able to:

- Analyze complex technical systems and processes within various sub-disciplines of mechanical engineering (e.g., automotive engineering, materials science, or manufacturing technology).
- Apply engineering principles and specialized methods to identify, formalize, and solve technical problems at an advanced level.
- Evaluate different technological approaches and their impact on industrial applications and production environments.
- Integrate interdisciplinary knowledge to address challenges at the intersection of engineering and management.

Content
This module provides a comprehensive overview of diverse subject areas within the field of mechanical engineering. Due to the wide range of elective courses, the specific content depends on the individual student's choice. The curriculum covers key areas such as:

- **Vehicle and Propulsion Technology**: Design, simulation, and analysis of automotive systems and power units.
- **Materials and Micro-Engineering**: Simulation of materials, microsystem technology, and advanced structural mechanics.
- **Manufacturing and Automation**: Robotics, quality management, and the digitalization of production processes.
- **Systems and Process Engineering**: Methods for modeling and managing complex engineering systems.

The module allows students to customize their technical profile by selecting from a versatile pool of courses offered by the KIT Department of Mechanical Engineering.

Additional Information
This module (Mechanical Engineering II) is content-wise identical to Mechanical Engineering I, as both modules comprise the same set of courses. Students should select Mechanical Engineering II if they intend to complete additional courses beyond the 9 credit points covered by Mechanical Engineering I.

Workload
The total workload for the module is at least 270 hours (calculated at 30 hours per credit point).

Due to the flexible selection of module components, the specific distribution of the workload varies depending on the chosen courses. In general, the workload for each course is divided into:

- Attendance time: Participation in lectures, tutorials, and seminars.
- Self-study: Preparation and follow-up of the course material, literature research, and completion of assignments or practical exercises.
- Exam preparation: Intensive study phase and participation in the respective examinations.

Detailed information on the specific workload for each course can be found in the individual module component descriptions.

Recommendations
As this module comprises a wide range of courses from the KIT Department of Mechanical Engineering, students are strongly advised to research specific course requirements in advance.

To ensure adequate prior knowledge for the selected module components, it is recommended to:

- Consult the official course catalog (Vorlesungsverzeichnis) at https://campus.studium.kit.edu/events/catalog.php for detailed course descriptions and logistics.
- Contact the respective lecturers at the KIT Department of Mechanical Engineering directly for guidance on necessary prerequisites or technical background.
- Review the specific syllabi or learning materials often provided on the institutes' websites or the ILIAS platform.

M 5.119 Module: Medical Image Processing for Guidance and Navigation [M-ETIT-106672]

Coordinators: Prof. Dr.-Ing. Maria Francesca Spadea
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Engineering and Informatics (Electrical Engineering and Information Technology)
Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each winter term	1 term	English	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-113425	Medical Image Processing for Guidance and Navigation	WT	9 CP	Spadea

Assessment
The examination takes place within the framework of an oral overall examination of approx. 30 minutes about the lecture including a presentation and discussion of the project developed during the course. The overall impression is rated.

Prerequisites
none

Competence Goal

- The students will be able to analyze, structure and formally describe problems in the field of image guided surgery and therapy.
- The students can apply the methods form medical image processing, surgical navigation, augmented reality for surgery and therapy, medical data science.
- The student will be able to communicate in English technical language.
- The students are able to perform calculations and use the necessary tools for this in a methodologically appropriate way.
- The students are able to critically evaluate them

Content

- This module is designed to provide students with the theoretical and practical aspects of image guidance for minimally invasive surgery and therapy
- This module gives an overview about current status of technology in operation rooms (OR) and advanced radiotherapy bunkers
- Furthermore, this module gives knowledge about image process for quantitative information extraction
- Table of contents
    - Introduction to the course: minimally invasive surgery and medical data science
    - Git introduction
    - Image characteristics
    - Basic point, histogram and masked based operations
    - Similarity metrics, projections
    - Planning imaging, Dicom format, pre processing pipeline
    - Case study: planning in radiotherapy
    - Path planning
    - Pixel based image segmentation: manual segmentation, threshold, region growing
    - Convolution based segmentation: edge detection, morphological filters
    - Case study: neurosurgery and tractography
    - Image registration
    - Atlas based segmentation: SABS, MABS, atlas selection
    - Rendering and computer graphics
    - In room imaging technology
    - Reference system, notation and transformation
    - Localizing systems, tracking and calibration
    - Case study: patient monitoring in radiotherapy, adaptive treatments
    - Lab demonstration
    - Point based registration
    - Surface registration
    - Image features and descriptors (example with SIFT SURF)
    - Radiomics Features
    - Deep Learning in image processing
    - The role of deep learning in radiotherapy
    - Augmentet reality

Module Grade Calculation
The module grade is the grade of the oral exam.

A bonus can be earned for submitting homework that will be provided during the lecture time.

The exact criteria for awarding a bonus will be announced at the beginning of the lecture period. If the grade in the oral exam is between 4.0 and 1.3, the bonus improves the grade by 0.3 or 0.4.

Bonus points do not expire and are retained for any examinations taken at a later date.

Additional Information
The course is limited to a number of 30 participants due to capacity reasons. If necessary, a selection procedure will be carried out. Places will be allocated taking into account the students' study program (students of “Biomedical Engineering” specialization will be preferred, students from Computer Science Program and interest in medical applications will be preferred) and academic progress. Details will be announced on the lecture website.

Workload
The workload includes:

1. attendance in lectures and exercises: 15*6 h = 90 h
2. preparation / follow-up: 15*8 h = 120 h
3. preparation of and attendance in examination: 60 h

A total of 270 h = 9 CR

Recommendations

- Basic knowledge in the field of medical imaging;
- Knowledge of basic programming concept;
- Familiarity with Linux environment;
- Basic knowledge of linear algebra (transformations);
- Attitude towards teamwork and code management in Git;
- It is recommended to have access to a personal computer or desktop

Teaching and Learning Methods
Lectures in "Medical Image Processing" (3 SWS), Seminars in "In room imaging modalities" (1 SWS), Tutorials/ Demostrations in Medical image processing and navigation (2 SWS)

M 5.120 Module: Microeconomic Theory (WW4VWL15) [M-WIWI-101500]

Coordinators: Prof. Dr. Clemens Puppe
Organisation: KIT Department of Business and Economics
Part of: Business and Economics
Electives (Economics)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	1 term	English	4	4

ID	Name	Term offered	Credits	Lecturers
Compulsory Elective Courses (Elective: at least 9 credits)				
T-WIWI-102609	Advanced Topics in Economic Theory	Irreg.	4,5 CP	Brumm, Mitusch
T-WIWI-102861	Advanced Game Theory	WT	4,5 CP	Ehrhart, Puppe, Reiß
T-WIWI-102613	Auction Theory	WT	4,5 CP	Ehrhart
T-WIWI-105781	Incentives in Organizations	ST	4,5 CP	Nieken
T-WIWI-113264	Matching Theory	see notes	4,5 CP	Puppe
T-WIWI-102859	Social Choice Theory	ST	4,5 CP	Puppe

Assessment
The assessment is carried out as partial exams (according to Section 4(2), 1 or 2 of the examination regulation) of the single courses of this module, whose sum of credits must meet the minimum requirement of credits of this module. The assessment procedures are described for each course of the module seperately.

The overall grade of the the module is the average of the grades for each course weighted by the credits and truncated after the first decimal.

Prerequisites
None

Competence Goal
Students

- are able to model practical microoeconomic problems mathematically and to analyze them with respect to positive and normative questions,
- understand individual incentives and social outcomes of different institutional designs.

Here is an example of a positive question: what firm decisions does a specific regulatory policy result in under imperfect competition? An example of a normative question would be: which voting rule has appealing properties?

Content
The module teaches advanced concepts and content in microeconomic theory. Thematically, it offers a formally rigorous treatment of game theory and exemplary applications, such as strategic interaction on markets and non-/cooperative bargaining ("Advanced Game Theory"), as well as specialized courses dedicated to auctions ("Auktionstheorie") and incentive systems in organizations ("Incentives in Organizations"). Moreover, it offers the opportunity to delve deeper into the mathematical theory of voting and collective decision making, i.e. the systematic aggregation of preferences and judgments ("Social Choice Theory").

Workload
Attendance time: 90 h

Self-study: approx. 100 h

Exam preparation: approx. 80 h

Total: 270 h

M 5.121 Module: Microfabrication (WW4INGMBIMT2) [M-MACH-101291]

Coordinators: Prof. Dr. Jan Gerrit Korvink
Organisation: KIT Department of Mechanical Engineering

Part of: Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	1 term	German	4	4

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-MACH-102166	Fabrication Processes in Microsystem Technology	WT+ST	4 CP	Brandner
Mikrofertigung (Ergänzungsbereich) (Elective: at least 6 credits)				
T-MACH-102164	Practical Training in Basics of Microsystem Technology	WT+ST	3 CP	Last
T-MACH-100530	Physics for Engineers	ST	5 CP	Nesterov-Müller
T-MACH-102191	Polymers in MEMS B: Physics, Microstructuring and Applications	WT	4 CP	Worgull
T-MACH-102192	Polymers in MEMS A: Chemistry, Synthesis and Applications	WT	4 CP	Worgull
T-MACH-102200	Polymers in MEMS C: Biopolymers and Bioplastics	ST	4 CP	Worgull
T-MACH-105556	Practical Course Polymers in MEMS	ST	2 CP	Worgull
T-MACH-109122	X-ray Optics	WT+ST	4,5 CP	Last
T-MACH-108312	Introduction to Microsystem Technology - Practical Course	WT+ST	3 CP	Last
T-MACH-114015	Energy Efficient and Sustainable Tribological Systems (in German)	ST	4 CP	Dienwiebel
T-MACH-114016	Energy Efficient and Sustainable Tribological Systems	WT	4 CP	Dienwiebel

Assessment
The assessment is carried out as partial exams

(according to Section 4 (2), 1-3 SPO) of the core course and further single courses of this module, whose sum of credits must meet the minimum requirement of credits of this module. The assessment procedures are described for each course of the module separately.

The overall grade of the module is the average of the grades for each course weighted by the credits and truncated after the first decimal.

Prerequisites
none

Competence Goal
The student

- gains advanced knowledge concerning fabrication techniques in micrometer scale
- aquires knowledge in up-to-date developing research
- can detect and use causal relation in microfabrication process chains.

Content
This engineering module allows the student to gain advanced knowledge in the area of microfabrication. Different manufacturing methods are described and analyzed in an advanced manner. Necessary interdisciplinary knowledge from physics, chemistry, materials science and also up-to-date developments (nano and x-ray optics) in micro fabrication is offered.

Workload
270 hours

M 5.122 Module: Microoptics (WW4INGMBIMT3) [M-MACH-101292]

Coordinators: Prof. Dr. Jan Gerrit Korvink
Organisation: KIT Department of Mechanical Engineering

Part of: Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	1 term	German	4	5

ID	Name	Term offered	Credits	Lecturers
Mikrooptik (Elective: at least 9 credits)				
T-MACH-102164	Practical Training in Basics of Microsystem Technology	WT+ST	3 CP	Last
T-MACH-101910	Microactuators	ST	4 CP	Kohl
T-ETIT-100741	Laser Physics	WT	4 CP	Eichhorn
T-MACH-109122	X-ray Optics	WT+ST	4,5 CP	Last
T-MACH-108312	Introduction to Microsystem Technology - Practical Course	WT+ST	3 CP	Last
T-ETIT-114418	Integrated Photonics		6 CP	Koos

Assessment
The assessment is carried out as partial exams

(according to Section 4 (2), 1-3 SPO) of the core course and further single courses of this module, whose sum of credits must meet the minimum requirement of credits of this module. The assessment procedures are described for each course of the module separately.

The overall grade of the module is the average of the grades for each course weighted by the credits and truncated after the first decimal.

Prerequisites
none

Competence Goal
The student

- basic knowlegde for the applications of microoptical systems
- understanding fabrication processes of microoptical elements & systems
- analyzing strengths and weaknesses of lithography processes
- knowledge on the basics of optical sources and detectors and their use in technical systems
- fundamental knowledge on different lasers and their design
- knowlegde on X-ray imaging methodes

Content
Optical imaging, measuring and sensor systems are a base for modern natural sciences. In particular life sciences and telecommunications have an intrinsic need for the application of optical technologies. Numerous fields of physics and engineering, e.g. astronomy and material sciences, require optical techniques. Micro optical systems are introduced in medical diagnostics and biological sensing as well as in products of the daily life.

In this module, an introduction to the basics of optics is provided; optical effects are presented with respect to their technical use.

Optical elements and instruments are presented. Fabrication processes of micro optical systems and elements, in particular lithography, are discussed.

In addition X-ray optics and X-ray imaging systems are presented as well as elements of optical telecommunication. A closer look on the physics behind lasers, being one of the most important technical light sources, is provided. As high end technology and clean room equipment is present in all the lectures of this module, the students will have a hands-on training with several experiments in micro optics.

Workload
270 hours

M 5.123 Module: Microsystem Technology (WW3INGMBIMT1) [M-MACH-101287]

Coordinators: Prof. Dr. Jan Gerrit Korvink
Organisation: KIT Department of Mechanical Engineering

Part of: Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	1 term	German	4	7

ID	Name	Term offered	Credits	Lecturers
Mikrosystemtechnik (Elective: at least 9 credits)				
T-MACH-100967	BioMEMS - Microsystems Technologies for Life-Sciences and Medicine II	ST	4 CP	Guber
T-MACH-100968	BioMEMS - Microsystems Technologies for Life-Sciences and Medicine III	ST	4 CP	Guber
T-MACH-111807	Introduction to Bionics	ST	4 CP	Hölscher
T-MACH-114100	Introduction to Microsystem Technology I	WT	4 CP	Badilita, Korvink
T-MACH-114101	Introduction to Microsystem Technology II	ST	4 CP	Badilita, Korvink
T-MACH-108312	Introduction to Microsystem Technology - Practical Course	WT+ST	3 CP	Last
T-MACH-101910	Microactuators	ST	4 CP	Kohl
T-MACH-102152	Novel Actuators and Sensors	WT	4 CP	Kohl, Sommer
T-ETIT-101907	Optoelectronic Components	ST	4 CP	Huber-Loyola
T-MACH-100530	Physics for Engineers	ST	5 CP	Nesterov-Müller

Assessment
The assessment is carried out as partial exams

(according to Section 4 (2), 1-3 SPO) of the single courses of this module, whose sum of credits must meet the minimum requirement of credits of this module. The assessment procedures are described for each course of the module separately.

The overall grade of the module is the average of the grades for each course weighted by the credits and truncated after the first decimal.

Prerequisites
none

Competence Goal
construction and production of e. g. mechanical, optical, fluidic and sensory microsystems.

Content
The module offers courses in microsystem technology. Knowledge is imparted in various fields like basics in construction and production of e. g. mechanical, optical, fluidic and sensory microsystems.

Workload
270 hours

M 5.124 Module: Mixed-Signal IC Design [M-ETIT-105893]

Coordinators: Prof. Dr.-Ing. Ahmet Cagri Ulusoy
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Engineering and Informatics (Electrical Engineering and Information Technology)
Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each summer term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-111845	Mixed-Signal IC Design	ST	3 CP	Ulusoy

Assessment
Success control takes place in form of an oral examination (approx. 30 min.)

Prerequisites
none

Competence Goal
- The students acquire the competencies to mixed-signal advanced microelectronics integrated circuits.
- They have a good understanding of circuit design with linear circuits and "switched-capacitor" circuit techniques.
- They can design a sample-and-hold and track-and-hold circuits and discuss how it can improve an A/D converter's performance.
- They can design an A/D or D/A converter to a given performance specification, choosing an overall architecture, number of stages and internal precision.
- They can design phase lock loop (PLL) circuits, including design details and benefits and disadvantages of each type.
- They are familiar with time-to-digital converters and applications.
- They are familiar with the design of low-power circuits.
- They are able to develop test procedures, test structures and test patterns (ATPG - Automatic Test Pattern Generation) for ASICs.
- They have the basic understanding of the printed circuit board design practices, die-attached and high-density interconnection technology in order to connect the final ASIC to other chips and measurement equipment.

Content
This course covers fundamentals of data converters, Nyquist-rate converters, discrete-time signal processing, central concept of oversampling and noise-shaping, and delta-sigma modulators, phasedlocked loops, assembly and testing procedures of such mixed-signal ICs. Intended for engineers working with digital and analog signals, seeking to learn more about mixed-signal (analog plus digital) circuit design, analysis, and application.

Module Grade Calculation
The module grade is the grade of the oral examination.

Workload
Each credit point corresponds to an approximately 25-30h of workload in average. Based on this, the amount of work for this lecture is calculated as follows:
1. Attendance to the lectures (15*2=30h)
3. Preparation to the lectures (15*2=30h)
4. Preparation to the oral exam (25h)
Total: 85h

Recommendations
Basic knowledge on analog and digital circuits are recommended.

Literature

1. CMOS Analog Integrated Circuits; Razavi; McGraw-Hill Education
2. Principles of Data Conversion System Design; Razavi; Wiley-IEEE Press
3. Time-to-Digital Converters; Stephan Henzler; Springer Series in Advanced Microelectronics
4. VLSI Technology; Sze; McGraw-Hill

5.125 Module: MMIC Design Laboratory [M-ETIT-105464]

Coordinators: Prof. Dr.-Ing. Ahmet Cagri Ulusoy
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Engineering and Informatics (Electrical Engineering and Information Technology)
Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-111006	MMIC Design Laboratory	WT+ST	6 CP	Ulusoy

Assessment
The written report (approx. 30-40 slides) and the oral presentation (approx. 40 min.) are used to mark the course. The overall impression is assessed.

Prerequisites
none

Competence Goal
The students have a comprehensive understanding on the design of monolithic microwave integrated circuits.

The students are able to deduce specifications of individual building blocks in a microwave system and are able to connect these with system level considerations.

They are familiar with various IC fabrication technologies, and are able to identify pros and cons of the various state of the art technologies that are available today.

The students are able to perform the design of a complete microwave sub-system from conception to schematic level design and layout design, and are able to apply high-level design verification methods.

The students can apply their theoretical knowledge on RF engineering using modern design tools.

Content
In this laboratory course, the students will be assigned an RF system and will propose a hardware solution that will meet the requirements of the assigned RF system. The students will then perform schematic level design and system-level simulations of the proposed hardware. The laboratory course will be finalized with a layout implementation and verification of the proposed hardware. The students will learn to use state of the art CAD tools for system level simulations, schematic design, electromagnetic simulations, and layout design and verification in modern IC process technologies. Each RF sub-system will be developed by a group of 3-4 students.

Module Grade Calculation
The written report and the oral presentation are used to mark the course. The overall impression is assessed.

Workload
Each credit point corresponds approximately to 30h of the student's workload. Here, the average student is expected to reach an average performance. This contains:

1. Attendance to the laboratory tutorials (10*(3)=30h)
2. Preparation to the laboratory tutorials (10*(2)=20h)
3. Implementation of assigned design tasks after each tutorial (10*(8)=80h)
4. Preparation of report and oral presentation (20h)

Total: 150h

Recommendations
Radio-Frequency Integrated Circuits and Systems, Modern Radio Systems Engineering, Microwave Engineering, Electromagnetics and Numerical Calculation of Fields

5.126 Module: Mobile Communications [M-ETIT-105971]

Coordinators: Prof. Dr.-Ing. Peter Rost
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Engineering and Informatics (Electrical Engineering and Information Technology)
Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-112127	Mobile Communications	WT	4 CP	Rost

Assessment
The success control takes place in the form of an oral examination lasting approx. 25 minutes. Before the examination, there is a preparation phase of 15 minutes in which preparatory tasks are solved.

Prerequisites
none

Competence Goal
Students are enabled to analyze and assess functionalities of mobile communication systems. They learn how to apply and implement fundamental methods of the lecture “Communications Engineering I” in mobile radio networks. Furthermore, students will be enabled to understand requirements and limitations of mobile applications.

Content
At the beginning, this course describes exemplary applications of mobile communications and elaborates on resulting requirements. Based on a solid understanding of those requirements, selected approaches and techniques will be presented that are solving the respective challenges in mobile communication systems. To this end, algorithms as well as system architectures are discussed in order to acquire solid knowledge on the radio network, the core network and the integration with applications and services.

Module Grade Calculation
Grade of the module corresponds to the grade of the oral exam.

Workload

1. Attendance time in lectures: 15 * 2 h = 30 h
2. Preparation and follow-up of lectures: 15 * 2 h = 30 h
3. Attendance time in excercises: 15 * 1 h = 15 h
4. Preparation and follow-up of excercises: 15 * 1 h = 15 h
5. Preparation for the oral exam: 30 h

In total: 120 h = 4 LP

Recommendations
Knowledge of basic engineering as well as basic knowledge of communications engineering and Previous attendance of the lecture "Communication Engineering I" is recommended. Sound English language skills are required.

M 5.127 Module: Mobile Communications II [M-ETIT-106244]

Coordinators: Prof. Dr.-Ing. Peter Rost
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Engineering and Informatics (Electrical Engineering and Information Technology)
Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each summer term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-112679	Mobile Communications II	ST	3 CP	Rost

Assessment
The success control takes place in the form of an oral examination lasting approx. 25 minutes. Before the examination, there is a preparation phase of 15 minutes in which preparatory tasks are solved.

Prerequisites
none

Competence Goal
Students are able to analyze and assess functionalities of mobile communication systems. They know how to apply and implement fundamental methods of the lecture “Communications Engineering” in mobile radio networks. Furthermore, students are able to understand requirements and limitations of mobile applications.

This lecture complements the contents of the lecture "Mobile Communications", which mainly deals with aspects of communications access networks. Building on this, the focus of this lecture is on mobile communication architectures, core networks, and specific application scenarios and relevant technologies.

Content
The subject of the lecture is to first introduce a basic mobile communication system architecture including core network and the integration into applications. Based on this, specific core network functions are explained in detail, e.g. user administration, security, quality of service. Finally, specific applications are introduced and it is explained how mobile communication services are integrated in, e.g. industrial networks, connected cars, wide-area IoT applications.

Module Grade Calculation
Grade of the module corresponds to the grade of the oral exam.

Workload

1. Attendance to the lecture: 15 * 2 h = 30 h
2. Preparation and review: 15 * 2 h = 30 h
3. Preparation for the exam: included in preparation and review = 30 h

In total: 90 h = 3 LP

Recommendations
Basic knowledge of communications engineering. For this purpose previous attendance of the modules "M-ETIT-102103 - Communication Engineering I" and “M-ETIT-105971 - Mobile Communications” is strongly recommended.

5.128 Module: Mobile Machines (WI4INGMB15) [M-MACH-101267]

Coordinators: Prof. Dr.-Ing. Marcus Geimer
Organisation: KIT Department of Mechanical Engineering

Part of: Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Level	Version
9 CP	graded	Each winter term	2 terms	4	6

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-MACH-105168	Mobile Machines	ST	8 CP	Geimer
Mobile Machines (Elective: at least 1 credit)				
T-MACH-105311	Design and Development of Mobile Machines	WT	4 CP	Geimer
T-MACH-108887	Design and Development of Mobile Machines - Advance	WT+ST	0 CP	Geimer, Siebert
T-MACH-111389	Fundamentals in the Development of Commercial Vehicles	see notes	4 CP	Weber
T-MACH-111821	Control of Mobile Machines	ST	4 CP	Becker, Geimer
T-MACH-111820	Control of Mobile Machines – Prerequisites	ST	0 CP	Becker, Geimer
T-MACH-113997	Vehicle Drive Technology	WT	4 CP	Geimer
T-MACH-113862	Simulation with Lumped Parameters	ST	3 CP	Geimer
T-MACH-113863	Tutorial Simulation with Lumped Parameters	ST	1 CP	Geimer
T-MACH-113912	Mathematical Methods in Hydraulics	WT+ST	4 CP	Geimer
T-MACH-113913	Tutorial Mathematical Methods in Hydraulics	WT	2 CP	Geimer

Assessment
The assessment is carried out as a general oral exam (duration approx. 60 mins) (according to Section 4(2), 2 of the examination regulation) of the single courses of this module, whose sum of credits must meet the minimum requirement of credits of this module.

The examination is offered every semester. Re-examinations are offered at every ordinary examination date.

The overall grade of the module is the grade of the oral examination.

The assessment may be carried out as partial oral exams (according to Section 4(2), 2 of the examination regulation) of the single courses of this module, whose sum of credits must meet the minimum requirement of credits of this module.In this case the overall grade of the module is the average of the grades for each course weighted by the credits and truncated after the first decimal.

The assessment procedures are described for each course of the module seperately.

Prerequisites
None

Competence Goal
The student

- knows and understands the basic structure of the machines
- masters the basic skills to develop the selected machines

Content
In the module of *Mobile Machines* [WI4INGMB15] the students will learn the structure of the machines and deepen the knowledge of the subject for developing the machines. After conclusion the module the student will know the latest developments in mobile machines and is able to evaluate the concepts and the trends of developments. The module is practically orientated and supported by industry partners.

Workload
The total workload for this module is approximately 270 hours. For further information see German version.

Recommendations
Knowledge of Fluid Power Systems are helpful, otherwise it is recommended to take the course *Fluid Power Systems* [2114093].

M 5.129 Module: Modeling the Dynamics of Financial Markets [M-WIWI-106660]

Coordinators: Prof. Dr. Maxim Ulrich
Organisation: KIT Department of Business and Economics
Part of: Business and Economics
Electives (Business Administration)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each summer term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-WIWI-113414	Modeling the Dynamics of Financial Markets	ST	9 CP	Ulrich

Assessment
The module examination takes the form of a one-hour written comprehensive examination on the courses "Dynamic Capital Market Theory", "Essentials for Dynamic Financial Machine Learning" and "Exercises, Python, Research Frontier in Dynamic Capital Markets".

Competence Goal
Dynamic Capital Market Theory:

Professional competence:

- Understanding of the principles of Dynamic Asset Pricing Theory
- Mastery of concepts such as stochastic calculus and dynamic modeling in discrete and continuous time
- Application of dynamic programming theory to portfolio and investment decisions
- Knowledge of pricing bonds, stocks, futures and options markets.

Interdisciplinary skills:

- Develop analytical skills for working on and solving complex problems in finance
- Ability to apply theoretical models to real financial market scenarios.

Essentials for Dynamic Financial Machine Learning:

Professional Competence:

- Competencies in Multivariate Time Series Modeling and Dynamic Volatility Modeling.
- Skills in dealing with big financial data.
- Knowledge in the estimation of risk premia and the application of Kalman Filtering.

Interdisciplinary skills:

- Analytical skills in applying machine learning algorithms to dynamic financial market data.
- Development of problem-solving skills through the practical application of Python in financial data analysis.

Content

Dynamic Capital Market Theory:

The course "Dynamic Capital Market Theory" offers an introduction to the modeling of dynamic capital markets. Portfolio holdings and asset prices move dynamically across time and states. This course teaches basic financial economic thinking to help understand why this is the case and how to optimally act in such environments.
Next to the asset pricing focus, the second focus of the course is on optimal portfolio choice (robo advisory). For that, this course develops the theory of dynamic programming in discrete and continuous time and applies it to solve portfolio choice and corporate investment decisions. These concepts are key for financial engineering and the machine learning branch of Reinforcement Learning.

Students obtain proficiency in the following topics:

- Dynamic Valuation and Optimal Dynamic Asset Allocation
- Dynamic modeling in discrete time and continuous time
- Stochastic Calculus
- Markov Decision Processes and Dynamic Programming in discrete time and continuous time
- Pricing of bonds, equity, futures and options

Lectures (2 SWS) develop all concepts on the whiteboard.

Essentials for Dynamic Financial Machine Learning:

The course "Essentials for Dynamic Financial Machine Learning" teaches students to work with financial data, algorithms and statistical concepts.
Students are exposed to algorithms to learn key quantities of dynamic capital markets, such as time-varying risk premia, time-varying volatility and unobserved realizations of random states. The course covers the following concepts:

- Multivariate time series modeling
- Dynamic volatility modeling
- Handling big financial data
- Estimating risk premia
- Kalman Filtering

Weekly lectures (2 SWS) develop all algorithmic material on the whiteboard.

Exercises, Python, Research Frontier in Dynamic Capital Markets:

This course provides hands-on experience in implementing concepts from dynamic capital market theory and financial machine learning using Python. Students will develop practical skills in coding and data analysis that complement the theoretical knowledge gained in the companion courses. The course covers:

- Introduction to Python for financial applications Data manipulation and visualization with pandas and matplotlib.
- Implementing dynamic portfolio optimization algorithms.
- Coding stochastic processes and simulations.
- Building and testing time series models.
- Applying machine learning techniques to financial data.
- Developing Reinforcement Learning algorithms for trading strategies.
- Implementing and backtesting option pricing models.
- Creating interactive financial dashboards

Weekly computer lab sessions (2 SWS) will guide students through coding exercises and problem sets that directly relate to topics covered in "Dynamic Capital Market Theory" and "Essentials for Dynamic Financial Machine Learning". Students will work on individual and group projects, applying their programming skills to real-world financial problems and current research questions in dynamic capital markets.
This course forms an integral part of the module, complementing the theoretical components with practical implementation skills essential for modern quantitative finance.

Workload

Total workload for 9 credit points: approx. 270 hours. The exact distribution is based on the credit points of the courses in the module:

- Dynamic Capital Market Theory: 3 CP
- Essentials for Dynamic Financial Machine Learning: 3 CP
- Exercises, Python, Research Frontier in Dynamic Capital Markets: 3 CP

The total number of hours per course is determined by the amount of time spent attending the lectures and tutorials, as well as the exam times and the time required to achieve the module's learning objectives for an average student for an average performance.

Recommendations

Recommendation: Knowledge in the fields of Advanced Statistics, Deep Learning, Financial Economics, Differential Equations, Optimization.

Teaching and Learning Methods
The module consists of two weekly lectures and respective tutorials:

1. **Dynamic Capital Market Theory** and
2. **Essentials for Dynamic Financial Machine Learning**.
3. **Exercises, Python, Research Frontier in Dynamic Capital Markets**

M 5.130 Module: Modern Mobility on Rails and Roads (WI3INGMB25) [M-MACH-106496]

Coordinators: Prof. Dr.-Ing. Martin Cichon
Organisation: KIT Department of Mechanical Engineering

Part of: Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	1 term	German	4	2

ID	Name	Term offered	Credits	Lecturers
Modern Mobility on Rail and Road (Elective: 2 items)				
T-MACH-113016	Digitization in the Railway System	WT	4 CP	Cichon
T-MACH-113069	Vehicle Systems for Urban Mobility		4 CP	Cichon
T-MACH-113068	Innovation and Project Management in Rail Vehicle Engineering	WT+ST	4 CP	Cichon

Competence Goal

- Students will have a basic understanding of train protection and its technical implementation in Germany, the functioning of the European Train Control System (ETCS) and its planning as well as Automated Train Operation. They can explain the knowledge they have learned (terms, relationships) in context and apply it to practical issues. Furthermore, students will be able to classify the operational and technical advantages and disadvantages in the context of the digitalization of the rail network in Germany and take future challenges into account. students will be able to discuss the technical aspects and areas of application of ETCS at the various levels and give a basic outline of balise planning for ETCS Level 2. Digital planning approaches such as PlanPro as well as measurement and test runs are known and can be categorized.
- Students gain a basic understanding of the key transport, transport policy and technological contexts of urban mobility. Based on this basic understanding, various public transport vehicle concepts in the urban and regional environment are analyzed, compared and the optimal range of applications discussed. In addition to established public transport systems, special attention is paid to innovative mobility solutions. In particular, the aim is to create an understanding of how sustainable, systemic mobility solutions should be designed depending on the individual application.
- In this course, students will learn the basics of innovation and project management in the context of rail vehicle development. Creativity techniques are applied to the challenges of the rail system in a practical way, such as aspects of sustainability. Students will also learn about the various organizational, systemic, economic and technological challenges of a project and project management.

M 5.131 Module: Nanotechnology (WW4INGMBIMT5) [M-MACH-101294]

Coordinators: Prof. Dr. Jan Gerrit Korvink
Organisation: KIT Department of Mechanical Engineering

Part of: Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	1 term	German	4	6

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-MACH-111814	Introduction to Nanotechnology	ST	4 CP	Hölscher
Nanotechnologie (Ergänzungsbereich) (Elective: at least 5 credits)				
T-MACH-111807	Introduction to Bionics	ST	4 CP	Hölscher
T-MACH-102152	Novel Actuators and Sensors	WT	4 CP	Kohl, Sommer
T-MACH-102164	Practical Training in Basics of Microsystem Technology	WT+ST	3 CP	Last
T-ETIT-100740	Quantum Functional Devices and Semiconductor Technology	ST	3 CP	Koos
T-MACH-105695	Selected topics of system integration for micro- and nanotechnology	WT	4 CP	Gengenbach, Hagenmeyer, Koker, Sieber
T-MACH-108809	Micro- and Nanosystem Integration for Medical, Fluidic and Optical Applications	WT	4 CP	Gengenbach, Koker, Sieber
T-MACH-111030	Micro- and nanotechnology in implant technology	WT	4 CP	Ahrens, Doll
T-PHYS-102282	Nano-Optics	WT	8 CP	Naber
T-PHYS-102504	Simulation of Nanoscale Systems, without Seminar	Irreg.	6 CP	Wenzel
T-MACH-108312	Introduction to Microsystem Technology - Practical Course	WT+ST	3 CP	Last
T-MACH-114552	System Integration in Micro- and Nanotechnology I	ST	4,5 CP	Gengenbach
T-MACH-114553	System Integration in Micro- and Nanotechnology II	WT	4,5 CP	Gengenbach
T-MACH-114015	Energy Efficient and Sustainable Tribological Systems (in German)	ST	4 CP	Dienwiebel
T-MACH-114016	Energy Efficient and Sustainable Tribological Systems	WT	4 CP	Dienwiebel

Assessment
The assessment is carried out as partial exams

(according to Section 4 (2), 1-3 SPO) of the core course and further single courses of this module, whose sum of credits must meet the minimum requirement of credits of this module. The assessment procedures are described for each course of the module separately.

The overall grade of the module is the average of the grades for each course weighted by the credits and truncated after the first decimal.

Prerequisites
none

Competence Goal
The student

- has detailed knowledge in the field of nanotechnology
- is able to evaluate the specific characteristics of nanosystems.

Content
The module deals with the most important principles and fundamentals of modern nanotechnology. The compulsory module “Introduction to Nanotechnology” introduces the basics of nanotechnology and nanoanalytics. The specific phenomena and properties found in nanoscale systems are the main topic of the module.

Workload
270 hours

Teaching and Learning Methods
Lectures

M 5.132 Module: Natural Sciences [M-WIWI-104904]

Coordinators: Professorenschaft des Fachbereichs Physik
Organisation: KIT Department of Business and Economics
Part of: Additional Examinations

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	pass/fail	Each term	1 term	German	4	1

ID	Name	Term offered	Credits	Lecturers
Elective Courses on Physics (Elective: at most 20 credits)				
T-PHYS-103117	Geological Hazards and Risks for External Students		4 CP	Gottschämmer
T-PHYS-101092	Climatology	ST	4 CP	Ginete Werner Pinto
T-PHYS-105594	Exam on Climatology		1 CP	Ginete Werner Pinto

M 5.133 Module: Network Economics (WW4VWL4) [M-WIWI-101406]

Coordinators: Prof. Dr. Kay Mitusch
Organisation: KIT Department of Business and Economics
Part of: Electives (Economics)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	1 term	German/English	4	3

ID	Name	Term offered	Credits	Lecturers
Compulsory Elective Courses (Elective: 9 credits)				
T-WIWI-100005	Competition in Networks	WT	4,5 CP	Mitusch
T-WIWI-100007	Transport Economics	ST	4,5 CP	Mitusch, Szimba
T-WIWI-102609	Advanced Topics in Economic Theory	Irreg.	4,5 CP	Brumm, Mitusch
T-WIWI-102712	Regulation Theory and Practice	see notes	4,5 CP	Mitusch
T-WIWI-113147	Telecommunications and Internet – Economics and Policy	WT	4,5 CP	Mitusch

Assessment
The assessment is carried out as partial exams (according to Section 4(2), 1 or 2 of the examination regulation) of the single courses of this module, whose sum of credits must meet the minimum requirement of credits of this module.

The exams are offered at the beginning of the recess period about the subject matter of the latest held lecture. Re-examinations are offered at every ordinary examination date. The assessment procedures are described for each course of the module separately.

The overall grade for the module is the average of the grades for each course weighted by the credits and truncated after the first decimal.

Prerequisites
None

Competence Goal
The students

- have acquired the basic knowledge for a future job in a network company or in a regulatory agency, ministry etc.
- recognize the specific characterizations of network sectors, know fundamental methods for an economic analysis of network sectors and recognize the interfaces for an interdisciplinary cooperation of economists, engineers and lawyers
- understand the interactions between infrastructures, control systems, and the users of networks, especially concerning their implications on investments, price setting and competitive behavior, and they can model or simulate exemplary applications
- can assess the necessity of regulation of natural monopolies and identify regulatory measures that are important for networks.

Content
The module is concerned with network or infrastructure industries in the economy, e.g. telecommunication, traffic and energy sectors. These sectors are characterized by close interdependencies of operators and users of infrastructure as well as on states. States intervene in various forms, by the public and regulation authorities, due to the importance of network industries and due to limited abilities of markets to work properly in these industries. The students are supposed to develop a broad knowledge of these sectors and of the political options available.

Workload
Total workload for 9 credit points: approx. 270 hours

The exact distribution is based on the credit points of the courses in the module.

Recommendations
Basics of microeconomics obtained within the undergraduate programme (B.Sc) of economics are required.

M 5.134 Module: Nonlinear Optics [M-ETIT-100430]

Coordinators: Prof. Dr.-Ing. Christian Koos
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Engineering and Informatics (Electrical Engineering and Information Technology)
Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each summer term	1 term	English	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-101906	Nonlinear Optics	ST	6 CP	Koos

Assessment
Type of Examination: oral exam

Duration of Examination: approx. 30 Minutes

Modality of Exam: The oral exam is offered continuously upon individual appointment.

Prerequisites
none

Competence Goal
The students

- understand and can mathematically describe the effect of basic nonlinear-optical phenomena using optical susceptibility tensors,
- understand and can mathematically describe wave propagation in nonlinear anisotropic materials,
- have an overview and can quantitatively describe common second-order nonlinear effects comprising the electro-optic effect, second-harmonic generation, sum- and difference frequency generation, parametric amplification and optical rectification,
- have an overview and can quantitatively describe the Kerr effect and other common third-order nonlinear effects, comprising self- and cross-phase modulation, four-wave mixing, self-focussing, and third-harmonic generation,
- have an overview and can describe nonlinear-optical interaction in active devices such as semiconductor optical amplifiers
- conceive the basic principles of various phase-matching techniques and can apply them to practical design problems,
- conceive the basic principles electro-optic modulators, can apply them to practical design problems, and have an overview on state-of-the art devices,
- conceive the basic principles third-order nonlinear signal processing and can apply them to practical design problems.

Content

1. The nonlinear optical susceptibility: Maxwell´s equations and constitutive relations, relation between electric field and polarization, formal definition and properties of the nonlinear optical susceptibility tensor,
2. Wave propagation in nonlinear anisotropic materials
3. Second-order nonlinear effects and devices: Linear electro-optic effect / Pockels effect, second-harmonic generation, sum- and difference-frequency generation, phase matching, parametric amplification, optical rectification
4. Third-order nonlinear effects and devices: Nonlinear refractive index and Kerr effect, self- and cross-phase modulation, four-wave mixing, self-focussing, third-harmonic generation
5. Nonlinear effects in active optical devices

Module Grade Calculation
The module grade is the grade of the oral exam.

There is a bonus system based on the problem sets that are solved during the tutorials: During the term, 3 problem sets will be collected in the tutorial and graded without prior announcement. If for each of these sets more than 70% of the problems have been solved correctly, a bonus of 0.3 grades will be granted on the final mark of the oral exam.

Workload
Approx. 180 h – 30 h lectures, 30 h exercises, 120 h homework and self-studies

Recommendations
Solid mathematical and physical background, basic knowledge in optics and photonics.

Literature

R. Boyd. Nonlinear Optics. Academic Press, New York, 1992.
E.H. Li S. Chiang Y. Guo, C.K. Kao. Nonlinear Photonics. Springer Verlag, 2002
G. Agrawal, Nonlinear Fiber Optics, Academic Press, San Diego, 1995.

5.135 Module: Operations Research [M-WIWI-104899]

Coordinators: Professorenschaft des Fachbereichs Operations Research
Organisation: KIT Department of Business and Economics
Part of: Additional Examinations

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	pass/fail	Each term	1 term	German	4	4

ID	Name	Term offered	Credits	Lecturers
Elective Courses on Operations Research (Elective: at most 40 credits)				
T-WIWI-102872	Challenges in Supply Chain Management	ST	4,5 CP	Mohr
T-WIWI-102758	Introduction to Operations Research I and II	see notes	9 CP	Nickel, Rebennack, Stein
T-WIWI-106546	Introduction to Stochastic Optimization	ST	4,5 CP	Rebennack
T-WIWI-102718	Discrete-Event Simulation in Production and Logistics	ST	4,5 CP	Spieckermann
T-WIWI-106548	Advanced Stochastic Optimization	Irreg.	4,5 CP	Rebennack
T-WIWI-102719	Mixed Integer Programming I	Irreg.	4,5 CP	Stein
T-WIWI-102720	Mixed Integer Programming II	Irreg.	4,5 CP	Stein
T-WIWI-102726	Global Optimization I	ST	4,5 CP	Stein
T-WIWI-102727	Global Optimization II	ST	4,5 CP	Stein
T-WIWI-103638	Global Optimization I and II	ST	9 CP	Stein
T-WIWI-102723	Graph Theory and Advanced Location Models	Irreg.	4,5 CP	Nickel
T-WIWI-102856	Convex Analysis	Irreg.	4,5 CP	Stein
T-WIWI-106549	Large-scale Optimization	ST	4,5 CP	Rebennack
T-WIWI-106199	Modeling and OR-Software: Introduction	ST	4,5 CP	Nickel
T-WIWI-106200	Modeling and OR-Software: Advanced Topics	WT	4,5 CP	Nickel
T-WIWI-111587	Multicriteria Optimization	see notes	4,5 CP	Stein
T-WIWI-103124	Multivariate Statistical Methods	Irreg.	4,5 CP	Grothe
T-WIWI-102724	Nonlinear Optimization I	WT	4,5 CP	Stein
T-WIWI-103637	Nonlinear Optimization I and II	WT	9 CP	Stein
T-WIWI-102725	Nonlinear Optimization II	WT	4,5 CP	Stein
T-WIWI-102884	Operations Research in Health Care Management	WT+ST	4,5 CP	Graß
T-WIWI-102715	Operations Research in Supply Chain Management	Irreg.	4,5 CP	Nickel
T-WIWI-106545	Optimization under Uncertainty	WT	4,5 CP	Rebennack
T-WIWI-102855	Parametric Optimization	Irreg.	4,5 CP	Stein
T-WIWI-102716	Practical Seminar: Health Care Management (with Case Studies)	WT+ST	4,5 CP	Nickel
T-WIWI-103488	Seminar in Operations Research (Bachelor)	WT+ST	3 CP	Nickel, Rebennack, Stein
T-WIWI-103481	Seminar in Operations Research A (Master)	WT+ST	3 CP	Nickel, Rebennack, Stein
T-WIWI-103482	Seminar in Operations Research B (Master)	WT+ST	3 CP	Nickel, Rebennack, Stein
T-WIWI-114109	Service Operations and Cyber Security	ST	4,5 CP	Mohr
T-WIWI-102704	Facility Location and Strategic Supply Chain Management	WT	4,5 CP	Nickel
T-WIWI-102714	Tactical and Operational Supply Chain Management	ST	4,5 CP	Nickel
T-WIWI-112109	Topics in Stochastic Optimization	WT	4,5 CP	Rebennack

M 5.136 Module: Operations Research in Supply Chain Management (WW4OR11) [M-WIWI-102832]

Coordinators: Prof. Dr. Stefan Nickel
Organisation: KIT Department of Business and Economics
Part of: Electives (Operations Research)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	2 terms	German/English	4	10

Election Notes
At least one of the courses "Operations Research in Supply Chain Management", "Graph Theory and Advanced Location Models", "Modeling and OR-Software: Advanced Topics" and"Special Topics of Stochastic Optimization (elective)" has to be taken.
Students who choose the module in the field "compulsory elective modules" may select any two courses of the module.

ID	Name	Term offered	Credits	Lecturers
Compulsory Elective Courses (Elective: between 1 and 2 items)				
T-WIWI-102723	Graph Theory and Advanced Location Models	Irreg.	4,5 CP	Nickel
T-WIWI-106200	Modeling and OR-Software: Advanced Topics	WT	4,5 CP	Nickel
T-WIWI-102715	Operations Research in Supply Chain Management	Irreg.	4,5 CP	Nickel
Supplementary Courses (Elective: at most 1 item)				
T-MACH-112213	Applied material flow simulation	WT	4,5 CP	Baumann
T-WIWI-106546	Introduction to Stochastic Optimization	ST	4,5 CP	Rebennack
T-WIWI-102718	Discrete-Event Simulation in Production and Logistics	ST	4,5 CP	Spieckermann
T-WIWI-102719	Mixed Integer Programming I	Irreg.	4,5 CP	Stein
T-WIWI-102720	Mixed Integer Programming II	Irreg.	4,5 CP	Stein
T-WIWI-106549	Large-scale Optimization	ST	4,5 CP	Rebennack
T-WIWI-111587	Multicriteria Optimization	see notes	4,5 CP	Stein
T-WIWI-114747	"Practical Seminar in Discrete Optimization: Implementing and Evaluating Solution Methods (with Python)"	Once	4,5 CP	Bakker
T-WIWI-112109	Topics in Stochastic Optimization	WT	4,5 CP	Rebennack

Assessment
The assessment is carried out as partial exams (according to § 4(2), 1 of the examination regulation) of the single courses of this module, whose sum of credits must meet the minimum requirement of credits of this module.

The assessment procedures are described for each course of the module seperately.

The overall grade of the module is the average of the grades for each course weighted by the credits and truncated after the first decimal.

Prerequisites
At least one of the courses "Operations Research in Supply Chain Management", "Graph Theory and Advanced Location Models", "Modeling and OR-Software: Advanced Topics" and "Special Topics of Stochastic Optimization (elective)" has to be taken.

Competence Goal
The student

- is familiar with basic concepts and terms of Supply Chain Management,
- knows the different areas of SCM and their respective optimization problems,
- is acquainted with classical location problem models (in planes, in networks and discrete) as well as fundamental methods for distribution and transport planning, inventory planning and management,
- is able to model practical problems mathematically and estimate their complexity as well as choose and adapt appropriate solution methods.

Content
Supply Chain Management is concerned with the planning and optimization of the entire, inter-company procurement, production and distribution process for several products taking place between different business partners (suppliers, logistics service providers, dealers). The main goal is to minimize the overall costs while taking into account several constraints including the satisfaction of customer demands.

This module considers several areas of SCM. On the one hand, the determination of optimal locations within a supply chain is addressed. Strategic decisions concerning the location of facilities as production plants, distribution centers or warehouses are of high importance for the rentability of Supply Chains. Thoroughly carried out, location planning tasks allow an efficient flow of materials and lead to lower costs and increased customer service. On the other hand, the planning of material transport in the context of supply chain management represents another focus of this module. By linking transport connections and different facilities, the material source (production plant) is connected with the material sink (customer). For given material flows or shipments, it is considered how to choose the optimal (in terms of minimal costs) distribution and transportation chain from the set of possible logistics chains, which asserts the compliance of delivery times and further constraints. Furthermore, this module offers the possibility to learn about different aspects of the tactical and operational planning level in Suppy Chain Mangement, including methods of scheduling as well as different approaches in procurement and distribution logistics. Finally, issues of warehousing and inventory management will be discussed.

Additional Information
Some lectures and courses are offered irregularly.

The planned lectures and courses for the next three years are announced online.

Workload
Total effort for 9 credits: ca. 270 hours

- Presence time: 84 hours
- Preparation/Wrap-up: 112 hours
- Examination and examination preparation: 74 hours

Recommendations
Basic knowledge as conveyed in the module *Introduction to Operations Research* is assumed.

5.137 Module: Optical Engineering and Machine Vision [M-ETIT-106974]

Coordinators: Prof. Dr.-Ing. Michael Heizmann
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Engineering and Informatics (Electrical Engineering and Information Technology)
Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each winter term	1 term	English	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-113941	Optical Engineering and Machine Vision	WT	6 CP	Heizmann

Assessment
The examination takes place in form of a written examination lasting 120 minutes.

Prerequisites
none

Competence Goal

- Students have a sound knowledge of the fundamentals (physical basics of optics, optical imaging, image sensors) and procedures of optical engineering and machine vision.
- Students are proficient in diverse methods for optical imaging, image acquisition, pre-processing and image evaluation and can characterize them based on their prerequisites, model assumptions and results.
- Students are able to analyze and structure optical engineering and machine vision tasks, synthesize possible solutions from optics principles and image processing methods and assess their suitability.

Content
Optical engineering and machine vision are collective terms for using optical signals to solve tasks of information retrieval for technical and other application. They comprise the propagation of light in optical systems, the acquisition of image signals using optical imaging and cameras, the processing of the recorded image signals using (digital) image processing and the evaluation of the image data to obtain useful information from the recorded images.

The module teaches the basics, procedures and exemplary applications of optical engineering and image processing.

The module include in detail:

- Optical Imaging
    - Imaging with a pinhole camera, central projection
    - Imaging using a (single) lens
- Color
    - Photometry
    - Color perception and color spaces
    - Filters
- Sensors for Image Acquisition
    - Point, line and area sensors
    - CCD, CMOS sensors
    - Line-scan cameras
    - Color sensors and color cameras
    - Infrared cameras
    - Quality criteria for image sensors
- Methods of Image Acquisition
    - Measuring optical properties
    - 3D shape capturing
- Image Signals
    - Mathematical model of image signals
    - Systems theory
    - Two-dimensional Fourier transform
    - Noise of digital imaging sensors (EMVA 1288)
    - Principal component analysis (Karhunen-Loève transform)

- Preprocessing and Image Enhancement
    - Simple image enhancement methods
    - Reduction of systematic errors
    - Attenuation of random disturbances

The following topics do not apply for KSOP students:

- Light
    - Light as an electromagnetic wave
    - Light as a quantum phenomenon
    - Interaction of light and matter
    - Light sources
- Segmentation
    - Region-based segmentation
    - Edge-oriented methods
- Morphological Image Processing
    - Binary morphology
    - Gray-scale morphology
- Texture analysis
    - Types of textures
    - Model-based texture analysis
    - Feature-based texture analysis
- Detection
    - Detection of known objects by linear filters
    - Detection of unknown objects (defects)
    - Detection of straight lines (Radon and Hough transform)

Module Grade Calculation
The module grade is the grade of the written examination.

Workload
The workload includes:

1. attendance in lectures and exercises: 15*4 h = 60 h
2. preparation / follow-up: 15*4 h = 60 h
3. preparation of and attendance in examination: 60 h

A total of 180 h = 6 CR

Recommendations
Basic knowledge of systems theory and signal processing (e.g. from the module “Signals and Systems”) as well as optics is helpful.

Teaching and Learning Methods
lecture (3 SWS) and exercise (1 SWS)

5.138 Module: Optical Transmitters and Receivers [M-ETIT-100436]

Coordinators: Prof.Dr.Dr.h.c. Wolfgang Freude
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Engineering and Informatics (Electrical Engineering and Information Technology)
Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each winter term	1 term	English	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-100639	Optical Transmitters and Receivers	WT	6 CP	Freude

Assessment
Oral examination (approx. 20 minutes). The individual dates for the oral examination are offered regularly.

Prerequisites
none

Competence Goal
The students

- understand the peculiarities of optical communications, and how optical signals are generated, transmitted and received,
- know about sampling, quantization and coding,
- learn the basics about noise on reception,
- understand the properties of a linear and a nonlinear optical fibre channel,grasp the idea of channel capacity and spectral efficiency,
- know about various forms of modulation,
- acquire knowledge of optical transmitter elements,
- understand the function of optical amplifiers,
- have a basic understanding of optical receivers,
- know the sensitivity limits of optical systems, and
- understand how these limits are measured.

Content
The course concentrates on basic optical communication concepts and connects them with the properties of physical components. The following topics are discussed:

- Advantages and limitations of optical communication systems
- Optical transmitters comprising lasers and modulators
- Optical receivers comprising direct and heterodyne reception
- Characterization of signal quality

Module Grade Calculation
The module grade is the grade of the oral exam.

Workload
Approx. 180 hours workload for the student. The amount of work is included:

30 h - Attendance times in lectures
30 h - Exercises
120 h - Preparation / revision phase

Recommendations
Knowledge of the physics of the pn-junction

Literature
Detailed textbook-style lecture notes can be downloaded from the IPQ lecture pages.
Grau, G.; Freude, W.: Optische Nachrichtentechnik, 3. Ed. Berlin: Springer-Verlag 1991. In German. Since 1997 out of print. Electronic version available via w.freude@kit.edu.
Kaminow, I. P.; Li, Tingye; Willner, A. E. (Eds.): Optical Fiber Telecommunications VI A: Components and Subsystems +VI B: Systems and Networks', 6th Ed. Elsevier (Imprint: Academic Press), Amsterdam 2013

M 5.139 Module: Optimal and Model Predictive Control [M-CIWVT-106317]

Coordinators: Prof. Dr.-Ing. Thomas Meurer
Organisation: KIT Department of Chemical and Process Engineering
Part of: Engineering and Informatics (Chemical and Process Engineering)
Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each summer term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-CIWVT-112825	Optimal and Model Predictive Control		6 CP	Meurer

Assessment
Learning control is an oral examination with a duration of about 45 minutes.

Prerequisites
none

Competence Goal
Students will gain an in-depth understanding of dynamic optimization with constraints, optimal control, and model predictive control. They will understand the underlying mathematical concepts and be able to apply them to new problems. They will gain a comprehensive understanding of optimization methods and be able to apply these methods independently to dynamic optimization problems. Students are familiar with various numerical solution approaches, understand how they work, and can implement them for optimization problems.

Content
Many problems in industry and economy rely on the determination of an optimal solution satisfying desired performance criteria and constraints. In mathematical terms this leads to the formulation of an optimization problem. Here it is in general distinguished between static and dynamic optimization with the latter involving a dynamical process. This lecture gives an introduction to the mathematical analysis and numerical solution of dynamic optimization problems with a particular focus on optimal control and model predictive control. The lecture addresses the following topics:

- Fundamentals of dynamic optimization problems
- Dynamic optimization without and with constraints
- Linear and nonlinear model predictive control
- Numerical methods

Selected examples are considered and solved in the exercises and dedicated computer exercises.

Module Grade Calculation
The grade of the module is the grade of the oral exam.

Workload
Attendance time: Lectures: 30 hrs. Exercises: 15 hrs.

Self-study: 60 hrs.

Exam preparation: 75 hrs.

Literature

- T. Meurer: Optimal and Model Predictive Control, Lecture Notes.
- D. G. Luenberger, Y. Ye: Linear and Nonlinear Programming, Springer, 2008.
- J. Nocedal, S.J. Wright: Numerical Optimization, Springer, 2006.
- M. Papageorgiou, M. Leibold, M. Buss: Optimierung, Springer, 2012.
- E. Camacho, C. Alba: Model Predictive Control, Springer, 2004
- L. Grüne, J. Pannek: Nonlinear Model Predictive Control: Theory and Algorithms, Springer, 2011.
- L. Wang: Model Predictive Control System Design and Implementation Using MATLAB, Springer, 2009.

M 5.140 Module: Optoelectronics and Optical Communication (WW4INGMBIMT6) [M-MACH-101295]

Coordinators: Prof. Dr. Jan Gerrit Korvink
Organisation: KIT Department of Mechanical Engineering

Part of: Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	1 term	German	4	3

ID	Name	Term offered	Credits	Lecturers
Optoelektronik und Optische Kommunikationstechnik (Kernbereich) (Elective: 1 item)				
T-ETIT-100639	Optical Transmitters and Receivers	WT	6 CP	Freude
Optoelektronik und Optische Kommunikationstechnik (Ergänzungsbereich) (Elective: at least 5 credits)				
T-MACH-102152	Novel Actuators and Sensors	WT	4 CP	Kohl, Sommer
T-ETIT-101938	Communication Systems and Protocols	ST	5 CP	Becker, Becker
T-ETIT-100741	Laser Physics	WT	4 CP	Eichhorn
T-ETIT-100740	Quantum Functional Devices and Semiconductor Technology	ST	3 CP	Koos
T-ETIT-114418	Integrated Photonics		6 CP	Koos

Assessment
The assessment is carried out as partial exams

(according to Section 4 (2), 1-3 SPO) of the core course and further single courses of this module, whose sum of credits must meet the minimum requirement of credits of this module. The assessment procedures are described for each course of the module separately.

The overall grade of the module is the average of the grades for each course weighted by the credits and truncated after the first decimal.

Prerequisites
none

Competence Goal
Student has basic knowledge of optical communication systems and related device and fabrication technologies.

- He/she can apply this knowledge to specific problems.

Content
This module covers practical and theoretical aspects in the areas of optical communications and optoelectronics. System aspects of communication networks are complemented by fundamental principles and device technologies of optoelectronics as well as and microsystem fabrication technologies.

Workload
270 hours

5.141 Module: Physical Foundations of Cryogenics [M-CIWVT-103068]

Coordinators: Prof. Dr.-Ing. Steffen Grohmann
Organisation: KIT Department of Chemical and Process Engineering
Part of: Engineering and Informatics (Chemical and Process Engineering)
Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each summer term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-CIWVT-106103	Physical Foundations of Cryogenics	ST	6 CP	Grohmann

Assessment
Learning control is an oral examination lasting approx. 30 minutes.

Prerequisites
None

Competence Goal
Understanding of the mechanisms of entropy generation, and the interaction of the first and the second law in thermodynamic cycles; understanding of cryogenic material properties; application, analysis and assessment of real gas models for classical helium I; understanding of quantum fluid properties of helium II based on Bose-Einstein condensation, understanding of cooling principles at lowest temperatures.

Content
Relation between energy and temperature, energy transformation on microscopic and on macroscopic scales, physical definitions of entropy and temperature, thermodynamic equlibria, reversibility of thermodynamic cycles, helium as classical and as quantum fluid, low-temperature material properties, cooling methods at temperatures below 1 K.

Module Grade Calculation
The grade of the oral examination is the module grade.

Workload

- Attendance time (Lecture): 45 h
- Homework: 45 h
- Exam Preparation: 90 h

Literature
Schroeder, D.V.: An introduction to thermal physics. Addison Wesley Longman (2000)

Pobell; F.: Matter and methods at low temperatures. 3rd edition, Springer (2007)

5.142 Module: Physics, Technology and Applications of Thin Films [M-ETIT-105608]

Coordinators: Dr. Konstantin Ilin
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Engineering and Informatics (Electrical Engineering and Information Technology)
Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-111237	Physics, Technology and Applications of Thin Films	WT	4 CP	Ilin

Assessment
Oral examination of approximately 20 minutes

Prerequisites
The modul "M-ETIT-102332 - Thin films: technology, physics and applications" and "Thin Films: Technology, Physics and Applications I" may neither be started nor completed.

Competence Goal
Students should be able to discuss interplay between growth conditions of thin films, physical and geometrical properties of nanostructure made of these films, and performance and suitable areas of application of detectors of radiation based on interaction of these nanostructures with electromagnetic power. The knowledge obtained by students should provide a theoretical basis for the most important steps in development of thin film nanoelectronic devices.

Content
Students will get practically oriented information about technology of thin films including different methods of deposition of thin films like magnetron sputtering, thermal evaporation, pulsed laser ablation, about basics of vacuum technology, and about mechanisms of growth of thin films of different materials at different conditions.

Patterning methods (photo- and e-beam lithography, reactive ion etching, ion milling, and lift-off techniques) suitable for nanometer scale features of electronic devices will be considered in details.

Experimental methods of characterization of material, geometrical, optical, physical, superconducting, electron and phonon properties of thin films, nanostructures made of these films, and devices based on these nanostructures will be discussed.

Consideration of technology and physics of thin film structures will be done on example of development of three types of fast and sensitive detectors of electro-magnetic radiation for applications in optical and THz spectral ranges: superconducting nanowire single-photon detector, hot-electron bolometer, and YBCO ps-fast detector of synchrotron emission. Dependence of detector's performance on their fabrication condition will be analyzed in frame of physical models which describe response mechanisms of the detectors to absorbed radiation.

Practical actualization of the knowledge is possible in frame of Praktikum Nanoelektronik (LVN 23669).

Module Grade Calculation
The module grade is the grade of the oral exam.

Workload
A workload of approx. 90 h is required for the successful completion of the module. This is composed as follows

1. attendance time in lecture/exercise 18 h
2. pre-/postprocessing of the lecture 24 h
3. preparation/attendance oral exam 48 h

M 5.143 Module: Power Electronic Systems [M-ETIT-107533]

Coordinators: Prof. Dr.-Ing. Marc Hiller
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Engineering and Informatics (Electrical Engineering and Information Technology)
Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-114751	Power Electronic Systems	WT	6 CP	Hiller

Assessment
Success control takes place in the form of an oral examination lasting approx. 25 minutes.

Prerequisites
none

Competence Goal
Students are familiar with the grid-connected and self-commutated power converter circuits relevant to energy technology applications.
They are able to select power converters for drive technology and grid applications (including high-voltage direct current transmission) and assess their operating characteristics.
They are familiar with the functionality as well as the advantages and disadvantages of the various multi-stage inverter circuits and are able to select the required power semiconductors depending on the electrical requirements and the type of cooling.
Students are also able to design the power semiconductors and passive components of a power converter circuit electrically and thermally.
They are familiar with the normative insulation requirements and can analyze and explain the requirements for the protection of a power converter.

Content
The electrical and thermal design and dimensioning of power converters in drive and energy technology are presented and discussed in detail in this lecture. Based on the terminal behavior of the various power converter topologies, the interactions with other system components are presented and evaluated.
The lecture provides an overview of possible measures to improve the system behavior and deals with the protection of power converter circuits.
The following topics are covered in detail:

- Introduction
- Grid-connected power converters under idealized and real conditions and their most important applications in power engineering
- Self-commutated multilevel converters: Neutral point clamped inverters, floating capacitor inverters, series cell inverters, modular multilevel converters, hybrid circuits, modulation methods
- Semiconductor components for grid-connected and self-commutated power converters, protective devices
- Heat dissipation concepts for power semiconductors and passive components, junction temperature calculations
- Load cycle stability of power semiconductors
- Short-circuit current design for mains and motor side
- Protection concepts
- Insulation coordination, standards
- Transformer, grid connection
- Mains and motor-side filters
- Cable models
- Interaction between inverter and machine (insulation, bearing currents)
- Reliability calculations
- Excursion to converter plant if necessary

The lecturer reserves the right to deviate from the content of the current lecture without special notice.

Module Grade Calculation
The module grade is the grade of the oral examination.

Workload

- 22x lecture and 8x exercise à 2 h = 60 h
- 22x follow-up of the lecture à 1 h = 22 h
- 8x preparation of the exercise à 2 h = 16 h
- Preparation for the exam = 81 h
- Examination time = 1 h

  Total = 180 h

Recommendations
Knowledge of the basics of power electronics and electrical machines is helpful, but not essential.

M 5.144 Module: Power Electronics [M-ETIT-104567]

Coordinators: Prof. Dr.-Ing. Marc Hiller
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Engineering and Informatics (Electrical Engineering and Information Technology)
Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each summer term	1 term	English	4	6

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-109360	Power Electronics	ST	6 CP	Hiller

Assessment
The examination takes place in form of a written examination lasting 120 minutes.

Prerequisites
None

Competence Goal
Students will be familiar with state-of-the-art power semiconductors including their application related features. Furthermore students will be familiar with the circuit topologies for DC/DC and DC/AC power conversion. They know the associated modulation and control methods and characteristics. They are able to analyze the circuit topologies with regard to harmonics and power losses. This also includes the thermal design of power electronic circuits. In addition, they are able to select and combine suitable circuits for given electrical energy conversion requirements.

Content
In the lecture, power electronic circuits for DC/DC and DC/AC power conversion using IGBTs and MOSFETs are presented and analyzed. First, the basic properties of self-commutated circuits under idealized
conditions are elaborated using the DC/DC converter as an example. Then, self-commutated power converters for three-phase applications are presented and analyzed with respect to modulation and their AC
and DC terminal behavior. Based on the real power semiconductor behavior in on- and off-state the device losses are calculated. Furthermore the thermal design of power converters is explained using thermal equivalent circuits of power devices and cooling equipment. The voltage and current stress on the power
semiconductors in switching operation is explained as well as protective snubber circuits allowing a reliable operation within the safe operating area of the devices.

In detail, the following topics are treated:

- Power Semiconductors
- Commutation principles
- DC/DC converters
- Self-commutated 1ph and 3ph DC/AC inverters
- Modulation methods (Fundamental frequency modulation, Pulse width modulation with 3rd harmonic injection, Space vector modulation)
- Multilevel inverters
- Switching behavior in hard and soft switching applications
- Loss calculation
- Thermal equivalent circuits, thermal design
- Snubber circuits.

The lecturer reserves the right to adapt the contents of the lecture to current needs without prior notice.

Module Grade Calculation
The module grade is the grade of the written exam.

Workload
14x lecture and 14x exercise à 2 h = 56 h
14x wrap-up of the lecture à 1 h = 14 h
14x preparation of the exercise à 2 h = 28 h
Preparation for the exam = 75 h
Examination time = 2 h
Total = approx. 175 h (corresponds to 6 LP)

5.145 Module: Power-to-X – Key Technology for the Energy Transition [M-CIWVT-105891]

Coordinators: Prof. Dr.-Ing. Roland Dittmeyer
Dr. Peter Holtappels
Organisation: KIT Department of Chemical and Process Engineering
Part of: Engineering and Informatics (Chemical and Process Engineering)
Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-CIWVT-111841	Power-to-X – Key Technology for the Energy Transition	WT+ST	4 CP	Dittmeyer, Holtappels
T-CIWVT-111842	Practical in Power-to-X: Key Technology for the Energy Transition	WT+ST	2 CP	Dittmeyer, Holtappels

Assessment
The learning control consists of two partial achievements:

1. Lab, completed coursework
2. Oral examination lastin approx. 30 minutes

Competence Goal
The students are familiar with the rationale and the basic concepts of Power-to-X conversion. They know the major routes and individual components and what can be expected in terms of performance metrics both on component and process level. They have developed a basic understanding of water and steam electrolysis as well as of plasma splitting of carbon dioxide. Moreover, they had a first encounter with real container plants for electrolysis and fuel synthesis in the Energy Lab 2.0 as well as modular setups for plasma splitting, fuel synthesis and fuel upgrading.

Content
The module will provide an introduction to Power-to-X technologies which are expected to play a major role in the future energy system. The rationale for converting renewable electrical energy into fuels and chemicals will be explained and substantiated with data from relevant studies. Concepts for central and distributed Power-to-X facilities will be described with a focus on modular technologies for distributed production. Different options for water and steam electrolysis as well as selective electrochemical reduction of carbon dioxide will be discussed with a view to technology readiness level, energy efficiency, and cost. The alternative concept of plasma-based activation of inert molecules will be introduced and the status of this technology will be assessed and compared to electrolysis. Basic process layouts for production of synthetic methane, liquid hydrocarbons, methanol and ammonia from renewable electrical energy, carbon dioxide and water will be described and assessed in terms of material and energy flows and options for process integration. Moreover, concepts for offshore Power-to-X production will be explained and current research in this area will be highlighted. Finally, industrial project initiatives in the field of Power-to-X will be presented and discussed. The practical will cover four days and will be done in larger groups of up to 15 persons. Participants will be introduced to the containerized Power-to-Liquid Plant and its infrastructure in the Energy Lab 2.0 at KIT Campus North. They will work at this site with a containerized water electrolyzer and steam electrolyzer for hydrogen production. Moreover, the group will be made familiar with an experimental setup for plasma splitting of carbon dioxide in the plasma lab jointly operated by IMVT and IHM and with the synthesis and upgrading of Fischer-Tropsch-Fuels in the synfuel lab at IMVT.

Module Grade Calculation
The module grade is the grade of the oral exam.

Additional Information
Practical course: Dates by arrangement, Location: IMVT, KIT Campus Nord, Energy Lab 2.0, Building 605.

Workload

- Attendance timet:
  lecture: 30 h,
  lab: 16 h (4 dates)
- Self-study: 90 h
- Exam preparation: 45 h

Literature

Florian Ausfelder, Hannah Dura, 3. Roadmap des Kopernikus-Projektes P2X Phase II, OPTIONEN FÜR EIN NACHHALTIGES ENERGIE- SYSTEM MIT POWER-TO-X- TECHNOLOGIEN, Transformation – Anwendungen – Potenziale, 2021 (https://www.kopernikus-projekte.de/aktuelles/news/p2x_roadmap_3_0)

M 5.146 Module: Practical Amplifier Design [M-ETIT-107156]

Coordinators: Prof. Dr.-Ing. Ahmet Cagri Ulusoy
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Engineering and Informatics (Electrical Engineering and Information Technology)
Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each summer term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-114179	Practical Amplifier Design	ST	3 CP	Ulusoy

Assessment
Success control takes place in the form of an oral examination of approx. 30 min.

Prerequisites
none

Competence Goal

- The students learn to apply theory of high-speed amplifier design into practical implementation
- They understand critical limitations such as interconnect parasitics, self-heating, power supply and stability, and learn methods how to handle these limitations in practice
- They are familiar with the required fabrication process and the components for the implementation of such amplifiers
- Students acquire hands-on experience on fabrication, assembly and characterization of high-speed amplifiers

Content
In this lecture, the student will be provided with material for theoretical understanding of high-speed amplifier design. The lectures will be organized around the design of a high-speed amplifier, that will be fabricated, assembled and characterized. Following topics will be covered:

- Transistor analysis, and characterization
- Schematic level amplifier design, steady-state analysis and important metrics of amplifier performance
- Layout design and practical limitations with respect to parasitics, interconnections
- Electro/thermal effects and self heating
- Impact of bias circuitry
- Stability analysis
- Assembly and characterization of high-speed amplfiiers

Module Grade Calculation
The module grade is the grade of the oral examination.

Workload
The total effort for this lecture is estimated as following:

1. Attendance to the lectures and exercises (15*(2)=30h)
2. Preparation to the lectures and exercises (15*(2)=30h)
3. Amplifier design, assembly, characterization (15h)
4. Preparation to the exam (15h)

Total: 90h

Recommendations
The contents of "M-ETIT-105124 – Radio-Frequency Electronics" is recommanded.

M 5.147 Module: Principles of Constrained Static Optimization [M-CIWVT-106313]

Coordinators: Dr.-Ing. Pascal Jerono
Prof. Dr.-Ing. Thomas Meurer

Organisation: KIT Department of Chemical and Process Engineering

Part of: Engineering and Informatics (Chemical and Process Engineering)
Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-CIWVT-112811	Principles of Constrained Static Optimization		4 CP	Jerono, Meurer

Assessment
Learning control is an oral exam with a duration of about 45 minutes.

Prerequisites
None

Competence Goal
Students have an in-depth understanding of static optimization with constraints. They understand the underlying mathematical concepts and can apply them to new problems. They have a comprehensive understanding of optimization methods and are able to apply these methods independently to static optimization problems. Students are familiar with various numerical solution approaches, understand how they work, and can implement them for optimization problems.

Content
Optimization problems arise in a broad variety in different scientific and engineering domains ranging from the fit of parameter based on a performance criterion to finding extreme values of an objective function and further extending to machine learning applications. While dynamic optimization (addressed on the module M-CIWVT-106317) involves dynamical systems in static optimization the minimization (maximization) of functions subject to equality and inequality constraints is considered. This module gives an introduction to the mathematical analysis and numerical solution of unconstrained and constrained static optimization problems. The lecture addresses the following topics:

- Fundamentals of static optimization problems
- Unconstrained static optimization
- Constrained static optimization
- Numerical methods

Selected examples are considered and solved in the exercises and dedicated computer exercises.

Module Grade Calculation
The grade of the module ist the grade of the oral exam.

Workload
Attendance time: Lectures: 15 hrs. exercises: 15 hrs.

Self-study: 50 hrs.

Exam praparation: 40 hrs.

Literature

- T. Meurer: Optimal and Model Predictive Control, Lecture Notes.
- D. G. Luenberger, Y. Ye: Linear and Nonlinear Programming, Springer, 2008.
- N. Nocedal, S.J. Wright: Numerical Optimization, Springer, 2006.
- M. Papageorgiou, M. Leibold, M. Buss: Optimierung, Springer, 2012.
- S. Boyd, L. Vandenberghe: Convex Optimization, Cambridge University Press, 2004.
- C.T. Kelley. Iterative Methods for Optimization. SIAM, 1999.

M 5.148 Module: Printed and Thin-Film Electronics [M-ETIT-107343]

Coordinators: Prof. Dr. Jasmin Aghassi-Hagmann
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Engineering and Informatics (Electrical Engineering and Information Technology)
Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
3 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-114417	Printed and Thin-Film Electronics	WT	3 CP	Aghassi-Hagmann, Hirtz

Assessment
The assessment takes place in form of an oral examination (approx. 20 minutes).

Prerequisites
none

Competence Goal
The students

- have a general overview over the topic of Printed Electronics and basic knowledge of the materials, manufacturing methods and applications
- can critically analyze the specific advantages and limits of different methods and approaches of manufacturing printed electronics and can thus evaluate the potential for certain applications in relation to classical electronics
- are able to design basic printed electronic devices and know how to characterize their electronic properties
- are enabled to follow current developments in printed electronics and utilize this knowledge in their own research projects
- are enabled to navigate in applications of multidisciplinary nature (physics, material science, medical engineering)
- have gained relevant expertise for responsible R&D positions in industries such as semiconductor, sensors, automation, instrumentation and measurement technology.

Content
This module gives the student an overview over theoretical and practical aspects of printed and thin-film electronics, including applications. It will introduce the basics of manufacturing methods such as 2D / 3D printing, thin-film methods as CVD, PLD, sputtering, nanolithography, e-beam lithography and lift-off processes. Different classes of materials (e.g. inorganic semiconductors, metals, biomaterials, ceramics) and their properties are discussed. Furthermore, the module will introduce the students to applications of printed electronics in a broad range of fields, as IoT, computing, medical wearables, hybrid sensor devices, bioelectronics, and soft robotics.

Module Grade Calculation
The module grade is the grade of the oral examination.

Workload

1. attendance time in lecture: 15*2 h = 30 h
2. preparation/follow-up of the same: 15*2 h = 30 h
3. exam preparation and attendance in the same: 30 h

Total: 90 h = 3 CP

Recommendations
Ideally, this module will be selected by students in combination with “Lab Course Printed and Flexible Electronics”, but this is optional. Basic knowledge in electronic devices, materials of electronics, introductory courses in physics and sensor systems will be beneficial.

M 5.149 Module: Private Business Law (WW4JURA5) [M-INFO-101216]

Coordinators: N.N.
Organisation: KIT Department of Informatics
Part of: Electives (Law or Sociology)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	2 terms	German	4	6

ID	Name	Term offered	Credits	Lecturers
Private Business Law (Elective: at least 1 item as well as at least 9 credits)				
T-INFO-111405	Seminar: Commercial and Corporate Law in the IT Industry	WT	3 CP	Nolte
T-INFO-101288	Corporate Compliance	WT	3 CP	Herzig
T-INFO-102036	Computer Contract Law	WT	3 CP	Menk
T-INFO-111436	Employment Law	ST	3 CP	Hoff
T-INFO-111437	Tax Law	ST	3 CP	Dietrich

Assessment
see partial achievements

Prerequisites
None

Competence Goal
The student

- has gained in-depth knowledge of German company law, commercial law and civil law;
- is able to analyze, evaluate and solve complex legal and economic relations and problems;
- is well grounded in individual labour law, collective labour law and commercial constitutional law, evaluates and critically assesses clauses in labour contracts;
- recognizes the significance of the parties to collective labour agreements within the economic system and has differentiated knowledge of labour disputes law and the law governing the supply of temporary workers and of social law;
- possesses detailed knowledge of national earnings and corporate tax law and is able to deal with provisions of tax law in a scientific manner and assesses the effect of these provisions on corporate decision-making.

Content
The module covers a range of special topics in corporate law, knowledge of which is essential in order to make sensible business decisions. Building on the knowledge previously acquired in private law, students gain practical insights into how contracts are drafted, as well as even more detailed knowledge of civil law and German commercial and company law. In addition, they will acquire solid knowledge of labor and tax law.

Workload
The total workload for this module is approximately 270 hours (9 credits). The allocation is based on the credits of the courses of the module. The workload for courses with 3 credits is about 90 hours.The total number of hours per course results from the effort required to attend the lectures as well as the examination times and the time required to achieve the learning objectives of the module for an average student for an average performance.

5.150 Module: Process Engineering in Construction (bauiEX402-VERFBB) [M-BGU-101110]

Coordinators: Prof. Dr.-Ing. Shervin Haghsheno
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each winter term	2 terms	German	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-BGU-101844	Process Engineering	WT+ST	3 CP	Schneider
Electives (Elective: between 2 and 3 items as well as between 6 and 7,5 credits)				
T-BGU-101845	Construction Equipment	WT+ST	3 CP	Gentes
T-BGU-101832	Operation Methods for Foundation and Marine Construction	WT+ST	1,5 CP	Schneider
T-BGU-101801	Operation Methods for Earthmoving	WT+ST	1,5 CP	Schlick
T-BGU-101846	Tunnel Construction and Blasting Engineering	WT+ST	3 CP	Haghsheno
T-BGU-101847	Project Studies	ST	3 CP	Gentes
T-BGU-101850	Disassembly Process Engineering	ST	3 CP	Gentes

Assessment
- module component T-BGU-101844 with written examination according to § 4 Par. 2 No. 1

according to selected course:

- module component T-BGU-101845 with written examination according to § 4 Par. 2 No. 1
- module component T-BGU-101832 with oral examination according to § 4 Par. 2 No. 2
- module component T-BGU-101801 with oral examination according to § 4 Par. 2 No. 2
- module component T-BGU-101846 with oral examination according to § 4 Par. 2 No. 2
- module component T-BGU-101847 with oral examination according to § 4 Par. 2 No. 2
- module component T-BGU-101850 with oral examination according to § 4 Par. 2 No. 2

details about the learning controls, see module components

Prerequisites
The course Verfahrenstechnik [6241704] is compulsory and must be examined.

Competence Goal
Students understand different processes and the related construction equipment, it's technology, capabilities and constraints. Students can define process solutions consisting of machinery and devices. They can evaluate existing processes through knowledge about process performance and operating conditions, and the can identify potential for improvement.

Content
Within the frame of this module, various construction und conditioning processes will be presented as well as performance calculations conducted. Students learn about the construction machinery and devices of these processes. Transmission, generation, conversion and controlling of power are explained with the help of various practical examples. Moreover, the module includes possibilities for an on-site familiarization.

Module Grade Calculation
grade of the module is CP weighted average of grades of the partial exams

Additional Information
None

Workload
contact hours (1 HpW = 1 h x 15 weeks):

- Process Engineering lecture: 30 h

according to the selected course and examination:

- Construction Equipment lecture: 30 h
- Operation Methods for Foundation and Marine Construction lecture: 15 h
- Operation Methods for Earthmoving lecture: 15 h
- Tunnel Construction and Blasting Engineering lecture: 30 h
- Project Studies lecture, exercise: 30 h
- Disassembly Process Engineering lecture, exercise: 30 h

independent study:

- preparation and follow-up lectures Process Engineering: 30 h
- examination preparation Process Engineering (partial examination, compulsory): 30 h

according to the selected course and examination:

- preparation and follow-up lectures Construction Equipment: 30 h
- examination preparation Construction Equipment (partial examination, selectable): 30 h
- preparation and follow-up lectures Operation Methods for Foundation and Marine Construction: 15 h
- examination preparation Operation Methods for Foundation and Marine Construction (partial examination, selectable): 15 h
- preparation and follow-up lectures Operation Methods for Earthmoving: 15 h
- examination preparation Operation Methods for Earthmoving (partial examination, selectable): 15 h
- preparation and follow-up lectures Tunnel Construction and Blasting Engineering: 30 h
- examination preparation Tunnel Construction and Blasting Engineering (partial examination, selectable): 30 h
- preparation and follow-up lectures, exercises Project Studies: 30 h
- examination preparation Project Studies (partial examination, selectable): 30 h
- preparation and follow-up lectures, exercises Disassembly Process Engineering: 30 h
- examination preparation Disassembly Process Engineering (partial examination, selectable): 30 h

total: 270 h

Recommendations
none

5.151 Module: Production Engineering [M-MACH-106590]

Coordinators: Prof. Dr.-Ing. Volker Schulze
Organisation: KIT Department of Mechanical Engineering

Part of: Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	2 terms	German	4	5

ID	Name	Term offered	Credits	Lecturers
Production Engineering (Elective: at least 9 credits)				
T-MACH-113985	Additive Manufacturing of Metallic Components	WT	4 CP	Zanger
T-MACH-113647	Digitalization from Product Concept to Production	WT	4,5 CP	Wawerla
T-MACH-114800	Global Production	WT	5 CP	Lanza
T-MACH-105188	Integrative Strategies in Production and Development of High Performance Cars	ST	4 CP	Schlichtenmayer
T-MACH-112115	Artificial Intelligence in Production	WT	4 CP	Fleischer
T-MACH-105783	Learning Factory “Global Production“	WT	6 CP	Lanza
T-MACH-108878	Laboratory Production Metrology	ST	4 CP	Benfer, Lanza
T-MACH-110318	Product- and Production-Concepts for Modern Automobiles	WT	4 CP	Kienzle, Steegmüller
T-MACH-110984	Production Technology for E-Mobility	ST	4 CP	Fleischer
T-MACH-113575	Project Course Additive Manufacturing: Design Optimization and Production of Metallic Components	ST	4 CP	Zanger
T-MACH-102107	Quality Management	WT	4 CP	Lanza
T-MACH-113031	Rapid Industrialization of Immature Products using the Example of Electric Mobility	WT	4 CP	Bauer
T-MACH-112121	Seminar Application of Artificial Intelligence in Production	ST	4 CP	Fleischer
T-MACH-105185	Control Technology	ST	4 CP	Gönnheimer
T-MACH-114969	Strategic Decision-Making in Global Production Networks through Optimization and Simulation	ST	4,5 CP	Lanza, Martin
T-MACH-105177	Metal Forming	ST	4 CP	Herlan
T-MACH-102148	Gear Cutting Technology	WT	4 CP	Hoffmeister, Klaiber

Assessment
Oral exams: duration approx. 5 min per credit point

Written exams: duration approx. 20 - 25 min per credit point

Amount, type and scope of the success control can vary according to the individually choice.

Prerequisites
The module M-MACH-101284 -Production Technology must not have been started.

Competence Goal
The students

- are able to apply the methods of production science to new problems.
- are able to analyze and evaluate the suitability of the methods, procedures and techniques for a specific problem.
- are able to use their knowledge target-oriented to achieve an efficient production technology.
- are able to analyze new situations and choose methods of production science target-oriented based on the analyses, as well as justifying their selection.
- are able to describe and compare complex production processes exemplarily.

Content
Within this module the students will get to know and learn about production science. Manifold lectures and excursions as part of several lectures provide specific insights into the field of production science.

Workload
The work load is about 270 hours, corresponding to 9 credit points.

Teaching and Learning Methods
Lectures, seminars, workshops, excursions

5.152 Module: Project Management in Construction (bauiEX403-PROJMAN) [M-BGU-101888]

Coordinators: Prof. Dr.-Ing. Shervin Haghsheno
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each winter term	2 terms	German	4	3

Election Notes
The course Project Management in Construction and Real Estate Industry II is only allowed to be selected if the selectable course Project Management in Construction and Real Estate Industry I has been passed.

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-BGU-103432	Project Management in Construction and Real Estate Industry I	WT	3 CP	Haghsheno
T-BGU-111921	Turnkey Construction	WT+ST	3 CP	Haghsheno
Electives (Elective: between 1 and 2 items as well as between 3 and 4,5 credits)				
T-BGU-103427	Site Management	ST	1,5 CP	Haghsheno
T-BGU-111211	Energetic Refurbishment	WT+ST	1,5 CP	Lennerts, Schneider
T-BGU-111922	Civil Engineering Structures and Regenerative Energies	WT+ST	3 CP	Haghsheno
T-BGU-103433	Project Management in Construction and Real Estate Industry II	WT	3 CP	Haghsheno

Assessment
- module component T-BGU-103432 with examination of another type according to § 4 Par. 2 No. 3
- module component T-BGU-111921 with written examination according to § 4 Par. 2 No. 1

according to selected course:

- module component T-BGU-103427 with oral examination according to § 4 Par. 2 No. 2
- module component T-BGU-111211 with oral examination according to § 4 Par. 2 No. 2
- module component T-BGU-111922 with written examination according to § 4 Par. 2 No. 1
- module component T-BGU-103433 with examination of another type according to § 4 Par. 2 No. 3

details about the learning controls, see module components

Prerequisites
none

Competence Goal
Students have basic and in-depth knowledge and skills in project management in the construction and real estate industry. They know the necessary skills for successful project management according to the ICB4 standard of the International Project Management Association (IPMA) in the areas of context, methods, personality and social issues. They will be able to apply selected project management content (especially project objectives and determination of requirements, project execution implementation strategies and award procedures, schedule management, cost management and quality management) and transfer project management methods to specific construction projects as part of a project setup. They are also familiar with the job description of project manager/project controller in the German construction and real estate industry and their corresponding tasks in a construction project. Furthermore, students will be able to collaborate effectively in teams and to present jointly developed content as a group (to a potential client) in the context of project work.

Content

- Project Management in Construction and Real Estate Industry I: In Part I, selected topics in the field of project management in the construction and real estate industry are examined in depth (including project preparation, procurement models, awarding processes, project organization, contract models, quality/schedule/cost management, risk management, project culture, digitalization).
- Turnkey Construction: In the subject area of turnkey construction besides the detailed design of shell construction, technical support and technical building services, there is also an explanation of the related basic knowledge in engineering. Also, basics of the technical support belong to the curriculum, e.g., heating installations, ventilation systems, A/C, electric installations. Most of all, there is a focus on regenerative energies. Furthermore, the explanation of the processes in turnkey construction, from design and construction permit to final acceptance of work, is part of the lecture.
- Site Management: The course site management presents the work of foreman, site manager, and project manager and contains significant aspects of management processes of the construction site. In addition to performance reporting, work costing and site management, the technical, legal and economic tasks of the site manager as well as communication and correspondence on the construction site will be highlighted. In addition, accident prevention regulations, active and passive protection measures as well as the organization of the labor protection during operation and on site are discussed.
- Energy Refurbishment: fundamentals of building physics, energy-related improvement of the building envelope, improvement of building services engineering and economic efficiency, energy sources and local energy conversion (e.g., photovoltaics, geothermal energy, heat pumps, etc.), calculation of the energy demands of buildings, energy performance assessment of buildings according to GEG.
- Civil Engineering Structures and Regenerative Energies: In the subject area of civil engineering structures and regenerative energies, besides basic knowledge in construction, there is also a focus on production methods for the construction and maintenance of the selected civil engineering structures. Adding to conventional construction methods there are topics like additive manufacturing in solid construction. This also includes the view on hydraulic constructions (e.g. water locks), waste disposal (e.g. waste disposal sites) and infrastructure constructions (e.g. steel composite bridge). Also, there is a focus on regenerative energies (e.g. wind power stations).
- Project Management in Construction and Real Estate Industry II: In part II, selected topics in the field of project management in the construction and real estate industry are examined in depth (including project preparation, procurement models, awarding processes, project organization, contract models, quality/deadline/cost management, risk management, project culture, digitization).

Module Grade Calculation
grade of the module is CP weighted average of grades of the partial exams

Additional Information
none

Workload
contact hours (1 HpW = 1 h x 15 weeks):

- Project Management in Construction and Real Estate Industry I lecture/exercise: 30 h
- Turnkey Construction lecture/exercise: 30 h

according to the selected course and examination:

- Site Management lecture: 15 h
- Energetic Refurbishment lecture: 15 h
- Civil Engineering Structures and Regenerative Energies lecture/exercise: 30 h
- Project Management in Construction and Real Estate Industry II lecture/exercise: 30 h

independent study:

- preparation and follow-up lecture/exercises Project Management in Construction and Real Estate Industry I: 30 h
- examination preparation Project Management in Construction and Real Estate Industry I (partial examination, compulsory): 30 h
- preparation and follow-up lecture/exercises Turnkey Construction: 30 h
- examination preparation Turnkey Construction (partial examination, compulsory): 30 h

according to the selected course and examination:

- preparation and follow-up lectures Site Management: 15 h
- examination preparation Site Management (selectable partial examination): 15 h
- preparation and follow-up lectures Energetic Refurbishment: 15 h
- examination preparation Energetic Refurbishment (selectable partial examination): 15 h
- preparation and follow-up lecture/exercises Civil Engineering Structures and Regenerative Energies: 30 h
- examination preparation Civil Engineering Structures and Regenerative Energies (selectable partial examination): 30 h
- preparation and follow-up lecture/exercises Project Management in Construction and Real Estate Industry II: 30 h
- examination preparation Project Management in Construction and Real Estate Industry II (selectable partial examination): 30 h

total: 270 h

Recommendations
see German version

Literature

Ahrens, H.; Bastian, K.; Muchowski, L. (Hrsg.) (2021): Handbuch – Projektsteurung, Baumanagement. Fraunhofer IRB Verlag.

Allison, M.; Ashcarft, H.; Cheng, R.; Klawens, S.; Pease, J. (2018): Integrated Project Delivery - An Action Guide for Leaders.

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Kochendörfer, B.; Liebchen, J. H.; Viering, M. G. (2018): Bau-Projekt-Management. Grundlagen und Vorgehensweisen. 5. Aufl. Leitfaden des Baubetriebs und der Bauwirtschaft. Wiesbaden, Springer Vieweg. ISBN: 978-3-8348-1823-2. DOI: 10.1007/978-3-8348-2245-1. URL: http: //dx.doi.org/10.1007/978-3-8348-2245-1.

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5.153 Module: Public Economic and Technology Law [M-INFO-106754]

Coordinators: TT-Prof. Dr. Frederike Zufall
Organisation: KIT Department of Informatics
Part of: Electives (Law or Sociology)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	2 terms	German/English	4	2

ID	Name	Term offered	Credits	Lecturers
Public Economic and Technology Law (Elective: at least 1 item as well as at least 9 credits)				
T-INFO-101309	Telecommunications Law	ST	3 CP	
T-INFO-101312	European and International Law	ST	3 CP	Brühann
T-INFO-111404	Seminar: IT- Security Law	WT	3 CP	Schallbruch
T-INFO-113381	Public International Law	ST	3 CP	Zufall
T-INFO-113887	EU Data Protection Law	WT	3 CP	Gil Gasiola

Assessment
see partial achievement

Prerequisites
see partial achievement

Competence Goal
Students
- have in-depth knowledge and understanding of selected legal areas of public commercial and technology law,
- understand the international and European dimensions of the legal system
- can make connections between technical and legal issues, classify them legally and evaluate them.

Content
The module covers a range of special topics in public commercial and technology law. In addition to the areas of telecommunications law and IT security law, this also includes an in-depth examination of the European and international legal framework. Current legal aspects of the platform economy, the digital single market and the regulation of artificial intelligence are addressed.

Workload
The total workload for this module is approx. 270 hours (9 credits). The distribution is based on the credit points of the courses in the module. The workload for courses with 3 credits is approx. 90 hours. The total number of work hours per course results from the time required to attend the lectures, the examination times and the time required to achieve the learning objectives of the module for an average student for an average performance.

Recommendations
see partial achievement

M 5.154 Module: Quantitative Core [M-WIWI-107187]

Coordinators: Studiendekan des KIT-Studienganges
Organisation: KIT Department of Business and Economics
Part of: Quantitative Core

Credits	Grading	Recurrence	Duration	Language	Level	Version
18 CP	graded	Each term	2 terms	English	4	1

ID	Name	Term offered	Credits	Lecturers
Economics (Elective: at most 1 item)				
T-WIWI-102861	Advanced Game Theory	WT	4,5 CP	Ehrhart, Puppe, Reiß
T-WIWI-109194	Dynamic Macroeconomics	WT	4,5 CP	Brumm
Business Administration (Elective: at most 1 item)				
T-WIWI-114979	Advanced Quantitative Finance	ST	4,5 CP	Thimme
T-WIWI-105777	Business Intelligence Systems	WT	4,5 CP	Mädche
T-WIWI-107720	Market Research	ST	4,5 CP	Klarmann
T-WIWI-114328	Navigating the Future of Work: Behavioral and Experiment Methods in the Workplace	WT	4,5 CP	Nieken
T-WIWI-114340	Quantitative Finance	WT	4,5 CP	Thimme
Operations Research (Elective: at most 1 item)				
T-WIWI-102726	Global Optimization I	ST	4,5 CP	Stein
T-WIWI-102727	Global Optimization II	ST	4,5 CP	Stein
T-WIWI-102724	Nonlinear Optimization I	WT	4,5 CP	Stein
T-WIWI-102725	Nonlinear Optimization II	WT	4,5 CP	Stein
T-WIWI-103123	Advanced Statistics	WT	4,5 CP	Grothe
Statistcs (Elective: at most 1 item)				
T-WIWI-111388	Applied Econometrics	WT	4,5 CP	Krüger
T-WIWI-110868	Predictive Modeling	Irreg.	4,5 CP	Krüger
T-WIWI-114325	Time Series Analysis	WT	4,5 CP	Schienle
Informatics (Elective: at most 1 item)				
T-WIWI-114211	Applied Informatics	ST	4,5 CP	Käfer, Lazarova-Molnar, Oberweis, Sack, Vinel, Volkamer, Zöllner

Assessment
One partial performance must be selected from each of four of five elective areas. For details on the performance assessments, see the respective partial performance.

The module grade is the LP-weighted average of the four components.

Competence Goal

- **Understanding**: Students should develop a fundamental understanding of quantitative methods and their application in the fields of engineering and management.
- **Application**: Students should be able to select and apply appropriate quantitative methods to solve complex problems in real-world situations.
- **Analysis**: Students should develop the ability to analyze and interpret data, and draw informed conclusions from it.
- **Decision-making**: Students should learn to use quantitative methods to support decision-making processes in companies.
- **Critical evaluation**: Students should be able to critically evaluate the results of quantitative analyses and recognize their limitations.

These qualification goals aim to equip students with the necessary skills to effectively use quantitative methods in their future careers.

Content

- **Fundamentals**:
    - Linear Algebra
    - Analysis
    - Statistics (descriptive and inferential)
    - Probability Theory
- **Specific Methods**:
    - Operations Research (e.g., linear programming, queuing theory)
    - Data Analysis and Visualization
    - Forecasting Methods
    - Simulation
    - Decision Theory
- **Applications**:
    - Supply Chain Management
    - Production Planning and Control
    - Quality Management
    - Project Management
    - Risk Management

Workload
Total workload for 18 credit points: approx. 540 hours.
Attendance time: approx. 180 hours
Preparation and follow-up: approx. 270 hours
Exam and exam preparation: approx. 90 hours
The exact distribution is based on the credit points of the courses in the module.

5.155 Module: Radar Systems Engineering [M-ETIT-100420]

Coordinators: Prof. Dr.-Ing. Marwan Younis
Prof. Dr.-Ing. Thomas Zwick

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: Engineering and Informatics (Electrical Engineering and Information Technology)
Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each winter term	1 term	English	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-100729	Radar Systems Engineering	WT	6 CP	Younis, Zwick

Assessment
Success control is carried out as part of a written overall examination (120 minutes) of the selected course, which in total meets the minimum requirement for LP.

Prerequisites
none

Competence Goal
Students can name the basic radar principles and explain how they work, their primary uses and their advantages and disadvantages. They are able to characterize the basic characteristics and mechanisms of propagation of electromagnetic waves and to apply the relevant equations. You can evaluate the influence of various system parameters on accuracy, resolution, false alarm rate, etc. and optimize systems. You can describe different radar system configurations (CW, FMCW, pulse, SAR) and apply the relevant radar signal processing methods. They are especially able to use and use the technologies and system configurations for the radars of the future for surveillance, automotive and industrial applications for research and development. In this lecture system technology is specifically taught.

Content
Based on electromagnetic field theory, the lecture teaches the basics of radar principles and their system technology. An insight into the system hardware is given and processing techniques are presented. All relevant, known radar systems (CW, FMCW, pulse and synthetic aperture radar) are described in detail. The system technology for the radars of the future is specifically dealt with. The reflective properties of radar targets are analyzed for their classification. In particular, polarimetry is taught. In this lecture, students learn how system technology contributes to the implementation of a radar system.

Module Grade Calculation
The module grade is the grade of the written exam.

Workload
Each credit point corresponds to approximately 25-30 hours of work (of the student). This is based on the average student who achieves an average performance. The workload includes:

Attendance study time lecture: 44 h

Attendance time computer exercise: 16 h

Self-study time including exam preparation: 120 h

A total of 180 h = 6 LP

M 5.156 Module: Radio Frequency Integrated Circuits and Systems [M-ETIT-105123]

Coordinators: Prof. Dr.-Ing. Ahmet Cagri Ulusoy
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Engineering and Informatics (Electrical Engineering and Information Technology)
Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each summer term	1 term	English	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-110358	Radio Frequency Integrated Circuits and Systems	ST	6 CP	Ulusoy

Assessment
The success criteria will be determined by an oral examination (approx. 20-30 min.)

Competence Goal

- The students acquire a comprehensive understanding in the design of monolithic integrated circuits for millimeter-wave frequencies, and they can apply the acquired knowledge using modern design tools.
- They have a good understanding of the critical performance parameters of high-frequency circuits such as stability, power gain and efficiency and reflection coefficient.
- They can describe the benefits and disadvantages of modern transistor technologies for millimeter-wave applications.T
- They can identify potential applications of integrated millimeter-wave circuits and understand the specific requirements of each application.
- They are familiar with basic elements of a high-frequency system, which consists of linear and non-linear circuits, low-noise and power amplifiers, as well as oscillators, switches and frequency converting circuits such as frequency multipliers and mixers.

Content
In this lecture the theory and the design methodology of monolithic integrated millimeter-wave circuits will be
studied in detail. The focus of the lecture is on the active linear and non-linear circuits in high-frequency frontends up to an application frequency of 300 GHz. In addition to this, fundamental topics such as impedance matching, stability, performance parameters of high-frequency transistors, and properties of active and passive circuit elements will be studied in detail. The operation principal of critical building blocks of a millimeter-wave system will be introduced including low-noise and power amplifiers, mixers, oscillators and switches. In the workshop, the students will have the chance to apply the acquired theoretical knowledge to design a millimeter-wave frontend using state-of-the art integrated circuit technology.

Module Grade Calculation
The module grade is the grade of the oral examination.

Workload
1. Attendance to the lectures (15*(2)=30h)
2. Attendance to the exercises (15*(2)=30h)
3. Preparation to the lectures and exercises (15*(2+2)=60h)
4. Preparation to the oral exam (40h)

Total: 160h

Recommendations
The lecture materials to „Grundlagen der Hochfrequenztechnik“ and „Halbleiterbauelemente“ are recommended.

M 5.157 Module: Radio-Frequency Electronics [M-ETIT-106955]

Coordinators: Prof. Dr.-Ing. Ahmet Cagri Ulusoy
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Engineering and Informatics (Electrical Engineering and Information Technology)
Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-113910	Radio-Frequency Electronics	WT	6 CP	Ulusoy

Assessment
The examination takes place in form of a written examination lasting 120 minutes.

Prerequisites
none

Competence Goal

- The students have a comprehensive understanding of the theory and the basic design methodology of RF and microwave circuits up to 300 GHz.
- They understand the limitations of active and passive circuit elements at high frequencies and their impact on the applications.
- They understand the limitations and how linear network theory is applied at higher frequencies.
- The students can apply the acquired theoretical knowledge to modern RF design problems.

Content
In this lecture, the theory and design methodology of RF electronic circuits will be studied in detail. The focus of the lecture is on the fundamentals of active and passive linear circuits. The important topics are:

- Phasor analysis and resonance,
- Electromagnetic theory, transmission lines and waveguides,
- Impedance matching networks,
- Two-port parameters of RF components and microwave network analysis,
- Feedback and stability analysis,
- High-frequency behavior of basic amplifier circuits, RF amplifiers design techniques,
- Microwave power dividers, couplers and filters

Module Grade Calculation
The module grade is the grade of the written examination.

Workload
The total effort for this lecture is estimated as following:

1. Attendance to the lectures (15*(3)=45h)
2. Attendance to the exercises (15*(1)=15h)
3. Preparation to the lectures and exercises (17*(3+1)=68h)
4. Preparation to the exam (52h)

A total of 180h

Recommendations
Basic knowledge of linear electrical networks and electronic circuits is recommended (e.g. M-ETIT-106417 – Lineare Elektrische Netze; M-ETIT-104465 – Elektronische Schaltungen).

M 5.158 Module: Rail System Technology (WI3INGMB25) [M-MACH-101274]

Coordinators: Prof. Dr.-Ing. Martin Cichon
Organisation: KIT Department of Mechanical Engineering

Part of: Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	1 term	German	4	5

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-MACH-102143	Rail System Technology	WT+ST	9 CP	Cichon

Assessment
written examination in German language
Duration: 120 minutes
No tools or reference materials may be used during the exam except calculator and dictionary

Competence Goal

- The students understand relations and interdependencies between rail vehicles, infrastructure and operation in a rail system.
- Based on operating requirements and legal framework they derive the requirements concerning a capable infrastructure and suitable concepts of rail vehicles.
- They recognize the impact of alignment, understand the important function of the wheel-rail-contact and estimate the impact of driving dynamics on the operating program.
- They evaluate the impact of operating concepts on safety and capacity of a rail system.
- They know the infrastructure to provide power supply to rail vehicles with different drive systems.
- The students learn the role of rail vehicles and understand their classification. They understand the basic structure und know the functions of the main systems. They understand the overall tasks of vehicle system technology.
- They learn functions and requirements of car bodies and jugde advantages and disadvantages of design principles. They know the functions of the car body’s interfaces.
- They know about the basics of running dynamics and bogies.
- The students learn about advantages and disadvantages of different types of traction drives and judge, which one fits best for each application.
- They understand brakes from a vehicular and an operational point of view. They assess the fitness of different brake systems.
- They know the basic setup of train control management system and understand the most important functions.
- They specify and define suitable vehicle concepts based on requirements for modern rail vehicles.

Content

1. Railway System: railway as system, subsystems and interdependencies, definitions, laws, rules, railway and environment, economic impact
2. Operation: Transportation, public transport, regional transport, long-distance transport, freight service, scheduling
3. Infrastructure: rail facilities, track alignment, railway stations, clearance diagram
4. Wheel-rail-contact: carrying of vehicle mass, adhesion, wheel guidance, current return
5. Vehicle dynamics: tractive and brake effort, driving resistance, inertial force, load cycles
6. Signaling and Control: operating procedure, succession of trains, European Train Control System, blocking period, automatic train control
7. Traction power supply: power supply of rail vehicles, comparison electric traction and diesel traction, dc and ac networks, system pantograph and contact wire, filling stations
8. Vehicle system technology: structure and main systems of rail vehicles
9. Car body: functions, requirements, design principles, crash elements, coupling, doors and windows
10. Bogies: forces, running gears, bogies, Jakobs-bogies, active components, connection to car body, wheel arrangement
11. Drives: priciples, electric drives (main components, asynchronous traction motor, inverter, with DC supply, with AC supply, without line supply, multisystem vehicles, dual mode vehicles, hybrid vehicles), non-electric drives
12. Brakes: basics, principles (wheel brakes, rail brakes, blending), brake control (requirements and operation modes, pneumatic brake, electropneumatic brake, emergency brake, parking brake)
13. Train control management system: definition of TCMS, bus systems, components, network architectures, examples, future trends
14. Vehicle concepts: trams, metros, regional trains, intercity trains, high speed trains, double deck vehicles, locomotives, freight wagons

Additional Information
A bibliography is available for download (Ilias-platform).

The lectures can be attended in the same term.

Workload

1. Regular attendance: 42 hours
2. Self-study: 42 hours
3. Exam and preparation: 186 hours

Teaching and Learning Methods
Lectures

5.159 Module: Reactor Modeling with CFD [M-CIWVT-106537]

Coordinators: Prof. Dr.-Ing. Gregor Wehinger
Organisation: KIT Department of Chemical and Process Engineering
Part of: Engineering and Informatics (Chemical and Process Engineering)
Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each summer term	1 term	English	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-CIWVT-113224	Reactor Modeling with CFD		6 CP	Wehinger

Assessment
Learning control is an examination of another type: presentation and term paper.

Prerequisites
None

Competence Goal
The students are able to

- describe and apply the mathematical and physical principles of computational fluid dynamics (CFD),
- use the commercial CFD software STAR-CCM+ independently and thoroughly (preprocessing, solving, postprocessing),
- develop a CFD reactor model for an unknown chemical process engineering problem and investigate alternative reactor designs based on this model,
- analyze and evaluate the results obtained, also using virtual reality (VR),
- identify and evaluate errors and uncertainties in CFD models,
- visualize, present, and critically discuss their CFD results in the form of a final report.

Content

1. Conservation laws for momentum, mass and energy
2. The Finite-Volume-Method, solution algorithms, and boundary conditions
3. Computational meshes
4. CFD- Modelling of chemical reactors
5. Use of virtual reality in CFD
6. Basics of writing a scientific paper

Module Grade Calculation
The module grade is the grade of the examination of another type.

Workload

- Attendance time: 45 h
- Self-study: 90 h
- Exam preparation: 45 h

Literature

- Ferziger, Perić: Numerische Strömungsmechanik; 2020 ; Springer
- Versteeg, Malalasekera; An Introduction to Computational Fluid Dynamics: The Finite Volume Method (2nd Edition); 2007; Pearson

5.160 Module: Robotics, Automation, and Systems Control Lab [M-ETIT-107524]

Coordinators: Prof. Dr.-Ing. Mike Barth
Prof. Dr.-Ing. Sören Hohmann

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: Engineering and Informatics (Electrical Engineering and Information Technology)
Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each summer term	1 term	English	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-114740	Robotics, Automation, and Systems Control Lab	ST	6 CP	Barth, Hohmann

Assessment
The examination takes place in the form of other types of examination. It consists of pitches after every thematic block, a final presentation and an oral examination in the amount of 20 minutes. The overall impression is evaluated.

Prerequisites
none

Competence Goal

- Students are enabled to develop approaches to various problems in a group, solve them, and present and defend their results. In addition, they are able to present their approach, thought processes, applied methods, and results in a comprehensible and scientifically precise style
- Students are able to familiarize themselves independently with a complex technical system and its components
- Students can discuss problem-solving strategies with team members in technical language and argue their preferred solution. Further they are able to reflect on the solution approaches used.
- Students can implement and validate automation solutions in various development environments (e.g. MATLAB/ Simulink, ROS, …). They are further able to apply their solutions across various system boundaries (e.g. automation concepts on Windows, ROS on Linux)
- Students gain insights into various engineering disciplines (e.g. mechanical engineering) and in rapid prototyping processes
- Students are able to use industrial bus systems (e.g. EtherCAT) in an automation context
- Students are able to apply methods that they have previously acquired in lectures on automation, control and robotics

Content
This module is designed to teach students the stated qualification objectives in the fields of automation, robotics and control by means of a complex automation task. The task not only covers the automation concept itself but also underlying aspects like hardware design, communication using an industrial bus system, drive system control and others. To do so, laboratory systems, namely laboratory scale robotic manipulator arms, are provided to the students.

Module Grade Calculation
The module grade results of the assessment of the pitches, presentation and the oral exam. Details will be given during the lecture at the beginning of the semester.

Additional Information
Due to capacity reasons there are only 25 places available. If necessary, a selection procedure will be carried out. Selection criteria are the progress in studies and in the recommended lectures. The application window is from March 1 for March 31 in Campus Plus (https://plus.campus.kit.edu/). For further information and current news, regularly visit the courses website (https://www.irs.kit.edu/Lectures_5404.php).

Workload

1. Introduction to the test stand (15h)
2. Introduction to single-axis motor control (15h)
3. Construction and print of end-effector components (30h)
4. Introduction to ROS (15h)
5. Implementation of hardware-interfaces in ROS (30h)
6. Automation, motion planning and control (30h)
7. Implementation on the real robot (15h)
8. Presentations and preparation (15h)
9. Oral exam and preparation (15h)

A total of 180 h = 6 CR

Recommendations
Knowledge in automation, control and robotics in general. Recommended, but not mandatory, lectures are:

- Practical Tools for Control Engineers – M-ETIT-106780
- Cyber Physical Modeling – M-ETIT-106953
- Seminar Industrial Process and Plant Engineering – M-ETIT-106970
- Optimization of Dynamic Systems – M-ETIT-100531
  (or successor Optimal Control – M-ETIT-107497)
- Regelung linearer Mehrgrößensysteme – M-ETIT-100374
  (or successor Multivariable Control Systems – M-ETIT-107515)
- Nichtlineare Regelungssysteme – M-ETIT-100371
  (or successor Nonlinear Control Systems - in preparation)

Teaching and Learning Methods
LACR – Laboratory Automation Control & Robotics (4 SWS)

5.161 Module: Safety, Computing and Law in Highway Engineering (bauiEX303-STRSER) [M-BGU-101066]

Coordinators: Dr.-Ing. Matthias Zimmermann
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each winter term	2 terms	German	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-BGU-101804	IT-Based Road Design	WT+ST	3 CP	Zimmermann
T-BGU-101674	Safety Management in Highway Engineering	WT+ST	3 CP	Zimmermann
T-BGU-106615	Laws concerning Traffic and Roads	WT+ST	3 CP	Hönig

Assessment
- module component T-BGU-101804 with oral examination according to § 4 Par. 2 No. 2
- module component T-BGU-101674 with oral examination according to § 4 Par. 2 No. 2
- module component T-BGU-106615 with written examination according to § 4 Par. 2 No. 1

details about the learning controls, see module components

Prerequisites
The examination "Design Basics in Highway Engineering" has to be passed. This can be taken either in the module "Design, Construction, Operation and Maintenance of Highways" (WI4INGBGU1) or can be approved from a previous study (e.g. Civil Engineering BSc at KIT).

Modeled Prerequisites
You have to fulfill one of 2 conditions:

1. The module component T-BGU-101670 - Design Basics in Highway Engineering must have been passed.
2. The module component T-BGU-106613 - Design Basics in Highway Engineering must have been passed.

Competence Goal
The student

- has in-depth knowledge of IT-based road design, traffic safety issues and aspects of road law.

Content
In this module, building on the design-relevant basics, the road design with special software is explained and practiced on the one hand, and on the other hand the concerns of traffic safety - also under (economic) economic aspects - are dealt with intensively in a lecture and a seminar. The module is rounded off with in-depth insights into specific planning, traffic and road law.

Module Grade Calculation
grade of the module is CP weighted average of grades of the partial exams

Additional Information
None

Workload
contact hours (1 HpW = 1 h x 15 weeks):

- IT-based Road Design lectures/exercises: 30 h
- Safety Management in Highway Engineering lectures/exercises: 30 h
- Laws concerning Traffic and Roads lecture: 30 h

independent study:

- preparation and follow-up lecture/exercises IT-based Road Design: 30 h
- examination preparation IT-based Road Design: 30 h
- preparation and follow-up lecture/exercises Safety Management in Highway Engineering: 15 h
- examination preparation Safety Management in Highway Engineering: 30 h
- preparation and follow-up lecture Laws concerning Traffic and Roads: 15 h
- examination preparation Infrastructure Management: 30 h

total: 270 h

Recommendations
None

5.162 Module: Seminar Industrial Process and Plant Engineering [M-ETIT-106970]

Coordinators: Prof. Dr.-Ing. Mike Barth
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Engineering and Informatics (Electrical Engineering and Information Technology)
Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each summer term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-113932	Seminar Industrial Process and Plant Engineering	ST	4 CP	Barth

Assessment
The examination will be the seminar presentation at the end of the semester. The criteria are:

- Live presentation of the created CAD and simulation models
- Poster design and usage within the presentation
- Answering the questions from the examiners
- Structure of the talk

Prerequisites
none

Competence Goal
The students:

- are able to create (concept and virtual realization) mechatronic plants and production facilities (e.g. robot cells).
- understand the Engineering-Lifecycle of unique production plants, machines and modules.
- are familiar with modern CAX-Engineering Methods.
- know advanced (cloud-based) CAX-Tool-Chains.
- know about advanced computer-aided design principles, e.g. model- and equation-based 3D-Design.
- are familiar with design definitions, patterns and features for semi-automated engineering tasks, e.g. automatic generation of variants.
- know CAX-information models, e.g. asset administration shell (AAS) or Parasolid.
- can implement new software-based Engineering features and interfaces.
- are able to perform the virtual commissioning of complex systems using advanced CAX-features.

Content

- This module is designed to teach students the theoretical and practical aspects of advanced model-based and computer-aided design of unique systems, e.g.
    - Industrial plants,
    - Production cells (e.g. robot),
    - Production modules and machines.
- This includes every lifecycle phase of an engineering project starting from 3D-CAD Design to advanced system simulations and concluding in virtual commissioning and optimization during operations.
- Introduction to advanced cloud-based 3D-Plant-Engineering (e.g. OnShape).
- Principals of model-based system design (3D-Modeling, Drawings, Feature-based Rule Set) including advanced kinematics and simulation.
- Configuration and use of VR-, AR-Setups for virtual commissioning and operator training setups.
- Concept and implementation of software-based feature scripts (e.g. C++, python) for advanced engineering in the used tool chain.
- Concept and implementation of interfaces to external engineering data-sources (e.g. data tables, Asset Administration Shell via REST-API, ROS, MQTT or OPC UA).

Module Grade Calculation
The module grade is the grade of the final presentation including the aspects named above.

Additional Information
The seminar is limited to a number of 20 participants due to capacity reasons. If necessary, a selection procedure will be carried out. Places will be allocated taking into account the students' academic progress. Details will be announced on the lecture website.

Workload

1. Attendance in Lecture Blocks for Engineering Theory: 10*2 h = 20 h
2. Guided Seminar Work Block: 80 h
3. Preparation of exam: 20 h

A total of 120 h = 4 CR

Recommendations
Enjoyment and interest in industrial engineering and Computer-aided-X Technologies like e.g. 3D CAD, Robotics, Kinematic Simulations, VR, MR and AR-Technologies.

M 5.163 Module: Seminar Module [M-WIWI-107204]

Coordinators: Studiendekan des KIT-Studienganges
Organisation: KIT Department of Business and Economics
Part of: Electives (mandatory)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	1 term	German/English	4	1

Election Notes
Students must complete two seminars from the module's course list, each worth 4 CP, offered by the KIT Department of Business and Economics. Both seminars may be taken at the same institute within the department.

One of the two required seminars may be replaced by a module component offered by a KIT Department of Engineering.

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-WIWI-114229	Supervised Profile Definition		1 CP	Studiendekan des KIT-Studienganges
Seminars of the Department of Economics and Management (Elective: at least 1 item)				
T-WIWI-103474	Seminar in Business Administration A (Master)	WT+ST	3 CP	Professorenschaft des Fachbereichs Betriebswirtschaftslehre
T-WIWI-103476	Seminar in Business Administration B (Master)	WT+ST	3 CP	Professorenschaft des Fachbereichs Betriebswirtschaftslehre
T-WIWI-103477	Seminar in Economics B (Master)	WT+ST	3 CP	Professorenschaft des Fachbereichs Volkswirtschaftslehre
T-WIWI-103478	Seminar in Economics A (Master)	WT+ST	3 CP	Professorenschaft des Fachbereichs Volkswirtschaftslehre
T-WIWI-103479	Seminar in Informatics A (Master)	WT+ST	3 CP	Professorenschaft des Instituts AIFB
T-WIWI-103480	Seminar in Informatics B (Master)	WT+ST	3 CP	Professorenschaft des Instituts AIFB
T-WIWI-103481	Seminar in Operations Research A (Master)	WT+ST	3 CP	Nickel, Rebennack, Stein
T-WIWI-103482	Seminar in Operations Research B (Master)	WT+ST	3 CP	Nickel, Rebennack, Stein
T-WIWI-103483	Seminar in Statistics A (Master)	WT+ST	3 CP	Grothe, Schienle
T-WIWI-103484	Seminar in Statistics B (Master)	WT+ST	3 CP	Grothe, Schienle
Engineering and Law Seminars (Elective: at most 1 item)				
T-WIWI-108763	Seminar in Engineering Science Master (approval)	WT+ST	3 CP	Fachvertreter ingenieurwissenschaftlicher Fakultäten

Assessment
The module examination is based on proof of two seminars and an approved profile definition worth one credit point. The individual performance assessments are described for each course in this module.

The overall grade for the module is calculated from the LP-weighted grades of the two seminars and cut off after the first decimal place.

Prerequisites
Course-specific prerequisites must be observed.

Students must complete two seminars from the module's course list, each worth 4 CP, offered by the KIT Department of Business and Economics. Both seminars may be taken at the same institute.

One of the two seminars may be replaced by a seminar at a KIT Department of Engineering, provided the content aligns with one of the engineering modules already completed. Coherence of content is deemed given if the seminar and the module are taken at the same institute. Otherwise, the module coordinator of a completed engineering module must certify that the seminar is compatible with the module's content. Seminars offered by the wbk (Institute of Production Science) and the IFL (Institute for Material Handling and Logistics) are exempt from this certification requirement.

The seminar must meet the performance standards of the KIT Department of Business and Economics (regular and active participation, a seminar paper on a specific aspect of the topic, a presentation, and a total workload of approx. 120 hours). Engineering seminars are generally subject to approval and must be applied for at the Examination Office of the KIT Department of Business and Economics. Applications must be submitted using the corresponding 'Engineering Seminar' form available on the Department's download page (wbk and IFL seminars are exempt from this approval process).

Competence Goal

- The students are in a position to independently handle current, research-based tasks according to scientific criteria.
- They are able to research, analyze, abstract and critically review the information.
- They can draw conclusions using their interdisciplinary knowledge from less structured information and selectively develop current research results.
- They can logically and systematically present the obtained results both orally and in written form in accordance with scientific guidelines (structuring, technical terminology, referencing). They can argue and defend the results professionally in a discussion.
- Students are familiar with the DFG´s Code of Conduct "Guidelines for Safeguarding Good Research Practice" and base their scientific work on it.

Content
The skills acquired in the seminar module are specifically designed to prepare students for their Master's thesis. Under the guidance of the respective examiners, students practice independent academic research by writing seminar papers and presenting their findings.

In addition to scientific research techniques, key qualifications (Schlüsselqualifikationen, SQ) are taught in an integrative manner through seminar participation. A detailed description of these integrated key qualifications can be found in the 'Key Qualifications' section of the module handbook.

The DFG Code 'Guidelines for Safeguarding Good Research Practice' is taught as part of the KIT Library's online course 'Good Scientific Practice,' which can be completed through self-study.

Additional Information
The seminar titles listed in the module handbook serve as placeholders. Within the Master's seminar module, it is possible to complete two seminars in the same subject area (e.g., 'Informatics'). For technical reasons, these placeholders are duplicated and offered in two versions: 'Seminar Informatics A' and 'Seminar Informatics B'. When registering online for your first seminar in a specific subject, please always select the 'A' version.

The seminars currently offered each semester are announced in the course catalog and on the respective institute websites. As a rule, seminar topics are published toward the end of the preceding semester. When planning your seminar module, please note that registration for certain seminars is required before the current semester ends. Available seminars and places offered by the KIT Department of Business and Economics are published via the WiWi portal (https://portal.wiwi.kit.edu) approximately at the same time as the course catalog.

Allocation of Seminar Places: A variety of individual seminars are offered across all subjects each semester. While demand for specific seminars or topics may exceed capacity, sufficient places are provided overall to meet total demand. Therefore, students typically apply for several seminars simultaneously.

Students with significant academic progress (based on earned credit points) who encounter difficulties securing a seminar place should contact the Examination Office with a note of urgency, ideally before the upcoming allocation round begins. The Examination Office provides support in obtaining a suitable placement, taking the student's preferred subject area—such as Business Administration, Economics, Operations Research, Informatics, Statistics, or Engineering—into account.

Selection criteria for the allocation of places and topics are announced on the respective course websites. Generally, places are allocated via the WiWi portal. Thematic assignment is based on student preferences and suitability, including professional experience, practical knowledge in the subject area, and, where applicable, language proficiency.

Workload
The total workload for this module is approximately 270 hours.

5.164 Module: Seminar New Power Electronic Systems and Technologies [M-ETIT-107588]

Coordinators: Prof. Dr.-Ing. Marc Hiller
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Engineering and Informatics (Electrical Engineering and Information Technology)
Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each term	1 term	German/English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-114862	Seminar New Power Electronic Systems and Technologies	WT+ST	4 CP	Hiller

Assessment
The success control takes place in the form of an examination of another type. It consists of a 15-minute final presentation followed by a discussion and a 2- to 3-page written summary paper. The overall impression is evaluated.

The following aspects are assessed:

Presentation:

- Quality of the presentation (form and content)
- Presentation style (structure, style, content)
- Behavior during the discussion

Paper with a summary of the main content:

- Format, spelling, linguistic style (academic/factual)
- Content, (graphical) presentation of the researched findings
- Relevance and quality of the references used, citation style

Prerequisites
none

Competence Goal
During the seminar, students learn to independently familiarize themselves with previously unknown technologies and applications in the field of power electronics. They will be able to independently research and select relevant scientific literature, which is usually in English, and interpret its content. In addition, students will be able to present the topic in a scientific lecture to an expert audience and summarize the essential content in a compact article in the style of a journal publication.

By critically evaluating and analyzing current scientific literature, students develop a deeper understanding of current development trends in technologies and applications in the field of power electronics. In addition, they develop skills in scientific writing and in presenting new scientific topics. These qualifications represent key competencies for the students' future academic and professional careers.

Content

The seminar "New Power Electronics Systems and Technologies" is aimed at students who are interested in the latest technologies and applications in the field of power electronics.

Participants in the seminar are expected to conduct independent studies on current topics in science and research. In addition to research, the selection of the most relevant findings and their presentation to an expert audience are key components of the seminar.

The specific topics are redefined each semester by the doctoral students and professors at ETI and are based on current developments in research and science. The topics cover a wide range and are presented in detail at the beginning of the seminar. Each participant can usually choose their "favorite topic."

In past seminars, topics from the following areas were offered, for example:

- Electrical energy storage
- Photovoltaic systems
- Power converter systems in energy transmission and distribution
- Control of power electronic systems
- Drive systems for cars and trucks
- Structure and properties of modern power semiconductors

If there is particular interest, topics suggested by the participating students can also be considered.

The seminar is aimed at master's students in electrical engineering, mechatronics, and biomedical engineering. During the seminar, students have the opportunity to familiarize themselves with current research results in the field of their chosen topic under the guidance of their supervisor and to discuss the topic during a presentation in the seminar.

Students should regularly attend group and individual meetings throughout the seminar. They will independently present the research topic in a 15-minute scientific lecture and prepare a short scientific summary (2-3 pages) based on the scientific literature on which the presentation is based.

The seminar includes participation in the following group and individual sessions:

- Introduction and presentation of possible seminar topics (all seminar participants)
- Discussion and assignment of topics (all seminar participants)
- Tips for literature research (all seminar participants)
- Helpful presentation techniques (all seminar participants)
- Presentation of the structure and content of the presentations (individually with the seminar topic supervisors)
- Discussion and feedback on the final presentation (individually with the supervisors of the seminar topic)
- Rehearsal presentation (individually with the supervisors of the seminar topic)
- Plenary seminar presentations (all seminar participants)

Module Grade Calculation

The module grade is based on the assessment of the final oral presentation and the written summary paper (each according to the criteria mentioned above). Further details will be provided at the beginning of the course.

If the candidate's grade falls between two grades, their participation in the preparatory meetings will be the deciding factor.

Additional Information

For capacity reasons, the number of participants in the seminar is limited to approximately eight students. If necessary, a selection procedure will be carried out. Places will be allocated taking into account the students' academic progress. Details will be announced on the seminar's website.

Presentations and summaries may be given/written in English or German.

Workload

The workload of approx. 110 hours (equivalent to 4 LP) includes:

- Participation in organizational and preparatory meetings: 14 hours
- Literature study on the chosen seminar topic: 40 hours
- Preparation of the seminar presentation, including rehearsal: 32 hours
- Preparation of the final documentation: 16 hours
- Participation in the final presentations of the seminar: 8 hours

5.165 Module: Sensor Technology I (WI4INGETIT3) [M-ETIT-101158]

Coordinators: TT-Prof. Dr. Johanna Schröder
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each summer term	1 term	German/English	4	5

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-101911	Sensors	ST	3 CP	Schröder
Compulsory Elective (Elective: at most 2 items as well as at least 6 credits)				
T-MACH-101910	Microactuators	ST	4 CP	Kohl
T-MACH-102164	Practical Training in Basics of Microsystem Technology	WT+ST	3 CP	Last
T-MACH-114100	Introduction to Microsystem Technology I	WT	4 CP	Badilita, Korvink
T-MACH-114101	Introduction to Microsystem Technology II	ST	4 CP	Badilita, Korvink

Assessment
The assessment is carried out as partial exams of the single courses of this module, whose sum of credits must meet the minimum requirement of credits of this module. The assessment procedures are described for each course of the module separately.

Prerequisites
none

Competence Goal
Mandatory

T-ETIT-101911 – Sensoren:
Students know the basic properties and functions of the most important industrially and commercially used sensors (temperature, pressure, gas, etc.). They have a basic understanding of the physical and chemical processes of signal formation and can use this knowledge to analyze problems, design and apply sensors and transfer it to other areas of their studies. They are able to communicate with specialists from related disciplines in the field of sensor technology and can actively contribute to the opinion-forming process in society with regard to scientific and technical issues.

Electives

T-MACH-101910 – Mikroaktorik:
Students can describe the actuator principles and their advantages and disadvantages and describe important manufacturing processes. They are able to construct microactuators and explain their function. They know important parameters (time constants, forces, actuating travel, etc.) and can calculate them and are able to create layouts based on requirement profiles.

T-MACH-102164 – Praktikum zu Grundlagen der Mikrosystemtechnik:
Students are familiar with modern sensory and microstructural technologies and can apply these in ten subject areas. They are able to set up and carry out experiments on the topics.

T-MACH-114100 – Introduction to Microsystem Technology I:
Students are familiar with the basic concepts of microsystems technology. Based on manufacturing processes from microelectronics, students are able to describe the most important technologies and materials used to manufacture microstructures and components. In particular, they will be able to describe silicon micromachining techniques and illustrate them using practical examples from microcomponents and microsystems. Students will have an understanding of the production of small, complex components that have a wide range of applications in sensor technology.

T-MACH-114101 – Introduction to Microsystem Technology II:
Students have basic knowledge of microsystems technology. They are familiar with various manufacturing methods, including lithographic processes, LIGA technology, micromechanical processing and structuring using lasers. They have a deeper understanding of these technologies through practical examples. In addition, students are able to describe the structure and connection technology of microcomponents and complete microsystems and are therefore able to understand and apply basic processes and techniques of microsystems technology.

Content
Mandatory

T-ETIT-101911 – Sensoren:
The lecture provides the most important basics for understanding commercially available sensors. In addition to the sensor effects, material aspects and the technical realization in components as well as the application of sensors in electrical circuits and systems are discussed. The following are covered: mechanical sensors, temperature sensors, optical sensors, magnetic sensors, ultrasonic sensors, gas sensors, chemical sensors.

Electives

T-MACH-101910 – Mikroaktorik:
The lecture includes amongst others the following topics:

- Basic knowledge in the material science of the actuation principles
- Layout and design optimization
- Fabrication technologies
- Selected developments
- Applications
- Microelectromechnical systems: linear actuators, microrelais, micromotors
- Medical technology and life sciences: Microvalves, micropumps, microfluidic systems
- Microrobotics: Microgrippers, polymer actuators (smart muscle)
- Information technology: Optical switches, mirror systems, read/write heads

T-MACH-102164 – Praktikum zu Grundlagen der Mikrosystemtechnik:

The practical course offers experiments on ten topics:

1. x-ray optics
2. UVL + SEM
3. mixer component
4. atomic force microscopy
5. 3D printing
6. light scattering on chrome masks
7. molding
8. SAW biosensor technology
9. nano3D printer - material transfer of ultra-thin layers
10. electrospinning

Each student can only take part in four experiments during the practical week.
The experiments are carried out at the real workstations at the IMT and supervised by IMT staff.

T-MACH-114100 – Introduction to Microsystem Technology I:
- Introduction in Nano- and Microtechnologies
- Silicon and processes for fabricating microelectronics circuits
- Basic physics background and crystal structure
- Materials for micromachining
- Processing technologies for microfabrication
- Silicon micromachining
- Examples

T-MACH-114101 – Introduction to Microsystem Technology II:
- Introduction in Nano- and Microtechnologies
- Lithography
- LIGA-technique
- Mechanical microfabrication
- Patterning with lasers
- Assembly and packaging
- Microsystems

Module Grade Calculation
The module grade is calculated from the credits-weighted grades of the module components and cut off after the first decimal place.

Workload
The workload includes:

1. attendance time in lectures, exercises: 15*6 h = 90 h

2. preparation/follow-up of the same: 15*6 h = 90 h

3. exam preparation: 90 h

Total workload for 9 credit points: approx. 270 hours

The exact distribution is based on the credit points of the courses of the module and may vary from the above description.

M 5.166 Module: Service Analytics (WW4BWLKSR1) [M-WIWI-101506]

Coordinators: Prof. Dr. Gerhard Satzger
Prof. Dr. Christof Weinhardt

Organisation: KIT Department of Business and Economics

Part of: Business and Economics
Electives (Business Administration)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	2 terms	English	4	11

ID	Name	Term offered	Credits	Lecturers
Compulsory Elective Courses (Elective: 9 credits)				
T-WIWI-108715	Artificial Intelligence in Service Systems	WT	4,5 CP	Satzger
T-WIWI-114209	Artificial Intelligence in Service Systems II: Generative AI Applications & Adoption	ST	4,5 CP	Satzger
T-WIWI-105777	Business Intelligence Systems	WT	4,5 CP	Mädche
T-WIWI-112152	Practical Seminar: Artificial Intelligence in Service Systems	Irreg.	4,5 CP	Satzger
T-WIWI-113725	Special Topics in Information Systems	WT+ST	4,5 CP	Weinhardt

Assessment
The assessment is carried out as partial exams (according to Section 4 (2), 1-3 SPO) of the core course and further single courses of this module, whose sum of credits must meet the minimum requirement of credits of this module. The assessment procedures are described for each course of the module separately.

The overall grade of the module is the average of the grades for each course weighted by the credits and truncated after the first decimal.

Prerequisites
None

Competence Goal
Students

- knows the theoretical bases and the key components of Business Intelligence systems,
- acquires the basic skills to make use of business intelligence and analytics software in the service context
- are introduced into various application scenarios of analytics in the service context
- are able to distinguish different analytics methods and apply them in context
- learn how to apply analytics software in the service context
- are trained for the structured compilation and solution of practice relevant problems with the help of commercial business intelligence software packages as well as analytics methods and tools

Content
The importance of services in modern economies is most evident – nearly 70% of gross value added are achieved in the tertiary sector and a growing number of industrial enterprises add customer specific services to their material goods or transform their business models fundamentally. The growing availability of data “Big Data” and their intelligent processing by applying analytic methods and business intelligence systems plays a key role.

It is the goal of the module to give students a comprehensive overview on the subject Business Intelligence & Analytics focusing on service issues. Various scenarios illustrate how the methods and systems introduced help to improve existing services or create innovative data-based services.

Workload
Total workload for 9 credit points: approx. 270 hours.

Attendance time: 90 hours

Preparation and follow-up: 100 hours

Exam and exam preparation: 80 hours

M 5.167 Module: Service Design Thinking (WW4BWLKSR2) [M-WIWI-101503]

Coordinators: Prof. Dr. Gerhard Satzger
Prof. Dr. Orestis Terzidis

Organisation: KIT Department of Business and Economics

Part of: Business and Economics
Electives (Business Administration)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each winter term	2 terms	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-WIWI-102849	Service Design Thinking	Irreg.	9 CP	Satzger, Terzidis

Assessment
The assessment is carried out as a general exam (according to Section 4(2), 3 of the examination regulation). The overall grade of the module is the grade of the examination (according to Section 4(2), 3 of the examination regulation).

Prerequisites
None

Competence Goal
Students

- Gain a comprehensive understanding of the globally recognized innovation approach “Design Thinking” as introduced and promoted by the Stanford University
- Apply the learned approach in the context of a real innovation project provided by a partner organization
- Conceive new, creative solutions through extensive need finding of relevant service users
  Develop prototypes early and independently, test them and improve them iteratively to solve the challenge provided by the partner organization
- Communicate, present and network in interdisciplinary and international environments.

Content
Course phases (roughly 4 weeks each):

Design Space Exploration:

- Exploring the problem space by questioning the given innovation challenge from practice.
- Familiarization with the topic area of the respective challenge.
- Gathering first impressions of the requirements and needs of people related to the problem.

Critical Function Prototype:

- Building an intensive understanding of the needs of the target group of the respective challenge.
- Deriving critical functions from the customer's perspective that could help solve the overall problem.
- Building prototypes for the critical functions and testing them in real customer situations.

Dark Horse Prototype:

- Reversal of assumptions and experiences made so far. The goal is to develop radically new and unconventional ideas.
- Implementation of the ideas into simple prototypes and subsequent testing.

Funky Prototype:

- Integration of the individual successfully tested functions from the critical function and dark horse phase into solution concepts. These are also tested and further developed.

Functional Prototype:

- Selection of successful funky prototypes and development of these towards high-resolution prototypes. The final solution approach for the project is written down in detail and feedback is obtained.

Final Prototype:

- Implementing the final prototype and presenting it to the practical partner as well as the SUGAR Network.

Additional Information
Due to practical project work as a component of the program, access is limited. The module (as well as the module component) spans two semesters. It starts in September every year and runs until end of June in the subsequent year. Entering the program is only possible at its beginning - after prior application in May/June. For more information on the application process and the program itself are provided in the module component description and the program's website (https://sdtkarlsruhe.de/). Furthermore, the lecturers provide an information event for applicants every year in May.

Workload
The workload for this module is approx. 2 days per week over a period of 9 months. The workload for this practical module is therefore comparatively high. The reason for this is that the participants work in international teams with students from other universities and partner organizations and solve real innovation challenges.

The workload of approx. 270 hours is spread over approx. 105 hours (3.5 CP) in the first semester and 165 hours (5.5 CP) in the second semester.

Recommendations
This course is held in English – proficiency in writing and communication is required.

Our past students recommend to take this course at the beginning of the masters program.

M 5.168 Module: Service Economics and Management (WW4BWLKSR3) [M-WIWI-102754]

Coordinators: Prof. Dr. Gerhard Satzger
Prof. Dr. Christof Weinhardt
Organisation: KIT Department of Business and Economics
Part of: Business and Economics
Electives (Business Administration)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	2 terms	English	4	6

ID	Name	Term offered	Credits	Lecturers
Compulsory Elective Courses (Elective: 9 credits)				
T-WIWI-112757	Digital Services: Innovation & Business Models	ST	4,5 CP	Satzger
T-WIWI-112823	Platform & Market Engineering: Commerce, Media, and Digital Democracy	ST	4,5 CP	Weinhardt

Assessment
The assessment is carried out as partial exams (according to Section 4 (2), 1-3 SPO), whose sum of credits must meet the minimum requirement of credits of this module. The assessment procedures are described for each course of the module separately.

The overall grade of the module is the average of the grades for each course weighted by the credits and truncated after the first decimal.

Prerequisites
None

Competence Goal
Students

- understand the scientific basics of the management of digital services and corresponding systems
- gain a comprehensive insight in the importance and the most important features of information systems as an central component of the digitalization of business processes, products and services
- know the most relevant concepts and theories to shape the digital transformation process of service systems successfully
- understand the OR methods in the sector of service management and apply them adequately
- are able to use large amounts of available data systematically for the planning, operation and improvement of complex service offers and to design and control information systems
- are able to develop market-oriented coordination mechanisms and apply service systems.

Content
This module provides the foundation for the management of digital services and corresponding systems. The courses in this module cover the major concepts for a successful management of service systems and their digital transformation. Current examples from the research and practice enhance the relevance of the discussed topics.

Additional Information
From summer semester 2023, the course Service Innovation will be offered with a revised course concept and content. The focus will be on the closer integration of the topics of service innovation and digitalization. Current foundational content (e.g., on service innovation challenges or human-centered innovation methods) will remain. New content will cover topics such as digital platforms and ecosystems, IoT and smart service innovation, and business models.

Workload
Total workload for 9 credit points: approx. 270 hours. The allocation is based on the credit points of the courses in the module.

Recommendations
None

M 5.169 Module: Service Innovation, Design & Engineering (WW4BWLKSR5) [M-WIWI-102806]

Coordinators: Prof. Dr. Alexander Mädche
Prof. Dr. Gerhard Satzger

Organisation: KIT Department of Business and Economics

Part of: Business and Economics
Electives (Business Administration)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	2 terms	English	4	6

ID	Name	Term offered	Credits	Lecturers
Compulsory Elective Courses (Elective: 9 credits)				
T-WIWI-112757	Digital Services: Innovation & Business Models	ST	4,5 CP	Satzger
T-WIWI-113460	Engineering Interactive Systems: AI & Wearables	WT	4,5 CP	Mädche
T-WIWI-114210	(Gen)AI-based Automation in Organizations	ST	4,5 CP	Mädche
T-WIWI-113459	Practical Seminar: Human-Centered Systems	WT+ST	4,5 CP	Mädche
T-WIWI-110887	Practical Seminar: Service Innovation	Irreg.	4,5 CP	Satzger

Assessment
The assessment is carried out as partial exams (according to Section 4 (2), 1-3 SPO), whose sum of credits must meet the minimum requirement of credits of this module. The assessment procedures are described for each course of the module separately.

The overall grade of the module is the average of the grades for each course weighted by the credits and truncated after the first decimal.

Prerequisites
Dependencies between courses:

The course Practical Seminar Service Innovation cannot be applied in combination with the course Practical Seminar Digital Service Design.

Competence Goal
Students

- know about the challenges, concepts, methods and tools of service innovation management and are able to use them successfully.
- have a profound comprehension of the development and design of innovative services and are able to apply suitable methods and tools on concrete and specific issues.
- are able to embed the concepts of innovation management, development and design of services into organisations
- are aware of the strategic importance of services, are able to present value creation in the context of services systems and to strategically exploit the possibilities of their digital transformation
- elaborate concrete and problem-solving solutions for practical tasks in teams.

Content
This module is designed to constitute the basis for the development of successful ICT supported innovations thus including the methods and tools for innovation management, for the design and the development of digital services and the implementation of new business models. Current examples from science and practice enhance the relevance of the topics addressed.

Additional Information
From summer semester 2023, the course Service Innovation will be offered with a revised course concept and content. The focus will be on the closer integration of the topics of service innovation and digitalization. Current foundational content (e.g., on service innovation challenges or human-centered innovation methods) will remain. New content will cover topics such as digital platforms and ecosystems, IoT and smart service innovation, and business models.

Workload
Total workload for 9 credit points: approx. 270 hours. The allocation is based on the credit points of the courses in the module.

Recommendations

Attending the course Practical Seminar Service Innovation [2595477] is recommended in combination with the course Service Innovation [2595468].

Attending the course Practical Seminar Digital Service Design [new] is recommended in combination with the course Digital Service Design [new].

M 5.170 Module: Service Management (WW4BWLISM6) [M-WIWI-101448]

Coordinators: Prof. Dr. Gerhard Satzger
Prof. Dr. Christof Weinhardt

Organisation: KIT Department of Business and Economics

Part of: Business and Economics
Electives (Business Administration)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	1 term	English	4	12

ID	Name	Term offered	Credits	Lecturers
Compulsory Elective Courses (Elective: 9 credits)				
T-WIWI-108715	Artificial Intelligence in Service Systems	WT	4,5 CP	Satzger
T-WIWI-114209	Artificial Intelligence in Service Systems II: Generative AI Applications & Adoption	ST	4,5 CP	Satzger
T-WIWI-112757	Digital Services: Innovation & Business Models	ST	4,5 CP	Satzger

Assessment
The assessment is carried out as partial exams, whose sum of credits must meet the minimum requirement of credits of this module. The assessment procedures are described for each course of the module separately.

The overall grade of the module is the average of the grades for each course weighted by the credits and truncated after the first decimal.

Competence Goal
The students

- understand the basics of developing and managing IT-based services,
- understand and apply OR methods in service management,
- systematically use vast amounts of available data for planning, operation, personalization and improvement of complex service offerings, and
- understand and analyze innovation processes in corporations.

Content
The module service management addresses the basics of developing and managing IT-based services. The lectures contained in this module teach the basics of developing and managing IT-based services and the application of OR methods in the field of service management. Moreover, students learn to systematically analyze vast amounts of data for planning, operation and improvement for complex service offerings. These tools enhance operational and strategic decision support and help to analyze and understand the overall innovation processes in corporations. Current examples from research and industry demonstrate the relevance of the topics discussed in this module.

Additional Information
From summer semester 2023, the course Service Innovation will be offered with a revised course concept and content. The focus will be on the closer integration of the topics of service innovation and digitalization. Current foundational content (e.g., on service innovation challenges or human-centered innovation methods) will remain. New content will cover topics such as digital platforms and ecosystems, IoT and smart service innovation, and business models.

Workload
Total workload for 9 credit points: approx. 270 hours. The allocation is based on the credit points of the courses in the module. 120-135 hours for the courses with 4.5 credits, 135-150 hours for the courses with 5 credits and 150-180 hours for the courses with 6 credits.

The total number of hours per course results from the time required to attend the lectures and exercises, as well as the examination times and the time required to achieve the learning objectives of the module for an average student for an average performance.

Recommendations
None

M 5.171 Module: Service Operations (WW4BWLKSR4) [M-WIWI-102805]

Coordinators: Prof. Dr. Stefan Nickel
Organisation: KIT Department of Business and Economics
Part of: Electives (Operations Research)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	1 term	German/English	4	9

Election Notes
At least one of the fourcourses Operations Research in Supply Chain Management, Operations Research in Health Care Management, Practical seminar: Health Care Management orDiscrete-Event Simulation in Production and Logistics has to be assigned.
Students who choose the module in the field "compulsory elective modules" may select any two courses of the module.

ID	Name	Term offered	Credits	Lecturers
Compulsory Elective Courses (Elective: at most 2 items)				
T-WIWI-102718	Discrete-Event Simulation in Production and Logistics	ST	4,5 CP	Spieckermann
T-WIWI-102884	Operations Research in Health Care Management	WT+ST	4,5 CP	Graß
T-WIWI-102715	Operations Research in Supply Chain Management	Irreg.	4,5 CP	Nickel
T-WIWI-102716	Practical Seminar: Health Care Management (with Case Studies)	WT+ST	4,5 CP	Nickel
Supplementary Courses (Elective: at most 1 item)				
T-MACH-112213	Applied material flow simulation	WT	4,5 CP	Baumann
T-WIWI-114109	Service Operations and Cyber Security	ST	4,5 CP	Mohr

Assessment
The assessment is carried out as partial exams (according to Section 4 (2), 1-3 SPO), whose sum of credits must meet the minimum requirement of credits of this module. The assessment procedures are described for each course of the module separately. The overall grade of the module is the average of the grades for each course weighted by the credits and truncated after the first decimal.

Prerequisites
At least one of the four courses "Operations Research in Supply Chain Management", "Operations Research in Health Care Management", "Practical seminar: Health Care Management" or "Discrete-Event Simulation in Production and Logistics" has to be assigned.

Competence Goal
Students

- knows the theoretical bases and the key components of Business Intelligence systems,
- acquires the basic skills to make use of business intelligence and analytics software in the service context
- are introduced into various application scenarios of analytics in the service context
- are able to distinguish different analytics methods and apply them in context
- learn how to apply analytics software in the service context
- are trained for the structured compilation and solution of practice relevant problems with the help of commercial business intelligence software packages as well as analytics methods and tools

Content
The importance of services in modern economies is most evident – nearly 70% of gross value added are achieved in the tertiary sector and a growing number of industrial enterprises add customer specific services to their material goods or transform their business models fundamentally. The growing availability of data “Big Data” and their intelligent processing by applying analytic methods and business intelligence systems plays a key role.

It is the goal of the module to give students a comprehensive overview on the subject Business Intelligence & Analytics focusing on service issues. Various scenarios illustrate how the methods and systems introduced help to improve existing services or create innovative data-based services.

Workload
Total workload for 9 credit points: approx. 270 hours. The allocation is based on the credit points of the courses in the module.

Recommendations
The course Practical Seminar Health Care should be combined with the course OR in Health Care Management.

M 5.172 Module: Single-Cell Technologies [M-CIWVT-106564]

Organisation: KIT Department of Chemical and Process Engineering

Part of: Engineering and Informatics (Chemical and Process Engineering)
Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each winter term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-CIWVT-113231	Single-Cell Technologies		4 CP	Grünberger

Assessment
The learning control is an oral examination lasting approx. 30 minutes.

Prerequisites
None

Competence Goal
Upon completion of the course, the students are able to:

- Know the fields and interdisciplinary nature of single-cell technologies
- Know basic methods in the field of single-cell technologies
- Are able to evaluate single-cell technologie
- Are able to choose single-cell platforms for specific biological questions
- Are aware of the complexity of the development of single-cell technologies

Content
While cell populations have historically been viewed as homogeneously behaving individuals, new research shows that cell-to-cell heterogeneity exists in all scales of biological systems. While most measurements are based on averages, individual cells can show dramatic differences in their properties such as growth, division and metabolic activity. Single-cell technologies have revolutionized our ability to delve into the intricacies of biological systems. By allowing the analysis of individual cells, these cutting-edge techniques provide insights into cellular heterogeneity, rare cell populations, and dynamic processes. Single-cell technologies range from single-cell microscopy, single-cell omics to single-cell cultivation, which all can be used to uncover hidden layers of complexity of a variety of cell types. These technologies have emerged in the last years and show a transformative, maybe revolutionizing, potential in many fields of basic and applied research of various scientific disciplines. This ranges from microbiology, biomedical research, drug discovery, biotechnology and bioprocess engineering.

The "Single-cell technologies" lecture aims to give an introduction and overview into single-cell technologies and provide students with a comprehensive understanding of the fundamental principles and practical applications of single-cell research. After a short introduction into the field, students will explore various single-cell technologies. Focus will be given on emerging field of microfluidic single-cell cultivation methods and their application. The characteristic features and functionality of selected systems are explained using current examples from science and research. Possibilities for applications in biotechnology and microbiology are discussed. The last part of the lecture provides an insight into single-cell data analysis and future challenges within the field. The course emphasizes the importance of uncovering cellular heterogeneity, and students will discover the role of these technologies in microbiology and biotechnology. They will stay updated on emerging trends and emerging application of this technically complex, but fast developing field. The interdisciplinary nature of single-cell technologies will be emphasized, fostering effective collaboration across fields. State of the art knowledge will be supported by insights into emerging fields and topics within the field. Upon completion, students will be well-prepared to contribute to cutting-edge research and innovations of single-cell technologies. The interdisciplinary and application-oriented lecture is aimed at technically interested students of molecular biotechnology, microbiology, biochemistry, bioprocess engineering, chemical engineering as well as all interested students of life sciences, chemistry, and physics.

Module Grade Calculation
The module grade is the grade of the oral exam.

Workload

- Attendance time: 30 hrs
- Self-study: 50 hrs
- Exam preparation: 40 hrs

Literature
No specific textbook is recommended.

5.173 Module: Single-Photon Detectors [M-ETIT-101971]

Coordinators: Dr. Konstantin Ilin
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Engineering and Informatics (Electrical Engineering and Information Technology)
Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each winter term	1 term	English	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-108390	Single-Photon Detectors	WT	4 CP	Ilin

Assessment
Type of Examination: Oral exam
Duration of Examination: approx. 20 minutes

Prerequisites
none

Competence Goal
After completing the module, students will get basic knowledges on various physical mechanisms underlying optical response of the currently available detectors with the ultimate sensitivity – the single-photon detectors (SPDs) – thereby will be able to explain their functionality in details. The grasp of these knowledges enables students to critically analyze advantages and limitations of different types of SPDs and to make a decision on development of the detection system for particular applications.

Content
The students will get an overview of the modern types of single-photon detectors already widely used in applications and currently developing as well. Basics of the response mechanisms of the detectors and particular areas of their application will be considered as well as the main directions of development and optimization of new types of SPDs and detection systems. In particular the following topics will be addressed:

- Applications of single-photon detectors (SPD)
- Detection system and light-matter interaction
- Basic characteristics of SPDs and experimental methods of their determination
- Photoelectric effect: photomultiplier tubes (PMT); microchannel plate (MCP)
- Semiconducting detectors: photoresistor, PIN photodiode, avalanche photodiode (APD), single-photon avalanche diode (SPAD), visible light photon counter (VLPC), quantum dot field effect transistor (QD-FET)
- Superconducting detectors: transition edge sensor (TES), superconducting tunnel junction (STJ), superconducting nanowire single-photon detector (SNSPD)
- Hybrid detection system

Module Grade Calculation
The module grade is the grade of the oral exam.

Workload

1. Lecture presence time in winter semester– 18 h
2. Exercises presence time – 9 h
3. Pre- /Post-preparation on lectures/exercises- 36 h
4. Preparation to and examination – 57 h

M 5.174 Module: Social Sciences [M-WIWI-104906]

Coordinators: Professorenschaft des Fachbereichs Geistes- und Sozialwissenschaften
Organisation: KIT Department of Business and Economics
Part of: Additional Examinations

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	pass/fail	Each term	1 term	German	4	1

ID	Name	Term offered	Credits	Lecturers
Elective Courses on Social Sciences (Elective: at most 18 credits)				
T-GEISTSOZ-104565	Computer Aided Data Analysis		0 CP	Nollmann
T-GEISTSOZ-109052	Application of Social Science Methods (WiWi)	WT+ST	9 CP	Nollmann
T-GEISTSOZ-109048	Social Science A (WiWi)	WT	3 CP	Nollmann
T-GEISTSOZ-109049	Social Science B (WiWi)	WT	3 CP	Nollmann
T-GEISTSOZ-109047	Analalysis of Social Structurs (WiWi)	WT	3 CP	Nollmann

M 5.175 Module: Sociology (WW4SOZ1) [M-GEISTSOZ-101169]

Coordinators: Prof. Dr. Gerd Nollmann
Organisation: KIT Department of Humanities and Social Sciences
Part of: Electives (Law or Sociology)

Credits	Grading	Recurrence	Duration	Level	Version
9 CP	graded	Each term	2 terms	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-GEISTSOZ-104565	Computer Aided Data Analysis		0 CP	Nollmann
T-GEISTSOZ-109052	Application of Social Science Methods (WiWi)	WT+ST	9 CP	Nollmann

Prerequisites
Students must pass three excersise sheets within the seminar "Computer based data analysis".

Competence Goal
The student

- Gains theoretical and methodical knowledge of social processes and structures,
- learns a script based data analysis tool (R, Stata, Python),
- gathers his/her data within an own framework and/or analyzes complex data,
- is able to present his/her work results in a precise and clear way.

Content
The Sociology module offers students the opportunity to learn a data analysis tool (R, Stata, Python) within the framework of a two-semester course and to independently transfer this tool to a content-related question. Both the tool and the contents are determined by the lecturers. The contents can refer to the analysis of large population surveys (SOEP, Microcensus, ALLBUS), to own experiments, to own field studies or to Big Data analyses.

Additional Information
Basic knowledge in multivariate regression and inference statistics is required.

5.176 Module: Specific Topics in Materials Science (WW4INGMB33) [M-MACH-101268]

Coordinators: Dr.-Ing. Wilfried Liebig
Organisation: KIT Department of Mechanical Engineering

Part of: Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	1 term	German	4	3

ID	Name	Term offered	Credits	Lecturers
Spezielle Werkstoffkunde (Elective: at least 9 credits)				
T-MACH-102141	Constitution and Properties of Wearresistant Materials	ST	4 CP	Ulrich
T-MACH-105150	Constitution and Properties of Protective Coatings	WT	4 CP	Ulrich
T-MACH-102099	Experimental Lab Class in Welding Technology, in Groups	WT	4 CP	Dietrich
T-MACH-105179	Functional Ceramics	WT	4 CP	Botros
T-MACH-105157	Foundry Technology	ST	4 CP	Günther, Klan
T-MACH-102111	Principles of Ceramic and Powder Metallurgy Processing	WT	4 CP	Schell
T-MACH-100287	Introduction to Ceramics	WT	6 CP	Pagnan Furlan
T-MACH-102182	Ceramic Processing Technology	ST	4 CP	Binder
T-MACH-105164	Laser in Automotive Engineering	ST	4 CP	Schneider
T-MACH-112763	Laser Material Processing	ST	4,5 CP	Schneider
T-MACH-102102	Physical Basics of Laser Technology	WT	5 CP	Schneider
T-MACH-102137	Polymer Engineering I	WT	4 CP	Liebig
T-MACH-102138	Polymer Engineering II	ST	4 CP	Liebig
T-MACH-102154	Laboratory Laser Materials Processing	WT+ST	4 CP	Schneider
T-MACH-105178	Practical Course Technical Ceramics	WT	4 CP	Ribas Gomes
T-MACH-102157	High Performance Powder Metallurgy Materials	ST	4 CP	Schell
T-MACH-105170	Welding Technology	WT	4 CP	Farajian
T-MACH-102170	Structural and Phase Analysis	WT	4 CP	Wagner
T-MACH-102103	Superhard Thin Film Materials	WT	4 CP	Ulrich
T-MACH-100531	Systematic Materials Selection	ST	4 CP	Dietrich, Schulze
T-MACH-102139	Failure of Structural Materials: Fatigue and Creep	WT	4 CP	Gruber, Gumbsch
T-MACH-102140	Failure of Structural Materials: Deformation and Fracture	WT	4 CP	Gumbsch, Weygand

Assessment
The assessment is carried out as partial exams of the single courses of this module, whose sum of credits must meet the minimum requirement of credits of this module. The assessment procedures are described for each course of the module seperately.

Prerequisites
None

Competence Goal
Students acquire special basic knowledge in selected areas of materials science and engineering and can apply them to technical problems. Specific teaching objectives are agreed with the respective coordinator of the course.

Content
See courses.

Module Grade Calculation
The overall grade of the module is the average of the grades for each course weighted by the credits and truncated after the first decimal.

Workload
The module requires an average workload of 270 hours.

Teaching and Learning Methods
Lecture, Tutorials.

M 5.177 Module: Statistics [M-WIWI-104902]

Coordinators: Professorenschaft des Fachbereichs Statistik
Organisation: KIT Department of Business and Economics
Part of: Additional Examinations

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	pass/fail	Each term	1 term	German	4	5

ID	Name	Term offered	Credits	Lecturers
Elective Courses on Statistics (Elective: at most 40 credits)				
T-WIWI-103063	Analysis of Multivariate Data	Irreg.	4,5 CP	Grothe
T-WIWI-111388	Applied Econometrics	WT	4,5 CP	Krüger
T-WIWI-114746	Selected Topics in Statistics and Analytics: Seminal Papers in Theory and Methods	Irreg.	4,5 CP	Grothe
T-WIWI-103064	Financial Econometrics	WT	4,5 CP	Schienle
T-WIWI-110939	Financial Econometrics II	ST	4,5 CP	Schienle
T-WIWI-111247	Mathematics for High Dimensional Statistics	Irreg.	4,5 CP	Grothe
T-WIWI-112153	Microeconometrics	Irreg.	4,5 CP	Krüger
T-WIWI-103126	Non- and Semiparametrics	Irreg.	4,5 CP	Schienle
T-WIWI-103127	Panel Data	ST	4,5 CP	Heller
T-WIWI-111941	Lab Practice Sessions in R for Statistics 1 and 2	WT	2 CP	Schienle
T-WIWI-103128	Portfolio and Asset Liability Management	ST	4,5 CP	Safarian
T-WIWI-110868	Predictive Modeling	Irreg.	4,5 CP	Krüger
T-WIWI-111387	Probabilistic Time Series Forecasting Challenge	Irreg.	4,5 CP	Krüger
T-WIWI-103489	Seminar in Statistics (Bachelor)	WT+ST	3 CP	Grothe, Schienle
T-WIWI-103483	Seminar in Statistics A (Master)	WT+ST	3 CP	Grothe, Schienle
T-WIWI-103484	Seminar in Statistics B (Master)	WT+ST	3 CP	Grothe, Schienle
T-WIWI-103123	Advanced Statistics	WT	4,5 CP	Grothe
T-WIWI-102737	Statistics I	ST	5 CP	Grothe, Schienle
T-WIWI-102738	Statistics II	WT	5 CP	Grothe, Schienle
T-WIWI-103065	Statistical Modeling of Generalized Regression Models	WT	4,5 CP	Heller
T-WIWI-103129	Stochastic Calculus and Finance	WT	4,5 CP	Safarian

M 5.178 Module: Stochastic Optimization [M-WIWI-103289]

Coordinators: Prof. Dr. Steffen Rebennack
Organisation: KIT Department of Business and Economics
Part of: Electives (Operations Research)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	1 term	German/English	4	11

ID	Name	Term offered	Credits	Lecturers
Compulsory Elective Courses (Elective: between 1 and 2 items)				
T-WIWI-106546	Introduction to Stochastic Optimization	ST	4,5 CP	Rebennack
T-WIWI-106548	Advanced Stochastic Optimization	Irreg.	4,5 CP	Rebennack
T-WIWI-106549	Large-scale Optimization	ST	4,5 CP	Rebennack
Supplementary Courses (Elective: at most 1 item)				
T-WIWI-102723	Graph Theory and Advanced Location Models	Irreg.	4,5 CP	Nickel
T-WIWI-102719	Mixed Integer Programming I	Irreg.	4,5 CP	Stein
T-WIWI-102720	Mixed Integer Programming II	Irreg.	4,5 CP	Stein
T-WIWI-111247	Mathematics for High Dimensional Statistics	Irreg.	4,5 CP	Grothe
T-WIWI-111587	Multicriteria Optimization	see notes	4,5 CP	Stein
T-WIWI-103124	Multivariate Statistical Methods	Irreg.	4,5 CP	Grothe
T-WIWI-102715	Operations Research in Supply Chain Management	Irreg.	4,5 CP	Nickel
T-WIWI-106545	Optimization under Uncertainty	WT	4,5 CP	Rebennack
T-WIWI-112109	Topics in Stochastic Optimization	WT	4,5 CP	Rebennack

Assessment
The assessment is carried out as partial exams (according to § 4(2), 1 of the examination regulation) of the single courses of this module, whose sum of credits must meet the minimum requirement of credits of this module.

The assessment procedures are described for each course of the module seperately.

The overall grade of the module is the average of the grades for each course weighted by the credits and truncated after the first decimal.

Prerequisites
At least one of the courses "Advanced Stochastic Optimization", "Large-scale Optimization" or "Introduction to Stochastic Optimization" has to be taken.

Competence Goal
The student

- names and describes basic notions for advanced stochastic optimization methods, in particular, ways to algorithmically exploit the special model structures,
- knows the indispensable methods and models for quantitative analysis of stochastic optimization problems,
- models and classifies stochastic optimization problems and chooses the appropriate solution methods to solve also challenging stochastic optimization problems independently and, if necessary, with the aid of a computer,
- validates, illustrates and interprets the obtained solutions,
- identifies drawbacks of the solution methods and, if necessary, is able to makes suggestions to adapt them to practical problems.

Content
The module focuses on the modeling as well as the imparting of theoretical principles and solution methods for optimization problems with special structure, which occur for example in the stochastic optimization.

Additional Information
The courses are sometimes offered irregularly. The curriculum, planned for three years in advance, can be found on the Internet at http://sop.ior.kit.edu/28.php.

Workload

The total workload for this module is approximately 270 hours (9 credits). The allocation is made according to the credit points of the courses of the module. The total number of hours per course is determined by the amount of time spent attending the lectures and exercises, as well as the exam times and the time required to achieve the module's learning objectives for an average student for an average performance.

Recommendations

It is recommended to listen to the lecture "Introduction to Stochastic Optimization" before the lecture "Advanced Stochastic Optimization" is visited.

5.179 Module: Strategic Design of Modern Production Systems [M-MACH-105455]

Coordinators: Prof. Dr.-Ing. Gisela Lanza
Organisation: KIT Department of Mechanical Engineering

Part of: Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	2 terms	German	4	5

ID	Name	Term offered	Credits	Lecturers
Strategic Design of Modern Production Systems (Elective: at least 9 credits)				
T-MACH-113647	Digitalization from Product Concept to Production	WT	4,5 CP	Wawerla
T-MACH-114800	Global Production	WT	5 CP	Lanza
T-MACH-105188	Integrative Strategies in Production and Development of High Performance Cars	ST	4 CP	Schlichtenmayer
T-MACH-105783	Learning Factory “Global Production“	WT	6 CP	Lanza
T-MACH-110318	Product- and Production-Concepts for Modern Automobiles	WT	4 CP	Kienzle, Steegmüller
T-MACH-102107	Quality Management	WT	4 CP	Lanza
T-MACH-114969	Strategic Decision-Making in Global Production Networks through Optimization and Simulation	ST	4,5 CP	Lanza, Martin

Assessment
Oral exams: duration approx. 5 min per credit point

Written exams: duration approx. 20 - 25 min per credit point

Amount, type and scope of the success control can vary according to the individually choice.

Prerequisites
none

Competence Goal
The students

- are able to apply the methods of the strategic design of modern production systems to new problems.
- are able to outline the underlying conditions and influencing factors of today's production and derive recommendations for action for an integrated strategy.
- are able to use their knowledge target-oriented to achieve an efficient production technology.
- are able to analyze new situations and choose methods of production science target-oriented based on the analyses, as well as justifying their selection.
- are able to describe and compare complex production processes exemplarily.

Content
Within this module the students will get to know and learn about methods for the strategic design of modern production systems. Manifold lectures and excursions as part of several lectures provide specific insights into the field of science.

Module Grade Calculation
The module grade is the LP-weighted average of the selected partial performances.

Workload
The work load is about 270 hours, corresponding to 9 credit points.

Teaching and Learning Methods
Lectures, seminars, workshops, excursions

M 5.180 Module: Student Innovation Lab (SIL) 1 [M-WIWI-105010]

Coordinators: Prof. Dr.-Ing. Sören Hohmann
Prof. Dr. Orestis Terzidis

Organisation: KIT Department of Business and Economics

Part of: Business and Economics
Electives (Business Administration)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each winter term	2 terms	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-WIWI-102864	Entrepreneurship	WT+ST	3 CP	Terzidis
T-WIWI-110166	SIL Entrepreneurship Project	WT	3 CP	Terzidis
T-WIWI-110287	SIL Entrepreneurship Emphasis	WT	3 CP	Terzidis

Assessment
The assessment of this module comprises a written examination of 60 minutes on the lecture contents of the lecture "Entrepreneurship" as well as two seminars. All examinations are graded. In both seminars the following tasks have to be fulfilled:

- "SIL Entrepreneurship Project": Presentation of the Value Profile & submission of the Business Plan
- "SIL Entrepreneurship Emphasis": Submission of price calculation, market potential analysis, competition analysis, financial plan, risk analysis, decision basis for funding and legal form

In addition, both courses provide for smaller, ungraded tasks to monitor progress.

The grade consists of 60 % of the written examination, 20 % of the examination "SIL Entrepreneurship Project" and 20 % of the examination "SIL Entrepreneurship Advanced".

Prerequisites
The module can only be completed together with the module M-WIWI-105011 "Student Innovation Lab 2".

An application is required for participation in the modules Student Innovation Lab (SIL) 1 and Student Innovation Lab (SIL) 2. Information about the application can be found at http://www.kit-student-innovation-lab.de/index.php/for-students/.

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Competence Goal
Personal competence

- Ability to reflect: Students can analyse certain elements of their actions in social interaction, critically assess them and develop alternative actions.
- Decision-making ability: Students can prepare a decision template in due time and provide the necessary factual arguments for alternative decisions and thus make timely decisions.
- Interdisciplinary cooperation: Students can recognise the limits of their domain competence and adjust to domains outside their subject area. The students are able to recognise missing (own) competences and to supplement them with complementary competences (of other persons in the team). Students can communicate their domain to others and develop a basic understanding of foreign domains.
- Value-based action: Students can use selected tools of psychology to recognize their own values. They can compare these values with other team members and critically reflect on whether their offers match these values.

Social competence

- Ability to cooperate: Students can analyse and assess their cooperation behaviour in the group.
  Communication skills: Students can present their information in a convincing, focused and target group-oriented way.
- Conflict ability: Students can recognise conflicts at an early stage, analyse conflict situations and name solution concepts.

Innovation and Entrepreneurship Competence

- Agile product development: Students can apply methods of agile product development such as Scrum.
  Methodical innovation finding: Students can perform user- or technology-centric innovation processes to develop sustainable value propositions for dedicated target groups (e.g. Design Thinking (DT), Technology Application Selection (TAS) process).
- Orientation on the management of new technology-based companies (NTBF): Students can name the central concepts of intellectual property and legal form. Students can name the most important tasks of entrepreneurial leadership. They can identify the relevant forms of business modelling and draw up a business plan. Students know the central approaches to building an organisation. Students will be able to identify the ownership structure of investments and how to develop a strategy. The students can name marketing concepts and create a business model.
- Create investment readiness: The students are able to create a rudimentary sales and cost planning. Furthermore, they are able to create a project plan for a company and derive an investment plan from it. The students can present the business plan to potential investors and develop investor empathy.
- Business model development competence: Students are able to use relevant tools for business modelling, e.g. the Business Model Canvas. Students can develop and evaluate alternative business models.
- Dealing with risks:Students can identify the basic risks in terms of desirability, technical feasibility and profitability. Students can use customer interaction methods to test desirability and willingness to pay. Students can draw up a rudimentary competitive analysis. Students can identify and identify risks and possible reactions.

Systemic technical competence

- Problem-solving competence: Students can analyse, assess and solve a technical problem in a structured way.
- Agile Methodology of System Development: Students can name the different system development processes and apply them appropriately.
- Validation in a volatile environment: Students can perform a technical and economic validation under volatile boundary conditions. For this purpose they can name the boundary conditions and interpret the results of the validation.
- Functional decomposition: Students are able to identify and interpret complex customer needs and derive functional requirements from them.
- Architecture development: The students are able to recognize correlations from the functional requirements and to derive a suitable system architecture.

Content
In a real laboratory, the module imparts professional, social and personal competences in entrepreneurship and in the respective technical domain. The aim is to prepare students in the best possible way for an entrepreneurial activity within or outside an established organisation. Our teaching is research-based and practice-oriented.

As an integral part, the lecture Entrepreneurship offers the theoretical basis and gives an overview of important theoretical concepts and empirical evidence. Current case studies and practical experiences of successful founders underline the theoretical and empirical contents. In order to operate a company on a long-term basis, important specialist knowledge is also of decisive importance. The content of the lecture therefore includes an introduction to Entrepreneurial Marketing and Leadership as well as the basics of Opportunity Recognition and Business Modeling. Customer-centric development methods, the lean start-up approach and methods for technology-oriented innovation are presented. Future founders must be able to develop and manage resources such as financial and human capital, infrastructure and intellectual property. Further aspects relate to the establishment of an organisation and the financing of one's own project.

The knowledge gained in the lecture Entrepreneurship will be applied in a practice-oriented seminar and in the labs. We use an action learning approach to complement the knowledge with skills and reflective attitudes. In five-member teams, the students experience their way from idea generation to the final investor pitch.

With regard to the labs, students have the following options:

- As an innovation platform, the Automation Innovation Lab offers flying robots for cooperative swarm solutions.
- The Industry 4.0 Innovation Lab enables innovations in the area of the next industrial revolution with mobile robot platforms.
- In the Internet of Things Innovation Lab, innovations in Assisted Living and Smart Housing are made possible by a comprehensive kit of mobile robots and sensors.
- The Computer Vision for Health Lab offers a selection of state-of-the-art imaging devices and powerful computing hardware for innovative image-based applications for medicine and healthcare.

The module also teaches methods of agile system development (Scrum) and the associated validation methods as well as methods of functional prototyping. Gate plans are applied within the module to determine project progress.

Methods for the reflection of individual & team work are treated and applied as well as group work specific knowledge about different roles of team members, solution of conflict situations and interdisciplinary teams are obtained.

Workload
Total effort for 9 credit points: approx. 270 hours. The distribution is based on the credit points of the courses of the module. The total number of hours per course results from the effort required to attend lectures and exercises, as well as the examination times and the time required to achieve the learning objectives of the module for an average student for an average performance.

M 5.181 Module: Superconducting Magnet Technology [M-ETIT-106684]

Coordinators: Prof. Dr. Tabea Arndt
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Engineering and Informatics (Electrical Engineering and Information Technology)
Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each summer term	1 term	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-113440	Superconducting Magnet Technology	ST	4 CP	Arndt

Assessment
The examination takes place in form of an oral exam (abt. 30 minutes).

Two timeslots (weeks) for examination dates will be announced (usually near end of lecture period & end of semester)

Prerequisites
none

Competence Goal

- The students have a solid knowledge of architecture and design aspects of applications in magnets, windings and coils in power engineering.
- For the most important magnet applications the students can apply the state of the art, choose between options and can reflect the main benefits.
- The students have a clear understanding of opportunities, benefits and limitations of superconducting windings and magnets.
- The students are able to perform the required design calculations and to solve fundamental design questions independently.

Content
As the materials become increasingly mature and powerful, using superconductivity in a variety of applications of electrical engineering is of rising interest and benefit, too. This module is focuses on Superconducting Magnet Technology:

Windings, coils and magnets may be used as a device by itself (providing high magnetic fields e.g. in MRI, NMR, accelerators, industry magnets, etc.) or as components for Power Systems.

This section will cover the following aspects:

- Unique selling points of superconducting windings.
- Basic approaches and tools to design superconducting windings.
- Discussion of winding architectures
- Criteria to design the appropriate operating temperatures, materials, conductors, cooling technology for the electromagnetic purpose.
- Limits and opportunities when preparing and operating superconducting windings.
- Measures for safe operation of superconducting magnets.
- High-Field Magnets
- Magnets for Fusion Technology
- 3D topologies (e.g. in dipole magnets or motors/ generators)
- New options potentially offered by widespread use of hydrogen.
- New winding topologies

In the exercises, selected magnets will be designed and calculated analytically and with some computational tools (e.g. dipole magnets and compact, cryogen free HTS-magnets)

The lecturer may change the details of the content without further notice. Materials will be offered on ILIAS.

Module Grade Calculation
The module grade is the grade of the oral exam.

Workload

1. attendance in lectures and exercises: 15*3 h = 45 h
2. preparation / follow-up: 15*3 h = 45 h
3. preparation of and attendance in examination: 30 h

A total of 120 h = 4 CR

Recommendations
Having knowledge in “Superconducting Materials” is beneficial, but not mandatory.

5.182 Module: Superconducting Materials [M-ETIT-105521]

Coordinators: Dr. Jens Hänisch
Prof. Dr. Bernhard Holzapfel

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: Engineering and Informatics (Electrical Engineering and Information Technology)
Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each winter term	2 terms	English	4	3

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-114729	Superconducting Materials Lecture	WT	3 CP	Hänisch, Holzapfel
T-ETIT-114730	Superconducting Materials Lab	ST	3 CP	Hänisch, Holzapfel

Assessment

The success contral takes place in two parts:

1. Part I (lecture): in the form of an oral examination lasting approx. 30 minutes.
2. Part II (lab): in the form of a written protocol (approx. 40 pages).

Prerequisites

none

Competence Goal

The students have a good knowledge and can describe and compare the properties of different superconducting materials including those currently employed in energy and electronic applications (niobium-based superconductors, oxocuprates, MgB2) and also promising recently discovered ones (pnictides), including their synthesis methods.

Students have a thorough understanding of the synthesis variations of superconducting materials in bulk, thin film and wire form as well as the close relationship between microstructural properties of superconductors and their current carrying capabilities. They are able to select the appropriate superconducting materials for the different application scenarios of superconductors.

The students are able to talk about topic-related aspects in English using the technical terminology of the field of study.

Content

This lecture series gives an overview on the basic properties of the known different classes of superconducting materials as well as their synthesis routes in bulk, thin film and wire form. Special emphasis s given to the close interaction of micro- and nanoscale microstructural properties and the superconducting electrical transport properties, which are the key to all large scale applications in power and magnet technology.
The lecture series will cover basic properties of superconductors, superconducting elements, classical metallic superconducting alloys and compounds, high temperature superconductors, Fe-based superconductors and some other “exotic” superconductors, synthesis of superconducting films and wires, superconducting critical currents and pinning in type II superconductors as well an overview on the most prominent applications of superconductors in electronics, medicine and power application.
The obligatory practical work covers a few experiments regarding the synthesis and characterization of superconducting materials.
The lecturer reserves the right to alter the contents of the course without prior notification.
Course material will be available on ILIAS. Up-to-date information will be available via the ITEP- homepage prior to the beginning of the semester.

Module Grade Calculation

The module grade is the average grade of the oral exam and the lab protocol.

Additional Information

WiSe: Superconducting Materials Part I - lecture

SoSe: Superconducting Materials Part II - lab

Workload
A workload of approx. 186h is required for the successful completion of the module. This is composed as follows:

- Attendance time in lectures: 30*2h = 60h
- Preparation and follow-up of lectures: 30*3h = 90h
- Preparation for the exam: 30h

Recommendations
It is recommanded to pass the lecture (part I) before the lab (part II).

M 5.183 Module: Superconducting Nanowire Detectors [M-ETIT-105609]

Coordinators: Dr. Konstantin Ilin
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: Engineering and Informatics (Electrical Engineering and Information Technology)
Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
4 CP	graded	Each summer term	1 term	English	4	2

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-111236	Superconducting Nanowire Detectors	ST	4 CP	Ilin

Assessment
Oral Exam (approx. 20 min.)

Prerequisites
Module "M-ETIT-102332 - Thin films: technology, physics and applications" + Thin Films: Technology, Physics and Applications II must not be started.

Competence Goal
Students should be able to discuss interplay between growth conditions of thin films, physical and geometrical properties of nanostructure made of these films, and performance and suitable areas of application of detectors of radiation based on interaction of these nanostructures with electromagnetic power. The knowledge obtained by students should provide a theoretical basis for the most important steps in development of thin film nanoelectronic devices.

Content
Students will get practically oriented information about technology of thin films including different methods of deposition of thin films like magnetron sputtering, thermal evaporation, pulsed laser ablation, about basics of vacuum technology, and about mechanisms of growth of thin films of different materials at different conditions.

Patterning methods (photo- and e-beam lithography, reactive ion etching, ion milling, and lift-off techniques) suitable for nanometer scale features of electronic devices will be considered in details.

Experimental methods of characterization of material, geometrical, optical, physical, superconducting, electron and phonon properties of thin films, nanostructures made of these films, and devices based on these nanostructures will be discussed.

Consideration of technology and physics of thin film structures will be done on example of development of three types of fast and sensitive detectors of electro-magnetic radiation for applications in optical and THz spectral ranges: superconducting nanowire single-photon detector, hot-electron bolometer, and YBCO ps-fast detector of synchrotron emission. Dependence of detector’s performance on their fabrication condition will be analyzed in frame of physical models which describe response mechanisms of the detectors to absorbed radiation.

Practical actualization of the knowledge is possible in frame of Praktikum Nanoelektronik (LVN 23669).

Module Grade Calculation
The module grade is the grade of the oral exam.

Workload
The workload in hours is broken down below:

1. Attendance time in lectures in the winter semester 15*3h = 45h
2. Preparation / follow-up of the same 15*3h = 45h
3. Exam preparation and attendance in the same 30h

Recommendations
Previous participation on Module "Physics, Technology and Applications of thin films " is recommanded.

M 5.184 Module: Superconductivity for Engineers [M-ETIT-105611]

Coordinators: Prof. Dr. Bernhard Holzapfel
Prof. Dr. Sebastian Kempf

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: Engineering and Informatics (Electrical Engineering and Information Technology)
Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
6 CP	graded	Each winter term	1 term	English	4	3

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-ETIT-111239	Superconductivity for Engineers	WT	6 CP	Holzapfel

Assessment
The assessment of success takes place in the form of a written examination lasting 120min.

Prerequisites
none

Competence Goal
Students know the physical fundamentals of superconductivity and can place various theoretical and practical aspects of superconductivity in the overall context. They understand the principles behind specific applications of superconductivity and are able to communicate with experts in the field.

Content
Superconductivity is one of the most fascinating and astonishing effects in solid state physics. It plays technologically an important role in many modern, scientific, medical and industrial applications. It establishes, for example, the basis of realizing high field electromagnets to be used in magnetic resonance imaging systems in healthcare or for guiding charged particle in modern particle accelerators such as the LHC. Moreover, it allows to build state-of-the-art energy systems as well as sensing devices such as magnetic field sensors or energy-dispersive single particle sensors. In addition, it is conceivable that superconductivity will be utilized in near future for energy and traffic engineering applications, e.g. for dissipationless power transmission over large distances or high-speed trains connecting major cities.

Within this context, this module gives a comprehensive introduction in the basics of superconductivity paving the way for the discussion of state-of-the-art applications of superconductivity. In particular, the module will cover the following topics:

- Historical remarks
- Overview of superconducting materials and applications of superconductivity
- Reminder of normal metals: free electron gas, Drude and Sommerfeld model, electrical and thermal properties, band structure
- Phenomena of superconductivity: zero electrical dc resistance, Meissner Ochsenfeld effect
- Thermodynamics and thermal properties of superconductors
- Phenomenological theories of superconductors: Two-fluid model, London theory, Pippard theory, Ginzburg-Landau theory
- Microscopic theory of conventional superconductors
- Type-I and type-II superconductivity
- Magnetic properties of type-I and type-II superconductors
- Irreversible magnetic properties, Bean model
- AC losses
- Electrical and thermal stabilization
- Energy gap and quasiparticle tunneling
- Unconventional superconductors
- High-frequency electrodynamics of superconductors
- Macroscopic quantum effects
- Overview of applications of superconductivity
- Superconducting magnets, especially for fusion
- Superconducting devices in power engineering
- Thermo-mechanical properties of relevant construction materials for superconducting applications

The tutorial is closely connected to the lecture and deepens important aspects from the field of superconductivity. Using exercises, important theories and effects as well as the realization of applications of superconductivity are discussed.

Module Grade Calculation
The module grade is the grade of the written examination.

Workload
A workload of approx. 180h is required for the successful completion of the module. This is composed as follows:

- Attendance time in lectures and exercises: 15*2*2h = 60h
- Preparation and follow-up of lectures: 22*3 h = 66h
- Preparation and follow-up of tutorials: 8*3h = 24h
- Preparation for the exam: 30h

Recommendations
None

5.185 Module: Supplementary Studies on Science, Technology and Society [M-FORUM-106753]

Coordinators: Dr. Christine Mielke
Christine Myglas

Organisation: General Studies. Forum Science and Society (FORUM)

Part of: Additional Examinations

Credits	Grading	Recurrence	Duration	Language	Level	Version
16 CP	graded	Each term	3 terms	German	4	1

Election Notes
Students have to self-record the achievements obtained in the Supplementary Studies on Science, Technology and Society in their study plan. FORUM (formerly ZAK) records the achievements as "non-assigned" under "ÜQ/SQ-Leistungen". Further instructions on self-recording of achievements can be found in the FAQ at https://campus.studium.kit.edu/ and on the FORUM homepage at https://www.forum.kit.edu/english/. The title of the examination and the amount of credits override the modules placeholders.

If you want to use FORUM achievements for both your Interdisciplinary Qualifications and for the Supplementary Studies, please record them in the Interdisciplinary Qualifications first. You can then get in contact with the FORUM study services (stg@forum.kit.edu) to also record them in your Supplementary Studies.

In the Advanced Unit you can choose examinations from three subject areas: "About Knowledge and Science", "Science in Society" and "Science in Social Debates". It is advised to complete courses from each of the three subject areas in the Advanced Unit.

To self-record achievements in the Advanced Unit, you have to select a free placeholder partial examination first. The placeholders' title do *not* affect which achievements the placeholder can be used for!

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-FORUM-113578	Lecture Series Supplementary Studies on Science, Technology and Society - Self Registration	ST	2 CP	Mielke, Myglas
T-FORUM-113579	Basic Seminar Supplementary Studies on Science, Technology and Society - Self Registration	ST	2 CP	Mielke, Myglas
Advanced Unit Supplementary Studies on Science, Technology and Society (Elective: at least 12 credits)				
T-FORUM-113580	Elective Specialization Supplementary Studies on Science, Technology and Society / About Knowledge and Science - Self-Registration	WT+ST	3 CP	Mielke, Myglas
T-FORUM-113581	Elective Specialization Supplementary Studies on Science, Technology and Society / Science in Society - Self-Registration	WT+ST	3 CP	Mielke, Myglas
T-FORUM-113582	Elective Specialization Supplementary Studies on Science, Technology and Society / Science in Public Debates - Self Registration	WT+ST	3 CP	Mielke, Myglas
Mandatory				
T-FORUM-113587	Registration for Certificate Issuance - Supplementary Studies on Science, Technology and Society	WT+ST	0 CP	Mielke, Myglas

Assessment
The monitoring is explained in the respective partial achievement.

They are composed of:
- Protocols
- Reflection reports
- Presentations
- Preparation of a project work
- An individual term paper
- An oral examination
- A written exam

Upon successful completion of the supplementary studies, graduates receive a graded report and a certificate issued by the FORUM.

Prerequisites

The course is offered during the course of study and does not have to be completed within a defined period. Enrollment is required for all assessments of the modules in the supplementary studies.

Participation in the supplementary studies is regulated by § 3 of the statutes. KIT students register for the supplementary studies by selecting this module in the student portal and booking a performance themselves. Registration for courses, assessments, and exams is regulated by § 8 of the statutes and is usually possible shortly before the start of the semester.

The course catalog, module description (module manual), statutes (study regulations), and guidelines for creating the various written performance requirements can be downloaded from the FORUM homepage at https://www.forum.kit.edu/begleitstudium-wtg.php.

Registration and exam modalities

PLEASE NOTE:

Registration on the FORUM, i.e. additionally via the module selection in the student portal, enables students to receive up-to-date information about courses or study modalities. In addition, registering on the FORUM ensures that you have proof of the credits you have earned. As it is currently (as of winter semester 24-25) not yet possible to continue additional credits acquired in the Bachelor's programme electronically in the Master's programme, we strongly advise you to digitally secure the credits you have earned by archiving the Bachelor's transcript of records yourself and by registering on FORUM.

In the event that a transcript of records of the Bachelor's certificate is no longer available - we can only assign the achievements of registered students and thus take them into account when issuing the certificate.

Competence Goal

Graduates of the Supplementary Studies on Science, Technology, and Society gain a solid foundation in understanding the interplay between science, the public, business, and politics. They develop practical skills essential for careers in media, political consulting, or research management. The program prepares them to foster innovation, influence social processes, and engage in dialogue with political and societal entities. Participants are introduced to interdisciplinary perspectives, encompassing social sciences and humanities, to enhance their understanding of science, technology, and society. The teaching objectives of this supplementary degree program include equipping participants with both subject-specific knowledge and insights from epistemological, economic, social, cultural, and psychological perspectives on scientific knowledge and its application in various sectors. Students are trained to critically assess and balance the implications of their actions at the intersection of science and society. This training prepares them for roles as students, researchers, future decision-makers, and active members of society.

Through the program, participants learn to contextualize in-depth content within broader frameworks, independently analyze and evaluate selected course materials, and communicate their findings effectively in both written and oral formats. Graduates are adept at analyzing social issues and problem areas, reflecting on them critically from a socially responsible and sustainable standpoint.

Content

The Supplementary Studies on Science, Technology and Society can be started in the 1st semester of the enrolled degree programme and is not limited in time. The wide range of courses offered by FORUM makes it possible to complete the program usually within three semesters. The supplementary studies comprises 16 or more credit points (LP). It consists of **two modules: the Basic Module (4 LP) and the Advanced Module (12 LP).**

The **basic Module** comprises the compulsory courses ‘Lecture Series Supplementary Studies on Science, Technology and Society’ and a basic seminar with a total of 4 LP.

The **Advanced Module** comprises courses totalling 12 LP in the humanities and social sciences subject areas ‘On Knowledge and Science’, ‘Science in Society’ and ‘Science in Public Debates’. The allocation of courses to the accompanying study programme can be found on the homepage https://www.forum.kit.edu/wtg-aktuelland in the printed FORUM course catalogue.

The 3 thematic subject areas:

Subject area 1: About Knowledge and Science

This is about the internal perspective of science: students explore the creation of knowledge, distinguishing between scientific and non-scientific statements (e.g., beliefs, pseudo-scientific claims, ideological statements), and examining the prerequisites, goals, and methods of knowledge generation. They investigate how researchers address their own biases, analyze the structure of scientific explanatory and forecasting models in various disciplines, and learn about the mechanisms of scientific quality assurance.

After completing courses in the "Knowledge and Science" area, students can critically reflect on the ideals and realities of contemporary science. They will be able to address questions such as: How robust is scientific knowledge? What are the capabilities and limitations of predictive models? How effective is quality assurance in science, and how can it be improved? What types of questions can science answer, and what questions remain beyond its scope?

Subject area 2: Science in Society

This focuses on the interactions between science and different areas of society, such as how scientific knowledge influences social decision-making and how social demands impact scientific research. Students learn about the specific functional logics of various societal sectors and, based on this understanding, estimate where conflicts of goals and actions might arise in transfer processes—for example, between science and business, science and politics, or science and journalism. Typical questions in this subject area include: How and under what conditions does an innovation emerge from a scientific discovery? How does scientific policy advice work? How do business and politics influence science, and when is this problematic? According to which criteria do journalists incorporate scientific findings into media reporting? Where does hostility towards science originate, and how can social trust in science be strengthened?

After completing courses in the "Sciene in Society" area, students can understand and assess the goals and constraints of actors in different societal sectors. This equips them to adopt various perspectives of communication and action partners in transfer processes and to act competently at various social interfaces with research in their professional lives.

Subject area 3: Science in Public Debates

The courses in this subject area provide insights into current debates on major social issues such as sustainability, digitalization, artificial intelligence, gender equality, social justice, and educational opportunities. Public debates on complex challenges are often polarized, leading to oversimplifications, defamation, or ideological thinking. This can hinder effective social solution-finding processes and alienate people from the political process and from science. Debates about sustainable development are particularly affected, as they involve a wide range of scientific and technological knowledge in both problem diagnosis (e.g., loss of biodiversity, climate change, resource consumption) and solution development (e.g., nature conservation, CCS, circular economy).

By attending courses in "Science in Public Debates," students are trained in an application-oriented way to engage in factual debates—exchanging arguments, addressing their own prejudices, and handling contradictory information. They learn that factual debates can often be conducted more deeply and with more nuance than is often seen in public discourse. This training enables them to handle specific factual issues in their professional lives independently of their own biases and to be open to differentiated, fact-rich arguments.

Supplementary credits:

Additional LP (supplementary work) totalling a maximum of 12 LP can also be acquired from the complementary study programme (see statutes for the WTG complementary study programme § 7). § 4 and § 5 of the statutes remain unaffected by this. These supplementary credits are not included in the overall grade of the accompanying study programme. At the request of the participant, the supplementary work will be included in the certificate of the accompanying study programme and marked as such. Supplementary coursework is listed with the grades provided for in § 9.

Module Grade Calculation

The overall grade of the supplementary course is calculated as a credit-weighted average of the grades that were achieved in the advanced module.

Additional Information
Climate change, biodiversity crisis, antibiotic resistance, artificial intelligence, carbon capture and storage, and gene editing are just a few areas where science and technology can diagnose and address numerous social and global challenges. The extent to which scientific findings are considered in politics and society depends on various factors, such as public understanding and trust, perceived opportunities and risks, and ethical, social, or legal considerations.
To enable students to use their expertise as future decision-makers in solving social and global challenges, we aim to equip them with the skills to navigate the interfaces between science, business, and politics competently and reflectively. In the Supplementary Studies, they acquire foundational knowledge about the interactions between science, technology, and society.

They learn:
- How reliable scientific knowledge is produced,
- how social expectations and demands influence scientific research, and
- how scientific knowledge is adopted, discussed, and utilized by society.

The program integrates essential insights from psychology, philosophy, economics, social sciences, and cultural studies into these topics. After completing the supplementary studies programme, students can place the content of their specialized studies within a broader social context. This prepares them, as future decision-makers, to navigate competently and reflectively at the intersections between science and various sectors of society, such as politics, business, or journalism, and to contribute effectively to innovation processes, public debates, or political decision-making.

Workload
The workload is made up of the number of hours of the individual modules:

- Basic Module approx. 120 hours
- Advanced Module approx. 360 hours

\> Total: approx. 480 hours

In the form of supplementary services, up to approximately 360 hours of work can be added.

Recommendations
It is recommended to complete the supplementary study program in three or more semesters, beginning with the lecture series on science, technology, and society in the summer semester. Alternatively, you can start with the basic seminar in the winter semester and then attend the lecture series in the summer semester.

Courses in the Advanced Module can be taken simultaneously. It is also advised to complete courses from each of the three subject areas in the advanced unit.

Teaching and Learning Methods
- Lectures
- Seminars/Project Seminars
- Workshops

M 5.186 Module: Sustainable Information Systems [M-WIWI-107649]

Coordinators: TT-Prof. Dr. Philipp Staudt
Organisation: KIT Department of Business and Economics
Part of: Business and Economics
Electives (Business Administration)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	2 terms	English	4	1

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-WIWI-114994	Digital Energy Systems	WT	4,5 CP	Staudt
T-WIWI-114995	Sustainable Information Systems	ST	4,5 CP	Staudt

Assessment
The module examination consists of partial examinations of the selected courses, which together meet the required credit point requirement. Details of the examination can be found in the respective partial performance description. The total grade of the module is formed from the grades of the partial examinations weighted by credit points (cut off after the first decimal place).

Prerequisites
None

Competence Goal
The student

- knows methods for investigating the use of sustainable information systems or information systems for promoting sustainability.
- is aware of the uses of digital technology for sustainability.
- understands how digital technology is changing the energy market and the energy system.
- is able to generate insights into the use of information systems with the help of scientific methods.
- is familiar with the economic and behavioral science theories underlying the digitization of the energy system.
- can make decisions on the use of digital technology for sustainability purposes or in the energy sector.

Content
The module deals with the use of digital technology for sustainability purposes from a general and an energy system perspective. The courses focus on context-specific as well as methodological and theoretical content. While Methods in Sustainable Information Systems focuses on the general perspective on sustainability and methodological knowledge, Digital Energy Systems focuses on the energy system context and economic and behavioral theory fundamentals.

Workload
Total workload for 9 credit points: approx. 270 hours.

- Attendance time: 51 hours
- Preparation and follow-up: 153 hours
- Exam and exam preparation: 66 hours

The exact distribution is based on the credit points of the courses in the module.

M 5.187 Module: Transport Infrastructure Policy and Regional Development (WW4VWL11) [M-WIWI-101485]

Coordinators: Prof. Dr. Kay Mitusch
Organisation: KIT Department of Business and Economics
Part of: Electives (Economics)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	2 terms	German/English	4	2

ID	Name	Term offered	Credits	Lecturers
Compulsory Elective Courses (Elective: 2 items)				
T-WIWI-103107	Spatial Economics	WT	4,5 CP	Ott
T-WIWI-100007	Transport Economics	ST	4,5 CP	Mitusch, Szimba

Assessment
The assessment is carried out as partial exams (according to Section 4(2), 1 or 2 of the examination regulation) of the single courses of this module, whose sum of credits must meet the minimum requirement of credits of this module. The exams are offered at the beginning of the recess period about the subject matter of the latest held lecture. Re-examinations are offered at every ordinary examination date. The assessment procedures are described for each course of the module separately. The overall grade for the module is the average of the grades for each course weighted by the credits and truncated after the first decimal.

Prerequisites
None

Competence Goal
The students

- understand the economic issues related to transport and regional development with a main focus on economic policy issues generated by the relationship of transport and regional development with the public sector
- are able to compare different considerations of politics, regulation and the private sector and to analyse and assess the respective decision problems both qualitatively and by applying appropriate methods from economic theory
- are prepared for careers in the public sector, particularly for public companies, politics, regulatory agencies, related consultancies, mayor construction companies or infrastructure project corporations

Content
The development infrastructure (e.g. transport, energy, telecommunications) has always been one of the most relevant factors for economic development and particularly influences the development of the regional economy. From the repertoire of state actions, investments into transport infrastructure are often regarded the most important measure to foster regional economic growth. Besides the direct effects of transport policy on passenger and freight transport, a variety of individual economic activities is significantly dependent on the available or potential transport options. Decisions on the planning, financing and realization of mayor infrastructure projects require a solid and far-reaching consideration of direct and indirect growth effects with the occurring costs.

Through its combination of lectures the module reflects the complex interdependencies between infrastructure policy, transport industry and regional policy and provides its participants with a comprehensive understanding of the functionalities of one of the most important sectors of the economy and its relevance for economic policy.

Additional Information
The courses *Assessment of Public Policies and Projects I* (winter term) and *Assessment of Public Policies and Projects II* (summer term) will no longer be part of this module. Student who have already had exams in this courses can integrate these exams in this module.

Workload
The total workload for this module is approximately 270 hours. The exact distribution is based on the credit points of the courses in the module. The total number of hours per course is determined by the amount of time spent attending the lectures and tutorials, as well as the exam times and the time required to achieve the module's learning objectives for an average student for an average performance.

M 5.188 Module: Transportation Modelling and Traffic Management (bauiEX306-VERMODMAN) [M-BGU-101065]

Coordinators: Prof. Dr.-Ing. Peter Vortisch
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	2 terms	German/English	4	8

Election Notes
Compulsory exams: 2 or 3 of the selectable module components must be selected

if 'Traffic Management and Transport Telematics' is selected, the 'Exercise Transportation Data Analysis' must be selected as examination prerequisite;

Compulsory elective: a maximum of one of the selectable module components must be selected;

if 'Transportation Data Analysis' is selected, the 'Exercise Transportation Data Analysis' must be selected as examination prerequisite;

ID	Name	Term offered	Credits	Lecturers
Compulsory Examination (Elective: between 2 and 3 items as well as between 6 and 9 credits)				
T-BGU-101797	Methods and Models in Transportation Planning	WT+ST	3 CP	Vortisch
T-BGU-101798	Traffic Engineering	WT+ST	3 CP	Vortisch
T-BGU-113971	Exercise Transportation Data Analysis	ST	0 CP	Vortisch
T-BGU-101799	Traffic Management and Transport Telematics	WT+ST	3 CP	Vortisch
T-BGU-101800	Traffic Flow Simulation	WT+ST	3 CP	Vortisch
Electives (Elective: between 0 and 3 credits)				
T-BGU-113671	Exercise Transportation Data Analysis	WT	0 CP	Kagerbauer
T-BGU-100010	Transportation Data Analysis	WT+ST	3 CP	Kagerbauer
T-BGU-106611	Freight Transport	WT+ST	3 CP	Szimba, Vortisch
T-BGU-106301	Long-Distance and Air Traffic	WT+ST	3 CP	Vortisch
T-BGU-101005	Tendering, Planning and Financing in Public Transport	WT+ST	3 CP	Vortisch
T-BGU-100014	Seminar in Transportation	WT+ST	3 CP	Kagerbauer, Vortisch
T-BGU-112552	Seminar on Modeling and Simulation in Transportation	WT	3 CP	Kagerbauer, Vortisch
T-BGU-103425	Mobility Services and New Forms of Mobility	WT+ST	3 CP	Kagerbauer
T-BGU-103426	Strategic Transport Planning	WT+ST	3 CP	Waßmuth
T-BGU-106608	Information Management for Public Mobility Services	WT	3 CP	Vortisch
T-BGU-111057	Sustainability in Mobility Systems	WT+ST	3 CP	Kagerbauer

Assessment
according to selected module component:

- module component T-BGU-101797 with oral examination according to § 4 Par. 2 No. 2
- module component T-BGU-101798 with oral examination according to § 4 Par. 2 No. 2
- module component T-BGU-113971 with not graded coursework according to § 4 Par. 3 as examination prerequisite for module component T-BGU-101799
- module component T-BGU-101799 with oral examination according to § 4 Par. 2 No. 2
- module component T-BGU-101800 with oral examination according to § 4 Par. 2 No. 2
- module component T-BGU-113671 with not graded coursework according to § 4 Par. 3 as examination prerequisite for module component T-BGU-100010
- module component T-BGU-100010 with oral examination according to § 4 Par. 2 No. 2
- module component T-BGU-106611 with written examination according to § 4 Par. 2 No. 1
- module component T-BGU-106301 with written examination according to § 4 Par. 2 No. 1
- module component T-BGU-101005 with oral examination according to § 4 Par. 2 No. 2
- module component T-BGU-100014 with examination of another type according to § 4 Par. 2 No. 3
- module component T-BGU-112552 with examination of another type according to § 4 Par. 2 No. 3
- module component T-BGU-103425 with oral examination according to § 4 Par. 2 No. 2
- module component T-BGU-103426 with oral examination according to § 4 Par. 2 No. 2
- module component T-BGU-106608 with examination of an other type according to § 4 Par. 2 No. 3
- module component T-BGU-111057 with written examination according to § 4 Par. 2 No. 1

details about the learning controls, see module components

Prerequisites
None

Competence Goal
The student has in-depth knowledge and can apply the essential tools in combination with the basic methodological knowledge as an industrial engineer/technical economist, depending on the chosen 'specialization',

- as a 'traffic engineer' (specialization in traffic engineering) AND/OR
- as a 'transport planner' (specialization in transport planning) AND/OR
- in the field of transport software (e.g. in transport modeling)
- or in similar professional fields

to work.

Content
This module deepens existing knowledge in the field of transportation. The choice of core courses determines the specialization - more in the direction of transport planning or more in the direction of traffic engineering and/or traffic simulation. The module is therefore an ideal continuation of the module Fundamentals of Transportation [WI4INGBGU15].

Module Grade Calculation
grade of the module is CP weighted average of grades of the partial exams

Additional Information
If you have already attended lectures that are no longer offered, they can still be examined within this module if the content is the same. In this case, please come to the office hours for coordination!

Workload
contact hours (1 HpW = 1 h x 15 weeks), according to selected course and exam:

- Methods and Models in Transportation Planning lecture/exercise: 30 h
- Traffic Engineering lecture/exercise: 30 h
- Traffic Management and Transport Telematics lecture/exercises: 30 h
- Traffic Flow Simulation lecture/exercise: 30 h
- Transportation Data Analysis lecture/exercise: 30 h
- Freight Transport lecture/exercise: 30 h
- Long-Distance and Air Traffic lecture: 30 h
- Tendering, Planning and Financing in Public Transport lecture: 30 h
- Seminar in Transportation: 30 h
- Seminar on Modeling and Simulation in Transportation: 30 h
- Mobility Services and New Forms of Mobility lecture/exercise: 30 h
- Strategic Transport Planning lecture: 30 h
- Information Management for Public Mobility Services lecture/exercise: 30 h
- Sustainability in Mobility Systems lecture: 30 h

independent study, according to selected course and exam:

- preparation and follow-up Methods and Models in Transportation Planning lecture/exercises: 30 h
- examination preparation Methods and Models in Transportation Planning (selectable partial examination): 30 h
- preparation and follow-up Traffic Engineering lecture/exercises: 30 h
- examination preparation Traffic Engineering (selectable partial examination): 30 h
- preparation and follow-up Traffic Management and Transport Telematics lecture/exercises: 30 h
- preparation of the Exercise Transportation Data Analysis (not graded examination prerequisite): 10 h
- examination preparation Traffic Management and Transport Telematics (selectable partial examination): 20 h
- preparation and follow-up Traffic Flow Simulation lecture/exercises: 30 h
- examination preparation Traffic Flow Simulation (selectable partial examination): 30 h
- preparation and follow-up Transportation Data Analysis lecture/exercises: 20 h
- preparation of the Exercise Transportation Data Analysis (not graded examination prerequisite): 10 h
- examination preparation Transportation Data Analysis (selectable partial examination): 30 h
- preparation and follow-up lecture/exercises Freight Transport: 30 h
- examination preparation Freight Transport (selectable partial examination): 30 h
- preparation and follow-up lectures Long-Distance and Air Traffic: 30 h
- examination preparation Long-Distance and Air Traffic (selectable partial examination): 30 h
- preparation and follow-up lectures Tendering, Planning and Financing in Public Transport: 30 h
- examination preparation Tendering, Planning and Financing in Public Transport (selectable partial examination): 30 h
- preparation of seminar paper in Transportation and presentation (selectable partial examination): 60 h
- work on a practical problem in the Seminar on Modeling and Simulation in Transportation (selectable partial examination): 60 h
- preparation and follow-up lecture/exercises Mobility Services and New Forms of Mobility: 30 h
- examination preparation Mobility Services and New Forms of Mobility (selectable partial examination): 30 h
- preparation and follow-up lectures Strategic Transport Planning: 30 h
- examination preparation Strategic Transport Planning (selectable partial examination): 30 h
- preparation and follow-up lecture/exercises Information Management for Public Mobility Services: 30 h
- preparation accompanying exercises Information Management for public Mobility Services (selectable partial examination): 30 h
- preparation and follow-up lectures Sustainability in Mobility Systems: 30 h
- examination preparation Sustainability in Mobility Systems (selectable partial examination): 30 h

total: 270 h

Recommendations
None

5.189 Module: Urban Water Technologies (bauiEX213-URBWATTEC) [M-BGU-104448]

Coordinators: Dr.-Ing. Mohammad Ebrahim Azari Najaf Abad
apl. Prof. Dr. Stephan Fuchs

Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences

Part of: Engineering and Informatics (Civil Engineering, Geo and Environmental Sciences)
Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each winter term	1 term	English	4	3

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-BGU-112365	Urban Water Technologies	WT+ST	9 CP	Azari Najaf Abad, Fuchs

Assessment
- module component T-BGU-112365 with oral examination according to § 4 Par. 2 No. 2

details about the learning control, see module component

Prerequisites
none

Competence Goal
Students acquire in-depth knowledge of the processes required to understand urban water management systems and their design. After completing the module, they should be familiar with the process engineering systems for wastewater and rainwater treatment and be able to explain the functional principles of the individual system components, assess their suitability for specific applications and apply basic design approaches.

Content
This module provides an in-depth grounding in the design and evaluation of urban water management systems. The chemical, physical and biological principles required for this are deepened. Based on the detailed consideration of individual elements, an overall understanding of the urban water management system and its interaction with surfaces is developed. To this end, the theoretical tools are developed and model approaches are presented.

Module Grade Calculation
grade of the module is grade of the exam

Additional Information
The number of participants in the course 6223901 Wastewater Treatment Technologies is limited to 30 persons. Registration takes place via ILIAS. Places will be allocated according to the progress of studies, primarily to students from *Water Science and Engineering*, then *Civil Engineering* and *Geoecology* and other degree programs.

Workload
contact hours (1 HpW = 1 h x 15 weeks):

- Urban Water Infrastructure and Management lecture/exercise: 60 h
- Wastewater Treatment Technologies lecture/exercise: 60 h

independent study:

- preparation and follow-up lecture/exercises Urban Water Infrastructure and Management: 30 h
- preparation and follow-up lecture/exercises Wastewater Treatment Technologies: 60 h
- examination preparation: 60 h

total: 270 h

Recommendations
none

5.190 Module: Vehicle Development (WI4INGMB14) [M-MACH-101265]

Coordinators: Prof. Dr.-Ing. Marcus Geimer
Organisation: KIT Department of Mechanical Engineering

Part of: Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each term	1 term	German/English	4	10

ID	Name	Term offered	Credits	Lecturers
Vehicle Development (Elective: at least 9 credits)				
T-MACH-102207	Tires and Wheel Development for Passenger Cars	ST	4 CP	Leister
T-MACH-111389	Fundamentals in the Development of Commercial Vehicles	see notes	4 CP	Weber
T-MACH-102156	Project Workshop: Automotive Engineering	WT+ST	6 CP	Frey, Gießler
T-MACH-110796	Python Algorithms for Vehicle Technology	ST	4 CP	Rhode
T-MACH-102148	Gear Cutting Technology	WT	4 CP	Hoffmeister, Klaiber
T-MACH-112126	Data-Driven Algorithms in Vehicle Technology	WT	4,5 CP	Scheubner
T-MACH-114075	Principles of Whole Vehicle Engineering (in German)	WT+ST	4 CP	Harrer
T-MACH-114095	Principles of Whole Vehicle Engineering	WT+ST	4,5 CP	Harrer
T-MACH-113862	Simulation with Lumped Parameters	ST	3 CP	Geimer
T-MACH-113863	Tutorial Simulation with Lumped Parameters	ST	1 CP	Geimer

Assessment
The assessment is carried out as partial exams.

The partial exams consists of a written exam (90 to 120 minutes) or an oral exam (duration 30 to 40 minutes).

Prerequisites
None

Competence Goal
The student

- knows and understands the procedures in automobile development,
- knows and understands the technical specifications at the development procedures,
- is aware of notable boundaries like legislation.

Content
By taking the module Vehicle Development the students get to know the methods and processes applied in the automobile industry. They learn the technical particularities which have to be considered during the vehicle development and it is shown how the numerous single components cooperate in a harmoniously balanced complete vehicle. There is also paid attention on special boundary conditions like legal requirements.

Workload
The total work load for this module is about 270 Hours (9 Credits). The partition of the work load is carried out according to the credit points of the courses of the module. The work load for courses with 6 credit points is about 180 hours, for courses with 4.5 credit points about 135 hours, for courses with 3 credit points about 90 hours, and for courses with 1.5 credit points about 45 hours. The total number of hours per course results from the time of visiting the lectures and exercises, as well as from the exam duration and the time that is required to achieve the objectives of the module as an average student with an average performance.

Recommendations
Knowledge of the content of the courses *Engineering Mechanics I* [2161238], *Engineering Mechanics II* [2162276] and *Basics of Automotive Engineering I* [2113805], *Basics of Automotive Engineering II* [2114835] is helpful.

Teaching and Learning Methods
The teaching and learning procedures (lecture, lab course, workshop) are described for each course of the module separately.

M 5.191 Module: Water Chemistry and Water Technology I (WW4INGCV6) [M-CIWVT-101121]

Coordinators: Prof. Dr. Harald Horn
Organisation: KIT Department of Chemical and Process Engineering
Part of: Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Language	Level	Version
9 CP	graded	Each winter term	1 term	German/English	4	3

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-CIWVT-106802	Water Technology	WT	6 CP	Horn
T-CIWVT-106840	Practical Course in Water Technology	WT	3 CP	Hille-Reichel, Horn

Prerequisites
none

Competence Goal
Students learn fundamental knowledge in water chemistry and how to apply it to processes in aquatic systems in general and in reactors for water treatment. Water treatment will be taught for drinking water and partly waste water. The students are able to apply physical, chemical and biochemical treatment for the respective removal of particulate and dissolved components in water. They are able to use the fundamental design parameters for the different types of unit operations.

Students can explain the most important processes in water treatment. They are able to do calculations, and to compare and interpret data. They learn how to use different methods, and to interpret different processes.

Content
Water cycle, different types of raw water (ground and surface water). Water as solvent, carbonate balance, differentiation between microbiological and chemical population. Unit operations: sieving, sedimentation, filtration, flocculation, flotation, ion exchange, aeration, oxidation, disinfection, adsorption). For all unit operations design parameters will be provided. Simple 1D models will be discussed for description of kinetics and retention time in reactors for water treatment.

6 different experiments out of: equilibrium study of the calcium carbonate system, flocculation, adsorption, oxidation, atomic absorption spectroscopy, ion chromatography, liquid chromatography, sum parameter, and an oral presentation of the student. In addition, excursions to two different treatment plants (waste water, drinking water)

Literature

- Crittenden, J. C. et al. (2012): Water treatment, principles and design. 3. Auflage, Wiley & Sons, Hoboken.
- Jekel, M., Czekalla, C. (Hrsg.) (2016). DVGW Lehr- und Handbuch der Wasserversorgung. Deutscher Industrieverlag.
- Harris, D. C., Lucy, C. A. (2019): Quantitative chemical analysis, 10. Auflage. W. H. Freeman and Company, New York.
- Patnaik, P., 2017: Handbook of environmental analysis: Chemical pollutants in air, water, soil, and solid wastes. CRC Press.
- Wilderer, P. (Ed., 2011): Treatise on water science, four-volume set, 1st edition, volume 3: Aquatic chemistry and biology. Elsevier, Oxford.

- Lecture notes will be provided in ILIAS

M 5.192 Module: Water Chemistry and Water Technology II (WW4INGCV7) [M-CIWVT-101122]

Coordinators: Prof. Dr. Harald Horn
Organisation: KIT Department of Chemical and Process Engineering
Part of: Electives (Engineering Sciences)

Credits	Grading	Recurrence	Duration	Level	Version
12 CP	graded	Each term	2 terms	4	3

ID	Name	Term offered	Credits	Lecturers
Mandatory				
T-CIWVT-113235	Exercises: Membrane Technologies	ST	1 CP	Horn, Saravia
T-CIWVT-113236	Membrane Technologies in Water Treatment	ST	5 CP	Horn, Saravia
T-CIWVT-106838	Fundamentals of Water Quality	WT	6 CP	Wagner

Prerequisites
The Module "Water Chemistry and Water Technology I" must be passed.

Competence Goal
The student

- has knowledge of types and sum of the water constituents and their interaction with each other and with the water molecules,
- is able to explain the interrelationships of the occurrence of geogenic and anthropogenic substances as well as of microorganisms in the different areas of the hydrological cycle and is able to select suitable analytical methods for their determination,
- knows about the different types of water treatment and water purification methods to convert, reduce or concentrate water constituents, especially for membrane processes,
- is able to use methodical tools, analyze the correlations and critically evaluate the critically evaluate the different procedures.

Content
The types of water, water law, basic terms of water chemical analysis, analysis quality, sampling, rapid test procedures and general investigation methods as well as summary parameters are dealt with. The analytical methods for main and secondary constituents as well as for organic and inorganic trace substances are discussed with examples for orientation.
The effects of the different treatment and purification methods are shown and it is explained how they can convert, reduce or concentrate water constituents.

6 Module components

T 6.1 Module component: "Practical Seminar in Discrete Optimization: Implementing and Evaluating Solution Methods (with Python)" [T-WIWI-114747]

Coordinators: Dr.-Ing. Hannah Bakker
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-102832 - Operations Research in Supply Chain Management

Type	Credits	Grading	Term offered	Version
Examination of another type	4,5 CP	graded	Once	1

Courses					
WT 25/26	2500066	Practical Seminar in Discrete Optimization: Implementing and Evaluating Solution Methods (with Python)	3 SWS	Others / 🗣	Bakker

Legend: Online, Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment is conducted as an alternative exam assessment in the form of a computational experiment. This includes the submission of a documented code project, a written report (6–8 pages), as well as an interim and a final presentation.

The final grade is composed of the graded and weighted assessment components: the code, the written report, and the interim and final presentations. Details will be announced at the beginning of the course.

Prerequisites
None

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Practical Seminar in Discrete Optimization: Implementing and Evaluating Solution Methods (with Python)

Others (sonst.)
On-Site

2500066, WS 25/26, 3 SWS, Language: German, Open in study portal

Content
This application-oriented seminar provides students with practical skills in working with heuristic methods for discrete optimization. Participants implement a given solution approach in Python and apply it to defined benchmark instances. The focus is not only on technical implementation but also on the systematic setup, execution, and evaluation of a computational experiment. Students practice presenting and assessing results in a structured manner. The aim of the seminar is to develop fundamental competencies in algorithmic implementation and empirical evaluation of heuristic methods, applied to a concrete optimization problem.

Organizational issues
Die Veranstaltung ist teilnahmebeschränkt. Die Anmeldung wird vom 15.08. bis zum 30.09. im Wiwi-Portal freigeschaltet sein.

T 6.2 Module component: (Gen)AI and Virtual Reality for Business [T-WIWI-114749]

Coordinators: Prof. Dr. Jella Pfeiffer
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-103117 - Data Science: Data-Driven Information Systems
M-WIWI-104080 - Designing Interactive Information Systems
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Examination of another type	4,5 CP	graded	Irregular	1

Assessment
Success is assessed in the form of an examination of another type. It consists of the creation of a podcast, poster or video and a final presentation. The overall impression will be assessed. Details on the structure of the assessment will be announced during the lecture.

Prerequisites
None

Recommendations
Basic knowledge of artificial intelligence and data science methods is an advantage.

Additional Information
There is a tutorial associated with the lecture, which takes place every two weeks.
This course component is expected to be offered firstly in the winter semester of 2026/27.

Workload:

- Lecture attendance time (2 SWS × 15 weeks): 30 h
- Attendance time tutorial (1 SWS × 15 weeks): 15 h
- Self-study lecture: 20 h
- Self-study tutorial: 50 h
- Exam preparation: 20 h
- Total: 135 h

Workload
135 hours

T 6.3 Module component: (Gen)AI-based Automation in Organizations [T-WIWI-114210]

Coordinators: Prof. Dr. Alexander Mädche
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-102806 - Service Innovation, Design & Engineering
M-WIWI-103117 - Data Science: Data-Driven Information Systems
M-WIWI-104068 - Information Systems in Organizations
M-WIWI-104080 - Designing Interactive Information Systems
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Examination of another type	4,5 CP	graded	Each summer term	1

Courses					
ST 2026	2500015	(Gen)AI-based Automation in Organizations	3 SWS	Lecture / Blended	Mädche, Benke

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
The alternative exam assessment is a single examination of another type that combines a written examination (duration ≈ 60 minutes) with the implementation of a capstone project. Both components are integrated into one overall assessment; the internal weighting of the two parts is defined by the teaching staff, but only one final grade is awarded for the whole examination. The final grade therefore reflects the total impression of the combined performance

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V (Gen)AI-based Automation in Organizations

2500015, SS 2026, 3 SWS, Language: English, Open in study portal

Lecture (V)
Blended (On-Site/Online)

Content

Content

The advent of generative artificial intelligence (GenAI) has received great attention in business and society due to its capabilities of content creation or decision making. Individuals started rapidly to use the capabilities of tools like ChatGPT, Claude or Google Gemini for text and image generation, personal recommendations, or decision support. At the same time, organizations are challenged to leverage GenAI and traditional AI technology within their business models, processes, and information systems. (Gen)AI technologies enable executing cognitive tasks which in the past were carried out manually by organizations' employees. Ultimately, this leads to an increase of automation in organizations. This digital transformation process to higher levels of automation, however, must be managed by organizations.

This course will address this organizational challenge and teach concepts about the automation of organizations and their employees leveraging (Gen)AI technologies. The course is not focusing on the technological foundations of GenAI but looks into the economic and social impact of the technology and its management.

Learning Objectives

As a result of attending this program, students will be able to:

- describe key concepts of (Gen)AI technologies enabling the increase of automation in organizations.
- understand the historical evolution and core concepts of automation to drive organizational efficiency and innovation.
- articulate (Gen)AI integration principles for effectively implementing automation in organizations.
- understand and apply automation management strategies and best practices for addressing challenges to ensure adoption of (Gen)AI capabilities for organizational automation from a socio-economic perspective.
- explore and prototype (Gen)AI-based applications to streamline individual tasks and workflows in the context of organizational automation.

Case-studies & No-code exercise

The course contains two applied elements:

- Case-study exercises in which the students apply the theoretical knowledge on integrating (Gen)AI-based services into an organization.
- No-code exercises in which the students will develop a no/low-code application using (Gen)AI services to implement automation.

Organization

The lecture kickoff will take place on **April 24, 2026** from 14:00 - 15:30 at Building 10.11 Room 213 (Plenarsaal).

More information about the schedule of the lecture, the exercises or any organizational aspect will be posted in advance in the ILIAS course of the lecture. For questions regarding the course please use the ILIAS Forum.

The lecture will be held by Dr. Ivo Benke and Till Carlo Schelhorn.

Lecture contact for organizational questions is Till Carlo Schelhorn (till.schelhorn∂kit.edu)

T 6.4 Module component: Access Control Systems: Models and Technology [T-INFO-112775]

Coordinators: Prof. Dr. Hannes Hartenstein
Organisation: KIT Department of Informatics
Part of: M-WIWI-104909 - Informatics (Department of Informatics)

Type	Credits	Grading	Term offered	Version
Written examination	5 CP	graded	Each summer term	1

Courses					
ST 2026	2400147	Access Control Systems: Models and Technology	3 SWS	Lecture / Practice / 🗣	Hartenstein, Hemken

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment is carried out as a written examination (§ 4 Abs. 2 Nr. 1 SPO) lasting 60 minutes.

Depending on the number of participants, it will be announced six weeks before the examination (§ 6 Abs. 3 SPO) whether the examination takes place

- in the form of an oral examination lasting 30 minutes pursuant to § 4 Abs. 2 Nr. 2 SPO or
- in the form of a written examination lasting 60 minutes in accordance with § 4 Abs. 2 Nr. 1 SPO.

Prerequisites
None.

Recommendations
Basics according to the lectures "Information Security" and "IT Security Management for Networked Systems" are recommended.

Below you will find excerpts from courses related to this module component:

V Access Control Systems: Models and Technology

2400147, SS 2026, 3 SWS, Language: English, Open in study portal

Lecture / Practice (VÜ)
On-Site

Content

Content:

Access control systems are everywhere and the backbone of secure services as they incorporate who is and who is not authorized: think of operating systems, information systems, banking, vehicles, robotics, cryptocurrencies, or decentralized applications as examples. The course starts with current challenges of access control in the era of hyperconnectivity, i.e., in cyber-physical or decentralized systems. Based on the derived needs for next generation access control, we first study how to specify access control and analyze strengths and weaknesses of various approaches. We then focus on up-to-date proposals, like IoT and AI access control. We look at current cryptographic access control aspects, blockchains and cryptocurrencies, and trusted execution environments. We also discuss the ethical dimension of access management. Students prepare for lecture and exercise sessions by studying previously announced literature and by preparation of exercises that are jointly discussed in the sessions.

Competency Goals:

· The student understands the challenges of access control in the era of hyperconnectivity.

· The student understands that an information security model defines access rights that express for a given system which subjects are allowed to perform which actions on which objects. The student understands that a system is said to be secure with respect to a given information security model, if it enforces the corresponding access rights.

· The student is able to derive suitable access control models from scenario requirements and is able to specify concrete access control systems. The student is able to decide which concrete architectures and protocols are technically suited for realizing a given access control model.

· The student knows access control protocols using cryptographic methods and is able to compare protocol realizations based on different cryptographic building blocks.

· The student is aware of the limits of access control models and systems with respect to their analyzability and performance and security characteristics. The student is able to identify the resulting tradeoffs.

· The student knows the state of the art with respect to current research endeavors, e.g., access control in the context of decentralized and distributed systems, Trusted Execution Environments, AI, robotics, or hash-chain based systems.

Workload:

1.Attendance time

Lecture: 2 SWS: 2,0h x 15 = 30h

Exercises: 1 SWS: 1,0h x 15 = 15h

2.Self-study (e.g., independent review of course material, work on homework assignments)

Weeklypreparation and follow-up ofthelecture: 15 x 1h x 3 = 45h

Weeklypreparation and follow-up oftheexercise: 15 x 2h = 30h

3.Preparation for the exam: 30h

Σ = 150h = 5 ECTS

Competency certificate:

Depending on the number of participants, it will be announced six weeks before the examination (§ 6 Abs. 3 SPO) whether the examination takes place

· in the form of an oral examination lasting 30 minutes pursuant to § 4 Abs. 2 Nr. 2 SPO or

· in the form of a written examination lasting 60 minutes in accordance with § 4 Abs. 2 Nr. 1 SPO.

Recommendations:

Basics according to the lectures "Information Security" and "IT Security Management for Networked Systems" are recommended.

Duration: One terms

T 6.5 Module component: Adaptive Optics [T-ETIT-107644]

Coordinators: Ph.D. Szymon Gladysz
Prof. Dr. Ulrich Lemmer

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: M-ETIT-103802 - Adaptive Optics

Type	Credits	Grading	Term offered	Version
Oral examination	3 CP	graded	Each winter term	1

Courses					
WT 25/26	2313724	Adaptive Optics	2 SWS	Lecture / 🗣	Gladysz

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Type of Examination: Oral examination

Duration of Examination: approx. 30 Minutes

Modality of Exam: The oral exam will be scheduled during the semester break.

The module grade is the grade of the oral exam.

Prerequisites
None.

Recommendations
Basic knowledge of statistics.

Workload
90 hours

T 6.6 Module component: Additive Manufacturing for Process Engineering - Examination [T-CIWVT-110902]

Coordinators: TT-Prof. Dr. Christoph Klahn
Organisation: KIT Department of Chemical and Process Engineering
Part of: M-CIWVT-105407 - Additive Manufacturing for Process Engineering

Type	Credits	Grading	Term offered	Version
Oral examination	5 CP	graded	Each summer term	1

Courses					
ST 2026	2241020	Additive Manufacturing for Process Engineering	2 SWS	Lecture / 🗣	Klahn

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, x Cancelled

Assessment
Oral examination lasting approx. 30 minutes.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-CIWVT-110903 - Practical in Additive Manufacturing for Process Engineering must have been passed.

T 6.7 Module component: Additive Manufacturing of Metallic Components [T-MACH-113985]

Coordinators: Prof. Dr.-Ing. Frederik Zanger
Organisation: KIT Department of Mechanical Engineering
Part of: M-MACH-106590 - Production Engineering

Type	Credits	Grading	Term offered	Expansion	Version
Oral examination	4 CP	graded	Each winter term	1 semesters	2

Courses					
WT 25/26	2149671	Additive Manufacturing of Metallic Components	2 SWS	Lecture / 🗣	Zanger
WT 26/27	2149671	Additive Manufacturing of Metallic Components	2 SWS	Lecture / 🗣	Zanger

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Oral examination, duration approx. 30 minutes

Prerequisites
The following module component may not have started:

- T-MACH-114019 - Additive Fertigung metallischer Bauteile: Designoptimierung und Herstellung

Additional Information
The course is offered in German.

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Additive Manufacturing of Metallic Components

Lecture (V)
On-Site

2149671, WS 25/26, 2 SWS, Language: German, Open in study portal

Content
The course “Additive Manufacturing of Metallic Components” teaches the theoretical fundamentals of metallic additive manufacturing, particularly using the example of the laser-based process powder bed fusion (PBF-LB/M). It thus provides the ideal prerequisite for the course “Project Internship Additive Manufacturing: Design Optimization and Manufacturing of Metallic Components.”

The course covers the underlying mechanisms and the necessary steps in the additive process chain for metallic components. Other processes for manufacturing metallic components using additive manufacturing methods are also discussed. Students will learn about the following topics:

- Structure and function of a PBF-LB/M system
- Influence of process parameters on the quality of components manufactured in the PBF-LB/M process
- Design for additive manufacturing
- Options for simulation assistance in the process chain
- Process monitoring and quality assurance in additive manufacturing
- Downstream processes for adjusting the required component state
- Quality control in additive manufacturing
- Fundamentals and possibilities of directed energy deposition, binder jetting, and bath-based photopolymerization processes.

Learning Outcomes:

Students ...

- are able to explain the basic principles of the laser-based additive manufacturing processes powder bed fusion and directed energy deposition, as well as the two processes binder jetting and bath-based photopolymerization.
- can describe the characteristics and areas of application of the four processes mentioned above.
- can describe the creation of a product along the entire additive process chain (CAD, simulation, construction job preparation, CAM) from the initial idea to production using the example of the powder bed fusion process.
- are able to discuss the development process for components that are optimized for additive manufacturing.
- can explain in detail the mechanisms behind the powder bed fusion process and derive measures for improving the process.
- can evaluate and select options for post-processing additively manufactured components based on their requirements.
- are able to name the most important methods for quality assurance in additive manufacturing and the relevant standards.

Workload:

regular attendance: 21 hours
self-study: 99 hours

Literature
Medien:
Skript zur Veranstaltung wird über ilias (https://ilias.studium.kit.edu/) bereitgestellt.

Media:
Lecture notes will be provided in ilias (https://ilias.studium.kit.edu/).

V	**Additive Manufacturing of Metallic Components** 2149671, WS 26/27, 2 SWS, Language: German, Open in study portal	**Lecture (V)** **On-Site**

Content
The course “Additive Manufacturing of Metallic Components” teaches the theoretical fundamentals of metallic additive manufacturing, particularly using the example of the laser-based process powder bed fusion (PBF-LB/M). It thus provides the ideal prerequisite for the course “Project Internship Additive Manufacturing: Design Optimization and Manufacturing of Metallic Components.”

The course covers the underlying mechanisms and the necessary steps in the additive process chain for metallic components. Other processes for manufacturing metallic components using additive manufacturing methods are also discussed. Students will learn about the following topics:

- Structure and function of a PBF-LB/M system
- Influence of process parameters on the quality of components manufactured in the PBF-LB/M process
- Design for additive manufacturing
- Options for simulation assistance in the process chain
- Process monitoring and quality assurance in additive manufacturing
- Downstream processes for adjusting the required component state
- Quality control in additive manufacturing
- Fundamentals and possibilities of directed energy deposition, binder jetting, and bath-based photopolymerization processes.

Learning Outcomes:

Students ...

- are able to explain the basic principles of the laser-based additive manufacturing processes powder bed fusion and directed energy deposition, as well as the two processes binder jetting and bath-based photopolymerization.
- can describe the characteristics and areas of application of the four processes mentioned above.
- can describe the creation of a product along the entire additive process chain (CAD, simulation, construction job preparation, CAM) from the initial idea to production using the example of the powder bed fusion process.
- are able to discuss the development process for components that are optimized for additive manufacturing.
- can explain in detail the mechanisms behind the powder bed fusion process and derive measures for improving the process.
- can evaluate and select options for post-processing additively manufactured components based on their requirements.
- are able to name the most important methods for quality assurance in additive manufacturing and the relevant standards.

Workload:

regular attendance: 21 hours
self-study: 99 hours

Literature
Medien:
Skript zur Veranstaltung wird über ilias (https://ilias.studium.kit.edu/) bereitgestellt.

Media:
Lecture notes will be provided in ilias (https://ilias.studium.kit.edu/).

T 6.8 Module component: Additive Manufacturing Processes [T-MACH-113570]

Coordinators: Prof. Dr.-Ing. Frederik Zanger
Organisation: KIT Department of Mechanical Engineering
Part of: M-MACH-101276 - Manufacturing Technology

Type	Credits	Grading	Term offered	Expansion	Version
Oral examination	4 CP	graded	Each summer term	1 semesters	1

Courses					
ST 2026	2150702	Additive Manufacturing Processes	3 SWS	Lecture / 🗣	Zanger

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Oral examination, duration approx. 20 minutes

Additional Information
The course is offered in German.

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Additive Manufacturing Processes

2150702, SS 2026, 3 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content

The course “Additive Manufacturing Processes” provides participants with in-depth knowledge of industrially relevant additive manufacturing processes and their process chains. During the course, students learn the theoretical principles and practical application aspects of the following processes:

MEX: Material extrusion
MJT: Material jetting
PBF: Powder bed fusion
BJT: Binder jetting
DED: Directed energy deposition
VPP: Vat photopolymerization.

In addition, the following topics are covered:

Holistic view of the process chain
Materials for additive manufacturing processes (production and characterisation)
Design methods and simulation in additive manufacturing
Future trends and innovative technologies.

The acquired knowledge is supplemented by practical lectures from industry, which offer valuable insights into current applications, challenges and innovations in additive manufacturing.

Learning Outcomes:

Students ...

can describe the principles and special features of industrially relevant additive manufacturing technologies
are able to explain the process of layer formation in additive manufacturing processes and evaluate it in terms of manufacturing possibilities
can describe the various methods of powder production and assess their advantages and disadvantages in relation to additive manufacturing
can name the essential materials for additive manufacturing, explain their properties and analyse their fields of application
are able to explain additive process chains, including heat treatment, post-processing and quality assurance, and to evaluate their significance for component quality.
can take into account the design requirements and process-specific features of additive manufacturing in component development
are able to classify current research and development topics in the field of additive manufacturing in the context of industrial applications

Workload:

regular attendance: 21 hours
self-study: 99 hours

Organizational issues

Bekanntgabe von möglichen Praxisvorlesungen erfolgt in der ersten Vorlesung

Literature

Medien:

Skript zur Veranstaltung wird über ilias (https://ilias.studium.kit.edu/) bereitgestellt.

Media:

Lecture notes will be provided in ilias (https://ilias.studium.kit.edu/).

T 6.9 Module component: Advanced Communications Engineering [T-ETIT-113676]

Coordinators: Dr.-Ing. Holger Jäkel
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-106815 - Advanced Communications Engineering

Type	Credits	Grading	Term offered	Version
Written examination	6 CP	graded	Each winter term	1

Courses					
WT 25/26	2310540	Advanced Communications Engineering	3 SWS	Lecture / 🗣💻	Jäkel
WT 25/26	2310541	Tutorial to 2310540 Advanced Communications Engineering	1 SWS	Practice / 🗣💻	Cil
WT 26/27	2310540	Advanced Communications Engineering	3 SWS	Lecture / 🗣	Jäkel

Legend: 🖥 Online, 🗣💻 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment takes place in the form of a written examination lasting 120 min.

The module grade is the grade of the written exam.

Prerequisites
none

T 6.10 Module component: Advanced Corporate Finance [T-WIWI-113469]

Coordinators: Prof. Dr. Martin Ruckes
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101453 - Applied Strategic Decisions
M-WIWI-101480 - Finance 3
M-WIWI-101483 - Finance 2
M-WIWI-101502 - Economic Theory and its Application in Finance
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each summer term	1

Courses					
ST 2026	2530214	Advanced Corporate Finance	2 SWS	Lecture / 🗣	Ruckes

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment of this course is a written examination (following §4(2), 1 SPO) of 60 mins.

The exam is offered each semester.

Below you will find excerpts from courses related to this module component:

V Advanced Corporate Finance

2530214, SS 2026, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content
The course covers the foundational principles of advanced topics of corporate finance, such as corporate governance, executive compensation, strategy & finance, mergers & acquisitions (M&A), and sustainable finance. Additionally, the course explores the respective institutional aspects within these areas of corporate finance. The approach is holistic, including both theoretical-conceptional aspects (e.g., moral hazard and the influence of asymmetric information) and empirical insights (e.g., the effects of financial decisions on firm value). Throughout, the course will emphasize both fundamental and current research findings.

Learning outcomes:

Upon successful completion of the course, students will possess profound knowledge and skills in advanced areas of corporate finance. These areas include topics such as corporate governance, executive compensation, strategy and finance, mergers and acquisitions (M&A), as well as key aspects of sustainable finance. Participants of this course will be able to describe and analyze the theoretical and conceptual foundations of the effects of information asymmetries and moral hazard on corporate financing behavior and assess their impact in corporate practice. Furthermore, upon completion of the course, participants will be familiar with the fundamental institutional elements in these areas and be able to discuss and solve advanced problems in corporate finance from both a theoretical and an empirical perspective. Moreover, students will acquire an advanced understanding of the central scientific findings in these topic areas, which will enable them to critically apply them in scientific and practical contexts.

Literature
Verschiedene Literaturquellen, u.a. Brealey/Myers/Allen/Edmans: Principles of Corporate Finance; Thomson/Conyon: Corporate Governance: Mechanisms and Systems; Larcker/Tayan: Corporate Governance Matters. Weitere Literatur wird in der Lehrveranstaltung bekannt gegeben.

Various source of literature, among others Brealey/Myers/Allen/Edmans: Principles of Corporate Finance; Thomson/Conyon: Corporate Governance: Mechanisms and Systems; Larcker/Tayan: Corporate Governance Matters. Additional reading materials will be introduced during the course.

T 6.11 Module component: Advanced Game Theory [T-WIWI-102861]

Coordinators: Prof. Dr. Karl-Martin Ehrhart
Prof. Dr. Clemens Puppe
Prof. Dr. Johannes Philipp Reiß

Organisation: KIT Department of Business and Economics

Part of: M-WIWI-101453 - Applied Strategic Decisions
M-WIWI-101500 - Microeconomic Theory
M-WIWI-101502 - Economic Theory and its Application in Finance
M-WIWI-104908 - Economics
M-WIWI-107187 - Quantitative Core

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each winter term	3

Courses					
WT 25/26	2521533	Advanced Game Theory	2 SWS	Lecture / 🗣	Reiß
WT 25/26	2521534	Übung zu Advanced Game Theory	1 SWS	Practice / 🗣	Reiß, Peters, Potarca

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment consists of a written exam (60 minutes) (following §4(2), 1 of the examination regulation). The exam takes place in every semester. Re-examinations are offered at every ordinary examination date.

Prerequisites
None

Recommendations
It is recommended that students have basic knowledge in mathematics and statistics.

Below you will find excerpts from courses related to this module component:

V Advanced Game Theory

Lecture (V)
On-Site

2521533, WS 25/26, 2 SWS, Language: English, Open in study portal

T 6.12 Module component: Advanced Lab Informatics (Bachelor) [T-WIWI-110541]

Coordinators: Professorenschaft des Instituts AIFB
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104901 - Informatics (KIT-Department of Economics and Management)

Type	Credits	Grading	Term offered	Version
Examination of another type	4,5 CP	graded	Each term	1

Courses					
WT 25/26	2512204	Lab Realisation of innovative services (Bachelor)	3 SWS	Practical course / Blended	Schiefer, Toussaint
WT 25/26	2512554	Praktikum Security, Usability and Society (Bachelor)	3 SWS	Practical course / Blended	Volkamer, Strufe, Berens, Fallahi, Ballreich, Hennig, Länge, Mossano, Hilt
ST 2026	2512204	Lab Realisation of innovative services (Bachelor)	3 SWS	Practical course / On-Site	Schiefer, Toussaint, Kruse
ST 2026	2512206	Lab Agentic AI im Smart IoT Office (Bachelor)	3 SWS	Practical course / On-Site	Oberweis, Rybinski, Ullrich
ST 2026	2512554	Practical Course: Security, Usability and Society	3 SWS	Practical course / Blended	Volkamer, Strufe, Hennig, Länge, Fallahi, Ballreich, Sikra, Hilt, Reimer, Algedri, Mack
WT 26/27	2512554	Praktikum Security, Usability and Society (Bachelor)	3 SWS	Practical course / Blended	Volkamer, Strufe, Berens, Fallahi, Ballreich, Hennig, Länge, Mossano, Hilt

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
The alternative exam assessment consists of:

- a practical work
- a presentation and
- a written seminar thesis

Practical work, presentation and written thesis are weighted according to the course.

Prerequisites
None

Additional Information
The title of this course is a generic one. Specific titles and the topics of offered seminars will be announced before the start of a semester in the internet at https://portal.wiwi.kit.edu.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Lab Realisation of innovative services (Bachelor)
2512204, WS 25/26, 3 SWS, Language: German, Open in study portal

Practical course (P)
Blended (On-Site/Online)

Content
As part of the lab, the participants should work together in small groups to realize innovative services (mainly for students).

Organizational issues
Informationen zu Themen und die Anmeldung erfolgt vor Praktikumsbeginn im Wiwi-Portal
https://portal.wiwi.kit.edu/ys

V **Praktikum Security, Usability and Society (Bachelor)**
2512554, WS 25/26, 3 SWS, Language: English, Open in study portal

Practical course (P)
Blended (On-Site/Online)

Content
In the lab course “Security, Usability and Society”, students explore practical and interdisciplinary topics in the field of IT security and privacy at the cutting edge of society. In addition to the programming of data-saving apps, the development or implementation of user studies can also be possible tasks in this course.

The course can be credited towards the KASTEL certificate. Further information about the KASTEL certificate can be found on the SECUSO website: https://secuso.aifb.kit.edu/Studium_und_Lehre.php

Prerequisites:

The internship is aimed at Bachelor's and Master's students from the Industrial Engineering and Management, Business Informatics and Computer Science degree programs as well as related degree programs.

Organization:

There are two **mandatory** attendance dates: The kick-off is scheduled for the first week of the lectures, and the final presentations will take place in the second-to-last week of lectures. Additional dates will be arranged individually with the supervisors. All in-person lectures will be held in English. The main components of the course are the work on the respective topic, a final presentation, and a final report. After consultation with the supervisor, all components can be completed in either German or English.

<u>Please note:</u> In-person attendance at the kick-off and final presentation is mandatory to be allocated a topic.

If you have any questions about the course or the registration, please contact contact@secuso.org.

Registration:

The topics for the course, as well as the registration, are organized via the WiWi Portal. To reserve a place and select a topic, students register for the course in the WiWi Portal. A description of the current topics as well as important dates and deadlines can also be found there.

Please note that the number of topics is limited, and topics are allocated in the order of registration.

V **Lab Realisation of innovative services (Bachelor)**
2512204, SS 2026, 3 SWS, Language: German, Open in study portal

Practical course (P)
On-Site

Content
As part of the lab, the participants should work together in small groups to realize innovative services (mainly for students).

Organizational issues
Informationen zu Themen und die Anmeldung erfolgt bis Mitte März im Wiwi-Portal
https://portal.wiwi.kit.edu/ys

V **Lab Agentic AI im Smart IoT Office (Bachelor)**
2512206, SS 2026, 3 SWS, Language: German, Open in study portal

Practical course (P)
On-Site

Content
As part of the lab, various topics on everyday automation are offered. During the lab, the participants will gain an insight into problem-solving oriented project work and work on a project together in small groups.

In case of questions, please contact fabian.rybinski@kit.edu.

Organizational issues
Informationen zu Themen und die Anmeldung erfolgt vor Praktikumsbeginn im Wiwi-Portal
https://portal.wiwi.kit.edu/ys

Bei Fragen bitte an fabian.rybinski@kit.edu wenden.

V **Practical Course: Security, Usability and Society**
2512554, SS 2026, 3 SWS, Language: English, Open in study portal

Practical course (P)
Blended (On-Site/Online)

Content
In the lab-course “Security, Usability and Society”, students deal with practical and interdisciplinary topics from the field of IT security and privacy at the cutting edge of society. In addition to the programming of data-saving apps, the development or implementation of user studies can also be possible tasks in this course.

The course can be credited towards the KASTEL certificate. Further information about the KASTEL certificate can be found on the SECUSO website: https://secuso.aifb.kit.edu/Studium_und_Lehre.php

Prerequisites:

The internship is aimed at Bachelor's and Master's students from the Industrial Engineering and Management, Business Informatics and Computer Science degree programs as well as related degree programs.

Organization:

There are two mandatory attendance dates: The kick-off is scheduled for the first week of the lectures, and the final presentations will take place in the second to last week of lectures. Additional dates will be arranged individually with the supervisors. All in-person lectures will be held in English. The main components of the course is the work on the respective topic, a final presentation and a final report. After consultation with the supervisor, all components can be either completed in German or English.

If you have any questions about the course or the registration, please contact contact@secuso.org.

Registration:

The topics for the course as well as the registration is organized via the WiWi-Portal. To reserve a place and choose a topic, students register for the course in the WiWi-Portal. A description of the current topics as well as important dates and deadlines can also be found there.

Please note that the number of topics is limited and topics are allocated in the order of registration.

V **Praktikum Security, Usability and Society (Bachelor)**
2512554, WS 26/27, 3 SWS, Language: English, Open in study portal

Practical course (P)
Blended (On-Site/Online)

Content
In the lab course “Security, Usability and Society”, students explore practical and interdisciplinary topics in the field of IT security and privacy at the cutting edge of society. In addition to the programming of data-saving apps, the development or implementation of user studies can also be possible tasks in this course.

The course can be credited towards the KASTEL certificate. Further information about the KASTEL certificate can be found on the SECUSO website: https://secuso.aifb.kit.edu/Studium_und_Lehre.php

Prerequisites:

The internship is aimed at Bachelor's and Master's students from the Industrial Engineering and Management, Business Informatics and Computer Science degree programs as well as related degree programs.

Organization:

There are two **mandatory** attendance dates: The kick-off is scheduled for the first week of the lectures, and the final presentations will take place in the second-to-last week of lectures. Additional dates will be arranged individually with the supervisors. All in-person lectures will be held in English. The main components of the course are the work on the respective topic, a final presentation, and a final report. After consultation with the supervisor, all components can be completed in either German or English.

Please note: In-person attendance at the kick-off and final presentation is mandatory to be allocated a topic.

If you have any questions about the course or the registration, please contact contact@secuso.org.

Registration:

The topics for the course, as well as the registration, are organized via the WiWi Portal. To reserve a place and select a topic, students register for the course in the WiWi Portal. A description of the current topics as well as important dates and deadlines can also be found there.

Please note that the number of topics is limited, and topics are allocated in the order of registration.

T 6.13 Module component: Advanced Lab Informatics (Master) [T-WIWI-110548]

Coordinators: Professorenschaft des Instituts AIFB
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101456 - Intelligent Systems and Services
M-WIWI-101472 - Informatics
M-WIWI-101477 - Development of Business Information Systems
M-WIWI-101628 - Emphasis in Informatics
M-WIWI-104901 - Informatics (KIT-Department of Economics and Management)
M-WIWI-105366 - Artificial Intelligence
M-WIWI-106803 - Advanced Topics in AI: Knowledge Graphs and the Web
M-WIWI-106804 - Advanced Topics in AI: Graph Neural Networks and Language Models

Type	Credits	Grading	Term offered	Version
Examination of another type	4,5 CP	graded	Each term	1

Courses					
WT 25/26	2512205	Lab Realisation of innovative services (Master)	3 SWS	Practical course / Blended	Schiefer, Toussaint
WT 25/26	2512314	Practical Course Linked Data and the Semantic Web (Master)	3 SWS	Practical course / On-Site	Käfer, Braun, Li, Kubelka
WT 25/26	2512501	Practical Course Cognitive automobiles and robots (Master)	3 SWS	Practical course / Blended	Zöllner, Schneider
WT 25/26	2512555	Praktikum Security, Usability and Society (Master)	3 SWS	Practical course / Blended	Volkamer, Strufe, Berens, Fallahi, Ballreich, Hennig, Länge, Mossano, Hilt
WT 25/26	2512600	Project lab Knowledge-driven Artificial Intelligence (Master)	3 SWS	Practical course / Blended	Sack, Tietz, Vafaie
ST 2026	2512205	Lab Realisation of innovative services (Master)	3 SWS	Practical course / On-Site	Schiefer, Toussaint, Kruse
ST 2026	2512207	Lab Agentic AI im Smart IoT Office (Master)	3 SWS	Practical course / On-Site	Oberweis, Rybinski, Ullrich
ST 2026	2512500	Project Lab Machine Learning	3 SWS	Practical course / On-Site	Zöllner, Schneider
ST 2026	2512555	Praktikum Security, Usability and Society (Master)	3 SWS	Practical course / Blended	Volkamer, Strufe, Hennig, Fallahi, Länge, Ballreich, Hilt, Sikra, Algedri, Reimer, Mack
WT 26/27	2512555	Praktikum Security, Usability and Society (Master)	3 SWS	Practical course / Blended	Volkamer, Strufe, Berens, Fallahi, Ballreich, Hennig, Länge, Mossano, Hilt

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
The alternative exam consists of a single examination of another type that combines three mandatory components:

- Practical work
- Presentation
- Written thesis

All three parts are integrated into one overall assessment. Their individual contributions are weighted according to the course-specific weighting scheme, but only one final grade is awarded for the whole examination. The final grade reflects the total impression of the combined performance.

Prerequisites
None

Additional Information
The title of this course is a generic one. Specific titles and the topics of offered seminars will be announced before the start of a semester in the internet at https://portal.wiwi.kit.edu.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V **Lab Realisation of innovative services (Master)**
2512205, WS 25/26, 3 SWS, Language: German, Open in study portal

Practical course (P)
Blended (On-Site/Online)

Content
As part of the lab, the participants should work together in small groups to realize innovative services (mainly for students).

Organizational issues
Informationen zu Themen und die Anmeldung erfolgt vor Praktikumsbeginn im Wiwi-Portal
https://portal.wiwi.kit.edu/ys

V **Practical Course Linked Data and the Semantic Web (Master)**
2512314, WS 25/26, 3 SWS, Language: German/English, Open in study portal

Practical course (P)
On-Site

Content
Linked Data is a way of publishing data on the web in a machine-understandable fashion. The aim of this practical seminar is to build applications and devise algorithms that consume, provide, or analyse Linked Data.

The Linked Data principles are a set of practices for data publishing on the web. Linked Data builds on the web architecture and uses HTTP for data access, and RDF for describing data, thus aiming towards web-scale data integration. There is a vast amount of data available published according to those principles: recently, 4.5 billion facts have been counted with information about various domains, including music, movies, geography, natural sciences. Linked Data is also used to make web-pages machine-understandable, corresponding annotations are considered by the big search engine providers. On a smaller scale, devices on the Internet of Things can also be accessed using Linked Data which makes the unified processing of device data and data from the web easy.

In this practical seminar, students will build prototypical applications and devise algorithms that consume, provide, or analyse Linked Data. Those applications and algorithms can also extend existing applications ranging from databases to mobile apps.

For the seminar, programming skills or knowledge about web development tools/technologies are highly recommended. Basic knowledge of RDF and SPARQL are also recommended, but may be acquired during the seminar. Students will work in groups. Seminar meetings will take place as 'Block-Seminar'.

Topics of interest include, but are not limited to:

- Travel Security
- Geo data
- Linked News
- Social Media

The exact dates and information for registration will be announced at the event page.

V **Practical Course Cognitive automobiles and robots (Master)**
2512501, WS 25/26, 3 SWS, Language: German/English, Open in study portal

Practical course (P)
Blended (On-Site/Online)

Content
The lab is intended as a practical supplement to courses such as "Machine Learning 1/2".
Scientific topics, mostly in the area of autonomous driving and robotics, will be addressed in joint work with ML/KI methods. The goal of the internship is for participants to design, develop, and evaluate ML Software system.

In addition to the scientific goals, such as the study and application of methods, the aspects of project-specific teamwork in research (from specification to presentation of results) are also worked on in this internship.

The individual projects require the analysis of the set task, selection of appropriate methods, specification and implementation and evaluation of the solution approach. Finally, the selected solution is to be documented and presented in a short lecture.

Learning Objectives:

- Students will be able to practically apply theoretical knowledge from lectures on machine learning to a selected area of current research.
- Students will be proficient in analyzing and solving thematic problems.
- Students will be able to evaluate, document, and present their concepts and results.

Recommendations:

- Theoretical knowledge of machine learning and/or AI.
- Python knowledge
- Initial experience with deep learning frameworks such as PyTorch/Jax/Tensorflow may be beneficial.

Workload:

The workload of 5 credit points consists of practical implementation of the selected solution, as well as time for literature research and planning/specification of the selected solution. In addition, a short report and presentation of the work performed will be prepared.

Organizational issues
Anmeldung und weitere Informationen sind im Wiwi-Portal zu finden.

Registration and further information can be found in the WiWi-portal.

V	**Praktikum Security, Usability and Society (Master)** 2512555, WS 25/26, 3 SWS, Language: English, Open in study portal	**Practical course (P)** **Blended (On-Site/Online)**

Content
In the lab course “Security, Usability and Society”, students explore practical and interdisciplinary topics in the field of IT security and privacy at the cutting edge of society. In addition to the programming of data-saving apps, the development or implementation of user studies can also be possible tasks in this course.

The course can be credited towards the KASTEL certificate. Further information about the KASTEL certificate can be found on the SECUSO website: https://secuso.aifb.kit.edu/Studium_und_Lehre.php

Prerequisites:

The internship is aimed at Bachelor's and Master's students from the Industrial Engineering and Management, Business Informatics and Computer Science degree programs as well as related degree programs.

Organization:

There are two **mandatory** attendance dates: The kick-off is scheduled for the first week of the lectures, and the final presentations will take place in the second-to-last week of lectures. Additional dates will be arranged individually with the supervisors. All in-person lectures will be held in English. The main components of the course are the work on the respective topic, a final presentation, and a final report. After consultation with the supervisor, all components can be completed in either German or English.

<u>Please note:</u> In-person attendance at the kick-off and final presentation is mandatory to be allocated a topic.

If you have any questions about the course or the registration, please contact contact@secuso.org.

Registration:

The topics for the course, as well as the registration, are organized via the WiWi Portal. To reserve a place and select a topic, students register for the course in the WiWi Portal. A description of the current topics as well as important dates and deadlines can also be found there.

Please note that the number of topics is limited, and topics are allocated in the order of registration.

V	**Project lab Knowledge-driven Artificial Intelligence (Master)** 2512600, WS 25/26, 3 SWS, Language: English, Open in study portal	**Practical course (P)** **Blended (On-Site/Online)**

Content

The **project course Knowledge-driven AI** is based on the summer semester lecture "Information Service Engineering". Goal of the project course is to work on a (real) research problem in small groups related to the ISE lecture topics, including:

- Large Language Models (LLM)
- Natural Language Processing (NLP),
- Knowledge Graphs (KG),
- Machine Learning (ML)

Previous project courses were based on real (funded) research projects and included project tasks, as e.g., leveraging (large) language models for OCR post-processing of scanned historic documents, knowledge extraction and knowledge graph creation from historic documents, entity typing based on knowledge graph embeddings and language models, knowledge graph completion, semantic and exploratory search based on knowledge graphs, and many more.

The project course topics this semester include:

- **"Telling Data Stories with Semantic Technologies and Generative AI"**.
  This topic addresses the challenge that large Knowledge Graphs are often overwhelming for non-technical users due to their complexity, making it difficult to understand the structures and contents in a clear and intuitive way. Data Stories are designed to help users explore data; they simplify the complex relationships within Knowledge Graphs, reveal "hidden" connections and patterns, and provide narrative summaries. In this course, we aim to conceptualize and implement methods for creating Data Stories from large and complex Knowledge Graphs. This includes the creation of engaging visualizations and the use of generative AI to bridge the gap between data creators and users. This topic is part of the currently running project NFDI4Culture. The course results will be published on the NFDI4Culture Portal.
- **"Leveraging Large Language Models to Suggest Heuristics for Ontology Reasoning".**
  Ontology reasoning is a key component of semantic technologies, but traditional reasoning engines often face scalability issues with large or complex ontologies. With the rise of Large Language Models (LLMs), there is growing interest in exploring their potential to support symbolic reasoning. This project presents a preliminary study on using LLMs to suggest simple heuristics for ontology reasoning optimization. Rather than replacing existing reasoners, LLMs will serve as heuristic advisors—for example, by identifying structural patterns or recommending strategies for small ontologies. The project will involve lightweight experiments on selected ontology datasets, comparing reasoning performance with and without LLM-guided heuristics. The goal is to provide an initial exploration of the feasibility of LLM-assisted reasoning, laying the foundation for more advanced research.
- **"Large Language Models for Ontology Alignment with Foundational Ontologies"**
  Ontology alignment with foundational ontologies like BFO is important but often complex and time-consuming. This topic explores how Large Language Models (LLMs) can support or automate this process by interpreting ontology classes, suggesting alignments, and explaining their reasoning.
- **"Information Extraction in the era of Large Language Models"**
  Large Language Models (LLMs) are known to produce inaccurate or even misleading answers. One approach to mitigate this gross limitation is to enhance their responses with verified, structured data from knowledge graphs. The process of extracting structured data from unstructured texts often employs task-specific language models. In this project, students will explore and implement cutting-edge techniques for Named Entity Recognition (NER) using LLMs, comparing different approaches and evaluating their effectiveness in transforming unstructured text into structured data.

In this course you have the chance to combine creativity and practical implementation tasks to develop solutions for real-world projects and problems. The project will be worked on in **teams of 3-4 students** each, guided by a tutor from the teaching staff.

The **kick-off meeting** will take place in person on Wednesday, October 29th in room 05-09 (AIFB building 05.20), 9:45am.

The project course will be held in a **hybrid format**. The kick-off session will take place in person, and the individual group meetings throughout the semester will take place online. Each group can decide together with their tutors on regular meeting times (30min each).

For questions, please reach out to Prof. Harald Sack (harald.sack@kit.edu) or Tabea Tietz (tabea.tietz@kit.edu).

Required coursework for each student team include:

- Mid term presentation (5-10 min)
- Final presentation (10-15 min)
- Course report (c. 16 pages)
- Participation and contribution of the students during the course
- Software development and delivery

Notes:

- If need be, the ISE project course can also be credited as a seminar.
- The project course will be restricted to overall 16 participants max.
- Participation in the lecture "Information Service Engineering" (summer semester) is not necessarily required, but highly beneficial (the lecture recordings are available on youtube).

Outcomes of previous project course editions:

- Hackathon Coding da Vinci 2022: Winners in the category "most FAIR application"
- Paper publications:

- ◦ M. Li, H. Yang, Z. Liu, M.M. Alam, E. Norouzi, H. Sack and G.A. Gesese: KGMistral: Towards Boosting the Performance of Large Language Models for Question Answering with Knowledge Graph Integration. In Workshop on Deep Learning and Large Language Models for Knowledge Graphs 2024
- ◦ O. Bruns, T. Tietz, M. B. Chaabane, M. Portz, F. Xiong, and H. Sack: The Nuremberg Address Knowledge Graph. In: Proceedings of the 17th Extended Semantic Web Conference, P&D Track 2021 [PDF]

Literature
ISE video channel on youtube: https://www.youtube.com/channel/UCjkkhNSNuXrJpMYZoeSBw6Q/

V Lab Realisation of innovative services (Master)

2512205, SS 2026, 3 SWS, Language: German, Open in study portal

Practical course (P)
On-Site

Content
As part of the lab, the participants should work together in small groups to realize innovative services (mainly for students).

Organizational issues
Informationen zu Themen und die Anmeldung erfolgt bis Mitte März im Wiwi-Portal
https://portal.wiwi.kit.edu/ys

V Lab Agentic AI im Smart IoT Office (Master)

2512207, SS 2026, 3 SWS, Language: German, Open in study portal

Practical course (P)
On-Site

Content
As part of the lab, various topics on everyday automation are offered. During the lab, the participants will gain an insight into problem-solving oriented project work and work on a project together in small groups.

In case of questions, please contact fabian.rybinski@kit.edu.

Organizational issues
Informationen zu Themen und die Anmeldung erfolgt vor Praktikumsbeginn im Wiwi-Portal
https://portal.wiwi.kit.edu/ys

Bei Fragen bitte an fabian.rybinski@kit.edu wenden.

V Project Lab Machine Learning

2512500, SS 2026, 3 SWS, Language: German/English, Open in study portal

Practical course (P)
On-Site

Content
The lab is intended as a practical supplement to lectures such as "Machine Learning". The theoretical basics are applied in the lab course. The aim of the lab course is that the participants work together to design, develop and evaluate a subsystem from the field of robotics and cognitive systems using one or more procedures from the field of AI/ML.

In addition to the scientific objectives involved in the investigation and application of the methods, aspects of project-specific teamwork in research (from specification to presentation of the results) are also developed in this practical course.

The individual projects require the analysis of the task at hand, selection of suitable procedures, specification and implementation and evaluation of the approach taken. Finally, the chosen solution has to be documented and presented in a short presentation.

Learning objectives:

- Students can practically apply knowledge from the Machine Learning lecture in a selected field of current research in robotics or cognitive automobiles.
- Students master the analysis and solution of corresponding problems in a team.
- Students can evaluate, document and present their concepts and results.

Recommendations:

Attendance of the lecture machine learning, C/C++ knowledge, Python knowledge

Workload:

The workload of 5 credit points consists of the time spent in the lab for practical implementation of the selected solution, as well as the time spent on literature research and planning/specifying the proposed solution. In addition, a short report and a presentation of the work carried out will be prepared.

Organizational issues
Anmeldung und weitere Informationen sind im Wiwi-Portal zu finden.

Registration and further information can be found in the WiWi-portal.

V **Praktikum Security, Usability and Society (Master)**
2512555, SS 2026, 3 SWS, Language: English, Open in study portal

Practical course (P)
Blended (On-Site/Online)

Content
In the lab-course "Security, Usability and Society", students deal with practical and interdisciplinary topics from the field of IT security and privacy at the cutting edge of society. In addition to the programming of data-saving apps, the development or implementation of user studies can also be possible tasks in this course.

The course can be credited towards the KASTEL certificate. Further information about the KASTEL certificate can be found on the SECUSO website: https://secuso.aifb.kit.edu/Studium_und_Lehre.php

Prerequisites:

The internship is aimed at Bachelor's and Master's students from the Industrial Engineering and Management, Business Informatics and Computer Science degree programs as well as related degree programs.

Organization:

There are two mandatory attendance dates: The kick-off is scheduled for the first week of the lectures, and the final presentations will take place in the second to last week of lectures. Additional dates will be arranged individually with the supervisors. All in-person lectures will be held in English. The main components of the course is the work on the respective topic, a final presentation and a final report. After consultation with the supervisor, all components can be either completed in German or English.

If you have any questions about the course or the registration, please contact contact@secuso.org.

Registration:

The topics for the course as well as the registration is organized via the WiWi-Portal. To reserve a place and choose a topic, students register for the course in the WiWi-Portal. A description of the current topics as well as important dates and deadlines can also be found there.

Please note that the number of topics is limited and topics are allocated in the order of registration.

V **Praktikum Security, Usability and Society (Master)**
2512555, WS 26/27, 3 SWS, Language: English, Open in study portal

Practical course (P)
Blended (On-Site/Online)

Content
In the lab course "Security, Usability and Society", students explore practical and interdisciplinary topics in the field of IT security and privacy at the cutting edge of society. In addition to the programming of data-saving apps, the development or implementation of user studies can also be possible tasks in this course.

The course can be credited towards the KASTEL certificate. Further information about the KASTEL certificate can be found on the SECUSO website: https://secuso.aifb.kit.edu/Studium_und_Lehre.php

Prerequisites:

The internship is aimed at Bachelor's and Master's students from the Industrial Engineering and Management, Business Informatics and Computer Science degree programs as well as related degree programs.

Organization:

There are two **mandatory** attendance dates: The kick-off is scheduled for the first week of the lectures, and the final presentations will take place in the second-to-last week of lectures. Additional dates will be arranged individually with the supervisors. All in-person lectures will be held in English. The main components of the course are the work on the respective topic, a final presentation, and a final report. After consultation with the supervisor, all components can be completed in either German or English.

Please note: In-person attendance at the kick-off and final presentation is mandatory to be allocated a topic.

If you have any questions about the course or the registration, please contact contact@secuso.org.

Registration:

The topics for the course, as well as the registration, are organized via the WiWi Portal. To reserve a place and select a topic, students register for the course in the WiWi Portal. A description of the current topics as well as important dates and deadlines can also be found there.

Please note that the number of topics is limited, and topics are allocated in the order of registration.

T 6.14 Module component: Advanced Lab Realization of Innovative Services (Bachelor) [T-WIWI-112915]

Coordinators: Prof. Dr. Andreas Oberweis
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104901 - Informatics (KIT-Department of Economics and Management)

Type	Credits	Grading	Term offered	Version
Examination of another type	4,5 CP	graded	Each term	1

Courses					
WT 25/26	2512204	Lab Realisation of innovative services (Bachelor)	3 SWS	Practical course / Blended	Schiefer, Toussaint
ST 2026	2512204	Lab Realisation of innovative services (Bachelor)	3 SWS	Practical course / On-Site	Schiefer, Toussaint, Kruse

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
The alternative exam assessment consists of:

- a practical work
- a presentation and
- a written seminar thesis

Practical work, presentation and written thesis are weighted according to the course.

Additional Information
As part of the lab, the participants should work together in small groups to produce innovative services (mainly for students).

Further information can be found on the ILIAS page of the lab.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Lab Realisation of innovative services (Bachelor)
2512204, WS 25/26, 3 SWS, Language: German, Open in study portal

Practical course (P)
Blended (On-Site/Online)

Content
As part of the lab, the participants should work together in small groups to realize innovative services (mainly for students).

Organizational issues
Informationen zu Themen und die Anmeldung erfolgt vor Praktikumsbeginn im Wiwi-Portal
https://portal.wiwi.kit.edu/ys

V Lab Realisation of innovative services (Bachelor)
2512204, SS 2026, 3 SWS, Language: German, Open in study portal

Practical course (P)
On-Site

Content
As part of the lab, the participants should work together in small groups to realize innovative services (mainly for students).

Organizational issues
Informationen zu Themen und die Anmeldung erfolgt bis Mitte März im Wiwi-Portal
https://portal.wiwi.kit.edu/ys

T 6.15 Module component: Advanced Lab Realization of Innovative Services (Master) [T-WIWI-112914]

Coordinators: Prof. Dr. Andreas Oberweis
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101472 - Informatics
M-WIWI-101477 - Development of Business Information Systems
M-WIWI-101628 - Emphasis in Informatics
M-WIWI-104901 - Informatics (KIT-Department of Economics and Management)

Type	Credits	Grading	Term offered	Version
Examination of another type	4,5 CP	graded	Each term	1

Courses					
WT 25/26	2512205	Lab Realisation of innovative services (Master)	3 SWS	Practical course / Blended	Schiefer, Toussaint
ST 2026	2512205	Lab Realisation of innovative services (Master)	3 SWS	Practical course / On-Site	Schiefer, Toussaint, Kruse

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
The alternative exam assessment consists of:

- a practical work
- a presentation and
- a written seminar thesis

Practical work, presentation and written thesis are weighted according to the course.

Additional Information
As part of the lab, the participants should work together in small groups to produce innovative services (mainly for students).
Further information can be found on the ILIAS page of the lab.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Lab Realisation of innovative services (Master)
2512205, WS 25/26, 3 SWS, Language: German, Open in study portal

Practical course (P)
Blended (On-Site/Online)

Content
As part of the lab, the participants should work together in small groups to realize innovative services (mainly for students).

Organizational issues
Informationen zu Themen und die Anmeldung erfolgt vor Praktikumsbeginn im Wiwi-Portal
https://portal.wiwi.kit.edu/ys

V Lab Realisation of innovative services (Master)
2512205, SS 2026, 3 SWS, Language: German, Open in study portal

Practical course (P)
On-Site

Content
As part of the lab, the participants should work together in small groups to realize innovative services (mainly for students).

Organizational issues
Informationen zu Themen und die Anmeldung erfolgt bis Mitte März im Wiwi-Portal
https://portal.wiwi.kit.edu/ys

T 6.16 Module component: Advanced Lab Security, Usability and Society [T-WIWI-108439]

Coordinators: Prof. Dr. Melanie Volkamer
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101472 - Informatics
M-WIWI-101628 - Emphasis in Informatics
M-WIWI-104520 - Human Factors in Security and Privacy
M-WIWI-104901 - Informatics (KIT-Department of Economics and Management)

Type	Credits	Grading	Term offered	Version
Examination of another type	4,5 CP	graded	Each term	3

Courses					
WT 25/26	2512554	Praktikum Security, Usability and Society (Bachelor)	3 SWS	Practical course / Blended (On-Site/Online)	Volkamer, Strufe, Berens, Fallahi, Ballreich, Hennig, Länge, Mossano, Hilt
WT 25/26	2512555	Praktikum Security, Usability and Society (Master)	3 SWS	Practical course / Blended (On-Site/Online)	Volkamer, Strufe, Berens, Fallahi, Ballreich, Hennig, Länge, Mossano, Hilt
ST 2026	2512554	Practical Course: Security, Usability and Society	3 SWS	Practical course / Blended (On-Site/Online)	Volkamer, Strufe, Hennig, Länge, Fallahi, Ballreich, Sikra, Hilt, Reimer, Algedri, Mack
ST 2026	2512555	Praktikum Security, Usability and Society (Master)	3 SWS	Practical course / Blended (On-Site/Online)	Volkamer, Strufe, Hennig, Fallahi, Länge, Ballreich, Hilt, Sikra, Algedri, Reimer, Mack
WT 26/27	2512554	Praktikum Security, Usability and Society (Bachelor)	3 SWS	Practical course / Blended (On-Site/Online)	Volkamer, Strufe, Berens, Fallahi, Ballreich, Hennig, Länge, Mossano, Hilt
WT 26/27	2512555	Praktikum Security, Usability and Society (Master)	3 SWS	Practical course / Blended (On-Site/Online)	Volkamer, Strufe, Berens, Fallahi, Ballreich, Hennig, Länge, Mossano, Hilt

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
Assessment takes the form of an alternative exam. It consists of a practical assignment, a presentation, and, if required, a written report. The lectures and the presentations will be in English. The written report can be written in either English or German after consultation with the supervisor.
Practical work, presentation, and written thesis are weighted according to the course.

Prerequisites
None

Recommendations
Knowledge from the lecture "Applied Informatics - Cybersecurity" is recommended.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Praktikum Security, Usability and Society (Bachelor)

2512554, WS 25/26, 3 SWS, Language: English, Open in study portal

Practical course (P)
Blended (On-Site/Online)

Content
In the lab course “Security, Usability and Society”, students explore practical and interdisciplinary topics in the field of IT security and privacy at the cutting edge of society. In addition to the programming of data-saving apps, the development or implementation of user studies can also be possible tasks in this course.

The course can be credited towards the KASTEL certificate. Further information about the KASTEL certificate can be found on the SECUSO website: https://secuso.aifb.kit.edu/Studium_und_Lehre.php

Prerequisites:

The internship is aimed at Bachelor's and Master's students from the Industrial Engineering and Management, Business Informatics and Computer Science degree programs as well as related degree programs.

Organization:

There are two **mandatory** attendance dates: The kick-off is scheduled for the first week of the lectures, and the final presentations will take place in the second-to-last week of lectures. Additional dates will be arranged individually with the supervisors. All in-person lectures will be held in English. The main components of the course are the work on the respective topic, a final presentation, and a final report. After consultation with the supervisor, all components can be completed in either German or English.

Please note: In-person attendance at the kick-off and final presentation is mandatory to be allocated a topic.

If you have any questions about the course or the registration, please contact contact@secuso.org.

Registration:

The topics for the course, as well as the registration, are organized via the WiWi Portal. To reserve a place and select a topic, students register for the course in the WiWi Portal. A description of the current topics as well as important dates and deadlines can also be found there.

Please note that the number of topics is limited, and topics are allocated in the order of registration.

V	**Praktikum Security, Usability and Society (Master)** 2512555, WS 25/26, 3 SWS, Language: English, Open in study portal	**Practical course (P)** **Blended (On-Site/Online)**

Content
In the lab course “Security, Usability and Society”, students explore practical and interdisciplinary topics in the field of IT security and privacy at the cutting edge of society. In addition to the programming of data-saving apps, the development or implementation of user studies can also be possible tasks in this course.

The course can be credited towards the KASTEL certificate. Further information about the KASTEL certificate can be found on the SECUSO website: https://secuso.aifb.kit.edu/Studium_und_Lehre.php

Prerequisites:

The internship is aimed at Bachelor's and Master's students from the Industrial Engineering and Management, Business Informatics and Computer Science degree programs as well as related degree programs.

Organization:

There are two **mandatory** attendance dates: The kick-off is scheduled for the first week of the lectures, and the final presentations will take place in the second-to-last week of lectures. Additional dates will be arranged individually with the supervisors. All in-person lectures will be held in English. The main components of the course are the work on the respective topic, a final presentation, and a final report. After consultation with the supervisor, all components can be completed in either German or English.

Please note: In-person attendance at the kick-off and final presentation is mandatory to be allocated a topic.

If you have any questions about the course or the registration, please contact contact@secuso.org.

Registration:

The topics for the course, as well as the registration, are organized via the WiWi Portal. To reserve a place and select a topic, students register for the course in the WiWi Portal. A description of the current topics as well as important dates and deadlines can also be found there.

Please note that the number of topics is limited, and topics are allocated in the order of registration.

V	**Practical Course: Security, Usability and Society** 2512554, SS 2026, 3 SWS, Language: English, Open in study portal	**Practical course (P)** **Blended (On-Site/Online)**

Content

In the lab-course “Security, Usability and Society”, students deal with practical and interdisciplinary topics from the field of IT security and privacy at the cutting edge of society. In addition to the programming of data-saving apps, the development or implementation of user studies can also be possible tasks in this course.

The course can be credited towards the KASTEL certificate. Further information about the KASTEL certificate can be found on the SECUSO website: https://secuso.aifb.kit.edu/Studium_und_Lehre.php

Prerequisites:

The internship is aimed at Bachelor's and Master's students from the Industrial Engineering and Management, Business Informatics and Computer Science degree programs as well as related degree programs.

Organization:

There are two mandatory attendance dates: The kick-off is scheduled for the first week of the lectures, and the final presentations will take place in the second to last week of lectures. Additional dates will be arranged individually with the supervisors. All in-person lectures will be held in English. The main components of the course is the work on the respective topic, a final presentation and a final report. After consultation with the supervisor, all components can be either completed in German or English.

If you have any questions about the course or the registration, please contact contact@secuso.org.

Registration:

The topics for the course as well as the registration is organized via the WiWi-Portal. To reserve a place and choose a topic, students register for the course in the WiWi-Portal. A description of the current topics as well as important dates and deadlines can also be found there.

Please note that the number of topics is limited and topics are allocated in the order of registration.

V **Praktikum Security, Usability and Society (Master)**
2512555, SS 2026, 3 SWS, Language: English, Open in study portal

Practical course (P)
Blended (On-Site/Online)

Content

In the lab-course “Security, Usability and Society”, students deal with practical and interdisciplinary topics from the field of IT security and privacy at the cutting edge of society. In addition to the programming of data-saving apps, the development or implementation of user studies can also be possible tasks in this course.

The course can be credited towards the KASTEL certificate. Further information about the KASTEL certificate can be found on the SECUSO website: https://secuso.aifb.kit.edu/Studium_und_Lehre.php

Prerequisites:

The internship is aimed at Bachelor's and Master's students from the Industrial Engineering and Management, Business Informatics and Computer Science degree programs as well as related degree programs.

Organization:

There are two mandatory attendance dates: The kick-off is scheduled for the first week of the lectures, and the final presentations will take place in the second to last week of lectures. Additional dates will be arranged individually with the supervisors. All in-person lectures will be held in English. The main components of the course is the work on the respective topic, a final presentation and a final report. After consultation with the supervisor, all components can be either completed in German or English.

If you have any questions about the course or the registration, please contact contact@secuso.org.

Registration:

The topics for the course as well as the registration is organized via the WiWi-Portal. To reserve a place and choose a topic, students register for the course in the WiWi-Portal. A description of the current topics as well as important dates and deadlines can also be found there.

Please note that the number of topics is limited and topics are allocated in the order of registration.

V **Praktikum Security, Usability and Society (Bachelor)**
2512554, WS 26/27, 3 SWS, Language: English, Open in study portal

Practical course (P)
Blended (On-Site/Online)

Content
In the lab course “Security, Usability and Society”, students explore practical and interdisciplinary topics in the field of IT security and privacy at the cutting edge of society. In addition to the programming of data-saving apps, the development or implementation of user studies can also be possible tasks in this course.

The course can be credited towards the KASTEL certificate. Further information about the KASTEL certificate can be found on the SECUSO website: https://secuso.aifb.kit.edu/Studium_und_Lehre.php

Prerequisites:

The internship is aimed at Bachelor's and Master's students from the Industrial Engineering and Management, Business Informatics and Computer Science degree programs as well as related degree programs.

Organization:

There are two **mandatory** attendance dates: The kick-off is scheduled for the first week of the lectures, and the final presentations will take place in the second-to-last week of lectures. Additional dates will be arranged individually with the supervisors. All in-person lectures will be held in English. The main components of the course are the work on the respective topic, a final presentation, and a final report. After consultation with the supervisor, all components can be completed in either German or English.

Please note: In-person attendance at the kick-off and final presentation is mandatory to be allocated a topic.

If you have any questions about the course or the registration, please contact contact@secuso.org.

Registration:

The topics for the course, as well as the registration, are organized via the WiWi Portal. To reserve a place and select a topic, students register for the course in the WiWi Portal. A description of the current topics as well as important dates and deadlines can also be found there.

Please note that the number of topics is limited, and topics are allocated in the order of registration.

V	**Praktikum Security, Usability and Society (Master)** 2512555, WS 26/27, 3 SWS, Language: English, Open in study portal	**Practical course (P)** **Blended (On-Site/Online)**

Content
In the lab course “Security, Usability and Society”, students explore practical and interdisciplinary topics in the field of IT security and privacy at the cutting edge of society. In addition to the programming of data-saving apps, the development or implementation of user studies can also be possible tasks in this course.

The course can be credited towards the KASTEL certificate. Further information about the KASTEL certificate can be found on the SECUSO website: https://secuso.aifb.kit.edu/Studium_und_Lehre.php

Prerequisites:

The internship is aimed at Bachelor's and Master's students from the Industrial Engineering and Management, Business Informatics and Computer Science degree programs as well as related degree programs.

Organization:

There are two **mandatory** attendance dates: The kick-off is scheduled for the first week of the lectures, and the final presentations will take place in the second-to-last week of lectures. Additional dates will be arranged individually with the supervisors. All in-person lectures will be held in English. The main components of the course are the work on the respective topic, a final presentation, and a final report. After consultation with the supervisor, all components can be completed in either German or English.

Please note: In-person attendance at the kick-off and final presentation is mandatory to be allocated a topic.

If you have any questions about the course or the registration, please contact contact@secuso.org.

Registration:

The topics for the course, as well as the registration, are organized via the WiWi Portal. To reserve a place and select a topic, students register for the course in the WiWi Portal. A description of the current topics as well as important dates and deadlines can also be found there.

Please note that the number of topics is limited, and topics are allocated in the order of registration.

T 6.17 Module component: Advanced Machine Learning [T-WIWI-109921]

Coordinators: Prof. Dr. Andreas Geyer-Schulz
Dr. Abdolreza Nazemi
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration
M-WIWI-105661 - Data Science: Intelligent, Adaptive, and Learning Information Services

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each summer term	1

Courses					
ST 2026	2540535	Advanced Machine Learning	2 SWS	Lecture	Nazemi
ST 2026	2540536	Exercise Advanced Machine Learning	1 SWS	Practice	Nazemi

Assessment
Written examination (60 minutes) according to §4(2), 1 SPO. The exam is considered passed if at least 50 out of a maximum of 100 possible points are achieved. The grades are graded in five steps (best grade 1.0 from 95 points). Details of the grade formation and scale will be announced in the course.
A bonus can be acquired through successful participation in the practice. If the grade of the written examination is between 4.0 and 1.3, the bonus improves the grade by one grade level (0.3 or 0.4). The exact criteria for awarding a bonus will be announced at the beginning of the course.

Please note: this exam will no longer be offered after August 2026.

Prerequisites
None

Additional Information
Please note: this course, exercises and exam will no longer be offered after August 2026.

Below you will find excerpts from courses related to this module component:

V Advanced Machine Learning

2540535, SS 2026, 2 SWS, Language: English, Open in study portal

Lecture (V)

Content
In recent years, the volume, variety, velocity, veracity, and variability of available data have increased due to improvements in computational and storage power. The rise of the Internet has made available large sets of data that allow us to use and merge them for different purposes. Data science helps us to extract knowledge from the continually-increasing large datasets. This course will introduce students to a wide range of machine learning and statistical techniques such as deep learning, LASSO, and support vector machine. You will get familiar with text mining, and the tools you need to analyze the various facets of data sets in practice. Students will learn theory and concepts with real data sets from different disciplines such as marketing, finance, and business.

Tentative Course Outline:

- Introduction
- Statistical Inference
- Shrinkage Methods
- Model Assessment and Selection
- Tree-based Machine Learning Algorithms
- Dimensionality Reduction
- Neural Networks and Deep Learning
- Natural Language Processing with Deep Learning
- Support Vector Machine

Time of attendance

- Attending the lecture: 13 x 90min = 19h 30m
- Attending the exercise classes: 7 x 90min = 10h 30m

The student will learn

- A wide range of machine learning algorithms and their weaknesses.
- The fundamental issues and challenges: data, high-dimension, train, model selection, etc.
- How to imply machine learning algorithms for real-world applications.
- The fundamentals of deep learning, main research activities, and on-going research in this field.

Literature

- Alpaydin, E. (2014). Introduction to Machine Learning. Third Edition, MIT Press.
- De Prado, M. L. (2018). Advances in Financial Machine Learning. John Wiley & Sons.
- Goodfellow, I., Bengio, Y., and A. Courville (2017). Deep Learning. MIT Press. (online available)
- Hastie, T., Tibshirani, R., and J. Friedman (2009). Elements of Statistical Learning. Second Edition. Springer. (online available)
- Leskovec, J., Rajaraman, A., Ullman, J. D., (2014). Mining of Massive Datasets. Cambridge University Press. (online available)
- Witten, I. H., Eibe, F., Hall, M. A., Pal, C. J. (2016). Data Mining: Practical Machine Learning Tools and Techniques. Morgan Kaufmann.

T 6.18 Module component: Advanced Machine Learning and Data Science [T-WIWI-111305]

Coordinators: Prof. Dr. Maxim Ulrich
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration
M-WIWI-105659 - Advanced Machine Learning and Data Science

Type	Credits	Grading	Term offered	Version
Examination of another type	9 CP	graded	Each term	5

Courses					
ST 2026	2500016	Advanced Machine Learning and Data Science	4 SWS	Project / 🗣	Ulrich

Legend: 🖥 Online, Blended (On-Site/Online), 🗣 On-Site, x Cancelled

Assessment
The assessment is carried out in an alternative form. The grade is evaluated based on the intermediate presentations during the project, the quality of the implementation, the final written thesis, and a final presentation.

Additional Information
The course is aimed at students majoring in data science and/or machine learning. It offers students the opportunity to gain practical knowledge of new developments in the fields of data science and machine learning.

Kick-off Meeting for Winter Term 2025/26:
We will introduce project topics designed for students of computer science, engineering, and mathematics. The meeting will provide an overview of the course structure, project topics, and expectations for the semester.

- **Date**: Tuesday, October 28, 2025, Time: 5:00 PM – 7:00 PM
- **Format**: Online (MS Teams link: Kick-Off (WS-2025)-Tuesday, October 28, 2025 | Meeting-Join | Microsoft Teams)
- **Ms Team Meeting link**: https://teams.microsoft.com/meet/3867172715541?p=GS2swv7ueF5gACIUEs
- **Form link**: https://docs.google.com/forms/d/e/1FAIpQLSek36NGe9l7JyIllc4iBnd-XEwbhexFS7Wn_P6Wga1-EKU8Qw/viewform?usp=dialog

After the meeting (within 48 hours): Please submit your Priority A and Priority B topic preferences via the form within 48 hours after the kick-off.

If you need more information, please visit https://risk.fbv.kit.edu/2307.php.

Workload
270 hours

Below you will find excerpts from courses related to this module component:

V Advanced Machine Learning and Data Science
2500016, SS 2026, 4 SWS, Language: English, Open in study portal

Project (PRO)
Blended (On-Site/Online)

Content
The course is targeted to students with a major in Data Science and/or Machine Learning. It offers students the opportunity to develop hands-on knowledge on new developments in data science and machine learning.

Organizational issues
Topics will be presented during the kick-off meeting in the first week.

- Kick-off Meeting: Online, Tuesday, April 21, 2026, 5:00 pm (first week of lectures). The meeting will be posted here soonFor more details about the module, please check the our website

We prepare topics for students of computer science, industrial engineering, and business mathematics.

Topics and student contributors will be matched after the kick-off meeting.

Literature
Literature and computer programs will be announced in the first lecture.

T 6.19 Module component: Advanced Management Accounting [T-WIWI-102885]

Coordinators: Prof. Dr. Marcus Wouters
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101510 - Cross-Functional Management Accounting
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Oral examination	4,5 CP	graded	Each winter term	2

Courses					
WT 25/26	2579907	Advanced Management Accounting	4 SWS	Lecture / 🗣	Wouters, Dickemann, Letmathe
WT 26/27	2579907	Advanced Management Accounting	4 SWS	Lecture / 🗣	Wouters, Letmathe, Binti Mohamad Riduan, Fimpel

Legend: 🖥 Online, ❃ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment consists of an oral exam (30 min) (according to §4 (2), 2 of the examination regulation). The exam takes place in every semester. Re-examinations are offered at every ordinary examination date.

Prerequisites
None.

Recommendations
The course requires significant prior knowledge of Management Accounting, similar to the content of the courses MA 1 and 2, although completion of these particular courses is not a formal requirement.

Additional Information
This course is held in English. Lectures and tutorials are integrated.

The course is compulsory and must be examined.

Students who are interested in attending this course should send an e-mail to Professor Wouters (marc.wouters∂kit.edu).

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Advanced Management Accounting

Lecture (V)
On-Site

2579907, WS 25/26, 4 SWS, Language: English, Open in study portal

Content
This course is held in English. Students who are interested in attending this course should send an e-mail to Professor Wouters (marc.wouters@kit.edu).

Inhalt:

- The course addresses several topics where management accounting is strongly related to marketing, finance, or organization and strategy, such as customer value propositions, financial performance measures, managing new product development, and technology investment decisions.

Learning objectives:

- Students will be able to consider advanced management accounting methods in an interdisciplinary way and to apply these to managerial decision-making problems in operations and innovation.
- They will also be able to identify relevant research results on such methods.

Examination:

- The assessment consists of an oral exam (30 min) taking place in the recess period (according to § 4 (2) No. 2 of the examination regulation).
- The exam takes place in every semester. Re-examinations are offered at every ordinary examination date.

Required prior Courses:

- The course is compulsory and must be examined.

Recommendations:

- The course requires significant prior knowledge of Management Accounting, similar to the content of the courses MA 1 and 2, although completion of these particular courses is not a formal requirement.

Workload:

- The total workload for this course is approximately 135 hours. For further information see German version.

Organizational issues
Die LV wird in englischer Sprache gehalten. Studierende, die Interesse haben, an dieser Lehrveranstaltung teilzunehmen, sollten bitte vorher eine E-Mail an Professor Wouters senden (marc.wouters@kit.edu).

Literature
Literature is mostly made available via ILIAS.

V	**Advanced Management Accounting** 2579907, WS 26/27, 4 SWS, Language: English, Open in study portal	**Lecture (V)** **On-Site**

Content
This course is held in English. Students who are interested in attending this course should send an e-mail to Professor Wouters (marc.wouters@kit.edu).

Inhalt:

- The course addresses several topics where management accounting is strongly related to marketing, finance, or organization and strategy, such as customer value propositions, financial performance measures, managing new product development, and technology investment decisions.

Learning objectives:

- Students will be able to consider advanced management accounting methods in an interdisciplinary way and to apply these to managerial decision-making problems in operations and innovation.
- They will also be able to identify relevant research results on such methods.

Examination:

- The assessment consists of an oral exam (30 min) taking place in the recess period (according to § 4 (2) No. 2 of the examination regulation).
- The exam takes place in every semester. Re-examinations are offered at every ordinary examination date.

Required prior Courses:

- The course is compulsory and must be examined.

Recommendations:

- The course requires significant prior knowledge of Management Accounting, similar to the content of the courses MA 1 and 2, although completion of these particular courses is not a formal requirement.

Workload:

- The total workload for this course is approximately 135 hours. For further information see German version.

Organizational issues
Die LV wird in englischer Sprache gehalten. Studierende, die Interesse haben, an dieser Lehrveranstaltung teilzunehmen, sollten bitte vorher eine E-Mail an Professor Wouters senden (marc.wouters@kit.edu).

Literature
Literature is mostly made available via ILIAS.

T 6.20 Module component: Advanced Programming - Application of Business Software [T-WIWI-102748]

Coordinators: Prof. Dr. Stefan Klink
Prof. Dr. Andreas Oberweis
Dr.-Ing. Clemens Schreiber
Dr.-Ing. Meike Ullrich

Organisation: KIT Department of Business and Economics

Part of: M-WIWI-104901 - Informatics (KIT-Department of Economics and Management)

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each winter term	3

Courses					
WT 25/26	2511026	Advanced Programming - Application of Business Software	2 SWS	Lecture / 🗣	Ullrich
WT 25/26	2511027	Exercises Advanced Programming - Application of Business Software	1 SWS	Practice / 🗣	Ullrich
WT 25/26	2511028	Computer lab Advanced Programming - Application of Business Software	2 SWS	Practice / 🖥	Ullrich

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment

The success control takes place in the form of a written examination. The duration of the exam is 60 minutes. The examination is offered every semester and can be repeated at any regular examination date.
The prerequisite for taking the exam is successful participation in a computer lab during the lecture in the winter semester. Attendance is compulsory for individual dates of the computer lab. More detailed information on registration to the computer lab and exercise sessions will be announced in the first lecture and on the lecture homepage on ILIAS.
Admission to take the exam can only be acquired in the winter semester and is valid indefinitely.

Prerequisites

This course cannot be taken together with *Advanced Programming - Java Network Programming.*

Modeled Prerequisites

The following conditions have to be fulfilled:

1. The module component T-WIWI-102747 - Advanced Programming - Java Network Programming must not have been started.

Recommendations

Knowledge of the course "Foundations of Informatics I und II" are helpful.

Workload

150 hours

Below you will find excerpts from courses related to this module component:

V Advanced Programming - Application of Business Software

2511026, WS 25/26, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content
Business (or Enterprise) Information Systems (BIS), which are often implemented as standard software, form the backbone of modern organizations. They enable, support, and accelerate digital business processes and new forms of collaboration. In an era of platform economies, data-driven decision-making, and hybrid work models, BIS constitute a central infrastructure for economic activity.

This course provides theoretical and practical foundations for the design, adaptation, and implementation of enterprise information systems. Through lectures, exercises, and practical computer sessions, students acquire in-depth knowledge in the following areas:

- Advantages and disadvantages of business standard software, as well as selection criteria
- Modeling and analysis of business processes with standard software
- Types and application areas of business information systems
- Methods and challenges related to the introduction of standard software
- Sustainability in the context of of business information systems

This course cannot be taken together with *Advanced Programming - Java Network Programming* [2511020].

Learning objectives:

Students

- explain fundamental concepts, criteria, and application areas of information systems,
- identify and describe types and components of information systems,
- apply standard software to model, analyze, and evaluate business processes using exemplary scenarios,
- analyze the implementation, use, and impact of business standard software.

Recommendations:

Knowledge of the courses "Grundlagen der Informatik I und II" are helpful.

Notes:

- No registration is required for the lecture
- An registration is required for the exercises for participation in the Computer Lab and the subsequent exam admission
- The registration phase for the exercises starts in the first week after lecture begin and ends with the first exercise session
- Important informations regarding the registration, exact dates and deadlines will be communicated on the lecture website (ILIAS)

Workload:

- Lecture 30h
- Exercise course 15h
- Review and preparation of lectures 23h
- Review and preparation of exercises 10h
- Computer Lab 30h
- Exam preparation 26h
- Exam 1h
- Total 135h
- Exercise courses are done by student tutors

Literature

- Schönthaler, Vossen, Oberweis, Karle: Business Processes for Business Communities: Modeling Languages, Methods, Tools. Springer 2012.
- Hasenkamp, Stahlknecht: Einführung in die Wirtschaftsinformatik. Springer 2012.
- Hansen, Neumann: Wirtschaftsinformatik I. Grundlagen betrieblicher Informationsverarbeitung. UTB 2009.
- Mertens et al.: Grundzüge der Wirtschaftsinformatik. Springer 2012.

Weitere Literatur wird in der Vorlesung bekannt gegeben.

T 6.21 Module component: Advanced Programming - Java Network Programming [T-WIWI-102747]

Coordinators: Prof. Dr. Dietmar Ratz
Prof. Dr.-Ing. Johann Marius Zöllner
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104901 - Informatics (KIT-Department of Economics and Management)

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each summer term	6

Courses					
ST 2026	2511020	Advanced Programming - Java Network Programming	2 SWS	Lecture / 🗣	Ratz
ST 2026	2511021	Übungen zu Programmierung kommerzieller Systeme - Anwendungen in Netzen mit Java	1 SWS	Practice / 🗣	Ratz, Stegmaier, Boualili
ST 2026	2511023	Rechnerpraktikum zu Programmierung kommerzieller Systeme - Anwendungen in Netzen mit Java	2 SWS	/ 🗣	Ratz, Stegmaier, Boualili

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
At the end of the lecture period, a written examination (90 min.) is offered (according to §4(2), 1 SPO), for which - through successful participation in the exercises during the semester - admission must be obtained. The exact details will be announced in the lecture. The examination is offered every semester and can be repeated at any regular examination date.

A bonus can be earned through successful participation in the exercises. If the grade of the written examination is between 4.0 and 1.3, the bonus improves the grade by one grade level (0.3 or 0.4). Details will be announced in the lecture

Prerequisites
This course cannot be taken together with Advanced Programming - Application of Business Software [2511026].

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-WIWI-102748 - Advanced Programming - Application of Business Software must not have been started.

Additional Information
The registration for the participation in the computer lab (precondition for the exam participation) already takes place in the first lecture week!

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Advanced Programming - Java Network Programming
2511020, SS 2026, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content
In the lecture, the exercises and computer labs to this course the practical handling with the programming language Java dominating within the range of economical applications is obtained. The basis for this is the current language standard. The knowledge from the lecture Introduction to Programming with Java will be deepened and extended. This is done, among other things, by addressing commercially relevant topics such as object-oriented modeling and programming, class hierarchy and inheritance, threads, applications and applets, AWT and Swing components for graphical user interfaces, exception and event processing, lambda expressions, input/output via streams, applications in networks, Internet communication, client and server programming, remote method invocation, servlets, Java Server Pages and Enterprise Java Beans.

This course cannot be taken together with *Advanced Programming - Application of Business Software* [2540886/2590886].

Learning objectives:

- Students learn the practical use of the object-oriented programming language Java and are enabled to design and implement component-based Internet applications using the latest technologies and tools.
- The ability to select and design these methods and systems appropriate to the situation and to use them for solving problems is imparted.
- Students are empowered to find strategic and creative answers in the search for solutions to well-defined, concrete and abstract problems.

Workload:

The total workload for this course is approximately 150 hours.

Organizational issues
Die Anmeldung zur Teilnahme am Rechnerpraktikum (Vorbedingung zur Klausurteilnahme) findet bereits in der ersten Vorlesungswoche statt!

Literature
Ratz, D. Schulmeister-Zimolong, D. Seese, J. Wiesenberger. Grundkurs Programmieren in Java. 9. Aktualisierte und erweiterte Auflage, Hanser 2024.

Weiterführende Literatur:

- S. Zakhour, S. Hommel, J. Royal. Das Java Tutorial. Addison Wesley 2007
- W. Eberling, J. Lessner. Enterprise JavaBeans 3. Hanser Verlag 2007.
- R. Oechsle. Parallele und verteilte Anwendungen. 2. Auflage. Hanser Verlag 2007.
- Weitere Literatur wird in der Vorlesung bekannt gegeben.

T 6.22 Module component: Advanced Quantitative Finance [T-WIWI-114979]

Coordinators: Prof. Dr. Julian Thimme
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101480 - Finance 3
M-WIWI-101483 - Finance 2
M-WIWI-104900 - Business Administration
M-WIWI-107187 - Quantitative Core

Type	Credits	Grading	Term offered	Version
Examination of another type	4,5 CP	graded	Each summer term	1

Assessment
The assessment of success in this course is carried out as a non exam assessment. The grade is made up as follows:

1. Written elaboration (weighting: 33 %).
2. Working on and presenting data analysis problems in small groups (weighting: 67 %).

Additional Information
New course from summer semester 2026

Workload
135 hours

T 6.23 Module component: Advanced Statistics [T-WIWI-103123]

Coordinators: Prof. Dr. Oliver Grothe
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101637 - Analytics and Statistics
M-WIWI-101639 - Econometrics and Statistics II
M-WIWI-104902 - Statistics
M-WIWI-107187 - Quantitative Core

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each winter term	1

Courses					
WT 25/26	2550552	Advanced Statistical Techniques, Including Multivariate and Simulation Methods	2 SWS	Lecture / 🗣	Grothe
WT 25/26	2550553	Exercises and Computer Labs in Advanced Statistical Techniques	2 SWS	Practice / 🗣	Grothe

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment of this course is a written examination (60 min) according to §4(2), 1 of the examination regulation. The exam is offered every semester. Re-examinations are offered only for repeaters.

Prerequisites
None

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Advanced Statistical Techniques, Including Multivariate and Simulation Methods

Lecture (V)
On-Site

2550552, WS 25/26, 2 SWS, Language: English, Open in study portal

Literature
Skript zur Vorlesung

T 6.24 Module component: Advanced Stochastic Optimization [T-WIWI-106548]

Coordinators: Prof. Dr. Steffen Rebennack
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101473 - Mathematical Programming
M-WIWI-103289 - Stochastic Optimization
M-WIWI-104899 - Operations Research

Type	Credits	Grading	Term offered	Version
Oral examination	4,5 CP	graded	Irregular	2

Courses					
WT 25/26	2500050	Advanced Stochastic Optimization	2 SWS	Lecture /	Rebennack
WT 25/26	2550468	Übung zu Advanced Stochastic Optimization	1 SWS	Practice /	Rebennack

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
The assessment consists of an oral exam (20 minutes). The exam is offered every semester.

Prerequisites
None.

Recommendations
It is recommended to attend the lecture "Introduction to Stochastic Optimization" before attending the lecture "Advanced Stochastic Optimization".

Additional Information
Lectures and tutorials are offered irregularly.

Workload
135 hours

T 6.25 Module component: Advanced Systems Engineering [T-MACH-113929]

Coordinators: Prof. Dr.-Ing. Tobias Düser
Organisation: KIT Department of Mechanical Engineering
Part of: M-WIWI-107718 - Mechanical Engineering I
M-WIWI-107730 - Mechanical Engineering II

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each summer term	3

Courses					
ST 2026	2146232	Advanced Systems Engineering	2 SWS	Lecture / 🗣	Düser

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, × Cancelled

Assessment
Written exam with duration of approx. 90 min

Prerequisites
None

Recommendations
None

Additional Information
The course is offered in English.

Students acquire the ability to analyze complex technical systems holistically and develop them systematically. They learn to apply model-based methods to design system architectures and to collect, structure, and implement requirements. In addition, they master basic methods of verification and validation to ensure the quality of systems. Students develop skills for interdisciplinary collaboration and cross-disciplinary communication. In addition, they expand their project management skills, particularly with regard to coordinating and controlling technical development processes and gain a basic understanding for using data management and AI in Engineering.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Advanced Systems Engineering

2146232, SS 2026, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content
Students acquire the ability to analyze complex technical systems holistically and develop them systematically. They learn to apply model-based methods to design system architectures and to collect, structure, and implement requirements. In addition, they master basic methods of verification and validation to ensure the quality of systems. Students develop skills for interdisciplinary collaboration and cross-disciplinary communication. In addition, they expand their project management skills, particularly with regard to coordinating and controlling technical development processes and gain a basic understanding for using data management and AI in engineering.

The following topics will be covered:

- Transformation of engineering
- Innovation management
- Development processes in industry
- MBSE
- DevOps
- Agility
- Verification and validation
- Artificial intelligence and GenAI in engineering
- Data management and interfaces

T 6.26 Module component: Advanced Topics in Digital Management [T-WIWI-111912]

Coordinators: Prof. Dr. Petra Nieken
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration
M-WIWI-105923 - Incentives, Interactivity & Decisions in Organizations

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Each summer term	1

Courses					
ST 2026	2573016	Advanced Topics in Digital Management	2 SWS	Colloquium / 🗣	Nieken, Mitarbeiter

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Alternative exam assessment. The following aspects are included:

- Regular and active participation in the course dates
- Presentation of a given research topic.

The point scheme for the assessment is determined by the lecturer of the respective course. It will be announced at the beginning of the course.

Recommendations
We recommend visiting the course Incentives in Organization before taking this course.
The course is strongly recommended for students interested in empirical research in the areas digital HRM, personnel economics, and leadership and those who are interest in an academic career path.

Workload
90 hours

Below you will find excerpts from courses related to this module component:

V Advanced Topics in Digital Management

Colloquium (KOL)
On-Site

2573016, SS 2026, 2 SWS, Language: English, Open in study portal

Content

The students will discuss and analyze selected research papers in the areas digital HRM, personnel economics, and leadership with a focus on digital management. The students will present research papers and discuss research methods and designs as well as content. They will develop an own research design on a predefined topic.

Aim

The student

- Looks into current research topics in the areas HRM, personnel economics, and leadership with a focus on digital management and AI.
- Analyzes research papers in detail and evaluates the research outcomes.
- Trains their presentation skills and discussion skills.
- Practices scientific debating.
- Learns to critically evaluate research methods and trains the scientific discussion culture.
- Gains deeper knowledge in the area of digital HRM and management.
- Learns to evaluate research designs and takes into account the ethical dimension of research.
- Learns how to develop an own research design and idea.

Notes

Due to the interactive nature of the course, the number of participants is limited. If you are interested, please contact Prof. Nieken by email.

Workload

The total workload for this course is approximately 90 hours.

Lecture: 30 hours

Preparation: 45 hours

Exam preparation: 15 hours

Literature

Selected research papers

Organizational issues

Geb. 05.20, Raum 2A-25, Termine werden bekannt gegeben

T 6.27 Module component: Advanced Topics in Digital Twins [T-WIWI-114951]

Coordinators: Prof. Dr.-Ing. Sanja Lazarova-Molnar
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101456 - Intelligent Systems and Services
M-WIWI-101472 - Informatics
M-WIWI-101628 - Emphasis in Informatics

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Once	1

Courses					
ST 2026	2511104	Advanced Topics in Digital Twins	2 SWS	Lecture / 🗣	Lazarova-Molnar
ST 2026	2511105	Exercises Advanced Topics in Digital Twins	1 SWS	Practice / 🗣	Lazarova-Molnar, Khodadadi, Zare, Jungmann, Mostafa, Lee, Mahmoud, Daniel, Deufel

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Written examination (60 minutes).

Prerequisites
None

Recommendations
Basic knowledge of modeling and simulation is helpful but not mandatory.
An interest in interdisciplinary research and applied innovation in cyber-physical systems is advantageous.

Additional Information
The course will be offered once in the summer semester 2026.
The course is conducted in-person only to encourage active participation and hands-on learning.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Advanced Topics in Digital Twins

2511104, SS 2026, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content

This lecture series provides an overview of current research in Digital Twins, delivered by members of the SYDSEN research group working on active research topics in the field. Each lecture focuses on a specific topic and covers state-of-the-art methods as well as practical applications. Topics include:

1. Intro to Modeling and Simulation (M&S) and Digital Twins (DTs)
2. Stochastic Petri Nets
3. Comprehensive Digital Twins
4. Integration of Expert Knowledge into Data-Driven Digital Twin Models
5. Human-in-the-Loop Digital Twins
6. Explainability in Digital Twins
7. Validation of Digital Twins for Production Systems
8. Automation of Simulation-Based Decision Support in Data-Driven Digital Twins
9. Digital Twins for Enhanced Reliability in Energy Systems
10. Data-driven Agent-Based Simulation: Towards Digital Twins of Multi-Agent Systems
11. Summary

By the end of the course, students will understand how Digital Twins are applied in different domains. They will be able to analyze research approaches, identify limitations, and assess the practical relevance of current work. Students will also be able to recognize emerging research directions and innovation opportunities.

Prerequisites

Basic programming experience and foundational knowledge in mathematics, modeling, and data-driven methods.

Learning Objectives

- Conduct critical analysis of Digital Twin research and its practical relevance.
- Evaluate limitations and opportunities for innovation in existing Digital Twin approaches.
- Identify emerging trends across application domains and research areas.
- Connect theoretical knowledge with hands-on experience through demonstrations and tool-based exercises.

Form of Instruction

Lectures covering theoretical foundations and research contributions, complemented by practical sessions including demonstrations, case studies, and tool-based exercises. A detailed course plan will be provided before the start of the semester.

Competence Certificate

The examination will be offered as a written exam (60 minutes).

T 6.28 Module component: Advanced Topics in Economic Theory [T-WIWI-102609]

Coordinators: Prof. Dr. Johannes Brumm
Prof. Dr. Kay Mitusch

Organisation: KIT Department of Business and Economics

Part of: M-WIWI-101406 - Network Economics
M-WIWI-101500 - Microeconomic Theory
M-WIWI-101502 - Economic Theory and its Application in Finance
M-WIWI-104908 - Economics
M-WIWI-107010 - Economics in a Connected World
M-WIWI-107011 - Economics of Innovation and Growth

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Irregular	1

Assessment
The assessment consists of a written exam (60min) (following §4(2), 1 of the examination regulation) at the end of the lecture period or at the beginning of the following semester.

Prerequisites
None

Recommendations
This course is designed for advanced Master students with a strong interest in economic theory and mathematical models. Bachelor students who would like to participate are free to do so, but should be aware that the level is much more advanced than in other courses of their curriculum.

T 6.29 Module component: Advanced Topics in Human Resource Management [T-WIWI-111913]

Coordinators: Prof. Dr. Petra Nieken
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration
M-WIWI-105923 - Incentives, Interactivity & Decisions in Organizations

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Irregular	1

Courses					
WT 25/26	2573014	Advanced Topics in Human Resource Management	2 SWS	Colloquium / 🗣	Nieken, Mitarbeiter

Legend: 🖥 Online, Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Alternative exam assessment.The following aspects are included:

- Regular and active participation in the course dates
- Presentation of a given research topic.

The point scheme for the assessment is determined by the lecturer of the respective course. It will be announced at the beginning of the course.

Recommendations
We recommend visiting the course Incentives in Organization before taking this course.
The course is strongly recommended for students interested in empirical research in the areas HRM, personnel economics, and leadership and those who are interest in an academic career path.

Additional Information
Teaching and learning format: Colloquium

Workload
90 hours

Below you will find excerpts from courses related to this module component:

V Advanced Topics in Human Resource Management

2573014, WS 25/26, 2 SWS, Language: German, Open in study portal

Colloquium (KOL)
On-Site

Content
The students will discuss and analyze selected research papers in the areas HRM, personnel economics, and leadership. The students will present research papers and discuss research methods and designs as well as content. They will develop an own research design on a predefined topic.

Aim

The student

- Looks into current research topics in the areas HRM, personnel economics, and leadership.
- Analyzes research papers in detail and evaluates the research outcomes.
- Trains their presentation skills and discussion skills.
- Practices scientific debating.
- Learns to critically evaluate research methods and trains the scientific discussion culture.
- Gains deeper knowledge in the area of HRM.
- Learns to evaluate research designs and takes into account the ethical dimension of research.
- Learns how to develop an own research design and idea.

Notes

Due to the interactive nature of the course, the number of participants is limited. If you are interested, please contact Prof. Nieken by email.

Workload

The total workload for this course is approximately 90 hours.

Lecture: 30 hours

Preparation: 45 hours

Exam preparation: 15 hours

Literature

Selected research papers

Organizational issues
siehe Homepage

T 6.30 Module component: Algorithms I [T-INFO-100001]

Coordinators: TT-Prof. Dr. Thomas Bläsius
Organisation: KIT Department of Informatics

Part of: M-WIWI-104909 - Informatics (Department of Informatics)

Type	Credits	Grading	Term offered	Version
Written examination	6 CP	graded	Each summer term	1

Courses					
ST 2026	24500	Algorithms I	4 SWS	Lecture / Practice / 🗣	Sanders, Uhl, Schimek, Laupichler, Borowitz

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, × Cancelled

Assessment
The performance assessment consists of a written final examination in accordance with Section 4 (2) No. 1 SPO lasting 120 minutes.

The lecturer may award a grade bonus of max. 0.4 (equivalent to one grade step) for good performance in the exercise for the Algorithmen I course.

This grade bonus is only valid for one examination in the same semester. After that, the grade bonus expires.

T 6.31 Module component: Algorithms II [T-INFO-102020]

Coordinators: Prof. Dr. Peter Sanders
Organisation: KIT Department of Informatics
Part of: M-WIWI-104909 - Informatics (Department of Informatics)

Type	Credits	Grading	Term offered	Version
Written examination	6 CP	graded	Each winter term	1

Assessment
The assessment is carried out as a written examination (§ 4 Abs. 2 No. 1 SPO) lasting 120 minutes.

Prerequisites
none.

T 6.32 Module component: Analalysis of Social Structurs (WiWi) [T-GEISTSOZ-109047]

Coordinators: Prof. Dr. Gerd Nollmann
Organisation: KIT Department of Humanities and Social Sciences
Part of: M-WIWI-104906 - Social Sciences

Type	Credits	Grading	Term offered	Version
Written examination	3 CP	graded	Each winter term	1

Courses					
WT 25/26	5011004	Analysis of Social Structures	2 SWS	Lecture / 🖥	Nollmann
WT 25/26	5011007	Analysis of Social Structures	2 SWS	Practice / 🖥	Nollmann

Legend: 🖥 Online, Blended (On-Site/Online), On-Site, × Cancelled

T 6.33 Module component: Analysis of Exhaust Gas and Lubricating Oil in Combustion Engines [T-MACH-105173]

Coordinators: Dr.-Ing. Marcus Gohl
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101303 - Combustion Engines II
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each summer term	1

Courses					
ST 2026	2134150	Gas, lubricating oil and operating media analysis in drive train development	2 SWS	Lecture / 🗣	Gohl

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, × Cancelled

Assessment
Oral examination, duration approx. 25 min, no aids

Prerequisites
none

Additional Information
The course is offered in German.

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Gas, lubricating oil and operating media analysis in drive train development

2134150, SS 2026, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Literature
Die Vorlesungsunterlagen werden vor jeder Veranstaltung an die Studenten verteilt.

T 6.34 Module component: Analysis of Multivariate Data [T-WIWI-103063]

Coordinators: Prof. Dr. Oliver Grothe
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104902 - Statistics

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Irregular	1

Assessment
The assessment of this course is a written examination (60 min) according to §4(2), 1 of the examination regulation. The exam is offered every semester. Re-examinations are offered only for repeaters.

Prerequisites
None

Recommendations
Attendance of the courses Statistics 1 [2600008] and Statistics 2 [2610020] is recommended.

Additional Information
The lecture is not offered regularly. The courses planned for three years in advance can be found online.

Workload
135 hours

T 6.35 Module component: Analysis Tools for Combustion Diagnostics [T-MACH-105167]

Coordinators: Jürgen Pfeil
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101303 - Combustion Engines II
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each summer term	1

Courses					
ST 2026	2134134	Analysis tools for combustion diagnostics	2 SWS	Lecture / 🗣	Pfeil

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
oral examination, Duration: 25 min., no auxiliary means

Prerequisites
none

Additional Information
The course is offered in German.

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Analysis tools for combustion diagnostics

2134134, SS 2026, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

T 6.36 Module component: Analytical CRM [T-WIWI-102596]

Coordinators: Prof. Dr. Andreas Geyer-Schulz
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each summer term	1

Assessment
The exam will be offered for first time writers for the last time in the summer semester 2020.

Written examination (60 minutes) according to §4(2), 1 SPO. The exam is considered passed if at least 50 out of a maximum of 100 possible points are achieved. The grades are graded in five steps (best grade 1.0 from 95 points). Details of the grade formation and scale will be announced in the course.
A bonus can be acquired through successful participation in the practice. If the grade of the written examination is between 4.0 and 1.3, the bonus improves the grade by one grade level (0.3 or 0.4). The exact criteria for awarding a bonus will be announced at the beginning of the course.

Prerequisites
None

Recommendations
We expect knowledge about data models and the UML modelling language concerning information systems.

T 6.37 Module component: Antennas and Beamforming [T-ETIT-113920]

Coordinators: Prof. Dr.-Ing. Marwan Younis
Prof. Dr.-Ing. Thomas Zwick
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-106956 - Antennas and Beamforming

Type	Credits	Grading	Term offered	Expansion	Version
Written examination	4 CP	graded	Each winter term	1 semesters	1

Courses					
WT 25/26	2308465	Antennas and Beamforming	2 SWS	Lecture / 🗣	Younis, Zwick
WT 25/26	2308466	Tutorial for 2308465 Antennas and Beamforming	1 SWS	Practice / 🗣	Younis

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The examination takes place in form of a written examination lasting 120 minutes.
The module grade is the grade of the written exam.

Prerequisites
none

Recommendations
Knowledge of the basics of radio frequency technology and some basic knowledge on communication and radar systems is recommended.

T 6.38 Module component: Application of Social Science Methods (WiWi) [T-GEISTSOZ-109052]

Coordinators: Prof. Dr. Gerd Nollmann
Organisation: KIT Department of Humanities and Social Sciences
Part of: M-GEISTSOZ-101169 - Sociology
M-WIWI-104906 - Social Sciences

Type	Credits	Grading	Term offered	Version
Examination of another type	9 CP	graded	Each term	2

Courses					
ST 2026	5011006	Gender Pay Gap	2 SWS	Seminar / 🖥	Nollmann
ST 2026	5011008	Decomposition and Regression Analysis	2 SWS	Seminar / 🖥	Nollmann

Legend: 🖥 Online, Blended (On-Site/Online), On-Site, ✕ Cancelled

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-GEISTSOZ-104565 - Computer Aided Data Analysis must have been passed.

T 6.39 Module component: Applied Data Science and Machine Learning in Engineering [T-MACH-115042]

Coordinators: Prof. Dr.-Ing. Anne Meyer
Organisation: KIT Department of Mechanical Engineering
Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Expansion	Version
Examination of another type	4 CP	graded	Each summer term	1 semesters	1

Courses					
ST 2026	2122352	Applied Data Science and Machine Learning in Engineering	4 SWS	Others / 🗣	Meyer

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, × Cancelled

Assessment
Graded examination of another type; for further information, see ILIAS.

Prerequisites
“Data Science Fundamentals in Engineering”, “Grundlagen der Künstlichen Intelligenz ” or a comparable course must be passed.

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Applied Data Science and Machine Learning in Engineering

Others (sonst.)
On-Site

2122352, SS 2026, 4 SWS, Language: English, Open in study portal

Content

Data-driven methods are transforming modern engineering, enabling smarter product design, more efficient production systems, and intelligent robotics. To prepare students for these challenges, this course emphasizes the applied, technical aspects of data-driven engineering.

Building on prior courses introducing data science and machine learning (see prerequisites), core content includes technical aspects of industrial data sources (e.g., database technologies, REST APIs), messaging technologies (e.g., MQTT), version control (e.g., git with GitLab), and server or cluster-based computations for scalable data processing (e.g., HPC at KIT). Students practice through guided exercises and work in small groups on applied machine learning case studies using real-world engineering data from our lab. The module also covers methods for model explainability and interpretability to support transparent decision-making in engineering applications.

The course is designed as an on-site project with regular in-person collaboration and close supervision, offering students continuous, structured, and individualized feedback on their code and technical implementation.

Prerequisite

“Data Science Fundamentals in Engineering”, “Grundlagen der Künstlichen Intelligenz ” or a comparable course must be passed.

Learning Objectives

After successful completion of the module, students are able to:

- **Apply** advanced data science and machine learning methods to engineering data sets, including data from industrial sources and streaming technologies.
- **Design and implement** data analysis and machine learning workflows using collaborative tools, version control, and cluster-based computing.
- **Analyze, interpret, and evaluate** the performance and limitations of machine learning models in applied engineering scenarios.

Additional Information

The number of participants is limited. Information on the application process is available in ILIAS.

Organizational Information

See further details on course organization at: lehre.imi.kit.edu and in ILIAS.

Workload

120 hours

Organizational issues

See ILIAS for time and location

T 6.40 Module component: Applied Econometrics [T-WIWI-111388]

Coordinators: Prof. Dr. Fabian Krüger
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101638 - Econometrics and Statistics I
M-WIWI-101639 - Econometrics and Statistics II
M-WIWI-104902 - Statistics
M-WIWI-107010 - Economics in a Connected World
M-WIWI-107011 - Economics of Innovation and Growth
M-WIWI-107187 - Quantitative Core

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each winter term	2

Courses					
WT 25/26	2520020	Applied Econometrics	2 SWS	Lecture / 🗣	Krüger, N.N.
WT 25/26	2520021	Tutorial in Applied Econometrics	2 SWS	Practice / 🗣	Krüger, N.N.

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment of this course is a written examination (90 min).

Prerequisites
None

Below you will find excerpts from courses related to this module component:

V Applied Econometrics
2520020, WS 25/26, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content
The course starts with a concise review of the linear regression model. It then presents methods for causal inference: The potential outcomes approach, methods for analyzing randomized controlled trials, and methods for analyzing observational data (e.g., regression discontinuity). Empirical examples and R code are used to illustrate the methodological concepts.

Learning goals

Students understand the properties of various econometric estimators and research designs, and can implement econometric estimators using R software.

Workload

The total workload for this course (4.5 credit points) is approximately 135 hours.

Literature
The following book is the main reference for the course:

Ding, P. (2024). A First Course in Causal Inference. Routledge.

Further literature will be announced in class.

T 6.41 Module component: Applied Informatics [T-WIWI-114211]

Coordinators: Dr.-Ing. Tobias Käfer
Prof. Dr.-Ing. Sanja Lazarova-Molnar
Prof. Dr. Andreas Oberweis
Prof. Dr. Harald Sack
Prof. Dr. Alexey Vinel
Prof. Dr. Melanie Volkamer
Prof. Dr.-Ing. Johann Marius Zöllner

Organisation: KIT Department of Business and Economics

Part of: M-WIWI-107187 - Quantitative Core

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each summer term	1

Assessment
The assessment of the lecture is a written examination (60 min) or an oral exam (30 min).

The exam takes place every semester and can be repeated at every regular examination date.

Prerequisites
None

Workload
135 hours

T 6.42 Module component: Applied Informatics – Applications of Artificial Intelligence [T-WIWI-110340]

Coordinators: Dr.-Ing. Tobias Käfer
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104901 - Informatics (KIT-Department of Economics and Management)

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each winter term	2

Courses					
WT 25/26	2511314	Applied Informatics - Applications of Artificial Intelligence	2 SWS	Lecture / Blended	Käfer, Kinder, Li
WT 25/26	2511315	Exercises to Applied Informatics - Applications of Artificial Intelligence	1 SWS	Practice / On-Site	Käfer, Kinder, Li

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
Written Examination (60 min) according to §4, Abs. 2, 1 of the examination regulations or oral examination of 20 minutes according to §4, Abs. 2, 2 of the examination regulations. The exam takes place every semester and can be repeated at every regular examination date.

Prerequisites
None.

Recommendations
Basics in logic, e.g. from lecture Foundations of Informatics 1 are important.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Applied Informatics - Applications of Artificial Intelligence

2511314, WS 25/26, 2 SWS, Language: German, Open in study portal

Lecture (V)
Blended (On-Site/Online)

Content
The lecture provides insights into the fundamentals of artificial intelligence. Basic methods of artificial intelligence and their applications in industry are presented.

Applications of the AI is a sub-area of computer science dealing with the automation of intelligent behavior. In general, it is a question of mapping human intelligence. Methods of artificial intelligence are presented in various areas such as, for example, question answering systems, speech recognition and image recognition.

The lecture gives an introduction to the basic concepts of artificial intelligence. Essential theoretical foundations, methods and their applications are presented and explained.

This lecture aims to provide students with a basic knowledge and understanding of the structure, analysis and application of selected methods and technologies on artificial intelligence. The topics include, among others, knowledge modeling, machine learning, text mining, uninformed search, and intelligent agents.

Learning objectives:

The students

- consider current research topics in the field of artificial intelligence and in particular learn about the topics of knowledge modeling, machine learning, text mining and uninformed search.
- interdisciplinary thinking.
- technological approaches to current problems.

Workload:

- The total workload for this course is approximately 135 hours
- Time of presentness: 45 hours
- Time of preperation and postprocessing: 60 hours
- Exam and exam preperation: 30 hours

V

Exercises to Applied Informatics - Applications of Artificial Intelligence

2511315, WS 25/26, 1 SWS, Language: German, Open in study portal

Practice (Ü)
On-Site

Content
The exercises are oriented on the lecture applications of AI.

Multiple exercises are held that capture the topics, held in the lecture Applications of AI and discuss them in detail. Thereby, practical examples are given to the students in order to transfer theoretical aspects into practical implementation.

This lecture aims to provide students with a basic knowledge and understanding of the structure, analysis and application of selected methods and technologies on artificial intelligence. The topics include, among others, knowledge modeling, machine learning, text mining, uninformed search, and intelligent agents.

Learning objectives:

The students

- consider current research topics in the field of artificial intelligence and in particular learn about the topics of knowledge modeling, machine learning, text mining and uninformed search.
- interdisciplinary thinking.
- technological approaches to current problems.

T 6.43 Module component: Applied Informatics – Cybersecurity [T-WIWI-114156]

Coordinators: Prof. Dr. Melanie Volkamer
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104901 - Informatics (KIT-Department of Economics and Management)

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each summer term	2

Courses					
ST 2026	2511550	Applied Informatics – Cybersecurity	2 SWS	Lecture / 🗣	Volkamer
ST 2026	2511551	Exercise Applied Informatics – Cybersecurity	1 SWS	Practice / 🗣	Volkamer, Berens, Ballreich

Legend: 🖥 Online, ✲ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment of this course is a written examination (60 min) according to §4(2), 1 of the examination regulation or an oral exam (30 min) following §4, Abs. 2, 2 of the examination regulation, for which admission must be obtained through successful participation in the exercise during the semester.

The exam takes place every semester and can be repeated at every regular examination date.

Additional Information
Competence Goal

The student

- can explain and apply the basics of information security
- knows appropriate measures to achieve different protection goals and can implement these measures
- can assess the quality of organizational protective measures, i.e. among other things
- knows what has to be taken intoaccount when using the individual measures
- understands the differences between information security in the enterprise and in the private context
- knows the areas of application of a variety of relevant standards and knows their weaknesses
- knows and can explain the problems of information security which may arise from human-machine interaction
- can assess messages about detected security problems in a critical way
- can structure a software project in the field of information security and explain and present results in oral and written form
- can use the techniques of Human Centred Security and Privacy by Design to create user-friendly software.

Content

- Basics and concepts of information security
- Understanding the protection objectives of information security and various attack models (including associated assumptions)
- introduction of measures to achieve the respective protection goals, taking into account different attack models
- Note: In contrast to the IT Security lecture, measures such as encryption algorithms are treated only abstractly, i.e. the idea of the measure, assumptions to the attacker and the deployment environment.
- Presentation and analysis of problems of information security arising from human-machine interaction and presentation of the Human Centered Security by Design approach.
- Introduction into organizational protective measures and standards to be observed for companies.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Applied Informatics – Cybersecurity
2511550, SS 2026, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content

- Basics and concepts of information security
- Understanding the protection objectives of information security and various attack models (including associated assumptions)
- introduction of measures to achieve the respective protection goals, taking into account different attack models
- Note: In contrast to the IT Security lecture, measures such as encryption algorithms are treated only abstractly, i. e. the idea of the measure, assumptions to the attacker and the deployment environment.
- Presentation and analysis of problems of information security arising from human-machine interaction and presentation of the Human Centered Security by Design approach.
- Introduction into organisational protective measures and standards to be observed for companies

Learing objectives:

The student

- can explain the basics of information security
- knows suitable measures to achieve different protection goals
- can assess the quality of organisational protective measures, i. e. among other things knows what has to be taken into account when using the individual measures
- understands the differences between information security in the organisational and in the private context
- knows the areas of application of different standards and knows their weaknesses
- knows and can explain the problems of information security that which arise from human-machine interaction
- is able to deal with messages concerning found security problems in a critical way.

This course can also be credited for the KASTEL certificate. Further information about obtaining the certificate can be found on the SECUSO website https://secuso.aifb.kit.edu/Studium_und_Lehre.php).

Literature

- P. Gerber, M. Ghiglieri, B. Henhapl, O. Kulyk, K. Marky, P. Mayer, B. Reinheimer, and M. Volkamer, *Human Factors in Security*. Springer, Jan. 2018, pp. 83–98.
- C. Eckert, *IT-Sicherheit: Konzepte-Verfahren-Protokolle*. Walter de Gruyter, 2013

V **Exercise Applied Informatics – Cybersecurity**
2511551, SS 2026, 1 SWS, Language: German, Open in study portal

Practice (Ü)
On-Site

Content

This course can also be credited for the KASTEL certificate. Further information about obtaining the certificate can be found on the SECUSO website https://secuso.aifb.kit.edu/Studium_und_Lehre.php).

T 6.44 Module component: Applied Informatics – Database Systems [T-WIWI-110341]

Coordinators: Prof. Dr. Andreas Oberweis
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104901 - Informatics (KIT-Department of Economics and Management)

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each summer term	2

Courses					
ST 2026	2511200	Applied Informatics - Database Systems	2 SWS	Lecture / 🗣	Sommer
ST 2026	2511201	Exercises Applied Informatics - Database Systems	1 SWS	Practice / 🗣	Sommer

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, × Cancelled

Assessment
The assessment consists of a written exam (60 minutes) in the first week after lecture period.

Additional Information
Replaces from summer semester 2020 T-WIWI-102660 "Database Systems".

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Applied Informatics - Database Systems

2511200, SS 2026, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content
As the foundation of business information systems, database systems (DBS) play a central role in companies. Internal and external data are stored and processed in the company's databases. Proper management and organization of this data helps resolve numerous issues, enables simultaneous queries by multiple users, and serves as the organizational and operational foundation for all of the company's workflows and processes. The course introduces students to the field of database theory, covers the fundamentals of database languages and database systems, examines basic concepts of object-oriented and XML databases, and teaches the principles of multi-user database control and physical data organization. In addition, it provides an overview of database problems frequently encountered in business practice, such as those related to data correctness (operational and semantic integrity), restoring a consistent database state, and synchronizing parallel transactions.

Learning objectives:

Students

- explain the concepts and principles of database models, languages, and systems, as well as their potential applications,
- design relational databases based on a solid theoretical foundation,
- create queries for relational database systems,
- gain an overview of advanced database issues in business practice.

Workload:

- Attendance time – Lecture (2 SWS × 15 weeks): 30 h
- Attendance time – Tutorial (1 SWS × 15 weeks): 15 h
- Self-study – Lecture: 45 h
- Self-study – Tutorial: 25 h
- Exam preparation: 20 h
- Total workload: 135 h

Literature

- Gunter Schlageter, Wolffried Stucky: Datenbanksysteme: Konzepte und Modelle. 2., neubearb. u. erw. Aufl. Teubner, Stuttgart 1983.
- Alfons Kemper, André Eickler: Datenbanksysteme: Eine Einführung. 10., erw. und aktual. Aufl. De Gruyter Oldenbourg, Berlin, 2015.
- Ramez Elmasri, Sham Navathe: Fundamentals of Database Systems. 7. ed. Pearson, Harlow, 2016.
- Gunter Saake, Kai-Uwe Sattler, Andreas Heuer: Datenbanken. Konzepte und Sprachen. 6. Aufl. mitp, Heidelberg u. a., 2018.
- Weitere Literatur wird in der Vorlesung bekannt gegeben. / Additional reading material will be announced during the lecture.

V **Exercises Applied Informatics - Database Systems**
2511201, SS 2026, 1 SWS, Language: German, Open in study portal

Practice (Ü)
On-Site

Content
As the foundation of business information systems, database systems (DBS) play a central role in companies. Internal and external data are stored and processed in the company's databases. Proper management and organization of this data helps resolve numerous issues, enables simultaneous queries by multiple users, and serves as the organizational and operational foundation for all of the company's workflows and processes. The course introduces students to the field of database theory, covers the fundamentals of database languages and database systems, examines basic concepts of object-oriented and XML databases, and teaches the principles of multi-user database control and physical data organization. In addition, it provides an overview of database problems frequently encountered in business practice, such as those related to data correctness (operational and semantic integrity), restoring a consistent database state, and synchronizing parallel transactions.

Learning objectives:

Students

- explain the concepts and principles of database models, languages, and systems, as well as their potential applications,
- design relational databases based on a solid theoretical foundation,
- create queries for relational database systems,
- gain an overview of advanced database issues in business practice.

Workload:

- Attendance time – Lecture (2 SWS × 15 weeks): 30 h
- Attendance time – Tutorial (1 SWS × 15 weeks): 15 h
- Self-study – Lecture: 45 h
- Self-study – Tutorial: 25 h
- Exam preparation: 20 h
- Total workload: 135 h

Literature

- Gunter Schlageter, Wolffried Stucky: Datenbanksysteme: Konzepte und Modelle. 2., neubearb. u. erw. Aufl. Teubner, Stuttgart 1983.
- Alfons Kemper, André Eickler: Datenbanksysteme: Eine Einführung. 10., erw. und aktual. Aufl. De Gruyter Oldenbourg, Berlin, 2015.
- Ramez Elmasri, Sham Navathe: Fundamentals of Database Systems. 7. ed. Pearson, Harlow, 2016.
- Gunter Saake, Kai-Uwe Sattler, Andreas Heuer: Datenbanken. Konzepte und Sprachen. 6. Aufl. mitp, Heidelberg u. a., 2018.
- Weitere Literatur wird in der Vorlesung bekannt gegeben. / Additional reading material will be announced during the lecture.

T 6.45 Module component: Applied Informatics – Mobile Computing [T-WIWI-113957]

Coordinators: Dr.-Ing. Gunther Schiefer
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104901 - Informatics (KIT-Department of Economics and Management)

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each summer term	1

Courses					
ST 2026	2511226	Applied Informatics - Mobile Computing	2 SWS	Lecture / 🗣	Schiefer, Kruse
ST 2026	2511227	Exercises Applied Informatics - Mobile Computing	1 SWS	Practice / 🗣	Schiefer

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success is assessed in the form of a written (60 min) or oral examination.

Prerequisites
None

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Applied Informatics - Mobile Computing

2511226, SS 2026, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content
The lecture covers the basics of mobile computing. These are interlinked with the economic background in Germany.

Contents are:

1. organizational matters
2. introduction & definitions
3. mobile devices
4. mobile radio technologies
5. mobile communications market
6. mobile applications
7. digital radio technologies
8. location & context

Note: The teaching units listed above each have a different scope.

Learning objectives:

If you are confronted with a question in your job which affects "Mobile Computing", you should be able to provide answers quickly and competently:

Market structures
technique
Possibilities for applications
lawsuits
issues

Workload:

The total workload for this course unit is approx. 135 hours (4.5 credit points).

Organizational issues
Vorlesung und Übung werden integriert angeboten.

Literature

- Jochen Schiller: Mobilkommunikation (2. Aufl. 2003)
  http://www.mi.fu-berlin.de/inf/groups/ag-tech/teaching/resources/ Mobile_Communications/course_Material/index.html
- Martin Sauter: Grundkurs Mobile Kommunikationssysteme (6. Aufl. 2015)
  http://link.springer.com/book/10.1007%2F978-3-658-08342-7
- Küpper, A.: Location-based Services. Fundamentals and Operation. Wiley & Sons, 2005.
- Roth, J.: Mobile Computing. Grundlagen, Technik, Konzepte. Dpunkt.verlag, 2. Auflage, 2005.
- Mansfeld, W.: Satellitenortung und Navigation:
  Grundlagen, Wirkungsweise und Anwendung globaler Satellitennavigationssysteme
- Dodel, H., Häupler, D.: Satellitennavigation

Einige relevante Informationen im Web

- Bundesnetzagentur http://www.bundesnetzagentur.de
  u.a. Jahresbericht und Marktbeobachtung
- VATM-Marktstudien
  http://www.vatm.de/vatm-marktstudien.html
- Verbände, bspw. BITKOM (bitkom.org), eco e.V. (eco.de)
- Presse, bspw. Teltarif, Heise, Golem, …
- Statistiken (Statista Lizenz des KIT)

T 6.46 Module component: Applied Informatics – Modelling [T-WIWI-110338]

Coordinators: Andreas Fritsch
Prof. Dr. Andreas Oberweis

Organisation: KIT Department of Business and Economics

Part of: M-WIWI-104901 - Informatics (KIT-Department of Economics and Management)

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each winter term	2

Courses					
WT 25/26	2511030	Applied Informatics - Modelling	2 SWS	Lecture / 🗣	Fritsch
WT 25/26	2511031	Exercises to Applied Informatics - Modelling	1 SWS	Practice / 🗣	Rybinski, Thompson

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment consists of a written examination (60 min) in the first week after lecture period (according to Section 4 (2),1 of the examination regulation).

Prerequisites
None

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Applied Informatics - Modelling

2511030, WS 25/26, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content
In the context of complex information systems, modelling is of central importance, e.g. – in the context of systems to be developed – for a better understanding of their functionality or in the context of existing systems for supporting maintenance and further development.

Modelling, in particular modelling of information systems, forms the core part of this lecture. The lecture is organized in two parts. The first part mainly covers the modelling of static aspectes, the second part covers the modelling of dynamic aspects of information systems.

The lecture sets out with a definition of modelling and the advantages of modelling. After that, advanced aspects of UML, the Entity Relationship model (ER model) and logics as a means of modelling static aspects will be explained. This will be complemented by the relational data model and the systematic design of databases based on ER models. For modelling dynamic aspects, different types of petri-nets together with their respective analysis techniques will be introduced.

Learning objectives:

Students

- explain the strengths and weaknesses of various modeling approachesfor Information Systems and choose an appropriate method for a given problem,
- create UML models, ER models and Petri nets for given problems,
- modelling given situations in propositional and predicate logic and can interpret them,
- analyze various properties in propositional and predicate logic,
- create and evaluate a relational database schema and express queries in relational algebra.

Workload:

- Total effort: 120-135 hours
- Presence time: 45 hours
- Self study: 75-90 hours

Literature

- Bernhard Rumpe. Modellierung mit UML, Springer-Verlag, 2004.
- R. Elmasri, S. B. Navathe. Fundamentals of Database Systems. Pearson Education 2009.
- W. Reisig. Petrinetze, Springer-Verlag, 2010.

Weiterführende Literatur:

- U. Kastens, H. Kleine Büning. Modellierung – Grundlagen und Formale Methoden. Carl Hanser Verlag, 2014
- J.L. Peterson. Petri Net Theory and Modeling of Systems, Prentice Hall, 1981.
- U. Schöning. Logik für Informatiker. Spektrum Akademischer Verlag,
  2000

V **Exercises to Applied Informatics - Modelling**
2511031, WS 25/26, 1 SWS, Language: German, Open in study portal

Practice (Ü)
On-Site

Content
In the context of complex information systems, modelling is of central importance, e.g. – in the context of systems to be developed – for a better understanding of their functionality or in the context of existing systems for supporting maintenance and further development.

Modelling, in particular modelling of information systems, forms the core part of this lecture. The lecture is organized in two parts. The first part mainly covers the modelling of static aspectes, the second part covers the modelling of dynamic aspects of information systems.

The lecture sets out with a definition of modelling and the advantages of modelling. After that, advanced aspects of UML, the Entity Relationship model (ER model) and logics as a means of modelling static aspects will be explained. This will be complemented by the relational data model and the systematic design of databases based on ER models. For modelling dynamic aspects, different types of petri-nets together with their respective analysis techniques will be introduced.

Learning objectives:

Students

- explain the strengths and weaknesses of various modeling approachesfor Information Systems and choose an appropriate method for a given problem,
- create UML models, ER models and Petri nets for given problems,
- modelling given situations in propositional and predicate logic and can interpret them,
- analyze various properties in propositional and predicate logic,
- create and evaluate a relational database schema and express queries in relational algebra.

Workload:

- Total effort: 120-135 hours
- Presence time: 45 hours
- Self study: 75-90 hours

Literature

-
    - Bernhard Rumpe. Modellierung mit UML, Springer-Verlag, 2004.
    - R. Elmasri, S. B. Navathe. Fundamentals of Database Systems. Pearson Education 2009.
    - W. Reisig. Petrinetze, Springer-Verlag, 2010.

  Weiterführende Literatur:

    - U. Kastens, H. Kleine Büning. Modellierung – Grundlagen und Formale Methoden. Carl Hanser Verlag, 2014
    - J.L. Peterson. Petri Net Theory and Modeling of Systems, Prentice Hall, 1981.
    - U. Schöning. Logik für Informatiker. Spektrum Akademischer Verlag,
      2000

T 6.47 Module component: Applied Informatics – Software Engineering [T-WIWI-110343]

Coordinators: Prof. Dr. Andreas Oberweis
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104901 - Informatics (KIT-Department of Economics and Management)

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	see Annotations	2

Assessment
The examination will be offered for the last time in the summer semester 2025 for first-time students. The last examination opportunity (only for repeaters) is in the winter semester 2025/2026. The assessment takes the form of a written examination (60 minutes) in accordance with §4(2), 1 SPO. It takes place in the first week after the lecture period.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-WIWI-100809 - Software Engineering must not have been started.

Additional Information
The lecture will no longer be offered from summer semester 2025. Parts of the lecture will be integrated into the new course "Applied Computer Science - Mobile Computing".

Workload
135 hours

T 6.48 Module component: Applied Information Theory [T-ETIT-114412]

Coordinators: Dr.-Ing. Holger Jäkel
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-107333 - Applied Information Theory

Type	Credits	Grading	Term offered	Version
Oral examination	6 CP	graded	Each winter term	1

Courses					
WT 25/26	2310537	Applied Information Theory	3 SWS	Lecture /	Jäkel
WT 25/26	2310539	Exercises for 2310537 Applied Information Theory	1 SWS	Practice /	Jäkel

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
The success control takes place in the form of an oral examination lasting approx. 25 minutes. Before the examination, there is a preparation phase of 15 minutes in which preparatory tasks are solved.

Prerequisites
none

T 6.49 Module component: Applied material flow simulation [T-MACH-112213]

Coordinators: Dr.-Ing. Marion Baumann
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101278 - Material Flow in Networked Logistic Systems
M-MACH-104888 - Advanced Module Logistics
M-WIWI-102805 - Service Operations
M-WIWI-102832 - Operations Research in Supply Chain Management

Type	Credits	Grading	Term offered	Version
Oral examination	4,5 CP	graded	Each winter term	1

Courses					
WT 25/26	2117054	Applied material flow simulation	3 SWS	Lecture / Practice / 🗣	Baumann

Legend: 🖥 Online, Blended (On-Site/Online), 🗣 On-Site, × Cancelled

Assessment
The assessment consists of an oral exam (20 min.) taking place in the recess period according to § 4 paragraph 2 Nr. 2 of the examination regulation.

Prerequisites
None

Recommendations

- Basic statistical knowledge and understanding
- Knowledge of a common programming language (Java, Python, ...)
- Recommended course: T-WIWI-102718 - Discrete Event Simulation in Production and Logistics

Additional Information
The course is offered in German.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Applied material flow simulation

Lecture / Practice (VÜ)
On-Site

2117054, WS 25/26, 3 SWS, Language: German, Open in study portal

Content
Learning Content:

- Methods of modeling a simulation such as:
  - Discrete-event simulation
  - Agent based simulation
- Design of a simulation model of a material flow system
- Data exchange in simulation models
- Verification and validation of simulation models
- Execution of simulation studies
- Statistical evaluation and parameter study

This is an application-oriented course in which the course contents are applied and deepened using the Anylogic software.

Learning Goals:
Students are able to:

- select the appropriate simulation modeling method depending on a modeling objective and build a suitable simulation model for material flow systems,
- extend a simulation model in a meaningful way with data import and export,
- verify and validate a simulation model,
- conduct a simulation study efficiently and with meaningful results, and
- design and conduct a parameter study and statistically analyze and evaluate the results.

Requirements:

- Basic knowledge of the Java programming language

Recommendations:

- Basic statistical skills
- Recommended course: T-WIWI-102718 - Discrete Event Simulation in Production and Logistics

Workload for 4,5 ECTS (135 h):

- regular attendance: 21 hours
  self-study: 114 hours

Organizational issues

- **Im Wintersemester 2025/2026 ist die Veranstaltung auf maximal 30 Teilnehmer beschränkt.**
- **Die Anmeldung ist durch Beitritt zum ILIAS-Kurs und Ausfüllen des Anmeldungsformulars (erforderliche Felder beim Beitritt zum ILIAS-Kurs) möglich.**
- **Die Anmeldung ist vom 01.09.2025 bis zum 30.09.2025 möglich.**

Literature
Borshev, A. (2022): The Big Book of Simulation Modeling - Multimethod Modeling with AnyLogic 8, https://www.anylogic.de/resources/books/big-book-of-simulation-modeling/.

Grigoryev, I. (2021): AnyLogic8 in Three Days, 5. Aufl., https://www.anylogic.de/resources/books/free-simulation-book-and-modeling-tutorials/.

Gutenschwager, K. et. al. (2017): Simulation in Produktion und Logistik, Springer Vieweg, Berlin.

VDI (2014): Simulation von Logistik-, Materialfluss- und Produktionssystemen - Grundlagen. VDI Richtlinie 3633, Blatt 1, VDI-Verlag, Düsseldorf.

VDI (2016): Simulation von Logistik-, Materialfluss- und Produktionssystemen - Simulation und Optimierung. VDI Richtlinie 3633, Blatt 12, VDI-Verlag, Düsseldorf

T 6.50 Module component: Applied Materials Simulation [T-MACH-110929]

Coordinators: Prof. Dr. Peter Gumbsch
Dr.-Ing. Johannes Schneider
Dr. Daniel Weygand

Organisation: KIT Department of Mechanical Engineering

Part of: M-WIWI-107718 - Mechanical Engineering I
M-WIWI-107730 - Mechanical Engineering II

Type	Credits	Grading	Term offered	Version
Oral examination	4,5 CP	graded	Each summer term	2

Courses					
ST 2026	2182616	Applied Materials Simulation	4 SWS	Lecture / Practice / 🗣	Gumbsch, Weygand

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
oral exam ca. 30 minutes

no tools or reference materials

Prerequisites
The successful participation in Exercises for Applied Materials Simulation is the condition for the admittance to the oral exam in Applied Materials Simulation.

T-MACH-107671 – Übungen zu Angewandte Werkstoffsimulation has not been started.

T-MACH-105527 – Angewandte Werkstoffsimulation has not been started.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-MACH-110928 - Exercises for Applied Materials Simulation must have been passed.

Additional Information
The course is offered in English.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Applied Materials Simulation

2182616, SS 2026, 4 SWS, Language: English, Open in study portal

Lecture / Practice (VÜ)
On-Site

Content
This lecture should give the students an overview of different simulation methods in the field of materials science and engineering. Numerical methods are presented and their use in different fields of application and size scales shown and discussed. On the basis of theoretical as well as practical aspects, a critical examination of the opportunities and challenges of numerical material simulation shall be carried out.

The student can

- define different numerical methods and distinguish their range of application
- approach issues by applying the finite element method and discuss the processes and results
- understand complex processes of metal forming and crash simulation and discuss the structural and material behavior
- define and apply the physical fundamentals of particle-based simulation techniques to applications of materials science
- illustrate the range of application of atomistic simulation methods and distinguish between different models

preliminary knowlegde in mathematics, physics and materials science recommended

regular attendance: 34 hours

exercise: 11 hours

self-study: 165 hours

oral exam ca. 35 minutes

no tools or reference materials

admission to the exam only with successful completion of the exercises

Literature

1. D. Frenkel, B. Smit: Understanding Molecular Simulation: From Algorithms to Applications, Academic Press, 2001
2. W. Kurz, D.J. Fisher: Fundamentals of Solidification, Trans Tech Publications, 1998
3. P. Haupt: Continuum Mechanics and Theory of Materials, Springer, 1999
4. M. P. Allen, D. J. Tildesley: Computer simulation of liquids, Clarendon Press, 1996

T 6.51 Module component: Artificial Intelligence in Production [T-MACH-112115]

Coordinators: Prof. Dr.-Ing. Jürgen Fleischer
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-106590 - Production Engineering

Type	Credits	Grading	Term offered	Version
Written examination	4 CP	graded	Each winter term	1

Courses					
WT 25/26	2149921	Artificial Intelligence in Production	3 SWS	Lecture / 🗣	Fleischer
WT 26/27	2149921	Artificial Intelligence in Production	3 SWS	Lecture / 🗣	Fleischer

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Written Exam (90 min)

Prerequisites
none

Additional Information
The course is offered in German.

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Artificial Intelligence in Production

2149921, WS 25/26, 3 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content
The module AI in Production is designed to teach students the practical, holistic integration of machine learning and artificial intelligence methods in production. The course is oriented towards the phases of the CRISP-DM process with the aim of developing a deep understanding of the necessary steps and content-related aspects (methods) within the individual phases. In addition to teaching the practical aspects of integrating the most important machine learning methods, the focus is primarily on the necessary steps for data generation and data preparation as well as the implementation and validation of the methods in an industrial environment.

The lecture"Artificial Intelligence in Production" deals with the theoretical basics in a practical context. Here, the six phases of the CRISP-DM process are run through sequentially and the necessary basics for the implementation of the respective phases are taught. The course first deals with the data sources that are prevalent in the production environment. Subsequently, possibilities for target-oriented data acquisition as well as data transfer and data storage are introduced. Possibilities for data filtering and data preprocessing are discussed and production-relevant aspects are pointed out. The course then covers in detail the necessary algorithms and procedures for implementing AI in production, before techniques and fundamentals for making the models permanent in production (deployment) are discussed.

Learning Outcomes:

The students

- understand the relevance for the application of AI in production and know the main drivers and challenges.
- will understand the CRISP-DM process for implementing AI projects in manufacturing. Students will be able to name the main data sources, data ingestion methods, communication architectures, models and methods for data processing.
- will understand the main machine learning techniques and be able to contrast and select them in the context of industrial issues.
- are able to assess whether a specific problem in the context of production can be solved in a target-oriented manner using machine learning methods, as well as what the necessary steps are for implementation.
- are able to assess the most important challenges and name possible approaches to solve them.
- are able to apply the phases of the CRISP-DM to a problem in production. Students will know the steps necessary to build a data pipeline and will be able to do so theoretically in the context of a real-world use case.
- are able to evaluate the results of common deep learning methods and, based on this, to theoretically elaborate and theoretically apply proposed solutions (from the field of machine learning).

Workload:

MACH:
regular attendance: 31,5 hours
self-study: 88,5 hours
WING:
regular attendance: 31,5 hours
self-study: 118,5 hours

Organizational issues
Vorlesungstermine freitags 14:00 Uhr, begleitet durch Online-Programmierübungen.

Zur Vertiefung des im Rahmen der Lehrveranstaltung erworbenen Wissens werden die theoretischen Vorlesungseinheiten durch Praxiseinheiten im Umfeld der Karlsruher Forschungsfabrik (https://www.karlsruher-forschungsfabrik.de) unterstützt.

The theoretical lectures are complemented by practical lectures in the Karlsruhe Research Factory (https://www.karlsruher-forschungsfabrik.de/en.html) to deepen the acquired knowledge.

Literature
Skript zur Veranstaltung wird über Ilias (https://ilias.studium.kit.edu/) bereitgestellt.

Lecture notes will be provided in Ilias (https://ilias.studium.kit.edu/).

V **Artificial Intelligence in Production** **Lecture (V)**
2149921, WS 26/27, 3 SWS, Language: German, Open in study portal **On-Site**

Content
The module AI in Production is designed to teach students the practical, holistic integration of machine learning and artificial intelligence methods in production. The course is oriented towards the phases of the CRISP-DM process with the aim of developing a deep understanding of the necessary steps and content-related aspects (methods) within the individual phases. In addition to teaching the practical aspects of integrating the most important machine learning methods, the focus is primarily on the necessary steps for data generation and data preparation as well as the implementation and validation of the methods in an industrial environment.

The lecture"Artificial Intelligence in Production" deals with the theoretical basics in a practical context. Here, the six phases of the CRISP-DM process are run through sequentially and the necessary basics for the implementation of the respective phases are taught. The course first deals with the data sources that are prevalent in the production environment. Subsequently, possibilities for target-oriented data acquisition as well as data transfer and data storage are introduced. Possibilities for data filtering and data preprocessing are discussed and production-relevant aspects are pointed out. The course then covers in detail the necessary algorithms and procedures for implementing AI in production, before techniques and fundamentals for making the models permanent in production (deployment) are discussed.

Learning Outcomes:

The students

- understand the relevance for the application of AI in production and know the main drivers and challenges.
- will understand the CRISP-DM process for implementing AI projects in manufacturing. Students will be able to name the main data sources, data ingestion methods, communication architectures, models and methods for data processing.
- will understand the main machine learning techniques and be able to contrast and select them in the context of industrial issues.
- are able to assess whether a specific problem in the context of production can be solved in a target-oriented manner using machine learning methods, as well as what the necessary steps are for implementation.
- are able to assess the most important challenges and name possible approaches to solve them.
- are able to apply the phases of the CRISP-DM to a problem in production. Students will know the steps necessary to build a data pipeline and will be able to do so theoretically in the context of a real-world use case.
- are able to evaluate the results of common deep learning methods and, based on this, to theoretically elaborate and theoretically apply proposed solutions (from the field of machine learning).

Workload:

MACH:
regular attendance: 31,5 hours
self-study: 88,5 hours
WING:
regular attendance: 31,5 hours
self-study: 118,5 hours

Organizational issues
Vorlesungstermine begleitet durch Online-Programmierübungen.

Zur Vertiefung des im Rahmen der Lehrveranstaltung erworbenen Wissens werden die theoretischen Vorlesungseinheiten durch Praxiseinheiten im Umfeld der Karlsruher Forschungsfabrik (https://www.karlsruher-forschungsfabrik.de) unterstützt.

The theoretical lectures are complemented by practical lectures in the Karlsruhe Research Factory (https://www.karlsruher-forschungsfabrik.de/en.html) to deepen the acquired knowledge.

Literature
Skript zur Veranstaltung wird über Ilias (https://ilias.studium.kit.edu/) bereitgestellt.

Lecture notes will be provided in Ilias (https://ilias.studium.kit.edu/).

T 6.52 Module component: Artificial Intelligence in Service Systems [T-WIWI-108715]

Coordinators: Prof. Dr. Gerhard Satzger
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101448 - Service Management
M-WIWI-101506 - Service Analytics
M-WIWI-103117 - Data Science: Data-Driven Information Systems
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each winter term	3

Courses					
WT 25/26	2595650	Artificial Intelligence in Service Systems	1.5 SWS	Lecture /	Spitzer, Holstein, Satzger
WT 25/26	2595651	Übung zu Artificial Intelligence in Service Systems	1.5 SWS	Practice /	Spitzer, Holstein, Hendriks, Satzger
WT 26/27	2595650	Artificial Intelligence in Service Systems	1.5 SWS	Lecture /	Spitzer, Holstein, Satzger
WT 26/27	2595651	Übung zu Artificial Intelligence in Service Systems	1.5 SWS	Practice /	Spitzer, Holstein, Hendriks, Satzger

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
The assessment consists of a written exam (60 min). Successful completion of the exercises is a prerequisite for admission to the written exam.

Prerequisites
None

Additional Information
The course will be offered in the form of a flipped classroom concept starting in winter semester 2022/2023. The lecture will be recorded in advance and made available online. During the exercise classes, the contents of the lecture will be discussed and applied as part of programming exercises.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Artificial Intelligence in Service Systems

2595650, WS 25/26, 1.5 SWS, Language: English, Open in study portal

Lecture (V)
Blended (On-Site/Online)

Content
Artificial Intelligence (AI) and the application of machine learning is becoming more and more popular to solve relevant business challenges — both within isolated entities but also within co-creating systems (like value chains). However, it is not only essential to be familiar with precise algorithms but rather a general understanding of the necessary steps with a holistic view—from real-world challenges to the successful deployment of an AI-based solution. As part of this course, we teach the complete lifecycle of an AI project focusing on supervised machine learning challenges. We do so by also introducing the use of Python and the required packages like scikit-learn, pytorch and langchain with exemplary data and use cases. We then take this knowledge to the more complex case of service systems with different entities (e.g., companies) who interact with each other and show possibilities on how to derive holistic insights. Apart from the technical aspects necessary when developing AI within service systems, we also shed light on the collaboration of humans and AI in such systems (e.g., with the support of XAI), topics of ethics and bias in AI, as well as AI's capabilities on being creative - like generative AI creating songs and images.Students of this course will be able to understand and implement the complete lifecycle of a typical Artificial Intelligence use case with supervised machine learning. Furthermore, they understand the importance and the means of applying AI within service systems, which allows multiple, independent entities to collaborate and derive insights. Besides technical aspects, they will gain an understanding of the broader challenges and aspects when dealing with AI. Students will be proficient with typical Python code for AI challenges.

Workload:

- Attendance time – Lecture (1.5 SWS × 15 weeks): 22.5 h
- Attendance time – Tutorial (1.5 SWS × 15 weeks): 22.5 h
- Self-study – Lecture: 35 h
- Self-study – Tutorial: 35 h
- Exam preparation: 20 h
- **Total workload: 135 h**

Organizational issues
The course will be offered in the form of a flipped classroom concept starting in winter semester 2022/2023. The lecture will be recorded in advance and made available online. During the exercise classes, the contents of the lecture will be discussed and applied as part of programming exercises.

Literature

- Baier, L., Kühl, N., & Satzger, G. (2019). How to cope with change?-preserving validity of predictive services over time. In Proceedings of the 52nd Hawaii International Conference on System Sciences.
- Cawley, G. C., & Talbot, N. L. (2010). On over-fitting in model selection and subsequent selection bias in performance evaluation. The Journal of Machine Learning Research, 11, 2079-2107.
- Fink, O., Netland, T., & Feuerriegel, S. (2021). Artificial intelligence across company borders. arXiv preprint arXiv:2107.03912.
- Gama, J., Žliobaitė, I., Bifet, A., Pechenizkiy, M., & Bouchachia, A. (2014). A survey on concept drift adaptation. ACM computing surveys (CSUR), 46(4), 1-37.
- Hemmer, P., Schemmer, M., Vössing, M., & Kühl, N. (2021). Human-AI Complementarity in Hybrid Intelligence Systems: A Structured Literature Review. PACIS 2021 Proceedings.
- Hirt, R., & Kühl, N. (2018). Cognition in the Era of Smart Service Systems: Inter-organizational Analytics through Meta and Transfer Learning. In 39th International Conference on Information Systems, ICIS 2018; San Francisco Marriott MarquisSan Francisco; United States; 13 December 2018 through 16 December 2018.
- Holstein, J., Spitzer, P., Hoell, M., Vössing, M., & Kühl, N. (2024). Understanding Data Understanding: A Framework to Navigate the Intricacies of Data Analytics. In European Conference on Information Systems (ECIS 2024), Paphos, Cyprus, 13-19 June, 2024.
- Kühl, N., Goutier, M., Hirt, R., & Satzger, G. (2019, January). Machine Learning in Artificial Intelligence: Towards a Common Understanding. In Proceedings of the 52nd Hawaii International Conference on System Sciences.
- Kühl, N., Hirt, R., Baier, L., Schmitz, B., & Satzger, G. (2021). How to Conduct Rigorous Supervised Machine Learning in Information Systems Research: The Supervised Machine Learning Report Card. Communications of the Association for Information Systems, 48(1), 46.
- Maleshkova, M., Kühl, N., & Jussen, P. (Eds.). (2020). Smart Service Management: Design Guidelines and Best Practices. Springer Nature.
- Martin, D., Hirt, R., & Kühl, N. (2019). Service Systems, Smart Service Systems and Cyber-Physical Systems—What's the difference? Towards a Unified Terminology. 14. Internationale Tagung Wirtschaftsinformatik 2019 (WI 2019), Siegen, Germany, February 24-27.
- Mehrabi, N., Morstatter, F., Saxena, N., Lerman, K., & Galstyan, A. (2019). A survey on bias and fairness in machine learning. arXiv preprint arXiv:1908.09635.
- Schemmer, M., Bartos, A., Spitzer, P., Hemmer, P., Kühl, N., Liebschner, J., & Satzger, G. (2023). Towards Effective Human-AI Decision-Making: The Role of Human Learning in Appropriate Reliance on AI Advice. In Proceedings of the 44th International Conference on Information Systems (ICIS2023), Hyderabad, India.
- Schöffer, J., Machowski, Y., & Kühl, N. (2021). A Study on Fairness and Trust Perceptions in Automated Decision Making. In Joint Proceedings of the ACM IUI 2021 Workshops, April 13–17, 2021, College Station, USA.
- Spitzer, P., Kühl, N., Goutier, M., Kaschura, M., & Satzger, G. (2024). Transferring Domain Knowledge with (X) AI-Based Learning Systems. In European Conference on Information Systems (ECIS 2024), Paphos, Cyprus, 13-19 June, 2024.
- Zahn, M. V., Feuerriegel, S., & Kühl, N. (2021). The cost of fairness in AI: Evidence from e-commerce. Business & information systems engineering.

V

Artificial Intelligence in Service Systems

2595650, WS 26/27, 1.5 SWS, Language: English, Open in study portal

Lecture (V)
Blended (On-Site/Online)

Content

Artificial Intelligence (AI) and the application of machine learning is becoming more and more popular to solve relevant business challenges — both within isolated entities but also within co-creating systems (like value chains). However, it is not only essential to be familiar with precise algorithms but rather a general understanding of the necessary steps with a holistic view— from real-world challenges to the successful deployment of an AI-based solution. As part of this course, we teach the complete lifecycle of an AI project focusing on supervised machine learning challenges. We do so by also introducing the use of Python and the required packages like scikit-learn, pytorch and langchain with exemplary data and use cases. We then take this knowledge to the more complex case of service systems with different entities (e.g., companies) who interact with each other and show possibilities on how to derive holistic insights. Apart from the technical aspects necessary when developing AI within service systems, we also shed light on the collaboration of humans and AI in such systems (e.g., with the support of XAI), topics of ethics and bias in AI, as well as AI's capabilities on being creative - like generative AI creating songs and images.Students of this course will be able to understand and implement the complete lifecycle of a typical Artificial Intelligence use case with supervised machine learning. Furthermore, they understand the importance and the means of applying AI within service systems, which allows multiple, independent entities to collaborate and derive insights. Besides technical aspects, they will gain an understanding of the broader challenges and aspects when dealing with AI. Students will be proficient with typical Python code for AI challenges.

Workload:

- Attendance time – Lecture (1.5 SWS × 15 weeks): 22.5 h
- Attendance time – Tutorial (1.5 SWS × 15 weeks): 22.5 h
- Self-study – Lecture: 35 h
- Self-study – Tutorial: 35 h
- Exam preparation: 20 h
- **Total workload: 135 h**

Organizational issues

The course will be offered in the form of a flipped classroom concept starting in winter semester 2022/2023. The lecture will be recorded in advance and made available online. During the exercise classes, the contents of the lecture will be discussed and applied as part of programming exercises.

Literature

- Baier, L., Kühl, N., & Satzger, G. (2019). How to cope with change?-preserving validity of predictive services over time. In Proceedings of the 52nd Hawaii International Conference on System Sciences.
- Buçinca, Z., Swaroop, S., Paluch, A. E., Doshi-Velez, F., & Gajos, K. Z. (2025). Contrastive explanations that anticipate human misconceptions can improve human decision-making skills. In Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems.
- Cawley, G. C., & Talbot, N. L. (2010). On over-fitting in model selection and subsequent selection bias in performance evaluation. The Journal of Machine Learning Research, 11, 2079-2107.
- Fink, O., Netland, T., & Feuerriegel, S. (2021). Artificial intelligence across company borders. arXiv preprint arXiv:2107.03912.
- Gama, J., Žliobaitė, I., Bifet, A., Pechenizkiy, M., & Bouchachia, A. (2014). A survey on concept drift adaptation. ACM computing surveys (CSUR), 46(4), 1-37.
- Hemmer, P., Schemmer, M., Vössing, M., & Kühl, N. (2021). Human-AI Complementarity in Hybrid Intelligence Systems: A Structured Literature Review. PACIS 2021 Proceedings.
- Hirt, R., & Kühl, N. (2018). Cognition in the Era of Smart Service Systems: Inter-organizational Analytics through Meta and Transfer Learning. In 39th International Conference on Information Systems, ICIS 2018; San Francisco Marriott MarquisSan Francisco; United States; 13 December 2018 through 16 December 2018.
- Holstein, J., Spitzer, P., Hoell, M., Vössing, M., & Kühl, N. (2024). Understanding Data Understanding: A Framework to Navigate the Intricacies of Data Analytics. In European Conference on Information Systems (ECIS 2024), Paphos, Cyprus, 13-19 June, 2024.
- Kühl, N., Goutier, M., Hirt, R., & Satzger, G. (2019, January). Machine Learning in Artificial Intelligence: Towards a Common Understanding. In Proceedings of the 52nd Hawaii International Conference on System Sciences.
- Kühl, N., Hirt, R., Baier, L., Schmitz, B., & Satzger, G. (2021). How to Conduct Rigorous Supervised Machine Learning in Information Systems Research: The Supervised Machine Learning Report Card. Communications of the Association for Information Systems, 48(1), 46.
- Maleshkova, M., Kühl, N., & Jussen, P. (Eds.). (2020). Smart Service Management: Design Guidelines and Best Practices. Springer Nature.
- Martin, D., Hirt, R., & Kühl, N. (2019). Service Systems, Smart Service Systems and Cyber-Physical Systems—What's the difference? Towards a Unified Terminology. 14. Internationale Tagung Wirtschaftsinformatik 2019 (WI 2019), Siegen, Germany, February 24-27.
- Mehrabi, N., Morstatter, F., Saxena, N., Lerman, K., & Galstyan, A. (2019). A survey on bias and fairness in machine learning. arXiv preprint arXiv:1908.09635.
- Schemmer, M., Bartos, A., Spitzer, P., Hemmer, P., Kühl, N., Liebschner, J., & Satzger, G. (2023). Towards Effective Human-AI Decision-Making: The Role of Human Learning in Appropriate Reliance on AI Advice. In Proceedings of the 44th International Conference on Information Systems (ICIS2023), Hyderabad, India.
- Schöffer, J., Machowski, Y., & Kühl, N. (2021). A Study on Fairness and Trust Perceptions in Automated Decision Making. In Joint Proceedings of the ACM IUI 2021 Workshops, April 13–17, 2021, College Station, USA.
- Spitzer, P., Kühl, N., Goutier, M., Kaschura, M., & Satzger, G. (2024). Transferring Domain Knowledge with (X) AI-Based Learning Systems. In European Conference on Information Systems (ECIS 2024), Paphos, Cyprus, 13-19 June, 2024.
- Von Zahn, M., Liebich, L., Jussupow, E., Hinz, O., & Bauer, K. (2025). Knowing (not) to know: Explainable artificial intelligence and human metacognition. *Information Systems Research*.
- Zahn, M. V., Feuerriegel, S., & Kühl, N. (2021). The cost of fairness in AI: Evidence from e-commerce. Business & information systems engineering.

T 6.53 Module component: Artificial Intelligence in Service Systems II: Generative AI Applications & Adoption [T-WIWI-114209]

Coordinators: Prof. Dr. Gerhard Satzger
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101448 - Service Management
M-WIWI-101506 - Service Analytics
M-WIWI-103117 - Data Science: Data-Driven Information Systems
M-WIWI-104900 - Business Administration
M-WIWI-105661 - Data Science: Intelligent, Adaptive, and Learning Information Services

Type	Credits	Grading	Term offered	Version
Examination of another type	4,5 CP	graded	Each summer term	1

Courses					
ST 2026	2595501	Artificial Intelligence in Service Systems - Generative AI Applications and Adoption	3 SWS	Lecture / 🗣	Spitzer, Holstein, Satzger

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success is assessed in the form of an examination of another type. The following aspects are included in the assessment:

- Collaborative development of a prototype as a group task
- Group presentation of the developed prototype
- Group report detailing the prototype and its development

Further details on the organization of the performance and the points system for the assessment will be announced in the lecture.

Additional Information
This course is admission restricted (see https://dsi.win.kit.edu/index.php). You can apply for this course via the Wiwi-Portal.
The course replaces T-WIWI-111219 "Artificial Intelligence in Service Systems - Applications in Computer Vision" as of summer semester 2025.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Artificial Intelligence in Service Systems - Generative AI Applications and Adoption

Lecture (V)
On-Site

2595501, SS 2026, 3 SWS, Language: English, Open in study portal

Content

---We renamed this course from "Artificial Intelligence in Service Systems - Applications in Computer Vision" to "Artificial Intelligence in Service Systems - Generative AI Applications and Adoption" ---

Learning objectives

This course provides deepens the students's theoretical knowledge and practical skills in developing AI-based services. It adds "state-of-the-art" generative AI technologies and the focus on integrating AI-based services into larger service systems and organizational workflows. Students will not only learn core theoretical concepts and frameworks, but also engage in team projects to gain hands-on experience in implementing and adapting these services for human adoption.

Description

This course builds on the course "Artificial Intelligence in Service Systems" (LV-Nr.: 2595650) and applies the "end-to-end" development of AI-based services to particular team projects with two key objectives: (1) capturing new Generative AI methods, but also (2) focus on the integration of the service in organizational workflows and the necessary adoption by humans. Starting with the fundamentals of generative AI, students work with Large Language Models (LLMs) and multimodal architectures to develop practical applications. Building on these implementations, the course investigates how to integrate these services into organizational workflows and information systems, focusing on user interaction, system transparency, and human-AI collaboration mechanisms.

Through a group project, students apply their learning by first implementing a technical artifact to address real-world challenges, then identifying and applying appropriate metrics to design and evaluate adoption while considering human factors such as user acceptance, trust, workflow integration, and ethical implications. This hands-on approach provides students with practical experience in both technical implementation and organizational integration of AI-based services.

Recommendations

The course is aimed at students in the Master's program with basic knowledge in statistics and applied programming in Python. Knowledge from the lecture Artificial Intelligence in Service Systems may be beneficial.

Additional information

- Group-based project work
- Flipped classroom format with pre-recorded lectures
- Three half-day block sessions for in-depth discussions and optional hands-on coding exercises

Workload:

- Attendance time (3 SWS × 15 weeks): 45 h
- Self-study: 60 h
- Exam preparation: 30 h
- **Total workload: 135 h**

Literature

- Baltrušaitis, T., Ahuja, C. and Morency, L.P., 2018. Multimodal machine learning: A survey and taxonomy. *IEEE transactions on pattern analysis and machine intelligence*, *41*(2), pp.423-443
- Chang, Y., Wang, X., Wang, J., Wu, Y., Yang, L., Zhu, K., Chen, H, Yi, X., Wang, C., Wang, Y., Ye, W., Zhang, Y., Chang, Y., Yu, P, Yang, Q., and Xie, X. (2024). A Survey on Evaluation of Large Language Models. ACM Trans. Intell. Syst. Technol. 15, 3, Article 39 (June 2024), 45 pages.
- Fourney, A., Bansal, G., Mozannar, H., Tan, C., Salinas, E., Niedtner, F., Proebsting, G., Bassman, G., Gerrits, J., Alber, J. and Chang, P., 2024. Magentic-one: A generalist multi-agent system for solving complex tasks. *arXiv preprint arXiv:2411.04468*.
- Hemmer, P., Schemmer, M., Kühl, N., Vössing, M., & Satzger, G. (2024). Complementarity in Human-AI Collaboration: Concept, Sources, and Evidence. *arXiv preprint arXiv:2404.00029*.
- Kreuzberger, D., Kühl, N. and Hirschl, S., 2023. Machine learning operations (mlops): Overview, definition, and architecture. *IEEE access*, *11*, pp.31866-31879.
- Schemmer, M., Kuehl, N., Benz, C., Bartos, A., & Satzger, G. (2023, March). Appropriate reliance on AI advice: Conceptualization and the effect of explanations. In *Proceedings of the 28th International Conference on Intelligent User Interfaces* (pp. 410-422).
- Zhang, Y., Li, Y., Cui, L., Cai, D., Liu, L., Fu, T., Huang, X., Zhao, E., Zhang, Y., Chen, Y., Wang, L., Luu, A.T., Bi, W., Shi, F., & Shi, S. (2023). Siren's Song in the AI Ocean: A Survey on Hallucination in Large Language Models. *ArXiv, abs/2309.01219*.

T 6.54 Module component: Asset Pricing [T-WIWI-102647]

Coordinators: Prof. Dr. Martin Ruckes
Prof. Dr. Marliese Uhrig-Homburg

Organisation: KIT Department of Business and Economics

Part of: M-WIWI-101480 - Finance 3
M-WIWI-101482 - Finance 1
M-WIWI-101483 - Finance 2
M-WIWI-101502 - Economic Theory and its Application in Finance
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each summer term	2

Courses					
ST 2026	2530555	Asset Pricing	2 SWS	Lecture / 🗣	Uhrig-Homburg, Thimme
ST 2026	2530556	Asset Pricing Exercices	1 SWS	Practice / 🗣	Böll, Uhrig-Homburg, Thimme

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success is assessed in the form of a written examination (60 minutes). The examination is offered every semester and can be repeated at any regular examination date.

A bonus can be earned for successful participation in the exercises by submitting correct solutions to at least 50% of the bonus exercises set. If the grade of the written examination is between 4.0 and 1.3, the bonus improves the grade by up to one grade level (0.3 or 0.4). Details will be announced in the lecture.

Prerequisites
None

Recommendations
We strongly recommend knowledge of the basic topics in investments (bachelor course), which will be necessary to be able to follow the course.

Below you will find excerpts from courses related to this module component:

V Asset Pricing Exercices

2530556, SS 2026, 1 SWS, Language: German, Open in study portal

Practice (Ü)
On-Site

Organizational issues
Die Übung findet erst ab der 2. Vorlesungswoche statt.

T 6.55 Module component: Auction Theory [T-WIWI-102613]

Coordinators: Prof. Dr. Karl-Martin Ehrhart
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101446 - Market Engineering
M-WIWI-101453 - Applied Strategic Decisions
M-WIWI-101500 - Microeconomic Theory
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each winter term	1

Courses					
WT 25/26	2520408	Auction Theory	2 SWS	Lecture	Ehrhart
WT 25/26	2520409	Auction Theory Exercise	1 SWS	Practice	Ehrhart

Assessment
The assessment of this course is a written examination (following §4(2), 1 SPO) of 60 mins.

The exam is offered each semester.

Prerequisites
None

Below you will find excerpts from courses related to this module component:

V Auction Theory
2520408, WS 25/26, 2 SWS, Open in study portal

Lecture (V)

Content
Workload:

- Attendance time lecture (2 SWS × 15 weeks): 30 h
- Exercise attendance time (1 SWS × 15 weeks): 15 h
- Self-study lecture: 45 h
- Self-study exercise: 25 h
- Exam preparation: 20 h
- **Total: 135 h**

Literature

- Ehrhart , K.-M. und S. Seifert: Auktionstheorie, Skript zur Vorlesung, KIT, 2011
- Krishna, V.: Auction Theory, Academic Press, Second Edition, 2010
- Milgrom, P.: Putting Auction Theory to Work, Cambridge University Press, 2004
- Ausubel, L.M. und P. Cramton: Demand Reduction and Inefficiency in Multi-Unit Auctions, University of Maryland, 1999

T 6.56 Module component: Automated Financial Advisory [T-WIWI-106495]

Coordinators: Prof. Dr. Maxim Ulrich
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Each summer term	1

Assessment
The grade consists of a written thesis and an oral presentation.

Prerequisites
There are two conditions for taking this course:

1. This course is only open for registered students of the module “Disruptive FinTech Innovations”.
2. Registered students do also attend in the same semester the lecture “Engineering FinTech Solutions” and the programming internship “Computational FinTech with Python and C++”.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-WIWI-106193 - Engineering FinTech Solutions must have been started.

Workload
90 hours

T 6.57 Module component: Automotive Engineering I [T-MACH-102203]

Coordinators: Dr.-Ing. Martin Gießler
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101266 - Automotive Engineering
M-WIWI-104907 - Engineering Sciences
M-WIWI-107718 - Mechanical Engineering I
M-WIWI-107730 - Mechanical Engineering II

Type	Credits	Grading	Term offered	Version
Written examination	6 CP	graded	Each winter term	4

Courses					
WT 25/26	2113809	Automotive Engineering I	4 SWS	Lecture / 🗣	Gießler
WT 26/27	2113809	Automotive Engineering I	4 SWS	Lecture / 🗣	Gießler

Legend: 🖥 Online, ❖ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Written examination

Duration: 90 minutes

Auxiliary means: none

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-MACH-100092 - Automotive Engineering I (in German) must not have been started.

Additional Information
The course is offered in English.

Workload
180 hours

Below you will find excerpts from courses related to this module component:

V Automotive Engineering I

2113809, WS 25/26, 4 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content
1. History and future of the automobile

2. Driving mechanics: driving resistances and driving performances, mechanics of longitudinal and lateral forces, active and passive safety

3. Drive systems: combustion engine, hybrid and electric drive systems

4. Transmission: clutches (e.g. friction clutch, visco clutch), transmission (e.g. mechanical transmission, hydraulic fluid transmission)

5.Power transmission and distribution: drive shafts, cardon joints, differentials

Learning Objectives:

The students know the movements and the forces at the vehicle and are familiar with active and passive safety. They have proper knowledge about operation of engines and alternative drives, the necessary transmission between engine and drive wheels and the power distribution. They have an overview of the components necessary for the drive and have the basic knowledge, to analyze, to evaluate, and to develop the complex system "vehicle".

Organizational issues
You will find the lecture material on ILIAS. To get the ILIAS password, KIT students refer to https://fast-web-01.fast.kit.edu/PasswoerterIlias/, students from eucor universities send an e-mail to martina.kaiser@kit.edu

Kann nicht mit LV Grundlagen der Fahrzeugtechnik I [2113805] kombiniert werden.

Can not be combined with lecture [2113805] Grundlagen der Fahrzeugtechnik I.

Literature
1. Robert Bosch GmbH: Automotive Handbook, 9th Edition, Wiley, Chichister 2015

2. Onori, S. / Serrao, L: / Rizzoni, G.: Hybrid Electric Vehicles - Energy Management Strategies, Springer London, Heidelberg, New York, Dordrecht 2016

3. Reif, K.: Brakes, Brake Control and Driver Assistance Systems - Function, Regulation and Components, Springer Vieweg, Wiesbaden 2015

4. Gauterin, F. / Gießler, M. / Gnadler, R.: Scriptum zur Vorlesung 'Automotive Engineering I', KIT, Institut für Fahrzeugsystemtechnik, Karlsruhe, jährlich aktualisiert

V	**Automotive Engineering I** 2113809, WS 26/27, 4 SWS, Language: English, Open in study portal	**Lecture (V)** **On-Site**

Content
1. History and future of the automobile

2. Driving mechanics: driving resistances and driving performances, mechanics of longitudinal and lateral forces, active and passive safety

3. Drive systems: combustion engine, hybrid and electric drive systems

4. Transmission: clutches (e.g. friction clutch, visco clutch), transmission (e.g. mechanical transmission, hydraulic fluid transmission)

5.Power transmission and distribution: drive shafts, cardon joints, differentials

Learning Objectives:

The students know the movements and the forces at the vehicle and are familiar with active and passive safety. They have proper knowledge about operation of engines and alternative drives, the necessary transmission between engine and drive wheels and the power distribution. They have an overview of the components necessary for the drive and have the basic knowledge, to analyze, to evaluate, and to develop the complex system "vehicle".

Organizational issues
You will find the lecture material on ILIAS. To get the ILIAS password, KIT students refer to https://fast-web-01.fast.kit.edu/PasswoerterIlias/, students from eucor universities send an e-mail to martina.kaiser@kit.edu

Kann nicht mit LV Grundlagen der Fahrzeugtechnik I [2113805] kombiniert werden.

Can not be combined with lecture [2113805] Grundlagen der Fahrzeugtechnik I.

Literature
1. Robert Bosch GmbH: Automotive Handbook, 9th Edition, Wiley, Chichister 2015

2. Onori, S. / Serrao, L: / Rizzoni, G.: Hybrid Electric Vehicles - Energy Management Strategies, Springer London, Heidelberg, New York, Dordrecht 2016

3. Reif, K.: Brakes, Brake Control and Driver Assistance Systems - Function, Regulation and Components, Springer Vieweg, Wiesbaden 2015

4. Gauterin, F. / Gießler, M. / Gnadler, R.: Scriptum zur Vorlesung 'Automotive Engineering I', KIT, Institut für Fahrzeugsystemtechnik, Karlsruhe, jährlich aktualisiert

T 6.58 Module component: Automotive Engineering I (in German) [T-MACH-100092]

Coordinators: Dr.-Ing. Martin Gießler
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101266 - Automotive Engineering
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Expansion	Language	Version
Written examination	8 CP	graded	Each winter term	1 semesters		4

Courses					
WT 25/26	2113805	Automotive Engineering I	4 SWS	Lecture / 🗣	Gießler
WT 26/27	2113805	Automotive Engineering I	4 SWS	Lecture / 🗣	Gießler

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Written examination

Duration: 120 minutes

Auxiliary means: none

Prerequisites
The module component "T-MACH-102203 - Automotive Engineering I" is not started or finished. The module components "T-MACH-100092 - Grundlagen der Fahrzeugtechnik I" and "T-MACH-102203 - Automotive Engineering I" can not be combined.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-MACH-102203 - Automotive Engineering I must not have been started.

Additional Information
The course is offered in German.

Workload
240 hours

Below you will find excerpts from courses related to this module component:

V Automotive Engineering I

2113805, WS 25/26, 4 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content
1. History and future of the automobile

2. Driving mechanics: driving resistances and driving performance, mechanics of longitudinal and lateral forces, active and passive safety

3. Drive systems: combustion engine, hybrid and electric drive systems

4. Transmission: clutches (e.g. friction clutch, visco clutch), transmission (e.g. mechanical transmission, hydraulic fluid transmission)

5. Power transmission and distribution: drive shafts, cardon joints, differentials

Learning Objectives:

The students know the movements and the forces at the vehicle and are familiar with active and passive safety. They have proper knowledge about operation of engines and alternative drives, the necessary transmission between engine and drive wheels and the power distribution. They have an overview of the components necessary for the drive and have the basic knowledge, to analyze, to evaluate, and to develop the complex system "vehicle".

Organizational issues
Das Vorlesungsmaterial wird auf ILIAS bereitgestellt. Das ILIAS-Passwort erhalten Sie unter https://fast-web-01.fast.kit.edu/PasswoerterIlias/

Kann nicht mit der Veranstaltung [2113809] kombiniert werden.

Can not be combined with lecture [2113809].

Literature
1. Mitschke, M. / Wallentowitz, H.: Dynamik der Kraftfahrzeuge, Springer Vieweg, Wiesbaden 2014

2. Pischinger, S. / Seiffert, U.: Handbuch Kraftfahrzeugtechnik, Springer Vieweg, Wiesbaden 2016

3. Gauterin, F. / Unrau, H.-J. / Gnadler, R.: Scriptum zur Vorlesung "Grundlagen der Fahrzeugtechnik I", KIT, Institut für Fahrzeugsystemtechnik, Karlsruhe, jährlich aktualisiert

V **Automotive Engineering I** — **Lecture (V) On-Site**
2113805, WS 26/27, 4 SWS, Language: German, Open in study portal

Content
1. History and future of the automobile

2. Driving mechanics: driving resistances and driving performance, mechanics of longitudinal and lateral forces, active and passive safety

3. Drive systems: combustion engine, hybrid and electric drive systems

4. Transmission: clutches (e.g. friction clutch, visco clutch), transmission (e.g. mechanical transmission, hydraulic fluid transmission)

5. Power transmission and distribution: drive shafts, cardon joints, differentials

Learning Objectives:

The students know the movements and the forces at the vehicle and are familiar with active and passive safety. They have proper knowledge about operation of engines and alternative drives, the necessary transmission between engine and drive wheels and the power distribution. They have an overview of the components necessary for the drive and have the basic knowledge, to analyze, to evaluate, and to develop the complex system "vehicle".

Organizational issues
Das Vorlesungsmaterial wird auf ILIAS bereitgestellt. Das ILIAS-Passwort erhalten Sie unter https://fast-web-01.fast.kit.edu/PasswoerterIlias/

Kann nicht mit der Veranstaltung [2113809] kombiniert werden.

Can not be combined with lecture [2113809].

Literature
1. Mitschke, M. / Wallentowitz, H.: Dynamik der Kraftfahrzeuge, Springer Vieweg, Wiesbaden 2014

2. Pischinger, S. / Seiffert, U.: Handbuch Kraftfahrzeugtechnik, Springer Vieweg, Wiesbaden 2016

3. Gauterin, F. / Unrau, H.-J. / Gnadler, R.: Scriptum zur Vorlesung "Grundlagen der Fahrzeugtechnik I", KIT, Institut für Fahrzeugsystemtechnik, Karlsruhe, jährlich aktualisiert

T 6.59 Module component: Automotive Engineering II [T-MACH-114435]

Coordinators: Dr.-Ing. Martin Gießler
Organisation: KIT Department of Mechanical Engineering

KIT Department of Business and Economics

Part of: M-MACH-101266 - Automotive Engineering
M-WIWI-104907 - Engineering Sciences
M-WIWI-107718 - Mechanical Engineering I
M-WIWI-107730 - Mechanical Engineering II

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each summer term	2

Courses					
ST 2026	2114855	Automotive Engineering II	2 SWS	Lecture / 🗣	Gießler

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, × Cancelled

Assessment
Written Examination

Duration: 90 minutes

Auxiliary means: none

Prerequisites
Only one out of the two moduls "M-MACH-102117 - Grundlagen der Fahrzeugtechnik II"and "M-MACH-114435 - Automotive Engineering II" is allowed.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-MACH-102117 - Automotive Engineering II (in German) must not have been started.

Additional Information
The course is offered in English.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Automotive Engineering II
2114855, SS 2026, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content

1. Chassis: Wheel suspensions (rear axles, front axles, kinematics of axles), tyres, springs, damping devices
2. Steering elements: Manual steering, servo steering, steer by wire
3. Brakes: Disc brake, drum brake, comparison of the designs

Learning Objectives:

The students have an overview of the modules which are necessary for the tracking of a motor vehicle and the power transmission between vehicle and roadway. They have knowledge of different wheel suspensions, tyres, steering elements, and brakes. They know different design versions, functions and the influence on driving and braking behavior. They are able to correctly develop the appropriate components. They are ready to analyze, to evaluate, and to optimize the complex interaction of the different components under consideration of boundary conditions.

Literature
Elective literature:

1. Robert Bosch GmbH: Automotive Handbook, 9th Edition, Wiley, Chichester 2015
2. Heißing, B. / Ersoy, M.: Chassis Handbook - fundamentals, driving dynamics, components, mechatronics, perspectives, Vieweg+Teubner, Wiesbaden 2011
3. Gießler, M. / Gnadler, R.: Script to the lecture "Automotive Engineering II", KIT, Institut of Vehicle System Technology, Karlsruhe, annual update

T 6.60 Module component: Automotive Engineering II (in German) [T-MACH-102117]

Coordinators: Dr.-Ing. Martin Gießler
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101266 - Automotive Engineering
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Written examination	4 CP	graded	Each summer term	3

Courses					
ST 2026	2114835	Automotive Engineering II	2 SWS	Lecture / 🗣	Gießler
ST 2026	2114855	Automotive Engineering II	2 SWS	Lecture / 🗣	Gießler

Legend: 📱 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Written Examination

Duration: 90 minutes

Auxiliary means: none

Prerequisites
Only one out of the two moduls "M-MACH-102117 - Grundlagen der Fahrzeugtechnik II"and "M-MACH-114435 - Automotive Engineering II" is allowed.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-MACH-114435 - Automotive Engineering II must not have been started.

Additional Information
The course is offered in German.

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Automotive Engineering II

2114835, SS 2026, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content
1. Chassis: Wheel suspensions (rear axles, front axles, kinematics of axles), tyres, springs, damping devices
2. Steering elements: Manual steering, servo steering, steer by wire
3. Brakes: Disc brake, drum brake, comparison of designs

Learning Objectives:

The students have an overview of the modules which are necessary for the tracking of a motor vehicle and the power transmission between vehicle bodywork and roadway. They have knowledge of different wheel suspensions, tyres, steering elements, and brakes. They know different design versions, functions and the influence on driving and braking behavior. They are able to correctly develop the appropriate components. They are ready to analyze, to evaluate, and to optimize the complex interaction of the different components under consideration of boundary conditions.

Organizational issues
Kann nicht mit der Veranstaltung [2114855] kombiniert werden.

Can not be combined with lecture [2114855]

Literature
1. Heißing, B. / Ersoy, M.: Fahrwerkhandbuch: Grundlagen, Fahrdynamik, Komponenten, Systeme, Mechatronik, Perspektiven, Springer Vieweg, Wiesbaden, 2013

2. Breuer, B. / Bill, K.-H.: Bremsenhandbuch: Grundlagen - Komponenten - Systeme - Fahrdynamik, Springer Vieweg, Wiesbaden, 2017

3. Unrau, H.-J. / Gnadler, R.: Scriptum zur Vorlesung 'Grundlagen der Fahrzeugtechnik II', KIT, Institut für Fahrzeugsystemtechnik, Karlsruhe, jährliche Aktualisierung

V **Automotive Engineering II**
2114855, SS 2026, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content

1. Chassis: Wheel suspensions (rear axles, front axles, kinematics of axles), tyres, springs, damping devices
2. Steering elements: Manual steering, servo steering, steer by wire
3. Brakes: Disc brake, drum brake, comparison of the designs

Learning Objectives:

The students have an overview of the modules which are necessary for the tracking of a motor vehicle and the power transmission between vehicle and roadway. They have knowledge of different wheel suspensions, tyres, steering elements, and brakes. They know different design versions, functions and the influence on driving and braking behavior. They are able to correctly develop the appropriate components. They are ready to analyze, to evaluate, and to optimize the complex interaction of the different components under consideration of boundary conditions.

Literature
Elective literature:

1. Robert Bosch GmbH: Automotive Handbook, 9th Edition, Wiley, Chichester 2015
2. Heißing, B. / Ersoy, M.: Chassis Handbook - fundamentals, driving dynamics, components, mechatronics, perspectives, Vieweg+Teubner, Wiesbaden 2011
3. Gießler, M. / Gnadler, R.: Script to the lecture "Automotive Engineering II", KIT, Institut of Vehicle System Technology, Karlsruhe, annual update

T 6.61 Module component: Automotive Vision [T-MACH-114149]

Coordinators: Dr. Martin Lauer
Prof. Dr.-Ing. Christoph Stiller

Organisation: KIT Department of Mechanical Engineering

Part of: M-WIWI-107718 - Mechanical Engineering I
M-WIWI-107730 - Mechanical Engineering II

Type	Credits	Grading	Term offered	Version
Written examination	6 CP	graded	Each summer term	3

Courses					
ST 2026	2138340	Automotive Vision	3 SWS	Lecture / 🗣	Lauer, Bätz

Legend: 🖥 Online, Blended (On-Site/Online), 🗣 On-Site, × Cancelled

Assessment
Type of Examination: written exam

Duration of Examination: 60 minutes

Prerequisites
none

Additional Information
The course is offered in English

Workload
180 hours

Below you will find excerpts from courses related to this module component:

V Automotive Vision

2138340, SS 2026, 3 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content
Lernziele (EN):

Machine perception and interpretation of the environment for the basis for the generation of intelligent behaviour. Especially visual perception opens the door to novel automotive applications. First driver assistance systems can already improve safety, comfort and efficiency in vehicles. Yet, several decades of research will be required to achieve an automated behaviour with a performance equivalent to a human operator. The lecture addresses students in mechanical engineering and related subjects who intend to get an interdisciplinary knowledge in a state-of-the-art technical domain. Machine vision, vehicle kinematics and advanced information processing techniques are presented to provide a broad overview on ßeeing vehicles'. Application examples from cutting-edge and future driver
assistance systems illustrate the discussed subjects.

Lehrinhalt (EN):

1. Driver assistance systems
2. Binocular vision
3. Feature point methods
4. Optical flow/tracking in images
5. Tracking and state estimation
6. Self-localization and mapping
7. Lane recognition
8. Behavior recognition

Nachweis: Written examination 60 minutes

Arbeitsaufwand (EN): 120 hours

Literature
Foliensatz zur Veranstaltung wird als kostenlose pdf-Datei bereitgestellt. Weitere Empfehlungen werden in der Vorlesung bekannt gegeben.

T 6.62 Module component: Autonomous Driving in Practice [T-MACH-115202]

Coordinators: Kevin Simon
Organisation: KIT Department of Mechanical Engineering
Part of: M-WIWI-107718 - Mechanical Engineering I
M-WIWI-107730 - Mechanical Engineering II

Type	Credits	Grading	Term offered	Version
Examination of another type	6 CP	graded	Each winter term	1

Courses					
WT 26/27	2113820	Autonomous Driving in Practice	3 SWS	Lecture / Practice / 🗣	Frey, Simon

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The final grade is based on: 2 written exams (duration about 30 minutes) and the final presentation of the practical task (duration 30 minutes max.), including the final report (2-3 pages). Submission of the homework is mandatory.

The overall impression will be evaluated.

Recommendations
Prior experience in autonomous driving, automotive engineering, and machine learning is a plus.

Additional Information
Due to capacity limitations, the lecture is limited to 16 students. If necessary, a selection process will be conducted. Places will be allocated based on the students' academic progress (semester level and subject-specific programming skills).

The course will be conducted in English.

Workload
180 hours

Below you will find excerpts from courses related to this module component:

V Autonomous Driving in Practice

2113820, WS 26/27, 3 SWS, Language: English, Open in study portal

Lecture / Practice (VÜ)
On-Site

Content
The main objective of this course is to implement the pipeline required for automated/autonomous driving functions using a real test vehicle. This includes environmental perception using various sensors, processing the collected sensor data, planning driving maneuvers, and finally executing the maneuvers via the actuators.

- Sensor data acquisition: Setup and data recording of the sensor systems on the test vehicle
- Perception: Data annotation, segmentation of sensor data, object detection
- Maneuver planning: Path and trajectory planning, behavior generation, etc.
- Maneuver execution: Vehicle control, implementation of the driving maneuver in the real test vehicle via actuators

Organizational issues
Begrenzte Teilnehmerzahl mit Auswahlverfahren, in englischer Sprache. Bewerbungen sind am Ende des vorhergehenden Semesters einzureichen.
Termin und Raum werden über die Institutshomepage bekannt gegeben.

Limited number of participants with selection procedure, in English language. Please send the application at the end of the previous semester

Date and room: see homepage of institute.

T 6.63 Module component: B2B Sales Management [T-WIWI-111367]

Coordinators: Prof. Dr. Martin Klarmann
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Examination of another type	4,5 CP	graded	Each winter term	1

Courses					
WT 25/26	2572187	B2B Sales Management	2 SWS	Lecture / 🗣	Klarmann
WT 25/26	2572188	Excercises B2B Sales Management	1 SWS	Practice / 🗣	Daumann, Weber
WT 26/27	2572187	B2B Sales Management	2 SWS	Lecture / 🗣	Klarmann

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment of success takes place through the preparation and presentation of a sales presentation based on a case study (max 30 points) and a written exam with additional aids in the sense of an open book exam (max. 60 points). In total, a maximum of 90 points can be achieved in the course. Further details will be announced during the lecture.

Prerequisites
None.

Additional Information
For further information, please contact Marketing and Sales Research Group (marketing.iism.kit.edu).

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V B2B Sales Management

2572187, WS 25/26, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content

Content

The event is designed to teach you taking on marketing responsibility in a very special business environment. This involves companies that sell and market their (often technically highly complex) products themselves to other companies, which is referred to as "business-to-business" (B2B) marketing and sales. Since traditional communication instruments (e.g. advertising) often hardly work in this environment and many projects lead to a long-term cooperation between supplier and customer, (personal) sales play a special role in marketing. Therefore, this event introduces marketing in B2B markets on the one hand and deals with questions of sales and distribution on the other hand.

Topics with regard to B2B sales management are:

- Basic aspects of B2B sales and B2B purchasing
- Understanding of marketing challenges in specific B2B business types (commodities, systems, solutions)
- Value pricing and value-based selling
- Organizational buying behavior
- Basics of B2B customer relationship management (e.g. key account management, reference customer management)
- Sales process (lead generation, sales presentations, customer-oriented selling, closing)
- Sales automation

Learning objectives

Students

- Are familiar with marketing and sales peculiarities and challenges in B2B environments
- Are able to identify different B2B business types and their marketing characteristics
- Are familiar with central theories of organizational buying behavior
- Are familiar with central objectives of Customer Relationship Management in B2B environments and are able to implement them with appropriate tools
- Are able to prioritize customers and calculate B2B Customer Lifetime Value
- Know how B2B sales presentations work and have also gained practical experience in this area
- Are able to determine value-based prices

Workload

The total workload for this course is approximately 135.0 hours.
Attendance time: 35.0 hours
Self-study: 100.0 hours

Organization

A detailed schedule will be announced.

Literature

Homburg, Christian (2016), Marketingmanagement, 6. Aufl., Wiesbaden.

V	**B2B Sales Management** 2572187, WS 26/27, 2 SWS, Language: German, Open in study portal	**Lecture (V)** **On-Site**

Content
Content

The event is designed to teach you taking on marketing responsibility in a very special business environment. This involves companies that sell and market their (often technically highly complex) products themselves to other companies, which is referred to as "business-to-business" (B2B) marketing and sales. Since traditional communication instruments (e.g. advertising) often hardly work in this environment and many projects lead to a long-term cooperation between supplier and customer, (personal) sales play a special role in marketing. Therefore, this event introduces marketing in B2B markets on the one hand and deals with questions of sales and distribution on the other hand.

Topics with regard to B2B sales management are:

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- Value pricing and value-based selling
- Organizational buying behavior
- Basics of B2B customer relationship management (e.g. key account management, reference customer management)
- Sales process (lead generation, sales presentations, customer-oriented selling, closing)
- Sales automation

Learning objectives

Students

- Are familiar with marketing and sales peculiarities and challenges in B2B environments
- Are able to identify different B2B business types and their marketing characteristics
- Are familiar with central theories of organizational buying behavior
- Are familiar with central objectives of Customer Relationship Management in B2B environments and are able to implement them with appropriate tools
- Are able to prioritize customers and calculate B2B Customer Lifetime Value
- Know how B2B sales presentations work and have also gained practical experience in this area
- Are able to determine value-based prices

Workload

The total workload for this course is approximately 135.0 hours.
Attendance time: 35.0 hours
Self-study: 100.0 hours

Organization

A detailed schedule will be announced.

Literature

Homburg, Christian (2016), Marketingmanagement, 6. Aufl., Wiesbaden.

T 6.64 Module component: Basic Principles of Economic Policy [T-WIWI-103213]

Coordinators: Prof. Dr. Ingrid Ott
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104908 - Economics

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	see Annotations	1

Courses					
ST 2026	2560280	Basic Principles of Economic Policy	2 SWS	Lecture / 🗣	Ott
ST 2026	2560281	Exercises of Basic Principles of Economic Policy	1 SWS	Practice / 🗣	Ghoniem

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success is assessed in the form of a written examination (60 minutes). The examination is offered every semester and can be repeated at any regular examination date.

Prerequisites
None

Recommendations
Basic knowledge of micro- and macroeconomics is assumed, as taught in the courses Economics I [2610012], and Economics II [2600014].

Additional Information
Please note that the lecture will not be held in summer semester 2026. The exam is offered.

Description:

Theory of general economic policy and discussion of current economic policy topics:

- Goals of economic policy,
- Instruments and institutions of economic policy,
- Triad of regional, national and European economic policies,
- special fields of economic policy, in particular growth, employment, provision of public infrastructure and climate policy.

Learning objectives:

Students learn:

- To apply basic concepts of micro- and macroeconomic theories to economic policy issues.
- to develop arguments on how state intervention in the market can be legitimized from a welfare economic perspective
- to derive theory-based policy recommendations.

Learning content:

- Market interventions: microeconomic perspective
- Market interventions: macroeconomic perspective
- Institutional economic aspects
- Economic policy and welfare economics
- Economic policy makers: Political-economic aspects

Workload:

- Total effort at 4.5 LP: approx. 135 hours
- Presence time: approx. 30 hours
- Self-study: approx. 105 hours

Media:

See course announcement

References:

See course announcement

Below you will find excerpts from courses related to this module component:

V **Basic Principles of Economic Policy** — **Lecture (V) On-Site**

2560280, SS 2026, 2 SWS, Language: German, Open in study portal

Content
The lecture deals with theories of general economic policy and discussion of current economic policy topics:

- Goals of economic policy,
- Instruments and institutions of economic policy,
- Triad of regional, national and European economic policies,
- special fields of economic policy, in particular growth, employment, provision of public infrastructure and climate policy.

Learning objectives:

Students shall be given the ability to

- apply basic concepts of micro- and macroeconomic theories to economic policy issues
- develop arguments on how state intervention in the market can be legitimized from a welfare economic perspective
- derive theory-based policy recommendations

Recommendations:

Basic micro- and macroeconomic knowledge is required, especially as taught in the courses Economics I [2610012] and Economics II [2600014].

Workload:

Total effort at 4.5 LP is approx. 135 hours and consists of:

- Presence time: approx. 30 hours
- Self-study: approx. 105 hours

Assessment:

The examination takes place in the form of a written examination (60min) (according to §4(2), 1 SPO). The examination is offered every semester and can be repeated at any regular examination date.

Organizational issues
Die Vorlesung wird im SoSe 2026 aufgrund eines Forschungssemesters nicht gelesen; Übung und Klausur finden statt.

Zugehörige Veranstaltung: Übungen zur Einführung in die Wirtschaftspolitik [2560281]
Vorbereitungsmaterialien finden Sie im Ilias.

Literature

- Klump, Rainer (2013): Wirtschaftspolitik. Pearson Studium
- Baldwin, Richard und Charles Wyplosz (2019): The Economics of European Integration, 6. Edition, McGraw-Hill Education, London
- Foliensatz zur Vorlesung
- Übungsaufgaben

V **Exercises of Basic Principles of Economic Policy** — **Practice (Ü) On-Site**

2560281, SS 2026, 1 SWS, Language: German, Open in study portal

Organizational issues
Zugehörige Veranstaltung: [2560280] Einführung in die Wirtschaftspolitik

Literature

- Klump, Rainer (2013): Wirtschaftspolitik. Pearson Studium
- Baldwin, Richard und Charles Wyplosz (2019): The Economics of European Integration, 6. Edition, McGraw-Hill Education, London
- Foliensatz zur Vorlesung
- Übungsaufgaben

T 6.65 Module component: Basic Seminar Supplementary Studies on Science, Technology and Society - Self Registration [T-FORUM-113579]

Coordinators: Dr. Christine Mielke
Christine Myglas

Organisation: General Studies. Forum Science and Society (FORUM)

Part of: M-FORUM-106753 - Supplementary Studies on Science, Technology and Society

Type	Credits	Grading	Term offered	Expansion	Version
Coursework	2 CP	pass/fail	Each summer term	1 semesters	1

Assessment
Study achievement in the form of a presentation or a term paper or project work in the selected course.

Prerequisites
None

Self Service Assignment of Supplementary Studies
This module component can be used for self service assignment of grades acquired from the following study providers:

- Studium Generale. Forum Wissenschaft und Gesellschaft (FORUM) (ehem. ZAK)
- FORUM (ehem. ZAK) Begleitstudium

Recommendations
It is recommended that the basic seminar be completed during the same semester as the lecture series "Science in Society". If it is not possible to attend the lecture series and the basic seminar in the same semester, the basic seminar can also be attended in the semesters before the lecture series.
However, attending courses in the advanced unit before attending the basic seminar should be avoided.

T 6.66 Module component: Basics of German Company Tax Law and Tax Planning [T-WIWI-108711]

Coordinators: Prof. Dr. Gerd Gutekunst
Prof. Dr. Berthold Wigger

Organisation: KIT Department of Business and Economics

Part of: M-WIWI-101511 - Advanced Topics in Public Finance
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each winter term	2

Courses					
WT 25/26	2560134	Basics of German Company Tax Law and Tax Planning	3 SWS	Lecture / 🗣	Wigger, Gutekunst

Legend: 🖥 Online, ❖ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success is assessed in the form of a written examination (90 minutes). The examination is offered every semester and can be repeated at any regular examination date.

Prerequisites
None

Recommendations
Knowledge of the collection of public revenues is assumed. Therefore it is recommended to attend the course “Öffentliche Einnahmen” beforehand.

Below you will find excerpts from courses related to this module component:

V Basics of German Company Tax Law and Tax Planning

2560134, WS 25/26, 3 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content
Workload:
The total workload for this course is approximately 135.0 hours. For further information see German version.

T 6.67 Module component: Basics of Production Automation [T-MACH-114049]

Coordinators: Prof. Dr.-Ing. Jürgen Fleischer
Organisation: KIT Department of Mechanical Engineering
Part of: M-MACH-101298 - Automated Manufacturing Systems
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Expansion	Version
Written examination	5 CP	graded	Each summer term	1 semesters	1

Courses					
ST 2026	2150703	Basics of Production Automation	3 SWS	Lecture / Practice / 🗣	Fleischer

Legend: 🖥 Online, ❋ Blended (On-Site/Online), 🗣 On-Site, x Cancelled

Assessment
written exam (60 minutes)

Prerequisites
none

Additional Information
The course is offered in German.

Workload
150 hours

Below you will find excerpts from courses related to this module component:

V Basics of Production Automation

2150703, SS 2026, 3 SWS, Language: German, Open in study portal

Lecture / Practice (VÜ)
On-Site

Content

The lecture provides an overview of the structure and functioning of automated production systems. The functioning and application of fundamental elements for the realisation of automated production systems are taught. These include:

- Drive and control technology
- Handling technology for the handling of workpieces and tools
- Industrial robots
- Quality assurance in automated production plants
- Structures and design of manufacturing and assembly systems
- Introduction to project planning of automated production plants

An interdisciplinary consideration of these sub-areas reveals interfaces to Industry 4.0 approaches. The theoretical foundations are supplemented by practical application examples and live demonstrations in the Karlsruhe Research Factory. Within exercises, the contents of the lecture are deepened and applied to specific problems and tasks.

Learning Outcomes:

The students ...

- are able to name and describe the automation tasks in production plants and the components required for implementation.
- can select components from the areas of ‘handling technique, ‘industrial robots’, ‘sensor technology’ and ‘control & drive technology’ for a given application.
- are able to compare different concepts of robotics and select the appropriate one for a given application.
- can assess the automation of production plants implemented on the basis of the examples given and apply them to new problems.
- know the process of planning automated production plants.

Workload:

MACH:
regular attendance: 32 hours
self-study: 88 hours
WING:
regular attendance: 32 hours
self-study: 118 hours

Organizational issues

Zur Vertiefung des im Rahmen der Lehrveranstaltung erworbenen Wissens werden die theoretischen Vorlesungseinheiten durch Praxiseinheiten im Umfeld der Karlsruher Forschungsfabrik (https://www.karlsruher-forschungsfabrik.de) unterstützt.

The theoretical lectures are complemented by practical lectures in the Karlsruhe Research Factory (https://www.karlsruher-forschungsfabrik.de/en.html) to deepen the acquired knowledge.

Literature

Medien:
Skript zur Veranstaltung wird über (https://ilias.studium.kit.edu/) bereitgestellt.

Media:
Lecture notes will be provided in Ilias (https://ilias.studium.kit.edu/).

T 6.68 Module component: Batteries, Fuel Cells, and Electrolysis [T-ETIT-113986]

Coordinators: Prof. Dr.-Ing. Ulrike Krewer
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-107005 - Batteries, Fuel Cells, and Electrolysis

Type	Credits	Grading	Term offered	Version
Written examination	5 CP	graded	Each winter term	3

Courses					
WT 25/26	2304240	Batteries, Fuel Cells and Electrolysis	2 SWS	Lecture /	Krewer
WT 25/26	2304241	Practical Exercise to 2304240 Batteries, Fuel Cells and Electrolysis	2 SWS	Practice /	Krewer, Sonder
WT 26/27	2304240	Batteries, Fuel Cells and Electrolysis	2 SWS	Lecture /	Krewer

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
Success control takes place in the form of a graded written examination lasting 120 minutes.

Prerequisites
The following module components **must not** have started:

- T-ETIT-100983 - Batterien und Brennstoffzellen
- T-ETIT-114097 - Batterien, Brennstoffzellen und ihre Systeme

The following module components **must** have started:

- T-ETIT-114957 - Batteries, Fuel Cells, and Electrolysis - Group Project

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-ETIT-114957 - Batteries, Fuel Cells, and Electrolysis - Group Project must have been started.

Additional Information
For details on content and qualification objectives see "M-ETIT-107005 - Batteries, Fuel Cells, and Electrolysis".

The course is offered in English.

Workload
150 hours

T 6.69 Module component: Batteries, Fuel Cells, and Electrolysis - Group Project [T-ETIT-114957]

Coordinators: Prof. Dr.-Ing. Ulrike Krewer
Organisation: KIT Department of Electrical Engineering and Information Technology
KIT Department of Mechanical Engineering
Part of: M-ETIT-107005 - Batteries, Fuel Cells, and Electrolysis

Type	Credits	Grading	Term offered	Version
Coursework	1 CP	pass/fail	Each winter term	1

Assessment
Success control takes place in the form of an ungraded written technical report (approx. 7-10 pages).

Prerequisites
none

Additional Information
For details on content and qualification objectives see "M-ETIT-107005 - Batteries, Fuel Cells, and Electrolysis".

The course is offered in English.

Workload
30 hours

T 6.70 Module component: Bayesian Statistics for Analyzing Data [T-WIWI-113471]

Coordinators: Prof. Dr. Benjamin Scheibehenne
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-103117 - Data Science: Data-Driven Information Systems
M-WIWI-104900 - Business Administration
M-WIWI-105714 - Consumer Research

Type	Credits	Grading	Term offered	Expansion	Version
Examination of another type	4,5 CP	graded	Irregular	1 semesters	1

Courses					
WT 25/26	2500025	Bayesian Statistics for Analyzing Data	2 SWS	Seminar	Scheibehenne
WT 26/27	2500025	Bayesian Statistics for Analyzing Data	2 SWS	Seminar	Scheibehenne

Assessment
Alternative exam assessment (assignments and active participation). Details will be communicated at the first day of class.

Additional Information
Participation is limited to 10 participants. Registration is required for the course. If too many students register, students in higher semesters are selected first.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Bayesian Statistics for Analyzing Data

Seminar (S)

2500025, WS 25/26, 2 SWS, Language: English, Open in study portal

Content
The goal of this class is to introduce Bayesian statistics as a viable alternative to conventional Null-Hypothesis significance testing (NHST) and the calculation of p-values. The class introduces the theoretical background of Bayesian statistics and its advantages over NHST. Based on this, students will work through hands-on approaches for analyzing various empirical data using Bayesian statistics. These analyses will mainly be conducted with the statistics software R and JASP. The class provides participants with the necessary skills to evaluate and interpret the results of published Bayesian analyses and to use the method for testing hypotheses and estimating model parameters based on empirical data. There will be regular reading and homework assignments.

Workload:

Attendance time (7 "double hours"): 45 h

Self-study: 90 h

Exam preparation: 0 h (performance assessment takes place during the semester)

Total: 135 h

Organizational issues
Beginn 29.10.25, 14:00 - 16:00 Uhr Seminarraum Geb. 01.87, R 5B 25

alle 2 Wochen mittwochs bis 21.2.26

V Bayesian Statistics for Analyzing Data

Seminar (S)

2500025, WS 26/27, 2 SWS, Language: English, Open in study portal

Content
The goal of this class is to introduce Bayesian statistics as a viable alternative to conventional Null-Hypothesis significance testing (NHST) and the calculation of p-values. The class introduces the theoretical background of Bayesian statistics and its advantages over NHST. Based on this, students will work through hands-on approaches for analyzing various empirical data using Bayesian statistics. These analyses will mainly be conducted with the statistics software R and JASP. The class provides participants with the necessary skills to evaluate and interpret the results of published Bayesian analyses and to use the method for testing hypotheses and estimating model parameters based on empirical data. There will be regular reading and homework assignments.

Workload:

Attendance time (7 "double hours"): 45 h

Self-study: 90 h

Exam preparation: 0 h (performance assessment takes place during the semester)

Total: 135 h

Organizational issues
Beginn 28.10.26, 14:00 - 16:00 Uhr

Seminarraum Geb. 01.87, R B5.25

alle 2 Wochen mittwochs bis 17.2.27

T 6.71 Module component: Behavioral Dimensions of Energy Transitions [T-WIWI-114980]

Coordinators: Prof. Dr. Wolf Fichtner
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101451 - Energy Economics and Energy Markets
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	3,5 CP	graded	Each summer term	1

Assessment
Written examination (60 minutes). The examination is offered every semester and can be repeated at any regular examination date.

Workload
105 hours

T 6.72 Module component: Behavioral Experiments in Action [T-WIWI-111393]

Coordinators: Prof. Dr. Benjamin Scheibehenne
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Expansion	Version
Examination of another type	4,5 CP	graded	Each summer term	1 semesters	1

Assessment
Gradings will be based on the quality of the experimental program, data, and the research report in Stage 2.

Prerequisites
Experimental design (either take the course in our module, or gain basic knowledge of experimental design by self-education)

Additional Information
In this course, students will gain first-hand experience into how to conduct an experimental study in the area of behavioral economics/psychology.

The course contains two stages. In Stage 1, students will learn how to plan, program, and run an experiment by attending to blocked lectures. In Stage 2, students will choose one classic experiment in the area of behavioral economics or psychology, conduct a replication of that experiment using the techniques acquired in Stage 1, and write a research report on the results of the replication.

The number of participants is limited. The registration will take place via the Wiwi-Portal.

Workload
135 hours

T 6.73 Module component: Behavioral Lab Exercise [T-WIWI-113095]

Coordinators: Prof. Dr. Petra Nieken
Prof. Dr. Benjamin Scheibehenne

Organisation: KIT Department of Business and Economics

Part of: M-WIWI-104900 - Business Administration
M-WIWI-105714 - Consumer Research
M-WIWI-105923 - Incentives, Interactivity & Decisions in Organizations

Type	Credits	Grading	Term offered	Expansion	Version
Examination of another type	4,5 CP	graded	Irregular	1 semesters	1

Courses					
WT 25/26	2500040	Behavioral Lab Exercise	4.5 SWS	Seminar / 🗣	Scheibehenne, Nieken

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Alternative exam assessment (presentation during the semester). Details will be communicated at the first day of class.

Recommendations
This class caters towards Master students who are interested in empirical research and in running lab experiments.

Additional Information
Due to the interactive nature of the class, the number of participants is limited. If you are interested, please contact the teachers directly via email.
In this class, students learn the core principles of psychological and economic experiments. The course covers topics ranging from design principles, to best-practices, preregistration, and analysis of the experimental data. Students will actively participate in the course by covering one selected topic in a talk. All students will discuss the topics together with the professors to develop solid knowledge about experimental design and analysis plans. In a second step, all students will develop a draft of an experimental design and analysis plan for their own topic and present it to the class. The students will get detailed feedback, enabling them to improve their drafts for future research.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Behavioral Lab Exercise

2500040, WS 25/26, 4.5 SWS, Language: English, Open in study portal

Seminar (S)
On-Site

Content
In this class, students learn the core principles of psychological and economic experiments. The course covers topics ranging from design principles, to best-practices, preregistration, and analysis of the experimental data. Students will actively participate in the course by covering one selected topic in a talk. All students will discuss the topics together with the professors to develop solid knowledge about experimental design and analysis plans. In a second step, all students will develop a draft of an experimental design and analysis plan for their own topic and present it to the class. The students will get detailed feedback enabling them to improve their drafts for future research.

T 6.74 Module component: Beyond Conventional Materials - Metamaterials & Architected Structures [T-MACH-114009]

Coordinators: Jun.-Prof. Dr. Jens Bauer
Organisation: KIT Department of Mechanical Engineering

Part of: M-WIWI-107718 - Mechanical Engineering I
M-WIWI-107730 - Mechanical Engineering II

Type	Credits	Grading	Term offered	Version
Oral examination	4,5 CP	graded	Each winter term	2

Courses					
WT 25/26	2186100	Beyond Conventional Materials - Metamaterials & Architected Structures	2 SWS	Lecture / Blended	Bauer

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
Oral examination, duration approx. 30 minutes

Prerequisites
none

Additional Information
The course is offered in English.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Beyond Conventional Materials - Metamaterials & Architected Structures

Lecture (V)
Blended (On-Site/Online)

2186100, WS 25/26, 2 SWS, Language: English, Open in study portal

Content
Conventional material design focuses on engineering the chemistry and microstructure of solids. Metamaterials go beyond these classical approaches. They are artificial materials that are built from spatially structured building blocks, like lattice-truss architectures. The integration of these rational architectures at the material level grants metamaterials unique unconventional properties which are inaccessible with classical material designs.

The course covers the fundamentals of the mechanics of different metamaterial architectures, discusses design principles and applicable fabrication techniques from the macro- to the nanoscale, as well as their interdependency, and considers emerging application scenarios in medicine, aerospace, microsystem technology, and mobility.

The students learn

- to design beam, shell and plate-based spatial architectures, such as for extreme strength & stiffness, programmable/adaptive behaviors and negative effective properties.
- to mathematically describe and predict the mechanical behavior of such architectural designs.
- the fundamentals of applicable fabrication techniques, including foaming, assembly and 3D-printing, and their design and material implications
- the relationship between architecture & size and how micro- and nanoscale architectures can leverage extreme physical size effects.

preliminary knowledge in mathematics, physics and materials science recommended

regular attendance: 22,5 hours

self-study: 97,5 hours

oral exam: ca. 30 minutes

no tools or reference materials

Literature

Gibson, L. J. & Ashby, M. F. Cellular Solids: Structure and properties. (Cambridge Univ. Pr., 2001).

Fleck, N. A., Deshpande, V. S. & Ashby, M. F. Micro-architectured materials: past, present and future. Proc. R. Soc. A Math. Phys. Eng. Sci. 466, 2495–2516 (2010).

Bauer, J. et al. Nanolattices: An Emerging Class of Mechanical Metamaterials. Adv. Mater. 29, 1701850 (2017).

Jiao, P., Mueller, J., Raney, J. R., Zheng, X. (Rayne) & Alavi, A. H. Mechanical metamaterials and beyond. Nat. Commun. 2023 141 14, 1–17 (2023).

T 6.75 Module component: Big Data Analytics [T-INFO-101305]

Coordinators: Prof. Dr.-Ing. Klemens Böhm
Organisation: KIT Department of Informatics
Part of: M-WIWI-104909 - Informatics (Department of Informatics)

Type	Credits	Grading	Term offered	Version
Oral examination	5 CP	graded	Each winter term	2

T 6.76 Module component: Biofilm Systems [T-CIWVT-106841]

Coordinators: Dr. Andrea Hille-Reichel
Dr. Michael Wagner
Organisation: KIT Department of Chemical and Process Engineering
Part of: M-CIWVT-103441 - Biofilm Systems

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each summer term	1

Courses					
ST 2026	2233820	Biofilm Systems	2 SWS	Lecture / 🗣	Hille-Reichel, Wagner

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, × Cancelled

Assessment
Oral exam, about 20 min.

T 6.77 Module component: Biologically Inspired Robots [T-MACH-113856]

Coordinators: Prof. Dr.-Ing. Arne Rönnau
Organisation: KIT Department of Mechanical Engineering
Part of: M-WIWI-107718 - Mechanical Engineering I
M-WIWI-107730 - Mechanical Engineering II

Type	Credits	Grading	Term offered	Expansion	Version
Oral examination	3 CP	graded	Each summer term	1 semesters	1

Courses					
ST 2026	2122330	Biologically Inspired Robots	2 SWS	Lecture / 🗣	Rönnau

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success is assessed in the form of an oral examination (approx. 15-20 minutes)

Prerequisites
none

Recommendations
It is recommended to listen to the course "Robotics I" beforehand .

Additional Information
The course is offered in English.

Workload
90 hours

Below you will find excerpts from courses related to this module component:

V Biologically Inspired Robots

2122330, SS 2026, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content

The lecture biologically inspired robots deals intensively with robots whose mechanical design, sensor concepts or control architecture were inspired by nature. In detail, we will look at solutions from nature (e.g. lightweight construction concepts using honeycomb structures, human muscles) and then at robot technologies that utilize these principles to solve similar tasks (lightweight 3D printed parts or artificial muscles in robotics).

After discussing these biologically inspired technologies, concrete robotic systems and applications from current research that successfully utilize these technologies will be presented. In particular, multi-legged walking robots, snake-like and humanoid robots are presented and their sensor and drive concepts are discussed.

The lecture focuses on the concepts of control and system architectures (e.g. behavior-based systems) of these robotic systems, with locomotion being the main focus. The lecture ends with an outlook on future developments and the development of commercial applications for these robots.

Learning Objectives

Students acquire knowledge of biologically inspired concepts in robotics and are able to systematically analyse, evaluate, and transfer them to technical, robotic solutions.

Specifically, the following skills are taught:

- Natural lightweight construction and material concepts and their significance for the energy efficiency of mobile robot systems
- Muscle types from biology, associated muscle models, and transfer to artificial muscles for robots
- Human senses, stimulus processing, information coding, and the equivalent technical sensor systems, especially for mobile walking robots
- Central pattern generators (CPG), neuro-oscillators, and their relevance for robotic multi-legged walking
- Types of locomotion, stability criteria, walking patterns, and the representation in gait diagrams
- Machine learning methods and their application in a robotic context
- Subsumption architecture, reactive and behaviour-based system architectures
- Mendelian laws, meiosis and mitosis, and genetic algorithms in robotics
- Challenges of humanoid robot systems and current approaches to solutions
- Applications of walking robots and other biologically motivated robots

Organizational issues

Siehe weitere Details zu der Organisation der Veranstaltung: https://lehre.imi.kit.edu sowie in ILIAS.

T 6.78 Module component: BioMEMS - Microsystems Technologies for Life-Sciences and Medicine I [T-MACH-100966]

Coordinators: Prof. Dr. Andreas Guber
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101290 - BioMEMS
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Written examination	4 CP	graded	Each winter term	3

Courses					
WT 25/26	2141864	BioMEMS - Microsystems Technologies for Life-Sciences and Medicine I	2 SWS	Lecture / 🗣	Guber, Ahrens

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
written exam (75 Min.)

Prerequisites
none

Additional Information
The course is offered in German

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V BioMEMS - Microsystems Technologies for Life-Sciences and Medicine I

2141864, WS 25/26, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Organizational issues
BioMEMS I-Klausur: Mo, 23.03.2026, 17:30 - 19:30; 10.21 Gottlieb-Daimler-Hörsaal

BioMEMS II-Klausur: Mo, 02.03.2026, 8:00 - 10:00; 10.21 Gottlieb-Daimler-Hörsaal

BioMEMS III-Klausur: Mo, 09.03.2026, 13:15 - 15:15; 10.21 Gottlieb-Daimler-Hörsaal

Literature
Menz, W., Mohr, J., O. Paul: Mikrosystemtechnik für Ingenieure, VCH-Verlag, Weinheim, 2005

M. Madou
Fundamentals of Microfabrication
Taylor & Francis Ltd.; Auflage: 3. Auflage. 2011

T 6.79 Module component: BioMEMS - Microsystems Technologies for Life-Sciences and Medicine II [T-MACH-100967]

Coordinators: Prof. Dr. Andreas Guber
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101287 - Microsystem Technology
M-MACH-101290 - BioMEMS
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Written examination	4 CP	graded	Each summer term	3

Courses					
ST 2026	2142883	BioMEMS - Microsystems Technologies for Life-Sciences and Medicine II	2 SWS	Lecture / 🗣	Guber, Ahrens

Legend: Online, Blended (On-Site/Online), 🗣 On-Site, × Cancelled

Assessment
Written exam (75 Min.)

Prerequisites
none

Additional Information
The course is offered in German

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V BioMEMS - Microsystems Technologies for Life-Sciences and Medicine II

2142883, SS 2026, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content
Examples of use in Life-Sciences and biomedicine: Microfluidic Systems:
LabCD, Protein Cristallisation
Microarrys
Tissue Engineering
Cell Chip Systems
Drug Delivery Systems
Micro reaction technology
Microfluidic Cells for FTIR-Spectroscopy
Microsystem Technology for Anesthesia, Intensive Care and Infusion
Analysis Systems of Person´s Breath
Neurobionics and Neuroprosthesis
Nano Surgery

Organizational issues
Zu jedem Vorlesungstermin werden via ILIAS die jeweiligen Folien im PDF-Format zur Verfügung gestellt.

schriftl. Prüfung: Mo, 07.09.2026, 8 - 10 Uhr; 10.21 Gottlieb-Daimler-Hörsaal

Literature
Menz, W., Mohr, J., O. Paul: Mikrosystemtechnik für Ingenieure, VCH-Verlag, Weinheim, 2005

Buess, G.: Operationslehre in der endoskopischen Chirurgie, Band I und II;
Springer-Verlag, 1994

M. Madou
Fundamentals of Microfabrication

T 6.80 Module component: BioMEMS - Microsystems Technologies for Life-Sciences and Medicine III [T-MACH-100968]

Coordinators: Prof. Dr. Andreas Guber
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101287 - Microsystem Technology
M-MACH-101290 - BioMEMS
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Written examination	4 CP	graded	Each summer term	3

Courses					
ST 2026	2142879	BioMEMS - Microsystems Technologies for Life-Sciences and Medicine III	2 SWS	Lecture / 🗣	Guber, Ahrens

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Written exam (75 Min.)

Prerequisites
none

Additional Information
The course is offered in German

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V BioMEMS - Microsystems Technologies for Life-Sciences and Medicine III

Lecture (V)
On-Site

2142879, SS 2026, 2 SWS, Language: German, Open in study portal

Content
Examples of use in minimally invasive therapy
Minimally invasive surgery (MIS)
Endoscopic neurosurgery
Interventional cardiology
NOTES
OP-robots and Endosystems
License of Medical Products and Quality Management

Organizational issues
Zu jedem Vorlesungstermin werden via ILIAS die jeweiligen Folien im PDF-Format zur Verfügung gestellt.

schriftl. Prüfung: Mo, 28.09.2026, 8:00 - 10:00 Uhr; 10.21 Gottlieb-Daimler.Hörsaal

Literature
Menz, W., Mohr, J., O. Paul: Mikrosystemtechnik für Ingenieure, VCH-Verlag, Weinheim, 2005

Buess, G.: Operationslehre in der endoskopischen Chirurgie, Band I und II;
Springer-Verlag, 1994

M. Madou
Fundamentals of Microfabrication

T 6.81 Module component: Bioprocess Scale-up [T-CIWVT-113712]

Coordinators: Prof. Dr.-Ing. Alexander Grünberger
Organisation: KIT Department of Chemical and Process Engineering
Part of: M-CIWVT-106837 - Bioprocess Scale-up

Type	Credits	Grading	Term offered	Version
Oral examination	6 CP	graded	Each winter term	2

Courses					
WT 25/26	2213040	Bioprocess Scale-Up	2 SWS	Lecture / 🗣	Grünberger
WT 25/26	2213041	Exercises to 2213040 Bioprocess Scale-Up	1 SWS	Practice / 🖥	Grünberger
WT 26/27	2213040	Bioprocess Scale-Up	2 SWS	Lecture / 🗣	Grünberger
WT 26/27	2213041	Exercises to 2213040 Bioprocess Scale-Up	2 SWS	Practice / 🖥	Grünberger

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, × Cancelled

T 6.82 Module component: Biosensors [T-CIWVT-113714]

Coordinators: Dr. Gözde Kabay
Organisation: KIT Department of Chemical and Process Engineering
Part of: M-CIWVT-106838 - Biosensors

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each term	1

Courses					
WT 25/26	2214810	Biosensors	2 SWS	Lecture / Blended	Kabay
ST 2026	2214810	Biosensors	2 SWS	Lecture / Blended	Kabay

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Prerequisites
None

T 6.83 Module component: Bond Markets [T-WIWI-110995]

Coordinators: Prof. Dr. Marliese Uhrig-Homburg
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101480 - Finance 3
M-WIWI-101483 - Finance 2
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each winter term	1

Courses					
WT 25/26	2530560	Bond Markets	3 SWS	Lecture / Practice / On-Site	Müller, Uhrig-Homburg, Molnar
WT 26/27	2530560	Bond Markets	3 SWS	Lecture / Practice / On-Site	Müller, Uhrig-Homburg, Molnar

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
The assessment consists of a written exam (75min.)

A bonus can be earned by correctly solving at least 50% of the posed bonus exercises. If the grade of the written examination is between 4.0 and 1.3, the bonus improves the grade by up to one level (0.3 or 0.4). The examination is offered in each semester and can be repeated at any regular examination date.

Additional Information
This course will be held in English.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Bond Markets
2530560, WS 25/26, 3 SWS, Language: English, Open in study portal

Lecture / Practice (VÜ)
On-Site

Content
The lecture "Bond Markets" deals with the national and international bond markets, which are an important source of financing for companies, as well as for the public sector. After an overview of the most important bond markets, different yield definitions are discussed. Based on this, the concept of the yield curve is presented. In addition, the theoretical and empirical relationships between ratings, default probabilities and spreads are analyzed. The focus will then be on questions regarding the valuation, measurement, management and control of credit risks.

The total workload for this course is approximately 135 hours (4.5 credits).

The assessment consists of a written exam (75min.) (according to §4(2), 1 SPO). A bonus can be earned by correctly solving at least 50% of the posed bonus exercices. If the grade of the written examination is between 4.0 and 1.3, the bonus improves the grade by up to one level (0.3 or 0.4). The examination is offered in each semester and can be repeated at any regular examination date.

Students deepen their knowledge of national and international bond markets. They gain knowledge of the traded instruments and their key figures for describing default risk such as ratings, default probabilities or credit spreads.

Organizational issues
Die Veranstaltung wird freitags in der ersten Semesterhälfte am Campus B (Geb. 09.21) im Raum 124 angeboten. Die Klausur findet bereits im Januar statt.

V Bond Markets
2530560, WS 26/27, 3 SWS, Language: English, Open in study portal

Lecture / Practice (VÜ)
On-Site

Content

The lecture "Bond Markets" deals with the national and international bond markets, which are an important source of financing for companies, as well as for the public sector. After an overview of the most important bond markets, different yield definitions are discussed. Based on this, the concept of the yield curve is presented. In addition, the theoretical and empirical relationships between ratings, default probabilities and spreads are analyzed. The focus will then be on questions regarding the valuation, measurement, management and control of credit risks.

The total workload for this course is approximately 135 hours (4.5 credits).

The assessment consists of a written exam (75min.) (according to §4(2), 1 SPO). A bonus can be earned by correctly solving at least 50% of the posed bonus exercices. If the grade of the written examination is between 4.0 and 1.3, the bonus improves the grade by up to one level (0.3 or 0.4). The examination is offered in each semester and can be repeated at any regular examination date.

Students deepen their knowledge of national and international bond markets. They gain knowledge of the traded instruments and their key figures for describing default risk such as ratings, default probabilities or credit spreads.

Organizational issues

Die Veranstaltung wird freitags in der ersten Semesterhälfte am Campus B (Geb. 09.21) im Raum 124 angeboten. Die Klausur findet bereits im Januar statt.

T 6.84 Module component: Bond Markets - Models & Derivatives [T-WIWI-110997]

Coordinators: Prof. Dr. Marliese Uhrig-Homburg
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101480 - Finance 3
M-WIWI-101483 - Finance 2
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Each winter term	1

Courses					
WT 25/26	2530565	Bond Markets - Models & Derivatives	2 SWS	Block / 🗣	Grauer, Uhrig-Homburg, Müller
WT 26/27	2530565	Bond Markets - Models & Derivatives	2 SWS	Block / 🗣	Grauer, Uhrig-Homburg, Müller

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment of success consists in equal parts of a written thesis and an oral exam including a discussion of one's own work. The main examination is offered once a year, re-examinations every semester.

Recommendations
Knowledge of "Bond Markets" and "Derivatives" courses is very helpful.

Additional Information
This course will be held in English.

Workload
90 hours

Below you will find excerpts from courses related to this module component:

V Bond Markets - Models & Derivatives
2530565, WS 25/26, 2 SWS, Language: English, Open in study portal

Block (B)
On-Site

Content

- **Competence Certificate:** The assessment of success consists in equal parts of a written thesis and an oral exam (according to §4(2), 3 SPO) including a discussion of one's own work. The main examination is offered once a year, re-examinations every semester.
- **Competence Goal:** Students deepen their knowledge of national and international bond markets. They are able to apply the knowledge they have gained about traded instruments and common valuation models for pricing derivative financial instruments.
- **Prerequisites:**
- **Content:** The lecture "Bond Markets – Models & Derivatives" deepens the content of the lecture "Bond Markets". The modelling of the dynamics of yield curves and the management of credit risks forms the theoretical foundation for the valuation of interest rate and credit derivatives to be discussed. In this course, students deal intensively with selected topics and acquire the relevant knowledge on their own.
- **Recommendation:** Knowledge of "Bond Markets" and "Derivatives" courses is very helpful.
- **Workload:** The total workload for this course is approximately 90 hours (3.0 credits).

Organizational issues
Die Veranstaltung mit Seminarcharakter und dem Ziel, ein selbstgewähltes Themenfeld in Form einer schriftlichen Ausarbeitung eigenständig zu erarbeiten, findet in der 2. Semesterhälfte statt.

V Bond Markets - Models & Derivatives
2530565, WS 26/27, 2 SWS, Language: English, Open in study portal

Block (B)
On-Site

Content

- **Competence Certificate:** The assessment of success consists in equal parts of a written thesis and an oral exam (according to §4(2), 3 SPO) including a discussion of one's own work. The main examination is offered once a year, re-examinations every semester.
- **Competence Goal:** Students deepen their knowledge of national and international bond markets. They are able to apply the knowledge they have gained about traded instruments and common valuation models for pricing derivative financial instruments.
- **Prerequisites:**
- **Content:** The lecture "Bond Markets – Models & Derivatives" deepens the content of the lecture "Bond Markets". The modelling of the dynamics of yield curves and the management of credit risks forms the theoretical foundation for the valuation of interest rate and credit derivatives to be discussed. In this course, students deal intensively with selected topics and acquire the relevant knowledge on their own.
- **Recommendation:** Knowledge of "Bond Markets" and "Derivatives" courses is very helpful.
- **Workload:** The total workload for this course is approximately 90 hours (3.0 credits).

Organizational issues

Die Veranstaltung mit Seminarcharakter und dem Ziel, ein selbstgewähltes Themenfeld in Form einer schriftlichen Ausarbeitung eigenständig zu erarbeiten, findet in der 2. Semesterhälfte statt.

T 6.85 Module component: Bond Markets - Tools & Applications [T-WIWI-110996]

Coordinators: Prof. Dr. Marliese Uhrig-Homburg
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101480 - Finance 3
M-WIWI-101483 - Finance 2
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Examination of another type	1,5 CP	graded	Each winter term	1

Courses					
WT 25/26	2530562	Bond Markets - Tools & Applications	1 SWS	Block / On-Site	Uhrig-Homburg, Grauer, Müller
WT 26/27	2530562	Bond Markets - Tools & Applications	1 SWS	Block / On-Site	Uhrig-Homburg, Grauer, Müller

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
The assessment consists of an empirical case study with written elaboration and presentation. The main examination is offered once a year, re-examinations every semester.

Recommendations
Knowledge of the "Bond Markets" course is very helpful.

Additional Information
This course will be held in English.

Workload
45 hours

Below you will find excerpts from courses related to this module component:

V Bond Markets - Tools & Applications

2530562, WS 25/26, 1 SWS, Language: English, Open in study portal

Block (B)
On-Site

Content

- **Competence Certificate:** The assessment consists of an empirical case study with written elaboration and presentation (according to $4(2), 3 SPO). The main examination is offered once a year, re-examinations every semester.
- **Competence Goal:** The students apply various methods in practice within the framework of a project-related case study. They are able to deal with empirical data and analyze them in a targeted manner.
- **Content:** The course "Bond Markets – Tools & Applications" includes a hands-on project in the field of national and international bond markets. Using empirical datasets, the students have to apply practical methods in order to analyze the data in a targeted manner.
- **Recommendation:** Knowledge of the "Bond Markets" course is very helpful.
- **Workload:** The total workload for this course is approximately 45 hours (1.5 credits).

Organizational issues
Die Veranstaltung findet in der ersten Semesterhälfte statt und beinhaltet eine eigenständige Projektarbeit im Umgang mit realen Bond Daten. Die Erfolgskontrolle erfolgt anhand einer schriftlichen Ausarbeitung und einer kurzen Präsentation.

V Bond Markets - Tools & Applications

2530562, WS 26/27, 1 SWS, Language: English, Open in study portal

Block (B)
On-Site

Content

- **Competence Certificate:** The assessment consists of an empirical case study with written elaboration and presentation (according to $4(2), 3 SPO). The main examination is offered once a year, re-examinations every semester.
- **Competence Goal:** The students apply various methods in practice within the framework of a project-related case study. They are able to deal with empirical data and analyze them in a targeted manner.
- **Content:** The course "Bond Markets – Tools & Applications" includes a hands-on project in the field of national and international bond markets. Using empirical datasets, the students have to apply practical methods in order to analyze the data in a targeted manner.
- **Recommendation:** Knowledge of the "Bond Markets" course is very helpful.
- **Workload:** The total workload for this course is approximately 45 hours (1.5 credits).

Organizational issues

Die Veranstaltung findet in der ersten Semesterhälfte statt und beinhaltet eine eigenständige Projektarbeit im Umgang mit realen Bond Daten. Die Erfolgskontrolle erfolgt anhand einer schriftlichen Ausarbeitung und einer kurzen Präsentation.

T 6.86 Module component: Boosting of Combustion Engines [T-MACH-105649]

Coordinators: Dr.-Ing. Johannes Kech
Dr. Nicolas Lachenmaier

Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101303 - Combustion Engines II

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each summer term	1

Courses					
ST 2026	2134153	Boosting of Combustion Engines	2 SWS	/ 🗣	Kech

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
oral exam, 20 min

Prerequisites
none

Additional Information
This course is offerd in German.

Workload
120 hours

T 6.87 Module component: Brand Management [T-WIWI-112156]

Coordinators: Prof. Dr. Ann-Kristin Kupfer
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Examination of another type	4,5 CP	graded	Each winter term	1

Courses					
WT 25/26	2572190	Brand Management	2 SWS	Lecture / 🗣	Kupfer
WT 25/26	2572191	Brand Management Exercise	1 SWS	Practice / 🗣	Cinar
WT 26/27	2572190	Brand Management	2 SWS	Lecture / 🗣	Kupfer
WT 26/27	2572191	Brand Management Exercise	1 SWS	Practice / 🗣	Cinar

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment of success will be done by the preparation and presentation of a case study as well as a written exam. Further details will be announced during the lecture.

Prerequisites
None

Recommendations
Students are highly encouraged to actively participate in class.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Brand Management
2572190, WS 25/26, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content
Students learn the theoretical foundations of brand management and its most important concepts. They learn both about the importance of brands for consumers as well as the importance of brands for firms. Special emphasis will be given to the development of brand strategies. Furthermore, students will learn how to evaluate and apply brand instruments. A tutorial offers the opportunity to apply the key learnings of the lecture using case studies.

The learning objectives are as follows:

- Getting to know the theoretical foundations of brand management
- Evaluating strategic branding options (e.g., relating to the development of the core of the brand and the brand architecture) and operative brand instruments (e.g., relating to the brand name and logo)
- Fostering critical and analytical thinking skills and the application of knowledge to marketing problems
- Improving English skills

Total time required for 4.5 credit points: approx. 135 hours

Attendance time: 30 hours

Self-study: 105 hours

V Brand Management
2572190, WS 26/27, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content
Students learn the theoretical foundations of brand management and its most important concepts. They learn both about the importance of brands for consumers as well as the importance of brands for firms. Special emphasis will be given to the development of brand strategies. Furthermore, students will learn how to evaluate and apply brand instruments. A tutorial offers the opportunity to apply the key learnings of the lecture using case studies.

The learning objectives are as follows:

- Getting to know the theoretical foundations of brand management
- Evaluating strategic branding options (e.g., relating to the development of the core of the brand and the brand architecture) and operative brand instruments (e.g., relating to the brand name and logo)
- Fostering critical and analytical thinking skills and the application of knowledge to marketing problems
- Improving English skills

Total time required for 4.5 credit points: approx. 135 hours

Attendance time: 30 hours

Self-study: 105 hours

T 6.88 Module component: Business Administration: Finance and Accounting [T-WIWI-102819]

Coordinators: Prof. Dr. Martin Ruckes
Prof. Dr. Marliese Uhrig-Homburg
Prof. Dr. Marcus Wouters

Organisation: KIT Department of Business and Economics

Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	4 CP	graded	Each winter term	1

Assessment

The assessment consists of a written exam (90 min.) according to Section 4(2), 1 of the examination regulation.

The assessment takes place in every semester. Re-examinations are offered at every ordinary examination date.

Prerequisites

None

T 6.89 Module component: Business Administration: Production Economics and Marketing [T-WIWI-102818]

Coordinators:	Prof. Dr. Wolf Fichtner Prof. Dr. Martin Klarmann Prof. Dr.-Ing. Thomas Lützkendorf Prof. Dr. Martin Ruckes Prof. Dr. Frank Schultmann
Organisation:	KIT Department of Business and Economics
Part of:	M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	4 CP	graded	Each summer term	1

Assessment
The assessment consists of a written exam (90 minutes) according to Section 4(2), 1 of the examination regulation.

Prerequisites
None

6.90 Module component: Business Administration: Strategic Management and Information Engineering and Management [T-WIWI-102817]

Coordinators: Prof. Dr. Petra Nieken
Prof. Dr. Martin Ruckes

Organisation: KIT Department of Business and Economics

Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	3 CP	graded	Each winter term	1

Assessment

The assessment consists of a written exam (90 min.) according to Section 4(2), 1 of the examination regulation.

The assessment takes place in every semester. Re-examinations are offered at every ordinary examination date.

Prerequisites

None

T 6.91 Module component: Business Data Strategy [T-WIWI-106187]

Coordinators: Prof. Dr. Christof Weinhardt
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-103117 - Data Science: Data-Driven Information Systems
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each winter term	1

Courses					
WT 25/26	2540484	Business Data Strategy	2 SWS	Lecture / 🗣	Weinhardt, van Dinther
WT 25/26	2540485	Übung zu Business Data Strategy	1 SWS	Practice / 🗣	Weinhardt, Schulz

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
The assessment consists of a written exam (60 min.) according to § 4 paragraph 2 Nr. 1 of the examination regulationand an alternative exam assessment according to § 4 paragraph 2 Nr. 3 of the examination regulation. The grade isdetermined by 2/3 through the written exam and by 1/3 through the alternative exam assessment (e.g., presentation).

Prerequisites
None

Recommendations
Students should be familiar with basic concepts of business organisations, information systems, and programming. However, all material will be introduced, so no formal pre-conditions are applied.

Additional Information
Limited number of participants.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Business Data Strategy

2540484, WS 25/26, 2 SWS, Language: German/English, Open in study portal

Lecture (V)
On-Site

Content
With new methods for capturing and using different types of data and industry's recognition that society's use of data is less than optimal, the need for comprehensive strategies is more important than ever before. Advances in cybersecurity and information sharing and the use of data in its raw form for decision making all add to the complexity of integrated processes, ownership, stewardship, and sharing. The life cycle of data in its entirety spans the infrastructure, system design, development, integration, and implementation of information-enabling solutions. This lecture focuses on teaching about these dynamics and tools to comprehend and manage them in organisation contexts. Given the increasing size and complexity of data, methods for the transformation and structured preparation are an important tool in the process of sense–making. Modern software solutions and programming languages provide frameworks for such tasks that form another part of this course ranging from conceptual systems modelling to data manipulation to automated generation of HTML reports and web-applications. Additionally, the disruptive factor of generative AI in the scope of data management is reflected upon.

Organizational issues
Application/Registration

Attendance will be limited. Application/registration is therefore preliminary. After the application deadline has passed, positions will be allocated, based on evaluation of the previous study records. Applications are accepted only through the Wiwi-Portal: https://portal.wiwi.kit.edu/ys/8946

Anmeldung

Die Teilnehmeranzahl ist begrenzt. Eine Anmeldung erfolgt deshalb zunächst unter Vorbehalt. Nach Ablauf der Anmeldefrist werden die Plätze zur Teilnahme, nach Einsicht der Vorleistungen im Studium vergeben. Die Anmeldung/Bewerbung erfolgt ausschließlich über das Wiwi-Portal: https://portal.wiwi.kit.edu/ys/8946

Dates / Termine

The lecture will be held as a block lecture, the dates and rooms will be published soon.

Die Vorlesung wird als Blockveranstaltung gehalten. Die Termine werden zeitnah bekanntgegeben.

T 6.92 Module component: Business Dynamics [T-WIWI-102762]

Coordinators: Prof. Dr. Andreas Geyer-Schulz
Dr Paul Glenn

Organisation: KIT Department of Business and Economics

Part of: M-WIWI-101409 - Electronic Markets
M-WIWI-104900 - Business Administration
M-WIWI-105661 - Data Science: Intelligent, Adaptive, and Learning Information Services

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each winter term	1

Courses					
WT 25/26	2540531	Business Dynamics	2 SWS	Lecture / 🗣	Geyer-Schulz, Glenn
WT 25/26	2540532	Exercise Business Dynamics	1 SWS	Practice / 🗣	Geyer-Schulz, Glenn

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Written examination (60 minutes) according to §4(2), 1 SPO. The exam is considered passed if at least 50 out of a maximum of 100 possible points are achieved. The grades are graded in five steps (best grade 1.0 from 95 points). Details of the grade formation and scale will be announced in the course.
A bonus can be acquired through successful participation in the practice. If the grade of the written examination is between 4.0 and 1.3, the bonus improves the grade by one grade level (0.3 or 0.4). The exact criteria for awarding a bonus will be announced at the beginning of the course.

Please note: this exam will no longer be offered after August 2026.

Prerequisites
None

Recommendations
None

Below you will find excerpts from courses related to this module component:

V Business Dynamics

2540531, WS 25/26, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Organizational issues
Termine:

Samstags von 9:00 - 19:00 Uhr

22.11.2025 => Termin 1

13.12.2025 => Termin 2

17.01.2026 => Termin 3 (dieser **Termin** wird **auf** den **24. Januar 2026 verschoben**)

24.01.2026 => Termin 3

14.02.2026 => Termin 4

Literature
John D. Sterman. Business Dynamics: Systems Thinking and Modeling for a Complex World. McGraw-Hill, 2000.

T 6.93 Module component: Business Innovation in Optics and Photonics [T-ETIT-104572]

Coordinators: Prof. Dr. Werner Nahm
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Examination of another type	4 CP	graded	Each winter term	2

Courses					
WT 25/26	2305742	Business Innovation in Optics and Photonics	2 SWS	Lecture / 🗣	Riedel, Nahm
WT 25/26	2305743	Erxercise for 2305742 Business Innovation in Optics and Photonics	1 SWS	Practice / 🗣	Riedel, Nahm

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Type of Examination: examination of another type

Duration of Examination: 4 group presentations à 20 minutes (approx.)

Modality of Exam: The exam consists of four group presentations. 2nd day: Technology Presentation. 3rd day: Development plan presentation. 4th day: Business Canvas presentation. Final presentation: Business pitch. The overall impression is rated.

Prerequisites
none

Recommendations
Good knowledge in optics & photonics. Personal motivation and interest for getting deeper into business development aspects, methods and tools. Commitment to active, regular and coninuous participation in the group work.

T 6.94 Module component: Business Intelligence Systems [T-WIWI-105777]

Coordinators: Prof. Dr. Alexander Mädche
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101506 - Service Analytics
M-WIWI-101510 - Cross-Functional Management Accounting
M-WIWI-103117 - Data Science: Data-Driven Information Systems
M-WIWI-104068 - Information Systems in Organizations
M-WIWI-104900 - Business Administration
M-WIWI-107187 - Quantitative Core

Type	Credits	Grading	Term offered	Version
Examination of another type	4,5 CP	graded	Each winter term	2

Courses					
WT 25/26	2540422	Business Intelligence Systems	3 SWS	Lecture / Blended	Mädche

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
Alternative exam assessment. The assessment consists of a one-hour exam and the implementation of a Capstone project. Details will be announced at the beginning of the course.

Prerequisites
None

Recommendations
Basic knowledge on database systems is helpful.

Below you will find excerpts from courses related to this module component:

V Business Intelligence Systems

2540422, WS 25/26, 3 SWS, Language: English, Open in study portal

Lecture (V)
Blended (On-Site/Online)

Content
In most modern enterprises, Business Intelligence & Analytics (BI&A) Systems represent a core enabler of decision-making in that they supply up-to-date and accurate information about all relevant aspects of a company's planning and operations: from stock levels to sales volumes, from process cycle times to key indicators of corporate performance. Modern BI&A systems leverage beyond reporting and dashboards also advanced analytical functions. Thus, today, they also play a major role in enabling data-driven products and services. This course aims to introduce theoretical foundations, concepts, tools, and current practice of BI&A Systems from a managerial and technical perspective.

The course is complemented by an engineering capstone project, where students work in a team with real-world use cases and data in order to create a prototypical Business intelligence & Analytics system using state-of-the-art technologies (e.g., scikit-learn in Python or Microsoft Power BI).

Learning objectives

- Understand the theoretical foundations of key Business Intelligence & Analytics concepts supporting decision-making
- Explore key capabilities of state-of-the-art Business Intelligence & Analytics Systems
- Learn how to successfully implement and run Business Intelligence & Analytics Systems from multiple perspectives, e.g. architecture, data management, consumption, analytics
- Get hands-on experience by working with Business Intelligence & Analytics Systems with real-world use cases and data

Prerequisites
This course is limited to 50 places. The capacity limitation is due to the format of the accompanying engineering capstone project. Strong analytical abilities and profound skills in SQL and Python are required. Students have to apply with their CVs and transcripts of records via the WiWi-Portal. The first lecture will present all organizational details and the underlying registration process for the lecture and the capstone project. The teaching language is English.

Die Erfolgskontrolle erfolgt in Form einer Prüfungsleistung anderer Art (Form) nach § 4 Abs. 2 Nr. 3 SPO. Die Leistungskontrolle erfolgt in Form einer einstündigen Klausur und durch Durchführung eines Capstone Projektes. Details zur Ausgestaltung der Erfolgskontrolle werden im Rahmen der Vorlesung bekannt gegeben.

Workload:

The total workload for this course is approximately 135 hours.

- Attendance Lecture (2 SWS × 15 weeks): 30 h
- Attendance Exercise (1 SWS × 15 weeks): 15 h
- Preparation and Wrap-up of the lecture: 45 h
- Preparation and Wrap-up of the exercise: 25 h
- Exam preparation: 20 h
- Total: 135 h (equivalent to 4.5 CP)

Literature

- Turban, E., Aronson, J., Liang T.-P., Sharda, R. 2008. "Decision Support and Business Intelligence Systems".
- Watson, H. J. 2014. "Tutorial: Big Data Analytics: Concepts, Technologies, and Applications," Communications of the Association for Information Systems (34), p. 24.
- Arnott, D., and Pervan, G. 2014. "A critical analysis of decision support systems research revisited: The rise of design science," Journal of Information Technology (29:4), Nature Publishing Group, pp. 269–293 (doi: 10.1057/jit.2014.16) .
- Carlo, V. (2009). "Business intelligence: data mining and optimization for decision making". Editorial John Wiley and Sons, 308-317.
- Chen, H., Chiang, R. H. L, and Storey, V. C. 2012. „Business Intelligence and Analytics: From Big Data to Big Impact," MIS Quarterly (36:4), pp. 1165-1188.
- Davenport, T. 2014. Big Data @ Work, Boston, MA: Harvard Business Review.
- Economist Intelligence Unit. 2015 "Big data evolution: Forging new corporate capabilities for the long term"
- Power, D. J. 2008. "Decision Support Systems: A Historical Overview," Handbook on Decision Support Systems, pp. 121–140 (doi: 10.1007/978-3-540-48713-5_7) .
- Sharma, R., Mithras, S., and Kankanhalli, A. 2014. „Transforming decision-making processes: a research agenda for understanding the impact of business analytics on organisations," European Journal of Information Systems (23:4), pp. 433-441.
- Silver, M. S. 1991. "Decisional Guidance for Computer-Based Decision Support," MIS Quarterly (15:1), pp. 105-122.

Further literature will be made available in the lecture.

T 6.95 Module component: Business Planning [T-WIWI-102865]

Coordinators: Prof. Dr. Orestis Terzidis
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Irregular	1

Courses					
WT 25/26	2545109	Business Planning for Founders - Startup CFO	2 SWS	Seminar / 🗣	Rosales Bravo, Wessendorf
ST 2026	2545109	Business Planning for Founders - Startup CFO	2 SWS	Seminar / 🗣	Wessendorf, Heidemann, Wessendorf

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Alternative exam assessment.

Prerequisites
None

Recommendations
None

Below you will find excerpts from courses related to this module component:

V Business Planning for Founders - Startup CFO

Seminar (S)
On-Site

2545109, WS 25/26, 2 SWS, Language: English, Open in study portal

Content
Content

Embark on a transformative journey into the dynamic realm of startup finance with our comprehensive course designed for Master's students interested in the task of aspiring to become future Chief Financial Officers (CFOs) or Chief Executive Officers (CEOs) in the startup. Particularly, students who previously attended classes on entrepreneurship or developed their business ideas in Design Thinking Seminars will work on the financial viability and, therefore, the potential for realizing their business ideas. The three-day seminar develops the financial literacy needed to start and operate an entrepreneurial venture, including analyzing and determining the cost and revenue structure of the firm and creating a financial strategy to execute the business plan successfully. Additionally, students will learn about the sources and conditions of different investment types and develop tailored fundraising strategies. The seminar is not restricted to the financial aspects but follows the Triple Bottom Line philosophy (3BL).

Throughout the course, real-world case studies and guest lectures, professional experts will provide valuable insights into the practical application of financial concepts.By the end of this course, you will be well-equipped to take on leadership roles in startups and startup ecosystems, armed with the managerial understanding required to drive success in dynamic and competitive markets.

Learning Objectives

Upon completion of this seminar, course participants will be able to

1. Analyze, forecast, and plan the cost structure and revenue streams of the venture project.
2. Reflect on the sustainability of a business based on the Triple Bottom Line theory.
3. Develop the essential financial statements for a startup.
4. Recall and reflect on investment strategies for startups.
5. Discover business stakeholders and prepare a tailored communication strategy.
6. Reflect on the role of information technology.
7. Apply negotiation techniques essential for securing favorable terms and agreements.
8. Have a brief overview of the related topic.

Credentials:

ONLY ONE of the two options - Business Planning for founders OR Business Planning for founders in the field of IT-Security - can be taken and credited under the in CAS mentioned partial credit, as they cover similar content. Registration must take place in the CAS for the respective examination.

Organizational issues
Registration is via the Wiwi-Portal.

In the seminar you will work on a project in teams of max. 5 persons. Team applications are welcome but not a prerequisite for participation. The seminars will be held in English.

V **Business Planning for Founders - Startup CFO**
2545109, SS 2026, 2 SWS, Language: English, Open in study portal

Seminar (S)
On-Site

Content
Content

Embark on a transformative journey into the dynamic realm of startup finance with our comprehensive course designed for Master's students interested in the task of aspiring to become future Chief Financial Officers (CFOs) or Chief Executive Officers (CEOs) in the startup. Particularly, students who previously attended classes on entrepreneurship or developed their business ideas in Design Thinking Seminars will work on the financial viability and, therefore, the potential for realizing their business ideas. The three-day seminar develops the financial literacy needed to start and operate an entrepreneurial venture, including analyzing and determining the cost and revenue structure of the firm and creating a financial strategy to execute the business plan successfully. Additionally, students will learn about the sources and conditions of different investment types and develop tailored fundraising strategies. The seminar is not restricted to the financial aspects but follows the Triple Bottom Line philosophy (3BL).

Throughout the course, real-world case studies and guest lectures, professional experts will provide valuable insights into the practical application of financial concepts.By the end of this course, you will be well-equipped to take on leadership roles in startups and startup ecosystems, armed with the managerial understanding required to drive success in dynamic and competitive markets.

Learning Objectives

Upon completion of this seminar, course participants will be able to

1. Analyze, forecast, and plan the cost structure and revenue streams of the venture project.
2. Reflect on the sustainability of a business based on the Triple Bottom Line theory.
3. Develop the essential financial statements for a startup.
4. Recall and reflect on investment strategies for startups.
5. Discover business stakeholders and prepare a tailored communication strategy.
6. Reflect on the role of information technology.
7. Apply negotiation techniques essential for securing favorable terms and agreements.
8. Have a brief overview of the related topic.

Organizational issues

Registration is via the Wiwi-Portal.

In the seminar you will work on a project in teams of max. 5 persons. Team applications are welcome but not a prerequisite for participation. The seminars will be held in English.

T 6.96 Module component: Business Planning for Founders [T-WIWI-115110]

Coordinators: Prof. Dr. Orestis Terzidis
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101488 - Entrepreneurship (EnTechnon)

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Irregular	1

Courses					
ST 2026	2545109	Business Planning for Founders - Startup CFO	2 SWS	Seminar / 🗣	Wessendorf, Heidemann, Wessendorf

Legend: 🖥 Online, ✣ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Alternative exam assessment.

Prerequisites
None

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-WIWI-102865 - Business Planning must not have been started.

Recommendations
None

Below you will find excerpts from courses related to this module component:

V Business Planning for Founders - Startup CFO

2545109, SS 2026, 2 SWS, Language: English, Open in study portal

Seminar (S)
On-Site

Content
Content

Embark on a transformative journey into the dynamic realm of startup finance with our comprehensive course designed for Master's students interested in the task of aspiring to become future Chief Financial Officers (CFOs) or Chief Executive Officers (CEOs) in the startup. Particularly, students who previously attended classes on entrepreneurship or developed their business ideas in Design Thinking Seminars will work on the financial viability and, therefore, the potential for realizing their business ideas. The three-day seminar develops the financial literacy needed to start and operate an entrepreneurial venture, including analyzing and determining the cost and revenue structure of the firm and creating a financial strategy to execute the business plan successfully. Additionally, students will learn about the sources and conditions of different investment types and develop tailored fundraising strategies. The seminar is not restricted to the financial aspects but follows the Triple Bottom Line philosophy (3BL).

Throughout the course, real-world case studies and guest lectures, professional experts will provide valuable insights into the practical application of financial concepts.By the end of this course, you will be well-equipped to take on leadership roles in startups and startup ecosystems, armed with the managerial understanding required to drive success in dynamic and competitive markets.

Learning Objectives

Upon completion of this seminar, course participants will be able to

1. Analyze, forecast, and plan the cost structure and revenue streams of the venture project.
2. Reflect on the sustainability of a business based on the Triple Bottom Line theory.
3. Develop the essential financial statements for a startup.
4. Recall and reflect on investment strategies for startups.
5. Discover business stakeholders and prepare a tailored communication strategy.
6. Reflect on the role of information technology.
7. Apply negotiation techniques essential for securing favorable terms and agreements.
8. Have a brief overview of the related topic.

Organizational issues
Registration is via the Wiwi-Portal.

In the seminar you will work on a project in teams of max. 5 persons. Team applications are welcome but not a prerequisite for participation. The seminars will be held in English.

T 6.97 Module component: Business Process Modelling [T-WIWI-102697]

Coordinators: Prof. Dr. Andreas Oberweis
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101472 - Informatics
M-WIWI-101628 - Emphasis in Informatics
M-WIWI-104901 - Informatics (KIT-Department of Economics and Management)

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each winter term	2

Courses					
WT 25/26	2511210	Business Process Modelling	2 SWS	Lecture / 🗣	Oberweis
WT 25/26	2511211	Exercise Business Process Modelling	1 SWS	Practice / 🗣	Kruse

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment of this course is a written examination (60 min) according to §4(2), 1 of the examination regulation in the first week after lecture period.

Prerequisites
None

Below you will find excerpts from courses related to this module component:

V Business Process Modelling

2511210, WS 25/26, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content
The proper modeling of relevant aspects of business processes is essential for an efficient and effective design and implementation of processes. This lecture presents different classes of modeling languages and discusses the respective advantages and disadvantages of using actual application scenarios. For that simulative and analytical methods for process analysis are introduced. In the accompanying exercise the use of process modeling tools is practiced.

Learning objectives:

Students

- describe goals of business process modeling and aplly different modeling languages,
- choose the appropriate modeling language according to a given context,
- use suitable tools for modeling business processes,
- apply methods for analysing and assessing process modells to evaluate specific quality characteristics of the process model.

Recommendations:

Knowledge of course Applied Informatics I - Modelling is expected.

Workload:

- Lecture 30h
- Exercise 15h
- Preparation of lecture 24h
- Preparation of exercises 25h
- Exam preparation 40h
- Exam 1h

Literature

- M. Weske: Business Process Management: Concepts, Languages, Architectures. Springer 2012.
- F. Schönthaler, G.Vossen, A. Oberweis, T. Karl: Business Processes for Business Communities: Modeling Languages, Methods, Tools. Springer 2012.

Weitere Literatur wird in der Vorlesung bekannt gegeben.

T 6.98 Module component: CAD Engineering Project for Intelligent Systems [T-MACH-113857]

Coordinators: Prof. Dr.-Ing. Arne Rönnau
Organisation: KIT Department of Mechanical Engineering
Part of: M-WIWI-107718 - Mechanical Engineering I
M-WIWI-107730 - Mechanical Engineering II

Type	Credits	Grading	Term offered	Expansion	Version
Examination of another type	6 CP	graded	Each term	1 semesters	2

Courses					
WT 25/26	2122331	CAD Engineering Project for Intelligent Systems	4 SWS	Project / 🗣	Rönnau
ST 2026	2122331	CAD Engineering Project for Intelligent Systems	4 SWS	Project / 🗣	Rönnau

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, × Cancelled

Assessment
Success control takes place in the form of another type of examination. Design project as well as written elaboration in a team and a final presentation. The overall impression is rated. Further details will be provided at the beginning of the course.

Prerequisites
none

Additional Information
The course is offered in English.

Workload
180 hours

Below you will find excerpts from courses related to this module component:

V CAD Engineering Project for Intelligent Systems
2122331, WS 25/26, 4 SWS, Language: English, Open in study portal

Project (PRO)
On-Site

Content
In this design project, students work in small groups using an agile approach to develop an innovative mechatronic component that meets previously defined requirements for an intelligent system.

To this end, students get to know a current CAD development environment and learn how to design the corresponding parts. The typical design and development process is followed from the idea to the finished model. The focus is on independent solution finding, teamwork, (robotic) functional fulfillment, 3D printing and manufacturing and biologically inspired design. The project results are presented at the end of the semester.

Organizational issues
See ILIAS for time and place

V CAD Engineering Project for Intelligent Systems
2122331, SS 2026, 4 SWS, Language: English, Open in study portal

Project (PRO)
On-Site

Content
The module *CAD Engineering Project for Intelligent Systems* is a project-based master's-level course that offers students the opportunity to gain and to deepen engineering competencies in the development of intelligent mechatronic systems through practical application. Working in interdisciplinary teams, students conceive, design, and realize a robotic prototype across a complete development cycle.

Starting from a defined problem statement, students develop a technical concept and implement it as a detailed mechanical design using CAD software (Autodesk Fusion 360). Particular emphasis is placed on the basics of rapid prototyping, function-oriented component design, correct parametrization, consideration of manufacturing constraints, and the integration of mechanical and electronic components as well as their software interfacing.

The designed components are predominantly manufactured using additive manufacturing methods (in particular 3D printing) and are supplemented by basic conventional manufacturing techniques. In parallel, students integrate suitable sensors, actuators, and additional electronic components into the system and develop the necessary software for control and regulation of the robotic prototype.

Finally, the developed system is assembled, commissioned, and tested and evaluated under realistic conditions. The results are subsequently documented and presented. In addition to technical expertise, the module particularly promotes project management skills, teamwork, interdisciplinary thinking, and a systematic, solution-oriented approach, as required for the development of intelligent technical systems in research and industry.

Prerequisites:
Basic knowledge of common CAD software solutions, ideally Fusion 360.

Students must hand in a small application, detailing motivation and prior experience. The application is done via Ilias (ilias.studium.kit.edu) - **Deadline: 16.04.2026 .**

Organizational issues
See ILIAS for time and place

T 6.99 Module component: CAD Workshop [T-MACH-114767]

Coordinators: Prof. Dr.-Ing. Arne Rönnau
Organisation: KIT Department of Mechanical Engineering
Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Expansion	Version
Examination of another type	4 CP	graded	Each term	1 semesters	2

Courses					
WT 25/26	2121330	CAD Workshop	4 SWS	Course / 🗣	Rönnau
ST 2026	2121330	CAD Workshop	4 SWS	Project / 🗣	Rönnau

Legend: 🖥 Online, ⚄ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Learning control is an examination of another type. The overall impression is evaluated. The following aspects are included in the grading:

- Completion of the project task and documentation
- 4 interim presentations
- Final presentation including live demonstration

The grading scale will be announced in the course.

Prerequisites
The Brick T-MACH-102153 - PLM-CAD Workshop must not have been started.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-MACH-102153 - PLM-CAD Workshop must not have been started.

Additional Information
Attendance is compulsory. The number of participants is limited. Information on the application process is available in ILIAS.

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V CAD Workshop
2121330, SS 2026, 4 SWS, Language: English, Open in study portal

Project (PRO)
On-Site

Content

The CAD Workshop is a project-based bachelor-level module that provides students with a hands-on introduction to the computer-aided development of a mechatronic prototype. The course has a particular focus on the development process from the initial idea to a functional prototype, including. team organisation, product management, and marketing.

Students work in small teams and operate under limited resources (time, budget, and personnel). Based on a task defined at the beginning of the workshop, the teams develop a product concept and implement it along a structured development process. The focus lies less on the design's complexity and more on the exemplary experience with typical tasks and challenges encountered throughout the development process, such as role and task allocation, workflow definition, bill of materials, version and data management, as well as project and time planning.

In parallel, students translate their concepts into mechanical designs using CAD software (e.g., Autodesk Fusion 360), with particular emphasis on creating assemblies. At least two components need to be manufactured using additive manufacturing methods (3D printing), while also providing basic insights into classical CAD design of individual parts. The designed components are assembled into a mechatronic prototype that may include basic electronic and software-based functionality. The control system of the prototype is implemented by the students.

The entire development process is documented, regularly reflected upon in milestone presentations, and finally presented in a final presentation, including a live demonstration of the prototype.

Learning Objectives

After successfully completing the module, students are able to:

- Work effectively in interdisciplinary teams and define roles, workflows, and responsibilities
- Develop a technical prototype under realistic constraints (time and resources)
- Create simple part designs with CAD, optimised for 3D printing
- Create CAD models of assemblies and integrate them into a coherent development process
- Conceive, build, and evaluate simple mechatronic systems
- Document and present project results in a clear and marketing-oriented manner

Additional Information

The number of participants is limited. Information on the application process is available in ILIAS.

Organizational Information

A project-based workshop with collaborative group work requiring regular attendance and active participation. Details regarding the schedule, assessment criteria, and milestones will be announced at the beginning of the course.

See further details on course organisation at: teaching.imi.kit.edu and in ILIAS.

Workload 120 hours

- regular attendance: 60 hours
- self-study and project work: 60 hours

Organizational issues

See ILIAS for time and location / Zeit und Ort siehe ILIAS

T 6.100 Module component: Case Studies Seminar: Innovation Management [T-WIWI-102852]

Coordinators: Prof. Dr. Marion Weissenberger-Eibl
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101488 - Entrepreneurship (EnTechnon)
M-WIWI-101507 - Innovation Management
M-WIWI-101507 - Innovation Management
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Each winter term	1

Courses					
WT 25/26	2545105	Case studies seminar: Innovation management	2 SWS	Seminar / 🗣	Weissenberger-Eibl
WT 26/27	2545105	Case studies seminar: Innovation management	2 SWS	Seminar / 🗣	Weissenberger-Eibl

Legend: 🖥 Online, Blended (On-Site/Online), 🗣 On-Site, × Cancelled

Assessment
Alternative exam assessments (§4(2), 3 SPO).

The grade is made up of 50% of the grade for the written paper * and 50% of the grade for the presentation.

* Scope according to the current requirements for recognition in the seminar module

Prerequisites
None

Recommendations
Prior attendance of the course Innovation Management is recommended.

Workload
90 hours

Below you will find excerpts from courses related to this module component:

V Case studies seminar: Innovation management

2545105, WS 25/26, 2 SWS, Language: German, Open in study portal

Seminar (S)
On-Site

Content
The objective of the seminar is to acquire selected concepts and methods of innovation management and to apply these in a practical way. The focus is on working on specific issues using a case study from business practice. Students work in groups and transfer the content taught, for example from the areas of scenario development, to entrepreneurial challenges in the context of fictitious case studies. The course follows a didactic interplay of theoretical input and direct application of what has been learned. At the end, the groups present their results in a plenary presentation, which is prepared by a short introduction to presentation techniques. The joint discussion promotes the transfer of knowledge to different contexts and strengthens the participants' ability to reflect.

As part of the seminar, students deal independently with current, research-oriented issues and systematically apply scientific criteria. They research relevant information, analyze, abstract and critically evaluate it. They draw conclusions independently from poorly structured data, incorporate interdisciplinary knowledge and contribute selectively to the further development of existing research perspectives. The results are validated and systematically prepared both orally and in writing using scientific standards - particularly with regard to structuring, terminology and correct citation of sources. In doing so, students argue in a technically sound manner and defend their findings in discourse with other participants and lecturers. They are also familiar with the DFG Code of Conduct “Guidelines for Safeguarding Good Scientific Practice” and apply these standards consistently in the preparation of their scientific work.

Literature
Werden in der ersten Veranstaltung bekannt gegeben.

V

Case studies seminar: Innovation management

2545105, WS 26/27, 2 SWS, Language: German, Open in study portal

Seminar (S)
On-Site

Content

The objective of the seminar is to acquire selected concepts and methods of innovation management and to apply these in a practical way. The focus is on working on specific issues using a case study from business practice. Students work in groups and transfer the content taught, for example from the areas of scenario development, to entrepreneurial challenges in the context of fictitious case studies. The course follows a didactic interplay of theoretical input and direct application of what has been learned. At the end, the groups present their results in a plenary presentation, which is prepared by a short introduction to presentation techniques. The joint discussion promotes the transfer of knowledge to different contexts and strengthens the participants' ability to reflect.

As part of the seminar, students deal independently with current, research-oriented issues and systematically apply scientific criteria. They research relevant information, analyze, abstract and critically evaluate it. They draw conclusions independently from poorly structured data, incorporate interdisciplinary knowledge and contribute selectively to the further development of existing research perspectives. The results are validated and systematically prepared both orally and in writing using scientific standards - particularly with regard to structuring, terminology and correct citation of sources. In doing so, students argue in a technically sound manner and defend their findings in discourse with other participants and lecturers. They are also familiar with the DFG Code of Conduct “Guidelines for Safeguarding Good Scientific Practice” and apply these standards consistently in the preparation of their scientific work.

Literature

Werden in der ersten Veranstaltung bekannt gegeben.

T 6.101 Module component: Ceramic Processing Technology [T-MACH-102182]

Coordinators: Dr. Joachim Binder
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101268 - Specific Topics in Materials Science
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each summer term	1

Courses					
WT 25/26	2126730	Advanced ceramics processing (EN)	2 SWS	Lecture /	Pagnan Furlan

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
The assessment consists of an oral exam (approx. 20 min) taking place at the agreed date.

Auxiliary means: none

The re-examination is offered upon agreement.

Prerequisites
none

Below you will find excerpts from courses related to this module component:

V Advanced ceramics processing (EN)

2126730, WS 25/26, 2 SWS, Language: English, Open in study portal

Lecture (V)
Blended (On-Site/Online)

Content
This course explores the engineering principles and techniques used to transform raw ceramic powders into high-performance advanced ceramics. Students learn how particle characteristics, forming strategies, and processing parameters influence microstructure and final properties. Emphasis is placed on understanding the interplay between materials design, processing control, and functional performance, supported by hands-on exercises and problem-solving sessions.

Organizational issues
Notice: the course starts on 04.11.2025!

Literature
1. (To refresh concepts from Fundamentals of Materials Engineering) Callister William D., Rethwisch David G., Fundamentals of materials science and engineering. Hoboken, N.J: Wiley. 2008. https://katalog.bibliothek.kit.edu/cgi-bin/koha/opac-detail.pl?biblionumber=309052&query_desc=callister

2. Richerson David W., Modern ceramic engineering. Boca Raton, FL: CRC Press. 2018. https://search.ebscohost.com/login.aspx?direct=true&db=nlebk&db=nlabk&AN=1802218 *

3. Bansal Narottam P. and Boccaccini Aldo R., Ceramics and composites processing methods. Hoboken, N.J: Wiley. 2012. https://ebookcentral.proquest.com/lib/karlsruhetech/detail.action?docID=817471 *

* Access possible via KIT Network.

T 6.102 Module component: Challenges in Supply Chain Management [T-WIWI-102872]

Coordinators: Esther Mohr
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104899 - Operations Research

Type	Credits	Grading	Term offered	Version
Examination of another type	4,5 CP	graded	Each summer term	2

Courses					
ST 2026	2550494	Service Operations and Cyber Security	3 SWS	Lecture /	Mohr

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
The assessment consists of a written paper and an oral exam of ca. 30-40 min.

Prerequisites
None

Recommendations
Basic knowledge as conveyed in the module "Introduction to Operations Research" is assumed.

Additional Information
The number of course participants is limited to 12 participants due to joint work in BASF project teams. Due to these capacity restrictions, registration before course start is required. For further information see the webpage of the course.

The course is offered irregularly. The planned lectures and courses for the next three years are announced online.

Below you will find excerpts from courses related to this module component:

V Service Operations and Cyber Security
2550494, SS 2026, 3 SWS, Language: German, Open in study portal

Lecture (V)
Blended (On-Site/Online)

Content
As part of the event, case studies on future challenges in Service Operations and Cyber Security will be addressed at the BSI (Federal Office for Information Security). The event aims to present, critically assess, and discuss current issues in Service Operations and Cyber Security.

The focus is not only on current trends but primarily on future challenges, particularly regarding their applicability in practical applications. The main part of the event consists of working on project-related case studies from the BSI in Bonn. Students are expected to scientifically address a practical issue: by delving into a scientific subtopic, students will become familiar with scientific literature while also learning critical argumentation techniques essential for practical applications. Furthermore, emphasis will be placed on a critical discussion of the approaches.

The content of the event covers forward-looking topics in Service Operations and Cyber Security. The exact topics will be announced at the beginning of each semester during an introductory meeting.

Organizational issues
Bewerbung über das Wiwi-Portal möglich:

https://portal.wiwi.kit.edu/ys/9111

(Bewerbungszeitraum: 01.03.2026 - 12.04.2026)

Literature
Wird in Abhängigkeit vom Thema in den Projektteams bekanntgegeben.

T 6.103 Module component: Characteristics of Transportation Systems [T-BGU-106609]

Coordinators: Prof. Dr.-Ing. Peter Vortisch
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: M-BGU-101064 - Fundamentals of Transportation
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Expansion	Version
Written examination	3 CP	graded	Each term	1 semesters	2

Courses					
ST 2026	6232806	Properties of Means of Transport	2 SWS	Lecture / 🗣	Vortisch

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, x Cancelled

Assessment
written exam, 60 min.

Prerequisites
None

Recommendations
None

Additional Information
None

Workload
90 hours

T 6.104 Module component: Chemical Hydrogen Storage [T-CIWVT-113234]

Coordinators: TT-Prof. Dr. Moritz Wolf
Organisation: KIT Department of Chemical and Process Engineering
Part of: M-CIWVT-106566 - Chemical Hydrogen Storage

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each winter term	1

Courses					
WT 25/26	2231420	Chemical Hydrogen Storage	2 SWS	Lecture / 🗣	Wolf, Sauer

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The learning control is an oral examination lasting approx. 20 minutes.

Prerequisites
None

T 6.105 Module component: Chemically Reacting Flows [T-MACH-113998]

Coordinators: Prof. Dr. Dr. Michael Fischlschweiger
Organisation: KIT Department of Mechanical Engineering

Part of: M-WIWI-107718 - Mechanical Engineering I
M-WIWI-107730 - Mechanical Engineering II

Type	Credits	Grading	Term offered	Version
Oral examination	9 CP	graded	Each term	3

Courses					
WT 25/26	2165540	Chemically Reacting Flows I	2 SWS	Lecture / 🗣	Shrotriya
WT 25/26	2165541	Chemically Reacting Flows I (Tutorial)	2 SWS	Practice / 🗣	Shrotriya
ST 2026	2166510	Chemically Reacting Flows II	2 SWS	Lecture / 🗣	Fischlschweiger, Yu
ST 2026	2166511	Chemically Reacting Flows II (Exercises)	2 SWS	Practice / 🗣	Fischlschweiger, Yu

Legend: Online, Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Oral exam, approx. 40 min

Prerequisites
T-MACH-114043 and T-MACH-114044 must not have been started.

T-MACH-105325 and T-MACH-105213 must not have been started.

Additional Information
The course is offered in English.

The module component consists of two lectures. Part 1 takes place in the winter semester, and Part 2 in the summer semester. It will be offered for the first time in the winter semester of 2025/2026.

Workload
270 hours

Below you will find excerpts from courses related to this module component:

V Chemically Reacting Flows I

2165540, WS 25/26, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content

- Fundamental concepts and phenomena
- Experimental analysis of flames
- Conservation equations for laminar flat flames
- Chemical reactions
- Chemical kinetics mechanisms
- Laminar premixed flames
- Laminar diffusion flames
- Ignition processes
- NOx formation
- Formation of hydrocarbons and soot

Organizational issues
This lecture will be offered in winter semester 2025/2026 the first time.

Literature
Vorlesungsskript,

Buch Verbrennung - Physikalisch-Chemische Grundlagen, Modellbildung, Schadstoffentstehung, Autoren: U. Maas, J. Warnatz, R.W. Dibble, Springer-Lehrbuch, Heidelberg 1996

V

Chemically Reacting Flows II (Exercises)

2166511, SS 2026, 2 SWS, Language: English, Open in study portal

Practice (Ü)
On-Site

Content
Calculation and Simulation of chemically reacting flows

Organizational issues
Nachteilsausgleiche bitte fristgerecht an den Dozenten schicken.
Please send the requests for compensations for disadvantages to the lecturer within the deadline

Literature
Skript Grundlagen der technischen Verbrennung (I+II) von Prof. Dr. rer. nat. habil. U. Maas
Buch Verbrennung - Physikalisch-Chemische Grundlagen, Modellbildung, Schadstoffentstehung, Autoren: U. Maas, J. Warnatz, R.W. Dibble, Springer-Lehrbuch, Heidelberg 1996

T 6.106 Module component: Circular Economy – Challenges and Potentials [T-WIWI-114057]

Coordinators: Prof. Dr. Frank Schultmann
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101412 - Industrial Production III
M-WIWI-101471 - Industrial Production II
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	3,5 CP	graded	Each summer term	1

Courses					
ST 2026	2581965	Circular Economy - Challenges and Potentials	2 SWS	Lecture / 🗣	Schultmann, Rudi

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, x Cancelled

Assessment
The assessment consists of an oral (30 minutes) or written exam (60 minutes) (following §4(2) of the examination regulation). The exam takes place in every semester. Re-examinations are offered at every ordinary examination date.

Workload
105 hours

Below you will find excerpts from courses related to this module component:

V Circular Economy - Challenges and Potentials

2581965, SS 2026, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content
Circular Economy (CE) is an economic system that on the one hand aims to minimize waste, emissions and resource consumption and on the other hand increase resource efficiency by keeping products and materials in use for as long as possible. Based on basic ideas and principles of CE this lecture tackles potentials and challenges for the design and operations of circular value chains and systems. Different research-orientied case studies reveal and illustrate the potential implementation as well as the limits and future needs of CE as a key element of sustainable industrial development.

Literature
Wird in der Lehrveranstaltung bekannt gegeben.

T 6.107 Module component: Circular Factory [T-MACH-114974]

Coordinators: Prof. Dr.-Ing. Gisela Lanza
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-107636 - Circular Factory
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Written examination	8 CP	graded	Each summer term	2

Courses					
ST 2026	2150620	Circular Factory	6 SWS	Lecture / Practice / 🗣	Lanza, Baumgärtner

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
written exam, duration 120 minutes

Prerequisites
T-MACH-113983 must not have been started.

Additional Information
The course is offered in English.

Workload
240 hours

Below you will find excerpts from courses related to this module component:

V Circular Factory

2150620, SS 2026, 6 SWS, Language: English, Open in study portal

Lecture / Practice (VÜ)
On-Site

Content
New, innovative economic systems are needed to decouple resource consumption from wealth. New, innovative economic systems are needed to decouple resource consumption from prosperity. In this lecture, students learn about the relevance of such systems of circular value creation. The main focus is on implementation by means of a so-called circular factory, which should enable a new life cycle for used products. Ideally, this is done today by upgrading the used product to the product generation currently in production. This shows the complex field of tension in which the topics of this course are embedded. The freedom of product development is restricted by the constraint of product upgrades and new paradigms are needed. The planning and control of products by the circular factory, including the associated logistics concepts, is complicated by the used condition of the returns. This must be measured and evaluated in order to enable a functional statement of the reprocessed product in a new life cycle. Not least, it requires appropriate (robotic) automation concepts for the disassembly of used products and their reprocessing, which must be specifically designed in high-wage countries in order to operate and establish the overall process economically.

The content of the course addresses and deals with the specific problems and solutions that arise in the realization of circular factories. The following topics are addressed in chronological order:

- An overview of the circular economy in general and different strategies to reduce the consumption of natural resources (R-strategies).
- Comprehensive overview of remanufacturing and presentation of challenges and solutions in the planning and control of disassembly and remanufacturing systems.
- Guidelines and constraints in the design, development and validation of sustainable products.
- Fundamentals of metrological assessment and evaluation of returning used products in the circular factory to enable a new life cycle.
- Methods for data-driven learning from human disassembly to enable automated robot-based automation.
- Processing technology solutions for the reprocessing of used products.
- Intralogistics and handling of recyclable products.

The lecture is complemented by practical items. Real-life industrial problems are presented and discussed in order to reinforce students' understanding of the relevance of the theoretical content and to enable a thorough practical orientation.

Learing Outcomes:

The students …

- can name dimensions of circularity and circular economy methods (e.g., repair, refurbish, recycle) and describe them in detail.
- can describe challenges in planning and control circular factories, including remanufacturing networks and disassembly systems.
- are able to apply guidelines for designing circular products.
- distinguish data acquisition techniques for metrologically assessing returned products and apply uncertainty-driven product modeling in circular production systems.
- have methodical knowledge on learning from human observation and disassembly automatization and apply this knowledge to new problem cases.
- can describe reprocessing methods, including reconditioning and material characterization.
- understand the challenges in intralogistics for circular products.

After completing this course, students are able to understand the challenges of establishing a circular economy. They are also able to evaluate possible solutions and assess them in relation to these challenges. In particular, students are ultimately able to understand circular production in a circular factory holistically and to relate it to existing concepts in industrial practice.

Workload:

regular attendance: 62 hours
self-study: 178 hours

Organizational issues
Lectures on Mondays at 3:45 p.m. and on Tuesdays at 3:45 p.m.,
Tutorials on Wednesdays at 3:45 pm. The tutorial dates will be announced in the first lecture.

Vorlesungstermine montags 15:45 Uhr und dienstags 15:45 Uhr,
Übungstermine mittwochs 15:45 Uhr. Bekanntgabe der konkreten Übungstermine erfolgt in der ersten Vorlesung.

Literature
Medien:
Skript zur Veranstaltung wird über (https://ilias.studium.kit.edu/) bereitgestellt.

Media:
Lecture notes will be provided in Ilias (https://ilias.studium.kit.edu/).

T 6.108 Module component: Civil Engineering Structures and Regenerative Energies [T-BGU-111922]

Coordinators: Prof. Dr.-Ing. Shervin Haghsheno
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: M-BGU-101884 - Lean Management in Construction
M-BGU-101888 - Project Management in Construction

Type	Credits	Grading	Term offered	Expansion	Version
Written examination	3 CP	graded	Each term	1 semesters	1

Courses					
ST 2026	6241810	Civil Engineering Structures and Regenerative Energies	2 SWS	Lecture / Practice / 🗣	Haghsheno, Mitarbeiter/innen

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
written exam, 60 min.

Prerequisites
none

Recommendations
none

Additional Information
none

Workload
90 hours

T 6.109 Module component: Civil Law for Beginners [T-INFO-103339]

Coordinators: Dr. Yvonne Matz
Organisation: KIT Department of Informatics
Part of: M-WIWI-104903 - Law

Type	Credits	Grading	Term offered	Version
Written examination	5 CP	graded	Each winter term	3

Courses					
WT 25/26	2424012	Civil Law for Beginners	4 SWS	Lecture / 🗣	Matz
WT 26/27	2424012	Civil Law for Beginners	4 SWS	Lecture / 🗣	Matz

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

T 6.110 Module component: Climatology [T-PHYS-101092]

Coordinators: Prof. Dr. Joaquim José Ginete Werner Pinto
Organisation: KIT Department of Physics
Part of: M-WIWI-104904 - Natural Sciences

Type	Credits	Grading	Term offered	Version
Coursework	4 CP	pass/fail	Each summer term	4

Courses					
ST 2026	4051111	Klimatologie	3 SWS	Lecture / 🗣	Ginete Werner Pinto
ST 2026	4051112	Übungen zu Klimatologie	1 SWS	Practice / 🗣	Ginete Werner Pinto, Christ, Dillerup, NN

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Prerequisites
none

Workload
120 hours

T 6.111 Module component: CO2-Neutral Combustion Engines and their Fuels I [T-MACH-111550]

Coordinators: Prof. Dr. Thomas Koch
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101275 - Combustion Engines I

Type	Credits	Grading	Term offered	Expansion	Version
Oral examination	4 CP	graded	Each winter term	1 semesters	2

Courses					
WT 25/26	2133113	CO2-neutral combustion engines and their fuels I	3 SWS	Lecture / Practice / 🗣	Koch

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
oral examination, Duration: 25 min., no auxiliary means

Prerequisites
none

Additional Information
This course is offered in German.

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V CO2-neutral combustion engines and their fuels I

2133113, WS 25/26, 3 SWS, Language: German, Open in study portal

Lecture / Practice (VÜ)
On-Site

Organizational issues
Übungstermine Donnerstags nach Bekanntgabe in der Vorlesung

T 6.112 Module component: CO2-Neutral Combustion Engines and their Fuels II [T-MACH-111560]

Coordinators: Prof. Dr. Thomas Koch
Organisation: KIT Department of Mechanical Engineering
Part of: M-MACH-101303 - Combustion Engines II

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each summer term	4

Courses					
ST 2026	2134151	CO2-neutral combustion engines and their fuels II	3 SWS	Lecture / Practice / 🗣	Koch

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
oral examination, duration: 25 minutes, no auxiliary means

Prerequisites
none

Recommendations
Fundamentals of Combustion Engines II helpful

Additional Information
This course is offered in German.

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V CO2-neutral combustion engines and their fuels II

Lecture / Practice (VÜ)
On-Site

2134151, SS 2026, 3 SWS, Language: German, Open in study portal

T 6.113 Module component: Cognitive Modeling [T-WIWI-111392]

Coordinators: Prof. Dr. Benjamin Scheibehenne
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Expansion	Version
Examination of another type	4,5 CP	graded	Irregular	1 semesters	1

Assessment
There will be 4 assignments during the course of the semester. Each will count 25% towards the final grade.

Prerequisites
Calculus, probability theory

Additional Information
The goal of this course is to help students develop a basic understanding of computational models in the study of human cognition and behavior.

In the first half of the semester, we will go over the following contents to prepare for the learning of cognitive modeling: basics of the R software, foundations of probability, and parameter estimation. In the second half, we will discuss the general ideas of modeling in behavioral science as well as some specific cognitive models. The class will take a biweekly lecture form. All lectures, materials, and assignments are in English.

The number of participants is limited. The registration will take place via the Wiwi-Portal.

Workload
135 hours

T 6.114 Module component: Collective Intelligence in Human Judgment and Decision Making [T-WIWI-114186]

Coordinators: Prof. Dr. Benjamin Scheibehenne
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration
M-WIWI-105714 - Consumer Research

Type	Credits	Grading	Term offered	Expansion	Version
Examination of another type	4,5 CP	graded	Each summer term	1 semesters	1

Courses					
ST 2026	2500026	Collective Intelligence in Human Judgment and Decision Making	2 SWS	Lecture / 🗣	Gradwohl

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The overall grade is derived from a weighted score of several assessment components (active participation, presentations, and written assignments). Details regarding the scoring system and the evaluation criteria will be announced at the start of the course.

Prerequisites
None

Recommendations
Willingness to engage with experimental research papers and basic mathematical models is required.

Additional Information
This course covers the key ideas of collective intelligence, asking when it emerges and when it fails. Students will engage with classical and contemporary research papers about how we can combine information from different individuals. They learn to understand and discuss under which conditions combining judgments and decisions of multiple individuals produces collective intelligence and when it creates the risk to run into collective madness and herding.

After the course students can define and describe important concepts of collective intelligence and can name and describe models, theories and experimental paradigms that are used to investigate it. They will practice their ability to read and understand research papers and to critically evaluate the claims in empirical and theoretical research papers.
Moreover, they can apply the concepts and ideas of collective intelligence to examples related to consumer behavior, management and their everyday lives.

All lectures, materials, and assignments will be in English.

The number of participants is limited. The registration will take place via the WiWi portal.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Collective Intelligence in Human Judgment and Decision Making

2500026, SS 2026, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content
This course covers the key ideas of collective intelligence, asking when it emerges and when it fails. Students will engage with classical and contemporary research papers about how we can combine information from different individuals. They learn to understand and discuss under which conditions combining judgments and decisions of multiple individuals produces collective intelligence and when it creates the risk to run into collective madness and herding.

After the course students can define and describe important concepts of collective intelligence and can name and describe models, theories and experimental paradigms that are used to investigate it. They will practice their ability to read and understand research papers and to critically evaluate the claims in empirical and theoretical research papers.

Moreover, they can apply the concepts and ideas of collective intelligence to examples related to consumer behavior, management and their everyday lives. All lectures, materials, and assignments will be in English.

A strong interest in human judgment and decision making, willingness to engage with scientific literature and empirical methods, as well as openness to engage with mathematical models are important prerequisites.

All lectures, materials, and assignments are in English. The number of participants is limited. The registration will take place via the Campus portal.

Grades will be determined based on different parts, including active participation and written assignments. Details will be discussed at the beginning of the course.

Organizational issues
Participation is limited to 15 participants. Registration via the WiWi portal is required for the course. If too many students register, students in higher semesters are selected first.

Literature
Kameda, T., Toyokawa, W., & Tindale, R. S. (2022). Information aggregation and collective intelligence beyond the wisdom of crowds. *Nature Reviews Psychology*, *1*(6), 345-357.

T 6.115 Module component: Collective Perception in Autonomous Driving [T-WIWI-113363]

Coordinators: Prof. Dr. Alexey Vinel
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101472 - Informatics
M-WIWI-101628 - Emphasis in Informatics
M-WIWI-104901 - Informatics (KIT-Department of Economics and Management)
M-WIWI-106631 - Cooperative Autonomous Vehicles

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each summer term	1

Courses					
ST 2026	2511456	Collective Perception in Autonomous Driving	2 SWS	Lecture / 🧩	Zhao, Bied, Vinel
ST 2026	2511457	Exercise Collective Perception in Autonomous Driving	1 SWS	Practice / 🧩	Flores Comeca, Arockiasamy, Bied, Zhao, Khoshkdahan

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The default assessment of this course is a written examination (60 min).

The exam takes place every semester and can be repeated at every regular examination date.

Prerequisites
None.

Workload
135 hours

6.116 Module component: Combustion Engines I [T-MACH-102194]

Coordinators: Prof. Dr. Thomas Koch
Dr.-Ing. Heiko Kubach

Organisation: KIT Department of Mechanical Engineering

Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each winter term	1

Courses					
WT 25/26	2133113	CO2-neutral combustion engines and their fuels I	3 SWS	Lecture / Practice / 🗣	Koch

Legend: 🖥 Online, ❃ Blended (On-Site/Online), 🗣 On-Site, × Cancelled

Assessment
oral examination, Duration: 25 min., no auxiliary means

Prerequisites
none

Below you will find excerpts from courses related to this module component:

V CO2-neutral combustion engines and their fuels I

Lecture / Practice (VÜ)
On-Site

2133113, WS 25/26, 3 SWS, Language: German, Open in study portal

Organizational issues
Übungstermine Donnerstags nach Bekanntgabe in der Vorlesung

T 6.117 Module component: Combustion Engines II [T-MACH-104609]

Coordinators: Dr.-Ing. Rainer Koch
Dr.-Ing. Heiko Kubach

Organisation: KIT Department of Mechanical Engineering

Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Oral examination	5 CP	graded	Each summer term	1

Courses					
ST 2026	2134151	CO2-neutral combustion engines and their fuels II	3 SWS	Lecture / Practice / 🗣	Koch

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
oral examination, duration: 25 minutes, no auxiliary means

Prerequisites
none

Recommendations
Fundamentals of Combustion Engines I helpful

Below you will find excerpts from courses related to this module component:

V CO2-neutral combustion engines and their fuels II

2134151, SS 2026, 3 SWS, Language: German, Open in study portal

Lecture / Practice (VÜ)
On-Site

T 6.118 Module component: Communication Systems and Protocols [T-ETIT-101938]

Coordinators: Dr.-Ing. Jens Becker
Prof. Dr.-Ing. Jürgen Becker
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-100539 - Communication Systems and Protocols
M-MACH-101295 - Optoelectronics and Optical Communication
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Written examination	5 CP	graded	Each summer term	1

Courses					
ST 2026	2311616	Communication Systems and Protocols	2 SWS	Lecture / 🗣	Becker, Becker
ST 2026	2311618	Tutorial for 2311616 Communication Systems and Protocols	1 SWS	Practice / 🗣	Stammler

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The examination consists of a written examination of 120 min.

Prerequisites
none

Recommendations
Knowledge of the basics from the lecture "Digitaltechnik"is helpful.

T 6.119 Module component: Competition in Networks [T-WIWI-100005]

Coordinators: Prof. Dr. Kay Mitusch
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101406 - Network Economics
M-WIWI-104908 - Economics

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each winter term	3

Courses					
WT 25/26	2561204	Competition in Networks	2 SWS	Lecture /	Mitusch
WT 25/26	2561205	Übung zu Wettbewerb in Netzen	1 SWS	Practice /	Mitusch, Corbo

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
Result of success is made by a 60 minutes written examination during the semester break (according to §4(2), 1 ERSC). Examination is offered every semester and can be retried at any regular examination date.

Prerequisites
None.

Recommendations
Basics of microeconomics obtained within the undergraduate programme (B.Sc) of economics are required.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Competition in Networks
2561204, WS 25/26, 2 SWS, Language: German, Open in study portal

Lecture (V)
Blended (On-Site/Online)

Content
Network or infrastructure industries like telecommunication, transport, and utilities form the backbone of modern economies. The lecture provides an overview of the economic characteristics of network industries. The planning of networks is complicated by the multitude of aspects involved (like spatial differentiation and the like). The interactions of different companies - competition or cooperation or both - are characterized by complex interdependencies within the networks: network effects, economies of scale, effects of vertical integration, switching costs, standardization, compatibility etc. appear increasingly in these sectors and even tend to appear in combination. Additionally, government interventions can often be observed, partly driven by the aims of competition policy and partly driven by the aims industrial policy. All these issues are brought up, analyzed formally (in part) and illustrated by several examples in the lecture

Literature
Literatur und Skripte werden in der Veranstaltung angegeben.

T 6.120 Module component: Computational Economics [T-WIWI-102680]

Coordinators: Prof. Dr. Pradyumn Kumar Shukla
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101472 - Informatics
M-WIWI-101628 - Emphasis in Informatics
M-WIWI-104901 - Informatics (KIT-Department of Economics and Management)

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each winter term	3

Courses					
WT 25/26	2590458	Computational Economics	2 SWS	Lecture / Blended	Shukla
WT 25/26	2590459	Excercises to Computational Economics	1 SWS	Practice / Blended	Shukla
WT 26/27	2590458	Computational Economics	2 SWS	Lecture / Blended	Shukla
WT 26/27	2590459	Excercises to Computational Economics	1 SWS	Practice / Blended	Shukla

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
Success is assessed in the form of a written (60 min) or oral examination.

The examination for this course is offered both in the semester in which the course takes place and in the following semester. In the following semester (summer term), participation in the exam is **exclusively reserved for students who did not pass the first attempt.** This regulation applies from winter semester 2025/2026.

Prerequisites
The prerequisite for admission to the examination "Computational Economics" is the successful completion of a computational assignment issued in mid-December; submissions are due in the third week of January or, alternatively, no later than one week before the examination registration.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Computational Economics

2590458, WS 25/26, 2 SWS, Language: English, Open in study portal

Lecture (V)
Blended (On-Site/Online)

Content
Examining complex economic problems with classic analytical methods usually requires making numerous simplifying assumptions, for example that agents behave rationally or homogeneously. Recently, widespread availability of computing power gave rise to a new field in economic research that allows the modeling of heterogeneity and forms of bounded rationality: Computational Economics. Within this new discipline, computer based simulation models are used for analyzing complex economic systems. In short, an artificial world is created which captures all relevant aspects of the problem under consideration. Given all exogenous and endogenous factors, the modelled economy evolves over time and different scenarios can be analyzed. Thus, the model can serve as a virtual testbed for hypothesis verification and falsification.

Learning objectives:

The student

- understands the methods of Computational Economics and applies them on practical issues,
- evaluates agent models considering bounded rational behaviour and learning algorithms,
- analyses agent models based on mathematical basics,
- knows the benefits and disadvantages of the different models and how to use them,
- examines and argues the results of a simulation with adequate statistical methods,
- is able to support the chosen solutions with arguments and can explain them.

Literature

- R. Axelrod: "Advancing the art of simulation in social sciences". R. Conte u.a., Simulating Social Phenomena, Springer, S. 21-40, 1997.
- R. Axtel: "Why agents? On the varied motivations for agent computing in the social sciences". CSED Working Paper No. 17, The Brookings Institution, 2000.
- K. Judd: "Numerical Methods in Economics". MIT Press, 1998, Kapitel 6-7.
- A. M. Law and W. D. Kelton: "Simulation Modeling and Analysis", McGraw-Hill, 2000.
- R. Sargent: "Simulation model verification and validation". Winter Simulation Conference, 1991.
- L. Tesfation: "Notes on Learning", Technical Report, 2004.
- L. Tesfatsion: "Agent-based computational economics". ISU Technical Report, 2003.

Weiterführende Literatur:

- Amman, H., Kendrick, D., Rust, J.: "Handbook of Computational Economics". Volume 1, Elsevier North-Holland, 1996.
- Tesfatsion, L., Judd, K.L.: "Handbook of Computational Economics". Volume 2: Agent-Based Computational Economics, Elsevier North-Holland, 2006.
- Marimon, R., Scott, A.: "Computational Methods for the Study of Dynamic Economies". Oxford University Press, 1999.
- Gilbert, N., Troitzsch, K.: "Simulation for the Social Scientist". Open University Press, 1999.

V Computational Economics

2590458, WS 26/27, 2 SWS, Language: English, Open in study portal

Lecture (V)
Blended (On-Site/Online)

Content

Examining complex economic problems with classic analytical methods usually requires making numerous simplifying assumptions, for example that agents behave rationally or homogeneously. Recently, widespread availability of computing power gave rise to a new field in economic research that allows the modeling of heterogeneity and forms of bounded rationality: Computational Economics. Within this new discipline, computer based simulation models are used for analyzing complex economic systems. In short, an artificial world is created which captures all relevant aspects of the problem under consideration. Given all exogenous and endogenous factors, the modelled economy evolves over time and different scenarios can be analyzed. Thus, the model can serve as a virtual testbed for hypothesis verification and falsification.

Learning objectives:

The student

- understands the methods of Computational Economics and applies them on practical issues,
- evaluates agent models considering bounded rational behaviour and learning algorithms,
- analyses agent models based on mathematical basics,
- knows the benefits and disadvantages of the different models and how to use them,
- examines and argues the results of a simulation with adequate statistical methods,
- is able to support the chosen solutions with arguments and can explain them.

Literature

- R. Axelrod: "Advancing the art of simulation in social sciences". R. Conte u.a., Simulating Social Phenomena, Springer, S. 21-40, 1997.
- R. Axtel: "Why agents? On the varied motivations for agent computing in the social sciences". CSED Working Paper No. 17, The Brookings Institution, 2000.
- K. Judd: "Numerical Methods in Economics". MIT Press, 1998, Kapitel 6-7.
- A. M. Law and W. D. Kelton: "Simulation Modeling and Analysis", McGraw-Hill, 2000.
- R. Sargent: "Simulation model verification and validation". Winter Simulation Conference, 1991.
- L. Tesfation: "Notes on Learning", Technical Report, 2004.
- L. Tesfatsion: "Agent-based computational economics". ISU Technical Report, 2003.

Weiterführende Literatur:

- Amman, H., Kendrick, D., Rust, J.: "Handbook of Computational Economics". Volume 1, Elsevier North-Holland, 1996.
- Tesfatsion, L., Judd, K.L.: "Handbook of Computational Economics". Volume 2: Agent-Based Computational Economics, Elsevier North-Holland, 2006.
- Marimon, R., Scott, A.: "Computational Methods for the Study of Dynamic Economies". Oxford University Press, 1999.
- Gilbert, N., Troitzsch, K.: "Simulation for the Social Scientist". Open University Press, 1999.

T 6.121 Module component: Computational FinTech with Python and C++ [T-WIWI-106496]

Coordinators: Prof. Dr. Maxim Ulrich
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Examination of another type	1,5 CP	graded	Each summer term	1

Assessment
The grade is based on a larger or several smaller programming exercises.

Prerequisites
There are two conditions for taking this course:

1. This course is only open for registered students of the module “Disruptive FinTech Innovations”.
2. Registered students do also attend in the same semester the lecture “Engineering FinTech Solutions” and the seminar “Automated Financial Advisory”.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-WIWI-106193 - Engineering FinTech Solutions must have been started.

Workload
45 hours

T 6.122 Module component: Computational Modelling of Judgments, Decisions and Cognition [T-WIWI-114185]

Coordinators: Prof. Dr. Benjamin Scheibehenne
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration
M-WIWI-105714 - Consumer Research

Type	Credits	Grading	Term offered	Expansion	Version
Examination of another type	4,5 CP	graded	Each summer term	1 semesters	1

Courses					
ST 2026	2500119	Computational Modelling of Judgments, Decisions and Cognition	2 SWS	Others / 🗣	Gradwohl

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The alternative exam assessment is a single examination of another type that combines four assignments performed during the semester. All four assignments are integrated into one overall assessment; their individual contributions are weighted equally (each ≈ 25 % of the total), but only one final grade is awarded for the whole examination.

Recommendations
Basic programming skills (first experience with R or Python is advantageous), willingness to engage with maths and mathematical notation, calculus, and probability theory are required.

Additional Information
This course enables students to develop a basic understanding of computational models in the study of human judgment, decision making and cognition. After this course students can describe why we use computational models of cognition and behavior, they will be able to list prominent models and some of their criticisms and weaknesses. Moreover, they will be capable of programming and understanding simple and more advanced computational models in mathematical notation and code.After covering the main ideas of using computational models and preparing technical prerequisites, we will learn how to interpret models with tools like simulations and how to fit models to data.

All lectures, materials, and assignments are in English. The number of participants is limited. The registration will take place via the WiWi portal.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Computational Modelling of Judgments, Decisions and Cognition

Others (sonst.)
On-Site

2500119, SS 2026, 2 SWS, Language: English, Open in study portal

Content
This course enables students to develop a basic understanding of computational models in the study of human judgment, decision making and cognition. After this course students can describe why we use computational models of cognition and behavior, they will be able to list prominent models and some of their criticisms and weaknesses. Moreover, they will be capable of programming and understanding simple and more advanced computational models in mathematical notation and code.

After covering the main ideas of using computational models and preparing technical prerequisites, we will learn how to interpret models with tools like simulations and how to fit models to data.

Grades will be determined based on 4 programming assignments during the semester.

All lectures, materials, and assignments are in English. The number of participants is limited. The registration will take place via the WiWi portal.

Organizational issues
Participation is limited to 8 participants. Registration via the WiWi portal is required for the course. If too many students register, students in higher semesters are selected first.

T 6.123 Module component: Computational Risk and Asset Management [T-WIWI-102878]

Coordinators: Prof. Dr. Maxim Ulrich
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-105032 - Data Science for Finance

Type	Credits	Grading	Term offered	Version
Examination of another type	4,5 CP	graded	Irregular	5

Assessment
The module examination takes the form of an alternative exam assessment.
The alternative exam assessment consists of a Python-based "Takehome Exam". At the end of the third week of January, the student is given a "Takehome Exam" which he processes and sends back independently within 4 hours using Python. Precise instructions will be announced at the beginning of the course. The alternative exam assessment can be repeated a maximum of once. A timely repeat option takes place at the end of the third week in March of the same year. More detailed instructions will be given at the beginning of the course.

Prerequisites
None.

Recommendations
Basic knowledge of capital markt theory.

Workload
135 hours

T 6.124 Module component: Computer Aided Data Analysis [T-GEISTSOZ-104565]

Coordinators: Prof. Dr. Gerd Nollmann
Organisation: KIT Department of Humanities and Social Sciences
Part of: M-GEISTSOZ-101169 - Sociology
M-WIWI-104906 - Social Sciences

Type	Credits	Grading	Version
Coursework	0 CP	pass/fail	1

Courses					
WT 25/26	5000058	Decompositions and regression methods	2 SWS	Course / 🖥	Nollmann
WT 25/26	5000059	The gender wage gap	2 SWS	Course / 🖥	Nollmann
ST 2026	5011018	Computational Social Science: Topics and positions in the German Parliament (Part 2)	2 SWS	Seminar / 🧩	Banisch

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Below you will find excerpts from courses related to this module component:

V Computational Social Science: Topics and positions in the German Parliament (Part 2)

5011018, SS 2026, 2 SWS, Language: German, Open in study portal

Seminar (S)
Blended (On-Site/Online)

Content
The course consists of two parts (5011018 and 5011002) that are ideally taken in parallel.

Organizational issues
The course consists of two parts (5011018 and 5011002) that are ideally taken in parallel.

See Ilias-course: 5011002 – Computational Social Science: Themen und Positionen im Deutschen Bundestag (Teil 1)

T 6.125 Module component: Computer Contract Law [T-INFO-102036]

Coordinators: Dr. Michael Menk
Organisation: KIT Department of Informatics
Part of: M-INFO-101216 - Private Business Law
M-WIWI-104903 - Law

Type	Credits	Grading	Term offered	Version
Written examination	3 CP	graded	Each winter term	2

Courses					
WT 25/26	2411604	Computer Contract Law	2 SWS	Lecture / 🗣	Menk

Legend: 🖥 Online, ✲ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-INFO-101316 - Law of Contracts must not have been started.

Below you will find excerpts from courses related to this module component:

V Computer Contract Law
2411604, WS 25/26, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content
The course deals with contracts from the following areas:

- Contracts of programming, licencing and maintaining software
- Contracts in the field of IT employment law
- IT projects and IT Outsourcing
- Internet Contracts

From these areas single contracts will be chosen and discussed (e.g. software maintenance, employment contract with a software engineer). Concerning the respective contract the technical features, the economic background and the subsumption in the national law of obligation (BGB-Schuldrecht) will be discussed. As a result different contractual clauses will be developed by the students. Afterwards typical contracts and conditions will be analysed with regard to their legitimacy as standard business terms (AGB). It is the aim to show the effects of the german law of standard business terms (AGB-Recht) and to point out that contracts are a means of drafting business concepts and market appearance.

It is the aim of this course to provide students with knowledge in the area of contract formation and formulation in practice that builds upon the knowledge the students have already acquired concerning the legal protection of computer programs. Students shall understand how the legal rules depend upon, and interact with, the economic background and the technical features of the subject. The contract drafts shall be prepared by the students and will be corporately completed during the lecture. It is the aim of the course that students will be able to formulate contracts by themselves.

Literature

- Langenfeld, Gerrit Vertragsgestaltung Verlag C.H.Beck, III. Aufl. 2004
- Heussen, Benno Handbuch Vertragsverhandlung und Vertragsmanagement Verlag C.H.Beck, II. Aufl. 2002
- Schneider, Jochen Handbuch des EDV-Rechts Verlag Dr. Otto Schmidt KG, III. Aufl. 2002

Weiterführende Literatur

Ergänzende Literatur wird in den Vorlesungsfolien angegeben.

T 6.126 Module component: Computer Organization [T-INFO-103531]

Coordinators: Prof. Dr. Wolfgang Karl
Organisation: KIT Department of Informatics
Part of: M-WIWI-104909 - Informatics (Department of Informatics)

Type	Credits	Grading	Version
Written examination	6 CP	graded	1

Courses					
WT 25/26	2424502	Computer Organization	3 SWS	Lecture	Karl, Lehmann
WT 25/26	2424505	Übungen zu Rechnerorganisation	2 SWS	Practice	Lehmann

T 6.127 Module component: Constitution and Properties of Protective Coatings [T-MACH-105150]

Coordinators: Prof. Sven Ulrich
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101268 - Specific Topics in Materials Science
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each winter term	1

Courses					
WT 25/26	2177601	Constitution and Properties of Protective Coatings	2 SWS	Lecture / 🗣	Ulrich
WT 26/27	2177601	Constitution and Properties of Protective Coatings	2 SWS	Lecture / 🗣	Ulrich

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
oral examination (about 30 min)

no tools or reference materials

Prerequisites
none

Additional Information
The course is offered in German.

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Constitution and Properties of Protective Coatings

Lecture (V)
On-Site

2177601, WS 25/26, 2 SWS, Language: German, Open in study portal

Content
oral examination (about 30 min); no tools or reference materials

Teaching Content:

introduction and overview

concepts of surface modification

coating concepts

coating materials

methods of surface modification

coating methods

characterization methods

state of the art of industrial coating of tools and components

new developments of coating technology

regular attendance: 22 hours
self-study: 98 hours

Transfer of the basic knowledge of surface engineering, of the relations between constitution, properties and performance, of the manifold methods of modification, coating and characterization of surfaces.

Recommendations: none

Organizational issues
Falls die Vorlesung online stattfinden muss, bitte um Anmeldung unter sven.ulrich@kit.edu bis zum 28.10.25.

Den entsprechenden MS Teams Link erhalten Sie dann per E-Mail am 29.10.25.

Literature
Bach, F.-W.: Modern Surface Technology, Wiley-VCH, Weinheim, 2006

Abbildungen und Tabellen werden verteilt; Copies with figures and tables will be distributed

V	**Constitution and Properties of Protective Coatings** 2177601, WS 26/27, 2 SWS, Language: German, Open in study portal	**Lecture (V) On-Site**

Content
oral examination (about 30 min); no tools or reference materials

Teaching Content:

introduction and overview

concepts of surface modification

coating concepts

coating materials

methods of surface modification

coating methods

characterization methods

state of the art of industrial coating of tools and components

new developments of coating technology

regular attendance: 22 hours
self-study: 98 hours

Transfer of the basic knowledge of surface engineering, of the relations between constitution, properties and performance, of the manifold methods of modification, coating and characterization of surfaces.

Recommendations: none

Organizational issues

Achtung: Die Vorlesung beginnt erst am 12.11.2026!

Falls die Vorlesung online stattfinden muss, bitte um Anmeldung unter sven.ulrich@kit.edu bis zum 28.10.26.

Den entsprechenden MS Teams Link erhalten Sie dann per E-Mail am 29.10.26.

Literature

Bach, F.-W.: Modern Surface Technology, Wiley-VCH, Weinheim, 2006

Abbildungen und Tabellen werden verteilt; Copies with figures and tables will be distributed

T 6.128 Module component: Constitution and Properties of Wearresistant Materials [T-MACH-102141]

Coordinators: Prof. Sven Ulrich
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101268 - Specific Topics in Materials Science
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each summer term	3

Courses					
ST 2026	2194643	Constitution and Properties of Wear resistant materials	2 SWS	Lecture / 🗣	Ulrich

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
oral examination (about 30 min)

no tools or reference materials

Prerequisites
none

Additional Information
The course is offered in German.

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Constitution and Properties of Wear resistant materials

2194643, SS 2026, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content
The assessment consists of an oral exam (ca. 30 min) taking place at the agreed date (according to Section 4(2), 2 of the examination regulation). The re-examination is offered upon agreement.

Teaching Content:

introduction

materials and wear

unalloyed and alloyed tool steels

high speed steels

stellites and hard alloys

hard materials

hard metals

ceramic tool materials

superhard materials

new developments

regular attendance: 22 hours
self-study: 98 hours

Basic understanding of constitution of wear-resistant materials, of the relations between constitution, properties and performance, of principles of increasing of hardness and toughness of materials as well as of the characteristics of the various groups of wear-resistant materials.

Recommendations: none

Organizational issues
Die Blockveranstaltung findet in folgendem Zeitraum statt:

26.05./27.05.2026: jeweils von 8:00-19:00 Uhr sowie

28.05.2026: 8:00-13:00 Uhr

Ort: KIT-CN, Geb. 681, Raum 214

Anmeldung verbindlich bis zum 24.05.2026 unter sven.ulrich@kit.edu.

Nach der Anmeldung wird Ihnen im Falle einer Online-Veranstaltung der Link zur Vorlesung per E-Mail am 25.05.2026 mitgeteilt.

Literature
Laska, R. Felsch, C.: Werkstoffkunde für Ingenieure, Vieweg Verlag, Braunschweig, 1981

Schedler, W.: Hartmetall für den Praktiker, VDI-Verlage, Düsseldorf, 1988

Schneider, J.: Schneidkeramik, Verlag moderne Industrie, Landsberg am Lech, 1995

Kopien der Abbildungen und Tabellen werden verteilt; Copies with figures and tables will be distributed

T 6.129 Module component: Construction Equipment [T-BGU-101845]

Coordinators: Prof. Dr.-Ing. Sascha Gentes
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: M-BGU-101110 - Process Engineering in Construction
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Expansion	Version
Written examination	3 CP	graded	Each term	1 semesters	1

Courses					
WT 25/26	6243701	Construction Equipment	2 SWS	Lecture / 🗣	Dörfler, Schneider

Legend: 🖥 Online, ❂ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
written exam, 60 min.

Prerequisites
none

Recommendations
none

Additional Information
none

Workload
90 hours

T 6.130 Module component: Construction Technology [T-BGU-101691]

Coordinators: Prof. Dr.-Ing. Shervin Haghsheno
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Expansion	Version
Written examination	6 CP	graded	Each term	1 semesters	1

Courses					
ST 2026	6200410	Construction Technology	3 SWS	Lecture / 🗣	Gentes, Haghsheno, Schneider
ST 2026	6200411	Exercises to Construction Technology	1 SWS	Practice / 🗣	Gentes, Haghsheno, Schneider, Waleczko

Legend: 🖥 Online, ❄ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
written exam with 75 minutes

Prerequisites
None

Recommendations
None

Additional Information
None

Workload
180 hours

T 6.131 Module component: Consumer Psychology [T-WIWI-114292]

Coordinators: Prof. Dr. Benjamin Scheibehenne
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Examination of another type	4,5 CP	graded	Each summer term	1

Assessment
The assessment of success takes the form of a presentation (weighting 30%) as part of the exercise, evaluations of weekly assignments throughout the semester (30%) and a written examination (90 minutes, weighting 40%). The details of the assessment will be announced at the beginning of the course.

Prerequisites
None.

Additional Information
Please note: Due to a research sabbatical of Prof. Scheibehenne, this course will not be offered in the summer semester 2026. The course is expected to be available again in the summer semester 2027.

Important information

Due to the interactive nature of the class, it is important to regularly participate in the weekly meetings and to regularly submit written assignments in preparation of these meetings. The assignments and active participation is part of the final grade. The 'Übung' associated with this course is mandatory: Students will be asked to do presentations in groups (introduce and discuss academic papers assigned by the lecturer). This will take place over one day (as a blocked event) during the semester (When and where will be announced at the beginning of the semester). The presentation also counts towards the final grade. There will be no weekly or bi-weekly Übung besides this event.

Goal

The goal of the class is to gain a better understanding of the psychological factors that influence consumer behavior. This includes situational, biological, cognitive, social, and evolutionary factors. We will address these questions from an interdisciplinary perspective. While the main focus is on theories from psychology, we will also incorporate relevant research from Marketing, Cognitive Science, Biology, and Economics.

Description

Just like Physics is an important foundation for Engineering, Psychology is an important basis for Economics, Management, and Marketing. Hence, getting a better understanding of the 'human condition' provides deeper insights into economic theory. This is particularly true for consumer behavior. Consumer decisions are ubiquitous in daily life, and they can have long-ranging and important consequences for individual (financial) well-being and health but also for societies and the planet as a whole. We will explore how consumer behavior is shaped by social influences, situational and cognitive constraints, as well as by emotions, motivations, evolutionary forces, neuronal processes, and individual differences. Across all topics covered in class, we will engage with basic theoretical work as well as with groundbreaking empirical research and current scientific debates. The class will be taught in English.

Grading

Grading is based on three pillars: Active participation during the semester (homework assignments, in-class participation, 30%), your presentation in the Übung (30%) and a written exam at the end of the semester (40%). The grading of the homework will be based on two randomly selected weeks throughout the semester. The exam will cover the content of the class. The exam questions will be in English. You are allowed to bring a language dictionary into the exam, but you are not allowed to bring notes. Details about the content and the format of the exam will be announced in the lecture. All part-grades and the final grade will be communicated by the end of the semester.

Workload

The total workload for this course is approximately 135 hours:

Presence time: 30 hours

Preparation and wrap-up of the course: 45 hours

Exam and exam preparation: 60 hours

For further information, please contact the research group Cognition and Consumer Behavior (http://cub.iism.kit.edu/).

Workload
135 hours

T 6.132 Module component: Context Sensitive Systems [T-INFO-107499]

Coordinators: Prof. Dr.-Ing. Michael Beigl
Organisation: KIT Department of Informatics
Part of: M-WIWI-104909 - Informatics (Department of Informatics)

Type	Credits	Grading	Term offered	Version
Oral examination	5 CP	graded	Each summer term	1

Courses					
ST 2026	2400099	Context Sensitive Systems	1 SWS	Practice / 🖥	Riedel
ST 2026	24658	Context Sensitive Systems	2 SWS	Lecture / 🗣	Riedel

Legend: 🖥 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

T 6.133 Module component: Control of Linear Multivariable Systems [T-ETIT-100666]

Coordinators: Dr.-Ing. Mathias Kluwe
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-101157 - Control Engineering II

Type	Credits	Grading	Term offered	Version
Written examination	6 CP	graded	Each winter term	1

Courses					
WT 25/26	2303193	Multivariable Control Systems	3 SWS	Lecture / 🗣	Kluwe
WT 25/26	2303194	Tutorial to 2303194 "Multivariable Control Systems"	1 SWS	Practice / 🗣	Fehn
WT 26/27	2303193	Multivariable Control Systems	3 SWS	Lecture / 🗣	Kluwe
WT 26/27	2303194	Tutorial to 2303194 "Multivariable Control Systems"	1 SWS	Practice / 🗣	Fehn

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success is checked as part of a written overall test (120 minutes) of the course.

Prerequisites
none

Recommendations
For a deeper understanding, basic knowledge of system dynamics and control technology is absolutely necessary, as taught in the ETIT Bachelor module "System Dynamics and Control Technology" M-ETIT-102181.

T 6.134 Module component: Control of Mobile Machines [T-MACH-111821]

Coordinators: Simon Becker
Prof. Dr.-Ing. Marcus Geimer

Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101266 - Automotive Engineering
M-MACH-101267 - Mobile Machines
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each summer term	3

Courses					
ST 2026	2114080	Control of Mobile Machines	2 SWS	Lecture / 🗣	Becker, Geimer

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment consists of an oral exam (approx. 20 min) taking place in the recess period. The exam takes place in every semester. Re-examinations are offered at every ordinary examination date.

Prerequisites
A prerequisite for participation in the examination is the preparation of a semester report. T-MACH-111820 must be passed.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-MACH-111820 - Control of Mobile Machines – Prerequisites must have been passed.

Additional Information
The course is offered in German.

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Control of Mobile Machines

2114080, SS 2026, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content

- Basics of sensors, controls and control architectures in mobile machines
- Basics and functionalities of data communication in mobile machines (CAN-Bus, PROFIBUS, Ethernet, ...)
- Legal aspects and requirements (SIL-level, ...)
- Requirements for sensors for use in mobile machines
- Introduction to machine learning methods and their application for the control of mobile machines
- Overview of current research and developments in the field of agricultural robotics
- Implementation of a specific task within the exercise lessons
- The results of the semester task will be summarized in a short report as a pre-requisite for the exam.

Learning objectives

The students learn the theoretical basics of data communication as well as the architecture of control systems in mobile machines. Furthermore, they will be able to identify influences and general conditions during usage and derive practical and legal requirements for sensors and control systems. The students will learn methods of machine learning for control tasks in mobile machines as well as their architecture and the handling of training data. After participating in the exercise, they will be able to implement, train and validate a control system for a specific task.

Recommendations

Basic knowledge of electrical engineering and computer science is recommended. Initial programming knowledge, preferably in Python, is required. The number of participants is limited as hardware will be provided for the exercise. Prior registration is required, details will be announced on the web pages of the Institute of Vehicle Systems Engineering / Department of Mobile Machinery. In case of high registration numbers exceeding the capacities, a selection among all interested persons will take place according to qualification.

regular attendance: 21 hours
total self-study: 92 hours

Literature

Etschberger, K.: Controller Area Network, Grundlagen, Protokolle, Bausteine, Anwendungen; München, Wien: Carl Hanser Verlag, 2002.
Engels, H.: CAN-Bus - CAN-Bus-Technik einfach, anschaulich und praxisnah dargestellt; Poing: Franzis Verlag, 2002.

CAN-Bus-Technik einfach, anschaulich und praxisnah dargestellt; Poing: Franzis Verlag, 2002.

6.135 Module component: Control of Mobile Machines – Prerequisites [T-MACH-111820]

Coordinators: Simon Becker
Prof. Dr.-Ing. Marcus Geimer

Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101266 - Automotive Engineering
M-MACH-101267 - Mobile Machines
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Coursework	0 CP	pass/fail	Each summer term	1

Courses					
ST 2026	2114081	Control of Mobile Machines	2 SWS	Practice / 🗣	Becker, Geimer

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Preparation of a report on the completion of the semester task

Prerequisites
none

Additional Information
The course is offered in German.

T 6.136 Module component: Control Technology [T-MACH-105185]

Coordinators: Dr.-Ing. Philipp Gönnheimer
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-106590 - Production Engineering
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Written examination	4 CP	graded	Each summer term	2

Courses					
ST 2026	2150683	Control Technology	2 SWS	Lecture / 🗣	Gönnheimer

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Written Exam (60 min)

Prerequisites
none

Additional Information
The course is offered in German.

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Control Technology

2150683, SS 2026, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content
The lecture control technology gives an integral overview of available control components within the field of industrial production systems.
The first part of the lecture deals with the fundamentals of signal processing and with control peripherals in the form of sensors and actors which are used in production systems for the detection and manipulation of process states.
The second part handles with the function of electric control systems in the production environment. The main focus in this chapter is laid on programmable logic controls, computerized numerical controls and robot controls. Finally the course ends with the topic of cross-linking and decentralization with the help of bus systems.
The lecture is very practice-oriented and illustrated with numerous examples from different branches.

The following topics will be covered:

- Signal processing
- Control peripherals
- Programmable logic controls
- Numerical controls
- Controls for industrial robots
- Distributed control systems
- Field bus
- Trends in the area of control technology

Learning Outcomes:

The students …

- are able to name the electrical controls which occur in the industrial environment and explain their function.
- can explain fundamental methods of signal processing. This involves in particular several coding methods, error protection methods and analog to digital conversion.
- are able to choose and to dimension control components, including sensors and actors, for an industrial application, particularly in the field of plant engineering and machine tools. Thereby, they can consider both, technical and economical issues.
- can describe the approach for projecting and writing software programs for a programmable logic control named Simatic S7 from Siemens. Thereby they can name several programming languages of the IEC 1131.

Workload:

regular attendance: 21 hours
self-study: 99 hours

Organizational issues
Zur Vertiefung des im Rahmen der Lehrveranstaltung erworbenen Wissens werden die theoretischen Vorlesungseinheiten durch Praxiseinheiten im Umfeld der Karlsruher Forschungsfabrik (https://www.karlsruher-forschungsfabrik.de) unterstützt.

The theoretical lectures are complemented by practical lectures in the Karlsruhe Research Factory (https://www.karlsruher-forschungsfabrik.de/en.html) to deepen the acquired knowledge.

Literature
Medien:
Skript zur Veranstaltung wird über ilias (https://ilias.studium.kit.edu/) bereitgestellt.

Media:
Lecture notes will be provided in ilias (https://ilias.studium.kit.edu/).

T 6.137 Module component: Convex Analysis [T-WIWI-102856]

Coordinators: Prof. Dr. Oliver Stein
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101473 - Mathematical Programming
M-WIWI-104899 - Operations Research

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Irregular	1

Assessment
The assessment of the lecture is a written examination (60 minutes) according to §4(2), 1 of the examination regulation. The successful completion of the exercises is required for admission to the written exam.

The examination is held in the semester of the lecture and in the following semester.

Prerequisites
None

Recommendations
It is strongly recommended to visit at least one lecture from the Bachelor program of this chair before attending this course.

Additional Information
The lecture is offered irregularly. The curriculum of the next three years is available online (www.ior.kit.edu).

T 6.138 Module component: Cooperative Autonomous Vehicles [T-WIWI-112690]

Coordinators: Prof. Dr. Alexey Vinel
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101472 - Informatics
M-WIWI-101628 - Emphasis in Informatics
M-WIWI-104901 - Informatics (KIT-Department of Economics and Management)
M-WIWI-106631 - Cooperative Autonomous Vehicles

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each summer term	1

Courses					
ST 2026	2511450	Cooperative Autonomous Vehicles	2 SWS	Lecture /	Vinel
ST 2026	2511451	Exercise Cooperative Autonomous Vehicles	1 SWS	Practice /	Vinel

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
The default assessment of this course is a written examination (60 min).

The exam takes place every semester and can be repeated at every regular examination date.

Prerequisites
None.

Workload
135 hours

T 6.139 Module component: Copyright [T-INFO-101308]

Coordinators: N.N.
Organisation: KIT Department of Informatics
Part of: M-INFO-101215 - Intellectual Property Law
M-WIWI-104903 - Law

Type	Credits	Grading	Term offered	Version
Written examination	3 CP	graded	Each winter term	1

Courses					
WT 25/26	24121	Copyright	2 SWS	Lecture / 🗣	Hartmann
WT 26/27	24121	Copyright	2 SWS	Lecture / 🗣	Sattler

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment is carried out as a written examination (§ 4 Abs. 2 No. 1 SPO) lasting 60 minutes.

Prerequisites
None.

Recommendations
None.

T 6.140 Module component: Corporate Compliance [T-INFO-101288]

Coordinators: Andreas Herzig
Organisation: KIT Department of Informatics
Part of: M-INFO-101216 - Private Business Law
M-WIWI-104903 - Law

Type	Credits	Grading	Term offered	Version
Written examination	3 CP	graded	Each winter term	1

Courses					
WT 25/26	2400087	Corporate Compliance	2 SWS	Lecture / 🧩	Herzig, Siddiq

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

T 6.141 Module component: Corporate Risk Management [T-WIWI-109050]

Coordinators: Prof. Dr. Martin Ruckes
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101480 - Finance 3
M-WIWI-101483 - Finance 2
M-WIWI-101502 - Economic Theory and its Application in Finance
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Irregular	2

Assessment
The success of the course (lecture and exercise) is assessed in the form of a written examination (60 min.) in accordance with § 4 para. 2 no. 1 SPO. If the number of participants registered for the written examination is low, we reserve the right to hold an oral examination instead of a written examination.
Please note that the examination is only offered in the semester of the lecture and in the following semester.

Prerequisites
None

Recommendations
None

Workload
135 hours

T 6.142 Module component: Current Directions in Consumer Psychology [T-WIWI-111100]

Coordinators: Prof. Dr. Benjamin Scheibehenne
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration
M-WIWI-105714 - Consumer Research

Type	Credits	Grading	Term offered	Expansion	Version
Examination of another type	4,5 CP	graded	Each term	1 semesters	2

Courses					
WT 25/26	2540441	Current Directions in Consumer Psychology	2 SWS	Seminar / 🗣	Scheibehenne

Legend: 🖥 Online, Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Alternative exam assessment. Grading will be based on a continuous basis throughout the semester.

Prerequisites
Strong interest in research. Students who wish to write a master's thesis at our department will be given priority in the allocation of places.

Additional Information
Please note: Due to a research sabbatical of Prof. Scheibehenne, this course will not be offered in the summer semester 2026. The course is expected to be available again in the summer semester 2027.

This class covers current research topics at the intersection between Psychology, Consumer Behavior, and Behavioral Economics. Based on weekly reading assignments of current scientific journal publications, students will get a first-hand experience of the ongoing topics and discussions at this exciting and dynamic area of research. The reading list will be announced at the first day of class and will be updated throughout the semester. Grades will be based on weekly participation throughout the semester including short oral presentation of papers in class, active engagement in discussions, and homework assignments. Due to the highly interactive format of this class the number of participants is limited.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Current Directions in Consumer Psychology

2540441, WS 25/26, 2 SWS, Language: English, Open in study portal

Seminar (S)
On-Site

Content
NOTE: sign-up required via the Campus+ Portal

This class covers current research topics at the intersection between Psychology, Consumer Behavior, and Behavioral Economics. Based on weekly reading assignments of current scientific journal publications, students will get a first-hand experience of the ongoing topics and discussions at this exciting and dynamic area of research. The reading list will be announced at the first day of class. Grades will be based on continuous participation throughout the semester including short oral presentation of papers in class, active engagement in discussions and homework assignments. This class will be taught in English.

Organizational issues
Participation is restricted to 6 participants.

If more than 6 students apply, we will assign places by motivation.

Achtung: Seminar findet im Seminarraum Geb 01.87, R 5B 25 statt.

Beginn Mittwoch 29.10.25 09:45 - 11:15 Uhr, dann immer mittwochs bis 18.2.2 gleiche Uhrzeit

T 6.143 Module component: Current Research Topics in Business Information Systems [T-WIWI-109819]

Coordinators: Prof. Dr. Alexander Mädche
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Examination of another type	1 CP	graded	see Annotations	1

Assessment
The type of success control has not yet been determined, but will be announced in good time before the start of the course.

Prerequisites
None

Recommendations
None

Additional Information
New course starting winter term 2019/2020.

Workload
135 hours

T 6.144 Module component: Current Topics on BioMEMS [T-MACH-102176]

Coordinators: Prof. Dr. Andreas Guber
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101290 - BioMEMS
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Examination of another type	4 CP	graded	Each term	2

Courses					
WT 25/26	2143873	Actual topics of BioMEMS	2 SWS	Seminar /	Guber, Ahrens
ST 2026	2143873	Actual topics of BioMEMS	2 SWS	Seminar /	Guber, Ahrens

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
active participation and own presentation (30 Min.)

Prerequisites
none

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Actual topics of BioMEMS
2143873, WS 25/26, 2 SWS, Language: German, Open in study portal

Seminar (S)
Blended (On-Site/Online)

Organizational issues
Aktuell werden im Rahmen dieses Seminars nur Vorträge zu Abschlussarbeiten gehalten. Neue Themen nur auf Anfrage.

V Actual topics of BioMEMS
2143873, SS 2026, 2 SWS, Language: German, Open in study portal

Seminar (S)
Blended (On-Site/Online)

Content

- Short introduction to the basics of BioMEMS
- Selected aspects of biomedical engineering and life sciences
- Possible micro technical manufacturing processes
- Selected application examples from research and industry

The seminar includes (bio)medical engineering as well as biological and biotechnological topics in the context of engineering sciences

- Use of microtechnical components and systems in innovative medical products
- Use of microfluidic chip systems in applied biology and biotechnology

Organizational issues
Aktuell werden im Rahmen dieses Seminars nur Vorträge zu Abschlussarbeiten gehalten. Neue Themen nur auf Anfrage.

T 6.145 Module component: Customer Relationship Management [T-WIWI-102595]

Coordinators: Prof. Dr. Andreas Geyer-Schulz
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each winter term	1

Assessment
Written examination (60 minutes) according to §4(2), 1 SPO. The exam is considered passed if at least 50 out of a maximum of 100 possible points are achieved. The grades are graded in five steps (best grade 1.0 from 95 points). Details of the grade formation and scale will be announced in the course.
A bonus can be acquired through successful participation in the practice. If the grade of the written examination is between 4.0 and 1.3, the bonus improves the grade by one grade level (0.3 or 0.4). The exact criteria for awarding a bonus will be announced at the beginning of the course.

Prerequisites
None

T 6.146 Module component: Cyber-Physical Modeling [T-ETIT-113908]

Coordinators: Prof. Dr.-Ing. Mike Barth
Prof. Dr.-Ing. Sören Hohmann
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-106953 - Cyber-Physical Modeling

Type	Credits	Grading	Version
Written examination	6 CP	graded	1

Courses					
ST 2026	2303310	Cyber Physical Modeling	3 SWS	Lecture / 🗣	Hohmann, Barth
ST 2026	2303311	Tutorial to 2303310 Cyber Physical Modeling	1 SWS	Practice / 🗣	Hohmann, Barth, Thömmes, Schicketanz

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The examination takes place in the form of a written examination lasting 90 min.
The module grade is the grade of the written exam.

Prerequisites
none

T 6.147 Module component: Data and Artificial Intelligence for Numerical Simulations [T-MACH-113926]

Coordinators: Dr. Anastasia August
Dr.-Ing. Arnd Hendrik Koeppe
Organisation: KIT Department of Mechanical Engineering
Part of: M-WIWI-107718 - Mechanical Engineering I
M-WIWI-107730 - Mechanical Engineering II

Type	Credits	Grading	Term offered	Version
Examination of another type	4,5 CP	graded	Each summer term	2

Courses					
ST 2026	2182222	Data and Artificial Intelligence for Numerical Simulations	4 SWS	Lecture / Practice /	Koeppe, August

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
Written report and oral examination/presentation (about 20 minutes)

Prerequisites
none

Additional Information
The course is offered in English.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Data and Artificial Intelligence for Numerical Simulations

2182222, SS 2026, 4 SWS, Language: English, Open in study portal

Lecture / Practice (VÜ)
Blended (On-Site/Online)

Content
Overview

Numerical simulations are essential for solving complex physical problems, and integrating data-driven methods with AI enables greater accuracy and efficiency. This course provides students the skills to apply AI techniques to scientific data and simulations, combining theoretical foundations with practical applications.

Learning Outcome

Students will master advanced data-driven modeling techniques and independently apply AI methods to real-world scientific problems through structured projects and challenges.

Learning Goals

- Students understand advanced methods of data-driven modeling.
- Students can apply AI techniques to scientific data and problems.
- Students work independently on self-organized machine-learning projects.

Teaching Methods

- Lectures: In presence and via videos/flipped classroom sessions.
- Individual Team Projects: Self-organized group challenges focusing on real-world applications.
- Consultation Hours: Regular mentoring and Q&A sessions.

Course Structure

The course is divided into two main parts, each culminating in a group project. A preliminary tutorial ensures all students have foundational skills in Python and machine learning.

Preliminary Tutorial: Python and Machine Learning Essentials

Objective: Establish a baseline in Python programming and machine learning.

Content: Data handling, basic ML workflows, and tools like scikit-learn and TensorFlow.

Part 1: Generating Structured Simulation Data

Objective: Learn systematic approaches to creating and managing simulation data for physical problems.

Key Topics:

- Data generation strategies for boundary and initial value problems.
- Active learning for efficient data collection.
- Structured data management and automated workflows.

Group Project/Challenge 1: Teams create and document structured datasets for a physical simulation problem.

Part 2: Deep Learning for Field- and Time-Dependent Data

Objective: Apply deep learning to dynamic and spatial data using data-driven and physics-informed approaches.

Key Topics:

- Deep learning models (e.g., CNNs, RNNs, GANs) for simulations.
- Hyperparameter tuning and optimization.
- Combining neural networks with physical laws for interpretability.

Group Project/Challenge 2: Teams develop a deep learning model to predict field- or time-dependent behavior in a physical system.

Examination

The course concludes with an oral examination and written reports based on group projects.

The course requires a *good foundation in (continuum) mechanics* or *numerical simulations*. **One of the following courses (or equivalent) must be completed successfully**:

- Angewandte Strömungsmechanik [T-MACH-114026]
  Computational Mechanics of Materials [T-MACH-113939]
- Experimentelle und numerische Strömungsmechanik [T-MACH-114025]
- Fluid Mechanics of Turbulent Flows [T-BGU-110841]
- Forschungsseminar Numerische Strömungsmechanik [T-MACH-114024]
- Grundlagen der nichtlinearen Kontinuumsmechanik [T-MACH-105324]
- Grundlagen der Phasenfeldmodellierung [T-MACH-114627]
- Mathematical Methods in Fluid Mechanics [T-MACH-113956]
- Mathematische Methoden der Strömungslehre [T-MACH-113955]
- Mathematische Methoden der Mikromechanik [T-MACH-111537]
- Microscale Fluid Mechanics [T-MACH-113144]
- Nonlinear Continuum Mechanics [T-MACH-111026]
- Numerische Strömungsmechanik [T-MACH-105338]
- Numerische Strömungsmechanik mit Forschungsseminar [T-MACH-114022]
- Numerische Strömungsmechanik mit PYTHON [T-MACH-110838]
- Phasenfeldmethode in der Thermomechanik [T-MACH-113694]
- Research Seminar in Continuum Mechanics [T-MACH-113992]

- Numerische Modellierung von Mehrphasenströmungen [T-MACH-105420]
- Numerische Simulation reagierender Zweiphasenströmungen [T-MACH-105339]
- Computational Elasticity [T-MACH-113989]
- Computational Inelasticity [T-MACH-113990]

Organizational issues

Due to high interest, please register within the first week of the semester through Campus+. There are 3 parallel sessions per week, each with a different group of students. Students are asked to only attend one session and not to switch between sessions.

6.148 Module component: Data Protection by Design [T-INFO-108405]

Coordinators: Prof. Dr. Oliver Raabe
Organisation: KIT Department of Informatics
Part of: M-WIWI-104903 - Law
M-WIWI-104909 - Informatics (Department of Informatics)

Type	Credits	Grading	Term offered	Version
Written examination	3 CP	graded	Irregular	2

T 6.149 Module component: Data Protection Law [T-INFO-101303]

Coordinators: Dr. Johannes Eichenhofer
Organisation: KIT Department of Informatics
Part of: M-WIWI-104903 - Law

Type	Credits	Grading	Term offered	Version
Written examination	3 CP	graded	Each winter term	1

T 6.150 Module component: Data Science and Artificial Intelligence for Energy Systems [T-INFO-113402]

Coordinators: TT-Prof. Dr. Benjamin Schäfer
Organisation: KIT Department of Informatics
Part of: M-WIWI-104909 - Informatics (Department of Informatics)

Type	Credits	Grading	Term offered	Version
Oral examination	6 CP	graded	Each summer term	1

Courses					
WT 25/26	2400079	Data Science and Artificial Intelligence for Energy Systems	2 SWS	Lecture / 🗣	Schäfer
WT 25/26	2400082	Data Science and Artificial Intelligence for Energy Systems	2 SWS	Practice / 🗣	Schäfer, Nikoltchovska, Sadric
ST 2026	2400098	Data science and Artificial Intelligence for Energy Systems	2 SWS	Lecture / 🗣	Schäfer
ST 2026	2400173	Data science and Artificial Intelligence for Energy Systems (Excercise)	2 SWS	Practice / 🗣	Schäfer

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment is carried out as an oral examination (§ 4 Abs. 2 Nr. 2 SPO) lasting about 30 minutes.

Prerequisites
None.

Recommendations
Knowledge of AI basics is very helpful.
Previous participation in “Energieinformatik 1” and/or “Energieinformatik 2” is beneficiary but not mandatory.
Knowledge of Python is highly recommended.

Below you will find excerpts from courses related to this module component:

V Data Science and Artificial Intelligence for Energy Systems
Lecture (V)
On-Site
2400079, WS 25/26, 2 SWS, Language: English, Open in study portal

Content
Artificial Intelligence (AI) is a key technology in many areas of society and research. Energy systems with the ongoing energy transition (“Energiewende”) make it a fascinating field for deploying AI methods. AI and machine learning algorithms can play a crucial role in improving energy efficiency, optimizing power generation and distribution or enhancing system stability while facilitating additional renewable energy integration. In this lecture, we review some mechanics of energy systems, their design and optimization questions and how to solve these using data-driven approaches. We will discuss deterministic dynamics, as well as stochastic aspects of energy systems and will explore fundamental AI algorithms and their applications in energy systems. We will cover both classical time series methods as well as state-of-the-art AI techniques, e.g. for optimization or forecasting.

V Data science and Artificial Intelligence for Energy Systems
Lecture (V)
On-Site
2400098, SS 2026, 2 SWS, Language: English, Open in study portal

Content
Artificial Intelligence (AI) is a key technology in many areas of society and research. Energy systems with the ongoing energy transition (“Energiewende”) make it a fascinating field for deploying AI methods. AI and machine learning algorithms can play a crucial role in improving energy efficiency, optimizing power generation and distribution or enhancing system stability while facilitating additional renewable energy integration. In this lecture, we review some mechanics of energy systems, their design and optimization questions and how to solve these using data-driven approaches. We will discuss deterministic dynamics, as well as stochastic aspects of energy systems and will explore fundamental AI algorithms and their applications in energy systems. We will cover both classical time series methods as well as state-of-the-art AI techniques, e.g. for optimization or forecasting.

T 6.151 Module component: Data Science for Business [T-WIWI-114089]

Coordinators: Prof. Dr. Jella Pfeiffer
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-103117 - Data Science: Data-Driven Information Systems
M-WIWI-103118 - Data Science: Data-Driven User Modeling
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Examination of another type	4,5 CP	graded	Each summer term	1

Courses					
WT 25/26	2540473	Business Data Analytics	2 SWS	Seminar / 🗣	Hariharan, Grote, Schulz, Motz
ST 2026	2540466	Data Science for Business	2 SWS	Lecture / 🗣	Pfeiffer
ST 2026	2540467	Exercise Data Science for Business	1 SWS	Practice / 🗣	Gutschow, Heßler

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success is monitored through ongoing elaborations and presentations of tasks and a written exam (60 minutes) at the end of the lecture period. Successful participation in the exercises is a prerequisite for admission to the written examination. The scoring scheme for the overall evaluation will be announced at the beginning of the course.

The number of participants is limited to 50, as this is the only way to ensure that the case study is supervised conscientiously.

Prerequisites
None

Recommendations
Knowledge of programming (particular python) and statistics is helpful.

Additional Information
The lecture and exercise are blocked. Three units take place consecutively in the first few weeks. The weeks after that can be used to work on the case (team project).

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Business Data Analytics
2540473, WS 25/26, 2 SWS, Language: German/English, Open in study portal

Seminar (S)
On-Site

Content
wird auf deutsch und englisch gehalten

Organizational issues
Blockveranstaltung, siehe WWW

V Data Science for Business
2540466, SS 2026, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content
In the course "Data Science for Business":

- You will learn about essential Data Science methods, including clustering and classification techniques (e.g., random forests, SVMs).
- You will understand process models such as CRISP-DM.
- You will explore different types of data, including eye-tracking data, click data, neurophysiological data, sales data, and other business-related data.
- You will gain skills in data visualization and evaluation using programming languages and software tools.

Workload:

- Attendance time Lecture (2 SWS × 15 weeks): 30 h
- Attendance time Tutorial (1 SWS × 15 weeks): 15 h
- Self-study Lecture: 20 h
- Self-study Tutorial: 50 h
- Exam preparation: 20 h
- Total: 135 h

Organizational issues

- ehemals Business Data Analytics: Applications and Tools
- formerly Business Data Analytics: Applications and Tools

V **Exercise Data Science for Business**
2540467, SS 2026, 1 SWS, Language: English, Open in study portal

Practice (Ü)
On-Site

Content
In the course "Data Science for Business":

- You will learn about essential Data Science methods, including clustering and classification techniques (e.g., random forests, SVMs).
- You will understand process models such as CRISP-DM.
- You will explore different types of data, including eye-tracking data, click data, neurophysiological data, sales data, and other business-related data.
- You will gain skills in data visualization and evaluation using programming languages and software tools.

Organizational issues

- ehemals Business Data Analytics: Applications and Tools
- formerly Business Data Analytics: Applications and Tools

T 6.152 Module component: Data Science Fundamentals in Engineering [T-MACH-114078]

Coordinators: Prof. Dr.-Ing. Anne Meyer
Organisation: KIT Department of Mechanical Engineering
Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Expansion	Version
Examination of another type	4 CP	graded	Each winter term	1 semesters	2

Courses					
WT 25/26	2121360	Data Science Fundamentals in Engineering	3 SWS	Lecture / Practice / 🗣	Meyer

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Graded examination of another type; for further information, see ILIAS.

Additional Information
The course is offered in English.

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Data Science Fundamentals in Engineering

2121360, WS 25/26, 3 SWS, Language: English, Open in study portal

Lecture / Practice (VÜ)
On-Site

Content

Data science and data-driven methods are an essential component of modern engineering and affect all stages of the product lifecycle, from data-supported product design to production, quality assurance, and intelligent or autonomous technical systems. This module introduces fundamental concepts, methods, and workflows of data science and machine learning with a strong focus on engineering applications.

Students gain an understanding of engineering-specific data characteristics and learn how to design and implement basic data analysis and machine learning pipelines. A central component of the course is practical group work on two data science and machine learning challenges using engineering data sets. Students apply state-of-the-art Python libraries to explore data, build models, and evaluate their performance.

The topics of this course include:

- Engineering data sources, data types, data quality, and data representation
- Data science workflows, exploratory data analysis, and visualization
- Statistical modeling and machine learning methods (regression, classification, clustering)
- Model training, evaluation, validation, and generalization
- Overfitting, model selection, and hyperparameter optimization
- Machine learning pipelines and practical implementation aspects
- Fundamentals of deep learning and neural networks for engineering applications

This module provides foundational competencies for subsequent courses in data-driven engineering, machine learning, and artificial intelligence.

Learning Objectives

After successful completion of the module, students are able to:

- **Explain** the role of data science, machine learning, and deep learning in engineering and their complementarity to simulation- and optimization-based approaches.
- **Analyze** engineering data sets with respect to data quality, structure, and suitability for data-driven, learning-based methods.
- **Apply** exploratory data analysis, statistical modeling, machine learning, and basic deep learning methods to engineering problems.
- **Design and implement** data analysis, machine learning, and deep learning pipelines using state-of-the-art Python libraries.
- **Evaluate** the performance, generalization capability, and limitations of machine learning and deep learning models using appropriate validation techniques.

Organizational Information

See further details on course organization at: lehre.imi.kit.edu and in ILIAS

Workload

120 hours

Organizational issues

The exercises take place on Wednesdays on specific days from 9:45 a.m. to 1:00 p.m. For further information, see ILIAS.

T 6.153 Module component: Data-Based Modeling and Control [T-CIWVT-112827]

Coordinators: Prof. Dr.-Ing. Thomas Meurer
Organisation: KIT Department of Chemical and Process Engineering
Part of: M-CIWVT-106319 - Data-Based Modeling and Control

Type	Credits	Grading	Version
Oral examination	6 CP	graded	1

Courses					
WT 25/26	2243070	Data-Based Modeling and Control	3 SWS	Lecture / Practice / 🗣	Meurer
WT 26/27	2243070	Data-Based Modeling and Control	3 SWS	Lecture / Practice / 🗣	Meurer

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

T 6.154 Module component: Data-Driven Algorithms in Vehicle Technology [T-MACH-112126]

Coordinators: Dr. Stefan Scheubner
Organisation: KIT Department of Mechanical Engineering
Part of: M-MACH-101265 - Vehicle Development
M-MACH-101266 - Automotive Engineering
M-WIWI-104907 - Engineering Sciences
M-WIWI-107718 - Mechanical Engineering I
M-WIWI-107730 - Mechanical Engineering II

Type	Credits	Grading	Term offered	Expansion	Version
Written examination	4,5 CP	graded	Each winter term	1 semesters	2

Courses					
WT 25/26	2113840	Data-Driven Algorithms in Vehicle Technology	2 SWS	Lecture / Blended	Scheubner
WT 26/27	2113840	Data-Driven Algorithms in Vehicle Technology	2 SWS	Lecture / Blended	Scheubner

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
Written Examination

Duration: 90 minutes

Additional Information
The course is offered in English.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Data-Driven Algorithms in Vehicle Technology

2113840, WS 25/26, 2 SWS, Language: English, Open in study portal

Lecture (V)
Blended (On-Site/Online)

Content
Course Syllabus: Data-Driven Algorithms in Vehicle Technology
Motivation for the Course: Nowadays, engineers often develop technical systems using a combination of hard- and software. This is true especially for modern passenger vehicle development. In a digitalized world, such developments are built on knowledge gained from relevant data sources, e.g. the vehicle sensors. Therefore, engineers in automobile technology need qualifications from data science to successfully create new functionalities in the cars. To prevent remaining purely theoretical, the algorithms in this course are explained using a real-world problem of “EV Routing”. Students have the opportunity to test methods in Python with frequent exercises presented.
Goal of the Course: Students have a basic understanding of data-driven algorithms such as Markov Models, Machine Learning or Monte-Carlo Methods. The approach for building data-driven models in automobile technology are known to students and they are able to test algorithms in the programming language “Python”. Furthermore, students have learnt how to analyse the algorithm performance.
Content:
1. Introduction to function development as well as the prerequisites for the course (e.g. Fundamentals for running Python code)
2. Fundamentals for EV Routing and relevant data sources
3. Parameter estimation and state classification algorithms to determine the current situation of the vehicle
4. Learning methods for driver behaviour
5. Forecast algorithms to predict future energy consumption of an electric vehicle

Organizational issues

Das Vorlesungsmaterial wird auf ILIAS bereitgestellt. Das ILIAS-Passwort erhalten Sie unter https://fast-web-01.fast.kit.edu/PasswoerterIlias/

Die erste VL am 12.11.2025 um 14:00 Uhr findet in Präsenz am Campus Ost, Geb. 70.04, Raum 220 statt.

Alle weiteren Vorlesungsinhalte werden als Videoaufzeichnungen in ILIAS bereit gestellt. In regelmäßigen Abständen wird es Sprechstunden geben. Die genauen Termine erfahren Sie dann über den entsprechenden ILIAS Kurs

V	**Data-Driven Algorithms in Vehicle Technology** 2113840, WS 26/27, 2 SWS, Language: English, Open in study portal	**Lecture (V)** **Blended (On-Site/Online)**

Content

Course Syllabus: Data-Driven Algorithms in Vehicle Technology
Motivation for the Course: Nowadays, engineers often develop technical systems using a combination of hard- and software. This is true especially for modern passenger vehicle development. In a digitalized world, such developments are built on knowledge gained from relevant data sources, e.g. the vehicle sensors. Therefore, engineers in automobile technology need qualifications from data science to successfully create new functionalities in the cars. To prevent remaining purely theoretical, the algorithms in this course are explained using a real-world problem of “EV Routing”. Students have the opportunity to test methods in Python with frequent exercises presented.
Goal of the Course: Students have a basic understanding of data-driven algorithms such as Markov Models, Machine Learning or Monte-Carlo Methods. The approach for building data-driven models in automobile technology are known to students and they are able to test algorithms in the programming language “Python”. Furthermore, students have learnt how to analyse the algorithm performance.
Content:

1. Introduction to function development as well as the prerequisites for the course (e.g. Fundamentals for running Python code)
2. Fundamentals for EV Routing and relevant data sources
3. Parameter estimation and state classification algorithms to determine the current situation of the vehicle
4. Learning methods for driver behaviour
5. Forecast algorithms to predict future energy consumption of an electric vehicle

Organizational issues

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Die erste VL am 12.11.2025 um 14:00 Uhr findet in Präsenz am Campus Ost, Geb. 70.04, Raum 220 statt.

Alle weiteren Vorlesungsinhalte werden als Videoaufzeichnungen in ILIAS bereit gestellt. In regelmäßigen Abständen wird es Sprechstunden geben. Die genauen Termine erfahren Sie dann über den entsprechenden ILIAS Kurs

T 6.155 Module component: Derivatives [T-WIWI-102643]

Coordinators: Prof. Dr. Marliese Uhrig-Homburg
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101480 - Finance 3
M-WIWI-101482 - Finance 1
M-WIWI-101483 - Finance 2
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each summer term	1

Courses					
ST 2026	2530550	Derivatives	2 SWS	Lecture / 🗣	Uhrig-Homburg
ST 2026	2530551	Übung zu Derivate	1 SWS	Practice / 🗣	Dinger, Uhrig-Homburg

Legend: 🖥 Online, Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success is assessed in the form of a written examination (60 minutes). The examination is offered every semester and can be repeated at any regular examination date.

A bonus can be earned for successful participation in the exercises by submitting correct solutions to at least 50% of the bonus exercises set. If the grade of the written examination is between 4.0 and 1.3, the bonus improves the grade by up to one grade level (0.3 or 0.4). Details will be announced in the lecture.

Prerequisites
None

Recommendations
None

Below you will find excerpts from courses related to this module component:

V Derivatives
2530550, SS 2026, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Literature

- Hull (2012): Options, Futures, & Other Derivatives, Prentice Hall, 8th Edition

Weiterführende Literatur:

Cox/Rubinstein (1985): Option Markets, Prentice Hall

T 6.156 Module component: Design and Development of Mobile Machines [T-MACH-105311]

Coordinators: Prof. Dr.-Ing. Marcus Geimer
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101267 - Mobile Machines

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each winter term	1

Courses					
WT 25/26	2113079	Design and Development of Mobile Machines	2 SWS	Lecture / 🗣	Geimer

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, x Cancelled

Assessment
The assessment consists of an oral exam (20 min) taking place in the recess period. The exam takes place in every semester. Re-examinations are offered at every ordinary examination date.

A registration is mandatory, the details will be announced on the webpages of the *Institute of Vehicle System Technology / Institute of Mobile Machines.* In case of too many applications, attendance will be granted based on pre-qualification.

The course will be replenished by interestung lectures of professionals from leading hydraulic companies.

Prerequisites
Required for the participation in the examination is the preparation of a report during the semester. T-MACH-108887 must have been passed.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-MACH-108887 - Design and Development of Mobile Machines - Advance must have been passed.

Recommendations
Prior attendance of a course in the field of fluid technology, for example Fluid Power[2113092] or Mathematical Methods of Hydraulics[2113090], is recommended."

Additional Information
After completion of the lecture, studens can:

- design working and travel drive train hydraulics of mobile machines and can derive characteristic key factors.
- choose and apply suitable state of the art designing methods succesfully
- analyse a mobile machines and break its structure down from a complex system to subsystems with reduced complexity
- identify and desrcibe interactions and links between subsystems of a mobile maschine
- present and document solutions of a technical problem according to R&D standards

The number of participants is limited.

Conent:

The working scenario of a mobile machine depends strongly on the machine itself. Highly specialised machines, e.g. pavers are also as common as universal machines with a wide range of applications, e.g. hydraulic excavators. In general, all mobile machines are required to do their intended work in an optimal way and satisfy various critera at the same time. This makes designing mobile machines to a great and interesting challenge. Nevertheless, usually key factors can be derived for every mobile machine, which affect all other machine parameters. During this lecture, those key factors and designing mobile machines accordingly will be adressed. To do so, an exemplary mobile machine will be discussed and designed in the lecture an as a semester project.

Literature:

See german recommendations

The course is offered in German.

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V **Design and Development of Mobile Machines**
2113079, WS 25/26, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content
Excavators and wheel loaders are highly specialized mobile machines. Their function is to loosen and pick up material and place/dump it again at a short distance.
The decisive factor for dimensioning is the capacity of the standard bucket. Using a wheel loader or excavator, the essential dimensioning steps for the design are worked through in this course. This includes, among other things

- Determining the size class and main dimensions,
- the dimensioning of an electric drive train,
- the design of the primary energy supply,
- determining the kinematics of the equipment,
- dimensioning the working hydraulics and
- strength calculations.

The entire sizing and design process for these machines is heavily influenced by the use of standards and guidelines. This aspect is also covered.

The course builds on knowledge from the fields of mechanics, strength of materials, machine elements, drive technology and fluid technology.
The course requires active participation and continuous involvement.

Recommendations:

Prior attendance of a course in the field of fluid technology, for example Fluid Power [2113092] or Mathematical Methods of Hydraulics [2113090], is recommended."

- Attendance time: 21 hours
- Self-study: 99 hours

Literature
Keine.

T 6.157 Module component: Design and Development of Mobile Machines - Advance [T-MACH-108887]

Coordinators: Prof. Dr.-Ing. Marcus Geimer
Jan Siebert

Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101267 - Mobile Machines

Type	Credits	Grading	Term offered	Version
Coursework	0 CP	pass/fail	Each term	1

Assessment
Preparation of semester report

Prerequisites
none

Additional Information
The course is offered in German.

T 6.158 Module component: Design and Operation of Industrial Plants and Processes [T-WIWI-114618]

Coordinators: Prof. Dr. Frank Schultmann
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101471 - Industrial Production II
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	5,5 CP	graded	Each winter term	1

Courses					
WT 25/26	2581952	Design and Operation of Industrial Plants and Processes	2 SWS	Lecture / 🗣	Schultmann, Rudi
WT 26/27	2581952	Design and Operation of Industrial Plants and Processes	2 SWS	Lecture / 🗣	Schultmann, Rudi

Legend: 🖥 Online, ❊ Blended (On-Site/Online), 🗣 On-Site, × Cancelled

Assessment
The assessment consists of a written exam (90 minutes) (following §4(2) of the examination regulation). The exam takes place in every semester. Re-examinations are offered at every ordinary examination date.

Prerequisites
None

Recommendations
None

Below you will find excerpts from courses related to this module component:

V Design and Operation of Industrial Plants and Processes

Lecture (V)
On-Site

2581952, WS 25/26, 2 SWS, Language: English, Open in study portal

Content
Industrial plant management incorporates a complex set of tasks along the entire life cycle of an industrial plant, starting with the initiation and erection up to operating and dismantling.

During this course students will get to know special characteristics of industrial plant management. Students will learn important methods to plan, realize and supervise the supply, start-up, maintenance, optimisation and shut-down of industrial plants. Alongside, students will have to handle the inherent question of choosing between technologies and evaluating each of them. This course pays special attention to the specific characteristics of plant engineering, commissioning and investment.

Literature
Wird in der Veranstaltung bekannt gegeben.

V Design and Operation of Industrial Plants and Processes

Lecture (V)
On-Site

2581952, WS 26/27, 2 SWS, Language: English, Open in study portal

Content
Industrial plant management incorporates a complex set of tasks along the entire life cycle of an industrial plant, starting with the initiation and erection up to operating and dismantling.

During this course students will get to know special characteristics of industrial plant management. Students will learn important methods to plan, realize and supervise the supply, start-up, maintenance, optimisation and shut-down of industrial plants. Alongside, students will have to handle the inherent question of choosing between technologies and evaluating each of them. This course pays special attention to the specific characteristics of plant engineering, commissioning and investment.

Literature
Wird in der Veranstaltung bekannt gegeben.

T 6.159 Module component: Design Basics in Highway Engineering [T-BGU-106613]

Coordinators: Dr.-Ing. Matthias Zimmermann
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: M-BGU-100998 - Design, Construction, Operation and Maintenance of Highways
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Expansion	Version
Oral examination	3 CP	graded	Each term	1 semesters	1

Courses					
ST 2026	6200408	Design Basics in Highway Engineering	2 SWS	Lecture / 🗣	Zimmermann, Stelzenmüller
ST 2026	6200409	Exercises to Design Basics in Highway Engineering	1 SWS	Practice / 🗣	Zimmermann, Stelzenmüller

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
oral exam, appr. 20 min.

Prerequisites
None

Recommendations
None

Additional Information
None

Workload
90 hours

T 6.160 Module component: Design of a Jet Engine Combustion Chamber [T-CIWVT-110571]

Coordinators: Dr.-Ing. Stefan Raphael Harth
Organisation: KIT Department of Chemical and Process Engineering
Part of: M-CIWVT-105206 - Design of a Jet Engine Combustion Chamber

Type	Credits	Grading	Term offered	Version
Examination of another type	6 CP	graded	Each winter term	1

Courses					
WT 25/26	2232310	Design of a Jet Engine Combustion Chamber	2 SWS	/ 🗣	Harth

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success control is an examination of another kind according to § 4 Abs. 2 Nr. 3 SPO.

Project: Participation and presentation as well as a final oral examination amounting to max. 30 minutes.

Prerequisites
None

T 6.161 Module component: Design Thinking [T-WIWI-102866]

Coordinators: Prof. Dr. Orestis Terzidis
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101488 - Entrepreneurship (EnTechnon)
M-WIWI-101507 - Innovation Management
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Irregular	1

Courses					
WT 25/26	2545008	Design Thinking (Track 1)	2 SWS	Seminar / Blended	Haller, Terzidis
ST 2026	2545008	Design Thinking (Track 1)	2 SWS	Seminar / On-Site	Haller, Haller

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
Alternative exam assessments (§4(2), 3 SPO).

Prerequisites
None

Recommendations
None

Additional Information
Please note that the number of participants is limited. Registration takes place via the Wiwi-Portal. The seminar contents will be published on the institute's homepage.

Below you will find excerpts from courses related to this module component:

V Design Thinking (Track 1)
2545008, WS 25/26, 2 SWS, Language: English, Open in study portal

Seminar (S)
Blended (On-Site/Online)

Content
Course Content:

Design Thinking is a user-centric method for innovation management. The iterative process first analyzes the problem space and builds a sound understanding of the future users. Subsequently, ideas for the solution are generated, prototypes created that get tested by the user group. The result is a validated concept of an innovation or product.

Learning Objectives

During the seminar, students learn basic procedures for achieving user-centric innovations. These are concrete methods focusing on actual users' needs and pain points. Design Thinking is problem-oriented and emphasizes specific customer situations. After attending the seminar, students have a clear understanding of the need to explore end-user needs and are able to independently apply the methods of Design Thinking for developing market-driven innovations at a basic level.

Credentials:

Registration is via the Wiwi portal.

ATTENTION: Creditability in the seminar module: The seminar is NOT credited in the seminar module! Crediting is only possible in the EXPERT MODULE ENTREPRENEURSHIP.

Organizational issues
Registration is via the Wiwi portal.

In the seminar you will work on a project in teams of 4-5 persons. The groups are formed in the seminar

V Design Thinking (Track 1)
2545008, SS 2026, 2 SWS, Language: English, Open in study portal

Seminar (S)
On-Site

Content
Content

Design Thinking is a user-centric method for innovation management. The iterative process first analyzes the problem space and builds a sound understanding of the future users. Subsequently, ideas for the solution are generated, prototypes created that get tested by the user group. The result is a validated concept of an innovation or product.

Learning Objectives

During the seminar, students learn basic procedures for achieving user-centric innovations. These are concrete methods focusing on actual users' needs and pain points. Design Thinking is problem-oriented and emphasizes specific customer situations. After attending the seminar, students have a clear understanding of the need to explore end-user needs and are able to independently apply the methods of Design Thinking for developing market-driven innovations at a basic level.

Credentials:

ATTENTION: Creditability in the seminar module: The seminar is NOT credited in the wiwi seminar module!

T 6.162 Module component: Design, Construction and Sustainability Assessment of Buildings I [T-WIWI-102742]

Coordinators: Prof. Dr.-Ing. Thomas Lützkendorf
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each winter term	1

Assessment
The examination offer has been discontinued. For all those who still have to complete the module examination with this exam, there is the last opportunity for an oral exam in WS 22/23. This must be arranged directly at the institute. If this is a repeat examination for which the first attempt was taken in writing, the approval of the examination board must be obtained in advance via the examination office.

Prerequisites
None

Recommendations
A combination with the module *Real Estate Management* and with engineering science modules in the area of building physics and structural design is recommended.

6.163 Module component: Design, Construction and Sustainability Assessment of Buildings II [T-WIWI-102743]

Coordinators: Prof. Dr.-Ing. Thomas Lützkendorf
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each summer term	1

Assessment
The examination offer has been discontinued. For all those who still have to complete the module examination with this exam, there is the last opportunity for an oral exam in WS 22/23. This must be arranged directly at the institute. If this is a repeat examination in which the first attempt was taken in writing, the approval of the examination board must be obtained in advance via the examination office.

Prerequisites
None

Recommendations
A combination with the module *Real Estate Management* and with engineering science modules from the areas building physics and structural designis recommended.

T 6.164 Module component: Designing Interactive Systems: Human-AI Interaction [T-WIWI-113465]

Coordinators: Prof. Dr. Alexander Mädche
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104068 - Information Systems in Organizations
M-WIWI-104080 - Designing Interactive Information Systems
M-WIWI-104900 - Business Administration
M-WIWI-105923 - Incentives, Interactivity & Decisions in Organizations

Type	Credits	Grading	Term offered	Version
Examination of another type	4,5 CP	graded	Each summer term	1

Courses					
ST 2026	2540558	Designing Interactive Systems: Human-AI Interaction	3 SWS	Lecture / 🧩	Mädche, Seitz Sänger

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Alternative exam assessment. The assessment consists of a one-hour exam and the implementation of a Capstone project. Details will be announced at the beginning of the course.

Additional Information
The course is held in english.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Designing Interactive Systems: Human-AI Interaction

Lecture (V)
Blended (On-Site/Online)

2540558, SS 2026, 3 SWS, Language: English, Open in study portal

Content
Description

The field of computing has transitioned from traditional batch processing toward the development of highly sophisticated interactive and intelligent systems. As artificial intelligence (AI) continues to advance, these systems have gained the capacity to learn, adapt to environmental stimuli, and simulate human cognitive processes. While the capacity of AI-based information technology to augment and automate human tasks has expanded, it simultaneously creates complex new demands for the design of interactive systems and necessitates a sophisticated understanding of the nuances within human-AI interaction.
This Master’s course explores the state-of-the-art scientific knowledge and practical applications required to navigate these complexities, specifically focusing on the design of AI-based interactive systems for personal productivity on the individual level and virtual collaboration on the team level within professional environments.

The course is divided into two phases: The first phase establishes a scientific foundation through a **flipped classroom model**. Students engage with conceptual foundations and theoretical frameworks independently, allowing in-person sessions to be dedicated to analysis and critical discussion of existing knowledge for designing interactive systems enabling effective human-AI interaction. The second phase transitions into the practical application of these concepts through a collaborative **capstone project** conducted in partnership with an industry partner. In this phase, students work in teams to apply knowledge acquired in the first phase culminating in the creation of a prototype of an interactive system. The project emphasizes the intricacies of human-AI interaction, requiring students to synthesize their theoretical knowledge into a functional and sophisticated design solution.

Learning objectives

- Evaluate existing theories and conceptual frameworks of interactive systems to assess their role in modern computing environments.
- Analyze the unique characteristics of human-AI interaction and determine their impact on future interactive systems design.
- Create a functional interactive prototype by following the human-centered design process and applying state-of-the-art knowledge within a collaborative capstone project to solve real-world problems.
- Get hands-on experience by applying lecture content in a design capstone project

Prerequisites

No specific prerequisites are required for the lecture

Literature

Further literature will be made available in the lecture.

Die Erfolgskontrolle erfolgt in Form einer Prüfungsleistung anderer Art (Form) nach § 4 Abs. 2 Nr. 3 SPO. Details zur Ausgestaltung der Erfolgskontrolle werden im Rahmen der Vorlesung bekannt gegeben.

Literature
Die Vorlesung basiert zu einem großen Teil auf

· Benyon, D. (2014). Designing interactive systems: A comprehensive guide to HCI, UX and interaction design (3. ed.). Harlow: Pearson.

Weiterführende Literatur wird in der Vorlesung bereitgestellt.

T 6.165 Module component: Development of Hybrid Drivetrains [T-MACH-110817]

Coordinators: Prof. Dr. Thomas Koch
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101303 - Combustion Engines II

Type	Credits	Grading	Term offered	Version
Written examination	4 CP	graded	Each summer term	1

Courses					
ST 2026	2134155	Development of Hybrid Powertrains	2 SWS	Lecture / 🗣	Koch, Doppelbauer

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
oral exam, 20 min.

Prerequisites
None

Additional Information
This course is offered in German.

Workload
120 hours

T 6.166 Module component: Development of Sustainable, Digital Business Models [T-WIWI-113663]

Coordinators: Prof. Dr. Marion Weissenberger-Eibl
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101507 - Innovation Management
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Irregular	1

Assessment
Non exam assessment. The final grade is composed 50% of the grade of the written paper (ca. 5 Pages /Person) and 50% of the presentation of the results.

Prerequisites
None

Recommendations
Prior attendance of the course Innovation Management is recommended.

Workload
90 hours

T 6.167 Module component: Digital Democracy [T-WIWI-113160]

Coordinators: Jonas Fegert
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101410 - Business & Service Engineering
M-WIWI-101446 - Market Engineering
M-WIWI-103117 - Data Science: Data-Driven Information Systems
M-WIWI-103118 - Data Science: Data-Driven User Modeling
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Expansion	Version
Examination of another type	4,5 CP	graded	Each winter term	1 semesters	1

Courses					
WT 25/26	00053	Übung zur Digital Democracy	1 SWS	Practice / Blended	Stein
WT 25/26	2500045	Digital Democracy - Challenges and Opportunities of the Digital Society	2 SWS	Seminar / Blended	Fegert, Stein, Bezzaoui
WT 25/26	2600052	Digital Democracy	2 SWS	Lecture / Blended	Fegert

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
Alternative exam assessment. The examination consists of two parts (presentation and oral exam). Details on the design of the exam will be announced at the beginning of the course.

Additional Information
Limited to 25 students. Application (cover letter) via the Wiwi-portal.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Digital Democracy

2600052, WS 25/26, 2 SWS, Language: English, Open in study portal

Lecture (V)
Blended (On-Site/Online)

Content
The "Digital Democracy" Lecture deals with opportunities and challenges of democracy and participation in a digitalized world. Social networks and other platforms have become a central place for human interaction.

These technologies open up many possibilities to connect people, promote societal discourse, and organize social movements. On the other hand, they are also used to undermine democracy by extremist forces.

One example is the spread of disinformation through social media, which can undermine trust in democratic institutions and exacerbate divisions in society. Big tech actors pursue their own economically driven interests, some of which run counter to societal ones.

So to what extent can Internet platforms help strengthen social discourse? And what measures can be taken to promote the quality and diversity of discourse in the digital world? What role do big tech players play in digital democracy and how can their interests be reconciled with democratic principles? These and many more questions will be explored in the lecture. The lecture introduces theoretical foundations and evidence-based research on digital democracy. It will address the following questions: What characterizes deliberative democracies, how do democracies change, and what can damage them? How does social polarization emerge and what drives it - off- and online. Accordingly, different platform types and phenomena of disinformation, such as clickbait, will be presented. The last part of the lecture series will deal with the search for approaches and alternatives to these problems.

The exercise session connected to this lecture is conducted in cooperation with an NGO and applies the lecture content in a practical context: The formulation of a data-based policy recommendation.

Organizational issues
Die Teilnahme am Kurs ist auf 25 Plätze beschränkt, diese erfolgt über das Wiwi-Portal.

Die Termine werden ebenfalls zeitnah bekannt gegeben.

T 6.168 Module component: Digital Energy Systems [T-WIWI-114994]

Coordinators: TT-Prof. Dr. Philipp Staudt
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-107649 - Sustainable Information Systems

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each winter term	1

Assessment
Written examination (60 minutes). The examination is offered every semester and can be repeated at any regular examination date.

Workload
135 hours

T 6.169 Module component: Digital Marketing [T-WIWI-112693]

Coordinators: Prof. Dr. Ann-Kristin Kupfer
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration
M-WIWI-105312 - Marketing and Sales Management
M-WIWI-106258 - Digital Marketing

Type	Credits	Grading	Term offered	Version
Examination of another type	4,5 CP	graded	Each summer term	1

Courses					
ST 2026	2571185	Digital Marketing	2 SWS	Lecture / 🗣	Kupfer
ST 2026	2571186	Digital Marketing Exercise	1 SWS	Practice / 🗣	Schmitt, Cinar, Kupfer

Legend: 🖥 Online, ✧ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success is assessed in the form of an examination of another type. The following aspects are included in the assessment:

- Elaboration and presentation of a group task
- Written exam

Further details on the organization of the performance and the points system for the assessment will be announced in the lecture.

Prerequisites
None

Recommendations
Students are highly encouraged to actively participate in class.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Digital Marketing

2571185, SS 2026, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content
Students learn the theoretical foundations of digital marketing and its most important concepts. They develop an understanding both for the digital consumer and the digital environment. Special emphasis will be given to digital marketing strategies and practices, such as content marketing and influencer marketing. A tutorial offers the opportunity to apply the key learnings of the lecture as part of a group work.

The learning objectives are as follows:

- Getting to know the theoretical foundations of digital marketing
- Evaluating digital marketing strategies and practices (e.g., in the context of content marketing and influencer marketing)
- Fostering critical and analytical thinking skills and the application of knowledge to marketing problems
- Improving English skills

Total time required for 4.5 credit points: approx. 135 hours
Attendance time: 30 hours
Self-study: 105 hours

Organizational issues
Termine werden bekannt gegeben.

T 6.170 Module component: Digital Marketing and Sales in B2B [T-WIWI-106981]

Coordinators: Prof. Dr. Martin Klarmann
Anja Konhäuser

Organisation: KIT Department of Business and Economics

Part of: M-WIWI-104900 - Business Administration
M-WIWI-105312 - Marketing and Sales Management
M-WIWI-106258 - Digital Marketing

Type	Credits	Grading	Term offered	Version
Examination of another type	1,5 CP	graded	Each summer term	1

Courses					
ST 2026	2571156	Digital Marketing and Sales in B2B	1 SWS	Others / 🗣	Konhäuser

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Alternative exam assessment according to § 4 paragraph 2 Nr. 3 of the examination regulation. (team presentation of a case study with subsequent discussion totalling 30 minutes).

Prerequisites
None.

Additional Information
This course will not take place in the summer term 2023, but is expected to be offered again on a regular basis starting in the summer term 2024.
Participation requires an application. The application period starts at the beginning of the semester. More information can be obtained on the website of the research group Marketing and Sales (marketing.iism.kit.edu). Access to this course is restricted. Typically all students will be granted the attendance of one course with 1.5 ECTS. Nevertheless attendance can not be guaranteed.For further information please contact Marketing and Sales Research Group (marketing.iism.kit.edu).Please note that only one of the 1.5-ECTS courses can be attended in this module.

Workload
45 hours

Below you will find excerpts from courses related to this module component:

V Digital Marketing and Sales in B2B

2571156, SS 2026, 1 SWS, Language: English, Open in study portal

Others (sonst.)
On-Site

Content
Learning Sessions:

The class gives insights into digital marketing strategies as well as the effects and potential of different channels (e.g., SEO, SEA, Social Media). After an overview of possible activities and leverages in the digital marketing field, including their advantages and limits, the focus will turn to the B2B markets. There are certain requirements in digital strategy specific to the B2B market, particularly in relation to the value chain, sales management and customer support. Therefore, certain digital channels are more relevant for B2B marketing than for B2C marketing.
Once the digital marketing and tactics for the B2B markets are defined, further insights will be given regarding core elements of a digital strategy: device relevance (mobile, tablet), usability concepts, website appearance, app decision, market research and content management. A major advantage of digital marketing is the possibility of being able to track many aspects of of user reactions and user behaviour. Therefore, an overview of key performance indicators (KPIs) will be discussed and relationships between these KPIs will be explained. To measure the effectiveness of digital activities, a digital report should be set up and connected to the performance numbers of the company (e.g. product sales) – within the course the setup of the KPI dashboard and combination of digital and non-digital measures will be shown to calculate the Return on Investment (RoI).

Presentation Sessions:

After the learning sessions, the students will form groups and work on digital strategies within a case study format. The presentation of the digital strategy will be in front of the class whereas the presentation will take 20 minutes followed by 10 minutes questions and answers.

- Understand digital marketing and sales approaches for the B2B sector
- Recognise important elements and understand how-to-setup of digital strategies
- Become familiar with the effectiveness and usage of different digital marketing channels
- Understand the effect of digital sales on sales management, customer support and value chain
- Be able to measure and interpret digital KPIs
- Calculate the Return on Investment (RoI) for digital marketing by combining online data with company performance data

time of presentness = 15 hrs.

private study = 30 hrs.

Organizational issues
Blockveranstaltung, Raum B5.26, Gebäude 01.87

6.5.26 11:00-17:00 Uhr

7.5.26 09:00-14:00 Uhr

Literature
-

T 6.171 Module component: Digital Markets and Market Design [T-WIWI-112228]

Coordinators: Prof. Dr. Adrian Hillenbrand
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each winter term	1

Courses					
WT 25/26	2500035	Digital Markets and Market Design	2 SWS	Lecture / 🗣	Hillenbrand
WT 25/26	2500036	Digital Markets and Market Design	1 SWS	Practice / 🗣	Hillenbrand
WT 26/27	2500035	Digital Markets and Market Design	2 SWS	Lecture / 🗣	Hillenbrand
WT 26/27	2500036	Digital Markets and Market Design	1 SWS	Practice / 🗣	Hillenbrand

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success is assessed in the form of a written examination (60 min.). The examination is only offered in the winter semester; two examination dates are offered during this period.

Prerequisites
None

Additional Information
The lecture will be held in English.

Below you will find excerpts from courses related to this module component:

V Digital Markets and Market Design
2500035, WS 25/26, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content
Online Markets determine our everyday lives. At the same time rapid technological advancements quickly change the landscape of online markets posing challenges for market design and consumer protection. In this course we apply theoretical economic models in the area of digital markets in order to make sense of current developments. Topics include consumer search, algorithmic pricing, recommender systems and steering, price discrimination and matching markets. We also discuss the potential effects of current policies like the Digital Markets Act and Digital Services Act on market outcomes.

V Digital Markets and Market Design
2500036, WS 25/26, 1 SWS, Language: English, Open in study portal

Practice (Ü)
On-Site

Content
Exercise Session for the course "Digital Markets and Market Design

Organizational issues
Jede zweite Woche eine Übung

V Digital Markets and Market Design
2500035, WS 26/27, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content
Online Markets determine our everyday lives. At the same time rapid technological advancements quickly change the landscape of online markets posing challenges for market design and consumer protection. In this course we apply theoretical economic models in the area of digital markets in order to make sense of current developments. Topics include consumer search, algorithmic pricing, recommender systems and steering, price discrimination and matching markets. We also discuss the potential effects of current policies like the Digital Markets Act and Digital Services Act on market outcomes.

V **Digital Markets and Market Design**
2500036, WS 26/27, 1 SWS, Language: English, Open in study portal

Practice (Ü)
On-Site

Content
Exercise Session for the course "Digital Markets and Market Design

Organizational issues
Jede zweite Woche eine Übung

T 6.172 Module component: Digital Markets and Mechanisms [T-WIWI-114598]

Coordinators: Dr. Frank Rosar
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104908 - Economics

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each summer term	1

Courses					
ST 2026	2560550	Digital Markets and Mechanisms	2 SWS	Lecture / 🗣	Rosar
ST 2026	2560551	Digital Markets and Mechanisms	1 SWS	Practice / 🗣	Rosar

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment consists of a written exam (60 minutes) (following §4(2), 1 of the examination regulation).
The exam takes place in every semester. Re-examinations are offered at every ordinary examination date.

A bonus can be earned through successful participation in the excercise. If the grade of the written examination is between 4.0 and 1.3, the bonus improves the grade by one grade level (0.3 or 0.4). The exact criteria for awarding a bonus will be announced at the beginning of the course.

Prerequisites
None

Recommendations
Basic knowledge of microeconomics (in particular game theory) as well as statistics (especially the use of probabilities and expected values) is desirable, but not required.

Below you will find excerpts from courses related to this module component:

V Digital Markets and Mechanisms

Lecture (V)
On-Site

2560550, SS 2026, 2 SWS, Language: German, Open in study portal

Content
Auctions are a central instrument of the digital economy—sometimes visible, as on eBay, but also operating behind the scenes, for example automatically in the background of every Google search or in the price formation of electricity markets. But which strategic incentives shape the behavior of bidders? And how can auction mechanisms be designed to maximize efficiency or revenue?

This course conveys the theoretical foundations and methodological tools for analyzing and designing auctions and mechanisms. Students will learn equilibrium concepts of strategic interaction in auction markets, compare auction formats in terms of efficiency and revenue, and become familiar with the principles of (Bayesian) mechanism design. Mechanism design can be understood as the deliberate design of games to achieve specific economic objectives. The concepts and methods taught in this course are also applicable in many other economic contexts.

The course is aimed at students interested in the game-theoretic modeling of markets as well as in the design of economic incentive mechanisms and strategic games. The necessary game-theoretic foundations will be introduced in the course.

V Digital Markets and Mechanisms

Practice (Ü)
On-Site

2560551, SS 2026, 1 SWS, Language: German, Open in study portal

Content
See course description.

T 6.173 Module component: Digital Real Time Simulations for Energy Technologies [T-ETIT-113449]

Coordinators: Prof. Dr.-Ing. Giovanni De Carne
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-106690 - Digital Real Time Simulations for Energy Technologies

Type	Credits	Grading	Version
Examination of another type	3 CP	graded	1

Courses					
ST 2026	2314013	Digital Real Time Simulations for Energy Technologies	2 SWS	Lecture / 🗣	De Carne

Legend: 🖥 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The examination takes place in form of other types of examination. It consists of an assessment from an exercise on HiL and an oral overall examination (approx. 15 minutes) explaining the exercise results. The overall impression is evaluated.
The module grade results of the assessment of an exercise and the oral exam. Details will be given during the lecture.

Prerequisites
none

T 6.174 Module component: Digital Services: Foundations [T-WIWI-111307]

Coordinators: TT-Prof. Dr. Maximilian Förster
Dr. Carsten Holtmann

Organisation: KIT Department of Business and Economics

Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each summer term	3

Courses					
ST 2026	2595466	Digital Services: Foundations	2 SWS	Lecture / 🗣	Förster, Holtmann
ST 2026	2595467	Exercise Digital Services: Foundations	1 SWS	Practice / 🗣	Förster, Schäfer

Legend: 📱 Online, Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment of success takes place through a written exam (60 minutes) with additional aids (according to § 4(2), 1 SPO). Further details regarding the assessment of success will be announced during the lecture.

A bonus can be acquired by successful completion of a a practically-oriented assignment (written report + short in-class presentation) during the semester. If the grade of the written examination is between 4.0 and 1.3, the bonus improves the grade by one grade level (0.3 or 0.4). Further details regarding the acquisition of a bonus will be announced at the beginning of the course.

Additional Information
The course will be offered as a pure in-person lecture starting in the summer semester of 2026.
Please check the announcements in the ILIAS forum.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Digital Services: Foundations
2595466, SS 2026, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content
The world has been moving towards "service-led" economies: In many developed countries, services already account for more than 70% of the gross domestic product. In order to design, engineer, and manage services, traditional "goods-oriented" business models are often inappropriate. At the same time, the rapid development of information and communication technology (ICT), such as Artificial Intelligence (AI), pushes "servitization" and the economic importance of digital services. This drives competition and opens up novel opportunities: Services can instantly be delivered anywhere across the globe; dynamic and scalable service ecosystems replace static value chains; increased interaction options allow "value co-creation" between providers and customers; and the analysis of big data enables service optimization and personalization.

Building on a systematic categorization of different types of services and on the general notion of "value co-creation", we cover concepts for managing, developing, and analyzing digital services, as well as current technological advances in digital services, allowing for further specialization in master level courses. Topics in this course include an introduction to digital services, human-centered development of services, and service analytics based on big data. Additionally, we cover emerging technologies and trends as well as essential design concepts for AI-based services, such as bias and fairness in AI, explainable AI, and sovereign AI. Finally, the lecture lays the practical foundations for implementing, analyzing, and managing digital services at scale. Besides those contents, the lecture entails first-hand research insights, exercises and discussion sessions, and guest lectures from practitioners that will illustrate the relevance of digital services in today's world.

Literature

- Bartelheimer, C., Heinz, D., Hönigsberg, S., et al. (2025). Conceptualizing hybrid intelligent service ecosystems. *Electronic Markets, 35*, Article 63.
- Böhmann, T., Leimeister, J. M., & Möslein, K. (2014). Service systems engineering. Business & Information Systems Engineering,6(2), 73-79.
- Brasse, J., Broder, H. R., Förster, M., Klier, M., & Sigler, I. (2023). Explainable artificial intelligence in information systems: A review of the status quo and future research directions. *Electronic Markets*, *33*(1), 26.
- Cardoso, J., Fromm, H., Nickel, S., Satzger, G., Studer, R., & Weinhardt, C. (Eds.). (2015). Fundamentals of service systems (Vol. 12). Heidelberg: Springer.
- Fromm, H., Habryn, F., & Satzger, G. (2012). Service analytics: Leveraging data across enterprise boundaries for competitive advantage. In Globalization of professional services (pp. 139-149). Springer, Berlin, Heidelberg.
- Ostrom, A. L., Parasuraman, A., Bowen, D. E., Patrício, L., & Voss, C. A. (2015). Service research priorities in a rapidly changing context. Journal of Service Research, 18(2), 127-159.
- Spohrer, J., Maglio, P. P., Bailey, J., & Gruhl, D. (2007). Steps toward a science of service systems. Computer, 40(1), 71-77.
- Satzger, G., Benz, C., Böhmann, T., & Roth, A. (2022). Servitization and digitalization as “Siamese Twins”: Concepts and research priorities. In B. Edvardsson & B. Tronvoll (Eds.), *The Palgrave Handbook of Service Management*. Palgrave Macmillan.

T 6.175 Module component: Digital Services: Innovation & Business Models [T-WIWI-112757]

Coordinators: Prof. Dr. Gerhard Satzger
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101410 - Business & Service Engineering
M-WIWI-101448 - Service Management
M-WIWI-102754 - Service Economics and Management
M-WIWI-102806 - Service Innovation, Design & Engineering
M-WIWI-102808 - Digital Service Systems in Industry
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each summer term	2

Courses					
ST 2026	2595468	Digital Services: Innovation & Business Models	1.5 SWS	Lecture / 🧩	Satzger, Benz, Schüritz, Heinz
ST 2026	2595469	Übung zu Digital Services: Innovation & Business Models	1.5 SWS	Practice / 🗣	Satzger, Benz, Schüritz, Heinz

Legend: 📱 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment consists of a written exam (60 min.).

Prerequisites
None

Recommendations
None

Additional Information
The course "Digital Services: Innovation & Business Models" replaces the course Service Innovation, based on a revised course concept and content. The focus will be on the closer integration of the topics of service innovation and digitalization. Previous foundational content (e.g., on service innovation challenges or human-centered innovation methods) will remain. New content will cover topics such as digital platforms and ecosystems, IoT and smart service innovation, and business models.

Below you will find excerpts from courses related to this module component:

V Digital Services: Innovation & Business Models

2595468, SS 2026, 1.5 SWS, Language: English, Open in study portal

Lecture (V)
Blended (On-Site/Online)

Content
Leveraging data and digital technologies for business success is a key challenge for organizations as they need to

- get aware of the newly arising potential
- develop suitable digital services that are user-centric and individualized
- “servitize” their offering portfolio and business model
- transform their organizations

This course will equip students with concepts and methods to tackle this challenge along two dimensions: First, we will cover innovation as a concept as well as apply contemporary innovation methods (like Design Thinking, Open Innovation) to the services space. Second, we deal with leveraging innovation to develop new business models (including multi-partner concepts in platforms or ecosystems), to servitize existing business models (e.g., via product-service-systems), and to accordingly transform the organization.

The course links innovation and business model theories with practical examples and exercises. Students are asked to actively engage in the discussion.

Organizational issues
The course will be offered in the form of a flipped classroom concept. The lecture will be recorded in advance and made available online. During the “in presence” sessions, the contents of the lecture will be applied and expanded on.

Literature

- Böhmann, T./ Leimeister, J.M./ Möslein, K. (2014), Service Systems Engineering, Business & Information Systems Engineering, Vol. 6, No.2, 73-79.
- Cardoso, J., Fromm, H., Nickel, S., Satzger, G., Studer, R., & Weinhardt, C. (Eds.) (2015). Fundamentals of service systems (Vol. 12). Heidelberg: Springer.
- Chesbrough, H. (2011). Open services innovation: Rethinking your business to grow and compete in a new era. John Wiley & Sons.
- Rogers, S. (2003). Diffusion of Innovations. 5. ed. New York: Free Press.
- Satzger, G., Benz, C., Böhmann, T., Roth, A. (2022). Servitization and Digitalization as Siamese Twins – Concepts and Research Agenda. Edvardsson/Tronvoll (eds.): The Palgrave Handbook of Service Management, 967-989.
- Uebernickel, F., Brenner, W., Pukall, B., Naef, T., & Schindlholzer, B. (2015). Design Thinking: Das Handbuch. Frankfurt am Main: Frankfurter Allgemeine Buch.
- Vargo, S.L., Lusch, R.F. (2017). Service-dominant logic 2025. Int. J. Res. Mark. 34, 46–67.
- Weill, P.; Woerner, S.L. (2018): “What‘s your Digital Business Model? – Six Questions to Help you Build the Next-Generation Enterprise“. Boston, Massachusetts: Harvard Business Review Press.

V **Übung zu Digital Services: Innovation & Business Models** **Practice (Ü) On-Site**
2595469, SS 2026, 1.5 SWS, Language: English, Open in study portal

Content
Leveraging data and digital technologies for business success is a key challenge for organizations as they need to

- get aware of the newly arising potential
- develop suitable digital services that are user-centric and individualized
- “servitize” their offering portfolio and business model
- transform their organizations

This course will equip students with concepts and methods to tackle this challenge along two dimensions: First, we will cover innovation as a concept as well as apply contemporary innovation methods (like Design Thinking, Open Innovation) to the services space. Second, we deal with leveraging innovation to develop new business models (including multi-partner concepts in platforms or ecosystems), to servitize existing business models (e.g., via product-service-systems), and to accordingly transform the organization.

The course links innovation and business model theories with practical examples and exercises. Students are asked to actively engage in the discussion.

Organizational issues
The course will be offered in the form of a flipped classroom concept. The lecture will be recorded in advance and made available online. During the “in presence” sessions, the contents of the lecture will be applied and expanded on.

Literature

- Böhmann, T./ Leimeister, J.M./ Möslein, K. (2014), Service Systems Engineering, Business & Information Systems Engineering, Vol. 6, No.2, 73-79.
- Cardoso, J., Fromm, H., Nickel, S., Satzger, G., Studer, R., & Weinhardt, C. (Eds.) (2015). Fundamentals of service systems (Vol. 12). Heidelberg: Springer.
- Chesbrough, H. (2011). Open services innovation: Rethinking your business to grow and compete in a new era. John Wiley & Sons.
- Rogers, S. (2003). Diffusion of Innovations. 5. ed. New York: Free Press.
- Satzger, G., Benz, C., Böhmann, T., Roth, A. (2022). Servitization and Digitalization as Siamese Twins – Concepts and Research Agenda. Edvardsson/Tronvoll (eds.): The Palgrave Handbook of Service Management, 967-989.
- Uebernickel, F., Brenner, W., Pukall, B., Naef, T., & Schindlholzer, B. (2015). Design Thinking: Das Handbuch. Frankfurt am Main: Frankfurter Allgemeine Buch.
- Vargo, S.L., Lusch, R.F. (2017). Service-dominant logic 2025. Int. J. Res. Mark. 34, 46–67.
- Weill, P.; Woerner, S.L. (2018): “What‘s your Digital Business Model? – Six Questions to Help you Build the Next-Generation Enterprise“. Boston, Massachusetts: Harvard Business Review Press.

T 6.176 Module component: Digital Transformation of Public Administration [T-WIWI-114977]

Coordinators: Philipp Bauer
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101507 - Innovation Management
M-WIWI-101507 - Innovation Management

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Irregular	1

Courses					
ST 2026	2500004	Digital transformation of public administration	2 SWS	Seminar / 🗣	Bauer
WT 26/27	2500057	Digital transformation of public administration	2 SWS	Seminar / 🗣	Bauer

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment of success in this course is carried out as a non exam assessment**.** The grade is made up as follows:

1. **Written elaboration**: (weighting: 75 %). The required scope depends on the current requirements of the degree program for recognition in the seminar module.
2. **Presentation of the final project**: (weighting: 25 %).

Prerequisites
None

Recommendations
Prior attendance of the course Innovation Management is recommended.

Workload
90 hours

Below you will find excerpts from courses related to this module component:

V Digital transformation of public administration

Seminar (S)
On-Site

2500004, SS 2026, 2 SWS, Language: German, Open in study portal

Content
Digital transformation poses major challenges for public administration, but also offers numerous opportunities. In the age of AI, big data, and digital networking, administrations are required to fundamentally rethink their processes, structures, and services in order to become more efficient, citizen-oriented, and future-proof. At the same time, they face complex questions: How can digital technologies be integrated in a meaningful way? What organizational and cultural hurdles must be overcome? And how can the failure of digitization projects be prevented?
The seminar “Digital Transformation of Public Administration” offers students the opportunity to address these key issues in a practical and scientifically sound manner. The aim of the seminar is to sensitize students to the importance of digital transformation in administration and to equip them with the skills to actively shape this change. A special focus is placed on the field of AI in public administration.
The seminar combines theory and practice: Students work in groups to examine municipal case studies from Baden-Württemberg. Through expert interviews with representatives from cities such as Freiburg, Ulm, and Karlsruhe, they analyze best practice examples of successful digitization projects. Among other things, students examine the following questions:

• Organization: How is the topic of digitization organized within the municipalities (centralized vs. decentralized)? Which departments are responsible for this and how do these departments work together?
• Digitization strategy: Which areas of action are described in the digitization strategy of the city under review? Do the municipalities have a structured digitization monitoring system for the areas of action and projects mentioned in the strategy? How (centralized/decentralized, portfolio management, etc.) and according to which criteria (cost-benefit analysis) are digitization projects prioritized within the city (e.g., internal process digitization vs. citizen-oriented services vs. business-oriented services)? How can the progress of digitization be measured in comparison to other municipalities (keyword: maturity model for the degree of digitization)?
• Online Access Act (OZG): What is the status of digitization in the context of OZG services? To what extent are there existing collaborations with the GovTech Campus to scale OZG services?
• Inter-municipal cooperation and network: What inter-municipal cooperation exists in the area of digitalization? Are there any applications/digital solutions that have been adopted by other municipalities? If so, which ones? What other networks exist in the area of digitalization and what contribution do they make to administrative digitalization?

Based on their analyses, the students develop practical recommendations for future transformation processes. In doing so, they critically reflect on the success factors and barriers of the digitization initiatives examined. How can the barriers and success factors be meaningfully clustered? What ideas do the municipalities examined have for accelerating digitalization? Where do they themselves see the greatest leverage? What do municipalities need from the federal government/state/population/industry in order for the digital transformation to succeed?

Prerequisites
- Bachelor's degree
- Attendance at the Innovation Management lecture is recommended

Organizational issues
Link: https://itm.entechnon.kit.edu/192_1675.php

V	**Digital transformation of public administration** 2500057, WS 26/27, 2 SWS, Language: German, Open in study portal	**Seminar (S)** **On-Site**

Content

Digital transformation poses major challenges for public administration, but also offers numerous opportunities. In the age of AI, big data, and digital networking, administrations are required to fundamentally rethink their processes, structures, and services in order to become more efficient, citizen-oriented, and future-proof. At the same time, they face complex questions: How can digital technologies be integrated in a meaningful way? What organizational and cultural hurdles must be overcome? And how can the failure of digitization projects be prevented?

The seminar “Digital Transformation of Public Administration” offers students the opportunity to address these key issues in a practical and scientifically sound manner. The aim of the seminar is to sensitize students to the importance of digital transformation in administration and to equip them with the skills to actively shape this change. A special focus is placed on the field of AI in public administration.

The seminar combines theory and practice: Students work in groups to examine municipal case studies from Baden-Württemberg. Through expert interviews with representatives from cities such as Freiburg, Ulm, and Karlsruhe, they analyze best practice examples of successful digitization projects. Among other things, students examine the following questions:

- Organization: How is the topic of digitization organized within the municipalities (centralized vs. decentralized)? Which departments are responsible for this and how do these departments work together?
- Digitization strategy: Which areas of action are described in the digitization strategy of the city under review? Do the municipalities have a structured digitization monitoring system for the areas of action and projects mentioned in the strategy? How (centralized/decentralized, portfolio management, etc.) and according to which criteria (cost-benefit analysis) are digitization projects prioritized within the city (e.g., internal process digitization vs. citizen-oriented services vs. business-oriented services)? How can the progress of digitization be measured in comparison to other municipalities (keyword: maturity model for the degree of digitization)?
- Online Access Act (OZG): What is the status of digitization in the context of OZG services? To what extent are there existing collaborations with the GovTech Campus to scale OZG services?
- Inter-municipal cooperation and network: What inter-municipal cooperation exists in the area of digitalization? Are there any applications/digital solutions that have been adopted by other municipalities? If so, which ones? What other networks exist in the area of digitalization and what contribution do they make to administrative digitalization?

Based on their analyses, the students develop practical recommendations for future transformation processes. In doing so, they critically reflect on the success factors and barriers of the digitization initiatives examined. How can the barriers and success factors be meaningfully clustered? What ideas do the municipalities examined have for accelerating digitalization? Where do they themselves see the greatest leverage? What do municipalities need from the federal government/state/population/industry in order for the digital transformation to succeed?

Prerequisites
- Bachelor's degree
- Attendance at the Innovation Management lecture is recommended

T 6.177 Module component: Digital Twin Engineering [T-ETIT-112224]

Coordinators: Prof. Dr.-Ing. Mike Barth
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-107673 - Automation and Control
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Examination of another type	4 CP	graded	Each winter term	2

Courses					
WT 25/26	2301486	Digital Twin Engineering	2 SWS	Lecture / 🗣	Barth, Witucki, Dorn

Legend: 🖥 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The examination takes place in form of other types of examination. It consists of a model library developed in the course of a semester-long project in the modeling language Modelica and a presentation of the library lasting 25 minutes. The quality of the model library is evaluated within the framework of the criteria: documentation, formal correctness, functionality, usability, HMI and modeling level of detail. The presentation is evaluated as an additional aspects. The overall impression is evaluated.

The assessment of the developed model library and the presentation of the library will be included in the module grade. More details will be given at the beginning of the course.

Prerequisites
none

T 6.178 Module component: Digitalization from Product Concept to Production [T-MACH-113647]

Coordinators: Dr.-Ing. Marc Wawerla
Organisation: KIT Department of Mechanical Engineering

Part of:
M-MACH-105455 - Strategic Design of Modern Production Systems
M-MACH-106590 - Production Engineering
M-WIWI-104907 - Engineering Sciences
M-WIWI-107718 - Mechanical Engineering I
M-WIWI-107730 - Mechanical Engineering II

Type	Credits	Grading	Term offered	Expansion	Version
Examination of another type	4,5 CP	graded	Each winter term	1 semesters	3

Courses					
WT 25/26	2149702	Digitalization from Product Concept to Production	2 SWS	Lecture / 🗣	Wawerla
WT 26/27	2149702	Digitalization from Product Concept to Production	2 SWS	Lecture / 🗣	Wawerla

Legend: 📱 Online, ❖ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Alternative test achievement (graded):
- Written processing of a case study (weighting 50%) and
- Presentation of the results (ca. 10 min.) followed by a colloquium (ca. 30 min.) (weighting 50%)

Prerequisites
T-MACH-110176 may not have started.

Additional Information
The course is offered in English.

For organizational reasons, the number of participants in the course is limited. As a result, a selection process will be carried out for registrations. Further information on the course and application can be found via https://www.wbk.kit.edu/studium-und-lehre.php.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Digitalization from Product Concept to Production

2149702, WS 25/26, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content
The lecture covers digitalization along the entire value chain in technology companies. The focus is on production and the supply chain. In this context, concepts, tools, methods, technologies, and specific applications in industry are presented. As part of the course, students have the opportunity to gain insights into the digitalization processes of a German high-tech company.

Main topics of the lecture:

- Concepts and methods such as disruptive innovation and agile project management
- Overview of available technologies
- Practical approaches to innovation
- Applications in industry
- Excursion to ZEISS IQS

Learning Outcomes:

The students are able to …

- describe and explain the methods and concepts presented in the course.
- analyze and evaluate the suitability of digitalization technologies for value creation processes in the manufacturing industry.
- assess the applicability of methods such as disruptive innovation and agile project management.
- assess the practical challenges of digitalization in industry.

Workload:

regular attendance: 21 hours
self-study: 99 hours

Organizational issues
Aus organisatorischen Gründen ist die Anzahl der Teilnehmenden der Lehrveranstaltung begrenzt. Infolgedessen findet ein Auswahlprozess der Anmeldungen statt. Weitere Informationen zur Lehrveranstaltung und Bewerbung sind unter https://www.wbk.kit.edu/studium-und-lehre.php zu finden.

For organizational reasons, the number of participants in the course is limited. As a result, a selection process will carried out for registrations. Further information on the course and application can be found via https://www.wbk.kit.edu/studium-und-lehre.php.

V

Digitalization from Product Concept to Production

2149702, WS 26/27, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content
The lecture covers digitalization along the entire value chain in technology companies. The focus is on production and the supply chain. In this context, concepts, tools, methods, technologies, and specific applications in industry are presented. As part of the course, students have the opportunity to gain insights into the digitalization processes of a German high-tech company.

Main topics of the lecture:

- Concepts and methods such as disruptive innovation and agile project management
- Overview of available technologies
- Practical approaches to innovation
- Applications in industry
- Excursion to ZEISS IQS

Learning Outcomes:

The students are able to …

- describe and explain the methods and concepts presented in the course.
- analyze and evaluate the suitability of digitalization technologies for value creation processes in the manufacturing industry.
- assess the applicability of methods such as disruptive innovation and agile project management.
- assess the practical challenges of digitalization in industry.

Workload:

regular attendance: 21 hours
self-study: 99 hours

Organizational issues

Aus organisatorischen Gründen ist die Anzahl der Teilnehmenden der Lehrveranstaltung begrenzt. Infolgedessen findet ein Auswahlprozess der Anmeldungen statt. Weitere Informationen zur Lehrveranstaltung und Bewerbung sind unter https://www.wbk.kit.edu/studium-und-lehre.php zu finden.

For organizational reasons, the number of participants in the course is limited. As a result, a selection process will carried out for registrations. Further information on the course and application can be found via https://www.wbk.kit.edu/studium-und-lehre.php.

T 6.179 Module component: Digitalization in Facility and Real Estate Management [T-BGU-108941]

Coordinators: Prof. Dr.-Ing. Kunibert Lennerts
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: M-BGU-105592 - Digitalization in Facility Management

Type	Credits	Grading	Term offered	Expansion	Version
Examination of another type	6 CP	graded	Each term	1 semesters	1

Courses					
WT 25/26	6242907	Digitalization in Facility- and Real Estate Management	4 SWS	Lecture / Practice / 🗣	Lennerts, Mitarbeiter/innen

Legend: 🖥 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
project work incl. report, appr. 15 pages, and presentation/colloquium, appr. 15 min

Prerequisites
none

Recommendations
none

Additional Information
none

Workload
180 hours

T 6.180 Module component: Digitization in the Railway System [T-MACH-113016]

Coordinators: Prof. Dr.-Ing. Martin Cichon
Organisation: KIT Department of Mechanical Engineering
Part of: M-MACH-106496 - Modern Mobility on Rails and Roads

Type	Credits	Grading	Term offered	Expansion	Version
Oral examination	4 CP	graded	Each winter term	1 semesters	2

Courses					
WT 25/26	2115920	Railway System Digitalisation	2 SWS	Lecture / 🗣	Jost, Cichon
WT 26/27	2115920	Railway System Digitalisation	2 SWS	Lecture / 🗣	Jost, Cichon

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Oral examination
Duration: approx. 20 minutes
No tools or reference material may be used during the exam.

Additional Information
The course is offered in German.

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Railway System Digitalisation

2115920, WS 26/27, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content
Qualification goals

Students have a basic understanding of train protection and its technical implementation in Germany, the functioning of the European Train Control System (ETCS) and its planning, Automated Train Operation. They can explain the knowledge they have learned (terms, relationships) in context and apply it to practical issues. Furthermore, students will be able to classify the operational and technical advantages and disadvantages in the context of the digitalization of the rail network in Germany and take future challenges into account.

Students will be able to discuss the technical aspects and areas of application of ETCS at the various levels and describe the main features of balise planning for ETCS Level 2. Digital planning approaches such as PlanPro as well as measurement and test runs are known and can be categorized.

Contents

1. introduction and motivation: organizational aspects; current developments in Germany, Europe

2. railroad system basics: terminology; interaction of vehicle, infrastructure and operation

3. securing train movements: overview of possibilities and areas of application; operational and technical aspects with a focus on Germany

4. fundamentals of signal boxes, control and safety elements: Train protection in Germany with PZB, LZB

5. safety and security: EN5012x, CENELEC, RAMS

6. European Train Control System (ETCS): specification; system components, braking curves; ETCS levels and modes, train integrity; vehicle and infrastructure interface, data exchange; infrastructure-side ETCS balise planning using the example of ETCS Level 2; route measurement, commissioning; digitalization of the planning process using the example of PlanPro

7. automatic train operation (ATO), communication-based train control (CBTC): system architecture, degree of automation (GoA); advantages and challenges of ATO; differences between CTBC and ETCS

8. future developments: Future Railway Mobile Communication System (FRMCS) as successor to GSM-R

Organizational issues
Die Vorlesung wird von unserem Lehrbeauftragten Herrn Dr.-Ing. Franz Jost gelesen.

weitere Informationen unter https://www.fast.kit.edu/bst/929_16400.php

Literature

- ETCS for Engineers, Stanley, 2011, ISBN 978-3-96245-034-2
- European Train Control System (ETCS), Schnieder, ISBN 978-3-662-66054-6
- Communications-Based Train Control (CBTC), Schnieder, ISBN 978-3-662-61012-1

T 6.181 Module component: Dimensioning of Material Flow Systems in Production and Logistics [T-MACH-113950]

Coordinators: Prof. Dr.-Ing. Kai Furmans
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101278 - Material Flow in Networked Logistic Systems
M-MACH-104888 - Advanced Module Logistics

Type	Credits	Grading	Term offered	Expansion	Version
Examination of another type	4 CP	graded	Each winter term	1 semesters	1

Courses					
WT 25/26	2117052	Dimensioning of material flow systems in production and logistics	3 SWS	Lecture / 🗣	Furmans

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, × Cancelled

Assessment
The assessment (Prüfungsleistung anderer Art) consists of an oral examination (approx. 20 minutes) and regular and active participation during the course.

Prerequisites
none

Recommendations

- The contents of the Bachelor course Material Flow Systems in Production and Logistics are assumed to be known and are necessary to follow the course.
- Basic statistical knowledge and understanding.
- Knowledge of a common programming language (Java, Python, ...).

Additional Information
As part of the inverted classroom model, the theoretical content and exercises are taught entirely online. The face-to-face events on campus are used exclusively to apply the knowledge learnt in realistic scenarios.

The course is offered in German.

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Dimensioning of material flow systems in production and logistics

2117052, WS 25/26, 3 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content
Course content:

- Material flow elements (conveyor line, branch, merge)
- Description of networked Material Flow models with graphs, matrices, etc.
- Queueing theory: Calculation of waiting times, utilisation rates, etc.
- Storage and picking
- Shuttle systems, automated storage and retrieval systems
- Value stream analysis
- Lean manufacturing topics

Learning objectives:

After successfully completing the course, you will be able to do the following independently and as part of a team:

- Describe a material flow system accurately in a conversation with experts.
- Model and parameterise the system load and the typical material flow elements.
- Design a material flow system for a specific task.
- Set the performance of a system depending on the requirements.
- Conceptually expand the limits of today's methods and system components as needed.

Description:

The course is divided into 6 thematic blocks, each of which is divided into the following phases and dates:

Off-campus:

- self-study
- exercise

On-campus:

- classroom sessions with practical application

Organizational issues
Anmerkungen: Im Rahmen des Inverted Classroom Modells erfolgt die Vermittlung der theoretischen Inhalte sowie der Übungen vollständig online. Die Präsenzveranstaltungen auf dem Campus dienen ausschließlich dazu, das erlernte Wissen in realitätsnahen Szenarien praktisch anzuwenden.

Erfolgskontrolle: Die Erfolgskontrolle erfolgt in Form einer Prüfungsleistung anderer Art. Die Bewertung setzt sich aus einer mündlichen Prüfung und der regelmäßigen und aktiven Teilnahme an den Kursterminen zusammen.

Empfehlungen:

- (von Vorteil): Statistische Grundkenntnisse und –verständnis.
- (von Vorteil): Kenntnisse in einer gängigen Programmiersprache (Java, Python, …).
- (von Vorteil): Da die Vorlesung auf der Veranstaltung **2118181 – Materialfluss in Produktion und Logistik** aufbaut, werden Kenntnisse aus dieser Veranstaltung vorausgesetzt.

Literature
Arnold, Dieter; Furmans, Kai: Materialfluss in Logistiksystemen; Springer-Verlag Berlin Heidelberg, 7. Auflage 2019

T 6.182 Module component: Disassembly Process Engineering [T-BGU-101850]

Coordinators: Prof. Dr.-Ing. Sascha Gentes
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: M-BGU-101110 - Process Engineering in Construction
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Expansion	Version
Oral examination	3 CP	graded	Each summer term	1 semesters	1

Courses					
ST 2026	6243803	Dismantling Techniques	2 SWS	Lecture / Practice / 🗣	Gentes

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
oral exam, appr. 20 min.

Prerequisites
none

Recommendations
none

Additional Information
none

Workload
90 hours

T 6.183 Module component: Discrete-Event Simulation in Production and Logistics [T-WIWI-102718]

Coordinators: Hon.-Prof. Dr. Sven Spieckermann
Organisation: KIT Department of Business and Economics
Part of: M-MACH-104888 - Advanced Module Logistics
M-WIWI-102805 - Service Operations
M-WIWI-102832 - Operations Research in Supply Chain Management
M-WIWI-104899 - Operations Research

Type	Credits	Grading	Term offered	Version
Examination of another type	4,5 CP	graded	Each summer term	2

Courses					
ST 2026	2550488	Ereignisdiskrete Simulation in Produktion und Logistik	3 SWS	Lecture / 🗣	Spieckermann

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment consists of a written paper and an oral exam of about 30-40 min (alternative exam assessment).

Prerequisites
None

Recommendations
Basic knowledge as conveyed in the module "Introduction to Operations Research" is assumed.

Additional Information
Due to capacity restrictions, registration before course start is required. For further information see the webpage of the course.

The course is planned to be held every summer term.

The planned lectures and courses for the next three years are announced online.

Below you will find excerpts from courses related to this module component:

V Ereignisdiskrete Simulation in Produktion und Logistik

2550488, SS 2026, 3 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content
Simulation of production and logistics systems is an interdisciplinary subject connecting expert knowledge from production management and operations research with mathematics/statistics as well as computer science and software engineering. With completion of this course, students know statistical foundations of discrete simulation, are able to classify and apply related software applications, and know the relation between simulation and optimization as well as a number of application examples. Furthermore, students are enabled to structure simulation studies and are aware of specific project scheduling issues.

Organizational issues
Den Bewerbungszeitraum finden Sie auf der Veranstaltungswebseite im Lehre-Bereich unter dol.ior.kit.edu

Literature

- Gutenschwager K., Rabe M., Spieckermann S. und S. Wenzel (2017): Simulation in Produktion und Logistik, Springer, Berlin.
- Banks J., Carson II J. S., Nelson B. L., Nicol D. M. (2010) Discrete-event system simulation, 5.Aufl., Pearson, Upper Saddle River.
- Eley, M. (2012): Simulation in der Logistik - Einführung in die Erstellung ereignisdiskreter Modelle unter Verwendung des Werkzeuges "Plant Simulation“, Springer, Berlin und Heidelberg
- Kosturiak, J. und M. Gregor (1995): Simulation von Produktionssystemen. Springer, Wien und New York.
- Law, A. M. (2015): Simulation Modeling and Analysis. 5th Edition, McGraw-Hill, New York usw.
- Liebl, F. (1995): Simulation. 2. Auflage, Oldenbourg, München.
- Noche, B. und S. Wenzel (1991): Marktspiegel Simulationstechnik. In: Produktion und Logistik. TÜV Rheinland, Köln.
- Pidd, M. (2004): Computer Simulation in Management Science. 5th Edition, Wiley, Chichester.
- Robinson S (2004) Simulation: the practice of model development and use. John Wiley & Sons, Chichester
- VDI (2014): Simulation von Logistik-, Materialfluß- und Produktionssystemen. VDI Richtlinie 3633, Blatt 1, VDI-Verlag, Düsseldorf.

T 6.184 Module component: Dynamic Macroeconomics [T-WIWI-109194]

Coordinators: Prof. Dr. Johannes Brumm
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104908 - Economics
M-WIWI-107010 - Economics in a Connected World
M-WIWI-107011 - Economics of Innovation and Growth
M-WIWI-107187 - Quantitative Core

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each winter term	4

Courses					
WT 25/26	2560402	Dynamic Macroeconomics	2 SWS	Lecture /	Brumm, Schang
WT 25/26	2560403	Übung zu Dynamic Macroeconomics	1 SWS	Practice /	Hußmann, Kissling

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
The assessment is a written exam (60 min.).

Prerequisites
None.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Dynamic Macroeconomics

2560402, WS 25/26, 2 SWS, Language: English, Open in study portal

Lecture (V)
Blended (On-Site/Online)

Content
This course addresses macroeconomic questions on an advanced level. The main focus of this course is on quantitative dynamic models and their fundamental role in modern macroeconomics in explaining observed phenomena and their use as a laboratory for policy exercises.

The first part of the course introduces seminal workhorse models in the representative agent framework. We will cover the Ramsey model for optimal growth, the Real Business Cycle (RBC) model for business cycle dynamics, and the New Keynesian (NK) model for analysing monetary economics. The second part expands these basic models to incorporate household heterogeneity and incomplete markets. We will analyse the Aiyagari model and the state-of-the-art Heterogeneous Agent New Keynesian (HANK) model to investigate more realistic economic inequality and analyse the distributional impact of policies along the income and wealth distribution.

Alongside the economic models, the course introduces advanced methods to numerically solve, simulate and estimate quantitative models in economics. The course pursues a hands-on approach: students will gain not only theoretical insights but also learn numerical tools to solve dynamic economic models using the programming language Python. The course will be an excellent preparation for writing a thesis at the Chair of Macroeconomics and for engaging with economic research more generally.

Literature
Literatur und Skripte werden in der Veranstaltung angegeben.

T 6.185 Module component: Dynamic Systems of Technical Logistics [T-MACH-112113]

Coordinators: Dr.-Ing. Martin Mittwollen
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-104888 - Advanced Module Logistics

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each summer term	2

Courses					
ST 2026	2148605	Dynamic Systems of Technical Logistics (DSTL)	3 SWS	Lecture / Practice / 🗣	Mittwollen

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment consists of an oral exam (approx. 20min) taking place in the recess period according to § 4 paragraph 2 Nr. 2 of the examination regulation.

Prerequisites
none

Recommendations
Knowledge out of **"Basics of Technical Logistics" (LV 2117057)** preconditioned.

Additional Information
The course is offered in German

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Dynamic Systems of Technical Logistics (DSTL)

2148605, SS 2026, 3 SWS, Language: German, Open in study portal

Lecture / Practice (VÜ)
On-Site

Content
Materials handling = motion = dynamics

Insight into the structure, mode of operation, dynamics and safety of materials handling equipment along the process chain of technical logistics from raw material extraction through processing, distribution, storage and order picking to shipping.

- Bulk material mining, transport, handling, storage
- Stability and tipping safety when turning, slewing, driving cranes
- Overhead cranes - structure, dynamics, safety
- Conveyors in material handling systems (belt, chain, AGV, EMS, ...)
- Elevators - structure, dynamics, safety
- Material flow systems - structure, basic elements, information flow
- Storage and racking systems - structure, dynamics, order picking
- Storage and retrieval systems - structure, dynamics, safety

Organizational issues
DSTL und DSTL-P sind zeitlich so gegliedert, dass zunächst ausschließlich der Vorlesungs- und Übungsteil bis ca. Anfang Juni gehalten wird. Der anschließende Zeitraum ist ausschließlich für die (optionale) Projektarbeit vorgesehen.

DSTL-P ist nur als Ergänzung zu DSTL belegbar, nicht als eigenständige Einzelleistung

T 6.186 Module component: Dynamic Systems of Technical Logistics - Project [T-MACH-112114]

Coordinators: Dr.-Ing. Martin Mittwollen
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-104888 - Advanced Module Logistics

Type	Credits	Grading	Term offered	Version
Examination of another type	2 CP	graded	Each summer term	2

Courses					
ST 2026	2148606	Dynamic Systems of Technical Logistics - Project (DSTL-P)	1 SWS	Project / Blended	Mittwollen

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
The success control takes place in the form of an examination performance of a different kind. This is composed as follows:

assessment of a project report in the form of a presentation with commentary
assessment of the presentation followed by a colloquium

Prerequisites
T-MACH-112113 (Dynamic Systems of Technical Logistics) must have been started.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-MACH-112113 - Dynamic Systems of Technical Logistics must have been started.

Additional Information
The course is offered in German

Workload
60 hours

Below you will find excerpts from courses related to this module component:

V Dynamic Systems of Technical Logistics - Project (DSTL-P)

2148606, SS 2026, 1 SWS, Language: German, Open in study portal

Project (PRO)
Blended (On-Site/Online)

Content
Conveyor technology = motion = dynamics
Course content:
The knowledge acquired in the lecture DSTL will be extended and deepened together with the previous knowledge from GTL in the context of an independent project work based on an application case from the current research and project work at IFL. Analyses, research, design work, calculations and simulations are used.

Organizational issues
DSTL und DSTL-P sind zeitlich so gegliedert, ausschließlich der Vorlesungs- und Übungsteil bis ca. Anfang Juni gehalten wird. Der anschließende Zeitraum ist ausschließlich für die (optionale) Projektarbeit vorgesehen.

DSTL-P ist nur als Ergänzung zu DSTL belegbar, nicht als eigenständige Einzelleistung

T 6.187 Module component: Economic Decision Making [T-WIWI-114174]

Coordinators: Prof. Dr. Benjamin Scheibehenne
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration
M-WIWI-105312 - Marketing and Sales Management
M-WIWI-105714 - Consumer Research
M-WIWI-105923 - Incentives, Interactivity & Decisions in Organizations
M-WIWI-106258 - Digital Marketing

Type	Credits	Grading	Term offered	Expansion	Version
Examination of another type	4,5 CP	graded	Each winter term	1 semesters	1

Courses					
WT 25/26	2500059	Economic Decision Making	3 SWS	Lecture / Practice / 🗣	Scheibehenne
WT 26/27	2500059	Economic Decision Making	3 SWS	Lecture / Practice / 🗣	Scheibehenne

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Alternative exam assessment. The course format consists of a lecture and a corresponding exercise. The grading includes the following aspects:

- A written exam (60 minutes)
- A presentation during the exercise
- Regular attendance
- In-class assignments

The scoring system for the grading will be announced at the beginning of the course.

Prerequisites
Registration via the WiWi Portal is required for participation in the Übung. The Übung is a prerequisite for the exam.

Additional Information
The judgments and decisions that we make can have long ranging and important consequences for our (financial) well-being and individual health. Hence, the goal of this lecture is to gain a better understanding of how people make judgments and decisions and the factors that influences their behavior. We will look into simple heuristics and mental shortcuts that decision makers use to navigate their environment, in particular so in an economic context. Following this, the lecture will provide an overview into social and emotional influences on decision making. In the second half of the semester we will look into some more specific topics including self-control, nudging, and food choice. The last part of the lecture will focus on risk communication and risk perception. We will address these questions from an interdisciplinary perspective at the intersection of Psychology, Behavioral Economics, Marketing, Cognitive Science, and Biology. Across all topics covered in class, we will engage with basic theoretical work as well as with groundbreaking empirical research and current scientific debates.
The workload of the class is 4.5 ECTS. This consists of 3 ETCS for the lecture and 1.5 ETCS for the Übung. Details about the Übung will be communicated at the first day of the class.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Economic Decision Making

Lecture / Practice (VÜ)
On-Site

2500059, WS 25/26, 3 SWS, Language: English, Open in study portal

Content

The judgments and decisions that we make can have long ranging and important consequences for our (financial) well-being and individual health. Hence, the goal of this lecture is to gain a better understanding of how people make judgments and decisions and the factors that influences their behavior. We will look into simple heuristics and mental shortcuts that decision makers use to navigate their environment, in particular so in an economic context. Following this, the lecture will provide an overview into social and emotional influences on decision making. In the second half of the semester we will look into some more specific topics including self-control, nudging, and food choice. The last part of the lecture will focus on risk communication and risk perception.
We will address these questions from an interdisciplinary perspective at the intersection of Psychology, Behavioral Economics, Marketing, Cognitive Science, and Biology. Across all topics covered in class, we will engage with basic theoretical work as well as with groundbreaking empirical research and current scientific debates.
The workload of the class is 4.5 ECTS. This consists of 3 ETCS for the lecture and 1.5 ETCS for the Übung. Details about the Übung will be communicated at the first day of the class

V	**Economic Decision Making** 2500059, WS 26/27, 3 SWS, Language: English, Open in study portal	**Lecture / Practice (VÜ)** **On-Site**

Content

The judgments and decisions that we make can have long ranging and important consequences for our (financial) well-being and individual health. Hence, the goal of this lecture is to gain a better understanding of how people make judgments and decisions and the factors that influences their behavior. We will look into simple heuristics and mental shortcuts that decision makers use to navigate their environment, in particular so in an economic context. Following this, the lecture will provide an overview into social and emotional influences on decision making. In the second half of the semester we will look into some more specific topics including self-control, nudging, and food choice. The last part of the lecture will focus on risk communication and risk perception.
We will address these questions from an interdisciplinary perspective at the intersection of Psychology, Behavioral Economics, Marketing, Cognitive Science, and Biology. Across all topics covered in class, we will engage with basic theoretical work as well as with groundbreaking empirical research and current scientific debates.
The workload of the class is 4.5 ECTS. This consists of 3 ETCS for the lecture and 1.5 ETCS for the Übung. Details about the Übung will be communicated at the first day of the class

Organizational issues

zusätzlich donnerstags, gleiche Uhrzeit, bis Weihnachten, Raum wird noch bekannt gegeben

T 6.188 Module component: Economics and Behavior [T-WIWI-102892]

Coordinators: Jun.-Prof. Dr. Wladislaw Mill
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104908 - Economics

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each winter term	1

Courses					
WT 25/26	2560137	Economics and Behavior	2 SWS	Lecture / 🗣	Mill
WT 25/26	2560138	Übung zu Economics and Behavior	1 SWS	Practice / 🗣	Mill

Legend: 🖥 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The exam takes place in every semester. Re-examinations are offered at every ordinary examination date.

Prerequisites
None

Recommendations
Basic knowledge of microeconomics and statistics are recommended. A background in game theory is helpful, but not absolutely necessary.

Additional Information
The lecture will be held in English.

Below you will find excerpts from courses related to this module component:

V Economics and Behavior

2560137, WS 25/26, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content
The course covers topics from behavioral economics with regard to contents and methods. In addition, the students gain insight into the design of economic experiments. Furthermore, the students will become acquainted with reading and critically evaluating current research papers in the field of behavioral economics.

The students

- gain insight into fundamental topics in behavioral economics;
- get to know different research methods in the field of behavioral economics;
- learn to critically evaluate experimental designs;
- get introduced to current research papers in behavioral economics;
- become acquainted with the technical terminology in English.

The assessment consists of a written exam (60 minutes) (following §4(2), 1 of the examination regulation).
The exam takes place in every semester. Re-examinations are offered at every ordinary examination date.

The grade will be determined in a final written exam. Students can earn a bonus to the final grade by successfully participating in the exercises.

The total workload for this course is approximately 135.0 hours. For further information see German version.

The lecture will be held in English.

Recommendations:

Basic knowledge of microeconomics and statistics are recommended. A background in game theory is helpful, but not absolutely necessary.

Literature
Kahnemann, Daniel: Thinking, Fast and Slow. Farrar, Straus and Giroux, 2011.

T 6.189 Module component: Economics I: Microeconomics [T-WIWI-102708]

Coordinators: Prof. Dr. Clemens Puppe
Prof. Dr. Johannes Philipp Reiß
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104908 - Economics

Type	Credits	Grading	Term offered	Version
Written examination	5 CP	graded	Each winter term	1

Courses					
WT 25/26	2610012	Economics I: Microeconomics	3 SWS	Lecture / 🗣	Puppe, Okulicz
WT 25/26	2610013		2 SWS	Tutorial / 🗣	Puppe, Okulicz

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
The assessment consists of a written exam (120 min) following §4, Abs. 2, 1 of the examination regulation.

The main exam takes place subsequent to the lectur. The re-examination is offered at the same examination period. As a rule, only repeating candidates are entitled for taking place the re-examination. For a detailed description on the exam regulations see the information of the respective chair.

Prerequisites
None

Below you will find excerpts from courses related to this module component:

V Economics I: Microeconomics

Lecture (V)
On-Site

2610012, WS 25/26, 3 SWS, Language: German, Open in study portal

Content
This course provides a solid grounding in microeconomic theory. The two main parts of the course deal with questions of microeconomic decision theory (household and firm decisions) and questions of market theory (equilibria and efficiency on competitive markets). The last part of the lecture deals with problems of imperfect competition (oligopoly markets) as well as the basics of game theory and welfare economics.

Learning objectives:
The main aim of the course is to teach students the basics of thinking in microeconomic models. In particular, students should be able to analyze goods markets and the determinants of market outcomes. In detail, students will learn

- to name and define the basic microeconomic terms.
- to explain the interrelationships in microeconomic models.
- to calculate the important parameters of microeconomic models.
- to judge the possible effects of economic policy measures on the behavior of economic agents (in simple decision problems) and possibly propose alternative measures.
- to analyze as a participant in a tutorial simple microeconomic problems by solving written exercises and presenting the results of the exercises on the blackboard.
- to become familiar with the basic literature on microeconomics.

In this way, students acquire the necessary basic knowledge

- to recognize the structure of economic problems on a microeconomic level and develop proposals for solutions.
- to provide active decision support for simple economic decision problems.

Workload:
Total workload for 5 credit points: approx. 150 hours
Attendance: 45 hours
Self-study: 105 hours

Literature

- Varian, H. R. 2016. *Grundzüge der Mikroökonomik*. 9. Auflage. De Gruyter Oldenburg Verlag.
- Pindyck, R. S. und Rubinfeld, D. L. 2015. *Mikroökonomie*. 8. Auflage. Pearson.
- Frank, R. H. 2006. *Microeconomics and Behavior*. 6. Auflage. McGraw-Hill/Irwin.

T 6.190 Module component: Economics II: Macroeconomics [T-WIWI-102709]

Coordinators: Prof. Dr. Berthold Wigger
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104908 - Economics

Type	Credits	Grading	Term offered	Version
Written examination	5 CP	graded	Each summer term	1

Courses					
ST 2026	2600014	Economics II: Macroeconomics	4 SWS	Lecture	Brumm
ST 2026	2660015	Economics II : Macroeconomics, Tutorial	2 SWS	Tutorial	Brumm, Kissling

Assessment
Success is assessed in the form of a written examination (120 minutes). The examination is offered every semester and can be repeated at any regular examination date.

Prerequisites
None

Below you will find excerpts from courses related to this module component:

V Economics II: Macroeconomics

Lecture (V)

2600014, SS 2026, 4 SWS, Language: German, Open in study portal

Content

Classical Theory of Macroeconomic Production

Chapter 1: Gross domestic product

Chapter 2: Money and Inflation

Chapter 3: Open Economy I

Chapter 4: Unemployment

Growth: The economy in the long term

Chapter 5: Growth I

Chapter 6: Growth II

Business cycle: The economy in the short term

Chapter 7: Economy and aggregate demand I

Chapter 8: Economy and aggregate demand II

Chapter 9: Open Economy II

Chapter 10: Macroeconomic supply

Advanced topics of macroeconomics

Chapter 11: Dynamic model of the economy as a whole

Chapter 12: Microeconomic foundations

Chapter 13: Macroeconomic economic policy

Learning goals:

The students. . .

- can name the basic indicators, technical terms and concepts of macroeconomics.
- can use models to reduce complex relationships to their basic components.
- can analyse economic policy debates and form their own opinion on them.

Workload:

Total effort for 5 credit points: approx. 150 hours

Presence time: 45 hours

Before and after the LV: 67.5 hours

Exam and exam preparation: 37.5 hours

Literature

Als Grundlage dieser Veranstaltung dient das bekannte Lehrbuch „Makroökonomik“ von Greg Mankiw vom Schäffer Poeschel Verlag in der aktuellen Fassung.

T 6.191 Module component: Economics of Innovation [T-WIWI-112822]

Coordinators: Prof. Dr. Ingrid Ott
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration
M-WIWI-107010 - Economics in a Connected World
M-WIWI-107011 - Economics of Innovation and Growth

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each summer term	1

Courses					
ST 2026	2560236	Economics of Innovation	2 SWS	Lecture / 🗣	Ott
ST 2026	2560237	Exercises of Economics of Innovation	1 SWS	Practice / 🗣	Ott, Mirzoyan

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success is assessed in the form of a written examination (60 minutes). The examination is offered every semester and can be repeated at any regular examination date.
A bonus can be earned by completing a short written assignment and presenting it in the tutorial. If the grade of the written examination is between 4.0 and 1.3, the bonus improves the grade by a maximum of one grade level (0.3 or 0.4). The exact criteria for awarding a bonus will be announced at the beginning of the lecture.

Prerequisites
None

Recommendations
Basic knowledge of micro- and macroeconomics is assumed, as taught in the courses Economics I [2600012], and Economics II [2600014]. In addition, an interest in quantitative-mathematical modeling is required.

Additional Information
Due to Prof. Dr. Ingrid Ott's research semester, the lectures for this course will not be offered in the summer semester 2026.

Below you will find excerpts from courses related to this module component:

V Economics of Innovation

Lecture (V)
On-Site

2560236, SS 2026, 2 SWS, Language: English, Open in study portal

Content
Learning objectives:

Students shall be given the ability to

- identify the importance of alternative incentive mechanisms for the emergence and dissemination of innovations
- understand the relationships between market structure and the development of innovation
- explain, in which situations market interventions by the state, for example taxes and subsidies, can be legitimized, and evaluate them in the light of economic welfare

Course content:

The course covers the following topics:

- Incentives for the emergence of innovations
- Patents
- Diffusion
- Impact of technological progress
- Innovation Policy

Recommendations:

Basic knowledge of micro- and macroeconomics is assumed, as taught in the courses Economics I [2600012], and Economics II [2600014]. In addition, an interest in quantitative-mathematical modeling is required.

Workload:

The total workload for this course is approximately 135.0 hours. For further information see German version.

Exam description:

The assessment consists of a written exam (60 min) according to Section 4(2), 1 of the examination regulation. The exam takes place in every semester. Re-examinations are offered at every ordinary examination date.

Students will be given the opportunity of writing and presenting a short paper during the lecture time to achieve a bonus on the exam grade. If the mandatory credit point exam is passed, the awarded bonus points will be added to the regular exam points. A deterioration is not possible by definition, and a grade does not necessarily improve, but is very likely to (not every additional point improves the total number of points, since a grade can not become better than 1). The voluntary elaboration of such a paper can not countervail a fail in the exam.

Organizational issues
Die Vorlesung wird im SoSe 2026 aufgrund eines Forschungssemesters nicht gelesen; Übung und Klausur finden statt.

Vorbereitungsmaterialien finden Sie im ILIAS.

Literature
Auszug:

- Aghion, P., Howitt, P. (2009), The Economics of Growth, MIT Press, Cambridge MA.
- de la Fuente, A. (2000), Mathematical Methods and Models for Economists. Cambridge University Press, Cambridge, UK.
- Klodt, H. (1995), Grundlagen der Forschungs- und Technologiepolitik. Vahlen, München.
- Linde, R. (2000), Allokation, Wettbewerb, Verteilung - Theorie, UNIBUCH Verlag, Lüneburg.
- Ruttan, V. W. (2001), Technology, Growth, and Development. Oxford University Press, Oxford.
- Scotchmer, S. (2004), Incentives and Innovation, MIT Press.
- Tirole, Jean (1988), The Theory of Industrial Organization, MIT Press, Cambridge MA.

T 6.192 Module component: Efficient Energy Systems and Electric Mobility [T-WIWI-102793]

Coordinators: Prof. Dr. Patrick Jochem
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101452 - Energy Economics and Technology
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	3,5 CP	graded	Each summer term	1

Courses					
ST 2026	2581006	Efficient Energy Systems and Electric Mobility	2 SWS	Lecture /	Jochem

Legend: Online, Blended (On-Site/Online), On-Site, ✕ Cancelled

Assessment
The assessment consists of a written exam (60 minutes) (following §4(2) of the examination regulation). The exam takes place in every semester. Re-examinations are offered at every ordinary examination date.

Prerequisites
None

Recommendations
Literature:

Masters, G. M. & K. F. Hsu (2023). Renewable and efficient electric power systems. John Wiley & Sons. (Chapter 9).
Viral, R., Tomar, A., Asija, D., Rao, U. M., & Sarwar, A. (Eds.). (2024). Smart Grids for Renewable Energy Systems, Electric Vehicles and Energy Storage Systems. CRC Press.

Below you will find excerpts from courses related to this module component:

V Efficient Energy Systems and Electric Mobility

2581006, SS 2026, 2 SWS, Language: German/English, Open in study portal

Lecture (V)
On-Site

Content
This lecture series gives you a systems-research view of how we can deliver the same (or better) services – heating, cooling, lighting, passenger- and freight-transport – while consuming less primary energy. The workload of the class is 3.5 ECTS (including homework). The course is split into two linked parts:

Part 1 – Energy-Efficient Systems

- **Why energy efficiency matters**: we start with the climate, economic and security arguments that make efficiency the "low-hanging fruit" of the energy transition.
- **Core concepts & analytical tools**: you will learn the definition of energy efficiency, the distinction between physical and monetary output, and how to set system boundaries (building, district, national) including core concepts such as **Kaya identity** and basic **Decomposition Analysis**.
- **Barriers and rebound effects**: financial, institutional and behavioural obstacles that slow investment are discussed, together with the paradox that efficiency gains can sometimes increase total energy use (backfire).
- **Policy landscape**: a short tour of the European Green Deal, Fit-for-55, the EU Energy Efficiency Directive, the EU Energy Performance of Buildings Directive and a broad theoretical overview of policy measures.
- **Technical options for the built environment**: heat-loss physics (U-values, heating degree days, temperature response functions), combined heat-and-power (CHP), heat-pump fundamentals, and the role of decentralized energy management.

By the end of Part 1 you will be able to quantify potential energy savings for a building, identify the most relevant barriers, and propose a realistic policy or technology package.

Part 2 – Electric Mobility

- **Transport's climate imprint**: growth trends for passenger and freight travel, projected CO_2 emissions to 2050, and the economic risks of Europe's heavy oil import dependence.
- **Vehicle-level efficiency**: limits of internal-combustion engines, well-to-wheel (WTW) efficiencies of ICEVs, battery-electric vehicles (BEVs) and fuel-cell vehicles (FCEVs), and the externalities (noise, air-pollution, accidents, congestion) that improve with electrification.
- **Analytical frameworks**: you will apply the **Kaya/IPAT** identity to transport and the **A-S-I-F** (Avoid -Shift-Improve-Finance) matrix to design mitigation pathways.
- **EV technology basics**: definitions of BEV, PHEV, HEV and FCEV; battery chemistries, performance metrics, ageing mechanisms, recycling and second-life concepts.
- **Market development & diffusion**: a brief history of electric cars, current global sales, the "Norway effect", and diffusion models (Bass curve, Multi-Level Perspective).
- **Charging infrastructure and grid interaction**: AC/DC chargers, home vs. public charging behaviour, load-profile impacts, smart-charging, vehicle-to-grid (V2G) services and the flexibility potential for electricity markets.
- **Policy toolbox for decarbonising transport**: taxes, subsidies, CO_2 fleet standards, ICEV bans, zero-emission-vehicle mandates, labeling, information campaigns, and cross-sectoral measures (urban planning, standardisation).
- **Life-cycle and WTW assessment**: a simplified LCA of a passenger car, focusing on production, battery manufacturing, recycling and the sensitivity to the electricity mix.

By the end of the course you will be equipped to analyse, design and communicate energy-efficient solutions for buildings and transport – a key skill set for the climate-neutral economy of the future.

Organizational issues
Termine: 08.05., 22.05., 05.06., 19.06., 26.06., 03.07., 17.07., 24.07.26

Literature
Wird in der Vorlesung bekanntgegeben.

T 6.193 Module component: eFinance: Information Systems for Securities Trading [T-WIWI-110797]

Coordinators: Prof. Dr. Christof Weinhardt
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101446 - Market Engineering
M-WIWI-101480 - Finance 3
M-WIWI-101483 - Finance 2
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each winter term	2

Courses					
WT 25/26	2540454	eFinance: Information Systems for Securities Trading	2 SWS	Lecture / 🗣	Weinhardt
WT 25/26	2540455	Übungen zu eFinance: Information Systems for Securities Trading	1 SWS	Practice / 🗣	Motz, Motz

Legend: 🖥 Online, ❖ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success is monitored by means of ongoing elaborations and presentations of tasks and an examination (60 minutes) at the end of the lecture period. The scoring scheme for the overall evaluation will be announced at the beginning of the course.

Additional Information
The course"eFinance: Information Systems for Securities Trading" covers different actors and their function in the securities industry in-depth, highlighting key trends in modern financial markets, such as Distributed Ledger Technology, Sustainable Finance, and Artificial Intelligence. Security prices evolve through a large number of bilateral trades, performed by market participants that have specific, well-regulated and institutionalized roles. Market microstructure is the subfield of financial economics that studies the price formation process. This process is significantly impacted by regulation and driven by technological innovation. Using the lens of theoretical economic models, this course reviews insights concerning the strategic trading behaviour of individual market participants, and models are brought market data. Analytical tools and empirical methods of market microstructure help to understand many puzzling phenomena in securities markets.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V eFinance: Information Systems for Securities Trading

2540454, WS 25/26, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Literature

- Picot, Arnold, Christine Bortenlänger, Heiner Röhrl (1996): "Börsen im Wandel". Knapp, Frankfurt
- Harris, Larry (2003): "Trading and Exchanges - Market Microstructure for Practitioners"". Oxford University Press, New York

Weiterführende Literatur:

- Gomber, Peter (2000): "Elektronische Handelssysteme - Innovative Konzepte und Technologien". Physika Verlag, Heidelberg
- Schwartz, Robert A., Reto Francioni (2004): "Equity Markets in Action - The Fundamentals of Liquidity, Market Structure and Trading". Wiley, Hoboken, NJ

T 6.194 Module component: Elective Specialization Supplementary Studies on Science, Technology and Society / About Knowledge and Science - Self-Registration [T-FORUM-113580]

Coordinators: Dr. Christine Mielke
Christine Myglas

Organisation: General Studies. Forum Science and Society (FORUM)

Part of: M-FORUM-106753 - Supplementary Studies on Science, Technology and Society

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Each term	1

Assessment
Another type of examination assessment under § 5, section 3 involves a presentation, term paper, or project work within the chosen course.

Prerequisites
None

Self Service Assignment of Supplementary Studies
This module component can be used for self service assignment of grades acquired from the following study providers:

- Studium Generale. Forum Wissenschaft und Gesellschaft (FORUM) (ehem. ZAK)
- FORUM (ehem. ZAK) Begleitstudium

Recommendations
The contents of the basic module are helpful. The basic module should be completed or attended in parallel, but not after the advanced module.
The reading recommendations for primary and specialist literature are determined individually by the respective lecturers according to the subject area and course.

Additional Information
This placeholder can be used for any achievement in the Advanced Unit of the Supplementary Studies.

In the Advanced Module, students can choose their own individual focus, e.g. sustainable development, data literacy, etc. The focus should be discussed with the module coordinator at the FORUM.

T 6.195 Module component: Elective Specialization Supplementary Studies on Science, Technology and Society / Science in Public Debates - Self Registration [T-FORUM-113582]

Coordinators: Dr. Christine Mielke
Christine Myglas
Organisation: General Studies. Forum Science and Society (FORUM)
Part of: M-FORUM-106753 - Supplementary Studies on Science, Technology and Society

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Each term	1

Assessment
Another type of examination assessment under § 5, section 3 involves a presentation, term paper, or project work within the chosen course.

Prerequisites
None

Self Service Assignment of Supplementary Studies
This module component can be used for self service assignment of grades acquired from the following study providers:

- Studium Generale. Forum Wissenschaft und Gesellschaft (FORUM) (ehem. ZAK)
- FORUM (ehem. ZAK) Begleitstudium

Recommendations
The contents of the basic module are helpful. The basic module should be completed or attended in parallel, but not after the advanced module.
The reading recommendations for primary and specialist literature are determined individually by the respective lecturers according to the subject area and course.

Additional Information
This placeholder can be used for any achievement in the Advanced Unit of the Supplementary Studies.

T 6.196 Module component: Elective Specialization Supplementary Studies on Science, Technology and Society / Science in Society - Self-Registration [T-FORUM-113581]

Coordinators: Dr. Christine Mielke
Christine Myglas

Organisation: General Studies. Forum Science and Society (FORUM)

Part of: M-FORUM-106753 - Supplementary Studies on Science, Technology and Society

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Each term	1

Assessment
Another type of examination assessment under § 5, section 3 involves a presentation, term paper, or project work within the chosen course.

Prerequisites
None

Self Service Assignment of Supplementary Studies
This module component can be used for self service assignment of grades acquired from the following study providers:

- Studium Generale. Forum Wissenschaft und Gesellschaft (FORUM) (ehem. ZAK)
- FORUM (ehem. ZAK) Begleitstudium

Recommendations
The contents of the basic module are helpful. The basic module should be completed or attended in parallel, but not after the advanced module.
The reading recommendations for primary and specialist literature are determined individually by the respective lecturers according to the subject area and course.

Additional Information
This placeholder can be used for any achievement in the Advanced Unit of the Supplementary Studies.

6.197 Module component: Electric Energy Systems [T-ETIT-112850]

Coordinators: Prof. Dr.-Ing. Marc Hiller
Prof. Dr.-Ing. Thomas Leibfried

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Written examination	6 CP	graded	Each summer term	1

Courses					
ST 2026	2306200	Electric Energy Systems	2 SWS	Lecture / 🗣	Hiller, Leibfried
ST 2026	2306201	Practice to Electric Energy Systems	2 SWS	Practice / 🗣	Hiller, Leibfried

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Prerequisites
none

T 6.198 Module component: Electric Power Transmission & Grid Control [T-ETIT-110883]

Coordinators: Prof. Dr.-Ing. Thomas Leibfried
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-101164 - Generation and Transmission of Renewable Power
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Expansion	Version
Written examination	6 CP	graded	Each summer term	1 semesters	2

Courses					
ST 2026	2307376	Electric Power Transmission & Grid Control	2 SWS	Lecture / 🗣	Leibfried
ST 2026	2307377	Tutorial for 2307376 Electric Power Transmission & Grid Control	2 SWS	Practice / 🗣	Weber

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The examination takes place in form of a written examination lasting 120 minutes. The module grade is the grade of the written exam.

Prerequisites
none

T 6.199 Module component: Electrical Engineering for Business Engineers, Part I [T-ETIT-100533]

Coordinators: Dr.-Ing. Matthias Brodatzki
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Written examination	3 CP	graded	Each winter term	1

Courses					
WT 25/26	2304223	Electrical Engineering for Business Engineers, Part I	2 SWS	Lecture / 🗣	Brodatzki, Beck
WT 25/26	2304225	Electrical Engineering for Business Engineers, Part I (Exercise to 2304223)	2 SWS	Practice / 🗣	Brodatzki

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, × Cancelled

T 6.200 Module component: Electrical Engineering for Business Engineers, Part II [T-ETIT-100534]

Coordinators: Dr.-Ing. Matthias Brodatzki
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Written examination	5 CP	graded	Each summer term	1

Courses					
ST 2026	2304224	Elektrotechnik II für Wirtschaftsingenieure	3 SWS	Lecture / 🗣	Brodatzki

Legend: 🖥 Online, ✪ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

T 6.201 Module component: Electromagnetic Energy in Process Engineering - Oral Exam [T-CIWVT-114829]

Coordinators: Dr. Alexander Navarrete Munoz
Organisation: KIT Department of Chemical and Process Engineering
Part of: M-CIWVT-107566 - Electromagnetic Energy in Process Engineering

Type	Credits	Grading	Version
Oral examination	5 CP	graded	1

Courses					
WT 25/26	2220120	Electromagnetic Energy in Process Engineering	2 SWS	Lecture / 🗣	Dittmeyer, Navarrete Munoz
WT 25/26	2220121	Exercise on 2220120 Electromagnetic Energy in Process Engineering	1 SWS	Practice / 🗣	Dittmeyer, Navarrete Munoz
ST 2026	2220120	Electromagnetic Energy in Process Engineering	2 SWS	Lecture / 🗣	Dittmeyer, Navarrete Munoz
ST 2026	2220121	Exercise on 2220120 Electromagnetic Energy in Process Engineering	1 SWS	Practice / 🗣	Dittmeyer, Navarrete Munoz

Legend: 🖥 Online, ❖ Blended (On-Site/Online), 🗣 On-Site, × Cancelled

Assessment
Learning control is an oral exam lasting approx. 30 minutes.

Prerequisites
None

6.202 Module component: Electronics and EMC [T-ETIT-100723]

Coordinators: Dr. Martin Sack
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-101163 - High-Voltage Technology

Type	Credits	Grading	Term offered	Version
Oral examination	3 CP	graded	Each summer term	1

Courses					
ST 2026	2307378	Electronics and EMC	2 SWS	Lecture / 🗣	Sack

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success is assessed in an overall oral examination (approx. 20 minutes).

Prerequisites
none

T 6.203 Module component: Elements and Systems of Technical Logistics [T-MACH-102159]

Coordinators: Dr.-Ing. Martin Mittwollen
Organisation: KIT Department of Mechanical Engineering

Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each winter term	1

Assessment
The assessment consists of an oral exam (20min) taking place in the recess period according to § 4 paragraph 2 Nr. 2 of the examination regulation.

Prerequisites
none

Recommendations
Knowledge out of "Basics of Technical Logistics I" (T-MACH-109919) preconditioned.

Workload
120 hours

T 6.204 Module component: Elements and Systems of Technical Logistics - Project [T-MACH-108946]

Coordinators: Dr.-Ing. Martin Mittwollen
Organisation: KIT Department of Mechanical Engineering

Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Examination of another type	2 CP	graded	Each winter term	1

Assessment
Presentation of performed project and defense (30min) according to $4 (2), No. 3 of the examination regulation

Prerequisites
T-MACH-102159 (Elements and Systems of Technical Logistics) must have been started

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-MACH-102159 - Elements and Systems of Technical Logistics must have been started.

Recommendations
Knowledge out of "Basics of Technical Logistics I" (T-MACH-109919) preconditioned.

Workload
60 hours

T 6.205 Module component: Emissions into the Environment [T-WIWI-102634]

Coordinators: Ute Karl
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101412 - Industrial Production III
M-WIWI-101471 - Industrial Production II
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	3,5 CP	graded	Each winter term	1

Courses					
WT 25/26	2581962	Emissions into the Environment	2 SWS	Lecture / 🗣	Karl
WT 26/27	2581962	Emissions into the Environment	2 SWS	Lecture / ⯐	Karl

Legend: 🖥 Online, ⯐ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment consists of an oral (30 minutes) or written exam (60 minutes) (following §4(2) of the examination regulation). The exam takes place in every semester. Re-examinations are offered at every ordinary examination date.

Recommendations
None

Workload
105 hours

Below you will find excerpts from courses related to this module component:

V Emissions into the Environment
2581962, WS 25/26, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content
Emission sources/emission monitoring/emission reduction: The lecture gives an overview of relevant emissions of air pollutants and greenhouse gases, emission monitoring and pollutant abatement options together with relevant legal regulations at national and international level. In addition, the fundamentals of circular economy, waste management and recycling are explained.

Structure:

Air pollution control

- Introduction, terms and definitions
- Sources of air pollutants
- Legal framework of air quality control
- Technical measures to reduce air pollutant emissions

Circular economy, recycling and waste management

- Waste collection and logistics
- Dual systems for packaging waste
- Recycling
- Thermal and biological waste treatment
- Final waste disposal

Literature
Wird in der Veranstaltung bekannt gegeben.

V Emissions into the Environment
2581962, WS 26/27, 2 SWS, Language: German, Open in study portal

Lecture (V)
Blended (On-Site/Online)

Content
Emission sources/emission monitoring/emission reduction: The lecture gives an overview of relevant emissions of air pollutants and greenhouse gases, emission monitoring and pollutant abatement options together with relevant legal regulations at national and international level. In addition, the fundamentals of circular economy, waste management and recycling are explained.

Air pollution control:

- Introduction, terms and definitions
- Sources of air pollutants
- Legal framework of air quality control
- Technical measures to reduce air pollutant emissions

Circular economy, recycling and waste management:

- Waste collection and logistics
- Dual systems for packaging waste
- Recycling
- Thermal and biological waste treatment
- Final waste disposal

Literature
Wird in der Veranstaltung bekannt gegeben.

T 6.206 Module component: Employment Law [T-INFO-111436]

Coordinators: Dr. Alexander Hoff
Organisation: KIT Department of Informatics
Part of: M-INFO-101216 - Private Business Law
M-WIWI-104903 - Law

Type	Credits	Grading	Term offered	Version
Written examination	3 CP	graded	Each summer term	2

Courses					
ST 2026	24668	Employment Law	2 SWS	Lecture / 🗣	Steg

Legend: 🖥 Online, ❂ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

T 6.207 Module component: Energetic Refurbishment [T-BGU-111211]

Coordinators: Prof. Dr.-Ing. Kunibert Lennerts
Dr.-Ing. Harald Schneider

Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences

Part of: M-BGU-101884 - Lean Management in Construction
M-BGU-101888 - Project Management in Construction
M-BGU-105592 - Digitalization in Facility Management
M-BGU-106453 - Facility Management in Hospitals for Industrial Engineering

Type	Credits	Grading	Term offered	Expansion	Version
Oral examination	1,5 CP	graded	Each term	1 semesters	1

Courses					
WT 25/26	6240903	Energetic Refurbishment	1 SWS	Lecture / 🗣	Lennerts, Mitarbeiter/innen

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
oral exam, appr. 20 min.

Prerequisites
none

Recommendations
none

Additional Information
none

Workload
45 hours

T 6.208 Module component: Energy and Environment [T-WIWI-102650]

Coordinators: Ute Karl
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101452 - Energy Economics and Technology
M-WIWI-101468 - Environmental Economics
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	3,5 CP	graded	Each summer term	2

Courses					
ST 2026	2581003	Energy and Environment	2 SWS	Lecture / 🗣	Karl

Legend: 🖥 Online, ❂ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment consists of a written exam (60 minutes) (following §4(2) of the examination regulation). The exam takes place in every semester. Re-examinations are offered at every ordinary examination date.

Prerequisites
None.

Workload
105 hours

Below you will find excerpts from courses related to this module component:

V Energy and Environment

2581003, SS 2026, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content
The lecture focuses on the environmental impacts arising from fossil fuels use and on the methods for the evaluation of such impacts. The first part of the lecture describes the environmental impacts of air pollutants and greenhouse gases as well as technical measures for emission control. The second part covers methods of impact assessment and their use in environmental communication as well as methods for the scientific support of emission control strategies.

The topics include:

- Fundamentals of energy conversion
- Formation of air pollutants during combustion
- Technical measures to control emissions from fossil-fuel combustion processes
- External effects of energy supply (life cycle analyses of selected energy systems)
- Environmental communication on energy services (e.g. electricity labelling, carbon footprint)
- Integrated Assessment Modelling to support the European Clean Air Strategy
- Cost-effectiveness analyses and cost-benefit analyses for emission control strategies
- Monetary valuation of external effects (external costs)

Literature
Die Literaturhinweise sind in den Vorlesungsunterlagen enthalten (vgl. ILIAS)

T 6.209 Module component: Energy and Process Technology I [T-MACH-102211]

Coordinators: Prof. Dr.-Ing. Hans-Jörg Bauer
Prof. Dr. Dr. Michael Fischlschweiger
Dr.-Ing. Corina Schwitzke
Dr. Amin Velji

Organisation: KIT Department of Mechanical Engineering

Institute of Thermal Turbomachinery

Part of: M-MACH-101296 - Energy and Process Technology I
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Written examination	9 CP	graded	Each winter term	1

Courses					
WT 25/26	2157961	Energy and Process Technology I	6 SWS	Lecture / Practice / 🗣	Bauer, Wirbser, Mitarbeiter, Wagner, Schwitzke, Reichel

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment consists of a written exam (120 minutes) (following §4(2), 1 of the examination regulation).

Prerequisites
none

Workload
270 hours

Below you will find excerpts from courses related to this module component:

V Energy and Process Technology I

2157961, WS 25/26, 6 SWS, Language: German, Open in study portal

Lecture / Practice (VÜ)
On-Site

Content
The last thrid of the lecture deals with the topic **Thermal Turbomachinery**. The basic principles, the functionality and the scope of application of gas and steam tubrines for the generation of electrical power and propulsion technology are addressed.

The students are able to:

- describe and calculate the basic physical-technical processes
- apply the mathematical and thermodynamical description
- reflect on and explain the diagrams and schematics
- comment on diagrams
- explain the functionality of gas and steam turbines and their components
- name the applications of thermal turbomachinery and their role in the field of electricity generation and propulsion technology

T 6.210 Module component: Energy and Process Technology II [T-MACH-102212]

Coordinators:	Prof. Dr. Dr. Michael Fischlschweiger Dr.-Ing. Corina Schwitzke
Organisation:	KIT Department of Mechanical Engineering Institute of Thermal Turbomachinery
Part of:	M-MACH-101297 - Energy and Process Technology II M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Written examination	9 CP	graded	Each summer term	1

Courses					
ST 2026	2170832	Energy and Process Technology II	6 SWS	Lecture / Practice / 🗣	Schwitzke, Pritz, Wirbser

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment consists of a written exam (120 minutes) (following §4(2), 1 of the examination regulation).

Prerequisites
none

Workload
270 hours

Below you will find excerpts from courses related to this module component:

V Energy and Process Technology II

2170832, SS 2026, 6 SWS, Language: German, Open in study portal

Lecture / Practice (VÜ)
On-Site

Content
Thermal Turbomaschinery - In the first part of the lecture deals with energy systems. Questions regarding global energy resources and their use, especially for the generation and provision of electrical energy, are addressed. Common fossile and nuclear power plants for the centralized supply with electrical power as well as concepts of power-heat cogeneration for the decentralized electrical power supply by means of block-unit heat and power plants, etc. are discussed. Moreover, the characteristics and the potential of renewable energy conversion concepts, such as wind and hydro-power, photovoltaics, solar heat, geothermal energy and fuel cells are compare and evaluated. The focus is on the description of the potentials, the risks and the economic feasibility of the different strategies aimed to protect resources and reduce CO2 emissions.

The students are able to:

- discuss and evaluate energy resources and reserves and their utility
- review the use of energy carriers for electrical power generation
- explain the concepts and properties of power-heat cogeneration, renewable energy conversion and fuel cells and their fields of application
- comment on and compare centralized and decentralized supply concepts
- calculate the potentials, riskis and economic feasibility of different strategies aiming at the protection of resources and the reduction of CO2 emissions
- name and judge on the options for solar energy utilization
- discuss the potential of geothermal energy and its utilization

T 6.211 Module component: Energy Conversion and Increased Efficiency in Internal Combustion Engines [T-MACH-105564]

Coordinators: Prof. Dr. Thomas Koch
Dr.-Ing. Heiko Kubach

Organisation: KIT Department of Mechanical Engineering

Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each winter term	1

Assessment
oral exam, 25 minutes, no auxillary means

Prerequisites
none

T 6.212 Module component: Energy Efficient and Sustainable Tribological Systems [T-MACH-114016]

Coordinators: Prof. Dr. Martin Dienwiebel
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101291 - Microfabrication
M-MACH-101294 - Nanotechnology
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each winter term	1

Assessment
oral exam, about 25 min

Prerequisites
Must not be used together with T-MACH-114015

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-MACH-114015 - Energy Efficient and Sustainable Tribological Systems (in German) must not have been started.

Additional Information
The course is offered in English.

Workload
120 hours

T 6.213 Module component: Energy Efficient and Sustainable Tribological Systems (in German) [T-MACH-114015]

Coordinators: Prof. Dr. Martin Dienwiebel
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101291 - Microfabrication
M-MACH-101294 - Nanotechnology
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each summer term	1

Courses					
ST 2026	2182100	Energy Efficient and Sustainable Tribological Systems	2 SWS	Lecture / 🗣	Dienwiebel

Legend: 🖥 Online, ✲ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
oral exam, about 25 min

Prerequisites
Must not be used together with T-MACH-114016

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-MACH-114016 - Energy Efficient and Sustainable Tribological Systems must not have been started.

Additional Information
The course is offered in German.

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Energy Efficient and Sustainable Tribological Systems

2182100, SS 2026, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content

Friction, lubrication, and wear significantly influence the energy efficiency and carbon footprint of machines and technical processes. The lecture will focus on new technologies for reducing friction that can drastically reduce energy losses caused by friction and wear. Regenerative and bio-based materials and lubricants will also be presented.

Topics

- Review of key tribology concepts
- In-depth study of fundamentals (surface interactions, tribochemistry)
- Low-friction systems: Superlubrication
    - Structural superlubrication
    - Liquid superlubrication
    - Applications
- Sustainable tribological systems
    - Substitution of critical substances
    - Biologically inspired solutions
    - Bio-based lubricants

Learning objectives

The student

- can describe basic tribochemical processes,
- is able to explain concepts and principles for achieving superlubricating systems in technology and nature,
- can name current approaches for substituting critical substances in tribological systems.

Further information

Basic knowledge of physics, chemistry, and materials science is required. Participation in the Tribology lecture is advantageous.

Attendance: 33.5 hours

Self-study: 116.5 hours

Success is assessed in the form of an approximately 30-minute oral exam (according to §4(2), 2 SPO).

T 6.214 Module component: Energy Efficient Intralogistic Systems [T-MACH-105151]

Coordinators: Dr.-Ing. Meike Kramer
Dr. Frank Schönung

Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101278 - Material Flow in Networked Logistic Systems
M-MACH-104888 - Advanced Module Logistics
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each winter term	1

Courses					
WT 25/26	2117500	Energy efficient intralogistic systems	2 SWS	Lecture / 🗣	Kramer, Schönung

Legend: 🖥 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Oral, 30 min. examination dates after the end of each lesson period.

Prerequisites
none

Recommendations
The content of course “Basics of Technical Logistics I” (T-MACH-109919) should be known.

Additional Information
Visit the IFL homepage of the course for the course dates and/or possible limitations of course participation.

The course is offered in German

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Energy efficient intralogistic systems

Lecture (V)
On-Site

2117500, WS 25/26, 2 SWS, Language: German, Open in study portal

Content
The content of course “Basics of Technical Logistics” should be knownn.

Literature
Keine.

T 6.215 Module component: Energy Market Engineering [T-WIWI-107501]

Coordinators: Prof. Dr. Christof Weinhardt
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101411 - Information Engineering
M-WIWI-101446 - Market Engineering
M-WIWI-101451 - Energy Economics and Energy Markets
M-WIWI-103720 - eEnergy: Markets, Services and Systems
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each summer term	1

Courses					
ST 2026	2540464	Energy Market Engineering	2 SWS	Lecture / 🗣	Semmelmann, Miskiw
ST 2026	2540465	Übung zu Energy Market Engineering	1 SWS	Practice / 🗣	Resch

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment consists of a written exam (60 min) (according to §4(2), 1 of the examination regulations). By successful completion of the exercises (§4 (2), 3 SPO 2007 respectively §4 (3) SPO 2015) a bonus can be obtained. If the grade of the written exam is at least 4.0 and at most 1.3, the bonus will improve it by one grade level (i.e. by 0.3 or 0.4).

Prerequisites
None

Recommendations
None

Additional Information
Former course title until summer term 2017: T-WIWI-102794 "eEnergy: Markets, Services, Systems".

The lecture has also been added in the IIP Module *Basics of Liberalised Energy Markets*.

Below you will find excerpts from courses related to this module component:

V Energy Market Engineering

2540464, SS 2026, 2 SWS, Language: German/English, Open in study portal

Lecture (V)
On-Site

Content
The lecture "Energy Market Engineering" addresses the design and analysis of energy markets considering current developments and challenges. A particular focus is on the integration of renewable energies and the associated market mechanisms and regulations.

Specifically, the following topics are covered:

- **Introduction to Market Engineering**: What design elements do markets and specifically auctions have in general, and what influence does this have on participant behavior.
- **Introduction to Energy Markets**: Fundamentals and current trends in the energy system, including climate change and the expansion of renewable energies.
- **Market Design and Products**: Various pricing models such as nodal pricing, zonal pricing, and the structure of capacity markets.
- **Grid Expansion, Distribution Networks, and Flexibility Markets**: Analysis of distribution network markets and the role of flexibility options like demand response and storage technologies.
- **Intermittent Generation and Grid Stability**: Challenges posed by fluctuating renewable energies and strategies to ensure grid stability.
- **Digitalization and Market Transparency**: The role of digitalization in improving market transparency and efficiency, including the use of smart metering systems and data-driven approaches.
- **Current Research Projects and Developments**: Presentation of ongoing research projects and their significance for the future design of energy markets.

Organizational issues
Die Vorlesung wird als zweiwöchige Blockveranstaltung gehalten. Die genauen Termine und Themenblöcke werden in der Auftaktveranstaltung sowie auf Ilias kommuniziert.

Literature

- Erdmann G, Zweifel P. *Energieökonomik, Theorie und Anwendungen*. Berlin Heidelberg: Springer; 2007.
- Grimm V, Ockenfels A, Zoettl G. Strommarktdesign: Zur Ausgestaltung der Auktionsregeln an der EEX *. *Zeitschrift für Energiewirtschaft*. 2008:147-161.
- Stoft S. *Power System Economics: Designing Markets for Electricity*. IEEE; 2002.,
- Ströbele W, Pfaffenberger W, Heuterkes M. *Energiewirtschaft: Einführung in Theorie und Politik*. 2nd ed. München: Oldenbourg Verlag; 2010:349.

T 6.216 Module component: Energy Networks and Regulation [T-WIWI-107503]

Coordinators: Prof. Dr. Christof Weinhardt
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101446 - Market Engineering
M-WIWI-103720 - eEnergy: Markets, Services and Systems
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Oral examination	4,5 CP	graded	Each winter term	2

Courses					
WT 25/26	2540494	Energy Networks and Regulation	2 SWS	Lecture / 🗣	Rogat, Miskiw
WT 25/26	2540495	Übung zu Energy Networks and Regulation	1 SWS	Practice / 🗣	Rogat, Miskiw

Legend: 🖥 Online, ❂ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success is assessed in the form of an oral examination (in accordance with §4(2), 1 SPO).
The examination is offered in the semester of the lecture.

Prerequisites
None

Recommendations
None

Below you will find excerpts from courses related to this module component:

V Energy Networks and Regulation

2540494, WS 25/26, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content
Learning Goals
The student,

- understands the business model of a network operator and knows its central tasks in the energy supply system,
- has a holistic overview of the interrelationships in the network economy,
- understands the regulatory and business interactions,
- is in particular familiar with the current model of incentive regulation with its essential components and understands its implications for the decisions of a network operator
- is able to analyse and assess controversial issues from the perspective of different stakeholders.

Content of teaching
The lecture “Energy Networks and Regulation” provides insights into the regulatory framework of electricity and gas. It touches upon the way the grids are operated and how regulation affects almost all grid activities. The lecture also addresses approaches of grid companies to cope with regulation on a managerial level. We analyze how the system influences managerial decisions and strategies such as investment or maintenance. Furthermore, we discuss how the system affects the operator’s abilities to deal with the massive challenges lying ahead (“Energiewende”, redispatch, European grid integration, electric vehicles etc.). Finally, we look at current developments and major upcoming challenges, e.g., the smart meter rollout. Covered topics include:

- Grid operation as a heterogeneous landscape: big vs. small, urban vs. rural, TSO vs. DSO
- Objectives of regulation: Fair price calculation and high standard access conditions
- The functioning of incentive regulation
- First major amendment to the incentive regulation: its merits, its flaws
- The revenue cap and how it is adjusted according to certain exogenous factors
- Grid tariffs: How are they calculated, what is the underlying rationale, do we need a reform (and which)?
- Exogenous costs shifted (arbitrarily?) into the grid, e.g. feed-in tariffs for renewable energy or decentralized supply.

Literature

Linnemann, M. (2024). Energiewirtschaft für (Quer-)Einsteiger: Einmaleins der Stromwirtschaft. Deutschland: Springer Fachmedien Wiesbaden.

Averch, H.; Johnson, L.L (1962). Behavior of the firm under regulatory constraint, in: American Economic Review, 52 (5), S. 1052 – 1069.

Bundesnetzagentur (2006): Bericht der Bundesnetzagentur nach § 112a EnWG zur Einführung der Anreizregulierung nach § 21a EnWG, http://www.bundesnetzagentur.de/SharedDocs/Downloads/DE/Sachgebiete/Energie/Unternehmen_Institutionen/Netzentgelte/Anreizregulierung/BerichtEinfuehrgAnreizregulierung.pdf?__blob=publicationFile&v=3.

Bundesnetzagentur (2015): Evaluierungsbericht nach § 33 Anreizregulierungsverordnung, https://www.bmwi.de/Redaktion/DE/Downloads/A/anreizregulierungsverordnung-evaluierungsbericht.pdf?__blob=publicationFile&v=1.

Filippini, M.; Wild, J.; Luchsinger, C. (2001) : Regulierung der Verteilnetzpreise zu Beginn der Marktöffnung. Erfahrungen in Norwegen und Schweden, Bundesamt für Energie, Bern, http://www.iaea.org/inis/collection/NCLCollectionStore/_Public/34/066/34066585.pdf.

Gómez, T. (2013): Monopoly Regulation, in: Pérez-Arriaga, I.J. (Hg.): Regulation of the Power Sector, S. 151 – 198, Springer-Verlag, London.

Gómez, T. (2013): Electricity Distribution, in: Pérez-Arriaga, I.J. (Hg.): Regulation of the Power Sector, S. 199 – 250, Springer-Verlag, London.

Pérez-Arriaga, I.J. (2013): Challenges in Power Sector Regulation, in: Pérez-Arriaga, I.J. (Hg.): Regulation of the Power Sector, S. 647 – 678, Springer-Verlag, London.

Rivier, M.; Pérez-Arriaga, I.J.; Olmos, L. (2013): Electricity Transmission, in: Pérez-Arriaga, I.J. (Hg.): Regulation of the Power Sector, S. 251 – 340, Springer-Verlag, London.

T 6.217 Module component: Energy Policy [T-WIWI-102607]

Coordinators: Prof. Dr. Martin Wietschel
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	3,5 CP	graded	Each summer term	3

Courses					
ST 2026	2581959	Energy Policy	2 SWS	Lecture / 🗣	Wietschel

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment consists of a written exam (60 minutes) (following §4(2) of the examination regulation). The exam takes place in every semester. Re-examinations are offered at every ordinary examination date.

Prerequisites
None.

Below you will find excerpts from courses related to this module component:

V Energy Policy

2581959, SS 2026, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content
The availability of cheap, environmentally friendly and secure energy is crucial for human welfare. However, the increasing scarcity of resources and increasing environmental pressures, with a particular focus on climate change, threaten human welfare through economic action. Energy contributes significantly to environmental pollution. The energy industry is characterised by high regulation and a significant influence of political decisions.

At the beginning of the lecture different perspectives on energy policy will be presented and the analysis of political decision-making processes will be discussed. Then the current energy policy challenges in the area of environmental pollution, regulation and the role of energy for households and industry will be discussed. Then the actors of energy policy and energy responsibilities in Europe will be discussed. The economic approaches from traditional environmental economics and sustainability as a new policy approach will then be discussed. Finally, energy policy instruments such as the promotion of renewable energies or energy efficiency are discussed in detail and how they can be evaluated.

The lecture emphasizes the relationship between theory and practice and presents some case studies.

Literature
Wird in der Vorlesung bekannt gegeben.

T 6.218 Module component: Energy Trading and Risk Management [T-WIWI-112151]

Coordinators: N.N.
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101451 - Energy Economics and Energy Markets
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	3,5 CP	graded	Each summer term	2

Courses					
ST 2026	2581020	Energy Trading and Risk Management	2 SWS	Lecture / 🗣	Kraft, Fichtner, Beranek

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The lecture "Energiehandel und Risikomanagement" will be held in English under the title "Energy Trading and Risk Management" from the summer semester 2022. The examination for the English-language lecture will be offered in English from the summer semester 2022.
The assessment consists of a written exam (60 minutes). The exam takes place in every semester. Re-examinations are offered at every ordinary examination date.

Prerequisites
None

Recommendations
None

Workload
105 hours

Below you will find excerpts from courses related to this module component:

V Energy Trading and Risk Management

2581020, SS 2026, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content

1. Introduction to Markets, Mechanisms and Interaction
2. Electricity Trading (platforms, products, mechanisms)
3. Balancing Energy Markets and Congestion Management
4. Coal Markets (reserves, supply, demand, and transport)
5. Investments and Capacity Markets
6. Oil and Gas Markets (supply, demand, trade, and players)
7. Trading Game
8. Risk Management in Energy Trading

T 6.219 Module component: Engine Measurement Techniques [T-MACH-105169]

Coordinators: Dr.-Ing. Sören Bernhardt
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101303 - Combustion Engines II
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each summer term	1

Courses					
ST 2026	2134137	Engine measurement techniques	2 SWS	Lecture / 🗣	Bernhardt

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
oral examination, Duration: 0,5 hours, no auxiliary means

Prerequisites
none

Recommendations
T-MACH-102194 Combustion Engines I

Additional Information
The course is offered in German.

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Engine measurement techniques

2134137, SS 2026, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Literature

1. Grohe, H.:Messen an Verbrennungsmotoren
2. Bosch: Handbuch Kraftfahrzeugtechnik
3. Veröffentlichungen von Firmen aus der Meßtechnik
4. Hoffmann, Handbuch der Meßtechnik
5. Klingenberg, Automobil-Meßtechnik, Band C

T 6.220 Module component: Engineering FinTech Solutions [T-WIWI-106193]

Coordinators: Prof. Dr. Maxim Ulrich
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Examination of another type	9 CP	graded	Each term	5

Assessment
The assessment is carried out in form of a written thesis based on the course "Engineering FinTech Solutions".

Workload
270 hours

T 6.221 Module component: Engineering Hydrology [T-BGU-108943]

Coordinators: Dr. rer. nat. Ralf Loritz
Dr. Franziska Villinger
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Expansion	Version
Written examination	3 CP	graded	Each term	1 semesters	2

Courses					
ST 2026	6200617	Engineering Hydrology	2 SWS	Lecture / Practice / 🗣	Villinger, Loritz

Legend: 🖥 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
written exam, 60 min.

Prerequisites
none

Recommendations
none

Additional Information
none

Workload
90 hours

T 6.222 Module component: Engineering Interactive Systems: AI & Wearables [T-WIWI-113460]

Coordinators: Prof. Dr. Alexander Mädche
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-102806 - Service Innovation, Design & Engineering
M-WIWI-104080 - Designing Interactive Information Systems
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Examination of another type	4,5 CP	graded	Each winter term	1

Courses					
WT 25/26	2540420	Engineering Interactive Systems: AI & Wearables	3 SWS	Lecture /	Mädche, Feick, Kretzer

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
Alternative exam assessment. The assessment consists of a one-hour exam and the implementation of a Capstone project. Details will be announced at the beginning of the course.

Prerequisites
None

Recommendations
None

Additional Information
The course is held in English.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Engineering Interactive Systems: AI & Wearables

2540420, WS 25/26, 3 SWS, Language: English, Open in study portal

Lecture (V)
Blended (On-Site/Online)

Literature
Siehe Englische Literatur

T 6.223 Module component: Enterprise Risk Management [T-WIWI-102608]

Coordinators: Prof. Dr. Ute Werner
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Oral examination	4,5 CP	graded	Each winter term	1

Assessment
The assessment consists of oral presentations (incl. papers) within the lecture (according to Section 4 (2), 3 of the examination regulation) and a final oral exam (according to Section 4 (2), 2 of the examination regulation).

The overall grade consists of the assessment of the oral presentations incl. term papers (50 percent) and the assessment of the oral exam (50 percent).

The examination will be offered latest until winter term 2017/2018 (beginners only).

Prerequisites
None

Recommendations
None

T 6.224 Module component: Enterprise Systems for Financial Accounting & Controlling [T-WIWI-113746]

Coordinators: Christian Fleig
Prof. Dr. Alexander Mädche

Organisation: KIT Department of Business and Economics

Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Examination of another type	4,5 CP	graded	Each winter term	2

Courses					
WT 25/26	2500060	Enterprise Systems for Financial Accounting & Controlling	3 SWS	Lecture / 🗣	Mädche, Fleig

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success is assessed in the form of an alternative exam assessment. It consists of a one-hour exam and the implementation of a capstone project.

The final grade is made up of 60% of the exam grade and 40% of the capstone project grade.

Details on the structure of the assessment will be announced during the lecture.

Prerequisites
Keine.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Enterprise Systems for Financial Accounting & Controlling

Lecture (V)
On-Site

2500060, WS 25/26, 3 SWS, Language: English, Open in study portal

Content

Enterprise Systems building on enterprise resource planning (ERP) packaged software such as SAP S/4HANA are information systems that target large-scale integration of business processes and data across a company's functional areas. These systems are crucial for financial accounting and controlling as they enable organizations to streamline and integrate their financial operations, ensuring accurate decision-making based on real-time financial data. Contemporary packaged ERP software provide modules that integrate core business processes in financial accounting including general ledger, accounts receivable, payable and asset accounting. The information generated in these processes serves as a major source of cost-related decision-making, reporting and data analyses in internal accounting ("controlling"). Packaged ERP software typically rely on industry best practices captured in the form of product software with a standardized structure of master data. Thereby, they also support regulatory compliance and analyzability of processes in approaches such as process mining which enhances overall business efficiency and competitiveness. However, implementing enterprise systems in practice imposes substantial challenges to organizations.

First, the B.Sc. lecture "Enterprise Systems for Financial Accounting & Controlling" introduces fundamental business processes and concepts in finance and controlling and explains how these processes are implemented in packaged ERP software such as SAP S/4HANA. Students learn the basic and most important terms and master data structures in the SAP FI/CO module. Second, students learn about the principles of packaged ERP software, gaining hands-on experience SAP S4/HANA. Third, the lecture introduces the challenges in enterprise system projects such as SAP S/4HANA implementations, Fourth, students actively apply their knowledge in collaborative team efforts when working with exemplary SAP data in Microsoft SQL Server to analyze finance and controlling master data processes (capstone project)

Learning Objectives:

The students …

- understand modern business concepts of financial accounting & controlling for large enterprises
- the importance of enterprise systems supporting the implemention of modern business concepts
- know the underlying principles of packaged software for enterprise resource planning and process intelligence
- Understand the opportunities and challenges of Enterprise Systems implementation at large enterprises
- Get hands-on knowledge about financial accounting & controlling with commercial product software (e.g., SAP S4/HANA)
- Apply their knowledge on enterprise systems implementation for financial accounting and controlling on real-world data in team effort

T 6.225 Module component: Entrepreneurship [T-WIWI-102864]

Coordinators: Prof. Dr. Orestis Terzidis
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101488 - Entrepreneurship (EnTechnon)
M-WIWI-101507 - Innovation Management
M-WIWI-104900 - Business Administration
M-WIWI-105010 - Student Innovation Lab (SIL) 1

Type	Credits	Grading	Term offered	Version
Written examination	3 CP	graded	Each term	2

Courses					
WT 25/26	2545001	Entrepreneurship	2 SWS	Lecture / 🧩	Malik, Terzidis, Dang
ST 2026	2545001	Entrepreneurship	2 SWS	Lecture / 🗣	Terzidis, Dang
WT 26/27	2545001	Entrepreneurship	2 SWS	Lecture / 🧩	Malik, Terzidis, Dang

Legend: 📱 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment consists of a written exam (60 minutes) (following §4(2), 1 of the examination regulation).

Students are offered the opportunity to earn a grade bonus through separate assignments. If the grade of the written exam is between 4.0 and 1.3, the bonus improves the grade by a maximum of one grade level (0.3 or 0.4). The exact criteria for awarding a bonus will be announced at the beginning of the lecture.

Prerequisites
None

Recommendations
None

Below you will find excerpts from courses related to this module component:

V Entrepreneurship
2545001, WS 25/26, 2 SWS, Language: English, Open in study portal

Lecture (V)
Blended (On-Site/Online)

Content
The lecture as an obligatory part of the module "Entrepreneurship" introduces the basic concepts of entrepreneurship. Important concepts and empirical facts are presented that relate to the conception and implementation of newly founded companies.

The focus here is on the introduction to methods for generating innovative business ideas, for transferring patents into business concepts and general principles of business modelling and business planning. In particular approaches such as Lean Startup and Effectuation as well as concepts for the financing of young enterprises are treated.

A "KIT Entrepreneurship Talk" is part of each session, in which experienced founder and entrepreneur personalities report on their experiences in practice of the establishment of an enterprise. Dates and speakers will be announced on the EnTechnon homepage.

Learning objectives:

The studentsare introduced to the topic Entrepreneurship. After successful attendance of the meeting they are to have an overview of the subranges of the Entrepreneurships and be able to understand basic concepts of the Entrepreneurships and apply key concepts.

Workload:

Total effort with 3 credit points: approx. 90 hours
Presence time: 30 hours
Pre- and postprocessing of the LV: 45.0 hours
Exam and exam preparation: 15.0 hours

Examination:

The assessment of success takes place in the form of a written examination (60 min.) (according to §4(2), 1 SPO). The grade is the grade of the written exam.

A grade bonus can be earned through successful participation in a case study in the Entrepreneurship lecture. If the grade of the written exam is between 4.0 and 1.3, the bonus improves the grade by up to 0.3 or 0.4.The bonus only applies if you have passed the exam with at least a 4.0. More details will be provided in the lecture. Participation in the case study is voluntary.

Exam date: tba

Organizational issues
VL findet jeweils Mo, 15:45 - 19:00 an folgenden Terminen statt:
27.10.2025
03.11.2025
10.11.2025
17.11.2025
24.11.2025
01.12.2025
08.12.2025 (inkl. Prep Session)

Literature
Aulet, Bill (2013): Disciplined Entrepreneurship. 24 Steps to a Successful Startup. Hoboken: Wiley.

R.C. Dorf, T.H. Byers: Technology Ventures – From Idea to Enterprise., (McGraw Hill 2008)

Füglistaller, Urs, Müller, Christoph and Volery, Thierry (2008): Entrepreneurship

Hisrich, Robert D.; Ramadani, Veland (2017): Effective entrepreneurial management. Strategy, planning, risk management, and organization. Cham, Switzerland: Springer.

Ries, Eric (2011): The Lean Startup.

Osterwalder, Alexander (2010): Business Model Generation.

V **Entrepreneurship**
2545001, SS 2026, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content

The lecture as a compulsory part of the module "Entrepreneurship" introduces the basic concepts of entrepreneurship. Important concepts and empirical facts are introduced, which relate to the conception and implementation of newly founded companies.

The focus here is on introducing methods for generating innovative business ideas, translating patents into business concepts, and general principles of business modeling and business planning. In particular, approaches such as Lean-Startup and Effectuation as well as concepts for financing young companies are covered.

A "KIT Entrepreneurship Talk" is part of each session, in which experienced founder and entrepreneur personalities report on their experiences in the practice of the establishment of an enterprise. Dates and speakers will be announced on the EnTechnon homepage.

Learning objectives:

The students will be introduced to the topic of entrepreneurship. After successful attendance of the course they should have an overview of the sub-areas of entrepreneurship and be able to understand basic concepts of entrepreneurship and apply key concepts.

Workload:

The total effort with 3 credit points: approx. 90 hours
Presence time: 30 hours
Pre- and postprocessing of the LV: 45.0 hours
Exam and exam preparation: 15.0 hours

Examination:

The assessment consists of a written exam (60 minutes) (following §4(2), 1 of the examination regulation)

A grade bonus can be earned by successfully participating in a case study as part of the Entrepreneurship lecture. If the grade of the written exam is between 4.0 and 1.3, the bonus improves the grade by up to 0.3 or 0.4. The bonus only applies if you have passed the exam with at least a 4.0. More details will be provided in the lecture. Participation in the case study is voluntary.

Exam dates: tbd

Organizational issues

VL findet jeweils Di, 15:45 - 19:00 an folgenden Terminen statt:

21.04.2026
28.04.2026
05.05.2026
12.05.2026
19.05.2026
02.06.2026
09.06.2026 (inkl. Prep Session)

16.06.2026 (Klausur)

Literature

Füglistaller, Urs, Müller, Christoph und Volery, Thierry (2008): Entrepreneurship

Ries, Eric (2011): The Lean Startup

Osterwalder, Alexander (2010): Business Model Generation

Aulet, Bill (2013): Disciplined Entrepreneurship. 24 Steps to a Successful Startup. Hoboken: Wiley.

R.C. Dorf, T.H. Byers: Technology Ventures – From Idea to Enterprise., (McGraw Hill 2008)

Hisrich, Robert D.; Ramadani, Veland (2017): Effective entrepreneurial management. Strategy, planning, risk management, and organization. Cham, Switzerland: Springer.

V **Entrepreneurship**
2545001, WS 26/27, 2 SWS, Language: English, Open in study portal

Lecture (V)
Blended (On-Site/Online)

Content
The lecture as an obligatory part of the module "Entrepreneurship" introduces the basic concepts of entrepreneurship. Important concepts and empirical facts are presented that relate to the conception and implementation of newly founded companies.

The focus here is on the introduction to methods for generating innovative business ideas, for transferring patents into business concepts and general principles of business modelling and business planning. In particular approaches such as Lean Startup and Effectuation as well as concepts for the financing of young enterprises are treated.

A "KIT Entrepreneurship Talk" is part of each session, in which experienced founder and entrepreneur personalities report on their experiences in practice of the establishment of an enterprise. Dates and speakers will be announced on the EnTechnon homepage.

Learning objectives:

The studentsare introduced to the topic Entrepreneurship. After successful attendance of the meeting they are to have an overview of the subranges of the Entrepreneurships and be able to understand basic concepts of the Entrepreneurships and apply key concepts.

Workload:

Total effort with 3 credit points: approx. 90 hours
Presence time: 30 hours
Pre- and postprocessing of the LV: 45.0 hours
Exam and exam preparation: 15.0 hours

Examination:

The assessment of success takes place in the form of a written examination (60 min.) (according to §4(2), 1 SPO). The grade is the grade of the written exam.

A grade bonus can be earned through successful participation in a case study in the Entrepreneurship lecture. If the grade of the written exam is between 4.0 and 1.3, the bonus improves the grade by up to 0.3 or 0.4.The bonus only applies if you have passed the exam with at least a 4.0. More details will be provided in the lecture. Participation in the case study is voluntary.

Exam date: tba

Organizational issues
VL findet jeweils Mo, 15:45 - 19:00 an folgenden Terminen statt:
26.10.2026
02.11.2026
09.11.2026
16.11.2026
23.11.2026
30.11.2026
07.12.2026 (inkl. Prep Session)

Literature
Aulet, Bill (2013): Disciplined Entrepreneurship. 24 Steps to a Successful Startup. Hoboken: Wiley.

R.C. Dorf, T.H. Byers: Technology Ventures – From Idea to Enterprise., (McGraw Hill 2008)

Füglistaller, Urs, Müller, Christoph and Volery, Thierry (2008): Entrepreneurship

Hisrich, Robert D.; Ramadani, Veland (2017): Effective entrepreneurial management. Strategy, planning, risk management, and organization. Cham, Switzerland: Springer.

Ries, Eric (2011): The Lean Startup.

Osterwalder, Alexander (2010): Business Model Generation.

T 6.226 Module component: Entrepreneurship Basics [T-WIWI-114637]

Coordinators: Prof. Dr. Orestis Terzidis
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Examination of another type	4,5 CP	graded	Each winter term	1

Courses					
WT 25/26	2545010	Entrepreneurship Basics (Track 1)	3 SWS	Seminar / Blended	Hirte
WT 25/26	2545011	Entrepreneurship Basics (Track 2)	3 SWS	Seminar / Blended	Lau
ST 2026	2545010	Entrepreneurship Basics (Track 1)	3 SWS	Seminar / On-Site	Hirte, Terzidis
WT 26/27	2545010	Entrepreneurship Basics (Track 1)	3 SWS	Seminar / Blended	Hirte

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
The assessment is conducted as an alternative exam assessment. The specific form of assessment depends on the course and may include, for example, the following components:

- **Project Report or Reflection Paper**
- **Mid-term and Final Presentation**
- **Group Work with Individual Contribution**
- **Active Participation and Mandatory Attendance**

The overall grade is determined by the weighted evaluation of these individual assessment components. The detailed weighting of these components for grade calculation will be communicated to students at the beginning of the course.

Prerequisites
None

Recommendations
An interest in entrepreneurial thinking and action, teamwork, and interdisciplinary project work is recommended. Basic knowledge in business administration and/or innovation methods is helpful but not required.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Entrepreneurship Basics (Track 1)
2545010, WS 25/26, 3 SWS, Language: English, Open in study portal

Seminar (S)
Blended (On-Site/Online)

Content
Course Content:

This seminar explains important factors for becoming an entrepreneur and guides you through a structured process from the first business idea to a pitch of your final business model. Therefore, a business idea will be developed in the context of the UN Sustainable Development Goals. In small teams you create, develop, validate and present your business model. It simulates the basics of a start-up process up to the investor pitch.

Learning Objectives

After completing this course, the course participants will be able to

- Reflect on and define your personal and team core values
- Reflect on and define your personal and team competencies
- Reflect on and recall a definition for business opportunity
- Define your field of interest for opportunity recognition using the UN SDGs
- Analyze a specific domain to identify business opportunities
- Develop a first draft for your business model by using the Business Model Canvas
- Pitch / present your business idea

Credentials:

Registration is via the Wiwi portal.

Exam:

Presentation + active participation + paper.

Target group:

Bachelor students

Organizational issues
Registration is via the Wiwi portal.

In the seminar you will work on a project in teams of max. 5 persons. The groups are formed in the seminar

V **Entrepreneurship Basics (Track 2)**
2545011, WS 25/26, 3 SWS, Language: English, Open in study portal

Seminar (S)
Blended (On-Site/Online)

Content
Course Content:

This seminar provides a hands-on introduction to entrepreneurial thinking and acting. Students will be guided through a structured, evidence-based process, from identifying a first business idea to designing a validated business model. This simulates the early stages of a start-up journey, culminating in an investor-style pitch. Working in interdisciplinary teams, participants will ideate, test, and refine their concepts using real-world methods and tools.

Starting from team formation and problem exploration, students will learn how to discover and validate customer needs. They will design and test hypotheses through customer interviews and rapid experiments, build Value Proposition and Business Model Canvases, and analyze unit economics and go-to-market options.

The seminar emphasizes teamwork, evidence-based decision-making, and communication skills. By the end, each team will deliver a tested business concept, a concise pitch deck, and an investor-style presentation.

Learning Objectives

After completing this seminar, students will have learned and actually practiced the whole business model development process. In particular this means that students will know:

- Develop core entrepreneurial skills through practice.
- Understand how to create and validate customer value.
- Learn to derive, prioritize, and test business hypotheses.
- Translate ideas into viable business models and economics.
- Build and test low-fidelity prototypes to generate evidence.
- Gain experience in pitching and convincing stakeholders with evidence.

Format & Deliverables:

- 2.5 days of in-class sessions (inputs, team sprints, coaching).
- Homework assignments between sessions (interviews, experiments, prototyping, deck iterations).
- Final deliverables: team pitch, pitch deck, evidence dossier, and reflection.

Target group:

Bachelor students

Credentials:

Registration is via the Wiwi portal.

Organizational issues
Registration is via the Wiwi portal.

In the seminar you will work on a project in teams of 4-5 persons. The groups are formed in the seminar.

V	**Entrepreneurship Basics (Track 1)** 2545010, SS 2026, 3 SWS, Language: English, Open in study portal	**Seminar (S)** **On-Site**

Content
Content

This seminar explains important factors for becoming an entrepreneur and guides you through a structured process from the first business idea to a pitch of your final business model. Therefore, a business idea will be developed in the context of the UN Sustainable Development Goals. In small teams you create, develop, validate and present your business model. It simulates the basics of a start-up process up to the investor pitch.

Learning Objectives

After completing this course, the course participants will be able to

- Reflect on and define your personal and team core values
- Reflect on and define your personal and team competencies
- Reflect on and recall a definition for business opportunity
- Define your field of interest for opportunity recognition using the UN SDGs
- Analyze a specific domain to identify business opportunities
- Develop a first draft for your business model by using the Business Model Canvas
- Pitch / present your business idea

Exam:

Presentation + active participation + paper.

Target group:

Bachelor students

Organizational issues
Registration is via the Wiwi-Portal.

In the seminar you will work on a project in teams of max. 5 persons. The groups are formed in the seminar.

V	**Entrepreneurship Basics (Track 1)** 2545010, WS 26/27, 3 SWS, Language: English, Open in study portal	**Seminar (S)** **Blended (On-Site/Online)**

Content
Course Content:

This seminar explains important factors for becoming an entrepreneur and guides you through a structured process from the first business idea to a pitch of your final business model. Therefore, a business idea will be developed in the context of the UN Sustainable Development Goals. In small teams you create, develop, validate and present your business model. It simulates the basics of a start-up process up to the investor pitch.

Learning Objectives

After completing this course, the course participants will be able to

- Reflect on and define your personal and team core values
- Reflect on and define your personal and team competencies
- Reflect on and recall a definition for business opportunity
- Define your field of interest for opportunity recognition using the UN SDGs
- Analyze a specific domain to identify business opportunities
- Develop a first draft for your business model by using the Business Model Canvas
- Pitch / present your business idea

Credentials:

Registration is via the Wiwi portal.

Exam:

Presentation + active participation + paper.

Target group:

Bachelor students

Organizational issues
Registration is via the Wiwi portal.

In the seminar you will work on a project in teams of max. 5 persons. The groups are formed in the seminar

T 6.227 Module component: Entrepreneurship Research [T-WIWI-102894]

Coordinators: Prof. Dr. Orestis Terzidis
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101488 - Entrepreneurship (EnTechnon)
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Each summer term	1

Courses					
ST 2026	2545002	Entrepreneurship Research	2 SWS	Seminar / 🗣	Kuschel

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The performance review is done via a so called other methods of performance review (term paper) (alternative exam assessment). The final grade is a result from both, the grade of the term paper and its presentation, as well as active participation during the seminar.

Prerequisites
None

Recommendations
None

Additional Information
The topics will be prepared in groups. The presentation of the results is done during a a block period seminar at the end of the semester. Students have to be present all day long during the seminar.

Below you will find excerpts from courses related to this module component:

V Entrepreneurship Research

2545002, SS 2026, 2 SWS, Language: English, Open in study portal

Seminar (S)
On-Site

Content
Content

In this course, the students choose from various relevant and current research topics in entrepreneurship and independently develop a topic that suits them in small teams. Initially, there is an introduction to standard methods such as systematic literature review, design science, qualitative and quantitative data analysis, and more. The seminar topic must be scientifically prepared and presented in 15-20 pages as part of a written elaboration. The seminar results are presented in a block event at the end of the semester (20 min + 10 min open discussion).

Learning Objectives

The foundations of independent scholarly work (literature review, argumentation + discussion, citation of literature sources, application of qualitative, quantitative, and simulation methods) are developed as part of the written elaboration. The competencies acquired in the seminar can be utilized in preparing for a potential master's thesis. Therefore, the seminar is mainly aimed at students who intend to write their thesis at the Chair of Entrepreneurship and Technology Management and wish to gain substantial experience in entrepreneurship research.

Organizational issues
Registration is via the Wiwi-Portal.

Seminar dates will probably be (as announced in Wiwi-Portal):

Wednesday, 29.04.2026, 10.00-16.00
Wednesday, 20.05.2026, 10.00-16.00
Wednesday, 24.06.2026, 09.00-12.00

Literature
Will be announced in the seminar.

T 6.228 Module component: Entrepreneurship Seasonal School [T-WIWI-113151]

Coordinators: Prof. Dr. Orestis Terzidis
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101488 - Entrepreneurship (EnTechnon)
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Irregular	1

Assessment
Alternative exam assessment. The grade is composed of the presentation and the written elaboration. Details on the design of the examination will be announced in the course.

Prerequisites
The Seasonal School is intended for advanced bachelor's and all master's students (all disciplines). Participation in the selection process is a prerequisite.

Recommendations
Basic knowledge of business administration, attendance of the lecture Entrepreneurship as well as openness and interest in intercultural exchange are recommended. Solid knowledge of the English language is an advantage.

Additional Information
Entrepreneurship Seasonal School

Workload
90 hours

T 6.229 Module component: Environmental and Resource Policy [T-WIWI-102616]

Coordinators: Rainer Walz
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101468 - Environmental Economics
M-WIWI-104908 - Economics

Type	Credits	Grading	Term offered	Version
Written examination	4 CP	graded	Each summer term	1

Courses					
ST 2026	2560548	Environmental and Ressource Policy	2 SWS	Lecture / Practice	Walz

Assessment
See German version

Recommendations
It is recommended to already have knowledge in the area of industrial organization and economic policy. This knowledge may be acquired in the courses*Introduction to Industrial Organization* [2520371] and*Economic Policy*[2560280].

Below you will find excerpts from courses related to this module component:

V Environmental and Ressource Policy

2560548, SS 2026, 2 SWS, Language: German, Open in study portal

Lecture / Practice (VÜ)

Literature
Weiterführende Literatur:

Michaelis, P.: Ökonomische Instrumente in der Umweltpolitik. Eine anwendungsorientierte Einführung, Heidelberg
OECD: Environmental Performance Review Germany, Paris

T 6.230 Module component: Environmental Economics and Sustainability [T-WIWI-102615]

Coordinators: Prof. Dr. Rainer Walz
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101468 - Environmental Economics
M-WIWI-104908 - Economics

Type	Credits	Grading	Term offered	Version
Written examination	3 CP	graded	Each winter term	2

Courses					
WT 25/26	2521547	Umweltökonomik und Nachhaltigkeit (mit Übung)	2 SWS	Lecture / Practice	Walz

Assessment
See German version

Prerequisites
None

Recommendations
It is recommended to already have knowledge in the area of macro- and microeconomics. This knowledge may be acquired in the courses *Economics I: Microeconomics* [2600012] and *Economics II: Macroeconomics* [2600014].

Workload
90 hours

T 6.231 Module component: Environmental Law [T-BGU-111102]

Coordinators: Dr. Ulrich Smeddinck
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: M-WIWI-101468 - Environmental Economics
M-WIWI-104903 - Law

Type	Credits	Grading	Term offered	Expansion	Version
Written examination	3 CP	graded	Each winter term	1 semesters	1

Courses					
WT 25/26	6111177	Environmental Law	2 SWS	Lecture / 🗣	Smeddinck

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, x Cancelled

Assessment
Written exam with 120 min

Prerequisites
None

Additional Information
None

Workload
90 hours

T 6.232 Module component: Estimator and Observer Design [T-CIWVT-112828]

Coordinators: Dr.-Ing. Pascal Jerono
Organisation: KIT Department of Chemical and Process Engineering
Part of: M-CIWVT-106320 - Estimator and Observer Design

Type	Credits	Grading	Version
Oral examination	6 CP	graded	1

Courses					
WT 25/26	2243110	Estimator and Observer Design	3 SWS	Lecture / Practice / 🗣	Jerono

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

T 6.233 Module component: EU Data Protection Law [T-INFO-113887]

Coordinators: Gustavo Gil Gasiola
Organisation: KIT Department of Informatics
Part of: M-INFO-106754 - Public Economic and Technology Law

Type	Credits	Grading	Term offered	Version
Written examination	3 CP	graded	Each winter term	1

Courses					
WT 25/26	2424019	EU Data Protection Law	2 SWS	Lecture / 🗣	Gil Gasiola

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment is carried out as a written examination (§ 4 Abs. 2 No. 1 SPO) lasting 60 minutes.

Prerequisites
None

Additional Information
Competency Goals:

Students are able to comprehend the EU data protection regulation, including the General Data Protection Regulation and related EU data regulations.

They know the foundations of data protection rules, including fundamental concepts (e.g., “personal data”, “processing”, “data subject”). They are also familiar with the principles of personal data processing (lawfulness, limited purpose, transparency, accountability) as well as the rights of the data subject.

They can identify the main obligations of the controller and the processor.

Students understand the conditions for the transfer of personal data to third countries.

They can identify the other regulations that govern data in the European Union.

Students are able to read and understand legal text related to data regulation.

They can understand and solve simple data protection cases.

Content:

The General Data Protection Regulation (GDPR) of the European Union is a milestone in protecting individuals from the unlawful use of their data. In a data-driven society, economy, and government, this protection has become essential to guarantee fundamental rights. In addition to its direct impact on the legal systems of all Member States, the GDPR has a major influence on third countries that have adopted similar regulations (e.g. Switzerland, Argentina, Brazil, South Africa, and many others). In this way, the EU Data Protection Regulation has established itself as the “gold standard” of data protection, providing guidance to address the challenges posed by new technologies and new ways of creating, using and sharing personal data. Understanding the structure of data protection in the EU is therefore essential to grasp its impact on individual rights, public administration, business models, and even technological development.
This lecture aims to provide a structured overview of the EU Data Protection Regulation, and to offer tools to understand the regulatory structure of the EU Data Regulation. The lecture will cover the following topics:

- Introduction to EU law
- Development of the EU data protection regulation
- Legal structure of data protection in the EU
- Role of national and sectoral laws
- Data protection as fundamental right
- Principles of data protection
- Lawfulness of personal data processing
- Anonymization and pseudonymization of personal data
- Special categories of personal data
- Rights of the data subject
- Transfer of personal data to third countries
- Responsibility of the controller and the processor
- Security of personal data and personal data breach
- Open Data Directive
- Data Governance Act
- Data Act

Workload

- Attendance time to the lectures = 15 x 90 min = 22 h 30 min
- Self-study during the semester = 47 h 30 min
- Preparation for the exam = 20 h
- Total = 90 h

Below you will find excerpts from courses related to this module component:

V	**EU Data Protection Law** 2424019, WS 25/26, 2 SWS, Language: English, Open in study portal	**Lecture (V)** **On-Site**

Content

The General Data Protection Regulation (GDPR) of the European Union is a milestone in protecting individuals from the unlawful use of their data. In a data-driven society, economy, and government, this protection has become essential to guarantee fundamental rights. In addition to its direct impact on the legal systems of all Member States, the GDPR has a major influence on third countries that have adopted similar regulations (e.g. Switzerland, Argentina, Brazil, South Africa, and many others). In this way, the EU Data Protection Regulation has established itself as the “gold standard” of data protection, providing guidance to address the challenges posed by new technologies and new ways of creating, using and sharing personal data. Understanding the structure of data protection in the EU is therefore essential to grasp its impact on individual rights, public administration, business models, and even technological development.

This lecture aims to provide a structured overview of the EU Data Protection Regulation, and to offer tools to understand the regulatory structure of the EU Data Regulation. The lecture will cover the following topics:

- Introduction to EU law
- Development of the EU data protection regulation
- Legal structure of data protection in the EU
- Role of national and sectoral laws
- Data protection as fundamental right
- Principles of data protection
- Lawfulness of personal data processing
- Anonymization and pseudonymization of personal data
- Special categories of personal data
- Rights of the data subject
- Transfer of personal data to third countries
- Responsibility of the controller and the processor
- Security of personal data and personal data breach
- Open Data Directive
- Data Governance Act
- Data Act

Organizational issues

Winter term 2025/26:

*****IMPORTANT*****

The sessions on 29.10., 5.11. and 12.11. will take place online via Zoom. All other sessions will take place in person in Seminar Room Nr. 303 at Vincenz-Prießnitz-Straße 3, in Karlsruhe.
Zoom link: https://kit-lecture.zoom-x.de/j/63132854819?pwd=6kyXgflG7H2SNwLX4qyerDasd56C5L.1

Wintersemester 2025/26:

*****WICHTIG*****

Die Sitzungen am 29.10., 5.11. und 12.11. finden online über Zoom statt. Alle anderen Sitzungen finden persönlich im Seminarraum Nr. 303 in der Vincenz-Prießnitz-Straße 3 in Karlsruhe statt.

Zoomlink: https://kit-lecture.zoom-x.de/j/63132854819?pwd=6kyXgflG7H2SNwLX4qyerDasd56C5L.1

T 6.234 Module component: European and International Law [T-INFO-101312]

Coordinators: Ulf Brühann
Organisation: KIT Department of Informatics
Part of: M-INFO-106754 - Public Economic and Technology Law
M-WIWI-104903 - Law

Type	Credits	Grading	Term offered	Version
Written examination	3 CP	graded	Each summer term	1

Courses					
ST 2026	24666	Europäisches und Internationales Recht	2 SWS	Lecture / 🗣	Brühann

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, × Cancelled

Below you will find excerpts from courses related to this module component:

V Europäisches und Internationales Recht

24666, SS 2026, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content
The course will be held in German.

The total workload for this course unit is 90 hours for 3 credit points, of which 22.5 hours are spent in attendance.

Organizational issues
Die drei folgenden Blockveranstaltungen finden jeweils im Seminarraum Nr. 313 (Geb. 07.08) statt:

- *Montag, den 27.04.2026, 09:30 - 17:00 (Mittagspause wird flexibel gehalten);*
- *Montag, den 01.06.2026, 09:30 - 17:00 (Mittagspause wird flexibel gehalten);*
- *Montag, den 29.06.2026, 09:30 - 17:00 Uhr (Mittagspause wird flexibel gehalten.*

(Stand per 13.11.2025)

Literature
Literatur wird in der Vorlesung angegeben.

Weiterführende Literatur

Erweiterte Literaturangaben werden in der Vorlesung bekannt gegeben.

T 6.235 Module component: Exam on Climatology [T-PHYS-105594]

Coordinators: Prof. Dr. Joaquim José Ginete Werner Pinto
Organisation: KIT Department of Physics
Part of: M-WIWI-104904 - Natural Sciences

Type	Credits	Grading	Version
Oral examination	1 CP	graded	5

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-PHYS-101092 - Climatology must have been passed.

T 6.236 Module component: Examination Prerequisite Project Management [T-BGU-113454]

Coordinators: Prof. Dr.-Ing. Shervin Haghsheno
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Expansion	Version
Coursework	0 CP	pass/fail	Each winter term	1 semesters	1

Courses					
WT 25/26	6200106	Project Management	2 SWS	Lecture / Practice / 🗣	Haghsheno, Schneider, Gloser

Legend: 🖥 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
4 online tests during the course with 20 questions und 20 min. duration each

Prerequisites
none

Recommendations
none

Additional Information
none

Workload
5 hours

Below you will find excerpts from courses related to this module component:

V Project Management
6200106, WS 25/26, 2 SWS, Language: German, Open in study portal

Lecture / Practice (VÜ)
On-Site

Content
This course provides a comprehensive introduction to (construction) project management. It takes a closer look at the organisation and delivery of a construction project from the client's perspective. In this context, a range of competences are presented that should be on hand for the successful execution of project management. In addition, we present a selection of project management methods for individual competences and illustrate them with case studies.

Organizational issues
Vorlesungen: Mittwochs vom 29.10.2025 bis 18.02.2026, jeweils 09:45 – 11:15 Uhr (hybrid)

Übungen: Asynchron ab 19.11.2025, 10.12.2025, 14.01.2026, 11.02.2026 (online)

Literature

- AHRENS, Hannsjörg; BASTIAN, Klemens; MUCHOWSKI, Lucian (Hrsg.) (2021) *Handbuch Projektsteuerung - Baumanagement: Ein praxisorientierter Leitfaden mit zahlreichen Hilfsmitteln und Arbeitsunterlagen*, 6. Auflage, Fraunhofer IRB Verlag, Stuttgart
- GPM Deutsche Gesellschaft für Projektmanagement e. V. (Hrsg.) (2017) *Individual Competence Baseline für Projektmanagement (Version 4.0)*, 1. Auflage, GPM Deutsche Gesellschaft für Projektmanagement e. V., Nürnberg
- HAGHSHENO, Shervin; JOHN, Paul Christian (2024) *Bauherrnseitige Projektmanagement-Dienstleistungen in Deutschland*, Forschungsbericht, DVP – Deutscher Verband für Projektmanagement in der Bau- und Immobilienwirtschaft e. V.
- KOCHENDÖRFER, Bernd; LIEBCHEN, Jens H.; VIERING, Markus G. (2021) *Bau-Projekt-Management: Grundlagen und Vorgehensweisen*, 6. Auflage, Springer Vieweg, Wiesbaden
- SCHULZ, Markus (2020) *Projektmanagement: Zielgerichtet. Effizient. Klar.*, 2. Auflage, UVK Verlag, Tübingen

T 6.237 Module component: Exercise Transportation Data Analysis [T-BGU-113671]

Coordinators: Prof. Dr.-Ing. Martin Kagerbauer
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: M-BGU-101065 - Transportation Modelling and Traffic Management
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Expansion	Version
Coursework	0 CP	pass/fail	Each winter term	1 semesters	1

Courses					
WT 25/26	6232901	Empirical Data in Transportation	2 SWS	Lecture / Practice /	Kagerbauer

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
Exercise to qualitative and quantitative analyses of travel surveys, appr. 2 pages

Prerequisites
none

Recommendations
none

Additional Information
none

Workload
10 hours

T 6.238 Module component: Exercise Transportation Data Analysis [T-BGU-113971]

Coordinators: Prof. Dr.-Ing. Peter Vortisch
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: M-BGU-101065 - Transportation Modelling and Traffic Management
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Expansion	Version
Coursework	0 CP	pass/fail	Each summer term	1 semesters	1

Courses					
ST 2026	6232802	Traffic Management and Telematics	2 SWS	Lecture / Practice / 🗣	Vortisch

Legend: 🖥 Online, ♻ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
programming exercise with Python

Prerequisites
none

Recommendations
none

Additional Information
none

Workload
10 hours

T 6.239 Module component: Exercises for Applied Materials Simulation [T-MACH-110928]

Coordinators: Prof. Dr. Peter Gumbsch
Dr.-Ing. Johannes Schneider
Dr. Daniel Weygand

Organisation: KIT Department of Mechanical Engineering

Part of: M-WIWI-107718 - Mechanical Engineering I
M-WIWI-107730 - Mechanical Engineering II

Type	Credits	Grading	Term offered	Version
Coursework	2 CP	pass/fail	Each summer term	1

Courses					
ST 2026	2182616	Applied Materials Simulation	4 SWS	Lecture / Practice / 🗣	Gumbsch, Weygand

Legend: 🖥 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
successful solving of all exercises

Prerequisites
T-MACH-107671 – Übungen zu Angewandte Werkstoffsimulation has not been started

Additional Information
The course is offered in English.

Workload
60 hours

Below you will find excerpts from courses related to this module component:

V Applied Materials Simulation

2182616, SS 2026, 4 SWS, Language: English, Open in study portal

Lecture / Practice (VÜ)
On-Site

Content
This lecture should give the students an overview of different simulation methods in the field of materials science and engineering. Numerical methods are presented and their use in different fields of application and size scales shown and discussed. On the basis of theoretical as well as practical aspects, a critical examination of the opportunities and challenges of numerical material simulation shall be carried out.

The student can

- define different numerical methods and distinguish their range of application
- approach issues by applying the finite element method and discuss the processes and results
- understand complex processes of metal forming and crash simulation and discuss the structural and material behavior
- define and apply the physical fundamentals of particle-based simulation techniques to applications of materials science
- illustrate the range of application of atomistic simulation methods and distinguish between different models

preliminary knowlegde in mathematics, physics and materials science recommended

regular attendance: 34 hours

exercise: 11 hours

self-study: 165 hours

oral exam ca. 35 minutes

no tools or reference materials

admission to the exam only with successful completion of the exercises

Literature

1. D. Frenkel, B. Smit: Understanding Molecular Simulation: From Algorithms to Applications, Academic Press, 2001
2. W. Kurz, D.J. Fisher: Fundamentals of Solidification, Trans Tech Publications, 1998
3. P. Haupt: Continuum Mechanics and Theory of Materials, Springer, 1999
4. M. P. Allen, D. J. Tildesley: Computer simulation of liquids, Clarendon Press, 1996

T 6.240 Module component: Exercises in Civil Law [T-INFO-102013]

Coordinators: Dr. Yvonne Matz
Organisation: KIT Department of Informatics
Part of: M-INFO-101191 - Commercial Law
M-WIWI-104903 - Law

Type	Credits	Grading	Term offered	Version
Examination of another type	9 CP	graded	Each term	3

Courses					
WT 25/26	2424011	Commercial and Corporate Law	2 SWS	Lecture / 🗣	Danek
WT 25/26	2424017	Exercises in Civil Law	2 SWS	Lecture / 🗣	Bress, Bosbach
ST 2026	24504	Advanced Civil Law	2 SWS	Lecture / 🗣	Matz
ST 2026	24506	Exercises in Civil Law	2 SWS	Lecture / 🗣	Sattler
ST 2026	24926	Case Studies in Civil Law	2 SWS	Practice / 🗣	Sattler, Bosbach
WT 26/27	2424017	Exercises in Civil Law	2 SWS	Lecture / 🗣	Sattler

Legend: 🖥 Online, ❁ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module M-INFO-101190 - Introduction to Civil Law must have been passed.

T 6.241 Module component: Exercises: Membrane Technologies [T-CIWVT-113235]

Coordinators: Prof. Dr. Harald Horn
Dr.-Ing. Florencia Saravia

Organisation: KIT Department of Chemical and Process Engineering

Part of: M-CIWVT-101122 - Water Chemistry and Water Technology II
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Coursework	1 CP	pass/fail	Each summer term	1

Courses					
ST 2026	2233011	Membrane Technologies in Water Treatment - Excercises	1 SWS	Practice / 🧩	Horn, Saravia, und Mitarbeitende

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, x Cancelled

Assessment
Learning control is a completed coursework: Submission of exercises, membrane design and short presentation (5 minutes, group work).

T 6.242 Module component: Experimental Design [T-WIWI-111395]

Coordinators: Prof. Dr. Benjamin Scheibehenne
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Expansion	Version
Examination of another type	4,5 CP	graded	Each term	1 semesters	1

Assessment
Alternative exam assessment. Details will be announced at the beginning of the course.

Additional Information
The course provides an overview of important methods of empirical research. Students learn basic theories and methods that are relevant in planning, conducting and evaluating experiments. They learn to analyze, critique, and independently develop experimental designs. The course covers, for example, the development of a research question, formulation of scientific hypotheses, sample selection, calculation of statistical power, the difference between correlative and causal relationships, and the relevance of experimental research to test the latter.

Exemplary studies from decision research are analyzed and discussed with respect to experimental design.

The workload of the course is 4.5 ECTS. This consists of exercises, smaller presentations by the students during the semester, as well as the preparation of the examination at the end of the semester.

The number of participants is limited. Places are allocated via the Wiwi-Portal. Course language is German.

Workload
135 hours

T 6.243 Module component: Experimental Lab Class in Welding Technology, in Groups [T-MACH-102099]

Coordinators: Dr.-Ing. Stefan Dietrich
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101268 - Specific Topics in Materials Science
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Coursework	4 CP	pass/fail	Each winter term	4

Courses					
WT 25/26	2173560	Welding Lab Course, in groupes	3 SWS	Practical course / 🗣	Dietrich, Schulze
WT 26/27	2173560	Welding Lab Course, in groupes	3 SWS	Practical course / 🗣	Dietrich, Schulze

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Work log

Additional Information
The course is offered in German.

The lab takes place at the beginning of the winter semester break once a year. The registration is possible during the lecture period via iam-wk-lehre@iam.kit.edu at the IAM – WK. The lab is carried out in the Handwerkskammer Karlsruhe.

If possible, please bring your own work shoes and long, disposable pants and tops, as we will get our hands dirty and be confronted with molten, flying metal.

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Welding Lab Course, in groupes

2173560, WS 25/26, 3 SWS, Language: German, Open in study portal

Practical course (P)
On-Site

Content
The lab takes place at the beginning of the winter semester break once a year. The registration is possible during the lecture period in the secretariat of the Institute of Applied Materials (IAM – WK). The lab is carried out in the Handwerkskammer Karlsruhe.

learning objectives: The students are capable to name a survey of current welding processes and their suitability for joining different metals. The students can evaluate the advantages and disadvantages of the individual procedures. The students have weld with different welding processes.

requirements:

You need sturdy shoes and long clothes!

workload:

regular attendance: 50 hours
preparation: 70 hours

Organizational issues
Die Anmeldung erfolgt durch den Beitritt in den ILIAS-Kurs.

Die Lehrveranstaltung "Experimentelles schweißtechnisches Praktikum" findet aufgrund der hohen Nachfrage nun in vier Gruppen statt:

Gruppe 1: 16.2.26 bis 18.2.26 Gruppe 2: 18.2.26 bis 20.2.26

Gruppe 3: 23.2.26 bis 25.2.26 Gruppe 4: 25.2.26 bis 27.2.26

Der Tag beginnt jeweils um 8 Uhr und endet um 16 Uhr bzw. freitags um 15 Uhr. Mittwochs um 13 Uhr findet der Gruppenwechsel statt.

Die Gruppeneinteilung erfolgt über selbstständigen Beitritt in eine der vier Gruppen.

Der Veranstaltungsort ist die

Bildungsakademie Handwerkskammer Karlsruhe
Hertzstr. 177
76187 Karlsruhe

Bitte bringen Sie, wenn möglich, ihre eigenen Arbeitsschuhe und lange und entbehrliche Hosen sowie Oberteile mit, da wir uns die Hände schmutzig machen und mit flüssigem, umherfliegendem Metall konfrontiert sein werden. Für die Mittagspause können Sie sich selbst versorgen oder auch in der Mensa der Bildungsakademie essen.

Literature
wird im Praktikum ausgegeben

V **Welding Lab Course, in groupes**
2173560, WS 26/27, 3 SWS, Language: German, Open in study portal

Practical course (P)
On-Site

Content
The lab takes place at the beginning of the winter semester break once a year. The registration is possible during the lecture period in the secretariat of the Institute of Applied Materials (IAM – WK). The lab is carried out in the Handwerkskammer Karlsruhe.

learning objectives: The students are capable to name a survey of current welding processes and their suitability for joining different metals. The students can evaluate the advantages and disadvantages of the individual procedures. The students have weld with different welding processes.

requirements:

You need sturdy shoes and long clothes!

workload:

regular attendance: 50 hours
preparation: 70 hours

Organizational issues
Die Anmeldung erfolgt durch den Beitritt in den ILIAS-Kurs.

Das Datum wird noch bekanntgegeben.

Der Veranstaltungsort ist die

Bildungsakademie Handwerkskammer Karlsruhe
Hertzstr. 177
76187 Karlsruhe

Bitte bringen Sie, wenn möglich, ihre eigenen Arbeitsschuhe und lange und entbehrliche Hosen sowie Oberteile mit, da wir uns die Hände schmutzig machen und mit flüssigem, umherfliegendem Metall konfrontiert sein werden. Für die Mittagspause können Sie sich selbst versorgen oder auch in der Mensa der Bildungsakademie essen.

Literature
wird im Praktikum ausgegeben

T 6.244 Module component: Experimental Research in Economics and Information Systems [T-WIWI-115061]

Coordinators: Prof. Dr. Christof Weinhardt

Organisation: KIT Department of Business and Economics

Part of: M-WIWI-101446 - Market Engineering
M-WIWI-101453 - Applied Strategic Decisions
M-WIWI-101505 - Experimental Economics
M-WIWI-103118 - Data Science: Data-Driven User Modeling
M-WIWI-104900 - Business Administration
M-WIWI-105923 - Incentives, Interactivity & Decisions in Organizations

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each winter term	1

Courses					
WT 25/26	2540489	Experimental Economics	2 SWS	Lecture / 🗣	Knierim
WT 25/26	2540493	Übung zu Experimental Economics	1 SWS	Practice / 🗣	del Puppo

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success is assessed in the form of a written examination (60 min).

Prerequisites
None

Additional Information
The language of the course is English.

Below you will find excerpts from courses related to this module component:

V Experimental Economics

2540489, WS 25/26, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Literature

- Strategische Spiele; S. Berninghaus, K.-M. Ehrhart, W. Güth; Springer Verlag, 2. Aufl. 2006.
- Handbook of Experimental Economics; J. Kagel, A. Roth; Princeton University Press, 1995.
- Experiments in Economics; J.D. Hey; Blackwell Publishers, 1991.
- Experimental Economics; D.D. Davis, C.A. Holt; Princeton University Press, 1993.
- Experimental Methods: A Primer for Economists; D. Friedman, S. Sunder; Cambridge University Press, 1994.

T 6.245 Module component: Extraordinary Additional Course in the Module Cross-Functional Management Accounting [T-WIWI-108651]

Coordinators: Prof. Dr. Marcus Wouters
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101510 - Cross-Functional Management Accounting

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each term	1

Assessment
The assessment depends on which extraordinary course becomes part of the module "Cross-Functional Management Accounting".

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Prerequisites
None

Additional Information
The pupose of this placeholder is to make it possible zu include an extraordinary course in the module "Cross-Functional Management Accounting". Proposals for specific courses have to be approved in advance by the module coordinator.

T 6.246 Module component: Fabrication Processes in Microsystem Technology [T-MACH-102166]

Coordinators: Prof. Dr.-Ing. Jürgen Brandner
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101291 - Microfabrication
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each term	1

Courses					
WT 25/26	2143882	Fabrication Processes in Microsystem Technology	2 SWS	Lecture / Blended	Bade
WT 26/27	2143882	Fabrication Processes in Microsystem Technology	2 SWS	Lecture / On-Site	Brandner

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
Oral examination, 20 minutes

Prerequisites
none

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Fabrication Processes in Microsystem Technology
2143882, WS 25/26, 2 SWS, Language: German, Open in study portal

Lecture (V)
Blended (On-Site/Online)

Literature
M. Madou
Fundamentals of Microfabrication

CRC Press, Boca Raton, 1997

W. Menz, J. Mohr, O. Paul

Mikrosystemtechnik für Ingenieure

Dritte Auflage, Wiley-VCH, Weinheim 2005

L.F. Thompson, C.G. Willson, A.J. Bowden
Introduction to Microlithography
2nd Edition, ACS, Washington DC, 1994

V Fabrication Processes in Microsystem Technology
2143882, WS 26/27, 2 SWS, Language: German/English, Open in study portal

Lecture (V)
On-Site

Content

The lecture provides an in-depth overview of manufacturing technologies for structure generation in microsystems engineering. A range of fundamental technologies is presented in detail, while related methods are introduced to enable independent further study. In addition to subtractive processes, particular emphasis is placed on modern additive structuring techniques. Various materials are considered, each with their specific properties and processing requirements.

A further focus lies on the design and application of manufacturing techniques for the fabrication of microscale devices. Relationships between process requirements (applications), materials, design, and manufacturing methods are systematically addressed. Aspects such as dimensional accuracy, structural quality, and surface properties are evaluated with respect to fabrication and functionality.

All process steps are explained in the context of applications from different areas of microsystems engineering, particularly microfluidics, in-situ and operando analytics, as well as micro process engineering. A solid understanding of the underlying mechanisms at the interface of engineering, physics, and chemistry is developed. Key concepts such as mass transport and kinetic control are used to derive relationships between process parameters and structural properties.

A particular focus is placed on the analysis of interactions within complex process chains and on cause–effect relationships between individual manufacturing steps. In addition, aspects such as surface homogeneity and defect control are discussed as key factors influencing the quality of microscale structures and their performance in applications.

Organizational issues

Die Vorlesung wird nur im Wintersemester angeboten.

Die Lehrveranstaltung wird in der Regel auf Englisch gehalten. Bei einer geringen Anzahl an englischsprachigen Studierenden kann die Durchführung auf Deutsch erfolgen. Die Lehrmaterialien werden in englischer Sprache zur Verfügung gestellt.

The lecture is offered exclusively in the winter semester.

The course is generally taught in English. Depending on the number of English-speaking students, it may be conducted in German. Course materials will be provided in English.

Literature

Menz, W., Mohr, J., O. Paul: Mikrosystemtechnik für Ingenieure, VCH-Verlag, Weinheim, 2005

M. Madou
Fundamentals of Microfabrication
Taylor & Francis Ltd.; Auflage: 3. Auflage. 2011

6.247 Module component: Facility and Real Estate Management II [T-BGU-111212]

Coordinators: Prof. Dr.-Ing. Kunibert Lennerts
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: M-BGU-105592 - Digitalization in Facility Management
M-BGU-106453 - Facility Management in Hospitals for Industrial Engineering

Type	Credits	Grading	Term offered	Expansion	Version
Oral examination	1,5 CP	graded	Each term	1 semesters	1

Courses					
WT 25/26	6242909	Facility and Real Estate Management II	1 SWS	Lecture / 🗣	Lennerts, Mitarbeiter/innen

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
oral exam, appr. 20 min.

Prerequisites
none

Recommendations
none

Additional Information
none

Workload
45 hours

T 6.248 Module component: Facility Location and Strategic Supply Chain Management [T-WIWI-102704]

Coordinators: Prof. Dr. Stefan Nickel
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104899 - Operations Research

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each winter term	4

Assessment
The assessment consists of a written exam (60 min) according to Section 4 (2), 1 of the examination regulation.
The exam takes place in every semester.
Prerequisite for admission to examination is the succesful completion of the online assessments.

Prerequisites
Prerequisite for admission to examination is the succesful completion of the online assessments.

Recommendations
None

Additional Information
The lecture is held in every winter term. The planned lectures and courses for the next three years are announced online at https://dol.ior.kit.edu/Lehrveranstaltungen.php.

Workload
135 hours

T 6.249 Module component: Facility Management in Hospitals [T-BGU-108004]

Coordinators: Prof. Dr.-Ing. Kunibert Lennerts
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: M-BGU-106453 - Facility Management in Hospitals for Industrial Engineering

Type	Credits	Grading	Term offered	Expansion	Version
Examination of another type	6 CP	graded	Each winter term	1 semesters	2

Courses					
WT 25/26	6242905	Facility Management in Hospitals	4 SWS	Lecture / Practice / 🗣	Lennerts, Faeghinezhad, Mitarbeiter/innen

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
term paper appr. 10 pages, with final presentation appr. 10 min.

Prerequisites
none

Recommendations
none

Additional Information
none

Workload
180 hours

T 6.250 Module component: Failure of Structural Materials: Deformation and Fracture [T-MACH-102140]

Coordinators: Prof. Dr. Peter Gumbsch
Dr. Daniel Weygand

Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101268 - Specific Topics in Materials Science
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each winter term	1

Courses					
WT 25/26	2181711	Failure of structural materials: deformation and fracture	3 SWS	Lecture / Practice / 🗣	Gumbsch, Weygand

Legend: 🖥 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
oral exam ca. 30 minutes

no tools or reference materials

Prerequisites
none

Recommendations
preliminary knowlegde in mathematics, mechanics and materials science

Additional Information
The course is offered in German.

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Failure of structural materials: deformation and fracture

Lecture / Practice (VÜ)
On-Site

2181711, WS 25/26, 3 SWS, Language: German, Open in study portal

Content

1. Introduction
2. linear elasticity
3. classification of stresses
4. Failure due to plasticity
    - tensile test
    - dislocations
    - hardening mechanisms
    - guidelines for dimensioning
5. composite materials
6. fracture mechanics
    - hypotheses for failure
    - linear elasic fracture mechanics
    - crack resitance
    - experimental measurement of fracture toughness
    - defect measurement
    - crack propagation
    - application of fracture mechanics
    - atomistics of fracture

The student

- has the basic understanding of mechanical processes to explain the relationship between externally applied load and materials strength.
- can explain the foundation of linear elastic fracture mechanics and is able to determine if this concept can be applied to a failure by fracture.
- can decribe the main empirical materials models for deformation and fracture and can apply them.
- has the physical understanding to describe and explain phenomena of failure.

preliminary knowlegde in mathematics, mechanics and materials science recommended

regular attendance: 22,5 hours
self-study: 97,5 hours

The assessment consists of an oral examination (ca. 30 min) according to Section 4(2), 2 of the examination regulation.

Organizational issues
Übungstermine werden in der Vorlesung bekannt gegeben!

Die Veranstaltung wird letztmals im Wintersemester 2025/2026 angeboten!

Literature

- Engineering Materials, M. Ashby and D.R. Jones (2nd Edition, Butterworth-Heinemann, Oxford, 1998); sehr lesenswert, relativ einfach aber dennoch umfassend, verständlich
- Mechanical Behavior of Materials, Thomas H. Courtney (2nd Edition, McGraw Hill, Singapur); Klassiker zu den mechanischen Eigenschaften der Werkstoffe, umfangreich, gut
- Bruchvorgänge in metallischen Werkstoffen, D. Aurich (Werkstofftechnische Verlagsgesellschaft Karlsruhe), relativ einfach aber dennoch umfassender Überblick für metallische Werkstoffe

T 6.251 Module component: Failure of Structural Materials: Fatigue and Creep [T-MACH-102139]

Coordinators: Dr. Patric Gruber
Prof. Dr. Peter Gumbsch

Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101268 - Specific Topics in Materials Science
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each winter term	1

Courses					
WT 25/26	2181715	Failure of Structural Materials: Fatigue and Creep	2 SWS	Lecture / 🗣	Gruber, Gumbsch

Legend: 🖥 Online, Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
oral exam ca. 30 minutes

no tools or reference materials

Prerequisites
none

Recommendations
preliminary knowlegde in mathematics, mechanics and materials science

Additional Information
The course is offered in German.

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Failure of Structural Materials: Fatigue and Creep

Lecture (V)
On-Site

2181715, WS 25/26, 2 SWS, Language: German, Open in study portal

Content
1 Fatigue
1.1 Introduction
1.2 Lifetime
1.3 Fatigue Mechanisms
1.4 Material Selection
1.5 Notches and Shape Optimization
1.6 Case Studies: ICE-Accidents

2 Creep
2.1 Introduction
2.2 High Temperature Plasticity
2.3 Phänomenological DEsciption of Creep
2.4 Creep Mechanisms
2.5 Alloying Effects

The student

- has the basic understanding of mechanical processes to explain the relationships between externally applied load and materials strength.
- can describe the main empirical materials models for fatigue and creep and can apply them.
- has the physical understanding to describe and explain phenomena of failure.
- can use statistical approaches for reliability predictions.
- can use its acquired skills, to select and develop materials for specific applications.

preliminary knowlegde in mathematics, mechanics and materials science recommended

regular attendance: 22,5 hours
self-study: 97,5 hours

The assessment consists of an oral examination (ca. 30 min) according to Section 4(2), 2 of the examination regulation.

Organizational issues
Die Veranstaltung wird letztmals im Wintersemester 2025/2026 angeboten!

Literature

- Engineering Materials, M. Ashby and D.R. Jones (2nd Edition, Butterworth-Heinemann, Oxford, 1998); sehr lesenswert, relativ einfach aber dennoch umfassend, verständlich
- Mechanical Behavior of Materials, Thomas H. Courtney (2nd Edition, McGraw Hill, Singapur); Klassiker zu den mechanischen Eigenschaften der Werkstoffe, umfangreich, gut
- Bruchvorgänge in metallischen Werkstoffen, D. Aurich (Werkstofftechnische Verlagsgesellschaft Karlsruhe), relativ einfach aber dennoch umfassender Überblick für metallische Werkstoffe
- Fatigue of Materials, Subra Suresh (2nd Edition, Cambridge University Press); Standardwerk über Ermüdung, alle Materialklassen, umfangreich, für Einsteiger und Fortgeschrittene

T 6.252 Module component: Field Propagation and Coherence [T-ETIT-100976]

Coordinators: Prof.Dr.Dr.h.c. Wolfgang Freude
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-100566 - Field Propagation and Coherence

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each winter term	1

Courses					
WT 25/26	2309466	Field Propagation and Coherence	2 SWS	Lecture / 🗣	Freude
WT 25/26	2309467	Tutorial for 2309466 Field Propagation and Coherence	1 SWS	Practice / 🗣	Freude, N.N.

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Type of Examination: oral exam

Duration of Examination: approx. 20 minutes

Modality of Exam: Oral examination, usually one examination day per month during the summer and winter terms. An extra questions-and-answers session will be held for preparation if students wish so.

Prerequisites
none

Recommendations
Minimal background required: Calculus, differential equations and Fourier transform theory. Electrodynamics and field calculations or a similar course on electrodynamics or optics is recommended.
Solution of the problems on the exercise sheet, which can be downloaded as homework each week, is highly recommended. Also, active participation in the problem classes and studying in learning groups are strongly advised.

T 6.253 Module component: Financial Accounting and Cost Accounting [T-WIWI-102816]

Coordinators: Dr. Jan-Oliver Strych
Organisation: KIT Department of Informatics
KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	4 CP	graded	Each winter term	1

Assessment
The assessment consists of a written exam following §4, Abs. 2, 1 of the examination regulation.

The examination takes place in every semester. Re-examinations are offered at eyery ordinary examination date.

Prerequisites
None

T 6.254 Module component: Financial Accounting for Global Firms [T-WIWI-107505]

Coordinators: Dr. Torsten Luedecke
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	see Annotations	1

Courses					
WT 25/26	2530242	Financial Accounting for Global Firms	2 SWS	Lecture / 🗣	Luedecke
WT 25/26	2530243	Übung zu Financial Accounting for Global Firms	1 SWS	Practice / 🗣	Luedecke

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The final assessment will be offered for the last time in August 2026. It will take the form of a written examination (60 min.) in accordance with § 4 (2) No. 1 of the Study Regulations. The grade is based on the result of the written examination.

Prerequisites
None

Recommendations
Basic knowledge in corporate finance and accounting.

Additional Information
This course will no longer be offered starting in the winter semester of 2026–2027.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Financial Accounting for Global Firms

2530242, WS 25/26, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Literature
Alexander, D. and C. Nobes (2017): Financial Accounting – An International Introduction, 6th ed., Pearson.

Coenenberg, A.G., Haller, A. und W. Schultze (2016): Jahresabschluss und Jahresabschlussanalyse, 24. Auflage. Schäffer-Poeschel Verlag Stuttgart.

T 6.255 Module component: Financial Analysis [T-WIWI-102900]

Coordinators: Dr. Torsten Luedecke
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101480 - Finance 3
M-WIWI-101483 - Finance 2
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	see Annotations	3

Courses					
ST 2026	2530205	Financial Analysis	2 SWS	Lecture / 🗣	Luedecke
ST 2026	2530206	Übungen zu Financial Analysis	2 SWS	Practice / 🗣	Luedecke

Legend: 🖥 Online, ❖ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The final assessment will be offered for the last time in March 2027. It will take the form of a 60-minute written examination during the semester break (pursuant to §4(2), 1 SPO).

The grade is based on the result of the written examination.

Prerequisites
None

Recommendations
Basic knowledge in corporate finance, accounting, and valuation is required.

Additional Information
The lecture will no longer be offered starting in the 2027 summer semester

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Financial Analysis

2530205, SS 2026, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Literature

- Alexander, D. and C. Nobes (2017): Financial Accounting – An International Introduction, 6th ed., Pearson.
- Penman, S.H. (2013): Financial Statement Analysis and Security Valuation, 5th ed., McGraw Hill.

T 6.256 Module component: Financial Data Science [T-WIWI-111238]

Coordinators: Prof. Dr. Maxim Ulrich
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Examination of another type	9 CP	graded	Each summer term	2

Courses					
ST 2026	2530371	Financial Data Science	4 SWS	Lecture / 🗣	Ulrich

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, × Cancelled

Assessment
The examination is structured as an alternative assessment.

Further details regarding submission deadlines, exam format, and retake opportunities will be announced in the first session.

Prerequisites
None.

Workload
270 hours

T 6.257 Module component: Financial Data Science in Practice [T-WIWI-114633]

Coordinators: Prof. Dr. Maxim Ulrich
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Examination of another type	9 CP	graded	Each term	1

Assessment
The examination is structured as an alternative exam assessment and consists of the following components:

- **Interim Presentations (Formative Milestones)**
- **Quality of Implementation** (Reproducibility, Code Documentation, Tests)
- **Final Written Paper** (15 – 20 pages, scientific standard)
- **Final Presentation** (approx. 30 minutes, public research talk)

The overall grade is determined by the weighted evaluation of the individual components of this assessment. The exact weighting of these components for grade calculation will be announced at the beginning of the course.

Prerequisites
The course T-WIWI-111238 "Financial Data Science" must be passed.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-WIWI-111238 - Financial Data Science must have been passed.

Workload
270 hours

T 6.258 Module component: Financial Econometrics [T-WIWI-103064]

Coordinators: Prof. Dr. Melanie Schienle
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101638 - Econometrics and Statistics I
M-WIWI-101639 - Econometrics and Statistics II
M-WIWI-104902 - Statistics

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each winter term	2

Courses					
WT 25/26	2520022	Financial Econometrics I	2 SWS	Lecture / 🗣	Schienle
WT 25/26	2520023	Übungen zu Financial Econometrics I	2 SWS	Practice / 🗣	Buse
WT 26/27	2520022	Financial Econometrics I	2 SWS	Lecture / 🗣	Schienle
WT 26/27	2520023	Übungen zu Financial Econometrics I	2 SWS	Practice / 🗣	Buse

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, × Cancelled

Assessment
The assessment consists of a written exam (90 minutes) (following §4(2), 1 of the examination regulation).

Prerequisites
None

Recommendations
It is recommended that students have substantive knowledge of the course “Economics III: Introduction to Econometrics”.

Additional Information
The course will be held in English.

Below you will find excerpts from courses related to this module component:

V Financial Econometrics I

2520022, WS 25/26, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content
Learning objectives:

The student

- shows a broad knowledge of fincancial econometric estimation and testing techniques
- is able to apply his/her technical knowledge using software in order to critically assess empirical problems

Content:

ARMA, ARIMA, ARFIMA, (non)stationarity, causality, cointegration, ARCH/GARCH, stochastic volatility models, computer based exercises

Requirements:

It is recommended to attend the course *Economics III: Introduction to Econometrics* [2520016] prior to this course.

Workload:

Total workload for 4.5 CP: approx. 135 hours

Attendance: 30 hours

Preparation and follow-up: 65 hours

Exam preparation: 40 hours

Literature

Taylor, S. J. (2005): "Asset Price Dynamics, Volatility, and Prediction", Princeton University Press.

Tsay, R. S. (2005): "Analysis of Financial Time Series: Financial Econometrics", Wiley, 2nd edition.

Cochrane, J. H. (2005): "Asset Pricing", revised edition, Princeton University Press.

Campbell, J. Y., A. W. Lo, and A. C. MacKinlay (1997): "The Econometrics of Financial Markets", Princeton University Press.

Hamilton, J. D. (1994): "Time Series Analysis", Princeton University Press.

Additional literature will be discussed in the lecture.

V **Financial Econometrics I**
2520022, WS 26/27, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content
Learning objectives:

The student

- shows a broad knowledge of fincancial econometric estimation and testing techniques
- is able to apply his/her technical knowledge using software in order to critically assess empirical problems

Content:

ARMA, ARIMA, ARFIMA, (non)stationarity, causality, cointegration, ARCH/GARCH, stochastic volatility models, computer based exercises

Requirements:

It is recommended to attend the course *Economics III: Introduction to Econometrics* [2520016] prior to this course.

Workload:

Total workload for 4.5 CP: approx. 135 hours

Attendance: 30 hours

Preparation and follow-up: 65 hours

Exam preparation: 40 hours

Literature

Taylor, S. J. (2005): "Asset Price Dynamics, Volatility, and Prediction", Princeton University Press.

Tsay, R. S. (2005): "Analysis of Financial Time Series: Financial Econometrics", Wiley, 2nd edition.

Cochrane, J. H. (2005): "Asset Pricing", revised edition, Princeton University Press.

Campbell, J. Y., A. W. Lo, and A. C. MacKinlay (1997): "The Econometrics of Financial Markets", Princeton University Press.

Hamilton, J. D. (1994): "Time Series Analysis", Princeton University Press.

Additional literature will be discussed in the lecture.

T 6.259 Module component: Financial Econometrics II [T-WIWI-110939]

Coordinators: Prof. Dr. Melanie Schienle
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101638 - Econometrics and Statistics I
M-WIWI-101639 - Econometrics and Statistics II
M-WIWI-104902 - Statistics

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each summer term	3

Courses					
ST 2026	2521302	Financial Econometrics II	2 SWS	Lecture / 🗣	Schienle, Buse
ST 2026	2521303	Übung zu Financial Econometrics II	2 SWS	Practice / 🗣	Buse, Schienle

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Written examination (90 minutes). If the number of participants is low, an oral examination will be held instead.

Prerequisites
None

Recommendations
It is recommended that students have substantive knowledge of the course **"Financial Econometrics."**

Additional Information
The course will be held in English.

Workload
135 hours

T 6.260 Module component: Financial Intermediation [T-WIWI-102623]

Coordinators: Prof. Dr. Martin Ruckes
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101453 - Applied Strategic Decisions
M-WIWI-101480 - Finance 3
M-WIWI-101483 - Finance 2
M-WIWI-101502 - Economic Theory and its Application in Finance
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each winter term	1

Courses					
WT 25/26	2530232	Financial Intermediation	2 SWS	Lecture / 🗣	Ruckes
WT 25/26	2530233	Übung zu Finanzintermediation	1 SWS	Practice	Ruckes, Benz

Legend: 🖥 Online, Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment of this course is a written examination (following §4(2), 1 SPO) of 60 mins.

The exam is offered each semester.

Prerequisites
None

Recommendations
None

Below you will find excerpts from courses related to this module component:

V Financial Intermediation

2530232, WS 25/26, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Organizational issues
Terminankündigungen des Instituts beachten

Literature
Weiterführende Literatur:

- Hartmann-Wendels/Pfingsten/Weber (2014): Bankbetriebslehre, 6. Auflage, Springer Verlag.
- Freixas/Rochet (2008): Microeconomics of Banking, 2. Auflage, MIT Press.

T 6.261 Module component: Financial Management [T-WIWI-102605]

Coordinators: Prof. Dr. Martin Ruckes
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each summer term	1

Courses					
ST 2026	2530216	Financial Management	2 SWS	Lecture / 🗣	Ruckes
ST 2026	2530217	Übung zu Financial Management	1 SWS	Practice / 🗣	Ruckes

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment consists of a written exam (60 min.) according to Section 4 (2), 1 of the examination regulation.
The exam takes place at every semester. Re-examinations are offered at every ordinary examination date.

Prerequisites
None

Recommendations
Knowledge of the content of the course Business Administration: Finance and Accounting [25026/25027] is recommended.

Below you will find excerpts from courses related to this module component:

V Financial Management

2530216, SS 2026, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Literature
Weiterführende Literatur:

- Ross, Westerfield, Jaffe, Jordan (2009): Modern Financial Management, McGraw-Hill International Edition
- Berk, De Marzo (2016): Corporate Finance, 4. Edition, Pearson Addison Wesley

T 6.262 Module component: FinTech [T-WIWI-112694]

Coordinators: Prof. Dr. Julian Thimme
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each winter term	1

Courses					
WT 25/26	2500032	FinTech	2 SWS	Lecture / 🗣	Thimme
WT 25/26	2500219	FinTech (Exercice)	1 SWS	Practice / 🗣	Meng, Thimme
WT 26/27	2500032	FinTech	2 SWS	Lecture / 🗣	Thimme
WT 26/27	2500219	FinTech (Exercice)	1 SWS	Practice / 🗣	Meng, Thimme

Legend: 🖥 Online, ⚭ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Written examination (90 minutes) during the lecture-free period of the semester (according to §4(2), 1 SPO).

The examination is offered every semester and can be repeated at any regular examination date.

Prerequisites
None

Recommendations
Knowledge of the course Business Administration: Finance and Accounting [25026/25027] is very helpful.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V FinTech — **Lecture (V) On-Site**
2500032, WS 25/26, 2 SWS, Language: English, Open in study portal

Content
Banks and other financial institutions are central entities that are crucial to the way capital markets are organized today. They have a variety of tasks: They lend to business and retail customers, thereby creating money, accept deposits, transfer money, advise customers, help companies issue stock, and much more. While all of these services are important for the smooth allocation of capital in an economy, recent technological advances allow companies called FinTechs to challenge banks' traditional business models. While FinTechs are still centralized entities, decentralized innovations based on blockchain technologies have the potential to fundamentally upend the system.

We start with a discussion of the problems with capital allocation and what financial institutions are doing to address these problems. The key question we are interested in is how efficient alternative solutions offered by traditional banks, FinTechs and Decentralized Finance applications are compared to each other. What are the advantages and disadvantages of the new technologies and how likely are they to prevail? In the different parts of the course, we will explore specific examples, including the status quo and possible ways forward.

V FinTech — **Lecture (V) On-Site**
2500032, WS 26/27, 2 SWS, Language: English, Open in study portal

Content

Banks and other financial institutions are central entities that are crucial to the way capital markets are organized today. They have a variety of tasks: They lend to business and retail customers, thereby creating money, accept deposits, transfer money, advise customers, help companies issue stock, and much more. While all of these services are important for the smooth allocation of capital in an economy, recent technological advances allow companies called FinTechs to challenge banks' traditional business models. While FinTechs are still centralized entities, decentralized innovations based on blockchain technologies have the potential to fundamentally upend the system.

We start with a discussion of the problems with capital allocation and what financial institutions are doing to address these problems. The key question we are interested in is how efficient alternative solutions offered by traditional banks, FinTechs and Decentralized Finance applications are compared to each other. What are the advantages and disadvantages of the new technologies and how likely are they to prevail? In the different parts of the course, we will explore specific examples, including the status quo and possible ways forward.

T 6.263 Module component: Flows with Chemical Reactions [T-MACH-105422]

Coordinators: apl. Prof. Dr. Andreas Class
Organisation: KIT Department of Mechanical Engineering

Part of: M-WIWI-107718 - Mechanical Engineering I
M-WIWI-107730 - Mechanical Engineering II

Type	Credits	Grading	Term offered	Version
Oral examination	4,5 CP	graded	Each winter term	2

Courses					
WT 25/26	2153406	Flows with chemical reactions	2 SWS	Lecture /	Class
WT 26/27	2153406	Flows with chemical reactions	2 SWS	Lecture /	Class

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
oral exam, duration 30 minutes

Auxiliary none

Prerequisites
none

Recommendations
Fluid Mechanics (T-MACH-105207)

Mathematical Methods in Fluid Mechanics (T-MACH-105295)

Additional Information
The course is offered in English.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Flows with chemical reactions

2153406, WS 25/26, 2 SWS, Language: English, Open in study portal

Lecture (V)
Blended (On-Site/Online)

Content
In this lecture, students acquire a solid understanding of fundamental phenomena in reacting flows. They learn how the high temperature sensitivity of chemical reactions leads to significant simplifications in modeling, particularly by concentrating the reaction on thin layers, known as free interfaces. The students are able to understand and apply simplified chemical models, with a focus on the fluid mechanical aspects. They develop the ability to use analytical methods to solve idealized problems and to simplify more complex issues so that efficient numerical methods can be applied. Furthermore, they learn to derive fluid mechanical equations in curvilinear coordinates and apply them to suitable problems. Overall, the students are empowered to formulate simplified models for reactive flows, critically analyze them, and solve them both analytically and numerically.

Literature
Vorlesungsskript

Buckmaster, J.D.; Ludford, G.S.S.: Lectures on Mathematical Combustion, SIAM 1983

V Flows with chemical reactions

2153406, WS 26/27, 2 SWS, Language: English, Open in study portal

Lecture (V)
Blended (On-Site/Online)

Content

In this lecture, students acquire a solid understanding of fundamental phenomena in reacting flows. They learn how the high temperature sensitivity of chemical reactions leads to significant simplifications in modeling, particularly by concentrating the reaction on thin layers, known as free interfaces. The students are able to understand and apply simplified chemical models, with a focus on the fluid mechanical aspects. They develop the ability to use analytical methods to solve idealized problems and to simplify more complex issues so that efficient numerical methods can be applied. Furthermore, they learn to derive fluid mechanical equations in curvilinear coordinates and apply them to suitable problems. Overall, the students are empowered to formulate simplified models for reactive flows, critically analyze them, and solve them both analytically and numerically.

Literature

Vorlesungsskript

Buckmaster, J.D.; Ludford, G.S.S.: Lectures on Mathematical Combustion, SIAM 1983

T 6.264 Module component: Foundations of Informatics I [T-WIWI-102749]

Coordinators: Dr.-Ing. Tobias Käfer
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104901 - Informatics (KIT-Department of Economics and Management)

Type	Credits	Grading	Term offered	Version
Written examination	5 CP	graded	Each summer term	2

Courses					
ST 2026	2511010	Foundations of Informatics I	2 SWS	Lecture / 🗣	Käfer
ST 2026	2511011	Exercises to Foundations of Informatics I		Practice / 🗣	Käfer, Noullet, Kinder

Legend: 🖥 Online, ⚄ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment consists of an 1h written exam according to Section 4 (2), 1 of the examination regulation.

The exam takes place every semester. Re-examinations are offered at every ordinary examination date.

Prerequisites
None

Below you will find excerpts from courses related to this module component:

V Foundations of Informatics I

2511010, SS 2026, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content
The lecture provides an introduction to basic concepts of computer science and software engineering. Essential theoretical foundations and problem-solving approaches, which are relevant in all areas of computer science, are presented and explained, as well as shown in practical implementations.

The following topics are covered:

- Object Oriented Modeling
- Logic (Propositional Calculus, Predicate Logic, Boolean Algebra)
- Algorithms and Their Properties
- Sort-and Search-Algorithms
- Complexity Theory
- Problem Specification
- Dynamic Data Structures

Learning objectives:
The student

- is able to formalise tasks in the domain of informatics and is able to identify solution methods
- knows the basic terminology of computer science and is capable of applying these terms to different problems.
- knows basic programming structures and is able to apply them (particularly simple data structures, object interaction and implementation of basic algorithms).

Workload:

- The total workload for this course is approximately 150 hours
- Time of presentness: 45 hours
- Time of preperation and postprocessing: 67.5 hours
- Exam and exam preperation: 37.5 hours

Literature

- H. Balzert. Lehrbuch Grundlagen der Informatik. Spektrum Akademischer Verlag 2004.
- U. Schöning. Logik für Informatiker. Spektrum Akademischer Verlag 2000.
- T. H. Cormen, C. E. Leiserson. Introduction to Algorithms, MIT Press 2001.

V

Exercises to Foundations of Informatics I

2511011, SS 2026, SWS, Language: German, Open in study portal

Practice (Ü)
On-Site

Content

The exercises are related to the lecture Foundations of Informatics I.

Multiple exercises are held that capture the topics, held in the lecture Foundations of Informatics I, and discuss them in detail. Thereby, practical examples are given to the students in order to transfer theoretical aspects into practical implementation.

The following topics are covered:

- Object Oriented Modeling
- Logic (Propositional Calculus, Predicate Logic, Boolean Algebra)
- Algorithms and Their Properties
- Sort-and Search-Algorithms
- Complexity Theory
- Problem Specification
- Dynamic Data Structures

Learning objectives:

The student

- is able to formalise tasks in the domain of informatics and is able to identify solution methods
- knows the basic terminology of computer science and is capable of applying these terms to different problems.
- knows basic programming structures and is able to apply them (particularly simple data structures, object interaction and implementation of basic algorithms).

Literature

- H. Balzert. Lehrbuch Grundlagen der Informatik. Spektrum Akademischer Verlag 2004.
- U. Schöning. Logik für Informatiker. Spektrum Akademischer Verlag 2000.
- T. H. Cormen, C. E. Leiserson. Introduction to Algorithms, MIT Press 2001.

T 6.265 Module component: Foundations of Informatics II [T-WIWI-102707]

Coordinators: Prof. Dr.-Ing. Sanja Lazarova-Molnar
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104901 - Informatics (KIT-Department of Economics and Management)

Type	Credits	Grading	Term offered	Version
Written examination	5 CP	graded	Each winter term	1

Courses					
WT 25/26	2511012	Foundations of Informatics II	3 SWS	Lecture /	Lazarova-Molnar
WT 25/26	2511013	Tutorien zu Grundlagen der Informatik II	1 SWS	Tutorial /	Lazarova-Molnar, Jungmann, Deufel
WT 26/27	2511012	Foundations of Informatics II	3 SWS	Lecture /	Lazarova-Molnar
WT 26/27	2511013	Tutorien zu Grundlagen der Informatik II	1 SWS	Tutorial /	Lazarova-Molnar, Jungmann, Deufel

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
The assessment takes the form of a written examination (in accordance with §4(2), 1 SPO), which can be completed in either German or English. The examination is offered every semester and can be retaken at any regular examination date.

The duration of the examination is 90 minutes.

Prerequisites
None

Recommendations
It is recommended to attend the course "Foundations of Informatics I" beforehand.

Active participation in the practical lessons is strongly recommended.

Below you will find excerpts from courses related to this module component:

V Foundations of Informatics II
2511012, WS 25/26, 3 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content
The course deals with formal models like finite automata, context-free grammars and push-down automata, offers formal descriptions of computational models of different complexity, and classifies language systems according to their expressive power; it investigates the limits of computability and theoretical complexity to distinguish efficiently solvable problems (P) from computationally intensive ones (NP-complete); at the same time it conveys the fundamentals of computer organization, including digital information formats (integer and floating-point), symbolic logic and the design of functional circuit components, followed by explanations of basic computer architectures, their instruction logic and the standardized fetch-decode-execute cycle, hierarchical storage concepts (cache levels, secondary memory), input-/output mechanisms, redundancy strategies, as well as methods for objective performance assessment and systematic optimization.

Learning objectives:

- Students acquire vast knowledge of methods and concepts in theoretical computer science and computer architectures.
- Based on the acquired knowledge and skills, students are capable of choosing and applying the appropriate methods and concepts for well-defined problem instances.
- Active participation in the tutorials enables students to acquire the necessary knowledge for developing appropriate solutions cooperatively.

Recommendations:

It is recommended to attend the course *Foundations of Informatics I* [2511010] beforehand.

Active participation in the practical lessons is strongly recommended.

Workload:

- Total workload for 5 credit points: approx. 150 hours
- Attendance time – Lecture: 30 h
- Attendance time – Tutorial: 15 h
- Preparation and follow-up: 67.5 hours
- Examination and exam preparation: 37.5 hours

Organizational issues
Die Vorlesung wird zu Beginn des Semesters 4-stündig und am Ende 2-stündig gelesen, um eine bessere Abdeckung des Inhalts in den Übungen zu gewährleisten.

Literature
Weiterführende Literatur:

Literatur wird in der Vorlesung bekannt gegeben.

V	**Foundations of Informatics II** 2511012, WS 26/27, 3 SWS, Language: English, Open in study portal	**Lecture (V)** **On-Site**

Content
The course deals with formal models like finite automata, context-free grammars and push-down automata, offers formal descriptions of computational models of different complexity, and classifies language systems according to their expressive power; it investigates the limits of computability and theoretical complexity to distinguish efficiently solvable problems (P) from computationally intensive ones (NP-complete); at the same time it conveys the fundamentals of computer organization, including digital information formats (integer and floating-point), symbolic logic and the design of functional circuit components, followed by explanations of basic computer architectures, their instruction logic and the standardized fetch-decode-execute cycle, hierarchical storage concepts (cache levels, secondary memory), input-/output mechanisms, redundancy strategies, as well as methods for objective performance assessment and systematic optimization.

Learning objectives:

- Students acquire vast knowledge of methods and concepts in theoretical computer science and computer architectures.
- Based on the acquired knowledge and skills, students are capable of choosing and applying the appropriate methods and concepts for well-defined problem instances.
- Active participation in the tutorials enables students to acquire the necessary knowledge for developing appropriate solutions cooperatively.

Recommendations:

It is recommended to attend the course *Foundations of Informatics I* [2511010] beforehand.

Active participation in the practical lessons is strongly recommended.

Workload:

- Total workload for 5 credit points: approx. 150 hours
- Attendance time – Lecture: 30 h
- Attendance time – Tutorial: 15 h
- Preparation and follow-up: 67.5 hours
- Examination and exam preparation: 37.5 hours

Organizational issues
Die Vorlesung wird zu Beginn des Semesters 4-stündig und am Ende 2-stündig gelesen, um eine bessere Abdeckung des Inhalts in den Übungen zu gewährleisten.

Literature
Weiterführende Literatur:

Literatur wird in der Vorlesung bekannt gegeben.

T 6.266 Module component: Foundations of Interactive Systems [T-WIWI-109816]

Coordinators: Prof. Dr. Alexander Mädche
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Examination of another type	4,5 CP	graded	Each summer term	3

Courses					
ST 2026	2540560	Foundations of Interactive Systems	3 SWS	Lecture / Blended	Mädche, Feick

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
Alternative exam assessment. The assessment is carried out in the form of a one-hour written examination and by carrying out a Capstone project.

Details on the assessment will be announced during the lecture.

Prerequisites
None

Recommendations
None

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Foundations of Interactive Systems

2540560, SS 2026, 3 SWS, Language: English, Open in study portal

Lecture (V)
Blended (On-Site/Online)

Content
Lecture Description

Computers have evolved from batch processors to highly interactive systems. This offers new possibilities besides challenges for designing a successful interaction between humans and computers. Interactive systems are socio-technical systems in which users perform tasks by interacting with technology in a specific context to achieve specified goals and outcomes.

This lecture introduces key concepts and principles of interactive systems from a human and computer perspective. From a human perspective, we discuss selected individual characteristics, cognitive processes, the interplay between cognition and activity, as well asmental models. From a computer perspective, we introduce established interaction technologies as well as contemporary multimodal technologies (e.g. augmented/mixed reality, eye-based interaction, etc.). We also introduce established principles and guidelines for designing user interfaces.. Furthermore, we describe the human-centered design process for interactive systems and supporting techniques & tools (e.g. personas, prototyping, user testing).

With this lecture, students acquire foundational knowledge to successfully **design the interaction between humans and computers** in business and private life. The course is complemented with a **Design Capstone Project**, where students in a team apply design methods & techniques to create an interactive prototype.

Learning Objectives

The students

- have a basic understanding of key conceptual and theoretical foundations of interactive systems from a human and computer perspective
- are aware of important design principles for the design of important classes of interactive systems
- know design processes and techniques for developing interactive systems
- know how to apply the knowledge and skills gathered in the lecture for a real-world problem (as part of design capstone project)

Prerequisites: No specific prerequisites are required for the lecture

Language of instruction: English

Bibliography

Alan Dix, Janet E. Finlay, Gregory D. Abowd, and Russell Beale. 2003. Human-Computer Interaction (3rd Edition). Prentice-Hall, Inc., USA.

Further literature will be made available in the lecture. In case of questions feel free to approach Siu Liu (siu.liu@kit.edu).

Die Erfolgskontrolle erfolgt in Form einer Prüfungsleistung anderer Art (Form) nach § 4 Abs. 2 Nr. 3 SPO. Die Leistungskontrolle erfolgt in Form einer einstündigen Klausur und der Durchführung eines Capstone Projektes. Details zur Ausgestaltung der Erfolgskontrolle werden im Rahmen der Vorlesung bekannt gegeben.

T 6.267 Module component: Foundry Technology [T-MACH-105157]

Coordinators: Dr.-Ing. Daniel Günther
Dr.-Ing. Steffen Klan
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101268 - Specific Topics in Materials Science
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Written examination	4 CP	graded	Each summer term	3

Courses					
ST 2026	2174575	Foundry Technology	2 SWS	Lecture / 🖥	Günther

Legend: 🖥 Online, Blended (On-Site/Online), On-Site, ✕ Cancelled

Assessment
The assessment is carried out as a written exam of 1 h.

Prerequisites
none

Recommendations
The lectures Materials Science I and Materials Science II should have been attended in advance.

Additional Information
The course is offered in German.

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Foundry Technology

2174575, SS 2026, 2 SWS, Language: German, Open in study portal

Lecture (V)
Online

Literature
Literaturhinweise werden in der Vorlesung gegeben

Reference to literature, documentation and partial lecture notes given in lecture

T 6.268 Module component: Freight Transport [T-BGU-106611]

Coordinators: Dr. Eckhard Szimba
Prof. Dr.-Ing. Peter Vortisch

Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences

Part of: M-BGU-101064 - Fundamentals of Transportation
M-BGU-101065 - Transportation Modelling and Traffic Management
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Expansion	Version
Written examination	3 CP	graded	Each term	1 semesters	2

Courses					
ST 2026	6232809	Freight Transport	2 SWS	Lecture / Practice / 🗣	Szimba

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
written exam, 60 min.

Prerequisites
none

Recommendations
none

Additional Information
none

Workload
90 hours

T 6.269 Module component: Fuels and Lubricants for Combustion Engines [T-MACH-105184]

Coordinators: Hon.-Prof. Dr. Bernhard Ulrich Kehrwald
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101303 - Combustion Engines II
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each winter term	1

Courses					
WT 25/26	2133108	Fuels and Lubricants for Combustion Engines	2 SWS	Lecture / 🗣	Kehrwald

Legend: 🖥 Online, Blended (On-Site/Online), 🗣 On-Site, x Cancelled

Assessment
oral examination, Duration: ca. 25 min., no auxiliary means

Prerequisites
none

Additional Information
The course is offered in German.

Below you will find excerpts from courses related to this module component:

V Fuels and Lubricants for Combustion Engines

2133108, WS 25/26, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Literature
Skript

T 6.270 Module component: Functional Ceramics [T-MACH-105179]

Coordinators: Dr. Miriam Botros
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101268 - Specific Topics in Materials Science

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each winter term	2

Courses					
WT 25/26	2126784	Functional Ceramics	2 SWS	Lecture / 🧩	Botros

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment consists of an oral exam (20 min) taking place at the agreed date.

Auxiliary means: none

The re-examination is offered upon agreement.

Prerequisites
T-MACH-114752 – Functional Ceramics must not be started yet.

Additional Information
The course is offered in German.

Workload
120 hours

T 6.271 Module component: Fundamentals for Design of Motor-Vehicle Bodies I [T-MACH-102116]

Coordinators: Eva-Maria Knoch
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101266 - Automotive Engineering
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Oral examination	2 CP	graded	Each winter term	2

Courses					
WT 25/26	2113814	Fundamentals for Design of Motor-Vehicles Bodies I	1 SWS	Lecture / 🗣	Knoch
WT 26/27	2113814	Fundamentals for Design of Motor-Vehicles Bodies I	1 SWS	Lecture / 🗣	Knoch

Legend: 🖥 Online, ❄ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Oral examination

Duration: approx. 30 minutes

Auxiliary means: none

Prerequisites
none

Additional Information
The course is offered in German.

Workload
60 hours

Below you will find excerpts from courses related to this module component:

V Fundamentals for Design of Motor-Vehicles Bodies I

2113814, WS 25/26, 1 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content
1. History and design

2. Aerodynamics

3. Design methods (CAD/CAM, FEM)

4. Manufacturing methods of body parts

5. Fastening technologie

6. Body in white / body production, body surface

Learning Objectives:

The students have an overview of the fundamental possibilities for design and manufacture of motor-vehicle bodies. They know the complete process, from the first idea, through the concept to the dimensioned drawings (e.g. with FE-methods). They have knowledge about the fundamentals and their correlations, to be able to analyze and to judge relating components as well as to develop them accordingly.

Organizational issues
Das Vorlesungsmaterial wird auf ILIAS bereitgestellt. Das ILIAS-Passwort erhalten Sie unter https://fast-web-01.fast.kit.edu/PasswoerterIlias/

Termine und nähere Informationen: siehe ILIAS oder Institutshomepage

Dates and further information will be published on the homepage of the institute

Literature
1. Automobiltechnische Zeitschrift ATZ, Friedr. Vieweg & Sohn Verlagsges. mbH, Wiesbaden
2. Automobil Revue, Bern (Schweiz)
3. Automobil Produktion, Verlag Moderne Industrie, Landsberg

V **Fundamentals for Design of Motor-Vehicles Bodies I** **Lecture (V)**
2113814, WS 26/27, 1 SWS, Language: German, Open in study portal **On-Site**

Content
1. History and design

2. Aerodynamics

3. Design methods (CAD/CAM, FEM)

4. Manufacturing methods of body parts

5. Fastening technologie

6. Body in white / body production, body surface

Learning Objectives:

The students have an overview of the fundamental possibilities for design and manufacture of motor-vehicle bodies. They know the complete process, from the first idea, through the concept to the dimensioned drawings (e.g. with FE-methods). They have knowledge about the fundamentals and their correlations, to be able to analyze and to judge relating components as well as to develop them accordingly.

Organizational issues
Termine und nähere Informationen: siehe ILIAS oder Institutshomepage

Dates and further information will be published on the homepage of the institute

Literature
1. Automobiltechnische Zeitschrift ATZ, Friedr. Vieweg & Sohn Verlagsges. mbH, Wiesbaden
2. Automobil Revue, Bern (Schweiz)
3. Automobil Produktion, Verlag Moderne Industrie, Landsberg

T 6.272 Module component: Fundamentals for Design of Motor-Vehicle Bodies II [T-MACH-102119]

Coordinators: Eva-Maria Knoch
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101266 - Automotive Engineering
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Oral examination	2 CP	graded	Each summer term	1

Courses					
WT 26/27	2114840	Fundamentals for Design of Motor-Vehicles Bodies II	1 SWS	Lecture / 🗣	Knoch

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Oral examination

Duration: approx. 30 minutes

Auxiliary means: none

Prerequisites
none

Additional Information
The course is offered in German.

Workload
60 hours

Below you will find excerpts from courses related to this module component:

V Fundamentals for Design of Motor-Vehicles Bodies II

2114840, WS 26/27, 1 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content
1. Body properties/testing procedures
2. External body-parts
3. Interior trim
4. Compartment air conditioning
5. Electric and electronic features
6. Crash tests
7. Project management aspects, future prospects

Learning Objectives:

The students know that, often the design of seemingly simple detail components can result in the solution of complex problems. They have knowledge in testing procedures of body properties. They have an overview of body parts such as bumpers, window lift mechanism and seats. They understand, as well as, parallel to the normal electrical system, about the electronic side of a motor vehicle. Based on this they are ready to analyze and to judge the relation of these single components. They are also able to contribute competently to complex development tasks by imparted knowledge in project management.

Organizational issues
Termine und nähere Informationen: siehe ILIAS oder Institutshomepage

Dates and further information will be published on the homepage of the institute

Literature
1. Automobiltechnische Zeitschrift ATZ, Friedr. Vieweg & Sohn Verlagsges. mbH, Wiesbaden
2. Automobil Revue, Bern (Schweiz)
3. Automobil Produktion, Verlag Moderne Industrie, Landsberg

T 6.273 Module component: Fundamentals for Financial -Quant and -Machine Learning Research [T-WIWI-111846]

Coordinators: Prof. Dr. Maxim Ulrich
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-105894 - Foundations for Advanced Financial -Quant and -Machine Learning Research

Type	Credits	Grading	Term offered	Version
Examination of another type	9 CP	graded	Irregular	1

Assessment
The examination is an alternative exam assessment with a maximum score of 100 points to be achieved. These points are distributed over 4 worksheets to be submitted during the semester. The worksheets cover the respective material of the module and are handed out, worked on and assessed in lecture weeks 3 (10 points), 6 (20 points), 9 (30 points) and 12 (40 points).

The module-wide exam (all 4 worksheets) must be taken in the same semester.

The worksheets are a mixture of analytical tasks and programming tasks with financial data.

Recommendations

- Strongly recommended to have good knowledge in financial econometrics (MLE, OLS, GLS, ARMA-GARCH), mathematics (differential equations, difference equations and optimization), investments (CAPM, factor models), asset pricing (SDF, SDF pricing), derivatives (Black-Scholes, risk-neutral pricing), and programming of statistical concepts (Java or R or Python or Matlab or C or ...)
- Strongly recommended to have a strong interest for interdisciplinary research work in statistics, programming, applied math and financial economics.
- Students lacking the prior knowledge might find the resources of the Chair helpful: www.youtube.com/c/cram-kit.

Additional Information
Teaching and learning format: Lecture and exercise.

Workload
270 hours

T 6.274 Module component: Fundamentals in the Development of Commercial Vehicles [T-MACH-111389]

Coordinators: Christof Weber
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101265 - Vehicle Development
M-MACH-101267 - Mobile Machines
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	see Annotations	2

Courses					
WT 25/26	2113812	Fundamentals in the Development of Commercial Vehicles I	1 SWS	Lecture / 🗣	Weber
ST 2026	2114844	Fundamentals in the Development of Commercial Vehicles II	1 SWS	Lecture / 🗣	Weber
WT 26/27	2113812	Fundamentals in the Development of Commercial Vehicles I	1 SWS	Lecture / 🗣	Weber

Legend: 🖥 Online, ⚄ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Oral group examination

Duration: appr. 30 minutes

Auxiliary means: none

Prerequisites
none

Additional Information
Fundamentals in the Development of Commercial Vehicles I, WT

Fundamentals in the Development of Commercial Vehicles II, ST

The course is offered in German.

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Fundamentals in the Development of Commercial Vehicles I

Lecture (V)
On-Site

2113812, WS 25/26, 1 SWS, Language: German, Open in study portal

Content

1. Introduction, definitions, history
2. Development tools
3. Complete vehicle
4. Cab, bodyshell work
5. Cab, interior fitting
6. Alternative drive systems
7. Drive train
8. Drive system diesel engine
9. Intercooled diesel engines

Learning Objectives:

The students have proper knowledge about the process of commercial vehicle development starting from the concept and the underlying original idea to the real design. They know that the customer requirements, the technical realisability, the functionality and the economy are important drivers.

The students are able to develop parts and components. Furthermore they have knowledge about different cab concepts, the interior and the interior design process. Consequently they are ready to analyze and to judge concepts of commercial vehicles as well as to participate competently in the commercial vehicle development.

Organizational issues

Das Vorlesungsmaterial wird auf ILIAS bereitgestellt. Das ILIAS-Passwort erhalten Sie unter https://fast-web-01.fast.kit.edu/PasswoerterIlias/

Termine und Nähere Informationen: siehe ILIAS oder Institutshomepage

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Literature

1. Marwitz, H., Zittel, S.: ACTROS -- die neue schwere Lastwagenbaureihe von Mercedes-Benz, ATZ 98, 1996, Nr. 9

2. Alber, P., McKellip, S.: ACTROS -- Optimierte passive Sicherheit, ATZ 98, 1996

3. Morschheuser, K.: Airbag im Rahmenfahrzeug, ATZ 97, 1995, S. 450 ff.

V	**Fundamentals in the Development of Commercial Vehicles II** 2114844, SS 2026, 1 SWS, Language: German, Open in study portal	**Lecture (V)** **On-Site**

Content

1. Gear boxes of commercial vehicles
2. Intermediate elements of the drive train
3. Axle systems
4. Front axles and driving dynamics
5. Chassis and axle suspension
6. Braking System
7. Systems
8. Excursion

Learning Objectives:

The students know the advantages and disadvantages of different drives. Furthermore they are familiar with components, such as transfer box, propeller shaft, powered and non-powered frontaxle etc. Beside other mechanical components, such as chassis, axle suspension and braking system, also electric and electronic systems are known. Consequently the student are able to analyze and to judge the general concepts as well as to adjust them precisely with the area of application.

Organizational issues

die Vorlesung findet an unregelmäßigen Terminen am Campus Ost statt. Genaue Termine sowie nähere Informationen und eventuelle Terminänderungen:

siehe Institutshomepage.

Literature

1.HILGERS, M.: Nutzfahrzeugtechnik lernen, Springer Vieweg, ISSN: 2510-1803

2.SCHITTLER, M.; HEINRICH, R.; KERSCHBAUM, W.: Mercedes-Benz Baureihe 500 – neue V-Motorengeneration für schwere Nutzfahrzeuge, MTZ 57 Nr. 9, S. 460 ff, 1996

3.Robert Bosch GmbH (Hrsg.): Bremsanalgen für Kraftfahrzeuge, VDI-Verlag, Düsseldorf, 1. Auflage, 1994

4.RUBI, V.; STRIFLER, P. (Hrsg. Institut für Kraftfahrwesen RWTH Aachen): Indiustrielle Nutzfahrzeugentwicklung, Schriftenreihe Automobiltechnik, 1993

5.TEUTSCH, R.; CHERUTI, R.; GASSER, R.; PEREIRA, M.; de SOUZA, A.; WEBER, C.: Fuel Efficiency Optimization of Market Specific Truck Applications, Proceedings of the 5th Commercial Vehicle Technology Symposium – CVT 2018

V **Fundamentals in the Development of Commercial Vehicles I**
2113812, WS 26/27, 1 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content

1. Introduction, definitions, history
2. Development tools
3. Complete vehicle
4. Cab, bodyshell work
5. Cab, interior fitting
6. Alternative drive systems
7. Drive train
8. Drive system diesel engine
9. Intercooled diesel engines

Learning Objectives:

The students have proper knowledge about the process of commercial vehicle development starting from the concept and the underlying original idea to the real design. They know that the customer requirements, the technical realisability, the functionality and the economy are important drivers.

The students are able to develop parts and components. Furthermore they have knowledge about different cab concepts, the interior and the interior design process. Consequently they are ready to analyze and to judge concepts of commercial vehicles as well as to participate competently in the commercial vehicle development.

Organizational issues

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Literature

1. Marwitz, H., Zittel, S.: ACTROS -- die neue schwere Lastwagenbaureihe von Mercedes-Benz, ATZ 98, 1996, Nr. 9

2. Alber, P., McKellip, S.: ACTROS -- Optimierte passive Sicherheit, ATZ 98, 1996

3. Morschheuser, K.: Airbag im Rahmenfahrzeug, ATZ 97, 1995, S. 450 ff.

T 6.275 Module component: Fundamentals of Catalytic Exhaust Gas Aftertreatment [T-MACH-105044]

Coordinators: Prof. Dr. Olaf Deutschmann
Prof. Dr. Jan-Dierk Grunwaldt
Dr.-Ing. Heiko Kubach
Hon.-Prof. Dr. Egbert Lox

Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101303 - Combustion Engines II
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each summer term	1

Courses					
ST 2026	2134138	Fundamentals of catalytic exhaust gas aftertreatment	2 SWS	Lecture / 🗣	Lox, Grunwaldt, Deutschmann

Legend: 🖥 Online, ✿ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
oral examination, Duration approx. 25 min., no auxiliary means

Prerequisites
none

Additional Information
The course is offered in German

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Fundamentals of catalytic exhaust gas aftertreatment

Lecture (V)
On-Site

2134138, SS 2026, 2 SWS, Language: German, Open in study portal

Organizational issues
Wenn Sie beabsichtigen, an der Vorlesung teilzunehmen, melden Sie sich bitte unter info@ifkm.kit.edu mit dem Betreff "Anmeldung Vorlesung Lox" an. Das hilft uns bei der Planung. Der Dozent hat eine sehr weite Anreise und wir möchten verhindern, dass die Mindestteilnehmerzahl nicht erreicht wird.

T 6.276 Module component: Fundamentals of National and International Group Taxation [T-WIWI-111304]

Coordinators: Prof. Dr. Berthold Wigger
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101511 - Advanced Topics in Public Finance
M-WIWI-104908 - Economics

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each summer term	1

Courses					
ST 2026	2560133	Fundamentals of National and International Group Taxation	3 SWS	Lecture / 🗣	Wigger, Gutekunst

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success is assessed in the form of a written (90 min.) examination (in accordance with §4(2), 1 SPO). The examination is offered every semester and can be repeated at any regular examination date.

Prerequisites
None

Recommendations
It is recommended to attend the course “Basics of German Company Tax Law and Tax Planning” beforehand.

Workload
135 hours

T 6.277 Module component: Fundamentals of Nuclear Energy and Radiation Protection [T-MACH-113627]

Coordinators: apl. Prof. Dr. Ron Dagan
Organisation: KIT Department of Mechanical Engineering

Part of: M-WIWI-107718 - Mechanical Engineering I
M-WIWI-107730 - Mechanical Engineering II

Type	Credits	Grading	Term offered	Version
Oral examination	4,5 CP	graded	Each winter term	2

Courses					
ST 2026	2100031	Fundamentals of Nuclear Energy and Radiation Protection	3 SWS	Lecture / 🗣	Dagan

Legend: 🖥 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
oral exam, duration approx. 30 minutes

Prerequisites
none

Workload
135 hours

T 6.278 Module component: Fundamentals of Probability and Statistics for Students of Computer Science [T-MATH-102244]

Coordinators: Prof. Dr. Nicole Bäuerle
Dr. rer. nat. Bruno Ebner
Prof. Dr. Vicky Fasen-Hartmann
Prof. Dr. Daniel Hug
PD Dr. Bernhard Klar
Prof. Dr. Günter Last
Prof. Dr. Mathias Trabs
PD Dr. Steffen Winter

Organisation: KIT Department of Mathematics

Part of: M-WIWI-104905 - Mathematics

Type	Credits	Grading	Version
Written examination	4,5 CP	graded	1

Courses					
WT 25/26	0133500	Grundlagen der Wahrscheinlichkeitstheorie und Statistik für Studierende der Informatik	2 SWS	Lecture / 🗣	Steffen
WT 25/26	0133600	Übungen zu 0133500 (Grundlagen der Wahrscheinlichkeitstheorie und Statistik für Studierende der Informatik)	1 SWS	Practice / 🗣	Steffen
WT 26/27	0133500	Grundlagen der Wahrscheinlichkeitstheorie und Statistik für Studierende der Informatik	2 SWS	Lecture / 🗣	Hug
WT 26/27	0133600	Übungen zu 0133500 (Grundlagen der Wahrscheinlichkeitstheorie und Statistik für Studierende der Informatik)	1 SWS	Practice / 🗣	Hug

Legend: 📱 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

T 6.279 Module component: Fundamentals of Production Management [T-WIWI-102606]

Coordinators: Prof. Dr. Frank Schultmann
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	5,5 CP	graded	Each summer term	1

Courses					
ST 2026	2581950	Fundamentals of Production Management	2 SWS	Lecture / 🗣	Schultmann
ST 2026	2581951	Übungen Grundlagen der Produktionswirtschaft	2 SWS	Practice / 🗣	Frank, Fuhg

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment consists of a written exam (90 minutes) (following §4(2) of the examination regulation). The exam takes place in every semester. Re-examinations are offered at every ordinary examination date.

Prerequisites
None

Below you will find excerpts from courses related to this module component:

V Fundamentals of Production Management

2581950, SS 2026, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content
This lecture focuses on strategic production management with respect to various economic aspects. Interdisciplinary approaches of systems theory will be used to describe the challenges of industrial production. This course will emphasize the importance of R&D as the central step in strategic corporate planning to ensure future long-term success. In the field of site selection and planning for firms and factories, attention will be drawn upon individual aspects of existing and greenfield sites as well as existing distribution and supply centres. Students will obtain knowledge in solving internal and external transport and storage problems.

Organizational issues
Blockveranstaltung, siehe Institutsaushang

Literature
Wird in der Veranstaltung bekannt gegeben.

T 6.280 Module component: Fundamentals of Water Quality [T-CIWVT-106838]

Coordinators: Dr. Michael Wagner
Organisation: KIT Department of Chemical and Process Engineering
Part of: M-CIWVT-101122 - Water Chemistry and Water Technology II
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Oral examination	6 CP	graded	Each winter term	2

Courses					
WT 25/26	2233230	Fundamentals of Water Quality	2 SWS	Lecture / 🗣	Horn, Wagner
WT 25/26	2233231	Fundamentals of Water Quality - Exercises	1 SWS	Practice / 🗣	Wagner, und Mitarbeitende

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Learning control ist an oral exam lasting approx. 20 minutes.

Prerequisites
None.

T 6.281 Module component: Gas Engines [T-MACH-102197]

Coordinators: Dr.-Ing. Rainer Golloch
Dr.-Ing. Heiko Kubach

Organisation: KIT Department of Mechanical Engineering

Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each summer term	1

Assessment
Oral examination, duration 25 min., no auxillary means

Prerequisites
none

T 6.282 Module component: Gear Cutting Technology [T-MACH-102148]

Coordinators: Dr.-Ing. Jürgen Hoffmeister
Hon.-Prof. Dr. Markus Klaiber

Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101265 - Vehicle Development
M-MACH-106590 - Production Engineering
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each winter term	1

Courses					
WT 25/26	2149655	Gear Technology	2 SWS	Lecture / 🗣	Klaiber
WT 26/27	2149655	Gear Technology	2 SWS	Lecture / 🗣	Hoffmeister

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Oral Exam (20 min)

Prerequisites
none

Additional Information
The course is offered in German.

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Gear Technology

2149655, WS 25/26, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content
Based on the gearing theory, manufacturing processes and machine technologies for producing gearings, the needs of modern gear manufacturing will be discussed in the lecture. For this purpose, various processes for various gear types are taught which represent the state of the art in practice today. A classification in soft and hard machining and furthermore in cutting and non-cutting technologies will be made. For comprehensive understanding the processes, machine technologies, tools and applications of the manufacturing of gearings will be introduced and the current developments presented. For assessment and classification of the applications and the performance of the technologies, the methods of mass production and manufacturing defects will be discussed. Sample parts, reports from current developments in the field of research and an excursion to a gear manufacturing company round out the lecture.

Learning Outcomes:

The students …

- can describe the basic terms of gearings and are able to explain the imparted basics of the gearwheel and gearing theory.
- are able to specify the different manufacturing processes and machine technologies for producing gearings. Furthermore they are able to explain the functional principles and the dis-/advantages of these manufacturing processes.
- can apply the basics of the gearing theory and manufacturing processes on new problems.
- are able to read and interpret measuring records for gearings. are able to make an appropriate selection of a process based on a given application
- can describe the entire process chain for the production of toothed components and their respective influence on the resulting workpiece properties.

Workload:

regular attendance: 21 hours
self-study: 99 hours

Literature
Medien:
Skript zur Veranstaltung wird über (https://ilias.studium.kit.edu/) bereitgestellt.

Media:
Lecture notes will be provided in Ilias (https://ilias.studium.kit.edu/).

V **Gear Technology** **Lecture (V)**
2149655, WS 26/27, 2 SWS, Language: German, Open in study portal **On-Site**

Content
Based on the gearing theory, manufacturing processes and machine technologies for producing gearings, the needs of modern gear manufacturing will be discussed in the lecture. For this purpose, various processes for various gear types are taught which represent the state of the art in practice today. A classification in soft and hard machining and furthermore in cutting and non-cutting technologies will be made. For comprehensive understanding the processes, machine technologies, tools and applications of the manufacturing of gearings will be introduced and the current developments presented. For assessment and classification of the applications and the performance of the technologies, the methods of mass production and manufacturing defects will be discussed. Sample parts, reports from current developments in the field of research and an excursion to a gear manufacturing company round out the lecture.

Learning Outcomes:

The students …

- can describe the basic terms of gearings and are able to explain the imparted basics of the gearwheel and gearing theory.
- are able to specify the different manufacturing processes and machine technologies for producing gearings. Furthermore they are able to explain the functional principles and the dis-/advantages of these manufacturing processes.
- can apply the basics of the gearing theory and manufacturing processes on new problems.
- are able to read and interpret measuring records for gearings. are able to make an appropriate selection of a process based on a given application
- can describe the entire process chain for the production of toothed components and their respective influence on the resulting workpiece properties.

Workload:

regular attendance: 21 hours
self-study: 99 hours

Literature
Medien:
Skript zur Veranstaltung wird über (https://ilias.studium.kit.edu/) bereitgestellt.

Media:
Lecture notes will be provided in Ilias (https://ilias.studium.kit.edu/).

T 6.283 Module component: Geological Hazards and Risks for External Students [T-PHYS-103117]

Coordinators: Dr. Ellen Gottschämmer
Organisation: KIT Department of Physics
Part of: M-WIWI-104904 - Natural Sciences

Type	Credits	Grading	Version
Oral examination	4 CP	graded	1

Courses					
WT 25/26	4060121	Geological Hazards and Risk	2 SWS	Lecture / 🗣	Schäfer, Rietbrock
WT 25/26	4060122	Exercises on Geological Hazards and Risk	2 SWS	Practice / 🗣	Schäfer, Rietbrock

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

T 6.284 Module component: Global Logistics [T-MACH-105379]

Coordinators: Prof. Dr.-Ing. Kai Furmans
Organisation: KIT Department of Mechanical Engineering

Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each summer term	1

Courses					
ST 2026	2118010	Global Logistics (Bachelor, MEI)	2 SWS	Block / 🗣	Seibold

Legend: 🖥 Online, ❂ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
oral exam (approx. 20 min)

Prerequisites
none

Additional Information
The course is offered in English.

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Global Logistics (Bachelor, MEI)
2118010, SS 2026, 2 SWS, Language: English, Open in study portal

Block (B)
On-Site

Content
Conveyor systems

- Basics of conveyor systems
- Key performance indicators
- Split elements
- Join elements

Queueing theory

- Basics of queueing theory
- Probability density functions and cumulative distribution functions
- Models M|M|1 and M|G|1

Storage and commissioning

- Automated storage and retrieval systems
- Single and dual command cycles
- Cycle times in single-deep storage systems
- Cycle times in multi-deep storage systems

Application for production logistics

- Location problame
- Distribution centers
- Inventory management

Organizational issues
Attendance during lecture is required. Admission to the exam is only possible when attending the lecture.

Literature
Arnold, D., & Furmans, K. (2019). *Materialfluss in Logistiksystemen* (Vol. 7). Berlin, Heidelberg, New York: Springer.

T 6.285 Module component: Global Logistics [T-MACH-111003]

Coordinators: Prof. Dr.-Ing. Kai Furmans
Organisation: KIT Department of Mechanical Engineering

Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Written examination	4 CP	graded	Each summer term	1

Courses					
ST 2026	2149600	Global Logistics (Master)	2 SWS	Lecture / 🗣	Furmans

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The success control takes place in form of a written examination (60 min) during the semester break (according to §4(2), 1 SPO). If the number of participants is low, an oral examination (according to §4 (2), 2 SPO) may also be offered.

Prerequisites
T-MACH-105159 - Global production and logistics - Part 2: Global logistics must not be started

Additional Information
The course is offered in German

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Global Logistics (Master)

2149600, SS 2026, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content
Content:

Characteristics of global trade

- Incoterms
- Customs clearance, documents and export control

Global transport and shipping

- Maritime transport, esp. container handling
- Air transport

Modeling of supply chains

- SCOR model
- Value stream analysis

Location planning in cross-border-networks

- Application of the Warehouse Location Problem
- Transport Planning

Inventory Management in global supply chains

- Stock keeping policies
- Inventory management considering lead time and shipping costs

Media:

presentations, black board

Workload:

regular attendance: 21 hours
self-study: 99 hours

Students are able to:

- assign basic problems of planning and operation of global supply chains and plan them with apropriate methods,
- describe requirements and characteristics of global trade and transport, and
- evaluate characteristics of the design from logistic chains regarding their suitability.

Exam:

The exam consists of a 60 minutes written examination (according to §4(2), 1 of the examination regulation).

The main exam is offered every summer semester. A second date for the exam is offered in winter semester only for students that did not pass the main exam.

Literature
Weiterführende Literatur:

- Arnold/Isermann/Kuhn/Tempelmeier. HandbuchLogistik, Springer Verlag, 2002 (Neuauflage in Arbeit)
- Domschke. Logistik, Rundreisen und Touren,Oldenbourg Verlag, 1982
- Domschke/Drexl. Logistik, Standorte, OldenbourgVerlag, 1996
- Gudehus. Logistik, Springer Verlag, 2007
- Neumann-Morlock. Operations-Research,Hanser-Verlag, 1993
- Tempelmeier. Bestandsmanagement in SupplyChains, Books on Demand 2006
- Schönsleben. IntegralesLogistikmanagement, Springer, 1998

T 6.286 Module component: Global Logistics [T-MACH-115017]

Coordinators: Prof. Dr.-Ing. Kai Furmans
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101278 - Material Flow in Networked Logistic Systems
M-MACH-101282 - Global Production and Logistics
M-MACH-104888 - Advanced Module Logistics
M-WIWI-107718 - Mechanical Engineering I
M-WIWI-107730 - Mechanical Engineering II

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each summer term	2

Courses					
ST 2026	2149600	Global Logistics (Master)	2 SWS	Lecture / 🗣	Furmans

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The success control takes place in form of a written examination (60 min) during the semester break (according to §4(2), 1 SPO). If the number of participants is low, an oral examination (according to §4 (2), 2 SPO) may also be offered.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Global Logistics (Master)

Lecture (V)
On-Site

2149600, SS 2026, 2 SWS, Language: English, Open in study portal

Content
Content:

Characteristics of global trade

- Incoterms
- Customs clearance, documents and export control

Global transport and shipping

- Maritime transport, esp. container handling
- Air transport

Modeling of supply chains

- SCOR model
- Value stream analysis

Location planning in cross-border-networks

- Application of the Warehouse Location Problem
- Transport Planning

Inventory Management in global supply chains

- Stock keeping policies
- Inventory management considering lead time and shipping costs

Media:

presentations, black board

Workload:

regular attendance: 21 hours
self-study: 99 hours

Students are able to:

- assign basic problems of planning and operation of global supply chains and plan them with apropriate methods,
- describe requirements and characteristics of global trade and transport, and
- evaluate characteristics of the design from logistic chains regarding their suitability.

Exam:

The exam consists of a 60 minutes written examination (according to §4(2), 1 of the examination regulation).

The main exam is offered every summer semester. A second date for the exam is offered in winter semester only for students that did not pass the main exam.

Literature
Weiterführende Literatur:

- Arnold/Isermann/Kuhn/Tempelmeier. HandbuchLogistik, Springer Verlag, 2002 (Neuauflage in Arbeit)
- Domschke. Logistik, Rundreisen und Touren,Oldenbourg Verlag, 1982
- Domschke/Drexl. Logistik, Standorte, OldenbourgVerlag, 1996
- Gudehus. Logistik, Springer Verlag, 2007
- Neumann-Morlock. Operations-Research,Hanser-Verlag, 1993
- Tempelmeier. Bestandsmanagement in SupplyChains, Books on Demand 2006
- Schönsleben. IntegralesLogistikmanagement, Springer, 1998

T 6.287 Module component: Global Manufacturing [T-WIWI-112103]

Coordinators: Dr. Henning Sasse
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101412 - Industrial Production III
M-WIWI-101471 - Industrial Production II
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	3,5 CP	graded	Each winter term	1

Courses					
WT 25/26	2581956	Global Manufacturing	2 SWS	Lecture /	Sasse
WT 26/27	2581956	Global Manufacturing	2 SWS	Lecture /	Sasse

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
The assessment consists of an oral (30 minutes) or written exam (60 minutes) (following §4(2) of the examination regulation). The exam takes place in every semester. Re-examinations are offered at every ordinary examination date.

Prerequisites
None

Recommendations
None

Workload
105 hours

Below you will find excerpts from courses related to this module component:

V Global Manufacturing
2581956, WS 25/26, 2 SWS, Language: English, Open in study portal

Lecture (V)
Blended (On-Site/Online)

Content

- Fundamentals of international business
- Forms of international cooperation and value creation
- Site selection
- Cost driven internationalization and site selection
- Sales and customer driven internationalization and site selection
- Challenges, risks and risk mitigation
- Management of international production sites
- Types and case studies of international production

Organizational issues
Blockveranstaltung, siehe Homepage

Literature
Wird in der Veranstaltung bekannt gegeben.

V Global Manufacturing
2581956, WS 26/27, 2 SWS, Language: English, Open in study portal

Lecture (V)
Blended (On-Site/Online)

Content

- Fundamentals of international business
- Forms of international cooperation and value creation
- Site selection
- Cost driven internationalization and site selection
- Sales and customer driven internationalization and site selection
- Challenges, risks and risk mitigation
- Management of international production sites
- Types and case studies of international production

Organizational issues
Blockveranstaltung, siehe Homepage

Literature
Wird in der Veranstaltung bekannt gegeben.

T 6.288 Module component: Global Optimization I [T-WIWI-102726]

Coordinators: Prof. Dr. Oliver Stein
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101473 - Mathematical Programming
M-WIWI-104899 - Operations Research
M-WIWI-107187 - Quantitative Core

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each summer term	1

Courses					
ST 2026	2550134	Global Optimization I	2 SWS	Lecture / 🗣	Stein

Legend: 🖥 Online, ❂ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success is in the form of a written examination in English (60 min.) (according to § 4(2), 1 SPO). The successful completion of the exercises is required for admission to the written exam.

The exam is offered in the lecture of semester and the following semester.

The success check can be done also with the success control for "Global optimization II". In this case, the duration of the written exam is 120 min.

Prerequisites
None

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-WIWI-103638 - Global Optimization I and II must not have been started.

Recommendations
None

Additional Information
Part I and II of the lecture are held consecutively in the **same** semester.

Below you will find excerpts from courses related to this module component:

V Global Optimization I

2550134, SS 2026, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content
In many optimization problems from economics, engineering and natural sciences, solution algorithms are only able to efficiently identify *local* optimizers, while it is much harder to find *globally* optimal points. This corresponds to the fact that by local search it is easy to find the summit of the closest mountain, but that the search for the summit of Mount Everest is rather elaborate.

The lecture treats methods for global optimization of convex functions under convex constraints. It is structured as follows:

- Introduction, examples, and terminology
- Existence results for optimal points
- Optimality in convex optimization
- Duality, bounds, and constraint qualifications
- Algorithms (Kelley's cutting plane method, Frank-Wolfe method, primal-dual interior point methods)

The lecture is accompanied by exercises which, amongst others, offers the opportunity to implement and to test some of the methods on practically relevant examples.

Remark:

The treatment of *nonconvex* optimization problems forms the contents of the lecture "Global Optimization II". The lectures "Global Optimization I" and "Global Optimization II" are held consecutively *in the same semester*.

Learning objectives:

The student

- knows and understands the fundamentals of deterministic global optimization in the convex case,
- is able to choose, design and apply modern techniques of deterministic global optimization in the convex case in practice.

Literature
O. Stein, Basic Concepts of Global Optimization, Springer, 2024

Weiterführende Literatur:

- W. Alt, Numerische Verfahren der konvexen, nichtglatten Optimierung, Teubner, 2004
- C.A. Floudas, Deterministic Global Optimization, Kluwer, 2000
- R. Horst, H. Tuy, Global Optimization, Springer, 1996
- A. Neumaier, Interval Methods for Systems of Equations, Cambridge University Press, 1990

T 6.289 Module component: Global Optimization I and II [T-WIWI-103638]

Coordinators: Prof. Dr. Oliver Stein
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101473 - Mathematical Programming
M-WIWI-104899 - Operations Research

Type	Credits	Grading	Term offered	Version
Written examination	9 CP	graded	Each summer term	1

Courses					
ST 2026	2550134	Global Optimization I	2 SWS	Lecture / 🗣	Stein
ST 2026	2550135	Exercise to Global Optimization I	1 SWS	Practice / 🗣	Stein, Huber
ST 2026	2550136	Global Optimization II	2 SWS	Lecture / 🗣	Stein

Legend: 📱 Online, Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment of the lecture is a written examination in English (120 minutes) according to §4(2), 1 of the examination regulation. The successful completion of the exercises is required for admission to the written exam.

The examination is held in the semester of the lecture and in the following semester.

Prerequisites
None

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-WIWI-102726 - Global Optimization I must not have been started.
2. The module component T-WIWI-102727 - Global Optimization II must not have been started.

Recommendations
None

Additional Information
Part I and II of the lecture are held consecutively in the **same** semester.

Below you will find excerpts from courses related to this module component:

V Global Optimization I
2550134, SS 2026, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content
In many optimization problems from economics, engineering and natural sciences, solution algorithms are only able to efficiently identify *local* optimizers, while it is much harder to find *globally* optimal points. This corresponds to the fact that by local search it is easy to find the summit of the closest mountain, but that the search for the summit of Mount Everest is rather elaborate.

The lecture treats methods for global optimization of convex functions under convex constraints. It is structured as follows:

- Introduction, examples, and terminology
- Existence results for optimal points
- Optimality in convex optimization
- Duality, bounds, and constraint qualifications
- Algorithms (Kelley's cutting plane method, Frank-Wolfe method, primal-dual interior point methods)

The lecture is accompanied by exercises which, amongst others, offers the opportunity to implement and to test some of the methods on practically relevant examples.

Remark:

The treatment of *nonconvex* optimization problems forms the contents of the lecture "Global Optimization II". The lectures "Global Optimization I" and "Global Optimization II" are held consecutively *in the same semester*.

Learning objectives:

The student

- knows and understands the fundamentals of deterministic global optimization in the convex case,
- is able to choose, design and apply modern techniques of deterministic global optimization in the convex case in practice.

Literature
O. Stein, Basic Concepts of Global Optimization, Springer, 2024

Weiterführende Literatur:

- W. Alt, Numerische Verfahren der konvexen, nichtglatten Optimierung, Teubner, 2004
- C.A. Floudas, Deterministic Global Optimization, Kluwer, 2000
- R. Horst, H. Tuy, Global Optimization, Springer, 1996
- A. Neumaier, Interval Methods for Systems of Equations, Cambridge University Press, 1990

V **Global Optimization II** — **Lecture (V) On-Site**
2550136, SS 2026, 2 SWS, Language: English, Open in study portal

Content
In many optimization problems from economics, engineering and natural sciences, solution algorithms are only able to efficiently identify *local* optimizers, while it is much harder to find *globally* optimal points. This corresponds to the fact that by local search it is easy to find the summit of the closest mountain, but that the search for the summit of Mount Everest is rather elaborate.

The lecture treats methods for global optimization of nonconvex functions under nonconvex constraints. It is structured as follows:

- Introduction and examples
- Convex relaxation
- Interval arithmetic
- Convex relaxation via alphaBB method
- Branch-and-bound methods
- Lipschitz optimization

The lecture is accompanied by exercises which, amongst others, offers the opportunity to implement and to test some of the methods on practically relevant examples.

Remark:

The treatment of *convex* optimization problems forms the contents of the lecture "Global Optimization I". The lectures "Global Optimization I" and "Global Optimization II" are held consecutively *in the same semester*.

Learning objectives:

The student

- knows and understands the fundamentals of deterministic global optimization in the nonconvex case,
- is able to choose, design and apply modern techniques of deterministic global optimization in the nonconvex case in practice.

Literature
O. Stein, Basic Concepts of Global Optimization, Springer, 2024

Weiterführende Literatur:

- W. Alt, Numerische Verfahren der konvexen, nichtglatten Optimierung, Teubner, 2004
- C.A. Floudas, Deterministic Global Optimization, Kluwer, 2000
- R. Horst, H. Tuy, Global Optimization, Springer, 1996
- A. Neumaier, Interval Methods for Systems of Equations, Cambridge University Press, 1990

T 6.290 Module component: Global Optimization II [T-WIWI-102727]

Coordinators: Prof. Dr. Oliver Stein
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101473 - Mathematical Programming
M-WIWI-104899 - Operations Research
M-WIWI-107187 - Quantitative Core

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each summer term	2

Courses					
ST 2026	2550136	Global Optimization II	2 SWS	Lecture / 🗣	Stein
ST 2026	2550137	Exercise to Global Optimization II	1 SWS	Practice / 🗣	Stein, Huber

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment of the lecture is a written examination in English (60 minutes) according to §4(2), 1 of the examination regulation. The successful completion of the exercises is required for admission to the written exam.

The examination is held in the semester of the lecture and in the following semester.

The examination can also be combined with the examination of "Global optimization I". In this case, the duration of the written examination takes 120 minutes.

Prerequisites
None

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-WIWI-103638 - Global Optimization I and II must not have been started.

Additional Information
Part I and II of the lecture are held consecutively in the **same** semester.

Below you will find excerpts from courses related to this module component:

V Global Optimization II
2550136, SS 2026, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content
In many optimization problems from economics, engineering and natural sciences, solution algorithms are only able to efficiently identify *local* optimizers, while it is much harder to find *globally* optimal points. This corresponds to the fact that by local search it is easy to find the summit of the closest mountain, but that the search for the summit of Mount Everest is rather elaborate.

The lecture treats methods for global optimization of nonconvex functions under nonconvex constraints. It is structured as follows:

- Introduction and examples
- Convex relaxation
- Interval arithmetic
- Convex relaxation via alphaBB method
- Branch-and-bound methods
- Lipschitz optimization

The lecture is accompanied by exercises which, amongst others, offers the opportunity to implement and to test some of the methods on practically relevant examples.

Remark:

The treatment of *convex* optimization problems forms the contents of the lecture "Global Optimization I". The lectures "Global Optimization I" and "Global Optimization II" are held consecutively *in the same semester*.

Learning objectives:

The student

- knows and understands the fundamentals of deterministic global optimization in the nonconvex case,
- is able to choose, design and apply modern techniques of deterministic global optimization in the nonconvex case in practice.

Literature
O. Stein, Basic Concepts of Global Optimization, Springer, 2024

Weiterführende Literatur:

- W. Alt, Numerische Verfahren der konvexen, nichtglatten Optimierung, Teubner, 2004
- C.A. Floudas, Deterministic Global Optimization, Kluwer, 2000
- R. Horst, H. Tuy, Global Optimization, Springer, 1996
- A. Neumaier, Interval Methods for Systems of Equations, Cambridge University Press, 1990

T 6.291 Module component: Global Production [T-MACH-114800]

Coordinators: Prof. Dr.-Ing. Gisela Lanza
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101282 - Global Production and Logistics
M-MACH-105455 - Strategic Design of Modern Production Systems
M-MACH-106590 - Production Engineering
M-WIWI-104907 - Engineering Sciences
M-WIWI-107718 - Mechanical Engineering I
M-WIWI-107730 - Mechanical Engineering II

Type	Credits	Grading	Term offered	Version
Written examination	5 CP	graded	Each winter term	1

Courses					
WT 25/26	2149613	Global Production	2 SWS	Lecture / Blended	Lanza
WT 26/27	2149613	Global Production	2 SWS	Lecture / Blended	Lanza

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
Written Exam (60 min)

A grade bonus can be earned for successful participation in online case studies. If the grade of the written examination is between 4.0 and 1.3, the bonus improves the grade by up to one grade level (0.3 or 0.4). Details will be announced in the course.

Prerequisites
T-MACH-108848 - Globale Produktion und Logistik - Teil 1: Globale Produktion must not be commenced.
T-MACH-105158 - Globale Produktion und Logistik - Teil 1: Globale Produktion must not be commenced.
T-MACH-110337 - Globale Produktion und Logistik must not be commenced.

Additional Information
The course is offered in English.

Workload
150 hours

Below you will find excerpts from courses related to this module component:

V Global Production

2149613, WS 25/26, 2 SWS, Language: English, Open in study portal

Lecture (V)
Blended (On-Site/Online)

Content
The course examines the planning, design and operation of global production networks of manufacturing companies. It gives an overview of the influencing factors and challenges of global production. In-depth knowledge of common methods and procedures for planning, designing and managing global production networks is imparted.

Therefore, the lecture first of all discusses the connections and interdependencies between the business strategy and the production strategy and illustrates necessary tasks for the definition of a production strategy. Methods for site selection, for the site-specific adaptation of product design and production technology as well as for the establishment of new production sites and for the adaptation of existing production networks to changing framework conditions are subsequently taught within the context of the design of the network footprint. With regard to the management of global production networks, the lecture addresses challenges associated with coordination, procurement and order management in global networks.

The lecture is complemented by an exercise that discusses the use of Industry 4.0 applications in the context of global production as well as current trends with regard to the planning, design and management of global production networks. The exercise is not relevant for Mechanical Engineering

The topics include:

- Basic conditions and influencing factors of global production (historical development, targets, chances and threats)
- Framework for planning, designing and managing global production networks
- Production strategies for global production networks
  - From business strategy to production strategy
  - Tasks of the production strategy (product portfolio management, circular economy, planning of production depth, production-related research and development)
- Design of global production networks
  - Basic types of network structures
  - Planning process for the design of the network footprint
  - Adaptation of the network footprint
  - Site selection
  - Location-specific adaptation of production technology and product design
- Management of global production networks
  - Network coordination
  - Procurement process
  - Order management

Exercise contents only for WiWis:
Exercise for Global Production coverring topics with respect to trends in resilience, sustainability and digitalisation in global production networks.

Learning Outcomes:

The students …

- can explain the challenges and enablers of global production networks and evaluate methods for decision support.
- are able to explain factors influencing decision-making in the design and operation of a global production network.
- are able to distinguish between corporate and business unit strategy as well as forms of growth and outsourcing.
- can explain the objectives of production and production network strategies as well as the ideal-typical forms of strategy
- are able to identify ideal-typical network structures of global production networks.
- can apply a network adjustment, taking into account the relevant aspects.
- can systematically analyse, evaluate and select new locations in the global production network.
- are able to apply challenges and solutions for location-specific manufacturing technology and product design.
- can evaluate the suitability of different forms of co-location of R&D and site roles for the sites in a production network.
- are able to explain the role distribution of sites, the management of external suppliers, order processing and demand planning in global production networks.

Workload MACH:
regular attendance: 21 hours
self-study: 99 hours

Workload WiWi:
regular attendance: 26 hours
self-study: 124 hours

Recommendations:
Combination with Strategic Decision-Making in Global Production Network Design: A Seminar on Optimisation and Simulation (76-T-MACH-113372) is recommended.

Literature
Medien
Skript zur Veranstaltung wird über (https://ilias.studium.kit.edu/) bereitgestellt

empfohlene Sekundärliteratur:
Abele, E. et al: Handbuch Globale Produktion, Hanser Fachbuchverlag, 2006 (deutsch)

Media
Lecture notes will be provided in Ilias (https://ilias.studium.kit.edu/)

recommended secondary literature:
Abele, E. et al: Global Production – A Handbook for Strategy and Implementation, Springer 2008 (english)

V	**Global Production** 2149613, WS 26/27, 2 SWS, Language: English, Open in study portal	**Lecture (V)** **Blended (On-Site/Online)**

Content

The course examines the planning, design and operation of global production networks of manufacturing companies. It gives an overview of the influencing factors and challenges of global production. In-depth knowledge of common methods and procedures for planning, designing and managing global production networks is imparted.

Therefore, the lecture first of all discusses the connections and interdependencies between the business strategy and the production strategy and illustrates necessary tasks for the definition of a production strategy. Methods for site selection, for the site-specific adaptation of product design and production technology as well as for the establishment of new production sites and for the adaptation of existing production networks to changing framework conditions are subsequently taught within the context of the design of the network footprint. With regard to the management of global production networks, the lecture addresses challenges associated with coordination, procurement and order management in global networks.

The lecture is complemented by an exercise that discusses the use of Industry 4.0 applications in the context of global production as well as current trends with regard to the planning, design and management of global production networks. The exercise is not relevant for Mechanical Engineering

The topics include:

- Basic conditions and influencing factors of global production (historical development, targets, chances and threats)
- Framework for planning, designing and managing global production networks
- Production strategies for global production networks
    - From business strategy to production strategy
    - Tasks of the production strategy (product portfolio management, circular economy, planning of production depth, production-related research and development)
- Design of global production networks
    - Basic types of network structures
    - Planning process for the design of the network footprint
    - Adaptation of the network footprint
    - Site selection
    - Location-specific adaptation of production technology and product design
- Management of global production networks
    - Network coordination
    - Procurement process
    - Order management

Exercise contents only for WiWis:

Exercise for Global Production coverring topics with respect to trends in resilience, sustainability and digitalisation in global production networks.

Learning Outcomes:

The students …

- can explain the challenges and enablers of global production networks and evaluate methods for decision support.
- are able to explain factors influencing decision-making in the design and operation of a global production network.
- are able to distinguish between corporate and business unit strategy as well as forms of growth and outsourcing.
- can explain the objectives of production and production network strategies as well as the ideal-typical forms of strategy
- are able to identify ideal-typical network structures of global production networks.
- can apply a network adjustment, taking into account the relevant aspects.
- can systematically analyse, evaluate and select new locations in the global production network.
- are able to apply challenges and solutions for location-specific manufacturing technology and product design.
- can evaluate the suitability of different forms of co-location of R&D and site roles for the sites in a production network.
- are able to explain the role distribution of sites, the management of external suppliers, order processing and demand planning in global production networks.

Workload MACH:

regular attendance: 21 hours
self-study: 99 hours

Workload WiWi:

regular attendance: 26 hours
self-study: 124 hours

Recommendations:

Combination with Strategic Decision-Making in Global Production Network Design: A Seminar on Optimisation and Simulation (76-T-MACH-113372) is recommended.

Literature

Medien

Skript zur Veranstaltung wird über (https://ilias.studium.kit.edu/) bereitgestellt

empfohlene Sekundärliteratur:
Abele, E. et al: Handbuch Globale Produktion, Hanser Fachbuchverlag, 2006 (deutsch)

Media

Lecture notes will be provided in Ilias (https://ilias.studium.kit.edu/)

recommended secondary literature:
Abele, E. et al: Global Production – A Handbook for Strategy and Implementation, Springer 2008 (english)

T 6.292 Module component: Graph Theory and Advanced Location Models [T-WIWI-102723]

Coordinators: Prof. Dr. Stefan Nickel
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101473 - Mathematical Programming
M-WIWI-102832 - Operations Research in Supply Chain Management
M-WIWI-103289 - Stochastic Optimization
M-WIWI-104899 - Operations Research

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Irregular	2

Courses					
ST 2026	2500033	Exercises for Graph Theory and Advanced Location Models	1.5 SWS	Practice / 🧩	Hoffmann
ST 2026	2500050	Graph Theory and Advanced Location Models	3 SWS	Lecture / 🗣	Nickel

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment of the course (lecture and exercise) is a 60-minute written examination (according to §4(2), 1 of the examination regulation).

The examination is held in the term of the lecture and the following lecture.

Prerequisites
None

Recommendations
It is recommended that students have knowledge of Operations Research, such as that taught in the course "Introduction to Operations Research".

Additional Information
The course is offered irregularly. Planned lectures for the next three years can be found in the internet at http://dol.ior.kit.edu/english/Courses.php.

T 6.293 Module component: Growth and Development [T-WIWI-112816]

Coordinators: Prof. Dr. Ingrid Ott
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104908 - Economics
M-WIWI-107010 - Economics in a Connected World
M-WIWI-107011 - Economics of Innovation and Growth

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each winter term	1

Courses					
WT 25/26	2561503	Growth and Development	2 SWS	Lecture / 🗣	Ott
WT 25/26	2561504	Exercise for Growth and Development	1 SWS	Practice / 🗣	Ott, Ghoniem
WT 26/27	2561503	Growth and Development	2 SWS	Lecture / 🗣	Ott
WT 26/27	2561504	Exercise for Growth and Development	1 SWS	Practice / 🗣	Ott, Ghoniem

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success is assessed in the form of a written examination (60 minutes). The examination is offered every semester and can be repeated at any regular examination date.

Prerequisites
None

Recommendations
It is recommended that students have basic micro- and macro-economic knowledge, as taught for example in the courses **Economics I** and **Economics II** . In addition, it is recommended that they have an interest in quantitative-mathematical modelling.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Growth and Development

2561503, WS 25/26, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content
This course is intended as an introduction to the field of advanced macroeconomics with a special focus on economic growth. Lectures aim to deal with the theoretical foundations of exogenous and endogenous growth models. The importance of growth for nations and discussion of some (well-known) growth theories together with the role of innovation, human capital and environment will therefore be primary focuses of this course.

Learning objective:

Students shall be given the ability to understand, analyze and evaluate selected models of endogenous growth theory.

Course content:

- Intertemporal consumption decision
- Growth models with exogenous saving rates: Solow
- Growth models with endogenous saving rates: Ramsey
- Growth and environmental resources
- Basic models of endogenous growth
- Human capital and economic growth
- Modelling of technological progress
- Diversity Models
- Schumpeterian growth
- Directional technological progress
- Diffusion of technologies

Recommendations:

Basic knowledge of micro- and macroeconomics is assumed, as taught in the courses Economics I [2600012], and Economics II [2600014]. In addition, an interest in quantitative-mathematical modeling is required.

Workload:

The total workload for this course is approximately 135.0 hours. For further information see German version.

Exam description:

The assessment consists of a written exam (60 min) according to Section 4(2), 1 of the examination regulation. The exam takes place in every semester. Re-examinations are offered at every ordinary examination date.

Students will be given the opportunity of writing and presenting a short paper during the lecture time to achieve a bonus on the exam grade. If the mandatory credit point exam is passed, the awarded bonus points will be added to the regular exam points. A deterioration is not possible by definition, and a grade does not necessarily improve, but is very likely to (not every additional point improves the total number of points, since a grade can not become better than 1). The voluntary elaboration of such a paper can not countervail a fail in the exam.

Literature
Auszug:

- Acemoglu, D. (2009): Introduction to modern economic growth. Princeton University Press, New Jersey.
- Aghion, P., Howitt, P. (2009): Economics of growth, MIT-Press, Cambridge/MA.
- Barro, R.J., Sala-I-Martin, X. (2003): Economic Growth. MIT-Press, Cambridge/MA.
- Sydsaeter, K., Hammond, P. (2008): Essential mathematics for economic analysis. Prentice Hall International, Harlow.
- Sydsæter, K., Hammond, P., Seierstad, A., Strom, A., (2008): Further Mathematics for Economic Analysis, Second Edition, Pearson Education Limited, Essex.

V	**Growth and Development** 2561503, WS 26/27, 2 SWS, Language: English, Open in study portal	**Lecture (V)** **On-Site**

Content
This course is intended as an introduction to the field of advanced macroeconomics with a special focus on economic growth. Lectures aim to deal with the theoretical foundations of exogenous and endogenous growth models. The importance of growth for nations and discussion of some (well-known) growth theories together with the role of innovation, human capital and environment will therefore be primary focuses of this course.

Learning objective:

Students shall be given the ability to understand, analyze and evaluate selected models of endogenous growth theory.

Course content:

- Intertemporal consumption decision
- Growth models with exogenous saving rates: Solow
- Growth models with endogenous saving rates: Ramsey
- Growth and environmental resources
- Basic models of endogenous growth
- Human capital and economic growth
- Modelling of technological progress
- Diversity Models
- Schumpeterian growth
- Directional technological progress
- Diffusion of technologies

Recommendations:

Basic knowledge of micro- and macroeconomics is assumed, as taught in the courses Economics I [2600012], and Economics II [2600014]. In addition, an interest in quantitative-mathematical modeling is required.

Workload:

The total workload for this course is approximately 135.0 hours. For further information see German version.

Exam description:

The assessment consists of a written exam (60 min) according to Section 4(2), 1 of the examination regulation. The exam takes place in every semester. Re-examinations are offered at every ordinary examination date.

Students will be given the opportunity of writing and presenting a short paper during the lecture time to achieve a bonus on the exam grade. If the mandatory credit point exam is passed, the awarded bonus points will be added to the regular exam points. A deterioration is not possible by definition, and a grade does not necessarily improve, but is very likely to (not every additional point improves the total number of points, since a grade can not become better than 1). The voluntary elaboration of such a paper can not countervail a fail in the exam.

Literature
Auszug:

- Acemoglu, D. (2009): Introduction to modern economic growth. Princeton University Press, New Jersey.
- Aghion, P., Howitt, P. (2009): Economics of growth, MIT-Press, Cambridge/MA.
- Barro, R.J., Sala-I-Martin, X. (2003): Economic Growth. MIT-Press, Cambridge/MA.
- Sydsaeter, K., Hammond, P. (2008): Essential mathematics for economic analysis. Prentice Hall International, Harlow.
- Sydsæter, K., Hammond, P., Seierstad, A., Strom, A., (2008): Further Mathematics for Economic Analysis, Second Edition, Pearson Education Limited, Essex.

T 6.294 Module component: Guide to Cruising through Informatics Program of Studies at KIT (eezi) [T-INFO-109862]

Coordinators: Dr. Ioana Gheta
Christine Glaubitz

Organisation: KIT Department of Informatics

Part of: M-WIWI-104910 - Interdisciplinary Qualifications

Type	Credits	Grading	Term offered	Version
Coursework	1 CP	pass/fail	Each winter term	1

Courses					
WT 25/26	2400037	Guide to navigating KIT Department of Informatics' Program of Studies		Others /	Glaubitz
WT 25/26	2411809	Tutorial for Guide to cruising through Informatics program of studies at KIT (eezi)		Tutorial /	Glaubitz
WT 26/27	2400037	Guide to navigating KIT Department of Informatics' Program of Studies		Others /	Glaubitz
WT 26/27	2411809	Tutorial for Guide to cruising through Informatics program of studies at KIT (eezi)		Tutorial /	Glaubitz

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
Success is assessed in the form of coursework in accordance with Section 4 (3) SPO.

Prerequisites
none.

Additional Information
The Module **eezi,** pronounced "easy", is a guide to navigating KIT Department of Informatics' program of studies. This Module is aimed at and recommended for all first-semester bachelor students majoring in informatics, information systems / Teaching Degree Program in Computer Science (Bachelor of Education) at KIT.

The intention of this Module is to support your transition and adjustment to the academic rigors your program of study and students life, by providing information and tools that you can apply to effectively plan and manage your studies in your first semester and beyond.

The module consists of:
- Three lectures
- Five tutorials and worksheet
- Check-in meeting with eezi tutor
- One eezi-Advising session with the module advisor

The course content is geared to fostering your smooth transition, acclimation, academic persistence and autonomy by providing information and resources at the relevant time in the semester. The focus is **heping students to help themselves**, by build on your self-reflection skills, conscious decision making, developing competencies, purpose and long-term strategies. Emphasis on interactive discussion, exchange of ideas, reflection and that social engagement and encouragement is also core part of this course.

Course topics include:
- Time, self, study, and stress management
- Learning techniques
- Exam preparation
- Tips and tricks for the first semester: “What I need to do to succeed in my studies”
- Study groups

The eezi-tutor check-in chat purpose is to provide a one-on-one personal exchange on equal terms between a first-semester student and a upper-semester student.

The eezi-advising session serves to answer students' questions about their studies and to assess their study situation. This includes reflection and evaluation of their own progress, as well as discussion of possible challenges and successes. (available in either German or English)

Students have the option of choosing either a tutorial in German or in English

Successful completion of the course is credited with **1 ECTS** as a key qualification/ general studies.

Below you will find excerpts from courses related to this module component:

V	**Guide to navigating KIT Department of Informatics' Program of Studies** 2400037, WS 25/26, SWS, Language: German/English, Open in study portal	**Others (sonst.)** **Blended (On-Site/Online)**

Content
The Module **eezi**, pronounced "easy", is a guide to navigating KIT Department of Informatics' program of studies. This Module is aimed at and recommended for all first-semester bachelor students majoring in informatics, information systems / Teaching Degree Program in Computer Science (Bachelor of Education) at KIT.

The intention of this Module is to support your transition and adjustment to the academic rigors your program of study and student life, by providing information and tools that you can apply to effectively plan and manage your studies in your first semester and beyond.

The module consists of:

- three lectures
- five tutorials and worksheet
- check-in chat with an eezi tutor
- one eezi-advising session with the module advisor

The course content is geared to fostering your smooth transition, acclimation, academic persistence and autonomy by providing information and resources at the relevant time in the semester. The focus is **helping students to help themselves**, by build on your self-reflection skills, conscious decision making, developing competencies, purpose and long-term strategies. Emphasis on interactive discussion, exchange of ideas, reflection and that social engagement and encouragement is also core part of this course.

Course topics include:

- Time, self, study, and stress management
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- Tips and tricks for the first semester: “What I need to do to succeed in my studies”
- Study groups

The eezi-tutor check-in chat purpose is to provide a one-on-one personal exchange on equal terms between a first-semester student and an upper-semester student.

The eezi-advising session serves to answer students' questions about their studies and to assess their study situation. This includes reflection and evaluation of their own progress, as well as discussion of possible challenges and successes. (available in either German or English)

Students have the option of choosing either a tutorial in German or in English

Successful completion of the course is credited with 1 ECTS as a key qualification/ general studies.

Audience: For first semester bachelor students in one of the following majors: Informatics / Information Systems / Teaching Degree Program in Computer Science at KIT.

V **Tutorial for Guide to cruising through Informatics program of studies at KIT (eezi)**
2411809, WS 25/26, SWS, Language: German/English, Open in study portal

Tutorial (Tu)
Blended (On-Site/Online)

Content
Tutorial for Guide to cruising through Informatics program of studies at KIT known as **eezi** ("easy") is recommended for all new bachelor students in the majors of Informatics / Information Systems / Teaching Degree Program in Informatics at KIT.

The focus of the tutorials is on helping students to help themselves in order to facilitate their transition into their major and building long-term strategies to effectively and successfully complete their studies.

Topics covered include: Time management, self-management, study management, stress management, study techniques, exam preparation, as well as tips to “How to survive the first semester” and “What do I need to do to manage my studies.”

The goal is to facilitate an eye level exchange between first semester students and students from a higher semester. The tutorial and tutor check-in meeting serves to answer students' questions about their studies and to assess their study situation. This includes reflecting on and evaluating their own progress, as well as discussing possible problems and successes.

The tutorial consists of 6 tutorials and 5 worksheets as well as a tutor check-in meeting. You can participate in the tutorial, but are not required to register for the credit point.

Students have the option of choosing either a tutorial in German or in English

V **Guide to navigating KIT Department of Informatics' Program of Studies**
2400037, WS 26/27, SWS, Language: German/English, Open in study portal

Others (sonst.)
Blended (On-Site/Online)

Content
The Module **eezi**, pronounced "easy", is a guide to navigating KIT Department of Informatics' program of studies. This Module is aimed at and recommended for all first-semester bachelor students majoring in informatics, information systems / Teaching Degree Program in Computer Science (Bachelor of Education) at KIT.

The intention of this Module is to support your transition and adjustment to the academic rigors your program of study and student life, by providing information and tools that you can apply to effectively plan and manage your studies in your first semester and beyond.

The module consists of:

- three lectures
- five tutorials and worksheet
- check-in chat with an eezi tutor
- one eezi-advising session with the module advisor

The course content is geared to fostering your smooth transition, acclimation, academic persistence and autonomy by providing information and resources at the relevant time in the semester. The focus is **helping students to help themselves**, by build on your self-reflection skills, conscious decision making, developing competencies, purpose and long-term strategies. Emphasis on interactive discussion, exchange of ideas, reflection and that social engagement and encouragement is also core part of this course.

Course topics include:

- Time, self, study, and stress management
- Learning techniques
- Exam preparation
- Tips and tricks for the first semester: “What I need to do to succeed in my studies”
- Study groups

The eezi-tutor check-in chat purpose is to provide a one-on-one personal exchange on equal terms between a first-semester student and an upper-semester student.

The eezi-advising session serves to answer students' questions about their studies and to assess their study situation. This includes reflection and evaluation of their own progress, as well as discussion of possible challenges and successes. (available in either German or English)

Students have the option of choosing either a tutorial in German or in English

Successful completion of the course is credited with 1 ECTS as a key qualification/ general studies.

Audience: For first semester bachelor students in one of the following majors: Informatics / Information Systems / Teaching Degree Program in Computer Science at KIT.

V **Tutorial for Guide to cruising through Informatics program of studies at KIT (eezi)**
2411809, WS 26/27, SWS, Language: German/English, Open in study portal

Tutorial (Tu)
Blended (On-Site/Online)

Content
Tutorial for Guide to cruising through Informatics program of studies at KIT known as **eezi** ("easy") is recommended for all new bachelor students in the majors of Informatics / Information Systems / Teaching Degree Program in Informatics at KIT.

The focus of the tutorials is on helping students to help themselves in order to facilitate their transition into their major and building long-term strategies to effectively and successfully complete their studies.

Topics covered include: Time management, self-management, study management, stress management, study techniques, exam preparation, as well as tips to “How to survive the first semester” and “What do I need to do to manage my studies.”

The goal is to facilitate an eye level exchange between first semester students and students from a higher semester. The tutorial and tutor check-in meeting serves to answer students' questions about their studies and to assess their study situation. This includes reflecting on and evaluating their own progress, as well as discussing possible problems and successes.

The tutorial consists of 6 tutorials and 5 worksheets as well as a tutor check-in meeting. You can participate in the tutorial, but are not required to register for the credit point.

Students have the option of choosing either a tutorial in German or in English

T 6.295 Module component: Handling Characteristics of Motor Vehicles I [T-MACH-105152]

Coordinators: Dr.-Ing. Fabian Weitz
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101264 - Handling Characteristics of Motor Vehicles
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each winter term	1

Courses					
WT 25/26	2113807	Handling Characteristics of Motor Vehicles I	2 SWS	Lecture / 🗣	Weitz
WT 26/27	2113807	Handling Characteristics of Motor Vehicles I	2 SWS	Lecture / 🗣	Weitz

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Verbally

Duration: 30 up to 40 minutes

Auxiliary means: none

Prerequisites
none

Additional Information
The course is offered in German.

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Handling Characteristics of Motor Vehicles I

Lecture (V)
On-Site

2113807, WS 25/26, 2 SWS, Language: German, Open in study portal

Content
1. Problem definition: Control loop driver - vehicle - environment (e.g. coordinate systems, modes of motion of the car body and the wheels)

2. Simulation models: Creation from motion equations (method according to D'Alembert, method according to Lagrange, programme packages for automatically producing of simulation equations), model for handling characteristics (task, motion equations)

3. Tyre behavior: Basics, dry, wet and winter-smooth roadway

Learning Objectives:

The students know the basic connections between drivers, vehicles and environment. They can build up a vehicle simulation model, with which forces of inertia, aerodynamic forces and tyre forces as well as the appropriate moments are considered. They have proper knowledge in the area of tyre characteristics, since a special meaning comes to the tire behavior during driving dynamics simulation. Consequently they are ready to analyze the most importent influencing factors on the driving behaviour and to contribute to the optimization of the handling characteristics.

Organizational issues
Die Vorlesung wird als Videostream zur Verfügung gestellt. Sie finden den Videostream und das Vorlesungsmaterial auf ILIAS. Das ILIAS-Passwort erhalten Sie unter https://fast-web-01.fast.kit.edu/PasswoerterIlias/

Literature

1. Willumeit, H.-P.: Modelle und Modellierungsverfahren in der Fahrzeugdynamik,
B. G. Teubner Verlag, 1998

2. Mitschke, M./Wallentowitz, H.: Dynamik von Kraftfahrzeugen, Springer-Verlag, Berlin, 2004

3. Gnadler, R.; Unrau, H.-J.: Umdrucksammlung zur Vorlesung Fahreigenschaften von Kraftfahrzeugen I

V **Handling Characteristics of Motor Vehicles I**
2113807, WS 26/27, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content

1. Problem definition: Control loop driver - vehicle - environment (e.g. coordinate systems, modes of motion of the car body and the wheels)

2. Simulation models: Creation from motion equations (method according to D'Alembert, method according to Lagrange, programme packages for automatically producing of simulation equations), model for handling characteristics (task, motion equations)

3. Tyre behavior: Basics, dry, wet and winter-smooth roadway

Learning Objectives:

The students know the basic connections between drivers, vehicles and environment. They can build up a vehicle simulation model, with which forces of inertia, aerodynamic forces and tyre forces as well as the appropriate moments are considered. They have proper knowledge in the area of tyre characteristics, since a special meaning comes to the tire behavior during driving dynamics simulation. Consequently they are ready to analyze the most importent influencing factors on the driving behaviour and to contribute to the optimization of the handling characteristics.

Organizational issues

Termine und Nähere Informationen: siehe ILIAS oder Institutshomepage

Dates and further information will be published on the homepage of the institute.

Die Vorlesung wird als Videostream zur Verfügung gestellt. Sie finden den Videostream und das Vorlesungsmaterial auf ILIAS. Das ILIAS-Passwort erhalten Sie unter https://fast-web-01.fast.kit.edu/PasswoerterIlias/

Literature

1. Willumeit, H.-P.: Modelle und Modellierungsverfahren in der Fahrzeugdynamik,
B. G. Teubner Verlag, 1998

2. Mitschke, M./Wallentowitz, H.: Dynamik von Kraftfahrzeugen, Springer-Verlag, Berlin, 2004

3. Gnadler, R.; Unrau, H.-J.: Umdrucksammlung zur Vorlesung Fahreigenschaften von Kraftfahrzeugen I

T 6.296 Module component: Handling Characteristics of Motor Vehicles II [T-MACH-105153]

Coordinators: Dr.-Ing. Fabian Weitz
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101264 - Handling Characteristics of Motor Vehicles
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each summer term	1

Courses					
ST 2026	2114838	Handling Characteristics of Motor Vehicles II	2 SWS	Lecture / 🗣	Weitz

Legend: 🖥 Online, ❁ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Oral Examination

Duration: 30 up to 40 minutes

Auxiliary means: none

Prerequisites
none

Additional Information
The course is offered in German.

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Handling Characteristics of Motor Vehicles II

2114838, SS 2026, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content
1. Vehicle handling: Bases, steady state cornering, steering input step, single sine, double track switching, slalom, cross-wind behavior, uneven roadway

2. stability behavior: Basics, stability conditions for single vehicles and for vehicles with trailer

Learning Objectives:

The students have an overview of common test methods, with which the handling of vehicles is gauged. They are able to interpret results of different stationary and transient testing methods. Apart from the methods, with which e.g. the driveability in curves or the transient behaviour from vehicles can be registered, also the influences from cross-wind and from uneven roadways on the handling characteristics are well known. They are familiar with the stability behavior from single vehicles and from vehicles with trailer. Consequently they are ready to judge the driving behaviour of vehicles and to change it by specific vehicle modifications.

Literature
1. Zomotor, A.: Fahrwerktechnik: Fahrverhalten, Vogel Verlag, 1991

2. Mitschke, M./Wallentowitz, H.: Dynamik von Kraftfahrzeugen, Springer-Verlag, Berlin, 2004

3. Gnadler, R.; Unrau, H.-J.: Umdrucksammlung zur Vorlesung Fahreigenschaften von Kraftfahrzeugen II

T 6.297 Module component: Hardware Modeling and Simulation [T-ETIT-100672]

Coordinators: Dr.-Ing. Jens Becker
Prof. Dr.-Ing. Jürgen Becker
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-100449 - Hardware Modeling and Simulation

Type	Credits	Grading	Term offered	Version
Written examination	4 CP	graded	Each winter term	2

Courses					
WT 25/26	2311608	Hardware Modeling and Simulation	2 SWS	Lecture / 🗣	Becker, Becker
WT 25/26	2311610	Tutorial for 2311608 Hardware Modeling and Simulation	1 SWS	Practice / 🗣	Sotiropoulos

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Achievement is examined in the form of a written examination lasting 120 minutes.

Prerequisites
none

T 6.298 Module component: Heat and Mass Transfer [T-MACH-114099]

Coordinators: Prof. Dr. Dr. Michael Fischlschweiger
Dr.-Ing. Chunkan Yu

Organisation: KIT Department of Mechanical Engineering

Part of: M-WIWI-107718 - Mechanical Engineering I
M-WIWI-107730 - Mechanical Engineering II

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each summer term	2

Courses					
ST 2026	3122512	Heat and Mass Transfer	2 SWS	Lecture / 🗣	Fischlschweiger, Yu
ST 2026	3122513	Heat and Mass Transfer (Excercises)	2 SWS	Practice / 🗣	Fischlschweiger, Yu

Legend: 🖥 Online, ❂ Blended (On-Site/Online), 🗣 On-Site, × Cancelled

Assessment
Written exam, 3 h

Prerequisites
none

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Heat and Mass Transfer

3122512, SS 2026, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content

- Steady and unsteady heat transfer in homogenous materials; Plates, pipe sections and sperical shells
- Molecular diffusion in gases; analogies between heat conduction and mass diffusion
- Convective, forced heat transfer in pipes/channels and around plates and profiles.
- Convective mass transfer, heat-/mass transfer analogy
- Multi phase convective heat transfer (ceondensation, evaporation)
- Radiative heat transfer

Organizational issues
Nachteilsausgleiche bitte fristgerecht an den Dozenten schicken.
Please send the requests for compensations for disadvantages to the lecturer within the deadline

Literature

- Maas ; Vorlesungsskript "Wärme- und Stoffübertragung"
- Baehr, H.-D., Stephan, K.: "Wärme- und Stoffübertragung" , Springer Verlag, 1993
- Incropera, F., DeWitt, F.: "Fundamentals of Heat and Mass Transfer" , John Wiley & Sons, 1996
- Bird, R., Stewart, W., Lightfoot, E.: "Transport Phenomena" , John Wiley & Sons, 1960

T 6.299 Module component: Heat and Mass Transfer [T-MACH-105292]

Coordinators: Prof. Dr. Dr. Michael Fischlschweiger
Dr.-Ing. Chunkan Yu

Organisation: KIT Department of Mechanical Engineering

Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Written examination	4 CP	graded	Each term	1

Courses					
WT 25/26	2165512	Heat and mass transfer	2 SWS	Lecture / 🗣	Yu, Schießl
WT 25/26	2165513	Heat and Mass Transfer (Tutorial)	2 SWS	Practice / 🗣	Yu, Schießl, Bykov
ST 2026	3122512	Heat and Mass Transfer	2 SWS	Lecture / 🗣	Fischlschweiger, Yu
ST 2026	3122513	Heat and Mass Transfer (Excercises)	2 SWS	Practice / 🗣	Fischlschweiger, Yu

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Written exam, approx. 3 h

Prerequisites
none

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Heat and mass transfer

2165512, WS 25/26, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content

- Steady and unsteady heat transfer in homogenous materials; Plates, pipe sections and sperical shells
- Molecular diffusion in gases; analogies between heat conduction and mass diffusion
- Convective, forced heat transfer in pipes/channels and around plates and profiles.
- Convective mass transfer, heat-/mass transfer analogy
- Multi phase convective heat transfer (ceondensation, evaporation)
- Radiative heat transfer

Literature

- Maas ; Vorlesungsskript "Wärme- und Stoffübertragung"
- Baehr, H.-D., Stephan, K.: "Wärme- und Stoffübertragung" , Springer Verlag, 1993
- Incropera, F., DeWitt, F.: "Fundamentals of Heat and Mass Transfer" , John Wiley & Sons, 1996
- Bird, R., Stewart, W., Lightfoot, E.: "Transport Phenomena" , John Wiley & Sons, 1960

V Heat and Mass Transfer (Tutorial)

2165513, WS 25/26, 2 SWS, Language: German, Open in study portal

Practice (Ü)
On-Site

V Heat and Mass Transfer

3122512, SS 2026, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content

- Steady and unsteady heat transfer in homogenous materials; Plates, pipe sections and sperical shells
- Molecular diffusion in gases; analogies between heat conduction and mass diffusion
- Convective, forced heat transfer in pipes/channels and around plates and profiles.
- Convective mass transfer, heat-/mass transfer analogy
- Multi phase convective heat transfer (ceondensation, evaporation)
- Radiative heat transfer

Organizational issues

Nachteilsausgleiche bitte fristgerecht an den Dozenten schicken.
Please send the requests for compensations for disadvantages to the lecturer within the deadline

Literature

- Maas ; Vorlesungsskript "Wärme- und Stoffübertragung"
- Baehr, H.-D., Stephan, K.: "Wärme- und Stoffübertragung" , Springer Verlag, 1993
- Incropera, F., DeWitt, F.: "Fundamentals of Heat and Mass Transfer" , John Wiley & Sons, 1996
- Bird, R., Stewart, W., Lightfoot, E.: "Transport Phenomena" , John Wiley & Sons, 1960

T 6.300 Module component: Heat Economy [T-WIWI-102695]

Coordinators: Prof. Dr. Wolf Fichtner
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101452 - Energy Economics and Technology
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	3,5 CP	graded	Each summer term	2

Courses					
ST 2026	2581001	Heat Economy	2 SWS	Lecture / 🗣	Fichtner

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment consists of a written (60 minutes) or oral exam (30 minutes) (following §4(2) of the examination regulation). The exam takes place in every semester. Re-examinations are offered at every ordinary examination date.

Prerequisites
None.

Recommendations
None

Additional Information
See German version.

Below you will find excerpts from courses related to this module component:

V Heat Economy
2581001, SS 2026, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Organizational issues
Block, Seminarraum Standort West - siehe Institutsaushang

T 6.301 Module component: High Performance Powder Metallurgy Materials [T-MACH-102157]

Coordinators: apl. Prof. Dr. Günter Schell
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101268 - Specific Topics in Materials Science
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each summer term	1

Courses					
ST 2026	2126749	Advanced powder metals	2 SWS	Lecture / Blended	Schell

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
oral exam, 20- 30 min

Prerequisites
none

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Advanced powder metals

2126749, SS 2026, 2 SWS, Language: German, Open in study portal

Lecture (V)
Blended (On-Site/Online)

Literature

- W. Schatt ; K.-P. Wieters ; B. Kieback. ".Pulvermetallurgie: Technologien und Werkstoffe", Springer, 2007
- R.M. German. "Powder metallurgy and particulate materials processing. Metal Powder Industries Federation, 2005
- F. Thümmler, R. Oberacker. "Introduction to Powder Metallurgy", Institute of Materials, 1993

T 6.302 Module component: High-Voltage Technology [T-ETIT-110266]

Coordinators: Prof. Dr.-Ing. Thomas Leibfried
Dr.-Ing. Michael Suriyah
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-101163 - High-Voltage Technology

Type	Credits	Grading	Term offered	Expansion	Version
Written examination	6 CP	graded	Each winter term	1 semesters	1

Courses					
WT 25/26	2307360	High-Voltage Technology	2 SWS	Lecture / 🗣	Suriyah
WT 25/26	2307362	Tutorial for 2307362High-Voltage Technology	2 SWS	Practice / 🗣	Badent, Wöhr

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success conraol takes place in the form of a written examination lasting 120 minutes.

Prerequisites
none

6.303 Module component: High-Voltage Technology I [T-ETIT-101913]

Coordinators: Dr.-Ing. Rainer Badent
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Written examination	4 CP	graded	Each winter term	1

Prerequisites
none

T 6.304 Module component: High-Voltage Technology II [T-ETIT-101914]

Coordinators: Dr.-Ing. Rainer Badent
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Written examination	4 CP	graded	Each summer term	1

Prerequisites
none

T 6.305 Module component: High-Voltage Test Technique [T-ETIT-101915]

Coordinators: Dr.-Ing. Rainer Badent
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-101164 - Generation and Transmission of Renewable Power
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each winter term	1

Courses					
WT 25/26	2307392	High-Voltage Test Technique	2 SWS	Lecture / 🗣	Badent
WT 25/26	2307394	Tutorial for 2307392 High-Voltage Test Technique	2 SWS	Practice / 🗣	Braun

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success is assessed in an overall oral examination (approx. 20 minutes) on the selected course.

Prerequisites
none

Recommendations
High voltage technology

T 6.306 Module component: Hot Research Topics in AI for Engineering Applications [T-MACH-113669]

Coordinators: Prof. Dr.-Ing. Anne Meyer
Organisation: KIT Department of Mechanical Engineering
Part of: M-MACH-107716 - Data Driven Engineering
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Expansion	Version
Examination of another type	4,5 CP	graded	Each winter term	1 semesters	3

Courses					
WT 25/26	2121341	Hot Research Topics in AI for Engineering Applications	3 SWS	Lecture / Practice / 🗣	Meyer, Dörr

Legend: 🖥 Online, Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The grade is determined by an examination of another type. This consists of an individual knowledge check after the lecture part, the continuous assessment of teamwork during the implementation task and a final presentation. The overall impression is assessed; in addition to the implementation task, the knowledge test and the final presentation are also taken into account.

Recommendations
Basic knowledge of artificial intelligence and machine learning, Programming experience, preferably in Python, English proficiency

Additional Information
The course is offered in English.

Limited number of participants. **Information on the application process is available in ILIAS.**

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Hot Research Topics in AI for Engineering Applications

Lecture / Practice (VÜ)
On-Site

2121341, WS 25/26, 3 SWS, Language: English, Open in study portal

Content
In " Hot Research Topics in AI for Engineering Applications", we explore the applicability of cutting-edge research findings in the fields of Machine Learning and Artificial Intelligence (e.g., LLM agents, Reinforcement Learning) to applications in engineering (e.g., optimization in production and logistics, creation of CAD models). Each year, we offer a different methodological focus (more on the IMI-homepage).

First, we provide the theoretical foundations and then move into a group work phase where students implement and analyze an application prototype. The event is aimed at students with prior knowledge in machine learning and programming.

The topics of this course:

- Theoretical foundations of the technologies considered in the course (e.g., Deep Learning, Transformers, LLM)
- Application possibilities of modern technologies in an industrial context
- Challenges in making current research findings usable for solving specific engineering problems and productive use
- Implementation of solutions to apply modern technologies to specified engineering problems (usually Python-based, using current frameworks)
- Independent execution of an implementation project with current, thematically relevant content (e.g., LLM agents for interaction with external systems such as robots, for algorithm construction, or for creating 3D CAD models, etc.)
- Technologies and applications are announced at the beginning of each semester

After the event, participants will be able to:

- Identify the technical and algorithmic foundations behind the relevant research topics and explain their functionalities
- Identify application possibilities of current research findings and related technologies in an industrial context, as well as the challenges that arise in the process
- Implement solutions proposed in recent publications using existing frameworks and codebases as prototypes
- Structure and execute programming projects in a team
- Clearly present the results of practical projects tailored to the audience

Participation Requirements

A foundational course in Machine Learning, Data Science, or Deep Learning, ideally including practical application in Python, must have been successfully completed.

Recommended courses include:

- T-MACH-114078 – Data Science Fundamentals in Engineering
- T-ETIT-109839 – Labor für angewandte Machine Learning Algorithmen
- M-INFO-106014 – Grundlagen der Künstlichen Intelligenz
- T-WIWI-110340 – Angewandte Informatik – Anwendungen der Künstlichen Intelligenz

Comparable completed courses may be accepted as proof of the required foundational knowledge.

Additional Information

The number of participants is limited. Information on the application process is available in ILIAS.

Organizational Information

See further details on course organization at: teaching.imi.kit.edu and in ILIAS.

Workload

120 hours

Organizational issues
Place and time of the course can be found in ILIAS, / Ort und Zeit der Lehrveranstaltung siehe ILIAS

T 6.307 Module component: HR-Management 1: HR Strategies in the Age of AI [T-WIWI-113745]

Coordinators: Prof. Dr. Petra Nieken
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each winter term	1

Courses					
WT 25/26	2573005	HR-Management 1: HR strategies in the age of AI	2 SWS	Lecture / 🗣	Nieken
WT 25/26	2573006	Übung zu HR-Management 1: HR Strategies in the age of AI	1 SWS	Practice / 🗣	Nieken, Mitarbeiter

Legend: 🖥 Online, Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment is conducted in the form of an oral (30 minutes) or written (60 minutes) examination (according to §4(2), 1 examination regulations). The exam is offered every semester and can be retaken at any regular examination date.

Prerequisites
None

Recommendations
Prior attendance of the Business Administration module is recommended.

Below you will find excerpts from courses related to this module component:

V HR-Management 1: HR strategies in the age of AI
2573005, WS 25/26, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content

In this course, students will acquire fundamental knowledge in the field of human resource management and delve deeply into the future of work. We explore not only classical topics but also the significance of artificial intelligence in the workplace, along with selected aspects related to sustainability and shaping the future of work. Drawing from microeconomic and behavioral economic approaches, we analyze various processes and tools in human resource management. We evaluate their alignment with corporate strategy. We investigate how we can design workplaces sustainably while considering the individual needs of employees. In addition, we look at how AI is transforming our work environment and the opportunities and challenges it presents.

Going beyond theoretical concepts, we validate our insights using real-world data from research papers and current events. Discussions are strongly encouraged!

Learning Outcomes

The student

- understands the processes and instruments of human resource management.
- analyzes different methods and evaluates their usefulness with a special focus on AI.
- analyzes different processes and evaluates the strengths and weaknesses.
- understands the challenges of human resource management and its link to corporate strategy with a special focus on AI and sustainability aspects.
- posses knowledge about the applicability and challenges of different scientific research methods and open science.

Workload

The total workload for this course is approximately 135 hours.

Lecture: 32 hours

Preparation of lecture: 52 hours

Exam preparation: 51 hours

Literature

- Personalmanagement, Stock-Homburg, 2019
- Personnel Economics, Kuhn, 2017
- Research papers and case studies (will be provided during the lecture)

T 6.308 Module component: HR-Management 2: Organization, Fairness & Leadership [T-WIWI-114178]

Coordinators: Prof. Dr. Petra Nieken
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each summer term	1

Courses					
ST 2026	2573001	HR-Management 2: Organization, Fairness & Leadership	2 SWS	Lecture / 🗣	Nieken
ST 2026	2573002	Übung zu HR-Management 2: Organization, Fairness & Leadership	1 SWS	Practice / 🗣	Nieken, Mitarbeiter, Gorny

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment of this course is a written examination of 1 hour. The exam takes place in every semester. Re-examinations are offered at every ordinary examination date.
In case of a small number of registrations, we might offer an oral exam instead of a written exam.

Prerequisites
None

Recommendations
Completion of module Business Administration is recommended.

Basic knowledge of microeconomics, game theory, and statistics is recommended.

Below you will find excerpts from courses related to this module component:

V HR-Management 2: Organization, Fairness & Leadership

Lecture (V)
On-Site

2573001, SS 2026, 2 SWS, Language: German, Open in study portal

Content

In the course, we explore central aspects of the working world. Students gain a deep understanding of the dynamics of wage and collective bargaining negotiations and critically engage with compensation structures within companies. A special focus lies in creating a sustainable workplace that meets both employees' needs and society's ecological and social demands. Additionally, we address topics related to diversity and inclusion. Students develop innovative approaches to leadership and new forms of work that are increasingly relevant in the modern work environment. Our analyses are based on microeconomic and behavioral economic approaches, evaluating their alignment with corporate strategy. We move beyond theoretical concepts, examining our insights using real-world data from research papers and current events. Discussions are explicitly encouraged!

Aim

The student

- understands the process and instruments of HR-Management with a focus on fair working conditions, sustainability, and leadership.
- analyzes various methods and evaluates their usefulness, particularly regarding fairness and leadership in organizations.
- analyzes various processes and assesses their strengths and weaknesses.
- evaluates the strengths and weaknesses of existing structures and regulations based on systematic criteria.
- possess knowledge about the applicability and challenges of different scientific research methods

Workload

The total workload for this course is approximately 135 hours.

Lecture 32 hours

Preparation of lecture 52 hours

Exam preparation 51 hours

Literature

- Arbeitsmarktökonomik, W. Franz, Springer, 2013
- The Nature of Leadership, Antonakis, J. Day, D. 2017

T 6.309 Module component: Human Factors in Autonomous Driving [T-WIWI-113059]

Coordinators: Prof. Dr. Alexey Vinel
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101472 - Informatics
M-WIWI-101628 - Emphasis in Informatics
M-WIWI-104901 - Informatics (KIT-Department of Economics and Management)
M-WIWI-106631 - Cooperative Autonomous Vehicles

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each winter term	1

Courses					
WT 25/26	2511452	Human Factors in Autonomous Driving	2 SWS	Lecture / 🧩	Vinel, Bied, Schrapel
WT 25/26	2511453	Exercises Human Factors in Autonomous Driving	1 SWS	Practice / 🧩	Vinel, Bied, Schrapel

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment of this course is a written examination (60 min) or an oral exam (approx. 20 min).

The exam takes place every semester and can be repeated at every regular examination date.

Workload
135 hours

T 6.310 Module component: Human Factors in Security and Privacy [T-WIWI-109270]

Coordinators: Prof. Dr. Melanie Volkamer
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101472 - Informatics
M-WIWI-101628 - Emphasis in Informatics
M-WIWI-104520 - Human Factors in Security and Privacy
M-WIWI-104901 - Informatics (KIT-Department of Economics and Management)

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Irregular	3

Assessment
The assessment of this course (lecture and exercise) is a written examination (60 min) according to §4(2), 1 of the examination regulation or an oral exam (30 min) following §4, Abs. 2, 2 of the examination regulation. Only those who have successfully participated in the exercises and the lecture will be admitted to the examination.

Prerequisites
Both need to be done:

- Pass Quiz on Paper for Graphical Passwords
- Presentation of Results Exercise 2

+ 9 of the following 11 need to be done:

- Submit ILIAS certificate until Oct 24
- Pass Quiz on InfoSec Lecture
- Active participation exercise 1 Part 1 - Evaluation and analyses methods
- Pass Quiz Paper Discussion 1 - User Behaviour and motivation theories
- Active participation exercise 1 Part 2
- Pass Quiz Paper Discussion 2 - User Behaviour and motivation theories
- Pass Quiz Paper Discussion 3 - Security Awareness
- Active participation exercise 1 Part 3
- Pass Quiz Paper Discussion 4 - Graphical Authentication
- Pass Quiz Paper Discussion 5 - Shoulder Surfing Authentication
- Active participation exercise 2

Recommendations
The prior attendance of the lecture "Information Security" is strongly recommended.

Additional Information
Please note: This course is only offered irregularly.

The lecture units are held partly in German and partly in English.

Workload
135 hours

T 6.311 Module component: Human-Computer-Interaction [T-INFO-114192]

Coordinators: Prof. Dr.-Ing. Michael Beigl
Organisation: KIT Department of Informatics
Part of: M-WIWI-104909 - Informatics (Department of Informatics)

Type	Credits	Grading	Term offered	Version
Written examination	6 CP	graded	Each summer term	1

Courses					
ST 2026	24659	Human-Computer-Interaction	2 SWS	Lecture /	Beigl

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
The assessment is carried out as a written exam (of approx. 60 minutes) as a digital exam according to §2 (3) of the Statute for the implementation of online examinations. The exam takes place ON SITE at KIT!

Prerequisites
Participation in the exercise is compulsory and the contents of the exercise are relevant for the examination.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-INFO-114193 - Human-Computer-Interaction Pass must have been passed.

Below you will find excerpts from courses related to this module component:

V Human-Computer-Interaction
24659, SS 2026, 2 SWS, Language: English, Open in study portal

Lecture (V)
Blended (On-Site/Online)

Content

Description:

The lecture introduces the fundamentals of human-computer communication. Upon completion of the module, students will have acquired basic knowledge of the field of human-computer interaction. They will have mastered the basic techniques for evaluating user interfaces, will be familiar with the basic rules and techniques for designing user interfaces, and will have knowledge of existing user interfaces and their functions. They will be able to apply these basic techniques, for example, to analyze user interfaces of computer systems and synthesize existing designs into alternative, better solutions.

Course content:

Topics covered include:

1. Human perception (physiological basics, human senses, Gestalt)
2. Human information processing (HIP models, psychological basics, action processes)
3. Design basics and design methods, principles, guidelines, and standards for designing user interfaces
4. Design analysis of human-computer interaction
5. Fundamentals and examples for the design of user interfaces and methods for modeling user interfaces
6. Studies: Evaluation of human-computer interaction systems (tools, evaluation methods, performance measurement, study design and implementation)
7. Practical application of the above fundamentals using practical examples and development of independent, new, and alternative user interfaces.

Workload

The total workload for this learning unit is approximately 180 hours (6.0 credits).

Activity

Workload

Attendance time: Attendance at lectures

15 x 90 min

22 h 30 min

Attendance time: Attendance at exercises

8 x 90 min

12 h 00 min

Preparation/follow-up for lectures

15 x 150 min

37 h 30 min

Preparation/follow-up for exercises

8 x 360 min

48 h 00 min

Reviewing the slide set/script 2 times

2 x 12 hours

24 hours

Preparing for the exam

36 hours

TOTAL

180 hours

Workload for the learning unit “Human-Machine Interaction”

Learning objectives

The lecture introduces the basics of human-machine communication. After completing the course, students will be able to

- demonstrate basic knowledge of the field of human-machine interaction
- name and apply basic techniques for analyzing user interfaces
- apply basic rules and techniques for designing user interfaces
- analyze and evaluate existing user interfaces and their functions

Language of instruction and language of examination:

The language of the lecture is English; exam questions are provided in English and German and may be answered in either English or German

Organizational issues

The lecture is a core module and is assessed in writing (written exam).

Literature

David Benyon: Designing Interactive Systems: A Comprehensive Guide to HCI and Interaction Design. Addison-Wesley Educational Publishers Inc; 2nd Revised edition; ISBN-13: 978-0321435330

Steven Heim: The Resonant Interface: HCI Foundations for Interaction Design. Addison Wesley; ISBN-13: 978-0321375964

T 6.312 Module component: Human-Computer-Interaction Pass [T-INFO-114193]

Coordinators: Prof. Dr.-Ing. Michael Beigl
Organisation: KIT Department of Informatics
Part of: M-WIWI-104909 - Informatics (Department of Informatics)

Type	Credits	Grading	Term offered	Version
Coursework	0 CP	pass/fail	Each summer term	1

Courses					
ST 2026	2400095	Human-Computer-Interaction	1 SWS	Practice / 🧩	Beigl

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO).

Exercise sheets must be handed in regularly to pass the course. The specific details will be announced in the lecture.

Prerequisites
None.

Additional Information
Participation in the exercise is compulsory and the contents of the exercise are relevant for the examination.

T 6.313 Module component: Hydraulic Engineering and Water Management [T-BGU-101667]

Coordinators: Prof. Dr. Mario Jorge Rodrigues Pereira da Franca
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Expansion	Version
Written examination	4 CP	graded	Each term	1 semesters	1

Courses					
WT 25/26	6200511	Hydraulic Engineering	2 SWS	Lecture / 🗣	Rodrigues Pereira da Franca
WT 25/26	6200512	Hydraulic Engineering - Excercise	1 SWS	Practice / 🗣	Seidel
WT 26/27	6200511	Hydraulic Engineering	2 SWS	Lecture / 🗣	Rodrigues Pereira da Franca
WT 26/27	6200512	Hydraulic Engineering - Excercise	1 SWS	Practice / 🗣	Seidel

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
written exam with 60 minutes

Prerequisites
None

Recommendations
None

Additional Information
None

Workload
120 hours

6.314 Module component: Hydrogen and reFuels - Energy Conversion in Combustion Engines [T-MACH-111585]

Coordinators: Dr.-Ing. Heiko Kubach
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101275 - Combustion Engines I

Type	Credits	Grading	Term offered	Expansion	Version
Oral examination	4 CP	graded	Each winter term	1 semesters	2

Courses					
WT 25/26	2134155	Hydrogen and reFuels - Energy Conversion in Combustion Engines	3 SWS	Lecture / Practice / 🗣	Koch

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
oral exam, appr. 25 minutes, no auxillary means

Prerequisites
T-MACH-113979 must not have been started.

Workload
120 hours

T 6.315 Module component: Hydrogen in Materials – Exercises and Lab Course [T-MACH-112159]

Coordinators: Dr. rer. nat. Stefan Wagner
Organisation: KIT Department of Mechanical Engineering
Part of: M-WIWI-107718 - Mechanical Engineering I
M-WIWI-107730 - Mechanical Engineering II

Type	Credits	Grading	Term offered	Expansion	Version
Coursework	4,5 CP	pass/fail	Each summer term	1 semesters	4

Courses					
ST 2026	2173584	Hydrogen in Materials – Exercises and Lab Course	2 SWS	Practice /	Wagner

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
Regular participation and participating in lab course, protocol included.

Prerequisites
T-MACH-112942 - Hydrogen in Materials - Exercises and lab course must not have started.

Recommendations
Participation is only possible parallel to the lecture.

Additional Information
The course is offered in English.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Hydrogen in Materials – Exercises and Lab Course

2173584, SS 2026, 2 SWS, Language: English, Open in study portal

Practice (Ü)
Blended (On-Site/Online)

Content
In this exercise with lab course the contents of the lecture “Hydrogen in Materials: from Energy Storage to Hydrogen Embrittlement” are deepened. The students know the differences in thermodynamics and kinetics of the hydrogen interaction with storage materials and construction materials. The students can describe the hydrogen interaction with microstructural defects in materials, and they know the resulting effects on the materials’ mechanical integrity. Based on this, the students can express the requirements of the respective materials classes and transfer them to engineering applications.

Utilizing proper experimental setups, the students can measure hydrogen induced stresses in materials as well as the hydrogens' diffusivity and its chemical potential. From the measurement data, the students can construct metal-hydrogen phase diagrams, and they can qualitatively assess the defect density in the metal.

Organizational issues
Participation is only possible parallel to the lecture.

T 6.316 Module component: Hydrogen in Materials: from Energy Storage to Hydrogen Embrittlement [T-MACH-110923]

Coordinators: Prof. Dr. Astrid Pundt
Organisation: KIT Department of Mechanical Engineering
Part of: M-WIWI-107718 - Mechanical Engineering I
M-WIWI-107730 - Mechanical Engineering II

Type	Credits	Grading	Term offered	Version
Oral examination	4,5 CP	graded	Each summer term	3

Courses					
ST 2026	2173588	Hydrogen in Materials: from Energy Storage to Hydrogen Embrittlement	2 SWS	Lecture / 🗣	Pundt, Wagner

Legend: 🖥 Online, 🔄 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Oral exam, about 25 minutes

Prerequisites
T-MACH-108853 - Wasserstoff in Materialien has not been started

T-MACH-110957 - Wasserstoff in Materialien: von der Energiespeicherung zur Materialversprödung has not been started

Additional Information
The course is offered in English.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Hydrogen in Materials: from Energy Storage to Hydrogen Embrittlement

Lecture (V)
On-Site

2173588, SS 2026, 2 SWS, Language: English, Open in study portal

Content
This lecture teaches physical and chemical basics of hydrogen adsorption and absorption of different materials. It trains the understanding of the specific lattice positions that hydrogen occupies within solids, and its impact on material properties. A thermodynamical approach yields Sievert's law, allowing the students to describe the different solubilities of hydrogen (and other gases) in solid materials. Further thermodynamic data can be obtained using van't Hoff plots of phase transformation pressures. The impact of ternary alloy components, as described by semi-empirical models, will be recognized. The specific mobility of hydrogen in materials will be understood, which divides into classical diffusion and quantum mechanical tunneling processes. The students can describe the interaction of hydrogen with defects in crystal lattices, which is of special interest for properties of nano-scale materials or for the hydrogen embrittlement of steels. Basic embrittlement models can be explained by the students. Actual hydrogen storage systems can be summarized.

learning objectives:

o Hydrogen as energy storage – the hydrogen cycle and safety issues

o methods for hydrogen charging of materials and hydrogen detection

o Hydrogen adsorption at and absorption in different solids, Sievert's law

o interstitial lattice sites and lattice expansion

o Hydrides, van't Hoff plots, phase transitions, M-H binary phase diagrams

o ternary alloy effects

o hydrogen mobility in materials: interstitial diffusion and quantum mechanical tunneling

o interaction of hydrogen with defects

o hydrogen embrittlement of steels, different embrittlement models

o hydrogen in nano-scale systems and new storage materials

Literature
Literaturhinweise und Unterlagen in der Vorlesung

T 6.317 Module component: Hydrology [T-BGU-101693]

Coordinators: Prof. Dr.-Ing. Erwin Zehe
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Expansion	Version
Written examination	4 CP	graded	Each term	1 semesters	2

Courses					
WT 25/26	6200513	Hydrology	2 SWS	Lecture / 🗣	Zehe, Wienhöfer
WT 25/26	6200514	Tutorial Hydrology	1 SWS	Practice / 🗣	Zehe, Wienhöfer

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
written exam, 60 min.

Prerequisites
None

Recommendations
None

Additional Information
None

Workload
120 hours

T 6.318 Module component: I4.0 Systems Platform [T-MACH-106457]

Coordinators: Dipl.-Ing. Thomas Maier
Prof. Dr.-Ing. Jivka Ovtcharova

Organisation: KIT Department of Mechanical Engineering

Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Examination of another type	4 CP	graded	Each term	3

Courses					
WT 25/26	2123900	I4.0 Systems platform	4 SWS	Course / 🗣	Meyer, Maier
ST 2026	2123900	I4.0 Systems platform	4 SWS	Project / 🗣	Meyer, Maier

Legend: 🖥 Online, ❂ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Learning control is an examination of another type.
The overall impression is evaluated. The following partial aspects are included in the grading:

- Achievement of the project objective
- Final presentation including live demonstration
- Documentation of the project implementation

The grading scale will be announced in the course.

Prerequisites
None

Additional Information
Limited number of participants.

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V I4.0 Systems platform

2123900, WS 25/26, 4 SWS, Language: German/English, Open in study portal

Course (Ku)
On-Site

Content
Industry 4.0, IT systems for fabrication (e.g.: CAx, PDM, CAM, ERP, MES), process modelling and execution, project work in teams, practice-relevant I4.0 problems, in automation, manufacturing industry and service.

Students can

- describe the fundamental concepts, challenges, and objectives of Industrie 4.0 and name the essential terms in context of information management
- explain the necessary information flow between the different IT systems. They get practically knowledge about using current IT systems in context of I4.0, from order to production.
- map and analyze processes in the context of Industry 4.0 with special methods of process modelling
- collaboratively grasp practical I4.0 issues using existing hardware and software and work out solutions for a continuous improvement process in a team
- prototypically implement the self-developed solution proposal with the given IT systems and the existing hardware equipment and finally present the results

Organizational issues
Zeit und Ort siehe ILIAS

Literature
Keine / None

V

I4.0 Systems platform

2123900, SS 2026, 4 SWS, Language: German, Open in study portal

Project (PRO)
On-Site

Content

Industry 4.0, IT systems for fabrication (e.g.: CAx, PDM, CAM, ERP, MES), process modelling and execution, project work in teams, practice-relevant I4.0 problems, in automation, manufacturing industry and service.

Students can

- describe the fundamental concepts, challenges, and objectives of Industrie 4.0 and name the essential terms in context of information management
- explain the necessary information flow between the different IT systems. They get practically knowledge about using current IT systems in context of I4.0, from order to production.
- map and analyze processes in the context of Industry 4.0 with special methods of process modelling
- collaboratively grasp practical I4.0 issues using existing hardware and software and work out solutions for a continuous improvement process in a team
- prototypically implement the self-developed solution proposal with the given IT systems and the existing hardware equipment and finally present the results

Organizational issues

Zeit und Ort siehe ILIAS

Literature

Keine / None

T 6.319 Module component: Ignition Systems [T-MACH-105985]

Coordinators: Dr.-Ing. Olaf Toedter
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101303 - Combustion Engines II

Type	Credits	Grading	Version
Oral examination	4 CP	graded	1

Courses					
WT 25/26	2133125	Ignition systems	3 SWS	Lecture / 🗣	Toedter

Legend: 🖥 Online, ✲ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
oral exam, 20 min

Prerequisites
none

Additional Information
This course is offered in German.

Workload
120 hours

T 6.320 Module component: Incentives in Organizations [T-WIWI-105781]

Coordinators: Prof. Dr. Petra Nieken
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101453 - Applied Strategic Decisions
M-WIWI-101500 - Microeconomic Theory
M-WIWI-101505 - Experimental Economics
M-WIWI-101510 - Cross-Functional Management Accounting
M-WIWI-104908 - Economics
M-WIWI-105923 - Incentives, Interactivity & Decisions in Organizations

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each summer term	1

Courses					
ST 2026	2573003	Incentives in Organizations	2 SWS	Lecture / 🗣	Nieken
ST 2026	2573004	Übung zu Incentives in Organizations	2 SWS	Practice / 🗣	Nieken, Mitarbeiter, Gorny

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment of this course is a written examination (60 min). The exam takesplace in every semester. Re-examinations are offered at every ordinaryexamination date. In case of a small number of registrations, we might offer an oral exam instead of a written exam.

Prerequisites
None

Recommendations
It is recommended that students have knowledge in microeconomics, game theory, and statistics.

Below you will find excerpts from courses related to this module component:

V Incentives in Organizations

Lecture (V)
On-Site

2573003, SS 2026, 2 SWS, Language: English, Open in study portal

Content

The students acquire profound knowledge about the design and the impact of different incentive and compensation systems. Topics covered are, for instance, performance based compensation, team work, intrinsic motivation, multitasking, and subjective performance evaluations. We will use microeconomic or behavioral models as well as empirical data to analyze incentive systems. We will investigate several widely used compensation schemes and their relationship with corporate strategy. Students will learn to develop practical implications which are based on the acquired knowledge of this course.

Aim

The student

- develops a strategic understanding about incentives systems and how they work.
- analyzes models from personnel economics.
- understands how econometric methods can be used to analyze performance and compensation data.
- knows incentive schemes that are used in companies and is able to evaluate them critically.
- can develop practical implications which are based on theoretical models and empirical data from companies.
- understands the challenges of managing incentive and compensation systems and their relationship with corporate strategy.

Workload

The total workload for this course is: approximately 135 hours.

Lecture: 32 hours

Preparation of lecture: 52 hours

Exam preparation: 51 hours

Literature

Slides, Additional case studies and research papers will be announced in the lecture.

Literature (complementary):

Managerial Economics and Organizantional Architecture, Brickley / Smith / Zimmerman, McGraw-Hill Education, 2015

Behavioral Game Theory, Camerer, Russel Sage Foundation, 2003

Personnel Economics in Practice, Lazear / Gibbs, Wiley, 2014

Introduction to Econometrics, Wooldridge, Andover, 2014

Econometric Analysis of Cross Section and Panel Data, Wooldridge, MIT Press, 2010

T 6.321 Module component: Industrial Mobile Robotics Lab [T-MACH-113701]

Coordinators: Prof. Dr.-Ing. Kai Furmans
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-104888 - Advanced Module Logistics
M-WIWI-107718 - Mechanical Engineering I
M-WIWI-107730 - Mechanical Engineering II

Type	Credits	Grading	Term offered	Version
Coursework	4,5 CP	pass/fail	Each term	3

Courses					
WT 25/26	2117073	Industrial Mobile Robotics Lab	2 SWS	Practical course / 🗣	Furmans
ST 2026	2117073	Industrial Mobile Robotics Lab	3 SWS	Practical course / 🗣	Furmans
WT 26/27	2117073	Industrial Mobile Robotics Lab	3 SWS	Practical course / 🗣	Furmans

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Certificate through colloquium with presentation, documentation of the work results and fulfilment of the attendance requirement.

Prerequisites
T-MACH-105230 must not be started.

Recommendations
Relevant prior experience in robotics, sensor and image processing, Python programming, and technical logistics

Additional Information
The number of participants is limited to 15 students.

- - Students in the Mechanical Engineering, Mechatronics, and Industrial Engineering and Management degree programs have priority over students in other degree programs
  - If prior knowledge is equal, priority is given based on academic progress
  - If academic progress is equal, based on waiting time
  - If waiting times are equal, selection is by lottery

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Industrial Mobile Robotics Lab

2117073, WS 25/26, 2 SWS, Language: English, Open in study portal

Practical course (P)
On-Site

Content
Mobile robots have become a standard in the industry, especially in the field of intralogistics. This course is designed to teach students how to operate and control mobile robotic systems. This course will give students a valuable practical experience in this area.

For self-study, videos on the various relevant topics will be made available before the start of the course. After, the students will work on implementing a robot control and fleet management system. A simulation environment will also be provided for the development process, allowing testing without hardware in the early phases of the project. The implementation will be based on a standardized communication interface - the VDA 5050 - which enables a uniform data exchange between the system participants. A Kickoff event, several Milestones and a final live demonstration of the impleemeted solution will give the students the needed input and feedback as well a a chance to present their devloped concepts.

Organizational issues
During the course, students will initially work independently on the assignment and then in teams. Regular consultation hours and additional input sessions will be offered. Progress will be presented in two interim milestones.

The number of participants is limited. Selection will be based on an application process. Applications can be submitted via the following link: https://plus.campus.kit.edu/signmeup/procedures/4859. Applications should include previous experience and motivation for the internship.

Basic programming skills in Python are required.

The course language is English.

Planned time frame for the course:
Start: November 5, 2025 - End: December 17, 2025

For questions please refer to pietro.schumacher@kit.edu.

Literature
VDA 5050: https://www.vda.de/en/topics/automotive-industry/vda-5050

V	**Industrial Mobile Robotics Lab** 2117073, SS 2026, 3 SWS, Language: English, Open in study portal	**Practical course (P)** **On-Site**

Content
Mobile robots have become a standard in the industry, especially in the field of intralogistics. This course is designed to teach students how to operate and control mobile robotic systems. This course will give students a valuable practical experience in this area. For self-study, videos on the various relevant topics will be made available before the start of the course. After, the students will work on implementing a robot control and fleet management system. A simulation environment will be provided for the development process, allowing testing without hardware in the early phases of the project. The implementation will be based on a standardized communication interface - the VDA 5050 - which enables a uniform data exchange between the system participants. An online and offline kickoff event, two milestones, consultation hours and a final live demonstration of the implemeted solution will give the students the needed input and feedback as well the chance to present their devloped concepts.

Organizational issues
During the course, students will initially work independently on the assignment and then in teams. Regular consultation hours and additional input sessions will be offered. Progress will be presented in two interim milestones.

Important information about the course:

- The number of participants is limited. Selection will be based on an application process. Applications can be submitted via the following link: https://plus.campus.kit.edu/signmeup/procedures/5878. Applications should include previous experience and motivation for the internship.
- In addition, make sure you can attend all the following **mandatory dates** as **attendance is required for participation in this course**. **Non-attendence will result in de-registration.** The dates are as follows:

  Kickoff: 13.05.26, 14:00 - 17:15
  Milestone 1: 20 min timeslot on the 03.06.26, 14:00 - 17:15
  Milestone 2: 10.06.26, 14:00 - 17:15
  Milestone 3: 24.06.26, 14:00 - 17:15

- Applicants will be inivited to an online info event where the course will be presented:
  Online info: 06.05.26, 14:00 - 15:30 (Teams-link will be sent to you before the event)

- Basic programming **skills in Python are required**.
- The course language is **English**.
- Planned time frame for the course:
  Start: May 13, 2026 - End: June 24, 2026

For questions please refer to pietro.schumacher@kit.edu.

Literature
VDA 5050: https://www.vda.de/en/topics/automotive-industry/vda-5050

V	**Industrial Mobile Robotics Lab** 2117073, WS 26/27, 3 SWS, Language: English, Open in study portal	**Practical course (P)** **On-Site**

Content
Mobile robots have become a standard in the industry, especially in the field of intralogistics. This course is designed to teach students how to operate and control mobile robotic systems. This course will give students a valuable practical experience in this area.

For self-study, videos on the various relevant topics will be made available before the start of the course. After, the students will work on implementing a robot control and fleet management system. A simulation environment will also be provided for the development process, allowing testing without hardware in the early phases of the project. The implementation will be based on a standardized communication interface - the VDA 5050 - which enables a uniform data exchange between the system participants. A Kickoff event, several Milestones and a final live demonstration of the impleemeted solution will give the students the needed input and feedback as well a a chance to present their devloped concepts.

Organizational issues
During the course, students will initially work independently on the assignment and then in teams. Regular consultation hours and additional input sessions will be offered. Progress will be presented in two interim milestones.

The number of participants is limited. Selection will be based on an application process. Applications can be submitted via the following link: (Link will be posted soon). Applications should include previous experience and motivation for the course.

Programming skills in Python beyond basics are required.

The course language is English.

There will be several appointments with mandatory attendence. The dates can be seen by clicking the link to the plus-campus portal.

Planned time frame for the course:
Start: *Coming soon* - End: *Coming Soon*

For questions please refer to pietro.schumacher@kit.edu.

Literature
VDA 5050: https://www.vda.de/en/topics/automotive-industry/vda-5050

T 6.322 Module component: Industrial Organization [T-WIWI-102844]

Coordinators: Prof. Dr. Johannes Philipp Reiß
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104908 - Economics

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Irregular	1

Courses					
ST 2026	2560238	Industrial Organization	2 SWS	Lecture / 🗣	Reiß
ST 2026	2560239	Übung zu Industrieökonomie	1 SWS	Practice / 🗣	Reiß, Potarca

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment consists of a written exam (60 minutes) (following §4(2), 1 of the examination regulation). The exam takes place in every semester. Re-examinations are offered at every ordinary examination date.

Prerequisites
None

Recommendations
Completion of the module Economics [WW1VWL] is assumed.

Additional Information
This course is not given in summer 2017.

Below you will find excerpts from courses related to this module component:

V Industrial Organization
2560238, SS 2026, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Literature
Verpflichtende Literatur:

H. Bester (2012): Theorie der Industrieökonomik, Springer-Verlag.

Ergänzende Literatur:

J. Tirole (1988): Theory of Industrial Organization, MIT Press.

D. Carlton / J. Perloff (2005): Modern Industrial Organization, Pearson.

P. Belleflamme / M. Peitz (2010): Industrial Organization

T 6.323 Module component: Information Engineering [T-MACH-102209]

Coordinators: Prof. Dr.-Ing. Anne Meyer
Prof. Dr.-Ing. Jivka Ovtcharova

Organisation: KIT Department of Mechanical Engineering

Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Each term	2

Courses					
WT 25/26	2121355	Information Engineering	2 SWS	Seminar / Blended	Meyer
ST 2026	2122014	Information Engineering	2 SWS	Seminar / Blended	Meyer, Rönnau

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
Alternative exam assessment (written composition and speech)

Prerequisites
None

Workload
90 hours

Below you will find excerpts from courses related to this module component:

V Information Engineering
2121355, WS 25/26, 2 SWS, Language: German/English, Open in study portal

Seminar (S)
Blended (On-Site/Online)

Content
Seminar papers on current research topics of the Institute for Information Management in Engineering. The respective topics are presented at the beginning of each semester.

Organizational issues
Ort und Zeit siehe ILIAS

Literature
Themenspezifische Literatur

V Information Engineering
2122014, SS 2026, 2 SWS, Language: German/English, Open in study portal

Seminar (S)
Blended (On-Site/Online)

Content
Seminar papers on current research topics of the Institute for Information Management in Engineering. The respective topics are presented at the beginning of each semester.

Organizational issues
Zeit, Ort und weitere Informationen siehe ILIAS / Time, place and further information see ILIAS

Literature
Themenspezifische Literatur

T 6.324 Module component: Information Management for Public Mobility Services [T-BGU-106608]

Coordinators: Prof. Dr.-Ing. Peter Vortisch
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: M-BGU-101064 - Fundamentals of Transportation
M-BGU-101065 - Transportation Modelling and Traffic Management
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Expansion	Version
Examination of another type	3 CP	graded	Each winter term	1 semesters	1

Courses					
WT 25/26	6232905	Information Management for Public Mobility Services	2 SWS	Block / 🗣	Vortisch

Legend: 🖥 Online, ✲ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
lecture accompanying exercises, appr. 5 pieces

Prerequisites
none

Recommendations
none

Additional Information
none

Workload
90 hours

6.325 Module component: Information Systems and Supply Chain Management [T-MACH-102128]

Coordinators: Dr.-Ing. Christoph Kilger
Organisation: KIT Department of Mechanical Engineering

Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Written examination	3 CP	graded	Each summer term	3

Assessment
The success control takes place in form of a written examination (60 min) during the semester break (according to §4(2), 1 SPO). If the number of participants is low, an oral examination (according to §4 (2), 2 SPO) may also be offered.

Prerequisites
none

Additional Information
The course is offered in German.

Workload
90 hours

T 6.326 Module component: Infrastructure Management [T-BGU-106300]

Coordinators: Dr.-Ing. Matthias Zimmermann
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: M-BGU-100998 - Design, Construction, Operation and Maintenance of Highways
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Expansion	Version
Written examination	6 CP	graded	Each term	1 semesters	1

Courses					
ST 2026	6233801	Design and Construction of Roads	2 SWS	Lecture / 🗣	Zimmermann, Stelzenmüller
ST 2026	6233802	Operation and Maintenance of Roads	2 SWS	Lecture / 🗣	Zimmermann, Hess, Stelzenmüller

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
written exam, 120 min.

Prerequisites
none

Recommendations
none

Additional Information
none

Workload
180 hours

T 6.327 Module component: Innovation and Project Management in Rail Vehicle Engineering [T-MACH-113068]

Coordinators: Prof. Dr.-Ing. Martin Cichon
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-106496 - Modern Mobility on Rails and Roads

Type	Credits	Grading	Term offered	Version
Examination of another type	4 CP	graded	Each term	5

Courses					
WT 25/26	2115921	Innovation and Project Management in Rail Vehicle Engineering	2 SWS	Lecture / 🗣	Lang, Cichon
ST 2026	2115921	Innovation and Project Management in Rail Vehicle Engineering	2 SWS	Lecture / 🗣	Lang, Cichon
WT 26/27	2115921	Innovation and Project Management in Rail Vehicle Engineering	2 SWS	Lecture / 🗣	Martinez Pina, Lang

Legend: 🖥 Online, ✲ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Graded examination:
2/3 of the examination: 20-minute oral examination on the content of the lecture
1/3 of the examination performance of another type: unit accompanying the lecture as part of a 10-minute presentation and a practical application from innovation and project management

Additional Information
The course is offered in German.

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Innovation and Project Management in Rail Vehicle Engineering
2115921, WS 25/26, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content
Learning objectives

The aim of this course is to familiarise students with the fundamentals of innovation and project management in the context of rail vehicle development. Creativity techniques are applied in a practical manner to the challenges faced by the rail system, such as aspects of sustainability. Students also learn about the various organisational, systemic, economic and technological challenges involved in a project and its management.

Course content

- Fundamentals of innovation management
- Challenges and aspects of sustainability in the railway system
- Autonomous testing of various creativity techniques
- Moderation of creativity workshops
- Techniques for idea generation and idea evaluation
- Fundamentals and methods of project management
- Practical challenges in project management
- Tools in project management
- Project team organisation and role distribution

Learning goals

Students will learn basic methods for identifying problem areas, developing new and creative solutions, and evaluating them.
They will be able to initiate an innovation workshop and moderate, conduct, document, and reflect on it in a targeted manner using creativity methods they have applied autonomously.
Students will be able to apply methods for structural planning, process planning, risk, cost, and quality management using examples.

Organizational issues

Benotete Prüfungsleistung:

2/3 der Prüfungsleistung: 20-minütige mündliche Prüfung über die Lehrinhalte der Vorlesung
1/3 der Prüfungsleistung anderer Art: vorlesungsbegleitende Einheit im Rahmen einer 10-minütigen Präsentation und einer praktischen Anwendung aus dem Innovations- und Projektmanagement

Literature

Eine Literaturliste steht den Studierenden auf der Ilias-Plattform zum Download zur Verfügung.

A bibliography is available for download (Ilias-platform).

V	**Innovation and Project Management in Rail Vehicle Engineering** 2115921, SS 2026, 2 SWS, Language: German, Open in study portal	**Lecture (V)** **On-Site**

Content
Learning objectives

The aim of this course is to familiarise students with the fundamentals of innovation and project management in the context of rail vehicle development. Creativity techniques are applied in a practical manner to the challenges faced by the rail system, such as aspects of sustainability. Students also learn about the various organisational, systemic, economic and technological challenges involved in a project and its management.

Course content

- Fundamentals of innovation management
- Challenges and aspects of sustainability in the railway system
- Autonomous testing of various creativity techniques
- Moderation of creativity workshops
- Techniques for idea generation and idea evaluation
- Fundamentals and methods of project management
- Practical challenges in project management
- Tools in project management
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Learning goals

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They will be able to initiate an innovation workshop and moderate, conduct, document, and reflect on it in a targeted manner using creativity methods they have applied autonomously.
Students will be able to apply methods for structural planning, process planning, risk, cost, and quality management using examples.

Organizational issues
Benotete Prüfungsleistung:

2/3 der Prüfungsleistung: 20-minütige mündliche Prüfung über die Lehrinhalte der Vorlesung
1/3 der Prüfungsleistung anderer Art: vorlesungsbegleitende Einheit im Rahmen einer 10-minütigen Präsentation und einer praktischen Anwendung aus dem Innovations- und Projektmanagement

Max. 12 Teilnehmer

Literature
Eine Literaturliste steht den Studierenden auf der Ilias-Plattform zum Download zur Verfügung.

A bibliography is available for download (Ilias-platform).

V **Innovation and Project Management in Rail Vehicle Engineering** — **Lecture (V) On-Site**
2115921, WS 26/27, 2 SWS, Language: English, Open in study portal

Content
Learning objectives

The aim of this course is to familiarise students with the fundamentals of innovation and project management in the context of rail vehicle development. Creativity techniques are applied in a practical manner to the challenges faced by the rail system, such as aspects of sustainability. Students also learn about the various organisational, systemic, economic and technological challenges involved in a project and its management.

Course content

- Fundamentals of innovation management
- Challenges and aspects of sustainability in the railway system
- Autonomous testing of various creativity techniques
- Moderation of creativity workshops
- Techniques for idea generation and idea evaluation
- Fundamentals and methods of project management
- Practical challenges in project management
- Tools in project management
- Project team organisation and role distribution

Learning goals

Students will learn basic methods for identifying problem areas, developing new and creative solutions, and evaluating them.
They will be able to initiate an innovation workshop and moderate, conduct, document, and reflect on it in a targeted manner using creativity methods they have applied autonomously.
Students will be able to apply methods for structural planning, process planning, risk, cost, and quality management using examples.

Organizational issues
For further information https://www.fast.kit.edu/bst/929_16399.php

Literature
Eine Literaturliste steht den Studierenden auf der Ilias-Plattform zum Download zur Verfügung.

A bibliography is available for download (Ilias-platform).

T 6.328 Module component: Innovation Management: Concepts, Strategies and Methods [T-WIWI-102893]

Coordinators: Prof. Dr. Marion Weissenberger-Eibl
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101488 - Entrepreneurship (EnTechnon)
M-WIWI-101507 - Innovation Management
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	3 CP	graded	Each summer term	1

Courses					
ST 2026	2545100	Innovation Management: Concepts, Strategies and Methods	2 SWS	Lecture / Blended	Weissenberger-Eibl

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
The assessment consists of a written exam (60 minutes). The exam takes place in every summer semester. Re-examinations are offered at every ordinary examination date.

Prerequisites
None

Recommendations
None

Below you will find excerpts from courses related to this module component:

V Innovation Management: Concepts, Strategies and Methods

2545100, SS 2026, 2 SWS, Language: German, Open in study portal

Lecture (V)
Blended (On-Site/Online)

Content
The course ‘Innovation Management: Concepts, Strategies and Methods’ offers scientific concepts which facilitate the understanding of the different phases of the innovation process and resulting strategies and appropriate methodologies suitable for application. The concepts refer to the entire innovation process so that an integrated perspective is made possible. This is the basis for the teaching of strategies and methods which fulfil the diverse demands of the complex innovation process. The course focuses particularly on the creation of interfaces between departments and between various actors in a company’s environment and the organisation of a company’s internal procedures. In this context a basic understanding of knowledge and communication is taught in addition to the specific characteristics of the respective actors. Subsequently methods are shown which are suitable for the profitable and innovation-led implementation of integrated knowledge.

Aim: Students develop a differentiated understanding of the different phases and concepts of the innovation process, different strategies and methods in innovation management.

Organizational issues
Wichtig! Bitte treten Sie dem **ILIAS-Kurs zur Vorlesung** bei, damit wir Ihnen weitere Informationen mitteilen können.

Literature
Eine ausführliche Literaturliste wird mit den Vorlesungsunterlagen zur Verfügung gestellt.

Eine Einführung bei: Vahs,D./Brem,A. (2013): Innovationsmanagement. Von der Idee zur erfolgreichen Vermarktung, 4. Auflage, Stuttgart 2013.

T 6.329 Module component: Innovation2Business – Innovation Strategy in the Industrial Corporate Practice [T-MACH-112882]

Coordinators: Prof. Dr.-Ing. Albert Albers
Organisation: KIT Department of Mechanical Engineering

Part of: M-WIWI-101488 - Entrepreneurship (EnTechnon)
M-WIWI-104907 - Engineering Sciences
M-WIWI-107718 - Mechanical Engineering I
M-WIWI-107730 - Mechanical Engineering II

Type	Credits	Grading	Term offered	Expansion	Version
Written examination	4,5 CP	graded	Each winter term	1 semesters	2

Courses					
WT 25/26	2145182	Innovation2Business – Innovation Strategy in the Industrial Corporate Practice	2 SWS	Lecture / 🗣	Albers
WT 26/27	2145182	Innovation2Business – Innovation Strategy in the Industrial Corporate Practice	2 SWS	Lecture / 🗣	Albers

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Written exam based on the lecture handout and materials, duration 90 minutes

Prerequisites
none

Recommendations
None

Additional Information
The course is offered in German.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Innovation2Business – Innovation Strategy in the Industrial Corporate Practice

Lecture (V)
On-Site

2145182, WS 25/26, 2 SWS, Language: English, Open in study portal

Content
lecture block at the Bühl & Herzogenaurach locations with plant tours & fireside evenings + exam-preparatory Q&A.
Exam: written, limited to 30 seats (recommended for: Master's degree; mechanical engineering, industrial engineering, electrical engineering, computer science) → see module manual for details.
In this lecture series, use Schaeffler as an example to learn how global companies continuously transform themselves to grow sustainably and become
maintain a leading position in the global market in the long term through business-oriented innovation.
Together we will go through the most important elements of the innovation and development process and learn about the successes and learnings based on
vivid examples from practice.
Join the fireside evenings with the speakers to discuss the lecture content and beyond in a relaxed atmosphere.
The event is limited to 30 students and is free for you (meals, bus transfers & accommodations).

Organizational issues
Vorlesung findet an Schaeffler-Standorten (Herzogenaurach und Bühl) statt.

Sprache: Unterlagen Englisch, Vortragssprache Englisch

V

Innovation2Business – Innovation Strategy in the Industrial Corporate Practice

2145182, WS 26/27, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content

lecture block at the Bühl & Herzogenaurach locations with plant tours & fireside evenings + exam-preparatory Q&A.
Exam: written, limited to 30 seats (recommended for: Master's degree; mechanical engineering, industrial engineering, electrical engineering, computer science) → see module manual for details.
In this lecture series, use Schaeffler as an example to learn how global companies continuously transform themselves to grow sustainably and become
maintain a leading position in the global market in the long term through business-oriented innovation.
Together we will go through the most important elements of the innovation and development process and learn about the successes and learnings based on
vivid examples from practice.
Join the fireside evenings with the speakers to discuss the lecture content and beyond in a relaxed atmosphere.
The event is limited to 30 students and is free for you (meals, bus transfers & accommodations).

Organizational issues

Vorlesung findet an Schaeffler-Standorten (Herzogenaurach und Bühl) statt.

Sprache: Unterlagen Englisch, Vortragssprache Englisch

T 6.330 Module component: Insurance Risk Management [T-WIWI-102636]

Coordinators: Harald Maser
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	2,5 CP	graded	Each summer term	1

Assessment
The assessment consists of a written or an oral exam (according to Section 4 (2), 1 or 2 of the examination regulation).

T-WIWI-102636 Insurance Risk Management will be offered as a seminar starting summer term 2017. The examination will be offered latest until summer term 2017 (beginners only).

Prerequisites
None

Recommendations
None

Additional Information
Block course. For organizational reasons, please register with the secretary of the chair: thomas.mueller3@kit.edu.

T 6.331 Module component: Integrated Photonics [T-ETIT-114418]

Coordinators: Prof. Dr.-Ing. Christian Koos
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-107344 - Integrated Photonics
M-MACH-101292 - Microoptics
M-MACH-101295 - Optoelectronics and Optical Communication
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Version
Oral examination	6 CP	graded	1

Courses					
WT 25/26	2309440	Integrated Photonics	2 SWS	Lecture / 🗣	Koos, N.N.
WT 25/26	2309441	Tutorial to Integrated Photonics	2 SWS	Practice / 🗣	Koos, N.N., Khairy

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment takes place in the form of an oral examination (approx. 25 minutes); appointments individually on demand.

Prerequisites
none

T 6.332 Module component: Integrated Product Development [T-MACH-105401]

Coordinators: Prof. Dr.-Ing. Tobias Düser
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-102626 - Major Field: Integrated Product Development
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Oral examination	16 CP	graded	Each winter term	4

Courses					
WT 25/26	2145156	Lecture: IP – Integrated Product Development	4 SWS	Lecture / 🗣	Albers
WT 25/26	2145157	Workshop: IP – Integrated Product Development	4 SWS	Practice / 🗣	Albers
WT 25/26	2145300	Project Work: IP - Integrated Product Development	2 SWS	Others / 🗣	Albers
WT 26/27	2145156	Lecture: IP – Integrated Product Development	2 SWS	Lecture / 🗣	Düser
WT 26/27	2145157	Workshop: IP – Integrated Product Development	4 SWS	Practice / 🗣	Düser
WT 26/27	2145300	Project Work: IP - Integrated Product Development	4 SWS	Others / 🗣	Düser

Legend: 📱 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
oral examination (approx. 60 minutes)

Prerequisites
none

Recommendations
Attendance of the lecture "Advanced Systems Engineering"

Additional Information
Due to organizational reasons, the number of participants is limited. Thus a selection has to be made. For registration to the selection process a standard form has to be used, that can be downloaded from IPEK hompage from April to July. The selection itself is made by the course's responsible in personal interviews. The criterion for selection is the progress of studies. In the event of equal progress, the decision is made by lot.

The course is offered in German.

Workload
480 hours

Below you will find excerpts from courses related to this module component:

V Lecture: IP – Integrated Product Development

2145156, WS 25/26, 4 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content
Registration required in the previous summer semester. The lecture starts in first week of October.

Prerequisites:

The participation in "Integrated Product Development" requires the concurrent participation in lectures (2145156), tutorials (2145157) and project work (2145300).

Due to organizational reasons, the number of participants is limited. Thus a selection has to be made. For registration to the selection process a standard form has to be used, that can be downloaded from IPEK hompage from april to july. The selection itself is made by Prof. Albers in personal interviews.

Recommendations:

none

Workload:

regular attendance: 84 h

self-study: 288 h

Examination:

oral examination (60 minutes)

combined examination of lectures, tutorials and project work

Course content:

organizational integration: integrated product engineering model, core team management and simultaneous engineering

informational integration: innovation management, cost management, quality management and knowledge management

personal integration: team coaching and leadership management

invited lectures

Learning objectives:

The Students are able to ...

- analyze and evaluate product development processes based on examples and their own experiences.
- plan, control and evaluate the working process systematically.
- choose and use suitable methods of product development, system analysis and innovation management under consideration of the particular situation.
- prove their results.
- develop complex technical solutions in a team and to present them to qualified persons as well as non-qualified persons
- to design overall product development processes under consideration of market-, customer- and company- aspects

Literature
Klaus Ehrlenspiel - Integrierte Produktentwicklung. Denkabläufe, Methodeneinsatz, Zusammenarbeit, Hanser Verlag, 2009

V **Workshop: IP – Integrated Product Development** — **Practice (Ü)** **On-Site**
2145157, WS 25/26, 4 SWS, Language: German, Open in study portal

Content
Prerequisites:

The participation in "Integrated Product Development" requires the concurrent participation in lectures (2145156), tutorials (2145157) and project work (2145300).

Due to organizational reasons, the number of participants is limited to 42 persons. Thus a selection has to be made. For registration to the selection process a standard form has to be used, that can be downloaded from IPEK hompage from april to july. The selection itself is made by Prof. Albers in personal interviews.

Recommendations:

none

Workload:

regular attendance: 84 h

self-study: 288 h

Examination:

lectures: 21 h

preparation to exam: 99 h

Course content:

problem solving: analysis techniques, creativity techniques and evaluation methods

professional skills: presentation techniques, moderation and teamcoaching

development tools: MS Project, Szenario-Manager & Pro/Engineer Wildfire

Learning objectives:

The theoretical background taught in the lecture, is deepened through methodworkshops, business games and case studies. The reflexion of the onself precedure allows for an applicability and practicability of the contents in the accompnying development project as well as for the career entry.

Literature
Klaus Ehrlenspiel - Integrierte Produktentwicklung. Denkabläufe, Methodeneinsatz, Zusammenarbeit, Hanser Verlag, 2009

V	**Project Work: IP - Integrated Product Development** 2145300, WS 25/26, 2 SWS, Language: German, Open in study portal	**Others (sonst.)** **On-Site**

Content
Participation only possible in combination with the lecture 2145156 'Integrated Product Development'.

Prerequisites:

The participation in "Integrated Product Development" requires the concurrent participation in lectures (2145156), tutorials (2145157) and project work (2145300).

Due to organizational reasons, the number of participants is limited to 42 persons. Thus a selection has to be made. For registration to the selection process a standard form has to be used, that can be downloaded from IPEK hompage from april to july. The selection itself is made by Prof. Albers in personal interviews.

Recommendations:

none

Workload:

regular attendance: 21 h

self-study: 99 h

Examination:

oral examination (60 minutes)

combined examination of lectures, tutorials and project work

Course content:

The project work begins with the early stages of product development, i.e. the identification of market trends and needs. Based on this information the students develop scenarios for future markets and create product profiles, which describe the customers and their demands without anticipating possible product solutions. After having passed several following milestones for ideas, concepts and designs, virtual prototypes and function prototypes are presented to an audience.
The project work is supported by coaching through skilled faculty staff. Additionally weekly tutorials, respectively workshops are given. For doing the project the teams gain access to team workspaces featuring IT-infrastructure and relevant software, such as office, CAD or FEA. Further on the teams learn how team cooperation and knowledge management can be supported in design project by using a wiki system.s

Learning objectives:

The center of "Integrated Product Development" constitutes itself in the development of a technical product within independent working student teams on the basis of the market situation up to virtual and real prototypes. Thereby the integrate treatment of the product development process is of importance. The project teams hereby represent development departments of medium sized companies, in which the presented methods and tools are field - experienced applied and ideas are transformed into concrete product models.
For the preparation of this development project the basics of 3D-CAD-modelling (Pro/ENGINEER) as well as different tools and methods of creative designing, of sketching and solution finding are mediated in workshops. Special events impart an insight of presentation techniques and the meaning of technical design.

V	**Lecture: IP – Integrated Product Development** 2145156, WS 26/27, 2 SWS, Language: German, Open in study portal	**Lecture (V)** **On-Site**

Content
Registration required in the previous summer semester. The course starts in first week of October.

Prerequisites:

The participation in "Integrated Product Development" requires the concurrent participation in lectures (2145156), tutorials (2145157) and product development project (2145300).

For organisational reasons, admission to the product development project is limited. Therefore, a selection process will take place. Registration for the selection process is done via a registration form, which is available annually from April to July on the IPEK homepage. The selection itself will then be made in personal interviews with Prof. Düser in accordance with §5 (4) SPO 2025 Master of Mechanical Engineering.

The course is offered in German.

Recommendations:

Optional: Participation in lecture *Advanced Systems Engineering*.

Workload:

regular attendance: 24 h

self-study: 72 h

Examination:

oral examination (60 minutes)

combined examination of lectures, tutorials and project work

Course content:

Product development, innovation management, competitiveness

Product management: fundamentals, strategic project management, operational project management, lifecycle management

Process models and project management

Cross-generational system development

Problem solving

Leadership and organization

Development management: operational development management, methodological development management, digitization in development management; idea generation; evaluation methods, reaching decisions; conceptualization / modeling principle and form; validation and verification, prototypes

Cost, quality and risk management, target costing

Implementation of corporate strategies

Portfolio management

Learning objectives:

The Students are able to ...

- analyze and evaluate product development processes based on examples and their own experiences.
- plan, control and evaluate the working process systematically.
- select and apply methods of product development, innovation management, development management and product management as appropriate to the situation and review the results of their work.
- prove their results.
- develop complex technical solutions in a team and to present them to qualified persons as well as non-qualified persons
- to design overall product development processes under consideration of market-, customer- and company- aspects

Literature
Klaus Ehrlenspiel - Integrierte Produktentwicklung. Denkabläufe, Methodeneinsatz, Zusammenarbeit, Hanser Verlag, 2009

V **Workshop: IP – Integrated Product Development**
2145157, WS 26/27, 4 SWS, Language: German, Open in study portal

Practice (Ü)
On-Site

Content
Registration required in the previous summer semester. The course starts in first week of October.

Prerequisites:

The participation in "Integrated Product Development" requires the concurrent participation in lectures (2145156), tutorials (2145157) and project development project (2145300).

For organisational reasons, admission to the product development project is limited. Therefore, a selection process will take place. Registration for the selection process is done via a registration form, which is available annually from April to July on the IPEK homepage. The selection itself will then be made in personal interviews with Prof. Düser in accordance with §5 (4) SPO 2025 Master of Mechanical Engineering.

The course is offered in German.

Workload:

workshops: 52 h

self-study: 52 h

Examination:

oral examination (60 minutes)

combined examination of lectures, tutorials and product development project

Course content:

problem solving: analysis techniques, creativity techniques and evaluation methods

professional skills: presentation techniques, patent research, MBSE – Model Based Systems Engineering, rapid prototyping and agile project management

Learning objectives:

The theoretical background taught in the lecture, is deepened through method workshops, business games and case studies. The reflexion of the onself procedure allows for applicability and practicability of the contents in the accompanying development project as well as for the career entry.

Literature
Klaus Ehrlenspiel - Integrierte Produktentwicklung. Denkabläufe, Methodeneinsatz, Zusammenarbeit, Hanser Verlag, 2009

V	**Project Work: IP - Integrated Product Development** 2145300, WS 26/27, 4 SWS, Language: German, Open in study portal	**Others (sonst.)** **On-Site**

Content
Registration required in the previous summer semester. The course starts in first week of October.

Participation only possible in combination with lecture 2145156 'Integrated Product Development'.

Prerequisites:

The participation in "Integrated Product Development" requires the concurrent participation in lecture (2145156), tutorial (2145157) and project work (2145300).

For organisational reasons, admission to the product development project is limited. Therefore, a selection process will take place. Registration for the selection process is done via a registration form, which is available annually from April to July on the IPEK homepage. The selection itself will then be made in personal interviews with Prof. Düser in accordance with §5 (4) SPO 2025 Master of Mechanical Engineering.

The course is offered in German.

Workload:

regular attendance: 52 h

self-study: 228 h

Examination:

oral examination (60 minutes)

combined examination of lectures, tutorials and project work

Course content:

Independent planning and execution of a real-world development task from industry

Practical implementation of the content taught in lectures and workshops

Identification of product profiles and product ideas based on customer needs and market potential

Modelling of principles and design using product development methods and tools

Presentation and defence of own solutions to industry partners

Construction and validation of virtual and real prototypes

Learning objectives:

The focus of the ‘Integrated Product Development’ course is the development of a technical product in independent student project teams, starting with the market situation and progressing to virtual and real prototypes. Particular emphasis is placed on a holistic view of the product development process. The project teams represent the development departments of medium-sized companies, in which the methods and tools presented are applied in a practical manner and ideas are translated into concrete product models and prototypes.

Special events provide insight into moderation techniques and the importance of technical design.

T 6.333 Module component: Integrated Production Planning in the Age of Industry 4.0 [T-MACH-109054]

Coordinators: Prof. Dr.-Ing. Gisela Lanza
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101272 - Integrated Production Planning
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Written examination	9 CP	graded	Each summer term	1

Courses					
ST 2026	2150660	Integrated Production Planning in the Age of Industry 4.0	6 SWS	Lecture / Practice / 🗣	Lanza

Legend: 🖥 Online, Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Written Exam (120 min)

Prerequisites
"T-MACH-108849 - Integrierte Produktionsplanung im Zeitalter von Industrie 4.0" as well as "T-MACH-102106 Integrierte Produktionsplanung" must not be commenced.

Additional Information
The course is offered in German.

Workload
270 hours

Below you will find excerpts from courses related to this module component:

V Integrated Production Planning in the Age of Industry 4.0

2150660, SS 2026, 6 SWS, Language: German, Open in study portal

Lecture / Practice (VÜ)
On-Site

Content

Integrated Production Planning in the age of Industry 4.0 will be taught in the context of this engineering science lecture. In addition to a comprehensive introduction to Industry 4.0, the following topics will be addressed at the beginning of the lecture:

- Basics, history and temporal development of production
- Integrated production planning and integrated digital engineering
- Principles of integrated production systems and further development with Industry 4.0

Building on this, the phases of integrated production planning are taught in accordance with VDI Guideline 5200, whereby special features of parts production and assembly are dealt with in the context of case studies:

- Factory planning system
- Definition of objectives
- Data collection and analysis
- Concept planning (structural development, structural dimensioning and rough layout)
- Detailed planning (PPS, process simulation as a validation tool, planning of conveyor technology and storage systems for linking production and IT systems in the I4.0 factory)
- Preparation and monitoring of implementation
- Start-up and series support

The lecture contents are complemented by numerous current practical examples with a strong Industry 4.0 reference. Aspects of sustainability are anchored in all units and thus basic knowledge of sustainable production planning is taught. Within the exercises the lecture contents are deepened and applied to specific problems and tasks.

Learning Outcomes:

The students …

- can discuss basic questions of production technology.
- are able to apply the methods of integrated production planning to new problems.
- are able to analyze and evaluate the suitability of the methods, procedures and techniques for a specific problem.
- can apply the learned methods of integrated production planning to new problems.
- can use their knowledge targeted for efficient production technology.
- know the basic features of sustainable production planning and can apply underlying knowledge.

Workload:

MACH:
regular attendance: 63 hours
self-study: 177 hours
WING:
regular attendance: 63 hours
self-study: 207 hours

Organizational issues

Vorlesungstermine dienstags 14.00 Uhr und donnerstags 14.00 Uhr, Übungstermine donnerstags 15.45 Uhr. Bekanntgabe der konkreten Übungstermine erfolgt in der ersten Vorlesung

Literature
Medien:
Skript zur Veranstaltung wird über (https://ilias.studium.kit.edu/) bereitgestellt.

Media:
Lecture notes will be provided in Ilias (https://ilias.studium.kit.edu/).

T 6.334 Module component: Integrative Strategies in Production and Development of High Performance Cars [T-MACH-105188]

Coordinators: Karl-Hubert Schlichtenmayer
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-105455 - Strategic Design of Modern Production Systems
M-MACH-106590 - Production Engineering
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Written examination	4 CP	graded	Each summer term	1

Courses					
ST 2026	2150601	Integrative Strategies in Production and Development of High Performance Cars	2 SWS	Lecture / 🗣	Schlichtenmayer

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Written Exam (60 min)

Prerequisites
none

Additional Information
The course is offered in German.

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Integrative Strategies in Production and Development of High Performance Cars

Lecture (V)
On-Site

2150601, SS 2026, 2 SWS, Language: German, Open in study portal

Content
The lecture deals with the technical and organizational aspects of integrated development and production of sports cars on the example of Porsche AG. The lecture begins with an introduction and discussion of social trends. The deepening of standardized development processes in the automotive practice and current development strategies follow. The management of complex development projects is a first focus of the lecture. The complex interlinkage between development, production and purchasing are a second focus. Methods of analysis of technological core competencies complement the lecture. The course is strongly oriented towards the practice and is provided with many current examples.

The main topics are:

- Introduction to social trends towards high performance cars
- Automotive Production Processes
- Integrative R&D strategies and holistic capacity management
- Management of complex projects
- Interlinkage between R&D, production and purchasing
- The modern role of manufacturing from a R&D perspective
- Global R&D and production
- Methods to identify core competencies

Learning Outcomes:

The students …

- are capable to specify the current technological and social challenges in automotive industry.
- are qualified to identify interlinkages between development processes and production systems.
- are able to explain challenges and solutions of global markets and global production of premium products.
- are able to explain modern methods to identify key competences of producing companies.

Workload:

regular attendance: 21 hours
self-study: 99 hours

Literature
Medien:
Skript zur Veranstaltung wird über (https://ilias.studium.kit.edu/) bereitgestellt.

Media:
Lecture notes will be provided in Ilias (https://ilias.studium.kit.edu/).

T 6.335 Module component: Intellectual Property Rights and Strategies in Industrial Companies [T-MACH-105442]

Coordinators: Dipl.-Ing. Frank Zacharias
Organisation: KIT Department of Mechanical Engineering

Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each term	1

Courses					
WT 25/26	2147161	Intellectual Property Rights and Strategies in Industrial Companies	2 SWS	Block / 🗣	Zacharias
ST 2026	2147160	Patents and Patentstrategies in innovative companies	2 SWS	/ 🗣	Zacharias

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
oral exam (ca. 20 min)

Prerequisites
none

Recommendations
None

Additional Information
The course is offered in German.

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Intellectual Property Rights and Strategies in Industrial Companies

2147161, WS 25/26, 2 SWS, Language: German, Open in study portal

Block (B)
On-Site

Content

Attendance at lectures (5 L): 24h

Personal preparation and follow-up of lecture and exercise: 5h

Preparation exam: 31h

The students understand and are able to describe the basics of intellectual property, particularly with regard to the filing and obtaining of property rights. They can name the criteria of project-integrated intellectual property management and strategic patenting in innovative companies. Students are also able to describe the key regulations of the law regarding employee invention and to illustrate the challenges of intellectual properties with reference to examples.

The lecture will describe the requirements to be fulfilled and how protection is obtained for patents, design rights and trademarks, with a particular focus on Germany, Europe and the EU. Active, project-integrated intellectual property management and the use of strategic patenting by technologically oriented companies will also be discussed. Furthermore, the significance of innovations and intellectual property for both business and industry will be demonstrated using practical examples, before going on to consider the international challenges posed by intellectual

property and current trends in the sector. Within the context of licensing and infringement, insight will be provided as to the relevance of communication, professional negotiations and dispute resolution procedures, such as mediation for example. The final item on the agenda will cover those aspects of corporate law that are relevant to intellectual property.

Lecture overview:

1. Introduction to intellectual property
2. The profession of the patent attorney
3. Filing and obtaining intellectual property rights
4. Patent literature as a source of knowledge and information
5. The law regarding employee inventions
6. Active, project-integrated intellectual property management
7. Strategic patenting
8. The significance of intellectual property
9. International challenges and trends
10. Professional negotiations and dispute resolution procedures
11. Aspects of corporate law

Organizational issues

Weitere Informationen siehe IPEK-Homepage.

https://www.ipek.kit.edu/2976_2858.php

V **Patents and Patentstrategies in innovative companies**
2147160, SS 2026, 2 SWS, Language: German, Open in study portal **On-Site**

Content
Attendance at lectures (5 L): 24h

Personal preparation and follow-up of lecture and exercise: 5h

Preparation exam: 31h

The students understand and are able to describe the basics of intellectual property, particularly with regard to the filing and obtaining of property rights. They can name the criteria of project-integrated intellectual property management and strategic patenting in innovative companies. Students are also able to describe the key regulations of the law regarding employee invention and to illustrate the challenges of intellectual properties with reference to examples.

The lecture will describe the requirements to be fulfilled and how protection is obtained for patents, design rights and trademarks, with a particular focus on Germany, Europe and the EU. Active, project-integrated intellectual property management and the use of strategic patenting by technologically oriented companies will also be discussed. Furthermore, the significance of innovations and intellectual property for both business and industry will be demonstrated using practical examples, before going on to consider the international challenges posed by intellectual

property and current trends in the sector. Within the context of licensing and infringement, insight will be provided as to the relevance of communication, professional negotiations and dispute resolution procedures, such as mediation for example. The final item on the agenda will cover those aspects of corporate law that are relevant to intellectual property.

Lecture overview:

1. Introduction to intellectual property
2. The profession of the patent attorney
3. Filing and obtaining intellectual property rights
4. Patent literature as a source of knowledge and information
5. The law regarding employee inventions
6. Active, project-integrated intellectual property management
7. Strategic patenting
8. The significance of intellectual property
9. International challenges and trends
10. Professional negotiations and dispute resolution procedures
11. Aspects of corporate law

T 6.336 Module component: Intelligent Agent Architectures [T-WIWI-111267]

Coordinators: Prof. Dr. Andreas Geyer-Schulz
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration
M-WIWI-105661 - Data Science: Intelligent, Adaptive, and Learning Information Services

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each winter term	1

Courses					
WT 25/26	2540525	Intelligent Agent Architectures	2 SWS	Lecture / 🗣	Geyer-Schulz
WT 25/26	2540526	Übung zu Intelligent Agent Architectures	1 SWS	Practice / 🗣	Geyer-Schulz

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Written examination (60 minutes) according to §4(2), 1 SPO. The exam is considered passed if at least 50 out of a maximum of 100 possible points are achieved. The grades are graded in five steps (best grade 1.0 from 95 points). Details of the grade formation and scale will be announced in the course.
A bonus can be acquired through successful participation in the practice. If the grade of the written examination is between 4.0 and 1.3, the bonus improves the grade by one grade level (0.3 or 0.4). The exact criteria for awarding a bonus will be announced at the beginning of the course.

Please note: this exam will no longer be offered after August 2026.

Prerequisites
None

Recommendations
It is recommended to additionally review the Bachelor-level lecture "Customer Relationship Management" from the module "CRM and Servicemanagement".

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Intelligent Agent Architectures

2540525, WS 25/26, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content
Course content:

The lecture is structured in three parts:

In the first part the methods used for architecture design are introduced (system analysis, UML, formal specification of interfaces, software and analysis patterns, and the separation in conceptual and IT-architectures. The second part is dedicated to learning architectures and machine learning methods. The third part presents examples of learning CRM-Architectures.

Workload:

The total workload for this course is approximately 135 hours (4.5 credits):

Time of attendance

- Attending the lecture: 15 x 90min = 22h 30m
- Attending the exercise classes: 7 x 90min = 10h 30m
- Examination: 1h 00m

Self-study

- Preparation and wrap-up of the lecture: 15 x 180min = 45h 00m
- Preparing the exercises: 25h 00m
- Preparation of the examination: 31h 00m

Sum: 135h 00m

Learning Goals:

Students have special knowledge of software architectures and of the methods which are used in their development (Systems analysis, formal methods for the specification of interfaces and algebraic semantic, UML, and, last but not least, the mapping of conceptual architectures to IT architectures.

Students know important architectural patterns and they can – based on their CRM knowledge – combine these patterns for innovative CRM applications.

Assessment:

The assessment consists of a written exam of 1-hour length following §4 (2), 1 of the examination regulation and by submitting written papers as part of the exercise following §4 (2), 3 of the examination regulation.

The course is considered successfully taken if at least 50 out of 100 points are acquired in the written exam. In this case, all additional points (up to 10) from exercise work will be added.

Grade: Minimum points

- 1,0: 95
- 1,3: 90
- 1,7: 85
- 2,0: 80
- 2,3: 75
- 2,7: 70
- 3,0: 65
- 3,3: 60
- 3,7: 55
- 4,0: 50
- 5,0: 0

Literature

- P. Clements *u. a.*, *Documenting Software Architectures. Views and Beyond*. Upper Saddle River: Addison-Wesley, 2011.
- Fowler, *Patterns of Enterprise Application Architecture*. Amsterdam: Addison-Wesley Longman, 2002.
- S. Russell und P. Norvig, *Artificial Intelligence: A Modern Approach*, 3. Aufl. Harlow Essex England: Pearson New International Edition, 2014.
- V. N. Vapnik, *The Nature of Statistical Learning Theory*. New York: Springer, 1995.

T 6.337 Module component: Intelligent Agents and Decision Theory [T-WIWI-110915]

Coordinators: Prof. Dr. Andreas Geyer-Schulz
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration
M-WIWI-105661 - Data Science: Intelligent, Adaptive, and Learning Information Services

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each summer term	1

Courses					
ST 2026	2540537	Intelligent Agents and Decision Theory	2 SWS	Lecture	Geyer-Schulz
ST 2026	2540538	Übung zu Intelligent Agents and Decision Theory	1 SWS	Practice	Bell

Assessment
Written examination (60 minutes). The exam is held in each semester and can be repeated at any regular examination date. Details of the grading system and any exam bonus that may be achieved from the practice are announced in the course.

Please note: this course, exercises and exam will be not longer be offered after August 2026.

Prerequisites
None

Recommendations
We assume knowledge in statistics, operations research and microeconomics as taught in the Bachelor program (VWL I, Operations Research I + II, Statistics I + II) and a familiarity with preferably the Python programming language.

Additional Information
Please note: this course, exercises and exam will be not longer be offered after August 2026.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Intelligent Agents and Decision Theory

2540537, SS 2026, 2 SWS, Language: English, Open in study portal

Lecture (V)

Content
The key assumption of this lecture is that the concept of artificial intelligence is inseparably linked to the economic concept of rationality of agents. We consider different classes of decision problems - decisions under certainty, risk and uncertainty - from an economic, managerial and AI-engineering perspective:

From an economic point of view, we analyze how to act rationally in these situations based on classic utility theory. In this regard, the course also introduces the relevant parts of decision theory for dealing with

- multiple conflicting objectives,
- incomplete, risky and uncertain information about the world,
- assessing utility functions, and
- quantifying the value of information ...

From an engineering perspective, we discuss how to develop practical solutions for these decision problems, using appropriate AI components. We introduce

- a general, agent-based design framework for AI systems,

as well as AI methods from the fields of

- search (for decisions under certainty),
- inference (for decions under risk) and
- learning (for decisions under uncertainty).

Where applicable, the course highlights the theoretical ties of these methods with decision theory.

We conclude with a discussion of ethical and philosophical issues concerning the development and use of AI.

Learning objectives

Students are able to design, analyze, implement, and evaluate intelligent agents.

Lecture Outline

1. Introduction: Artificial intelligence and the economic concept of rationality
2. Intelligent Agents: A general, agent-based design framework for AI systems
3. Decision under certainty: Assessing utility functions for decisions with multiple objectives
4. Search: Linear programming for decisions under certainty
5. Decisions under risk: The expected utility principle
6. Information systems: Improving economic decisions under risk
7. Inference: Bayesian networks for decisions under risk
8. Learning: Bayesian Networks (Basics)
9. Learning: Bayesian Networks (Algorithms I)
10. Learning: Bayesian Networks (Algorithms II)

Note: This rough outline may be subject to change.

Literature
Bamberg, Coenenberg & Krapp (2019). Betriebswirtschaftliche Entscheidungslehre (16th ed.). Verlag Franz Vahlen GmbH.

Fishburn (1988). Nonlinear preference and utility theory. Baltimore: Johns Hopkins University Press.

Keeney & Raiffa (1993). Decisions with multiple objectives: preferences and value trade-offs. Cambridge University Press.

Nickel, S., Stein, O., & Waldmann, K.-H. (2014). Operations Research (2nd ed.). Springer Berlin Heidelberg.

Russell & Norvig (2016). Artificial Intelligence: A Modern Approach (3rd Global Edition). Pearson.

Koller, D., & Friedman, N. (2009). Probabilistic graphical models: principles and techniques. MIT Press.

Sutton & Barto (2018). Reinforcement learning: An introduction. Cambridge: MIT press.

T 6.338 Module component: International Entrepreneurship and Brand Development [T-WIWI-114893]

Coordinators: Prof. Dr. Ann-Kristin Kupfer
Prof. Dr. Orestis Terzidis

Organisation: KIT Department of Business and Economics

Part of: M-WIWI-101488 - Entrepreneurship (EnTechnon)
M-WIWI-104900 - Business Administration
M-WIWI-105312 - Marketing and Sales Management
M-WIWI-106258 - Digital Marketing

Type	Credits	Grading	Term offered	Version
Examination of another type	6 CP	graded	see Annotations	1

Courses					
WT 25/26	2572189	International Entrepreneurship and Brand Development	4 SWS	Block / 🗣	Terzidis, Kupfer
WT 26/27	2572189	International Entrepreneurship and Brand Development	4 SWS	Block / 🗣	Terzidis, Kupfer

Legend: 🖥 Online, Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Non exam assessment. The grade is based on the presentation, the subsequent discussion and the written elaboration.

Additional Information
Please contact the Marketing and Sales Research Group for further information.

Workload
180 hours

Below you will find excerpts from courses related to this module component:

V International Entrepreneurship and Brand Development
2572189, WS 25/26, 4 SWS, Language: English, Open in study portal

Block (B)
On-Site

Content
This course is offered as part of the EUCOR programme in cooperation with EM Strasbourg. Max. 10 students of KIT and max. 10 students of EM Strasbourg will develop a sales presentation in tandems (teams of 2). This is based on the value proposition of a business model.

- An application is required to participate in this event. The application phase usually takes place at the beginning of the lecture period. Further information on the application process can be found on the website of the Institute for Entrepreneurship and Technology and the Institute for Customer Insights (CIN) shortly before the start of the lecture period.

Total workload for 6 ECTS: about 180 hours.

V International Entrepreneurship and Brand Development
2572189, WS 26/27, 4 SWS, Language: English, Open in study portal

Block (B)
On-Site

Content
This course is offered as part of the EUCOR programme in cooperation with EM Strasbourg. Max. 10 students of KIT and max. 10 students of EM Strasbourg will develop a sales presentation in tandems (teams of 2). This is based on the value proposition of a business model.

- An application is required to participate in this event. The application phase usually takes place at the beginning of the lecture period. Further information on the application process can be found on the website of the Institute for Entrepreneurship and Technology and the Institute for Customer Insights (CIN) shortly before the start of the lecture period.

Total workload for 6 ECTS: about 180 hours.

Organizational issues
Raum, Datum/Uhrzeit wird bekannt gegeben

T 6.339 Module component: International Finance [T-WIWI-102646]

Coordinators: Prof. Dr. Marliese Uhrig-Homburg
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101480 - Finance 3
M-WIWI-101483 - Finance 2
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	3 CP	graded	see Annotations	1

Courses					
ST 2026	2530570	International Finance	2 SWS	Lecture / 🗣	Walter, Uhrig-Homburg

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The success control takes place in form of a written examination (60 min). If the number of participants is low, an oral examination may also be offered. The examination is offered every semester and can be repeated at any regular examination date.

Prerequisites
None

Recommendations
None

Additional Information
The course is offered as a 14-day or block course.

Below you will find excerpts from courses related to this module component:

V International Finance
2530570, SS 2026, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Organizational issues
Kickoff am Mittwoch, 29.04.26, 16:00 - 19:15 Uhr im Raum 320 im Geb. 09.21 (Blücherstr. 17). Die Veranstaltung wird samstags als Blockveranstaltung angeboten (nach dem Kickoff nach Absprache).

Literature
Weiterführende Literatur:

- Eiteman, D. et al., Multinational Business Finance, 13. Auflage, 2012.
- Solnik, B. und D. McLeavey, Global Investments, 6. Auflage, 2008.

T 6.340 Module component: International Marketing [T-WIWI-102807]

Coordinators: Dr. Sven Feurer
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	1,5 CP	graded	see Annotations	1

Assessment
The examination will be open to first-time writers for the last time in the 2021 summer semester.

The assessment consists of a written exam (60 minutes) (following §4(2), 1 of the examination regulation).

Prerequisites
None

Additional Information
The course "International Marketing" will be offered for the last time in winter semester 2020/21.
For further information please contact the Marketing & Sales Research Group (marketing.iism.kit.edu).

T 6.341 Module component: Internet Law [T-INFO-101307]

Coordinators: N.N.
Organisation: KIT Department of Informatics
Part of: M-INFO-101215 - Intellectual Property Law
M-WIWI-104903 - Law

Type	Credits	Grading	Term offered	Version
Written examination	3 CP	graded	Each winter term	2

Courses					
WT 25/26	2424354	Internet Law	2 SWS	Lecture / 🔀	Idelberger
WT 26/27	2424354	Internet Law	2 SWS	Lecture / 🔀	Sattler

Legend: 🖥 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment is carried out as a written examination (§ 4 Abs. 2 No. 1 SPO) lasting 120 minutes.

Prerequisites
The course **Ausgewählte Rechtsfragen des Internetrechts T-INFO-108462** may not have started.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-INFO-108462 - Selected Legal Issues of Internet Law must not have been started.

Recommendations
None.

Additional Information
Lecture (with written exam) **Internet Law** T-INFO-101307 is offered in the winter semester.

Colloquium (other type of examination) **Selected Legal Issues in Internet Law** T-INFO-108462 is offered in the summer semester.

T 6.342 Module component: Internet of Everything [T-INFO-101337]

Coordinators: Prof. Dr. Martina Zitterbart
Organisation: KIT Department of Informatics
Part of: M-WIWI-104909 - Informatics (Department of Informatics)

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each winter term	1

Courses					
WT 25/26	2424104	Internet of Everything	2 SWS	Lecture / 🗣	Zitterbart, Hildenbrand, Mahrt, Neumeister
WT 26/27	2424104	Internet of Everything	2 SWS	Lecture / 🗣	Zitterbart, Hildenbrand, Mahrt, Neumeister

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment is carried out as an oral examination (§ 4 Abs. 2 Nr. 2 SPO) lasting 20 minutes.

Depending on the number of participants, it will be announced six weeks before the examination (Section 6 (3) SPO) whether the assessment will take the form of an oral examination of approx.

- in the form of an oral examination of approx. 30 minutes in accordance with § 4 Para. 2 No. 2 SPO or

- in the form of a written examination in accordance with § 4 Para. 2 No. 1 SPO

takes place.

Prerequisites
None.

Recommendations
The contents of the lecture Introduction to Computer Networks are assumed to be known. Attendance of the lecture Telematics is strongly recommended, as the contents are an important basis for understanding and classifying the material.

Additional Information
Offered for the last time in WS 26/27!!!!!

T 6.343 Module component: Introduction to Automotive and Industrial Lidar Technology [T-ETIT-111011]

Coordinators: Prof. Dr. Wilhelm Stork
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-105461 - Introduction to Automotive and Industrial Lidar Technology

Type	Credits	Grading	Term offered	Expansion	Version
Examination of another type	3 CP	graded	Each winter term	1 semesters	2

Courses					
WT 25/26	2311604	Introduction to automotive and industrial Lidar technology	2 SWS	Lecture / 🗣	Heußner

Legend: 🖥 Online, ⚄ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

T 6.344 Module component: Introduction to Bionics [T-MACH-111807]

Coordinators: apl. Prof. Dr. Hendrik Hölscher
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101287 - Microsystem Technology
M-MACH-101290 - BioMEMS
M-MACH-101294 - Nanotechnology
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Written examination	4 CP	graded	Each summer term	3

Courses					
ST 2026	2142151	Introduction to Biomimetics	2 SWS	Lecture / 🗣	Hölscher

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
written exam (duration: 60 minutes)

Prerequisites
none

Additional Information
Module component T-MACH-102172 may not be started

Below you will find excerpts from courses related to this module component:

V Introduction to Biomimetics

2142151, SS 2026, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content
Bionics focuses on the design of technical products following the example of nature. For this purpose we have to learn from nature and to understand its basic design rules. Therefore, the lecture focuses on the analysis of the fascinating effects used by many plants and animals. Possible implementations into technical products are discussed in the end.

The students should be able analyze, judge, plan and develop biomimetic strategies and products.

Basic knowledge in physics and chemistry

The successfull attandence of the lecture is controlled by a written examination.

Organizational issues
Im ILIAS werden Materialien (Videos, Originalliteratur, Übungen) zur Vertiefung zur Verfügung gestellt.

Für die schriftliche Klausur werden zwei Termine angeboten (zweite Woche nach Vorlesungsende im Sommersemester und eine Woche vor Vorlesungsbeginn im Wintersemester).

Literature
Folien und Literatur werden in ILIAS zur Verfügung gestellt.

T 6.345 Module component: Introduction to Ceramics [T-MACH-100287]

Coordinators: Prof. Dr. Kaline Pagnan Furlan
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101268 - Specific Topics in Materials Science
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Oral examination	6 CP	graded	Each winter term	1

Courses					
WT 25/26	2125757	Introduction to Ceramics	3 SWS	Lecture /	Pagnan Furlan, Wagner, Ribas Gomes

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
The assessment consists of an oral exam (30 min) taking place at a specific date.

The re-examination is offered at a specific date.

Prerequisites
None

Workload
180 hours

Below you will find excerpts from courses related to this module component:

V Introduction to Ceramics
2125757, WS 25/26, 3 SWS, Language: German, Open in study portal

Lecture (V)
Blended (On-Site/Online)

Content
This course provides a comprehensive introduction to ceramic materials, covering chemical bonding, crystal structures, crystal defects, phases, and phase diagrams. Students learn about powder preparation, shaping, and sintering (solid- and liquid-phase), as well as microstructure development and grain growth. The course explores mechanical, electrical, and other functional properties, linking structure and processing to performance. Lectures are complemented by exercises to apply theory to practical problems and design considerations.

Literature

- Callister William D., Rethwisch David G., Scheffler Michael: *Materialwissenschaften und Werkstofftechnik*, Wiley-VCH Verlag, Weinheim, 2013.
  Online verfügbar: http://textbooks.wiley-vch.de/book/callister0072/ *
- Salmang H., Scholze H., Telle R.: *Keramik*, Springer Berlin Heidelberg, 2007.
  Online verfügbar: https://link.springer.com/book/10.1007/978-3-540-49469-0 *
- Kollenberg Wolfgang: *Technische Keramik*, Vulkan-Verlag, Essen, 2004.
  Online-Katalog: Link zur Bibliothek
- Richerson David W.: *Modern ceramic engineering*, CRC Press, Boca Raton, FL, 2018.
  Verfügbar im Netz des KIT: https://search.ebscohost.com/login.aspx?direct=true&db=nlebk&db=nlabk&AN=1802218 *

** Zugang möglich im Netz des KIT.*

T 6.346 Module component: Introduction to Digital Economics [T-WIWI-112722]

Coordinators: Prof. Dr. Johannes Brumm
Prof. Dr. Johannes Philipp Reiß
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104908 - Economics

Type	Credits	Grading	Term offered	Version
Written examination	6 CP	graded	Each term	1

Courses					
WT 25/26	2610001	The Digital Economy: Cases and Models	2 SWS	Lecture	Reiß, Potarca, Peters
ST 2026	2500163	The Digital Economy: Micro and Macro Perspective	2 SWS	Lecture / 🗣	Brumm, Reiß

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The module examination takes the form of an overall examination on the two courses "The Digital Economy: Cases and Models" and "The Digital Economy: Micro and Macro Perspective" lasting 120 minutes. The exam is offered every semester and can be repeated at any regular exam date. The module grade corresponds to the exam grade.

Prerequisites
None

Workload
180 hours

T 6.347 Module component: Introduction to Econometrics [T-WIWI-114622]

Coordinators: Prof. Dr. Melanie Schienle
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104908 - Economics

Type	Credits	Grading	Term offered	Version
Written examination	5 CP	graded	Each summer term	1

Courses					
ST 2026	2520016	Introduction to Econometrics	2 SWS	Lecture / 🗣	Schienle, Bracher
ST 2026	2520017	Übungen zu Einführung in die Ökonometrie	2 SWS	Practice	Schienle, Rüter, Bracher, Leimenstoll

Legend: 🖥 Online, Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success is assessed in the form of a written examination lasting 1 hour.

Grade Bonus through Bonus Exercises: By achieving qualified participation in at least 80% of the bonus exercises accompanying the lectures, students can earn a grade bonus for the examinations directly following the semester. This bonus of three points (out of a maximum of 90 points for the exam) will only be credited if the examination is considered passed without the application of this bonus. The grade bonus can thus result in an improvement of up to one grade increment.

Prerequisites
None

Recommendations
For successful participation, sound knowledge of statistics is helpful. It is recommended to have completed the courses Statistics I and Statistics II beforehand.

Below you will find excerpts from courses related to this module component:

V Introduction to Econometrics
2520016, SS 2026, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content
Learning objectives:

- Familiarity with the basic concepts and methods of econometrics
- Preparation of simple econometric surveys

Content:

- Simple and multiple linear regression (estimating parameters, confidence interval, testing, prognosis, testing assumptions)
- Model assessment

Requirements:

Knowledge of the lectures Statistics I + II is required.

Workload:

Total workload for 5 CP: approx. 150 hours

Attendance: 30 hours

Preparation and follow-up: 120 hours

Literature
Von Auer: Ökonometrie ISBN 3-540-00593-5

Goldberger: A course in Econometrics ISBN 0-674-17544-1

Gujarati. Basic Econometrics ISBN 0-07-113964-8

Schneeweiß: Ökonometrie ISBN 3-7908-0008-2

T 6.348 Module component: Introduction to Energy Economics [T-WIWI-102746]

Coordinators: Prof. Dr. Wolf Fichtner
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	5,5 CP	graded	Each summer term	7

Courses					
ST 2026	2581010	Introduction to Energy Economics	2 SWS	Lecture / 🗣	Fichtner
ST 2026	2581011	Übungen zu Einführung in die Energiewirtschaft	2 SWS	Practice / 🗣	Layer, Fichtner, Raab

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment consists of a written exam (90 minutes) (following §4(2) of the examination regulation). The exam takes place in every semester. Re-examinations are offered at every ordinary examination date.

Prerequisites
None.

Below you will find excerpts from courses related to this module component:

V Introduction to Energy Economics

2581010, SS 2026, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content

1. Introduction: terms, units, conversions
2. The energy carrier gas (reserves, resources, technologies)
3. The energy carrier oil (reserves, resources, technologies)
4. The energy carrier hard coal (reserves, resources, technologies)
5. The energy carrier lignite (reserves, resources, technologies)
6. The energy carrier uranium (reserves, resources, technologies)
7. The final carrier source electricity
8. The final carrier source heat
9. Other final energy carriers (cooling energy, hydrogen, compressed air)

The student is able to

- characterize and judge the different energy carriers and their peculiarities,
- understand contexts related to energy economics.

Literature
Weiterführende Literatur:

Pfaffenberger, Wolfgang. Energiewirtschaft. ISBN 3-486-24315-2

Feess, Eberhard. Umweltökonomie und Umweltpolitik. ISBN 3-8006-2187-8

Müller, Leonhard. Handbuch der Elektrizitätswirtschaft. ISBN 3-540-67637-6

Stoft, Steven. Power System Economics. ISBN 0-471-15040-1

Erdmann, Georg. Energieökonomik. ISBN 3-7281-2135-5

6.349 Module component: Introduction to Engineering Geology [T-BGU-101500]

Coordinators: Prof. Dr. Philipp Blum
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Written examination	5 CP	graded	Each winter term	1

Courses					
WT 25/26	6339057	Introduction to Engineering Geology	4 SWS	Lecture / Practice	Blum, Menberg, Steger

Prerequisites
none

Workload
150 hours

T 6.350 Module component: Introduction to Engineering Mechanics I: Statics and Strength of Materials [T-MACH-102208]

Coordinators: Prof. Dr.-Ing. Alexander Fidlin
Organisation: KIT Department of Mechanical Engineering

Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Written examination	5 CP	graded	Each summer term	2

Courses					
ST 2026	2162265	Introduction to Engineering Mechanics I: Statics and Strength of Materials (Tutorial)	2 SWS	Practice / 🗣	Fidlin, Ströhle, Thasan
ST 2026	2162274	Introduction to Engineering Mechanics I: Statics and Strength of Materials	2 SWS	Lecture / 🗣	Fidlin, Ströhle

Legend: 🖥 Online, Blended (On-Site/Online), 🗣 On-Site, × Cancelled

Assessment
The assessment consists of a written examination (120 min) taking place in the recess period (according to Section 4(2), 1 of the examination regulation). The examination takes place in every semester. Re-examinations are offered at eyery ordinary examination date.

For students of economics the assessement consists of a written examination (Statics - 75 min.)

Permitted utilities: non-programmable calculator

Prerequisites
None

Additional Information
The course is offered in German.

Workload
150 hours

T 6.351 Module component: Introduction to Engineering Mechanics II : Dynamics [T-MACH-102210]

Coordinators: Prof. Dr.-Ing. Alexander Fidlin
Organisation: KIT Department of Mechanical Engineering

Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Written examination	5 CP	graded	Each winter term	1

Courses					
WT 25/26	2161276	Introduction to Engineering Mechanics II : Dynamics	2 SWS	Lecture / 🗣	Fidlin, Ströhle

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment consists of a written examination (75 min) taking place in the recess period (according to Section 4(2), 1 of the examination regulation). The examination is offered every semester. Re-examinations are offered at every ordinary examination date.

Permitted utilities: non-programmable calculator, literature.

Prerequisites
None

Additional Information
The course is offered in German.

Below you will find excerpts from courses related to this module component:

V Introduction to Engineering Mechanics II : Dynamics — **Lecture (V) On-Site**
2161276, WS 25/26, 2 SWS, Language: German, Open in study portal

T 6.352 Module component: Introduction to Finance and Accounting [T-WIWI-112820]

Coordinators: Dr. Torsten Luedecke
Prof. Dr. Martin Ruckes
Prof. Dr. Marcus Wouters

Organisation: KIT Department of Business and Economics

Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	5 CP	graded	Each summer term	2

Courses					
ST 2026	2500002	Jahresabschluss und Bewertung	1 SWS	Lecture	Ruckes, Luedecke
ST 2026	2500025	Tutorial Introduction to Finance and Accounting	2 SWS	Tutorial	Wouters, Ruckes, Assistenten, Kohl
ST 2026	2610026	Introduction to Finance and Accounting	2 SWS	Lecture / 🗣	Ruckes, Wouters, Thimme

Legend: Online, Blended (On-Site/Online), 🗣 On-Site, × Cancelled

Assessment
Written examination (2.5 hours) takes place as part of the examination course "Financing and Accounting" and covers the content of the two courses "Introduction to Finance and Accounting" and "Annual Financial Statements and Valuation". The examination is offered at the beginning of each lecture-free period. Repeat examinations are possible on every regular examination date.

Workload
150 hours

Below you will find excerpts from courses related to this module component:

V Introduction to Finance and Accounting

Lecture (V)
On-Site

2610026, SS 2026, 2 SWS, Language: German, Open in study portal

Content
The lecture covers the following topics:

- Investment and Finance
  - Valuation of Bonds and Stocks
  - Capital Budgeting
  - Portfolio Theory
- Financial Accounting
- Management Accounting

Literature
Ausführliche Literaturhinweise werden in den Materialen zur Vorlesung gegeben.

T 6.353 Module component: Introduction to Game Theory [T-WIWI-102850]

Coordinators: Prof. Dr. Clemens Puppe
Prof. Dr. Johannes Philipp Reiß
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104908 - Economics

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each summer term	3

Courses					
ST 2026	2520525	Introduction to Game Theory	2 SWS	Lecture / 🗣	Rosar, Puppe, Ammann
ST 2026	2520526	Übungen zu Einführung in die Spieltheorie	1 SWS	Practice / 🗣	Puppe, Rosar, Ammann

Legend: 📱 Online, ✲ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment consists of a written exam (60 minutes) according to Section 4(2),1 of the examination regulation.

The exam takes place in the recess period and can be repeated at every ordinary examination date.

Recommendations
Knowledge from the lecture "Economics I: Microeconomics" is recommended. Furthermore, basic knowledge of mathematics and statistics is assumed.

Below you will find excerpts from courses related to this module component:

V Introduction to Game Theory
Lecture (V)
On-Site
2520525, SS 2026, 2 SWS, Language: German, Open in study portal

Content
The course focusses on non-cooperative game theory. It discusses models, solution concepts, and applications for simultaneous games as well as sequential games. Various solution concepts, e.g., Nash equilibrium and subgame-perfect equilibrium, are introduced along with more advanced concepts.

This course offers an introduction to the theoretical analysis of strategic interaction situations. At the end of the course, students shall be able to analyze situations of strategic interaction systematically and to use game theory to predict outcomes and give advice in applied economics settings.

Literature
Verpflichtende Literatur:

Gibbons (1992): A Primer in Game Theory, Harvester-Wheatsheaf.

Ergänzende Literatur:

Berninghaus/Ehrhart/Güth (2010): Strategische Spiele, Springer Verlag.

Binmore (1991): Fun and Games, DC Heath.

Fudenberg/Tirole (1991): Game Theory, MIT Press.

Heifetz (2012): Game Theory, Cambridge Univ. Press.

T 6.354 Module component: Introduction to GIS for Students of Natural, Engineering and Geo Sciences [T-BGU-101681]

Coordinators: Dr.-Ing. Sven Wursthorn
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Written examination	3 CP	graded	Each winter term	5

Courses					
WT 25/26	6071101	Introduction to GIS for Students of Natural Sciences, Engineering and Geosciences, L+E	4 SWS	Lecture / Practice / 🗣	Wursthorn

Legend: 🖥 Online, ✿ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
written exam, 90 min.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-BGU-103541 - Introduction to GIS for Students of Natural, Engineering and Geo Sciences, Prerequisite must have been passed.

Workload
90 hours

T 6.355 Module component: Introduction to GIS for Students of Natural, Engineering and Geo Sciences, Prerequisite [T-BGU-103541]

Coordinators: Dr.-Ing. Sven Wursthorn
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Expansion	Version
Coursework	3 CP	pass/fail	Each winter term	1 semesters	5

Courses					
WT 25/26	6071101	Introduction to GIS for Students of Natural Sciences, Engineering and Geosciences, L+E	4 SWS	Lecture / Practice / 🗣	Wursthorn

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The achievement control takes place via accepted exercises.

Prerequisites
none

Recommendations
none

Additional Information
none

Workload
90 hours

T 6.356 Module component: Introduction to Hydrogeology [T-BGU-101499]

Coordinators: Prof. Dr. Nico Goldscheider
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Written examination	5 CP	graded	Each winter term	2

Courses					
WT 25/26	6339050	Introduction to Hydrogeology	4 SWS	Lecture / Practice /	Goldscheider

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
Written exam with 90 minutes

Prerequisites
none

Workload
150 hours

6.357 Module component: Introduction to Machine Learning [T-WIWI-111028]

Coordinators: Prof. Dr. Andreas Geyer-Schulz
Dr. Abdolreza Nazemi
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Expansion	Version
Written examination	4,5 CP	graded	Each winter term	1 semesters	1

Courses					
WT 25/26	2540539	Introduction to Machine Learning	2 SWS	Lecture / 🗣	Nazemi
WT 25/26	2540540	Übung zu Introduction to Machine Learning	1 SWS	Practice / 🗣	Nazemi

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Written examination (60 minutes) according to §4(2), 1 SPO. The exam is considered passed if at least 50 out of a maximum of 100 possible points are achieved. The grades are graded in five-point-steps (best grade 1.0 from 95 points). Details of the grade formation and scale will be announced in the course.
A bonus can be acquired through successful participation in the practice. If the grade of the written examination is between 4.0 and 1.3, the bonus improves the grade by one grade level (0.3 or 0.4). The exact criteria for awarding a bonus will be announced at the beginning of the course.

Please note: this exam will no longer be offered after August 2026.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Introduction to Machine Learning
2540539, WS 25/26, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content

- Introduction
- Data Cleaning
- Data Visualization
- Linear Regression
- Logistic Regression
- Tree-based Algorithms
- Support Vector Machine
- Shrinkage Models
- Dimensionality Reduction
- Clustering

Literature

- Alpaydin, E. (2014). *Introduction to Machine Learning*. Third Edition, MIT Press.
- Hall, J. (2020). *Machine Learning in Business: An Introduction to the World of Data Science.*Independently published.
- James, G., Witten, D., Hastie, T., and R. Tibshirani (2013). *An Introduction to Statistical Learning: with Applications in R*. Springer.
- Tan, P. N., Steinbach, M., Karpatne, A., & Kumar, V. (2018). *Introduction to data mining*. Pearson

T 6.358 Module component: Introduction to Microsystem Technology - Practical Course [T-MACH-108312]

Coordinators: Dr. Arndt Last
Organisation: KIT Department of Mechanical Engineering

Part of:
M-MACH-101287 - Microsystem Technology
M-MACH-101290 - BioMEMS
M-MACH-101291 - Microfabrication
M-MACH-101292 - Microoptics
M-MACH-101294 - Nanotechnology
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Coursework	3 CP	pass/fail	Each term	3

Courses					
WT 25/26	2143877	Introduction to Microsystem Technology - Practical Course	2 SWS	Practical course / 🗣	Last
ST 2026	2143877	Introduction to Microsystem Technology - Practical Course	2 SWS	Practical course / 🗣	Last

Legend: 📱 Online, Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
This practical course concludes with a written examination (duration 1 h). The assessment is not graded and may be repeated until it is passed.

Prerequisites
none

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-MACH-102164 - Practical Training in Basics of Microsystem Technology must not have been started.

Workload
90 hours

Below you will find excerpts from courses related to this module component:

V Introduction to Microsystem Technology - Practical Course
Practical course (P)
On-Site

2143877, WS 25/26, 2 SWS, Language: German, Open in study portal

Organizational issues
Das Praktikum findet in den Laboren des IMT am CN statt. Treffpunkt: Bau 301, vor dem Eingang.

Das "Laborpraktikum" ist das selbe wie das "Praktikum", nur unbenotet! Beide mit schriftlicher Klausur.

Wer teilnehmen möchte, muss sich ab 19.1.2026, 8h00 über das Campussystem unter Veranstaltungen (nicht unter Prüfungen!) auf die Warteliste setzen. Die endgültige Entscheidung, ob man teilnehmen kann, erfolgt erst 10.2.2026.

Literature
Menz, W., Mohr, J.: Mikrosystemtechnik für Ingenieure, VCH-Verlag, Weinheim, 1997
Unterlagen zum Praktikum zur Vorlesung ' Grundlagen der Mikrosystemtechnik'

V Introduction to Microsystem Technology - Practical Course
Practical course (P)
On-Site

2143877, SS 2026, 2 SWS, Language: German, Open in study portal

Content
In the practical training includes nine experiments:

1. X-ray optics
2. UVL + REM
3. Micromixer
4. Atomic force microscopy
5. 3D-Printing
6. Light dirffraction at Chromium masks
7. Moulding
8. SAW-bio-sensors
9. Nano3D-printer - material transfer of thin foils
10. Electro spinning

Each student takes part in only four experiments.
The experiments are carried out at real workstations at the IMT and coached by IMT-staff.

Organizational issues
Das Praktikum findet 14.-18.9.2026 an vier halben Tagen in den Laboren des IMT am KIT-CN statt. Treffpunkt: Eingang Bau 301.

Das "Laborpraktikum" ist das selbe wie das "Praktikum", nur unbenotet! Beide mit schriftlicher Klausur in der Woche danach, voraussichtlich am 24.9.2026.

Wer teilnehmen möchte, muss sich ab 6.7.2026, 8h00 über das Campussystem unter Veranstaltungen (nicht unter Prüfungen!) auf die Warteliste setzen. Die endgültige Entscheidung, ob man teilnehmen kann, erfolgt erst 31.8.2026.

Literature
Menz, W., Mohr, J.: Mikrosystemtechnik für Ingenieure, VCH-Verlag, Weinheim, 1997
Unterlagen zum Praktikum zur Vorlesung ' Grundlagen der Mikrosystemtechnik'

T 6.359 Module component: Introduction to Microsystem Technology I [T-MACH-114100]

Coordinators: Dr. Vlad Badilita
Prof. Dr. Jan Gerrit Korvink

Organisation: KIT Department of Mechanical Engineering

Part of: M-ETIT-101158 - Sensor Technology I
M-MACH-101287 - Microsystem Technology
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Written examination	4 CP	graded	Each winter term	1

Courses					
WT 25/26	2141861	Introduction to Microsystem Technology I	2 SWS	Lecture / 🗣	Korvink, Badilita

Legend: 🖥 Online, Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
written examination (60 min)

Prerequisites
T-MACH-114035 and T-MACH-105182 must not have started

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Introduction to Microsystem Technology I

2141861, WS 25/26, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Literature
Mikrosystemtechnik für Ingenieure, W. Menz und J. Mohr, VCH Verlagsgesellschaft, Weinheim 2005

M. Madou
Fundamentals of Microfabrication
Taylor & Francis Ltd.; Auflage: 3. Auflage. 2011

T 6.360 Module component: Introduction to Microsystem Technology II [T-MACH-114101]

Coordinators: Dr. Vlad Badilita
Prof. Dr. Jan Gerrit Korvink

Organisation: KIT Department of Mechanical Engineering

Part of: M-ETIT-101158 - Sensor Technology I
M-MACH-101287 - Microsystem Technology
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Written examination	4 CP	graded	Each summer term	1

Courses					
ST 2026	2142874	Introduction to Microsystem Technology II	2 SWS	Lecture / 🗣	Korvink, Badilita

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
written examination (60 min)

Prerequisites
T-MACH-114035 and T-MACH-105183 must not have started

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Introduction to Microsystem Technology II

Lecture (V)
On-Site

2142874, SS 2026, 2 SWS, Language: English, Open in study portal

Content
- Introduction in Nano- and Microtechnologies
- Lithography
- LIGA-technique
- Mechanical microfabrication
- Patterning with lasers
- Assembly and packaging
- Microsystems

Organizational issues
Topic: Grundlagen der Mikrosystemtechnik II (MST II) SS 21
Time: Thursdays 14:00 - 15:30

10.91 Redtenbacher-Hörsaal

Literature
Menz, W., Mohr, J., O. Paul: Mikrosystemtechnik für Ingenieure, VCH-Verlag, Weinheim, 2005

M. Madou
Fundamentals of Microfabrication
Taylor & Francis Ltd.; Auflage: 3. Auflage. 2011

T 6.361 Module component: Introduction to Nanotechnology [T-MACH-111814]

Coordinators: apl. Prof. Dr. Hendrik Hölscher
Organisation: KIT Department of Mechanical Engineering
KIT Department of Business and Economics
Part of: M-MACH-101294 - Nanotechnology
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Written examination	4 CP	graded	Each summer term	2

Courses					
ST 2026	2142152	Introduction to Nanotechnology	2 SWS	Lecture / 🗣	Hölscher

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, x Cancelled

Assessment
written exam 90 min

Prerequisites
none

Additional Information
T-MACH-111814 may not be started

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Introduction to Nanotechnology
2142152, SS 2026, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content
Nanotechnology deals with the fabrication and analysis of nanostructures. The topics of the lecture include

- the most common measurement principles of nanotechnology especially scanning probe methods
- the analysis of physical and chemical properties of surfaces
- interatomic forces and their influence on nanostructures
- methods of micro- and nanofabrication and lithography
- basic models of contact mechanics and nanotribology
- important functional characteristics of nanodevices

Basic knowledge in mathematics and physics is assumed

The successfull attandence of the lecture is controlled by a 30 minutes oral exam.

Organizational issues
Es werden im ILIAS Materialien (Videos, Originalliteratur, Übungen) zum Vertiefung zur Verfügung gestellt.

Für die mündlichen Prüfungen werden zwei Termine angeboten (zweite Woche nach Vorlesungsende im Sommersemester und eine Woche vor Vorlesungsbeginn im Wintersemester).

Literature
Alle Folien und Originalliteratur werden auf ILIAS zur Verfügung gestellt.

T 6.362 Module component: Introduction to Neural Networks and Genetic Algorithms [T-WIWI-111029]

Coordinators: Prof. Dr. Andreas Geyer-Schulz
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Expansion	Version
Written examination	4,5 CP	graded	Each summer term	1 semesters	1

Courses					
ST 2026	2540541	Introduction to Neural Networks and Genetic Algorithms	2 SWS	Lecture	Geyer-Schulz
ST 2026	2540542	Übung Introduction to Neural Networks and Genetic Algorithms	1 SWS	Practice	Bell

Assessment
Written examination (60 minutes) according to §4(2), 1 SPO. The exam is considered passed if at least 50 out of a maximum of 100 possible points are achieved. The grades are graded in five-point-steps (best grade 1.0 from 95 points). Details of the grade formation and scale will be announced in the course.
A bonus can be acquired through successful participation in the practice. If the grade of the written examination is between 4.0 and 1.3, the bonus improves the grade by one grade level (0.3 or 0.4). The exact criteria for awarding a bonus will be announced at the beginning of the course.

Please note: this exam will no longer be offered after August 2026.

Additional Information
Please note: this course, exercises and exam will no longer be offered after August 2026.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Introduction to Neural Networks and Genetic Algorithms

2540541, SS 2026, 2 SWS, Language: English, Open in study portal

Lecture (V)

Content
The course consists of a short introduction and two parts:

1. In the introduction, the biological mechanisms of neural and genetic methods are presented.
Furthermore, a common framework for the learning performance evaluation of these methods in applications is introduced.
2. In the field of genetic methods, simple genetic algorithms and their variants are introduced, analyzed, and applied.
3. In the area of neural methods, the basic algorithms are presented (e.g., backpropagation) as well as their applications in data science.

Learning Objectives:

The student knows the essential algorithms, learning procedures, and methods for neural networks and genetic algorithms. They can apply these methods (e.g. in R) and evaluate their quality.

Literature

- Goldberg, David E. (2001)
Genetic Algorithms in Search, Optimization and Machine Learning.
Addison-Wesley, New York.
- Bishop, Christopher M. (2006)
Pattern Recognition and Machine Learning.
Springer, New York.
- Goodfellow, Ian; Bengio, Yoshua; Courville, Aaron (2016)
Deep Learning.
MIT Press. Cambridge.

T 6.363 Module component: Introduction to Numerical Simulation of Reacting Flows [T-CIWVT-113436]

Coordinators: Prof. Dr. Oliver Thomas Stein
Organisation: KIT Department of Chemical and Process Engineering
Part of: M-CIWVT-106676 - Introduction to Numerical Simulation of Reacting Flows

Type	Credits	Grading	Version
Oral examination	3 CP	graded	2

Courses					
WT 25/26	2232130	Introduction to Numerical Simulation of Reacting Flows	2 SWS	Lecture / 🗣	Stein
WT 25/26	2232131	Introduction to Numerical Simulation of Reacting Flows - Exercises	2 SWS	Practice / 🗣	Stein

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The learning control ist an oral examination lasting approx. 30 minutes.

Prerequisites
The prerequisite must be passed before taking the oral examination.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-CIWVT-113435 - Introduction to Numerical Simulation of Reacting Flows - Prerequisite must have been passed.

T 6.364 Module component: Introduction to Numerical Simulation of Reacting Flows - Prerequisite [T-CIWVT-113435]

Coordinators: Prof. Dr. Oliver Thomas Stein
Organisation: KIT Department of Chemical and Process Engineering
Part of: M-CIWVT-106676 - Introduction to Numerical Simulation of Reacting Flows

Type	Credits	Grading	Version
Coursework	5 CP	pass/fail	2

Courses					
WT 25/26	2232130	Introduction to Numerical Simulation of Reacting Flows	2 SWS	Lecture / 🗣	Stein
WT 25/26	2232131	Introduction to Numerical Simulation of Reacting Flows - Exercises	2 SWS	Practice / 🗣	Stein

Legend: 🖥 Online, ⚄ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The learning control is a completed coursework: Reports on the tutorials documenting the processed task, the data generated and their analysis.

Prerequisites
None

T 6.365 Module component: Introduction to Operations Research I and II [T-WIWI-102758]

Coordinators: Prof. Dr. Stefan Nickel
Prof. Dr. Steffen Rebennack
Prof. Dr. Oliver Stein

Organisation: KIT Department of Business and Economics

Part of: M-WIWI-104899 - Operations Research

Type	Credits	Grading	Term offered	Version
Written examination	9 CP	graded	see Annotations	2

Courses					
WT 25/26	2500030	Computer Exercises on Introduction to Operations Research II	1 SWS	Tutorial / Online	Dunke
WT 25/26	2530043	Introduction to Operations Reseasrch II	2 SWS	Lecture / On-Site	Stein
WT 25/26	2530044		2 SWS	Tutorial / On-Site	Dunke
ST 2026	2500008	Computer Exercises on Introduction to Operations Research I	1 SWS	Tutorial / Online	Dunke
ST 2026	2550040	Introduction to Operations Research I	2 SWS	Lecture / On-Site	Rebennack
ST 2026	2550043	Tutorials on Introduction to Operations Research I	2 SWS	Tutorial / On-Site	Dunke
WT 26/27	2500030	Computer Exercises on Introduction to Operations Research II	1 SWS	Tutorial / Online	Dunke
WT 26/27	2530043	Introduction to Operations Reseasrch II	2 SWS	Lecture / Blended	Rebennack
WT 26/27	2530044		2 SWS	Tutorial / On-Site	Dunke

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
The assessment of the module is carried out by a written examination (120 minutes) according to Section 4(2), 1 of the examination regulation.
In each term (usually in March and August), one examination is held for both courses.
The overall grade of the module is the grade of the written examination.

Prerequisites
None

Recommendations
Knowledge of Mathematics I and II is recommended, as well as programming knowledge for the software laboratory.
It is strongly recommended to attend the course Introduction to Operations Research I
[2550040] before attending the courseIntroduction to Operations Research II
[2530043].

Workload
270 hours

Below you will find excerpts from courses related to this module component:

V Introduction to Operations Reseasrch II

2530043, WS 25/26, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content
Integer and combinatorial optimization: basic concepts, cutting plane methods, branch-and-bound methods, branch-and-cut methods, heuristic methods.

Nonlinear optimization: basic concepts, optimality conditions, solution methods for convex and nonconvex optimization problems.

Dynamic and stochastic models and methods: Dynamic optimization, Bellman methods, lot-sizing models and dynamic and stochastic models of inventory, queues.

Learning Objectives:

The student

- knows and describes the basic concepts of integer and combinatorial optimization, nonlinear optimization and dynamic optimization,
- knows the methods and models indispensable for a quantitative analysis,
- models and classifies optimization problems and selects appropriate solution procedures to solve simple optimization problems independently,
- validates, illustrates and interprets obtained solutions.

Literature

- Nickel, Rebennack, Stein, Waldmann: Operations Research, 3. Auflage, Springer, 2022
- Hillier, Lieberman: Introduction to Operations Research, 8th edition. McGraw-Hill, 2005
- Murty: Operations Research. Prentice-Hall, 1995
- Neumann, Morlock: Operations Research, 2. Auflage. Hanser, 2006
- Winston: Operations Research - Applications and Algorithms, 4th edition. PWS-Kent, 2004

V **Introduction to Operations Research I** **Lecture (V)** **On-Site**
2550040, SS 2026, 2 SWS, Language: German, Open in study portal

Content
Examples for typical OR problems.

Linear Programming: Basic notions, simplex method, duality, special versions of the simplex method (dual simplex method, three phase method), sensitivity analysis, parametric optimization, game theory.

Graphs and Networks: Basic notions of graph theory, shortest paths in networks, project scheduling, maximal and minimal cost flows in networks.

Learning objectives:

The student

- names and describes basic notions of linear programming as well as graphs and networks,
- knows the indispensable methods and models for quantitative analysis,
- models and classifies optimization problems and chooses the appropriate solution methods to solve optimization problems independently,
- validates, illustrates and interprets the obtained solutions.

Literature

- Nickel, Rebennack, Stein, Waldmann: Operations Research, 3. Auflage, Springer, 2022
- Hillier, Lieberman: Introduction to Operations Research, 8th edition. McGraw-Hill, 2005
- Murty: Operations Research. Prentice-Hall, 1995
- Neumann, Morlock: Operations Research, 2. Auflage. Hanser, 2006
- Winston: Operations Research - Applications and Algorithms, 4th edition. PWS-Kent, 2004

V **Introduction to Operations Reseasrch II** **Lecture (V)** **Blended (On-Site/Online)**
2530043, WS 26/27, 2 SWS, Language: German, Open in study portal

Content
Integer and combinatorial optimization: basic concepts, cutting plane methods, branch-and-bound methods, branch-and-cut methods, heuristic methods.

Nonlinear optimization: basic concepts, optimality conditions, solution methods for convex and nonconvex optimization problems.

Dynamic and stochastic models and methods: Dynamic optimization, Bellman methods, lot-sizing models and dynamic and stochastic models of inventory, queues.

Learning Objectives:

The student

- knows and describes the basic concepts of integer and combinatorial optimization, nonlinear optimization and dynamic optimization,
- knows the methods and models indispensable for a quantitative analysis,
- models and classifies optimization problems and selects appropriate solution procedures to solve simple optimization problems independently,
- validates, illustrates and interprets obtained solutions.

Literature

- Nickel, Rebennack, Stein, Waldmann: Operations Research, 3. Auflage, Springer, 2022
- Hillier, Lieberman: Introduction to Operations Research, 8th edition. McGraw-Hill, 2005
- Murty: Operations Research. Prentice-Hall, 1995
- Neumann, Morlock: Operations Research, 2. Auflage. Hanser, 2006
- Winston: Operations Research - Applications and Algorithms, 4th edition. PWS-Kent, 2004

T 6.366 Module component: Introduction to Programming with Java [T-WIWI-102735]

Coordinators: Prof. Dr.-Ing. Johann Marius Zöllner
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104901 - Informatics (KIT-Department of Economics and Management)

Type	Credits	Grading	Term offered	Version
Written examination	5 CP	graded	Each winter term	2

Courses					
WT 25/26	2511000	Introduction to Programming with Java	3 SWS	Lecture / 🗣	Zöllner
WT 25/26	2511002	Tutorien zu Programmieren I: Java	1 SWS	Tutorial	Zöllner, Stegmaier, Boualili
WT 25/26	2511003	Computer lab Introduction to Programming with Java	2 SWS		Zöllner, Stegmaier, Boualili

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, × Cancelled

Assessment
The assessment consists of a written resp. computer-based exam (60 min) according to Section 4 (2),1 of the examination regulation.

The successful completion of the compulsory tests in the computer lab is prerequisited for admission to the written resp. computer-based exam.

The examination takes place every semester. Re-examinations are offered at every ordinary examination date.

Additional Information
see german version

Below you will find excerpts from courses related to this module component:

V Introduction to Programming with Java
2511000, WS 25/26, 3 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content
The lecture "Introduction to Programming with Java " introduces systematic programming and provides essential practical basics for all advanced computer science lectures.

Based on considerations of the structured and systematic design of algorithms, the most important constructs of modern higher programming languages as well as programming methods are explained and illustrated with examples. One focus of the lecture is on teaching the concepts of object-oriented Programming. Java is used as the programming language. Knowledge of this language is required in advanced computer science lectures.

At the end of the lecture period, a written examination will be held for which admission must be granted during the semester after successful participation in the practices. The exact details will be announced in the lecture.

Learning objectives:

- Knowledge of the fundamentals, methods and systems of computer science.
- The students acquire the ability to independently solve algorithmic problems in the programming language Java, which dominates in business applications.
- In doing so, they will be able to find strategic and creative answers in finding solutions to well-defined, concrete and abstract problems.

Workload:

The total workload for this course is approximately 150 hours. For further information see German version.

Literature
Ratz, D. Schulmeister-Zimolong, D. Seese, J. Wiesenberger. Grundkurs Programmieren in Java. 8. Aktualisierte und erweiterte Auflage, Hanser 2018

T 6.367 Module component: Introduction to Public Finance [T-WIWI-102877]

Coordinators: Prof. Dr. Berthold Wigger
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104908 - Economics

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each winter term	1

Courses					
WT 25/26	2560131	Introduction to Public Finance	3 SWS	Lecture /	Wigger

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
Success is assessed in the form of a written examination (60 minutes). The examination is offered every semester and can be repeated at any regular examination date.

Prerequisites
None

Below you will find excerpts from courses related to this module component:

V Introduction to Public Finance

2560131, WS 25/26, 3 SWS, Language: German, Open in study portal

Lecture (V)
Blended (On-Site/Online)

Content
The course *Introduction to Public Finance* provides an overview of the fundamental issues in public economics. The first part of the course deals with normative theories about the economic role of the state in a market economy. Welfare economics theory is offered as a base model, with which alternative normative theories are compared and contrasted. Within this theoretical framework, arguments concerning efficiency and equity are developed as justification for varying degrees of economic intervention by the state. The second part of the course deals with the positivist theory of public economics. Processes of public decision making are examined and the conditions that lead to market failures resulting from collective action problems are discussed. The third part of the course examines a variety of public spending programs, including social security systems, the public education system, and programs aimed at reducing poverty. The fifth part of the course addresses the key theoretical and political issues associated with fiscal federalism.

Learning goals:

Students are able to:

- critically assess the economic role of the state in a market economy
- explain and discuss key concepts in public finance, including: public goods; economic externalities; and market failure
- explain and critically discuss competing theoretical approaches to public finance, including welfare economics and public choice theory
- explain the theory of bureaucracy according to Weber and critically assess its strengths and weaknesses
- evaluate the incentives inherent in the bureaucratic model, as well as the more recent introduction of market-oriented incentives associated with public-sector reform

Workload:

The total workload for this course is approximately 135.0 hours. For further information see German version.

Literature
Literatur:

Wigger, B. U. 2006. *Grundzüge der Finanzwissenschaft.* Springer: Berlin.

T 6.368 Module component: Introduction to Quantum Computing (IQC) [T-INFO-112344]

Coordinators: Prof. Dr. Bernhard Beckert
Prof. Dr.-Ing. Ina Schaefer
Organisation: KIT Department of Informatics
Part of: M-WIWI-104909 - Informatics (Department of Informatics)

Type	Credits	Grading	Term offered	Version
Written examination	3 CP	graded	Each winter term	1

Courses					
WT 25/26	2400073	Introduction to Quantum Computing	2 SWS	Lecture / 🗣	Schaefer
WT 26/27	2400073	Introduction to Quantum Computing	2 SWS	Lecture / 🗣	Schaefer

Legend: 🖥 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

T 6.369 Module component: Introduction to Stochastic Optimization [T-WIWI-106546]

Coordinators: Prof. Dr. Steffen Rebennack
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-102832 - Operations Research in Supply Chain Management
M-WIWI-103289 - Stochastic Optimization
M-WIWI-104899 - Operations Research

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each summer term	3

Courses					
ST 2026	2550470	Introduction to Stochastic Optimization	2 SWS	Lecture / 🖥	Rebennack
ST 2026	2550471	Übung zur Einführung in die Stochastische Optimierung	1 SWS	Practice / 🗣	Rebennack, Kandora
ST 2026	2550474	Rechnerübung zur Einführung in die Stochastische Optimierung	2 SWS	Others	Rebennack, Kandora

Legend: 🖥 Online, Blended (On-Site/Online), 🗣 On-Site, × Cancelled

Assessment
The assessment consists of a written exam (60 minutes). The exam takes place in every semester.

Prerequisites
None.

Workload
135 hours

T 6.370 Module component: Investments [T-WIWI-102604]

Coordinators: Prof. Dr. Marliese Uhrig-Homburg
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each summer term	1

Courses					
ST 2026	2530575	Investments	2 SWS	Lecture / 🗣	Uhrig-Homburg, Müller
ST 2026	2530576	Übung zu Investments	1 SWS	Practice / 🗣	Uhrig-Homburg, Kargus, Müller

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success is assessed in the form of a written examination (60 minutes). The examination is offered every semester and can be repeated at any regular examination date.

A bonus can be earned for successful participation in the exercises by submitting correct solutions to at least 50% of the bonus exercises set. If the grade of the written examination is between 4.0 and 1.3, the bonus improves the grade by up to one grade level (0.3 or 0.4). Details will be announced in the lecture.

Prerequisites
None

Recommendations
Knowledge of Business Administration: Finance and Accounting [2610026] is recommended.

Below you will find excerpts from courses related to this module component:

V Investments

2530575, SS 2026, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Literature
Weiterführende Literatur:

Bodie/Kane/Marcus (2010): Essentials of Investments, 8. Aufl., McGraw-Hill Irwin, Boston

T 6.371 Module component: IoT Platform for Engineering [T-MACH-106743]

Coordinators: Prof. Dr.-Ing. Jivka Ovtcharova
Organisation: KIT Department of Mechanical Engineering

Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Examination of another type	4 CP	graded	Each term	2

Courses					
WT 25/26	2123352	IoT platform for engineering	3 SWS	Course / 🗣	Meyer, Maier
ST 2026	2123352	IoT platform for engineering	3 SWS	Project / 🗣	Meyer, Maier

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Assessment of another type (graded), Group teaching project on Industry 4.0 consisting of: Conception, implementation, accompanying documentation and final presentation.

Below you will find excerpts from courses related to this module component:

V IoT platform for engineering

2123352, WS 25/26, 3 SWS, Language: German/English, Open in study portal

Course (Ku)
On-Site

Content
Industry 4.0, IT systems for fabrication and assembly, process modelling and execution, project work in teams, practice-relevant I4.0 problems, in automation, manufacturing industry and service.

Students can

- map and analyze processes in the context of Industry 4.0 with special methods of process modelling
- collaboratively grasp practical I4.0 issues using existing hardware and software and work out solutions for a continuous improvement process in a team
- prototypically implement the self-developed solution proposal with the given IT systems and the existing hardware equipment and finally present the results

Organizational issues
Zeit und Ort siehe ILIAS

Literature
Keine / None

V IoT platform for engineering

2123352, SS 2026, 3 SWS, Language: German, Open in study portal

Project (PRO)
On-Site

Content
Industry 4.0, IT systems for fabrication and assembly, process modelling and execution, project work in teams, practice-relevant I4.0 problems, in automation, manufacturing industry and service.

Students can

- map and analyze processes in the context of Industry 4.0 with special methods of process modelling
- collaboratively grasp practical I4.0 issues using existing hardware and software and work out solutions for a continuous improvement process in a team
- prototypically implement the self-developed solution proposal with the given IT systems and the existing hardware equipment and finally present the results

Organizational issues
Zeit und Ort siehe ILIAS

Literature
Keine / None

T 6.372 Module component: IT/OT-Security Seminar [T-ETIT-113648]

Coordinators: Prof. Dr.-Ing. Mike Barth
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-106789 - IT/OT-Security Seminar

Type	Credits	Grading	Term offered	Expansion	Version
Oral examination	4 CP	graded	Each winter term	1 semesters	1

Courses					
WT 25/26	2303201	IT/OT-Security Seminar	2 SWS	Seminar /	Barth, Madsen

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
The examination takes place in the form of an oral examinationof approx. 25 min.

The module grade is the grade of the oral exam.

Prerequisites
none

T 6.373 Module component: IT-Based Road Design [T-BGU-101804]

Coordinators: Dr.-Ing. Matthias Zimmermann
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: M-BGU-101066 - Safety, Computing and Law in Highway Engineering
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Expansion	Version
Oral examination	3 CP	graded	Each term	1 semesters	1

Courses					
WT 25/26	6233901	IT-based Road Design	2 SWS	Lecture / Practice / 🗣	Zimmermann

Legend: 🖥 Online, Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
oram exam with 15 minutes

Prerequisites
None

Recommendations
None

Additional Information
None

Workload
90 hours

T 6.374 Module component: IT-Fundamentals of Logistics [T-MACH-105187]

Coordinators: Prof. Dr.-Ing. Frank Thomas
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101278 - Material Flow in Networked Logistic Systems
M-MACH-104888 - Advanced Module Logistics
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each summer term	4

Assessment
The assessment consists of an oral exam (20 min.) taking place in the recess period according to § 4 paragraph 2 Nr. 2 of the examination regulation.

Prerequisites
none

Additional Information
The course is offered in German

Workload
120 hours

T 6.375 Module component: IT-Security Management for Networked Systems [T-INFO-101323]

Coordinators: Prof. Dr. Hannes Hartenstein
Organisation: KIT Department of Informatics
Part of: M-WIWI-104909 - Informatics (Department of Informatics)

Type	Credits	Grading	Term offered	Version
Oral examination	5 CP	graded	Each winter term	1

Courses					
WT 26/27	2424149	IT-Security Management for Networked Systems	3 SWS	Lecture / Practice /	Hartenstein, Grundmann

Legend: Online, Blended (On-Site/Online), On-Site, ✕ Cancelled

T 6.376 Module component: Joint Entrepreneurship Summer School [T-WIWI-109064]

Coordinators: Prof. Dr. Orestis Terzidis
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101488 - Entrepreneurship (EnTechnon)
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Examination of another type	6 CP	graded	Irregular	1

Courses					
ST 2026	2545021	Joint Entrepreneurship School China	4 SWS	Seminar / 🗣	Kleinn, Terzidis

Legend: 🖥 Online, ⚄ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The learning control of the program (Summer School) consists of two parts:
A) **Investor Pitch**:
Based on a presentation (investor pitch) in front of a jury, the insights gained and developed during the course of the event are presented and the business idea presented. Among other things, the presentation performance of the team, the structured content and the logical consistency of the business idea are evaluated. The exact evaluation criteria will be announced in the course.

B) **Written elaboration:**
The second part of the assessment is a written report. The iterative knowledge gain of the entire event is systematically logged and can be further supplemented by the contents of the presentation. The report documents key action steps, applied methods, findings, market analyzes and interviews and prepares them in writing. The exact structure and requirements will be announced in the course.

The grade consists of 50% presentation performance and 50% written preparation. The points system for the assessment is determined by the lecturer of the course. It will be announced at the beginning of the course.

Prerequisites
The Summer School is aimed at master students of KIT. Prerequisite is the participation in the selection process.

Recommendations
We recommend basic business knowledge, the lecture Entrepreneurship as well as openness and interest in intercultural exchange. Solid knowledge of the English language is an advantage.

Additional Information
The working language during the Summer School is English. A one-week stay in China is part of the Summer School.

Below you will find excerpts from courses related to this module component:

V Joint Entrepreneurship School China

2545021, SS 2026, 4 SWS, Language: English, Open in study portal

Seminar (S)
On-Site

Content
During the Summer School in Shanghai and Karlsruhe, students develop a business model of technologies and patents developed at KIT in workshops in German-Chinese tandems over the period of two weeks.

Click on our website for detailed information and a video: https://etm.entechnon.kit.edu/english/1095.php

Organizational issues
Dates:

- Briefing: April / May
- Karlsruhe: Presumably: July /August 2026
- Shanghai: Presumably: September 2026
- Deliverables: November 2026

T 6.377 Module component: KD²Lab Hands-On Research Course: New Ways and Tools in Experimental Economics [T-WIWI-111109]

Coordinators: Prof. Dr. Christof Weinhardt

Organisation: KIT Department of Business and Economics

Part of: M-WIWI-101446 - Market Engineering
M-WIWI-103118 - Data Science: Data-Driven User Modeling
M-WIWI-104080 - Designing Interactive Information Systems
M-WIWI-104900 - Business Administration
M-WIWI-105714 - Consumer Research
M-WIWI-105923 - Incentives, Interactivity & Decisions in Organizations

Type	Credits	Grading	Term offered	Expansion	Version
Examination of another type	4,5 CP	graded	Irregular	1 semesters	1

Assessment

The assessment of the success of the lecture is carried out as a non-exam assessment. Grading will be based on a continuous basis throughout the semester. The assessment consists of:

- A written paper, and
- a group presentation with subsequent discussion and question and answer session of 30 minutes.

For particularly active and constructive participation in the discussions of other papers during the final presentation, a bonus of one grade level (0.3 or 0.4) can be achieved on the passed exam. Details on the grading will be announced at the beginning of the event.

Additional Information

The number of participants is limited due to laboratory capacity and to ensure optimal supervision of the project groups. Places are allocated on the basis of preferences and suitability for the topics. Previous knowledge in the field of experimental economic research is particularly important.

The course cannot be offered in the summer semester 2024.

Workload

135 hours

T 6.378 Module component: Knowledge Discovery [T-WIWI-102666]

Coordinators: Dr.-Ing. Tobias Käfer
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101456 - Intelligent Systems and Services
M-WIWI-101472 - Informatics
M-WIWI-101628 - Emphasis in Informatics
M-WIWI-104901 - Informatics (KIT-Department of Economics and Management)
M-WIWI-105366 - Artificial Intelligence
M-WIWI-106804 - Advanced Topics in AI: Graph Neural Networks and Language Models

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each winter term	4

Courses					
WT 25/26	2511303	Knowledge Discovery, Graph Neural Networks, and Language Models	3 SWS	Lecture / Practice /	Käfer, Shao, Noullet, Kubelka

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
The Knowledge Discovery course is assessed through a written examination with a duration of 60 minutes. Depending on the number of participants, the examiner may replace the written exam with an oral examination lasting approximately 30 minutes

Prerequisites
None

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Knowledge Discovery, Graph Neural Networks, and Language Models

Lecture / Practice (VÜ)
Blended (On-Site/Online)

2511303, WS 25/26, 3 SWS, Language: English, Open in study portal

Content
This course represents an advanced, methodologically oriented course in the field of Artificial Intelligence. The course offers an in-depth examination of modern methods for knowledge extraction and processing at the interface of graph-based machine learning and natural language processing.

In the area of text-based methods, the lecture focuses on large language models (LLMs) such as ChatGPT, LLAMA, and DeepSeek, examining their underlying principles, training methods, and applications. Additional language processing techniques from text mining are also covered extensively.

In the area of graph-based methods, the lecture is dedicated to graph representation learning, which focuses on forming useful representations of graph data. The mathematical foundations of graph and geometric deep learning are covered, and applications in areas such as explainable artificial intelligence are highlighted.

Learning obectives:

Students

- Explain the core concepts of text mining, large language models (LLMs), and graph neural networks (GNNs).
- Compare and select different neural architectures and graph models in terms of structure, performance, and interpretability.
- Implement corresponding systems in practice, including data preprocessing, model selection, and evaluation.

Workload:

- The total workload for this course is approximately 135 hours
- Time of presentness: 45 hours
- Time of preperation and postprocessing: 60 hours
- Exam and exam preperation: 30 hours

Literature

- T. Hastie, R. Tibshirani, J. Friedman. The Elements of Statistical Learning: Data Mining, Inference, and Prediction (http://www-stat.stanford.edu/~tibs/ElemStatLearn/)
- T. Mitchell. Machine Learning. 1997
- M. Berhold, D. Hand (eds). Intelligent Data Analysis - An Introduction. 2003
- P. Tan, M. Steinbach, V. Kumar: Introduction to Data Mining, 2005, Addison Wesley

T 6.379 Module component: Knowledge-Driven Artificial Intelligence [T-WIWI-114863]

Coordinators: Prof. Dr. Harald Sack
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101456 - Intelligent Systems and Services
M-WIWI-101472 - Informatics
M-WIWI-101628 - Emphasis in Informatics
M-WIWI-104901 - Informatics (KIT-Department of Economics and Management)

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each summer term	1

Courses					
ST 2026	2511606	Knowledge-driven Artificial Intelligence	2 SWS	Lecture / 🗣	Sack, Malakzadeh Mahani, Patronilo, Nahani, Yargan, Gader
ST 2026	2511607	Exercises to Knowledge-driven Artificial Intelligence	1 SWS	Practice / 🗣	Sack, Tietz, Malekzadeh Mahani, Ondraszek

Legend: 📱 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment of this course is a written examination (60 min) according to §4(2), 1 of the examination regulation or an oral exam (20 min) following §4, Abs. 2, 2 of the examination regulation.

The exam takes place every semester and can be repeated at every regular examination date.

Prerequisites
None

Workload
150 hours

Below you will find excerpts from courses related to this module component:

V Knowledge-driven Artificial Intelligence

2511606, SS 2026, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content
- The Art of Understanding

- From Numbers to Insights
- Data, Information, and Knowledge
- Natural Language and Successful Communication
- The Art of Understanding
- The Principles of Learning

- Basic Machine Learning

- Machine Learning Fundamentals
- How to evaluate a Machine Learning Experiment?
- Evaluation and Generalisation Problems
- Supervised and Unsupervised ML
- Basic ML Algorithms
- Linear Regression
- Decision Trees
- k-means Clustering
- Neural Networks and Deep Learning

- Natural Language Processing

- NLP and Basic Linguistic Knowledge
- NLP Applications, Techniques and Challenges
- Regular Expressions, Tokenisation and Word Normalisation
- Statistical Language Models (N-Gram Model)
- Naive Bayes Text Classification
- Distributional Semantics and Neural Language Models
- Word Embeddings

- Knowledge Graphs

- Knowledge Representations and Ontologies
- Resource Description Framework (RDF)
- Modeling with RDFS
- Querying RDF(S) with SPARQL
- Popular Knowledge Graphs - Wikidata and DBpedia
- Ontologies with the Web Ontology Language (OWL)
- Linked Data Quality Assurance with SHACL
- The Graph in Knowledge Graphs

- Neurosymbolic AI

- Symbolic and Subsymbolic AI
- Knowledge Graph Embeddings and KG Completion
- The Limits of AI
- KDAI Master Thesis Topics

Learning objectives:

- The students know the fundamentals and measures of information theory and are able to apply those in the context of Knowledge-driven AI.
- The students have basic skills of natural language processing and are enabled to apply natural language processing technology to solve and evaluate simple text analysis tasks.
- The students have fundamental skills of knowledge representation with ontologies as well as basic knowledge of Semantic Web and Linked Data technologies. The students are able to apply these skills for simple representation and analysis tasks.
- The students know the fundamentals of basic machine learning technologies. The students are able to apply these skills for simple prediction and classification tasks.
- The students know the basic principles of neurosymbolic AI.

Literature

- Machine Learning:
  G. Rebala, A. Ravi, S. Churiwala, An Introduction to Machine Learning, Springer, 2019. (available via KIT network)
- Natural Language Processing:
  D. Jurafsky, J.H. Martin, Speech and Language Processing, 3rd ed. Draft, 2023.
- Knowledge Graphs:
  A. Hogan, The Web of Data, Springer, 2020. (available via KIT network)

T 6.380 Module component: Lab Practice Sessions in R for Statistics 1 and 2 [T-WIWI-111941]

Coordinators: Prof. Dr. Melanie Schienle
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104902 - Statistics

Type	Credits	Grading	Term offered	Version
Coursework (practical)	2 CP	pass/fail	Each winter term	1

Courses					
WT 25/26	2610022	PC-Praktikum zu Statistik II	2 SWS		Krüger, Koster
ST 2026	2600010	PC-Praktikum zu Statistik I	2 SWS	Block / Online	Grothe, Becker

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
The assessment consists of two projects - one for Statistic 1 and one for Statistic 2. Both must be passed to pass the course.

Workload
60 hours

T 6.381 Module component: Laboratory Laser Materials Processing [T-MACH-102154]

Coordinators: Dr.-Ing. Johannes Schneider
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101268 - Specific Topics in Materials Science
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Coursework	4 CP	pass/fail	Each term	2

Courses					
WT 25/26	2183640	Laboratory "Laser Materials Processing"	3 SWS	Practical course /	Schneider, Pfleging
ST 2026	2183640	Laboratory "Laser Materials Processing"	3 SWS	Practical course /	Schneider, Pfleging

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
The assessment consists of a colloquium for every single experiment and an overall final colloquium incl. an oral presentation of 20 min.

Prerequisites
None

Recommendations
Basic knowledge of physics, chemistry and material science is assumed.

Additional Information
The course is offered in German.

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Laboratory "Laser Materials Processing"

2183640, WS 25/26, 3 SWS, Language: German, Open in study portal

Practical course (P)
Blended (On-Site/Online)

Content
The laboratory compromises 8 half-day experiments, which address the following laser processing topics of metals, ceramics and polymers:

• safety aspects

• surface hardening and remelting

• melt and reactive cutting

• surface modification by dispersing or alloying

• welding

• surface texturing

• metrology

There are used CO2-, excimer-, Nd:YAG- and high power diode-laser sources within the laboratory.

The student

- can describe the influence of laser, material and process parameters and can choose suitable parameters for the most important methods of laser-based processing in automotive engineering.
- can explain the requirements for safe handling of laser radiation and for the design of safe laser systems.

Basic knowledge of physics, chemistry and material science is assumed.

The attendance to one of the courses Physical Basics of Laser Technology (2181612) or Laser Application in Automotive Engineering (2182642) is strongly recommended.

regular attendance: 34 hours
self-study: 86 hours

The assessment consists of a colloquium for every single experiment and an overall final colloquium incl. an oral presentation of 20 min.

Organizational issues
Maximal 16 Teilnehmer/innen!
Es sind nur noch wenige Plätze frei (Stand 31.05.2025)! Registrierung für die Nachrückliste möglich per Email an johannes.schneider@kit.edu
Praktikum findet in Kleingruppen semesterbegleitend (dienstags bzw. mittwochs, halbtägig) auf dem Campus Nord am IAM-AWP (Geb. 681) und auf dem Campus Süd am IAM-CMS (Geb. 30.48) statt!

Termine werden mit den Teilnehmern/innen direkt abgestimmt.

Literature

F. K. Kneubühl, M. W. Sigrist: Laser, 2008, Vieweg+Teubner

T. Graf: Laser - Grundlagen der Laserstrahlquellen, 2009, Vieweg-Teubner Verlag

R. Poprawe: Lasertechnik für die Fertigung, 2005, Springer

H. Hügel, T. Graf: Laser in der Fertigung, 2009, Vieweg+Teubner

J. Eichler, H.-J. Eichler: Laser - Bauformen, Strahlführung, Anwendungen, 2006, Springer

V **Laboratory "Laser Materials Processing"**
2183640, SS 2026, 3 SWS, Language: German, Open in study portal

Practical course (P)
Blended (On-Site/Online)

Content
The laboratory compromises 8 half-day experiments, which address the following laser processing topics of metals, ceramics and polymers:

• safety aspects

• surface hardening and remelting

• melt and reactive cutting

• surface modification by dispersing or alloying

• welding

• surface texturing

• metrology

There are used CO2-, excimer-, Nd:YAG- and high power diode-laser sources within the laboratory.

The student

- can describe the influence of laser, material and process parameters and can choose suitable parameters for the most important methods of laser-based processing in automotive engineering.
- can explain the requirements for safe handling of laser radiation and for the design of safe laser systems.

Basic knowledge of physics, chemistry and material science is assumed.

The attendance to one of the courses Physical Basics of Laser Technology (2181612) or Laser Application in Automotive Engineering (2182642) is strongly recommended.

regular attendance: 34 hours
self-study: 86 hours

The assessment consists of a colloquium for every single experiment and an overall final colloquium incl. an oral presentation of 20 min.

Organizational issues
Anmeldung (Aktuell nur noch für die Nachrückliste) per Email an johannes.schneider@kit.edu

Das Praktikum findet semesterbegleitend in 4 Kleingruppen (Di-Vormittag, Di-Nachmittag, Mi-Vormittag, Mi-Nachmittag) am IAM-ZM (CS) bzw. IAM-AWP (CN) statt!

Die Termine werden zu Beginn des Semesters bekannt gegeben.

Literature
F. K. Kneubühl, M. W. Sigrist: Laser, 2008, Vieweg+Teubner

T. Graf: Laser - Grundlagen der Laserstrahlquellen, 2009, Vieweg-Teubner Verlag

R. Poprawe: Lasertechnik für die Fertigung, 2005, Springer

H. Hügel, T. Graf: Laser in der Fertigung, 2009, Vieweg+Teubner

J. Eichler, H.-J. Eichler: Laser - Bauformen, Strahlführung, Anwendungen, 2006, Springer

W.T. Silfvast: Laser Fundamentals, 2008, Cambrigde University Press

W.M. Steen: Laser Materials Processing, 2010, Springer

T 6.382 Module component: Laboratory Production Metrology [T-MACH-108878]

Coordinators: Dr.-Ing. Martin Benfer
Prof. Dr.-Ing. Gisela Lanza

Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-106590 - Production Engineering
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Examination of another type	4 CP	graded	Each summer term	5

Courses					
ST 2026	2150550	Laboratory Production Metrology	3 SWS	Practical course / 🗣	Lanza, Benfer

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Alternative Test Achievement: Group presentation of 15 min at the beginning of each experiment and evaluation of the participation during the experiments

and

Oral Exam (15 min)

Prerequisites
T-MACH-114827 must not have been started.

Additional Information
The course is offered in German.

For organizational reasons the number of participants for the course is limited. Hence al selection process will take place. Applications are made via the homepage of wbk (http://www.wbk.kit.edu/studium-und-lehre.php).

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Laboratory Production Metrology

Practical course (P)
On-Site

2150550, SS 2026, 3 SWS, Language: German, Open in study portal

Content
During this course, students get to know measurement systems that are used in a production system. In the age of Industry 4.0, sensors are becoming more important. Therefore, the application of in-line measurement technology such as machine vision and non-destructive testing is focussed. Additionally, laboratory based measurement technologies such as computed tomography are addressed. The students learn the theoretical background as well as practical applications for industrial examples. The students use sensors by themselves during the course. Additionally, they are trained on how to integrate sensors in production processes and how to analyze measurement data with suitable software.

The following topics are addressed:

- Classification and examples for different measurement technologies in a production environment
- Machine vision with optical sensors
- Information fusion based on optical measurements
- Robot-based optical measurements
- Non-destructive testing by means of acoustic measurements
- Coodinate measurement technology
- Industrial computed tomography
- Measurement uncertainty evaluation
- Analysis of production data by means of data mining

Learning Outcomes:

The students …

- are able to name, describe and mark out different measurement technologies that are relevant in a production environment.
- are able to conduct measurements with the presented in-line and laboratory based measurement systems.
- are able to analyze measurement results and asses the measurement uncertainty of these.
- are able to deduce whether a work piece fulfills quality relevant specifications by analysing measurement results.
- are able to use the presented measurement technologies for a new task.

Workload:

regular attendance: 31,5 hours
self-study: 88,5 hours

Organizational issues
Aus organisatorischen Gründen ist die Teilnehmerzahl für die Lehrveranstaltung begrenzt. Infolgedessen wird ein Auswahlprozess stattfinden. Die Bewerbung erfolgt über die Homepage des wbk (http://www.wbk.kit.edu/studium-und-lehre.php).

For organizational reasons the number of participants for the course is limited. Hence a selection process will take place. Applications are made via the homepage of wbk (KIT - wbk Institute of Production Science Education).

Literature
Skript zur Veranstaltung wird über (https://ilias.studium.kit.edu/) bereitgestellt. Ebenso wird auf gängie Fachliteratur verwiesen.

Lecture notes will be provided in Ilias (https://ilias.studium.kit.edu/). Additional reference to literature will be provided, as well.

T 6.383 Module component: Laboratory Work Water Chemistry [T-CIWVT-103351]

Coordinators: Dr. Gudrun Abbt-Braun
Prof. Dr. Harald Horn

Organisation: KIT Department of Chemical and Process Engineering

Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Version
Examination of another type	4 CP	graded	1

Courses					
WT 25/26	2233032	Practical Course: Water Quality and Water Assessment	2 SWS	Practical course / 🗣	Horn, Hille-Reichel, und Mitarbeitende

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Prerequisites
none

T 6.384 Module component: Large Diesel and Gas Engines for Ship Propulsions [T-MACH-110816]

Coordinators: Dr. German Weisser
Organisation:
Part of: M-MACH-101303 - Combustion Engines II

Type	Credits	Grading	Term offered	Expansion	Version
Oral examination	4 CP	graded	Each summer term	1 semesters	1

Courses					
ST 2026	2134154	Large Diesel and Gas Engines for Ship Propulsions	2 SWS	Lecture / 🗣	Weisser

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
oral exam, 20 minutes

Prerequisites
None

Additional Information
This course is offered in German.

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Large Diesel and Gas Engines for Ship Propulsions

Lecture (V)
On-Site

2134154, SS 2026, 2 SWS, Language: German, Open in study portal

T 6.385 Module component: Large-scale Optimization [T-WIWI-106549]

Coordinators: Prof. Dr. Steffen Rebennack
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101473 - Mathematical Programming
M-WIWI-102832 - Operations Research in Supply Chain Management
M-WIWI-103289 - Stochastic Optimization
M-WIWI-104899 - Operations Research

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each summer term	3

Assessment
The assessment consists of a written exam (60 minutes). The exam takes place in every semester.

Prerequisites
None.

Workload
135 hours

T 6.386 Module component: Laser in Automotive Engineering [T-MACH-105164]

Coordinators: Dr.-Ing. Johannes Schneider
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101268 - Specific Topics in Materials Science

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each summer term	3

Courses					
ST 2026	2182642	Laser Material Processing	2 SWS	Lecture / 🗣	Schneider

Legend: 🖥 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
oral examination (30 min)

no tools or reference materials

Prerequisites
It is not possible, to combine this module component with Laser Material Processing [T-MACH-112763], Physical Basics of Laser Technology [T-MACH-109084] and Physical Basics of Laser Technology [T-MACH-102102]

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-MACH-102102 - Physical Basics of Laser Technology must not have been started.
2. The module component T-MACH-112763 - Laser Material Processing must not have been started.

Recommendations
preliminary knowlegde in mathematics, physics and materials science

Additional Information
The course is offered in German.

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Laser Material Processing

2182642, SS 2026, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content
Based on a short description of the physical basics of laser technology the lecture reviews the most important high power lasers and their various applications in automotive engineering. Furthermore the application of laser light in metrology and safety aspects will be addressed.

- physical basics of laser technology
- laser beam sources (Nd:YAG-, CO2-, high power diode-laser)
- beam properties, guiding and shaping
- basics of materials processing with lasers
- laser applications in material processing
- savety aspects

The student

- can explain the principles of light generation, the conditions for light amplification as well as the basic structure and function of Nd:YAG-, CO2- and high power diode-laser sources.
- can describe the most important methods of laser-based processing in automotive engineering and illustrate the influence of laser, material and process parameters
- can analyse manufacturing problems and is able to choose a suitable laser source and process parameters.
- can explain the requirements for safe handling of laser radiation and for the design of safe laser systems.

Basic knowledge of physics, chemistry and material science is assumed.

It is not possible, to combine this lecture with the lecture *Physical basics of laser technology* [2181612].

regular attendance: 22,5 hours
self-study: 97,5 hours

oral examination (ca. 30 min)

no tools or reference materials

Organizational issues
Die Vorlesung ersetzt die bisherige Vorlesung "Lasereinsatz im Automobilbau" und wird jetzt auf Englisch angeboten!

The lecture replaces the previous lecture "Laser Application in Automotive Engineering" and is now offered in English!

Literature
W. T. Silvast: Laser Fundamentals, 2004, Cambridge University Press

J. Eichler, H.-J. Eichler: Laser - Basics, Advances, Applications, 2018, Springer

P. Poprawe: Tailored Light 1, 2018, Springer

K. F. Renk: Basics of Laser Physics, 2017, Springer

M. W. Sigrist: Laser: Theorie, Typen und Anwendungen, 2018, Springer-Spektrum

H. Hügel, T. Graf: Materialbearbeitung mit Laser, 2022, Springer Vieweg

T. Graf: Laser - Grundlagen der Laserstrahlquellen, 2009, Vieweg-Teubner Verlag

R. Poprawe: Lasertechnik für die Fertigung, 2005, Springer

T 6.387 Module component: Laser Material Processing [T-MACH-112763]

Coordinators: Dr.-Ing. Johannes Schneider
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101268 - Specific Topics in Materials Science
M-WIWI-104907 - Engineering Sciences
M-WIWI-107718 - Mechanical Engineering I
M-WIWI-107730 - Mechanical Engineering II

Type	Credits	Grading	Term offered	Version
Oral examination	4,5 CP	graded	Each summer term	2

Courses					
ST 2026	2182642	Laser Material Processing	2 SWS	Lecture / 🗣	Schneider

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, × Cancelled

Assessment
oral examination (30 min)

no tools or reference materials

Prerequisites
It is not possible, to combine this module component with Laser in Automotive Engineering [T-MACH-105164], module component Physical Basics of Laser Technology [T-MACH-109084] and module component Physical Basics of Laser Technology [T-MACH-102102].

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-MACH-102102 - Physical Basics of Laser Technology must not have been started.
2. The module component T-MACH-105164 - Laser in Automotive Engineering must not have been started.

Recommendations
preliminary knowlegde in mathematics, physics and materials science

Additional Information
The course is offered in English.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Laser Material Processing

2182642, SS 2026, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content
Based on a short description of the physical basics of laser technology the lecture reviews the most important high power lasers and their various applications in automotive engineering. Furthermore the application of laser light in metrology and safety aspects will be addressed.

- physical basics of laser technology
- laser beam sources (Nd:YAG-, CO2-, high power diode-laser)
- beam properties, guiding and shaping
- basics of materials processing with lasers
- laser applications in material processing
- savety aspects

The student

- can explain the principles of light generation, the conditions for light amplification as well as the basic structure and function of Nd:YAG-, CO2- and high power diode-laser sources.
- can describe the most important methods of laser-based processing in automotive engineering and illustrate the influence of laser, material and process parameters
- can analyse manufacturing problems and is able to choose a suitable laser source and process parameters.
- can explain the requirements for safe handling of laser radiation and for the design of safe laser systems.

Basic knowledge of physics, chemistry and material science is assumed.

It is not possible, to combine this lecture with the lecture *Physical basics of laser technology* [2181612].

regular attendance: 22,5 hours
self-study: 97,5 hours

oral examination (ca. 30 min)

no tools or reference materials

Organizational issues
Die Vorlesung ersetzt die bisherige Vorlesung "Lasereinsatz im Automobilbau" und wird jetzt auf Englisch angeboten!

The lecture replaces the previous lecture "Laser Application in Automotive Engineering" and is now offered in English!

Literature
W. T. Silvast: Laser Fundamentals, 2004, Cambridge University Press

J. Eichler, H.-J. Eichler: Laser - Basics, Advances, Applications, 2018, Springer

P. Poprawe: Tailored Light 1, 2018, Springer

K. F. Renk: Basics of Laser Physics, 2017, Springer

M. W. Sigrist: Laser: Theorie, Typen und Anwendungen, 2018, Springer-Spektrum

H. Hügel, T. Graf: Materialbearbeitung mit Laser, 2022, Springer Vieweg

T. Graf: Laser - Grundlagen der Laserstrahlquellen, 2009, Vieweg-Teubner Verlag

R. Poprawe: Lasertechnik für die Fertigung, 2005, Springer

T 6.388 Module component: Laser Metrology [T-ETIT-100643]

Coordinators: Prof. Dr. Marc Eichhorn
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-100434 - Laser Metrology

Type	Credits	Grading	Term offered	Version
Oral examination	3 CP	graded	Each summer term	1

Courses					
ST 2026	2303200	Laser Metrology	2 SWS	Lecture / 🗣	Eichhorn

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, x Cancelled

Assessment
The exam will be taken as an oral examination (about 20 minutes). The individual appointments for examination are offered at two previously determined dates.

Prerequisites
none

Below you will find excerpts from courses related to this module component:

V Laser Metrology

2303200, SS 2026, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content
Current time schedule can be found in ILIAS

Organizational issues
Beginn am Do. 30. April 2026, 9:45 - 13:15

Seminarraum IRS, Raum 119 Geb. 30.33.

Weitere Details werden in ILIAS bekannt gegeben. Prüfungen werden ebenfalls über ILIAS organisiert

Starting on Thursday, 30th April, 9:45 - 13:15

Room 119, Building 30.33

Further details are annouced in ILIAS. Exam registration will also be organised via ILIAS.

T 6.389 Module component: Laser Physics [T-ETIT-100741]

Coordinators: Prof. Dr. Marc Eichhorn
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-MACH-101292 - Microoptics
M-MACH-101295 - Optoelectronics and Optical Communication
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each winter term	1

Courses					
WT 25/26	2301480	Laserphysics	2 SWS	Lecture / 🗣	Eichhorn
WT 25/26	2301481	Exercise for 2301480 Laserphysics	1 SWS	Practice / 🗣	Eichhorn

Legend: 🖥 Online, ❀ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The exam will be taken as an oral examination (about 20 minutes). The individual appointments for examination are offered regularly at two previously determined dates.

Prerequisites
none

T 6.390 Module component: Laser-Assisted Methods and Their Application for Energy Storage Materials [T-MACH-106739]

Coordinators: Prof. Wilhelm Pfleging
Organisation: KIT Department of Mechanical Engineering

Part of: M-WIWI-107718 - Mechanical Engineering I
M-WIWI-107730 - Mechanical Engineering II

Type	Credits	Grading	Term offered	Version
Oral examination	4,5 CP	graded	Each term	3

Courses					
WT 25/26	2193013	Laser-assisted methods and their application for energy storage materials	2 SWS	Lecture / On-Site	Pfleging
ST 2026	2193013	Laser-assisted methods and their application for energy storage materials	2 SWS	Lecture / Blended	Pfleging
WT 26/27	2193013	Laser-assisted methods and their application for energy storage materials	2 SWS	Lecture / On-Site	Pfleging

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
oral exam (about 30 min)

Prerequisites
none

Recommendations
Fundamentals of solid state physics and optics

Additional Information
The course is in English.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Laser-assisted methods and their application for energy storage materials

2193013, WS 25/26, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content
Registration via ILIAS or by e-mail to: pfleging@kit.edu

consulting-hour: Wednesdays after the lecture, 4 - 5 p.m.; Campus South, building 10.50, room 603.2

Oral Examination: ca. 30 min

Teaching Content:

- Optics and beam shaping
- Laser-induced plasma
- Thermal-assisted laser materials processing
- Functionalization of surfaces
- Self-organized processes
- Fundamental aspects of battery technology
- Laser processes in battery manufacturing
- Advanced concepts for high energy and high power batteries
- Laser-based post-mortem analytics

Recommendations: Basics of Solid State Physics and Optics

- Attendance in Lecture: 18 Stunden
- Extra Requirements: 98 Stunden

Laser technology is a cutting-edge field with a wide range of applications. This course covers innovative laser processes, including cutting, welding, and structuring, at the micro and nanometer scale. It also discusses different laser beam sources and their integration into battery production. The students are equipped with comprehensive tools to independently evaluate, design, and optimize a process. The laser group at KIT is the only one that provides such extensive training on the use of state-of-the-art beam sources in battery production in an application-oriented manner.

Organizational issues
You will receive the lecture material and further information via ILIAS

Literature

- Laser in der Fertigung, Grundlagen der Strahlquellen, Systeme, Fertigungsverfahren, Autoren: Hügel, Helmut, Graf, Thomas, ISBN 978-3-8348-1817-1, Springer Verlag, 2014
- Laser Processing and Chemistry, Autor: Bäuerle, Dieter W., ISBN 978-3-642-17613-5, Springer, 2011
- Handbuch Lithium-Ionen-Batterien, Korthauer, Reiner (Hrsg.), ISBN 978-3-642-30653-2, Springer Verlag, 2013
- Lithium-ion Battery Materials and Engineering, Autoren: Malgorzata K. Gulbinska, ISBN 978-1-4471-6548-4, Springer Verlag, 2014
- Laser-Induced Breakdown Spectroscopy, Theory and Applications, Autoren: Sergio Musazzi, Umberto Perini, Springer Series in Optical Sciences, ISBN 978-3-642-45084-6, 2007

V **Laser-assisted methods and their application for energy storage materials**
2193013, SS 2026, 2 SWS, Language: English, Open in study portal

Lecture (V)
Blended (On-Site/Online)

Content
Oral Examination: ca. 30 min

Teaching Content:

- Optics and beam shaping
- Laser-induced plasma
- Thermal-assisted laser materials processing
- Functionalization of surfaces
- Self-organized processes
- Fundamental aspects of battery technology
- Laser processes in battery manufacturing
- Advanced concepts for high energy and high power batteries
- Laser-based post-mortem analytics

Recommendations: Basics of Solid State Physics and Optics

- Attendance in Lecture: 18 Stunden
- Extra Requirements: 98 Stunden

The students will get an in-depth insight into the various aspects of modern laser technology and laser beam-material interactions. They will get knowledge about the use of laser radiation for functionalization of modern energy storage materials for batteries. They get used handling of scientific methods for describing the physical processes which is communicated in an application-oriented manner.

Organizational issues
The lecture will take place in building 30.28, room R220
The lecture can possibly take place online. Find out more on ILIAS.
Register if possible by April 14, 2026 by email to pfleging@kit.edu or via ILIAS.

Literature

- Laser in der Fertigung, Grundlagen der Strahlquellen, Systeme, Fertigungsverfahren, Autoren: Hügel, Helmut, Graf, Thomas, ISBN 978-3-8348-1817-1, Springer Verlag, 2014
- Laser Processing and Chemistry, Autor: Bäuerle, Dieter W., ISBN 978-3-642-17613-5, Springer, 2011
- Handbuch Lithium-Ionen-Batterien, Korthauer, Reiner (Hrsg.), ISBN 978-3-642-30653-2, Springer Verlag, 2013
- Lithium-ion Battery Materials and Engineering, Autoren: Malgorzata K. Gulbinska, ISBN 978-1-4471-6548-4, Springer Verlag, 2014
- Laser-Induced Breakdown Spectroscopy, Theory and Applications, Autoren: Sergio Musazzi, Umberto Perini, Springer Series in Optical Sciences, ISBN 978-3-642-45084-6, 2007

V **Laser-assisted methods and their application for energy storage materials** — **Lecture (V) On-Site**
2193013, WS 26/27, 2 SWS, Language: English, Open in study portal

Content
Registration via ILIAS or by e-mail to: pfleging@kit.edu

consulting-hour: Wednesdays after the lecture, 4 - 5 p.m.; Campus South, building 10.50, room 603.2

Oral Examination: ca. 30 min

Teaching Content:

- Optics and beam shaping
- Laser-induced plasma
- Thermal-assisted laser materials processing
- Functionalization of surfaces
- Self-organized processes
- Fundamental aspects of battery technology
- Laser processes in battery manufacturing
- Advanced concepts for high energy and high power batteries
- Laser-based post-mortem analytics

Recommendations: Basics of Solid State Physics and Optics

- Attendance in Lecture: 18 Stunden
- Extra Requirements: 98 Stunden

Laser technology is a cutting-edge field with a wide range of applications. This course covers innovative laser processes, including cutting, welding, and structuring, at the micro and nanometer scale. It also discusses different laser beam sources and their integration into battery production. The students are equipped with comprehensive tools to independently evaluate, design, and optimize a process. The laser group at KIT is the only one that provides such extensive training on the use of state-of-the-art beam sources in battery production in an application-oriented manner.

Organizational issues
You will receive the lecture material and further information via ILIAS

Literature

- Laser in der Fertigung, Grundlagen der Strahlquellen, Systeme, Fertigungsverfahren, Autoren: Hügel, Helmut, Graf, Thomas, ISBN 978-3-8348-1817-1, Springer Verlag, 2014
- Laser Processing and Chemistry, Autor: Bäuerle, Dieter W., ISBN 978-3-642-17613-5, Springer, 2011
- Handbuch Lithium-Ionen-Batterien, Korthauer, Reiner (Hrsg.), ISBN 978-3-642-30653-2, Springer Verlag, 2013
- Lithium-ion Battery Materials and Engineering, Autoren: Malgorzata K. Gulbinska, ISBN 978-1-4471-6548-4, Springer Verlag, 2014
- Laser-Induced Breakdown Spectroscopy, Theory and Applications, Autoren: Sergio Musazzi, Umberto Perini, Springer Series in Optical Sciences, ISBN 978-3-642-45084-6, 2007

T 6.391 Module component: Law of Contracts [T-INFO-101316]

Coordinators: Dr. Alexander Hoff
Organisation: KIT Department of Informatics
Part of: M-WIWI-104903 - Law

Type	Credits	Grading	Term offered	Version
Written examination	3 CP	graded	Each summer term	2

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-INFO-102036 - Computer Contract Law must not have been started.

T 6.392 Module component: Laws concerning Traffic and Roads [T-BGU-106615]

Coordinators: Hon.-Prof. Dr. Dietmar Hönig
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: M-BGU-101066 - Safety, Computing and Law in Highway Engineering
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Expansion	Version
Written examination	3 CP	graded	Each term	1 semesters	1

Courses					
ST 2026	6233803	Laws Concerning Traffic and Roads	2 SWS	Lecture / 🗣	Hönig

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
written exam, 60 min.

Prerequisites
None

Recommendations
None

Additional Information
None

Workload
90 hours

T 6.393 Module component: Lean Construction [T-BGU-108000]

Coordinators: Prof. Dr.-Ing. Shervin Haghsheno
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: M-BGU-101884 - Lean Management in Construction
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Expansion	Version
Written examination	4,5 CP	graded	Each term	1 semesters	1

Courses					
WT 25/26	6241901	Lean Construction	4 SWS	Lecture / Practice / 🗣	Haghsheno, Mitarbeiter/innen

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
written exam, 70 min.

Prerequisites
none

Recommendations
none

Additional Information
The number of participants is limited to 50. Registration modalities will be published in time on the institute's homepage or in the corresponding ILIAS course. Students of Profile 2 'Project Management and Lean Construction' in the degree program 'Technology and Management in Construction Management' always receive a confirmation of participation (basic module). Any necessary selection for the remaining places will be made with priority given to students from Civil Engineering (study focus module), Technology and Management in Construction Management (specialization module) and Industrial Engineering (elective module in Engineering Sciences), taking into account the degree program and the student's progress.

Workload
140 hours

Below you will find excerpts from courses related to this module component:

V Lean Construction

6241901, WS 25/26, 4 SWS, Language: German, Open in study portal

Lecture / Practice (VÜ)
On-Site

Content
At the beginning of this module, the theoretical foundations of Lean Philosophy and Lean Construction are presented and deepened through learning simulations and exercises. Subsequently, industry and specialist lectures examine methods such as takt planning and takt control, the Last Planner System™, value stream analysis, and cooperative contract forms in both theory and practice. Furthermore, general project management aspects in construction projects such as construction site logistics, cost and quality management are addressed from a Lean perspective.

As part of the project work, students work in small groups on selected problem areas, present the results, and prepare a written report. This written report, together with the group presentation, is evaluated as a partial module assessment.

Organizational issues
Die Anmeldung zum Kurs findet über das in CAS hinterlegte Anmeldeverfahren statt.

Die Teilnehmerzahl ist auf 50 beschränkt.

Erste Vorlesung: 27.10.2025. Weitere Termine: siehe zugehöriger ILIAS-Kurs

Literature
Gehbauer, F. (2013) *Lean Management Im Bauwesen*. Skript des Instituts für Technologie und Management im Baubetrieb, Karlsruher Institut für Technologie (KIT).
Liker, J. & Meier, D. (2007) *Praxisbuch, der Toyota Weg: für jedes Unternehmen*. Finanzbuch Verlag.
Rother, M., Shook, J., & Wiegand, B. (2006). *Sehen lernen: mit Wertstromdesign die Wertschöpfung erhöhen und Verschwendung beseitigen*. Lean Management Institut.

T 6.394 Module component: Learning Factory “Global Production“ [T-MACH-105783]

Coordinators: Prof. Dr.-Ing. Gisela Lanza
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-105455 - Strategic Design of Modern Production Systems
M-MACH-106590 - Production Engineering
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Examination of another type	6 CP	graded	Each winter term	5

Courses					
WT 25/26	2149612	Learning Factory “Global Production“	4 SWS	/	Lanza
WT 26/27	2149612	Learning Factory “Global Production“	4 SWS	/	Lanza

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
Alternative test achievement (graded):

- Knowledge acquisition in the context of the seminar (4 achievements approx. 20 min each) with weighting 35%.
- Interaction between participants with weighting 20%.
- Scientific colloquium (in groups of 3 students approx. 45 min each) with weighting 45%.

Prerequisites
none

Additional Information
The course is offered in German.

For organisational reasons, the number of participants for the course is limited to 20. As a result, a selection process will take place. Applications must be submitted via the wbk homepage (http://www.wbk.kit.edu/lernfabrik.php).

Due to the limited number of participants, advance registration is required.

Students should have previous knowledge in at least one of the following areas:

- Integrated Production Planning
- Global Production and Logistics
- Quality Management

Workload
180 hours

Below you will find excerpts from courses related to this module component:

V Learning Factory “Global Production“

2149612, WS 25/26, 4 SWS, Language: German, Open in study portal **Blended (On-Site/Online)**

Content

The Learning Factory Global Production serves as a modern teaching environment for the challenges of global production. These challenges are brought to life using the example of electric motor manufacturing under real production conditions. The course is divided into e-learning units and face-to-face sessions. The e-learning units serve to teach essential basics and deepen specific topics (e.g., site selection, Industry 4.0, and production network planning). The face-to-face sessions focus on the case-specific application of relevant methods for planning and controlling global production systems. The Lean & Industry 4.0 unit examines classic methods and tools for designing lean production systems (e.g., Kanban and JIT/JIS, line balancing). A Six Sigma project is used to teach essential methods for quality assurance in complex production systems and to provide practical experience. The “Agile Global Production Networks” unit examines the operation and design of global production networks. Participants are familiarized with the basics of global production and immediately apply the methods they have learned to the assembly of electric motors. The Circular Production module offers insights into the theoretical and practical complexities and solution strategies of remanufacturing.

Main focus of the lecture:

- Site selection
- Lean & Industry 4.0
- Six Sigma 4.0 – Quality management in production
- Agile global production networks
- Circular production

Learning Outcomes:

Students are able to …

- evaluate and select location alternatives using appropriate methods and procedures.
- apply lean management methods and tools to plan and control location-appropriate production systems.
- use the Six Sigma system in a targeted manner and are capable of goal-oriented process management.
- depending on company-specific circumstances, apply methods for planning global production networks, outline a suitable network, and classify and evaluate it based on specific criteria.
- apply methods of circular production and demonstrate how products can be designed for circularity.
- apply the methods and approaches learned for problem solving in a global production environment and reflect on their effectiveness.

Workload:

e-Learning: ~ 36 h
regular attendence: ~ 64 h
self-study: ~ 80 h

Organizational issues

Termine werden über die Institutshomepage bekanntgegeben.

Aus organisatorischen Gründen ist die Teilnehmerzahl für die Lehrveranstaltung auf 20 Teilnehmer begrenzt. Infolgedessen wird ein Auswahlprozess stattfinden. Die Bewerbung erfolgt über die Homepage des wbk (http://www.wbk.kit.edu/studium-und-lehre.php)

Aufgrund der begrenzten Teilnehmerzahl ist eine Voranmeldung erforderlich.

Die Studierenden sollten Vorkenntnisse in mindestens einem der folgenden Bereiche haben:

- Integrierte Produktionsplanung
- Globale Produktion und Logistik
- Qualitätsmanagement

For organisational reasons, the number of participants for the course is limited to 20. As a result, a selection process will take place. Applications must be submitted via the wbk homepage (http://www.wbk.kit.edu/studium-und-lehre.php).

Due to the limited number of participants, advance registration is required.

Students should have previous knowledge in at least one of the following areas:

- Integrated Production Planning
- Global Production and Logistics
- Quality Management

Literature

Medien:

E-Learning Plattform ilias, Powerpoint, Fotoprotokoll. Die Medien werden über ilias (https://ilias.studium.kit.edu/) bereitgestellt.

Media:

E-learning platform ilias, powerpoint, photo protocol. The media are provided through ilias (https://ilias.studium.kit.edu/).

V

Learning Factory “Global Production“

2149612, WS 26/27, 4 SWS, Language: German, Open in study portal

Blended (On-Site/Online)

Content

The Learning Factory Global Production serves as a modern teaching environment for the challenges of global production. These challenges are brought to life using the example of electric motor manufacturing under real production conditions. The course is divided into e-learning units and face-to-face sessions. The e-learning units serve to teach essential basics and deepen specific topics (e.g., site selection, Industry 4.0, and production network planning). The face-to-face sessions focus on the case-specific application of relevant methods for planning and controlling global production systems. The Lean & Industry 4.0 unit examines classic methods and tools for designing lean production systems (e.g., Kanban and JIT/JIS, line balancing). A Six Sigma project is used to teach essential methods for quality assurance in complex production systems and to provide practical experience. The “Agile Global Production Networks” unit examines the operation and design of global production networks. Participants are familiarized with the basics of global production and immediately apply the methods they have learned to the assembly of electric motors. The Circular Production module offers insights into the theoretical and practical complexities and solution strategies of remanufacturing.

Main focus of the lecture:

- Site selection
- Lean & Industry 4.0
- Six Sigma 4.0 – Quality management in production
- Agile global production networks
- Circular production

Learning Outcomes:

Students are able to …

- evaluate and select location alternatives using appropriate methods and procedures.
- apply lean management methods and tools to plan and control location-appropriate production systems.
- use the Six Sigma system in a targeted manner and are capable of goal-oriented process management.
- depending on company-specific circumstances, apply methods for planning global production networks, outline a suitable network, and classify and evaluate it based on specific criteria.
- apply methods of circular production and demonstrate how products can be designed for circularity.
- apply the methods and approaches learned for problem solving in a global production environment and reflect on their effectiveness.

Workload:

e-Learning: ~ 36 h
regular attendence: ~ 64 h
self-study: ~ 80 h

Organizational issues

Termine werden über die Institutshomepage bekanntgegeben.

Aus organisatorischen Gründen ist die Teilnehmerzahl für die Lehrveranstaltung auf 20 Teilnehmer begrenzt. Infolgedessen wird ein Auswahlprozess stattfinden. Die Bewerbung erfolgt über die Homepage des wbk (http://www.wbk.kit.edu/studium-und-lehre.php)

Aufgrund der begrenzten Teilnehmerzahl ist eine Voranmeldung erforderlich.

Die Studierenden sollten Vorkenntnisse in mindestens einem der folgenden Bereiche haben:

- Integrierte Produktionsplanung
- Globale Produktion und Logistik
- Qualitätsmanagement
- Circular Factory

For organisational reasons, the number of participants for the course is limited to 20. As a result, a selection process will take place. Applications must be submitted via the wbk homepage (http://www.wbk.kit.edu/studium-und-lehre.php).

Due to the limited number of participants, advance registration is required.

Students should have previous knowledge in at least one of the following areas:

- Integrated Production Planning
- Global Production and Logistics
- Quality Management
- Circular Factory

Literature
Medien:
E-Learning Plattform ilias, Powerpoint, Fotoprotokoll. Die Medien werden über ilias (https://ilias.studium.kit.edu/) bereitgestellt.

Media:
E-learning platform ilias, powerpoint, photo protocol. The media are provided through ilias (https://ilias.studium.kit.edu/).

T 6.395 Module component: Lecture Series Supplementary Studies on Science, Technology and Society - Self Registration [T-FORUM-113578]

Coordinators: Dr. Christine Mielke
Christine Myglas
Organisation: General Studies. Forum Science and Society (FORUM)
Part of: M-FORUM-106753 - Supplementary Studies on Science, Technology and Society

Type	Credits	Grading	Term offered	Expansion	Version
Coursework	2 CP	pass/fail	Each summer term	1 semesters	1

Courses					
ST 2026	1130716	Lecture series Science in Society	2 SWS	Lecture / 🗣	Post, Mielke

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Active participation, learning protocols, if applicable.

Prerequisites
None

Self Service Assignment of Supplementary Studies
This module component can be used for self service assignment of grades acquired from the following study providers:

- Studium Generale. Forum Wissenschaft und Gesellschaft (FORUM) (ehem. ZAK)
- FORUM (ehem. ZAK) Begleitstudium

Recommendations
It is recommended that you complete the lecture series "Science in Society" before attending events in the advanced module and in parallel with attending the basic seminar.
If it is not possible to attend the lecture series and the basic seminar in the same semester, the lecture series can also be attended after attending the basic seminar.
However, attending events in the advanced module before attending the lecture series should be avoided.

Additional Information
The basic module consists of the lecture series "Science in Society" and the basic seminar. The lecture series is only offered during the summer semester.
The basic seminar can be attended in the summer or winter semester.

Below you will find excerpts from courses related to this module component:

V Lecture series Science in Society

Lecture (V)
On-Site

1130716, SS 2026, 2 SWS, Language: German, Open in study portal

Content
The lectures are also expected to be available in English via Lecture Translator.

Organizational issues
Anmeldung erforderlich unter: https://plus.campus.kit.edu/signmeup/procedures/6078

T 6.396 Module component: Liberalised Power Markets [T-WIWI-107043]

Coordinators: Prof. Dr. Wolf Fichtner
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101451 - Energy Economics and Energy Markets
M-WIWI-102808 - Digital Service Systems in Industry
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	5,5 CP	graded	Each winter term	4

Courses					
WT 25/26	2581998	Liberalised Power Markets	2 SWS	Lecture / 🗣	Fichtner
WT 25/26	2581999	Übungen zu Liberalised Power Markets	2 SWS	Practice / 🗣	Signer, Fichtner, Beranek
WT 26/27	2581998	Liberalised Power Markets	2 SWS	Lecture / 🗣	Fichtner
WT 26/27	2581999	Übungen zu Liberalised Power Markets	2 SWS	Practice / 🗣	Fichtner

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, × Cancelled

Assessment
The assessment consists of a written exam (60 minutes) (following §4(2) of the examination regulation). The exam takes place in every semester. Re-examinations are offered at every ordinary examination date.

Recommendations
None

Workload
165 hours

Below you will find excerpts from courses related to this module component:

V Liberalised Power Markets

2581998, WS 25/26, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content

1. Power markets in the past, now and in future

2. Designing liberalised power markets

2.1. Unbundling Dimensions of liberalised power markets
2.2. Central dispatch versus markets without central dispatch
2.3. The short-term market model
2.4. The long-term market model
2.5. Market flaws and market failure
2.6. Regulation in liberalised markets

3. The power (sub)markets

3.1 Day-ahead market
3.2 Intraday market
3.3 (Long-term) Forwards and futures markets
3.4 Emission rights market
3.5 Market for ancillary services
3.6 The “market” for renewable energies
3.7 Future market segments

4. Grid operation and congestion management

4.1. Grid operation
4.2. Congestion management

5. Market power

5.1. Defining market power
5.2. Indicators of market power
5.3. Reducing market power

6. Future market structures in the electricity value chain

1. Power markets in the past, now and in future

2. Designing liberalised power markets

2.2. Unbundling Dimensions of liberalised power markets
2.3. Central dispatch versus markets without central dispatch
2.4. The short-term market model
2.5. The long-term market model
2.6. Market flaws and market failure
2.7. Regulation in liberalised markets

3. The power (sub)markets

3.1 Day-ahead market
3.2 Intraday market
3.3 (Long-term) Forwards and futures markets
3.4 Emission rights market
3.5 Market for ancillary services
3.6 The “market” for renewable energies
3.7 Future market segments

4. Grid operation and congestion management

4.1. Grid operation
4.2. Congestion management

5. Market power

5.1. Defining market power
5.2. Indicators of market power
5.3. Reducing market power

6. Future market structures in the electricity value chain

Literature
Weiterführende Literatur:

Power System Economics; Steven Stoft, IEEE Press/Wiley-Interscience Press, 0-471-15040-1

V Liberalised Power Markets

2581998, WS 26/27, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content

1. Power markets in the past, now and in future

2. Designing liberalised power markets

2.1. Unbundling Dimensions of liberalised power markets
2.2. Central dispatch versus markets without central dispatch
2.3. The short-term market model
2.4. The long-term market model
2.5. Market flaws and market failure
2.6. Regulation in liberalised markets

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3.4 Emission rights market
3.5 Market for ancillary services
3.6 The “market” for renewable energies
3.7 Future market segments

4. Grid operation and congestion management

4.1. Grid operation
4.2. Congestion management

5. Market power

5.1. Defining market power
5.2. Indicators of market power
5.3. Reducing market power

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1. Power markets in the past, now and in future

2. Designing liberalised power markets

2.2. Unbundling Dimensions of liberalised power markets
2.3. Central dispatch versus markets without central dispatch
2.4. The short-term market model
2.5. The long-term market model
2.6. Market flaws and market failure
2.7. Regulation in liberalised markets

3. The power (sub)markets

3.1 Day-ahead market
3.2 Intraday market
3.3 (Long-term) Forwards and futures markets
3.4 Emission rights market
3.5 Market for ancillary services
3.6 The “market” for renewable energies
3.7 Future market segments

4. Grid operation and congestion management

4.1. Grid operation
4.2. Congestion management

5. Market power

5.1. Defining market power
5.2. Indicators of market power
5.3. Reducing market power

6. Future market structures in the electricity value chain

Literature
Weiterführende Literatur:

Power System Economics; Steven Stoft, IEEE Press/Wiley-Interscience Press, 0-471-15040-1

T 6.397 Module component: Life Cycle Assessment – Basics and Application Possibilities in an Industrial Context [T-WIWI-113107]

Coordinators: Prof. Dr. Frank Schultmann
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101412 - Industrial Production III
M-WIWI-101471 - Industrial Production II
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	3,5 CP	graded	Each winter term	1

Courses					
WT 25/26	2581995	Life Cycle Assessment - Basics and Application Possibilities in an Industrial Context	2 SWS	Lecture / 🗣	Treml, Schultmann, Wolf
WT 26/27	2581995	Life Cycle Assessment - Basics and Application Possibilities in an Industrial Context	2 SWS	Lecture / 🗣	Schultmann, Wolf

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment consists of an oral (approx. 30 minutes) or written exam (60 minutes) (following §4(2) of the examination regulation). The exam takes place in every semester. Re-examinations are offered at every ordinary examination date.

Prerequisites
None.

Recommendations
None

Below you will find excerpts from courses related to this module component:

V Life Cycle Assessment - Basics and Application Possibilities in an Industrial Context
Lecture (V)
On-Site

2581995, WS 25/26, 2 SWS, Language: German, Open in study portal

Content
The lecture focuses on the analysis of the environmental impacts of products and processes using Life Cycle Assessment (short: LCA). Structure and steps are conveyed in detail and selected further developments are shown. In order to record the methodology and classify potential environmental impacts, the practical development of what has been learned is also focused on using LCA software and interactive formats.

Topics include:

- Significance and areas of application
- Calculation models
- Attributional/Consequential LCA
- Life Cycle Sustainability Assessment, Social LCA and Life Cycle Costing
- Limitations
- Development of a Case Study

Literature
werden in der Veranstaltung bekannt gegeben

V Life Cycle Assessment - Basics and Application Possibilities in an Industrial Context
Lecture (V)
On-Site

2581995, WS 26/27, 2 SWS, Language: German, Open in study portal

Content

The lecture focuses on the analysis of the environmental impacts of products and processes using Life Cycle Assessment (short: LCA). Structure and steps are conveyed in detail and selected further developments are shown. In order to record the methodology and classify potential environmental impacts, the practical development of what has been learned is also focused on using LCA software and interactive formats.

Topics include:

- Significance and areas of application
- Calculation models
- Attributional/Consequential LCA
- Life Cycle Sustainability Assessment, Social LCA and Life Cycle Costing
- Limitations
- Development of a Case Study

Literature

werden in der Veranstaltung bekannt gegeben

T 6.398 Module component: Logistics and Supply Chain Management [T-MACH-110771]

Coordinators: Prof. Dr.-Ing. Kai Furmans
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-105298 - Logistics and Supply Chain Management

Type	Credits	Grading	Term offered	Version
Examination of another type	9 CP	graded	Each summer term	6

Courses					
ST 2026	2118078	Logistics and Supply Chain Management	6 SWS	Lecture / 🗣	Seibold, Furmans, Alicke

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The success control takes place in the form of an examination performance of a different kind. This is composed as follows:

- 50% assessment of a written examination (60 min) during the semester break
- 50% assessment of an oral examination (approx. 20 min) during the semester break

To pass the examination, both examination performances must be passed.

Prerequisites
None

Additional Information
The module component cannot be taken if one of the module components "T-MACH-102089 – Logistics - Organisation, Design and Control of Logistic Systems" and "T-MACH-105181 – Supply Chain Management" have been taken.

The course is offered in English.

Workload
270 hours

Below you will find excerpts from courses related to this module component:

V Logistics and Supply Chain Management

2118078, SS 2026, 6 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content

In the lecture "Logistics and Supply Chain Management", comprehensive and well-founded fundamentals of crucial issues in logistics and supply chain management are presented. Furthermore, the interaction of different design elements of supply chains is emphasized. For this purpose, both qualitative and quantitative models are presented and applied. Additionally, methods for mapping and evaluating logistics systems and supply chains are described. The contents of the lecture are deepened in exercises and case studies and comprehension is partially reviewed in case studies. The contents will be illustrated, among other things, on the basis of supply chains in the automotive industry.

Among others, the following topics are covered:

- Inventory Management
- Forecasting
- Bullwhip Effect
- Supply Chain Segmentation and Collaboration
- Key Performance Indicators
- Supply Chain Risk Management
- Production Logistics
- Location Planning
- Route Planning

It is intended to provide an interactive format in which students can also contribute (and work alone or in groups). Since logistics and supply chain management requires working in an international environment and therefore many terms are derived from English, the lecture will be held in English.

Plenary: The plenary sessions take place on Mondays from 09:45 - 13:00 and from 14:00 - 17:15.

Exercises: There are a total of five exercise sessions, which take place on Thursdays from 2:00 p.m. to 3:30 p.m. The scheduling can be found in Ilias from 20 April 2026.
Examination dates: This is a different type of examination. The date of the written exam will be announced on ILIAS as soon as defined. The oral exams are scheduled for the two weeks before that. An oral exam lasts 20 minutes.
Contact persons: In the summer semester 2026, the contact persons for organisational matters are Etienne Hoffmann and Alexander Ernst. Please contact us at
log-scm∂ifl.kit.edu

T 6.399 Module component: Logistics and Supply Chain Management [T-WIWI-102870]

Coordinators: Prof. Dr. Frank Schultmann
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	3,5 CP	graded	Each summer term	2

Courses					
ST 2026	2581996	Logistics and Supply Chain Management	2 SWS	Lecture / 🗣	Schultmann, Rudi

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment consists of an oral (30 minutes) or written exam (60 minutes) (following §4(2) of the examination regulation). The exam takes place in every semester. Re-examinations are offered at every ordinary examination date.

Below you will find excerpts from courses related to this module component:

V Logistics and Supply Chain Management

Lecture (V)
On-Site

2581996, SS 2026, 2 SWS, Language: English, Open in study portal

Content
Students are introduced to the methods and tools of logistics and supply chain management. They students learn the key terms and components of supply chains together with key economic trade-offs. In detail, students gain knowledge of decisions in supply chain management, such as facility location, supply chain planning, inventory management, pricing and supply chain cooperation. In this manner, students will gain knowledge in analyzing, designing and steering of decisions in the domain of logistics and supply chain management.

- Introduction: Basic terms and concepts
- Facility location and network optimization
- Supply chain planning I: flexibility
- Supply chain planning II: forecasting
- Inventory management & pricing
- Supply chain coordination I: the Bullwhip-effect
- Supply chain coordination II: double marginalization
- Supply chain risk management

Literature
Wird in der Veranstaltung bekannt gegeben.

T 6.400 Module component: Long-Distance and Air Traffic [T-BGU-106301]

Coordinators: Prof. Dr.-Ing. Peter Vortisch
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: M-BGU-101064 - Fundamentals of Transportation
M-BGU-101065 - Transportation Modelling and Traffic Management
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Expansion	Version
Written examination	3 CP	graded	Each term	1 semesters	1

Courses					
WT 25/26	6232904	Long-distance and Air Transportation	2 SWS	Lecture /	Magdolen, Dozenten

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
written exam, 60 min.

Prerequisites
none

Recommendations
none

Additional Information
none

Workload
90 hours

T 6.401 Module component: Low Power Design [T-INFO-101344]

Coordinators: Prof. Dr.-Ing. Jörg Henkel
Organisation: KIT Department of Informatics
Part of: M-WIWI-104909 - Informatics (Department of Informatics)

Type	Credits	Grading	Term offered	Version
Oral examination	3 CP	graded	Each summer term	1

Courses					
ST 2026	2424672	Low Power Design	2 SWS	Lecture / 🗣	Henkel, Nassar, Khdr

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment is carried out as an oral examination lasting 25-30 minutes, in accordance with Section 4 (2) No. 2 SPO.

Prerequisites
None.

Recommendations
- Basic knowledge from the modules "Design and Architectures of Embedded Systems (ESII)" and "Optimization and Synthesis of Embedded Systems (ESI)" are helpful but not essential for understanding of this lecture.
- The lecture is equally suitable for students from both computer science as well as electrical engineering department.
- The Lab of "Low Power Design and Embedded Systems" enables students to apply some of the theoretical knowledge gained from the lecture in practice.

Below you will find excerpts from courses related to this module component:

V Low Power Design

2424672, SS 2026, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content
Smart embedded devices driven by advances in fields as diverse as automotive smart home, to high-tech like lithography or battery technology for IoT devices are now omnipresent in our lives. Today's consumers have very high expectations from the embedded devices they own. Many emerging technologies such as virtual reality, robotics and artificial intelligence are limited in scope only by the performance of the underlying embedded devices. Unfortunately, performance of embedded devices is inherently constrained both by their limited cost, size as well as heat dissipating capacity and their limited on-board battery. The fact that all contemporary smartphones have multi-core chips running at low frequencies instead of single-core chips running at high frequencies can be attributed directly to the power consumption constraints imposed on them.

The constraints mandate highly optimized hardware-software co-design techniques for embedded devices that allows extraction of maximum performance with minimal power consumption. A good low power design requires all three building blocks of an embedded device – hardware, software and operating system – to work together synergistically. The lectures cover all the three aspects alongside their interactions from a low power design perspective in depth.

The lecture provides an overview of design methods, synthesis tools, estimation models, software techniques, operating system strategies, scheduling algorithms, etc., with the aim of minimizing the power consumption of embedded devices without compromising their performance. Both the research-relevant and industry-prevalent topics at different level of abstractions (from circuit to system) are discussed in this lecture.

Recommendations: Module "Entwurf und Architekturen für eingebettete Systeme". Basic knowledge from the module "Optimierung und Synthese Eingebetteter Systeme" is helpful but not essential for understanding of this lecture. The lecture is equally suitable for students from both computer science as well as electrical engineering department.

Students are made aware of various low power design optimizations employed in state-of-the-art embedded devices. At the end of the lecture, the students will be able to recognize the challenges involved in crafting efficient low power designs and how to tackle them.

T 6.402 Module component: Machine Dynamics [T-MACH-105210]

Coordinators: Prof. Dr.-Ing. Carsten Proppe
Organisation: KIT Department of Mechanical Engineering

Part of: M-WIWI-107718 - Mechanical Engineering I
M-WIWI-107730 - Mechanical Engineering II

Type	Credits	Grading	Term offered	Version
Written examination	5 CP	graded	Each summer term	1

Courses					
WT 25/26	2161224	Machine Dynamics	2 SWS	Lecture / Online	Proppe
ST 2026	2161224	Machine Dynamics	2 SWS	Lecture / On-Site	Qi, Proppe
ST 2026	2161225	Machine Dynamics (Tutorial)	1 SWS	Practice / On-Site	Qi, Proppe

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
written exam, 180 min.

Prerequisites
none

Additional Information
The course is offered in English.

Workload
150 hours

Below you will find excerpts from courses related to this module component:

V Machine Dynamics
2161224, WS 25/26, 2 SWS, Language: English, Open in study portal

Lecture (V)
Online

Content
1. Introduction

2. Machine as mechatronic system
3. Rigid rotors: equations of motion, transient and stationary motion, balancing
4. Flexible rotors: Laval rotor (equations of motion, transient and stationary behavior, critical speed, secondary effects), refined models)
5. Slider-crank mechanisms: kinematics, equations of motion, mass and power balancing

Organizational issues
Für die Vorlesung werden Unterlagen bereitgestellt.

Literature
Biezeno, Grammel: Technische Dynamik, 2. Aufl., 1953

Holzweißig, Dresig: Lehrbuch der Maschinendynamik, 1979

Dresig, Vulfson: Dynamik der Mechanismen, 1989

V Machine Dynamics
2161224, SS 2026, 2 SWS, Language: German/English, Open in study portal

Lecture (V)
On-Site

Content
1. Introduction
2. Machine as mechatronic system
3. Rigid rotors: equations of motion, transient and stationary motion, balancing
4. Flexible rotors: Laval rotor (equations of motion, transient and stationary behavior, critical speed, secondary effects), refined models)
5. Slider-crank mechanisms: kinematics, equations of motion, mass and power balancing

Literature

Biezeno, Grammel: Technische Dynamik, 2. Aufl., 1953

Holzweißig, Dresig: Lehrbuch der Maschinendynamik, 1979

Dresig, Vulfson: Dynamik der Mechanismen, 1989

V **Machine Dynamics (Tutorial)** **Practice (Ü)**
2161225, SS 2026, 1 SWS, Language: English, Open in study portal **On-Site**

Content
Exercises related to the lecture

T 6.403 Module component: Machine Learning 1 - Basic Methods [T-WIWI-106340]

Coordinators: Prof. Dr.-Ing. Johann Marius Zöllner
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101472 - Informatics
M-WIWI-101628 - Emphasis in Informatics
M-WIWI-103356 - Machine Learning
M-WIWI-104901 - Informatics (KIT-Department of Economics and Management)

Type	Credits	Grading	Term offered	Version
Written examination	5 CP	graded	Each winter term	4

Courses					
WT 25/26	2511500	Machine Learning 1 - Fundamental Methods	2 SWS	Lecture / 🗣	Zöllner
WT 25/26	2511501	Exercises to Machine Learning 1 - Fundamental Methods	1 SWS	Practice / 🗣	Zöllner, Polley, Fechner

Legend: 🖥 Online, ❖ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success is assessed in the form of a written examination (60 minutes). The examination is offered every semester and can be repeated at any regular examination date.

A grade bonus can be earned by successfully completing exercises. If the grade of the written examination is between 4.0 and 1.3, the bonus improves the grade by up to one grade level (0.3 or 0.4). Details will be announced in the lecture.

Prerequisites
None.

Workload
150 hours

Below you will find excerpts from courses related to this module component:

V Machine Learning 1 - Fundamental Methods

Lecture (V)
On-Site

2511500, WS 25/26, 2 SWS, Language: German, Open in study portal

Content
The course prepares students for the rapidly evolving field of machine learning by providing a solid foundation, covering core concepts and techniques to get started in the field. Students delve into different methods in supervised, unsupervised, and reinforcement learning, as well as various model types, ranging from basic linear classifiers to more complex methods, such as deep neural networks. Topics include general learning theory, support vector machines, decision trees, neural network fundamentals, convolutional neural networks, recurrent neural networks, unsupervised learning, reinforcement learning, and Bayesian learning.

The course is accompanied by a corresponding exercise, where students gain hands-on experience by implementing and experimenting with different machine learning algorithms, helping them to apply machine learning algorithms on real world problems.

By the end of the course, students will have acquired a solid foundation in machine learning, enabling them to apply state-of-the-art algorithms to solve complex problems, contribute to research efforts, and explore advanced topics in the field.

Learning obectives:

- Students acquire knowledge of the fundamental methods in the field of machine learning.
- Students can classify, formally describe and evaluate methods of machine learning.
- Students can use their knowledge to select suitable models and methods for selected problems in the field of of machine learning.

Literature
Die Foliensätze sind als PDF verfügbar

Weiterführende Literatur

- Machine Learning - Tom Mitchell
- Deep Learning - Ian Goodfellow, Yoshua Bengio, Aaron Courville
- Pattern Recognition and Machine Learning - Christopher M. Bishop
- Artificial Intelligence: A Modern Approach - Peter Norvig and Stuart J. Russell
- Reinforcement Learning: An Introduction - Richard S. Sutton and Andrew G. Barto

Weitere (spezifische) Literatur zu einzelnen Themen wird in der Vorlesung angegeben.

T 6.404 Module component: Machine Learning 2 - Advanced Methods [T-WIWI-106341]

Coordinators: Prof. Dr.-Ing. Johann Marius Zöllner
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101472 - Informatics
M-WIWI-101628 - Emphasis in Informatics
M-WIWI-101637 - Analytics and Statistics
M-WIWI-103356 - Machine Learning
M-WIWI-104901 - Informatics (KIT-Department of Economics and Management)

Type	Credits	Grading	Term offered	Version
Written examination	5 CP	graded	Each summer term	5

Courses					
ST 2026	2511502	Machine Learning 2 - Advanced Methods	2 SWS	Lecture /	Zöllner, Fechner, Polley, Stegmaier
ST 2026	2511503	Exercises for Machine Learning 2 - Advanced Methods	1 SWS	Practice /	Zöllner, Fechner, Polley, Stegmaier

Legend: Online, Blended (On-Site/Online), On-Site, ✕ Cancelled

Assessment
Success is assessed in the form of a written examination (70 minutes). The examination is offered every semester and can be repeated at any regular examination date.

Prerequisites
None.

Workload
150 hours

Below you will find excerpts from courses related to this module component:

V Machine Learning 2 - Advanced Methods
2511502, SS 2026, 2 SWS, Language: German/English, Open in study portal

Lecture (V)
Blended (On-Site/Online)

Content
Machine learning has become a central driver of innovation in science, engineering, and industry. Recent advances in learning algorithms and model architectures have enabled capabilities that were previously out of reach, such as high-quality image, video, and text generation, robust perception in complex environments, and intelligent decision-making under uncertainty. These developments are transforming how intelligent systems interact with the world and are increasingly deployed in large-scale, safety-critical, and autonomous applications.

This lecture provides an in-depth overview of the methods that enable these capabilities. It covers state-of-the-art network architectures, including transformers and graph neural networks, as well as modern generative models such as GANs and diffusion models. In addition, the lecture introduces model-free and model-based reinforcement learning for sequential decision-making and addresses practical aspects such as large-scale training, uncertainty-aware learning, active learning, and advanced computer vision. Through this integrated perspective, students gain a solid understanding of the principles, strengths, and limitations of modern machine learning methods and learn how to apply them in complex real-world systems.

Learning objectives:

- **Students have advanced knowledge of state-of-the-art machine learning methods and architectures**, including modern deep learning models, generative approaches, and reinforcement learning, as well as their theoretical foundations, assumptions, and application domains.
- **Students are able to formally describe, classify, and analyze advanced machine learning methods and architectures**, and to apply these methods within complex inference and decision-making systems while accounting for practical constraints such as data availability, scalability, robustness, and uncertainty.
- **Students are capable of critically evaluating advanced machine learning methods and architectures** and of selecting and justifying appropriate models and learning strategies for given problems in machine learning, based on problem characteristics, methodological trade-offs, and application requirements.

Recommendations: Attending the lecture Machine Learning 1 or a comparable lecture is very helpful in understanding this lecture.

Notes:

- The lecture will be taught in German with English slides. The exam can be taken in either German or English.
- The time of the exam is 70 minutes.

Literature
Weiterführende Literatur

- Probabilistic Machine Learning: An Introduction - Kevin P. Murphy
- Probabilistic Machine Learning: Advanced Topics - Kevin P. Murphy
- Deep Learning - Ian Goodfellow, Yoshua Bengio, Aaron Courville
- Reinforcement Learning: An Introduction - Richard S. Sutton and Andrew G. Barto

Weitere (spezifische) Literatur zu einzelnen Themen wird in der Vorlesung angegeben.

T 6.405 Module component: Machine Learning and Optimization in Energy Systems [T-WIWI-113073]

Coordinators: Prof. Dr. Wolf Fichtner
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101452 - Energy Economics and Technology
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	4 CP	graded	Each winter term	4

Courses					
WT 25/26	2581050	Machine Learning and Optimization in Energy Systems	3 SWS	Lecture / Practice / 🗣	Dengiz, Yilmaz, Kleinebrahm
WT 26/27	2581050	Machine Learning and Optimization in Energy Systems	3 SWS	Lecture / Practice / 🗣	Dengiz, Yilmaz, Kleinebrahm

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment of this course is a written examination (60 min) or an oral exam (30 min) depending on the number of participants.

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Machine Learning and Optimization in Energy Systems

Lecture / Practice (VÜ)
On-Site

2581050, WS 25/26, 3 SWS, Language: English, Open in study portal

Content
Goals:
Participants should know about the most common optimization and machine learning approaches for the application in energy systems. They should understand the basic principles of the methods and should be able to apply them for solving important problems of future energy systems with high shares of renewable energy sources.

Content:
In the beginning, the essential transition of the energy system into a smart grid and the need for methods from the field of optimization and machine learning are explained. The course can be subdivided into an optimization part and a larger machine learning part. In the optimization part, the basics of optimization approaches that are used in energy systems are shown. Further, heuristic methods and approaches from the field of multiobjective optimization are introduced. In the machine learning part, the most important methods from the field of unsupervised learning, supervised learning and reinforcement learning are introduced and their application in future energy systems are investigated.

Amongst the considered applications are power plant dispatch, intelligent heating with heat pumps, charging strategies for electric vehicles, clustering of energy data for energy system models and electricity demand and renewable generation forecasting.

We also offer a voluntary computer exercise that deepens the understanding of the methods and applications covered in the lecture. The students will have the opportunity to solve problems from the energy domain by using optimization and machine learning approaches implemented in the programming language Python.

The course's general focus is on the application of the methods in the energy field and not on the mathematical details of the different approaches.

The total workload for this course is approximately 105 hours:

- Attendance: 30 hours
- Self-study: 30 hours
- Exam preparation: 45 hours

V Machine Learning and Optimization in Energy Systems

Lecture / Practice (VÜ)
On-Site

2581050, WS 26/27, 3 SWS, Language: English, Open in study portal

Content

Goals:

Participants should know about the most common optimization and machine learning approaches for the application in energy systems. They should understand the basic principles of the methods and should be able to apply them for solving important problems of future energy systems with high shares of renewable energy sources.

Content:

In the beginning, the essential transition of the energy system into a smart grid and the need for methods from the field of optimization and machine learning are explained. The course can be subdivided into an optimization part and a larger machine learning part. In the optimization part, the basics of optimization approaches that are used in energy systems are shown. Further, heuristic methods and approaches from the field of multiobjective optimization are introduced. In the machine learning part, the most important methods from the field of unsupervised learning, supervised learning and reinforcement learning are introduced and their application in future energy systems are investigated.

Amongst the considered applications are power plant dispatch, intelligent heating with heat pumps, charging strategies for electric vehicles, clustering of energy data for energy system models and electricity demand and renewable generation forecasting.

We also offer a voluntary computer exercise that deepens the understanding of the methods and applications covered in the lecture. The students will have the opportunity to solve problems from the energy domain by using optimization and machine learning approaches implemented in the programming language Python.

The course's general focus is on the application of the methods in the energy field and not on the mathematical details of the different approaches.

The total workload for this course is approximately 105 hours:

- Attendance: 30 hours
- Self-study: 30 hours
- Exam preparation: 45 hours

T 6.406 Module component: Machine Tools and High-Precision Manufacturing Systems [T-MACH-110963]

Coordinators: Prof. Dr.-Ing. Jürgen Fleischer
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101286 - Machine Tools and Industrial Handling
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Oral examination	9 CP	graded	Each winter term	2

Courses					
WT 25/26	2149910	Machine Tools and High-Precision Manufacturing Systems	6 SWS	Lecture / Practice / 🗣	Fleischer
WT 26/27	2149910	Machine Tools and High-Precision Manufacturing Systems	6 SWS	Lecture / Practice / 🗣	Fleischer

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Oral exam (approx. 45 minutes)

Prerequisites
T-MACH-102158 - Machine Tools and Industrial Handling must not be commenced.
T-MACH-109055 - Machine Tools and Industrial Handling must not be commenced.
T-MACH-110962 - Machine Tools and High-Precision Manufacturing Systems must not be commenced.

Additional Information
The course is offered in German.

Workload
270 hours

Below you will find excerpts from courses related to this module component:

V Machine Tools and High-Precision Manufacturing Systems

Lecture / Practice (VÜ)
On-Site

2149910, WS 25/26, 6 SWS, Language: German, Open in study portal

Content
The lecture gives an overview of the construction, use and application of machine tools and high-precision manufacturing systems. In the course of the lecture a well-founded and practice-oriented knowledge for the selection, design and evaluation of machine tools and high-precision manufacturing systems is conveyed. First, the main components of the systems are systematically explained and their design principles as well as the integral system design are discussed. Subsequently, the use and application of machine tools and high-precision manufacturing systems will be demonstrated using typical machine examples. Based on examples from current research and industrial applications, the latest developments are discussed, especially concerning the implementation of Industry 4.0 and artificial intelligence.
Guest lectures from industry round off the lecture with insights into practice.

The individual topics are:

- Structural components of dynamic manufacturing Systems
- Feed axes: High-precision positioning
- Spindles of cutting machine Tools
- Peripheral Equipment
- Machine control unit
- Metrological Evaluation
- Maintenance strategies and condition Monitoring
- Process Monitoring
- Development process for machine tools and high-precision manufacturing Systems
- Machine examples

Learning Outcomes:

The students …

- are able to assess the use and application of machine tools and high-precision manufacturing systems and to differentiate between them in terms of their characteristics and design.
- can describe and discuss the essential elements of machine tools and high-precision manufacturing systems (frame, main spindle, feed axes, peripheral equipment, control unit).
- are able to select and dimension the essential components of machine tools and high-precision manufacturing systems.
- are capable of selecting and evaluating machine tools and high-precision manufacturing systems according to technical and economic criteria.

Workload:

MACH:
regular attendance: 63 hours
self-study: 177 hours
WING/TVWL:
regular attendance: 63 hours
self-study: 207 hours

Organizational issues
Vorlesungstermine montags und mittwochs, Übungstermine donnerstags.
Bekanntgabe der konkreten Übungstermine erfolgt in der ersten Vorlesung.

Lectures on Mondays and Wednesdays, tutorial on Thursdays.
The tutorial dates will announced in the first lecture.

Zur Vertiefung des im Rahmen der Lehrveranstaltung erworbenen Wissens werden die theoretischen Vorlesungseinheiten durch Praxiseinheiten im Umfeld der Karlsruher Forschungsfabrik (https://www.karlsruher-forschungsfabrik.de) unterstützt.

The theoretical lectures are complemented by practical lectures in the Karlsruhe Research Factory (https://www.karlsruher-forschungsfabrik.de/en.html) to deepen the acquired knowledge.

Literature
Medien:
Skript zur Veranstaltung wird über Ilias (https://ilias.studium.kit.edu/) bereitgestellt.

Media:
Lecture notes will be provided in Ilias (https://ilias.studium.kit.edu/).

V	**Machine Tools and High-Precision Manufacturing Systems** 2149910, WS 26/27, 6 SWS, Language: German, Open in study portal	**Lecture / Practice (VÜ)** **On-Site**

Content
The lecture gives an overview of the construction, use and application of machine tools and high-precision manufacturing systems. In the course of the lecture a well-founded and practice-oriented knowledge for the selection, design and evaluation of machine tools and high-precision manufacturing systems is conveyed. First, the main components of the systems are systematically explained and their design principles as well as the integral system design are discussed. Subsequently, the use and application of machine tools and high-precision manufacturing systems will be demonstrated using typical machine examples. Based on examples from current research and industrial applications, the latest developments are discussed, especially concerning the implementation of Industry 4.0 and artificial intelligence.
Guest lectures from industry round off the lecture with insights into practice.

The individual topics are:

- Structural components of dynamic manufacturing Systems
- Feed axes: High-precision positioning
- Spindles of cutting machine Tools
- Peripheral Equipment
- Machine control unit
- Metrological Evaluation
- Maintenance strategies and condition Monitoring
- Process Monitoring
- Development process for machine tools and high-precision manufacturing Systems
- Machine examples

Learning Outcomes:

The students …

- are able to assess the use and application of machine tools and high-precision manufacturing systems and to differentiate between them in terms of their characteristics and design.
- can describe and discuss the essential elements of machine tools and high-precision manufacturing systems (frame, main spindle, feed axes, peripheral equipment, control unit).
- are able to select and dimension the essential components of machine tools and high-precision manufacturing systems.
- are capable of selecting and evaluating machine tools and high-precision manufacturing systems according to technical and economic criteria.

Workload:

MACH:
regular attendance: 63 hours
self-study: 177 hours
WING/TVWL:
regular attendance: 63 hours
self-study: 207 hours

Organizational issues
Vorlesungstermine montags und mittwochs, Übungstermine donnerstags.
Bekanntgabe der konkreten Übungstermine erfolgt in der ersten Vorlesung.

Lectures on Mondays and Wednesdays, tutorial on Thursdays.
The tutorial dates will announced in the first lecture.

Zur Vertiefung des im Rahmen der Lehrveranstaltung erworbenen Wissens werden die theoretischen Vorlesungseinheiten durch Praxiseinheiten im Umfeld der Karlsruher Forschungsfabrik (https://www.karlsruher-forschungsfabrik.de) unterstützt.

The theoretical lectures are complemented by practical lectures in the Karlsruhe Research Factory (https://www.karlsruher-forschungsfabrik.de/en.html) to deepen the acquired knowledge.

Literature
Medien:
Skript zur Veranstaltung wird über Ilias (https://ilias.studium.kit.edu/) bereitgestellt.

Media:
Lecture notes will be provided in Ilias (https://ilias.studium.kit.edu/).

T 6.407 Module component: Machine Vision [T-MACH-105223]

Coordinators: Dr. Martin Lauer
Prof. Dr.-Ing. Christoph Stiller

Organisation: KIT Department of Mechanical Engineering

Part of: M-WIWI-107718 - Mechanical Engineering I
M-WIWI-107730 - Mechanical Engineering II

Type	Credits	Grading	Term offered	Version
Written examination	9 CP	graded	Each winter term	4

Courses					
WT 25/26	2137308	Machine Vision	4 SWS	Lecture / Practice / 🗣	Lauer, Merkert, Truetsch
WT 26/27	2137308	Machine Vision	4 SWS	Lecture / Practice / 🗣	Lauer, Merkert, Truetsch

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Type of Examination: written exam

Duration of Examination: 60 minutes

Prerequisites
None

Additional Information
The course is offered in English

Workload
270 hours

Below you will find excerpts from courses related to this module component:

V Machine Vision

2137308, WS 25/26, 4 SWS, Language: English, Open in study portal

Lecture / Practice (VÜ)
On-Site

Content
Lernziele (EN):

Machine vision (or *computer vision*) describes all kind of techniques that can be used to extract information from camera images in an automated way. Considerable improvements of machine vision techniques throughout recent years, e.g. by the advent of deep learning, have caused growing interest in these techniques and enabled applications in various domains, e.g. robotics, autonomous driving, gaming, production control, visual inspection, medicine, surveillance systems, and augmented reality.

The participants should gain an overview over the basic techniques in machine vision and obtain hands-on experience.

Nachweis: written exam, 60 min.

Arbeitsaufwand: 240 hours

Voraussetzungen: none

Literature
Foliensatz zur Veranstaltung wird als kostenlose pdf-Datei bereitgestellt.Weitere Empfehlungen
werden in der Vorlesung bekannt gegeben.

V Machine Vision

2137308, WS 26/27, 4 SWS, Language: English, Open in study portal

Lecture / Practice (VÜ)
On-Site

Content
Lernziele (EN):

Machine vision (or *computer vision*) describes all kind of techniques that can be used to extract information from camera images in an automated way. Considerable improvements of machine vision techniques throughout recent years, e.g. by the advent of deep learning, have caused growing interest in these techniques and enabled applications in various domains, e.g. robotics, autonomous driving, gaming, production control, visual inspection, medicine, surveillance systems, and augmented reality.

The participants should gain an overview over the basic techniques in machine vision and obtain hands-on experience.

Nachweis: written exam, 60 min.

Arbeitsaufwand: 240 hours

Voraussetzungen: none

Literature
Foliensatz zur Veranstaltung wird als kostenlose pdf-Datei bereitgestellt.Weitere Empfehlungen
werden in der Vorlesung bekannt gegeben.

T 6.408 Module component: Macroeconomic Theory [T-WIWI-109121]

Coordinators: Prof. Dr. Johannes Brumm
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104908 - Economics

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each winter term	4

Courses					
WT 25/26	2560404	Macroeconomic Theory	2 SWS	Lecture / 🗣	Brumm, Pegorari
WT 25/26	2560405	Übung zu Macroeconomic Theory	1 SWS	Practice / 🗣	Pegorari

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment consists of a written exam (60 min.) according to § 4 paragraph 2 Nr. 1 of the examination regulation.

Prerequisites
None.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Macroeconomic Theory

2560404, WS 25/26, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content
This course introduces a modern approach to macroeconomics by building on microeconomic principles. To be able to rigorously address key macroeconomic questions a general framework based on intertemporal decision making is introduced. Starting by the principles of consumer and firm behavior, this framework is successively expanded by introducing market imperfections, monetary factors as well as international trade. With this framework at hand students are able to analyze labor market policies, government deficits, monetary policy, trade policy, and other important macroeconomic problems. Throughout the course, we not only point out the power of theory but also its limitations.

Literature
Literatur und Skripte werden in der Veranstaltung angegeben.

T 6.409 Module component: Management Accounting 1 [T-WIWI-102800]

Coordinators: Prof. Dr. Marcus Wouters
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101498 - Management Accounting
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	see Annotations	2

Courses					
ST 2026	2579900	Management Accounting 1	2 SWS	Lecture / Online	Wouters
ST 2026	2579901	Tutorial Management Accounting 1 (Bachelor)	2 SWS	Practice / On-Site	Dickemann
ST 2026	2579902	Tutorial Management Accounting 1 (Master)	2 SWS	Practice / On-Site	Dickemann

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
The assessment consists of a written exam (120 min.) according to § 4 paragraph 2 Nr. 1 of the examination regulation.

Recommendations
We recommend that you take part in our exercise for the lecture.

Additional Information
The course "Management Accounting 1" and the corresponding assessment for first-time examinees will be offered for the last time in the **summer semester 2026**.

The exercise is offered separately for Bachelor's students as well as for students in the Master's transfer and Master's program.

Note for exam registration:

- Bachelor students: 79-2579900-B Management Accounting 1 (Bachelor)
- Students in the Master's transfer and Master's program: 79-2579900-M Management Accounting 1 (Master's transfer and Master)

Below you will find excerpts from courses related to this module component:

V Management Accounting 1

2579900, SS 2026, 2 SWS, Language: English, Open in study portal

Lecture (V)
Online

Content
The course covers topics in management accounting in a decision-making framework. Some of these topics in the course MA1 are: short-term planning, investment decisions, budgeting and activity-based costing.

We will use international material written in English.

We will approach these topics primarily from the perspective of the users of financial information (not so much from the controller who prepares the information).

The course builds on an introductory level of understanding of accounting concepts from Business Administration courses in the core program. The course is intended for students in Industrial Engineering.

Learning objectives:

- Students have an understanding of theory and applications of management accounting topics.
- They can use financial information for various purposes in organizations.

Examination:

- The assessment consists of a written exam (120 minutes) at the end of each semester (following § 4 (2) No. 1 of the examination regulation).

Workload:

- The total workload for this course is approximately 135.0 hours. For further information see German version.

Organizational issues
Bitte beachten Sie:
- die Vorlesung Management Accounting 1 wird letztmalig im SoS 2026 und
- die Vorlesung Management Accounting 2 wird letztmalig im WS 2026/27 angeboten.

Literature

- Marc Wouters, Frank H. Selto, Ronald W. Hilton, Michael W. Maher: Cost Management – Strategies for Business Decisions, 2012, Publisher: McGraw-Hill Higher Education (ISBN-13 9780077132392 / ISBN-10 0077132394)
- In addition, several papers that will be available on ILIAS.

V **Tutorial Management Accounting 1 (Bachelor)**
2579901, SS 2026, 2 SWS, Language: English, Open in study portal

Practice (Ü)
On-Site

Content
see Module Handbook

Organizational issues
Bitte beachten Sie:
- die Übung Management Accounting 1 wird letztmalig im SoS 2026 und
- die Übung Management Accounting 2 wird letztmalig im WS 2026/27 angeboten.

V **Tutorial Management Accounting 1 (Master)**
2579902, SS 2026, 2 SWS, Language: English, Open in study portal

Practice (Ü)
On-Site

Content
see Module Handbook

Organizational issues
Bitte beachten Sie:
- die Übung Management Accounting 1 wird letztmalig im SoS 2026 und
- die Übung Management Accounting 2 wird letztmalig im WS 2026/27 angeboten.

T 6.410 Module component: Management Accounting 2 [T-WIWI-102801]

Coordinators: Prof. Dr. Marcus Wouters
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101498 - Management Accounting
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	see Annotations	2

Courses					
WT 25/26	2579903	Management Accounting 2	2 SWS	Lecture / 🖥	Wouters
WT 25/26	2579904	Tutorial Management Accounting 2 (Bachelor)	2 SWS	Practice / 🗣	Letmathe
WT 25/26	2579905	Tutorial Management Accounting 2 (Master)	2 SWS	Practice / 🗣	Letmathe
WT 26/27	2579903	Management Accounting 2	2 SWS	Lecture / 🖥	Wouters
WT 26/27	2579904	Tutorial Management Accounting 2 (Bachelor)	2 SWS	Practice / 🗣	Fimpel, Binti Mohamad Riduan
WT 26/27	2579905	Tutorial Management Accounting 2 (Master)	2 SWS	Practice / 🗣	Fimpel, Binti Mohamad Riduan

Legend: 🖥 Online, Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment consists of a written exam (120 min.) according to § 4 paragraph 2 Nr. 1 of the examination regulation.

Prerequisites
None

Recommendations
It is recommended:

- to take part in the course "Management Accounting1" before this course
- participation in the exercise for the lecture "Management Accounting 2"

Additional Information
The course "Management Accounting 2" and the corresponding assessment for first-time examinees will be offered for the last time in the **winter semester 2026/2027**.

The exercise for the lecture is offered separately for Bachelor's students as well as for students in the Master's transfer and Master's program.

Note for exam registration: Bachelor students:

- 79-2579903-B Management Accounting 2 (Bachelor)
- Students in the Master's transfer and Master's program: 79-2579903-M Management Accounting 2 (Master's transfer and Master)

Below you will find excerpts from courses related to this module component:

V Management Accounting 2

Lecture (V)
Online

2579903, WS 25/26, 2 SWS, Language: English, Open in study portal

Content
The course covers topics in management accounting in a decision-making framework. Some of these topics in the course MA2 are: cost estimation, product costing and cost allocation, financial performance measures, transfer pricing, strategic performance measurement systems.

We will use international material written in English.

We will approach these topics primarily from the perspective of the users of financial information (not so much from the controller who prepares the information).

The course builds on an introductory level of understanding of accounting concepts from Business Administration courses in the core program. The course is intended for students in Industrial Engineering.

Learning objectives:

- Students have an understanding of theory and applications of management accounting topics. They can use financial information for various purposes in organizations.

Recommendations:

- It is recommended to take part in the course "Management Accounting 1" before this course.

Examination:

- The assessment consists of a written exam (120 min) at the end of each semester (following § 4 (2) No. 1 of the examination regulation).

Workload:

- The total workload for this course is approximately 135.0 hours. For further information see German version.

Literature

- Marc Wouters, Frank H. Selto, Ronald W. Hilton, Michael W. Maher: Cost Management – Strategies for Business Decisions, 2012, Verlag: McGraw-Hill Higher Education (ISBN-13 9780077132392 / ISBN-10 0077132394)
- Zusätzlich werden Artikel auf ILIAS zur Vergügung gestellt.

V	**Tutorial Management Accounting 2 (Bachelor)** 2579904, WS 25/26, 2 SWS, Language: English, Open in study portal	**Practice (Ü)** **On-Site**

Content
see ILIAS

V	**Tutorial Management Accounting 2 (Master)** 2579905, WS 25/26, 2 SWS, Language: English, Open in study portal	**Practice (Ü)** **On-Site**

Content
see ILIAS

V	**Management Accounting 2** 2579903, WS 26/27, 2 SWS, Language: English, Open in study portal	**Lecture (V)** **Online**

Content
The course covers topics in management accounting in a decision-making framework. Some of these topics in the course MA2 are: cost estimation, product costing and cost allocation, financial performance measures, transfer pricing, strategic performance measurement systems.

We will use international material written in English.

We will approach these topics primarily from the perspective of the users of financial information (not so much from the controller who prepares the information).

The course builds on an introductory level of understanding of accounting concepts from Business Administration courses in the core program. The course is intended for students in Industrial Engineering.

Learning objectives:

- Students have an understanding of theory and applications of management accounting topics. They can use financial information for various purposes in organizations.

Recommendations:

- It is recommended to take part in the course "Management Accounting 1" before this course.

Examination:

- The assessment consists of a written exam (120 min) at the end of each semester (following § 4 (2) No. 1 of the examination regulation).

Workload:

- The total workload for this course is approximately 135.0 hours. For further information see German version.

Organizational issues
Das derzeitige Modul „Management Accounting“ (MA) läuft mit dem Wintersemester 2026/27 aus, und das neue Modul „Environmental Management Accounting“ (EMA) beginnt mit dem Sommersemester 2027. Siehe dazu die Hinweise auf unserer Homepage

Literature

- Marc Wouters, Frank H. Selto, Ronald W. Hilton, Michael W. Maher: Cost Management – Strategies for Business Decisions, 2012, Verlag: McGraw-Hill Higher Education (ISBN-13 9780077132392 / ISBN-10 0077132394)
- Zusätzlich werden Artikel auf ILIAS zur Vergügung gestellt.

V **Tutorial Management Accounting 2 (Bachelor)**
2579904, WS 26/27, 2 SWS, Language: English, Open in study portal

Practice (Ü)
On-Site

Content
see ILIAS

Organizational issues
Das derzeitige Modul „Management Accounting“ (MA) läuft mit dem Wintersemester 2026/27 aus, und das neue Modul „Environmental Management Accounting“ (EMA) beginnt mit dem Sommersemester 2027. Siehe dazu die Hinweise auf unserer Homepage

V **Tutorial Management Accounting 2 (Master)**
2579905, WS 26/27, 2 SWS, Language: English, Open in study portal

Practice (Ü)
On-Site

Content
see ILIAS

Organizational issues
Das derzeitige Modul „Management Accounting“ (MA) läuft mit dem Wintersemester 2026/27 aus, und das neue Modul „Environmental Management Accounting“ (EMA) beginnt mit dem Sommersemester 2027. Siehe dazu die Hinweise auf unserer Homepage

T 6.411 Module component: Management and Marketing [T-WIWI-111594]

Coordinators: Prof. Dr. Martin Klarmann
Prof. Dr. Hagen Lindstädt
Prof. Dr. Petra Nieken
Prof. Dr. Orestis Terzidis

Organisation: KIT Department of Business and Economics

Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	5 CP	graded	Each winter term	2

Courses					
WT 25/26	2600023	Management	2 SWS	Lecture / 🗣	Nieken, Lindstädt, Terzidis
WT 25/26	2610026	Marketing	2 SWS	Lecture / 🗣	Klarmann
WT 26/27	2600023	Management	2 SWS	Lecture / 🗣	Nieken, Lindstädt, Terzidis
WT 26/27	2610026	Marketing	2 SWS	Lecture / 🗣	Klarmann

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Written exam (90 min) on the two courses "Management" and "Marketing". The examination is offered at the beginning of each lecture-free period. Repeat examinations are possible at any regular examination date.

Prerequisites
None

Workload
150 hours

Below you will find excerpts from courses related to this module component:

V Marketing
2610026, WS 25/26, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Literature
Ausführliche Literaturhinweise werden in den Materialen zur Vorlesung gegeben.

V Marketing
2610026, WS 26/27, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Literature
Ausführliche Literaturhinweise werden in den Materialen zur Vorlesung gegeben.

T 6.412 Module component: Management of IT-Projects [T-WIWI-113968]

Coordinators: Prof. Dr.-Ing. Sascha Alpers
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101472 - Informatics
M-WIWI-101477 - Development of Business Information Systems
M-WIWI-101628 - Emphasis in Informatics
M-WIWI-104901 - Informatics (KIT-Department of Economics and Management)

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each summer term	1

Courses					
ST 2026	2511214	IT Project Management	2 SWS	Lecture / 🗣	Alpers
ST 2026	2511215	Exercise IT Project Management	1 SWS	Practice / 🗣	Rybinski

Legend: 🖥 Online, ✲ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success is assessed in the form of a written examination (written exam) lasting 60 minutes.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V IT Project Management

Lecture (V)
On-Site

2511214, SS 2026, 2 SWS, Language: German, Open in study portal

Content
Contents:

The lecture deals with the general framework, impact factors and methods for planning, handling, and controlling of IT projects. Especially following topics are addressed:

- project environment
- project organisation
- project planning including the following items:
    - plan of the project structure
    - flow chart
    - project schedule
    - plan of resources
- effort estimation
- project infrastructur
- project controlling
- risk management
- feasibility studies
- decision processes, conduct of negotiations, time management.

Learning objectives:

Students

- explain the terminology of IT project management and typical used methods for planning, handling and controlling,
- apply methods appropiate to current project phases and project contexts,
- consider organisational and social impact factors.

Recommendations:

Knowledge about Software Engineering is helpful.

Workload:

- Lecture 30h
- Exercise 15h
- Preparation of lecture 24h
- Preparation of exercises 25h
- Exam preparation 40h
- Exam 1h

Literature

- B. Hindel, K. Hörmann, M. Müller, J. Schmied. Basiswissen Software-Projektmanagement. dpunkt.verlag 2004
- Project Management Institute Standards Committee. A Guide to the Project Management Body of Knowledge (PMBoK guide). Project Management Institute. Four Campus Boulevard. Newton Square. PA 190733299. U.S.A.

V **Exercise IT Project Management**
2511215, SS 2026, 1 SWS, Language: German, Open in study portal

Practice (Ü)
On-Site

Content
The general conditions, influencing factors and methods in the planning, execution and control of IT projects are dealt with. In particular, the following topics will be dealt with: Project environment, project organization, project structure plan, effort estimation, project infrastructure, project control, decision-making processes, negotiation, time management.

T 6.413 Module component: Managing Organizations [T-WIWI-102630]

Coordinators: Prof. Dr. Hagen Lindstädt
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each winter term	5

Courses					
WT 25/26	2577902	Managing Organizations	2 SWS	Lecture / 🗣	Lindstädt
WT 26/27	2577902	Managing Organizations	2 SWS	Lecture / 🗣	Lindstädt

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment will consist of a written exam (60 min) taking place at the beginning of the recess period (according to Section 4 (2), 2 of the examination regulation). The exam takes place in every semester. Re-examinations are offered at every ordinary examination date.

Prerequisites
None

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Managing Organizations

2577902, WS 25/26, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content
This course enables participants to make a sound assessment of existing organizational structures and regulations. Students learn concepts and models for designing organizational structures, regulating organizational processes, and managing organizational change.

Through intensive exposure to real-world case studies, students are encouraged to learn and apply strategic actions in real-world business settings. The course features an action-oriented approach and provides students with a realistic understanding of the possibilities and limitations of rational design approaches.

Content in Keywords:

- Fundamentals of organizational management: fundamental concepts and theoretical background knowledge
- Management of organizational structures and processes: Corporate headquarters, departmental organization, instruction structure and incentive systems
- Ideal organizational structures: organic vs. mechanistic, Mintzberg's types, relationship to strategy and 7S model
- Management of organizational change (change management): Change processes within an organization, management of revolutionary change

Structure:

Lectures in the course are available to students online as recordings, while class dates are reserved for active discussion of real-world case studies.

Learning Objectives:

Upon completion of the course, students will be able to,

- critically evaluate existing organizational structures and regulations
- compare alternative structural options in a practical setting and evaluate and interpret their effectiveness and efficiency
- analyze and evaluate change processes in organizational management
- apply theoretical knowledge in practical situations

Recommendations:

None.

Workload:

- Total workload for 3.5 credit points: approx. 105 hours
- Attendance time: 30 hours
- Self-study: 75 hours

Competence Certificate:

The assessment of success takes place in the form of a written examination (60min.) (according to §4(2), 1 SPO) at the beginning of the lecture-free period of the semester. The examination is offered every semester and can be repeated at any regular examination date.
A bonus can be earned through successful participation in the exercise. If the grade on the written exam is between 4.0 and 1.3, the bonus improves the grade by one grade level (0.3 or 0.4). The exact criteria for awarding a bonus will be announced at the beginning of the lecture.

Literature

- Laux, H.; Liermann, F.: *Grundlagen der Organisation*, Springer. 6. Aufl. Berlin 2005.
- Lindstädt, H.: *Organisation*, in Scholz, C. (Hrsg.): Vahlens Großes Personallexikon, Verlag Franz Vahlen. 1. Aufl. München, 2009.
- Schreyögg, G.: *Organisation. Grundlagen moderner Organisationsgestaltung*, Gabler. 4. Aufl. Wiesbaden 2003.

Die relevanten Auszüge und zusätzlichen Quellen werden in der Veranstaltung bekannt gegeben.

V **Managing Organizations** — **Lecture (V) On-Site**
2577902, WS 26/27, 2 SWS, Language: German, Open in study portal

Content
This course enables participants to make a sound assessment of existing organizational structures and regulations. Students learn concepts and models for designing organizational structures, regulating organizational processes, and managing organizational change.

Through intensive exposure to real-world case studies, students are encouraged to learn and apply strategic actions in real-world business settings. The course features an action-oriented approach and provides students with a realistic understanding of the possibilities and limitations of rational design approaches.

Content in Keywords:

- Fundamentals of organizational management: fundamental concepts and theoretical background knowledge
- Management of organizational structures and processes: Corporate headquarters, departmental organization, instruction structure and incentive systems
- Ideal organizational structures: organic vs. mechanistic, Mintzberg's types, relationship to strategy and 7S model
- Management of organizational change (change management): Change processes within an organization, management of revolutionary change

Structure:

Lectures in the course are available to students online as recordings, while class dates are reserved for active discussion of real-world case studies.

Learning Objectives:

Upon completion of the course, students will be able to,

- critically evaluate existing organizational structures and regulations
- compare alternative structural options in a practical setting and evaluate and interpret their effectiveness and efficiency
- analyze and evaluate change processes in organizational management
- apply theoretical knowledge in practical situations

Recommendations:

None.

Workload:

- Total workload for 3.5 credit points: approx. 105 hours
- Attendance time: 30 hours
- Self-study: 75 hours

Competence Certificate:

The assessment of success takes place in the form of a written examination (60min.) (according to §4(2), 1 SPO) at the beginning of the lecture-free period of the semester. The examination is offered every semester and can be repeated at any regular examination date.
A bonus can be earned through successful participation in the exercise. If the grade on the written exam is between 4.0 and 1.3, the bonus improves the grade by one grade level (0.3 or 0.4). The exact criteria for awarding a bonus will be announced at the beginning of the lecture.

Literature

- Laux, H.; Liermann, F.: *Grundlagen der Organisation*, Springer. 6. Aufl. Berlin 2005.
- Lindstädt, H.: *Organisation*, in Scholz, C. (Hrsg.): Vahlens Großes Personallexikon, Verlag Franz Vahlen. 1. Aufl. München, 2009.
- Schreyögg, G.: *Organisation. Grundlagen moderner Organisationsgestaltung*, Gabler. 4. Aufl. Wiesbaden 2003.

Die relevanten Auszüge und zusätzlichen Quellen werden in der Veranstaltung bekannt gegeben.

T 6.414 Module component: Managing the Marketing Mix [T-WIWI-102805]

Coordinators: Prof. Dr. Martin Klarmann
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Examination of another type	4,5 CP	graded	Each summer term	2

Courses					
ST 2026	2571152	Managing the Marketing Mix	2 SWS	Lecture / 🗣	Klarmann
ST 2026	2571153	Übung zu Marketing Mix (Bachelor)	1 SWS	Practice / 🗣	Daumann, Weber

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment of success takes place through the preparation and presentation of a case study (max. 30 points) as well as a written exam with additional aids in the sense of an open book exam (max. 60 points). In total, a maximum of 90 points can be achieved in the course. Further details will be announced during the lecture.

Prerequisites
None

Additional Information
The course is compulsory in the module "Foundations of Marketing".
For further information please contact Marketing & Sales Research Group (marketing.iism.kit.edu).

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Managing the Marketing Mix

2571152, SS 2026, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content
The content of this course concentrates on the elements of the marketing mix. Therefore the main chapters are brand management, pricing, promotion and sales management.

For further information please contact Marketing & Sales Research Group (marketing.iism.kit.edu).

This course is compulsory within or the module "Foundations of Marketing" and must be examined.

Learning objectives:

student

- know the meaning of the branding, the brand positioning and the possibilities of the brand value calculation
- understand the price behavior of customers and can apply this knowledge to the practice
  know different methods for price determination (conjoint analysis, cost-plus determination, target costing, customer surveys, bidding procedures) and price differentiation
- are able to name and explain the relevant communication theories
- can identify crisis situations and formulate appropriate response strategies
- can name and judge different possibilities of the Intermediaplanung
- know various design elements of advertising communication
- understand the measurement of advertising impact and can apply it
- know the basics of sales organization
- are able to evaluate basic sales channel decisions

Workload:

The total workload for this course is approximately 135.0 hours.

Organizational issues
Achtung Raumwechsel, neuer Raum Gebäude 10.50 Raum 701.3 ab 11.5.26

Literature

Homburg, Christian (2016), Marketingmanagement, 6. Aufl., Wiesbaden.

T 6.415 Module component: Market Research [T-WIWI-107720]

Coordinators: Prof. Dr. Martin Klarmann
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101510 - Cross-Functional Management Accounting
M-WIWI-101647 - Data Science: Evidence-based Marketing
M-WIWI-104900 - Business Administration
M-WIWI-105312 - Marketing and Sales Management
M-WIWI-105714 - Consumer Research
M-WIWI-106258 - Digital Marketing
M-WIWI-107187 - Quantitative Core

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each summer term	3

Courses					
ST 2026	2571150	Market Research	2 SWS	Lecture / 🗣	Klarmann
ST 2026	2571151	Market Research Tutorial	1 SWS	Practice / 🗣	Klarmann

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment of success takes place through a written exam (70 minutes) with additional aids in the sense of an open book exam. Further details will be announced during the lecture.

Prerequisites
None

Recommendations
None

Additional Information
Please note that this course has to be completed successfully by students interested in master thesis positions at the Marketing & Sales Research Group.

Below you will find excerpts from courses related to this module component:

V Market Research — **Lecture (V) On-Site**
2571150, SS 2026, 2 SWS, Language: English, Open in study portal

Content
Within the lecture, essential statistical methods for measuring customer attitudes (e.g. satisfaction measurement), understanding customer behavior and making strategic decisions will be discussed. The practical use as well as the correct handling of different survey methods will be taught, such as experiments and surveys. To analyze the collected data, various analysis methods are presented, including hypothesis tests, factor analyses, cluster analyses, variance and regression analyses. Building on this, the interpretation of the results will be discussed.

Topics addressed in this course are for example:

- Theoretical foundations of market research
- Statistical foundations of market research
- Measuring customer attitudes
- Understanding customer reactions
- Strategical decision making

The aim of this lecture is to give an overview of essential statistical methods. In the lecture students learn the practical use as well as the correct handling of different statistical survey methods and analysis procedures. In addition, emphasis is put on the interpretation of the results after the application of an empirical survey. The derivation of strategic options is an important competence that is required in many companies in order to react optimally to customer needs.

The assessment is carried out (according to §4(2), 3 SPO) in the form of a written open book exam.

The total workload for this course is approximately 135.0 hours.

Presence time: 30 hours

Preparation and wrap-up of the course: 45.0 hours

Exam and exam preparation: 60.0 hours

Please note that this course has to be completed successfully by students interested in master thesis positions at the chair of marketing.

Literature
Homburg, Christian (2016), Marketingmanagement, 6. Aufl., Wiesbaden.

T 6.416 Module component: Marketing Analytics [T-WIWI-103139]

Coordinators: Prof. Dr. Martin Klarmann
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101647 - Data Science: Evidence-based Marketing
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Examination of another type	4,5 CP	graded	Each winter term	6

Courses					
WT 25/26	2572170	Marketing Analytics	2 SWS	Lecture / 🗣	Klarmann
WT 25/26	2572171		1 SWS	Practice / 🗣	Jacob
WT 26/27	2572170	Marketing Analytics	2 SWS	Lecture / 🗣	Klarmann

Legend: 📱 Online, ✲ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Alternative (according to §4(2), 3 of the examination regulation) exam assessment (working on tasks in groups during the lecture).

Prerequisites
None

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-WIWI-107720 - Market Research must have been started.

Recommendations
It is strongly recommended to complete the course "Market Research" prior to taking the "Marketing Analytics" course.

Additional Information
"Marketing Analytics" is offered as a block course with an alternative exam assessment.
Starting in the winter semester 22/23, the course will be scheduled to be completed after two thirds of the semester. For further information, please contact the Marketing and Sales Research Group (marketing.iism.kit.edu). Exchange students can bypass the requirement of passing Market Research if they can prove that they possess sufficient statistical knowledge based on courses attended at their home institution. This will be examined individually by the Marketing and Sales Research Group.

Below you will find excerpts from courses related to this module component:

V Marketing Analytics

Lecture (V)
On-Site

2572170, WS 25/26, 2 SWS, Language: English, Open in study portal

Content
In this course various relevant market research questions are addressed, as for example measuring and understanding customer attitudes, preparing strategic decisions and sales forecasting. In order to analyze these questions, students learn to handle social media data, panel data, nested observations and experimental design. To analyze the data, advanced methods, as for example multilevel modeling and return on marketing models are taught. Also, problems of causality are addressed in-depth. The lecture is accompanied by a computer-based exercise, in the course of which the methods are applied practically.

Students

- receive based on the course market research an overview of advanced empirical methods
- learn in the course of the lecture to handle advanced data collection and data analysis methods
- are based on the acquired knowledge able to interpret results and derive strategic implications

Total workload for 4.5 ECTS: ca. 135 hours.

In order to attend Marketing Analytics, students are required to have passed the course Market Research.

Exchange students can bypass the requirement of passing Market Research if they can prove that they possess sufficient statistical knowledge based on courses attended at their home institution. This will be examined individually by the Marketing & Sales Research Group.

For further information please contact the Marketing and Sales Research Group (marketing.iism.kit.edu).

Literature

- Hanssens, Dominique M., Parsons, Leonard J., Schultz, Randall L. (2003), Market response models: Econometric and time series analysis, 2nd ed, Boston.
- Gelman, Andrew, Hill, Jennifer (2006), Data analysis using regression and multilevel/hierarchical models, New York.
- Cameron, A. Colin, Trivedi, Pravin K. (2005), Microeconometrics: methods and applications, New York.
- Chapman, Christopher, Feit, Elea M. (2015), R for Marketing Research and Analytics, Cham.
- Ledolter, Johannes (2013), Data mining and business analytics with R, New York.

V

2572171, WS 25/26, 1 SWS, Language: English, Open in study portal

Practice (Ü)
On-Site

Content
Tasks parallel to the lecture to work on in a group of students.

Organizational issues
Blockveranstaltung: genaue Uhrzeiten und Raum werden noch bekannt gegeben

V

Marketing Analytics
2572170, WS 26/27, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content
In this course various relevant market research questions are addressed, as for example measuring and understanding customer attitudes, preparing strategic decisions and sales forecasting. In order to analyze these questions, students learn to handle social media data, panel data, nested observations and experimental design. To analyze the data, advanced methods, as for example multilevel modeling and return on marketing models are taught. Also, problems of causality are addressed in-depth. The lecture is accompanied by a computer-based exercise, in the course of which the methods are applied practically.

Students

- receive based on the course market research an overview of advanced empirical methods
- learn in the course of the lecture to handle advanced data collection and data analysis methods
- are based on the acquired knowledge able to interpret results and derive strategic implications

Total workload for 4.5 ECTS: ca. 135 hours.

In order to attend Marketing Analytics, students are required to have passed the course Market Research.

Exchange students can bypass the requirement of passing Market Research if they can prove that they possess sufficient statistical knowledge based on courses attended at their home institution. This will be examined individually by the Marketing & Sales Research Group.

For further information please contact the Marketing and Sales Research Group (marketing.iism.kit.edu).

Literature

- Hanssens, Dominique M., Parsons, Leonard J., Schultz, Randall L. (2003), Market response models: Econometric and time series analysis, 2nd ed, Boston.
- Gelman, Andrew, Hill, Jennifer (2006), Data analysis using regression and multilevel/hierarchical models, New York.
- Cameron, A. Colin, Trivedi, Pravin K. (2005), Microeconometrics: methods and applications, New York.
- Chapman, Christopher, Feit, Elea M. (2015), R for Marketing Research and Analytics, Cham.
- Ledolter, Johannes (2013), Data mining and business analytics with R, New York.

T 6.417 Module component: Master's Thesis [T-WIWI-115109]

Coordinators: Studiendekan des KIT-Studienganges
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-107719 - Master's Thesis

Type	Credits	Grading	Language	Version
Final Thesis	30 CP	graded	D	1

Assessment
see module description

Prerequisites
see module description

Final Thesis
This module component represents a final thesis. The following periods have been supplied:

Submission deadline	6 months
Maximum extension period	3 months
Correction period	8 weeks

T 6.418 Module component: Matching Theory [T-WIWI-113264]

Coordinators: Prof. Dr. Clemens Puppe
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101500 - Microeconomic Theory
M-WIWI-104908 - Economics

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	see Annotations	1

Courses					
WT 25/26	2500042	Matching Theory	3 SWS	Lecture / Practice / 🗣	Okulicz

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The examination will be offered for the last time in the summer semester 2026. Success is assessed in the form of a written examination lasting 90 minutes.

Additional Information
The course is no longer offered.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Matching Theory

2500042, WS 25/26, 3 SWS, Language: English, Open in study portal

Lecture / Practice (VÜ)
On-Site

Content
How should we organize recruitment of students to schools? Could we improve the placement of doctors to hospitals? Why there always seems to be a better roommate to the one you currently have? Matching Theory answers all these questions and more. During the course we will formally study mathematical systems of allocating goods and people, and see their many real life applications from organizing kidney exchange to improving dating apps. The course will cover three main topics in Matching Theory and Market Design: (1) assignment problems (e.g., allocation of social housing), (2) two-sided matching (e.g., allocation of children to schools), (3) transferable-utility matching (e.g., labor market).

The students are expected to:

1. Understand the mathematical properties of allocations and commonly used mechanism
2. Understand the connection between Matching Theory and real-life allocation systems
3. Be able to use their knowledge to propose solutions for novel real-life problems

Workload:

- Attendance time – Lecture (2 SWS × 15 weeks): 30 h
- Attendance time – Tutorial (1 SWS × 15 weeks): 15 h
- Self-study – Lecture: 45 h
- Self-study – Tutorial: 25 h
- Exam preparation: 20 h
- **Total workload: 135 h**

T 6.419 Module component: Material Science II for Business Engineers [T-MACH-102079]

Coordinators: Dr.-Ing. Susanne Wagner
Organisation: KIT Department of Mechanical Engineering

Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Written examination	5 CP	graded	Each summer term	2

Courses					
ST 2026	2126782	Materials Science II for Industrial Engineers	2 SWS	Lecture / 🗣	Wagner

Legend: 🖥 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment consists of a written examination (150 min) taking place in the recess period (according to Section 4(2), 1 of the examination regulation). The examination takes place every semester. Re-examinations are offered at every ordinary examination date. The examination at the end of the winter term is carried out by a written or oral exam.

Prerequisites
T-MACH-102078 „Werkstoffkunde I für Wirtschaftsingenieure" has to be completed beforehand.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-MACH-102078 - Materials Science I must have been passed.

Workload
150 hours

Below you will find excerpts from courses related to this module component:

V Materials Science II for Industrial Engineers

2126782, SS 2026, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Literature
Weiterführende Literatur:

- Werkstoffwissenschaften - Eigenschaften, Vorgänge, Technologien, B. Ilscher, Springer – Verlag, Berlin Heidelberg New York, ISBN 3-540-10725-5
- Werkstoffwissenschaften, Schatt, Werner / Worch, Hartmut (Hrsg.) Wiley-VCH, Weinheim, ISBN-10: 3-527-30535-1
- Metallkunde für das Maschinenwesen I/II, K.G. Schmitt-Thomas, Springer-Verlag, ISBN 3-540-51913-0
- Materials Science and Engineering – An Introduction, William D. Callister (Jr.), John Wiley & Son, ISBN-10: 978-0-471-73696-7

T 6.420 Module component: Materials Science I [T-MACH-102078]

Coordinators: Dr.-Ing. Susanne Wagner
Organisation: KIT Department of Mechanical Engineering

Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Written examination	3 CP	graded	Each winter term	1

Courses					
WT 25/26	2125760	Materials Science I	2 SWS	Lecture /	Pagnan Furlan, Wagner

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
The assessment consists of a written examination (150 min) taking place in the recess period (according to Section 4(2), 1 of the examination regulation). The examination takes place every semester. Re-examinations are offered at every ordinary examination date. The examination at the end of the summer term is carried out by a written or oral exam.

Prerequisites
None

Workload
90 hours

Below you will find excerpts from courses related to this module component:

V Materials Science I

2125760, WS 25/26, 2 SWS, Language: German, Open in study portal

Lecture (V)
Blended (On-Site/Online)

Literature
Weiterführende Literatur:

Werkstoffwissenschaften - Eigenschaften, Vorgänge, Technologien, B. Ilscher, Springer – Verlag, Berlin Heidelberg New York, ISBN 3-540-10725-5
Werkstoffwissenschaften, Schatt, Werner / Worch, Hartmut (Hrsg.) Wiley-VCH, Weinheim, ISBN-10: 3-527-30535-1
Metallkunde für das Maschinenwesen I/II, K.G. Schmitt-Thomas, Springer-Verlag, ISBN 3-540-51913-0
Materials Science and Engineering – An Introduction, William D. Callister (Jr.), John Wiley & Son, ISBN-10: 978-0-471-73696-7 .

T 6.421 Module component: Mathematical Methods for Production Systems [T-MACH-113914]

Coordinators: Dr.-Ing. Marion Baumann
Prof. Dr.-Ing. Kai Furmans

Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101278 - Material Flow in Networked Logistic Systems
M-WIWI-107718 - Mechanical Engineering I
M-WIWI-107730 - Mechanical Engineering II

Type	Credits	Grading	Term offered	Version
Oral examination	6 CP	graded	Each summer term	1

Courses					
ST 2026	2118081	Mathematical methods for Production Systems	4 SWS	Lecture / 🗣	Baumann

Legend: Online, Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment consists of an oral exam (20 min.) taking place in the recess period according to § 4 paragraph 2 Nr. 2 of the examination regulation.

Prerequisites
none

Additional Information
The course is offered in English.

Workload
180 hours

Below you will find excerpts from courses related to this module component:

V Mathematical methods for Production Systems

2118081, SS 2026, 4 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content
Media:

lecture notes, presentations

Learning Content:

- single server systems: M/M/1, M/G/1: priority rules, model of failures
- networks: open and closed approximations, exact solutions and approximations
- discrete-time modeling of queuing systems

Learning Goals:

Students are able to:

- Describe queueing systems with analytical solvable stochastic models,
- Derive approches for modeling and controlling material flow and production systems based on models of queueing theory,
- Use simulation and exakt methods.

Recommendations:

- Basic knowledge of statistic
- recommended compusory optional subject: Stochastics

Workload:

regular attendance: 42 hours
self-study: 198 hours

Organizational issues

- **Im Sommersemester 2026 ist die Veranstaltung auf maximal 30 Teilnehmer beschränkt.**
- **Die Anmeldung erfolgt durch Beitritt zum ILIAS-Kurs und Ausfüllen des Anmeldungsformulars (erforderliche Felder beim Beitritt zum ILIAS-Kurs).**
- **Die Anmeldung ist vom 01.03.2026 bis zum 31.03.2026 möglich. Die verfügbaren Plätze werden anschließend vergeben.**

Literature

Ronald W. Wolff (1989) Stochastic Modeling and the Theory of Queues, Englewood Cliffs, NJ : Prentice-Hall.
John A. Buzacott, J. George Shanthikumar (1993) Stochastic Models of Manufacturing Systems, Upper Saddle River, NJ : Prentice Hall.

T 6.422 Module component: Mathematical Methods in Hydraulics [T-MACH-113912]

Coordinators: Prof. Dr.-Ing. Marcus Geimer
Organisation: KIT Department of Mechanical Engineering
Part of: M-MACH-101266 - Automotive Engineering
M-MACH-101267 - Mobile Machines

Type	Credits	Grading	Term offered	Expansion	Version
Oral examination	4 CP	graded	Each term	1 semesters	3

Courses					
WT 25/26	2113090	Mathematical Methods of Fluid Power	2 SWS	Lecture / 🗣	Geimer

Legend: 🖥 Online, Blended (On-Site/Online), 🗣 On-Site, × Cancelled

Assessment
Oral examination, duration approx. 30min

Prerequisites
T-MACH-113913 - Exercises for Mathematical Methods of Fluid Power must be passed.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-MACH-113913 - Tutorial Mathematical Methods in Hydraulics must have been passed.

Additional Information
The course is offered in German.

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Mathematical Methods of Fluid Power

2113090, WS 25/26, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content

- Fundamentals of hydraulics,
- Description of components for converting mechanical energy into hydraulic energy and vice versa (pumps and motors),
- Presentation of components for controlling pressure and volume flow (pressure and flow valves)
- Illustration and description of hydraulic systems
- Solution methods for the pressure curve over time
- Simulation approaches for hydraulic networks

T 6.423 Module component: Mathematical Models and Methods for Production Systems [T-MACH-105189]

Coordinators: Dr.-Ing. Marion Baumann
Prof. Dr.-Ing. Kai Furmans

Organisation: KIT Department of Mechanical Engineering

Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Oral examination	6 CP	graded	Each summer term	2

Courses					
ST 2026	2118081	Mathematical methods for Production Systems	4 SWS	Lecture / 🗣	Baumann

Legend: 🖥 Online, ✿ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment consists of an oral exam (20 min.) taking place in the recess period according to § 4 paragraph 2 Nr. 2 of the examination regulation.

Prerequisites
none

Additional Information
The next course will take place in the summer semester of 2026.

The course is offered in English

Workload
180 hours

Below you will find excerpts from courses related to this module component:

V Mathematical methods for Production Systems

2118081, SS 2026, 4 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content
Media:

lecture notes, presentations

Learning Content:

- single server systems: M/M/1, M/G/1: priority rules, model of failures
- networks: open and closed approximations, exact solutions and approximations
- discrete-time modeling of queuing systems

Learning Goals:

Students are able to:

- Describe queueing systems with analytical solvable stochastic models,
- Derive approches for modeling and controlling material flow and production systems based on models of queueing theory,
- Use simulation and exakt methods.

Recommendations:

- Basic knowledge of statistic
- recommended compusory optional subject: Stochastics

Workload:

regular attendance: 42 hours
self-study: 198 hours

Organizational issues

- **Im Sommersemester 2026 ist die Veranstaltung auf maximal 30 Teilnehmer beschränkt.**
- **Die Anmeldung erfolgt durch Beitritt zum ILIAS-Kurs und Ausfüllen des Anmeldungsformulars (erforderliche Felder beim Beitritt zum ILIAS-Kurs).**
- **Die Anmeldung ist vom 01.03.2026 bis zum 31.03.2026 möglich. Die verfügbaren Plätze werden anschließend vergeben.**

Literature

Ronald W. Wolff (1989) Stochastic Modeling and the Theory of Queues, Englewood Cliffs, NJ : Prentice-Hall.
John A. Buzacott, J. George Shanthikumar (1993) Stochastic Models of Manufacturing Systems, Upper Saddle River, NJ : Prentice Hall.

T 6.424 Module component: Mathematical Models and Methods of the Theory of Thermochemical Processes [T-MACH-114062]

Coordinators: Dr. Viatcheslav Bykov
Organisation: KIT Department of Mechanical Engineering

Part of: M-WIWI-107718 - Mechanical Engineering I
M-WIWI-107730 - Mechanical Engineering II

Type	Credits	Grading	Term offered	Version
Oral examination	4,5 CP	graded	Each winter term	2

Courses					
WT 25/26	2165522	Mathematical Models and Methods of the Theory of Thermochemical Processes	2 SWS	Lecture / 🗣	Bykov

Legend: 🖥 Online, ❖ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Oral exam, approx. 30 min

Prerequisites
T-MACH-113942 and T-MACH-105419 must not be started

Additional Information
The course is offered in English.

Workload
135 hours

T 6.425 Module component: Mathematics for High Dimensional Statistics [T-WIWI-111247]

Coordinators: Prof. Dr. Oliver Grothe
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101473 - Mathematical Programming
M-WIWI-101637 - Analytics and Statistics
M-WIWI-103289 - Stochastic Optimization
M-WIWI-104902 - Statistics

Type	Credits	Grading	Term offered	Version
Oral examination	4,5 CP	graded	Irregular	1

Assessment
The assessment of the course (lecture and exercise) consists of an oral exam (approx. 30 min.) taking place in the recess period.

Prerequisites
None

Recommendations
Basic knowledge of mathematics and statistics is assumed.
Knowledge in multivariate statistics is an advantage, but not necessary for the course.

T 6.426 Module component: Mathematics I - Final Exam [T-MATH-111493]

Coordinators: Prof. Dr. Daniel Hug
Dr. Franz Nestmann
PD Dr. Steffen Winter

Organisation: KIT Department of Mathematics

Part of: M-WIWI-104905 - Mathematics

Type	Credits	Grading	Version
Written examination	5 CP	graded	1

Courses					
WT 26/27	0135000	Mathematik 1 für die Fachrichtung Wirtschaftswissenschaften	6 SWS	Lecture	Nestmann

T 6.427 Module component: Mathematics I - Final Exam for WiIng B.Sc. 2026 [T-MATH-115205]

Coordinators: Prof. Dr. Daniel Hug
Dr. Franz Nestmann
PD Dr. Steffen Winter

Organisation: KIT Department of Mathematics

Part of: M-WIWI-104905 - Mathematics

Type	Credits	Grading	Version
Written examination	5 CP	graded	1

T 6.428 Module component: Mathematics I - Midterm Exam [T-MATH-111492]

Coordinators: Prof. Dr. Daniel Hug
Dr. Franz Nestmann
PD Dr. Steffen Winter

Organisation: KIT Department of Mathematics

Part of: M-WIWI-104905 - Mathematics

Type	Credits	Grading	Version
Written examination	5 CP	graded	1

Courses					
WT 26/27	0135000	Mathematik 1 für die Fachrichtung Wirtschaftswissenschaften	6 SWS	Lecture	Nestmann

T 6.429 Module component: Mathematics I - Midterm Exam for WiIng B.Sc. 2026 [T-MATH-115204]

Coordinators: Prof. Dr. Daniel Hug
Dr. Franz Nestmann
PD Dr. Steffen Winter

Organisation: KIT Department of Mathematics

Part of: M-WIWI-104905 - Mathematics

Type	Credits	Grading	Version
Written examination	5 CP	graded	1

T 6.430 Module component: Mathematics II - Final Exam [T-MATH-111496]

Coordinators: Prof. Dr. Daniel Hug
Dr. Franz Nestmann
PD Dr. Steffen Winter

Organisation: KIT Department of Mathematics

Part of: M-WIWI-104905 - Mathematics

Type	Credits	Grading	Version
Written examination	3,5 CP	graded	1

T 6.431 Module component: Mathematics II - Final Exam for WiIng B.Sc. 2026 [T-MATH-115207]

Coordinators: Prof. Dr. Daniel Hug
Dr. Franz Nestmann
PD Dr. Steffen Winter

Organisation: KIT Department of Mathematics

Part of: M-WIWI-104905 - Mathematics

Type	Credits	Grading	Version
Written examination	3,5 CP	graded	1

T 6.432 Module component: Mathematics II - Midterm Exam [T-MATH-111495]

Coordinators: Prof. Dr. Daniel Hug
Dr. Franz Nestmann
PD Dr. Steffen Winter

Organisation: KIT Department of Mathematics

Part of: M-WIWI-104905 - Mathematics

Type	Credits	Grading	Version
Written examination	3,5 CP	graded	1

T 6.433 Module component: Mathematics II - Midterm Exam for WiIng B.Sc. 2026 [T-MATH-115206]

Coordinators: Prof. Dr. Daniel Hug
Dr. Franz Nestmann
PD Dr. Steffen Winter

Organisation: KIT Department of Mathematics

Part of: M-WIWI-104905 - Mathematics

Type	Credits	Grading	Version
Written examination	3,5 CP	graded	1

T 6.434 Module component: Mathematics III - Final Exam [T-MATH-111498]

Coordinators: Prof. Dr. Daniel Hug
Dr. Franz Nestmann
PD Dr. Steffen Winter

Organisation: KIT Department of Mathematics

Part of: M-WIWI-104905 - Mathematics

Type	Credits	Grading	Version
Written examination	4 CP	graded	1

T 6.435 Module component: Mathematics III - Final Exam for WiIng B.Sc. 2026 [T-MATH-115208]

Coordinators: Prof. Dr. Daniel Hug
Dr. Franz Nestmann
PD Dr. Steffen Winter

Organisation: KIT Department of Mathematics

Part of: M-WIWI-104905 - Mathematics

Type	Credits	Grading	Version
Written examination	4 CP	graded	1

T 6.436 Module component: Measurement and Control Technology [T-ETIT-112852]

Coordinators: Prof. Dr.-Ing. Michael Heizmann
Prof. Dr.-Ing. Sören Hohmann

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Expansion	Version
Written examination	6 CP	graded	Each summer term	1 semesters	1

Courses					
ST 2026	2302300	Measurement and Control Technology	2 SWS	Lecture / 🗣	Heizmann, Hohmann, Piscol, Baßler
ST 2026	2302301	Practice to 2302300 Measurement and Control Technology	2 SWS	Practice / 🗣	Heizmann, Hohmann, Piscol, Baßler

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment of success takes place in the form of a written examination lasting 120 minutes. The module grade is the grade of the written examination.

Prerequisites
none

Recommendations
Kenntnisse aus „Signale und Systeme“ sind hilfreich.

T 6.437 Module component: Mechanical Design I and II - CIW [T-MACH-104739]

Coordinators: Prof. Dr.-Ing. Sven Matthiesen
Organisation: KIT Department of Mechanical Engineering

Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Written examination	6 CP	graded	Each winter term	2

Assessment
Written Exam (90min) on the topics of MKLI and MKLII for CIW.

Prerequisites
The bricks "T-MACH-102132 - Maschinenkonstruktionslehre I, Vorleistung" and "T-MACH-102133 - Maschinenkonstruktionslehre II, Vorleistung" must be passed successfully.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-MACH-102133 - Mechanical Design II, Tutorial must have been passed.
2. The module component T-MACH-102132 - Mechanical Design I, Tutorial must have been passed.

Additional Information
The course is offered in German.

Workload
90 hours

T 6.438 Module component: Mechanical Design I, Tutorial [T-MACH-102132]

Coordinators: Prof. Dr.-Ing. Sven Matthiesen
Organisation: KIT Department of Mechanical Engineering

Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Coursework	1 CP	pass/fail	Each winter term	2

Assessment
To pass the preliminary work, attendance at 3 workshop sessions of the MKL1 transmission workshop and the passing of a colloquium at the beginning of each workshop are prerequisites. In addition, participation in an online test is a prerequisite

Prerequisites
None

Additional Information
The course is offered in German.

Workload
30 hours

T 6.439 Module component: Mechanical Design II, Tutorial [T-MACH-102133]

Coordinators: Prof. Dr.-Ing. Sven Matthiesen
Organisation: KIT Department of Mechanical Engineering

Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Coursework	1 CP	pass/fail	Each summer term	2

Assessment
IP-MATH-CIW-NWT: For passing the prerequisite it is necessary that a design task is successfully completed as a technical hand drawing

MIT: To pass the preliminary examination, attendance at workshop sessions and a colloquium at the beginning of each workshop are required.

Prerequisites
None

Additional Information
The course is offered in German.

Workload
30 hours

T 6.440 Module component: Mechatronical Systems and Products [T-MACH-112647]

Coordinators: Prof. Dr.-Ing. Sören Hohmann
Prof. Dr.-Ing. Sven Matthiesen

Organisation: KIT Department of Mechanical Engineering

Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Written examination	4 CP	graded	Each winter term	2

Courses					
ST 2026	2303003	Exercises for 2303161 Mechatronical Systems and Products	1 SWS	Practice / 🗣	Matthiesen, Hohmann, Teltschik
ST 2026	2303161	Mechatronical Systems and Products	2 SWS	Lecture / 🧩	Matthiesen, Hohmann

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
written exam (60 min)

Additional Information
The course is offered in German.

Workload
120 hours

T 6.441 Module component: Media Management [T-WIWI-112711]

Coordinators: Prof. Dr. Ann-Kristin Kupfer
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration
M-WIWI-106258 - Digital Marketing

Type	Credits	Grading	Term offered	Version
Examination of another type	4,5 CP	graded	Each winter term	1

Courses					
WT 25/26	2572192	Media Management	2 SWS	Lecture / 🗣	Kupfer
WT 25/26	2572193	Media Management Exercise	1 SWS	Practice / 🗣	Kupfer, Mitarbeiter, Schmitt
WT 26/27	2572192	Media Management	2 SWS	Lecture / 🗣	Kupfer
WT 26/27	2572193	Media Management Exercise	1 SWS	Practice / 🗣	Schmitt

Legend: 📱 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success is assessed in the form of an examination of another type. The following aspects are included in the assessment:

- Elaboration and presentation of a group task
- Written exam

Further details on the organization of the performance and the points system for the assessment will be announced in the lecture.

Prerequisites
None

Recommendations
Students are highly encouraged to actively participate in class.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Media Management
2572192, WS 25/26, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content
Students learn the theoretical foundations of media management and its most important concepts. They learn both about the key characteristics of both media products and media markets. They further get to know essential business models of media markets. Special emphasis will be given to understanding media consumers and the marketing mix of media products. A tutorial offers the opportunity to apply the key learnings of the lecture.

The learning objectives are as follows:

- Getting to know the theoretical foundations of media management
- Evaluating strategies for media products and services as media-specific marketing mix instruments
- Fostering critical and analytical thinking skills and the application of knowledge to marketing problems
- Improvement of skills and competences in the area of project management within the framework of group work
- Improvement of foreign language skills (business English)

Total time required for 4.5 credit points: approx. 135 hours

Attendance time: 30 hours

Self-study: 105 hours

Organizational issues
Appointments to be announced.

V **Media Management** Lecture (V)
2572192, WS 26/27, 2 SWS, Language: English, Open in study portal On-Site

Content
Students learn the theoretical foundations of media management and its most important concepts. They learn both about the key characteristics of both media products and media markets. They further get to know essential business models of media markets. Special emphasis will be given to understanding media consumers and the marketing mix of media products. A tutorial offers the opportunity to apply the key learnings of the lecture.

The learning objectives are as follows:

- Getting to know the theoretical foundations of media management
- Evaluating strategies for media products and services as media-specific marketing mix instruments
- Fostering critical and analytical thinking skills and the application of knowledge to marketing problems
- Improvement of skills and competences in the area of project management within the framework of group work
- Improvement of foreign language skills (business English)

Total time required for 4.5 credit points: approx. 135 hours

Attendance time: 30 hours

Self-study: 105 hours

T 6.442 Module component: Medical Image Processing for Guidance and Navigation [T-ETIT-113425]

Coordinators: Prof. Dr.-Ing. Maria Francesca Spadea
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-106672 - Medical Image Processing for Guidance and Navigation

Type	Credits	Grading	Term offered	Version
Oral examination	9 CP	graded	Each winter term	2

Courses					
WT 25/26	2305297	Medical Image Processing for Guidance and Navigation	6 SWS	Lecture / Practice / 🗣	Spadea, Raggio, Riggio, Hopp

Legend: 🖥 Online, ✇ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The examination takes place within the framework of an oral overall examination of approx. 30 minutes about the lecture including a presentation and discussion of the project developed during the course. The overall impression is rated.

The module grade is the grade of the oral exam.

A bonus can be earned for submitting homework that will be provided during the lecture time.

The exact criteria for awarding a bonus will be announced at the beginning of the lecture period. If the grade in the oral exam is between 4.0 and 1.3, the bonus improves the grade by 0.3 or 0.4.

Bonus points do not expire and are retained for any examinations taken at a later date.

Prerequisites
none

T 6.443 Module component: Membrane Technologies in Water Treatment [T-CIWVT-113236]

Coordinators: Prof. Dr. Harald Horn
Dr.-Ing. Florencia Saravia

Organisation: KIT Department of Chemical and Process Engineering

Part of: M-CIWVT-101122 - Water Chemistry and Water Technology II
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Written examination	5 CP	graded	Each summer term	1

Courses					
ST 2026	2233010	Membrane Technologies in Water Treatment	2 SWS	Lecture / 🗣	Horn, Saravia
ST 2026	2233011	Membrane Technologies in Water Treatment - Excercises	1 SWS	Practice / 🧩	Horn, Saravia, und Mitarbeitende

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Learning control is an written examination lasting 90 minutes.

Prerequisites
Prerequisite: Submission of exercises, membrane design and short presentation (5 minutes, group work).

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-CIWVT-113235 - Exercises: Membrane Technologies must have been passed.

T 6.444 Module component: Metal Forming [T-MACH-105177]

Coordinators: Prof. Dr.-Ing. Thomas Herlan
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-106590 - Production Engineering
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each summer term	2

Courses					
ST 2026	2150681	Metal Forming	2 SWS	Lecture / 🗣	Herlan

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Oral Exam (20 min)

Prerequisites
none

Additional Information
The course is offered in German.

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Metal Forming

2150681, SS 2026, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content

At the beginning of the lecture the basics of metal forming are briefly introduced. The focus of the lecture is on massive forming (forging, extrusion, rolling) and sheet forming (car body forming, deep drawing, stretch drawing). This includes the systematic treatment of the appropriate metal forming Machines and the corresponding tool technology. Aspects of tribology, as well as basics in material science and aspects of production planning are also discussed briefly. The plastic theory is presented to the extent necessary in order to present the numerical simulation method and the FEM computation of forming processes or tool design. The lecture will be completed by product samples from the forming technology.

The topics are as follows:

- Introduction and basics
- Hot forming
- Metal forming machines
- Tools
- Metallographic fundamentals
- Plastic theory
- Tribology
- Sheet forming
- Extrusion
- Numerical simulation
- Sustainability in forming technology
- I4.0 and AI applications
- Production planning and quality assurance

Learning Outcomes:

The students …

- are able to reflect the basics, forming processes, tools, Machines and equipment of metal forming in an integrated and systematic way.
- are capable to illustrate the differences between the forming processes, tools, machines and equipment with concrete examples and are qualified to analyze and assess them in terms of their suitability for the particular application.
- are also able to transfer and apply the acquired knowledge to other metal forming problems.

Workload:

regular attendance: 21 hours
self-study: 99 hours

Organizational issues

Vorlesungstermine freitags, wöchentlich.
Die konkreten Termine werden in der ersten Vorlesung bekannt gegeben und auf der Institutshomepage und ILIAS veröffentlicht.

Zur Vertiefung des im Rahmen der Lehrveranstaltung erworbenen Wissens werden die theoretischen Vorlesungseinheiten durch Praxiseinheiten im Umfeld der Karlsruher Forschungsfabrik (https://www.karlsruher-forschungsfabrik.de) unterstützt.

The theoretical lectures are complemented by practical lectures in the Karlsruhe Research Factory (https://www.karlsruher-forschungsfabrik.de/en.html) to deepen the acquired knowledge.

Literature
Medien:
Skript zur Veranstaltung wird über (https://ilias.studium.kit.edu/) bereitgestellt.

Media:
Lecture notes will be provided in Ilias (https://ilias.studium.kit.edu/)

T 6.445 Module component: Methods and Models in Transportation Planning [T-BGU-101797]

Coordinators: Prof. Dr.-Ing. Peter Vortisch
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: M-BGU-101065 - Transportation Modelling and Traffic Management
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Expansion	Version
Oral examination	3 CP	graded	Each term	1 semesters	1

Courses					
WT 25/26	6232701	Calculation Methods and Models in Traffic Planning	2 SWS	Lecture / Practice / 🗣	Vortisch, Mitarbeiter/innen

Legend: 🖥 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
oral exam, appr. 20 min.

Prerequisites
None

Recommendations
None

Additional Information
None

Workload
90 hours

T 6.446 Module component: Methods in Economics [T-WIWI-114054]

Coordinators: Prof. Dr. Ingrid Ott
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104908 - Economics
M-WIWI-107011 - Economics of Innovation and Growth

Type	Credits	Grading	Term offered	Version
Examination of another type	1,5 CP	graded	Each summer term	1

Courses					
ST 2026	2560240	Methods in Economics	1 SWS	Lecture / 🗣	Ott

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, × Cancelled

Assessment
The assessment of success of the lecture will be in the form of an examination of another type.

Prerequisites
None

Recommendations
Basic knowledge of micro- and macroeconomics is assumed, as taught in the courses Economics I [2600012] and Economics II [2600014]. Further, it is assumed that students have interest in using quantiative-mathematical methods.

Additional Information
Due to Prof. Dr. Ingrid Ott's research semester, the lectures for this course will not be offered in the summer semester 2026.

Below you will find excerpts from courses related to this module component:

V Methods in Economics

2560240, SS 2026, 1 SWS, Language: German/English, Open in study portal

Lecture (V)
On-Site

Content
The economic exploitation of inventions is an important part of innovation economics. Intellectual property rights such as patents or trademarks play a central role. Within this workshop, the recording, processing and analysis of such intellectual property rights will be deepened, e.g. considering specific technologies. Students will learn how to work with relational databases, the econometric evaluation of recorded data, and methods for visualising them.

Learning objectives:

The student

- learns to query data sources.
- is able to analyse data with statistical methods.
- visualises and interprets data evaluations (e.g. using dashboards or methods of network analysis).

Recommendations:

An interest in working with data, basic knowledge on databases as well as basic knowledge in economics and statistics are advantageous.

Workload:

The total workload for this course is approximately 45 hours.

- Classes: ca. 5 h
- Self-study: ca. 40 h

Assessment:

Non exam assessment according to § 4 paragraph 3 of the examination regulation (SPO 2015).

Organizational issues
Due to research semester the course will not be offered in the summer semester 2026

The course is structured along three meetings. The first of which is getting an individual assignment, the second meeting to guide/direct the students, whereas the third meeting is assignment as a group project.

Students are offered the opportunity to participate in this course jointly with the course “Seminar in Economic Policy”, within the module “Economics of Innovation and Growth”. The work in both courses will be strongly related to each other, as students will work on the same topic from two different perspectives. Students in the course “Seminar in Economic Policy” will be provided with the opportunity to write a paper that addresses the results found by the students in the course “Methods in Economics”. Taking both courses together will enable the students to earn 4.5 ECTS.

Literature
Relevante Literatur wird in der Vorlesung bekanntgegeben.
(Relevant literature will be announced in the lecture.)

T 6.447 Module component: Methods in Innovation Management [T-WIWI-110263]

Coordinators: Prof. Dr. Marion Weissenberger-Eibl
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101507 - Innovation Management
M-WIWI-101507 - Innovation Management

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Irregular	1

Assessment
The assessment is an alternative exam assessment consisting of a presentation (25%) and a written paper (75%). The points system for the assessment is determined by the lecturer of the course. It will be announced at the beginning of the course.

Prerequisites
None.

Recommendations
Prior attendance of the course “Innovation Management: Concepts, Strategies and Methods” is recommended.

Additional Information
Teaching and learning format: Seminar

Workload
90 hours

T 6.448 Module component: Micro- and Nanosystem Integration for Medical, Fluidic and Optical Applications [T-MACH-108809]

Coordinators: apl. Prof. Dr. Ulrich Gengenbach
Dr. Liane Koker
apl. Prof. Dr. Ingo Sieber

Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101294 - Nanotechnology

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each winter term	1

Assessment
Oral exam (Duration: 30min)

Prerequisites
T-MACH-105695 "Selected topics of system integration for micro- and nanotechnology" must not be started.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-MACH-105695 - Selected topics of system integration for micro- and nanotechnology must not have been started.

Workload
120 hours

T 6.449 Module component: Micro- and nanotechnology in implant technology [T-MACH-111030]

Coordinators: Dr. Ralf Ahrens
Dr. Patrick Wolfgang Doll

Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101294 - Nanotechnology

Type	Credits	Grading	Term offered	Expansion	Version
Oral examination	4 CP	graded	Each winter term	1 semesters	1

Courses					
WT 25/26	2141871	Micro- and nanotechnology in implant technology	2 SWS	Lecture / 🗣	Doll, Ahrens, Guber
ST 2026	2141871	Micro- and nanotechnology in implant technology	2 SWS	Lecture / 🗣	Doll, Ahrens, Guber

Legend: Online, Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Oral Exam (20 min.)

Prerequisites
None

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Micro- and nanotechnology in implant technology

Lecture (V)
On-Site

2141871, WS 25/26, 2 SWS, Language: German, Open in study portal

Content
siehe oben

V Micro- and nanotechnology in implant technology

Lecture (V)
On-Site

2141871, SS 2026, 2 SWS, Language: German, Open in study portal

Content
siehe oben

T 6.450 Module component: Microactuators [T-MACH-101910]

Coordinators: Prof. Dr. Manfred Kohl
Organisation: KIT Department of Mechanical Engineering

Part of: M-ETIT-101158 - Sensor Technology I
M-MACH-101287 - Microsystem Technology
M-MACH-101290 - BioMEMS
M-MACH-101292 - Microoptics
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Written examination	4 CP	graded	Each summer term	3

Courses					
ST 2026	2142881	Microactuators	2 SWS	Lecture / 🗣	Kohl

Legend: 🖥 Online, ❂ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
written exam, 60 min.

Prerequisites
T-MACH-114036 must not be started

Additional Information
The course is offered in German

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Microactuators

2142881, SS 2026, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content
- Basic knowledge in the material science of the actuation principles
- Layout and design optimization
- Fabrication technologies
- Selected developments
- Applications

The lecture includes amongst others the following topics:

- Microelectromechnical systems: linear actuators, microrelais, micromotors
- Medical technology and life sciences: Microvalves, micropumps, microfluidic systems
- Microrobotics: Microgrippers, polymer actuators (smart muscle)
- Information technology: Optical switches, mirror systems, read/write heads

Literature
- Folienskript "Mikroaktorik"
- D. Jendritza, Technischer Einsatz Neuer Aktoren: Grundlagen, Werkstoffe, Designregeln und Anwendungsbeispiele, Expert-Verlag, 3. Auflage, 2008
- M. Kohl, Shape Memory Microactuators, M. Kohl, Springer-Verlag Berlin, 2004
- N.TR. Nguyen, S.T. Wereley, Fundamentals and applications of Microfluidics, Artech House, Inc. 2002
- H. Zappe, Fundamentals of Micro-Optics, Cambride University Press 2010

T 6.451 Module component: Microeconometrics [T-WIWI-112153]

Coordinators: Prof. Dr. Fabian Krüger
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104902 - Statistics

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Irregular	1

Courses					
ST 2026	2500019	Tutorial in Microeconometrics	2 SWS	Practice / 🗣	Krüger
ST 2026	2500032	Microeconometrics	2 SWS	Lecture / 🗣	Krüger

Legend: 🖥 Online, ✿ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment consists of a written examination (90 minutes) according to § 4(2), 1 of the examination regulation.

A bonus can be acquired by successful completion of an assignment (written report + short in-class presentation) during the semester. If the grade of the written examination is between 4.0 and 1.3, the bonus improves the grade by one grade level (0.3 or 0.4).

Prerequisites
None

Recommendations
Students are expected to have a good working knowledge of the linear regression model (e.g. by having attended the course `Volkswirtschaftslehre III: Einführung in die Ökonometrie', or attending it in the same semester as `Microeconometrics').

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Microeconometrics

Lecture (V)
On-Site

2500032, SS 2026, 2 SWS, Language: English, Open in study portal

Content
Microeconometrics is concerned with modeling data from an individual (`micro') unit like a person, household or firm. The response variables of interest are often discrete. For example, a person's type of employment may be coded as a binary variable (e.g. working in IT sector versus not working in IT sector), and a person's choice of transportation mode can be cast as a multinomial variable (e.g. bike, train, car, or other). These dependent variables often require nonlinear regression modeling.

The course first introduces maximum likelihood estimation which is particularly useful in microeconometrics. We then discuss econometric models for various types of response variables, as well as methods for estimation and model evaluation. Throughout the course, implementation via R software plays an important role.

Prerequisites: Course participants are expected to have a good working knowledge of the linear regression model (e.g. by having attended the course `Einführung in die Ökonometrie', or attending it in the same semester as `Microeconometrics').

Literature
Winkelmann, R., Boes, S. (2006): Analysis of Microdata. Springer.

T 6.452 Module component: Microenergy Technologies [T-MACH-105557]

Coordinators: Prof. Dr. Manfred Kohl
Dr. Jingyuan Xu

Organisation: KIT Department of Mechanical Engineering

Part of: M-WIWI-104907 - Engineering Sciences
M-WIWI-107718 - Mechanical Engineering I
M-WIWI-107730 - Mechanical Engineering II

Type	Credits	Grading	Term offered	Version
Oral examination	4,5 CP	graded	Each summer term	2

Courses					
ST 2026	2142897	Microenergy Technologies	2 SWS	Lecture / 🗣	Xu

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Oral examination (approx 30 min.)

Prerequisites
none

Additional Information
The course is offered in English.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Microenergy Technologies

2142897, SS 2026, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content
- Basic physical principles of energy conversion

- Layout and design optimization

- Technologies

- Selected devices

- Applications

The lecture includes amongst others the following topics:

- Micro energy harvesting of vibrations using different conversion principles (piezo, electrostatic, electromagnetic, etc.)
- Thermoelectric energy generation
- Novel thermal energy conversion principles (thermomagnetic, pyroelectric)
- Miniature scale solar devices
- RF energy harvesting
- Miniature scale heat pumping
- Solid-state cooling technologies (magneto-, electro-, mechanocalorics)
- Power management
- Energy storage technologies (microbatteries, supercapacito4rs, fuel cells)

Literature
- Folienskript "Micro Energy Technologies"

- Stephen Beeby, Neil White, Energy Harvesting for Autonomous Systems, Artech House, 2010

- Shashank Priya, Daniel J. Inman, Energy Harvesting Technologies, Springer, 2009

T 6.453 Module component: Mixed Integer Programming I [T-WIWI-102719]

Coordinators: Prof. Dr. Oliver Stein
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101473 - Mathematical Programming
M-WIWI-102832 - Operations Research in Supply Chain Management
M-WIWI-103289 - Stochastic Optimization
M-WIWI-104899 - Operations Research

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Irregular	1

Courses					
WT 25/26	2550138	Mixed-integer Programming I	2 SWS	Lecture / 🗣	Stein
WT 25/26	2550139	Exercises Mixed Integer Programming I	2 SWS	Practice / 🗣	Stein, Beck, Neussel
ST 2026	2550140	Mixed-integer Programming II	2 SWS	Lecture / 🗣	Stein

Legend: 🖥 Online, Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment of the lecture is a written examination (60 minutes) according to §4(2), 1 of the examination regulation. The successful completion of the exercises is required for admission to the written exam.

The examination is held in the semester of the lecture and in the following semester.

The examination can also be combined with the examination of *Mixed Integer Programming II* [25140]. In this case, the duration of the written examination takes 120 minutes.

Prerequisites
None

Recommendations
It is strongly recommended to visit at least one lecture from the Bachelor program of this chair before attending this course.

Additional Information
The lecture is offered irregularly. The curriculum of the next three years is available online (kop.ior.kit.edu).

Below you will find excerpts from courses related to this module component:

V Mixed-integer Programming I

Lecture (V)
On-Site

2550138, WS 25/26, 2 SWS, Language: German, Open in study portal

Content

Many optimization problems from economics, engineering and natural sciences are modeled with continuous as well as with discrete variables. Examples are the energy minimal design of a chemical process in which several reactors may be switched on or off, and portfolio optimization with limitations on the number of securities. For the algorithmic identification of optimal points of such problems an interaction of ideas from discrete as well as continuous optimization is necessary.

The lecture focusses on mixed-integer *linear* optimization problems and is structured as follows:

- Introduction, solvability, and basic concepts
- LP relaxation and error bounds for roundings
- Branch-and-bound method
- Gomory's cutting plane method
- Benders decomposition

The lecture is accompanied by exercises which, amongst others, offers the opportunity to implement and to test some of the methods on practically relevant examples.

Remark:

The treatment of mixed-integer *nonlinear* optimization problems forms the contents of the lecture "Mixed-integer Programming II".

Learning objectives:

The student

- knows and understands the fundamentals of linear mixed integer programming,
- is able to choose, design and apply modern techniques of linear mixed integer programming in practice.

Literature

- C.A. Floudas, Nonlinear and Mixed-Integer Optimization: Fundamentals and Applications, Oxford University Press, 1995
- J. Kallrath: Gemischt-ganzzahlige Optimierung, Vieweg, 2002
- D. Li, X. Sun: Nonlinear Integer Programming, Springer, 2006
- G.L. Nemhauser, L.A. Wolsey, Integer and Combinatorial Optimization, Wiley, 1988
- M. Tawarmalani, N.V. Sahinidis, Convexification and Global Optimization in Continuous and Mixed-Integer Nonlinear Programming, Kluwer, 2002.

V **Mixed-integer Programming II** — **Lecture (V)** **On-Site**

2550140, SS 2026, 2 SWS, Language: German, Open in study portal

Content

Many optimization problems from economics, engineering and natural sciences are modeled with continuous as well as with discrete variables. Examples are the energy minimal design of a chemical process in which several reactors may be switched on or off, portfolio optimization with limitations on the number of securities, the choice of locations to serve customers at minimum cost, and the optimal design of vote allocations in election procedures. For the algorithmic identification of optimal points of such problems an interaction of ideas from discrete as well as continuous optimization is necessary.

The lecture focusses on mixed-integer *nonlinear* optimization problems and is structured as follows:

- Continuous relaxation and error bounds for roundings
- Branch-and-Bound for convex and nonconvex problems
- Generalized Benders decomposition
- Outer approximation methods
- Lagrange relaxation
- Dantzig-Wolfe decomposition
- Heuristics

The lecture is accompanied by exercises which, amongst others, offers the opportunity to implement and to test some of the methods on practically relevant examples.

Remark:

The treatment of mixed-integer *linear* optimization problems forms the contents of the lecture "Mixed-integer Programming I".

Learning objectives:

The student

- knows and understands the fundamentals of nonlinear mixed integer programming,
- is able to choose, design and apply modern techniques of nonlinear mixed integer programming in practice.

Literature

- C.A. Floudas, Nonlinear and Mixed-Integer Optimization: Fundamentals and Applications, Oxford University Press, 1995
- J. Kallrath: Gemischt-ganzzahlige Optimierung, Vieweg, 2002
- D. Li, X. Sun: Nonlinear Integer Programming, Springer, 2006
- G.L. Nemhauser, L.A. Wolsey, Integer and Combinatorial Optimization, Wiley, 1988
- M. Tawarmalani, N.V. Sahinidis, Convexification and Global Optimization in Continuous and Mixed-Integer Nonlinear Programming, Kluwer, 2002.

T 6.454 Module component: Mixed Integer Programming II [T-WIWI-102720]

Coordinators: Prof. Dr. Oliver Stein
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101473 - Mathematical Programming
M-WIWI-102832 - Operations Research in Supply Chain Management
M-WIWI-103289 - Stochastic Optimization
M-WIWI-104899 - Operations Research

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Irregular	1

Courses					
ST 2026	2550140	Mixed-integer Programming II	2 SWS	Lecture / 🗣	Stein
ST 2026	2550141	Exercise to Mixed-integer Programming II	1 SWS	Practice / 🗣	Stein, Schwarze

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment of the lecture is a written examination (60 minutes) according to §4(2), 1 of the examination regulation. The successful completion of the exercises is required for admission to the written exam.
The examination is held in the semester of the lecture and in the following semester.
The examination can also be combined with the examination of *Mixed Integer Programming I* [2550138]. In this case, the duration of the written examination takes 120 minutes.

Prerequisites
None

Recommendations
It is strongly recommended to visit at least one lecture from the Bachelor program of this chair before attending this course.

Additional Information
The lecture is offered irregularly. The curriculum of the next three years is available online (kop.ior.kit.edu).

Below you will find excerpts from courses related to this module component:

V Mixed-integer Programming II

Lecture (V)
On-Site

2550140, SS 2026, 2 SWS, Language: German, Open in study portal

Content

Many optimization problems from economics, engineering and natural sciences are modeled with continuous as well as with discrete variables. Examples are the energy minimal design of a chemical process in which several reactors may be switched on or off, portfolio optimization with limitations on the number of securities, the choice of locations to serve customers at minimum cost, and the optimal design of vote allocations in election procedures. For the algorithmic identification of optimal points of such problems an interaction of ideas from discrete as well as continuous optimization is necessary.

The lecture focusses on mixed-integer *nonlinear* optimization problems and is structured as follows:

- Continuous relaxation and error bounds for roundings
- Branch-and-Bound for convex and nonconvex problems
- Generalized Benders decomposition
- Outer approximation methods
- Lagrange relaxation
- Dantzig-Wolfe decomposition
- Heuristics

The lecture is accompanied by exercises which, amongst others, offers the opportunity to implement and to test some of the methods on practically relevant examples.

Remark:

The treatment of mixed-integer *linear* optimization problems forms the contents of the lecture "Mixed-integer Programming I".

Learning objectives:

The student

- knows and understands the fundamentals of nonlinear mixed integer programming,
- is able to choose, design and apply modern techniques of nonlinear mixed integer programming in practice.

Literature

- C.A. Floudas, Nonlinear and Mixed-Integer Optimization: Fundamentals and Applications, Oxford University Press, 1995
- J. Kallrath: Gemischt-ganzzahlige Optimierung, Vieweg, 2002
- D. Li, X. Sun: Nonlinear Integer Programming, Springer, 2006
- G.L. Nemhauser, L.A. Wolsey, Integer and Combinatorial Optimization, Wiley, 1988
- M. Tawarmalani, N.V. Sahinidis, Convexification and Global Optimization in Continuous and Mixed-Integer Nonlinear Programming, Kluwer, 2002.

T 6.455 Module component: Mixed-Signal IC Design [T-ETIT-111845]

Coordinators: Prof. Dr.-Ing. Ahmet Cagri Ulusoy
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-105893 - Mixed-Signal IC Design

Type	Credits	Grading	Term offered	Version
Oral examination	3 CP	graded	Each summer term	1

Courses					
ST 2026	2308443	Mixed-Signal IC Design	2 SWS	Lecture / 🗣	Ulusoy

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, × Cancelled

Assessment
Success control takes place in form of an oral examination (approx. 30 min.)

T 6.456 Module component: MMIC Design Laboratory [T-ETIT-111006]

Coordinators: Prof. Dr.-Ing. Ahmet Cagri Ulusoy
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-105464 - MMIC Design Laboratory

Type	Credits	Grading	Term offered	Expansion	Version
Examination of another type	6 CP	graded	Each term	1 semesters	1

Courses					
WT 25/26	2308438	MMIC Design Laboratory	4 SWS	Practical course /	Ulusoy
ST 2026	2308423	MMIC Design Laboratory	4 SWS	Practical course /	Ulusoy

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
The written report (approx. 30-40 slides) and the oral presentation (approx. 40 min.) are used to mark the course. The overall impression is assessed.

T 6.457 Module component: Mobile Communications [T-ETIT-112127]

Coordinators: Prof. Dr.-Ing. Peter Rost
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-105971 - Mobile Communications

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each winter term	1

Courses					
WT 25/26	2310523	Mobile Communications	2 SWS	Lecture /	Rost
WT 25/26	2310524	Exercises for 2310523 Mobile Communications	1 SWS	Practice /	Rost

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
The success control takes place in the form of an oral examination lasting approx. 25 minutes. Before the examination, there is a preparation phase of 15 minutes in which preparatory tasks are solved.

Prerequisites
none

T 6.458 Module component: Mobile Communications II [T-ETIT-112679]

Coordinators: Prof. Dr.-Ing. Peter Rost
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-106244 - Mobile Communications II

Type	Credits	Grading	Term offered	Expansion	Version
Oral examination	3 CP	graded	Each summer term	1 semesters	1

Courses					
ST 2026	2310514	Mobile Communications II	2 SWS	Lecture / 🗘	Rost

Legend: 🖥 Online, 🗘 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The success control takes place in the form of an oral examination lasting approx. 25 minutes. Before the examination, there is a preparation phase of 15 minutes in which preparatory tasks are solved.

Prerequisites
none

T 6.459 Module component: Mobile Machines [T-MACH-105168]

Coordinators: Prof. Dr.-Ing. Marcus Geimer
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101267 - Mobile Machines

Type	Credits	Grading	Term offered	Version
Oral examination	8 CP	graded	Each summer term	3

Courses					
ST 2026	2114073	Mobile Machines	4 SWS	Lecture / 🗣	Geimer

Legend: 🖥 Online, Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment consists of an oral exam (45 min) taking place in the recess period. The exam takes place in every semester. Re-examinations are offered at every ordinary examination date.

Prerequisites
none

Recommendations
Knowledge of the mathematical method of hydraulics is assumed. Prior attendance of the course "Mathematical Methods of Hydraulics"[2113090] is recommended.

Additional Information
Learning objectives:

After successful participation in the course:

- the student will be able to name the wide range of mobile machinery
- know the possible applications and operating sequences of the most important mobile machines
- be able to describe selected subsystems and components

Content:

- Presentation of the components used and the most important mobile machines
- Basics and structure of the machines
- Practical insights into the development of the machines

Media:

Downloadable set of slides for the lecture

Book "Grundlagen mobiler Arbeitsmaschinen", Karlsruhe series of publications on vehicle systems technology, Volume 22, KIT Scientific Publishing

The course is offered in German.

Workload
240 hours

Below you will find excerpts from courses related to this module component:

V **Mobile Machines** — **Lecture (V) On-Site**
2114073, SS 2026, 4 SWS, Language: German, Open in study portal

Content

- Introduction of the required components and machines
- Basics of the structure of the whole system
- Practical insight in the development techniques

Knowledge in Fluid Power is required.

Recommendations:

It is recommended to attend the course *Fluid Power* [2113092] or *Mathematische Methoden der Hydraulik* [2113090] beforehand.

- regular attendance: 42 hours
- self-study: 184 hours

T 6.460 Module component: Mobility Services and New Forms of Mobility [T-BGU-103425]

Coordinators: Prof. Dr.-Ing. Martin Kagerbauer
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: M-BGU-101064 - Fundamentals of Transportation
M-BGU-101065 - Transportation Modelling and Traffic Management
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Expansion	Version
Oral examination	3 CP	graded	Each term	1 semesters	1

Courses					
ST 2026	6232811	Mobility Services and New Forms of Mobility	2 SWS	Lecture / 🗣	Kagerbauer

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
oral exam, appr. 20 min.

Prerequisites
None

Recommendations
None

Additional Information
None

Workload
90 hours

T 6.461 Module component: Model Based Application Methods [T-MACH-102199]

Coordinators: Dr. Frank Kirschbaum
Organisation: KIT Department of Mechanical Engineering

Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each summer term	1

Assessment
take-home exam, short presentation with oral examination

Prerequisites
none

T 6.462 Module component: Model Reduction Methods for Modeling and Simulation of Reacting Flows [T-MACH-114061]

Coordinators: Dr. Viatcheslav Bykov
Organisation: KIT Department of Mechanical Engineering

Part of: M-WIWI-107718 - Mechanical Engineering I
M-WIWI-107730 - Mechanical Engineering II

Type	Credits	Grading	Term offered	Version
Oral examination	4,5 CP	graded	Each summer term	2

Courses					
ST 2026	2166512	Model Reduction Methods for Modeling and Simulation of Reacting Flows	2 SWS	Lecture / 🗣	Bykov

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Oral exam, approx. 20 min

Prerequisites
T-MACH-114060 and T-MACH-105421 must not be started

Additional Information
The course is offered in English.

Workload
135 hours

T 6.463 Module component: Modeling and OR-Software: Advanced Topics [T-WIWI-106200]

Coordinators: Prof. Dr. Stefan Nickel
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-102808 - Digital Service Systems in Industry
M-WIWI-102832 - Operations Research in Supply Chain Management
M-WIWI-104899 - Operations Research

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each winter term	4

Courses					
WT 25/26	2550490	Modellieren und OR-Software: Fortgeschrittene Themen	3 SWS	Practical course /	Pomes, Linner, Nickel
WT 26/27	2550490	Modellieren und OR-Software: Fortgeschrittene Themen	3 SWS	Practical course /	Pomes, Linner, Nickel

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
The assessment is a written examination. The examination is held in every semester. The prerequisite can only be obtained in semesters in which the course exercises are offered.

Prerequisites
Prerequisite for admission to the exam is the successful participation in the exercises. This includes the processing and presentation of exercises.

Recommendations
Basic knowledge as conveyed in the module *Introduction to Operations Research* is assumed.

Successful completion of the course *Modeling and OR-Software: Introduction*.

Additional Information
Due to the limited number of participants, please register in advance. Further information can be found on the website of the course. Registration takes place via the Wiwi-Portal.The course is offered every winter semester. The planned lectures and courses for the next three years are announced online at https://dol.ior.kit.edu/Lehrveranstaltungen.php.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Modellieren und OR-Software: Fortgeschrittene Themen
2550490, WS 25/26, 3 SWS, Language: German, Open in study portal

Practical course (P)
Blended (On-Site/Online)

Content
The advanced course is designated for Master students that already attended the introductory course or gained equivalent experience elsewhere, e.g. during a seminar or bachelor thesis. We will work on advanced topics and methods in OR, among others, cutting planes, column generation and constraint programming. The Software used for the exercises is IBM ILOG CPLEX Optimization Studio. The associated modelling programming language is Python.

Please note that the second exam in the summer semester 26 will only be offered to students in their second try.

V Modellieren und OR-Software: Fortgeschrittene Themen
2550490, WS 26/27, 3 SWS, Language: German, Open in study portal

Practical course (P)
Blended (On-Site/Online)

Content

The advanced course is designated for Master students that already attended the introductory course or gained equivalent experience elsewhere, e.g. during a seminar or bachelor thesis. We will work on advanced topics and methods in OR, among others, cutting planes, column generation and constraint programming. The Software used for the exercises is IBM ILOG CPLEX Optimization Studio. The associated modelling programming language is Python.

Please note that the second exam in the summer semester 27 will only be offered to students in their second try.

Further information is available on our homepage. Registration takes place via the Wiwi portal, where the registration period can also be found.

T 6.464 Module component: Modeling and OR-Software: Introduction [T-WIWI-106199]

Coordinators: Prof. Dr. Stefan Nickel
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104899 - Operations Research

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each summer term	4

Courses					
ST 2026	2550490	Modellieren und OR-Software: Einführung	3 SWS	Practical course / 🗣	Nickel, Linner, Pomes

Legend: 🖥 Online, 🗣 Blended (On-Site/Online), 🗣 On-Site, x Cancelled

Assessment
The assessment is a written examination (60 min.). The successful completion of the exercises is required for admission to the written exam.
The examination is held in every semester

Recommendations
Firm knowledge of the contents from the lecture *Introduction to Operations Research I* [2550040] of the module *Operations Research.*

Additional Information
Due to capacity restrictions, registration before course start is required. For further information, see the webpage of the course. Registration is possible in the Wiwi portal. The lecture is offered in every term. The planned lectures and courses for the next three years are announced online at https://dol.ior.kit.edu/Lehrveranstaltungen.php.
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Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Modellieren und OR-Software: Einführung
2550490, SS 2026, 3 SWS, Language: German, Open in study portal

Practical course (P)
Blended (On-Site/Online)

Content
After an introduction to general concepts of modelling tools (implementation, data handling, result interpretation, …) CPLEX in combination with Python will be discussed which can be used to solve OR problems on a computer-aided basis. Subsequently, a broad range of exercises will be discussed. The main goals of the exercises from literature and practical applications are to learn the process of modeling optimization problems as linear or mixed-integer programs, to efficiently utilize the presented tools for solving these optimization problems and to implement heuristic solution procedures for mixed-integer programs.

Organizational issues
Die Teilnehmerzahl für diese Veranstaltung ist begrenzt.

Die Bewerbung erfolgt über das Wiwi-Portal.

Die Nachklausur zu dieser Veranstaltung wird nur für Nachschreibende angeboten.

T 6.465 Module component: Modeling and Simulation [T-WIWI-112685]

Coordinators: Prof. Dr.-Ing. Sanja Lazarova-Molnar
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101456 - Intelligent Systems and Services
M-WIWI-101472 - Informatics
M-WIWI-101628 - Emphasis in Informatics
M-WIWI-104901 - Informatics (KIT-Department of Economics and Management)

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each summer term	1

Courses					
ST 2026	2511100	Modeling and Simulation	2 SWS	Lecture / 🗣	Lazarova-Molnar
ST 2026	2511101	Exercises Modeling and Simulation	1 SWS	Practice / 🗣	Lazarova-Molnar, Mostafa

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Depending on the number of participants in the course, the exam will be offered either as an oral exam (20 min), or as a written exam (60 min).

The exam takes place every semester and can be repeated at every regular examination date.

Prerequisites
None

Recommendations
Some experience in programming and knowledge of basic mathematics and statistics.

Additional Information
Instruction is in the form of lectures and exercises. A detailed course schedule will be published before the start of the semester.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Modeling and Simulation
2511100, SS 2026, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content

Modeling and Simulation is the most widely used operations research / systems engineering technique for designing new systems and optimizing the performance of existing systems. In one way or another, just about every engineering or scientific field uses simulation as an exploration, modeling, or analysis technique. The course is designed to provide students with basic knowledge of modeling and simulation approaches and to provide them with first experience of using a simulation package. The course will focus on modeling and simulation of real-world discrete event systems. Examples of discrete events are customer arrivals at a queue of a service desk, machine failures in manufacturing systems, telephone calls in a call center, etc. Moreover, continuous and hybrid models will be also discussed. Topics include Discrete-Event Simulation, Input Modeling, Output Analysis, Random Number Generation, Verification and Validation, Stochastic Petri Nets and Markov Chains.

Competence Certificate

Depending on the number of participants in the course, the exam will be offered either as an oral exam (20 min), or as a written exam (60 min).

The exam takes place every semester and can be repeated at every regular examination date.

Learning Objectives

Knowledge:

- Demonstrate knowledge about general and specific theories, challenges, algorithms, methods, technologies, and tools related to modelling and simulation
- Demonstrate knowledge of two important classes of simulation:
    - Discrete-event Monte-Carlo simulation,
    - Continuous simulation with ODEs
- Demonstrate knowledge of algorithms necessary to build a simulator

Skills:

- Analyse suitability of an approach/tool for a given modelling problem
- Understand simulation models of various types
- Demonstrate methods and techniques to overcome common challenges in modelling and simulation
- Model simulation input data
- Analyse and model discrete stochastic systems
- Analyse and interpret simulation results

Competences:

- Use different methods to conduct simulation-based analysis of real-world data
- Build and simulate stochastic models
- Use simulation software

Prerequisites

Some experience in programming and knowledge of basic mathematics and statistics

Form of instruction

Lectures and exercises. A detailed course plan will be published before the semester start.

Literature

Discrete-Event System Simulation, 5th Edition

Jerry Banks, John S. Carson, II, Barry L. Nelson and David M. Nicol

T 6.466 Module component: Modeling the Dynamics of Financial Markets [T-WIWI-113414]

Coordinators: Prof. Dr. Maxim Ulrich
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration
M-WIWI-106660 - Modeling the Dynamics of Financial Markets

Type	Credits	Grading	Term offered	Version
Written examination	9 CP	graded	Each summer term	1

Assessment
The examination takes the form of a one-hour written comprehensive examination on the courses "Dynamic Capital Marke Theory", "Essentials for Dynamic Financial Machine Learning" and "Exercises, Python, Resesearch Frontier in Dynamic Capital Markets".

Recommendations
Recommendation: Knowledge in the fields of Advanced Statistics, Deep Learning, Financial Economics, Differential Equations, Optimization.

Workload
270 hours

T 6.467 Module component: Modelling and Identification [T-ETIT-100699]

Coordinators: Prof. Dr.-Ing. Sören Hohmann
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each winter term	1

Prerequisites
none

T 6.468 Module component: Morphodynamics [T-BGU-101859]

Coordinators: Prof. Dr. Mario Jorge Rodrigues Pereira da Franca
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Expansion	Version
Oral examination	3 CP	graded	Each term	1 semesters	1

Courses					
ST 2026	6222805	Landscape and River Morphology	2 SWS	Lecture / Practice / 🗣	Rodrigues Pereira da Franca, Vanzo

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Prerequisites
none

Workload
90 hours

6.469 Module component: Multicriteria Optimization [T-WIWI-111587]

Coordinators: Prof. Dr. Oliver Stein
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101473 - Mathematical Programming
M-WIWI-102832 - Operations Research in Supply Chain Management
M-WIWI-103289 - Stochastic Optimization
M-WIWI-104899 - Operations Research

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	see Annotations	1

Courses					
WT 26/27	2550155	Multicriteria Optimization	2 SWS	Lecture / 🗣	Stein
WT 26/27	2550156	Exercises to Multicriteria Optimization	1 SWS	Practice / 🗣	Stein, Neussel

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment of the lecture is a written examination (60 minutes) according to §4(2), 1 of the examination regulation. The successful completion of the exercises is required for admission to the written exam.
The examination is held in the semester of the lecture and in the following semester.

Prerequisites
None

Recommendations
It is strongly recommended to visit at least one lecture from the Bachelor program of this chair before attending this course.

Additional Information
The course is offered every second winter semester (starting WiSe 22/23). The curriculum of the next three years is available online (www.ior.kit.edu).

Contents:

Multicriteria optimization deals with optimization problems with multiple objective functions. In practice, the minimization or maximization of several objectives often conflict with each other, such as weight and stability of mechanical components, return and risk of stock portfolios, or cost and duration of transports. Various scalarization approaches allow one to formulate single-objective problems that can be solved using nonlinear or global optimization techniques, and whose optimal points have a reasonable interpretation for the underlying multicriteria problem.

However, some seemingly obvious scalarization approaches suffer from various drawbacks, so that regardless of scalarization approaches, it is necessary to clarify what is meant by the solution of a multicriteria optimization problem in the first place. For such Pareto-optimal points, optimality conditions and solution procedures based on them can be formulated. From the usually non-unique Pareto set, decision makers finally choose an alternative based on their subjective preferences.

The lecture gives a mathematically sound introduction to multicriteria optimization and is structured as follows:

- Introductory examples and terminology
- Solution concepts
- Methods for the determination of the Pareto set
- Selection of Pareto-optimal points under subjective preferences

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Multicriteria Optimization

2550155, WS 26/27, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content

Multicriteria optimization deals with optimization problems with multiple objective functions. In practice, the minimization or maximization of several objectives often conflict with each other, such as weight and stability of mechanical components, return and risk of stock portfolios, or cost and duration of transports. Various scalarization approaches allow one to formulate single-objective problems that can be solved using nonlinear or global optimization techniques, and whose optimal points have a reasonable interpretation for the underlying multicriteria problem.

However, some seemingly obvious scalarization approaches suffer from various drawbacks, so that regardless of scalarization approaches, it is necessary to clarify what is meant by the solution of a multicriteria optimization problem in the first place. For such Pareto-optimal points, optimality conditions and solution procedures based on them can be formulated. From the usually non-unique Pareto set, decision makers finally choose an alternative based on their subjective preferences.

The lecture gives a mathematically sound introduction to multicriteria optimization and is structured as follows:

- Introductory examples and terminology
- Solution concepts
- Methods for the determination of the Pareto set
- Selection of Pareto-optimal points under subjective preferences

Learning objectives:

The student

- knows and understands the fundamentals of multicriteria optimization,
- is able to choose, design and apply modern techniques of multicriteria optimization in practice.

Literature

- M. Ehrgott, Multicriteria Optimization, Second Edition, Springer, Berlin, 2005
- J. Jahn, Vector Optimization, Second Edition, Springer, Berlin, 2011
- K. Miettinen, Nonlinear Multiobjective Optimization, Springer, New York, 2004
- Y. Sawaragi, H. Nakayama, T. Tanino, Theory of Multiobjective Optimization, Academic Press, Orlando, FL, 1985

T 6.470 Module component: Multivariable Control Systems [T-ETIT-114727]

Coordinators: Prof. Dr.-Ing. Sören Hohmann
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-107673 - Automation and Control
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Version
Written examination	6 CP	graded	1

Courses					
WT 25/26	2303193	Multivariable Control Systems	3 SWS	Lecture / 🗣	Kluwe
WT 25/26	2303194	Tutorial to 2303194 "Multivariable Control Systems"	1 SWS	Practice / 🗣	Fehn
WT 26/27	2303193	Multivariable Control Systems	3 SWS	Lecture / 🗣	Kluwe
WT 26/27	2303194	Tutorial to 2303194 "Multivariable Control Systems"	1 SWS	Practice / 🗣	Fehn

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The examination takes place in form of a written examination lasting 120 minutes.

The module grade is the grade of the written exam.

Prerequisites
none

T 6.471 Module component: Multivariate Statistical Methods [T-WIWI-103124]

Coordinators: Prof. Dr. Oliver Grothe
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101473 - Mathematical Programming
M-WIWI-101637 - Analytics and Statistics
M-WIWI-101639 - Econometrics and Statistics II
M-WIWI-103289 - Stochastic Optimization
M-WIWI-104899 - Operations Research

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Irregular	1

Courses					
ST 2026	2550554	Multivariate Methods for Data Analysis	2 SWS	Lecture / 🗣	Grothe
ST 2026	2550555	Übung zu Multivariate Verfahren	2 SWS	Practice / 🗣	Vetter

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Witten examination lasting 60 minutes.

The examination is offered during the examination period of the lecture semester. Only repeaters (and not first-time writers) are admitted to the repeat examination in the examination period of the following semester.

Prerequisites
None

Recommendations
Due to the strong quantitative focus and the treatment of advanced statistical methods, in-depth prior knowledge of statistics (comparable to the content of the 'Advanced Statistics' course) is very advantageous for successful completion of the course. Students without such prior knowledge are advised to prepare intensively before the start of the course, as a significantly higher level of familiarization is to be expected.

In addition, prior attendance of the Bachelor course 'Analysis of multivariate data' is recommended. Alternatively, the lecture notes for this course can be made available to students to help them prepare independently.

Additional Information
The course (lecture and exercise) is offered irregularly. Detailed information can be found on the chair's website.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Multivariate Methods for Data Analysis

2550554, SS 2026, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Literature
Skript zur Vorlesung

T 6.472 Module component: Nano-Optics [T-PHYS-102282]

Coordinators: PD Dr. Andreas Naber
Organisation: KIT Department of Physics
Part of: M-MACH-101294 - Nanotechnology

Type	Credits	Grading	Term offered	Version
Oral examination	8 CP	graded	Each winter term	2

Courses					
WT 25/26	4020021	Nano-Optics	3 SWS	Lecture / 🗣	Naber
WT 25/26	4020022	Exercises to Nano-Optics	1 SWS	Practice / 🗣	Naber

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Prerequisites
none

T 6.473 Module component: Nature-Inspired Optimization Methods [T-WIWI-102679]

Coordinators: Prof. Dr. Pradyumn Kumar Shukla
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101472 - Informatics
M-WIWI-101628 - Emphasis in Informatics
M-WIWI-104901 - Informatics (KIT-Department of Economics and Management)

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each summer term	2

Courses					
ST 2026	2511106	Nature-Inspired Optimization Methods	2 SWS	Lecture /	Shukla
ST 2026	2511107	Übungen zu Nature-Inspired Optimization Methods	1 SWS	Practice /	Shukla

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
Success is assessed in the form of a written (60 min) or oral examination.

The examination for this course is offered both in the semester in which the course takes place and in the following semester. In the following semester (winter term), participation in the exam is **exclusively reserved for students who did not pass the first attempt.** This regulation applies from summer semester 2025.

Prerequisites
None

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Nature-Inspired Optimization Methods

2511106, SS 2026, 2 SWS, Language: English, Open in study portal

Lecture (V)
Blended (On-Site/Online)

Content
Many optimization problems are too complex to be solved to optimality. A promising alternative is to use stochastic heuristics, based on some fundamental principles observed in nature. Examples include evolutionary algorithms, ant algorithms, or simulated annealing. These methods are widely applicable and have proven very powerful in practice. During the course, such optimization methods based on natural principles are presented, analyzed and compared. Since the algorithms are usually quite computational intensive, possibilities for parallelization are also investigated.

Learning objectives:

Students learn:

- Different nature-inspired methods: local search, simulated annealing, tabu search, evolutionary algorithms, ant colony optimization, particle swarm optimization
- Different aspects and limitation of the methods
- Applications of such methods
- Multi-objective optimization methods
- Constraint handling methods
- Different aspects in parallelization and computing platforms

Literature
* E. L. Aarts and J. K. Lenstra: 'Local Search in Combinatorial Optimization'. Wiley, 1997 * D. Corne and M. Dorigo and F. Glover: 'New Ideas in Optimization'. McGraw-Hill, 1999 * C. Reeves: 'Modern Heuristic Techniques for Combinatorial Optimization'. McGraw-Hill, 1995 * Z. Michalewicz, D. B. Fogel: How to solve it: Modern Heuristics. Springer, 1999 * E. Bonabeau, M. Dorigo, G. Theraulaz: 'Swarm Intelligence'. Oxford University Press, 1999 * A. E. Eiben, J. E. Smith: 'Introduction to Evolutionary Computation'. * M. Dorigo, T. Stützle: 'Ant Colony Optimization'. Bradford Book, 2004 Springer, 2003

T 6.474 Module component: Navigating the Future of Work: Behavioral and Experiment Methods in the Workplace [T-WIWI-114328]

Coordinators: Prof. Dr. Petra Nieken
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-107187 - Quantitative Core

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each winter term	1

Assessment
The assessment of this course is a written examination of 1 hour. The exam takes place in every semester. Re-examinations are offered at every ordinary examination date.
In case of a small number of registrations, we might offer an oral exam instead of a written exam.

Prerequisites
None

Workload
135 hours

T 6.475 Module component: Network Security: Architectures and Protocols [T-INFO-101319]

Coordinators: Prof. Dr. Martina Zitterbart
Organisation: KIT Department of Informatics
Part of: M-WIWI-104909 - Informatics (Department of Informatics)

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each summer term	1

Courses					
ST 2026	24601	Network Security: Architectures and Protocols	2 SWS	Lecture / 🗣	Baumgart, Bless, Zitterbart

Legend: 🖥 Online, ✲ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment is carried out as an oral examination (§ 4 Abs. 2 Nr. 2 SPO) lasting 20 minutes.

Depending on the number of participants, it will be announced six weeks before the examination (Section 6 (3) SPO) whether the assessment will take the form of an oral examination of approx.

- in the form of an oral examination of approx. 30 minutes in accordance with § 4 Para. 2 No. 2 SPO or

- in the form of a written examination in accordance with § 4 Para. 2 No. 1 SPO

takes place.

Prerequisites
None.

Recommendations
The contents of the lecture Introduction to Computer Networks are assumed to be known. Attendance of the lecture Telematics is strongly recommended, as the contents are an important basis for understanding and classifying the material.

Below you will find excerpts from courses related to this module component:

V Network Security: Architectures and Protocols
24601, SS 2026, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content
Content

The lecture "Network Security: Architectures and Protocols" looks at challenges and techniques in the design of secure communication protocols as well as data protection and privacy issues. Complex systems such as Kerberos are examined in detail and their design decisions with regard to security aspects are highlighted. Special focus is placed on PKI fundamentals, infrastructures and specific PKI formats. Further emphasis is placed on the common security protocols IPSec and TLS/SSL as well as protocols for infrastructure protection.

Contents of the lecture "Einführung in Rechnernetze" are assumed to be known.

Furthermore it is highly recommended to visit the Telematics lecture, because its contents build an important base to understand and cathegorize the knowledge of the Network Security lecture.

Learning Objective

Students

- know basic challenges, protection goals and cryptographic building blocks that are relevant for the design of secure communication systems
- are proficient in security-relevant communication protocols (e.g. Kerberos, TLS, IPSec) and can identify and explain basic security mechanisms
- have the ability to analyze and evaluate communication protocols from a security perspective
- have the ability to assess and evaluate the quality of security mechanisms in relation to the required security objectives

In particular, students are familiar with typical attack techniques such as eavesdropping, interception or replaying and can explain these using examples. In addition, students are familiar with cryptographic primitives such as symmetric and asymmetric encryption, digital signatures, message authentication codes and can apply these in particular for the design of secure communication services.

Students are familiar with the Kerberos distributed authentication service and can explain the protocol flow in their own words and name basic concepts (e.g. tickets). In addition, students are familiar with relevant communication protocols for protecting communication on the Internet (e.g. IPsec, TLS) and can explain these and analyze and evaluate their security properties.

Students know different methods for network access protection and can explain and compare common authentication methods (e.g. CHAP, PAP, EAP). Furthermore, students are proficient in methods for protecting wireless access networks and can analyze and evaluate methods such as WEP, WPA and WPA2.

Students master different trust models and can explain and apply basic technical concepts (e.g. digital certificates, PKI) in their own words. In addition, students develop an understanding of data protection aspects in communication networks and can explain and apply technical procedures to protect privacy.

Organizational issues
Netzsicherheit: Architekturen und Protokolle

Literature
Roland Bless et al. Sichere Netzwerkkommunikation. Springer-Verlag, Heidelberg, Juni 2005.

Weiterführende Literatur

- Charlie Kaufman, Radia Perlman und Mike Speciner. Network Security: Private Communication in a Public World. 2nd Edition. Prentice Hall, New Jersey, 2002.
- Carlisle Adams und Steve Lloyd. Understanding PKI. Addison Wesley, 2003
- Rolf Oppliger. Secure Messaging with PGP and S/MIME. Artech House, Norwood, 2001.
- Sheila Frankel. Demystifiying the IPsec Puzzle. Artech House, Norwood, 2001.
- Thomas Hardjono und Lakshminath R. Dondeti. Security in Wireless LANs and MANs. Artech House, Norwood, 2005.
- Eric Rescorla. SSL and TLS: Designing and Building Secure Systems. Addison Wesley, Indianapolis, 2000.

T 6.476 Module component: Non- and Semiparametrics [T-WIWI-103126]

Coordinators: Prof. Dr. Melanie Schienle
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101638 - Econometrics and Statistics I
M-WIWI-101639 - Econometrics and Statistics II
M-WIWI-104902 - Statistics

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Irregular	1

Courses					
WT 25/26	2521300	Non- and Semiparametrics	2 SWS	Lecture	Schienle
WT 25/26	2521301		2 SWS	Practice	Rüter
WT 26/27	2521300	Non- and Semiparametrics	2 SWS	Lecture	Schienle
WT 26/27	2521301		2 SWS	Practice	Rüter

Assessment
The assessment consists of a written exam (90 minutes) (following §4(2), 1 of the examination regulation).

Prerequisites
None

Recommendations
Knowledge of the contents covered by the course *"Applied Econometrics"* [2520020]

Additional Information
The course takes place every second winter semester: 2018/19 then 2020/21

Below you will find excerpts from courses related to this module component:

V Non- and Semiparametrics — Lecture (V)
2521300, WS 25/26, 2 SWS, Language: German, Open in study portal

Content
Learning objectives:

The student

- has profound knowledge of non- and semiparametric estimation methods
- is capable of implementing these methods using statistical software and using them to assess empirical problems

Content:

Kernel density estimation, local constant and local linear regression, bandwidth choice, series and sieve estimators, additive models, semiparametric models

Requirements:

It is recommended to attend the course *Applied Econometrics* prior to this course.

Workload:

Total workload for 4.5 CP: approx. 135 hours

Attendance: 30 hours

Preparation and follow-up: 65 hours

Exam preparation: 40 hours

Literature
Li, Racine: Nonparametric Econometrics: Theory and Practice. Princeton University Press, 2007.

V Non- and Semiparametrics — Lecture (V)
2521300, WS 26/27, 2 SWS, Language: German, Open in study portal

Content
Learning objectives:

The student

- has profound knowledge of non- and semiparametric estimation methods
- is capable of implementing these methods using statistical software and using them to assess empirical problems

Content:

Kernel density estimation, local constant and local linear regression, bandwidth choice, series and sieve estimators, additive models, semiparametric models

Requirements:

It is recommended to attend the course *Applied Econometrics* prior to this course.

Workload:

Total workload for 4.5 CP: approx. 135 hours

Attendance: 30 hours

Preparation and follow-up: 65 hours

Exam preparation: 40 hours

Literature
Li, Racine: Nonparametric Econometrics: Theory and Practice. Princeton University Press, 2007.

T 6.477 Module component: Non-ferrous Metals and Alloys [T-MACH-114956]

Coordinators: Prof. Dr.-Ing. Martin Heilmaier
Dr. Emma White

Organisation: KIT Department of Mechanical Engineering

Part of: M-WIWI-107718 - Mechanical Engineering I
M-WIWI-107730 - Mechanical Engineering II

Type	Credits	Grading	Term offered	Version
Oral examination	4,5 CP	graded	Each summer term	2

Courses					
ST 2026	2174555	Non-ferrous Metals and alloys	3 SWS	Lecture / 🗣	Heilmaier, White

Legend: 🖥 Online, ⚄ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
oral exam (about 25 min.)

Prerequisites
none

Additional Information
The course is offered in English.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Non-ferrous Metals and alloys

2174555, SS 2026, 3 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content
This lecture gives an introduction in the material physics of non-ferrous metals and alloys. Focus is placed on:

- Synthesis and manufacturing
- Constitution (phase diagrams)
- Microstructure
- Mechanical and physical properties

which determine their respective applications. Since the students get an overview of the potentials and limitations of non-ferrous metals and alloys, they will receive the expertise to assess and decide about their different possible fields of applications.

Literature
Materialkunde der Nichteisenmetalle und Legierungen, J. Freudenberger und M. Heilmaier, Wiley-VCH 2020 (auf Deutsch)

T 6.478 Module component: Nonlinear Control Systems [T-ETIT-100980]

Coordinators: Dr.-Ing. Mathias Kluwe
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-101157 - Control Engineering II

Type	Credits	Grading	Term offered	Version
Written examination	3 CP	graded	Each summer term	1

Courses					
ST 2026	2303173	Nichtlineare Regelungssysteme	2 SWS	Lecture / ✕	Kluwe

Legend: Online, Blended (On-Site/Online), On-Site, ✕ Cancelled

Prerequisites
none

T 6.479 Module component: Nonlinear Optics [T-ETIT-101906]

Coordinators: Prof. Dr.-Ing. Christian Koos
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-100430 - Nonlinear Optics

Type	Credits	Grading	Term offered	Version
Oral examination	6 CP	graded	Each summer term	2

Courses					
ST 2026	2309468	Nonlinear Optics	2 SWS	Lecture / 🗣	Koos
ST 2026	2309469	Nonlinear Optics (Tutorial)	2 SWS	Practice / 🗣	Koos

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, x Cancelled

Assessment
Type of Examination: oral exam

Duration of Examination: approx. 30 Minutes

Modality of Exam: The oral exam is offered continuously upon individual appointment.

Prerequisites
none

Recommendations
Solid mathematical and physical background, basic knowledge in optics and photonics.

Additional Information
The module grade is the grade of the oral exam.

There is a bonus system based on the problem sets that are solved during the tutorials: During the term, 3 problem sets will be collected in the tutorial and graded without prior announcement. If for each of these sets more than 70% of the problems have been solved correctly, a bonus of 0.3 grades will be granted on the final mark of the oral exam.

T 6.480 Module component: Nonlinear Optimization I [T-WIWI-102724]

Coordinators: Prof. Dr. Oliver Stein
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101473 - Mathematical Programming
M-WIWI-104899 - Operations Research
M-WIWI-107187 - Quantitative Core

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each winter term	4

Courses					
WT 25/26	2550111	Nonlinear Optimization I	2 SWS	Lecture / 🗣	Stein
WT 25/26	2550112	Exercises Nonlinear Optimization I	1 SWS	Practice / 🗣	Stein, Schwarze, Neussel
WT 26/27	2550111	Nonlinear Optimization I	2 SWS	Lecture / 🗣	Stein
WT 26/27	2550112	Exercises Nonlinear Optimization I	1 SWS	Practice / 🗣	Stein, Schwarze, Huber

Legend: 📱 Online, ❖ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment

The assessment consists of a written exam (60 minutes) according to Section 4(2), 1 of the examination regulation. The successful completion of the exercises is required for admission to the written exam. The exam takes place in the semester of the lecture and in the following semester.
The examination can also be combined with the examination of Nonlinear Optimization II [2550113]. In this case, the duration of the written examination takes 120 minutes.

Prerequisites

The module component exam T-WIWI-103637 "Nonlinear Optimization I and II" may not be selected.

Additional Information

Part I and II of the lecture are held consecutively in the *same* semester.

Below you will find excerpts from courses related to this module component:

V Nonlinear Optimization I

Lecture (V)
On-Site

2550111, WS 25/26, 2 SWS, Language: English, Open in study portal

Content

The lecture treats the minimization of smooth nonlinear functions without constraints. For such problems, which occur very often in economics, engineering, and natural sciences, optimality conditions are derived and, based on them, solution algorithms are developed. The lecture is structured as follows:

- Introduction, examples, and terminology
- Existence results for optimal points
- First and second order optimality condtions
- Algorithms (line search, steepest descent method, variable metric methods, Newton method, Quasi Newton methods, CG method, trust region method)

The lecture is accompanied by exercises which, amongst others, offers the opportunity to implement and to test some of the methods on practically relevant examples.

Remark:

The treatment of optimization problems *with* constraints forms the contents of the lecture "Nonlinear Optimization II". The lectures "Nonlinear Optimization I" and "Nonlinear Optimization II" are held consecutively *in the same semester*.

Learning objectives:

The student

- knows and understands fundamentals of unconstrained nonlinear optimization,
- is able to choose, design and apply modern techniques of unconstrained nonlinear optimization in practice.

Literature
O. Stein, Basic Concepts of Nonlinear Optimization, Springer, Berlin, 2024

Additional Literature:

- W. Alt, Nichtlineare Optimierung, Vieweg, 2002
- M.S. Bazaraa, H.D. Sherali, C.M. Shetty, Nonlinear Programming, Wiley, 1993
- O. Güler, Foundations of Optimization, Springer, 2010
- H.Th. Jongen, K. Meer, E. Triesch, Optimization Theory, Kluwer, 2004
- J. Nocedal, S. Wright, Numerical Optimization, Springer, 2000

V **Nonlinear Optimization I** **Lecture (V)**
2550111, WS 26/27, 2 SWS, Language: English, Open in study portal **On-Site**

Content
The lecture treats the minimization of smooth nonlinear functions without constraints. For such problems, which occur very often in economics, engineering, and natural sciences, optimality conditions are derived and, based on them, solution algorithms are developed. The lecture is structured as follows:

- Introduction, examples, and terminology
- Existence results for optimal points
- First and second order optimality condtions
- Algorithms (line search, steepest descent method, variable metric methods, Newton method, Quasi Newton methods, CG method, trust region method)

The lecture is accompanied by exercises which, amongst others, offers the opportunity to implement and to test some of the methods on practically relevant examples.

Remark:

The treatment of optimization problems *with* constraints forms the contents of the lecture "Nonlinear Optimization II". The lectures "Nonlinear Optimization I" and "Nonlinear Optimization II" are held consecutively *in the same semester*.

Learning objectives:

The student

- knows and understands fundamentals of unconstrained nonlinear optimization,
- is able to choose, design and apply modern techniques of unconstrained nonlinear optimization in practice.

Literature
O. Stein, Basic Concepts of Nonlinear Optimization, Springer, Berlin, 2024

Additional Literature:

- W. Alt, Nichtlineare Optimierung, Vieweg, 2002
- M.S. Bazaraa, H.D. Sherali, C.M. Shetty, Nonlinear Programming, Wiley, 1993
- O. Güler, Foundations of Optimization, Springer, 2010
- H.Th. Jongen, K. Meer, E. Triesch, Optimization Theory, Kluwer, 2004
- J. Nocedal, S. Wright, Numerical Optimization, Springer, 2000

T 6.481 Module component: Nonlinear Optimization I and II [T-WIWI-103637]

Coordinators: Prof. Dr. Oliver Stein
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101473 - Mathematical Programming
M-WIWI-104899 - Operations Research

Type	Credits	Grading	Term offered	Version
Written examination	9 CP	graded	Each winter term	6

Courses					
WT 25/26	2550111	Nonlinear Optimization I	2 SWS	Lecture / 🗣	Stein
WT 25/26	2550112	Exercises Nonlinear Optimization I	1 SWS	Practice / 🗣	Stein, Schwarze, Neussel
WT 25/26	2550113	Nonlinear Optimization II	2 SWS	Lecture / 🗣	Stein
WT 26/27	2550111	Nonlinear Optimization I	2 SWS	Lecture / 🗣	Stein
WT 26/27	2550112	Exercises Nonlinear Optimization I	1 SWS	Practice / 🗣	Stein, Schwarze, Huber
WT 26/27	2550113	Nonlinear Optimization II	2 SWS	Lecture / 🗣	Stein

Legend: 📱 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment consits of a written exam in English (120 minutes) according to Section 4(2), 1 of the examination regulation. The successful completion of the exercises is required for admission to the written exam.

The exam takes place in the semester of the lecture and in the following semester.

Prerequisites
None.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-WIWI-102724 - Nonlinear Optimization I must not have been started.
2. The module component T-WIWI-102725 - Nonlinear Optimization II must not have been started.

Additional Information
Part I and II of the lecture are held consecutively in the **same** semester.

Below you will find excerpts from courses related to this module component:

V Nonlinear Optimization I

2550111, WS 25/26, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content
The lecture treats the minimization of smooth nonlinear functions without constraints. For such problems, which occur very often in economics, engineering, and natural sciences, optimality conditions are derived and, based on them, solution algorithms are developed. The lecture is structured as follows:

- Introduction, examples, and terminology
- Existence results for optimal points
- First and second order optimality condtions
- Algorithms (line search, steepest descent method, variable metric methods, Newton method, Quasi Newton methods, CG method, trust region method)

The lecture is accompanied by exercises which, amongst others, offers the opportunity to implement and to test some of the methods on practically relevant examples.

Remark:

The treatment of optimization problems *with* constraints forms the contents of the lecture "Nonlinear Optimization II". The lectures "Nonlinear Optimization I" and "Nonlinear Optimization II" are held consecutively *in the same semester*.

Learning objectives:

The student

- knows and understands fundamentals of unconstrained nonlinear optimization,
- is able to choose, design and apply modern techniques of unconstrained nonlinear optimization in practice.

Literature
O. Stein, Basic Concepts of Nonlinear Optimization, Springer, Berlin, 2024

Additional Literature:

- W. Alt, Nichtlineare Optimierung, Vieweg, 2002
- M.S. Bazaraa, H.D. Sherali, C.M. Shetty, Nonlinear Programming, Wiley, 1993
- O. Güler, Foundations of Optimization, Springer, 2010
- H.Th. Jongen, K. Meer, E. Triesch, Optimization Theory, Kluwer, 2004
- J. Nocedal, S. Wright, Numerical Optimization, Springer, 2000

V

Nonlinear Optimization II

2550113, WS 25/26, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content
The lecture treats the minimization of smooth nonlinear functions under nonlinear constraints. For such problems, which occur very often in economics, engineering, and natural sciences, optimality conditions are derived and, based on them, solution algorithms are developed. The lecture is structured as follows:

- Topology and first order approximations of the feasible set
- Theorems of the alternative, first and second order optimality conditions
- Algorithms (penalty method, multiplier method, barrier method, interior point method, SQP method, quadratic optimization)

The lecture is accompanied by exercises which, amongst others, offers the opportunity to implement and to test some of the methods on practically relevant examples.

Remark:

The treatment of optimization problems *without* constraints forms the contents of the lecture "Nonlinear Optimization I". The lectures "Nonlinear Optimization I" and "Nonlinear Optimization II" are held consecutively *in the same semester*.

Learning objectives:

The student

- knows and understands fundamentals of constrained nonlinear optimization,
- is able to choose, design and apply modern techniques of constrained nonlinear optimization in practice.

Literature
O. Stein, Basic Concepts of Nonlinear Optimization, Springer, Berlin, 2024

Additional Literature:

- W. Alt, Nichtlineare Optimierung, Vieweg, 2002
- M.S. Bazaraa, H.D. Sherali, C.M. Shetty, Nonlinear Programming, Wiley, 1993
- O. Güler, Foundations of Optimization, Springer, 2010
- H.Th. Jongen, K. Meer, E. Triesch, Optimization Theory, Kluwer, 2004
- J. Nocedal, S. Wright, Numerical Optimization, Springer, 2000

V

Nonlinear Optimization I

2550111, WS 26/27, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content

The lecture treats the minimization of smooth nonlinear functions without constraints. For such problems, which occur very often in economics, engineering, and natural sciences, optimality conditions are derived and, based on them, solution algorithms are developed. The lecture is structured as follows:

- Introduction, examples, and terminology
- Existence results for optimal points
- First and second order optimality condtions
- Algorithms (line search, steepest descent method, variable metric methods, Newton method, Quasi Newton methods, CG method, trust region method)

The lecture is accompanied by exercises which, amongst others, offers the opportunity to implement and to test some of the methods on practically relevant examples.

Remark:

The treatment of optimization problems *with* constraints forms the contents of the lecture "Nonlinear Optimization II". The lectures "Nonlinear Optimization I" and "Nonlinear Optimization II" are held consecutively *in the same semester*.

Learning objectives:

The student

- knows and understands fundamentals of unconstrained nonlinear optimization,
- is able to choose, design and apply modern techniques of unconstrained nonlinear optimization in practice.

Literature

O. Stein, Basic Concepts of Nonlinear Optimization, Springer, Berlin, 2024

Additional Literature:

- W. Alt, Nichtlineare Optimierung, Vieweg, 2002
- M.S. Bazaraa, H.D. Sherali, C.M. Shetty, Nonlinear Programming, Wiley, 1993
- O. Güler, Foundations of Optimization, Springer, 2010
- H.Th. Jongen, K. Meer, E. Triesch, Optimization Theory, Kluwer, 2004
- J. Nocedal, S. Wright, Numerical Optimization, Springer, 2000

V

Nonlinear Optimization II

2550113, WS 26/27, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content

The lecture treats the minimization of smooth nonlinear functions under nonlinear constraints. For such problems, which occur very often in economics, engineering, and natural sciences, optimality conditions are derived and, based on them, solution algorithms are developed. The lecture is structured as follows:

- Topology and first order approximations of the feasible set
- Theorems of the alternative, first and second order optimality conditions
- Algorithms (penalty method, multiplier method, barrier method, interior point method, SQP method, quadratic optimization)

The lecture is accompanied by exercises which, amongst others, offers the opportunity to implement and to test some of the methods on practically relevant examples.

Remark:

The treatment of optimization problems *without* constraints forms the contents of the lecture "Nonlinear Optimization I". The lectures "Nonlinear Optimization I" and "Nonlinear Optimization II" are held consecutively *in the same semester*.

Learning objectives:

The student

- knows and understands fundamentals of constrained nonlinear optimization,
- is able to choose, design and apply modern techniques of constrained nonlinear optimization in practice.

Literature
O. Stein, Basic Concepts of Nonlinear Optimization, Springer, Berlin, 2024

Additional Literature:

- W. Alt, Nichtlineare Optimierung, Vieweg, 2002
- M.S. Bazaraa, H.D. Sherali, C.M. Shetty, Nonlinear Programming, Wiley, 1993
- O. Güler, Foundations of Optimization, Springer, 2010
- H.Th. Jongen, K. Meer, E. Triesch, Optimization Theory, Kluwer, 2004
- J. Nocedal, S. Wright, Numerical Optimization, Springer, 2000

T 6.482 Module component: Nonlinear Optimization II [T-WIWI-102725]

Coordinators: Prof. Dr. Oliver Stein
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101473 - Mathematical Programming
M-WIWI-104899 - Operations Research
M-WIWI-107187 - Quantitative Core

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each winter term	3

Courses					
WT 25/26	2550112	Exercises Nonlinear Optimization I	1 SWS	Practice / 🗣	Stein, Schwarze, Neussel
WT 25/26	2550113	Nonlinear Optimization II	2 SWS	Lecture / 🗣	Stein
WT 25/26	2550114	Exercise to Nonlinear Optimization II	1 SWS	Practice / 🗣	Stein, Schwarze, Neussel
WT 26/27	2550112	Exercises Nonlinear Optimization I	1 SWS	Practice / 🗣	Stein, Schwarze, Huber
WT 26/27	2550113	Nonlinear Optimization II	2 SWS	Lecture / 🗣	Stein
WT 26/27	2550114	Exercise to Nonlinear Optimization II	1 SWS	Practice / 🗣	Stein, Schwarze, Huber

Legend: 📱 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment consits of a written exam (60 minutes) in English according to Section 4(2), 1 of the examination regulation. The successful completion of the exercises is required for admission to the written exam.

The exam takes place in the semester of the lecture and in the following semester.

The exam can also be combined with the examination of *Nonlinear Optimization I* [2550111]. In this case, the duration of the written exam takes 120 minutes.

Prerequisites
None.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-WIWI-103637 - Nonlinear Optimization I and II must not have been started.

Additional Information
Part I and II of the lecture are held consecutively in the same semester.

Below you will find excerpts from courses related to this module component:

V Nonlinear Optimization II
2550113, WS 25/26, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content
The lecture treats the minimization of smooth nonlinear functions under nonlinear constraints. For such problems, which occur very often in economics, engineering, and natural sciences, optimality conditions are derived and, based on them, solution algorithms are developed. The lecture is structured as follows:

- Topology and first order approximations of the feasible set
- Theorems of the alternative, first and second order optimality conditions
- Algorithms (penalty method, multiplier method, barrier method, interior point method, SQP method, quadratic optimization)

The lecture is accompanied by exercises which, amongst others, offers the opportunity to implement and to test some of the methods on practically relevant examples.

Remark:

The treatment of optimization problems *without* constraints forms the contents of the lecture "Nonlinear Optimization I". The lectures "Nonlinear Optimization I" and "Nonlinear Optimization II" are held consecutively *in the same semester*.

Learning objectives:

The student

- knows and understands fundamentals of constrained nonlinear optimization,
- is able to choose, design and apply modern techniques of constrained nonlinear optimization in practice.

Literature
O. Stein, Basic Concepts of Nonlinear Optimization, Springer, Berlin, 2024

Additional Literature:

- W. Alt, Nichtlineare Optimierung, Vieweg, 2002
- M.S. Bazaraa, H.D. Sherali, C.M. Shetty, Nonlinear Programming, Wiley, 1993
- O. Güler, Foundations of Optimization, Springer, 2010
- H.Th. Jongen, K. Meer, E. Triesch, Optimization Theory, Kluwer, 2004
- J. Nocedal, S. Wright, Numerical Optimization, Springer, 2000

V **Nonlinear Optimization II** — **Lecture (V) On-Site**
2550113, WS 26/27, 2 SWS, Language: English, Open in study portal

Content
The lecture treats the minimization of smooth nonlinear functions under nonlinear constraints. For such problems, which occur very often in economics, engineering, and natural sciences, optimality conditions are derived and, based on them, solution algorithms are developed. The lecture is structured as follows:

- Topology and first order approximations of the feasible set
- Theorems of the alternative, first and second order optimality conditions
- Algorithms (penalty method, multiplier method, barrier method, interior point method, SQP method, quadratic optimization)

The lecture is accompanied by exercises which, amongst others, offers the opportunity to implement and to test some of the methods on practically relevant examples.

Remark:

The treatment of optimization problems *without* constraints forms the contents of the lecture "Nonlinear Optimization I". The lectures "Nonlinear Optimization I" and "Nonlinear Optimization II" are held consecutively *in the same semester*.

Learning objectives:

The student

- knows and understands fundamentals of constrained nonlinear optimization,
- is able to choose, design and apply modern techniques of constrained nonlinear optimization in practice.

Literature
O. Stein, Basic Concepts of Nonlinear Optimization, Springer, Berlin, 2024

Additional Literature:

- W. Alt, Nichtlineare Optimierung, Vieweg, 2002
- M.S. Bazaraa, H.D. Sherali, C.M. Shetty, Nonlinear Programming, Wiley, 1993
- O. Güler, Foundations of Optimization, Springer, 2010
- H.Th. Jongen, K. Meer, E. Triesch, Optimization Theory, Kluwer, 2004
- J. Nocedal, S. Wright, Numerical Optimization, Springer, 2000

T 6.483 Module component: Novel Actuators and Sensors [T-MACH-102152]

Coordinators: Prof. Dr. Manfred Kohl
Dr. Martin Sommer

Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101287 - Microsystem Technology
M-MACH-101294 - Nanotechnology
M-MACH-101295 - Optoelectronics and Optical Communication
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Written examination	4 CP	graded	Each winter term	4

Courses					
WT 25/26	2141865	Novel actuators and sensors	2 SWS	Lecture / 🗣	Kohl, Sommer

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
written exam, 60 minutes

Prerequisites
T-MACH-114036 must not be started

Additional Information
The course is offered in German

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Novel actuators and sensors

2141865, WS 25/26, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Literature
- Vorlesungsskript "Neue Aktoren" und Folienskript "Sensoren"
- Donald J. Leo, Engineering Analysis of Smart Material Systems, John Wiley & Sons, Inc., 2007
- "Sensors Update", Edited by H.Baltes, W. Göpel, J. Hesse, VCH, 1996, ISBN: 3-527-29432-5
- "Multivariate Datenanalyse – Methodik und Anwendungen in der Chemie", R. Henrion, G. Henrion, Springer 1994, ISBN 3-540-58188-X

6.484 Module component: Nuclear Power and Reactor Technology [T-MACH-110332]

Coordinators: Dr. Aurelian Florin Badea
Organisation: KIT Department of Mechanical Engineering

KIT Department of Business and Economics

Part of: M-WIWI-107718 - Mechanical Engineering I
M-WIWI-107730 - Mechanical Engineering II

Type	Credits	Grading	Expansion	Version
Oral examination	4,5 CP	graded	1 semesters	2

Courses					
WT 25/26	2189921	Nuclear Power and Reactor Technology	3 SWS	Lecture / 🗣	Badea
WT 26/27	2189921	Nuclear Power and Reactor Technology	3 SWS	Lecture / 🗣	Badea

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
oral exam, approx. 20 min.

Prerequisites
None

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Nuclear Power and Reactor Technology

2189921, WS 25/26, 3 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content
This lecture is dedicated to Master students of mechanical engineering and other engineering studies. Goal of the lecture is the understanding of reactor technology and of the major physical processes in converting nuclear power into electrical energy. The students acquire comprehensive knowledge on the physics of nuclear fission reactors: neutron flux, cross sections, fission, breeding processes, chain reaction, critical size of a nuclear system, moderation, reactor dynamics, transport- and diffusion-equation for the neutron flux distribution, power density distributions in reactor, one-group, two-group and multi-group theories for the neutron spectrum. Students are able to analyze and understand the obtained results. The students are capable of understanding the advantages and disadvantages of different reactor technologies - LWR, heavy water reactors, nuclear power systems of generation IV -by using the delivered knowledge on reactor physics, thermal-hydraulics, reactor design, control, safety and requirements of the front-end and back-end of the fuel cycle. The students are qualified for further training in nuclear energy and safety field and for (also research-related) professional activity in the nuclear industry.

- nuclear fission & fusion,
- radioactive decay, neutron excess, fission, fast and thermal neutrons, fissile and fertile nuclei, enrichment, neutron flux, cross section, reaction rate, mean free path,
- chain reaction, critical size, moderation,
- reactor dynamics,
- transport- and diffusion-equation for the neutron flux distribution,
- power distributions in reactor,
- one-group and two-group theories,
- light-water reactors,
- reactor safety,
- design of nuclear reactors,
- breeding processes,
- nuclear power systems of generation IV

V

Nuclear Power and Reactor Technology

2189921, WS 26/27, 3 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content

This lecture is dedicated to Master students of mechanical engineering and other engineering studies. Goal of the lecture is the understanding of reactor technology and of the major physical processes in converting nuclear power into electrical energy. The students acquire comprehensive knowledge on the physics of nuclear fission reactors: neutron flux, cross sections, fission, breeding processes, chain reaction, critical size of a nuclear system, moderation, reactor dynamics, transport- and diffusion-equation for the neutron flux distribution, power density distributions in reactor, one-group, two-group and multi-group theories for the neutron spectrum. Students are able to analyze and understand the obtained results. The students are capable of understanding the advantages and disadvantages of different reactor technologies - LWR, heavy water reactors, nuclear power systems of generation IV -by using the delivered knowledge on reactor physics, thermal-hydraulics, reactor design, control, safety and requirements of the front-end and back-end of the fuel cycle. The students are qualified for further training in nuclear energy and safety field and for (also research-related) professional activity in the nuclear industry.

- nuclear fission & fusion,
- radioactive decay, neutron excess, fission, fast and thermal neutrons, fissile and fertile nuclei, enrichment, neutron flux, cross section, reaction rate, mean free path,
- chain reaction, critical size, moderation,
- reactor dynamics,
- transport- and diffusion-equation for the neutron flux distribution,
- power distributions in reactor,
- one-group and two-group theories,
- light-water reactors,
- reactor safety,
- design of nuclear reactors,
- breeding processes,
- nuclear power systems of generation IV

T 6.485 Module component: Nuclear Power Plant and Fusion Technologies [T-MACH-113977]

Coordinators: Dr. Aurelian Florin Badea
Prof. Dr.-Ing. Xu Cheng

Organisation: KIT Department of Mechanical Engineering

KIT Department of Business and Economics

Part of: M-WIWI-107718 - Mechanical Engineering I
M-WIWI-107730 - Mechanical Engineering II

Type	Credits	Grading	Term offered	Version
Oral examination	9 CP	graded	Each term	2

Courses					
WT 25/26	2189920	Nuclear Fusion Technology	2 SWS	Lecture / 🗣	Badea
ST 2026	2170460	Nuclear Power Plant Technology	2 SWS	Lecture / 🗣	Cheng, Schulenberg
WT 26/27	2189920	Nuclear Fusion Technology	2 SWS	Lecture / 🗣	Badea

Legend: 🖥 Online, ❋ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
oral exam, Duration: approximately 30 minutes

no tools or reference materials may be used during the exam

Prerequisites
T-MACH-105411 must not be started.

Additional Information
The course is offered in English.

Workload
270 hours

Below you will find excerpts from courses related to this module component:

V Nuclear Fusion Technology

2189920, WS 25/26, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content
This lecture is dedicated to Master students of mechanical engineering and other engineering studies. Goal of the lecture is the understanding of the physics of fusion, the components of a fusion reactor and their functions. The technological requirements for using fusion technology for future commercial production of electricity and the related environmental impact are also addressed. The students are capable of giving technical assessment of the usage of the fusion energy with respect to its safety and sustainability. The students are qualified for further training in fusion energy field and for research-related professional activity.

- nuclear fission & fusion
- neutronics for fusion
- fuel cycles, cross sections
- gravitational, magnetic and inertial confinement
- fusion experimental devices
- energy balance for fusion systems; Lawson criterion and Q-factor
- materials for fusion reactors
- plasma physics, confinement
- plasma heating
- timeline of the fusion technology
- ITER, DEMO
- safety and waste management

V

Nuclear Power Plant Technology

2170460, SS 2026, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content
The training objective of the course is the qualification for a research-related professional activity in nuclear power plant engineering. The participants can describe the most important components of nuclear power plants and their function. You can design or modify nuclear power plants independently and creatively. They have acquired a broad knowledge of this power plant technology, including specific knowledge of core design, design of primary and secondary systems, and of nuclear safety technologies. Based on the acquired knowledge in thermodynamics and neutron physics, they can describe and analyze the specific behavior of the nuclear power plant components and assess risks. Participants of the lecture have a trained analytical thinking and judgment in the design of nuclear power plants.

Power plants with pressurized water reactors:

Design of the pressurized water reactor

- Fuel assemblies
- Control rods and drives
- Core instrumentation
- Reactor pressure vessel and its internals

Components of the primary system

- Primary coolant pumps
- Pressurizer
- Steam generator
- Water make-up system

Secondary system:

- Turbines
- Reheater
- Feedwater system
- Cooling systems

Containment

- Containment design
- Components of safety systems
- Components of residual heat removal systems

Control of a nuclear power plant with PWR

Power plants with boiling water reactors:

Design of the boiling water reactor

- Fuel assemblies
- Control elements and drives
- Reactor pressure vessel and its internals

Containment and components of safety systems

Control of a nuclear power plant with boiling water reactors

Literature
Vorlesungsmanuskript

V

Nuclear Fusion Technology

2189920, WS 26/27, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content
This lecture is dedicated to Master students of mechanical engineering and other engineering studies. Goal of the lecture is the understanding of the physics of fusion, the components of a fusion reactor and their functions. The technological requirements for using fusion technology for future commercial production of electricity and the related environmental impact are also addressed. The students are capable of giving technical assessment of the usage of the fusion energy with respect to its safety and sustainability. The students are qualified for further training in fusion energy field and for research-related professional activity.

- nuclear fission & fusion
- neutronics for fusion
- fuel cycles, cross sections
- gravitational, magnetic and inertial confinement
- fusion experimental devices
- energy balance for fusion systems; Lawson criterion and Q-factor
- materials for fusion reactors
- plasma physics, confinement
- plasma heating
- timeline of the fusion technology
- ITER, DEMO
- safety and waste management

T 6.486 Module component: Online Concepts for Karlsruhe City Retailers [T-WIWI-111848]

Coordinators: Prof. Dr. Ann-Kristin Kupfer
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101510 - Cross-Functional Management Accounting
M-WIWI-104900 - Business Administration
M-WIWI-105312 - Marketing and Sales Management
M-WIWI-106258 - Digital Marketing

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Each summer term	2

Courses					
ST 2026	2571184	Online concepts for Karlsruhe city retailers	2 SWS	Others / 🗣	Kupfer

Legend: 📱 Online, Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Alternative exam assessment:

- presentations in teams (in each case to the extent of approx. 15 minutes per team with subsequent discussio)
- delivery of a written elaboration per team.

Additional Information
Please note that an application is required to participate in this workshop. The application phase usually takes place at the beginning of the lecture period in the summer semester. More information on the application process is usually available on the Marketing and Sales Research Group website (marketing.iism.kit.edu) shortly before the start of the lecture period in the summer semester.

Workload
90 hours

Below you will find excerpts from courses related to this module component:

V Online concepts for Karlsruhe city retailers

2571184, SS 2026, 2 SWS, Language: German, Open in study portal

Others (sonst.)
On-Site

Content
As part of a practical project in cooperation with the city marketing department of KME Karlsruhe Marketing und Event GmbH, students will have the opportunity to directly interact with retailers in Karlsruhe. Challenges of the digitalization of brick-and-mortar retailing will be analyzed and solutions will be developed and implemented.

In a theoretical part at the beginning of the event, students will gain an insight into the theoretical foundations of specific online marketing instruments. In cooperation with Karlsruhe City Marketing, students are taught application-oriented skills in online marketing tools, such as content management systems, social media platforms, search engine optimization or Google Ads campaigns.

In the practical part of the course, student teams cooperate with a real retailer in Karlsruhe's city center and learn how to analyze and optimize online presences and digital solutions based on key performance indicators. Possible use cases range from social media communication and website optimization to the introduction of innovative pricing and payment methods. In this way, students are given the tools for developing, maintaining and optimizing individual websites and digital solutions in stationary retailing.

Learning objectives result accordingly as follows:
- Learning of theoretical basics of central, application-oriented tools of online marketing
- Application and practical deep-dive of the acquired knowledge in a real case
- Concise and structured presentation of results

Total time required for 3 credit points: approx. 90.0 hours

Attendance time: 12 hours

Preparation and wrap-up of the course: 58 hours

Exam and exam preparation: 20 hours

Organizational issues

Kick off Montag 11.5.26 Gebäude 01.87 Seminarraum 025

T 6.487 Module component: Operation Methods for Earthmoving [T-BGU-101801]

Coordinators: Dr.-Ing. Heinrich Schlick
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: M-BGU-101110 - Process Engineering in Construction
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Expansion	Version
Oral examination	1,5 CP	graded	Each term	1 semesters	1

Courses					
WT 25/26	6241905	Earthwork	1 SWS	Lecture / 🗣	Haghsheno, Waleczko

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
oral exam, appr. 20 min.

Prerequisites
none

Recommendations
none

Additional Information
none

Workload
45 hours

6.488 Module component: Operation Methods for Foundation and Marine Construction [T-BGU-101832]

Coordinators: Dr.-Ing. Harald Schneider
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: M-BGU-101110 - Process Engineering in Construction
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Expansion	Version
Oral examination	1,5 CP	graded	Each term	1 semesters	1

Courses					
WT 25/26	6241904	Underground Construction	1 SWS	Lecture / 🗣	Haghsheno, Schneider

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
oral exam, appr. 20 min.

Prerequisites
none

Recommendations
none

Additional Information
none

Workload
45 hours

T 6.489 Module component: Operations Research in Health Care Management [T-WIWI-102884]

Coordinators: Jun.-Prof. Dr. Emilia Graß
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-102805 - Service Operations
M-WIWI-104899 - Operations Research

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each term	4

Courses					
WT 25/26	2550495	Operations Research in Health Care Management	2 SWS	Lecture / 🗣	Graß
WT 25/26	2550496	Übungen zu OR im Health Care Management	1 SWS	Practice	Graß
ST 2026	2550495	Operations Research in Health Care Management	2 SWS	Lecture / 🗣	Graß
ST 2026	2550496	Übungen zu OR im Health Care Management	1 SWS	Practice / 🖥	Graß

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success is assessed in English in the form of a 60-minute written examination (in accordance with §4(2), 1 SPO).

The examination is offered every semester.

Prerequisites
None

Recommendations
Basic knowledge as co nveyed in the module "Introduction to Operations Research" is assumed.

Additional Information
Lectures and examinations are held in English.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Operations Research in Health Care Management
2550495, WS 25/26, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Literature
Elective literature:

- Fleßa: Grundzüge der Krankenhausbetriebslehre, Oldenbourg, 2007
- Fleßa: Grundzüge der Krankenhaussteuerung, Oldenbourg, 2008
- Hall: Patient flow: reducing delay in healthcare delivery, Springer, 2006

V Operations Research in Health Care Management
2550495, SS 2026, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Literature
Weiterführende Literatur:

- Fleßa: Grundzüge der Krankenhausbetriebslehre, Oldenbourg, 2007
- Fleßa: Grundzüge der Krankenhaussteuerung, Oldenbourg, 2008
- Hall: Patient flow: reducing delay in healthcare delivery, Springer, 2006

T 6.490 Module component: Operations Research in Supply Chain Management [T-WIWI-102715]

Coordinators: Prof. Dr. Stefan Nickel
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101473 - Mathematical Programming
M-WIWI-102805 - Service Operations
M-WIWI-102832 - Operations Research in Supply Chain Management
M-WIWI-103289 - Stochastic Optimization
M-WIWI-104899 - Operations Research

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Irregular	2

Courses					
WT 26/27	2550480	Operations Research in Supply Chain Management	2 SWS	Lecture / 🗣	Nickel
WT 26/27	2550481	Übungen zu OR in Supply Chain Management	1 SWS	Practice / ⯐	Hoffmann

Legend: 📱 Online, ⯐ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment is a 60 minutes written examination (according to §4(2), 1 of the examination regulation).

The examination is held in the term of the lecture and the following lecture.

Prerequisites
None

Recommendations
Basic knowledge as conveyed in the module Introduction to Operations Research and in the lectures Facility Location and Strategic SCM,Tactical and operational SCMis assumed.

Additional Information
The course is offered irregularly. Planned lectures for the next three years can be found in the internet at http://dol.ior.kit.edu/english/Courses.php.

Below you will find excerpts from courses related to this module component:

V Operations Research in Supply Chain Management

2550480, WS 26/27, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content
Supply Chain Management constitutes a general tool for logistics process planning in supply networks. To an increasing degree quantitative decision support is provided by methods and models from Operations Research. The lecture "OR in Supply Chain Management" conveys concepts and approaches for solving practical problems and presents an insight to current research topics. The lecture's focus is set on modeling and solution methods for applications originating in different domains of a supply chain. The emphasis is put on mathematical methods like mixed integer programming, valid inequalities or column generation, and the derivation of optimal solution strategies.

In form and content, the lecture addresses multiple areas of Supply Chain Management: After a short introduction, inventory models, scheduling, assembly line balancing as well as cutting and packing will be discussed. Another main focus of the lecture is the application of methods from online optimization. This optimization discipline has gained more and more importance in the optimization of supply chains over the several past years due to an increasing amount of dynamic data flows.

Literature

- Simchi-Levi, D.; Chen, X.; Bramel, J.: The Logic of Logistics: Theory, Algorithms, and Applications for Logistics and Supply Chain Management, 2nd edition, Springer, 2005
- Simchi-Levi, D.; Kaminsky, P.; Simchi-Levi, E.: Designing and Managing the Supply Chain: Concepts, Strategies, and Case Studies, McGraw-Hill, 2000
- Silver, E. A.; Pyke, D. F.; Peterson, R.: Inventory Management and Production Planning and Scheduling, 3rd edition, Wiley, 1998
- Blazewicz, J.: Handbook on Scheduling - From Theory to Applications, Springer, 2007
- Pinedo, M. L.: Scheduling - Theory, Algorithms, and Systems (3rd edition), Springer, 2008
- Dyckhoff, H.; Finke, U.: Cutting and Packing in Production and Distribution - A Typology and Bibliography, Physica-Verlag, 1992
- Borodin, A.; El-Yaniv, R.: Online Computation and Competitive Analysis, Cambridge University Press, 2005
- Francis, R. L.; McGinnis, L. F.; White, A.: Facility Layout and Location: An Analytical Approach, 2nd edition, Prentice-Hall, 1992

T 6.491 Module component: Operative CRM [T-WIWI-102597]

Coordinators: Prof. Dr. Andreas Geyer-Schulz
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each winter term	1

Assessment
Written examination (60 minutes) according to §4(2), 1 SPO. The exam is considered passed if at least 50 out of a maximum of 100 possible points are achieved. The grades are graded in five steps (best grade 1.0 from 95 points). Details of the grade formation and scale will be announced in the course.
A bonus can be acquired through successful participation in the practice. If the grade of the written examination is between 4.0 and 1.3, the bonus improves the grade by one grade level (0.3 or 0.4). The exact criteria for awarding a bonus will be announced at the beginning of the course.

Prerequisites
None

Recommendations
The attendance of courses Customer Relationship Management and Analytical CRM is advised.

T 6.492 Module component: Optical Engineering and Machine Vision [T-ETIT-113941]

Coordinators: Prof. Dr.-Ing. Michael Heizmann
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-106974 - Optical Engineering and Machine Vision

Type	Credits	Grading	Term offered	Version
Written examination	6 CP	graded	Each winter term	2

Courses					
WT 25/26	2302150	Optical Engineering and Machine Vision	3 SWS	Lecture / 🗣	Heizmann
WT 25/26	2302151	Tutorial to 2302150 Optical Engineering and Machine Vision	1 SWS	Practice / 🗣	Leyer

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The examination takes place in form of a written examination lasting 120 (KSOP/O&P: 90) minutes.

The module grade is the grade of the written examination.

Prerequisites
none

T 6.493 Module component: Optical Transmitters and Receivers [T-ETIT-100639]

Coordinators: Prof.Dr.Dr.h.c. Wolfgang Freude
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-100436 - Optical Transmitters and Receivers
M-MACH-101295 - Optoelectronics and Optical Communication
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Oral examination	6 CP	graded	Each winter term	2

Courses					
WT 25/26	2309460	Optical Transmitters and Receivers	2 SWS	Lecture / 🗣	Freude
WT 25/26	2309461	Tutorial for 2309460 Optical Transmitters and Receivers	2 SWS	Practice / 🗣	Freude, N.N.

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Access controll takes placein form of an overall oral examination (approx. 20 minutes). The individual dates for the oral examination are offered regularly.

Prerequisites
none

Recommendations
Knowledge of the physics of the pn-transition.

T 6.494 Module component: Optimal and Model Predictive Control [T-CIWVT-112825]

Coordinators: Prof. Dr.-Ing. Thomas Meurer
Organisation: KIT Department of Chemical and Process Engineering
Part of: M-CIWVT-106317 - Optimal and Model Predictive Control

Type	Credits	Grading	Version
Oral examination	6 CP	graded	1

Courses					
ST 2026	2243030	Optimal and Model Predictive Control	2 SWS	Lecture / 🗣	Meurer
ST 2026	2243031	Optimal and Model Predictive Control - Exercises	1 SWS	Practice / 🗣	Meurer

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

T 6.495 Module component: Optimization under Uncertainty [T-WIWI-106545]

Coordinators: Prof. Dr. Steffen Rebennack
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-103289 - Stochastic Optimization
M-WIWI-104899 - Operations Research

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each winter term	3

Courses					
WT 25/26	2550464	Optimization Under Uncertainty	2 SWS	Lecture / 🔄	Rebennack, Kandora
WT 25/26	2550465	Übungen zu Optimierungsansätze unter Unsicherheit	1 SWS	Practice / 🗣	Rebennack
WT 25/26	2550466		2 SWS	Others	Rebennack

Legend: 🖥 Online, 🔄 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment consists of a written exam (60 minutes) according to Section 4(2), 1 of the examination regulation. The exam takes place in every the semester.

Prerequisites
None.

Workload
135 hours

T 6.496 Module component: Optoelectronic Components [T-ETIT-101907]

Coordinators: TT-Prof. Dr. Tobias Huber-Loyola
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-MACH-101287 - Microsystem Technology
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each summer term	1

Courses					
ST 2026	2309486	Optoelectronic Components	2 SWS	Lecture /	Huber-Loyola, Gablinger, Sauter
ST 2026	2309487	Optoelectronic Components (Tutorial)	1 SWS	Practice /	Huber-Loyola

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
Type of Examination: oral exam

Duration of Examination: approx. 30 minutes

Modality of Exam: Oral examination, usually one examination day per month during the Summer and Winter terms. An extra questions-and-answers session will be held if students wish so.

Prerequisites
none

Recommendations
Minimal background required: Calculus, differential equations, Fourier transforms and p-n junction physics.

T 6.497 Module component: Overview of Manufacturing Technologies für Industrial Engineers [T-MACH-114686]

Coordinators: Prof. Dr.-Ing. Volker Schulze
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101276 - Manufacturing Technology

Type	Credits	Grading	Term offered	Expansion	Version
Coursework	1 CP	pass/fail	Each winter term	1 semesters	1

Courses					
WT 25/26	2149674	Overview of ManufacturingTechnologies for Industrial Engineers	1 SWS	Lecture / 🗣	Schulze
WT 26/27	2149674	Overview of ManufacturingTechnologies for Industrial Engineers	1 SWS	Lecture / 🗣	Schulze

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Control of success is evaluated on the basis of regular attendance.
Regular attendance is achieved when the student has actively participated in at least 80 % of the course hours that have taken place.
If attendance is less than 80 %, the examiner may decide on a case-by-case basis what additional tasks must be completed in order to still achieve the learning objectives of the course. In the case of absence for an important reason (e.g., illness with a doctor's note), regular attendance is usually recognized.

Additional Information
The course is offered in German.

Workload
30 hours

Below you will find excerpts from courses related to this module component:

V Overview of ManufacturingTechnologies for Industrial Engineers

Lecture (V)
On-Site

2149674, WS 25/26, 1 SWS, Language: German, Open in study portal

Content
The lecture provides an overview of the various manufacturing processes and the systematics of their classification according to DIN 8580. The focus is on processes that are not covered in the other courses of the module.

The aim of the lecture is to place manufacturing technology within the context of production engineering and to provide an overview of manufacturing processes. To this end, the lecture conveys fundamental knowledge of manufacturing technology and covers the main types of manufacturing processes.

The specific topics include:

- Primary shaping (casting, plastics engineering, sintering, additive manufacturing processes)
- Forming (sheet-metal forming, massive forming)
- Cutting (machining with defined and undefined cutting edges, separating, ablating)
- Joining
- Coating
- Chaning material properties

Learning Outcomes:

The students ...

- are able to classify the manufacturing processes according to their fundamental mode of operation based on the six main groups defined in DIN 8580.
- are capable of naming the essential manufacturing processes of the six main groups (DIN 8580) and explaining their functions.
- are able to select suitable manufacturing processes for given components.

Workload:

regular attendance: 6 hours
self-study: 24 hours

Literature
Medien:
Skript zur Veranstaltung wird über ilias (https://ilias.studium.kit.edu/)
bereitgestellt.

Media:
Lecture notes will be provided in ilias (https://ilias.studium.kit.edu/).

V **Overview of ManufacturingTechnologies for Industrial Engineers**
2149674, WS 26/27, 1 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content

The lecture provides an overview of the various manufacturing processes and the systematics of their classification according to DIN 8580. The focus is on processes that are not covered in the other courses of the module.

The aim of the lecture is to place manufacturing technology within the context of production engineering and to provide an overview of manufacturing processes. To this end, the lecture conveys fundamental knowledge of manufacturing technology and covers the main types of manufacturing processes.

The specific topics include:

- Primary shaping (casting, plastics engineering, sintering, additive manufacturing processes)
- Forming (sheet-metal forming, massive forming)
- Cutting (machining with defined and undefined cutting edges, separating, ablating)
- Joining
- Coating
- Chaning material properties

Learning Outcomes:

The students ...

- are able to classify the manufacturing processes according to their fundamental mode of operation based on the six main groups defined in DIN 8580.
- are capable of naming the essential manufacturing processes of the six main groups (DIN 8580) and explaining their functions.
- are able to select suitable manufacturing processes for given components.

Workload:

regular attendance: 6 hours
self-study: 24 hours

Literature
Medien:
Skript zur Veranstaltung wird über ilias (https://ilias.studium.kit.edu/)
bereitgestellt.

Media:
Lecture notes will be provided in ilias (https://ilias.studium.kit.edu/).

T 6.498 Module component: Panel Data [T-WIWI-103127]

Coordinators: apl. Prof. Dr. Wolf-Dieter Heller
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101638 - Econometrics and Statistics I
M-WIWI-101639 - Econometrics and Statistics II
M-WIWI-104902 - Statistics

Type	Credits	Grading	Term offered	Version
Examination of another type	4,5 CP	graded	Each summer term	2

Courses					
ST 2026	2520320	Panel Data	2 SWS	Lecture	Heller
ST 2026	2520321	Übungen zu Paneldaten	2 SWS	Practice	Heller

Assessment
The assessment is carried out as an examination of another type in the form of a one-hour exam that includes both a written and an oral part. The exam may be taken as an individual exam or in pairs. The individual components are integrated into a single overall grade that reflects the overall impression of the combined performance

Prerequisites
None

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Panel Data
2520320, SS 2026, 2 SWS, Language: German, Open in study portal

Lecture (V)

Content
Content:

Fixed-Effects-Models, Random-Effects-Models, Time-Demeaning

Workload:

Total workload for 4.5 CP: approx. 135 hours

Attendance: 30 hours

Preparation and follow-up: 65 hours

Exam preparation: 40 hours

Exam preparation: 40 hours

Literature
Wooldridge, J. M. (2002). *Econometric analysis of cross section and panel data*. Cambridge and London: MIT Press.

Wooldridge, J. M. (2009). *Introductory Econometrics: A Modern Approach* (5th ed.). Mason, Ohio: South-Western Cengage Learning.

T 6.499 Module component: Parallel Computer Systems and Parallel Programming [T-INFO-101345]

Coordinators: Prof. Dr. Achim Streit
Organisation: KIT Department of Informatics
Part of: M-WIWI-104909 - Informatics (Department of Informatics)

Type	Credits	Grading	Term offered	Version
Written examination	4 CP	graded	Each summer term	2

Courses					
ST 2026	24617	Parallel computer systems and parallel programming	2 SWS	Lecture	Streit, Caspart, Baer

T 6.500 Module component: Parametric Optimization [T-WIWI-102855]

Coordinators: Prof. Dr. Oliver Stein
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101473 - Mathematical Programming
M-WIWI-104899 - Operations Research

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Irregular	1

Assessment
The assessment of the lecture is a written examination (60 minutes) according to §4(2), 1 of the examination regulation. The successful completion of the exercises is required for admission to the written exam.

The examination is held in the semester of the lecture and in the following semester.

Prerequisites
None

Recommendations
It is strongly recommended to visit at least one lecture from the Bachelor program of this chair before attending this course.

Additional Information
The lecture is offered irregularly. The curriculum of the next three years is available online (www.ior.kit.edu).

T 6.501 Module component: Particle Dynamics and Atomistic Simulation [T-MACH-114129]

Coordinators: Prof. Dr. Peter Gumbsch
Dr.-Ing. Johannes Schneider
Dr. Daniel Weygand

Organisation: KIT Department of Mechanical Engineering

Part of: M-WIWI-107718 - Mechanical Engineering I
M-WIWI-107730 - Mechanical Engineering II

Type	Credits	Grading	Term offered	Version
Oral examination	4,5 CP	graded	Each summer term	2

Courses					
ST 2026	2181740	Particle Dynamics and Atomistic Simulation	3 SWS	Lecture / Practice / 🗣	Weygand, Gumbsch

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Oral exam: approximately 30 minutes

Prerequisites
none

Recommendations
Recommended prerequisites: Mathematics, Physics, and Materials Science

Additional Information
The course is offered in English.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Particle Dynamics and Atomistic Simulation

2181740, SS 2026, 3 SWS, Language: English, Open in study portal

Lecture / Practice (VÜ)
On-Site

Content
Particle-based methods are numerical techniques used to simulate and analyse systems consisting of many discrete particles. They are particularly useful in fields where traditional continuum mechanical approaches are insufficient, such as granular materials, complex fluids, and defects in solids. In the lecture, the Discrete Element Method (DEM) for particles and Molecular Dynamics (MD) for the atomistic description of material behaviour will be covered. These methods span different length and time scales.

1. Introduction to Particle-Based Methods
   a) origin and application
   b) classification of particle-based methods
2. Fundamentals of Particle Dynamics
   a) Newtonian mechanics and conservation laws
   b) contact mechanics and friction laws
   c) kinematics and dynamics of particles
3. Discrete Element Method (DEM)
   a) principles and fundamentals
   b) numerical implementation: discretizing space and time
   c) particle detection and contact modelling
   d) application examples
4. Atomistic Methods: Molecular Dynamics (MD) and Statics (MS)
   a) fundamentals of atomistic models
   b) interaction: interatomic potentials
      i. pair potentials and their limits
      ii. many-body potentials
   c) integration methods (e.g., Verlet, Leap-Frog)
   d) periodic boundary conditions and neighbour lists
   e) applications in materials science
5. Structural Analysis:
   a) classification of neighbourhoods, distribution functions
   b) defect energy
   c) stresses, strains
6. Statistical Aspects of Atomistic Models
   a) phase space
   b) physical ensembles: microcanonical, canonical, grand canonical
   c) control of temperature, pressure, stresses: thermostats and barostats
   d) fluctuations and physical properties

The lecture covers both fundamental and advanced aspects of particle-based methods, with a particular focus on simple atomistic approaches. The accompanying computer exercises are designed to deepen and complement the lecture content through practical examples using the freely available particle simulation tool "LAMMPS" and to serve as a forum for detailed questions from students.

Objective: The student will be able to

- explain the physical principles of particle-based simulations,
- describe the application areas of particle-based simulation methods,
- apply particle-based simulation methods to address problems in materials science, materials engineering, and process engineering.

Recommended Prerequisites: mathematics, physics, and materials science

Lecture: 22.5 hours

Exercises: 12 hours

Self-study: 85.5 hours

Oral exam: approximately 30 minutes

Organizational issues
Die Vorlesung wird auf Englisch angeboten!

Literature

1. Understanding Molecular Simulation: From Algorithms to Applications, Daan Frenkel and Berend Smit (Academic Press, 2001) wie alle guten MD Bücher stark aus dem Bereich der physikalischen Chemie motiviert und auch aus diesem Bereich mit Anwendungsbeispielen gefüllt, trotzdem für mich das beste Buch zum Thema!
2. Computer simulation of liquids, M. P. Allen and Dominic J. Tildesley (Clarendon Press, Oxford, 1996) Immer noch der Klassiker zu klassischen MD Anwendungen. Weniger stark im Bereich der Nichtgleichgewichts-MD.
3. Computational Granular Dynamics. T. Pöschel, T. Schwager, Springer, 2005. Diskrete Element Methoden.
4. Lecture Slides and Exercises.

T 6.502 Module component: Patent Law [T-INFO-101310]

Coordinators: Patric Straßburger
Organisation: KIT Department of Informatics
Part of: M-INFO-101215 - Intellectual Property Law
M-WIWI-104903 - Law

Type	Credits	Grading	Term offered	Version
Written examination	3 CP	graded	Each summer term	3

Courses					
ST 2026	24656	Patent Law	2 SWS	Lecture / 🗣	Straßburger

Legend: 🖥 Online, ❂ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment is carried out as a written examination (§ 4 Abs. 2 No. 1 SPO) lasting 60 minutes.

Prerequisites
None.

Recommendations
None.

T 6.503 Module component: Perceiving, Understanding and Communicating Risk and Uncertainty [T-WIWI-114750]

Coordinators: Dr. rer. nat. Nico Gradwohl
Prof. Dr. Benjamin Scheibehenne

Organisation: KIT Department of Business and Economics

Part of: M-WIWI-104900 - Business Administration
M-WIWI-105714 - Consumer Research

Type	Credits	Grading	Term offered	Expansion	Version
Examination of another type	4,5 CP	graded	Each winter term	1 semesters	1

Courses					
WT 25/26	2500007	Perceiving, Understanding and Communicating Risk and Uncertainty	2 SWS	Block / 🗣	Gradwohl

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The course will be assessed via an examination of another type, comprising:

- Presentation of a research topic
- Active participation
- Homework assignments
- Designing and presenting an intervention to improve the understanding or communication of risk and uncertainty (e.g., online tool, a visualization for a specific topic, training program)

The overall impression will be assessed. Further details on the assessment's specific design will be provided during the course.

Prerequisites
None

Recommendations
A strong interest in research on human cognition and consumer behavior, as well as interest to engage with basic mathematical models and probability theory.

Additional Information
How do people perceive the risks of natural disasters, new technologies, or medical treatments like vaccines? And how can we communicate such risks clearly and transparently to support informed decision-making among consumers?

This course explores the science behind risk perception and communication. We will examine different types of risk and uncertainty, how individuals interpret them, and how to communicate them effectively. Topics include experimental methods, key research findings, common misconceptions, and the cognitive processes that shape our understanding of risk.

By the end of the course, students will be able to:

- Define key concepts related to risk and uncertainty.
- Describe experimental paradigms used to study risk perception.
- Critically analyze empirical and theoretical research.
- Apply insights to real-world challenges involving public communication of risk.
- Design strategies for improving how risk-related information is conveyed and understood.

This class will be taught in English. The number of participants is limited. The registration will take place via the Campus portal.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Perceiving, Understanding and Communicating Risk and Uncertainty

2500007, WS 25/26, 2 SWS, Language: English, Open in study portal

Block (B)
On-Site

Content

How do people perceive the risks of natural disasters, new technologies, or medical treatments like vaccines? And how can we communicate such risks clearly and transparently to support informed decision-making among consumers?

This course explores the science behind risk perception and communication. We will examine different types of risk and uncertainty, how individuals interpret them, and how to communicate them effectively. Topics include experimental methods, key research findings, common misconceptions, and the cognitive processes that shape our understanding of risk.

By the end of the course, students will be able to:

- Define key concepts related to risk and uncertainty.
- Describe experimental paradigms used to study risk perception, like decisions from description and decision from experience.
- Critically analyze empirical and theoretical research.
- Apply insights to involving public communication of risk.
- Design strategies for improving how risk-related information is conveyed and understood in real-world challenges

Competence Certificate

· Presentation of a research topic

· Active participation

· Homework assignments

· Designing and presenting an intervention to improve the understanding or communication of risk and uncertainty (e.g., an online tool, a visualization for a specific topic, or training programme)

Organizational issues

This class will be taught in English. The number of participants is limited to 12 participants.

The registration will take place via the WiWi portal. If too many students register, students in higher semesters are selected first. Further selection will take place on a first-come, first served basis.

Literature

Dhami, M. K., & Mandel, D. R. (2022). Communicating uncertainty using words and numbers. Trends in Cognitive Sciences, 26(6), 514-526. 10.1016/j.tics.2022.03.002

Slovic, P. (1987). Perception of risk. *Science*, *236*(4799), 280–285. https://doi.org/10.1126/science.3563507

Tannenbaum, D., Fox, C. R., & Ülkümen, G. (2017). Judgment extremity and accuracy under epistemic vs. aleatory uncertainty. *Management Science*, *63*(2), 497-518. https://doi.org/10.1287/mnsc.2015.2344

Van Der Bles, A. M., van der Linden, S., Freeman, A. L., & Spiegelhalter, D. J. (2020). The effects of communicating uncertainty on public trust in facts and numbers. *Proceedings of the National Academy of Sciences*, *117*(14), 7672-7683. 10.1073/pnas.1913678117

T 6.504 Module component: PH APL-ING-TL01 [T-WIWI-106291]

Organisation: University
Part of: M-WIWI-101404 - Extracurricular Module in Engineering

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Once	1

T 6.505 Module component: PH APL-ING-TL02 [T-WIWI-106292]

Organisation: University
Part of: M-WIWI-101404 - Extracurricular Module in Engineering

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Once	1

T 6.506 Module component: PH APL-ING-TL03 [T-WIWI-106293]

Organisation: University
Part of: M-WIWI-101404 - Extracurricular Module in Engineering

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Once	1

6.507 Module component: PH APL-ING-TL04 ub [T-WIWI-106294]

Organisation: University

Part of: M-WIWI-101404 - Extracurricular Module in Engineering

Type	Credits	Grading	Term offered	Version
Coursework	0 CP	pass/fail	Once	1

6.508 Module component: PH APL-ING-TL05 ub [T-WIWI-106295]

Organisation: University
Part of: M-WIWI-101404 - Extracurricular Module in Engineering

Type	Credits	Grading	Term offered	Version
Coursework	0 CP	pass/fail	Once	1

T 6.509 Module component: PH APL-ING-TL06 ub [T-WIWI-106296]

Organisation: University
Part of: M-WIWI-101404 - Extracurricular Module in Engineering

Type	Credits	Grading	Term offered	Version
Coursework	0 CP	pass/fail	Once	1

T 6.510 Module component: PH APL-ING-TL07 [T-WIWI-108384]

Organisation: University

Part of: M-WIWI-101404 - Extracurricular Module in Engineering

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Once	1

T 6.511 Module component: Photovoltaic System Design [T-ETIT-100724]

Coordinators: Dipl.-Ing. Robin Grab
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Written examination	3 CP	graded	Each summer term	1

Courses					
ST 2026	2307380	Photovoltaic System Design	2 SWS	Lecture / 🗣	Grab

Legend: 🖥 Online, ✲ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Prerequisites
none

T 6.512 Module component: Physical Basics of Laser Technology [T-MACH-102102]

Coordinators: Dr.-Ing. Johannes Schneider
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101268 - Specific Topics in Materials Science
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Oral examination	5 CP	graded	Each winter term	5

Courses					
WT 25/26	2181612	Physical basics of laser technology	3 SWS	Lecture / Practice / 🗣	Schneider

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
oral examination (ca. 25-30 min)

no tools or reference materials

Prerequisites
It is not possible, to combine this module component with Laser Material Processing [T-MACH-112763], Laser Application in Automotive Engineering [T-MACH-105164] or Physical Basics of Laser Technology [T-MACH-109084].

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-MACH-112763 - Laser Material Processing must not have been started.
2. The module component T-MACH-105164 - Laser in Automotive Engineering must not have been started.

Recommendations
Basic knowledge of physics, chemistry and material science

Additional Information
The course is offered in German.

Workload
150 hours

Below you will find excerpts from courses related to this module component:

V Physical basics of laser technology

2181612, WS 25/26, 3 SWS, Language: German, Open in study portal

Lecture / Practice (VÜ)
On-Site

Content
Based on the description of the physical basics about the formation and the properties of laser light the lecture goes through the different types of laser beam sources used in industry these days. The lecture focuses on the usage of lasers especially in materials engineering. Other areas like measurement technology or medical applications are also mentioned.

- physical basics of laser technology
- laser beam sources (solid state, diode, gas, liquid and other lasers)
- beam properties, guiding and shaping
- lasers in materials processing
- lasers in measurement technology
- lasers for medical applications
- savety aspects

The lecture is complemented by a tutorial.

The student

- can explain the principles of light generation, the conditions for light amplification as well as the basic structure and function of different laser sources.
- can describe the influence of laser, material and process parameters for the most important methods of laser-based materials processing and choose laser sources suitable for specific applications.
- can illustrate the possible applications of laser sources in measurement and medicine technology
- can explain the requirements for safe handling of laser radiation and for the design of safe laser systems.

Basic knowledge of physics, chemistry and material science is assumed.

regular attendance: 33,5 hours
self-study: 116,5 hours

The assessment consists of an oral exam (ca. 30 min) taking place at the agreed date (according to Section 4(2), 2 of the examination regulation). The re-examination is offered upon agreement.

It is allowed to select only one of the lectures "Laser in automotive engineering" (2182642) or "Physical basics of laser technology" (2181612) during the Bachelor and Master studies.

Organizational issues
Termine für die Übung werden in der Vorlesung bekannt gegeben!

Literature
M. W. Sigrist: Laser: Theorie, Typen und Anwendungen, 2018, Springer Spektrum

T. Graf: Laser - Grundlagen der Laserstrahlerzeugung 2015, Springer Vieweg

R. Poprawe: Lasertechnik für die Fertigung, 2005, Springer

H. Hügel, T. Graf: Materialbearbeitung mit Laser, 2023, Springer Vieweg

J. Eichler, H.-J. Eichler: Lasers - Basics, Advances and Applications, 2018, Springer

W. T. Silfvast: Laser Fundamentals, 2008, Cambridge University Press

W. M. Steen: Laser Material Processing, 2010, Springer

R. Poprawe, et al.: Tailored Light 1 - High Power Lasers for Production, 2018, Springer

R. Poprawe, et al.: Tailored Light 2 - Laser Applications, 2024, Springer

T 6.513 Module component: Physical Foundations of Cryogenics [T-CIWVT-106103]

Coordinators: Prof. Dr.-Ing. Steffen Grohmann
Organisation: KIT Department of Chemical and Process Engineering
Part of: M-CIWVT-103068 - Physical Foundations of Cryogenics

Type	Credits	Grading	Term offered	Version
Oral examination	6 CP	graded	Each summer term	1

Courses					
ST 2026	2250130	Physical Foundations of Cryogenics	2 SWS	Lecture / 🗣	Grohmann
ST 2026	2250131	Exercises on 2250130 Physical Foundations of Cryogenics	1 SWS	Practice / 🗣	Grohmann

Legend: 🖥 Online, ✣ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Learning control is an oral examination lasting approx. 30 minutes.

Prerequisites
None

T 6.514 Module component: Physics for Engineers [T-MACH-100530]

Coordinators: apl. Prof. Dr. Alexander Nesterov-Müller
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101287 - Microsystem Technology
M-MACH-101291 - Microfabrication
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Written examination	5 CP	graded	Each summer term	1

Courses					
ST 2026	2142890	Physics for Engineers	4 SWS	Lecture / Practice / 🗣	Nesterov-Müller

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
written exam 90 min

Prerequisites
none

Additional Information
The course is offered in German.

Workload
150 hours

Below you will find excerpts from courses related to this module component:

V Physics for Engineers

2142890, SS 2026, 4 SWS, Language: German, Open in study portal

Lecture / Practice (VÜ)
On-Site

Content
1) Foundations of solid state physics

- Wave particle dualism
- Tunnelling
- Schrödinger equation
- H-atom

2) Electrical conductivity of solids

- solid state: periodic potentials
- Pauli Principle
- band structure
- metals, semiconductors and isolators
- p-n junction / diode

3) Optics

- quantum mechanical principles of the laser
- linear optics
- non-linear optics

Exercises are used for complementing and deepening the contents of the lecture as well as for answering more extensive questions raised by the students and for testing progress in learning of the topics.

The student

- has the basic understanding of the physical foundations to explain the relationship between the quantum mechanical principles and the optical as well as electrical properties of materials
- can describe the fundamental experiments, which allow the illustration of these principles

regular attendance: 22,5 hours (lecture) and 22,5 hours (excerises)
self-study: 105 hours

The assessment consists of a written exam (90 minutes) (following §4(2), 1 of the examination regulation).

Literature

- Tipler und Mosca: Physik für Wissenschaftler und Ingenieure, Elsevier, 2004
- Haken und Wolf: Atom- und Quantenphysik. Einführung in die experimentellen und theoretischen Grundlagen, 7. Aufl., Springer, 2000
- Harris, Moderne Physik, Pearson Verlag, 2013

T 6.515 Module component: Physics, Technology and Applications of Thin Films [T-ETIT-111237]

Coordinators: Dr. Konstantin Ilin
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-105608 - Physics, Technology and Applications of Thin Films

Type	Credits	Grading	Term offered	Expansion	Version
Oral examination	4 CP	graded	Each winter term	1 semesters	1

Courses					
WT 25/26	2312710	Physics, Technology and Application of Thin Films	2 SWS	Lecture / 🗣	Ilin
WT 25/26	2312711	Exercise for 2312710 Physics, Technology and Application of Thin Films	1 SWS	Practice / 🗣	Ilin
WT 26/27	2312710	Physics, Technology and Application of Thin Films	2 SWS	Lecture / 🗣	Ilin
WT 26/27	2312711	Exercise for 2312710 Physics, Technology and Application of Thin Films	1 SWS	Practice / 🗣	Ilin

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The success control takes place within the framework of an oral overall examination of approx. 20 minutes.

T 6.516 Module component: Pioneering Leadership in German SMEs [T-WIWI-114184]

Coordinators: Prof. Dr. Marion Weissenberger-Eibl
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101507 - Innovation Management
M-WIWI-101507 - Innovation Management
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Irregular	1

Assessment
Non exam assessment consisting of a presentation of the results and a seminar paper (written in the group).

The grade is composed of 70% of the grade for the written work and 30% of the grade for the presentation.

Prerequisites
None

Recommendations
Prior attendance of the course Innovation Management is recommended.

Workload
90 hours

T 6.517 Module component: Platform & Market Engineering: Commerce, Media, and Digital Democracy [T-WIWI-112823]

Coordinators: Prof. Dr. Christof Weinhardt
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101409 - Electronic Markets
M-WIWI-101411 - Information Engineering
M-WIWI-101446 - Market Engineering
M-WIWI-101453 - Applied Strategic Decisions
M-WIWI-102754 - Service Economics and Management
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each summer term	2

Courses					
ST 2026	2540460	Platform & Market Engineering: Commerce, Media, and Digital Democracy	2 SWS	Lecture / 🗣	Weinhardt, Fegert
ST 2026	2540461	Übungen zu Platform & Market Engineering: Commerce, Media, and Digital Democracy	1 SWS	Practice / 🗣	Fegert, Stano, Schick

Legend: 🖥 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment consists of a written exam (60 min) (according to §4(2), 1 of the examination regulations). By successful completion of the exercises (§4 (2), 3 SPO 2007 respectively §4 (3) SPO 2015) up to 6 bonus points can be obtained. If the grade of the written examination is between 4.0 and 1.3, the bonus improves the grade by max. one grade level (0.3 or 0.4).

Prerequisites
None

Below you will find excerpts from courses related to this module component:

V Platform & Market Engineering: Commerce, Media, and Digital Democracy
Lecture (V)
On-Site

2540460, SS 2026, 2 SWS, Language: English, Open in study portal

Content
This lecture introduces an *innovative learning format* centered around *dynamic, AI-assisted self-study*, complemented by *in-person discussion sessions* to deepen content understanding. Instead of traditional lecture-based teaching (Frontalunterricht), we are piloting a type of flipped-classroom approach. The course is designed to be *co-developed by students and lecturers*—for example, through the creation of podcasts using AI tools like NotebookLM in small groups. This not only reinforces your grasp of the material but also builds your AI literacy.

The course will be graded with 80 points total. 60 of them can be earned in the exam at the end of the semester. The remaining 20 points can be earned in the exercise, where a podcast of a part of the lecture content has to be generated. For outstanding podcast submissions, you can receive up to *10 additional bonus* points.

Digital platforms and markets play an increasingly vital role in modern economies and societies. Understanding how to engineer these systems for efficiency, fairness, and societal benefit is crucial for shaping the digital future. By combining economic theory, engineering principles, and hands-on applications, this course prepares you to address real-world challenges in eCommerce, digital media, and digital democracy.
This lecture provides an in-depth exploration of the theoretical foundations, practical applications, and engineering principles essential for understanding and designing modern markets and digital platforms.

We aim to:

- Equip students with the ability to analyze, design, and evaluate digital markets and platforms.
- Provide an understanding of market mechanisms, economic principles, and the role of digital infrastructure in shaping economic and social interactions.
- Explore the influence of digital platforms on media, democracy, and citizen participation.
- Explore ethical implications of digital platforms and online market mechanisms.
- Apply generative AI tools to analyze, structure, and communicate topics from the lecture.
- Develop skills in critical evaluation of AI-generated content.

Course Structure:

1. Foundations of Platform & Market Engineering
    1. Market Engineering and Institutional Economics
    2. The "House of Market Engineering"
    3. Key concepts: efficiency, fairness, incentive compatibility, market convergence
2. Applications and Principles of Markets
    1. Market Engineering and Institutional Economics
    2. Economic theories in digital markets and platforms
3. Market Engineering Microstructure and Infrastructure
    1. Game Theory
    2. Mechanism Design
    3. Trust and Enforcement
    4. Auctions (single-item, combinatorial)
    5. IT & Business Infrastructure
    6. Evaluating Market Engineering: Experimental Economics
4. Digital Platforms and the Media
5. Digital Democracy:
    1. Online Polarization and Disinformation
    2. Digital Participation Engineering
    3. Digital Citizen Science Engineering
6. Ethical Implications

Organizational issues
ehemals: "**Market Engineering: Information in Institutions"**

Literature

- Roth, A., The Economist as Engineer: Game Theory, Experimental Economics and Computation as Tools for Design Economics. Econometrica 70(4): 1341-1378, 2002.
- Weinhardt, C. ,Holtmann, C., Neumann, D., Market Engineering. Wirtschaftsinformatik, 2003.
- Wolfstetter, E., Topics in Microeconomics - Industrial Organization, Auctions, and Incentives. Cambridge, Cambridge University Press, 1999.
- Smith, V. "Theory, Experiments and Economics", The Journal of Economic Perspectives,Vol. 3, No. 1, 151-69 1989

T 6.518 Module component: Platform Economy [T-WIWI-107506]

Coordinators: Prof. Dr. Christof Weinhardt
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Examination of another type	4,5 CP	graded	Each winter term	4

Courses					
WT 25/26	2540468	Platform Economy	2 SWS	Lecture / 🗣	Weinhardt, Fegert
WT 25/26	2540469	Übung zu Platform Economy	1 SWS	Practice / 🗣	Stano

Legend: 🖥 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Alternative exam assessment. The assessment is carried out in the form of a one-hour written examination and by carrying out a case study. Details on the assessment will be announced during the lecture.

Prerequisites
see below

Recommendations
None

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Platform Economy

2540468, WS 25/26, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content
The date for the lecture kick-off this Semester will be published soon.

Lecture and Exercise

The "Platform Economy" lecture provides a broad range of knowledge related to online platforms and their business models, examining their significance for users, operators, and society as a whole. The course is structured into 8 topical blocks, each exploring a different aspect of the platform economy in depth. Each block is led by a different lecturer who is an expert in the respective topic. The key topics covered in the lecture include:

Network Effects and Two-Sided Markets

- Business Models and Auctions
- Energy Market Engineering
- Digital Involvement: Crowd X & Citizen Science
- Digital Democracy and Social Media
- Analyzing User Behavior
- Trust and Reputation in Digital Platforms
- Ethical Considerations in the Platform Economy

To reinforce the lecture material, each block is accompanied by interactive exercises that encourage a deeper understanding of the topics. In these exercises, students will engage in discussions and explore practical examples that illustrate the theoretical concepts introduced during the lectures. The lecture and exercise also offer a chance to get an idea of the lectures offered during the master's program at our chair.

Case Study

In addition to the lectures, you will work on a case study in small groups. Your task will be to develop a business model for an innovative and novel online platform, which will be presented to you by one of our experts, either from the academic team or the industry. This case study offers a chance to gain deeper insights into current trends in the platform economy and to apply the knowledge acquired throughout the course in a practical, hands-on way.

Literature

- Bundesministerium für Wirtschaft und Energie (2017). „Kompetenzen für eine digitale Sourveränität“(abrufbar unter https://www.bmwi.de/Redaktion/DE/Publikationen/Studien/kompetenzen-fuer-eine-digitale-souveraenitaet.html)
- Bundesministerium für Wirtschaft und Energie (2017). „Weißbuch Digitale Plattformen.“ (abrufbar unter https://www.bmwi.de/Redaktion/DE/Publikationen/Digitale-Welt/weissbuch-digitale-plattformen.pdf?__blob=publicationFile&v=8)
- Easley, D., and Kleinberg, J. 2010. “Network Effects,” in Networks, Crowds, and Markets: Reasoning about a Highly Connected World, Cambridge University Press, pp. 509–542.
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- Gassmann, O., Frankenberger, K., and Csik, M. 2013. Geschäftsmodelle entwickeln: 55 innovative Konzepte mit dem St. Galler Business Model Navigator, Hanser.
- Wattenhofer, R. 2016. “The science of the blockchain.” CreateSpace Independent Publishing Platform.
- Roth, A. 2002. “The Economist as Engineer: Game Theory, Experimental Economics and Computation as Tools for Design Economics,” Econometrica 70(4): 1341-1378, 2002.
- Weinhardt, C. ,Holtmann, C., Neumann, D., Market Engineering. Wirtschaftsinformatik, 2003.
- Wolfstetter, E., 1999. “Topics in Microeconomics - Industrial Organization, Auctions, and Incentives,” Cambridge, Cambridge University Press.
- Teubner, T., and Hawlitschek, F. (in press). “The economics of P2P online sharing,” in The Sharing Economy: Possibilities, Challenges, and the way forward, Praeger Publishing.

T 6.519 Module component: PLM-CAD Workshop [T-MACH-102153]

Coordinators: Prof. Dr.-Ing. Jivka Ovtcharova
Organisation: KIT Department of Mechanical Engineering

Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Examination of another type	4 CP	graded	Each term	4

Courses					
WT 25/26	2121357	PLM-CAD Workshop	4 SWS	Project / 🗣	Rönnau
ST 2026	2121357	PLM-CAD Workshop	4 SWS	Project / 🗣	Rönnau, Meyer

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Learning control is an examination of another type. The overall impression is evaluated. The following aspects are included in the grading:

- Completion of the project task and documentation
- 4 interim presentations
- Final presentation including live demonstration

The grading scale will be announced in the course.

Prerequisites
None

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-MACH-114767 - CAD Workshop must not have been started.

Additional Information
Number of participants is limited, compulsory attendance

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V PLM-CAD Workshop
2121357, WS 25/26, 4 SWS, Language: German/English, Open in study portal

Project (PRO)
On-Site

Content
The aim of the workshop is to demonstrate the benefits of collaborative product development using PLM methods and to emphasize their added value compared to classical CAD development.

Students learn how to develop and produce a prototype with the help of modern PLM and CAx systems.

Organizational issues
Zeit und Ort siehe ILIAS

Literature
Workshop-Unterlagen / workshop materials

V PLM-CAD Workshop
2121357, SS 2026, 4 SWS, Language: German, Open in study portal

Project (PRO)
On-Site

Content

The aim of the workshop is to demonstrate the benefits of collaborative product development using PLM methods and to emphasize their added value compared to classical CAD development.

Students learn how to develop and produce a prototype with the help of modern PLM and CAx systems.

Organizational issues

Zeit und Ort siehe ILIAS

Literature

Workshop-Unterlagen / workshop materials

T 6.520 Module component: Polymer Engineering I [T-MACH-102137]

Coordinators: Dr.-Ing. Wilfried Liebig
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101268 - Specific Topics in Materials Science
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each winter term	2

Courses					
WT 25/26	2173590	Polymer Engineering I	2 SWS	Lecture / 🗣	Liebig
WT 26/27	2173590	Polymer Engineering I	2 SWS	Lecture / 🗣	Liebig

Legend: 🖥 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Oral exam, about 25 minutes

Prerequisites
T-MACH-114007 must not have been started

Additional Information
The course is offered in German.

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Polymer Engineering I

2173590, WS 25/26, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content
1. Economical aspects of polymers
2. Introductiom of mechanical,
chemical end electrical properties
3. Processing of polymers
(introduction)
4. Material science of polymers
5. Synthesis

learning objectives:

The field of Polymer Engineering includes synthesis, material science, processing, construction, design, tool engineering, production technology, surface engineering and recycling. The aim is, to equip the students with knowledge and technical skills, and to use the material "polymer" meeting its requirements in an economical and ecological way.

The students

- are able to describe and classify polymers
  based on the fundamental synthesis processing techniques
- can find practical applications for state-of-the-art polymers and manufacturing technologies
- are able to apply the processing techniques, the application of polymers and polymer composites regarding to the basic principles of material science
- can describe the special mechanical, chemical and elctrical prooperties of polymers and correlate these properties to the chemical bindings.
- can define application areas and the limitation in the use of polymers

requirements:
none

workload:

regular attendance: 21 hours
self-study: 99 hours

Literature
Literaturhinweise, Unterlagen und Teilmanuskript werden in der Vorlesung ausgegeben.

V **Polymer Engineering I** **Lecture (V)**
2173590, WS 26/27, 2 SWS, Language: German, Open in study portal **On-Site**

Content
1. Economical aspects of polymers
2. Introductiom of mechanical,
chemical end electrical properties
3. Processing of polymers
(introduction)
4. Material science of polymers
5. Synthesis

learning objectives:

The field of Polymer Engineering includes synthesis, material science, processing, construction, design, tool engineering, production technology, surface engineering and recycling. The aim is, to equip the students with knowledge and technical skills, and to use the material "polymer" meeting its requirements in an economical and ecological way.

The students

- are able to describe and classify polymers
  based on the fundamental synthesis processing techniques
- can find practical applications for state-of-the-art polymers and manufacturing technologies
- are able to apply the processing techniques, the application of polymers and polymer composites regarding to the basic principles of material science
- can describe the special mechanical, chemical and elctrical prooperties of polymers and correlate these properties to the chemical bindings.
- can define application areas and the limitation in the use of polymers

requirements:
none

workload:

regular attendance: 21 hours
self-study: 99 hours

Literature
Literaturhinweise, Unterlagen und Teilmanuskript werden in der Vorlesung ausgegeben.

T 6.521 Module component: Polymer Engineering II [T-MACH-102138]

Coordinators: Dr.-Ing. Wilfried Liebig
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101268 - Specific Topics in Materials Science
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each summer term	2

Courses					
ST 2026	2174596	Polymer Engineering II	2 SWS	Lecture / 🗣	Liebig

Legend: 🖥 Online, Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Oral exam, about 25 minutes

Prerequisites
T-MACH-114007 must not be started.

Recommendations
Knowledge in Polymerengineering I

Additional Information
The course is offered in German.

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Polymer Engineering II

2174596, SS 2026, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content
1. Processing of polymers
2. Properties of polymer components
Based on practical examples and components
2.1 Selection of material
2.2 Component design
2.3 Tool engineering
2.4 Production technology
2.5 Surface engineering
2.6 Sustainability, recycling

learning objectives:

The field of Polymer Engineering includes synthesis, material science, processing, construction, design, tool engineering, production technology, surface engineering and recycling. The aim is, that the students gather knowledge and technical skills to use the material "polymer" meeting its requirements in an economical and ecological way.

The students

- can describe and classify different processing techniques
  and can exemplify mould design principles based on technical parts.
- know about practical applications and processing of polymer parts
- are able to design polymer parts according to given restrictions
- can choose appropriate polymers based on the technical requirements
- can decide how to use polymers regarding the production, economical and ecological requirements

requirements:
Polymerengineering I
workload:

The workload for the lecture Polymerengineering II is 120 h per semester and consists of the presence during the lecture (21 h) as well as preparation and rework time at home (99 h).

Literature
Literaturhinweise, Unterlagen und Teilmanuskript werden in der Vorlesung ausgegeben.

Recommended literature and selected official lecture notes are provided in the lecture.

T 6.522 Module component: Polymers in MEMS A: Chemistry, Synthesis and Applications [T-MACH-102192]

Coordinators: Dr.-Ing. Matthias Worgull
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101291 - Microfabrication
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each winter term	1

Courses					
WT 25/26	2141853	Polymers in MEMS A: Chemistry, Synthesis and Applications	2 SWS	/	Worgull

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
Oral examination

Prerequisites
none

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Polymers in MEMS A: Chemistry, Synthesis and Applications

2141853, WS 25/26, 2 SWS, Language: German, Open in study portal

Blended (On-Site/Online)

Organizational issues
Findet als Blockveranstaltung am Semesterende statt.

T 6.523 Module component: Polymers in MEMS B: Physics, Microstructuring and Applications [T-MACH-102191]

Coordinators: Dr.-Ing. Matthias Worgull
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101291 - Microfabrication
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each winter term	1

Courses					
WT 25/26	2141854	Polymers in MEMS B: Physics, Microstructuring and Applications	2 SWS	Lecture /	Worgull

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
Oral examination

Prerequisites
none

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Polymers in MEMS B: Physics, Microstructuring and Applications

2141854, WS 25/26, 2 SWS, Language: German, Open in study portal

Lecture (V)
Blended (On-Site/Online)

T 6.524 Module component: Polymers in MEMS C: Biopolymers and Bioplastics [T-MACH-102200]

Coordinators: Dr.-Ing. Matthias Worgull
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101291 - Microfabrication
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each summer term	1

Courses					
ST 2026	2142855	Polymers in MEMS C - Biopolymers and Bioplastics	2 SWS	/ 🖥	Worgull

Legend: 🖥 Online, Blended (On-Site/Online), On-Site, ✕ Cancelled

Assessment
Oral examination

Prerequisites
none

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Polymers in MEMS C - Biopolymers and Bioplastics

2142855, SS 2026, 2 SWS, Language: German, Open in study portal

Online

Content
Polymers are ubiquitous in everyday life: from packaging materials all the way to specialty products in medicine and medical engineering. Today it is difficult to find a product which does not (at least in parts) consist of polymeric materials. The question of how these materials can be improved with respect to their disposal and consumption of (natural) resources during manufacturing is often raised. Today polymers must be fully recycled in Germany and many other countries due to the fact that they do not (or only very slowly) decompose in nature. Furthermore significant reductions of crude oil consumption during synthesis are of increasing importance in order to improve the sustainability of this class of materials. With respect to disposal polymers which do not have to be disposed by combustion but rather allow natural decomposition (composting) are of increasing interest. Polymers from renewable sources are also of interest for modern microelectromechanical systems (MEMS) especially if the systems designed are intended as single-use products.

This lecture will introduce the most important classes of these so-called biopolymers and bioplastics. It will also discuss and highlight polymers which are created from naturally created analogues (e.g. via fermentation) to petrochemical polymer precursors and describe their technical processing. Numerous examples from MEMS as well as everyday life will be given.

Some of the topics covered are:

- What are biopolyurethanes and how can you produce them from castor oil?
- What are "natural glues" and how are they different from chemical glues?
- How do you make tires from natural rubbers?
- What are the two most important polymers for life on earth?
- How can you make polymers from potatoes?
- Can wood be formed by injection molding?
- How do you make buttons from milk?
- Can you play music on biopolymers?
- Where and how do you use polymers for tissue engineering?
- How can you built LEGO with DNA?

The lecture will be given in German language unless non-German speaking students attend. In this case, the lecture will be given in English (with some German translations of technical vocabulary). The lecture slides are in English language and will be handed out for taking notes. Additional literature is not required.

For further details, please contact the lecturer, PD Dr.-Ing. Matthias Worgull (matthias.worgull@kit.edu). Preregistration is not necessary.

Organizational issues
Für weitere Rückfragen, wenden Sie sich bitte an PD Dr.-Ing- Matthias Worgull (matthias.worgull@kit.edu). Eine Voranmeldung ist nicht notwendig.

Literature
Zusätzliche vorlesungsbegleitende Literatur ist nicht notwendig.

T 6.525 Module component: Portfolio and Asset Liability Management [T-WIWI-103128]

Coordinators: Dr. Mher Safarian
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101639 - Econometrics and Statistics II
M-WIWI-104902 - Statistics

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each summer term	1

Courses					
ST 2026	2520357	Portfolio and Asset Liability Management	2 SWS	Lecture	Safarian
ST 2026	2520358	Übungen zu Portfolio and Asset Liability Management	2 SWS	Practice	Safarian

Assessment
The assessment of this course consists of a written examination (following §4(2), 1 SPOs, 180 min.).

Prerequisites
None

Below you will find excerpts from courses related to this module component:

V Portfolio and Asset Liability Management

2520357, SS 2026, 2 SWS, Language: English, Open in study portal

Lecture (V)

Content
Learning objectives:

Knowledge of various portfolio management techniques in the financial industry.

Content:

Portfolio theory: principles of investment, Markowitz- portfolio analysis, Modigliani-Miller theorems and absence of arbitrage, efficient markets, capital asset pricing model (CAPM), multi factorial CAPM, arbitragepricing theory (APT), arbitrage and hedging, multi factorial models, equity-portfolio management, passive strategies, active investment

Asset liability: statistical portfolio analysis in stock allocation, measures of success, dynamic multi seasonal models, models in building scenarios, stochastic programming in bond and liability management, optimal investment strategies, integrated asset liability management

Workload:

Total workload for 4.5 CP: approx. 135 hours

Attendance: 30 hours

Preparation and follow-up: 65 hours

Exam preparation: 40 hours

Exam preparation: 40 hours

Organizational issues
Blockveranstaltung, Termine werden über Ilias bekanntgegeben

Literature
To be announced in the lecture

T 6.526 Module component: Power Electronic Systems [T-ETIT-114751]

Coordinators: Prof. Dr.-Ing. Marc Hiller
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-107533 - Power Electronic Systems

Type	Credits	Grading	Term offered	Version
Oral examination	6 CP	graded	Each winter term	1

Courses					
WT 25/26	2306357	Power Electronic Systems	3 SWS	Lecture / 🗣	Hiller
WT 25/26	2306358	Practical Exercise to 2306357 Power Electronic Systems	1 SWS	Practice / 🧩	Hiller, Pfeffer

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success control takes place in the form of an oral examination lasting approx. 25 minutes.

The module grade is the grade of the oral examination.

Prerequisites
none

T 6.527 Module component: Power Electronics [T-ETIT-109360]

Coordinators: Prof. Dr.-Ing. Marc Hiller
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-104567 - Power Electronics

Type	Credits	Grading	Term offered	Expansion	Version
Written examination	6 CP	graded	Each summer term	1 semesters	6

Courses					
ST 2026	2300004	Ausweich- und Praktikumstermin für ETI-Vorlesungen	2 SWS	Practical course / 🧩	Hiller, Thönelt
ST 2026	2306323	Power Electronics	2 SWS	Lecture / 🧩	Hiller
ST 2026	2306324	Tutorial for 2306385 Power Electronics	2 SWS	Practice / 🧩	Hiller, Thönelt, Becher

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The examination takes place in form of a written examination lasting 120 minutes.

Prerequisites
none

6.528 Module component: Power Generation [T-ETIT-101924]

Coordinators: Dr.-Ing. Bernd Hoferer
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Oral examination	3 CP	graded	Each summer term	3

Courses					
WT 25/26	2307356	Power Generation	2 SWS	Lecture / ✕	Hoferer
ST 2026	2307356	Power Generation	2 SWS	Lecture / 🗣	Hoferer

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Prerequisites
none

T 6.529 Module component: Power Network [T-ETIT-114244]

Coordinators: Prof. Dr.-Ing. Thomas Leibfried
Organisation: KIT Department of Electrical Engineering and Information Technology
KIT Department of Mechanical Engineering
Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Written examination	6 CP	graded	Each winter term	1

Courses					
WT 25/26	2307371	Power Network	2 SWS	Lecture / 🗣	Leibfried
WT 25/26	2307373	Tutorial for 2307371 Power Network	2 SWS	Practice / 🗣	Leibfried, Mader

Legend: 🖥 Online, ❁ Blended (On-Site/Online), 🗣 On-Site, × Cancelled

T 6.530 Module component: Power Network [T-ETIT-100830]

Coordinators: Prof. Dr.-Ing. Thomas Leibfried
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Written examination	5 CP	graded	Each winter term	4

Courses					
WT 25/26	2307371	Power Network	2 SWS	Lecture / 🗣	Leibfried
WT 25/26	2307373	Tutorial for 2307371 Power Network	2 SWS	Practice / 🗣	Leibfried, Mader

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

T 6.531 Module component: Power-to-X – Key Technology for the Energy Transition [T-CIWVT-111841]

Coordinators: Prof. Dr.-Ing. Roland Dittmeyer
Dr. Peter Holtappels

Organisation: KIT Department of Chemical and Process Engineering

Part of: M-CIWVT-105891 - Power-to-X – Key Technology for the Energy Transition

Type	Credits	Grading	Term offered	Expansion	Version
Oral examination	4 CP	graded	Each term	1 semesters	1

Courses					
WT 25/26	2220110	Power-to-X – Key Technology for the Energy Transition	2 SWS	Lecture / 🗣	Holtappels
ST 2026	2220110	Power-to-X: Key Technology for the Energy Transition	2 SWS	Lecture / 🗣	Holtappels

Legend: 🖥 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Oral examination lastin approx. 30 minutes.

Prerequisites
None.

T 6.532 Module component: Practical Amplifier Design [T-ETIT-114179]

Coordinators: Prof. Dr.-Ing. Ahmet Cagri Ulusoy
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-107156 - Practical Amplifier Design

Type	Credits	Grading	Term offered	Version
Oral examination	3 CP	graded	Each summer term	1

Courses					
ST 2026	2308460	Practical Amplifier Design	1 SWS	Lecture / 🗣	Ulusoy
ST 2026	2308461	Exercise for 2308460 Practical Amplifier Design	1 SWS	Practice / ⚙	Ulusoy

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success control takes place in the form of an oral examination of approx. 30 min.

The module grade is the grade of the oral examination.

Prerequisites
none

T 6.533 Module component: Practical Course in Water Technology [T-CIWVT-106840]

Coordinators: Dr. Andrea Hille-Reichel
Prof. Dr. Harald Horn

Organisation: KIT Department of Chemical and Process Engineering

Part of: M-CIWVT-101121 - Water Chemistry and Water Technology I
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Each winter term	4

Courses					
WT 25/26	2233032	Practical Course: Water Quality and Water Assessment	2 SWS	Practical course / 🗣	Horn, Hille-Reichel, und Mitarbeitende

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The learning control is an examination of another type:
6 Experiments including entrance test, protocol; presentation about a selected experiment (about 15 minutes); final test.

Prerequisites
Participation in two excursions, report.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module M-CIWVT-103407 - Water Technology must have been started.
2. The module component T-CIWVT-110866 - Excursions: Water Supply must have been passed.

T 6.534 Module component: Practical Course Polymers in MEMS [T-MACH-105556]

Coordinators: Dr.-Ing. Matthias Worgull
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101291 - Microfabrication
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Coursework	2 CP	pass/fail	Each summer term	1

Assessment
The practical course will close with an oral examination.There will be only passed and failed results, no grades.

Prerequisites
none

T 6.535 Module component: Practical Course Technical Ceramics [T-MACH-105178]

Coordinators: Diego Ribas Gomes
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101268 - Specific Topics in Materials Science
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Coursework	4 CP	pass/fail	Each winter term	2

Assessment
Colloquium and laboratory report for the respective experiments.

Prerequisites
none

Workload
120 hours

T 6.536 Module component: Practical in Additive Manufacturing for Process Engineering [T-CIWVT-110903]

Coordinators: TT-Prof. Dr. Christoph Klahn
Organisation: KIT Department of Chemical and Process Engineering
Part of: M-CIWVT-105407 - Additive Manufacturing for Process Engineering

Type	Credits	Grading	Version
Coursework (practical)	1 CP	pass/fail	1

Courses					
ST 2026	2241021	Practical in Additive Manufacturing for Process Engineering	1 SWS	Practical course / 🗣	Klahn

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Learning control is an ungraded completed coursework: Lab, 8 experiments.

T 6.537 Module component: Practical in Power-to-X: Key Technology for the Energy Transition [T-CIWVT-111842]

Coordinators: Prof. Dr.-Ing. Roland Dittmeyer
Dr. Peter Holtappels
Organisation: KIT Department of Chemical and Process Engineering
Part of: M-CIWVT-105891 - Power-to-X – Key Technology for the Energy Transition

Type	Credits	Grading	Term offered	Expansion	Version
Coursework (practical)	2 CP	pass/fail	Each term	1 semesters	1

Courses					
WT 25/26	2220111	Practical in Power-to-X: Key Technology for the Energy Transition	1 SWS	Practical course / 🗣	Holtappels
ST 2026	2220111	Practical in Power-to-X: Key Technology for the Energy Transition	1 SWS	Practical course / 🗣	Holtappels

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Ungraded lab: Participation in all four experiments.

Prerequisites
None

Additional Information
Dates by arrangement, Location: IMVT, KIT Campus Nord, Energy Lab 2.0, Building 605.

T 6.538 Module component: Practical on Electromagnetic Energy in Process Engineering [T-CIWVT-114830]

Coordinators: Dr. Alexander Navarrete Munoz
Organisation: KIT Department of Chemical and Process Engineering
Part of: M-CIWVT-107566 - Electromagnetic Energy in Process Engineering

Type	Credits	Grading	Version
Coursework (practical)	1 CP	pass/fail	1

Courses					
WT 25/26	2220122	Practical on 2220120 Electromagnetic Energy in Process Engineering	1 SWS	Practical course / 🗣	Dittmeyer, Navarrete Munoz
ST 2026	2220122	Practical on 2220120 Electromagnetic Energy in Process Engineering	1 SWS	Practical course / 🗣	Dittmeyer, Navarrete Munoz

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Learning control is an completed coursework: Practical.

Prerequisites
None.

T 6.539 Module component: Practical Seminar Digital Service Systems [T-WIWI-106563]

Coordinators: Prof. Dr. Gerhard Satzger
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-102808 - Digital Service Systems in Industry
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Examination of another type	4,5 CP	graded	Irregular	1

Assessment
The assessment is carried out as a single examination of another type that combines three mandatory components:

1. written documentation
2. presentation of the results of the practical components
3. active participation in the discussions

A practical component—e.g. conducting a survey or implementing an application—may also be part of the regular workload and is described in the specific assignment description of the course.

The individual components are integrated into an overall assessment that reflects the overall impression of the combined performance

Prerequisites
None

Recommendations
None

6.540 Module component: Practical Seminar Interaction [T-WIWI-109935]

Coordinators: Prof. Dr. Alexander Mädche
Prof. Dr. Christof Weinhardt

Organisation: KIT Department of Business and Economics

Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Examination of another type	4,5 CP	graded	Each term	2

Assessment
The assessment of this course is according to §4(2), 3 SPO in form of a written documentation, a presentation of the outcome of the conducted practical components and an active participation in class. Please take into account that, beside the written documentation, also a practical component (e.g. implementation of a prototype) is part of the course. Please examine the course description for the particular tasks. The final mark is based on the graded and weighted attainments (such as the written documentation, presentation, practical work and an active participation in class). In the winter terms, the course is only offered as a seminar.

Prerequisites
None.

T 6.541 Module component: Practical Seminar Platforms [T-WIWI-109937]

Coordinators: Prof. Dr. Gerhard Satzger
Prof. Dr. Christof Weinhardt

Organisation: KIT Department of Business and Economics

Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Examination of another type	4,5 CP	graded	Each term	2

Assessment

The assessment of this course is according to §4(2), 3 SPO in form of a written documentation, a presentation of the outcome of the conducted practical components and an active participation in class. Please take into account that, beside the written documentation, also a practical component (e.g. implementation of a prototype) is part of the course. Please examine the course description for the particular tasks. The final mark is based on the graded and weighted attainments (such as the written documentation, presentation, practical work and an active participation in class). In the winter terms, the course is only offered as a seminar.

Prerequisites

None.

T 6.542 Module component: Practical Seminar Servitization [T-WIWI-109939]

Coordinators: Prof. Dr. Alexander Mädche
Prof. Dr. Gerhard Satzger

Organisation: KIT Department of Business and Economics

Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Examination of another type	4,5 CP	graded	Each term	1

Assessment

The assessment of this course is according to §4(2), 3 SPO in form of a written documentation, a presentation of the outcome of the conducted practical components and an active participation in class. Please take into account that, beside the written documentation, also a practical component (e.g. implementation of a prototype) is part of the course. Please examine the course description for the particular tasks. The final mark is based on the graded and weighted attainments (such as the written documentation, presentation, practical work and an active participation in class). In the winter terms, the course is only offered as a seminar.

Prerequisites

None.

T 6.543 Module component: Practical Seminar: Advanced Analytics [T-WIWI-108765]

Coordinators:	Prof. Dr. Christof Weinhardt
Organisation:	KIT Department of Business and Economics
Part of:	M-WIWI-103118 - Data Science: Data-Driven User Modeling M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Examination of another type	4,5 CP	graded	Each term	1

Assessment
The assessment consists of practical work in the field of advanced analytics, a seminar paper, a presentation of the results and the contribution to the discussion (according to §4(2), 3 of the examination regulation). The final grade is based on the evaluation of each component (seminar paper, oral presentation, and active participation).

Prerequisites
None

Recommendations
At least one module offered by the institute should have been chosen before attending this seminar.

Additional Information
The course is held in English. The course is not offered regularly.

T 6.544 Module component: Practical Seminar: AI & Data Science [T-WIWI-114892]

Coordinators: Prof. Dr. Alexander Mädche
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Examination of another type	4,5 CP	graded	Irregular	1

Courses					
WT 25/26	2500019	Practical Seminar: AI & Data Science	3 SWS	Lecture / 🔀	Mädche
ST 2026	2500027	Practical Seminar: AI & Data Science	3 SWS	Lecture / 🔀	Mädche
ST 2026	2500073	Practical Seminar: AI-based Mental Healthcare	3 SWS	Lecture / 🔀	Kuhlmeier, Mädche

Legend: Online, Blended (On-Site/Online), On-Site, ✕ Cancelled

Assessment
Success is assessed in the form of an alternative exam assessment. Success is assessed by carrying out a practical component, preparing written documentation and actively participating in the discussions. A total of 60 points can be achieved, of which

- a maximum of 25 points for the written documentation
- a maximum of 25 points for the practical component
- a maximum of 10 points for active participation in the discussions

At least 30 points must be achieved to pass the performance review. Please note that a practical component such as conducting a survey or implementing an application is also part of the regular scope of the course in addition to the written documentation. The respective tasks can be found in the announcement on the institute's website https://h-lab.win.kit.edu.

Prerequisites
None

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Practical Seminar: AI-based Mental Healthcare

2500073, SS 2026, 3 SWS, Language: English, Open in study portal

Lecture (V)
Blended (On-Site/Online)

Content

Mental health disorders affect a significant share of the global population. Meeting the resulting demand for care is a challenge on two fronts: patients face long waiting times and limited access to support, while mental health practitioners work under significant time pressure with processes that are often still manual and fragmented. Two developments in AI offer complementary responses: AI-based process digitalization can make clinical and therapeutic workflows more efficient for practitioners, while conversational AI agents can extend therapeutic support directly to those who need it. This seminar engages students with both developments: working on real, ongoing projects at the h-lab and its clinical partners, students design, build, and evaluate prototypes of AI-based applications for mental health and acquire the domain knowledge needed to do so responsibly.

Three project briefs are available, each rooted in an active h-lab research project. Students may also propose their own project in the mental health and AI space.

1. **AI-assisted ADHD assessment system** in collaboration with LWL-Klinikum Marsberg, centered on the analysis of primary school reports as a source of retrospective diagnostic evidence. Example tasks include building an OCR and LLM pipeline for extracting and analyzing free-text reports, or evaluating the diagnostic utility of LLM-generated assessments against clinician judgments.
2. **AI-enhanced CBT group therapy platform** in collaboration with Rethink Wellbeing. Example tasks include building a faciliator feedback system that monitors group dynamics and recommends facilitator actions, automating participant or facilitator selection workflows, or evaluating the accuracy and clinical acceptability of AI-generated recommendations.
3. **AI-based Conversational Mental Health Agents**, in collaboration with the Chair of Clinical Psychology and Psychotherapy at the University of Greifswald. The h-lab and its partners developed an LLM-based chatbot that delivers a single-session behavioral activation session for young people with depressive symptoms. Example tasks include improving the behavioral activation module, implementing additional therapeutic modules (cognitive restructuring, sleep hygiene, emotion regulation, social skills) as LLM-based multi-agent systems, or evaluating agent outputs for therapeutic fidelity and safety.

The seminar is structured around **three phases**:

1. I**ntroductory session:** Introduction to designing and evaluating AI-based mental health and the project briefs.
2. **Working phase:** Students design, develop, and evaluate their prototypes (individually or in groups), with ad hoc support from h-lab assistants.
3. **Final presentation**: Groups present their projects, including a demonstration of their working prototype and a discussion of their evaluation approach and preliminary findings. Participants provide feedback to each other's projects. Technical documentation and a brief evaluation report are due at the end of the semester.

Learning Objectives: Upon completion, students will be able to:

- Understand the mental health domain sufficiently to make informed design decisions
- Build functional AI-based application prototypes addressing a concrete need in an active mental health research or practice context
- Design and apply appropriate evaluation strategies for AI-based mental health applications, assessing feasibility, effectiveness, and safety of their prototypes

T 6.545 Module component: Practical Seminar: Artificial Intelligence in Service Systems [T-WIWI-112152]

Coordinators: Prof. Dr. Gerhard Satzger
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101506 - Service Analytics
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Examination of another type	4,5 CP	graded	Irregular	1

Assessment
The assessment of this course is in form of a written documentation, a presentation of the outcome of the conducted practical components and an active participation in class.
Please take into account that, beside the written documentation, also a practical component (such as a survey or an implementation of an application) is part of the course. Please examine the course description for the particular tasks.

The final mark is based on the graded and weighted attainments (such as the written documentation, presentation, practical work and an active participation in class).

Prerequisites
None.

Recommendations
Knowledge in the field of Artificial Intelligence in Service Systems is assumed. Therefore, it is recommended to attend the course Artificial Intelligence in Service Systems [2595650] beforehand.

Workload
135 hours

T 6.546 Module component: Practical Seminar: Data-Driven Information Systems [T-WIWI-106207]

Coordinators: Prof. Dr. Gerhard Satzger
Prof. Dr. Christof Weinhardt

Organisation: KIT Department of Business and Economics

Part of: M-WIWI-103117 - Data Science: Data-Driven Information Systems
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Examination of another type	4,5 CP	graded	Irregular	1

Assessment
The assessment consists of a seminar paper, a presentation of the results and the contribution to the discussion (according to §4(2), 3 of the examination regulation). The final grade is based on the evaluation of each component (seminar paper, oral presentation, and active participation).

Prerequisites
None

Recommendations
At least one module offered by the institute should have been chosen before attending this seminar.

Additional Information
The course is held in english. The course is not offered regularly.

T 6.547 Module component: Practical Seminar: Digital Services [T-WIWI-110888]

Coordinators: Prof. Dr. Gerhard Satzger
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Examination of another type	4,5 CP	graded	Each summer term	1

Assessment
The assessment consists of a seminar paper, a presentation of the results and the contribution to the discussion. In the seminar, a maximum score of 60 points can be achieved, consisting of

- maximum 25 points for the documentation (written examination)
- maximum 25 points for the practical assessment
- maximum 10 points for the participation during the discussion sessions

The practical seminar is passed when at least a score of 30 points is achieved.

Prerequisites
None

Recommendations
None

Additional Information
The current range of seminar topics is announced on the following Website:
www.dsi.iism.kit.edu.

T 6.548 Module component: Practical Seminar: Health Care Management (with Case Studies) [T-WIWI-102716]

Coordinators: Prof. Dr. Stefan Nickel
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-102805 - Service Operations
M-WIWI-104899 - Operations Research

Type	Credits	Grading	Term offered	Version
Examination of another type	4,5 CP	graded	Each term	3

Courses					
WT 25/26	2500008	Practical seminar: Health Care Management	3 SWS	Others / 🗣	Nickel, Mitarbeiter

Legend: 🖥 Online, ❂ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment of this module is an examination of another type and consists of the following components:

- a case study to be worked on,
- a seminar paper and
- a final oral examination.

The overall grade results from the weighted evaluation of the case study, the seminar paper and the presentation. The exact weighting of these individual components for the grade will be announced at the beginning of the course.

Prerequisites
None.

Recommendations
Basic knowledge as conveyed in the module *Introduction toOperations Research* is assumed.

Additional Information
The lecture is offered every term.

The planned lectures and courses for the next three years are announced online.

Workload
135 hours

T 6.549 Module component: Practical Seminar: Human-Centered Systems [T-WIWI-113459]

Coordinators: Prof. Dr. Alexander Mädche
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-102806 - Service Innovation, Design & Engineering
M-WIWI-103117 - Data Science: Data-Driven Information Systems
M-WIWI-104068 - Information Systems in Organizations
M-WIWI-104080 - Designing Interactive Information Systems
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Examination of another type	4,5 CP	graded	Each term	1

Courses					
WT 25/26	2540554	Practical Seminar: Human-Centered Systems	3 SWS	Lecture / 🔀	Mädche
ST 2026	2500073	Practical Seminar: AI-based Mental Healthcare	3 SWS	Lecture / 🔀	Kuhlmeier, Mädche
ST 2026	2540554	Practical Seminar: Human-Centered Systems	3 SWS	Lecture / 🔀	Mädche

Legend: 🖥 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment of this course is in the form of a different type of examination. The assessment is carried out by a practical component, preparing written documentation and actively participating in the discussions. A total of 60 points can be achieved, of which:

- a maximum of 25 points for the written documentation
- a maximum of 25 points for the practical component
- a maximum of 10 points for active participation in the discussions

At least 30 points must be achieved to pass the performance assessment. Please note that a practical component such as conducting a survey or implementing an application is also part of the regular scope of the course in addition to the written documentation. The respective tasks can be found in the announcement on the institute's website https://h-lab.iism.kit.edu.

Below you will find excerpts from courses related to this module component:

V Practical Seminar: Human-Centered Systems
2540554, WS 25/26, 3 SWS, Language: English, Open in study portal

Lecture (V)
Blended (On-Site/Online)

V Practical Seminar: AI-based Mental Healthcare
2500073, SS 2026, 3 SWS, Language: English, Open in study portal

Lecture (V)
Blended (On-Site/Online)

Content

Mental health disorders affect a significant share of the global population. Meeting the resulting demand for care is a challenge on two fronts: patients face long waiting times and limited access to support, while mental health practitioners work under significant time pressure with processes that are often still manual and fragmented. Two developments in AI offer complementary responses: AI-based process digitalization can make clinical and therapeutic workflows more efficient for practitioners, while conversational AI agents can extend therapeutic support directly to those who need it. This seminar engages students with both developments: working on real, ongoing projects at the h-lab and its clinical partners, students design, build, and evaluate prototypes of AI-based applications for mental health and acquire the domain knowledge needed to do so responsibly.

Three project briefs are available, each rooted in an active h-lab research project. Students may also propose their own project in the mental health and AI space.

1. **AI-assisted ADHD assessment system** in collaboration with LWL-Klinikum Marsberg, centered on the analysis of primary school reports as a source of retrospective diagnostic evidence. Example tasks include building an OCR and LLM pipeline for extracting and analyzing free-text reports, or evaluating the diagnostic utility of LLM-generated assessments against clinician judgments.
2. **AI-enhanced CBT group therapy platform** in collaboration with Rethink Wellbeing. Example tasks include building a faciliator feedback system that monitors group dynamics and recommends facilitator actions, automating participant or facilitator selection workflows, or evaluating the accuracy and clinical acceptability of AI-generated recommendations.
3. **AI-based Conversational Mental Health Agents**, in collaboration with the Chair of Clinical Psychology and Psychotherapy at the University of Greifswald. The h-lab and its partners developed an LLM-based chatbot that delivers a single-session behavioral activation session for young people with depressive symptoms. Example tasks include improving the behavioral activation module, implementing additional therapeutic modules (cognitive restructuring, sleep hygiene, emotion regulation, social skills) as LLM-based multi-agent systems, or evaluating agent outputs for therapeutic fidelity and safety.

The seminar is structured around **three phases**:

1. I**ntroductory session:** Introduction to designing and evaluating AI-based mental health and the project briefs.
2. **Working phase:** Students design, develop, and evaluate their prototypes (individually or in groups), with ad hoc support from h-lab assistants.
3. **Final presentation**: Groups present their projects, including a demonstration of their working prototype and a discussion of their evaluation approach and preliminary findings. Participants provide feedback to each other's projects. Technical documentation and a brief evaluation report are due at the end of the semester.

Learning Objectives: Upon completion, students will be able to:

- Understand the mental health domain sufficiently to make informed design decisions
- Build functional AI-based application prototypes addressing a concrete need in an active mental health research or practice context
- Design and apply appropriate evaluation strategies for AI-based mental health applications, assessing feasibility, effectiveness, and safety of their prototypes

V	**Practical Seminar: Human-Centered Systems** 2540554, SS 2026, 3 SWS, Language: English, Open in study portal	**Lecture (V)** **Blended (On-Site/Online)**

Content

In this practical seminar, students get an individual assignment and develop a running software prototype. Beside the software prototype, the students also deliver a written documentation.

Please find the current open offerings on our website: https://h-lab.iism.kit.edu/thesis.php

Prerequisites

Profound skills in software development are required

Literature

Further literature will be made available in the seminar.

T 6.550 Module component: Practical Seminar: Interactive Systems [T-WIWI-111914]

Coordinators: Prof. Dr. Alexander Mädche
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Examination of another type	4,5 CP	graded	Each term	2

Courses					
WT 25/26	2540555	Practical Seminar: Interactive Systems	3 SWS	Lecture /	Mädche
ST 2026	2500073	Practical Seminar: AI-based Mental Healthcare	3 SWS	Lecture /	Kuhlmeier, Mädche
ST 2026	2540555	Practical Seminar: Interactive Systems	3 SWS	Lecture /	Mädche

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
Alternative exam assessment.

The assessment of this course consists of the implementation of a practical component, the preparation of a written documentation, and active participation in the discussions.

A total of 60 points can be achieved, of which:

- maximum 25 points for the written documentation
- maximum 25 points for the practical component
- maximum 10 points for active participation in the discussions

A minimum of 30 points must be achieved to pass this course.

Please note that a practical component, such as conducting a survey or implementing an application, is also part of the course. Please refer to the institute website issd.iism.kit.edu for the current offer of practical seminar theses.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Practical Seminar: AI-based Mental Healthcare

2500073, SS 2026, 3 SWS, Language: English, Open in study portal

Lecture (V)
Blended (On-Site/Online)

Content

Mental health disorders affect a significant share of the global population. Meeting the resulting demand for care is a challenge on two fronts: patients face long waiting times and limited access to support, while mental health practitioners work under significant time pressure with processes that are often still manual and fragmented. Two developments in AI offer complementary responses: AI-based process digitalization can make clinical and therapeutic workflows more efficient for practitioners, while conversational AI agents can extend therapeutic support directly to those who need it. This seminar engages students with both developments: working on real, ongoing projects at the h-lab and its clinical partners, students design, build, and evaluate prototypes of AI-based applications for mental health and acquire the domain knowledge needed to do so responsibly.

Three project briefs are available, each rooted in an active h-lab research project. Students may also propose their own project in the mental health and AI space.

1. **AI-assisted ADHD assessment system** in collaboration with LWL-Klinikum Marsberg, centered on the analysis of primary school reports as a source of retrospective diagnostic evidence. Example tasks include building an OCR and LLM pipeline for extracting and analyzing free-text reports, or evaluating the diagnostic utility of LLM-generated assessments against clinician judgments.
2. **AI-enhanced CBT group therapy platform** in collaboration with Rethink Wellbeing. Example tasks include building a faciliator feedback system that monitors group dynamics and recommends facilitator actions, automating participant or facilitator selection workflows, or evaluating the accuracy and clinical acceptability of AI-generated recommendations.
3. **AI-based Conversational Mental Health Agents**, in collaboration with the Chair of Clinical Psychology and Psychotherapy at the University of Greifswald. The h-lab and its partners developed an LLM-based chatbot that delivers a single-session behavioral activation session for young people with depressive symptoms. Example tasks include improving the behavioral activation module, implementing additional therapeutic modules (cognitive restructuring, sleep hygiene, emotion regulation, social skills) as LLM-based multi-agent systems, or evaluating agent outputs for therapeutic fidelity and safety.

The seminar is structured around **three phases**:

1. I**ntroductory session:** Introduction to designing and evaluating AI-based mental health and the project briefs.
2. **Working phase:** Students design, develop, and evaluate their prototypes (individually or in groups), with ad hoc support from h-lab assistants.
3. **Final presentation**: Groups present their projects, including a demonstration of their working prototype and a discussion of their evaluation approach and preliminary findings. Participants provide feedback to each other's projects. Technical documentation and a brief evaluation report are due at the end of the semester.

Learning Objectives: Upon completion, students will be able to:

- Understand the mental health domain sufficiently to make informed design decisions
- Build functional AI-based application prototypes addressing a concrete need in an active mental health research or practice context
- Design and apply appropriate evaluation strategies for AI-based mental health applications, assessing feasibility, effectiveness, and safety of their prototypes

V	**Practical Seminar: Interactive Systems** 2540555, SS 2026, 3 SWS, Language: English, Open in study portal	**Lecture (V)** **Blended (On-Site/Online)**

Content

In this practical seminar, students get an individual assignment and develop a running software prototype. Beside the software prototype, the students also deliver a written documentation.

Please find the current open offerings on our website: https://h-lab.iism.kit.edu/thesis.php

6.551 Module component: Practical Seminar: Platform Economy [T-WIWI-112154]

Coordinators: Prof. Dr. Christof Weinhardt
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Examination of another type	4,5 CP	graded	Each term	2

Assessment
The assessment of this course is in form of a written documentation, a presentation of the outcome of the conducted practical components and an active participation in class. Please take into account that, beside the written documentation, also a practical component (e.g. implementation of a prototype) is part of the course. Please examine the course description for the particular tasks. The final mark is based on the graded and weighted attainments (such as the written documentation, presentation, practical work and an active participation in class).

Prerequisites
None.

Additional Information
Teaching and learning format: Seminar

Workload
135 hours

T 6.552 Module component: Practical Seminar: Service Innovation [T-WIWI-110887]

Coordinators: Prof. Dr. Gerhard Satzger
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101410 - Business & Service Engineering
M-WIWI-102806 - Service Innovation, Design & Engineering
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Examination of another type	4,5 CP	graded	Irregular	1

Assessment
Success is assessed through the preparation of written documentation, a presentation of the results of the practical components carried out and active participation in the discussions (in accordance with §4(2), 3 SPO).

Please note that a practical component such as conducting a survey or implementing an application is also part of the regular scope of the course in addition to the written documentation. Please refer to the course description for the respective tasks.

The overall grade is made up of the weighted components (e.g. documentation, oral presentation, practical work and active participation). The weighting of these components for the grade will be announced at the beginning of the course.

Recommendations
Knowledge of service innovation methods is strongly recommended.

Additional Information
Due to the project work, the number of participants is limited and participation requires knowledge about models, concepts and approaches. Details for registration will be announced on the web pages for this course.
The seminar is not offered regularly.

Workload
135 hours

6.553 Module component: Practical Training in Basics of Microsystem Technology [T-MACH-102164]

Coordinators: Dr. Arndt Last
Organisation: KIT Department of Mechanical Engineering

Part of: M-ETIT-101158 - Sensor Technology I
M-MACH-101290 - BioMEMS
M-MACH-101291 - Microfabrication
M-MACH-101292 - Microoptics
M-MACH-101294 - Nanotechnology

Type	Credits	Grading	Term offered	Version
Written examination	3 CP	graded	Each term	3

Courses					
WT 25/26	2143875	Introduction to Microsystem Technology - Practical Course	2 SWS	Practical course / 🗣	Last
ST 2026	2143875	Introduction to Microsystem Technology - Practical Course	2 SWS	Practical course / 🗣	Last

Legend: 📱 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success is assessed in the form of a written examination lasting approx. 60 minutes.

Prerequisites
none;
Mutual exclusion to T-MACH-108312 - Laboratory practical course on fundamentals of microsystems technology (ungraded)

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-MACH-108312 - Introduction to Microsystem Technology - Practical Course must not have been started.

Workload
90 hours

Below you will find excerpts from courses related to this module component:

V Introduction to Microsystem Technology - Practical Course

2143875, WS 25/26, 2 SWS, Language: German, Open in study portal

Practical course (P)
On-Site

Organizational issues
Das Praktikum findet in den Laboren des IMT am CN statt. Treffpunkt: Bau 301, vor dem Eingang.

Das "Praktikum" ist das selbe wie das "Laborpraktikum", nur benotet! Beide mit schriftlicher Klausur.

Wer teilnehmen möchte, muss sich ab 19.1.2026, 8h00 über das Campussystem unter Veranstaltungen (nicht unter Prüfungen!) auf die Warteliste setzen. Die endgültige Entscheidung, ob man teilnehmen kann, erfolgt erst 10.2.2026.

Die Teilnehmer und Teilnehmerinnen machen in Gruppen zu 3-5 Personen vier Versuche, jeder ~4 h Dauer mit Themen des Instituts aus diesen Bereichen (keine Auswahl möglich):

Röntgenoptik
UV-Lithographie am Beispiel des Photoresists AZ 4533
Fluidische Komponenten aus Polymerwerkstoffen am Beispiel eines Mischerbauteils
Rasterkraftmikroskopie
3D-Printing
Lichtstreuung an Chrommasken
Heißprägen von Kunststoff-Mikrostrukturen
Grundlagen der SAW-Biosensorik
Nano3D-Drucker - Materialtransfer dünnster Schichten
Electrospinning technology for 3D additive manufacturing
Nuclear magnetic resonance imaging
Introduction to Two-Photon Lithography

Literature
Menz, W., Mohr, J.: Mikrosystemtechnik für Ingenieure, VCH-Verlag, Weinheim, 1997
Unterlagen zum Praktikum zur Vorlesung ' Grundlagen der Mikrosystemtechnik'

V	**Introduction to Microsystem Technology - Practical Course** 2143875, SS 2026, 2 SWS, Language: German, Open in study portal	**Practical course (P)** **On-Site**

Content
In the practical training includes ten experiments:

1. X-ray optics
2. UV-lithography + SEM
3. Micro fluid mixer
4. AFM
5. 3D-printing
6. Scattering at chromium-masks
7. Micro moulding
8. SAW-bio sensorics
9. Nano3D-printer - material transfer in thin layers
10. Electro spinning

Each student takes part in only four experiments.
The experiments are carried out at real workstations at the IMT and coached by IMT-staff.

Organizational issues
Das Praktikum findet 14.-18.9.2026 an vier halben Tagen in den Laboren des IMT am CN statt. Treffpunkt: Bau 301, vor dem Eingang.

Das "Praktikum" ist das selbe wie das "Laborraktikum", nur benotet! Beide mit schriftlicher Klausur in der Woche nach dem Praktikum, voraussichtlich am 24.9.2026.

Wer teilnehmen möchte, muss sich ab 6.7.2026, 8h00 über das Campussystem unter Veranstaltungen (nicht unter Prüfungen!) auf die Warteliste setzen. Die endgültige Entscheidung, ob man teilnehmen kann, erfolgt erst 31.8.2026.

Literature
Menz, W., Mohr, J.: Mikrosystemtechnik für Ingenieure, VCH-Verlag, Weinheim, 1997
Unterlagen zum Praktikum zur Vorlesung ' Grundlagen der Mikrosystemtechnik'

T 6.554 Module component: Predictive Mechanism and Market Design [T-WIWI-102862]

Coordinators: Prof. Dr. Johannes Philipp Reiß
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101453 - Applied Strategic Decisions
M-WIWI-101505 - Experimental Economics
M-WIWI-104908 - Economics

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Irregular	1

Courses					
ST 2026	2500014	Predictive Mechanism and Market Design	2 SWS	Lecture / 🗣	Reiß
ST 2026	2500037	Übung zu Predictive Mechanism and Market Design	1 SWS	Practice / 🖥	Reiß, Peters

Legend: 🖥 Online, Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment consists of a written exam (60 minutes) (following §4(2), 1 of the examination regulation).

Prerequisites
None

Additional Information
The retake exam is given in the summer term subsequent to the fall term where the course (lecture and final exam) is given.

T 6.555 Module component: Predictive Modeling [T-WIWI-110868]

Coordinators: Prof. Dr. Fabian Krüger
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101638 - Econometrics and Statistics I
M-WIWI-101639 - Econometrics and Statistics II
M-WIWI-104902 - Statistics
M-WIWI-107187 - Quantitative Core

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Irregular	2

Courses					
ST 2026	2521311	Predictive Modeling	2 SWS	Lecture / 🗣	Krüger, Siefert
ST 2026	2521312	Predictive Modeling (Tutorial)	2 SWS	Practice / 🗣	Siefert, Krüger

Legend: 🖥 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment of this course is a written examination (90 minutes) according to §4(2), 1 of the examination regulation. A bonus can be acquired by successful completion of an assignment (written report + short in-class presentation) during the semester. If the grade of the written examination is between 4.0 and 1.3, the bonus improves the grade by one grade level (0.3 or 0.4).

Prerequisites
None

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Predictive Modeling
2521311, SS 2026, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content
Contents

This course presents methods for making and evaluating statistical predictions based on data. We consider various types of predictions (mean, probability, quantile, and full distribution), all of which are practically relevant. In each case, we discuss selected modeling approaches and their implementation using R software. We consider various economic case studies. Furthermore, we present methods for absolute evaluation (assessing whether a given model is compatible with the data) and relative evaluation (comparing the predictive performance of alternative models).

Learning objectives
Students have a good conceptual understanding of statistical prediction methods. They are able to implement these methods using statistical software, and can assess which method is suitable in a given situation.

Prerequisites

Students should know econometrics on the level of the course `Applied Econometrics' [2520020]

Literature

- Elliott, G., und A. Timmermann (Hrsg.): "Handbook of Economic Forecasting", vol. 2A und 2B, 2013.
- Gneiting, T., und M. Katzfuss: "Probabilistic Forecasting", Annual Review of Statistics and Its Application 1, 125-151, 2014.
- Hastie, T., Tibshirani, R., and J. Friedman: "The Elements of Statistical Learning", 2. Ausgabe, Springer, 2009.
- Weitere Literatur wird in der Vorlesung bekanntgegeben.

V Predictive Modeling (Tutorial)
2521312, SS 2026, 2 SWS, Language: English, Open in study portal

Practice (Ü)
On-Site

T 6.556 Module component: Price Management [T-WIWI-105946]

Coordinators: Prof. Dr. Andreas Geyer-Schulz
Dr Paul Glenn

Organisation: KIT Department of Business and Economics

Part of: M-WIWI-101409 - Electronic Markets
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each summer term	1

Courses					
ST 2026	2540529	Price Management	2 SWS	Lecture / 🗣	Glenn
ST 2026	2540530	Exercise Price Management	1 SWS	Practice / 🗣	Glenn

Legend: 📱 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Written examination (60 minutes) according to §4(2), 1 SPO. The exam is considered passed if at least 50 out of a maximum of 100 possible points are achieved. The grades are graded in five steps (best grade 1.0 from 95 points). Details of the grade formation and scale will be announced in the course.
A bonus can be acquired through successful participation in the practice. If the grade of the written examination is between 4.0 and 1.3, the bonus improves the grade by one grade level (0.3 or 0.4). The exact criteria for awarding a bonus will be announced at the beginning of the course.

Please note: this exam will no longer be offered after August 2026.

Prerequisites
None

Recommendations
None

Additional Information
Please note: this course, exercises and exam will no longer be offered after August 2026.

Below you will find excerpts from courses related to this module component:

V Price Management

2540529, SS 2026, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Literature

- H. Simon and M. Fassnacht, *Preismanagement*, vol. 4. Wiesbaden: Springer Gabler, 2016.
- T. T. Nagle, J. E. Hogan, und J. Zalee, *The Strategy and Tactics of Pricing: A guide to growing more profitably*. New Jersey: Prentice Hall, 2010.

T 6.557 Module component: Pricing [T-WIWI-102883]

Coordinators: Prof. Dr. Martin Klarmann
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration
M-WIWI-105312 - Marketing and Sales Management

Type	Credits	Grading	Term offered	Version
Examination of another type	4,5 CP	graded	Each winter term	3

Courses					
WT 25/26	2572199	Pricing	3 SWS	Block / 🗣	Klarmann, Bill, Schröder
WT 26/27	2572199	Pricing	3 SWS	Block / 🗣	Klarmann, Bill, Schröder

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Alternative exam assessment. The examination (and thus the grade) is composed of three parts:

1. The design and execution of your own small experimental study around the topic of behavioral pricing (as group work).
2. The processing and presentation of a case study on pricing (as group work).
3. The execution of a simulated price negotiation based on a systematic preparation (usually in teams of two).

Prerequisites
Since the earlier course (a) "Pricing Excellence" and (b) "Price Negotiations and Sales Presentations" become parts of the Pricing course, Pricing cannot be taken if (a) and/or (b) have already been completed.

Recommendations
Students are highly encouraged to actively participate in class.

Additional Information
A small application is required for participation in this class. The application phase usually takes place at the beginning of the lecture period in the winter semester. More information on the application process will be made available on the Marketing and Sales Research Group website (marketing.iism.kit.edu) shortly before the start of the winter semester lecture period. This course is limited to 24 participants.

Below you will find excerpts from courses related to this module component:

V Pricing

2572199, WS 25/26, 3 SWS, Language: English, Open in study portal

Block (B)
On-Site

Content
At the Pricing lecture, students learn about current research and best practices in price management. Delivered in workshop format, the lecture has three key elements:

1. "Behavioral Pricing" workshop
   In this part of the course, central concepts and findings from behavioral pricing research (e.g. price information processing, reference prices, price fairness and mental accounting) are presented and discussed on the basis of important behavioral theories (e.g. prospect theory and information economics). After a brief introduction to experimental research, participants will then conduct their own small experimental study in the form of group work on a hypothesis they have developed on pricing behavior, analyze the data, and present it.

2. "Pricing Excellence" workshop
   In a theory section at the beginning of the course, students are taught theoretical principles of pricing. This includes an introduction to (1) pricing of product prices as well as (2) pricing of net customer prices (development of discount systems). Furthermore, theoretical basics of price enforcement and price monitoring are discussed. This will be followed by a practical application of what has been learned by working on a case study in small groups with a concluding presentation.

3. "Price Negotiation" workshop
   After an introduction to key theories and concepts of negotiation, students prepare and then conduct a simulated price negotiation in small groups with guidance.

Learning Objectives:

Students...

- are familiar with central theories explaining behavioral phenomena regarding consumers dealing with prices
- are able to describe and explain central phenomena of behavioral science with regard to price behavior and derive implications from them
- can formulate their own hypotheses on price behavior and design, conduct and evaluate a suitable experimental study for this purpose
- learn theoretical basics of pricing behavior
- learn the theoretical basics of price enforcement and price monitoring
- apply the acquired knowledge in a practical case study
- know important conceptual basics on the subject of price negotiations
- can prepare and competently conduct price negotiations
- present the results of their group work in a concise and structured manner

All events will take place in presence with compulsory attendance at all dates.

Total time required for 4.5 credit points: approx. 135 hours

Attendance time: 30 hours

Self-study: 105 hours

Organizational issues
Dates will be announced.

V	**Pricing** 2572199, WS 26/27, 3 SWS, Language: English, Open in study portal	**Block (B)** **On-Site**

Content

At the Pricing lecture, students learn about current research and best practices in price management. Delivered in workshop format, the lecture has three key elements:

1. "Behavioral Pricing" workshop
   In this part of the course, central concepts and findings from behavioral pricing research (e.g. price information processing, reference prices, price fairness and mental accounting) are presented and discussed on the basis of important behavioral theories (e.g. prospect theory and information economics). After a brief introduction to experimental research, participants will then conduct their own small experimental study in the form of group work on a hypothesis they have developed on pricing behavior, analyze the data, and present it.

2. "Pricing Excellence" workshop
   In a theory section at the beginning of the course, students are taught theoretical principles of pricing. This includes an introduction to (1) pricing of product prices as well as (2) pricing of net customer prices (development of discount systems). Furthermore, theoretical basics of price enforcement and price monitoring are discussed. This will be followed by a practical application of what has been learned by working on a case study in small groups with a concluding presentation.

3. "Price Negotiation" workshop
   After an introduction to key theories and concepts of negotiation, students prepare and then conduct a simulated price negotiation in small groups with guidance.

Learning Objectives:

Students...

- are familiar with central theories explaining behavioral phenomena regarding consumers dealing with prices
- are able to describe and explain central phenomena of behavioral science with regard to price behavior and derive implications from them
- can formulate their own hypotheses on price behavior and design, conduct and evaluate a suitable experimental study for this purpose
- learn theoretical basics of pricing behavior
- learn the theoretical basics of price enforcement and price monitoring
- apply the acquired knowledge in a practical case study
- know important conceptual basics on the subject of price negotiations
- can prepare and competently conduct price negotiations
- present the results of their group work in a concise and structured manner

All events will take place in presence with compulsory attendance at all dates.

Total time required for 4.5 credit points: approx. 135 hours

Attendance time: 30 hours

Self-study: 105 hours

Organizational issues

Prof. Dr. Martin Klarmann:

Freitag, 13.11.2026, 10 Uhr bis 17 Uhr Lecture

Freitag, 20.11.2026, 10 Uhr bis 17 Uhr Lecture

Freitag, 11.12.2026, 10 Uhr bis 13:30 Uhr Presentation

Dr. Fabian Bill:

Mittwoch, 16.12. 2026 von 09 Uhr bis 15 Uhr Lecture

Mittwoch, 20.01.2027 von 09 Uhr bis 15 Uhr. Presentation Session

Mark Schröder:

Mittwoch 20.01. 2027 von 15-17h Theorieteil

Donnerstag 21.01.2027 von 09:00-17:00h Preisverhandlungen

Alles im Gebäude 01.87

Rüppurrer Straße 1a, House B

5. OG, Room B5.25

D-76137 Karlsruhe

T 6.558 Module component: Principles of Ceramic and Powder Metallurgy Processing [T-MACH-102111]

Coordinators: apl. Prof. Dr. Günter Schell
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101268 - Specific Topics in Materials Science
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each winter term	1

Courses					
WT 25/26	2193010	Basic principles of powder metallurgical and ceramic processing	2 SWS	Lecture /	Schell

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
The assessment consists of an oral exam (20-30 min) taking place at the agreed date. The re-examination is offered upon agreement.

Prerequisites
none

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Basic principles of powder metallurgical and ceramic processing

2193010, WS 25/26, 2 SWS, Language: German, Open in study portal

Lecture (V)
Blended (On-Site/Online)

Literature

- R.J. Brook: Processing of Ceramics I+II, VCH Weinheim, 1996
- M.N. Rahaman: Cermamic Processing and Sintering, 2nd Ed., Marcel Dekker, 2003
- W. Schatt ; K.-P. Wieters ; B. Kieback. ".Pulvermetallurgie: Technologien und Werkstoffe", Springer, 2007
- R.M. German. "Powder metallurgy and particulate materials processing. Metal Powder Industries Federation, 2005
- F. Thümmler, R. Oberacker. "Introduction to Powder Metallurgy", Institute of Materials, 1993

T 6.559 Module component: Principles of Constrained Static Optimization [T-CIWVT-112811]

Coordinators: Dr.-Ing. Pascal Jerono
Prof. Dr.-Ing. Thomas Meurer

Organisation: KIT Department of Chemical and Process Engineering

Part of: M-CIWVT-106313 - Principles of Constrained Static Optimization

Type	Credits	Grading	Version
Oral examination	4 CP	graded	1

Courses					
WT 25/26	2243060	Principles of Constrained Static Optimization	2 SWS	Lecture / Practice / 🗣	Meurer, Jerono
WT 26/27	2243060	Principles of Constrained Static Optimization	2 SWS	Lecture / Practice / 🗣	Meurer, Jerono

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

T 6.560 Module component: Principles of Food Process Engineering [T-CIWVT-101874]

Coordinators: PD Dr. Volker Gaukel
Organisation: KIT Department of Chemical and Process Engineering
Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Version
Oral examination	9 CP	graded	1

Prerequisites
none

T 6.561 Module component: Principles of Insurance Management [T-WIWI-102603]

Coordinators: Prof. Dr. Ute Werner
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Oral examination	4,5 CP	graded	Each summer term	1

Assessment
The assessment consists of oral presentations (incl. papers) within the lecture (according to Section 4 (2), 3 of the examination regulation) and a final oral exam (according to Section 4 (2), 2 of the examination regulation).

The overall grade consists of the assessment of the oral presentations incl. papers (50 percent) and the assessment of the oral exam (50 percent).

The examination will be offered latest until summer term 2017 (beginners only).

Prerequisites
None

Recommendations
None

T 6.562 Module component: Principles of Whole Vehicle Engineering [T-MACH-114095]

Coordinators: Dr. Manfred Harrer
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101265 - Vehicle Development
M-WIWI-107718 - Mechanical Engineering I
M-WIWI-107730 - Mechanical Engineering II

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each term	3

Courses					
WT 25/26	2113851	Principles of Whole Vehicle Engineering I	1 SWS	Lecture / 🗣	Harrer
ST 2026	2114860	Principles of Whole Vehicle Engineering II	1 SWS	/ 🗣	Harrer

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Written examination

Duration: 90 minutes

Auxiliary means: none

Prerequisites
T-MACH-114075 – Grundsätze der PKW-Entwicklung must not be started.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-MACH-114075 - Principles of Whole Vehicle Engineering (in German) must not have been started.

Additional Information
The course is offered in English.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Principles of Whole Vehicle Engineering I

Lecture (V)
On-Site

2113851, WS 25/26, 1 SWS, Language: English, Open in study portal

Content
1. Process of automobile development
2. Conceptual dimensioning and design of an automobile
3. Laws and regulations – National and international boundary conditions
4. Aero dynamical dimensioning and design of an automobile I
5. Aero dynamical dimensioning and design of an automobile II
6. Thermo-management in the conflict of objectives between styling, aerodynamic and packaging guidelines I
7. Thermo-management in the conflict of objectives between styling, aerodynamic and packaging guidelines II

Learning Objectives:

The students have an overview of the fundamentals of the development of automobiles. They know the development process, the national and the international legal requirements that are to be met. They have knowledge about the thermo-management, aerodynamics and the design of an automobile. They are ready to judge goal conflicts in the field of automobile development and to work out approaches to solving a problem.

Organizational issues
You will find the lecture material on ILIAS. To get the ILIAS password, KIT students refer to https://fast-web-01.fast.kit.edu/PasswoerterIlias/

Termine und nähere Informationen finden Sie auf der Institutshomepage.

Dats and further information will be published on the homepage of the institute.

Kann nicht mit Lehrveranstaltung 2113810 kombiniert werden

Cannot be combined with lecture 2113810.

Literature
Skript zur Vorlesung wird zu Beginn des Semesters ausgegeben

The scriptum will be provided during the first lessons

V **Principles of Whole Vehicle Engineering II**
2114860, SS 2026, 1 SWS, Language: English, Open in study portal **On-Site**

Content
1. Application-oriented material and production technology I
2. Application-oriented material and production technology II
3. Overall vehicle acoustics in the automobile development
4. Drive train acoustics in the automobile development
5. Testing of the complete vehicle
6. Properties of the complete automobile

Learning Objectives:

The students are familiar with the selection of appropriate materials and the choice of adequate production technology. They have knowledge of the acoustical properties of the automobiles, covering both the interior sound and exterior noise. They have an overview of the testing procedures of the automobiles. They know in detail the evaluation of the properties of the complete automobile. They are ready to participate competently in the development process of the complete vehicle.

Organizational issues
Dates and further information will be published on the homepage of the institute.

Veranstaltung findet als Blockvorlesung statt.

Kann nicht mit der Veranstaltung [2114842] kombiniert werden.

Cannot be combined with lecture [2114842].

Literature
Das Skript zur Vorlesung ist über ILIAS verfügbar.

T 6.563 Module component: Principles of Whole Vehicle Engineering (in German) [T-MACH-114075]

Coordinators: Dr. Manfred Harrer
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101265 - Vehicle Development
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Written examination	4 CP	graded	Each term	2

Courses					
WT 25/26	2113810	Fundamentals of Automobile Development I	1 SWS	Lecture / 🗣	Harrer
ST 2026	2114842	Principles of Whole Vehicle Engineering II	1 SWS	Block / 🗣	Harrer

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Written examination

Duration: 90 minutes

Auxiliary means: none

Prerequisites
T-MACH-114095 - Fundamentals of Automobile Development must not be started.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-MACH-114095 - Principles of Whole Vehicle Engineering must not have been started.

Additional Information
The course is offered in German.

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Fundamentals of Automobile Development I

2113810, WS 25/26, 1 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content
1. Process of automobile development
2. Conceptual dimensioning and design of an automobile
3. Laws and regulations – National and international boundary conditions
4. Aero dynamical dimensioning and design of an automobile I
5. Aero dynamical dimensioning and design of an automobile II
6. Thermo-management in the conflict of objectives between styling, aerodynamic and packaging guidelines I
7. Thermo-management in the conflict of objectives between styling, aerodynamic and packaging guidelines II

Learning Objecitves:

The students have an overview of the fundamentals of the development of automobiles. They know the development process, the national and the international legal requirements that are to be met. They have knowledge about the thermo-management, aerodynamics and the design of an automobile. They are ready to judge goal conflicts in the field of automobile development and to work out approaches to solving a problem.

Organizational issues
Das Vorlesungsmaterial wird auf ILIAS bereitgestellt. Das ILIAS-Passwort erhalten Sie unter https://fast-web-01.fast.kit.edu/PasswoerterIlias/

Termine und nähere Informationen finden Sie auf der Institutshomepage.

Kann nicht mit Lehrveranstaltung 2113851 kombiniert werden.

Date and further information will be published on the homepage of the institute.

Cannot be combined with lecture 2113851.

Literature
Skript zur Vorlesung wird zu Beginn des Semesters ausgegeben

The scriptum will be provided during the first lessons

V **Principles of Whole Vehicle Engineering II**
2114842, SS 2026, 1 SWS, Language: German, Open in study portal

Block (B)
On-Site

Content
1. Application-oriented material and production technology I
2. Application-oriented material and production technology II
3. Overall vehicle acoustics in the automobile development
4. Drive train acoustics in the automobile development
5. Testing of the complete vehicle
6. Properties of the complete automobile

Learning Objectives:

The students are familiar with the selection of appropriate materials and the choice of adequate production technology. They have knowledge of the acoustical properties of the automobiles, covering both the interior sound and exterior noise. They have an overview of the testing procedures of the automobiles. They know in detail the evaluation of the properties of the complete automobile. They are ready to participate competently in the development process of the complete vehicle.

Organizational issues
Vorlesung findet als Blockvorlesung statt. Termine werden über die Homepage des Instituts bekannt gegeben.

Kann nicht mit der Veranstaltung [2114860] kombiniert werden.

Cannot be combined with lecture [2114860].

Literature
Skript zur Vorlesung ist über ILIAS verfügbar.

T 6.564 Module component: Printed and Thin-Film Electronics [T-ETIT-114417]

Coordinators: Prof. Dr. Jasmin Aghassi-Hagmann
Dr. Dr. Michael Hirtz

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: M-ETIT-107343 - Printed and Thin-Film Electronics

Type	Credits	Grading	Term offered	Version
Oral examination	3 CP	graded	Each winter term	1

Courses					
WT 25/26	2308480	Printed and Thin-Film Electronics	2 SWS	Lecture /	Aghassi-Hagmann, Hirtz

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
The assessment takes place in form of an oral examination (approx. 20 minutes).

Prerequisites
none

6.565 Module component: Probabilistic Time Series Forecasting Challenge [T-WIWI-111387]

Coordinators: Prof. Dr. Fabian Krüger
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101638 - Econometrics and Statistics I
M-WIWI-101639 - Econometrics and Statistics II
M-WIWI-104902 - Statistics

Type	Credits	Grading	Term offered	Version
Examination of another type	4,5 CP	graded	Irregular	2

Assessment
Alternative exam assessment. Necessary conditions to pass the course:

- Weekly submission of statistical forecasts during the semester (excluding the Christmas break),
- Presentation (ca. 20 minutes) during the semester,
- Submission of a final report (5-10 pages) around the end of the semester.

Grading is based on the presentation (30%) and the final report (70%).

Prerequisites
Good methodological knowledge in statistics and data science.
Good knowledge in applied data analysis, incl. programming skills in R, Python or similar.
Knowledge of time series analysis is helpful, but not required.

Additional Information
The course is limited in participation. Participants will be selected via the WIWI portal.

Workload
135 hours

T 6.566 Module component: Problem Solving, Communication and Leadership [T-WIWI-102871]

Coordinators: Prof. Dr. Hagen Lindstädt
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	2 CP	graded	Each summer term	2

Courses					
WT 25/26	2577910	Problem solving, communication and leadership	1 SWS	Lecture / 🗣	Lindstädt

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment consists of a written exam (30 minutes) (following §4(2), 1 of the examination regulation). The exam takes place in every semester. Re-examinations are offered at every ordinary examination date.

Prerequisites
None

Workload
60 hours

Below you will find excerpts from courses related to this module component:

V Problem solving, communication and leadership
2577910, WS 25/26, 1 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content

The course highlights the aspects of problem solving and communication by first providing a structured look at how problem solving processes work. Participants will be empowered to identify, structure, analyze and communicate problems effectively. In addition, they are introduced to precise concepts for systematically structuring problem-solving processes. They learn how to apply and analyze structured communication in diagrams and presentations.

In addition, the course teaches key leadership concepts and frameworks that address the influence of situation, leadership personality and characteristics of those being led. Driven by current and practical perspectives, the course thus aims to teach cross-disciplinary skills.

In addition, through intensive interaction via selected case studies, participants are prepared for the practical application of what they have learned in various professional contexts.

Structure

The lectures of the course are available to students online as recordings, while the course dates are reserved for active discussion of practice-relevant case studies.

Learning Objectives

Upon completion of the course, students will be able to,

- structure problem-solving processes,
- apply the principles of goal-oriented communication in diagrams and presentations,
- Understand leadership decisions and place them in the context of situation and personality.

Recommendations:

None.

Workload:

- Total workload for 2 credit hours: approximately 30*2 hours.
- Thereof attendance time: 12-14 hours
- Remainder for preparation and post-processing as well as exam preparation.

Competence Certificate:

Success is assessed in the form of a written examination (30 min.) (in accordance with §4(2), 1 SPO) at the beginning of the lecture-free period of the semester. The examination is offered every semester and can be repeated at any regular examination date.

Literature

Verpflichtende Literatur:

Die relevanten Auszüge und zusätzlichen Quellen werden in der Veranstaltung bekannt gegeben.

Ergänzende Literatur:

- Hungenberg, Harlad: Problemlösung und Kommunikation, 3. Aufl. München 2010

- Zelazny, Gene; Delker, Christel: Wie aus zahlen Bilder werden, 6. Aufl. Wiesbaden 2008

- Minto, Barbara: Das Prinzip der Pyramide: Ideen klar, verständlich und erfolgreich kommunizieren. 2005

T 6.567 Module component: Procedures of Remote Sensing [T-BGU-103542]

Coordinators: Dr.-Ing. Uwe Weidner
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Version
Oral examination	3 CP	graded	2

Courses					
ST 2026	6020243	Procedures of Remote Sensing	2 SWS	Lecture / 🗣	Weidner

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-BGU-101638 - Procedures of Remote Sensing, Prerequisite must have been passed.

T 6.568 Module component: Procedures of Remote Sensing, Prerequisite [T-BGU-101638]

Coordinators: Dr.-Ing. Uwe Weidner
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Coursework	1 CP	pass/fail	Each summer term	2

Courses					
ST 2026	6020244	Procedures of Remote Sensing, Excercise	1 SWS	Practice / 🗣	Weidner

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Prerequisites
None

Recommendations
None

Additional Information
None

Workload
30 hours

T 6.569 Module component: Process Design for Machining of Metallic Components [T-MACH-114058]

Coordinators: Prof. Dr.-Ing. Volker Schulze
Organisation: KIT Department of Mechanical Engineering
Part of: M-MACH-101276 - Manufacturing Technology

Type	Credits	Grading	Term offered	Expansion	Version
Written examination	4 CP	graded	Each winter term	1 semesters	1

Courses					
WT 25/26	2149672	Process Design for Machining of Metallic Components	3 SWS	Lecture / 🗣	Schulze
WT 26/27	2149672	Process Design for Machining of Metallic Components	3 SWS	Lecture / 🗣	Schulze

Legend: 🖥 Online, ⚄ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
written exam, duration 60 minutes

Additional Information
The course is offered in German.

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Process Design for Machining of Metallic Components

Lecture (V)
On-Site

2149672, WS 25/26, 3 SWS, Language: German, Open in study portal

Content

The objective of the course “Process Development for Machining Metallic Components” is to enable students to descripe essential machining processes for metallic parts. Based on mechanical and materials fundamentals, the course covers all machining methods.

Within this course, students learn theoretical and application aspects of the following topics:

- Orthogonal cutting (cutting forces, rake angle, friction, tool wear)
- Turning and drilling (tool geometry, chip formation, setup times, productivity)
- Milling (face, contour, profile milling; high-speed milling)
- Grinding (wheel composition, thermal effects, surface integrity)
- Honing, lapping, polishing (tolerances, microstructure, surface quality)
- Process chains (raw part → rough machining → finish; handover parameters)
- Modeling and simulation (FEM-based modeling; empirical models)
- Chip formation simulation (numerical prediction of chip shape and breakage)
- Large-scale modeling (wear modeling; sensor integration)
- CNC machining and machine dynamics (G-/M-Codes; axis dynamics; chatter avoidance).

The acquired knowledge is supplemented by practical lectures from industry, which offer valuable insights into current applications, challenges and innovations in machining methods.

Learning Outcomes:

The students ...

- can describe the fundamental mechanisms and characteristic features of individual machining processes (orthogonal cutting, turning, drilling, milling, grinding, honing, lapping, polishing).
- are able to classify machining methods by their operating principle (geometry-defined vs. geometry-indeterminate) and categorize them based on technical parameters (cutting force, feed rate, material removal rate).
- can select an appropriate process for given component requirements (material, geometry, tolerance) and justify their choice.
- are capable of identifying interactions between individual steps in a multi-stage process chain and integrating them effectively.
- can evaluate machining processes with respect to technical factors (material behavior, tool wear, surface integrity) and economic considerations (setup time, productivity, cost metrics) and develop an optimized process strategy.
- are able to apply basic modeling and simulation methods (FEM-based models, empirical approaches) to estimate cutting forces, temperatures, and tool life.
- can interpret numerical chip-formation simulations and use them for process optimization and tool design.
- understand the fundamentals of CNC programming (G- and M-codes) and can analyze machine-dynamic effects (axis dynamics, vibration phenomena, chatter) to ensure stable machining.

Workload:

regular attendance: 21 hours
self-study: 99 hours

Literature
Medien:
Skript zur Veranstaltung wird über ilias (https://ilias.studium.kit.edu/)
bereitgestellt.

Media:
Lecture notes will be provided in ilias (https://ilias.studium.kit.edu/).

V	**Process Design for Machining of Metallic Components** 2149672, WS 26/27, 3 SWS, Language: German, Open in study portal	**Lecture (V)** **On-Site**

Content
The objective of the course “Process Development for Machining Metallic Components” is to enable students to descripe essential machining processes for metallic parts. Based on mechanical and materials fundamentals, the course covers all machining methods.

Within this course, students learn theoretical and application aspects of the following topics:

- Orthogonal cutting (cutting forces, rake angle, friction, tool wear)
- Turning and drilling (tool geometry, chip formation, setup times, productivity)
- Milling (face, contour, profile milling; high-speed milling)
- Grinding (wheel composition, thermal effects, surface integrity)
- Honing, lapping, polishing (tolerances, microstructure, surface quality)
- Process chains (raw part → rough machining → finish; handover parameters)
- Modeling and simulation (FEM-based modeling; empirical models)
- Chip formation simulation (numerical prediction of chip shape and breakage)
- Large-scale modeling (wear modeling; sensor integration)
- CNC machining and machine dynamics (G-/M-Codes; axis dynamics; chatter avoidance).

The acquired knowledge is supplemented by practical lectures from industry, which offer valuable insights into current applications, challenges and innovations in machining methods.

Learning Outcomes:

The students ...

- can describe the fundamental mechanisms and characteristic features of individual machining processes (orthogonal cutting, turning, drilling, milling, grinding, honing, lapping, polishing).
- are able to classify machining methods by their operating principle (geometry-defined vs. geometry-indeterminate) and categorize them based on technical parameters (cutting force, feed rate, material removal rate).
- can select an appropriate process for given component requirements (material, geometry, tolerance) and justify their choice.
- are capable of identifying interactions between individual steps in a multi-stage process chain and integrating them effectively.
- can evaluate machining processes with respect to technical factors (material behavior, tool wear, surface integrity) and economic considerations (setup time, productivity, cost metrics) and develop an optimized process strategy.
- are able to apply basic modeling and simulation methods (FEM-based models, empirical approaches) to estimate cutting forces, temperatures, and tool life.
- can interpret numerical chip-formation simulations and use them for process optimization and tool design.
- understand the fundamentals of CNC programming (G- and M-codes) and can analyze machine-dynamic effects (axis dynamics, vibration phenomena, chatter) to ensure stable machining.

Workload:

regular attendance: 21 hours
self-study: 99 hours

Organizational issues
Bekanntgabe von möglichen Praxisvorlesungen erfolgt in der ersten Vorlesung.

Literature
Medien:
Skript zur Veranstaltung wird über ilias (https://ilias.studium.kit.edu/)
bereitgestellt.

Media:
Lecture notes will be provided in ilias (https://ilias.studium.kit.edu/).

T 6.570 Module component: Process Engineering [T-BGU-101844]

Coordinators: Dr.-Ing. Harald Schneider
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: M-BGU-101110 - Process Engineering in Construction
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Expansion	Version
Written examination	3 CP	graded	Each term	1 semesters	1

Courses					
WT 25/26	6241703	Process Engineering	2 SWS	Lecture / 🗣	Schneider, Waleczko

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
written exam, 60 min.

Prerequisites
none

Recommendations
none

Additional Information
none

Workload
90 hours

T 6.571 Module component: Process Fundamentals by the Example of Food Production [T-CIWVT-106058]

Coordinators: PD Dr. Volker Gaukel
Organisation: KIT Department of Chemical and Process Engineering
Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Version
Written examination	3 CP	graded	2

Prerequisites
none

T 6.572 Module component: Process Mining [T-WIWI-109799]

Coordinators: Dr.-Ing. Clemens Schreiber
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101472 - Informatics
M-WIWI-101628 - Emphasis in Informatics
M-WIWI-104901 - Informatics (KIT-Department of Economics and Management)

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each summer term	2

Courses					
ST 2026	2511204	Process Mining	2 SWS	Lecture / 🗣	Schreiber
ST 2026	2511205	Exercise Process Mining	1 SWS	Practice / 🗣	Rybinski

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment of this course is a written examination (60 min) according to §4(2), 1 of the examination regulation in the first week after lecture period.

Prerequisites
None

Additional Information
Former name (up to winter semester 2018/1019) "Workflow Management".

Below you will find excerpts from courses related to this module component:

V Process Mining

2511204, SS 2026, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content
The area of process mining covers approaches which aim at deducting new knowledge on the basis of logfiles generated by information systems. Such information systems are e.g., workflow-management-systems which are used for an efficient control of processes in enterprises and organisations. The lecture introduces the foundations of processes and respective modeling and analysis techniques. In the following, the foundations of process mining and the three classical types of approaches - discovery, conformance and enhancement - will be taught. In addition to the theoretical basics, tools, application scenarios in practice and open research questions are covered as well.

Learning objectives:

Students

- understand the concepts and approaches of process mining and know how they are applied,
- create and evaluate business process models,
- analyze static and dynamic properties of workflows,
- apply approaches and tools of process mining.

Recommendations:

Knowledge of course Applied Informatics - Modelling is expected.

Workload:

- Lecture 30h
- Exercise 15h
- Preparation of lecture 24h
- Preparation of exercises 25h
- Exam preparation 40h
- Exam 1h

Literature

- W. van der Aalst, H. van Kees: Workflow Management: Models, Methods and Systems, Cambridge, The MIT Press, 2002.
- W. van der Aalst: Process Mining: Data Science in Action. Springer, 2016.
- J. Carmona, B. van Dongen, A. Solti, M. Weidlich: Conformance Checking: Relating Processes and Models. Springer, 2018.
- A. Drescher, A. Koschmider, A. Oberweis: Modellierung und Analyse von Geschäftsprozessen: Grundlagen und Übungsaufgaben mit Lösungen. De Gruyter Studium, 2017.
- A. Oberweis: Modellierung und Ausführung von Workflows mit Petri-Netzen. Teubner-Reihe Wirtschaftsinformatik, B.G. Teubner Verlag, 1996.
- R. Peters, M. Nauroth: Process-Mining: Geschäftsprozesse: smart, schnell und einfach, Springer, 2019.
- F. Schönthaler, G.Vossen, A. Oberweis, T. Karle: Business Processes for Business Communities: Modeling Languages, Methods, Tools. Springer, 2012.
- M. Weske: Business Process Management: Concepts, Languages, Architectures. Springer, 2012.

Weitere Literatur wird in der Vorlesung bekannt gegeben.

T 6.573 Module component: Product- and Production-Concepts for Modern Automobiles [T-MACH-110318]

Coordinators: Dr. Stefan Kienzle
Dr. Dieter Steegmüller

Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-105455 - Strategic Design of Modern Production Systems
M-MACH-106590 - Production Engineering
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each winter term	1

Courses					
WT 25/26	2149670	Product- and Production-Concepts for modern Automobiles	2 SWS	Lecture /	Steegmüller, Kienzle
WT 26/27	2149670	Product- and Production-Concepts for modern Automobiles	2 SWS	Lecture /	Steegmüller, Kienzle

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
Oral Exam (20 min)

Prerequisites
T-MACH-105166 - Materials and Processes for Body Leightweight Construction in the Automotive Industry must not have been started.

Additional Information
The course is offered in German.

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Product- and Production-Concepts for modern Automobiles

2149670, WS 25/26, 2 SWS, Language: German, Open in study portal

Lecture (V)
Blended (On-Site/Online)

Content

The lecture illuminates the practical challenges of modern automotive engineering. As former leaders of the automotive industry, the lecturers refer to current aspects of automotive product development and production.

The aim is to provide students with an overview of technological trends in the automotive industry. In this context, the course also focuses on changes in requirements due to new vehicle concepts, which may be caused by increased demands for individualisation, digitisation and sustainability. The challenges that arise in this context will be examined from both a production technology and product development perspective and will be illustrated with practical examples thanks to the many years of industrial experience of both lecturers.

The topics covered are:

- General conditions for vehicle and body development
- Integration of new drive technologies
- Functional requirements (crash safety etc.), also for electric vehicles
- Development Process at the Interface Product & Production, CAE/Simulation
- Energy storage and supply infrastructure
- Aluminium and lightweight steel construction
- FRP and hybrid parts
- Battery, fuel cell and electric motor production
- Joining technology in modern car bodies
- Modern factories and production processes, Industry 4.0.

Learning Outcomes:

The students …

- are able to name the presented general conditions of vehicle development and are able to discuss their influences on the final product using practical examples.
- are able to name the various lightweight approaches and identify possible areas of application.
- are able to identify the different production processes for manufacturing lightweight structures and explain their functions.
- are able to perform a process selection based on the methods and their characteristics.

Workload:

regular attendance: 25 hours
self-study: 95 hours

Organizational issues

Termine werden über Ilias bekannt gegeben.

Bei der Vorlesung handelt es sich um eine Blockveranstaltung. Eine Anmeldung über Ilias ist erforderlich.

Zur Vertiefung des im Rahmen der Lehrveranstaltung erworbenen Wissens werden die theoretischen Vorlesungseinheiten durch Praxiseinheiten im Umfeld der Karlsruher Forschungsfabrik (https://www.karlsruher-forschungsfabrik.de) unterstützt.

The lecture is a block course. An application in Ilias is mandatory.

The theoretical lectures are complemented by practical lectures in the Karlsruhe Research Factory (https://www.karlsruher-forschungsfabrik.de/en.html) to deepen the acquired knowledge.

Literature

Medien:

Skript zur Veranstaltung wird über (https://ilias.studium.kit.edu/) bereitgestellt.

Media:

Lecture notes will be provided in Ilias (https://ilias.studium.kit.edu/).

V	**Product- and Production-Concepts for modern Automobiles** 2149670, WS 26/27, 2 SWS, Language: German, Open in study portal	**Lecture (V)** **Blended (On-Site/Online)**

Content

The lecture illuminates the practical challenges of modern automotive engineering. As former leaders of the automotive industry, the lecturers refer to current aspects of automotive product development and production.

The aim is to provide students with an overview of technological trends in the automotive industry. In this context, the course also focuses on changes in requirements due to new vehicle concepts, which may be caused by increased demands for individualisation, digitisation and sustainability. The challenges that arise in this context will be examined from both a production technology and product development perspective and will be illustrated with practical examples thanks to the many years of industrial experience of both lecturers.

The topics covered are:

- General conditions for vehicle and body development
- Integration of new drive technologies
- Functional requirements (crash safety etc.), also for electric vehicles
- Development Process at the Interface Product & Production, CAE/Simulation
- Energy storage and supply infrastructure
- Aluminium and lightweight steel construction
- FRP and hybrid parts
- Battery, fuel cell and electric motor production
- Joining technology in modern car bodies
- Modern factories and production processes, Industry 4.0.

Learning Outcomes:

The students …

- are able to name the presented general conditions of vehicle development and are able to discuss their influences on the final product using practical examples.
- are able to name the various lightweight approaches and identify possible areas of application.
- are able to identify the different production processes for manufacturing lightweight structures and explain their functions.
- are able to perform a process selection based on the methods and their characteristics.

Workload:

regular attendance: 25 hours
self-study: 95 hours

Organizational issues

Termine werden über Ilias bekannt gegeben.

Bei der Vorlesung handelt es sich um eine Blockveranstaltung. Eine Anmeldung über Ilias ist erforderlich.

Zur Vertiefung des im Rahmen der Lehrveranstaltung erworbenen Wissens werden die theoretischen Vorlesungseinheiten durch Praxiseinheiten im Umfeld der Karlsruher Forschungsfabrik (https://www.karlsruher-forschungsfabrik.de) unterstützt.

The lecture is a block course. An application in Ilias is mandatory.

The theoretical lectures are complemented by practical lectures in the Karlsruhe Research Factory (https://www.karlsruher-forschungsfabrik.de/en.html) to deepen the acquired knowledge.

Literature

Medien:

Skript zur Veranstaltung wird über (https://ilias.studium.kit.edu/) bereitgestellt.

Media:

Lecture notes will be provided in Ilias (https://ilias.studium.kit.edu/).

T 6.574 Module component: Production and Logistics [T-WIWI-111632]

Coordinators: Prof. Dr. Wolf Fichtner
Prof. Dr. Stefan Nickel
Prof. Dr. Frank Schultmann

Organisation: KIT Department of Business and Economics

Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	3 CP	graded	Each winter term	1

Courses					
WT 26/27	2600005	Produktion und Logistik	2 SWS	Lecture / 🗣	Fichtner, Nickel, Schultmann

Legend: 🖥 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Written examination (60 min) on the course "Production and Logistics". The exam is offered at the beginning of each lecture-free period. Repeat examinations are possible at any regular examination date.

Prerequisites
None

T 6.575 Module component: Production and Logistics Management [T-WIWI-102632]

Coordinators: Prof. Dr. Frank Schultmann
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101412 - Industrial Production III
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	5,5 CP	graded	Each summer term	1

Courses					
ST 2026	2581954	Production and Logistics Management	2 SWS	Lecture / 🗣	Schultmann, Rosenberg
ST 2026	2581955	Production and Logistics Managment	2 SWS	Practice / 🗣	Schultmann

Legend: 📱 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment consists of a written exam (90 minutes) (following §4(2) of the examination regulation). The exam takes place in every semester. Re-examinations are offered at every ordinary examination date.

Prerequisites
None

Recommendations
None

Below you will find excerpts from courses related to this module component:

V Production and Logistics Management

2581954, SS 2026, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content
This course covers central tasks and challenges of operative production and logistics management. Students get to know the set-up and mode of planning systems such as production planning and control systems, enterprise resource planning systems and advanced planning systems to cope with the accompanying planning tasks in supply chain management. Methods to solve these tasks from the field of operational research will be explored with respect to manufacturing program planning, material requirement planning, lot size problems and scheduling. Alongside to MRP II (Manufacturing Resources Planning), students will be introduced to integrated supply chain management approaches. Finally, commercially available planning systems will be presented and discussed.

Literature
Wird in der Veranstaltung bekannt gegeben.

T 6.576 Module component: Production Economics and Sustainability [T-WIWI-102820]

Coordinators: Prof. Dr. Frank Schultmann
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	3,5 CP	graded	Each winter term	1

Courses					
WT 25/26	2581960	Production Economics and Sustainability	2 SWS	Lecture / 🗣	Schultmann, Steins
WT 26/27	2581960	Production Economics and Sustainability	2 SWS	Lecture / 🗣	Schultmann, Steins

Legend: 🖥 Online, ⧉ Blended (On-Site/Online), 🗣 On-Site, × Cancelled

Assessment
The assessment consists of an oral (30 minutes) or written exam (60 minutes) (following §4(2) of the examination regulation). The exam takes place in every semester. Re-examinations are offered at every ordinary examination date.

Workload
105 hours

Below you will find excerpts from courses related to this module component:

V Production Economics and Sustainability
2581960, WS 25/26, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content
The focus of the lecture is on the development of production systems towards more sustainable production. The lecture concentrates on the assessment of economically and ecologically reasonable measures and their potential implementation in companies. Relevant methods (including material flow analysis, life cycle assessment, eco-chart of accounts, investment planning) are also presented in the lecture.

Topics:
- Resources
- Material flow analysis
- Life cycle assessment
- Carbon footprint and water footprint
- Environmental management and environmental controlling
- Investment planning
- Circular planning
- Circular economy

Literature
wird in der Veranstaltung bekannt gegeben

V Production Economics and Sustainability
2581960, WS 26/27, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content

The focus of the lecture is on the development of production systems towards more sustainable production. The lecture concentrates on the assessment of economically and ecologically reasonable measures and their potential implementation in companies. Relevant methods (including material flow analysis, life cycle assessment, eco-chart of accounts, investment planning) are also presented in the lecture.

Topics:

- Resources
- Material flow analysis
- Life cycle assessment
- Carbon footprint and water footprint
- Environmental management and environmental controlling
- Investment planning
- Circular planning
- Circular economy

Literature

wird in der Veranstaltung bekannt gegeben

T 6.577 Module component: Production Technology for E-Mobility [T-MACH-110984]

Coordinators: Prof. Dr.-Ing. Jürgen Fleischer
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-106590 - Production Engineering
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Written examination	4 CP	graded	Each summer term	2

Courses					
ST 2026	2150605	Production Technology for E-Mobility	3 SWS	Lecture / 🗣	Fleischer

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Written Exam (60 min)

Prerequisites
none

Additional Information
The course is offered in German.

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Production Technology for E-Mobility
2150605, SS 2026, 3 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content
In the lecture Production Engineering for Electromobility the students should be enabled to design, select and develop production processes for the production of the components of an electric drive train (electric motor, battery cells, fuel cells) by using research-oriented teaching.

Learning Outcomes:

The students are able to:

- describe the structure and function of a fuel cell, an electric traction drive and a batterysystem.
- reproduce the process chains for the production of the components fuel cell, battery and electric traction drive.
- apply methodical tools to solve problems along the process chain.
- derive the challenges in the production of electric drives for electric mobility.
- describe the factors influencing the individual process steps on each other using the process chain of Li-ion battery cells.
- enumerate or describe the necessary process parameters to counteract the influencing factors of the process steps in Li-ion battery cell production.
- apply methodical tools to solve problems along the process chain for the production of Li-ion battery cells.
- derive the challenge of mounting and dismounting battery modules.
- derive the challenges in the production of fuel cells for use in mobility.
- develop solutions to overcome challenges in the production of fuel cells.

Workload:

regular attendance: 42 hours
self-study: 78 hours

Literature
Skript zur Veranstaltung wird über Ilias (https://ilias.studium.kit.edu/) bereitgestellt.

Lecture notes will be provided in Ilias (https://ilias.studium.kit.edu/)

T 6.578 Module component: Production, Logistics and Information Systems [T-WIWI-111602]

Coordinators: Prof. Dr. Wolf Fichtner
Prof. Dr. Andreas Geyer-Schulz
Prof. Dr. Alexander Mädche
Prof. Dr. Stefan Nickel
Prof. Dr. Frank Schultmann
Prof. Dr. Christof Weinhardt

Organisation: KIT Department of Business and Economics

Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	5 CP	graded	Each winter term	2

Courses					
WT 25/26	2600004		2 SWS	Lecture	Mädche, Feick
WT 25/26	2600005	Produktion und Logistik	2 SWS	Lecture / 🗣	Fichtner, Nickel, Schultmann
WT 25/26	2610029		2 SWS	Tutorial	Ruckes
WT 26/27	2600004		2 SWS	Lecture	Mädche, Feick
WT 26/27	2600005	Produktion und Logistik	2 SWS	Lecture / 🗣	Fichtner, Nickel, Schultmann
WT 26/27	2610029		2 SWS	Tutorial	Ruckes

Legend: 🖥 Online, Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Written Exam (90 min). The examination is offered at the beginning of each lecture-free period. Repeat examinations are possible at any regular examination date.

Workload
150 hours

Below you will find excerpts from courses related to this module component:

V **Lecture (V)**
2600004, WS 25/26, 2 SWS, Open in study portal

V **Lecture (V)**
2600004, WS 26/27, 2 SWS, Open in study portal

T 6.579 Module component: Programming [T-INFO-101531]

Coordinators: Prof. Dr.-Ing. Anne Koziolek
Prof. Dr. Ralf Reussner

Organisation: KIT Department of Informatics

Part of: M-WIWI-104909 - Informatics (Department of Informatics)

Type	Credits	Grading	Term offered	Version
Examination of another type	5 CP	graded	Each winter term	1

Courses					
WT 25/26	2424004	Programming	4 SWS	Lecture / Practice	Koziolek
ST 2026	2400083	Programming Exercise	0 SWS	Practice / 🗣	Koziolek
WT 26/27	2424004	Programming	4 SWS	Lecture / Practice	Koziolek

Legend: 🖥 Online, ✲ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-INFO-101967 - Programming Pass must have been passed.

T 6.580 Module component: Programming Pass [T-INFO-101967]

Coordinators: Prof. Dr.-Ing. Anne Koziolek
Prof. Dr. Ralf Reussner

Organisation: KIT Department of Informatics

Part of: M-WIWI-104909 - Informatics (Department of Informatics)

Type	Credits	Grading	Term offered	Version
Coursework	0 CP	pass/fail	Each term	1

Courses					
WT 25/26	2424004	Programming	4 SWS	Lecture / Practice	Koziolek
ST 2026	2400083	Programming Exercise	0 SWS	Practice / 🗣	Koziolek
WT 26/27	2424004	Programming	4 SWS	Lecture / Practice	Koziolek

Legend: 📱 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

T 6.581 Module component: Project Course Additive Manufacturing: Design Optimization and Production of Metallic Components [T-MACH-113575]

Coordinators: Prof. Dr.-Ing. Frederik Zanger
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-106590 - Production Engineering

Type	Credits	Grading	Term offered	Version
Examination of another type	4 CP	graded	Each summer term	3

Courses					
ST 2026	2150704	Project Course Additive Manufacturing: Design Optimization and Production of Metallic Components	3 SWS	Practical course / 🗣	Zanger, Frey

Legend: 📱 Online, Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Assessment of another type of examination (graded)

The competence certificate is a project work; alternative test achievement according to § 4 Abs. 2 No. 3 of the SPO. Here, the project work, the milestone-based presentation of the results in presentation form (10 min each) and a final oral examination (15 min) are included in the assessment.

Prerequisites
The following module component may not have started:

- T-MACH-114019 - Additive Fertigung metallischer Bauteile: Designoptimierung und Herstellung
- T-MACH-110983 - Projektpraktikum Additive Fertigung: Entwicklung und Fertigung eines additiven Bauteils
- T-MACH-114624 - Projektpraktikum Additive Fertigung: Designoptimierung und Herstellung metallischer Bauteile
- T-MACH-110960 - Projektpraktikum Additive Fertigung: Entwicklung und Fertigung eines additiven Bauteils

Additional Information
The course is offered in German.

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Project Course Additive Manufacturing: Design Optimization and Production of Metallic Components

Practical course (P)
On-Site

2150704, SS 2026, 3 SWS, Language: German, Open in study portal

Content

The course “Project Course Additive Manufacturing: Design Optimization and Production of Metallic Components” combines the fundamentals of powder bed fusion laser beam melting (PBF-LB/M) with a development project in collaboration with an industry-relevant application case.

Students learn the basics of the following topics:

- Influence of various process variables on the component quality of parts manufactured using the PBF-LB/M process
- Preparation and simulation of the PBF-LB/M process
- Manufacturing of additive metal components
- Process monitoring and quality assurance in additive manufacturing
- Topology optimization
- Computer-aided manufacturing (CAM) for machining rework

The topics are demonstrated in practice in various workshops on the individual topics and transferred to the development task in teamwork. Finally, the results of the work are additively manufactured and machined. The results are then presented, reflected upon, and discussed.

Attendance of the course “Additive Fertigung metallischer Bauteile” is recommended.

Learning Outcomes:

Students ...

- can describe the characteristics and areas of application of the additive manufacturing process powder bed fusion of metals (PBF-LB/M).
- can describe and implement the creation of a product along the entire additive process chain (CAD, simulation, construction job preparation, CAM) from the initial idea to production.
- are able to discuss the development process for components that are optimized for additive manufacturing.
- are able to perform topology optimization of a component for manufacturing using additive manufacturing.
- are able to simulate the additive process, compensate for process-related distortion, and determine the ideal alignment on the build platform.
- are able to create the necessary support structures for the additive process and derive a build job file.
- are able to create a CAM model for the machining post-processing of additive components and to machine the component.
- can select suitable powder fractions and process variables for the additive manufacturing of high-quality components based on existing test results.

Workload:

regular attendance: 14 hours
self-study: 106 hours

Organizational issues
Das Praktikum wird erstmals im Sommersemester 2026 angeboten.

Unregelmäßige Termine, siehe Zeitplan auf wbk-Homepage.
Irregular dates, see schedule on the wbk homepage.

Literature

Skript zur Veranstaltung wird über (https://ilias.studium.kit.edu/) bereitgestellt.

Lecture notes will be provided in Ilias (https://ilias.studium.kit.edu/).

T 6.582 Module component: Project in Applied Remote Sensing [T-BGU-101814]

Coordinators: Prof. Dr.-Ing. Stefan Hinz
Dr.-Ing. Uwe Weidner
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Version
Coursework	1 CP	pass/fail	1

Courses					
ST 2026	6020245	Project Exercise Applied Remote Sensing	2 SWS	Practice / 🗣	Hinz, Weidner, Wursthorn

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-BGU-101638 - Procedures of Remote Sensing, Prerequisite must have been passed.

Workload
30 hours

T 6.583 Module component: Project Lab Cognitive Automobiles and Robots [T-WIWI-109985]

Coordinators: Prof. Dr.-Ing. Johann Marius Zöllner
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101472 - Informatics
M-WIWI-101628 - Emphasis in Informatics
M-WIWI-103356 - Machine Learning
M-WIWI-104901 - Informatics (KIT-Department of Economics and Management)

Type	Credits	Grading	Term offered	Version
Examination of another type	5 CP	graded	Each winter term	3

Courses					
WT 25/26	2512501	Practical Course Cognitive automobiles and robots (Master)	3 SWS	Practical course / 🧩	Zöllner, Schneider

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The alternative exam assessment consists of:

- a practical work
- a presentation and
- a written seminar thesis

Details of the grade formation will be announced at the beginning of the course.

Prerequisites
None

Workload
150 hours

Below you will find excerpts from courses related to this module component:

V Practical Course Cognitive automobiles and robots (Master)

2512501, WS 25/26, 3 SWS, Language: German/English, Open in study portal

Practical course (P)
Blended (On-Site/Online)

Content
The lab is intended as a practical supplement to courses such as "Machine Learning 1/2".
Scientific topics, mostly in the area of autonomous driving and robotics, will be addressed in joint work with ML/KI methods. The goal of the internship is for participants to design, develop, and evaluate ML Software system.

In addition to the scientific goals, such as the study and application of methods, the aspects of project-specific teamwork in research (from specification to presentation of results) are also worked on in this internship.

The individual projects require the analysis of the set task, selection of appropriate methods, specification and implementation and evaluation of the solution approach. Finally, the selected solution is to be documented and presented in a short lecture.

Learning Objectives:

- Students will be able to practically apply theoretical knowledge from lectures on machine learning to a selected area of current research.
- Students will be proficient in analyzing and solving thematic problems.
- Students will be able to evaluate, document, and present their concepts and results.

Recommendations:

- Theoretical knowledge of machine learning and/or AI.
- Python knowledge
- Initial experience with deep learning frameworks such as PyTorch/Jax/Tensorflow may be beneficial.

Workload:

The workload of 5 credit points consists of practical implementation of the selected solution, as well as time for literature research and planning/specification of the selected solution. In addition, a short report and presentation of the work performed will be prepared.

Organizational issues

Anmeldung und weitere Informationen sind im Wiwi-Portal zu finden.

Registration and further information can be found in the WiWi-portal.

T 6.584 Module component: Project Lab Machine Learning [T-WIWI-109983]

Coordinators: Prof. Dr.-Ing. Johann Marius Zöllner
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101472 - Informatics
M-WIWI-101628 - Emphasis in Informatics
M-WIWI-103356 - Machine Learning
M-WIWI-104901 - Informatics (KIT-Department of Economics and Management)

Type	Credits	Grading	Term offered	Version
Examination of another type	5 CP	graded	Each summer term	3

Courses					
ST 2026	2512500	Project Lab Machine Learning	3 SWS	Practical course / 🗣	Zöllner, Schneider

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The alternative exam assessment consists of:

- a practical work
- a presentation and
- a written seminar thesis

Details of the grade formation will be announced at the beginning of the course.

Prerequisites
None

Workload
150 hours

Below you will find excerpts from courses related to this module component:

V Project Lab Machine Learning

2512500, SS 2026, 3 SWS, Language: German/English, Open in study portal

Practical course (P)
On-Site

Content
The lab is intended as a practical supplement to lectures such as "Machine Learning". The theoretical basics are applied in the lab course. The aim of the lab course is that the participants work together to design, develop and evaluate a subsystem from the field of robotics and cognitive systems using one or more procedures from the field of AI/ML.

In addition to the scientific objectives involved in the investigation and application of the methods, aspects of project-specific teamwork in research (from specification to presentation of the results) are also developed in this practical course.

The individual projects require the analysis of the task at hand, selection of suitable procedures, specification and implementation and evaluation of the approach taken. Finally, the chosen solution has to be documented and presented in a short presentation.

Learning objectives:

- Students can practically apply knowledge from the Machine Learning lecture in a selected field of current research in robotics or cognitive automobiles.
- Students master the analysis and solution of corresponding problems in a team.
- Students can evaluate, document and present their concepts and results.

Recommendations:

Attendance of the lecture machine learning, C/C++ knowledge, Python knowledge

Workload:

The workload of 5 credit points consists of the time spent in the lab for practical implementation of the selected solution, as well as the time spent on literature research and planning/specifying the proposed solution. In addition, a short report and a presentation of the work carried out will be prepared.

Organizational issues

Anmeldung und weitere Informationen sind im Wiwi-Portal zu finden.

Registration and further information can be found in the WiWi-portal.

T 6.585 Module component: Project Management [T-WIWI-103134]

Coordinators: Prof. Dr. Frank Schultmann
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101412 - Industrial Production III
M-WIWI-101471 - Industrial Production II
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	3,5 CP	graded	Each winter term	1

Courses					
WT 25/26	2581963	Project Management	2 SWS	Lecture / 🗣	Schultmann
WT 26/27	2581963	Project Management	2 SWS	Lecture / 🗣	Schultmann

Legend: 🖥 Online, Blended (On-Site/Online), 🗣 On-Site, × Cancelled

Assessment
The assessment consists of a written exam (60 minutes) (following §4(2) of the examination regulation). The exam takes place in every semester. Re-examinations are offered at every ordinary examination date.

Prerequisites
None

Recommendations
None

Below you will find excerpts from courses related to this module component:

V Project Management
2581963, WS 25/26, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content

1. Introduction
2. Principles of Project Management
3. Project Scope Management
4. Time Management and Resource Scheduling
5. Cost Management
6. Quality Management
7. Risk Management
8. Stakeholder
9. Communication, Negotiation and Leadership
10. Project Controlling
11. Agile Project Management

Literature
Wird in der Veranstaltung bekannt gegeben.

V Project Management
2581963, WS 26/27, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content

1. Introduction
2. Principles of Project Management
3. Project Scope Management
4. Time Management and Resource Scheduling
5. Cost Management
6. Quality Management
7. Risk Management
8. Stakeholder
9. Communication, Negotiation and Leadership
10. Project Controlling
11. Agile Project Management

Literature
Wird in der Veranstaltung bekannt gegeben.

T 6.586 Module component: Project Management [T-BGU-113720]

Coordinators: Prof. Dr.-Ing. Shervin Haghsheno
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Expansion	Version
Examination of another type	3 CP	graded	Each winter term	1 semesters	1

Courses					
WT 25/26	6200106	Project Management	2 SWS	Lecture / Practice / 🗣	Haghsheno, Schneider, Gloser

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, x Cancelled

Assessment
preparation of a term paper, appr. 5 pages, with presentation, appr. 10 min., and online test, 45 min.

Prerequisites
The Examination Prerequisite Project Management (T-BGU-113454) has to be passed.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-BGU-113454 - Examination Prerequisite Project Management must have been passed.

Recommendations
participation in the online mock test in the last lecture of the winter semester

Additional Information
None

Workload
90 hours

Below you will find excerpts from courses related to this module component:

V Project Management

6200106, WS 25/26, 2 SWS, Language: German, Open in study portal

Lecture / Practice (VÜ)
On-Site

Content
This course provides a comprehensive introduction to (construction) project management. It takes a closer look at the organisation and delivery of a construction project from the client's perspective. In this context, a range of competences are presented that should be on hand for the successful execution of project management. In addition, we present a selection of project management methods for individual competences and illustrate them with case studies.

Organizational issues
Vorlesungen: Mittwochs vom 29.10.2025 bis 18.02.2026, jeweils 09:45 – 11:15 Uhr (hybrid)

Übungen: Asynchron ab 19.11.2025, 10.12.2025, 14.01.2026, 11.02.2026 (online)

Literature

- AHRENS, Hannsjörg; BASTIAN, Klemens; MUCHOWSKI, Lucian (Hrsg.) (2021) *Handbuch Projektsteuerung - Baumanagement: Ein praxisorientierter Leitfaden mit zahlreichen Hilfsmitteln und Arbeitsunterlagen*, 6. Auflage, Fraunhofer IRB Verlag, Stuttgart
- GPM Deutsche Gesellschaft für Projektmanagement e. V. (Hrsg.) (2017) *Individual Competence Baseline für Projektmanagement (Version 4.0)*, 1. Auflage, GPM Deutsche Gesellschaft für Projektmanagement e. V., Nürnberg
- HAGHSHENO, Shervin; JOHN, Paul Christian (2024) *Bauherrnseitige Projektmanagement-Dienstleistungen in Deutschland*, Forschungsbericht, DVP – Deutscher Verband für Projektmanagement in der Bau- und Immobilienwirtschaft e. V.
- KOCHENDÖRFER, Bernd; LIEBCHEN, Jens H.; VIERING, Markus G. (2021) *Bau-Projekt-Management: Grundlagen und Vorgehensweisen*, 6. Auflage, Springer Vieweg, Wiesbaden
- SCHULZ, Markus (2020) *Projektmanagement: Zielgerichtet. Effizient. Klar.*, 2. Auflage, UVK Verlag, Tübingen

T 6.587 Module component: Project Management in Construction and Real Estate Industry I [T-BGU-103432]

Coordinators: Prof. Dr.-Ing. Shervin Haghsheno
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: M-BGU-101884 - Lean Management in Construction
M-BGU-101888 - Project Management in Construction
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Expansion	Version
Examination of another type	3 CP	graded	Each winter term	1 semesters	2

Courses					
WT 25/26	6241701	Construction Project Management	4 SWS	Lecture / Practice / 🗣	Haghsheno, Mitarbeiter/innen

Legend: 🖥 Online, ✪ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
report of case study in the first half of the semester, appr. 8 pages;
presentation of results and colloquium in the middle of the semester, appr. 10 min. each

Prerequisites
none

Recommendations
none

Additional Information
none

Workload
90 hours

Below you will find excerpts from courses related to this module component:

V Construction Project Management

6241701, WS 25/26, 4 SWS, Language: German, Open in study portal

Lecture / Practice (VÜ)
On-Site

Content
This course provides a comprehensive and in-depth introduction to construction project management. It takes a closer look at the organisation and delivery of a construction project from the client's perspective. Selected project management content is taught and applied in teamwork within the context of four practice-based case studies. At the end of the semester, the results are presented in a role play (acquisition meeting with a client).

Organizational issues
genaue Termine: siehe zugehöriger ILIAS-Kurs

Literature

- AHRENS, Hannsjörg; BASTIAN, Klemens; MUCHOWSKI, Lucian (Hrsg.) (2021): *Handbuch Projektsteuerung - Baumanagement: Ein praxisorientierter Leitfaden mit zahlreichen Hilfsmitteln und Arbeitsunterlagen*, 6. Auflage, Fraunhofer IRB Verlag, Stuttgart
- FEWINGS, Peter; HENJEWELE, Christian (2019): *Construction Project Management – An Integrated Approach*, 3. Auflage, Routledge, New York (USA)
- GPM Deutsche Gesellschaft für Projektmanagement e. V. (Hrsg.) (2017): *Individual Competence Baseline für Projektmanagement (Version 4.0)*, 1. Auflage, GPM Deutsche Gesellschaft für Projektmanagement e. V., Nürnberg
- HAGHSHENO, Shervin; JOHN, Paul Christian (2024): *Bauherrnseitige Projektmanagement-Dienstleistungen in Deutschland*, Forschungsbericht, DVP – Deutscher Verband für Projektmanagement in der Bau- und Immobilienwirtschaft e. V.
- HUEMANN, Martina; TURNER, J. Rodney (Hrsg.) (2024): *The Handbook of Project Management*, 6. Auflage, Routledge, New York (USA)
- KOCHENDÖRFER, Bernd; LIEBCHEN, Jens H.; VIERING, Markus G. (2021): *Bau-Projekt-Management: Grundlagen und Vorgehensweisen*, 5. Auflage, Springer Vieweg, Wiesbaden
- SCHULZ, Markus (2020): *Projektmanagement: Zielgerichtet. Effizient. Klar.*, 2. Auflage, UVK Verlag, Tübingen

T 6.588 Module component: Project Management in Construction and Real Estate Industry II [T-BGU-103433]

Coordinators: Prof. Dr.-Ing. Shervin Haghsheno
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: M-BGU-101884 - Lean Management in Construction
M-BGU-101888 - Project Management in Construction
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Expansion	Version
Examination of another type	3 CP	graded	Each winter term	1 semesters	2

Courses					
WT 25/26	6241701	Construction Project Management	4 SWS	Lecture / Practice / 🗣	Haghsheno, Mitarbeiter/innen

Legend: 🖥 Online, Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
report of case study in the seond half of the semester, appr. 8 pages;
presentation of results and colloquium in the end of the semester, appr. 10 min. each

Prerequisites
Project Management in Construction and Real Estate Industry I (T-BGU-103432) has to be passed.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-BGU-103432 - Project Management in Construction and Real Estate Industry I must have been passed.

Recommendations
none

Additional Information
none

Workload
90 hours

Below you will find excerpts from courses related to this module component:

V Construction Project Management

6241701, WS 25/26, 4 SWS, Language: German, Open in study portal

Lecture / Practice (VÜ)
On-Site

Content
This course provides a comprehensive and in-depth introduction to construction project management. It takes a closer look at the organisation and delivery of a construction project from the client's perspective. Selected project management content is taught and applied in teamwork within the context of four practice-based case studies. At the end of the semester, the results are presented in a role play (acquisition meeting with a client).

Organizational issues
genaue Termine: siehe zugehöriger ILIAS-Kurs

Literature

- AHRENS, Hannsjörg; BASTIAN, Klemens; MUCHOWSKI, Lucian (Hrsg.) (2021): *Handbuch Projektsteuerung - Baumanagement: Ein praxisorientierter Leitfaden mit zahlreichen Hilfsmitteln und Arbeitsunterlagen*, 6. Auflage, Fraunhofer IRB Verlag, Stuttgart
- FEWINGS, Peter; HENJEWELE, Christian (2019): *Construction Project Management – An Integrated Approach*, 3. Auflage, Routledge, New York (USA)
- GPM Deutsche Gesellschaft für Projektmanagement e. V. (Hrsg.) (2017): *Individual Competence Baseline für Projektmanagement (Version 4.0)*, 1. Auflage, GPM Deutsche Gesellschaft für Projektmanagement e. V., Nürnberg
- HAGHSHENO, Shervin; JOHN, Paul Christian (2024): *Bauherrnseitige Projektmanagement-Dienstleistungen in Deutschland*, Forschungsbericht, DVP – Deutscher Verband für Projektmanagement in der Bau- und Immobilienwirtschaft e. V.
- HUEMANN, Martina; TURNER, J. Rodney (Hrsg.) (2024): *The Handbook of Project Management*, 6. Auflage, Routledge, New York (USA)
- KOCHENDÖRFER, Bernd; LIEBCHEN, Jens H.; VIERING, Markus G. (2021): *Bau-Projekt-Management: Grundlagen und Vorgehensweisen*, 5. Auflage, Springer Vieweg, Wiesbaden
- SCHULZ, Markus (2020): *Projektmanagement: Zielgerichtet. Effizient. Klar.*, 2. Auflage, UVK Verlag, Tübingen

T 6.589 Module component: Project Paper Lean Construction [T-BGU-101007]

Coordinators: Prof. Dr.-Ing. Shervin Haghsheno
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: M-BGU-101884 - Lean Management in Construction
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Expansion	Version
Examination of another type	1,5 CP	graded	Each winter term	1 semesters	1

Courses					
WT 25/26	6241901	Lean Construction	4 SWS	Lecture / Practice / 🗣	Haghsheno, Mitarbeiter/innen

Legend: 🖥 Online, Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
project:

report, appr. 10 pages, and
presentation, appr. 10 min.

Prerequisites
none

Recommendations
none

Additional Information
The number of participants is limited to 50. Registration modalities will be published in time on the institute's homepage or in the corresponding ILIAS course. Students of Profile 2 'Project Management and Lean Construction' in the degree program 'Technology and Management in Construction Management' always receive a confirmation of participation (basic module). Any necessary selection for the remaining places will be made with priority given to students from Civil Engineering (study focus module), Technology and Management in Construction Management (specialization module) and Industrial Engineering (elective module in Engineering Sciences), taking into account the degree program and the student's progress.

Workload
40 hours

Below you will find excerpts from courses related to this module component:

V Lean Construction

6241901, WS 25/26, 4 SWS, Language: German, Open in study portal

Lecture / Practice (VÜ)
On-Site

Content
At the beginning of this module, the theoretical foundations of Lean Philosophy and Lean Construction are presented and deepened through learning simulations and exercises. Subsequently, industry and specialist lectures examine methods such as takt planning and takt control, the Last Planner System™, value stream analysis, and cooperative contract forms in both theory and practice. Furthermore, general project management aspects in construction projects such as construction site logistics, cost and quality management are addressed from a Lean perspective.

As part of the project work, students work in small groups on selected problem areas, present the results, and prepare a written report. This written report, together with the group presentation, is evaluated as a partial module assessment.

Organizational issues
Die Anmeldung zum Kurs findet über das in CAS hinterlegte Anmeldeverfahren statt.

Die Teilnehmerzahl ist auf 50 beschränkt.

Erste Vorlesung: 27.10.2025. Weitere Termine: siehe zugehöriger ILIAS-Kurs

Literature

Gehbauer, F. (2013) *Lean Management Im Bauwesen*. Skript des Instituts für Technologie und Management im Baubetrieb, Karlsruher Institut für Technologie (KIT).
Liker, J. & Meier, D. (2007) *Praxisbuch, der Toyota Weg: für jedes Unternehmen*. Finanzbuch Verlag.
Rother, M., Shook, J., & Wiegand, B. (2006). *Sehen lernen: mit Wertstromdesign die Wertschöpfung erhöhen und Verschwendung beseitigen.* Lean Management Institut.

T 6.590 Module component: Project Studies [T-BGU-101847]

Coordinators: Prof. Dr.-Ing. Sascha Gentes
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: M-BGU-101110 - Process Engineering in Construction
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Expansion	Version
Oral examination	3 CP	graded	Each summer term	1 semesters	1

Courses					
ST 2026	6243801	Project Studies	2 SWS	Lecture / Practice /	Gentes

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
oral exam, appr. 20 min.

Prerequisites
none

Recommendations
none

Additional Information
none

Workload
90 hours

T 6.591 Module component: Project Workshop: Automotive Engineering [T-MACH-102156]

Coordinators: Dr.-Ing. Michael Frey
Dr.-Ing. Martin Gießler

Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101264 - Handling Characteristics of Motor Vehicles
M-MACH-101265 - Vehicle Development
M-MACH-101266 - Automotive Engineering
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Oral examination	6 CP	graded	Each term	1

Courses					
WT 25/26	2115817	Project Workshop: Automotive Engineering	3 SWS	Lecture / 🗣	Gießler, Frey
ST 2026	2115817	Project Workshop: Automotive Engineering	3 SWS	Lecture / 🗣	Gießler, Frey
WT 26/27	2115817	Project Workshop: Automotive Engineering	3 SWS	Lecture / 🗣	Gießler, Frey

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Oral examination

Duration: duration 30 up to 40 minutes

Auxiliary means: none

Prerequisites
none

Workload
180 hours

Below you will find excerpts from courses related to this module component:

V Project Workshop: Automotive Engineering

Lecture (V)
On-Site

2115817, WS 25/26, 3 SWS, Language: German, Open in study portal

Content
During the Project Workshop Automotive Engineering a team of six persons will work on a task given by an German industrial partner using the instruments of project management. The task is relevant for the actual business and the results are intended to be industrialized after the completion of the project workshop.

The team will generate approaches in its own responsibility and will develop solutions for practical application. Coaching will be supplied by both, company and institute.

At the beginning in a start-up meeting goals and structure of the project will be specified. During the project workshop there will be weekly team meetings. Also a milestone meeting will be held together with persons from the industrial company. In a final presentation the project results will be presented to the company management and to institute representatives.

Learning Objectives:

During the Project Workshop Automotive Engineering a team of six persons will work on a task given by an German industrial partner using the instruments of project management. The task is relevant for the actual business and the results are intended to be industrialized after the completion of the project workshop.

The team will generate approaches in its own responsibility and will develop solutions for practical application. Coaching will be supplied by both, company and institute.

At the beginning in a start-up meeting goals and structure of the project will be specified. During the project workshop there will be weekly team meetings. Also a milestone meeting will be held together with persons from the industrial company. In a final presentation the project results will be presented to the company management and to institute representatives.

Organizational issues
Begrenzte Teilnehmerzahl mit Auswahlverfahren, in deutscher Sprache. Bewerbungen sind am Ende des vorhergehenden Semesters einzureichen.
Termin und Raum: siehe Institutshomepage.

Limited number of participants with selection procedure, in German language. Please send the application at the end of the previous semester

Date and room: see homepage of institute.

Literature
Steinle, Claus; Bruch, Heike; Lawa, Dieter (Hrsg.), Projektmanagement, Instrument moderner Innovation, FAZ Verlag, Frankfurt a. M., 2001, ISBN 978-3929368277

Skripte werden beim Start-up Meeting ausgegeben.

The scripts will be supplied in the start-up meeting.

V	**Project Workshop: Automotive Engineering** 2115817, SS 2026, 3 SWS, Language: German, Open in study portal	**Lecture (V)** **On-Site**

Content
During the Project Workshop Automotive Engineering a team of six persons will work on a task given by an German industrial partner using the instruments of project management. The task is relevant for the actual business and the results are intended to be industrialized after the completion of the project workshop.

The team will generate approaches in its own responsibility and will develop solutions for practical application. Coaching will be supplied by both, company and institute.

At the beginning in a start-up meeting goals and structure of the project will be specified. During the project workshop there will be weekly team meetings. Also a milestone meeting will be held together with persons from the industrial company. In a final presentation the project results will be presented to the company management and to institute representatives.

Learning Objectives:

The students are familiar with typical industrial development processes and working style. They are able to apply knowledge gained at the university to a practical task. They are able to analyze and to judge complex relations. They are ready to work self-dependently, to apply different development methods and to work on approaches to solve a problem, to develop practice-oriented products or processes.

Organizational issues
Begrenzte Teilnehmerzahl mit Auswahlverfahren, die Bewerbungen sind am Ende des vorhergehenden Semesters einzureichen.

Raum und Termine: s. Aushang bzw. Homepage

Literature
Steinle, Claus; Bruch, Heike; Lawa, Dieter (Hrsg.), Projektmanagement, Instrument moderner Innovation, FAZ Verlag, Frankfurt a. M., 2001, ISBN 978-3929368277

Skripte werden beim Start-up Meeting ausgegeben.

V	**Project Workshop: Automotive Engineering** 2115817, WS 26/27, 3 SWS, Language: German, Open in study portal	**Lecture (V)** **On-Site**

Content
During the Project Workshop Automotive Engineering a team of six persons will work on a task given by an German industrial partner using the instruments of project management. The task is relevant for the actual business and the results are intended to be industrialized after the completion of the project workshop.

The team will generate approaches in its own responsibility and will develop solutions for practical application. Coaching will be supplied by both, company and institute.

At the beginning in a start-up meeting goals and structure of the project will be specified. During the project workshop there will be weekly team meetings. Also a milestone meeting will be held together with persons from the industrial company. In a final presentation the project results will be presented to the company management and to institute representatives.

Learning Objectives:

During the Project Workshop Automotive Engineering a team of six persons will work on a task given by an German industrial partner using the instruments of project management. The task is relevant for the actual business and the results are intended to be industrialized after the completion of the project workshop.

The team will generate approaches in its own responsibility and will develop solutions for practical application. Coaching will be supplied by both, company and institute.

At the beginning in a start-up meeting goals and structure of the project will be specified. During the project workshop there will be weekly team meetings. Also a milestone meeting will be held together with persons from the industrial company. In a final presentation the project results will be presented to the company management and to institute representatives.

Organizational issues
Begrenzte Teilnehmerzahl mit Auswahlverfahren, in deutscher Sprache. Bewerbungen sind am Ende des vorhergehenden Semesters einzureichen.
Termin und Raum: siehe Institutshomepage.

Limited number of participants with selection procedure, in German language. Please send the application at the end of the previous semester

Date and room: see homepage of institute.

Literature
Steinle, Claus; Bruch, Heike; Lawa, Dieter (Hrsg.), Projektmanagement, Instrument moderner Innovation, FAZ Verlag, Frankfurt a. M., 2001, ISBN 978-3929368277

Skripte werden beim Start-up Meeting ausgegeben.

The scripts will be supplied in the start-up meeting.

T 6.592 Module component: Project: Data-Driven Engineering [T-MACH-115045]

Coordinators: Prof. Dr.-Ing. Anne Meyer
Organisation: KIT Department of Mechanical Engineering
Part of: M-MACH-107716 - Data Driven Engineering
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Expansion	Version
Examination of another type	4,5 CP	graded	Each term	1 semesters	2

Courses					
ST 2026	2123330	Project: Data-Driven Engineering	4 SWS	Project / 🗣	Meyer

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Graded examination of another type - See further details on course organization at: lehre.imi.kit.edu and in ILIAS.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Project: Data-Driven Engineering

2123330, SS 2026, 4 SWS, Language: English, Open in study portal

Project (PRO)
On-Site

Content
Data-driven methods are reshaping advanced engineering practice and research across domains such as intelligent product design, production systems, robotics, and other technical fields, requiring the ability to integrate machine learning, optimization, and simulation in complex systems.

In this project, students work independently in small teams on a complex, data-driven engineering problem. The focus is on identifying and formulating a challenging research- or application-oriented task, selecting and combining advanced methods from machine learning, optimization, and simulation, and critically evaluating the results. Students are expected to incorporate and build upon the research results in their methods and solutions.

Using modern software frameworks and computational resources (e.g., Isaac Sim, PyTorch, scikit-learn, distributed or cluster computing environments), students design, implement, and evaluate sophisticated data-driven engineering solutions for one of the above-mentioned domains. Results are documented in an appropriate format (e.g., a written report or comparable documentation) and presented through oral presentations, complemented by short project videos or pitch formats.

The course is conducted primarily on-site to support collaboration, supervision, and access to technical resources. Good Python programming skills are expected. Students should be prepared to independently work with and extend their knowledge of operating systems and tools such as Linux and Docker.

Learning Objectives

After successful completion of the project, students will be able to:

- **Identify, formulate, and analyze** complex engineering problems as data-driven tasks, and justify the selection of methods and tools.
- **Design, implement, and optimize** data-driven solutions using advanced machine learning, optimization, and simulation techniques.
- **Critically evaluate and defend** findings and methodological choices effectively, including reports, presentations, and multimedia formats.

Additional Information

The number of participants is limited. Information on the application process is available in ILIAS.

Organizational Information

See further details on course organization at: lehre.imi.kit.edu and in ILIAS

Workload

120 hours

Organizational issues
Time and Location see ILIAS

T 6.593 Module component: Project: Data-Driven Engineering Fundamentals [T-MACH-115010]

Coordinators: Prof. Dr.-Ing. Anne Meyer
Organisation: KIT Department of Mechanical Engineering
Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Expansion	Version
Examination of another type	4 CP	graded	Each term	1 semesters	1

Courses					
ST 2026	2122354	Project: Data-Driven Engineering Fundamentals	4 SWS	Project / 🗣	Meyer

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Graded examination of aother type. The following aspects are included in the grade:

- Documentation of the project implementation
- Achievement of the project goals
- Intermediate presentations and final presentation including live demonstration

The overall impression is assessed. The grading scale will be announced in the course.

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Project: Data-Driven Engineering Fundamentals

2122354, SS 2026, 4 SWS, Language: English, Open in study portal

Project (PRO)
On-Site

Content
Modern engineering increasingly relies on data-driven solutions across domains such as product design, production systems, logistics, and robotics. To address these developments, this project-based course enables students to work in small teams on a data-driven engineering problem in one of the domains.

The focus is on collaboratively structuring and solving an open-ended task, organizing project work, and selecting or combining suitable methods from machine learning, optimization, and simulation. Students use state-of-the-art tools and software libraries for data-driven engineering applications (depending on the task, e.g., Isaac Sim, PyTorch, ROS2, Gymnasium) to develop, implement, and evaluate a prototypical solution. Students document their work and communicate their results through presentations and suitable media formats.

The course is designed as an on-site project with regular in-person collaboration and supervision. Familiarity with Python programming is strongly recommended. Students should be open to learning and working with operating systems and tools such as Linux, Docker, and Git.

Learning Objectives

After successful completion of the course, students will be able to:

- **Analyze and formulate** an open-ended engineering problem as a data-driven task and select appropriate methods from machine learning, optimization, and simulation.
- **Design, implement, and manage** a team-based project workflow, including task allocation, milestone planning, and effective team communication, using modern tools and software libraries for data-driven engineering applications.
- **Evaluate and communicate** project results effectively to different audiences using appropriate formats, including technical presentations, pitch-style formats, and short project videos.

Additional Information

The number of participants is limited. Information on the application process is available in ILIAS.

Organizational Information

See further details on course organization at: lehre.imi.kit.edu and in ILIAS.

Workload

120 hours

Organizational issues
See ILIAS for time and location / Zeit und Ort siehe ILIAS

T 6.594 Module component: Public International Law [T-INFO-113381]

Coordinators: TT-Prof. Dr. Frederike Zufall
Organisation: KIT Department of Informatics
Part of: M-INFO-106754 - Public Economic and Technology Law

Type	Credits	Grading	Term offered	Version
Written examination	3 CP	graded	Each summer term	2

Courses					
ST 2026	2400172	Public International Law with an Economic Law Focus	2 SWS	Lecture / 🗣	Kasper-Schöniger

Legend: 🖥 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment is carried out as a written examination (§ 4 Abs. 2 No. 1 SPO) lasting 60 minutes.

Depending on the number of participants, it will be announced six weeks before the examination (§ 6 (3) SPO) whether the performance assessment is carried out
- as an oral examination (duration approx. 20 mins.) (§ 4 Abs. 2 Nr. 2 SPO) or
- as a written examination (lasting 60 mins.) (§ 4 Abs. 2 No. 1 SPO).

Prerequisites
None.

Recommendations
- General knowledge of (public) law (eg, through participating in public law or EU law modules) is helpful but not necessary.
- Interest in international affairs and politics is welcomed.

Additional Information
Competency Goals:
- Participating students will be able to navigate the plethora of multilateral treaties to detect relevant international law for specific cases.
- They can develop solutions for legal problems based on case law of international courts and tribunals.
- Students will be able to read and comprehend international treaties and case law.
- They will have a fundamental understand of the interplay between various subfields of public international law.
- Students can identify and explain current issues in public international law.

Content:
The lecture is designed to provide participating students with a general understanding of the foundations, subjects, and sources of public international law, its interplay with national legal regimes, and more detailed knowledge of particular subfields of public international law.

Since the lecture targets students of information systems, particular focus will be given to economic topics in international law, such as investment and trade law aspects. Due to the general importance of climate change for todays (economic) law, international climate change law and environmental law will form further focus areas.

In addition, a concise overview on human rights law, the law on State responsibility, and the peaceful settlement of disputes will be provided.

Throughout the lecture, important case law will be referenced and students are expected to read relevant cases in part to facilitate a discussion of such cases and their relevance for a subject field. Although the United Nations, including its principal judicial organ, the International Court of Justice, is one of the, if not the, key international organization in public international law, further international organizations (eg, Council of Europe, World Trade Organization) and their respective law(s) will also be touched.

Students are advised to have a statute book at hand that includes the most important international treaties and conventions (eg, Evans, Blackstone's International Law Documents, currently 15th ed 2021).

Conducting the lecture in English intends to facilitate students to link their ideas and arguments to current debates in international law.

Below you will find excerpts from courses related to this module component:

V Public International Law with an Economic Law Focus
2400172, SS 2026, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content

The lecture is designed to provide participating students with a general understanding of the foundations, subjects, and sources of public international law with an economic law focus, its interplay with national legal regimes, and more detailed knowledge of particular subfields of public international law.

Since the lecture targets students of information systems, particular focus will be given to economic topics in international law, such as investment and trade law aspects. Due to the general importance of climate change for todays (economic) law, international climate change law and environmental law will form further focus areas.

In addition, a concise overview on human rights law, the law on State responsibility, and the peaceful settlement of disputes will be provided.

Throughout the lecture, important case law will be referenced and students are expected to read relevant cases in part to facilitate a discussion of such cases and their relevance for a subject field. Although the United Nations, including its principal judicial organ, the International Court of Justice, is one of the, if not the, key international organization in public international law, further international organizations (eg, Council of Europe, World Trade Organization) and their respective law(s) will also be touched.

Students are advised to have a statute book at hand that includes the most important international treaties and conventions (eg, Evans, Blackstone's International Law Documents, currently 15th ed 2021).

Conducting the lecture in English intends to facilitate students to link their ideas and arguments to current debates in international law.

Organizational issues

Lecture Dates:

- Friday, 26th June 2026, 9 - 17 h
- Saturday, 27th June 2026, 9-17 h
- Saturday, 25th July 2026, 9-17 h

The room will be announced in good time in the Campus System (as per 5th January 2026).

Lageplan:

https://www.kit.edu/campusplan/

T 6.595 Module component: Public Law I & II [T-INFO-110300]

Coordinators: TT-Prof. Dr. Frederike Zufall
Organisation: KIT Department of Informatics
Part of: M-WIWI-104903 - Law

Type	Credits	Grading	Term offered	Version
Written examination	6 CP	graded	Each summer term	1

Courses					
WT 25/26	2424016	Öffentliches Recht I - Grundlagen	2 SWS	Lecture / 🗣	Zufall
ST 2026	24520	Öffentliches Recht II - Öffentliches Wirtschaftsrecht	2 SWS	Lecture / 🗣	Zufall

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

T 6.596 Module component: Public Management [T-WIWI-102740]

Coordinators: Prof. Dr. Berthold Wigger
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101504 - Collective Decision Making
M-WIWI-101511 - Advanced Topics in Public Finance
M-WIWI-104908 - Economics

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each winter term	1

Courses					
WT 25/26	2561127	Public Management	3 SWS	Lecture / Practice /	Wigger

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
Success is assessed in the form of a written examination (90 minutes). The examination is offered every semester and can be repeated at any regular examination date.

Prerequisites
None

Recommendations
Basic knowledge of Public Finance is required.

Below you will find excerpts from courses related to this module component:

V Public Management

2561127, WS 25/26, 3 SWS, Language: German, Open in study portal

Lecture / Practice (VÜ)
Blended (On-Site/Online)

Literature
Weiterführende Literatur:

- Damkowski, W. und C. Precht (1995): Public Management; Kohlhammer
- Richter, R. und E.G. Furubotn (2003): Neue Institutionenökonomik; 3. Auflage, Mohr
- Schedler, K. und I. Proeller (2003): New Public Management; 2. Auflage; UTB
- Mueller, D.C. (2009): Public Choice III; Cambridge University Press
- Wigger, B.U. (2006): Grundzüge der Finanzwissenschaft; 2. Auflage; Springer

T 6.597 Module component: Public Media Law [T-INFO-101311]

Organisation: KIT Department of Informatics
Part of: M-WIWI-104903 - Law

Type	Credits	Grading	Term offered	Version
Written examination	3 CP	graded	Each winter term	1

T 6.598 Module component: Public Revenues [T-WIWI-102739]

Coordinators: Prof. Dr. Berthold Wigger
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101511 - Advanced Topics in Public Finance
M-WIWI-104908 - Economics

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each summer term	1

Courses					
ST 2026	2560120	Public Revenues	2 SWS	Lecture / 🗣	Wigger
ST 2026	2560121	Übung zu Öffentliche Einnahmen	1 SWS	Practice / 🗣	Wigger, Schmelzer

Legend: 🖥 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success is assessed in the form of a written examination (60 minutes). The examination is offered every semester and can be repeated at any regular examination date.

Prerequisites
None

Recommendations
Basic knowledge of Public Finance is required.

Below you will find excerpts from courses related to this module component:

V Public Revenues
2560120, SS 2026, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content
The *Public Revenues* lecture is concerned with the theory and policy of taxation and public dept. In the first chapter, fundamental concepts of taxation theory are introduced, whereas the second chapter deals with key elements of the German taxation system. The allocative and distributive effects of different taxation types are examined in chapter three and four. Chapter five integrates both allocative and distributive components in order to derive a theory of optimal taxation. The core of the sixth chapter is represented by international aspects of taxation. The debt part begins with a description of the extent and structure of public dept in chapter seven. In the following chapter, macroeconomic theories of national dept are evolved, while chapter nine is concerned with its long term consequences when employed as a regular instrument of budgeting. Finally, the tenth chapter deals with constitutional limits to public debt-incurring.

Learning goals:

See German version.

Workload:

The total workload for this course is approximately 135.0 hours. For further information see German version.

Literature
Literatur:

- Homburg, S.(2000): *Allgemeine Steuerlehre,* Vahlen
- Rosen, H.S.(1995): *Public Finance*; 4. Aufl., Irwin
- Wellisch, D.(2000): *Finanzwissenschaft I* und *Finanzwissenschaft III*, Vahlen
- Wigger, B. U.(2006): *Grundzüge der Finanzwissenschaft*; 2. Aufl., Springer

T 6.599 Module component: Python Algorithms for Vehicle Technology [T-MACH-110796]

Coordinators: Stephan Rhode
Organisation:
Part of: M-MACH-101265 - Vehicle Development
M-MACH-101266 - Automotive Engineering
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Written examination	4 CP	graded	Each summer term	1

Courses					
ST 2026	2114862	Python Algorithms for Automotive Engineering	2 SWS	Lecture / 🔀	Rhode

Legend: 🖥 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Written Examination

Duration: 90 minutes

Prerequisites
none

Additional Information
The course is offered in German.

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Python Algorithms for Automotive Engineering

2114862, SS 2026, 2 SWS, Language: German, Open in study portal

Lecture (V)
Blended (On-Site/Online)

Content
Teaching content:

- Introduction to Python and useful tools and libraries for creating algorithms, graphical representation, optimization, symbolic arithmetic and machine learning
  - Anaconda, Pycharm, Jupyter
  - NumPy, Matplotlib, SymPy, Scikit-Learn
- Methods and tools for creating software
  - Version management GitHub, git
  - Testing software pytest, Pylint
  - Documentation Sphinx
  - Continuous Integration (CI) Travis CI
  - Workflows in Open Source and Inner Source, Kanban, Scrum
- Practical programming projects to:
  - Road sign recognition
  - Vehicle state estimation
  - Calibration of vehicle models by mathematical optimization
  - Data-based modelling of the powertrain of an electric vehicle

Objectives:

The students have an overview of the programming language Python and important Python libraries to solve automotive engineering problems with computer programs. The students know current tools around Python to create algorithms, to apply them and to interpret and visualize their results. Furthermore, the students know
basics in the creation of software to be used in later programming projects in order to develop high-quality software solutions in teamwork. Through practical programming projects (road sign recognition, vehicle state estimation, calibration, data-based modelling), the students can perform future complex tasks from the area of driver assistance systems.

Organizational issues
Die Vorlesung beginnt mit zwei Kick-Off Veranstaltung in Präsenz am Campus Ost, Geb.70.04, Raum 219. Die entsprechenden Termine finden Sie auf der Homepage des Instituts. Die restlichen Termine finden überwiegend digital statt. Weitere Infos über ILIAS.

Literature

- A Whirlwind Tour of Python, Jake VanderPlas, Publisher: O'Reilly Media, Inc. Release Date: August 2016, ISBN: 9781492037859 link
- Scientific Computing with Python 3, Olivier Verdier, Jan Erik Solem, Claus Führer, Publisher: Packt Publishing, Release Date: December 2016, ISBN: 9781786463517 link
- Introduction to Machine Learning with Python, Sarah Guido, Andreas C. Müller, Publisher: O'Reilly Media, Inc., Release Date: October 2016, ISBN: 9781449369880, link
- Clean Code, Robert C. Martin, Publisher: Prentice Hall, Release Date: August 2008, ISBN: 9780136083238, link

T 6.600 Module component: Python for Computational Risk and Asset Management [T-WIWI-110213]

Coordinators: Prof. Dr. Maxim Ulrich
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-105032 - Data Science for Finance

Type	Credits	Grading	Term offered	Version
Examination of another type	4,5 CP	graded	Each winter term	3

Assessment
The examination takes the form of an alternative exam assessment.
The alternative exam assessment consists of a Python-based "Takehome Exam". At the end of the third week of January, the student is given a "Takehome Exam" which he processes and sends back independently within 4 hours using Python. Precise instructions will be announced at the beginning of the course. The alternative exam assessment can be repeated a maximum of once. A timely repeat option takes place at the end of the third week in March of the same year. More detailed instructions will be given at the beginning of the course.

Prerequisites
None.

Recommendations
Good knowledge of statistics and basic programming skills

Workload
135 hours

T 6.601 Module component: Python-Basics Course [T-MACH-115104]

Coordinators: Prof. Dr.-Ing. Anne Meyer
Organisation: KIT Department of Mechanical Engineering
Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Expansion	Version
Coursework	1 CP	pass/fail	Each winter term	1 semesters	1

Courses					
ST 2026	2122379	Python-Basics Course	1 SWS	Block / 🖥	Meyer

Legend: 🖥 Online, Blended (On-Site/Online), On-Site, x Cancelled

Assessment
Successful participation in a colloquium in individual performance at the end of the Python course.

Additional Information
Self-study course with on-site assessment

Workload
30 hours

Below you will find excerpts from courses related to this module component:

V Python-Basics Course

2122379, SS 2026, 1 SWS, Language: English, Open in study portal

Block (B)
Online

Content
Practice-oriented self-study course on Python programming, Python syntax, programming environments, Python libraries for data science in engineering

Organizational issues
Self-paced course with on-site assessment

T 6.602 Module component: Quality Management [T-MACH-102107]

Coordinators: Prof. Dr.-Ing. Gisela Lanza
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-105455 - Strategic Design of Modern Production Systems
M-MACH-106590 - Production Engineering
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Written examination	4 CP	graded	Each winter term	4

Courses					
WT 25/26	2149667	Quality Management	2 SWS	Lecture /	Lanza, Benfer
WT 26/27	2149667	Quality Management	2 SWS	Lecture /	Lanza, Benfer

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
Written Exam (60 min)

Prerequisites
It is not possible to combine this module component with module component Quality Management [T-MACH-112586].

Additional Information
The course is offered in German.

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Quality Management
2149667, WS 25/26, 2 SWS, Language: German, Open in study portal

Lecture (V)
Blended (On-Site/Online)

Content

Based on the quality philosophies Total Quality Management (TQM) and Six Sigma, the lecture deals with the requirements of modern quality management. Within this context, the process concept of a modern enterprise and the process-specific fields of application of quality assurance methods are presented. The lecture covers the current state of the art in preventive and non-preventive quality management methods in addition to manufacturing metrology, statistical methods and service related quality management. The content is completed with the presentation of certification possibilities and legal quality aspects.

Main topics of the lecture:

- The term "Quality"
- Total Quality Management (TQM) and Six Sigma
- Universal methods and tools
- QM during early product stages – product denition
- QM during product development and in procurement
- QM in production – manufacturing metrology
- QM in production – statistical methods
- QM in service
- Quality management systems
- Legal aspects of QM

Learning Outcomes:

The students …

- are capable to comment on the content covered by the lecture.
- are capable of substantially quality philosophies.
- are able to apply the QM tools and methods they have learned about in the lecture to new problems from the context of the lecture.
- are able to analyze and evaluate the suitability of the methods, procedures and techniques they have learned about in the lecture for a specific problem.

Workload:

regular attendance: 21 hours
self-study: 99 hours

Organizational issues

Vorlesungstermine montags 09:45 Uhr
Übung erfolgt während der Vorlesung

Literature

Medien:
Skript zur Veranstaltung wird über (https://ilias.studium.kit.edu/) bereitgestellt:

Media:
Lecture notes will be provided in Ilias (https://ilias.studium.kit.edu/).

V	**Quality Management** 2149667, WS 26/27, 2 SWS, Language: English, Open in study portal	**Lecture (V)** **Blended (On-Site/Online)**

Content
Based on the quality philosophies Total Quality Management (TQM) and Six Sigma, the lecture deals with the requirements of modern quality management. Within this context, the process concept of a modern enterprise and the process-specific fields of application of quality assurance methods are presented. The lecture covers the current state of the art in preventive and non-preventive quality management methods in addition to manufacturing metrology, statistical methods and service related quality management. The content is completed with the presentation of certification possibilities and legal quality aspects.

Main topics of the lecture:

- The term "Quality"
- Total Quality Management (TQM) and Six Sigma
- Universal methods and tools
- QM during early product stages – product denition
- QM during product development and in procurement
- QM in production – manufacturing metrology
- QM in production – statistical methods
- QM in service
- Quality management systems
- Legal aspects of QM

Learning Outcomes:

The students …

- are capable to comment on the content covered by the lecture.
- are capable of substantially quality philosophies.
- are able to apply the QM tools and methods they have learned about in the lecture to new problems from the context of the lecture.
- are able to analyze and evaluate the suitability of the methods, procedures and techniques they have learned about in the lecture for a specific problem.

Workload:

regular attendance: 21 hours
self-study: 99 hours

Organizational issues
Vorlesungstermine montags 09:45 Uhr
Übung erfolgt während der Vorlesung

Literature
Medien:
Skript zur Veranstaltung wird über (https://ilias.studium.kit.edu/) bereitgestellt:

Media:
Lecture notes will be provided in Ilias (https://ilias.studium.kit.edu/).

T 6.603 Module component: Quality Management [T-MACH-115056]

Coordinators: Prof. Dr.-Ing. Gisela Lanza
Organisation: KIT Department of Mechanical Engineering

Part of: M-WIWI-107718 - Mechanical Engineering I
M-WIWI-107730 - Mechanical Engineering II

Type	Credits	Grading	Term offered	Version
Written examination	4 CP	graded	Each winter term	1

Courses					
WT 26/27	2149667	Quality Management	2 SWS	Lecture /	Lanza, Benfer

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
Written Exam (60 min)

Prerequisites
It is not possible to combine this module component with module component Quality Management [T-MACH-112586].

Additional Information
The course is offered in English.

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Quality Management

2149667, WS 26/27, 2 SWS, Language: English, Open in study portal

Lecture (V)
Blended (On-Site/Online)

Content
Based on the quality philosophies Total Quality Management (TQM) and Six Sigma, the lecture deals with the requirements of modern quality management. Within this context, the process concept of a modern enterprise and the process-specific fields of application of quality assurance methods are presented. The lecture covers the current state of the art in preventive and non-preventive quality management methods in addition to manufacturing metrology, statistical methods and service related quality management. The content is completed with the presentation of certification possibilities and legal quality aspects.

Main topics of the lecture:

- The term "Quality"
- Total Quality Management (TQM) and Six Sigma
- Universal methods and tools
- QM during early product stages – product denition
- QM during product development and in procurement
- QM in production – manufacturing metrology
- QM in production – statistical methods
- QM in service
- Quality management systems
- Legal aspects of QM

Learning Outcomes:

The students …

- are capable to comment on the content covered by the lecture.
- are capable of substantially quality philosophies.
- are able to apply the QM tools and methods they have learned about in the lecture to new problems from the context of the lecture.
- are able to analyze and evaluate the suitability of the methods, procedures and techniques they have learned about in the lecture for a specific problem.

Workload:

regular attendance: 21 hours
self-study: 99 hours

Organizational issues
Vorlesungstermine montags 09:45 Uhr
Übung erfolgt während der Vorlesung

Literature
Medien:
Skript zur Veranstaltung wird über (https://ilias.studium.kit.edu/) bereitgestellt:

Media:
Lecture notes will be provided in Ilias (https://ilias.studium.kit.edu/).

T 6.604 Module component: Quantitative Finance [T-WIWI-114340]

Coordinators: Prof. Dr. Julian Thimme
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration
M-WIWI-107187 - Quantitative Core

Type	Credits	Grading	Term offered	Version
Examination of another type	4,5 CP	graded	Each winter term	1

Courses					
WT 25/26	2500065	Quantitative Finance	2 SWS	Lecture / 🗣	Thimme, Meng
WT 26/27	2500065	Quantitative Finance	2 SWS	Lecture / 🗣	Thimme, Meng

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment of success in this course is carried out as a non exam assessment. The grade is made up as follows:

1. Written elaboration (weighting: 33 %).
2. Working on and presenting data analysis problems in small groups (weighting: 67 %).

Prerequisites
None

Recommendations
Knowledge from the course Business Administration: Finance and Accounting [2610026] is very helpful.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Quantitative Finance

2500065, WS 25/26, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content
In Quantitative Finance, participants gain in-depth knowledge of analyzing corporate and financial market data to methodically explore key questions in finance. The course begins with a brief introduction to the Python programming language and the development environment used throughout the course. Following this, seven thematically structured modules cover both fundamental issues in finance and the associated empirical analytical methods. In three hackathons, participants work in small teams with real datasets to apply the theoretical concepts to concrete problems and subsequently present their findings.

Organizational issues
findet in der ersten Semesterhälfte statt

V Quantitative Finance

2500065, WS 26/27, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content
In Quantitative Finance, participants gain in-depth knowledge of analyzing corporate and financial market data to methodically explore key questions in finance. The course begins with a brief introduction to the Python programming language and the development environment used throughout the course. Following this, seven thematically structured modules cover both fundamental issues in finance and the associated empirical analytical methods. In three hackathons, participants work in small teams with real datasets to apply the theoretical concepts to concrete problems and subsequently present their findings.

Organizational issues
findet in der ersten Semesterhälfte statt

T 6.605 Module component: Quantitative Methods in Energy Economics [T-WIWI-107446]

Coordinators: Patrick Plötz
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101451 - Energy Economics and Energy Markets
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Oral examination	3,5 CP	graded	Each winter term	4

Courses					
WT 25/26	2581007	Quantitative Methods in Energy Economics	2 SWS	Lecture / 🗣	Plötz
WT 25/26	2581008	Übungen zu Quantitative Methods in Energy Economics	1 SWS	Practice / 🗣	Plötz, Britto
WT 26/27	2581007	Quantitative Methods in Energy Economics	2 SWS	Lecture / 🗣	Plötz
WT 26/27	2581008	Übungen zu Quantitative Methods in Energy Economics	1 SWS	Practice / 🗣	Plötz

Legend: 🖥 Online, ❂ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment consists of an oral (app. 30 minutes) exam (following §4(2) of the examination regulation). The exam takes place in every semester. Re-examinations are offered at every ordinary examination date.

Prerequisites
None

Recommendations
None

Below you will find excerpts from courses related to this module component:

V Quantitative Methods in Energy Economics

Lecture (V)
On-Site

2581007, WS 25/26, 2 SWS, Language: English, Open in study portal

Content
Energy economics makes use of many quantitative methods in exploration and analysis of data as well as in simulations and modelling. This lecture course aims at introducing students of energy economics into the application of quantitative methods and techniques as taught in elementary courses to real problems in energy economics. The focus is mainly on regression, simulation, time series analysis and related statistical methods as applied in energy economics.

Learning Goals:

The student

- knows and understands selected quantitative methods of energy economics
- is able to use selected quantitative methods of energy economics
- understands they range of usage, limits and is autonomously able to adress new problems by them.

Literature
Wird in der Vorlesung bekannt gegeben.

V Quantitative Methods in Energy Economics

Lecture (V)
On-Site

2581007, WS 26/27, 2 SWS, Language: English, Open in study portal

Content

Energy economics makes use of many quantitative methods in exploration and analysis of data as well as in simulations and modelling. This lecture course aims at introducing students of energy economics into the application of quantitative methods and techniques as taught in elementary courses to real problems in energy economics. The focus is mainly on regression, simulation, time series analysis and related statistical methods as applied in energy economics.

Learning Goals:

The student

- knows and understands selected quantitative methods of energy economics
- is able to use selected quantitative methods of energy economics
- understands they range of usage, limits and is autonomously able to adress new problems by them.

Literature

Wird in der Vorlesung bekannt gegeben.

T 6.606 Module component: Quantum Functional Devices and Semiconductor Technology [T-ETIT-100740]

Coordinators: Prof. Dr.-Ing. Christian Koos
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-MACH-101294 - Nanotechnology
M-MACH-101295 - Optoelectronics and Optical Communication
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Oral examination	3 CP	graded	Each summer term	1

Prerequisites
none

T 6.607 Module component: Radar Systems Engineering [T-ETIT-100729]

Coordinators: Prof. Dr.-Ing. Marwan Younis
Prof. Dr.-Ing. Thomas Zwick

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: M-ETIT-100420 - Radar Systems Engineering

Type	Credits	Grading	Term offered	Version
Written examination	6 CP	graded	Each winter term	2

Courses					
WT 25/26	2308454	Radar Systems Engineering	3 SWS	Lecture / 🗣	Zwick, Younis
WT 25/26	2308455	Exercise to 2308454 Radar Systems Engineering	1 SWS	Practice / 🧩	Zwick, Younis

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, x Cancelled

Prerequisites
none

T 6.608 Module component: Radio Frequency Integrated Circuits and Systems [T-ETIT-110358]

Coordinators: Prof. Dr.-Ing. Ahmet Cagri Ulusoy
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-105123 - Radio Frequency Integrated Circuits and Systems

Type	Credits	Grading	Term offered	Version
Oral examination	6 CP	graded	Each summer term	2

Courses					
ST 2026	2308419	Radio Frequency Integrated Circuits and Systems	2 SWS	Lecture / 🗣	Ulusoy
ST 2026	2308421	Workshop for 2308421 Radio Frequency Integrated Circuits and Systems	2 SWS	Practice / 🗣	Ulusoy, Tsai

Legend: 🖥 Online, ❖ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The success criteria will be determined by an oral examination (approx. 20-30 min.)

Recommendations
The lecture materials to „Grundlagen der Hochfrequenztechnik“ and „Halbleiterbauelemente“ are recommended.

Below you will find excerpts from courses related to this module component:

V Radio Frequency Integrated Circuits and Systems
2308419, SS 2026, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content

- Understanding of RF-IC technologies
- Analysis of active and passive components in IC-design
- Overview of system level approach to RF-ICs and transceiver topologies
- Theory and design of basic building blocks of an integrated transceiver including low-noise amplifiers (LNAs), mixers, voltage-controlled oscillators (VCOs) and power amplifiers (PAs)

T 6.609 Module component: Radio-Frequency Electronics [T-ETIT-113910]

Coordinators: Prof. Dr.-Ing. Ahmet Cagri Ulusoy
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-106955 - Radio-Frequency Electronics

Type	Credits	Grading	Term offered	Version
Written examination	6 CP	graded	Each winter term	1

Courses					
WT 25/26	2308503	Radio-Frequency Electronics	3 SWS	Lecture / 🗣	Ulusoy, Pauli
WT 25/26	2308504	Exercise for 2308503 Radio-Frequency Electronics	1 SWS	Practice / 🧩	Kuo

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The examination takes place in form of a written examination lasting 120 minutes.

The module grade is the grade of the written examination.

Prerequisites
none

T 6.610 Module component: Rail System Technology [T-MACH-106424]

Coordinators: Prof. Dr.-Ing. Martin Cichon
Organisation: KIT Department of Mechanical Engineering

Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Written examination	4 CP	graded	Each term	5

Courses					
WT 25/26	2115919	Rail System Technology	2 SWS	Lecture / 🗣	Cichon
ST 2026	2115919	Rail System Technology	2 SWS	Lecture / 🗣	Cichon
WT 26/27	2115919	Rail System Technology	2 SWS	Lecture / 🗣	Cichon

Legend: 📱 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
writen examination in German language
Duration: 60 minutes
No tools or reference materials may be used during the exam except calculator and dictionary

Prerequisites
none

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The following conditions have to be fulfilled:

Additional Information
The course is offered in German.

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Rail System Technology

2115919, WS 25/26, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content

1. Railway System: railway as system, subsystems and interdependencies, definitions, laws, rules, railway and environment, economic impact
2. Operation: Transportation, public transport, regional transport, long-distance transport, freight service, scheduling
3. Infrastructure: rail facilities, track alignment, railway stations, clearance diagram
4. Wheel-rail-contact: carrying of vehicle mass, adhesion, wheel guidance, current return
5. Vehicle dynamics: tractive and brake effort, driving resistance, inertial force, load cycles
6. Signaling and Control: operating procedure, succession of trains, European Train Control System, blocking period, automatic train control
7. Traction power supply: power supply of rail vehicles, comparison electric traction and diesel traction, dc and ac networks, system pantograph and contact wire, filling stations

Literature
Eine Literaturliste steht den Studierenden auf der Ilias-Plattform zum Download zur Verfügung.

A bibliography is available for download (Ilias-platform).

V Rail System Technology

2115919, SS 2026, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content

1. Railway System: railway as system, subsystems and interdependencies, definitions, laws, rules, railway and environment, economic impact
2. Operation: Transportation, public transport, regional transport, long-distance transport, freight service, scheduling
3. Infrastructure: rail facilities, track alignment, railway stations, clearance diagram
4. Wheel-rail-contact: carrying of vehicle mass, adhesion, wheel guidance, current return
5. Vehicle dynamics: tractive and brake effort, driving resistance, inertial force, load cycles
6. Signaling and Control: operating procedure, succession of trains, European Train Control System, blocking period, automatic train control
7. Traction power supply: power supply of rail vehicles, comparison electric traction and diesel traction, dc and ac networks, system pantograph and contact wire, filling stations

Organizational issues
schriftliche Prüfung am 5.8.2026, 10.00-11.00 Uhr, CO, 70.04, der Raum steht noch nicht fest

Literature
Eine Literaturliste steht den Studierenden auf der Ilias-Plattform zum Download zur Verfügung.

A bibliography is available for download (Ilias-platform).

V **Rail System Technology** **Lecture (V) On-Site**
2115919, WS 26/27, 2 SWS, Language: German, Open in study portal

Content

1. Railway System: railway as system, subsystems and interdependencies, definitions, laws, rules, railway and environment, economic impact
2. Operation: Transportation, public transport, regional transport, long-distance transport, freight service, scheduling
3. Infrastructure: rail facilities, track alignment, railway stations, clearance diagram
4. Wheel-rail-contact: carrying of vehicle mass, adhesion, wheel guidance, current return
5. Vehicle dynamics: tractive and brake effort, driving resistance, inertial force, load cycles
6. Signaling and Control: operating procedure, succession of trains, European Train Control System, blocking period, automatic train control
7. Traction power supply: power supply of rail vehicles, comparison electric traction and diesel traction, dc and ac networks, system pantograph and contact wire, filling stations

Organizational issues
weitere Informationen unter https://www.fast.kit.edu/bst/929_1870.php

Literature
Eine Literaturliste steht den Studierenden auf der Ilias-Plattform zum Download zur Verfügung.

A bibliography is available for download (Ilias-platform).

T 6.611 Module component: Rail System Technology [T-MACH-102143]

Coordinators: Prof. Dr.-Ing. Martin Cichon
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101274 - Rail System Technology
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Written examination	9 CP	graded	Each term	6

Courses					
WT 25/26	2115919	Rail System Technology	2 SWS	Lecture / 🗣	Cichon
WT 25/26	2115996	Rail Vehicle Technology	2 SWS	Lecture / 🗣	Cichon
ST 2026	2115919	Rail System Technology	2 SWS	Lecture / 🗣	Cichon
ST 2026	2115996	Rail Vehicle Technology	2 SWS	Lecture / 🗣	Cichon
WT 26/27	2115919	Rail System Technology	2 SWS	Lecture / 🗣	Cichon
WT 26/27	2115996	Rail Vehicle Technology	2 SWS	Lecture / 🗣	Cichon

Legend: 🖥 Online, ✨ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
writen examination in German language
Duration. 120 minutes
No tools or reference materials may be used during the exam except calculator and dictionary

Prerequisites
none

Additional Information
The course is offered in German.

Workload
270 hours

Below you will find excerpts from courses related to this module component:

V Rail System Technology

2115919, WS 25/26, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content

1. Railway System: railway as system, subsystems and interdependencies, definitions, laws, rules, railway and environment, economic impact
2. Operation: Transportation, public transport, regional transport, long-distance transport, freight service, scheduling
3. Infrastructure: rail facilities, track alignment, railway stations, clearance diagram
4. Wheel-rail-contact: carrying of vehicle mass, adhesion, wheel guidance, current return
5. Vehicle dynamics: tractive and brake effort, driving resistance, inertial force, load cycles
6. Signaling and Control: operating procedure, succession of trains, European Train Control System, blocking period, automatic train control
7. Traction power supply: power supply of rail vehicles, comparison electric traction and diesel traction, dc and ac networks, system pantograph and contact wire, filling stations

Literature
Eine Literaturliste steht den Studierenden auf der Ilias-Plattform zum Download zur Verfügung.

A bibliography is available for download (Ilias-platform).

V Rail Vehicle Technology

2115996, WS 25/26, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content

1. Vehicle system technology: structure and main systems of rail vehicles
2. Car body: functions, requirements, design principles, crash elements, coupling, doors and windows
3. Bogies: forces, running gears, bogies, Jakobs-bogies, active components, connection to car body, wheel arrangement
4. Drives: priciples, electric drives (main components, asynchronous traction motor, inverter, with DC supply, with AC supply, without line supply, multisystem vehicles, dual mode vehicles, hybrid vehicles), non-electric drives
5. Brakes: basics, principles (wheel brakes, rail brakes, blending), brake control (requirements and operation modes, pneumatic brake, electropneumatic brake, emergency brake, parking brake)
6. Train control management system: definition of TCMS, bus systems, components, network architectures, examples, future trends
7. Vehicle concepts: trams, metros, regional trains, intercity trains, high speed trains, double deck vehicles, locomotives, freight wagons

Literature
Eine Literaturliste steht den Studierenden auf der Ilias-Plattform zum Download zur Verfügung.

A bibliography is available for download (Ilias-platform).

V **Rail System Technology**
2115919, SS 2026, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content

1. Railway System: railway as system, subsystems and interdependencies, definitions, laws, rules, railway and environment, economic impact
2. Operation: Transportation, public transport, regional transport, long-distance transport, freight service, scheduling
3. Infrastructure: rail facilities, track alignment, railway stations, clearance diagram
4. Wheel-rail-contact: carrying of vehicle mass, adhesion, wheel guidance, current return
5. Vehicle dynamics: tractive and brake effort, driving resistance, inertial force, load cycles
6. Signaling and Control: operating procedure, succession of trains, European Train Control System, blocking period, automatic train control
7. Traction power supply: power supply of rail vehicles, comparison electric traction and diesel traction, dc and ac networks, system pantograph and contact wire, filling stations

Organizational issues
schriftliche Prüfung am 5.8.2026, 10.00-11.00 Uhr, CO, 70.04, der Raum steht noch nicht fest

Literature
Eine Literaturliste steht den Studierenden auf der Ilias-Plattform zum Download zur Verfügung.

A bibliography is available for download (Ilias-platform).

V **Rail Vehicle Technology**
2115996, SS 2026, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content

1. Vehicle system technology: structure and main systems of rail vehicles
2. Car body: functions, requirements, design principles, crash elements, coupling, doors and windows
3. Bogies: forces, running gears, bogies, Jakobs-bogies, active components, connection to car body, wheel arrangement
4. Drives: priciples, electric drives (main components, asynchronous traction motor, inverter, with DC supply, with AC supply, without line supply, multisystem vehicles, dual mode vehicles, hybrid vehicles), non-electric drives
5. Brakes: basics, principles (wheel brakes, rail brakes, blending), brake control (requirements and operation modes, pneumatic brake, electropneumatic brake, emergency brake, parking brake)
6. Train control management system: definition of TCMS, bus systems, components, network architectures, examples, future trends
7. Vehicle concepts: trams, metros, regional trains, intercity trains, high speed trains, double deck vehicles, locomotives, freight wagons

Organizational issues
schriftliche Prüfung am 5.8.2026, 11.30-12.30 Uhr, CO, 70.04, der Raum steht noch nicht fest

Literature
Eine Literaturliste steht den Studierenden auf der Ilias-Plattform zum Download zur Verfügung.

A bibliography is available for download (Ilias-platform).

V Rail System Technology

2115919, WS 26/27, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content

1. Railway System: railway as system, subsystems and interdependencies, definitions, laws, rules, railway and environment, economic impact
2. Operation: Transportation, public transport, regional transport, long-distance transport, freight service, scheduling
3. Infrastructure: rail facilities, track alignment, railway stations, clearance diagram
4. Wheel-rail-contact: carrying of vehicle mass, adhesion, wheel guidance, current return
5. Vehicle dynamics: tractive and brake effort, driving resistance, inertial force, load cycles
6. Signaling and Control: operating procedure, succession of trains, European Train Control System, blocking period, automatic train control
7. Traction power supply: power supply of rail vehicles, comparison electric traction and diesel traction, dc and ac networks, system pantograph and contact wire, filling stations

Organizational issues

weitere Informationen unter https://www.fast.kit.edu/bst/929_1870.php

Literature

Eine Literaturliste steht den Studierenden auf der Ilias-Plattform zum Download zur Verfügung.

A bibliography is available for download (Ilias-platform).

V Rail Vehicle Technology

2115996, WS 26/27, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content

1. Vehicle system technology: structure and main systems of rail vehicles
2. Car body: functions, requirements, design principles, crash elements, coupling, doors and windows
3. Bogies: forces, running gears, bogies, Jakobs-bogies, active components, connection to car body, wheel arrangement
4. Drives: priciples, electric drives (main components, asynchronous traction motor, inverter, with DC supply, with AC supply, without line supply, multisystem vehicles, dual mode vehicles, hybrid vehicles), non-electric drives
5. Brakes: basics, principles (wheel brakes, rail brakes, blending), brake control (requirements and operation modes, pneumatic brake, electropneumatic brake, emergency brake, parking brake)
6. Train control management system: definition of TCMS, bus systems, components, network architectures, examples, future trends
7. Vehicle concepts: trams, metros, regional trains, intercity trains, high speed trains, double deck vehicles, locomotives, freight wagons

Organizational issues

weitere Informationen unter https://www.fast.kit.edu/bst/929_7522.php

Literature

Eine Literaturliste steht den Studierenden auf der Ilias-Plattform zum Download zur Verfügung.

A bibliography is available for download (Ilias-platform).

6.612 Module component: Rail Vehicle Technology [T-MACH-105353]

Coordinators: Prof. Dr.-Ing. Martin Cichon
Organisation: KIT Department of Mechanical Engineering

Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Written examination	4 CP	graded	Each term	5

Courses					
WT 25/26	2115996	Rail Vehicle Technology	2 SWS	Lecture / 🗣	Cichon
ST 2026	2115996	Rail Vehicle Technology	2 SWS	Lecture / 🗣	Cichon
WT 26/27	2115996	Rail Vehicle Technology	2 SWS	Lecture / 🗣	Cichon

Legend: 🖥 Online, ❖ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
writen examination in German language
Duration: approx 60 minutes
No tools or reference materials may be used during the exam except calculator and dictionary

Prerequisites
none

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The following conditions have to be fulfilled:

Additional Information
The course is offered in German.

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Rail Vehicle Technology
2115996, WS 25/26, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content

1. Vehicle system technology: structure and main systems of rail vehicles
2. Car body: functions, requirements, design principles, crash elements, coupling, doors and windows
3. Bogies: forces, running gears, bogies, Jakobs-bogies, active components, connection to car body, wheel arrangement
4. Drives: priciples, electric drives (main components, asynchronous traction motor, inverter, with DC supply, with AC supply, without line supply, multisystem vehicles, dual mode vehicles, hybrid vehicles), non-electric drives
5. Brakes: basics, principles (wheel brakes, rail brakes, blending), brake control (requirements and operation modes, pneumatic brake, electropneumatic brake, emergency brake, parking brake)
6. Train control management system: definition of TCMS, bus systems, components, network architectures, examples, future trends
7. Vehicle concepts: trams, metros, regional trains, intercity trains, high speed trains, double deck vehicles, locomotives, freight wagons

Literature
Eine Literaturliste steht den Studierenden auf der Ilias-Plattform zum Download zur Verfügung.

A bibliography is available for download (Ilias-platform).

V

Rail Vehicle Technology

2115996, SS 2026, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content

1. Vehicle system technology: structure and main systems of rail vehicles
2. Car body: functions, requirements, design principles, crash elements, coupling, doors and windows
3. Bogies: forces, running gears, bogies, Jakobs-bogies, active components, connection to car body, wheel arrangement
4. Drives: priciples, electric drives (main components, asynchronous traction motor, inverter, with DC supply, with AC supply, without line supply, multisystem vehicles, dual mode vehicles, hybrid vehicles), non-electric drives
5. Brakes: basics, principles (wheel brakes, rail brakes, blending), brake control (requirements and operation modes, pneumatic brake, electropneumatic brake, emergency brake, parking brake)
6. Train control management system: definition of TCMS, bus systems, components, network architectures, examples, future trends
7. Vehicle concepts: trams, metros, regional trains, intercity trains, high speed trains, double deck vehicles, locomotives, freight wagons

Organizational issues
schriftliche Prüfung am 5.8.2026, 11.30-12.30 Uhr, CO, 70.04, der Raum steht noch nicht fest

Literature
Eine Literaturliste steht den Studierenden auf der Ilias-Plattform zum Download zur Verfügung.

A bibliography is available for download (Ilias-platform).

V

Rail Vehicle Technology

2115996, WS 26/27, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content

1. Vehicle system technology: structure and main systems of rail vehicles
2. Car body: functions, requirements, design principles, crash elements, coupling, doors and windows
3. Bogies: forces, running gears, bogies, Jakobs-bogies, active components, connection to car body, wheel arrangement
4. Drives: priciples, electric drives (main components, asynchronous traction motor, inverter, with DC supply, with AC supply, without line supply, multisystem vehicles, dual mode vehicles, hybrid vehicles), non-electric drives
5. Brakes: basics, principles (wheel brakes, rail brakes, blending), brake control (requirements and operation modes, pneumatic brake, electropneumatic brake, emergency brake, parking brake)
6. Train control management system: definition of TCMS, bus systems, components, network architectures, examples, future trends
7. Vehicle concepts: trams, metros, regional trains, intercity trains, high speed trains, double deck vehicles, locomotives, freight wagons

Organizational issues
weitere Informationen unter https://www.fast.kit.edu/bst/929_7522.php

Literature
Eine Literaturliste steht den Studierenden auf der Ilias-Plattform zum Download zur Verfügung.

A bibliography is available for download (Ilias-platform).

T 6.613 Module component: Rapid Industrialization of Immature Products using the Example of Electric Mobility [T-MACH-113031]

Coordinators: Dr. Jörg Bauer
Organisation: KIT Department of Mechanical Engineering
Part of: M-MACH-106590 - Production Engineering

Type	Credits	Grading	Term offered	Expansion	Version
Written examination	4 CP	graded	Each winter term	1 semesters	1

Courses					
WT 25/26	2149621	Rapid Industrialization of Immature Products using the Example of Electric Mobility	2 SWS	Lecture / 🗣	Bauer
WT 26/27	2149621	Rapid Industrialization of Immature Products using the Example of Electric Mobility	2 SWS	Lecture / 🗣	Bauer

Legend: 🖥 Online, Blended (On-Site/Online), 🗣 On-Site, × Cancelled

Assessment
Written Exam (60 min)

Prerequisites
none

Additional Information
The course is offered in German.

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Rapid Industrialization of Immature Products using the Example of Electric Mobility

Lecture (V)
On-Site

2149621, WS 25/26, 2 SWS, Language: German, Open in study portal

Content

The lecture "Rapid Industrialization of Immature Products using the Example of Electric Mobility" deals with production engineering methods for the robust and cost-effective production of technologically novel, so-called "immature" products. In this context, approaches for solving the central challenges resulting from the tension triangle of product development, industrialization and production are identified and discussed.

Based on the motivation for rapid market entry, the current approach involving stakeholders and other participants is explained. On this basis, key enablers for rapid and targeted industrialization are derived and discussed. For example, robust industrial processes based on flexible equipment are an essential core element for cost-effective production. Against this background, industry-relevant concepts for the automation and flexibilization of production processes are presented in the lecture in order to be able to deal efficiently and effectively with product-specific changes on the production side. Therefore, the main goal of an industrialization process is to develop production technologies and processes that enable robust, resource-efficient and cost-effective manufacturing of established and innovative products.

The lecture is structured as follows:

1. Motivation for rapid industrialization (complex market requirements, shortened development and product cycles, decreasing quantities per variant, ...).
2. Industrialization methods (simultaneous engineering, releases, frozen zones, high volumes, ...)
3. Key enablers to accelerate industrialization (simulation and digitalization, flexible and digital production equipment
4. Supply chains and suppliers
5. Testing and deployment
6. Ramp-up

Learning Outcomes:

- The students are familiar with the essential elements of simultaneous engineering and industrialization (motivation, processes, fields of action, challenges).
- The Students know the key enablers for the rapid industrialization of immature products (digitization, flexible production equipment, rapid manufacturing processes for primary production).
- The Students are familiar with the basic principles, methods and procedures of the main enablers. The understanding is deepened through theory, case and practical examples.
- The toolbox of key enablers described in the lecture allows students to select and independently apply the enablers in the context of future challenges.
- The Students are able to disseminate and to apply the knowledge acquired during the lecture in their future working lives.

Workload:

regular attendance: 21 hours
self-study: 99 hours

Organizational issues

Blockvorlesung vom 03.11.2025 bis 07.11.2025, täglich von 14:00 bis 19:00 Uhr in der Karlsruher Forschungsfabrik 70.41. Die theoretischen Inhalte der Vorlesung werden durch praktische Aufbauten in der Karlsruher Forschungsfabrik veranschaulicht. Vorlesungsunterlagen und weitere Informationen werden über ILIAS (https://ilias.studium.kit.edu/) zur Verfügung gestellt.

Block course from 03.11.2025 to 07.11.2025, daily from 14:00 to 19:00 in the Karlsruhe Research Factory 70.41. The theoretical contents of the lecture will be illustrated by practical setups in the Karlsruhe Research Factory. Lecture notes and further information will be made available via ILIAS (https://ilias.studium.kit.edu/).

Literature

Foliensatz zur Veranstaltung wird über Ilias (https://ilias.studium.kit.edu/) bereitgestellt.

Lecture notes will be provided in Ilias (https://ilias.studium.kit.edu/).

V	**Rapid Industrialization of Immature Products using the Example of Electric Mobility** 2149621, WS 26/27, 2 SWS, Language: German, Open in study portal	**Lecture (V) On-Site**

Content

The lecture "Rapid Industrialization of Immature Products using the Example of Electric Mobility" deals with production engineering methods for the robust and cost-effective production of technologically novel, so-called "immature" products. In this context, approaches for solving the central challenges resulting from the tension triangle of product development, industrialization and production are identified and discussed.

Based on the motivation for rapid market entry, the current approach involving stakeholders and other participants is explained. On this basis, key enablers for rapid and targeted industrialization are derived and discussed. For example, robust industrial processes based on flexible equipment are an essential core element for cost-effective production. Against this background, industry-relevant concepts for the automation and flexibilization of production processes are presented in the lecture in order to be able to deal efficiently and effectively with product-specific changes on the production side. Therefore, the main goal of an industrialization process is to develop production technologies and processes that enable robust, resource-efficient and cost-effective manufacturing of established and innovative products.

The lecture is structured as follows:

1. Motivation for rapid industrialization (complex market requirements, shortened development and product cycles, decreasing quantities per variant, ...).
2. Industrialization methods (simultaneous engineering, releases, frozen zones, high volumes, ...)
3. Key enablers to accelerate industrialization (simulation and digitalization, flexible and digital production equipment
4. Supply chains and suppliers
5. Testing and deployment
6. Ramp-up

Learning Outcomes:

- The students are familiar with the essential elements of simultaneous engineering and industrialization (motivation, processes, fields of action, challenges).
- The Students know the key enablers for the rapid industrialization of immature products (digitization, flexible production equipment, rapid manufacturing processes for primary production).
- The Students are familiar with the basic principles, methods and procedures of the main enablers. The understanding is deepened through theory, case and practical examples.
- The toolbox of key enablers described in the lecture allows students to select and independently apply the enablers in the context of future challenges.
- The Students are able to disseminate and to apply the knowledge acquired during the lecture in their future working lives.

Workload:

regular attendance: 21 hours
self-study: 99 hours

Organizational issues

Blockvorlesung vom 16.11.2026 bis 20.11.2026, täglich von 13:45 bis 18:45 Uhr in der Karlsruher Forschungsfabrik 70.41. Die theoretischen Inhalte der Vorlesung werden durch praktische Aufbauten in der Karlsruher Forschungsfabrik veranschaulicht. Vorlesungsunterlagen und weitere Informationen werden über ILIAS (https://ilias.studium.kit.edu/) zur Verfügung gestellt.

Block course from 16.11.2026 to 20.11.2026, daily from 13:45 to 18:45 in the Karlsruhe Research Factory 70.41. The theoretical contents of the lecture will be illustrated by practical setups in the Karlsruhe Research Factory. Lecture notes and further information will be made available via ILIAS (https://ilias.studium.kit.edu/).

Literature

Foliensatz zur Veranstaltung wird über Ilias (https://ilias.studium.kit.edu/) bereitgestellt.

Lecture notes will be provided in Ilias (https://ilias.studium.kit.edu/).

T 6.614 Module component: Reactor Modeling with CFD [T-CIWVT-113224]

Coordinators: Prof. Dr.-Ing. Gregor Wehinger
Organisation: KIT Department of Chemical and Process Engineering
Part of: M-CIWVT-106537 - Reactor Modeling with CFD

Type	Credits	Grading	Version
Examination of another type	6 CP	graded	2

Courses					
ST 2026	2220060	Reactor Modeling with CFD	1 SWS	Lecture / 🗣	Wehinger, Schulz
ST 2026	2220061	Exercise Reactor Modeling with CFD	2 SWS	Practice / 🗣	Wehinger, und Mitarbeitende, Schulz

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Prerequisites
None.

T 6.615 Module component: Real Estate Economics and Sustainability Part 1: Basics and Valuation [T-WIWI-102838]

Coordinators: Prof. Dr David Lorenz
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each winter term	1

Assessment
The examination for the courses generally consist of a 60 minute written exam. A 20 minute oral exam is only offered after the second failure of the written exam. The exams for the respective parts (Part 1: Basics and Valuation and Part 2: Reporting and Rating) happen in the same semester in which the lectures take place.

Therefore, Part I currently only takes place in the winter semester and Part II takes place in the summer semester. In each semester there are two alternative dates for the exam and exams can be re-sat at any regular exam date.

Prerequisites
None

Recommendations
A combination with courses in the area of

- Finance
- Insurance
- Civil engineering and architecture

is recommended.

Particularly recommended is the successful completion of the following Bachelor-Modules:

- Real Estate Management I and II
- Design, Construction and Assessment of Green Buildings I and II

T 6.616 Module component: Real Estate Economics and Sustainability Part 2: Reporting and Rating [T-WIWI-102839]

Coordinators: Prof. Dr David Lorenz
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each summer term	1

Assessment
It is currently unclear whether the course "Real Estate Economics and Sustainability Part 2: Reporting and Rating" can be offered in summer term 2018. It must therefore be expected that the corresponding module M-WIWI-101508 "Real Estate Management and Sustainability" can not be completed according to schedule.

The examination for the courses generally consist of a 60 minute written exam. A 20 minute oral exam is only offered after the second failure of the written exam. The exams for the respective parts (Part 1: Basics and Valuation and Part 2: Reporting and Rating) happen in the same semester in which the lectures take place.

Therefore, Part I currently only takes place in the winter semester and Part II takes place in the summer semester. In each semester there are two alternative dates for the exam and exams can be re-sat at any regular exam date.

Prerequisites
None

Recommendations
A combination with courses in the area of

- Finance
- Insurance
- Civil engineering and architecture

is recommended.

Particularly recommended is the successful completion of the following Bachelor-Modules:

- Real Estate Management I and II
- Design, Construction and Assessment of Green Buildings I and II

T 6.617 Module component: Real Estate Management I [T-WIWI-102744]

Coordinators: Prof. Dr.-Ing. Thomas Lützkendorf
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each winter term	1

Assessment
The examination offer has been discontinued. For all those who still have to complete the module examination with this exam, there is the last opportunity for an oral exam in WS 22/23. This must be arranged directly at the institute. If this is a repeat examination in which the first attempt was taken in writing, the approval of the examination board must be obtained in advance via the examination office.

Prerequisites
None

Additional Information
The course is replenished by excursions and guest lectures by practicioners out of the real estate business.

T 6.618 Module component: Real Estate Management II [T-WIWI-102745]

Coordinators: Prof. Dr.-Ing. Thomas Lützkendorf
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each summer term	1

Assessment
The examination offer has been discontinued. For all those who still have to complete the module examination with this exam, there is the last opportunity for an oral exam in WS 22/23. This must be arranged directly at the institute. If this is a repeat examination in which the first attempt was taken in writing, the approval of the examination board must be obtained in advance via the examination office.

Prerequisites
None

Recommendations
A combination with the module *Design Construction and Assessment of Green Buildings I* is recommended.

Furthermore it is recommeded to choose courses of the following fields

- Finance and Banking
- Insurance
- Civil Engineering and Architecture (building physics, structural design, facility management)

Additional Information
The course is replenished by excursions and guest lectures by practicioners out of the real estate business.

T 6.619 Module component: Recommender Systems [T-WIWI-102847]

Coordinators: Prof. Dr. Andreas Geyer-Schulz
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101410 - Business & Service Engineering
M-WIWI-104900 - Business Administration
M-WIWI-105661 - Data Science: Intelligent, Adaptive, and Learning Information Services

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each winter term	1

Courses					
WT 25/26	2540506	Recommender Systems	2 SWS	Lecture / 🗣	Geyer-Schulz
WT 25/26	2540507	Exercise Recommender Systems	1 SWS	Practice / 🗣	Geyer-Schulz, Nazemi

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Written examination (60 minutes) according to §4(2), 1 SPO. The exam is considered passed if at least 50 out of a maximum of 100 possible points are achieved. The grades are graded in five steps (best grade 1.0 from 95 points). Details of the grade formation and scale will be announced in the course.A bonus can be acquired through successful participation in the practice. If the grade of the written examination is between 4.0 and 1.3, the bonus improves the grade by one grade level (0.3 or 0.4). The exact criteria for awarding a bonus will be announced at the beginning of the course.

Attention: Exams will not be offered after August 2026.

Prerequisites
None

Recommendations
None

Below you will find excerpts from courses related to this module component:

V Recommender Systems

2540506, WS 25/26, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content

At first, an overview of general aspects and concepts of recommender systems and its relevance for service providers and customers is given. Next, different categories of recommender systems are discussed. This includes explicit recommendations like customer reviews as well as implicit services based on behavioral data. Furthermore, the course gives a detailed view of the current research on recommender systems at the Chair of Information Services and Electronic Markets.

Learning objectives:

The student

- is proficient in different statistical, data-mining, and game theory methods of computing implicit and explicit recommendations
- evaluates recommender systems and compares these with related services

Workload:

The total workload for this course is approximately 135 hours (4.5 credits):

Time of attendance

- Attending the lecture: 15 x 90min = 22h 30m
- Attending the exercise classes: 7 x 90min = 10h 30m
- Examination: 1h 00m

Self-study

- Preparation and wrap-up of the lecture: 15 x 180min = 45h 00m
- Preparing the exercises: 25h 00m
- Preparation of the examination: 31h 00m

Sum: 135h 00m

Exam:

Assessment consists of a written exam of 1 hour length following §4 (2), 1 of the examination regulation and by submitting written papers as part of the exercise following §4 (2), 3 of the examination regulation.

The course is considered successfully taken, if at least 50 out of 100 points are acquired in the written exam. In this case, all additional points (up to 10) from excersise work will be added.

Grade: Minimum points

- 1,0: 95
- 1,3: 90
- 1,7: 85
- 2,0: 80
- 2,3: 75
- 2,7: 70
- 3,0: 65
- 3,3: 60
- 3,7: 55
- 4,0: 50
- 5,0: 0

Organizational issues

Geb. 10.11, Raum 223

Literature

Rakesh Agrawal, Tomasz Imielinski, and Arun Swami. Mining association rules between sets of items in large databases. In Sushil Jajodia Peter Buneman, editor, Proceedings of the ACM SIGMOD International Conference on Management of Data, volume 22, Washington, D.C., USA, Jun 1993. ACM, ACM Press.

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Richard O. Duda, Peter E. Hart, and David G. Stork. Pattern Classification. Wiley-Interscience, New York, 2 edition, 2001.

Andreas Geyer-Schulz, Michael Hahsler, and Maximilian Jahn. A customer purchase incidence model applied to recommender services. In R. Kohavi et al., editor, Proceedings of the WebKDD 2001 – Mining log data across all customer touchpoints, volume 2356 of Lecture Notes in Artificial Intelligence LNAI, pages 25–47, Berlin, 2002. ACM, Springer-Verlag.

Jon M. Kleinberg. Authoritative sources in a hyperlinked environment. JACM, 46(5):604–632, sep 1999.

Joseph Konstan, Bradley Miller, David Maltz, Jonathan Herlocker, Lee Gordon, and John Riedl. Grouplens: Applying Collaborative Filtering to Usernet News. Communications of the ACM, 40(3):77 – 87, Mar 1997.

Paul Resnick, Neophytos Iacovou, Peter Bergstrom, and John Riedl. Grouplens: An open architecture for collaborative filtering of netnews. In Proceedings of the conference on Computer supported cooperative work, pages 175 – 186. ACM Press, 1994.

Weiterführende Literatur:

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Thomas J. Blischok. Every transaction tells a story. Chain Store Age Executive with Shopping Center Age, 71(3):50–56, Mar 1995.

Hans Hermann Bock. Automatische Klassifikation. Vandenhoeck und Ruprecht, Göttingen, 1974.

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Wolfgang Gaul, Andreas Geyer-Schulz, Michael Hahsler, and Lars Schmidt-Thieme. eMarketing mittels Recommendersystemen. Marketing ZFP, 24:47 – 55, 2002.

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Andreas Mild and Martin Natter. Collaborative filtering or regression models for Internet recommendation systems? Journal of Targeting, Measurement and Analysis for Marketing, 10(4):304 – 313, Jan 2002.

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J. Ben Schafer, Joseph Konstan, and Jon Riedl. Recommender Systems in E-commerce. In Proceedings of the 1st ACM conference on Electronic commerce, pages 158 – 166, Denver, Colorado, USA, Nov 1999. ACM.

Upendra Shardanand and Patti Maes. Social information filtering: Algorithms for automating "word of mouth". In Proceedings of ACM SIGCHI, volume 1 of Papers: Using the Information of Others, pages 210 – 217. ACM, 1995.

T 6.620 Module component: Registration for Certificate Issuance - Supplementary Studies on Science, Technology and Society [T-FORUM-113587]

Coordinators: Dr. Christine Mielke
Christine Myglas

Organisation: General Studies. Forum Science and Society (FORUM)

Part of: M-FORUM-106753 - Supplementary Studies on Science, Technology and Society

Type	Credits	Grading	Term offered	Version
Coursework	0 CP	pass/fail	Each term	1

Prerequisites

In order to register, it is mandatory that the basic module and the advanced module have been completed and that the grades for the partial performances in the advanced module are available.

Registration as a partial achievement means the issue of a certificate.

T 6.621 Module component: Regulation Theory and Practice [T-WIWI-102712]

Coordinators: Prof. Dr. Kay Mitusch
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101406 - Network Economics
M-WIWI-101451 - Energy Economics and Energy Markets
M-WIWI-104908 - Economics

Type	Credits	Grading	Term offered	Version
Oral examination	4,5 CP	graded	see Annotations	2

Assessment
The lecture is not offered for an indefinite period of time.
Result of success is made by a 20-30 minutes oral examination. Examination is offered every semester and can be retried at any regular examination date.

Prerequisites
None

Recommendations
Basic knowledge and skills of microeconomics from undergraduate studies (bachelor's degree) are expected.

Particularly helpful but not necessary: Industrial Economics and Principal-Agent- or Contract theories. Prior attendance of the lecture*Competition in Networks* [26240] is helpful in any case but not considered a formal precondition.

Additional Information
The lecture is not offered for an indefinite period of time.

T 6.622 Module component: Reinforcement Learning [T-INFO-111255]

Coordinators: TT-Prof. Dr. Rudolf Lioutikov
Prof. Dr. Gerhard Neumann
Organisation: KIT Department of Informatics
Part of: M-WIWI-104909 - Informatics (Department of Informatics)

Type	Credits	Grading	Term offered	Version
Written examination	6 CP	graded	Each winter term	2

Courses					
WT 25/26	2400163	Reinforcement Learning		Lecture / Practice /	Neumann, Lioutikov

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
The success control takes place in the form of a written exam, usually 90 minutes in length, according to § 4 Abs. 2 Nr. 1 SPO.

A bonus can be acquired through successful participation in the exercise as a success control of a different kind (§4(2), 3 SPO 2008) or study performance (§4(3) SPO 2015). The exact criteria for awarding a bonus will be announced at the beginning of the lecture. If the grade of the written examination is between 4.0 and 1.3, the bonus improves the grade by one grade level (0.3 or 0.4). The bonus is only valid for the main and post exams of the semester in which it was earned. After that, the grade bonus expires.

Prerequisites
None.

Recommendations

- It is recommended that students are familiar with the content of the "Foundations of Artificial Intelligence" lecture.
- Good Python knowledge is recommended.
- Good mathematical background knowledge is recommended.

Below you will find excerpts from courses related to this module component:

V Reinforcement Learning — **Lecture / Practice (VÜ) On-Site**
2400163, WS 25/26, SWS, Language: English, Open in study portal

Content
Reinforcement Learning (RL) is a sub-field of machine learning in which an artificial agent has to interact with its environment and learn how to improve its behaviour by trial and error. For doing so, the agent is provided with an evaluative feedback signal, called reward, that he perceives for each action performed in its environment. RL is one of the hardest machine learning problems, as, in contrast to standard supervised learning, we do not know the targets (i.e. the optimal actions) for our inputs (i.e. the state of the environment) and we also need to consider the long-term effects of the agent's actions on the state of the environment. Due to recent successes, RL has gained a lot of popularity with applications in robotics, automation, health care, trading and finance, natural language processing, autonomous driving and computer games. This lecture will introduce the concepts and theory of RL and review current state of the art methods with a particular focus on RL applications in robotics. An exemplary list of topics is given below:

- Primer in Machine Learning and Deep Learning
- Supervised Learning of Behaviour
- Introduction in Reinforcement Learning
- Dynamic Programming
- Value Based Methods
- Policy Optimization and Trust Regions
- Episodic Reinforcement Learning and Skill Learning
- Bayesian Optimization
- Variational Inference, Max-Entropy RL and Versatility
- Model-based Reinforcement Learning
- Offline Reinforcement Learning
- Inverse Reinforcement Learning
- Hierarchical Reinforcement Learning
- Exploration and Artificial Curiosity
- Meta Reinforcement Learning

Lernziele:

- Students are able to understand the RL problem and challenges.

- Students can differentiate between different RL algorithm and understand their underlying theory

- Students will know the mathematical tools necessary to understand RL algorithms

- Students can implement RL algorithms for various tasks

- Students understand current research questions in RL

Empfehlungen:

- Der Vorlesungsinhalt von Maschinelles Lernen – Grundverfahren wird vorausgesetzt
- Gute Python Kenntnisse erforderlich
- Gute mathematische Grundkenntnisse

Erfolgskontrolle: Siehe Modulhandbuch!

Arbeitsaufwand:

180h, aufgeteilt in:

- ca 45h Vorlesungsbesuch
- ca 15h Übungsbesuch
- ca 90h Nachbearbeitung und Bearbeitung der Übungsblätter

ca 30h Prüfungsvorbereitung

Organizational issues
6 ECTS

Vorlesungs-und Übungsturnus: Siehe ILIAS

T 6.623 Module component: Remote Sensing, Exam [T-BGU-101636]

Coordinators: Prof. Dr. Jan Cermak
Prof. Dr.-Ing. Stefan Hinz
Dr.-Ing. Uwe Weidner

Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences

Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each term	2

Courses					
ST 2026	6020241	Remote Sensing Systems	1 SWS	Lecture / 🗣	Hinz, Cermak
ST 2026	6020242	Remote Sensing Systems, Excercise	1 SWS	Practice / 🗣	Pauli
ST 2026	6020243	Procedures of Remote Sensing	2 SWS	Lecture / 🗣	Weidner
ST 2026	6020244	Procedures of Remote Sensing, Excercise	1 SWS	Practice / 🗣	Weidner

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-BGU-101637 - Systems of Remote Sensing, Prerequisite must have been passed.
2. The module component T-BGU-101638 - Procedures of Remote Sensing, Prerequisite must have been passed.

Recommendations
None

Workload
120 hours

T 6.624 Module component: Renewable Energy-Resources, Technologies and Economics [T-WIWI-100806]

Coordinators: Prof. Dr. Patrick Jochem
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	3,5 CP	graded	Each winter term	8

Courses					
WT 25/26	2581012	Renewable Energy – Resources, Technologies and Economics	2 SWS	Lecture / 🗣	Jochem

Legend: 🖥 Online, ⚄ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment consists of a written exam (60 minutes, in English, answers are possible in German or English) (following §4(2) of the examination regulation). The exam takes place in every semester. Re-examinations are offered at every ordinary examination date.

Prerequisites
None.

Below you will find excerpts from courses related to this module component:

V Renewable Energy – Resources, Technologies and Economics

2581012, WS 25/26, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content
1. General introduction: Motivation, Global situation

2. Basics of renewable energies: Energy balance of the earth, potential definition

3. Hydro

4. Wind

5. Solar

6. Biomass

7. Geothermal

8. Other renewable energies

9. Promotion of renewable energies

10. Interactions in systemic context

11. Excursion to the "Energieberg" in Mühlburg

Learning Goals:

The student

- understands the motivation and the global context of renewable energy resources.
- gains detailed knowledge about the different renewable resources and technologies as well as their potentials.
- understands the systemic context and interactions resulting from the increased share of renewable power generation.
- understands the important economic aspects of renewable energies, including electricity generation costs, political promotion and marketing of renewable electricity.
- is able to characterize and where required calculate these technologies.

Literature
Weiterführende Literatur:

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T 6.625 Module component: Requirements Engineering [T-INFO-101300]

Coordinators: Prof. Dr.-Ing. Anne Koziolek
Organisation: KIT Department of Informatics
Part of: M-WIWI-104909 - Informatics (Department of Informatics)

Type	Credits	Grading	Term offered	Version
Written examination	3 CP	graded	Each summer term	2

Recommendations
Das Modul Softwaretechnik II wird empfohlen.

T 6.626 Module component: Responsible Artificial Intelligence [T-WIWI-111385]

Coordinators: Prof. Dr. Jella Pfeiffer
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-103117 - Data Science: Data-Driven Information Systems
M-WIWI-103118 - Data Science: Data-Driven User Modeling
M-WIWI-104900 - Business Administration
M-WIWI-105923 - Incentives, Interactivity & Decisions in Organizations

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each winter term	2

Courses					
WT 25/26	2545164	Responsible Artificial Intelligence	3 SWS	Lecture / Practice / 🗣	Hoffmann, Gutschow, Bennardo
WT 26/27	2545164	Responsible Artificial Intelligence	3 SWS	Lecture / Practice / 🗣	Pfeiffer, Bennardo

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment consists of a written exam (60 min.) according to § 4 paragraph 2 Nr. 1 of the examination regulation.

Prerequisites
Prior to the start of the lecture, introductory materials will be provided for self-study. The lecture has a limitation of participants. Therefore, prior registration via the Wiwi-Portal is mandatory.

Additional Information
Can a technology really be trustworthy or even responsible? Since the success of LLMs at the latest, this question has been increasingly asked in society. With the increasing use of artificial intelligence, terms such as "Trustworthy AI", "Responsible AI" or "Ethical AI" are therefore gaining in importance. But what exactly is behind them? Technology is only ever used by people for specific purposes. So if we want to "trust" an AI solution, we need to understand how the people and organizations involved develop AI responsibly. According to the European Commission's HLEG AI, trustworthy AI must be lawful, ethical and robust.

This lecture sheds light on all these areas and thus provides an answer to the question of what a responsible and thus sustainable approach to AI can look like. After an introduction to AI and data, various approaches will be discussed with which actions and technology applications can be morally evaluated. The aim of this ethical reflection is to find out what we should do with AI instead of limiting ourselves to what we can do with AI.

In the context of robustness, vulnerabilities of AI and measures to address them will be discussed. The lecture will cover other topics such as bias, adversarial attacks, transparency, privacy and human-computer interaction. Current developments in regulatory requirements at European level will also be discussed. Guest lectures and continuous insights into business practice complement the foundations laid.

After successfully completing the course, students should be able

- to classify and evaluate the scientific discussion on ethics in artificial intelligence systems,
- understand the concept of trust and responsibility in the context of artificial intelligence and apply the relevant knowledge to change processes in companies,
- shape the social and entrepreneurial discussion on the use of AI themselves and
- know the legal requirements for AI and implement them in the corporate context.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Responsible Artificial Intelligence
2545164, WS 25/26, 3 SWS, Language: German, Open in study portal

Lecture / Practice (VÜ)
On-Site

Content
Can a technology really be trustworthy or even responsible? Since the success of LLMs at the latest, this question has been increasingly asked in society. With the increasing use of artificial intelligence, terms such as "Trustworthy AI", "Responsible AI" or "Ethical AI" are therefore gaining in importance. But what exactly is behind them? Technology is only ever used by people for specific purposes. So if we want to "trust" an AI solution, we need to understand how the people and organizations involved develop AI responsibly. According to the European Commission's HLEG AI, trustworthy AI must be lawful, ethical and robust.

This lecture sheds light on all these areas and thus provides an answer to the question of what a responsible and thus sustainable approach to AI can look like. After an introduction to AI and **data,** various approaches will be discussed with which actions and technology applications can be morally evaluated. The aim of this ethical reflection is to find out what we should do with AI instead of limiting ourselves to what we can do with AI.

In the context of robustness, vulnerabilities of AI and measures to address them will be discussed. The lecture will cover other topics such as bias, adversarial attacks, transparency, privacy and human-computer interaction. Current developments in regulatory requirements at European level will also be discussed. Guest lectures and continuous insights into business practice complement the foundations laid.

After successfully completing the course, students should be able to:

- classify and evaluate the scientific discussion on ethics in artificial intelligence systems,
- understand the concept of trust and responsibility in the context of artificial intelligence and apply the relevant knowledge to change processes in companies,
- shape the social and entrepreneurial discussion on the use of AI themselves and
- know the legal requirements for AI and implement them in the corporate context.

Organizational issues
Die Vorlesung und die Übung wechseln sich im zweiwöchentlichen Rhythmus ab: In den Vorlesungswochen liegt die Veranstaltung von 15:45 – 19:00 Uhr (2 SWS), in den Übungswochen von 15:45 – 17:15 Uhr (1 SWS).

Die Übung findet planmäßig online statt.

V	**Responsible Artificial Intelligence** 2545164, WS 26/27, 3 SWS, Language: German, Open in study portal	**Lecture / Practice (VÜ)** **On-Site**

Content
Can a technology really be trustworthy or even responsible? Since the success of LLMs at the latest, this question has been increasingly asked in society. With the increasing use of artificial intelligence, terms such as "Trustworthy AI", "Responsible AI" or "Ethical AI" are therefore gaining in importance. But what exactly is behind them? Technology is only ever used by people for specific purposes. So if we want to "trust" an AI solution, we need to understand how the people and organizations involved develop AI responsibly. According to the European Commission's HLEG AI, trustworthy AI must be lawful, ethical and robust.

This lecture sheds light on all these areas and thus provides an answer to the question of what a responsible and thus sustainable approach to AI can look like. After an introduction to AI and **data,** various approaches will be discussed with which actions and technology applications can be morally evaluated. The aim of this ethical reflection is to find out what we should do with AI instead of limiting ourselves to what we can do with AI.

In the context of robustness, vulnerabilities of AI and measures to address them will be discussed. The lecture will cover other topics such as bias, adversarial attacks, transparency, privacy and human-computer interaction. Current developments in regulatory requirements at European level will also be discussed. Guest lectures and continuous insights into business practice complement the foundations laid.

After successfully completing the course, students should be able to:

- classify and evaluate the scientific discussion on ethics in artificial intelligence systems,
- understand the concept of trust and responsibility in the context of artificial intelligence and apply the relevant knowledge to change processes in companies,
- shape the social and entrepreneurial discussion on the use of AI themselves and
- know the legal requirements for AI and implement them in the corporate context.

Organizational issues
Die Vorlesung und die Übung wechseln sich im zweiwöchentlichen Rhythmus ab: In den Vorlesungswochen liegt die Veranstaltung von 15:45 – 19:00 Uhr (2 SWS), in den Übungswochen von 15:45 – 17:15 Uhr (1 SWS).

Die Übung ist lediglich als optionale Fragerunde gedacht und findet planmäßig online statt.

T 6.627 Module component: Risk Communication [T-WIWI-102649]

Coordinators: Prof. Dr. Ute Werner
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Oral examination	4,5 CP	graded	Each winter term	1

Assessment
The assessment consists of oral presentations (incl. papers) within the lecture (according to Section 4 (2), 3 of the examination regulation) and a final oral exam (30 min.) according to Section 4 (2), 2 of the examination regulation.

The overall grade consists of the assessment of the oral presentations incl. papers (50 percent) and the assessment of the oral exam (50 percent).

Prerequisites
None

Recommendations
None

T 6.628 Module component: Risk Management in Industrial Supply Networks [T-WIWI-102826]

Coordinators: Prof. Dr. Frank Schultmann
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101412 - Industrial Production III
M-WIWI-101471 - Industrial Production II
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	3,5 CP	graded	Each winter term	1

Courses					
WT 25/26	2581992	Risk Management in Industrial Supply Networks	2 SWS	Lecture / 🗣	Schultmann, Rosenberg
WT 26/27	2581992	Risk Management in Industrial Supply Networks	2 SWS	Lecture / 🗣	Schultmann, Rosenberg

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment consists of an oral (30 minutes) or written exam (60 minutes) (following §4(2) of the examination regulation). The exam takes place in every semester. Re-examinations are offered at every ordinary examination date.

Prerequisites
None

Recommendations
None

Below you will find excerpts from courses related to this module component:

V Risk Management in Industrial Supply Networks

Lecture (V)
On-Site

2581992, WS 25/26, 2 SWS, Language: English, Open in study portal

Content
Students learn methods and tools to manage risks in complex and dynamically evolving supply chain networks. In the first part of the lectures, students are introduced to the key terms and concepts of risk management and decision theory for industrial application. Based on the theoretic prerequisites, students are able to determine and analyze risk diversification, risk pooling and insurance mechanisms in supply chain network management. Lastly the lectures cover the differences and connection between risk management and resilience in industrial networks.

Literature
Wird in der Veranstaltung bekannt gegeben.

V Risk Management in Industrial Supply Networks

Lecture (V)
On-Site

2581992, WS 26/27, 2 SWS, Language: English, Open in study portal

Content
Students learn methods and tools to manage risks in complex and dynamically evolving supply chain networks. In the first part of the lectures, students are introduced to the key terms and concepts of risk management and decision theory for industrial application. Based on the theoretic prerequisites, students are able to determine and analyze risk diversification, risk pooling and insurance mechanisms in supply chain network management. Lastly the lectures cover the differences and connection between risk management and resilience in industrial networks.

Literature
Wird in der Veranstaltung bekannt gegeben.

T 6.629 Module component: Robotics, Automation, and Systems Control Lab [T-ETIT-114740]

Coordinators: Prof. Dr.-Ing. Mike Barth
Prof. Dr.-Ing. Sören Hohmann
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-107524 - Robotics, Automation, and Systems Control Lab

Type	Credits	Grading	Term offered	Version
Examination of another type	6 CP	graded	Each summer term	2

Courses					
WT 25/26	2303195	Robotics, Automation, and Systems Control Lab	4 SWS	Practical course / 🗣	Breisacher, Wagemann, Witucki
ST 2026	2303195	Robotics, Automation, and Systems Control Lab	4 SWS	Practical course / 🗣	Breisacher, Wagemann, Witucki

Legend: 🖥 Online, Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The examination takes place in the form of other types of examination. It consists of pitches after every thematic block, a final presentation and an oral examination in the amount of 20 minutes. The overall impression is evaluated.

The module grade results of the assessment of the pitches, presentation and the oral exam. Details will be given during the lecture at the beginning of the semester.

Prerequisites
none

Recommendations
Knowledge in automation, control and robotics in general. Recommended, but not mandatory, lectures are:

- Practical Tools for Control Engineers, PTCE – M-ETIT-106780
- Cyber Physical Modeling – M-ETIT-106953
- Seminar Industrial Process and Plant Engineering – M-ETIT-106970
- Optimal Control – M-ETIT-107497
- Multivariable Control Systems – M-ETIT-107515
- Nonlinear Control Systems - in preparation

Additional Information
Due to capacity reasons there are only 25 places available. If necessary, a selection procedure will be carried out. Selection criteria are the progress in studies and in the recommended lectures. The application window is from March 1 for March 31 in Campus Plus (https://plus.campus.kit.edu/). For further information and current news, regularly visit the courses website (https://www.irs.kit.edu/Lectures_5404.php).

T 6.630 Module component: Safety Engineering [T-MACH-105171]

Coordinators: Hans-Peter Kany
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101278 - Material Flow in Networked Logistic Systems
M-MACH-104888 - Advanced Module Logistics
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each winter term	2

Courses					
WT 25/26	2117061	Safety Engineering	2 SWS	Lecture / 🗣	Kany

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment consists of an oral exam (20 min.) taking place in the recess period according to § 4 paragraph 2 Nr. 2 of the examination regulation.

Prerequisites
none

Additional Information
The course is offered in German

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Safety Engineering

2117061, WS 25/26, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content
Media
Presentations

Learning content
The course provides basic knowledge of safety engineering. In particular the basics of health at the working place, job safety in Germany, national and European safety rules and the basics of safe machine design are covered. The implementation of these aspects will be illustrated by examples of material handling and storage technology. This course focuses on: basics of safety at work, safety regulations, basic safety principles of machine design, protection devices, system security with risk analysis, electronics in safety engineering, safety engineering for storage and material handling technique, electrical dangers and ergonomics. So, mainly, the technical measures of risk reduction in specific technical circumstances are covered.

Learning goals
The students are able to:

- Name and describe relevant safety concepts of safety engineering,
- Discuss basics of health at work and labour protection in Germany,
- Evaluate the basics for the safe methods of design of machinery with the national and European safety regulations and
- Realize these objectives by using examples in the field of storage and material handling systems.

Recommendations
None

Workload
Regular attendance: 21 hours
Self-study: 99 hours

Organizational issues
Termine: siehe ILIAS.

Literature
Defren/Wickert: Sicherheit für den Maschinen- und Anlagenbau, Druckerei und Verlag: H. von Ameln, Ratingen

T 6.631 Module component: Safety Management in Highway Engineering [T-BGU-101674]

Coordinators: Dr.-Ing. Matthias Zimmermann
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: M-BGU-101066 - Safety, Computing and Law in Highway Engineering
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Expansion	Version
Oral examination	3 CP	graded	Each term	1 semesters	1

Courses					
ST 2026	6233906	Safety Management in Highway Engineering	2 SWS	Lecture / Practice / 🗣	Zimmermann

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
oral exam with 15 minutes

Prerequisites
None

Recommendations
None

Additional Information
None

Workload
90 hours

T 6.632 Module component: Science Communication in the Innovation Process [T-WIWI-114748]

Coordinators: Dr. Mara Schwind
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101507 - Innovation Management
M-WIWI-101507 - Innovation Management
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Irregular	1

Courses					
WT 25/26	2500055	Science communication in the innovation process	2 SWS	Seminar / 🗣	Schwind, Weissenberger-Eibl, Schwind

Legend: 🖥 Online, ❂ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Alternative exam assessments. The grade consists of the following weighted components:

- 25% workshop
- 25% final presentation
- 50% seminar paper

Details will be announced at the beginning of the course.

Prerequisites
None

Recommendations
Prior attendance of the course Innovation Management [2545015] is recommended.

Additional Information
Teaching and learning format: Seminar

Workload
90 hours

Below you will find excerpts from courses related to this module component:

V Science communication in the innovation process

2500055, WS 25/26, 2 SWS, Language: German, Open in study portal

Seminar (S)
On-Site

Content
This seminar explores the role of science communication in the innovation process. Students will learn the basic definitions, concepts and models of science communication and critically examine them. They also explore the landscape of science communication actors, the distinction between communication for the common good and strategic communication, and the role of science communication in the acceptance of radical innovations. Through the critical analysis and reflection of science communication formats and their modes of action, students gain an understanding of proven and innovative communication strategies, learning how to adapt different formats to specific target groups and innovation phases.

As part of the seminar, students examine current, research-oriented issues independently and on the basis of scientific criteria. They research, analyze, abstract and reflect on relevant information. They draw their own conclusions using their interdisciplinary knowledge and develop current research results further in certain areas. They know how to validate the results obtained and present them logically and systematically in written and oral form, taking into account scientific methods (structuring, specialist terminology, citing sources). In doing so, they are able to argue professionally and defend the results in discussions with experts. In addition, students are familiar with the DFG Code of Conduct “Guidelines for Safeguarding Good Scientific Practice” and successfully apply these guidelines in the preparation of their scientific work.

T 6.633 Module component: Selected Applications of Technical Logistics [T-MACH-102160]

Coordinators: Dr.-Ing. Martin Mittwollen
Organisation: KIT Department of Mechanical Engineering

Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each summer term	1

Assessment
The assessment consists of an oral exam (20 min.) taking place in the recess period according to § 4 paragraph 2 Nr. 2 of the examination regulation.

Prerequisites
none

Recommendations
Knowledge out of Basics of Technical Logistics I (T-MACH-109919) / Elements and Systems of Technical Logistics (T-MACH-102159) preconditioned.

Workload
120 hours

T 6.634 Module component: Selected Applications of Technical Logistics - Project [T-MACH-108945]

Coordinators: Dr.-Ing. Martin Mittwollen
Organisation: KIT Department of Mechanical Engineering

Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Examination of another type	2 CP	graded	Each summer term	1

Assessment
presentation of performed project and defense (30min) according to $4 (2), No. 3 of the examination regulation

Prerequisites
T-MACH-102160 (selected applications of technical logistics) must have been started

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-MACH-102160 - Selected Applications of Technical Logistics must have been started.

Recommendations
Knowledge out of Basics of Technical Logistics I (T-MACH-109919) / Elements and Systems of Technical Logistics (T-MACH-102159) preconditioned.

Workload
60 hours

T 6.635 Module component: Selected Legal Issues of Internet Law [T-INFO-108462]

Coordinators: N.N.
Organisation: KIT Department of Informatics
Part of: M-INFO-101215 - Intellectual Property Law
M-WIWI-104903 - Law

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Each summer term	1

Courses					
ST 2026	24821	Selected legal issues of Internet law	2 SWS	Colloquium / 🗣	Sattler

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment is carried out as an examination of another type (§ 4 Abs. 2 No. 3 SPO).

The overall impression is evaluated. The following partial aspects are included in the grading: oral presentation and discussion.

Prerequisites
The course **Internet Law T-INFO-101307** must not have started.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-INFO-101307 - Internet Law must not have been started.

Recommendations
Keine.

Additional Information
Lecture (with written exam) **Internet Law** T-INFO-101307 is offered in the winter semester.

Colloquium (other type of examination) **Selected Legal Issues of Internet Law** T-INFO-108462 offered in the summer semester

T 6.636 Module component: Selected Topics in Statistics and Analytics: Seminal Papers in Theory and Methods [T-WIWI-114746]

Coordinators: Prof. Dr. Oliver Grothe
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101637 - Analytics and Statistics
M-WIWI-104902 - Statistics

Type	Credits	Grading	Term offered	Version
Examination of another type	4,5 CP	graded	Irregular	1

Courses					
WT 25/26	2500005	Selected Topics in Statistics and Analytics: Seminal Papers in Theory and Methods	3 SWS	Lecture / Practice / 🗣	Grothe

Legend: 🖥 Online, Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success is assessed in the form of an alternative exam assessment. The grade is composed of 30% active participation, 30% preparation of the sessions, and 40% written contributions during the semester (e.g. questions or reflections on the assigned papers).
Details on the structure of the performance assessment will be announced at the beginning of the course.

Prerequisites
None.

Recommendations
We recommend a sound statistical basis, such as that taught in the Advanced Statistics course.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Selected Topics in Statistics and Analytics: Seminal Papers in Theory and Methods
Lecture / Practice (VÜ)
On-Site

2500005, WS 25/26, 3 SWS, Language: German/English, Open in study portal

Content
In this course, 6–8 key scientific papers ("seminal works") from various areas of statistics and data analysis are read, analyzed, discussed, and optionally implemented. The goal is to develop and reflect on both core statistical concepts and scientific reading and writing skills.
The course is intended for advanced students with a solid background in statistics.

T 6.637 Module component: Selected topics of system integration for micro- and nanotechnology [T-MACH-105695]

Coordinators: apl. Prof. Dr. Ulrich Gengenbach
Prof. Dr. Veit Hagenmeyer
Dr. Liane Koker
apl. Prof. Dr. Ingo Sieber

Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101294 - Nanotechnology

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each winter term	2

Assessment
oral exam (Duration: 30min)

Prerequisites
T-MACH-108809 "Micro- and nanosystem integration for medical, fluidic and optical applications" must not be started.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-MACH-108809 - Micro- and Nanosystem Integration for Medical, Fluidic and Optical Applications must not have been started.

Additional Information
The course is offered in German.

Workload
120 hours

T 6.638 Module component: Selected Topics on Optics and Microoptics for Mechanical Engineers [T-MACH-102165]

Coordinators: Dr. Timo Mappes
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101290 - BioMEMS

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each term	1

Assessment
Oral examination

Prerequisites
none

T 6.639 Module component: Self-Booking-HOC-SPZ-FORUM-STK-Graded [T-WIWI-111439]

Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104910 - Interdisciplinary Qualifications

Type	Credits	Grading	Version
Examination of another type	2 CP	graded	2

Self Service Assignment of Supplementary Studies
This module component can be used for self service assignment of grades acquired from the following study providers:

- House of Competence
- Sprachenzentrum
- Studium Generale. Forum Wissenschaft und Gesellschaft (FORUM) (ehem. ZAK)
- Studienkolleg

Additional Information
Placeholder for self-booking of a graded interdisciplinary qualification, which was provided at the House of Competence, the "Sprachenzentrum" or the Center for Applied Cultural Studies and Studium Generale.

T 6.640 Module component: Self-Booking-HOC-SPZ-FORUM-STK-Graded [T-WIWI-113354]

Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104910 - Interdisciplinary Qualifications

Type	Credits	Grading	Version
Examination of another type	3 CP	graded	1

Self Service Assignment of Supplementary Studies
This module component can be used for self service assignment of grades acquired from the following study providers:

- House of Competence
- Sprachenzentrum
- Studium Generale. Forum Wissenschaft und Gesellschaft (FORUM) (ehem. ZAK)
- Studienkolleg

Additional Information
Placeholder for self-booking of a graded interdisciplinary qualification, which was provided at the House of Competence, the "Sprachenzentrum" or the Center for Applied Cultural Studies and Studium Generale.

T 6.641 Module component: Self-Booking-HOC-SPZ-FORUM-STK-Graded [T-WIWI-111440]

Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104910 - Interdisciplinary Qualifications

Type	Credits	Grading	Version
Examination of another type	3 CP	graded	2

Self Service Assignment of Supplementary Studies
This module component can be used for self service assignment of grades acquired from the following study providers:

- House of Competence
- Sprachenzentrum
- Studium Generale. Forum Wissenschaft und Gesellschaft (FORUM) (ehem. ZAK)
- Studienkolleg

Additional Information
Placeholder for self-booking of a graded interdisciplinary qualification, which was provided at the House of Competence, the "Sprachenzentrum" or the Center for Applied Cultural Studies and Studium Generale.

T 6.642 Module component: Self-Booking-HOC-SPZ-FORUM-STK-Graded [T-WIWI-111438]

Organisation: KIT Department of Business and Economics

Part of: M-WIWI-104910 - Interdisciplinary Qualifications

Type	Credits	Grading	Version
Examination of another type	1 CP	graded	2

Self Service Assignment of Supplementary Studies
This module component can be used for self service assignment of grades acquired from the following study providers:

- House of Competence
- Sprachenzentrum
- Studium Generale. Forum Wissenschaft und Gesellschaft (FORUM) (ehem. ZAK)
- Studienkolleg

Additional Information
Placeholder for self-booking of a graded interdisciplinary qualification, which was provided at the House of Competence, the "Sprachenzentrum" or the Center for Applied Cultural Studies and Studium Generale.

T 6.643 Module component: Self-Booking-HOC-SPZ-FORUM-STK-Graded [T-WIWI-113353]

Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104910 - Interdisciplinary Qualifications

Type	Credits	Grading	Version
Examination of another type	2 CP	graded	1

Self Service Assignment of Supplementary Studies
This module component can be used for self service assignment of grades acquired from the following study providers:

- House of Competence
- Sprachenzentrum
- Studium Generale. Forum Wissenschaft und Gesellschaft (FORUM) (ehem. ZAK)
- Studienkolleg

Additional Information
Placeholder for self-booking of a graded interdisciplinary qualification, which was provided at the House of Competence, the "Sprachenzentrum" or the Center for Applied Cultural Studies and Studium Generale.

T 6.644 Module component: Self-Booking-HOC-SPZ-FORUM-STK-Graded [T-WIWI-113352]

Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104910 - Interdisciplinary Qualifications

Type	Credits	Grading	Version
Examination of another type	1 CP	graded	1

Self Service Assignment of Supplementary Studies
This module component can be used for self service assignment of grades acquired from the following study providers:

- House of Competence
- Sprachenzentrum
- Studium Generale. Forum Wissenschaft und Gesellschaft (FORUM) (ehem. ZAK)
- Studienkolleg

Additional Information
Placeholder for self-booking of a graded interdisciplinary qualification, which was provided at the House of Competence, the "Sprachenzentrum" or the Center for Applied Cultural Studies and Studium Generale.

T 6.645 Module component: Self-Booking-HOC-SPZ-FORUM-STK-Ungraded [T-WIWI-113357]

Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104910 - Interdisciplinary Qualifications

Type	Credits	Grading	Version
Coursework	3 CP	pass/fail	1

Self Service Assignment of Supplementary Studies
This module component can be used for self service assignment of grades acquired from the following study providers:

- House of Competence
- Sprachenzentrum
- Studium Generale. Forum Wissenschaft und Gesellschaft (FORUM) (ehem. ZAK)
- Studienkolleg

Additional Information
Placeholder for self-booking of a ungraded interdisciplinary qualification, which was provided at the House of Competence, the "Sprachenzentrum" or the Center for Applied Cultural Studies and Studium Generale.

6.646 Module component: Self-Booking-HOC-SPZ-FORUM-STK-Ungraded [T-WIWI-113356]

Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104910 - Interdisciplinary Qualifications

Type	Credits	Grading	Version
Coursework	2 CP	pass/fail	1

Self Service Assignment of Supplementary Studies
This module component can be used for self service assignment of grades acquired from the following study providers:

- House of Competence
- Sprachenzentrum
- Studium Generale. Forum Wissenschaft und Gesellschaft (FORUM) (ehem. ZAK)
- Studienkolleg

Additional Information
Placeholder for self-booking of a ungraded interdisciplinary qualification, which was provided at the House of Competence, the "Sprachenzentrum" or the Center for Applied Cultural Studies and Studium Generale.

T 6.647 Module component: Self-Booking-HOC-SPZ-FORUM-STK-Ungraded [T-WIWI-113355]

Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104910 - Interdisciplinary Qualifications

Type	Credits	Grading	Version
Coursework	1 CP	pass/fail	1

Self Service Assignment of Supplementary Studies
This module component can be used for self service assignment of grades acquired from the following study providers:

- House of Competence
- Sprachenzentrum
- Studium Generale. Forum Wissenschaft und Gesellschaft (FORUM) (ehem. ZAK)
- Studienkolleg

Additional Information
Placeholder for self-booking of a ungraded interdisciplinary qualification, which was provided at the House of Competence, the "Sprachenzentrum" or the Center for Applied Cultural Studies and Studium Generale.

T 6.648 Module component: Self-Booking-HOC-SPZ-FORUM-STK-Ungraded [T-WIWI-111443]

Organisation: KIT Department of Business and Economics

Part of: M-WIWI-104910 - Interdisciplinary Qualifications

Type	Credits	Grading	Version
Coursework	3 CP	pass/fail	1

Self Service Assignment of Supplementary Studies

This module component can be used for self service assignment of grades acquired from the following study providers:

- House of Competence
- Sprachenzentrum
- Studium Generale. Forum Wissenschaft und Gesellschaft (FORUM) (ehem. ZAK)
- Studienkolleg

Additional Information

Placeholder for self-booking of a ungraded interdisciplinary qualification, which was provided at the House of Competence, the "Sprachenzentrum" or the Center for Applied Cultural Studies and Studium Generale.

T 6.649 Module component: Self-Booking-HOC-SPZ-FORUM-STK-Ungraded [T-WIWI-111442]

Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104910 - Interdisciplinary Qualifications

Type	Credits	Grading	Version
Coursework	2 CP	pass/fail	1

Self Service Assignment of Supplementary Studies
This module component can be used for self service assignment of grades acquired from the following study providers:

- House of Competence
- Sprachenzentrum
- Studium Generale. Forum Wissenschaft und Gesellschaft (FORUM) (ehem. ZAK)
- Studienkolleg

Additional Information
Placeholder for self-booking of a ungraded interdisciplinary qualification, which was provided at the House of Competence, the "Sprachenzentrum" or the Center for Applied Cultural Studies and Studium Generale.

6.650 Module component: Self-Booking-HOC-SPZ-FORUM-STK-Ungraded [T-WIWI-111441]

Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104910 - Interdisciplinary Qualifications

Type	Credits	Grading	Version
Coursework	1 CP	pass/fail	1

Self Service Assignment of Supplementary Studies
This module component can be used for self service assignment of grades acquired from the following study providers:

- House of Competence
- Sprachenzentrum
- Studium Generale. Forum Wissenschaft und Gesellschaft (FORUM) (ehem. ZAK)
- Studienkolleg

Additional Information
Placeholder for self-booking of a ungraded interdisciplinary qualification, which was provided at the House of Competence, the "Sprachenzentrum" or the Center for Applied Cultural Studies and Studium Generale.

T 6.651 Module component: Semantic Web Technologies [T-WIWI-110848]

Coordinators: Dr.-Ing. Tobias Käfer
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101456 - Intelligent Systems and Services
M-WIWI-101472 - Informatics
M-WIWI-101628 - Emphasis in Informatics
M-WIWI-104901 - Informatics (KIT-Department of Economics and Management)
M-WIWI-105366 - Artificial Intelligence
M-WIWI-106803 - Advanced Topics in AI: Knowledge Graphs and the Web

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each summer term	1

Courses					
ST 2026	2511310	Semantic Web Technologies	2 SWS	Lecture / 🗣	Käfer, Braun, Kubelka
ST 2026	2511311	Exercises to Semantic Web Technologies	1 SWS	Practice / 🗣	Käfer, Braun, Kubelka

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment consists of an 1h written exam following §4, Abs. 2, 1 of the examination regulation or of an oral exam (20 min) following §4, Abs. 2, 2 of the examination regulation.

The exam takes place every semester and can be repeated at every regular examination date.

Prerequisites
None

Recommendations
It is recommended that students have previously attended computer-science lectures from the Bachelor programmes Information Systems or Industrial Engineering and Management in semesters 1 – 4, or have taken equivalent courses.

Below you will find excerpts from courses related to this module component:

V Semantic Web Technologies

2511310, SS 2026, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content
The aim of the Semantic Web is to make the meaning (semantics) of data on the web usable in intelligent systems, e.g. in e-commerce and internet portals

Central concepts are the representation of knowledge in form of RDF and ontologies, the access via Linked Data, as well as querying the data by using SPARQL. This lecture provides the foundations of knowledge representation and processing for the corresponding technologies and presents example applications.

The following topics are covered:

- Resource Description Framework (RDF) and RDF Schema (RDFS)
- Web Architecture and Linked Data
- Web Ontology Language (OWL)
- Query language SPARQL
- Rule languages
- Applications

Learning objectives:

The student

- understands the motivation and foundational ideas behind Semantic Web and Linked Data technologies, and is able to analyse and realise systems
- demonstrates basic competency in the areas of data and system integration on the web
- masters advanced knowledge representation scenarios involving ontologies

Recommendations:

Lectures on Informatics of the Bachelor on Information Systems (Semester 1-4) or equivalent are required. Knowledge of modeling with UML is required.

Workload:

- The total workload for this course is approximately 135 hours
- Time of presentness: 45 hours
- Time of preperation and postprocessing: 60 hours
- Exam and exam preperation: 30 hours

Literature

- Pascal Hitzler, Markus Krötzsch, Sebastian Rudolph, York Sure: Semantic Web – Grundlagen. Springer, 2008.
- John Domingue, Dieter Fensel, James A. Hendler (Editors). Handbook of Semantic Web Technologies. Springer, 2011.

Weitere Literatur

- S. Staab, R. Studer (Editors). Handbook on Ontologies. International Handbooks in Information Systems. Springer, 2003.
- Tim Berners-Lee. Weaving the Web. Harper, 1999 geb. 2000 Taschenbuch.
- Ian Jacobs, Norman Walsh. Architecture of the World Wide Web, Volume One. W3C Recommendation 15 December 2004. http://www.w3.org/TR/webarch/
- Dean Allemang. Semantic Web for the Working Ontologist: Effective Modeling in RDFS and OWL. Morgan Kaufmann, 2008.
- Tom Heath and Chris Bizer. Linked Data: Evolving the Web into a Global Data Space. Synthesis Lectures on the Semantic Web: Theory and Technology, 2011.

V **Exercises to Semantic Web Technologies**
2511311, SS 2026, 1 SWS, Language: English, Open in study portal

Practice (Ü)
On-Site

Content
The exercises are related to the lecture Semantic Web Technologies.

Multiple exercises are held that capture the topics, held in the lecture Semantic Web Technologies, and discuss them in detail. Thereby, practical examples are given to the students in order to transfer theoretical aspects into practical implementation.

The following topics are covered:

- Resource Description Framework (RDF) and RDF Schema (RDFS)
- Web Architecture and Linked Data
- Web Ontology Language (OWL)
- Query language SPARQL
- Rule languages
- Applications

Learning objectives:

The student

- understands the motivation and foundational ideas behind Semantic Web and Linked Data technologies, and is able to analyse and realise systems
- demonstrates basic competency in the areas of data and system integration on the web
- masters advanced knowledge representation scenarios involving ontologies

Recommendations:

Lectures on Informatics of the Bachelor on Information Systems (Semester 1-4) or equivalent are required. Knowledge of modeling with UML is required.

Literature

- Pascal Hitzler, Markus Krötzsch, Sebastian Rudolph, York Sure: Semantic Web – Grundlagen. Springer, 2008.
- John Domingue, Dieter Fensel, James A. Hendler (Editors). Handbook of Semantic Web Technologies. Springer, 2011.

Weitere Literatur

- S. Staab, R. Studer (Editors). Handbook on Ontologies. International Handbooks in Information Systems. Springer, 2003.
- Tim Berners-Lee. Weaving the Web. Harper, 1999 geb. 2000 Taschenbuch.
- Ian Jacobs, Norman Walsh. Architecture of the World Wide Web, Volume One. W3C Recommendation 15 December 2004. http://www.w3.org/TR/webarch/
- Dean Allemang. Semantic Web for the Working Ontologist: Effective Modeling in RDFS and OWL. Morgan Kaufmann, 2008.
- Tom Heath and Chris Bizer. Linked Data: Evolving the Web into a Global Data Space. Synthesis Lectures on the Semantic Web: Theory and Technology, 2011.

T 6.652 Module component: Seminar Application of Artificial Intelligence in Production [T-MACH-112121]

Coordinators: Prof. Dr.-Ing. Jürgen Fleischer
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-106590 - Production Engineering

Type	Credits	Grading	Term offered	Version
Examination of another type	4 CP	graded	Each summer term	5

Courses					
ST 2026	2150910	Seminar Application of Artificial Intelligence in Production	2 SWS	Seminar / 🗣	Fleischer

Legend: 🖥 Online, ✥ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Alternative test achievement (graded):

- Presentation of the results (approx. 20 min) followed by a colloquium (approx. 15 min) with weighting 25%
- Written processing of the results with weighting 75%

Prerequisites
none

Recommendations
Previous participation in the lecture 2149921 "Artificial Intelligence in Production" or advanced knowledge of Python.

Additional Information
The course is offered in German.

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Seminar Application of Artificial Intelligence in Production

2150910, SS 2026, 2 SWS, Language: German, Open in study portal

Seminar (S)
On-Site

Content

The module AI in Production is designed to teach students the practical, holistic integration of machine learning methods and the application of artificial intelligence in production. The course is based on the phases of the CRISP-DM process with the aim of developing a deep understanding of the necessary steps and content aspects (methods) within the individual phases. In addition to teaching the practice-relevant aspects for integrating the most important methods of machine learning, the focus here is primarily on the necessary steps for data generation and data preparation, as well as the implementation and safeguarding of the methods in an industrial environment.

The lecture “Seminar on the Application of Artificial Intelligence in Production” aims at the practical integration of current methods of machine learning using realistic industrial use cases. The content of the course is based on the holistic, practical implementation of an AI project in production. In doing so, students solve a problem from a production context using methods of data analysis, processing and machine learning.

Learning Outcomes:

The students

- are able to independently analyze a practical problem in production with regard to the application of machine learning methods.
- will be able to independently apply common deep learning algorithms to practical data sets, validate them, and analyze the results.
- understand the challenges of using deep learning methods in production.
- will know the main action areas and open research questions for the successful implementation of AI in production and for the implementation of autonomous machines.
- are able to evaluate the results of current deep learning methods and, based on these, to develop and practically apply proposed solutions (from the field of machine learning).

Workload:

regular attendance: 21 hours
self-study: 99 hours

Organizational issues

Auftaktveranstaltung am 24.04.2026, 8:00 - 9:30 Uhr im Vorlesungsbereich der Karlsruher Forschungsfabrik (Geb. 70.41).

Alle nachfolgenden Termine werden über Ilias (https://ilias.studium.kit.edu/) bekanntgegeben.

Literature

Materialien zur Lehrveranstaltung werden über Ilias (https://ilias.studium.kit.edu/) bereitgestellt.

Course materials will be provided onIlias (https://ilias.studium.kit.edu/).

T 6.653 Module component: Seminar Creating a Patent Specification [T-ETIT-100754]

Coordinators: Prof. Dr. Wilhelm Stork
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Coursework	3 CP	pass/fail	Each summer term	1

Courses					
ST 2026	2311633	Seminar Creating a Patent Specification	2 SWS	Seminar / 🗣	Schmid, Stork

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Prerequisites
none

T 6.654 Module component: Seminar Data-Mining in Production [T-MACH-108737]

Coordinators: Prof. Dr.-Ing. Gisela Lanza
Organisation: KIT Department of Mechanical Engineering

Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Each winter term	3

Courses					
WT 25/26	2151643	Seminar Data Mining in Production	2 SWS	Seminar / 🗣	Lanza
WT 26/27	2151643	Seminar Data Mining in Production	2 SWS	Seminar / 🗣	Lanza

Legend: 🖥 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
alternative test achievement (graded):

- written elaboration (workload of at least 80 h)
- oral presentation (approx. 30 min)

Prerequisites
none

Additional Information
The course is offered in German.

The number of students is limited to twelve. Dates and deadlines for the seminar will be announced at https://www.wbk.kit.edu/studium-und-lehre.php.

Workload
90 hours

Below you will find excerpts from courses related to this module component:

V Seminar Data Mining in Production
2151643, WS 25/26, 2 SWS, Language: German, Open in study portal

Seminar (S)
On-Site

Content
In the age of Industry 4.0, large amounts of production data are generated by the global production networks and value chains. Their analysis enables valuable conclusions about production and lead to an increasing process efficiency. The aim of the seminar is to get to know production data analysis as an important component of future industrial projects. The students get to know the data mining tool KNIME and use it for analyses. A specific industrial use case with real production data enables practical work and offers direct references to industrial applications. The participants learn selected methods of data mining and apply them to the production data. The work within the seminar takes place in small groups on the computer. Subsequently, presentations on specific data mining methods have to be prepared.

Learning Outcomes:

The students …

- can name, describe and distinguish between different methods, procedures and techniques of production data analysis.
- can perform basic data analyses with the data mining tool KNIME.
- can analyze and evaluate the results of data analyses in the production environment.
- are able to derive suitable recommendations for action.
- are able to explain and apply the CRISP-DM model.

Workload:

regular attendance: 10 hours
self-study: 80 hours

Organizational issues
Die Teilnehmerzahl ist auf zwölf Studierende begrenzt. Termine und Fristen zur Veranstaltung werden unter https://www.wbk.kit.edu/studium-und-lehre.php bekanntgegeben.

The number of students is limited to twelve. Dates and deadlines for the seminar will be announced at https://www.wbk.kit.edu/studium-und-lehre.php.

Literature
Medien:
KNIME Analytics Platform

Media:
KNIME Analytics Platform

V Seminar Data Mining in Production

2151643, WS 26/27, 2 SWS, Language: German, Open in study portal

Seminar (S)
On-Site

Content
In the age of Industry 4.0, large amounts of production data are generated by the global production networks and value chains. Their analysis enables valuable conclusions about production and lead to an increasing process efficiency. The aim of the seminar is to get to know production data analysis as an important component of future industrial projects. The students get to know the data mining tool KNIME and use it for analyses. A specific industrial use case with real production data enables practical work and offers direct references to industrial applications. The participants learn selected methods of data mining and apply them to the production data. The work within the seminar takes place in small groups on the computer. Subsequently, presentations on specific data mining methods have to be prepared.

Learning Outcomes:

The students …

- can name, describe and distinguish between different methods, procedures and techniques of production data analysis.
- can perform basic data analyses with the data mining tool KNIME.
- can analyze and evaluate the results of data analyses in the production environment.
- are able to derive suitable recommendations for action.
- are able to explain and apply the CRISP-DM model.

Workload:

regular attendance: 10 hours
self-study: 80 hours

Organizational issues
Die Teilnehmerzahl ist auf zwölf Studierende begrenzt. Termine und Fristen zur Veranstaltung werden unter https://www.wbk.kit.edu/studium-und-lehre.php bekanntgegeben.

The number of students is limited to twelve. Dates and deadlines for the seminar will be announced at https://www.wbk.kit.edu/studium-und-lehre.php.

Literature
Medien:
KNIME Analytics Platform

Media:
KNIME Analytics Platform

T 6.655 Module component: Seminar Development of Automated Production Systems [T-MACH-113999]

Coordinators: Prof. Dr.-Ing. Jürgen Fleischer
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101298 - Automated Manufacturing Systems
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Examination of another type	4 CP	graded	Each winter term	1

Courses					
WT 25/26	2149911	Seminar Development of Automated Production Systems	2 SWS	Seminar / 🗣	Fleischer
WT 26/27	2149911	Seminar Development of Automated Production Systems	2 SWS	Seminar / 🗣	Fleischer

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Alternative test achievement (graded):

- Presentation of the results (approx. 20 min) followed by a colloquium (approx. 15 min) with weighting 25%
- Written processing of the results with weighting 75%

Prerequisites
T-MACH-108844 - Automated production systems must not be started

Additional Information
The course is offered in German.

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Seminar Development of Automated Production Systems

2149911, WS 25/26, 2 SWS, Language: German, Open in study portal

Seminar (S)
On-Site

Content
The course "Seminar Development of Automated Production Systems" aims at the practical project planning of automated production systems based on realistic industrial use cases. The content framework of the course results from the holistic, practical project planning task of an automated production plant. Firstly, the basics of production automation are taught as an introduction. The aspects of multi-machine systems and project planning are then examined in depth. An interdisciplinary approach to these sub-areas results in interfaces to Industry 4.0 approaches. The core of the course is the project planning of a use case based on the procedure taught. In doing so, the students should apply the methods taught in a problem-related and results-orientated manner and thus develop an automation solution.

Learning Outcomes:

The students...

- are able to name and describe the automation tasks in production plants and the components required for implementation.
- understand the challenges that can arise when using automation solutions in production.
- are able to independently analyse a practical problem in production with regard to the application of automation.
- are able to assess the results of automation problems and, based on this, develop and apply practical solutions.

Workload:

regular attendance: 21 hours
self-study: 99 hours

Literature
Medien:
Skript zur Veranstaltung wird über ilias (https://ilias.studium.kit.edu/)
bereitgestellt.

Media:
Lecture notes will be provided in ilias (https://ilias.studium.kit.edu/).

V **Seminar Development of Automated Production Systems**
2149911, WS 26/27, 2 SWS, Language: German, Open in study portal

Seminar (S)
On-Site

Content
The course "Seminar Development of Automated Production Systems" aims at the practical project planning of automated production systems based on realistic industrial use cases. The content framework of the course results from the holistic, practical project planning task of an automated production plant. Firstly, the basics of production automation are taught as an introduction. The aspects of multi-machine systems and project planning are then examined in depth. An interdisciplinary approach to these sub-areas results in interfaces to Industry 4.0 approaches. The core of the course is the project planning of a use case based on the procedure taught. In doing so, the students should apply the methods taught in a problem-related and results-orientated manner and thus develop an automation solution.

Learning Outcomes:

The students...

- are able to name and describe the automation tasks in production plants and the components required for implementation.
- understand the challenges that can arise when using automation solutions in production.
- are able to independently analyse a practical problem in production with regard to the application of automation.
- are able to assess the results of automation problems and, based on this, develop and apply practical solutions.

Workload:

regular attendance: 21 hours
self-study: 99 hours

Literature
Medien:
Skript zur Veranstaltung wird über ilias (https://ilias.studium.kit.edu/)
bereitgestellt.

Media:
Lecture notes will be provided in ilias (https://ilias.studium.kit.edu/).

T 6.656 Module component: Seminar in Business Administration (Bachelor) [T-WIWI-103486]

Coordinators: Professorenschaft des Fachbereichs Betriebswirtschaftslehre
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Each term	1

Courses					
WT 25/26	2500045	Digital Democracy - Challenges and Opportunities of the Digital Society	2 SWS	Seminar /	Fegert, Stein, Bezzaoui
WT 25/26	2500061	Special Topics in Transportation Strategy	2 SWS	Seminar /	Müller
WT 25/26	2500067	Seminar in Electrogastrogram and Economic Decision Making	2 SWS	Seminar /	Dharmapalan, Scheibehenne
WT 25/26	2500125	Human-Centered Systems Seminar: Engineering	2 SWS	Seminar /	Mädche
WT 25/26	2530580	Seminar in Finance (Bachelor)	2 SWS	Seminar /	Uhrig-Homburg, Thimme
WT 25/26	2530586		2 SWS	Seminar /	Uhrig-Homburg, Molnar
WT 25/26	2530610	Seminar in Financial Economics (Bachelor)	2 SWS	Seminar /	Thimme
WT 25/26	2540471	Bachelor Seminar (Information Systems III) – Sustainable Economies / More Human-like Conversational Agents	2 SWS	Seminar /	Bennardo, Gutschow, Heßler
WT 25/26	2540473	Business Data Analytics	2 SWS	Seminar /	Hariharan, Grote, Schulz, Motz
WT 25/26	2540475	Positive Information Systems	2 SWS	Seminar /	Knierim, del Puppo
WT 25/26	2540478	Smart Grids and Energy Markets	2 SWS	Seminar /	Weinhardt, Semmelmann, Miskiw, Speck
WT 25/26	2540524	Bachelor Seminar in Data Science and Machine Learning	2 SWS	Seminar	Geyer-Schulz, Nazemi
WT 25/26	2540557	Human-Centered Systems Seminar: Research	2 SWS	Seminar /	Mädche
WT 25/26	2571180	Seminar in Marketing and Sales (Bachelor)	2 SWS	Seminar /	Klarmann
WT 25/26	2573010	Seminar: Human Resources and Organizations (Bachelor)	2 SWS	Seminar /	Nieken, Mitarbeiter
WT 25/26	2573011	Seminar: Human Resource Management (Bachelor)	2 SWS	Seminar /	Nieken, Mitarbeiter
WT 25/26	2579919	Seminar Management Accounting - Sustainability Topics	2 SWS	Seminar /	Wouters, Dickemann
WT 25/26	2581030	Seminar in Energy Economics	2 SWS	Seminar /	Fichtner, Sloot
WT 25/26	2581032	Seminar Energy Economics VI	2 SWS	Seminar /	Fichtner, Slednev
WT 25/26	2581976	Seminar in Production and Operations Management I	2 SWS	Seminar /	Schultmann, Rudi
WT 25/26	2581977	Seminar in Production and Operations Management II	2 SWS	Seminar /	Schultmann, Steins
WT 25/26	2581978	Seminar in Production and Operations Management	2 SWS	Seminar /	Schultmann, Rosenberg
WT 25/26	2581979	Seminar in Energy Economics	2 SWS	Seminar /	Fichtner, Kleinebrahm
WT 25/26	2581980	Seminar in Energy Economics	2 SWS	Seminar /	Fichtner, Sandmeier

WT 25/26	2581981	Seminar in Energy Economics	2 SWS	Seminar /	Ardone, Fichtner
ST 2026	2500020	Digital Democracy - Challenges and opportunities of the digital society	2 SWS	Seminar /	Fegert
ST 2026	2500038	AI in Human Domains: Decision-Making, Oversight, and Accountability	2 SWS	Seminar /	Pfeiffer, Hüholt, Bennardo
ST 2026	2500061	Special Topics in Transportation Strategy	2 SWS	Seminar /	Müller
ST 2026	2500125	Human-Centered Systems Seminar: Engineering	3 SWS	Seminar /	Mädche
ST 2026	2530293	Seminar in Finance (Bachelor, Prof. Ruckes)	2 SWS	Seminar /	Ruckes, Luedecke, Benz, Kohl, Sarac, Fkyerat
ST 2026	2530610	Seminar Financial Economics	2 SWS	Seminar /	Thimme
ST 2026	2540468	Bachelor Seminar: Intelligent & Responsible Information Systems	2 SWS	Seminar /	Pfeiffer, Schlaich
ST 2026	2540473	Applied Responsible AI	2 SWS	Seminar	Schulz, Weinhardt
ST 2026	2540475	Positive Information Systems	2 SWS	Seminar	Knierim
ST 2026	2540478	Smart Grid Economics & Energy Markets	2 SWS	Seminar	Weinhardt
ST 2026	2540524	Bachelor Seminar in Data Science and Machine Learning	2 SWS	Seminar	Geyer-Schulz
ST 2026	2540553	Biosignal-Adaptive Systems Seminar	2 SWS	Seminar /	Mädche, Beigl
ST 2026	2540557	Human-Centered Systems Seminar: Research	3 SWS	Seminar /	Mädche
ST 2026	2571187	Seminar Digital Marketing (Bachelor)	2 SWS	Seminar /	Kupfer
ST 2026	2573010	Seminar Human Resources and Organizations (Bachelor)	2 SWS	Seminar /	Nieken, Mitarbeiter
ST 2026	2573011	Seminar Human Resource Management (Bachelor)	2 SWS	Seminar /	Nieken, Mitarbeiter, Gorny
ST 2026	2581030	Seminar Energiewirtschaft IV: Gerechtigkeitsaspekte in der Transformation des Energiesystems	2 SWS	Seminar /	Fichtner, Sloot
ST 2026	2581031	Seminar Energiewirtschaft V	2 SWS	Seminar /	Plötz
ST 2026	2581032	Seminar Energiewirtschaft VI	2 SWS	Seminar /	Slednev, Fichtner
ST 2026	2581978	Seminar Produktionswirtschaft und Logistik III	2 SWS	Seminar /	Schultmann, Rosenberg
ST 2026	2581979	Seminar Energiewirtschaft I	2 SWS	Seminar /	Fichtner, Kleinebrahm
ST 2026	2581980	Seminar Energiewirtschaft II: Sustainable Transformation of Energy Infrastructures	2 SWS	Seminar /	Fichtner, Sandmeier
ST 2026	2581981	Seminar Energiewirtschaft III	2 SWS	Seminar /	Ardone, Fichtner
WT 26/27	2500061	Special Topics in Transportation Strategy	2 SWS	Seminar /	Müller
WT 26/27	2530580	Seminar in Finance (Bachelor)	2 SWS	Seminar /	Uhrig-Homburg, Thimme
WT 26/27	2530586		2 SWS	Seminar /	Uhrig-Homburg, Molnar
WT 26/27	2530610	Seminar in Financial Economics (Bachelor)	2 SWS	Seminar /	Thimme
WT 26/27	2540471	Bachelor Seminar: Intelligent & Responsible Information Systems	2 SWS	Seminar /	Bennardo, Gutschow, Heßler
WT 26/27	2581030	Seminar in Energy Economics	2 SWS	Seminar /	Fichtner, Sloot
WT 26/27	2581032	Seminar Energy Economics VI	2 SWS	Seminar /	Fichtner, Slednev
WT 26/27	2581976	Seminar in Production and Operations Management I	2 SWS	Seminar /	Schultmann, Rudi

WT 26/27	2581977	Seminar in Production and Operations Management II	2 SWS	Seminar / 🗣	Schultmann, Steins
WT 26/27	2581978	Seminar in Production and Operations Management	2 SWS	Seminar / 🗣	Schultmann, Rosenberg
WT 26/27	2581979	Seminar in Energy Economics	2 SWS	Seminar / 🗣	Fichtner, Kleinebrahm
WT 26/27	2581980	Seminar in Energy Economics	2 SWS	Seminar / 🗣	Fichtner, Sandmeier
WT 26/27	2581981	Seminar in Energy Economics	2 SWS	Seminar / 🗣	Ardone, Fichtner

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Alternative exam assessment (§ 4(2), 3 SPO 2015). The following aspects are included:

- Regular participation in the seminar dates
- Preparation of a seminar paper on a partial aspect of the seminar topic according to scientific methods
- Lecture on the topic of the seminar paper.

The point scheme for the assessment is determined by the lecturer of the respective course. It will be announced at the beginning of the course.

Prerequisites
None.

Recommendations
See seminar description in the course catalogue of the KIT (https://campus.kit.edu/)

Additional Information
The listed seminar titles are placeholders. Currently offered seminars of each semester will be published on the websites of the institutes and in the course catalogue of the KIT. In general, the current seminar topics of each semester are already announced at the end of the previous semester. Furthermore for some seminars there is an application required.

The available places are listed on the internet: https://portal.wiwi.kit.edu.

Workload
90 hours

Below you will find excerpts from courses related to this module component:

V **Seminar in Electrogastrogram and Economic Decision Making** **Seminar (S) On-Site**
2500067, WS 25/26, 2 SWS, Language: English, Open in study portal

Content

The gut is often referred to as the "second brain" due to its substantial influence on cognition, emotion regulation, and mental health. This concept is grounded in the gut-brain axis (GBA), the bidirectional communication network between the central nervous system (CNS) and the enteric nervous system (ENS). These systems interact through the autonomic nervous system (ANS), and the hypothalamic-pituitary-adrenal (HPA) axis, along with a complex network of neuroimmune and neuroendocrine mediators (The Gut-Brain Axis: Influence of Microbiota on Mood and Mental Health, 2018).

While much of the existing literature emphasizes the role of gut microbiota, which provides retrospective insight into gut function and mental health, real-time physiological measures of gut activity remain relatively underexplored. In this context, electrogastrography (EGG) — a non-invasive method for recording gastric myoelectric activity — offers a novel tool to investigate the dynamic interplay between gut physiology and mental states, including economic decision-making.

The use of EGG in cognitive neuroscience and behavioral science is still emerging, providing space for critical inquiry and methodological development. This seminar is designed to explore the landscape of EGG research from technical, theoretical, and applied perspectives.

Seminar Structure
The seminar is organized around five core thematic areas. Each student will choose one topic and develop a focused literature review that generates both technical knowledge and real-world application.

Topics
1. Gut-Brain Axis and the Basics of EGG
Goal: Introduce the physiological, theoretical, and conceptual background for why we want to measure gut signals.

This topic offers a deep dive into the anatomy and physiology of the gut-brain axis — including the enteric and autonomic nervous systems, interoceptive processing, and affective-body theories such as the somatic marker hypothesis, embodied emotion, and homeostasis. Students will also learn the basics of gastric myoelectric activity (slow waves, frequency bands) and what exactly EGG measures, in contrast to pressure sensors, motility tracking, or microbiota analysis.

Key Readings:

Mayer et al. (2015) – Gut-brain axis
Herbert & Pollatos (2012) – Interoception
Damasio (1996) – Somatic marker hypothesis
Koch & Stern (2004) – Basics of EGG

2. EGG Sensors, Signal Acquisition, and Processing
Goal: Understand the technical underpinnings of how EGG data is captured and analyzed.

This topic covers the sensor and hardware ecosystem for EGG, including types of electrodes, placements, and configurations. It explores wearable vs. lab-based systems, amplifier design, signal acquisition standards, and detailed signal processing techniques (e.g., filtering, artifact rejection, power analysis, and normogastria). The emphasis is on methodological rigor and the challenges of working with noisy physiological data.

Key Readings:

Gopu et al. (2010) – Signal processing in EGG
Stern et al. (2007)

3. EGG in Behavioral, Affective, and Clinical Research
Goal: Explore how EGG is used in real-world studies across behavioral and medical domains.

This topic focuses on empirical applications of EGG in behavioral science, affective research, and clinical diagnostics. Topics include EGG studies on emotion regulation, interoception, and cognitive tasks, as well as clinical research in IBS, gastroparesis (Hasler et al., 2004), nausea, and psychiatry. Students will critically assess limitations in sample sizes, interpretability, and translational potential.

Key Readings:

Vianna & Tranel (2006)
Wolpert et al. (2020)

4. EGG in Multimodal Biosignal Contexts and Future Directions
Goal: Position EGG among other biosignals and critically reflect on the field’s limitations and future.

This module explores how EGG signal acquisition and processing compares to other biosignals such as ECG (heart rate variability), EDA (skin conductance), EEG (brain activity), and respiration in behavioural studies. Students will examine challenges of multimodal integration, signal synchrony, and interpretation. The topics include the use of wearables, AI-based classification, biofeedback, and the potential of open physiological datasets.

Key Readings:

Critchley et al. (2004)
Kim et al. (2021)
Wearable gut sensor papers - Vujic et al. (2020)

5. Multimodal Biosignals, Decision-Making, and Future Directions
Goal: Understand how EGG could capture aspects of decision-making in various contexts.

This topic explores the behavioral and cognitive implications of gut signals in decision-making tasks. Students will investigate studies linking gastric activity to risk-taking, delay discounting, and intuition, and how EGG data may reflect the visceral foundations of judgment. It also touches on EGG's potential in affective computing, real-time monitoring, and neuroeconomic models.

Key Readings:

Dunn et al. (2010)

Learning Outcomes
Throughout the course, students will sharpen key academic skills, including literature review and synthesis, critical analysis of research methods and findings, scientific writing, and clear communication of complex concepts. These skills will be developed through an in-depth exploration of electrogastrography (EGG), with a focus on its physiological and technological foundations, its relationship to gut-brain interactions, and its role in research on emotion, cognition, and decision-making. The course encourages students to engage critically with how EGG has been used so far, how it compares to other biosignals like ECG, EDA, and EEG, and how it might be applied in behavioural science and affective research. By the end of the seminar, students will contribute to a shared understanding of the field's current state, and reflect on future directions, challenges, and possibilities for applying EGG in both basic and applied research contexts.

Organizational issues
Application for this Seminar takes place exclusively via the WiWi portal during the specified timeframe. Please refrain from requests until the end of the application pahse.

There will be a kick-off and a final presentation.

Central criteria for evaluating your performance are the identification of relevant scientific sources, a systematization of literature, the highlighting of central findings, a clear, concise and intelligible presentation of results, and the identification of interesting research gaps.

- Di 4.11.2025, 15:45-17:15 -- Kick-off
- Di 10.02.2026, 9:00-12:15 – Abschlusspräsentationen
  Seminarraum Geb. 01.87 5B 25-26

V 2530586, WS 25/26, 2 SWS, Language: German, Open in study portal — **Seminar (S) On-Site**

Content
Within this seminar eLearning videos are produced to different topics out of the contents of our lectures. The student gets in touch with scientific work. Through profound working on a specific scientific topic the student is meant to learn the foundations of scientific research and reasoning in particular in finance.Through conduction of the video the student becomes familiar with the fundamental techniques for presentations and foundations of scientific reasoning. In addition, the student earns rhetorical skills.

The success is monitored by the development of an eLearning video and by the writing of a project report (according to §4(2), 3 SPO).

The overall grade is made up of these partial performances.

Recommendations:

Knowledge of the content of the modules *Essentials of Finance* [WW3BWLFBV1] (for bachelor students) and *F1 (Finance)* [WW4BWLFBV1] (for master students) is assumed.

The total workload for this course is approximately 90 hours. For further information see German version.

Organizational issues
Kickoff am 27.10.25 um 16 Uhr, Zwischenpräsentation am 09.12.25, 16 Uhr und Abschlusspräsentation am 27.01.26, 17:30 Uhr am Campus B (Geb. 09.21), Raum 209

V **Bachelor Seminar (Information Systems III) – Sustainable Economies / More Human-like Conversational Agents** — **Seminar (S) On-Site**
2540471, WS 25/26, 2 SWS, Language: English, Open in study portal

Content
This seminar is offered by the Information Systems III research group, which has been based at the Institute for Information Systems (WIN) since January 2025 and is led by Prof. Dr. Jella Pfeiffer. Further information about us can be found on our website (WIN - Information Systems III).

V **Business Data Analytics** — **Seminar (S) On-Site**
2540473, WS 25/26, 2 SWS, Language: German/English, Open in study portal

Content
wird auf deutsch und englisch gehalten

Organizational issues
Blockveranstaltung, siehe WWW

V	**Bachelor Seminar in Data Science and Machine Learning** 2540524, WS 25/26, 2 SWS, Language: German, Open in study portal	**Seminar (S)**

Literature
Weiterführende Literatur:

- W. Thomson. A Guide for the Young Economist. The MIT Press, 2001
- D.J. Brauner, H.-U. Vollmer. Erfolgreiches wissenschaftliches Arbeiten. Verlag Wissenschaft & Praxis, 2004
- University of Chicago Press. The Chicago Manual of Style. University of Chicago Press, 13th ed., 1982
- American Psychological Association. Concise of Rules of APA Style. American Psychological Association, 2005
- American Psychological Association. Publication Manual of the American Psychological Association. American Psychological Association, 2001

V	**Seminar: Human Resources and Organizations (Bachelor)** 2573010, WS 25/26, 2 SWS, Language: German, Open in study portal	**Seminar (S)** **On-Site**

Content
The topics are redefined each semester on basis of current research topics.The topics will be announced on the website of the Wiwi-Portal.

Aim

The student

- looks critically into current research topics in the fields of human resources and organizations.
- trains his / her presentation skills.
- learns to get his / her ideas and insights across in a focused and concise way, both in oral and written form, and to sum up the crucial facts.
- cultivates the discussion of research approaches.

Workload

The total workload for this course is: approximately 90 hours.

Lecture: 30h
Preparation of lecture: 45h
Exam preparation: 15h

Literature

Selected journal articles and books.

Organizational issues
Blockveranstaltung siehe Homepage

V	**Seminar: Human Resource Management (Bachelor)** 2573011, WS 25/26, 2 SWS, Language: German, Open in study portal	**Seminar (S)** **On-Site**

Content
The topics are redefined each semester on basis of current research topics. The topics will be announced on the website of the Wiwi-Portal.

Aim

The student

- looks critically into current research topics in the fields of Human Resource Management and Personnel Economics.
- trains his / her presentation skills.
- learns to get his / her ideas and insights across in a focused and concise way, both in oral and written form, and to sum up the crucial facts.
- cultivates the discussion of research approaches.

Workload

The total workload for this course is: approximately 90 hours.

Lecture: 30h
Preparation of lecture: 45h
Exam preparation: 15h

Literature

Selected journal articles and books.

Organizational issues
Blockveranstaltung siehe Homepage

V	**Seminar Management Accounting - Sustainability Topics** 2579919, WS 25/26, 2 SWS, Language: English, Open in study portal	**Seminar (S)** **On-Site**

Content
The course will be a mix of lectures, discussions, and student presentations. Students will write a paper in small groups, and present this in the final week. Topics are selectively prediscibed. The seminar course is concentrated in several meetings that are spread throughout the semester.

Learning objectives:

- Students are largely independently able to identify a distinct topic in Management Accounting,
- Students are capable to research the topic, analyze the information, to conceptualize and deduct fundamental principles and relationships from relatively unstructured information,
- Students can afterwards logically and systematically present the results in writing and as an oral presentation, following a scientific approach (structuring, terminology, sources.

Examination:

- The performance review is carried out in the form of a “Prüfungsleistung anderer Art” (following § 4 (2) No. 3 of the examination regulation), which in this case is an essay the seminar participants prepare in group work.
- The final grade is made up of the grade of the seminar paper, the presentation and the contributions in the seminar sessions.

Required prior Courses:

- The course requires a basic knowledge of finance and accounting.

Workload:

- The total workload for this course is approximately 90 hours. For further information see German version.

Note:

- Maximum of 8 students.

Organizational issues
Ort und Zeit werden noch bekannt gegeben bzw. über ILIAS

Literature
Will be announced in the course.

V	**Special Topics in Transportation Strategy** 2500061, SS 2026, 2 SWS, Language: English, Open in study portal	**Seminar (S)** **On-Site**

Content
Course Content:

This course covers a range of important strategic questions across different transportation industries. Using suitable models from the areas of strategy and management, the participants will learn to evaluate structures, market forces and dynamics across different transportation industries and their impact on corporate strategy. On the basis of this, they will derive precise and well-founded recommendations for action.

This course offers students the opportunity to engage with current strategic issues and sharpen their skills in strategic analysis and evaluation. Through intensive collaboration and the practical application of the knowledge they have learned, students are optimally preparing for the requirements and challenges of modern corporate management.

The seminar is offered by the Chair of Management (Prof. Dr. Hagen Lindstädt) and will be held by Dr. Jürgen Müller. Dr. Müller worked at McKinsey & Company for 20 years, where he worked in different transportation and infrastructure industries (rail, road, sea, air) on an international level. Since 2015, he is Executive Vice President for Strategy and Rolling stock at the Danish State Railways.

Topics:

1. **Deregulation**: Deregulation in the European passenger air and rail markets has resulted in very different market outcomes. Describe, compare and evaluate the underlying structures, the process of deregulation and its outcomes in the two industries and develop recommendations for the future.
2. **Concentration:** Why is the combined global market share of the top 5 players in container shipping vs. logistics so different? Describe and evaluate the causes and most important implications from the past. How will the topic develop in the future?
3. **Modal share:** Why ist there such a big difference in modal share between rail freight in the US vs. rail freight in Europe? Describe and evaluate the causes and most important implications from the past. What development do you foresee for the European market?
4. **Cyclicaty:** Why is there such a high cyclicaty in the global container shipping market? Describe and evaluate the causes and most important implications from the past. How will the topic develop in the future?
5. V**ertical integration:** Large logistics companies have chosen different strategies with regards to vertical integration between container shipping and land-based logistics (e.g. Maersk /K&N vs. DB Schenker / DSV). Describe and evaluate the resprective rationales and most important implications so far. How will the topic develop in the future?
6. **Profitability (I):** Why is average industry profitability of freight carriers so different across transport modes (road, rail, air and sea)? Assess the differences. Describe and evaluate the causes and most important implications from the past. How will the topic develop in the future?
7. **Pfrofitability (II):** "The quickest way to become a millionaire in the airline business is to start out as a billionaire" (Richard Branson). Analyze and discuss the truth of this statement and identify the underlying reasons. On this basis, derive recommendations for new entrants into the airline business.

Structure:

The course begins with an overarching introduction. Topics can be evaluated after the kickoff-meeting until 26.04.2026. Based on this, topics are assigned to groups of two. The main part of the course consists of writing and presenting a seminar paper - including a discussion of the results.

Target group: Bachelor students

Dates:

1. Kickoff: 24.04.2026,13:00 - 16:00 (remote via MS Teams)
2. Presentation Day 1: 22.06.2026, 14:00 - 18:00 (in person at IBU, building 05.20, room 2A-12.1)
3. Presentation Day 2: 23.06.2026, 09:00 - 12:00 (in person at IBU, building 05.20, room 2A-12.1)

Attendance is mandatory on all dates.

After completion of the course, the students are able to

- ... analyze complex industry structures and company situations, think strategically and derive well-founded recommendations.
- ... produce well-structured and convincing written papers that present the analysis and recommendations.
- ... present and discuss the results in an engaging and convincing manner and actively participate in discussions.

V	**Human-Centered Systems Seminar: Engineering** 2500125, SS 2026, 3 SWS, Language: English, Open in study portal	**Seminar (S)** **Blended (On-Site/Online)**

Content

Formerly known as "Current Topics in Digital Transformation"

With this seminar, we aim to provide students with the possibility to independently work on state-of-the-art research topics in addition to the knowledge gained in the lectures of the human-centered systems lab (Prof. Mädche). Students will work on a dedicated topic in the context of human-centered systems and apply a pre-defined research method. A broad spectrum of topics is offered every semester, topics may range from creating an experimental design, analyzing collected data, or systematically comparing existing software prototypes in a specific field of interest.

V	**Biosignal-Adaptive Systems Seminar** 2540553, SS 2026, 2 SWS, Language: English, Open in study portal	**Seminar (S)** **Blended (On-Site/Online)**

Content
Biosignal-adaptive systems collect and analyze biosignals from users to recognize user states as a basis for adaptation. Thermic, mechanical, electric, acoustic, and optical signals are collected using sensors which are integrated in wearables, e.g. glasses, earphones, belts, or bracelets. The collected data is processed with analytics and machine learning techniques in order to determine short-term, evolving over time, and long-term user states in the form of user characteristics, affective-cognitive states, or behavior. Finally, the recognized user states are leveraged for realizing user-centric adaptations.

In this seminar, interdisciplinary teams of students design, develop, and evaluate a biosignal-adaptive system prototype leveraging state-of-the-art hard- and software. This seminar follows an interdisciplinary approach. Students from the fields of computer science, information systems and industrial engineering & management collaborate in the prototype design, development, and evaluation.

The seminar is carried out in cooperation between Teco/Chair of Pervasive Computing Systems (Prof. Beigl) and the Institute of Information Systems and Marketing (h-lab, Prof. Mädche). It is offered as part of the DFG-funded graduate school “KD2School: Designing Adaptive Systems for Economic Decisions“ (https://kd2school.info/)

Learning objectives of the seminar

- Explain what a biosignal-adaptive system is and how it can be conceptualized
- Suggest and evaluate different design solutions for addressing the identified problem
- Build a biosignal-adaptive system prototype using state-of-the-art hard- and software
- Perform a user-centric evaluation of the biosignal-adaptive system prototype

Prerequisites

Strong analytical abilities and profound software development skills are required.

Organizational issues
Termine werden bekannt gegeben

Literature
Required literature will be made available in the seminar.

V	**Human-Centered Systems Seminar: Research** 2540557, SS 2026, 3 SWS, Language: English, Open in study portal	**Seminar (S)** **Blended (On-Site/Online)**

Content

Formerly known as "Information Systems and Service Design Seminar"

With this seminar, we aim to provide students with the possibility to independently work on state-of-the-art research topics in addition to the knowledge gained in the lectures of the research group IS I (Prof. Mädche). The research group “Information Systems I” (IS I) headed by Prof. Mädche focuses in research, education, and innovation on designing interactive intelligent systems. It is positioned at the intersection of Information Systems and Human-Computer Interaction (HCI).

In the seminar, participants will get deeper insights in a contemporary research topic in the field of information systems, specifically interactive intelligent systems.

The actual seminar topics will be derived from current research activities of the research group. Our research assistants offer a rich set of topics from our research clusters (digital experience and participation, intelligent enterprise systems, or digital services design & innovation). Students can select among these topics individually depending on their personal interests. The seminar is carried out in the form of a literature-based thesis project. In the seminar, students will acquire the important methodological skills of running a systematic literature review.

Learning Objectives

- focus on a contemporary topic at the intersection of Information Systems and Human-Computer Interaction (HCI), specifically interactive intelligent systems
- carry out a structured literature search for a given topic
- aggregate the collected information in a suitable way to present and extract knowledge
- write a seminar thesis following academic writing standards
- deliver a presentation in a scientific context in front of an auditorium

Prerequisites

No specific prerequisites are required for the seminar.

Literature

Further literature will be made available in the seminar.

Organizational issues

Termine werden bekannt gegeben

V **Seminar Human Resources and Organizations (Bachelor)** — **Seminar (S) On-Site**

2573010, SS 2026, 2 SWS, Language: German, Open in study portal

Content

The topics are redefined each semester on basis of current research topics.The topics will be announced on the website of the Wiwi-Portal.

Aim

The student

- looks critically into current research topics in the fields of human resources and organizations.
- trains his / her presentation skills.
- learns to get his / her ideas and insights across in a focused and concise way, both in oral and written form, and to sum up the crucial facts.
- cultivates the discussion of research approaches.

Workload

The total workload for this course is: approximately 90 hours.

Lecture: 30h
Preparation of lecture: 45h
Exam preparation: 15h

Literature

Selected journal articles and books.

Organizational issues

Geb. 05.20, Raum 2A-12.1, Termine werden bekannt gegeben

V	**Seminar Human Resource Management (Bachelor)** 2573011, SS 2026, 2 SWS, Language: German, Open in study portal	**Seminar (S)** **On-Site**

Content
The topics are redefined each semester on basis of current research topics. The topics will be announced on the website of the Wiwi-Portal.

Aim

The student

- looks critically into current research topics in the fields of Human Resource Management and Personnel Economics.
- trains his / her presentation skills.
- learns to get his / her ideas and insights across in a focused and concise way, both in oral and written form, and to sum up the crucial facts.
- cultivates the discussion of research approaches.

Workload

The total workload for this course is: approximately 90 hours.

Lecture: 30h
Preparation of lecture: 45h
Exam preparation: 15h

Literature

Selected journal articles and books.

Organizational issues
Geb. 05.20, Raum 2A-12.1, Termine werden bekannt gegeben

V	2530586, WS 26/27, 2 SWS, Language: German, Open in study portal	**Seminar (S)** **On-Site**

Content
Within this seminar eLearning videos are produced to different topics out of the contents of our lectures. The student gets in touch with scientific work. Through profound working on a specific scientific topic the student is meant to learn the foundations of scientific research and reasoning in particular in finance.Through conduction of the video the student becomes familiar with the fundamental techniques for presentations and foundations of scientific reasoning. In addition, the student earns rhetorical skills.

The success is monitored by the development of an eLearning video and by the writing of a project report (according to §4(2), 3 SPO).

The overall grade is made up of these partial performances.

Recommendations:

Knowledge of the content of the modules *Essentials of Finance* [WW3BWLFBV1] (for bachelor students) and *F1 (Finance)* [WW4BWLFBV1] (for master students) is assumed.

The total workload for this course is approximately 90 hours. For further information see German version.

Organizational issues
Kickoff am XX.10.26 um XX Uhr, Zwischenpräsentation am XX.12.26, XX Uhr und Abschlusspräsentation am XX.01.27, XX:XX Uhr am Campus B (Geb. 09.21), Raum 209

V	**Bachelor Seminar: Intelligent & Responsible Information Systems** 2540471, WS 26/27, 2 SWS, Language: English, Open in study portal	**Seminar (S)** **On-Site**

Content
This seminar is offered by the Information Systems III research group, which has been based at the Institute for Information Systems (WIN) since January 2025 and is led by Prof. Dr. Jella Pfeiffer. Further information about us can be found on our website (WIN - Information Systems III).

T 6.657 Module component: Seminar in Business Administration A (Master) [T-WIWI-103474]

Coordinators: Professorenschaft des Fachbereichs Betriebswirtschaftslehre
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration
M-WIWI-107204 - Seminar Module

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Each term	1

Courses					
WT 25/26	2500045	Digital Democracy - Challenges and Opportunities of the Digital Society	2 SWS	Seminar /	Fegert, Stein, Bezzaoui
WT 25/26	2500049	Seminar in Cognition and Consumer Behavior	2 SWS	Seminar /	Gradwohl
WT 25/26	2500055	Science communication in the innovation process	2 SWS	Seminar /	Schwind, Weissenberger-Eibl, Schwind
WT 25/26	2500125	Human-Centered Systems Seminar: Engineering	2 SWS	Seminar /	Mädche
WT 25/26	2530293		2 SWS	Seminar /	Ruckes, Luedecke, Kohl, Sarac, Fkyerat
WT 25/26	2530586		2 SWS	Seminar /	Uhrig-Homburg, Molnar
WT 25/26	2540472	Master Seminar (Information Systems III) – Future of Commerce / AI and Fairness	2 SWS	Seminar /	Bennardo, Gutschow, Heßler
WT 25/26	2540473	Business Data Analytics	2 SWS	Seminar /	Hariharan, Grote, Schulz, Motz
WT 25/26	2540475	Positive Information Systems	2 SWS	Seminar /	Knierim, del Puppo
WT 25/26	2540478	Smart Grids and Energy Markets	2 SWS	Seminar /	Weinhardt, Semmelmann, Miskiw, Speck
WT 25/26	2540510	Master Seminar in Data Science and Machine Learning	2 SWS	Seminar /	Geyer-Schulz, Nazemi
WT 25/26	2540557	Human-Centered Systems Seminar: Research	2 SWS	Seminar /	Mädche
WT 25/26	2545105	Case studies seminar: Innovation management	2 SWS	Seminar /	Weissenberger-Eibl
WT 25/26	2571181	Seminar Digital Marketing (Master)	2 SWS	Seminar /	Kupfer, Schmitt, Cinar
WT 25/26	2573012	Seminar Human Resource Management (Master)	2 SWS	Seminar /	Nieken, Mitarbeiter
WT 25/26	2573013	Seminar Human Resources and Organizations (Master)	2 SWS	Seminar /	Nieken, Mitarbeiter
WT 25/26	2579919	Seminar Management Accounting - Sustainability Topics	2 SWS	Seminar /	Wouters, Dickemann
WT 25/26	2581030	Seminar in Energy Economics	2 SWS	Seminar /	Fichtner, Sloot
WT 25/26	2581032	Seminar Energy Economics VI	2 SWS	Seminar /	Fichtner, Slednev
WT 25/26	2581976	Seminar in Production and Operations Management I	2 SWS	Seminar /	Schultmann, Rudi
WT 25/26	2581977	Seminar in Production and Operations Management II	2 SWS	Seminar /	Schultmann, Steins
WT 25/26	2581978	Seminar in Production and Operations Management	2 SWS	Seminar /	Schultmann, Rosenberg
WT 25/26	2581979	Seminar in Energy Economics	2 SWS	Seminar /	Fichtner, Kleinebrahm
WT 25/26	2581980	Seminar in Energy Economics	2 SWS	Seminar /	Fichtner, Sandmeier

WT 25/26	2581981	Seminar in Energy Economics	2 SWS	Seminar /	Ardone, Fichtner
ST 2026	2500018	Successful transformation through innovation	2 SWS	Seminar /	Busch
ST 2026	2500020	Digital Democracy - Challenges and opportunities of the digital society	2 SWS	Seminar /	Fegert
ST 2026	2500038	AI in Human Domains: Decision-Making, Oversight, and Accountability	2 SWS	Seminar /	Pfeiffer, Hüholt, Bennardo
ST 2026	2500125	Human-Centered Systems Seminar: Engineering	3 SWS	Seminar /	Mädche
ST 2026	2530580	Seminar in Finance (Master)	2 SWS	Seminar /	Uhrig-Homburg
ST 2026	2540469	Master Seminar: Intelligent & Responsible Information Systems	2 SWS	Seminar /	Heßler, Bennardo, Gutschow
ST 2026	2540473	Applied Responsible AI	2 SWS	Seminar	Schulz, Weinhardt
ST 2026	2540475	Positive Information Systems	2 SWS	Seminar	Knierim
ST 2026	2540478	Smart Grid Economics & Energy Markets	2 SWS	Seminar	Weinhardt
ST 2026	2540493	Data Science for Industrial Applications	2 SWS	Seminar /	Spitzer, Holstein, Hendriks
ST 2026	2540510	Master Seminar in Data Science and Machine Learning	2 SWS	Seminar	Geyer-Schulz
ST 2026	2540553	Biosignal-Adaptive Systems Seminar	2 SWS	Seminar /	Mädche, Beigl
ST 2026	2540557	Human-Centered Systems Seminar: Research	3 SWS	Seminar /	Mädche
ST 2026	2545002	Entrepreneurship Research	2 SWS	Seminar /	Kuschel
ST 2026	2571180	Seminar in Marketing and Sales (Master)	2 SWS	Seminar /	Klarmann, Mitarbeiter
ST 2026	2571182	Seminar “The Future of Marketing” (Master)	2 SWS	Seminar /	Kupfer
ST 2026	2573012	Seminar Human Resource Management (Master)	2 SWS	Seminar /	Nieken, Mitarbeiter, Gorny
ST 2026	2573013	Seminar Human Resources and Organizations (Master)	2 SWS	Seminar /	Nieken, Mitarbeiter
ST 2026	2581030	Seminar Energiewirtschaft IV: Gerechtigkeitsaspekte in der Transformation des Energiesystems	2 SWS	Seminar /	Fichtner, Sloot
ST 2026	2581031	Seminar Energiewirtschaft V	2 SWS	Seminar /	Plötz
ST 2026	2581032	Seminar Energiewirtschaft VI	2 SWS	Seminar /	Slednev, Fichtner
ST 2026	2581978	Seminar Produktionswirtschaft und Logistik III	2 SWS	Seminar /	Schultmann, Rosenberg
ST 2026	2581979	Seminar Energiewirtschaft I	2 SWS	Seminar /	Fichtner, Kleinebrahm
ST 2026	2581980	Seminar Energiewirtschaft II: Sustainable Transformation of Energy Infrastructures	2 SWS	Seminar /	Fichtner, Sandmeier
ST 2026	2581981	Seminar Energiewirtschaft III	2 SWS	Seminar /	Ardone, Fichtner
WT 26/27	2530293		2 SWS	Seminar /	Ruckes, Kohl, Sarac, Fkyerat, Liu
WT 26/27	2530586		2 SWS	Seminar /	Uhrig-Homburg, Molnar
WT 26/27	2540472	Master Seminar: Intelligent & Responsible Information Systems	2 SWS	Seminar /	Bennardo, Gutschow, Heßler
WT 26/27	2545105	Case studies seminar: Innovation management	2 SWS	Seminar /	Weissenberger-Eibl
WT 26/27	2581030	Seminar in Energy Economics	2 SWS	Seminar /	Fichtner, Sloot
WT 26/27	2581032	Seminar Energy Economics VI	2 SWS	Seminar /	Fichtner, Slednev

WT 26/27	2581976	Seminar in Production and Operations Management I	2 SWS	Seminar / 🗣	Schultmann, Rudi
WT 26/27	2581977	Seminar in Production and Operations Management II	2 SWS	Seminar / 🗣	Schultmann, Steins
WT 26/27	2581978	Seminar in Production and Operations Management	2 SWS	Seminar / 🗣	Schultmann, Rosenberg
WT 26/27	2581979	Seminar in Energy Economics	2 SWS	Seminar / 🗣	Fichtner, Kleinebrahm
WT 26/27	2581980	Seminar in Energy Economics	2 SWS	Seminar / 🗣	Fichtner, Sandmeier
WT 26/27	2581981	Seminar in Energy Economics	2 SWS	Seminar / 🗣	Ardone, Fichtner

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Alternative exam assessment (§ 4(2), 3 SPO 2015). The following aspects are included:

- Regular participation in the seminar dates
- Preparation of a seminar paper on a partial aspect of the seminar topic according to scientific methods
- Lecture on the topic of the seminar paper.

The point scheme for the assessment is determined by the lecturer of the respective course. It will be announced at the beginning of the course.

Prerequisites
None.

Recommendations
See seminar description in the course catalogue of the KIT (https://campus.kit.edu/)

Additional Information
The listed seminar titles are placeholders. Currently offered seminars of each semester will be published on the websites of the institutes and in the course catalogue of the KIT. In general, the current seminar topics of each semester are already announced at the end of the previous semester. Furthermore for some seminars there is an application required.

The available places are listed on the internet: https://portal.wiwi.kit.edu.

Workload
90 hours

Below you will find excerpts from courses related to this module component:

V **Science communication in the innovation process**
2500055, WS 25/26, 2 SWS, Language: German, Open in study portal
Seminar (S)
On-Site

Content
This seminar explores the role of science communication in the innovation process. Students will learn the basic definitions, concepts and models of science communication and critically examine them. They also explore the landscape of science communication actors, the distinction between communication for the common good and strategic communication, and the role of science communication in the acceptance of radical innovations. Through the critical analysis and reflection of science communication formats and their modes of action, students gain an understanding of proven and innovative communication strategies, learning how to adapt different formats to specific target groups and innovation phases.

As part of the seminar, students examine current, research-oriented issues independently and on the basis of scientific criteria. They research, analyze, abstract and reflect on relevant information. They draw their own conclusions using their interdisciplinary knowledge and develop current research results further in certain areas. They know how to validate the results obtained and present them logically and systematically in written and oral form, taking into account scientific methods (structuring, specialist terminology, citing sources). In doing so, they are able to argue professionally and defend the results in discussions with experts. In addition, students are familiar with the DFG Code of Conduct “Guidelines for Safeguarding Good Scientific Practice” and successfully apply these guidelines in the preparation of their scientific work.

V
2530586, WS 25/26, 2 SWS, Language: German, Open in study portal
Seminar (S)
On-Site

Content
Within this seminar eLearning videos are produced to different topics out of the contents of our lectures. The student gets in touch with scientific work. Through profound working on a specific scientific topic the student is meant to learn the foundations of scientific research and reasoning in particular in finance.Through conduction of the video the student becomes familiar with the fundamental techniques for presentations and foundations of scientific reasoning. In addition, the student earns rhetorical skills.

The success is monitored by the development of an eLearning video and by the writing of a project report (according to §4(2), 3 SPO).

The overall grade is made up of these partial performances.

Recommendations:

Knowledge of the content of the modules *Essentials of Finance* [WW3BWLFBV1] (for bachelor students) and *F1 (Finance)* [WW4BWLFBV1] (for master students) is assumed.

The total workload for this course is approximately 90 hours. For further information see German version.

Organizational issues
Kickoff am 27.10.25 um 16 Uhr, Zwischenpräsentation am 09.12.25, 16 Uhr und Abschlusspräsentation am 27.01.26, 17:30 Uhr am Campus B (Geb. 09.21), Raum 209

V	**Master Seminar (Information Systems III) – Future of Commerce / AI and Fairness** 2540472, WS 25/26, 2 SWS, Language: English, Open in study portal	**Seminar (S)** **On-Site**

Content
This seminar is offered by the Information Systems III research group, which has been based at the Institute for Information Systems (WIN) since January 2025 and is led by Prof. Dr. Jella Pfeiffer. Further information about us can be found on our website (WIN - Information Systems III).

V	**Business Data Analytics** 2540473, WS 25/26, 2 SWS, Language: German/English, Open in study portal	**Seminar (S)** **On-Site**

Content
wird auf deutsch und englisch gehalten

Organizational issues
Blockveranstaltung, siehe WWW

V	**Master Seminar in Data Science and Machine Learning** 2540510, WS 25/26, 2 SWS, Language: German, Open in study portal	**Seminar (S)** **Blended (On-Site/Online)**

V	**Case studies seminar: Innovation management** 2545105, WS 25/26, 2 SWS, Language: German, Open in study portal	**Seminar (S)** **On-Site**

Content
The objective of the seminar is to acquire selected concepts and methods of innovation management and to apply these in a practical way. The focus is on working on specific issues using a case study from business practice. Students work in groups and transfer the content taught, for example from the areas of scenario development, to entrepreneurial challenges in the context of fictitious case studies. The course follows a didactic interplay of theoretical input and direct application of what has been learned. At the end, the groups present their results in a plenary presentation, which is prepared by a short introduction to presentation techniques. The joint discussion promotes the transfer of knowledge to different contexts and strengthens the participants' ability to reflect.

As part of the seminar, students deal independently with current, research-oriented issues and systematically apply scientific criteria. They research relevant information, analyze, abstract and critically evaluate it. They draw conclusions independently from poorly structured data, incorporate interdisciplinary knowledge and contribute selectively to the further development of existing research perspectives. The results are validated and systematically prepared both orally and in writing using scientific standards - particularly with regard to structuring, terminology and correct citation of sources. In doing so, students argue in a technically sound manner and defend their findings in discourse with other participants and lecturers. They are also familiar with the DFG Code of Conduct “Guidelines for Safeguarding Good Scientific Practice” and apply these standards consistently in the preparation of their scientific work.

Literature
Werden in der ersten Veranstaltung bekannt gegeben.

V	**Seminar Human Resource Management (Master)** 2573012, WS 25/26, 2 SWS, Language: German, Open in study portal	**Seminar (S)** **On-Site**

Content
The topics are redefined each semester on basis of current research topics. The topics will be announced on the website of the Wiwi-Portal.

Aim

The student

- looks critically into current research topics in the fields of Human Resource Management and Personnel Economics.
- trains his / her presentation skills.
- learns to get his / her ideas and insights across in a focused and concise way, both in oral and written form, and to sum up the crucial facts.
- cultivates the discussion of research approaches.

Workload

The total workload for this course is: approximately 90 hours.

Lecture: 30h
Preparation of lecture: 45h
Exam preparation: 15h

Literature

Selected journal articles and books.

Organizational issues
Blockveranstaltung siehe Homepage

V	**Seminar Human Resources and Organizations (Master)** 2573013, WS 25/26, 2 SWS, Language: German, Open in study portal	**Seminar (S)** **On-Site**

Content
The topics are redefined each semester on basis of current research topics. The topics will be announced on the website of the Wiwi-Portal.

Aim

The student

- looks critically into current research topics in the fields of human resources and organizations.
- trains his / her presentation skills.
- learns to get his / her ideas and insights across in a focused and concise way, both in oral and written form, and to sum up the crucial facts.
- cultivates the discussion of research approaches.

Workload

The total workload for this course is: approximately 90 hours.

Lecture: 30h
Preparation of lecture: 45h
Exam preparation: 15h

Literature

Selected journal articles and books.

Organizational issues
Blockveranstaltung siehe Homepage

V	**Seminar Management Accounting - Sustainability Topics** 2579919, WS 25/26, 2 SWS, Language: English, Open in study portal	**Seminar (S)** **On-Site**

Content
The course will be a mix of lectures, discussions, and student presentations. Students will write a paper in small groups, and present this in the final week. Topics are selectively prediscibed. The seminar course is concentrated in several meetings that are spread throughout the semester.

Learning objectives:

- Students are largely independently able to identify a distinct topic in Management Accounting,
- Students are capable to research the topic, analyze the information, to conceptualize and deduct fundamental principles and relationships from relatively unstructured information,
- Students can afterwards logically and systematically present the results in writing and as an oral presentation, following a scientific approach (structuring, terminology, sources.

Examination:

- The performance review is carried out in the form of a “Prüfungsleistung anderer Art” (following § 4 (2) No. 3 of the examination regulation), which in this case is an essay the seminar participants prepare in group work.
- The final grade is made up of the grade of the seminar paper, the presentation and the contributions in the seminar sessions.

Required prior Courses:

- The course requires a basic knowledge of finance and accounting.

Workload:

- The total workload for this course is approximately 90 hours. For further information see German version.

Note:

- Maximum of 8 students.

Organizational issues
Ort und Zeit werden noch bekannt gegeben bzw. über ILIAS

Literature
Will be announced in the course.

V Successful transformation through innovation

2500018, SS 2026, 2 SWS, Language: German, Open in study portal

Seminar (S)
On-Site

Content
This seminar uses strategic innovation management and concepts such as organizational ambidexterity, boundary spanning and stakeholder approaches to shed light on how companies can increase their innovative capacity through innovation. Students will use a core paper to understand the steps a company takes to become an innovative organization. The aim is to understand how medium-sized companies can develop into innovation-driven organizations in the context of organizational inertia and path dependency with the help of the aforementioned concepts. Part of the seminar will be to analyze the role of different stakeholders and how companies can become part of innovation ecosystems.

Based on the impulses and the core paper, students will apply the concepts they have learned to selected companies and present their results. These are also incorporated into academic seminar papers. As part of this work process, students deal independently with a current, research-oriented question and apply scientific criteria consistently. They research relevant information, analyze and abstract it critically and draw their own conclusions on this basis, reflecting their interdisciplinary knowledge and selectively developing current research findings.

The results obtained are systematically presented in written and oral form in compliance with academic standards - including structured presentation, correct specialist terminology and comprehensible references. In doing so, the students argue in a technically sound manner and are able to defend their positions in scientific discourse with experts. In addition, they are familiar with the DFG Code of Conduct “Guidelines for Safeguarding Good Scientific Practice” and successfully apply these guidelines in the preparation of their scientific work.

Organizational issues
Weblink: https://itm.entechnon.kit.edu/192_1281.phphttps://itm.entechnon.kit.edu/192_1673.php

V Human-Centered Systems Seminar: Engineering

2500125, SS 2026, 3 SWS, Language: English, Open in study portal

Seminar (S)
Blended (On-Site/Online)

Content
Formerly known as "Current Topics in Digital Transformation"

With this seminar, we aim to provide students with the possibility to independently work on state-of-the-art research topics in addition to the knowledge gained in the lectures of the human-centered systems lab (Prof. Mädche). Students will work on a dedicated topic in the context of human-centered systems and apply a pre-defined research method. A broad spectrum of topics is offered every semester, topics may range from creating an experimental design, analyzing collected data, or systematically comparing existing software prototypes in a specific field of interest.

V **Data Science for Industrial Applications** — **Seminar (S)** **On-Site**
2540493, SS 2026, 2 SWS, Language: English, Open in study portal

Content
Learning Objectives

This seminar will require you to screen, select, and apply information systems theories and methodologies to solve contemporary challenges in the manufacturing and adjacent industries. This will include both critical reviews of the literature state-of-the-art [1-2] as well as the systematic conduct of design science research and machine learning methods [3-4]. You will identify key problems in real-world use cases, derive relevant research questions, and systematically gather, choose, and apply academic knowledge to develop solutions in the form of proof-of-concepts or prototypes.

Course Credits

The seminar can be credited as Seminar Betriebswirtschaftslehre A [T-WIWI-103474], Seminar Betriebswirtschaftslehre B [T-WIWI-103476] or Seminar Wirtschaftsinformatik [T-WIWI-109827] (3 ECTS). Other courses may be credited upon request.

Seminar Description

The Internet of Things (IoT) is significantly transforming industries such as automotive, healthcare, and energy. With the rise of ubiquitous computing power, connectivity/internet access, and the economic application of sensors [5], physical products are providing vast amounts of data, enabling the development of smart services [6]. While such IoT use cases are projected to open a market potential valued at $3.3 billion in 2030 [7], the industry is still far from exploiting its full capabilities. To solve this challenge, cutting-edge academic knowledge in information systems and machine learning is key to generating valuable insights from machine data.

The seminar is held in cooperation with international industry partners, who provide real-world datasets and ongoing access to subject matter experts. Students will work in teams of 2-4 on different topics and datasets. The assignments will be handed out in a joint kick-off event – to be scheduled once participating students have been selected. Attendance at this kick-off event is mandatory and a prerequisite for participation. Students are required to submit a seminar paper of 12-15 pages on an individual basis.

Expertise in Python and Data Science / Machine Learning as well as successful participation in the course “Artificial Intelligence in Service Systems” (T-WIWI-108715) are strongly recommended.

Contact

Daniel Hendricks – daniel.hendriks@kit.edu

Philipp Spitzer - philipp.spitzer@kit.edu

Joshua Holstein – joshua.holstein@kit.edu

The practical seminar will be held in English. Application documents can be handed in in English or German.

[1] Webster, J., Watson, R. T. (2002). Analyzing the Past to Prepare for the Future: Writing a Literature Review. MIS Quarterly, 26 (2) xiii-xxiii.

[2] Brocke, J. v. et al. (2009), Reconstructing the Giant: On the Importance of Rigour in Documenting the Literature Search Process. *Proceedings of the European Conference on Information Systems*, paper 161.

[3] Wirth, R., Hipp, J. (2000). CRISP-DM: Towards a Standard Process Model for Data Mining. Proceedings of the 4th International Conference on the Practical Applications of Knowledge Discovery and Data Mining, 29-40.

[4] Peffers, K., Tuunanen, T., Rothenberger, M., Chatterjee, S. (2008). A Design Science Research Methodology for Information Systems Research. Journal of Management Information Systems, 24 (3) 45–78.

[5] Martin, D.; Kühl, N.; Satzger, G. (2021). Virtual Sensors. Business & Information Systems Engineering, 63 (3) 315-323.

[6] Hunke, F., Heinz, D. Satzger, G. (2022). Creating customer value from data: foundations and archetypes of analytics-based services. Electronic Markets, 32, 503–521.

[7] Chui, M., Collins, M., Patel, M. (2021). IoT value set to accelerate through 2030: Where and how to capture it. McKinsey & Company. URL: https://www.mckinsey.com/capabilities/mckinsey-digital/our-insights/iot-value-set-to-accelerate-through-2030-where-and-how-to-capture-it

V **Master Seminar in Data Science and Machine Learning** — **Seminar (S)**
2540510, SS 2026, 2 SWS, Language: German/English, Open in study portal

V **Biosignal-Adaptive Systems Seminar**
2540553, SS 2026, 2 SWS, Language: English, Open in study portal

Seminar (S)
Blended (On-Site/Online)

Content
Biosignal-adaptive systems collect and analyze biosignals from users to recognize user states as a basis for adaptation. Thermic, mechanical, electric, acoustic, and optical signals are collected using sensors which are integrated in wearables, e.g. glasses, earphones, belts, or bracelets. The collected data is processed with analytics and machine learning techniques in order to determine short-term, evolving over time, and long-term user states in the form of user characteristics, affective-cognitive states, or behavior. Finally, the recognized user states are leveraged for realizing user-centric adaptations.

In this seminar, interdisciplinary teams of students design, develop, and evaluate a biosignal-adaptive system prototype leveraging state-of-the-art hard- and software. This seminar follows an interdisciplinary approach. Students from the fields of computer science, information systems and industrial engineering & management collaborate in the prototype design, development, and evaluation.

The seminar is carried out in cooperation between Teco/Chair of Pervasive Computing Systems (Prof. Beigl) and the Institute of Information Systems and Marketing (h-lab, Prof. Mädche). It is offered as part of the DFG-funded graduate school “KD2School: Designing Adaptive Systems for Economic Decisions“ (https://kd2school.info/)

Learning objectives of the seminar

- Explain what a biosignal-adaptive system is and how it can be conceptualized
- Suggest and evaluate different design solutions for addressing the identified problem
- Build a biosignal-adaptive system prototype using state-of-the-art hard- and software
- Perform a user-centric evaluation of the biosignal-adaptive system prototype

Prerequisites
Strong analytical abilities and profound software development skills are required.

Organizational issues
Termine werden bekannt gegeben

Literature
Required literature will be made available in the seminar.

V **Human-Centered Systems Seminar: Research**
2540557, SS 2026, 3 SWS, Language: English, Open in study portal

Seminar (S)
Blended (On-Site/Online)

Content
Formerly known as "Information Systems and Service Design Seminar"

With this seminar, we aim to provide students with the possibility to independently work on state-of-the-art research topics in addition to the knowledge gained in the lectures of the research group IS I (Prof. Mädche). The research group “Information Systems I” (IS I) headed by Prof. Mädche focuses in research, education, and innovation on designing interactive intelligent systems. It is positioned at the intersection of Information Systems and Human-Computer Interaction (HCI).

In the seminar, participants will get deeper insights in a contemporary research topic in the field of information systems, specifically interactive intelligent systems.

The actual seminar topics will be derived from current research activities of the research group. Our research assistants offer a rich set of topics from our research clusters (digital experience and participation, intelligent enterprise systems, or digital services design & innovation). Students can select among these topics individually depending on their personal interests. The seminar is carried out in the form of a literature-based thesis project. In the seminar, students will acquire the important methodological skills of running a systematic literature review.

Learning Objectives

- focus on a contemporary topic at the intersection of Information Systems and Human-Computer Interaction (HCI), specifically interactive intelligent systems
- carry out a structured literature search for a given topic
- aggregate the collected information in a suitable way to present and extract knowledge
- write a seminar thesis following academic writing standards
- deliver a presentation in a scientific context in front of an auditorium

Prerequisites

No specific prerequisites are required for the seminar.

Literature

Further literature will be made available in the seminar.

Organizational issues
Termine werden bekannt gegeben

V	**Entrepreneurship Research** 2545002, SS 2026, 2 SWS, Language: English, Open in study portal	**Seminar (S)** **On-Site**

Content
Content

In this course, the students choose from various relevant and current research topics in entrepreneurship and independently develop a topic that suits them in small teams. Initially, there is an introduction to standard methods such as systematic literature review, design science, qualitative and quantitative data analysis, and more. The seminar topic must be scientifically prepared and presented in 15-20 pages as part of a written elaboration. The seminar results are presented in a block event at the end of the semester (20 min + 10 min open discussion).

Learning Objectives

The foundations of independent scholarly work (literature review, argumentation + discussion, citation of literature sources, application of qualitative, quantitative, and simulation methods) are developed as part of the written elaboration. The competencies acquired in the seminar can be utilized in preparing for a potential master's thesis. Therefore, the seminar is mainly aimed at students who intend to write their thesis at the Chair of Entrepreneurship and Technology Management and wish to gain substantial experience in entrepreneurship research.

Organizational issues
Registration is via the Wiwi-Portal.

Seminar dates will probably be (as announced in Wiwi-Portal):

Wednesday, 29.04.2026, 10.00-16.00
Wednesday, 20.05.2026, 10.00-16.00
Wednesday, 24.06.2026, 09.00-12.00

Literature
Will be announced in the seminar.

V	**Seminar Human Resource Management (Master)** 2573012, SS 2026, 2 SWS, Language: German, Open in study portal	**Seminar (S)** **On-Site**

Content
The topics are redefined each semester on basis of current research topics. The topics will be announced on the website of the Wiwi-Portal.

Aim

The student

- looks critically into current research topics in the fields of Human Resource Management and Personnel Economics.
- trains his / her presentation skills.
- learns to get his / her ideas and insights across in a focused and concise way, both in oral and written form, and to sum up the crucial facts.
- cultivates the discussion of research approaches.

Workload

The total workload for this course is: approximately 90 hours.

Lecture: 30h
Preparation of lecture: 45h
Exam preparation: 15h

Literature

Selected journal articles and books.

Organizational issues
Geb. 05.20, Raum 2A-12.1, Termine werden bekannt gegeben

V **Seminar Human Resources and Organizations (Master)** — **Seminar (S) On-Site**
2573013, SS 2026, 2 SWS, Language: German, Open in study portal

Content
The topics are redefined each semester on basis of current research topics. The topics will be announced on the website of the Wiwi-Portal.

Aim

The student

- looks critically into current research topics in the fields of human resources and organizations.
- trains his / her presentation skills.
- learns to get his / her ideas and insights across in a focused and concise way, both in oral and written form, and to sum up the crucial facts.
- cultivates the discussion of research approaches.

Workload

The total workload for this course is: approximately 90 hours.

Lecture: 30h
Preparation of lecture: 45h
Exam preparation: 15h

Literature

Selected journal articles and books.

Organizational issues
Geb. 05.20, Raum 2A-12.1, Termine werden bekannt gegeben

V — **Seminar (S) On-Site**
2530586, WS 26/27, 2 SWS, Language: German, Open in study portal

Content
Within this seminar eLearning videos are produced to different topics out of the contents of our lectures. The student gets in touch with scientific work. Through profound working on a specific scientific topic the student is meant to learn the foundations of scientific research and reasoning in particular in finance.Through conduction of the video the student becomes familiar with the fundamental techniques for presentations and foundations of scientific reasoning. In addition, the student earns rhetorical skills.

The success is monitored by the development of an eLearning video and by the writing of a project report (according to §4(2), 3 SPO).

The overall grade is made up of these partial performances.

Recommendations:

Knowledge of the content of the modules *Essentials of Finance* [WW3BWLFBV1] (for bachelor students) and *F1 (Finance)* [WW4BWLFBV1] (for master students) is assumed.

The total workload for this course is approximately 90 hours. For further information see German version.

Organizational issues
Kickoff am XX.10.26 um XX Uhr, Zwischenpräsentation am XX.12.26, XX Uhr und Abschlusspräsentation am XX.01.27, XX:XX Uhr am Campus B (Geb. 09.21), Raum 209

V **Master Seminar: Intelligent & Responsible Information Systems** — **Seminar (S) On-Site**
2540472, WS 26/27, 2 SWS, Language: English, Open in study portal

Content
This seminar is offered by the Information Systems III research group, which has been based at the Institute for Information Systems (WIN) since January 2025 and is led by Prof. Dr. Jella Pfeiffer. Further information about us can be found on our website (WIN - Information Systems III).

V **Case studies seminar: Innovation management** — **Seminar (S) On-Site**
2545105, WS 26/27, 2 SWS, Language: German, Open in study portal

Content
The objective of the seminar is to acquire selected concepts and methods of innovation management and to apply these in a practical way. The focus is on working on specific issues using a case study from business practice. Students work in groups and transfer the content taught, for example from the areas of scenario development, to entrepreneurial challenges in the context of fictitious case studies. The course follows a didactic interplay of theoretical input and direct application of what has been learned. At the end, the groups present their results in a plenary presentation, which is prepared by a short introduction to presentation techniques. The joint discussion promotes the transfer of knowledge to different contexts and strengthens the participants' ability to reflect.

As part of the seminar, students deal independently with current, research-oriented issues and systematically apply scientific criteria. They research relevant information, analyze, abstract and critically evaluate it. They draw conclusions independently from poorly structured data, incorporate interdisciplinary knowledge and contribute selectively to the further development of existing research perspectives. The results are validated and systematically prepared both orally and in writing using scientific standards - particularly with regard to structuring, terminology and correct citation of sources. In doing so, students argue in a technically sound manner and defend their findings in discourse with other participants and lecturers. They are also familiar with the DFG Code of Conduct “Guidelines for Safeguarding Good Scientific Practice” and apply these standards consistently in the preparation of their scientific work.

Literature
Werden in der ersten Veranstaltung bekannt gegeben.

T 6.658 Module component: Seminar in Business Administration B (Master) [T-WIWI-103476]

Coordinators: Professorenschaft des Fachbereichs Betriebswirtschaftslehre
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration
M-WIWI-107204 - Seminar Module

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Each term	1

Courses					
WT 25/26	2500045	Digital Democracy - Challenges and Opportunities of the Digital Society	2 SWS	Seminar /	Fegert, Stein, Bezzaoui
WT 25/26	2500055	Science communication in the innovation process	2 SWS	Seminar /	Schwind, Weissenberger-Eibl, Schwind
WT 25/26	2500125	Human-Centered Systems Seminar: Engineering	2 SWS	Seminar /	Mädche
WT 25/26	2530293		2 SWS	Seminar /	Ruckes, Luedecke, Kohl, Sarac, Fkyerat
WT 25/26	2530586		2 SWS	Seminar /	Uhrig-Homburg, Molnar
WT 25/26	2540472	Master Seminar (Information Systems III) – Future of Commerce / AI and Fairness	2 SWS	Seminar /	Bennardo, Gutschow, Heßler
WT 25/26	2540473	Business Data Analytics	2 SWS	Seminar /	Hariharan, Grote, Schulz, Motz
WT 25/26	2540475	Positive Information Systems	2 SWS	Seminar /	Knierim, del Puppo
WT 25/26	2540478	Smart Grids and Energy Markets	2 SWS	Seminar /	Weinhardt, Semmelmann, Miskiw, Speck
WT 25/26	2540510	Master Seminar in Data Science and Machine Learning	2 SWS	Seminar /	Geyer-Schulz, Nazemi
WT 25/26	2540557	Human-Centered Systems Seminar: Research	2 SWS	Seminar /	Mädche
WT 25/26	2545105	Case studies seminar: Innovation management	2 SWS	Seminar /	Weissenberger-Eibl
WT 25/26	2571181	Seminar Digital Marketing (Master)	2 SWS	Seminar /	Kupfer, Schmitt, Cinar
WT 25/26	2573012	Seminar Human Resource Management (Master)	2 SWS	Seminar /	Nieken, Mitarbeiter
WT 25/26	2573013	Seminar Human Resources and Organizations (Master)	2 SWS	Seminar /	Nieken, Mitarbeiter
WT 25/26	2579919	Seminar Management Accounting - Sustainability Topics	2 SWS	Seminar /	Wouters, Dickemann
WT 25/26	2581030	Seminar in Energy Economics	2 SWS	Seminar /	Fichtner, Sloot
WT 25/26	2581032	Seminar Energy Economics VI	2 SWS	Seminar /	Fichtner, Slednev
WT 25/26	2581976	Seminar in Production and Operations Management I	2 SWS	Seminar /	Schultmann, Rudi
WT 25/26	2581977	Seminar in Production and Operations Management II	2 SWS	Seminar /	Schultmann, Steins
WT 25/26	2581978	Seminar in Production and Operations Management	2 SWS	Seminar /	Schultmann, Rosenberg
WT 25/26	2581979	Seminar in Energy Economics	2 SWS	Seminar /	Fichtner, Kleinebrahm
WT 25/26	2581980	Seminar in Energy Economics	2 SWS	Seminar /	Fichtner, Sandmeier
WT 25/26	2581981	Seminar in Energy Economics	2 SWS	Seminar /	Ardone, Fichtner

ST 2026	2500018	Successful transformation through innovation	2 SWS	Seminar /	Busch
ST 2026	2500020	Digital Democracy - Challenges and opportunities of the digital society	2 SWS	Seminar /	Fegert
ST 2026	2500038	AI in Human Domains: Decision-Making, Oversight, and Accountability	2 SWS	Seminar /	Pfeiffer, Hüholt, Bennardo
ST 2026	2500125	Human-Centered Systems Seminar: Engineering	3 SWS	Seminar /	Mädche
ST 2026	2530580	Seminar in Finance (Master)	2 SWS	Seminar /	Uhrig-Homburg
ST 2026	2540469	Master Seminar: Intelligent & Responsible Information Systems	2 SWS	Seminar /	Heßler, Bennardo, Gutschow
ST 2026	2540473	Applied Responsible AI	2 SWS	Seminar	Schulz, Weinhardt
ST 2026	2540475	Positive Information Systems	2 SWS	Seminar	Knierim
ST 2026	2540478	Smart Grid Economics & Energy Markets	2 SWS	Seminar	Weinhardt
ST 2026	2540493	Data Science for Industrial Applications	2 SWS	Seminar /	Spitzer, Holstein, Hendriks
ST 2026	2540510	Master Seminar in Data Science and Machine Learning	2 SWS	Seminar	Geyer-Schulz
ST 2026	2540553	Biosignal-Adaptive Systems Seminar	2 SWS	Seminar /	Mädche, Beigl
ST 2026	2540557	Human-Centered Systems Seminar: Research	3 SWS	Seminar /	Mädche
ST 2026	2545002	Entrepreneurship Research	2 SWS	Seminar /	Kuschel
ST 2026	2571180	Seminar in Marketing and Sales (Master)	2 SWS	Seminar /	Klarmann, Mitarbeiter
ST 2026	2571182	Seminar “The Future of Marketing” (Master)	2 SWS	Seminar /	Kupfer
ST 2026	2573012	Seminar Human Resource Management (Master)	2 SWS	Seminar /	Nieken, Mitarbeiter, Gorny
ST 2026	2573013	Seminar Human Resources and Organizations (Master)	2 SWS	Seminar /	Nieken, Mitarbeiter
ST 2026	2581030	Seminar Energiewirtschaft IV: Gerechtigkeitsaspekte in der Transformation des Energiesystems	2 SWS	Seminar /	Fichtner, Sloot
ST 2026	2581031	Seminar Energiewirtschaft V	2 SWS	Seminar /	Plötz
ST 2026	2581032	Seminar Energiewirtschaft VI	2 SWS	Seminar /	Slednev, Fichtner
ST 2026	2581978	Seminar Produktionswirtschaft und Logistik III	2 SWS	Seminar /	Schultmann, Rosenberg
ST 2026	2581979	Seminar Energiewirtschaft I	2 SWS	Seminar /	Fichtner, Kleinebrahm
ST 2026	2581980	Seminar Energiewirtschaft II: Sustainable Transformation of Energy Infrastructures	2 SWS	Seminar /	Fichtner, Sandmeier
ST 2026	2581981	Seminar Energiewirtschaft III	2 SWS	Seminar /	Ardone, Fichtner
WT 26/27	2530293		2 SWS	Seminar /	Ruckes, Kohl, Sarac, Fkyerat, Liu
WT 26/27	2530586		2 SWS	Seminar /	Uhrig-Homburg, Molnar
WT 26/27	2540472	Master Seminar: Intelligent & Responsible Information Systems	2 SWS	Seminar /	Bennardo, Gutschow, Heßler
WT 26/27	2545105	Case studies seminar: Innovation management	2 SWS	Seminar /	Weissenberger-Eibl
WT 26/27	2581030	Seminar in Energy Economics	2 SWS	Seminar /	Fichtner, Sloot
WT 26/27	2581032	Seminar Energy Economics VI	2 SWS	Seminar /	Fichtner, Slednev
WT 26/27	2581976	Seminar in Production and Operations Management I	2 SWS	Seminar /	Schultmann, Rudi

WT 26/27	2581977	Seminar in Production and Operations Management II	2 SWS	Seminar / 🗣	Schultmann, Steins
WT 26/27	2581978	Seminar in Production and Operations Management	2 SWS	Seminar / 🗣	Schultmann, Rosenberg
WT 26/27	2581979	Seminar in Energy Economics	2 SWS	Seminar / 🗣	Fichtner, Kleinebrahm
WT 26/27	2581980	Seminar in Energy Economics	2 SWS	Seminar / 🗣	Fichtner, Sandmeier
WT 26/27	2581981	Seminar in Energy Economics	2 SWS	Seminar / 🗣	Ardone, Fichtner

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Alternative exam assessment (§ 4(2), 3 SPO 2015). The following aspects are included:

- Regular participation in the seminar dates
- Preparation of a seminar paper on a partial aspect of the seminar topic according to scientific methods
- Lecture on the topic of the seminar paper.

The point scheme for the assessment is determined by the lecturer of the respective course. It will be announced at the beginning of the course.

Prerequisites
None.

Recommendations
See seminar description in the course catalogue of the KIT (https://campus.kit.edu/)

Additional Information
The listed seminar titles are placeholders. Currently offered seminars of each semester will be published on the websites of the institutes and in the course catalogue of the KIT. In general, the current seminar topics of each semester are already announced at the end of the previous semester. Furthermore for some seminars there is an application required.

The available places are listed on the internet: https://portal.wiwi.kit.edu.

Workload
90 hours

Below you will find excerpts from courses related to this module component:

V **Science communication in the innovation process** — **Seminar (S) On-Site**
2500055, WS 25/26, 2 SWS, Language: German, Open in study portal

Content
This seminar explores the role of science communication in the innovation process. Students will learn the basic definitions, concepts and models of science communication and critically examine them. They also explore the landscape of science communication actors, the distinction between communication for the common good and strategic communication, and the role of science communication in the acceptance of radical innovations. Through the critical analysis and reflection of science communication formats and their modes of action, students gain an understanding of proven and innovative communication strategies, learning how to adapt different formats to specific target groups and innovation phases.

As part of the seminar, students examine current, research-oriented issues independently and on the basis of scientific criteria. They research, analyze, abstract and reflect on relevant information. They draw their own conclusions using their interdisciplinary knowledge and develop current research results further in certain areas. They know how to validate the results obtained and present them logically and systematically in written and oral form, taking into account scientific methods (structuring, specialist terminology, citing sources). In doing so, they are able to argue professionally and defend the results in discussions with experts. In addition, students are familiar with the DFG Code of Conduct “Guidelines for Safeguarding Good Scientific Practice” and successfully apply these guidelines in the preparation of their scientific work.

V — **Seminar (S) On-Site**
2530586, WS 25/26, 2 SWS, Language: German, Open in study portal

Content
Within this seminar eLearning videos are produced to different topics out of the contents of our lectures. The student gets in touch with scientific work. Through profound working on a specific scientific topic the student is meant to learn the foundations of scientific research and reasoning in particular in finance.Through conduction of the video the student becomes familiar with the fundamental techniques for presentations and foundations of scientific reasoning. In addition, the student earns rhetorical skills.

The success is monitored by the development of an eLearning video and by the writing of a project report (according to §4(2), 3 SPO).

The overall grade is made up of these partial performances.

Recommendations:

Knowledge of the content of the modules *Essentials of Finance* [WW3BWLFBV1] (for bachelor students) and *F1 (Finance)* [WW4BWLFBV1] (for master students) is assumed.

The total workload for this course is approximately 90 hours. For further information see German version.

Organizational issues
Kickoff am 27.10.25 um 16 Uhr, Zwischenpräsentation am 09.12.25, 16 Uhr und Abschlusspräsentation am 27.01.26, 17:30 Uhr am Campus B (Geb. 09.21), Raum 209

V	**Master Seminar (Information Systems III) – Future of Commerce / AI and Fairness** 2540472, WS 25/26, 2 SWS, Language: English, Open in study portal	**Seminar (S)** **On-Site**

Content
This seminar is offered by the Information Systems III research group, which has been based at the Institute for Information Systems (WIN) since January 2025 and is led by Prof. Dr. Jella Pfeiffer. Further information about us can be found on our website (WIN - Information Systems III).

V	**Business Data Analytics** 2540473, WS 25/26, 2 SWS, Language: German/English, Open in study portal	**Seminar (S)** **On-Site**

Content
wird auf deutsch und englisch gehalten

Organizational issues
Blockveranstaltung, siehe WWW

V	**Master Seminar in Data Science and Machine Learning** 2540510, WS 25/26, 2 SWS, Language: German, Open in study portal	**Seminar (S)** **Blended (On-Site/Online)**

V	**Case studies seminar: Innovation management** 2545105, WS 25/26, 2 SWS, Language: German, Open in study portal	**Seminar (S)** **On-Site**

Content
The objective of the seminar is to acquire selected concepts and methods of innovation management and to apply these in a practical way. The focus is on working on specific issues using a case study from business practice. Students work in groups and transfer the content taught, for example from the areas of scenario development, to entrepreneurial challenges in the context of fictitious case studies. The course follows a didactic interplay of theoretical input and direct application of what has been learned. At the end, the groups present their results in a plenary presentation, which is prepared by a short introduction to presentation techniques. The joint discussion promotes the transfer of knowledge to different contexts and strengthens the participants' ability to reflect.

As part of the seminar, students deal independently with current, research-oriented issues and systematically apply scientific criteria. They research relevant information, analyze, abstract and critically evaluate it. They draw conclusions independently from poorly structured data, incorporate interdisciplinary knowledge and contribute selectively to the further development of existing research perspectives. The results are validated and systematically prepared both orally and in writing using scientific standards - particularly with regard to structuring, terminology and correct citation of sources. In doing so, students argue in a technically sound manner and defend their findings in discourse with other participants and lecturers. They are also familiar with the DFG Code of Conduct “Guidelines for Safeguarding Good Scientific Practice” and apply these standards consistently in the preparation of their scientific work.

Literature
Werden in der ersten Veranstaltung bekannt gegeben.

V
Seminar Human Resource Management (Master)
2573012, WS 25/26, 2 SWS, Language: German, Open in study portal

Seminar (S)
On-Site

Content
The topics are redefined each semester on basis of current research topics. The topics will be announced on the website of the Wiwi-Portal.

Aim

The student

- looks critically into current research topics in the fields of Human Resource Management and Personnel Economics.
- trains his / her presentation skills.
- learns to get his / her ideas and insights across in a focused and concise way, both in oral and written form, and to sum up the crucial facts.
- cultivates the discussion of research approaches.

Workload

The total workload for this course is: approximately 90 hours.

Lecture: 30h
Preparation of lecture: 45h
Exam preparation: 15h

Literature

Selected journal articles and books.

Organizational issues
Blockveranstaltung siehe Homepage

V
Seminar Human Resources and Organizations (Master)
2573013, WS 25/26, 2 SWS, Language: German, Open in study portal

Seminar (S)
On-Site

Content
The topics are redefined each semester on basis of current research topics. The topics will be announced on the website of the Wiwi-Portal.

Aim

The student

- looks critically into current research topics in the fields of human resources and organizations.
- trains his / her presentation skills.
- learns to get his / her ideas and insights across in a focused and concise way, both in oral and written form, and to sum up the crucial facts.
- cultivates the discussion of research approaches.

Workload

The total workload for this course is: approximately 90 hours.

Lecture: 30h
Preparation of lecture: 45h
Exam preparation: 15h

Literature

Selected journal articles and books.

Organizational issues
Blockveranstaltung siehe Homepage

V
Seminar Management Accounting - Sustainability Topics
2579919, WS 25/26, 2 SWS, Language: English, Open in study portal

Seminar (S)
On-Site

Content
The course will be a mix of lectures, discussions, and student presentations. Students will write a paper in small groups, and present this in the final week. Topics are selectively prediscibed. The seminar course is concentrated in several meetings that are spread throughout the semester.

Learning objectives:

- Students are largely independently able to identify a distinct topic in Management Accounting,
- Students are capable to research the topic, analyze the information, to conceptualize and deduct fundamental principles and relationships from relatively unstructured information,
- Students can afterwards logically and systematically present the results in writing and as an oral presentation, following a scientific approach (structuring, terminology, sources.

Examination:

- The performance review is carried out in the form of a "Prüfungsleistung anderer Art" (following § 4 (2) No. 3 of the examination regulation), which in this case is an essay the seminar participants prepare in group work.
- The final grade is made up of the grade of the seminar paper, the presentation and the contributions in the seminar sessions.

Required prior Courses:

- The course requires a basic knowledge of finance and accounting.

Workload:

- The total workload for this course is approximately 90 hours. For further information see German version.

Note:

- Maximum of 8 students.

Organizational issues
Ort und Zeit werden noch bekannt gegeben bzw. über ILIAS

Literature
Will be announced in the course.

V	**Successful transformation through innovation** 2500018, SS 2026, 2 SWS, Language: German, Open in study portal	**Seminar (S)** **On-Site**

Content
This seminar uses strategic innovation management and concepts such as organizational ambidexterity, boundary spanning and stakeholder approaches to shed light on how companies can increase their innovative capacity through innovation. Students will use a core paper to understand the steps a company takes to become an innovative organization. The aim is to understand how medium-sized companies can develop into innovation-driven organizations in the context of organizational inertia and path dependency with the help of the aforementioned concepts. Part of the seminar will be to analyze the role of different stakeholders and how companies can become part of innovation ecosystems.

Based on the impulses and the core paper, students will apply the concepts they have learned to selected companies and present their results. These are also incorporated into academic seminar papers. As part of this work process, students deal independently with a current, research-oriented question and apply scientific criteria consistently. They research relevant information, analyze and abstract it critically and draw their own conclusions on this basis, reflecting their interdisciplinary knowledge and selectively developing current research findings.

The results obtained are systematically presented in written and oral form in compliance with academic standards - including structured presentation, correct specialist terminology and comprehensible references. In doing so, the students argue in a technically sound manner and are able to defend their positions in scientific discourse with experts. In addition, they are familiar with the DFG Code of Conduct "Guidelines for Safeguarding Good Scientific Practice" and successfully apply these guidelines in the preparation of their scientific work.

Organizational issues
Weblink: https://itm.entechnon.kit.edu/192_1281.phphttps://itm.entechnon.kit.edu/192_1673.php

V	**Human-Centered Systems Seminar: Engineering** 2500125, SS 2026, 3 SWS, Language: English, Open in study portal	**Seminar (S)** **Blended (On-Site/Online)**

Content
Formerly known as "Current Topics in Digital Transformation"

With this seminar, we aim to provide students with the possibility to independently work on state-of-the-art research topics in addition to the knowledge gained in the lectures of the human-centered systems lab (Prof. Mädche). Students will work on a dedicated topic in the context of human-centered systems and apply a pre-defined research method. A broad spectrum of topics is offered every semester, topics may range from creating an experimental design, analyzing collected data, or systematically comparing existing software prototypes in a specific field of interest.

V	**Data Science for Industrial Applications** 2540493, SS 2026, 2 SWS, Language: English, Open in study portal	**Seminar (S)** **On-Site**

Content
Learning Objectives

This seminar will require you to screen, select, and apply information systems theories and methodologies to solve contemporary challenges in the manufacturing and adjacent industries. This will include both critical reviews of the literature state-of-the-art [1-2] as well as the systematic conduct of design science research and machine learning methods [3-4]. You will identify key problems in real-world use cases, derive relevant research questions, and systematically gather, choose, and apply academic knowledge to develop solutions in the form of proof-of-concepts or prototypes.

Course Credits

The seminar can be credited as Seminar Betriebswirtschaftslehre A [T-WIWI-103474], Seminar Betriebswirtschaftslehre B [T-WIWI-103476] or Seminar Wirtschaftsinformatik [T-WIWI-109827] (3 ECTS). Other courses may be credited upon request.

Seminar Description

The Internet of Things (IoT) is significantly transforming industries such as automotive, healthcare, and energy. With the rise of ubiquitous computing power, connectivity/internet access, and the economic application of sensors [5], physical products are providing vast amounts of data, enabling the development of smart services [6]. While such IoT use cases are projected to open a market potential valued at $3.3 billion in 2030 [7], the industry is still far from exploiting its full capabilities. To solve this challenge, cutting-edge academic knowledge in information systems and machine learning is key to generating valuable insights from machine data.

The seminar is held in cooperation with international industry partners, who provide real-world datasets and ongoing access to subject matter experts. Students will work in teams of 2-4 on different topics and datasets. The assignments will be handed out in a joint kick-off event – to be scheduled once participating students have been selected. Attendance at this kick-off event is mandatory and a prerequisite for participation. Students are required to submit a seminar paper of 12-15 pages on an individual basis.

Expertise in Python and Data Science / Machine Learning as well as successful participation in the course “Artificial Intelligence in Service Systems” (T-WIWI-108715) are strongly recommended.

Contact

Daniel Hendricks – daniel.hendriks@kit.edu

Philipp Spitzer - philipp.spitzer@kit.edu

Joshua Holstein – joshua.holstein@kit.edu

The practical seminar will be held in English. Application documents can be handed in in English or German.

[1] Webster, J., Watson, R. T. (2002). Analyzing the Past to Prepare for the Future: Writing a Literature Review. MIS Quarterly, 26 (2) xiii-xxiii.

[2] Brocke, J. v. et al. (2009), Reconstructing the Giant: On the Importance of Rigour in Documenting the Literature Search Process. *Proceedings of the European Conference on Information Systems*, paper 161.

[3] Wirth, R., Hipp, J. (2000). CRISP-DM: Towards a Standard Process Model for Data Mining. Proceedings of the 4th International Conference on the Practical Applications of Knowledge Discovery and Data Mining, 29-40.

[4] Peffers, K., Tuunanen, T., Rothenberger, M., Chatterjee, S. (2008). A Design Science Research Methodology for Information Systems Research. Journal of Management Information Systems, 24 (3) 45–78.

[5] Martin, D.; Kühl, N.; Satzger, G. (2021). Virtual Sensors. Business & Information Systems Engineering, 63 (3) 315-323.

[6] Hunke, F., Heinz, D. Satzger, G. (2022). Creating customer value from data: foundations and archetypes of analytics-based services. Electronic Markets, 32, 503–521.

[7] Chui, M., Collins, M., Patel, M. (2021). IoT value set to accelerate through 2030: Where and how to capture it. McKinsey & Company. URL: https://www.mckinsey.com/capabilities/mckinsey-digital/our-insights/iot-value-set-to-accelerate-through-2030-where-and-how-to-capture-it

V	**Master Seminar in Data Science and Machine Learning** 2540510, SS 2026, 2 SWS, Language: German/English, Open in study portal	**Seminar (S)**

V	**Biosignal-Adaptive Systems Seminar** 2540553, SS 2026, 2 SWS, Language: English, Open in study portal	**Seminar (S)** **Blended (On-Site/Online)**

Content
Biosignal-adaptive systems collect and analyze biosignals from users to recognize user states as a basis for adaptation. Thermic, mechanical, electric, acoustic, and optical signals are collected using sensors which are integrated in wearables, e.g. glasses, earphones, belts, or bracelets. The collected data is processed with analytics and machine learning techniques in order to determine short-term, evolving over time, and long-term user states in the form of user characteristics, affective-cognitive states, or behavior. Finally, the recognized user states are leveraged for realizing user-centric adaptations.

In this seminar, interdisciplinary teams of students design, develop, and evaluate a biosignal-adaptive system prototype leveraging state-of-the-art hard- and software. This seminar follows an interdisciplinary approach. Students from the fields of computer science, information systems and industrial engineering & management collaborate in the prototype design, development, and evaluation.

The seminar is carried out in cooperation between Teco/Chair of Pervasive Computing Systems (Prof. Beigl) and the Institute of Information Systems and Marketing (h-lab, Prof. Mädche). It is offered as part of the DFG-funded graduate school “KD2School: Designing Adaptive Systems for Economic Decisions“ (https://kd2school.info/)

Learning objectives of the seminar

- Explain what a biosignal-adaptive system is and how it can be conceptualized
- Suggest and evaluate different design solutions for addressing the identified problem
- Build a biosignal-adaptive system prototype using state-of-the-art hard- and software
- Perform a user-centric evaluation of the biosignal-adaptive system prototype

Prerequisites

Strong analytical abilities and profound software development skills are required.

Organizational issues
Termine werden bekannt gegeben

Literature
Required literature will be made available in the seminar.

V	**Human-Centered Systems Seminar: Research** 2540557, SS 2026, 3 SWS, Language: English, Open in study portal	**Seminar (S)** **Blended (On-Site/Online)**

Content
Formerly known as "Information Systems and Service Design Seminar"

With this seminar, we aim to provide students with the possibility to independently work on state-of-the-art research topics in addition to the knowledge gained in the lectures of the research group IS I (Prof. Mädche). The research group "Information Systems I" (IS I) headed by Prof. Mädche focuses in research, education, and innovation on designing interactive intelligent systems. It is positioned at the intersection of Information Systems and Human-Computer Interaction (HCI).

In the seminar, participants will get deeper insights in a contemporary research topic in the field of information systems, specifically interactive intelligent systems.

The actual seminar topics will be derived from current research activities of the research group. Our research assistants offer a rich set of topics from our research clusters (digital experience and participation, intelligent enterprise systems, or digital services design & innovation). Students can select among these topics individually depending on their personal interests. The seminar is carried out in the form of a literature-based thesis project. In the seminar, students will acquire the important methodological skills of running a systematic literature review.

Learning Objectives

- focus on a contemporary topic at the intersection of Information Systems and Human-Computer Interaction (HCI), specifically interactive intelligent systems
- carry out a structured literature search for a given topic
- aggregate the collected information in a suitable way to present and extract knowledge
- write a seminar thesis following academic writing standards
- deliver a presentation in a scientific context in front of an auditorium

Prerequisites

No specific prerequisites are required for the seminar.

Literature

Further literature will be made available in the seminar.

Organizational issues
Termine werden bekannt gegeben

V	**Entrepreneurship Research** 2545002, SS 2026, 2 SWS, Language: English, Open in study portal	**Seminar (S)** **On-Site**

Content
Content

In this course, the students choose from various relevant and current research topics in entrepreneurship and independently develop a topic that suits them in small teams. Initially, there is an introduction to standard methods such as systematic literature review, design science, qualitative and quantitative data analysis, and more. The seminar topic must be scientifically prepared and presented in 15-20 pages as part of a written elaboration. The seminar results are presented in a block event at the end of the semester (20 min + 10 min open discussion).

Learning Objectives

The foundations of independent scholarly work (literature review, argumentation + discussion, citation of literature sources, application of qualitative, quantitative, and simulation methods) are developed as part of the written elaboration. The competencies acquired in the seminar can be utilized in preparing for a potential master's thesis. Therefore, the seminar is mainly aimed at students who intend to write their thesis at the Chair of Entrepreneurship and Technology Management and wish to gain substantial experience in entrepreneurship research.

Organizational issues
Registration is via the Wiwi-Portal.

Seminar dates will probably be (as announced in Wiwi-Portal):

Wednesday, 29.04.2026, 10.00-16.00
Wednesday, 20.05.2026, 10.00-16.00
Wednesday, 24.06.2026, 09.00-12.00

Literature
Will be announced in the seminar.

V	**Seminar Human Resource Management (Master)** 2573012, SS 2026, 2 SWS, Language: German, Open in study portal	**Seminar (S)** **On-Site**

Content
The topics are redefined each semester on basis of current research topics. The topics will be announced on the website of the Wiwi-Portal.

Aim

The student

- looks critically into current research topics in the fields of Human Resource Management and Personnel Economics.
- trains his / her presentation skills.
- learns to get his / her ideas and insights across in a focused and concise way, both in oral and written form, and to sum up the crucial facts.
- cultivates the discussion of research approaches.

Workload

The total workload for this course is: approximately 90 hours.

Lecture: 30h
Preparation of lecture: 45h
Exam preparation: 15h

Literature

Selected journal articles and books.

Organizational issues
Geb. 05.20, Raum 2A-12.1, Termine werden bekannt gegeben

V	**Seminar Human Resources and Organizations (Master)** 2573013, SS 2026, 2 SWS, Language: German, Open in study portal	**Seminar (S)** **On-Site**

Content
The topics are redefined each semester on basis of current research topics. The topics will be announced on the website of the Wiwi-Portal.

Aim

The student

- looks critically into current research topics in the fields of human resources and organizations.
- trains his / her presentation skills.
- learns to get his / her ideas and insights across in a focused and concise way, both in oral and written form, and to sum up the crucial facts.
- cultivates the discussion of research approaches.

Workload

The total workload for this course is: approximately 90 hours.

Lecture: 30h
Preparation of lecture: 45h
Exam preparation: 15h

Literature

Selected journal articles and books.

Organizational issues
Geb. 05.20, Raum 2A-12.1, Termine werden bekannt gegeben

V	2530586, WS 26/27, 2 SWS, Language: German, Open in study portal	**Seminar (S)** **On-Site**

Content
Within this seminar eLearning videos are produced to different topics out of the contents of our lectures. The student gets in touch with scientific work. Through profound working on a specific scientific topic the student is meant to learn the foundations of scientific research and reasoning in particular in finance.Through conduction of the video the student becomes familiar with the fundamental techniques for presentations and foundations of scientific reasoning. In addition, the student earns rhetorical skills.

The success is monitored by the development of an eLearning video and by the writing of a project report (according to §4(2), 3 SPO).

The overall grade is made up of these partial performances.

Recommendations:

Knowledge of the content of the modules *Essentials of Finance* [WW3BWLFBV1] (for bachelor students) and *F1 (Finance)* [WW4BWLFBV1] (for master students) is assumed.

The total workload for this course is approximately 90 hours. For further information see German version.

Organizational issues
Kickoff am XX.10.26 um XX Uhr, Zwischenpräsentation am XX.12.26, XX Uhr und Abschlusspräsentation am XX.01.27, XX:XX Uhr am Campus B (Geb. 09.21), Raum 209

V	**Master Seminar: Intelligent & Responsible Information Systems** 2540472, WS 26/27, 2 SWS, Language: English, Open in study portal	**Seminar (S)** **On-Site**

Content
This seminar is offered by the Information Systems III research group, which has been based at the Institute for Information Systems (WIN) since January 2025 and is led by Prof. Dr. Jella Pfeiffer. Further information about us can be found on our website (WIN - Information Systems III).

V	**Case studies seminar: Innovation management** 2545105, WS 26/27, 2 SWS, Language: German, Open in study portal	**Seminar (S)** **On-Site**

Content
The objective of the seminar is to acquire selected concepts and methods of innovation management and to apply these in a practical way. The focus is on working on specific issues using a case study from business practice. Students work in groups and transfer the content taught, for example from the areas of scenario development, to entrepreneurial challenges in the context of fictitious case studies. The course follows a didactic interplay of theoretical input and direct application of what has been learned. At the end, the groups present their results in a plenary presentation, which is prepared by a short introduction to presentation techniques. The joint discussion promotes the transfer of knowledge to different contexts and strengthens the participants' ability to reflect.

As part of the seminar, students deal independently with current, research-oriented issues and systematically apply scientific criteria. They research relevant information, analyze, abstract and critically evaluate it. They draw conclusions independently from poorly structured data, incorporate interdisciplinary knowledge and contribute selectively to the further development of existing research perspectives. The results are validated and systematically prepared both orally and in writing using scientific standards - particularly with regard to structuring, terminology and correct citation of sources. In doing so, students argue in a technically sound manner and defend their findings in discourse with other participants and lecturers. They are also familiar with the DFG Code of Conduct “Guidelines for Safeguarding Good Scientific Practice” and apply these standards consistently in the preparation of their scientific work.

Literature
Werden in der ersten Veranstaltung bekannt gegeben.

T 6.659 Module component: Seminar in Economic Policy [T-WIWI-102789]

Coordinators: Prof. Dr. Ingrid Ott
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104908 - Economics
M-WIWI-107011 - Economics of Innovation and Growth

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Each term	1

Assessment
The assessment is carried out through a term paper within the range of 12 to 15 pages, a presentation of the results of the work in a seminar meeting, and active participation in the discussions of the seminar meeting (§ 4 (2), 3 SPO).

The final grade is composed of the weighted scored examinations (Essay 50%, 40% oral presentation, active participation 10%).

Prerequisites
None

Recommendations
At least one of the lectures "Theory of Endogenous Growth" or "Innovation Theory and Policy" should be attended in advance, if possible.

T 6.660 Module component: Seminar in Economics (Bachelor) [T-WIWI-103487]

Coordinators: Professorenschaft des Fachbereichs Volkswirtschaftslehre
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104908 - Economics

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Each term	1

Courses					
WT 25/26	2500052	Seminar on Topics in Digital Economics	2 SWS	Seminar /	Reiß, Hillenbrand, Potarca
WT 25/26	2500063	Economics of Crime	2 SWS	Seminar /	Mill
WT 25/26	2520561	Wirtschaftstheoretisches Seminar I (Bachelor)	2 SWS	Seminar /	Puppe, Ammann, Kretz, Okulicz
WT 25/26	2520562	Wirtschaftstheoretisches Seminar II (Bachelor)	2 SWS	Seminar /	Puppe, Ammann, Kretz
WT 25/26	2521310	Topics in Econometrics	2 SWS	Seminar	Schienle, Krüger, Bracher, Buse, Koster, Rüter, Leimenstoll, Renner, Pohle
WT 25/26	2560130	Seminar Public Finance	2 SWS	Seminar /	Wigger, Setio, Schmelzer
WT 25/26	2560140	Seminar in Digital Economics (Bachelor)	2 SWS	Seminar /	Rosar
WT 25/26	2560141	Seminar Game Theory and Behavioral Economics (Bachelor)	2 SWS	Seminar /	Rau
WT 25/26	2560400	Seminar in Macroeconomics I	2 SWS	Seminar /	Brumm, Pegorari, Kissling, Schang
WT 25/26	2561208	Selected aspects of European transport planning and -modelling	2 SWS	Seminar	Szimba, Sand
ST 2026	2500009	Seminar in Economic Theory IV	2 SWS	Seminar /	Ammann, Kretz, Okulicz
ST 2026	2500012	Making and Evaluating Predictions	2 SWS	Seminar	Schienle, Allen, Zhang
ST 2026	2500035	Efficiency analyses in network sectors	2 SWS	Seminar /	Mitusch, Latsch
ST 2026	2520367	Strategische Entscheidungen	2 SWS	Seminar /	Ehrhart
ST 2026	2520535	Seminar in Economic Theory I	2 SWS	Seminar /	Ammann, Kretz, Okulicz
ST 2026	2560130	Seminar Public Finance	2 SWS	Block /	Wigger, Schmelzer
ST 2026	2560259	Organisation and Management of Development Projects	2 SWS	Seminar /	Sieber
ST 2026	2560400	Seminar in Macroeconomics I	2 SWS	Seminar /	Brumm, Frank
ST 2026	2560555	Seminar Lying and Cheating in Economic Decision Situations (Bachelor)	2 SWS	Seminar /	Rau
WT 26/27	2521310	Topics in Econometrics	2 SWS	Seminar	Schienle, Krüger, Bracher, Buse, Koster, Rüter, Leimenstoll, Renner

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
Alternative exam assessment (§ 4(2), 3 SPO 2015). The following aspects are included:

- Regular participation in the seminar dates
- Preparation of a seminar paper on a partial aspect of the seminar topic according to scientific methods
- Lecture on the topic of the seminar paper.

The point scheme for the assessment is determined by the lecturer of the respective course. It will be announced at the beginning of the course.

Prerequisites
None.

Recommendations
See seminar description in the course catalogue of the KIT (https://campus.kit.edu/)

Additional Information
The listed seminar titles are placeholders. Currently offered seminars of each semester will be published on the websites of the institutes and in the course catalogue of the KIT. In general, the current seminar topics of each semester are already announced at the end of the previous semester. Furthermore for some seminars there is an application required.

The available places are listed on the internet: https://portal.wiwi.kit.edu.

Workload
90 hours

Below you will find excerpts from courses related to this module component:

V **Economics of Crime** Seminar (S) On-Site
2500063, WS 25/26, 2 SWS, Open in study portal

Content
For Bachelor students of the fields Industrial Engineering and Management, Information Engineering and Management, Economics Engineering or Economathematics.

Organizational issues
Application is possible via https://portal.wiwi.kit.edu/Seminare

Kick-off: 29.10.2025, 11.00 - 12.30 h (Raum KD2Lab tbc)

Presentations: 16.01.2026 09.00 - 18.00 h, 01.85, KD2 Lab (1. floor über Außentreppe), Team Room

V **Topics in Econometrics** Seminar (S)
2521310, WS 25/26, 2 SWS, Language: German, Open in study portal

Organizational issues
Wöchentliche Veranstaltung, Donnerstags 13:00-14:30 Uhr in B17 - R320

V **Seminar in Digital Economics (Bachelor)** Seminar (S) On-Site
2560140, WS 25/26, 2 SWS, Language: English, Open in study portal

Content
The seminar addresses issues in the digital economy from a game-theoretic perspective. All information about the seminar will be published on the Wiwi-Portal.

V **Seminar Game Theory and Behavioral Economics (Bachelor)** Seminar (S) On-Site
2560141, WS 25/26, 2 SWS, Language: English, Open in study portal

Content
For Bachelor students of the fields Industrial Engineering and Management, Information Engineering and Management, Economics Engineering or Economathematics.

The student develops an own idea for an economic experiment in this research direction. Students work in groups. Changing topics each semester. For current topics, see https://polit.econ.kit.edu or https://portal.wiwi.kit.edu/Seminare

Seminar Papers of 8–10 pages are to be handed in.

Recommendation: Knowledge in the field of experimental economic research or behavioral economics as well as in the field of microeconomics and game theory may be helpful.

Organizational issues
Application is possible via https://portal.wiwi.kit.edu/Seminare

Kick-off: 29.10.2025, 14.00 - 15.30 h, Geb. 01.85, KD2Lab Team room (über Außentreppe)

Presentations: 19.01.2026, 08.00 - 13.00 h, Geb. 01.85, KD2Lab Team room (über Außentreppe)

V	**Making and Evaluating Predictions** 2500012, SS 2026, 2 SWS, Language: German/English, Open in study portal	**Seminar (S)**

Content
A key goal of statistics and data science is to make predictions. Good predictions are important for decision-making, allowing us to balance risks and mitigate the impacts of potentially damaging events. In this seminar, we will study theoretical questions regarding the evaluation of predictions, such as “What does it mean for a prediction to be good?”, before introducing statistical and machine learning methods to generate both point-valued and probabilistic predictions. We will discuss applications to economics, meteorology, and energy forecasting.

Format: The course will start with a short block course at the beginning of the lecture period. Evaluation will be based on a seminar thesis, which students are required to hand in at the end of the lecture period, and a short presentation. The presentations will also take place towards the end of the lecture period.

Organizational issues
Blockveranstaltung, Termine werden bekannt gegeben

V	**Seminar Public Finance** 2560130, SS 2026, 2 SWS, Language: German, Open in study portal	**Block (B)** **Blended (On-Site/Online)**

Content
See German version.

Organizational issues
Termine werden bekannt gegeben.

Literature
Literatur wird zu Beginn des jeweiligen Seminars vorgestellt.

V	**Seminar Lying and Cheating in Economic Decision Situations (Bachelor)** 2560555, SS 2026, 2 SWS, Language: English, Open in study portal	**Seminar (S)** **On-Site**

Content
Objective of the Seminar: The student develops an own idea for an economic experiment in this research direction. Students work in groups. Changing topics each semester. For current topics, see http://polit.econ.kit.edu or https://portal.wiwi.kit.edu/Seminare

The acceptance of students for the seminar is based on preferences and suitability for the topics. This includes theoretical and practical experience with Behavioral Economics as well as English skills.

Seminar Papers of 12–15 pages are to be handed in.

Students' grades will be based on the quality of presentations in the seminar (40%) and the seminar paper (60%). There may be a bonus on the grade for actively participating in the discussions of the presentations.

Recommendation: Knowledge in the field of experimental economic research or behavioral economics as well as in the field of microeconomics and game theory may be helpful.

Organizational issues
Obligatory: Application via WiWi-Portal during the seminar registration period

Seminar Kick-off: 28.04.2026, 14.00 - 14.45 h

Seminar Presentations: 30.06.2026 9.30 - 18.00 h

Hand-in Seminar Papers: 02.08.2026

V	**Topics in Econometrics** 2521310, WS 26/27, 2 SWS, Language: German, Open in study portal	**Seminar (S)**

Organizational issues
Wöchentliche Veranstaltung, Donnerstags 13:00-14:30 Uhr in B17 - R320

T 6.661 Module component: Seminar in Economics A (Master) [T-WIWI-103478]

Coordinators: Professorenschaft des Fachbereichs Volkswirtschaftslehre
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104908 - Economics
M-WIWI-107204 - Seminar Module

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Each term	1

Courses					
WT 25/26	25000111	Statistics and Epidemics	2 SWS	Seminar /	Bracher, Nemcova
WT 25/26	2500024	Wirtschaftstheoretisches Seminar IV (Master)	2 SWS	Seminar /	Puppe, Kretz, Ammann, Okulicz
WT 25/26	2500052	Seminar on Topics in Digital Economics	2 SWS	Seminar /	Reiß, Hillenbrand, Potarca
WT 25/26	2500064	Economics of Crime	2 SWS	Seminar /	Mill
WT 25/26	2520500	Workshop on Economics, Finance and Statistics	2 SWS	Seminar	Puppe, Brumm, Nieken, Ott, Reiß, Ruckes, Schienle, Uhrig-Homburg, Wigger, Krüger
WT 25/26	2520563	Wirtschaftstheoretisches Seminar III (Master)	2 SWS	Seminar /	Ammann, Kretz
WT 25/26	2521310	Topics in Econometrics	2 SWS	Seminar	Schienle, Krüger, Bracher, Buse, Koster, Rüter, Leimenstoll, Renner, Pohle
WT 25/26	2560130	Seminar Public Finance	2 SWS	Seminar /	Wigger, Setio, Schmelzer
WT 25/26	2560142	Seminar in Digital Economics (Master)	2 SWS	Seminar /	Rosar
WT 25/26	2560143	Seminar Game Theory and Behavioral Economics (Master)	2 SWS	Seminar /	Rau
WT 25/26	2560282	Seminar in Economic Policy	2 SWS	Seminar /	Ott, Assistenten
WT 25/26	2560400	Seminar in Macroeconomics I	2 SWS	Seminar /	Brumm, Pegorari, Kissling, Schang
WT 25/26	2561208	Selected aspects of European transport planning and -modelling	2 SWS	Seminar	Szimba, Sand
ST 2026	2500012	Making and Evaluating Predictions	2 SWS	Seminar	Schienle, Allen, Zhang
ST 2026	2500035	Efficiency analyses in network sectors	2 SWS	Seminar /	Mitusch, Latsch
ST 2026	2520367	Strategische Entscheidungen	2 SWS	Seminar /	Ehrhart
ST 2026	2520536	Seminar in Economic Theory II	2 SWS	Seminar /	Ammann, Kretz, Okulicz
ST 2026	2520563	Seminar in Economic Theory III	2 SWS	Seminar /	Ammann, Kretz, Okulicz
ST 2026	2521310	Advanced Topics in Econometrics	2 SWS	Seminar	Schienle, Krüger, Bracher, Buse, Koster, Rüter, Leimenstoll, Renner
ST 2026	2560130	Seminar Public Finance	2 SWS	Block /	Wigger, Schmelzer
ST 2026	2560282	Seminar in economic policy	2 SWS	Seminar /	Ott, Assistenten, Ghoniem
ST 2026	2560400	Seminar in Macroeconomics I	2 SWS	Seminar /	Brumm, Frank

ST 2026	2560554	Seminar Lying and Cheating in Economic Decision Situations (Master)	2 SWS	Seminar / 🗣	Rau
WT 26/27	2521310	Topics in Econometrics	2 SWS	Seminar	Schienle, Krüger, Bracher, Buse, Koster, Rüter, Leimenstoll, Renner
WT 26/27	2560282	Seminar in Economic Policy	2 SWS	Seminar / 🗣	Ott, Assistenten

Legend: 🖥 Online, ❖ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Alternative exam assessment (§ 4(2), 3 SPO 2015). The following aspects are included:

- Regular participation in the seminar dates
- Preparation of a seminar paper on a partial aspect of the seminar topic according to scientific methods
- Lecture on the topic of the seminar paper.

The point scheme for the assessment is determined by the lecturer of the respective course. It will be announced at the beginning of the course.

Prerequisites
None

Recommendations
See seminar description in the course catalogue of the KIT (https://campus.kit.edu/)

Additional Information
Note on Prof. Ingrid Ott's seminars: Due to a research semester, no seminars will be offered in the summer semester 2026.

The listed seminar titles are placeholders. Currently offered seminars of each semester will be published on the websites of the institutes and in the course catalogue of the KIT. In general, the current seminar topics of each semester are already announced at the end of the previous semester. Furthermore for some seminars there is an application required.

The available places are listed on the internet: https://portal.wiwi.kit.edu.

Workload
90 hours

Below you will find excerpts from courses related to this module component:

V **Statistics and Epidemics** **Seminar (S)**
25000111, WS 25/26, 2 SWS, Language: English, Open in study portal **On-Site**

Content
Motivation
Infectious disease epidemiology gives rise to a large variety of real-time data streams. During the COVID-19 pandemic, the interpretation and statistical analysis of these data has proven crucial, but also highly challenging. In this seminar, students will get to know central concepts of infectious disease surveillance and modelling from a statistical perspective. Following an overview of various aspects in the form of blocked lectures, students will choose a more specific topic for their seminar thesis.

Learning Goals

Students develop an understanding of central modeling tasks and methods, including

- estimation of reproductive numbers

- compartment models of disease spread

- nowcasting and short-term forecasting of disease spread

- detection of outbreaks

- diagnostic testing

Moreover, they get to know various data types commonly used in the analysis of disease spread.

Logistics

The project seminar is worth 4.5 credit points (Leistungspunkte). There will be three blocked lectures (approx. 135 minutes each) in the beginning of the lecture period. For the various topics covered, subjects for seminar theses will be proposed (and students are allowed to propose their own topics). Towards the end of the semester, students present their progress on the chosen topics to the group. Grades will be based on this presentation (25%) and the final report (75%).

Organizational issues
Prerequisites

Students should have a very good working knowledge of statistics, including proficiency in a programming language for applied data analysis. The lecture VWL3 Introduction to Econometrics is a prerequisite for the project seminar. Most available software in the field is in R, but in principle Python can be used as well. Advanced knowledge of biology, medicine or epidemiology is not required.

Application Procedure
Please submit a transcript of records as well as a short letter of motivation (roughly 200 words) via WIWI-Portal: https://portal.wiwi.kit.edu/ys/8902

Application time frame: September 5th, 2025 to October, 31th, 2025.

V	**Economics of Crime** 2500064, WS 25/26, 2 SWS, Open in study portal	**Seminar (S)** **On-Site**

Content
For Bachelor students of the fields Industrial Engineering and Management, Information Engineering and Management, Economics Engineering or Economathematics.

Organizational issues
Application is possible via https://portal.wiwi.kit.edu/Seminare

Kick-off: 29.10.2025, 11.00 - 12.30 h 01.85, KD2 Lab (1. floor über Außentreppe), Team Room(Raum tbc)

Presentations: 17.01.2026 09.00 - 18.00 h, 01.85, KD2 Lab (1. floor über Außentreppe), Team Room

V	**Topics in Econometrics** 2521310, WS 25/26, 2 SWS, Language: German, Open in study portal	**Seminar (S)**

Organizational issues
Wöchentliche Veranstaltung, Donnerstags 13:00-14:30 Uhr in B17 - R320

V	**Seminar in Digital Economics (Master)** 2560142, WS 25/26, 2 SWS, Language: English, Open in study portal	**Seminar (S)** **On-Site**

Content
The seminar addresses issues in the digital economy from a game-theoretic perspective. All information about the seminar will be published on the Wiwi-Portal.

V	**Seminar Game Theory and Behavioral Economics (Master)** 2560143, WS 25/26, 2 SWS, Language: English, Open in study portal	**Seminar (S)** **On-Site**

Content
For Master students of the fields Industrial Engineering and Management, Information Engineering and Management, Economics Engineering or Economathematics.

The student develops an own idea for an economic experiment in this research direction. Students work in groups. Changing topics each semester. For current topics, see https://polit.econ.kit.edu or https://portal.wiwi.kit.edu/Seminare

Seminar Papers of 8–10 pages are to be handed in.

Recommendation: Knowledge in the field of experimental economic research or behavioral economics as well as in the field of microeconomics and game theory may be helpful.

Organizational issues
Application is possible via https://portal.wiwi.kit.edu/Seminare

Kick-off: 29.10.2025, 14.00 - 15.30 h, Geb. 01.85, KD2Lab Team room (über Außentreppe)

Presentations: 19.01.2026, 14.00 - 18.00 h, Geb. 01.85, KD2Lab Team room (über Außentreppe)

V	**Making and Evaluating Predictions** 2500012, SS 2026, 2 SWS, Language: German/English, Open in study portal	**Seminar (S)**

Content

A key goal of statistics and data science is to make predictions. Good predictions are important for decision-making, allowing us to balance risks and mitigate the impacts of potentially damaging events. In this seminar, we will study theoretical questions regarding the evaluation of predictions, such as “What does it mean for a prediction to be good?”, before introducing statistical and machine learning methods to generate both point-valued and probabilistic predictions. We will discuss applications to economics, meteorology, and energy forecasting.

Format: The course will start with a short block course at the beginning of the lecture period. Evaluation will be based on a seminar thesis, which students are required to hand in at the end of the lecture period, and a short presentation. The presentations will also take place towards the end of the lecture period.

Organizational issues

Blockveranstaltung, Termine werden bekannt gegeben

V	**Advanced Topics in Econometrics** 2521310, SS 2026, 2 SWS, Language: German/English, Open in study portal	**Seminar (S)**

Organizational issues

wöchentlich, Donnerstags 13:00-14:30, B17, R320

V	**Seminar Public Finance** 2560130, SS 2026, 2 SWS, Language: German, Open in study portal	**Block (B)** **Blended (On-Site/Online)**

Content

See German version.

Organizational issues

Termine werden bekannt gegeben.

Literature

Literatur wird zu Beginn des jeweiligen Seminars vorgestellt.

V	**Seminar Lying and Cheating in Economic Decision Situations (Master)** 2560554, SS 2026, 2 SWS, Language: English, Open in study portal	**Seminar (S)** **On-Site**

Content

Objective of the seminar: The student develops an own idea for an economic experiment in this research direction. Students work in groups. Changing topics each semester. For current topics, see http://polit.econ.kit.edu or https://portal.wiwi.kit.edu/Seminare

The acceptance of students for the seminar is based on preferences and suitability for the topics. This includes theoretical and practical experience with Behavioral Economics as well as English skills.

Seminar Papers of 12–15 pages are to be handed in.

Students' grades will be based on the quality of presentations in the seminar (40%) and the seminar paper (60%). There may be a bonus on the grade for actively participating in the discussions of the presentations.

Recommendation: Knowledge in the field of experimental economic research or behavioral economics as well as in the field of microeconomics and game theory may be helpful.

Organizational issues

Obligatory: Application via WiWi-Portal during the seminar registration period

Seminar Kick-off: 28.04.2026, 14.00 - 14.45 h

Seminar Presentations: 30.06.2026 9.30 - 18.00 h

Hand-in Seminar Papers: 02.08.2026

V	**Topics in Econometrics** 2521310, WS 26/27, 2 SWS, Language: German, Open in study portal	**Seminar (S)**

Organizational issues

Wöchentliche Veranstaltung, Donnerstags 13:00-14:30 Uhr in B17 - R320

T 6.662 Module component: Seminar in Economics B (Master) [T-WIWI-103477]

Coordinators: Professorenschaft des Fachbereichs Volkswirtschaftslehre
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104908 - Economics
M-WIWI-107204 - Seminar Module

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Each term	1

Courses					
WT 25/26	25000111	Statistics and Epidemics	2 SWS	Seminar /	Bracher, Nemcova
WT 25/26	2500024	Wirtschaftstheoretisches Seminar IV (Master)	2 SWS	Seminar /	Puppe, Kretz, Ammann, Okulicz
WT 25/26	2500052	Seminar on Topics in Digital Economics	2 SWS	Seminar /	Reiß, Hillenbrand, Potarca
WT 25/26	2500064	Economics of Crime	2 SWS	Seminar /	Mill
WT 25/26	2520500	Workshop on Economics, Finance and Statistics	2 SWS	Seminar	Puppe, Brumm, Nieken, Ott, Reiß, Ruckes, Schienle, Uhrig-Homburg, Wigger, Krüger
WT 25/26	2520563	Wirtschaftstheoretisches Seminar III (Master)	2 SWS	Seminar /	Ammann, Kretz
WT 25/26	2521310	Topics in Econometrics	2 SWS	Seminar	Schienle, Krüger, Bracher, Buse, Koster, Rüter, Leimenstoll, Renner, Pohle
WT 25/26	2560130	Seminar Public Finance	2 SWS	Seminar /	Wigger, Setio, Schmelzer
WT 25/26	2560142	Seminar in Digital Economics (Master)	2 SWS	Seminar /	Rosar
WT 25/26	2560143	Seminar Game Theory and Behavioral Economics (Master)	2 SWS	Seminar /	Rau
WT 25/26	2560282	Seminar in Economic Policy	2 SWS	Seminar /	Ott, Assistenten
WT 25/26	2560400	Seminar in Macroeconomics I	2 SWS	Seminar /	Brumm, Pegorari, Kissling, Schang
WT 25/26	2561208	Selected aspects of European transport planning and -modelling	2 SWS	Seminar	Szimba, Sand
ST 2026	2500012	Making and Evaluating Predictions	2 SWS	Seminar	Schienle, Allen, Zhang
ST 2026	2500035	Efficiency analyses in network sectors	2 SWS	Seminar /	Mitusch, Latsch
ST 2026	2520367	Strategische Entscheidungen	2 SWS	Seminar /	Ehrhart
ST 2026	2520536	Seminar in Economic Theory II	2 SWS	Seminar /	Ammann, Kretz, Okulicz
ST 2026	2520563	Seminar in Economic Theory III	2 SWS	Seminar /	Ammann, Kretz, Okulicz
ST 2026	2521310	Advanced Topics in Econometrics	2 SWS	Seminar	Schienle, Krüger, Bracher, Buse, Koster, Rüter, Leimenstoll, Renner
ST 2026	2560130	Seminar Public Finance	2 SWS	Block /	Wigger, Schmelzer
ST 2026	2560259	Organisation and Management of Development Projects	2 SWS	Seminar /	Sieber
ST 2026	2560282	Seminar in economic policy	2 SWS	Seminar /	Ott, Assistenten, Ghoniem
ST 2026	2560400	Seminar in Macroeconomics I	2 SWS	Seminar /	Brumm, Frank

ST 2026	2560554	Seminar Lying and Cheating in Economic Decision Situations (Master)	2 SWS	Seminar / 🗣	Rau
WT 26/27	2521310	Topics in Econometrics	2 SWS	Seminar	Schienle, Krüger, Bracher, Buse, Koster, Rüter, Leimenstoll, Renner
WT 26/27	2560282	Seminar in Economic Policy	2 SWS	Seminar / 🗣	Ott, Assistenten

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Alternative exam assessment (§ 4(2), 3 SPO 2015). The following aspects are included:

- Regular participation in the seminar dates
- Preparation of a seminar paper on a partial aspect of the seminar topic according to scientific methods
- Lecture on the topic of the seminar paper.

The point scheme for the assessment is determined by the lecturer of the respective course. It will be announced at the beginning of the course.

Prerequisites
None.

Recommendations
See seminar description in the course catalogue of the KIT (https://campus.kit.edu/)

Additional Information
The listed seminar titles are placeholders. Currently offered seminars of each semester will be published on the websites of the institutes and in the course catalogue of the KIT. In general, the current seminar topics of each semester are already announced at the end of the previous semester. Furthermore for some seminars there is an application required.

The available places are listed on the internet: https://portal.wiwi.kit.edu.

Workload
90 hours

Below you will find excerpts from courses related to this module component:

V **Statistics and Epidemics** **Seminar (S) On-Site**
25000111, WS 25/26, 2 SWS, Language: English, Open in study portal

Content
Motivation
Infectious disease epidemiology gives rise to a large variety of real-time data streams. During the COVID-19 pandemic, the interpretation and statistical analysis of these data has proven crucial, but also highly challenging. In this seminar, students will get to know central concepts of infectious disease surveillance and modelling from a statistical perspective. Following an overview of various aspects in the form of blocked lectures, students will choose a more specific topic for their seminar thesis.

Learning Goals

Students develop an understanding of central modeling tasks and methods, including

- estimation of reproductive numbers

- compartment models of disease spread

- nowcasting and short-term forecasting of disease spread

- detection of outbreaks

- diagnostic testing

Moreover, they get to know various data types commonly used in the analysis of disease spread.

Logistics

The project seminar is worth 4.5 credit points (Leistungspunkte). There will be three blocked lectures (approx. 135 minutes each) in the beginning of the lecture period. For the various topics covered, subjects for seminar theses will be proposed (and students are allowed to propose their own topics). Towards the end of the semester, students present their progress on the chosen topics to the group. Grades will be based on this presentation (25%) and the final report (75%).

Organizational issues
Prerequisites

Students should have a very good working knowledge of statistics, including proficiency in a programming language for applied data analysis. The lecture VWL3 Introduction to Econometrics is a prerequisite for the project seminar. Most available software in the field is in R, but in principle Python can be used as well. Advanced knowledge of biology, medicine or epidemiology is not required.

Application Procedure
Please submit a transcript of records as well as a short letter of motivation (roughly 200 words) via WIWI-Portal: https://portal.wiwi.kit.edu/ys/8902

Application time frame: September 5th, 2025 to October, 31th, 2025.

V	**Economics of Crime** 2500064, WS 25/26, 2 SWS, Open in study portal	**Seminar (S)** **On-Site**

Content
For Bachelor students of the fields Industrial Engineering and Management, Information Engineering and Management, Economics Engineering or Economathematics.

Organizational issues
Application is possible via https://portal.wiwi.kit.edu/Seminare

Kick-off: 29.10.2025, 11.00 - 12.30 h 01.85, KD2 Lab (1. floor über Außentreppe), Team Room(Raum tbc)

Presentations: 17.01.2026 09.00 - 18.00 h, 01.85, KD2 Lab (1. floor über Außentreppe), Team Room

V	**Topics in Econometrics** 2521310, WS 25/26, 2 SWS, Language: German, Open in study portal	**Seminar (S)**

Organizational issues
Wöchentliche Veranstaltung, Donnerstags 13:00-14:30 Uhr in B17 - R320

V	**Seminar in Digital Economics (Master)** 2560142, WS 25/26, 2 SWS, Language: English, Open in study portal	**Seminar (S)** **On-Site**

Content
The seminar addresses issues in the digital economy from a game-theoretic perspective. All information about the seminar will be published on the Wiwi-Portal.

V	**Seminar Game Theory and Behavioral Economics (Master)** 2560143, WS 25/26, 2 SWS, Language: English, Open in study portal	**Seminar (S)** **On-Site**

Content
For Master students of the fields Industrial Engineering and Management, Information Engineering and Management, Economics Engineering or Economathematics.

The student develops an own idea for an economic experiment in this research direction. Students work in groups. Changing topics each semester. For current topics, see https://polit.econ.kit.edu or https://portal.wiwi.kit.edu/Seminare

Seminar Papers of 8–10 pages are to be handed in.

Recommendation: Knowledge in the field of experimental economic research or behavioral economics as well as in the field of microeconomics and game theory may be helpful.

Organizational issues
Application is possible via https://portal.wiwi.kit.edu/Seminare

Kick-off: 29.10.2025, 14.00 - 15.30 h, Geb. 01.85, KD2Lab Team room (über Außentreppe)

Presentations: 19.01.2026, 14.00 - 18.00 h, Geb. 01.85, KD2Lab Team room (über Außentreppe)

V	**Making and Evaluating Predictions** 2500012, SS 2026, 2 SWS, Language: German/English, Open in study portal	**Seminar (S)**

Content
A key goal of statistics and data science is to make predictions. Good predictions are important for decision-making, allowing us to balance risks and mitigate the impacts of potentially damaging events. In this seminar, we will study theoretical questions regarding the evaluation of predictions, such as “What does it mean for a prediction to be good?”, before introducing statistical and machine learning methods to generate both point-valued and probabilistic predictions. We will discuss applications to economics, meteorology, and energy forecasting.

Format: The course will start with a short block course at the beginning of the lecture period. Evaluation will be based on a seminar thesis, which students are required to hand in at the end of the lecture period, and a short presentation. The presentations will also take place towards the end of the lecture period.

Organizational issues
Blockveranstaltung, Termine werden bekannt gegeben

V	**Advanced Topics in Econometrics** 2521310, SS 2026, 2 SWS, Language: German/English, Open in study portal	**Seminar (S)**

Organizational issues
wöchentlich, Donnerstags 13:00-14:30, B17, R320

V	**Seminar Public Finance** 2560130, SS 2026, 2 SWS, Language: German, Open in study portal	**Block (B)** **Blended (On-Site/Online)**

Content
See German version.

Organizational issues
Termine werden bekannt gegeben.

Literature
Literatur wird zu Beginn des jeweiligen Seminars vorgestellt.

V	**Seminar Lying and Cheating in Economic Decision Situations (Master)** 2560554, SS 2026, 2 SWS, Language: English, Open in study portal	**Seminar (S)** **On-Site**

Content
Objective of the seminar: The student develops an own idea for an economic experiment in this research direction. Students work in groups. Changing topics each semester. For current topics, see http://polit.econ.kit.edu or https://portal.wiwi.kit.edu/Seminare

The acceptance of students for the seminar is based on preferences and suitability for the topics. This includes theoretical and practical experience with Behavioral Economics as well as English skills.

Seminar Papers of 12–15 pages are to be handed in.

Students' grades will be based on the quality of presentations in the seminar (40%) and the seminar paper (60%). There may be a bonus on the grade for actively participating in the discussions of the presentations.

Recommendation: Knowledge in the field of experimental economic research or behavioral economics as well as in the field of microeconomics and game theory may be helpful.

Organizational issues
Obligatory: Application via WiWi-Portal during the seminar registration period

Seminar Kick-off: 28.04.2026, 14.00 - 14.45 h

Seminar Presentations: 30.06.2026 9.30 - 18.00 h

Hand-in Seminar Papers: 02.08.2026

V	**Topics in Econometrics** 2521310, WS 26/27, 2 SWS, Language: German, Open in study portal	**Seminar (S)**

Organizational issues
Wöchentliche Veranstaltung, Donnerstags 13:00-14:30 Uhr in B17 - R320

T 6.663 Module component: Seminar in Engineering Science Master (approval) [T-WIWI-108763]

Coordinators: Fachvertreter ingenieurwissenschaftlicher Fakultäten
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104907 - Engineering Sciences
M-WIWI-107204 - Seminar Module

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Each term	3

Assessment
See German version.

Prerequisites
See module description.

Recommendations
None

T 6.664 Module component: Seminar in Informatics (Bachelor) [T-WIWI-103485]

Coordinators: Professorenschaft des Instituts AIFB
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104901 - Informatics (KIT-Department of Economics and Management)

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Each term	1

Courses					
WT 25/26	2513214	Seminar Information security and Data protection (Bachelor)	2 SWS	Seminar /	Volkamer, Raabe, Schiefer, Werner
WT 25/26	2513312	Seminar Linked Data and the Semantic Web (Bachelor)	3 SWS	Seminar /	Käfer, Braun, Li, Kubelka
WT 25/26	2513314	Seminar Real-World Challenges in Data Science and Analytics (Bachelor)	3 SWS	/	Hoellig, Käfer, Thoma
WT 25/26	2513315	Seminar Real-World Challenges in Data Science and Analytics (Master)	3 SWS	/	Hoellig, Käfer, Thoma
ST 2026	2513220	Seminar Intelligente Fahrzeugsysteme der nächsten Generation: Kognition, Personalisierung und adaptive Mobilität (Bachelor)	2 SWS	Seminar	Ullrich, Schreiber
ST 2026	2513308	Seminar Knowledge Discovery and Data Mining (Bachelor)	2 SWS	Seminar /	Käfer, Noullet, Kinder, Li
ST 2026	2513310	Seminar Data Science & Real-time Big Data Analytics (Bachelor)	2 SWS	Seminar /	Käfer, Thoma, Slobodyanik
WT 26/27	2513214	Seminar Information security and Data protection (Bachelor)	2 SWS	Seminar /	Volkamer, Raabe, Schiefer, Werner

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
Alternative exam assessment (§ 4(2), 3 SPO 2015). The following aspects are included:

- Regular participation in the seminar dates
- Preparation of a seminar paper on a partial aspect of the seminar topic according to scientific methods
- Lecture on the topic of the seminar paper.

The point scheme for the assessment is determined by the lecturer of the respective course. It will be announced at the beginning of the course.

Prerequisites
None.

Recommendations
See seminar description in the course catalogue of the KIT (https://campus.kit.edu/)

Additional Information
Placeholder for seminars offered by the Institute AIFB. Currently offered seminars of each semester will be published on the websites of the institutes and in the course catalogue of the KIT. In general, the current seminar topics of each semester are already announced at the end of the previous semester. Furthermore for some seminars there is an application required.

The available places are listed on the internet: https://portal.wiwi.kit.edu.

Workload
90 hours

Below you will find excerpts from courses related to this module component:

V **Seminar Linked Data and the Semantic Web (Bachelor)**
2513312, WS 25/26, 3 SWS, Language: German/English, Open in study portal

Seminar (S)
On-Site

Content
Linked Data is a way of publishing data on the web in a machine-understandable fashion. The aim of this practical seminar is to build applications and devise algorithms that consume, provide, or analyse Linked Data.

The Linked Data principles are a set of practices for data publishing on the web. Linked Data builds on the web architecture and uses HTTP for data access, and RDF for describing data, thus aiming towards web-scale data integration. There is a vast amount of data available published according to those principles: recently, 4.5 billion facts have been counted with information about various domains, including music, movies, geography, natural sciences. Linked Data is also used to make web-pages machine-understandable, corresponding annotations are considered by the big search engine providers. On a smaller scale, devices on the Internet of Things can also be accessed using Linked Data which makes the unified processing of device data and data from the web easy.

In this practical seminar, students will build prototypical applications and devise algorithms that consume, provide, or analyse Linked Data. Those applications and algorithms can also extend existing applications ranging from databases to mobile apps.

For the seminar, programming skills or knowledge about web development tools/technologies are highly recommended. Basic knowledge of RDF and SPARQL are also recommended, but may be acquired during the seminar. Students will work in groups. Seminar meetings will take place as 'Block-Seminar'.

Topics of interest include, but are not limited to:

- Travel Security
- Geo data
- Linked News
- Social Media

The exact dates and information for registration will be announced at the event page.

V **Seminar Real-World Challenges in Data Science and Analytics (Bachelor)**
2513314, WS 25/26, 3 SWS, Language: German/English, Open in study portal

On-Site

Content
In the seminar, various Real-World Challenges in Data Science and Analytics will be worked on.

During this seminar, groups of students work on a case challenge with data provided. Here, the typical process of a data science project is depicted: integration of data, analysis of these, modeling of the decisions and visualization of the results.

During the seminar, solution concepts are worked out, implemented as a software solution and presented in an intermediate and final presentation. The seminar "Real-World Challenges in Data Science and Analytics" is aimed at students in master's programs.

The exact dates and information for registration will be announced at the course page.

V **Seminar Real-World Challenges in Data Science and Analytics (Master)**
2513315, WS 25/26, 3 SWS, Language: German/English, Open in study portal

On-Site

Content
In the seminar, various Real-World Challenges in Data Science and Analytics will be worked on.

During this seminar, groups of students work on a case challenge with data provided. Here, the typical process of a data science project is depicted: integration of data, analysis of these, modeling of the decisions and visualization of the results.

During the seminar, solution concepts are worked out, implemented as a software solution and presented in an intermediate and final presentation. The seminar "Real-World Challenges in Data Science and Analytics" is aimed at students in master's programs.

The exact dates and information for registration will be announced at the course page.

V **Seminar Knowledge Discovery and Data Mining (Bachelor)**
2513308, SS 2026, 2 SWS, Language: English, Open in study portal

Seminar (S)
On-Site

Content
In this seminar different machine learning and data mining methods are implemented.

The seminar includes different methods of machine learning and data mining. Participants of the seminar should have basic knowledge of machine learning and programming skills.

Domains of interest include, but are not limited to:

- Medicine
- Social Media
- Finance Market
- Scientific Publications

Further Information: https://aifb.kit.edu/web/Lehre/Praktikum_Knowledge_Discovery_and_Data_Science

The exact dates and information for registration will be announced at the event page.

Organizational issues
Die Anmeldung erfolgt über das WiWi-Portal https://portal.wiwi.kit.edu/ .

Literature
Detaillierte Referenzen werden zusammen mit den jeweiligenThemen angegeben. Allgemeine Hintergrundinformationen ergeben sich z.B.aus den folgenden Lehrbüchern:

- Mitchell, T.; Machine Learning
- McGraw Hill, Cook, D.J. and Holder, L.B. (Editors) Mining Graph Data, ISBN:0-471-73190-0
- Wiley, Manning, C. and Schütze, H.; Foundations of Statistical NLP, MIT Press, 1999.

V

Seminar Data Science & Real-time Big Data Analytics (Bachelor)

2513310, SS 2026, 2 SWS, Language: English, Open in study portal

Seminar (S)
On-Site

Content
In this seminar, students will design applications in teams that use meaningful and creative Event Processing methods. Thereby, students have access to an existing record.

Event processing and real-time data are everywhere: financial market data, sensors, business intelligence, social media analytics, logistics. Many applications collect large volumes of data in real time and are increasingly faced with the challenge of being able to process them quickly and react promptly. The challenges of this real-time processing are currently also receiving a great deal of attention under the term "Big Data". The complex processing of real-time data requires both knowledge of methods for data analysis (data science) and their processing (real-time analytics). Seminar papers are offered on both of these areas as well as on interface topics, the input of own ideas is explicitly desired.

Further information to the seminar is given under the following Link:
http://seminar-cep.fzi.de

Questions are answered via the e-mail address sem-ep@fzi.de.

Organizational issues
Die Anmeldung erfolgt über das WiWi-Portal https://portal.wiwi.kit.edu/ .

T 6.665 Module component: Seminar in Informatics A (Master) [T-WIWI-103479]

Coordinators: Professorenschaft des Instituts AIFB
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104901 - Informatics (KIT-Department of Economics and Management)
M-WIWI-107204 - Seminar Module

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Each term	1

Courses					
WT 25/26	2400125	Security and Privacy Awareness	2 SWS	Seminar /	Seidel-Saul, Volkamer, Boehm, Ballreich, Sikra, Veit
WT 25/26	2512101	Seminar: From Physical Models to Digital Twins: A Data-Driven Simulation Workshop (Master)	2 SWS	Seminar /	Lazarova-Molnar, Khodadadi, Daniel
WT 25/26	2513103	Seminar: Applications of Digital Twins (Master)	2 SWS	Seminar /	Lazarova-Molnar, Lee, Mahmoud
WT 25/26	2513105	Seminar: New Trends in Artificial Intelligence Techniques for Noise Prediction (Master)	2 SWS	Seminar /	Lazarova-Molnar, Demetgül
WT 25/26	2513107	Seminar: Data-Driven Reliability Modeling of Energy Systems (Master)	2 SWS	Seminar /	Lazarova-Molnar, Mostafa
WT 25/26	2513313	Seminar Linked Data and the Semantic Web (Master)	3 SWS	Seminar /	Käfer, Braun, Li, Kubelka
WT 25/26	2513314	Seminar Real-World Challenges in Data Science and Analytics (Bachelor)	3 SWS	/	Hoellig, Käfer, Thoma
WT 25/26	2513315	Seminar Real-World Challenges in Data Science and Analytics (Master)	3 SWS	/	Hoellig, Käfer, Thoma
WT 25/26	2513451	Seminar Cooperative Autonomous Vehicles (Master)	2 SWS	Seminar /	Vinel
WT 25/26	2513458	Seminar Artificial Intelligence for Autonomous Driving (Master)	2 SWS	Seminar /	Zhao, Vinel
WT 25/26	2513460	Seminar Cooperative Robots (Master)	2 SWS	Seminar /	Vinel
WT 25/26	2513500	Seminar Cognitive Automobiles and Robots (Master)	2 SWS	Seminar /	Zöllner, Schneider
ST 2026	2512101	Seminar: From Physical Models to Digital Twins: A Data-Driven Simulation Workshop (Seminar/ Master)	2 SWS	Seminar /	Lazarova-Molnar, Khodadadi, Lee, Daniel
ST 2026	2513103	Seminar: Applications of Digital Twins (Master)	2 SWS	Seminar /	Lazarova-Molnar, Lee, Mahmoud
ST 2026	2513109	Seminar: Data-driven Agent-based Modeling and Simulation (Master)	2 SWS	Seminar /	Lazarova-Molnar, Lee
ST 2026	2513211	Seminar Business Information Systems (Master)	2 SWS	Seminar /	Oberweis, Forell, Fritsch, Rybinski, Schreiber, Ullrich, König
ST 2026	2513221	Seminar Intelligente Fahrzeugsysteme der nächsten Generation: Kognition, Personalisierung und adaptive Mobilität (Master)	2 SWS	Seminar	Ullrich, Schreiber

ST 2026	2513309	Seminar Knowledge Discovery and Data Mining (Master)	2 SWS	Seminar / 🗣	Käfer, Noullet, Kinder, Li
ST 2026	2513311	Seminar Data Science & Real-time Big Data Analytics (Master)	2 SWS	Seminar / 🗣	Käfer, Thoma, Slobodyanik
ST 2026	2513455	Seminar Machine Learning in Autonomous Driving (Master)	2 SWS	Seminar / 🧩	Zhao, Vinel
ST 2026	2513459	Seminar Vulnerable Road User Technologies (Master)	2 SWS	Seminar / 🧩	Schrapel, Vinel
ST 2026	2513500	Cognitive Automobiles and Robots	2 SWS	Seminar / 🗣	Zöllner, Schneider
ST 2026	2513607	Seminar Knowledge Graphs and Large Language Models (Master)	2 SWS	Seminar / 🧩	Sack, Gesese
WT 26/27	2400125	Security and Privacy Awareness	2 SWS	Seminar / 🧩	Volkamer, Boehm, Ballreich, Algedri
WT 26/27	2512101	Seminar: From Physical Models to Digital Twins: A Data-Driven Simulation Workshop (Master)	2 SWS	Seminar / 🗣	Lazarova-Molnar, Khodadadi, Daniel, Mahmoud
WT 26/27	2513113	Seminar: From Agent-based Systems to Digital Twins: Data-Driven Agent-based Modeling Workshop (Master)	2 SWS	Seminar	Lazarova-Molnar, Lee
WT 26/27	2513115	Seminar: From System Design to Data-Driven Reliability Modeling in Renewable Energy Systems (Master)	2 SWS	Seminar	Lazarova-Molnar, Mostafa

Legend: 📱 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Alternative exam assessment (§ 4(2), 3 SPO 2015). The following aspects are included:

- Regular participation in the seminar dates
- Preparation of a seminar paper on a partial aspect of the seminar topic according to scientific methods
- Lecture on the topic of the seminar paper.

The point scheme for the assessment is determined by the lecturer of the respective course. It will be announced at the beginning of the course.

Prerequisites
None.

Recommendations
See seminar description in the course catalogue of the KIT (https://campus.kit.edu/)

Additional Information
Placeholder for seminars offered by the Institute AIFB.

Currently offered seminars of each semester will be published on the websites of the institutes and in the course catalogue of the KIT. In general, the current seminar topics of each semester are already announced at the end of the previous semester. Furthermore for some seminars there is an application required.

The available places are listed on the internet: https://portal.wiwi.kit.edu.

Workload
90 hours

Below you will find excerpts from courses related to this module component:

V **Security and Privacy Awareness**
2400125, WS 25/26, 2 SWS, Language: German/English, Open in study portal

Seminar (S)
Blended (On-Site/Online)

Content
Within the framework of this interdisciplinary seminar, the topics security awareness and privacy awareness are to be considered from different perspectives. It deals with legal, information technology, psychological, social as well as philosophical aspects.

Important notes:

- The seminar language is English. This means that English is the standard for the presentation and the seminar paper, and German is only permitted for the paper after consultation with the supervisor, for example, if the topic of the thesis requires it.
- The seminar is only for MASTER students (or Mastervorzug)
- The link to enrol is for every student, regardless of the study background

Topics:

The advertised topics can be found in the wiwi portal.

V	**Seminar: Applications of Digital Twins (Master)** 2513103, WS 25/26, 2 SWS, Language: English, Open in study portal	**Seminar (S)** **On-Site**

Content

Seminar Name: Applications of Digital Twins

Size: 10 students (with 10 different topics)

Workload:

- 2 Lectures:
  - o Introduction to Digital Twins and topic distribution
  - o "How to Give Effective Presentations" lecture
- 10 student presentations (each 45 minutes in total)
- 10 student reports

Responsible Person: Hui Min Lee, Sanja Lazarova-Molnar

Deliverables for Grade:

· 1 Report per student and topic (8 pages, including references, IEEE Template, compulsory usage of Reference Manager – Zotero or EndNote)

· 25 mins presentation per student plus 20 min discussion (focus on the presentation topic + presentation skills) = 45 minutes for each student

Credits: 3 credits = 90 hours

Format/ Structure of the Seminar (Draft):

· 2 Lectures at the beginning of the semester

· Students have 1 week time to provide a priority list of 5 topics, distribution will be decided based on first come – first serve, ensuring that core topics are covered

· Q&As can be asked and answered over mails or ad-hoc appointments

· Students have time to work on the report and presentation during the semester

· Submission of all reports will be required 2 months after the intro lecture for ensuring fairness

· Presentations are done in blocks of 2 students per class, starting mid-June, presentations will be submitted at the day of the scheduled presentation

Approximate Time Consumption for Students (Draft):

· Lectures: 3 hours

· Student Presentations: 7.5 hours

· Topic Subscription: 1 hour

· Presentation Preparation: 15 hours

· Paper Writing and Literature Review: 63.5 hours

Description:

The seminar focuses on applications of Digital Twins and data-driven modeling, with an additional goal of improving scientific research and presentation skills for Master students. The seminar covers the diverse applications and use cases of Digital Twins in different domains such as manufacturing, energy systems, healthcare and many more, offering students an in-depth understanding of the role of Digital Twins in transforming the industries.

The seminar is structured as a literature review seminar. Each student can select a topic out of a predefined set, conduct further research and then write a comprehensive research paper. Students will also deliver presentations, synthesizing insights from both the provided starting reference literature and their own additional research.

By the end of the course, students will not only have a solid understanding of the current applications of Digital Twins and emerging trends but also be well-prepared to present their findings in an academic setting.

Topics:

1. Digital Twins for Manufacturing Systems

References:

· Zhang, Chenyuan, et al. "A reconfigurable modeling approach for digital twin-based manufacturing system." *Procedia Cirp* 83 (2019): 118-125. (96 citations)

· Kritzinger, Werner, et al. "Digital Twin in manufacturing: A categorical literature review and classification." *Ifac-PapersOnline* 51.11 (2018): 1016-1022. (1934 citations)

· Jaensch, Florian, et al. "Digital twins of manufacturing systems as a base for machine learning." *2018 25th International conference on mechatronics and machine vision in practice (M2VIP)*. IEEE, 2018. (73 citations)

2. Digital Twins for Energy Systems

References:

· Steindl, Gernot, et al. "Generic digital twin architecture for industrial energy systems." *Applied Sciences* 10.24 (2020): 8903. (78 citations)

· Granacher, Julia, et al. "Overcoming decision paralysis—A digital twin for decision making in energy system design." *Applied Energy* 306 (2022): 117954. (33 citations) -> focus on interactive digital twins

· Palensky, Peter, et al. "Digital twins and their use in future power systems." *Digital Twin* 1 (2022): 4. (37 citations)

3. Digital Twins in Healthcare

References:

· Alazab, Mamoun, et al. "Digital twins for healthcare 4.0-recent advances, architecture, and open challenges." *IEEE Consumer Electronics Magazine* (2022). (26 citations)

· Croatti, Angelo, et al. "On the integration of agents and digital twins in healthcare." *Journal of Medical Systems* 44 (2020): 1-8. (163 citations)

· Erol, Tolga, Arif Furkan Mendi, and Dilara Doğan. "The digital twin revolution in healthcare." *2020 4th international symposium on multidisciplinary studies and innovative technologies (ISMSIT)*. IEEE, 2020. (106 citations)

4. Digital Twins of City Infrastructures (in Smart Cities)

References:

· Deren, Li, Yu Wenbo, and Shao Zhenfeng. "Smart city based on digital twins." *Computational Urban Science* 1 (2021): 1-11. (110 citations)

· Deng, Tianhu, Keren Zhang, and Zuo-Jun Max Shen. "A systematic review of a digital twin city: A new pattern of urban governance toward smart cities." *Journal of Management Science and Engineering* 6.2 (2021): 125-134. (192 citations)

· Mylonas, Georgios, et al. "Digital twins from smart manufacturing to smart cities: A survey." *Ieee Access* 9 (2021): 143222-143249. (99 citations)

5. Digital Twins in Logistics

References:

· Moshood, Taofeeq D., et al. "Digital twins driven supply chain visibility within logistics: A new paradigm for future logistics." *Applied System Innovation* 4.2 (2021): 29. (71 citations)

· Agalianos, K., et al. "Discrete event simulation and digital twins: review and challenges for logistics." *Procedia Manufacturing* 51 (2020): 1636-1641. (74 citations)

· Korth, Benjamin, Christian Schwede, and Markus Zajac. "Simulation-ready digital twin for realtime management of logistics systems." *2018 IEEE international conference on big data (big data)*. IEEE, 2018. (64 citations)

6. Cognitive Digital Twins

References:

· Al Faruque, Mohammad Abdullah, et al. "Cognitive digital twin for manufacturing systems." *2021 Design, Automation & Test in Europe Conference & Exhibition (DATE)*. IEEE, 2021. (28 citations)

· Zhang, Nan, Rami Bahsoon, and Georgios Theodoropoulos. "Towards engineering cognitive digital twins with self-awareness." *2020 IEEE International Conference on Systems, Man, and Cybernetics (SMC)*. IEEE, 2020. (22 citations)

· Zheng, Xiaochen, Jinzhi Lu, and Dimitris Kiritsis. "The emergence of cognitive digital twin: vision, challenges and opportunities." *International Journal of Production Research* 60.24 (2022): 7610-7632. (92 citations)

7. Fusing Data and Human Expert Knowledge in Digital Twins

References:

· Kulkarni, Vinay, Souvik Barat, and Tony Clark. "Towards adaptive enterprises using digital twins." *2019 winter simulation conference (WSC)*. IEEE, 2019. (22 citations)

· Vogel-Heuser, Birgit, et al. "Potential for combining semantics and data analysis in the context of digital twins." *Philosophical Transactions of the Royal Society A* 379.2207 (2021): 20200368. (16 citations)

· Todorovski, Ljupčo, and Sašo Džeroski. "Integrating knowledge-driven and data-driven approaches to modeling." *ecological modelling* 194.1-3 (2006): 3-13. (80 citations)

8. Digital Twins for Multi-agent / Complex Systems

References:

· Pretel, Elena, Alejandro Moya, Elena Navarro, Víctor López-Jaquero, and Pascual González. "Analysing the synergies between Multi-agent Systems and Digital Twins: A systematic literature review." *Information and Software Technology* (2024): 107503. (2 citations)

· Mariani, S., Picone, M., Ricci, A. (2022). About Digital Twins, Agents, and Multiagent Systems: A Cross-Fertilisation Journey. In: Melo, F.S., Fang, F. (eds) *Autonomous Agents and Multiagent Systems. Best and Visionary Papers. AAMAS 2022*. Lecture Notes in Computer Science, vol 13441. (11 citations)

· Marah H, Challenger M. Adaptive hybrid reasoning for agent-based digital twins of distributed multi-robot systems. *SIMULATION.* 2024;100(9):931-957. (0 citations – new articles)

9. Digital Twins for Energy Systems

References:

· Kabir, Md Rafiul, Dipal Halder, and Sandip Ray. "Digital Twins for IoT-driven Energy Systems: A Survey." *IEEE Access* (2024). (0 citations – new articles)

· Brosinsky, Christoph, Rainer Krebs, and Dirk Westermann. "Embedded Digital Twins in future energy management systems: paving the way for automated grid control." *at-Automatisierungstechnik* 68, no. 9 (2020): 750-764.(21 citations)

· Song, Zhao, Christoph M. Hackl, Abhinav Anand, Andre Thommessen, Jonas Petzschmann, Omar Kamel, Robert Braunbehrens, Anton Kaifel, Christian Roos, and Stefan Hauptmann. "Digital twins for the future power system: An overview and a future perspective." *Sustainability* 15, no. 6 (2023): 5259. .(32 citations)

· Mostafa, Omar & Lazarova-Molnar, Sanja. (2024). Enhancing Reliability of Energy Systems with Digital Twins: Challenges and Opportunities. (0 citations – new articles)

10. Digital Twins in Transportation and Automotive

References:

· Schwarz, Chris, and Ziran Wang. "The role of digital twins in connected and automated vehicles." *IEEE Intelligent Transportation Systems Magazine* 14, no. 6 (2022): 41-51. (91 citations)

· Bhatti, Ghanishtha, Harshit Mohan, and R. Raja Singh. "Towards the future of smart electric vehicles: Digital twin technology." *Renewable and Sustainable Energy Reviews* 141 (2021): 110801. (422 citations)

· Almeaibed, Sadeq, Saba Al-Rubaye, Antonios Tsourdos, and Nicolas P. Avdelidis. "Digital twin analysis to promote safety and security in autonomous vehicles." *IEEE Communications Standards Magazine* 5, no. 1 (2021): 40-46. (135 citations)

11. Digital Twins for Environment and Sustainability

References:

· Tzachor, Asaf, Soheil Sabri, Catherine E. Richards, Abbas Rajabifard, and Michele Acuto. "Potential and limitations of digital twins to achieve the sustainable development goals." *Nature Sustainability* 5, no. 10 (2022): 822-829. (110 citations)

· Corrado, Casey R., Suzanne M. DeLong, Emily G. Holt, Edward Y. Hua, and Andreas Tolk. "Combining green metrics and digital twins for sustainability planning and governance of smart buildings and cities." *Sustainability* 14, no. 20 (2022): 12988. (36 citations)

· Kim, Byungmo, Jaewon Oh, and Cheonhong Min. "Development of a simulation model for digital twin of an oscillating water column wave power generator structure with ocean environmental effect." *Sensors* 23, no. 23 (2023): 9472. (3 citations)

12. Digital Twins in Agriculture

References:

· Peladarinos, Nikolaos, Dimitrios Piromalis, Vasileios Cheimaras, Efthymios Tserepas, Radu Adrian Munteanu, and Panagiotis Papageorgas. "Enhancing smart agriculture by implementing digital twins: A comprehensive review." *Sensors* 23, no. 16 (2023): 7128. (60 citations)

· Escribà-Gelonch, Marc, Shu Liang, Pieter van Schalkwyk, Ian Fisk, Nguyen Van Duc Long, and Volker Hessel. "Digital Twins in Agriculture: Orchestration and Applications." *Journal of Agricultural and Food Chemistry* 72, no. 19 (2024): 10737-10752. (12 citations)

· Verdouw, Cor, Bedir Tekinerdogan, Adrie Beulens, and Sjaak Wolfert. "Digital twins in smart farming." *Agricultural Systems* 189 (2021): 103046. (494 citations)

V

Seminar: New Trends in Artificial Intelligence Techniques for Noise Prediction (Master)

2513105, WS 25/26, 2 SWS, Language: English, Open in study portal

Seminar (S)
On-Site

Content

Noise, especially in urban areas, is a major environmental issue that impacts quality of life and health, contributing to stress, sleep disturbances, and cardiovascular problems. Traffic noise, primarily from tire-road interactions, has become more prominent as electric vehicles reduce engine noise. Tackling this issue involves both passive methods, like noise barriers, and active solutions such as noise cancellation technologies.

In recent years, artificial intelligence (AI) has emerged as a powerful tool for managing noise. AI-based systems can classify noise sources, create noise maps, and develop control strategies. Advanced AI techniques, including Generative Adversarial Networks (GANs), AutoEncoders, Bi-Long Short-Term Memory (LSTM), and Bi-Gated Recurrent Units (GRUs), Graphical Convolutional Networks (GCN), Physics-informed neural networks, YOLO, Transformer, show great potential for reducing noise. Additionally, many computer vision techniques are used to improve noise conditions. This seminar will explore these AI methods and their role in enhancing conditions safety, minimizing environmental noise, and supporting intelligent transportation systems.

In this seminar, we try to understand Noise through data analysis and other techniques. We discuss current approaches to noise prediction and innovative AI approaches based on data science and machine learning.

Topics:

Introduction to Noise and Tire-Road Noise

Overview on Noise and Tire-Road Noise

Time Series Analysis and Image Analysis

Data Exploration and Visualization

Noise Feature Extraction and Analysis

Machine learning and Deep Learning Approach for Tire-Road Noise

Who are we looking for:

We are looking for students who want to expand their specialist knowledge and practical experience in artificial intelligence, signal processing, computer vision and road-tire noise. Participation provides the opportunity to actively participate in shaping the future of using artificial intelligence, computer vision and signal processing to reduce road-tire and traffic noise.

What we offer:

We provide you with tyre-road noise data. With this data, you can apply many signal processing, computer vision and artificial intelligence algorithms. This is where you can let your creativity run free and implement innovative solutions with our guidance.

Literature

- Stansfeld, S., Clark, C., Smuk, M., Gallacher, J., & Babisch, W. (2021). Road traffic noise, noise sensitivity, noise annoyance, psychological and physical health and mortality. Environmental Health, 20, 1-15.
- Lan, Y., Roberts, H., Kwan, M. P., & Helbich, M. (2020). Transportation noise exposure and anxiety: A systematic review and meta-analysis. Environmental research, 191, 110118.
- WHO. (2018) Environmental Noise Guidelines for the European Region.
- EU. (2019) Regulation (EU) No 540/2014 of the European Parliament and of the Council of 16 April 2014 on the Sound Level of Motor Vehicles and of Replacement Silencing Systems, and Amending Directive 2007/46/EC and Repealing Directive 70/157/EEC.

V **Seminar Linked Data and the Semantic Web (Master)** **Seminar (S) On-Site**

2513313, WS 25/26, 3 SWS, Language: German/English, Open in study portal

Content
Linked Data is a way of publishing data on the web in a machine-understandable fashion. The aim of this practical seminar is to build applications and devise algorithms that consume, provide, or analyse Linked Data.

The Linked Data principles are a set of practices for data publishing on the web. Linked Data builds on the web architecture and uses HTTP for data access, and RDF for describing data, thus aiming towards web-scale data integration. There is a vast amount of data available published according to those principles: recently, 4.5 billion facts have been counted with information about various domains, including music, movies, geography, natural sciences. Linked Data is also used to make web-pages machine-understandable, corresponding annotations are considered by the big search engine providers. On a smaller scale, devices on the Internet of Things can also be accessed using Linked Data which makes the unified processing of device data and data from the web easy.

In this practical seminar, students will build prototypical applications and devise algorithms that consume, provide, or analyse Linked Data. Those applications and algorithms can also extend existing applications ranging from databases to mobile apps.

For the seminar, programming skills or knowledge about web development tools/technologies are highly recommended. Basic knowledge of RDF and SPARQL are also recommended, but may be acquired during the seminar. Students will work in groups. Seminar meetings will take place as 'Block-Seminar'.

Topics of interest include, but are not limited to:

- Travel Security
- Geo data
- Linked News
- Social Media

The exact dates and information for registration will be announced at the event page.

V Seminar Real-World Challenges in Data Science and Analytics (Bachelor)
2513314, WS 25/26, 3 SWS, Language: German/English, Open in study portal **On-Site**

Content
In the seminar, various Real-World Challenges in Data Science and Analytics will be worked on.

During this seminar, groups of students work on a case challenge with data provided. Here, the typical process of a data science project is depicted: integration of data, analysis of these, modeling of the decisions and visualization of the results.

During the seminar, solution concepts are worked out, implemented as a software solution and presented in an intermediate and final presentation. The seminar "Real-World Challenges in Data Science and Analytics" is aimed at students in master's programs.

The exact dates and information for registration will be announced at the course page.

V Seminar Real-World Challenges in Data Science and Analytics (Master)
2513315, WS 25/26, 3 SWS, Language: German/English, Open in study portal **On-Site**

Content
In the seminar, various Real-World Challenges in Data Science and Analytics will be worked on.

During this seminar, groups of students work on a case challenge with data provided. Here, the typical process of a data science project is depicted: integration of data, analysis of these, modeling of the decisions and visualization of the results.

During the seminar, solution concepts are worked out, implemented as a software solution and presented in an intermediate and final presentation. The seminar "Real-World Challenges in Data Science and Analytics" is aimed at students in master's programs.

The exact dates and information for registration will be announced at the course page.

V Seminar Cognitive Automobiles and Robots (Master)
2513500, WS 25/26, 2 SWS, Language: German/English, Open in study portal **Seminar (S)** **Blended (On-Site/Online)**

Content

The seminar is intended as a theoretical supplement to lectures such as "Machine Learning". The theoretical basics will be deepened in the seminar. The aim of the seminar is that the participants work individually to analyze a subsystem from the field of robotics and cognitive systems using one or more procedures from the field of AI/ML.

The individual projects require the analysis of the task at hand, selection of suitable procedures, specification and theoretical evaluation of the approach taken. Finally, the chosen solution has to be documented and presented in a short presentation.

Learning objectives:

- Students can apply knowledge from the Machine Learning lecture in a selected field of current research in robotics or cognitive automobiles for theoretical analysis.
- Students can evaluate, document and present their concepts and results.

Recommendations:

Attendance of the lecture machine learning

Workload:

The workload of 3 credit points consists of the time spent on literature research and planning/specifying the proposed solution. In addition, a short report and a presentation of the work carried out will be prepared.

Organizational issues

Anmeldung und weitere Informationen sind im Wiwi-Portal zu finden.

Registration and further information can be found in the WiWi-portal.

V **Seminar: From Physical Models to Digital Twins: A Data-Driven Simulation Workshop (Seminar/Master)**

2512101, SS 2026, 2 SWS, Language: English, Open in study portal

Seminar (S)
On-Site

Content

This seminar focuses on the data-driven discovery of simulation models in industrial settings, providing a hands-on approach to understanding and optimizing production processes.

Students will start by designing and constructing production lines using Lego Spike and similar modular systems. This activity will include developing comprehensive data-capturing pipelines to collect detailed event-logging raw data from their production lines.

Next, the seminar will explore advanced techniques for transforming this raw data into simulation models, e.g., Petri nets. Participants will learn and apply data-driven model extraction methods, such as process mining to extract workflow processes; statistical methods to fit probability distributions and analyze trends, and machine learning algorithms to model complex behaviors within the production process. Through these techniques, students will extract simulation models that reflect the real-world dynamics of their production lines. The seminar will then guide participants on how to validate the extracted simulation models to ensure their accuracy.

By the end of the seminar, students will be equipped with the skills to build model production lines, collect event logging data from them, transform event log data into actionable simulation models and use these models to drive efficiency and innovation in industrial production settings.

Grading Scheme:

Report - 50%

Presentations - 40%

Implementation - 10%

V **Seminar: Applications of Digital Twins (Master)**

2513103, SS 2026, 2 SWS, Language: English, Open in study portal

Seminar (S)
On-Site

Content

Seminar Name: Applications of Digital Twins

Size: 10 students (with 10 different topics)

Workload:

- 2 Lectures:
  - o Introduction to Digital Twins and topic distribution
  - o "How to Give Effective Presentations" lecture
- 10 student presentations (each 45 minutes in total)
- 10 student reports

Responsible Person: Hui Min Lee, Sanja Lazarova-Molnar

Deliverables for Grade:

· 1 Report per student and topic (8 pages, including references, IEEE Template, compulsory usage of Reference Manager – Zotero or EndNote)

· 25 mins presentation per student plus 20 min discussion (focus on the presentation topic + presentation skills) = 45 minutes for each student

Credits: 3 credits = 90 hours

Format/ Structure of the Seminar (Draft):

· 2 Lectures at the beginning of the semester

· Students have 1 week time to provide a priority list of 5 topics, distribution will be decided based on first come – first serve, ensuring that core topics are covered

· Q&As can be asked and answered over mails or ad-hoc appointments

· Students have time to work on the report and presentation during the semester

· Submission of all reports will be required 2 months after the intro lecture for ensuring fairness

· Presentations are done in blocks of 2 students per class, starting mid-June, presentations will be submitted at the day of the scheduled presentation

Approximate Time Consumption for Students (Draft):

· Lectures: 3 hours

· Student Presentations: 7.5 hours

· Topic Subscription: 1 hour

· Presentation Preparation: 15 hours

· Paper Writing and Literature Review: 63.5 hours

Description:

The seminar focuses on applications of Digital Twins and data-driven modeling, with an additional goal of improving scientific research and presentation skills for Master students. The seminar covers the diverse applications and use cases of Digital Twins in different domains such as manufacturing, energy systems, healthcare and many more, offering students an in-depth understanding of the role of Digital Twins in transforming the industries.

The seminar is structured as a literature review seminar. Each student can select a topic out of a predefined set, conduct further research and then write a comprehensive research paper. Students will also deliver presentations, synthesizing insights from both the provided starting reference literature and their own additional research.

By the end of the course, students will not only have a solid understanding of the current applications of Digital Twins and emerging trends but also be well-prepared to present their findings in an academic setting.

Topics:

1. Digital Twins for Manufacturing Systems

References:

· Zhang, Chenyuan, et al. "A reconfigurable modeling approach for digital twin-based manufacturing system." *Procedia Cirp* 83 (2019): 118-125. (96 citations)

· Kritzinger, Werner, et al. "Digital Twin in manufacturing: A categorical literature review and classification." *Ifac-PapersOnline* 51.11 (2018): 1016-1022. (1934 citations)

· Jaensch, Florian, et al. "Digital twins of manufacturing systems as a base for machine learning." *2018 25th International conference on mechatronics and machine vision in practice (M2VIP)*. IEEE, 2018. (73 citations)

2. Digital Twins for Energy Systems

References:

· Steindl, Gernot, et al. "Generic digital twin architecture for industrial energy systems." *Applied Sciences* 10.24 (2020): 8903. (78 citations)

· Granacher, Julia, et al. "Overcoming decision paralysis—A digital twin for decision making in energy system design." *Applied Energy* 306 (2022): 117954. (33 citations) -> focus on interactive digital twins

· Palensky, Peter, et al. "Digital twins and their use in future power systems." *Digital Twin* 1 (2022): 4. (37 citations)

3. Digital Twins in Healthcare

References:

· Alazab, Mamoun, et al. "Digital twins for healthcare 4.0-recent advances, architecture, and open challenges." *IEEE Consumer Electronics Magazine* (2022). (26 citations)

· Croatti, Angelo, et al. "On the integration of agents and digital twins in healthcare." *Journal of Medical Systems* 44 (2020): 1-8. (163 citations)

· Erol, Tolga, Arif Furkan Mendi, and Dilara Doğan. "The digital twin revolution in healthcare." *2020 4th international symposium on multidisciplinary studies and innovative technologies (ISMSIT)*. IEEE, 2020. (106 citations)

4. Digital Twins of City Infrastructures (in Smart Cities)

References:

· Deren, Li, Yu Wenbo, and Shao Zhenfeng. "Smart city based on digital twins." *Computational Urban Science* 1 (2021): 1-11. (110 citations)

· Deng, Tianhu, Keren Zhang, and Zuo-Jun Max Shen. "A systematic review of a digital twin city: A new pattern of urban governance toward smart cities." *Journal of Management Science and Engineering* 6.2 (2021): 125-134. (192 citations)

· Mylonas, Georgios, et al. "Digital twins from smart manufacturing to smart cities: A survey." *Ieee Access* 9 (2021): 143222-143249. (99 citations)

5. Digital Twins in Logistics

References:

· Moshood, Taofeeq D., et al. "Digital twins driven supply chain visibility within logistics: A new paradigm for future logistics." *Applied System Innovation* 4.2 (2021): 29. (71 citations)

· Agalianos, K., et al. "Discrete event simulation and digital twins: review and challenges for logistics." *Procedia Manufacturing* 51 (2020): 1636-1641. (74 citations)

· Korth, Benjamin, Christian Schwede, and Markus Zajac. "Simulation-ready digital twin for realtime management of logistics systems." *2018 IEEE international conference on big data (big data)*. IEEE, 2018. (64 citations)

6. Cognitive Digital Twins

References:

· Al Faruque, Mohammad Abdullah, et al. "Cognitive digital twin for manufacturing systems." *2021 Design, Automation & Test in Europe Conference & Exhibition (DATE)*. IEEE, 2021. (28 citations)

· Zhang, Nan, Rami Bahsoon, and Georgios Theodoropoulos. "Towards engineering cognitive digital twins with self-awareness." *2020 IEEE International Conference on Systems, Man, and Cybernetics (SMC)*. IEEE, 2020. (22 citations)

· Zheng, Xiaochen, Jinzhi Lu, and Dimitris Kiritsis. "The emergence of cognitive digital twin: vision, challenges and opportunities." *International Journal of Production Research* 60.24 (2022): 7610-7632. (92 citations)

7. Fusing Data and Human Expert Knowledge in Digital Twins

References:

· Kulkarni, Vinay, Souvik Barat, and Tony Clark. "Towards adaptive enterprises using digital twins." *2019 winter simulation conference (WSC)*. IEEE, 2019. (22 citations)

· Vogel-Heuser, Birgit, et al. "Potential for combining semantics and data analysis in the context of digital twins." *Philosophical Transactions of the Royal Society A* 379.2207 (2021): 20200368. (16 citations)

· Todorovski, Ljupčo, and Sašo Džeroski. "Integrating knowledge-driven and data-driven approaches to modeling." *ecological modelling* 194.1-3 (2006): 3-13. (80 citations)

8. Digital Twins for Multi-agent / Complex Systems

References:

· Pretel, Elena, Alejandro Moya, Elena Navarro, Víctor López-Jaquero, and Pascual González. "Analysing the synergies between Multi-agent Systems and Digital Twins: A systematic literature review." *Information and Software Technology* (2024): 107503. (2 citations)

· Mariani, S., Picone, M., Ricci, A. (2022). About Digital Twins, Agents, and Multiagent Systems: A Cross-Fertilisation Journey. In: Melo, F.S., Fang, F. (eds) *Autonomous Agents and Multiagent Systems. Best and Visionary Papers. AAMAS 2022*. Lecture Notes in Computer Science, vol 13441. (11 citations)

· Marah H, Challenger M. Adaptive hybrid reasoning for agent-based digital twins of distributed multi-robot systems. *SIMULATION.* 2024;100(9):931-957. (0 citations – new articles)

9. Digital Twins for Energy Systems

References:

· Kabir, Md Rafiul, Dipal Halder, and Sandip Ray. "Digital Twins for IoT-driven Energy Systems: A Survey." *IEEE Access* (2024). (0 citations – new articles)

· Brosinsky, Christoph, Rainer Krebs, and Dirk Westermann. "Embedded Digital Twins in future energy management systems: paving the way for automated grid control." *at-Automatisierungstechnik* 68, no. 9 (2020): 750-764.(21 citations)

· Song, Zhao, Christoph M. Hackl, Abhinav Anand, Andre Thommessen, Jonas Petzschmann, Omar Kamel, Robert Braunbehrens, Anton Kaifel, Christian Roos, and Stefan Hauptmann. "Digital twins for the future power system: An overview and a future perspective." *Sustainability* 15, no. 6 (2023): 5259. .(32 citations)

· Mostafa, Omar & Lazarova-Molnar, Sanja. (2024). Enhancing Reliability of Energy Systems with Digital Twins: Challenges and Opportunities. (0 citations – new articles)

10. Digital Twins in Transportation and Automotive

References:

· Schwarz, Chris, and Ziran Wang. "The role of digital twins in connected and automated vehicles." *IEEE Intelligent Transportation Systems Magazine* 14, no. 6 (2022): 41-51. (91 citations)

· Bhatti, Ghanishtha, Harshit Mohan, and R. Raja Singh. "Towards the future of smart electric vehicles: Digital twin technology." *Renewable and Sustainable Energy Reviews* 141 (2021): 110801. (422 citations)

· Almeaibed, Sadeq, Saba Al-Rubaye, Antonios Tsourdos, and Nicolas P. Avdelidis. "Digital twin analysis to promote safety and security in autonomous vehicles." *IEEE Communications Standards Magazine* 5, no. 1 (2021): 40-46. (135 citations)

11. Digital Twins for Environment and Sustainability

References:

· Tzachor, Asaf, Soheil Sabri, Catherine E. Richards, Abbas Rajabifard, and Michele Acuto. "Potential and limitations of digital twins to achieve the sustainable development goals." *Nature Sustainability* 5, no. 10 (2022): 822-829. (110 citations)

· Corrado, Casey R., Suzanne M. DeLong, Emily G. Holt, Edward Y. Hua, and Andreas Tolk. "Combining green metrics and digital twins for sustainability planning and governance of smart buildings and cities." *Sustainability* 14, no. 20 (2022): 12988. (36 citations)

· Kim, Byungmo, Jaewon Oh, and Cheonhong Min. "Development of a simulation model for digital twin of an oscillating water column wave power generator structure with ocean environmental effect." *Sensors* 23, no. 23 (2023): 9472. (3 citations)

12. Digital Twins in Agriculture

References:

· Peladarinos, Nikolaos, Dimitrios Piromalis, Vasileios Cheimaras, Efthymios Tserepas, Radu Adrian Munteanu, and Panagiotis Papageorgas. "Enhancing smart agriculture by implementing digital twins: A comprehensive review." *Sensors* 23, no. 16 (2023): 7128. (60 citations)

· Escribà-Gelonch, Marc, Shu Liang, Pieter van Schalkwyk, Ian Fisk, Nguyen Van Duc Long, and Volker Hessel. "Digital Twins in Agriculture: Orchestration and Applications." *Journal of Agricultural and Food Chemistry* 72, no. 19 (2024): 10737-10752. (12 citations)

· Verdouw, Cor, Bedir Tekinerdogan, Adrie Beulens, and Sjaak Wolfert. "Digital twins in smart farming." *Agricultural Systems* 189 (2021): 103046. (494 citations)

V

Seminar Knowledge Discovery and Data Mining (Master)

2513309, SS 2026, 2 SWS, Language: English, Open in study portal

Seminar (S)
On-Site

Content

In this seminar different machine learning and data mining methods are implemented.

The seminar includes different methods of machine learning and data mining. Participants of the seminar should have basic knowledge of machine learning and programming skills.

Domains of interest include, but are not limited to:

- Medicine
- Social Media
- Finance Market
- Scientific Publications

Further Information: https://aifb.kit.edu/web/Lehre/Praktikum_Knowledge_Discovery_and_Data_Science

The exact dates and information for registration will be announced at the event page.

Organizational issues

Die Anmeldung erfolgt über das WiWi-Portal https://portal.wiwi.kit.edu/ .

Literature

Detaillierte Referenzen werden zusammen mit den jeweiligenThemen angegeben. Allgemeine Hintergrundinformationen ergeben sich z.B.aus den folgenden Lehrbüchern:

- Mitchell, T.; Machine Learning
- McGraw Hill, Cook, D.J. and Holder, L.B. (Editors) Mining Graph Data, ISBN:0-471-73190-0
- Wiley, Manning, C. and Schütze, H.; Foundations of Statistical NLP, MIT Press, 1999.

V

Seminar Data Science & Real-time Big Data Analytics (Master)

2513311, SS 2026, 2 SWS, Language: English, Open in study portal

Seminar (S)
On-Site

Content
In this seminar, students will design applications in teams that use meaningful and creative Event Processing methods. Thereby, students have access to an existing record.

Event processing and real-time data are everywhere: financial market data, sensors, business intelligence, social media analytics, logistics. Many applications collect large volumes of data in real time and are increasingly faced with the challenge of being able to process them quickly and react promptly. The challenges of this real-time processing are currently also receiving a great deal of attention under the term "Big Data". The complex processing of real-time data requires both knowledge of methods for data analysis (data science) and their processing (real-time analytics). Seminar papers are offered on both of these areas as well as on interface topics, the input of own ideas is explicitly desired.

Further information to the practical seminar is given under the following Link:
http://seminar-cep.fzi.de

Questions are answered via the e-mail address sem-ep@fzi.de.

Organizational issues
Die Anmeldung erfolgt über das WiWi-Portal https://portal.wiwi.kit.edu/ .

V **Cognitive Automobiles and Robots**
2513500, SS 2026, 2 SWS, Language: German/English, Open in study portal
Seminar (S)
On-Site

Content
The seminar is intended as a theoretical supplement to lectures such as "Machine Learning". The theoretical basics will be deepened in the seminar. The aim of the seminar is that the participants work individually to analyze a subsystem from the field of robotics and cognitive systems using one or more procedures from the field of AI/ML.

The individual projects require the analysis of the task at hand, selection of suitable procedures, specification and theoretical evaluation of the approach taken. Finally, the chosen solution has to be documented and presented in a short presentation.

Learning objectives:

- Students can apply knowledge from the Machine Learning lecture in a selected field of current research in robotics or cognitive automobiles for theoretical analysis.
- Students can evaluate, document and present their concepts and results.

Recommendations:

Attendance of the lecture machine learning

Workload:

The workload of 3 credit points consists of the time spent on literature research and planning/specifying the proposed solution. In addition, a short report and a presentation of the work carried out will be prepared.

Organizational issues
Anmeldung und weitere Informationen sind im Wiwi-Portal zu finden.

Registration and further information can be found in the WiWi-portal.

V **Seminar Knowledge Graphs and Large Language Models (Master)**
2513607, SS 2026, 2 SWS, Language: English, Open in study portal
Seminar (S)
Blended (On-Site/Online)

Content
Large language models (LLMs) such as GPT-4 have shown remarkable capabilities in transforming various natural language processing (NLP) tasks across different domains. However, LLMs often generate incorrect answers, known as hallucinations, posing significant challenges to their usability and reliability. Additionally, LLMs operate as black boxes, making it difficult to understand how they arrive at specific conclusions, leading to transparency and explainability issues. Combining LLMs with KGs creates a powerful synergy that significantly enhances the capabilities of artificial intelligence across various tasks. This integration leverages the strengths of both technologies, with LLMs excelling at understanding and generating human-like text, and KGs providing structured, reliable information about entities and their relationships. Together, they offer a robust approach to problem-solving across diverse domains.

This seminar will focus on the intersection of LLMs and KGs, covering areas of interest including, but not limited to:

- KG completion using LLMs
- Question answering with KGs and LLMs
- Explainability of LLMs with KG integration
- Reasoning with LLMs and KGs
- Enhanced prompt engineering using KGs

Contributions of the students:

Each student will be assigned one paper on the topic, which could be a research paper discussing a novel approach or a resource paper presenting datasets, tools, etc. The student will be responsible for the following tasks:

1. **Report Writing**: Read the assigned paper thoroughly and write a 15-page seminar report explaining the methods and findings in their own words.
2. **Presenting**: Prepare and deliver a seminar presentation to share insights from the paper with other seminar participants.
3. **Conducting Experiments**: If the authors provide code, re-implement it for small-scale experiments using Google Colab or make the implementation available via GitHub.

V **Security and Privacy Awareness**
2400125, WS 26/27, 2 SWS, Language: German/English, Open in study portal
Seminar (S)
Blended (On-Site/Online)

Content
Within the framework of this interdisciplinary seminar, the topics security awareness and privacy awareness are to be considered from different perspectives. It deals with legal, information technology, psychological, social as well as philosophical aspects.

Important notes:

- The seminar language is English. This means that English is the standard for the presentation and the seminar paper, and German is only permitted for the paper after consultation with the supervisor, for example, if the topic of the thesis requires it.
- The seminar is only for MASTER students (or Mastervorzug)
- The link to enrol is for every student, regardless of the study background

Topics:

The advertised topics can be found in the wiwi portal.

V **Seminar: From Physical Models to Digital Twins: A Data-Driven Simulation Workshop (Master)**
2512101, WS 26/27, 2 SWS, Language: English, Open in study portal
Seminar (S)
On-Site

Content

This seminar focuses on the data-driven discovery of simulation models in industrial settings, providing a hands-on approach to understanding and optimizing production processes.

Students will start by designing and constructing production lines using Lego Spike and similar modular systems. This activity will include developing comprehensive data-capturing pipelines to collect detailed event-logging raw data from their production lines. Next, the seminar will explore advanced techniques for transforming this raw data into simulation models, e.g., Petri nets. Participants will learn and apply data-driven model extraction methods, such as process mining to extract workflow processes; statistical methods to fit probability distributions and analyze trends, and machine learning algorithms to model complex behaviors within the production process. Through these techniques, students will extract simulation models that reflect the real-world dynamics of their production lines. The seminar will then guide participants on how to validate the extracted simulation models to ensure their accuracy.

By the end of the seminar, students will be equipped with the skills to build model production lines, collect event logging data from them, transform event log data into actionable simulation models and use these models to drive efficiency and innovation in industrial production settings.

T 6.666 Module component: Seminar in Informatics B (Master) [T-WIWI-103480]

Coordinators: Professorenschaft des Instituts AIFB
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104901 - Informatics (KIT-Department of Economics and Management)
M-WIWI-107204 - Seminar Module

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Each term	1

Courses					
WT 25/26	2400125	Security and Privacy Awareness	2 SWS	Seminar /	Seidel-Saul, Volkamer, Boehm, Ballreich, Sikra, Veit
WT 25/26	2512101	Seminar: From Physical Models to Digital Twins: A Data-Driven Simulation Workshop (Master)	2 SWS	Seminar /	Lazarova-Molnar, Khodadadi, Daniel
WT 25/26	2513103	Seminar: Applications of Digital Twins (Master)	2 SWS	Seminar /	Lazarova-Molnar, Lee, Mahmoud
WT 25/26	2513105	Seminar: New Trends in Artificial Intelligence Techniques for Noise Prediction (Master)	2 SWS	Seminar /	Lazarova-Molnar, Demetgül
WT 25/26	2513107	Seminar: Data-Driven Reliability Modeling of Energy Systems (Master)	2 SWS	Seminar /	Lazarova-Molnar, Mostafa
WT 25/26	2513313	Seminar Linked Data and the Semantic Web (Master)	3 SWS	Seminar /	Käfer, Braun, Li, Kubelka
WT 25/26	2513314	Seminar Real-World Challenges in Data Science and Analytics (Bachelor)	3 SWS	/	Hoellig, Käfer, Thoma
WT 25/26	2513315	Seminar Real-World Challenges in Data Science and Analytics (Master)	3 SWS	/	Hoellig, Käfer, Thoma
WT 25/26	2513451	Seminar Cooperative Autonomous Vehicles (Master)	2 SWS	Seminar /	Vinel
WT 25/26	2513458	Seminar Artificial Intelligence for Autonomous Driving (Master)	2 SWS	Seminar /	Zhao, Vinel
WT 25/26	2513460	Seminar Cooperative Robots (Master)	2 SWS	Seminar /	Vinel
WT 25/26	2513500	Seminar Cognitive Automobiles and Robots (Master)	2 SWS	Seminar /	Zöllner, Schneider
ST 2026	2512101	Seminar: From Physical Models to Digital Twins: A Data-Driven Simulation Workshop (Seminar/ Master)	2 SWS	Seminar /	Lazarova-Molnar, Khodadadi, Lee, Daniel
ST 2026	2513103	Seminar: Applications of Digital Twins (Master)	2 SWS	Seminar /	Lazarova-Molnar, Lee, Mahmoud
ST 2026	2513109	Seminar: Data-driven Agent-based Modeling and Simulation (Master)	2 SWS	Seminar /	Lazarova-Molnar, Lee
ST 2026	2513211	Seminar Business Information Systems (Master)	2 SWS	Seminar /	Oberweis, Forell, Fritsch, Rybinski, Schreiber, Ullrich, König
ST 2026	2513221	Seminar Intelligente Fahrzeugsysteme der nächsten Generation: Kognition, Personalisierung und adaptive Mobilität (Master)	2 SWS	Seminar	Ullrich, Schreiber

ST 2026	2513309	Seminar Knowledge Discovery and Data Mining (Master)	2 SWS	Seminar / 🗣	Käfer, Noullet, Kinder, Li
ST 2026	2513311	Seminar Data Science & Real-time Big Data Analytics (Master)	2 SWS	Seminar / 🗣	Käfer, Thoma, Slobodyanik
ST 2026	2513455	Seminar Machine Learning in Autonomous Driving (Master)	2 SWS	Seminar / 🧩	Zhao, Vinel
ST 2026	2513459	Seminar Vulnerable Road User Technologies (Master)	2 SWS	Seminar / 🧩	Schrapel, Vinel
ST 2026	2513500	Cognitive Automobiles and Robots	2 SWS	Seminar / 🗣	Zöllner, Schneider
ST 2026	2513607	Seminar Knowledge Graphs and Large Language Models (Master)	2 SWS	Seminar / 🧩	Sack, Gesese
WT 26/27	2400125	Security and Privacy Awareness	2 SWS	Seminar / 🧩	Volkamer, Boehm, Ballreich, Algedri
WT 26/27	2512101	Seminar: From Physical Models to Digital Twins: A Data-Driven Simulation Workshop (Master)	2 SWS	Seminar / 🗣	Lazarova-Molnar, Khodadadi, Daniel, Mahmoud
WT 26/27	2513113	Seminar: From Agent-based Systems to Digital Twins: Data-Driven Agent-based Modeling Workshop (Master)	2 SWS	Seminar	Lazarova-Molnar, Lee
WT 26/27	2513115	Seminar: From System Design to Data-Driven Reliability Modeling in Renewable Energy Systems (Master)	2 SWS	Seminar	Lazarova-Molnar, Mostafa

Legend: 📱 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Alternative exam assessment (§ 4(2), 3 SPO 2015). The following aspects are included:

- Regular participation in the seminar dates
- Preparation of a seminar paper on a partial aspect of the seminar topic according to scientific methods
- Lecture on the topic of the seminar paper.

The point scheme for the assessment is determined by the lecturer of the respective course. It will be announced at the beginning of the course.

Prerequisites
None.

Recommendations
See seminar description in the course catalogue of the KIT (https://campus.kit.edu/)

Additional Information
Placeholder for seminars offered by the Institute AIFB.

The listed seminar titles are placeholders. Currently offered seminars of each semester will be published on the websites of the institutes and in the course catalogue of the KIT. In general, the current seminar topics of each semester are already announced at the end of the previous semester. Furthermore for some seminars there is an application required.

The available places are listed on the internet: https://portal.wiwi.kit.edu.

Workload
90 hours

Below you will find excerpts from courses related to this module component:

V **Security and Privacy Awareness**
2400125, WS 25/26, 2 SWS, Language: German/English, Open in study portal

Seminar (S)
Blended (On-Site/Online)

Content
Within the framework of this interdisciplinary seminar, the topics security awareness and privacy awareness are to be considered from different perspectives. It deals with legal, information technology, psychological, social as well as philosophical aspects.

Important notes:

- The seminar language is English. This means that English is the standard for the presentation and the seminar paper, and German is only permitted for the paper after consultation with the supervisor, for example, if the topic of the thesis requires it.
- The seminar is only for MASTER students (or Mastervorzug)
- The link to enrol is for every student, regardless of the study background

Topics:

The advertised topics can be found in the wiwi portal.

V	**Seminar: Applications of Digital Twins (Master)** 2513103, WS 25/26, 2 SWS, Language: English, Open in study portal	**Seminar (S)** **On-Site**

Content

Seminar Name: Applications of Digital Twins

Size: 10 students (with 10 different topics)

Workload:

- 2 Lectures:
  - o Introduction to Digital Twins and topic distribution
  - o "How to Give Effective Presentations" lecture
- 10 student presentations (each 45 minutes in total)
- 10 student reports

Responsible Person: Hui Min Lee, Sanja Lazarova-Molnar

Deliverables for Grade:

· 1 Report per student and topic (8 pages, including references, IEEE Template, compulsory usage of Reference Manager – Zotero or EndNote)

· 25 mins presentation per student plus 20 min discussion (focus on the presentation topic + presentation skills) = 45 minutes for each student

Credits: 3 credits = 90 hours

Format/ Structure of the Seminar (Draft):

· 2 Lectures at the beginning of the semester

· Students have 1 week time to provide a priority list of 5 topics, distribution will be decided based on first come – first serve, ensuring that core topics are covered

· Q&As can be asked and answered over mails or ad-hoc appointments

· Students have time to work on the report and presentation during the semester

· Submission of all reports will be required 2 months after the intro lecture for ensuring fairness

· Presentations are done in blocks of 2 students per class, starting mid-June, presentations will be submitted at the day of the scheduled presentation

Approximate Time Consumption for Students (Draft):

· Lectures: 3 hours

· Student Presentations: 7.5 hours

· Topic Subscription: 1 hour

· Presentation Preparation: 15 hours

· Paper Writing and Literature Review: 63.5 hours

Description:

The seminar focuses on applications of Digital Twins and data-driven modeling, with an additional goal of improving scientific research and presentation skills for Master students. The seminar covers the diverse applications and use cases of Digital Twins in different domains such as manufacturing, energy systems, healthcare and many more, offering students an in-depth understanding of the role of Digital Twins in transforming the industries.

The seminar is structured as a literature review seminar. Each student can select a topic out of a predefined set, conduct further research and then write a comprehensive research paper. Students will also deliver presentations, synthesizing insights from both the provided starting reference literature and their own additional research.

By the end of the course, students will not only have a solid understanding of the current applications of Digital Twins and emerging trends but also be well-prepared to present their findings in an academic setting.

Topics:

1. Digital Twins for Manufacturing Systems

References:

· Zhang, Chenyuan, et al. "A reconfigurable modeling approach for digital twin-based manufacturing system." *Procedia Cirp* 83 (2019): 118-125. (96 citations)

· Kritzinger, Werner, et al. "Digital Twin in manufacturing: A categorical literature review and classification." *Ifac-PapersOnline* 51.11 (2018): 1016-1022. (1934 citations)

· Jaensch, Florian, et al. "Digital twins of manufacturing systems as a base for machine learning." *2018 25th International conference on mechatronics and machine vision in practice (M2VIP)*. IEEE, 2018. (73 citations)

2. Digital Twins for Energy Systems

References:

· Steindl, Gernot, et al. "Generic digital twin architecture for industrial energy systems." *Applied Sciences* 10.24 (2020): 8903. (78 citations)

· Granacher, Julia, et al. "Overcoming decision paralysis—A digital twin for decision making in energy system design." *Applied Energy* 306 (2022): 117954. (33 citations) -> focus on interactive digital twins

· Palensky, Peter, et al. "Digital twins and their use in future power systems." *Digital Twin* 1 (2022): 4. (37 citations)

3. Digital Twins in Healthcare

References:

· Alazab, Mamoun, et al. "Digital twins for healthcare 4.0-recent advances, architecture, and open challenges." *IEEE Consumer Electronics Magazine* (2022). (26 citations)

· Croatti, Angelo, et al. "On the integration of agents and digital twins in healthcare." *Journal of Medical Systems* 44 (2020): 1-8. (163 citations)

· Erol, Tolga, Arif Furkan Mendi, and Dilara Doğan. "The digital twin revolution in healthcare." *2020 4th international symposium on multidisciplinary studies and innovative technologies (ISMSIT)*. IEEE, 2020. (106 citations)

4. Digital Twins of City Infrastructures (in Smart Cities)

References:

· Deren, Li, Yu Wenbo, and Shao Zhenfeng. "Smart city based on digital twins." *Computational Urban Science* 1 (2021): 1-11. (110 citations)

· Deng, Tianhu, Keren Zhang, and Zuo-Jun Max Shen. "A systematic review of a digital twin city: A new pattern of urban governance toward smart cities." *Journal of Management Science and Engineering* 6.2 (2021): 125-134. (192 citations)

· Mylonas, Georgios, et al. "Digital twins from smart manufacturing to smart cities: A survey." *Ieee Access* 9 (2021): 143222-143249. (99 citations)

5. Digital Twins in Logistics

References:

· Moshood, Taofeeq D., et al. "Digital twins driven supply chain visibility within logistics: A new paradigm for future logistics." *Applied System Innovation* 4.2 (2021): 29. (71 citations)

· Agalianos, K., et al. "Discrete event simulation and digital twins: review and challenges for logistics." *Procedia Manufacturing* 51 (2020): 1636-1641. (74 citations)

· Korth, Benjamin, Christian Schwede, and Markus Zajac. "Simulation-ready digital twin for realtime management of logistics systems." *2018 IEEE international conference on big data (big data)*. IEEE, 2018. (64 citations)

6. Cognitive Digital Twins

References:

· Al Faruque, Mohammad Abdullah, et al. "Cognitive digital twin for manufacturing systems." *2021 Design, Automation & Test in Europe Conference & Exhibition (DATE)*. IEEE, 2021. (28 citations)

· Zhang, Nan, Rami Bahsoon, and Georgios Theodoropoulos. "Towards engineering cognitive digital twins with self-awareness." *2020 IEEE International Conference on Systems, Man, and Cybernetics (SMC)*. IEEE, 2020. (22 citations)

· Zheng, Xiaochen, Jinzhi Lu, and Dimitris Kiritsis. "The emergence of cognitive digital twin: vision, challenges and opportunities." *International Journal of Production Research* 60.24 (2022): 7610-7632. (92 citations)

7. Fusing Data and Human Expert Knowledge in Digital Twins

References:

· Kulkarni, Vinay, Souvik Barat, and Tony Clark. "Towards adaptive enterprises using digital twins." *2019 winter simulation conference (WSC)*. IEEE, 2019. (22 citations)

· Vogel-Heuser, Birgit, et al. "Potential for combining semantics and data analysis in the context of digital twins." *Philosophical Transactions of the Royal Society A* 379.2207 (2021): 20200368. (16 citations)

· Todorovski, Ljupčo, and Sašo Džeroski. "Integrating knowledge-driven and data-driven approaches to modeling." *ecological modelling* 194.1-3 (2006): 3-13. (80 citations)

8. Digital Twins for Multi-agent / Complex Systems

References:

· Pretel, Elena, Alejandro Moya, Elena Navarro, Víctor López-Jaquero, and Pascual González. "Analysing the synergies between Multi-agent Systems and Digital Twins: A systematic literature review." *Information and Software Technology* (2024): 107503. (2 citations)

· Mariani, S., Picone, M., Ricci, A. (2022). About Digital Twins, Agents, and Multiagent Systems: A Cross-Fertilisation Journey. In: Melo, F.S., Fang, F. (eds) *Autonomous Agents and Multiagent Systems. Best and Visionary Papers. AAMAS 2022*. Lecture Notes in Computer Science, vol 13441. (11 citations)

· Marah H, Challenger M. Adaptive hybrid reasoning for agent-based digital twins of distributed multi-robot systems. *SIMULATION.* 2024;100(9):931-957. (0 citations – new articles)

9. Digital Twins for Energy Systems

References:

· Kabir, Md Rafiul, Dipal Halder, and Sandip Ray. "Digital Twins for IoT-driven Energy Systems: A Survey." *IEEE Access* (2024). (0 citations – new articles)

· Brosinsky, Christoph, Rainer Krebs, and Dirk Westermann. "Embedded Digital Twins in future energy management systems: paving the way for automated grid control." *at-Automatisierungstechnik* 68, no. 9 (2020): 750-764.(21 citations)

· Song, Zhao, Christoph M. Hackl, Abhinav Anand, Andre Thommessen, Jonas Petzschmann, Omar Kamel, Robert Braunbehrens, Anton Kaifel, Christian Roos, and Stefan Hauptmann. "Digital twins for the future power system: An overview and a future perspective." *Sustainability* 15, no. 6 (2023): 5259. .(32 citations)

· Mostafa, Omar & Lazarova-Molnar, Sanja. (2024). Enhancing Reliability of Energy Systems with Digital Twins: Challenges and Opportunities. (0 citations – new articles)

10. Digital Twins in Transportation and Automotive

References:

· Schwarz, Chris, and Ziran Wang. "The role of digital twins in connected and automated vehicles." *IEEE Intelligent Transportation Systems Magazine* 14, no. 6 (2022): 41-51. (91 citations)

· Bhatti, Ghanishtha, Harshit Mohan, and R. Raja Singh. "Towards the future of smart electric vehicles: Digital twin technology." *Renewable and Sustainable Energy Reviews* 141 (2021): 110801. (422 citations)

· Almeaibed, Sadeq, Saba Al-Rubaye, Antonios Tsourdos, and Nicolas P. Avdelidis. "Digital twin analysis to promote safety and security in autonomous vehicles." *IEEE Communications Standards Magazine* 5, no. 1 (2021): 40-46. (135 citations)

11. Digital Twins for Environment and Sustainability

References:

· Tzachor, Asaf, Soheil Sabri, Catherine E. Richards, Abbas Rajabifard, and Michele Acuto. "Potential and limitations of digital twins to achieve the sustainable development goals." *Nature Sustainability* 5, no. 10 (2022): 822-829. (110 citations)

· Corrado, Casey R., Suzanne M. DeLong, Emily G. Holt, Edward Y. Hua, and Andreas Tolk. "Combining green metrics and digital twins for sustainability planning and governance of smart buildings and cities." *Sustainability* 14, no. 20 (2022): 12988. (36 citations)

· Kim, Byungmo, Jaewon Oh, and Cheonhong Min. "Development of a simulation model for digital twin of an oscillating water column wave power generator structure with ocean environmental effect." *Sensors* 23, no. 23 (2023): 9472. (3 citations)

12. Digital Twins in Agriculture

References:

· Peladarinos, Nikolaos, Dimitrios Piromalis, Vasileios Cheimaras, Efthymios Tserepas, Radu Adrian Munteanu, and Panagiotis Papageorgas. "Enhancing smart agriculture by implementing digital twins: A comprehensive review." *Sensors* 23, no. 16 (2023): 7128. (60 citations)

· Escribà-Gelonch, Marc, Shu Liang, Pieter van Schalkwyk, Ian Fisk, Nguyen Van Duc Long, and Volker Hessel. "Digital Twins in Agriculture: Orchestration and Applications." *Journal of Agricultural and Food Chemistry* 72, no. 19 (2024): 10737-10752. (12 citations)

· Verdouw, Cor, Bedir Tekinerdogan, Adrie Beulens, and Sjaak Wolfert. "Digital twins in smart farming." *Agricultural Systems* 189 (2021): 103046. (494 citations)

V

Seminar: New Trends in Artificial Intelligence Techniques for Noise Prediction (Master)

2513105, WS 25/26, 2 SWS, Language: English, Open in study portal

Seminar (S)
On-Site

Content

Noise, especially in urban areas, is a major environmental issue that impacts quality of life and health, contributing to stress, sleep disturbances, and cardiovascular problems. Traffic noise, primarily from tire-road interactions, has become more prominent as electric vehicles reduce engine noise. Tackling this issue involves both passive methods, like noise barriers, and active solutions such as noise cancellation technologies.

In recent years, artificial intelligence (AI) has emerged as a powerful tool for managing noise. AI-based systems can classify noise sources, create noise maps, and develop control strategies. Advanced AI techniques, including Generative Adversarial Networks (GANs), AutoEncoders, Bi-Long Short-Term Memory (LSTM), and Bi-Gated Recurrent Units (GRUs), Graphical Convolutional Networks (GCN), Physics-informed neural networks, YOLO, Transformer, show great potential for reducing noise. Additionally, many computer vision techniques are used to improve noise conditions. This seminar will explore these AI methods and their role in enhancing conditions safety, minimizing environmental noise, and supporting intelligent transportation systems.

In this seminar, we try to understand Noise through data analysis and other techniques. We discuss current approaches to noise prediction and innovative AI approaches based on data science and machine learning.

Topics:

Introduction to Noise and Tire-Road Noise

Overview on Noise and Tire-Road Noise

Time Series Analysis and Image Analysis

Data Exploration and Visualization

Noise Feature Extraction and Analysis

Machine learning and Deep Learning Approach for Tire-Road Noise

Who are we looking for:

We are looking for students who want to expand their specialist knowledge and practical experience in artificial intelligence, signal processing, computer vision and road-tire noise. Participation provides the opportunity to actively participate in shaping the future of using artificial intelligence, computer vision and signal processing to reduce road-tire and traffic noise.

What we offer:

We provide you with tyre-road noise data. With this data, you can apply many signal processing, computer vision and artificial intelligence algorithms. This is where you can let your creativity run free and implement innovative solutions with our guidance.

Literature

- Stansfeld, S., Clark, C., Smuk, M., Gallacher, J., & Babisch, W. (2021). Road traffic noise, noise sensitivity, noise annoyance, psychological and physical health and mortality. Environmental Health, 20, 1-15.
- Lan, Y., Roberts, H., Kwan, M. P., & Helbich, M. (2020). Transportation noise exposure and anxiety: A systematic review and meta-analysis. Environmental research, 191, 110118.
- WHO. (2018) Environmental Noise Guidelines for the European Region.
- EU. (2019) Regulation (EU) No 540/2014 of the European Parliament and of the Council of 16 April 2014 on the Sound Level of Motor Vehicles and of Replacement Silencing Systems, and Amending Directive 2007/46/EC and Repealing Directive 70/157/EEC.

V **Seminar Linked Data and the Semantic Web (Master)** **Seminar (S) On-Site**

2513313, WS 25/26, 3 SWS, Language: German/English, Open in study portal

Content
Linked Data is a way of publishing data on the web in a machine-understandable fashion. The aim of this practical seminar is to build applications and devise algorithms that consume, provide, or analyse Linked Data.

The Linked Data principles are a set of practices for data publishing on the web. Linked Data builds on the web architecture and uses HTTP for data access, and RDF for describing data, thus aiming towards web-scale data integration. There is a vast amount of data available published according to those principles: recently, 4.5 billion facts have been counted with information about various domains, including music, movies, geography, natural sciences. Linked Data is also used to make web-pages machine-understandable, corresponding annotations are considered by the big search engine providers. On a smaller scale, devices on the Internet of Things can also be accessed using Linked Data which makes the unified processing of device data and data from the web easy.

In this practical seminar, students will build prototypical applications and devise algorithms that consume, provide, or analyse Linked Data. Those applications and algorithms can also extend existing applications ranging from databases to mobile apps.

For the seminar, programming skills or knowledge about web development tools/technologies are highly recommended. Basic knowledge of RDF and SPARQL are also recommended, but may be acquired during the seminar. Students will work in groups. Seminar meetings will take place as 'Block-Seminar'.

Topics of interest include, but are not limited to:

- Travel Security
- Geo data
- Linked News
- Social Media

The exact dates and information for registration will be announced at the event page.

V **Seminar Real-World Challenges in Data Science and Analytics (Bachelor)**
2513314, WS 25/26, 3 SWS, Language: German/English, Open in study portal
On-Site

Content
In the seminar, various Real-World Challenges in Data Science and Analytics will be worked on.

During this seminar, groups of students work on a case challenge with data provided. Here, the typical process of a data science project is depicted: integration of data, analysis of these, modeling of the decisions and visualization of the results.

During the seminar, solution concepts are worked out, implemented as a software solution and presented in an intermediate and final presentation. The seminar "Real-World Challenges in Data Science and Analytics" is aimed at students in master's programs.

The exact dates and information for registration will be announced at the course page.

V **Seminar Real-World Challenges in Data Science and Analytics (Master)**
2513315, WS 25/26, 3 SWS, Language: German/English, Open in study portal
On-Site

Content
In the seminar, various Real-World Challenges in Data Science and Analytics will be worked on.

During this seminar, groups of students work on a case challenge with data provided. Here, the typical process of a data science project is depicted: integration of data, analysis of these, modeling of the decisions and visualization of the results.

During the seminar, solution concepts are worked out, implemented as a software solution and presented in an intermediate and final presentation. The seminar "Real-World Challenges in Data Science and Analytics" is aimed at students in master's programs.

The exact dates and information for registration will be announced at the course page.

V **Seminar Cognitive Automobiles and Robots (Master)**
2513500, WS 25/26, 2 SWS, Language: German/English, Open in study portal
Seminar (S)
Blended (On-Site/Online)

Content
The seminar is intended as a theoretical supplement to lectures such as "Machine Learning". The theoretical basics will be deepened in the seminar. The aim of the seminar is that the participants work individually to analyze a subsystem from the field of robotics and cognitive systems using one or more procedures from the field of AI/ML.

The individual projects require the analysis of the task at hand, selection of suitable procedures, specification and theoretical evaluation of the approach taken. Finally, the chosen solution has to be documented and presented in a short presentation.

Learning objectives:

- Students can apply knowledge from the Machine Learning lecture in a selected field of current research in robotics or cognitive automobiles for theoretical analysis.
- Students can evaluate, document and present their concepts and results.

Recommendations:

Attendance of the lecture machine learning

Workload:

The workload of 3 credit points consists of the time spent on literature research and planning/specifying the proposed solution. In addition, a short report and a presentation of the work carried out will be prepared.

Organizational issues
Anmeldung und weitere Informationen sind im Wiwi-Portal zu finden.

Registration and further information can be found in the WiWi-portal.

V **Seminar: From Physical Models to Digital Twins: A Data-Driven Simulation Workshop (Seminar/Master)** **Seminar (S) On-Site**
2512101, SS 2026, 2 SWS, Language: English, Open in study portal

Content
This seminar focuses on the data-driven discovery of simulation models in industrial settings, providing a hands-on approach to understanding and optimizing production processes.
Students will start by designing and constructing production lines using Lego Spike and similar modular systems. This activity will include developing comprehensive data-capturing pipelines to collect detailed event-logging raw data from their production lines.
Next, the seminar will explore advanced techniques for transforming this raw data into simulation models, e.g., Petri nets. Participants will learn and apply data-driven model extraction methods, such as process mining to extract workflow processes; statistical methods to fit probability distributions and analyze trends, and machine learning algorithms to model complex behaviors within the production process. Through these techniques, students will extract simulation models that reflect the real-world dynamics of their production lines. The seminar will then guide participants on how to validate the extracted simulation models to ensure their accuracy.
By the end of the seminar, students will be equipped with the skills to build model production lines, collect event logging data from them, transform event log data into actionable simulation models and use these models to drive efficiency and innovation in industrial production settings.

Grading Scheme:

Report - 50%

Presentations - 40%

Implementation - 10%

V **Seminar: Applications of Digital Twins (Master)** **Seminar (S) On-Site**
2513103, SS 2026, 2 SWS, Language: English, Open in study portal

Content

Seminar Name: Applications of Digital Twins

Size: 10 students (with 10 different topics)

Workload:

- 2 Lectures:
  - o Introduction to Digital Twins and topic distribution
  - o "How to Give Effective Presentations" lecture
- 10 student presentations (each 45 minutes in total)
- 10 student reports

Responsible Person: Hui Min Lee, Sanja Lazarova-Molnar

Deliverables for Grade:

· 1 Report per student and topic (8 pages, including references, IEEE Template, compulsory usage of Reference Manager – Zotero or EndNote)

· 25 mins presentation per student plus 20 min discussion (focus on the presentation topic + presentation skills) = 45 minutes for each student

Credits: 3 credits = 90 hours

Format/ Structure of the Seminar (Draft):

· 2 Lectures at the beginning of the semester

· Students have 1 week time to provide a priority list of 5 topics, distribution will be decided based on first come – first serve, ensuring that core topics are covered

· Q&As can be asked and answered over mails or ad-hoc appointments

· Students have time to work on the report and presentation during the semester

· Submission of all reports will be required 2 months after the intro lecture for ensuring fairness

· Presentations are done in blocks of 2 students per class, starting mid-June, presentations will be submitted at the day of the scheduled presentation

Approximate Time Consumption for Students (Draft):

· Lectures: 3 hours

· Student Presentations: 7.5 hours

· Topic Subscription: 1 hour

· Presentation Preparation: 15 hours

· Paper Writing and Literature Review: 63.5 hours

Description:

The seminar focuses on applications of Digital Twins and data-driven modeling, with an additional goal of improving scientific research and presentation skills for Master students. The seminar covers the diverse applications and use cases of Digital Twins in different domains such as manufacturing, energy systems, healthcare and many more, offering students an in-depth understanding of the role of Digital Twins in transforming the industries.

The seminar is structured as a literature review seminar. Each student can select a topic out of a predefined set, conduct further research and then write a comprehensive research paper. Students will also deliver presentations, synthesizing insights from both the provided starting reference literature and their own additional research.

By the end of the course, students will not only have a solid understanding of the current applications of Digital Twins and emerging trends but also be well-prepared to present their findings in an academic setting.

Topics:

1. Digital Twins for Manufacturing Systems

References:

· Zhang, Chenyuan, et al. "A reconfigurable modeling approach for digital twin-based manufacturing system." *Procedia Cirp* 83 (2019): 118-125. (96 citations)

· Kritzinger, Werner, et al. "Digital Twin in manufacturing: A categorical literature review and classification." *Ifac-PapersOnline* 51.11 (2018): 1016-1022. (1934 citations)

· Jaensch, Florian, et al. "Digital twins of manufacturing systems as a base for machine learning." *2018 25th International conference on mechatronics and machine vision in practice (M2VIP)*. IEEE, 2018. (73 citations)

2. Digital Twins for Energy Systems

References:

· Steindl, Gernot, et al. "Generic digital twin architecture for industrial energy systems." *Applied Sciences* 10.24 (2020): 8903. (78 citations)

· Granacher, Julia, et al. "Overcoming decision paralysis—A digital twin for decision making in energy system design." *Applied Energy* 306 (2022): 117954. (33 citations) -> focus on interactive digital twins

· Palensky, Peter, et al. "Digital twins and their use in future power systems." *Digital Twin* 1 (2022): 4. (37 citations)

3. Digital Twins in Healthcare

References:

· Alazab, Mamoun, et al. "Digital twins for healthcare 4.0-recent advances, architecture, and open challenges." *IEEE Consumer Electronics Magazine* (2022). (26 citations)

· Croatti, Angelo, et al. "On the integration of agents and digital twins in healthcare." *Journal of Medical Systems* 44 (2020): 1-8. (163 citations)

· Erol, Tolga, Arif Furkan Mendi, and Dilara Doğan. "The digital twin revolution in healthcare." *2020 4th international symposium on multidisciplinary studies and innovative technologies (ISMSIT)*. IEEE, 2020. (106 citations)

4. Digital Twins of City Infrastructures (in Smart Cities)

References:

· Deren, Li, Yu Wenbo, and Shao Zhenfeng. "Smart city based on digital twins." *Computational Urban Science* 1 (2021): 1-11. (110 citations)

· Deng, Tianhu, Keren Zhang, and Zuo-Jun Max Shen. "A systematic review of a digital twin city: A new pattern of urban governance toward smart cities." *Journal of Management Science and Engineering* 6.2 (2021): 125-134. (192 citations)

· Mylonas, Georgios, et al. "Digital twins from smart manufacturing to smart cities: A survey." *Ieee Access* 9 (2021): 143222-143249. (99 citations)

5. Digital Twins in Logistics

References:

· Moshood, Taofeeq D., et al. "Digital twins driven supply chain visibility within logistics: A new paradigm for future logistics." *Applied System Innovation* 4.2 (2021): 29. (71 citations)

· Agalianos, K., et al. "Discrete event simulation and digital twins: review and challenges for logistics." *Procedia Manufacturing* 51 (2020): 1636-1641. (74 citations)

· Korth, Benjamin, Christian Schwede, and Markus Zajac. "Simulation-ready digital twin for realtime management of logistics systems." *2018 IEEE international conference on big data (big data)*. IEEE, 2018. (64 citations)

6. Cognitive Digital Twins

References:

· Al Faruque, Mohammad Abdullah, et al. "Cognitive digital twin for manufacturing systems." *2021 Design, Automation & Test in Europe Conference & Exhibition (DATE)*. IEEE, 2021. (28 citations)

· Zhang, Nan, Rami Bahsoon, and Georgios Theodoropoulos. "Towards engineering cognitive digital twins with self-awareness." *2020 IEEE International Conference on Systems, Man, and Cybernetics (SMC)*. IEEE, 2020. (22 citations)

· Zheng, Xiaochen, Jinzhi Lu, and Dimitris Kiritsis. "The emergence of cognitive digital twin: vision, challenges and opportunities." *International Journal of Production Research* 60.24 (2022): 7610-7632. (92 citations)

7. Fusing Data and Human Expert Knowledge in Digital Twins

References:

· Kulkarni, Vinay, Souvik Barat, and Tony Clark. "Towards adaptive enterprises using digital twins." *2019 winter simulation conference (WSC)*. IEEE, 2019. (22 citations)

· Vogel-Heuser, Birgit, et al. "Potential for combining semantics and data analysis in the context of digital twins." *Philosophical Transactions of the Royal Society A* 379.2207 (2021): 20200368. (16 citations)

· Todorovski, Ljupčo, and Sašo Džeroski. "Integrating knowledge-driven and data-driven approaches to modeling." *ecological modelling* 194.1-3 (2006): 3-13. (80 citations)

8. Digital Twins for Multi-agent / Complex Systems

References:

· Pretel, Elena, Alejandro Moya, Elena Navarro, Víctor López-Jaquero, and Pascual González. "Analysing the synergies between Multi-agent Systems and Digital Twins: A systematic literature review." *Information and Software Technology* (2024): 107503. (2 citations)

· Mariani, S., Picone, M., Ricci, A. (2022). About Digital Twins, Agents, and Multiagent Systems: A Cross-Fertilisation Journey. In: Melo, F.S., Fang, F. (eds) *Autonomous Agents and Multiagent Systems. Best and Visionary Papers. AAMAS 2022*. Lecture Notes in Computer Science, vol 13441. (11 citations)

· Marah H, Challenger M. Adaptive hybrid reasoning for agent-based digital twins of distributed multi-robot systems. *SIMULATION.* 2024;100(9):931-957. (0 citations – new articles)

9. Digital Twins for Energy Systems

References:

· Kabir, Md Rafiul, Dipal Halder, and Sandip Ray. "Digital Twins for IoT-driven Energy Systems: A Survey." *IEEE Access* (2024). (0 citations – new articles)

· Brosinsky, Christoph, Rainer Krebs, and Dirk Westermann. "Embedded Digital Twins in future energy management systems: paving the way for automated grid control." *at-Automatisierungstechnik* 68, no. 9 (2020): 750-764.(21 citations)

· Song, Zhao, Christoph M. Hackl, Abhinav Anand, Andre Thommessen, Jonas Petzschmann, Omar Kamel, Robert Braunbehrens, Anton Kaifel, Christian Roos, and Stefan Hauptmann. "Digital twins for the future power system: An overview and a future perspective." *Sustainability* 15, no. 6 (2023): 5259. .(32 citations)

· Mostafa, Omar & Lazarova-Molnar, Sanja. (2024). Enhancing Reliability of Energy Systems with Digital Twins: Challenges and Opportunities. (0 citations – new articles)

10. Digital Twins in Transportation and Automotive

References:

· Schwarz, Chris, and Ziran Wang. "The role of digital twins in connected and automated vehicles." *IEEE Intelligent Transportation Systems Magazine* 14, no. 6 (2022): 41-51. (91 citations)

· Bhatti, Ghanishtha, Harshit Mohan, and R. Raja Singh. "Towards the future of smart electric vehicles: Digital twin technology." *Renewable and Sustainable Energy Reviews* 141 (2021): 110801. (422 citations)

· Almeaibed, Sadeq, Saba Al-Rubaye, Antonios Tsourdos, and Nicolas P. Avdelidis. "Digital twin analysis to promote safety and security in autonomous vehicles." *IEEE Communications Standards Magazine* 5, no. 1 (2021): 40-46. (135 citations)

11. Digital Twins for Environment and Sustainability

References:

· Tzachor, Asaf, Soheil Sabri, Catherine E. Richards, Abbas Rajabifard, and Michele Acuto. "Potential and limitations of digital twins to achieve the sustainable development goals." *Nature Sustainability* 5, no. 10 (2022): 822-829. (110 citations)

· Corrado, Casey R., Suzanne M. DeLong, Emily G. Holt, Edward Y. Hua, and Andreas Tolk. "Combining green metrics and digital twins for sustainability planning and governance of smart buildings and cities." *Sustainability* 14, no. 20 (2022): 12988. (36 citations)

· Kim, Byungmo, Jaewon Oh, and Cheonhong Min. "Development of a simulation model for digital twin of an oscillating water column wave power generator structure with ocean environmental effect." *Sensors* 23, no. 23 (2023): 9472. (3 citations)

12. Digital Twins in Agriculture

References:

· Peladarinos, Nikolaos, Dimitrios Piromalis, Vasileios Cheimaras, Efthymios Tserepas, Radu Adrian Munteanu, and Panagiotis Papageorgas. "Enhancing smart agriculture by implementing digital twins: A comprehensive review." *Sensors* 23, no. 16 (2023): 7128. (60 citations)

· Escribà-Gelonch, Marc, Shu Liang, Pieter van Schalkwyk, Ian Fisk, Nguyen Van Duc Long, and Volker Hessel. "Digital Twins in Agriculture: Orchestration and Applications." *Journal of Agricultural and Food Chemistry* 72, no. 19 (2024): 10737-10752. (12 citations)

· Verdouw, Cor, Bedir Tekinerdogan, Adrie Beulens, and Sjaak Wolfert. "Digital twins in smart farming." *Agricultural Systems* 189 (2021): 103046. (494 citations)

V

Seminar Knowledge Discovery and Data Mining (Master)

2513309, SS 2026, 2 SWS, Language: English, Open in study portal

Seminar (S)
On-Site

Content

In this seminar different machine learning and data mining methods are implemented.

The seminar includes different methods of machine learning and data mining. Participants of the seminar should have basic knowledge of machine learning and programming skills.

Domains of interest include, but are not limited to:

- Medicine
- Social Media
- Finance Market
- Scientific Publications

Further Information: https://aifb.kit.edu/web/Lehre/Praktikum_Knowledge_Discovery_and_Data_Science

The exact dates and information for registration will be announced at the event page.

Organizational issues

Die Anmeldung erfolgt über das WiWi-Portal https://portal.wiwi.kit.edu/ .

Literature

Detaillierte Referenzen werden zusammen mit den jeweiligenThemen angegeben. Allgemeine Hintergrundinformationen ergeben sich z.B.aus den folgenden Lehrbüchern:

- Mitchell, T.; Machine Learning
- McGraw Hill, Cook, D.J. and Holder, L.B. (Editors) Mining Graph Data, ISBN:0-471-73190-0
- Wiley, Manning, C. and Schütze, H.; Foundations of Statistical NLP, MIT Press, 1999.

V

Seminar Data Science & Real-time Big Data Analytics (Master)

2513311, SS 2026, 2 SWS, Language: English, Open in study portal

Seminar (S)
On-Site

Content
In this seminar, students will design applications in teams that use meaningful and creative Event Processing methods. Thereby, students have access to an existing record.

Event processing and real-time data are everywhere: financial market data, sensors, business intelligence, social media analytics, logistics. Many applications collect large volumes of data in real time and are increasingly faced with the challenge of being able to process them quickly and react promptly. The challenges of this real-time processing are currently also receiving a great deal of attention under the term "Big Data". The complex processing of real-time data requires both knowledge of methods for data analysis (data science) and their processing (real-time analytics). Seminar papers are offered on both of these areas as well as on interface topics, the input of own ideas is explicitly desired.

Further information to the practical seminar is given under the following Link:
http://seminar-cep.fzi.de

Questions are answered via the e-mail address sem-ep@fzi.de.

Organizational issues
Die Anmeldung erfolgt über das WiWi-Portal https://portal.wiwi.kit.edu/ .

V	**Cognitive Automobiles and Robots** 2513500, SS 2026, 2 SWS, Language: German/English, Open in study portal	**Seminar (S)** **On-Site**

Content
The seminar is intended as a theoretical supplement to lectures such as "Machine Learning". The theoretical basics will be deepened in the seminar. The aim of the seminar is that the participants work individually to analyze a subsystem from the field of robotics and cognitive systems using one or more procedures from the field of AI/ML.

The individual projects require the analysis of the task at hand, selection of suitable procedures, specification and theoretical evaluation of the approach taken. Finally, the chosen solution has to be documented and presented in a short presentation.

Learning objectives:

- Students can apply knowledge from the Machine Learning lecture in a selected field of current research in robotics or cognitive automobiles for theoretical analysis.
- Students can evaluate, document and present their concepts and results.

Recommendations:

Attendance of the lecture machine learning

Workload:

The workload of 3 credit points consists of the time spent on literature research and planning/specifying the proposed solution. In addition, a short report and a presentation of the work carried out will be prepared.

Organizational issues
Anmeldung und weitere Informationen sind im Wiwi-Portal zu finden.

Registration and further information can be found in the WiWi-portal.

V	**Seminar Knowledge Graphs and Large Language Models (Master)** 2513607, SS 2026, 2 SWS, Language: English, Open in study portal	**Seminar (S)** **Blended (On-Site/Online)**

Content
Large language models (LLMs) such as GPT-4 have shown remarkable capabilities in transforming various natural language processing (NLP) tasks across different domains. However, LLMs often generate incorrect answers, known as hallucinations, posing significant challenges to their usability and reliability. Additionally, LLMs operate as black boxes, making it difficult to understand how they arrive at specific conclusions, leading to transparency and explainability issues. Combining LLMs with KGs creates a powerful synergy that significantly enhances the capabilities of artificial intelligence across various tasks. This integration leverages the strengths of both technologies, with LLMs excelling at understanding and generating human-like text, and KGs providing structured, reliable information about entities and their relationships. Together, they offer a robust approach to problem-solving across diverse domains.

This seminar will focus on the intersection of LLMs and KGs, covering areas of interest including, but not limited to:

- KG completion using LLMs
- Question answering with KGs and LLMs
- Explainability of LLMs with KG integration
- Reasoning with LLMs and KGs
- Enhanced prompt engineering using KGs

Contributions of the students:
Each student will be assigned one paper on the topic, which could be a research paper discussing a novel approach or a resource paper presenting datasets, tools, etc. The student will be responsible for the following tasks:

1. **Report Writing**: Read the assigned paper thoroughly and write a 15-page seminar report explaining the methods and findings in their own words.
2. **Presenting**: Prepare and deliver a seminar presentation to share insights from the paper with other seminar participants.
3. **Conducting Experiments**: If the authors provide code, re-implement it for small-scale experiments using Google Colab or make the implementation available via GitHub.

V	**Security and Privacy Awareness** 2400125, WS 26/27, 2 SWS, Language: German/English, Open in study portal	**Seminar (S)** **Blended (On-Site/Online)**

Content
Within the framework of this interdisciplinary seminar, the topics security awareness and privacy awareness are to be considered from different perspectives. It deals with legal, information technology, psychological, social as well as philosophical aspects.

Important notes:

- The seminar language is English. This means that English is the standard for the presentation and the seminar paper, and German is only permitted for the paper after consultation with the supervisor, for example, if the topic of the thesis requires it.
- The seminar is only for MASTER students (or Mastervorzug)
- The link to enrol is for every student, regardless of the study background

Topics:
The advertised topics can be found in the wiwi portal.

V	**Seminar: From Physical Models to Digital Twins: A Data-Driven Simulation Workshop (Master)** 2512101, WS 26/27, 2 SWS, Language: English, Open in study portal	**Seminar (S)** **On-Site**

Content

This seminar focuses on the data-driven discovery of simulation models in industrial settings, providing a hands-on approach to understanding and optimizing production processes.

Students will start by designing and constructing production lines using Lego Spike and similar modular systems. This activity will include developing comprehensive data-capturing pipelines to collect detailed event-logging raw data from their production lines. Next, the seminar will explore advanced techniques for transforming this raw data into simulation models, e.g., Petri nets. Participants will learn and apply data-driven model extraction methods, such as process mining to extract workflow processes; statistical methods to fit probability distributions and analyze trends, and machine learning algorithms to model complex behaviors within the production process. Through these techniques, students will extract simulation models that reflect the real-world dynamics of their production lines. The seminar will then guide participants on how to validate the extracted simulation models to ensure their accuracy.

By the end of the seminar, students will be equipped with the skills to build model production lines, collect event logging data from them, transform event log data into actionable simulation models and use these models to drive efficiency and innovation in industrial production settings.

T 6.667 Module component: Seminar in Mathematics (Bachelor) [T-MATH-102265]

Coordinators: Prof. Dr. Daniel Hug
Dr. Franz Nestmann
PD Dr. Steffen Winter

Organisation: KIT Department of Mathematics

Part of: M-WIWI-104905 - Mathematics

Type	Credits	Grading	Version
Examination of another type	3 CP	graded	2

6.668 Module component: Seminar in Operations Research (Bachelor) [T-WIWI-103488]

Coordinators: Prof. Dr. Stefan Nickel
Prof. Dr. Steffen Rebennack
Prof. Dr. Oliver Stein

Organisation: KIT Department of Business and Economics

Part of: M-WIWI-104899 - Operations Research

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Each term	1

Courses					
WT 25/26	2500056	Seminar: Modern OR and Innovative Logistics	2 SWS	Seminar / Blended	Nickel, Mitarbeiter, Pomes, Hoffmann
WT 25/26	2550131	Seminar on Methodical Foundations of Operations Research (B)	2 SWS	Seminar / On-Site	Stein, Beck, Schwarze, Neussel
WT 25/26	2550461	Seminar on Trending Topics in Optimization and Machine Learning (Bachelor)	2 SWS	Seminar / Blended	Rebennack, Kandora
WT 25/26	2550472	Seminar on Energy and Power Systems Optimization (Bachelor)	2 SWS	Seminar / Blended	Rebennack, Kandora
ST 2026	2500028	Seminar: Modern OR and Innovative Logistics	2 SWS	Seminar / Blended	Nickel, Mitarbeiter, Pomes
ST 2026	2550131	Seminar on Methodical Foundations of Operations Research (BA)	2 SWS	Seminar / On-Site	Stein, Schwarze, Neussel, Huber
ST 2026	2550132	Seminar on Mathematical Optimization (MA)	2 SWS	Seminar / On-Site	Stein, Schwarze, Neussel, Huber
ST 2026	2550461	Seminar: Trending Topics in Machine Learning and Optimization (Bachelor)	2 SWS	Seminar / Online	Rebennack, Kandora
ST 2026	2550472	Seminar: Energy and Power Systems Optimization (Bachelor)	2 SWS	Seminar / Blended	Rebennack, Kandora
WT 26/27	2500056	Seminar: Modern OR and Innovative Logistics	2 SWS	Seminar / Blended	Nickel, Mitarbeiter, Pomes, Hoffmann
WT 26/27	2550131	Seminar on Methodical Foundations of Operations Research (B)	2 SWS	Seminar / On-Site	Stein, Huber, Neussel, Schwarze

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
Alternative exam assessment (§ 4(2), 3 SPO 2015). The following aspects are included:

- Regular participation in the seminar dates
- Preparation of a seminar paper on a partial aspect of the seminar topic according to scientific methods
- Lecture on the topic of the seminar paper.

The point scheme for the assessment is determined by the lecturer of the respective course. It will be announced at the beginning of the course.

Prerequisites
None.

Recommendations
See seminar description in the course catalogue of the KIT (https://campus.kit.edu/)

Additional Information
The listed seminar titles are placeholders. Currently offered seminars of each semester will be published on the websites of the institutes and in the course catalogue of the KIT. In general, the current seminar topics of each semester are already announced at the end of the previous semester. Furthermore for some seminars there is an application required.

The available places are listed on the internet: https://portal.wiwi.kit.edu.

Workload
90 hours

Below you will find excerpts from courses related to this module component:

V	**Seminar: Modern OR and Innovative Logistics** 2500056, WS 25/26, 2 SWS, Language: German, Open in study portal	**Seminar (S)** **Blended (On-Site/Online)**

Content
This seminar presents current issues in the field of Operations Research and Logistics, critically evaluates them, and discusses them using examples. The focus of the seminar lies on optimization models and algorithms, with particular attention to their practical applicability (especially in supply chain and health care management). All participants are required to write a seminar paper and give a presentation. Depending on the topic, an implementation of the models or heuristics using standard software (e.g., IBM CPLEX or Java) is expected. Further details can be found in the information sheet on Prof. Nickel's website. All topics have the potential to be developed into a thesis.

The seminar topics will be assigned during a preliminary meeting at the beginning of the semester. Attendance is mandatory for the preliminary meeting as well as for all seminar presentations.

In addition, students will receive an introduction to academic writing, literature research, and scientific presenting in three sessions.

Organizational issues
Die Anmeldung erfolgt über das Wiwi-Portal. Nähere Informationen erhalten Sie auf der Homepage dol.ior.kit.edu bzw. im WiWi-Portal sowie bei der Vorbesprechung.

Literature
Die Literatur und die relevanten Quellen werden zu Beginn des Seminars bekannt gegeben.

V	**Seminar on Methodical Foundations of Operations Research (B)** 2550131, WS 25/26, 2 SWS, Language: German, Open in study portal	**Seminar (S)** **On-Site**

Content
The seminar aims at describing, evaluating, and discussing recent as well as classical topics in continuous optimization. The focus is on the treatment of optimization models and algorithms, also with respect to their practical application.

Bachelor studenst are introduced to the style of scientific work. By focussed treatment of a scientific topic they deal with the basics of scientific investigation and reasoning.

For further development of a scientific work style, master students are particularly expected to critically question the seminar topics.

With regard to the oral presentations the students become acquainted with presentation techniques and basics of scientifc reasoning. Also rethoric abilities may be improved.

Remarks:

Attendance at all oral presentations is compulsory.

Preferably at least one module offered by the Institute of Operations Research should have been chosen before attending this seminar.

Assessment:

The assessment is composed of a 15-20 page paper as well as a 40-60 minute oral presentation according to §4(2), 3 of the examination regulation.The grade is composed of the equally weighted assessments of the paper and the oral presentation.

The seminar is appropriate for bachelor as well as for master students. Their differentiation results from different assessment criteria for the seminar paper and the oral presentation.

Workload:

The total workload for this course is approximately 90 hours. For further information see German version.

Literature
Die Literaur und die relevanten Quellen werden gegen Ende des vorausgehenden Semesters im Wiwi-Portal und in einer Seminarvorbesprechung bekannt gegeben.

References and relevant sources are announced at the end of the preceding semester in the Wiwi-Portal and in a prepatory meeting.

V	**Seminar: Modern OR and Innovative Logistics** 2500028, SS 2026, 2 SWS, Language: German, Open in study portal	**Seminar (S)** **Blended (On-Site/Online)**

Content
The seminar aims at the presentation, critical evaluation and exemplary discussion of recent questions in discrete optimization. The focus lies on optimization models and algorithms, also with regard to their applicability in practical cases (especially in Supply Chain and Health Care Management). The students get in touch with scientific working: The in-depth work with a special scientific topic makes the students familiar with scientific literature research and argumentation methods. As a further aspect of scientific work, especially for Master students the emphasis is put on a critical discussion of the seminar topic. Regarding the seminar presentations, the students will be familiarized with basic presentational and rhetoric skills.

Organizational issues
Anmeldung erfolgt über das Wiwi-Portal. Nähere Informationen hierzu finden Sie hier zu einem späteren Zeitpunkt.

Literature
Die Literatur und die relevanten Quellen werden zu Beginn des Seminars bekannt gegeben.

V	**Seminar on Methodical Foundations of Operations Research (BA)** 2550131, SS 2026, 2 SWS, Language: German, Open in study portal	**Seminar (S)** **On-Site**

Content
The seminar aims at describing, evaluating, and discussing recent as well as classical topics in continuous optimization. The focus is on the treatment of optimization models and algorithms, also with respect to their practical application.

Bachelor studenst are introduced to the style of scientific work. By focussed treatment of a scientific topic they deal with the basics of scientific investigation and reasoning.

For further development of a scientific work style, master students are particularly expected to critically question the seminar topics.

With regard to the oral presentations the students become acquainted with presentation techniques and basics of scientifc reasoning. Also rethoric abilities may be improved.

Remarks:

Attendance at all oral presentations is compulsory.

Preferably at least one module offered by the Institute of Operations Research should have been chosen before attending this seminar.

Assessment:

The assessment is composed of a 15-20 page paper as well as a 40-60 minute oral presentation according to §4(2), 3 of the examination regulation.The grade is composed of the equally weighted assessments of the paper and the oral presentation.

The seminar is appropriate for bachelor as well as for master students. Their differentiation results from different assessment criteria for the seminar paper and the oral presentation.

Workload:

The total workload for this course is approximately 90 hours. For further information see German version.

Literature
Die Literaur und die relevanten Quellen werden gegen Ende des vorausgehenden Semesters im Wiwi-Portal und in einer Seminarvorbesprechung bekannt gegeben.

References and relevant sources are announced at the end of the preceding semester in the Wiwi-Portal and in a prepatory meeting.

V	**Seminar: Modern OR and Innovative Logistics** 2500056, WS 26/27, 2 SWS, Language: German, Open in study portal	**Seminar (S)** **Blended (On-Site/Online)**

Content
This seminar presents current issues in the field of Operations Research and Logistics, critically evaluates them, and discusses them using examples. The focus of the seminar lies on optimization models and algorithms, with particular attention to their practical applicability (especially in supply chain and health care management). All participants are required to write a seminar paper and give a presentation. Depending on the topic, an implementation of the models or heuristics using standard software (e.g., IBM CPLEX or Java) is expected. Further details can be found in the information sheet on Prof. Nickel's website. All topics have the potential to be developed into a thesis.

The seminar topics will be assigned during a preliminary meeting at the beginning of the semester. Attendance is mandatory for the preliminary meeting as well as for all seminar presentations.

In addition, students will receive an introduction to academic writing, literature research, and scientific presenting in three sessions.

Organizational issues
Die Anmeldung erfolgt über das Wiwi-Portal. Nähere Informationen erhalten Sie auf der Homepage dol.ior.kit.edu bzw. im WiWi-Portal sowie bei der Vorbesprechung.

Literature
Die Literatur und die relevanten Quellen werden zu Beginn des Seminars bekannt gegeben.

V **Seminar on Methodical Foundations of Operations Research (B)**
2550131, WS 26/27, 2 SWS, Language: English, Open in study portal

Seminar (S)
On-Site

Content
The seminar aims at describing, evaluating, and discussing recent as well as classical topics in continuous optimization. The focus is on the treatment of optimization models and algorithms, also with respect to their practical application.

Bachelor studenst are introduced to the style of scientific work. By focussed treatment of a scientific topic they deal with the basics of scientific investigation and reasoning.

For further development of a scientific work style, master students are particularly expected to critically question the seminar topics.

With regard to the oral presentations the students become acquainted with presentation techniques and basics of scientifc reasoning. Also rethoric abilities may be improved.

Remarks:

Attendance at all oral presentations is compulsory.

Preferably at least one module offered by the Institute of Operations Research should have been chosen before attending this seminar.

Assessment:

The assessment is composed of a 15-20 page paper as well as a 40-60 minute oral presentation according to §4(2), 3 of the examination regulation.The grade is composed of the equally weighted assessments of the paper and the oral presentation.

The seminar is appropriate for bachelor as well as for master students. Their differentiation results from different assessment criteria for the seminar paper and the oral presentation.

Workload:

The total workload for this course is approximately 90 hours. For further information see German version.

Literature
Die Literaur und die relevanten Quellen werden gegen Ende des vorausgehenden Semesters im Wiwi-Portal und in einer Seminarvorbesprechung bekannt gegeben.

References and relevant sources are announced at the end of the preceding semester in the Wiwi-Portal and in a prepatory meeting.

T 6.669 Module component: Seminar in Operations Research A (Master) [T-WIWI-103481]

Coordinators: Prof. Dr. Stefan Nickel
Prof. Dr. Steffen Rebennack
Prof. Dr. Oliver Stein

Organisation: KIT Department of Business and Economics

Part of: M-WIWI-104899 - Operations Research
M-WIWI-107204 - Seminar Module

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Each term	1

Courses					
WT 25/26	2500056	Seminar: Modern OR and Innovative Logistics	2 SWS	Seminar / Blended	Nickel, Mitarbeiter, Pomes, Hoffmann
WT 25/26	2550131	Seminar on Methodical Foundations of Operations Research (B)	2 SWS	Seminar / On-Site	Stein, Beck, Schwarze, Neussel
WT 25/26	2550132	Seminar zur Mathematischen Optimierung (MA)	2 SWS	Seminar / On-Site	Stein, Beck, Schwarze, Neussel
WT 25/26	2550462	Seminar on Trending Topics in Optimization and Machine Learning (Master)	2 SWS	Seminar / Blended	Rebennack, Kandora
WT 25/26	2550473	Seminar on Energy and Power Systems Optimization (Master)	2 SWS	Seminar / Blended	Rebennack, Kandora
ST 2026	2500028	Seminar: Modern OR and Innovative Logistics	2 SWS	Seminar / Blended	Nickel, Mitarbeiter, Pomes
ST 2026	2550131	Seminar on Methodical Foundations of Operations Research (BA)	2 SWS	Seminar / On-Site	Stein, Schwarze, Neussel, Huber
ST 2026	2550132	Seminar on Mathematical Optimization (MA)	2 SWS	Seminar / On-Site	Stein, Schwarze, Neussel, Huber
ST 2026	2550462	Seminar: Trending Topics in Machine Learning and Optimization (Master)	2 SWS	Seminar / Online	Rebennack, Kandora
ST 2026	2550473	Seminar: Energy and Power Systems Optimization (Master)	2 SWS	Seminar / Blended	Rebennack, Kandora
WT 26/27	2500056	Seminar: Modern OR and Innovative Logistics	2 SWS	Seminar / Blended	Nickel, Mitarbeiter, Pomes, Hoffmann
WT 26/27	2550131	Seminar on Methodical Foundations of Operations Research (B)	2 SWS	Seminar / On-Site	Stein, Huber, Neussel, Schwarze
WT 26/27	2550132	Seminar on Mathematical Optimization (MA)	2 SWS	Seminar / On-Site	Stein, Huber, Neussel, Schwarze

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
Alternative exam assessment (§ 4(2), 3 SPO 2015). The following aspects are included:

- Regular participation in the seminar dates
- Preparation of a seminar paper on a partial aspect of the seminar topic according to scientific methods
- Lecture on the topic of the seminar paper.

The point scheme for the assessment is determined by the lecturer of the respective course. It will be announced at the beginning of the course.

Prerequisites
None.

Recommendations
See seminar description in the course catalogue of the KIT (https://campus.kit.edu/)

Additional Information
The listed seminar titles are placeholders. Currently offered seminars of each semester will be published on the websites of the institutes and in the course catalogue of the KIT. In general, the current seminar topics of each semester are already announced at the end of the previous semester. Furthermore for some seminars there is an application required.

The available places are listed on the internet: https://portal.wiwi.kit.edu.

Workload
90 hours

Below you will find excerpts from courses related to this module component:

V	**Seminar: Modern OR and Innovative Logistics** 2500056, WS 25/26, 2 SWS, Language: German, Open in study portal	**Seminar (S)** **Blended (On-Site/Online)**

Content
This seminar presents current issues in the field of Operations Research and Logistics, critically evaluates them, and discusses them using examples. The focus of the seminar lies on optimization models and algorithms, with particular attention to their practical applicability (especially in supply chain and health care management). All participants are required to write a seminar paper and give a presentation. Depending on the topic, an implementation of the models or heuristics using standard software (e.g., IBM CPLEX or Java) is expected. Further details can be found in the information sheet on Prof. Nickel's website. All topics have the potential to be developed into a thesis.

The seminar topics will be assigned during a preliminary meeting at the beginning of the semester. Attendance is mandatory for the preliminary meeting as well as for all seminar presentations.

In addition, students will receive an introduction to academic writing, literature research, and scientific presenting in three sessions.

Organizational issues
Die Anmeldung erfolgt über das Wiwi-Portal. Nähere Informationen erhalten Sie auf der Homepage dol.ior.kit.edu bzw. im WiWi-Portal sowie bei der Vorbesprechung.

Literature
Die Literatur und die relevanten Quellen werden zu Beginn des Seminars bekannt gegeben.

V	**Seminar on Methodical Foundations of Operations Research (B)** 2550131, WS 25/26, 2 SWS, Language: German, Open in study portal	**Seminar (S)** **On-Site**

Content
The seminar aims at describing, evaluating, and discussing recent as well as classical topics in continuous optimization. The focus is on the treatment of optimization models and algorithms, also with respect to their practical application.

Bachelor studenst are introduced to the style of scientific work. By focussed treatment of a scientific topic they deal with the basics of scientific investigation and reasoning.

For further development of a scientific work style, master students are particularly expected to critically question the seminar topics.

With regard to the oral presentations the students become acquainted with presentation techniques and basics of scientifc reasoning. Also rethoric abilities may be improved.

Remarks:

Attendance at all oral presentations is compulsory.

Preferably at least one module offered by the Institute of Operations Research should have been chosen before attending this seminar.

Assessment:

The assessment is composed of a 15-20 page paper as well as a 40-60 minute oral presentation according to §4(2), 3 of the examination regulation.The grade is composed of the equally weighted assessments of the paper and the oral presentation.

The seminar is appropriate for bachelor as well as for master students. Their differentiation results from different assessment criteria for the seminar paper and the oral presentation.

Workload:

The total workload for this course is approximately 90 hours. For further information see German version.

Literature
Die Literaur und die relevanten Quellen werden gegen Ende des vorausgehenden Semesters im Wiwi-Portal und in einer Seminarvorbesprechung bekannt gegeben.

References and relevant sources are announced at the end of the preceding semester in the Wiwi-Portal and in a prepatory meeting.

V	**Seminar: Modern OR and Innovative Logistics** 2500028, SS 2026, 2 SWS, Language: German, Open in study portal	**Seminar (S)** **Blended (On-Site/Online)**

Content
The seminar aims at the presentation, critical evaluation and exemplary discussion of recent questions in discrete optimization. The focus lies on optimization models and algorithms, also with regard to their applicability in practical cases (especially in Supply Chain and Health Care Management). The students get in touch with scientific working: The in-depth work with a special scientific topic makes the students familiar with scientific literature research and argumentation methods. As a further aspect of scientific work, especially for Master students the emphasis is put on a critical discussion of the seminar topic. Regarding the seminar presentations, the students will be familiarized with basic presentational and rhetoric skills.

Organizational issues
Anmeldung erfolgt über das Wiwi-Portal. Nähere Informationen hierzu finden Sie hier zu einem späteren Zeitpunkt.

Literature
Die Literatur und die relevanten Quellen werden zu Beginn des Seminars bekannt gegeben.

V	**Seminar on Methodical Foundations of Operations Research (BA)** 2550131, SS 2026, 2 SWS, Language: German, Open in study portal	**Seminar (S)** **On-Site**

Content
The seminar aims at describing, evaluating, and discussing recent as well as classical topics in continuous optimization. The focus is on the treatment of optimization models and algorithms, also with respect to their practical application.

Bachelor studenst are introduced to the style of scientific work. By focussed treatment of a scientific topic they deal with the basics of scientific investigation and reasoning.

For further development of a scientific work style, master students are particularly expected to critically question the seminar topics.

With regard to the oral presentations the students become acquainted with presentation techniques and basics of scientifc reasoning. Also rethoric abilities may be improved.

Remarks:

Attendance at all oral presentations is compulsory.

Preferably at least one module offered by the Institute of Operations Research should have been chosen before attending this seminar.

Assessment:

The assessment is composed of a 15-20 page paper as well as a 40-60 minute oral presentation according to §4(2), 3 of the examination regulation.The grade is composed of the equally weighted assessments of the paper and the oral presentation.

The seminar is appropriate for bachelor as well as for master students. Their differentiation results from different assessment criteria for the seminar paper and the oral presentation.

Workload:

The total workload for this course is approximately 90 hours. For further information see German version.

Literature
Die Literaur und die relevanten Quellen werden gegen Ende des vorausgehenden Semesters im Wiwi-Portal und in einer Seminarvorbesprechung bekannt gegeben.

References and relevant sources are announced at the end of the preceding semester in the Wiwi-Portal and in a prepatory meeting.

V	**Seminar: Modern OR and Innovative Logistics** 2500056, WS 26/27, 2 SWS, Language: German, Open in study portal	**Seminar (S)** **Blended (On-Site/Online)**

Content
This seminar presents current issues in the field of Operations Research and Logistics, critically evaluates them, and discusses them using examples. The focus of the seminar lies on optimization models and algorithms, with particular attention to their practical applicability (especially in supply chain and health care management). All participants are required to write a seminar paper and give a presentation. Depending on the topic, an implementation of the models or heuristics using standard software (e.g., IBM CPLEX or Java) is expected. Further details can be found in the information sheet on Prof. Nickel's website. All topics have the potential to be developed into a thesis.

The seminar topics will be assigned during a preliminary meeting at the beginning of the semester. Attendance is mandatory for the preliminary meeting as well as for all seminar presentations.

In addition, students will receive an introduction to academic writing, literature research, and scientific presenting in three sessions.

Organizational issues
Die Anmeldung erfolgt über das Wiwi-Portal. Nähere Informationen erhalten Sie auf der Homepage dol.ior.kit.edu bzw. im WiWi-Portal sowie bei der Vorbesprechung.

Literature
Die Literatur und die relevanten Quellen werden zu Beginn des Seminars bekannt gegeben.

V **Seminar on Methodical Foundations of Operations Research (B)** **Seminar (S) On-Site**
2550131, WS 26/27, 2 SWS, Language: English, Open in study portal

Content
The seminar aims at describing, evaluating, and discussing recent as well as classical topics in continuous optimization. The focus is on the treatment of optimization models and algorithms, also with respect to their practical application.

Bachelor studenst are introduced to the style of scientific work. By focussed treatment of a scientific topic they deal with the basics of scientific investigation and reasoning.

For further development of a scientific work style, master students are particularly expected to critically question the seminar topics.

With regard to the oral presentations the students become acquainted with presentation techniques and basics of scientifc reasoning. Also rethoric abilities may be improved.

Remarks:

Attendance at all oral presentations is compulsory.

Preferably at least one module offered by the Institute of Operations Research should have been chosen before attending this seminar.

Assessment:

The assessment is composed of a 15-20 page paper as well as a 40-60 minute oral presentation according to §4(2), 3 of the examination regulation.The grade is composed of the equally weighted assessments of the paper and the oral presentation.

The seminar is appropriate for bachelor as well as for master students. Their differentiation results from different assessment criteria for the seminar paper and the oral presentation.

Workload:

The total workload for this course is approximately 90 hours. For further information see German version.

Literature
Die Literaur und die relevanten Quellen werden gegen Ende des vorausgehenden Semesters im Wiwi-Portal und in einer Seminarvorbesprechung bekannt gegeben.

References and relevant sources are announced at the end of the preceding semester in the Wiwi-Portal and in a prepatory meeting.

V **Seminar on Mathematical Optimization (MA)** **Seminar (S) On-Site**
2550132, WS 26/27, 2 SWS, Language: English, Open in study portal

Content
The seminar aims at describing, evaluating, and discussing recent as well as classical topics in continuous optimization. The focus is on the treatment of optimization models and algorithms, also with respect to their practical application.

Bachelor studenst are introduced to the style of scientific work. By focussed treatment of a scientific topic they deal with the basics of scientific investigation and reasoning.

For further development of a scientific work style, master students are particularly expected to critically question the seminar topics.

With regard to the oral presentations the students become acquainted with presentation techniques and basics of scientifc reasoning. Also rethoric abilities may be improved.

Remarks:

Attendance at all oral presentations is compulsory.

Preferably at least one module offered by the Institute of Operations Research should have been chosen before attending this seminar.

Assessment:

The assessment is composed of a 15-20 page paper as well as a 40-60 minute oral presentation according to §4(2), 3 of the examination regulation.The grade is composed of the equally weighted assessments of the paper and the oral presentation.

The seminar is appropriate for bachelor as well as for master students. Their differentiation results from different assessment criteria for the seminar paper and the oral presentation.

Workload:

The total workload for this course is approximately 90 hours. For further information see German version.

T 6.670 Module component: Seminar in Operations Research B (Master) [T-WIWI-103482]

Coordinators: Prof. Dr. Stefan Nickel
Prof. Dr. Steffen Rebennack
Prof. Dr. Oliver Stein

Organisation: KIT Department of Business and Economics

Part of: M-WIWI-104899 - Operations Research
M-WIWI-107204 - Seminar Module

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Each term	1

Courses					
WT 25/26	2500056	Seminar: Modern OR and Innovative Logistics	2 SWS	Seminar /	Nickel, Mitarbeiter, Pomes, Hoffmann
WT 25/26	2550131	Seminar on Methodical Foundations of Operations Research (B)	2 SWS	Seminar /	Stein, Beck, Schwarze, Neussel
WT 25/26	2550132	Seminar zur Mathematischen Optimierung (MA)	2 SWS	Seminar /	Stein, Beck, Schwarze, Neussel
WT 25/26	2550462	Seminar on Trending Topics in Optimization and Machine Learning (Master)	2 SWS	Seminar /	Rebennack, Kandora
WT 25/26	2550473	Seminar on Energy and Power Systems Optimization (Master)	2 SWS	Seminar /	Rebennack, Kandora
ST 2026	2500028	Seminar: Modern OR and Innovative Logistics	2 SWS	Seminar /	Nickel, Mitarbeiter, Pomes
ST 2026	2550131	Seminar on Methodical Foundations of Operations Research (BA)	2 SWS	Seminar /	Stein, Schwarze, Neussel, Huber
ST 2026	2550132	Seminar on Mathematical Optimization (MA)	2 SWS	Seminar /	Stein, Schwarze, Neussel, Huber
ST 2026	2550462	Seminar: Trending Topics in Machine Learning and Optimization (Master)	2 SWS	Seminar /	Rebennack, Kandora
ST 2026	2550473	Seminar: Energy and Power Systems Optimization (Master)	2 SWS	Seminar /	Rebennack, Kandora
WT 26/27	2500056	Seminar: Modern OR and Innovative Logistics	2 SWS	Seminar /	Nickel, Mitarbeiter, Pomes, Hoffmann
WT 26/27	2550131	Seminar on Methodical Foundations of Operations Research (B)	2 SWS	Seminar /	Stein, Huber, Neussel, Schwarze
WT 26/27	2550132	Seminar on Mathematical Optimization (MA)	2 SWS	Seminar /	Stein, Huber, Neussel, Schwarze

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
Alternative exam assessment (§ 4(2), 3 SPO 2015). The following aspects are included:

- Regular participation in the seminar dates
- Preparation of a seminar paper on a partial aspect of the seminar topic according to scientific methods
- Lecture on the topic of the seminar paper.

The point scheme for the assessment is determined by the lecturer of the respective course. It will be announced at the beginning of the course.

Prerequisites
None.

Recommendations
See seminar description in the course catalogue of the KIT (https://campus.kit.edu/)

Additional Information
The listed seminar titles are placeholders. Currently offered seminars of each semester will be published on the websites of the institutes and in the course catalogue of the KIT. In general, the current seminar topics of each semester are already announced at the end of the previous semester. Furthermore for some seminars there is an application required.

The available places are listed on the internet: https://portal.wiwi.kit.edu.

Workload
90 hours

Below you will find excerpts from courses related to this module component:

V	**Seminar: Modern OR and Innovative Logistics** 2500056, WS 25/26, 2 SWS, Language: German, Open in study portal	**Seminar (S)** **Blended (On-Site/Online)**

Content
This seminar presents current issues in the field of Operations Research and Logistics, critically evaluates them, and discusses them using examples. The focus of the seminar lies on optimization models and algorithms, with particular attention to their practical applicability (especially in supply chain and health care management). All participants are required to write a seminar paper and give a presentation. Depending on the topic, an implementation of the models or heuristics using standard software (e.g., IBM CPLEX or Java) is expected. Further details can be found in the information sheet on Prof. Nickel's website. All topics have the potential to be developed into a thesis.

The seminar topics will be assigned during a preliminary meeting at the beginning of the semester. Attendance is mandatory for the preliminary meeting as well as for all seminar presentations.

In addition, students will receive an introduction to academic writing, literature research, and scientific presenting in three sessions.

Organizational issues
Die Anmeldung erfolgt über das Wiwi-Portal. Nähere Informationen erhalten Sie auf der Homepage dol.ior.kit.edu bzw. im WiWi-Portal sowie bei der Vorbesprechung.

Literature
Die Literatur und die relevanten Quellen werden zu Beginn des Seminars bekannt gegeben.

V	**Seminar on Methodical Foundations of Operations Research (B)** 2550131, WS 25/26, 2 SWS, Language: German, Open in study portal	**Seminar (S)** **On-Site**

Content
The seminar aims at describing, evaluating, and discussing recent as well as classical topics in continuous optimization. The focus is on the treatment of optimization models and algorithms, also with respect to their practical application.

Bachelor studenst are introduced to the style of scientific work. By focussed treatment of a scientific topic they deal with the basics of scientific investigation and reasoning.

For further development of a scientific work style, master students are particularly expected to critically question the seminar topics.

With regard to the oral presentations the students become acquainted with presentation techniques and basics of scientifc reasoning. Also rethoric abilities may be improved.

Remarks:

Attendance at all oral presentations is compulsory.

Preferably at least one module offered by the Institute of Operations Research should have been chosen before attending this seminar.

Assessment:

The assessment is composed of a 15-20 page paper as well as a 40-60 minute oral presentation according to §4(2), 3 of the examination regulation.The grade is composed of the equally weighted assessments of the paper and the oral presentation.

The seminar is appropriate for bachelor as well as for master students. Their differentiation results from different assessment criteria for the seminar paper and the oral presentation.

Workload:

The total workload for this course is approximately 90 hours. For further information see German version.

Literature
Die Literaur und die relevanten Quellen werden gegen Ende des vorausgehenden Semesters im Wiwi-Portal und in einer Seminarvorbesprechung bekannt gegeben.

References and relevant sources are announced at the end of the preceding semester in the Wiwi-Portal and in a prepatory meeting.

V **Seminar: Modern OR and Innovative Logistics**
2500028, SS 2026, 2 SWS, Language: German, Open in study portal
Seminar (S)
Blended (On-Site/Online)

Content
The seminar aims at the presentation, critical evaluation and exemplary discussion of recent questions in discrete optimization. The focus lies on optimization models and algorithms, also with regard to their applicability in practical cases (especially in Supply Chain and Health Care Management). The students get in touch with scientific working: The in-depth work with a special scientific topic makes the students familiar with scientific literature research and argumentation methods. As a further aspect of scientific work, especially for Master students the emphasis is put on a critical discussion of the seminar topic. Regarding the seminar presentations, the students will be familiarized with basic presentational and rhetoric skills.

Organizational issues
Anmeldung erfolgt über das Wiwi-Portal. Nähere Informationen hierzu finden Sie hier zu einem späteren Zeitpunkt.

Literature
Die Literatur und die relevanten Quellen werden zu Beginn des Seminars bekannt gegeben.

V **Seminar on Methodical Foundations of Operations Research (BA)**
2550131, SS 2026, 2 SWS, Language: German, Open in study portal
Seminar (S)
On-Site

Content
The seminar aims at describing, evaluating, and discussing recent as well as classical topics in continuous optimization. The focus is on the treatment of optimization models and algorithms, also with respect to their practical application.

Bachelor studenst are introduced to the style of scientific work. By focussed treatment of a scientific topic they deal with the basics of scientific investigation and reasoning.

For further development of a scientific work style, master students are particularly expected to critically question the seminar topics.

With regard to the oral presentations the students become acquainted with presentation techniques and basics of scientifc reasoning. Also rethoric abilities may be improved.

Remarks:

Attendance at all oral presentations is compulsory.

Preferably at least one module offered by the Institute of Operations Research should have been chosen before attending this seminar.

Assessment:

The assessment is composed of a 15-20 page paper as well as a 40-60 minute oral presentation according to §4(2), 3 of the examination regulation.The grade is composed of the equally weighted assessments of the paper and the oral presentation.

The seminar is appropriate for bachelor as well as for master students. Their differentiation results from different assessment criteria for the seminar paper and the oral presentation.

Workload:

The total workload for this course is approximately 90 hours. For further information see German version.

Literature
Die Literaur und die relevanten Quellen werden gegen Ende des vorausgehenden Semesters im Wiwi-Portal und in einer Seminarvorbesprechung bekannt gegeben.

References and relevant sources are announced at the end of the preceding semester in the Wiwi-Portal and in a prepatory meeting.

V **Seminar: Modern OR and Innovative Logistics**
2500056, WS 26/27, 2 SWS, Language: German, Open in study portal
Seminar (S)
Blended (On-Site/Online)

Content
This seminar presents current issues in the field of Operations Research and Logistics, critically evaluates them, and discusses them using examples. The focus of the seminar lies on optimization models and algorithms, with particular attention to their practical applicability (especially in supply chain and health care management). All participants are required to write a seminar paper and give a presentation. Depending on the topic, an implementation of the models or heuristics using standard software (e.g., IBM CPLEX or Java) is expected. Further details can be found in the information sheet on Prof. Nickel's website. All topics have the potential to be developed into a thesis.

The seminar topics will be assigned during a preliminary meeting at the beginning of the semester. Attendance is mandatory for the preliminary meeting as well as for all seminar presentations.

In addition, students will receive an introduction to academic writing, literature research, and scientific presenting in three sessions.

Organizational issues
Die Anmeldung erfolgt über das Wiwi-Portal. Nähere Informationen erhalten Sie auf der Homepage dol.ior.kit.edu bzw. im WiWi-Portal sowie bei der Vorbesprechung.

Literature
Die Literatur und die relevanten Quellen werden zu Beginn des Seminars bekannt gegeben.

V **Seminar on Methodical Foundations of Operations Research (B)**
2550131, WS 26/27, 2 SWS, Language: English, Open in study portal
Seminar (S)
On-Site

Content
The seminar aims at describing, evaluating, and discussing recent as well as classical topics in continuous optimization. The focus is on the treatment of optimization models and algorithms, also with respect to their practical application.

Bachelor studenst are introduced to the style of scientific work. By focussed treatment of a scientific topic they deal with the basics of scientific investigation and reasoning.

For further development of a scientific work style, master students are particularly expected to critically question the seminar topics.

With regard to the oral presentations the students become acquainted with presentation techniques and basics of scientifc reasoning. Also rethoric abilities may be improved.

Remarks:

Attendance at all oral presentations is compulsory.

Preferably at least one module offered by the Institute of Operations Research should have been chosen before attending this seminar.

Assessment:

The assessment is composed of a 15-20 page paper as well as a 40-60 minute oral presentation according to §4(2), 3 of the examination regulation.The grade is composed of the equally weighted assessments of the paper and the oral presentation.

The seminar is appropriate for bachelor as well as for master students. Their differentiation results from different assessment criteria for the seminar paper and the oral presentation.

Workload:

The total workload for this course is approximately 90 hours. For further information see German version.

Literature
Die Literaur und die relevanten Quellen werden gegen Ende des vorausgehenden Semesters im Wiwi-Portal und in einer Seminarvorbesprechung bekannt gegeben.

References and relevant sources are announced at the end of the preceding semester in the Wiwi-Portal and in a prepatory meeting.

V **Seminar on Mathematical Optimization (MA)**
2550132, WS 26/27, 2 SWS, Language: English, Open in study portal
Seminar (S)
On-Site

Content
The seminar aims at describing, evaluating, and discussing recent as well as classical topics in continuous optimization. The focus is on the treatment of optimization models and algorithms, also with respect to their practical application.

Bachelor studenst are introduced to the style of scientific work. By focussed treatment of a scientific topic they deal with the basics of scientific investigation and reasoning.

For further development of a scientific work style, master students are particularly expected to critically question the seminar topics.

With regard to the oral presentations the students become acquainted with presentation techniques and basics of scientifc reasoning. Also rethoric abilities may be improved.

Remarks:

Attendance at all oral presentations is compulsory.

Preferably at least one module offered by the Institute of Operations Research should have been chosen before attending this seminar.

Assessment:

The assessment is composed of a 15-20 page paper as well as a 40-60 minute oral presentation according to §4(2), 3 of the examination regulation.The grade is composed of the equally weighted assessments of the paper and the oral presentation.

The seminar is appropriate for bachelor as well as for master students. Their differentiation results from different assessment criteria for the seminar paper and the oral presentation.

Workload:

The total workload for this course is approximately 90 hours. For further information see German version.

T 6.671 Module component: Seminar in Statistics (Bachelor) [T-WIWI-103489]

Coordinators: Prof. Dr. Oliver Grothe
Prof. Dr. Melanie Schienle
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104902 - Statistics

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Each term	1

Courses					
WT 25/26	25000111	Statistics and Epidemics	2 SWS	Seminar / 🗣	Bracher, Nemcova
WT 25/26	2500018		2 SWS	Seminar / 🗣	Grothe, Liu
WT 25/26	2521310	Topics in Econometrics	2 SWS	Seminar	Schienle, Krüger, Bracher, Buse, Koster, Rüter, Leimenstoll, Renner, Pohle
ST 2026	2500012	Making and Evaluating Predictions	2 SWS	Seminar	Schienle, Allen, Zhang
ST 2026	2521310	Advanced Topics in Econometrics	2 SWS	Seminar	Schienle, Krüger, Bracher, Buse, Koster, Rüter, Leimenstoll, Renner
ST 2026	2550560	Spezielle Themen zu Statistik, Datenanalyse und maschinellem Lernen	2 SWS	Seminar / 🗣	Grothe, Liu
WT 26/27	2521310	Topics in Econometrics	2 SWS	Seminar	Schienle, Krüger, Bracher, Buse, Koster, Rüter, Leimenstoll, Renner

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Alternative exam assessment (§ 4(2), 3 SPO 2015). The following aspects are included:

- Regular participation in the seminar dates
- Preparation of a seminar paper on a partial aspect of the seminar topic according to scientific methods
- Lecture on the topic of the seminar paper.

The point scheme for the assessment is determined by the lecturer of the respective course. It will be announced at the beginning of the course.

Prerequisites
None.

Recommendations
See seminar description in the course catalogue of the KIT (https://campus.kit.edu/)

Additional Information
The listed seminar titles are placeholders. Currently offered seminars of each semester will be published on the websites of the institutes and in the course catalogue of the KIT. In general, the current seminar topics of each semester are already announced at the end of the previous semester. Furthermore for some seminars there is an application required.

The available places are listed on the internet: https://portal.wiwi.kit.edu.

Workload
90 hours

Below you will find excerpts from courses related to this module component:

V Statistics and Epidemics

25000111, WS 25/26, 2 SWS, Language: English, Open in study portal

Seminar (S)
On-Site

Content
Motivation
Infectious disease epidemiology gives rise to a large variety of real-time data streams. During the COVID-19 pandemic, the interpretation and statistical analysis of these data has proven crucial, but also highly challenging. In this seminar, students will get to know central concepts of infectious disease surveillance and modelling from a statistical perspective. Following an overview of various aspects in the form of blocked lectures, students will choose a more specific topic for their seminar thesis.

Learning Goals

Students develop an understanding of central modeling tasks and methods, including

- estimation of reproductive numbers
- compartment models of disease spread
- nowcasting and short-term forecasting of disease spread
- detection of outbreaks
- diagnostic testing

Moreover, they get to know various data types commonly used in the analysis of disease spread.

Logistics

The project seminar is worth 4.5 credit points (Leistungspunkte). There will be three blocked lectures (approx. 135 minutes each) in the beginning of the lecture period. For the various topics covered, subjects for seminar theses will be proposed (and students are allowed to propose their own topics). Towards the end of the semester, students present their progress on the chosen topics to the group. Grades will be based on this presentation (25%) and the final report (75%).

Organizational issues
Prerequisites

Students should have a very good working knowledge of statistics, including proficiency in a programming language for applied data analysis. The lecture VWL3 Introduction to Econometrics is a prerequisite for the project seminar. Most available software in the field is in R, but in principle Python can be used as well. Advanced knowledge of biology, medicine or epidemiology is not required.

Application Procedure
Please submit a transcript of records as well as a short letter of motivation (roughly 200 words) via WIWI-Portal: https://portal.wiwi.kit.edu/ys/8902

Application time frame: September 5th, 2025 to October, 31th, 2025.

V **Topics in Econometrics** **Seminar (S)**
2521310, WS 25/26, 2 SWS, Language: German, Open in study portal

Organizational issues
Wöchentliche Veranstaltung, Donnerstags 13:00-14:30 Uhr in B17 - R320

V **Making and Evaluating Predictions** **Seminar (S)**
2500012, SS 2026, 2 SWS, Language: German/English, Open in study portal

Content
A key goal of statistics and data science is to make predictions. Good predictions are important for decision-making, allowing us to balance risks and mitigate the impacts of potentially damaging events. In this seminar, we will study theoretical questions regarding the evaluation of predictions, such as “What does it mean for a prediction to be good?”, before introducing statistical and machine learning methods to generate both point-valued and probabilistic predictions. We will discuss applications to economics, meteorology, and energy forecasting.

Format: The course will start with a short block course at the beginning of the lecture period. Evaluation will be based on a seminar thesis, which students are required to hand in at the end of the lecture period, and a short presentation. The presentations will also take place towards the end of the lecture period.

Organizational issues
Blockveranstaltung, Termine werden bekannt gegeben

V **Advanced Topics in Econometrics** **Seminar (S)**
2521310, SS 2026, 2 SWS, Language: German/English, Open in study portal

Organizational issues
wöchentlich, Donnerstags 13:00-14:30, B17, R320

V

Topics in Econometrics

2521310, WS 26/27, 2 SWS, Language: German, Open in study portal

Seminar (S)

Organizational issues
Wöchentliche Veranstaltung, Donnerstags 13:00-14:30 Uhr in B17 - R320

T 6.672 Module component: Seminar in Statistics A (Master) [T-WIWI-103483]

Coordinators: Prof. Dr. Oliver Grothe
Prof. Dr. Melanie Schienle
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104902 - Statistics
M-WIWI-107204 - Seminar Module

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Each term	1

Courses					
WT 25/26	25000111	Statistics and Epidemics	2 SWS	Seminar / 🗣	Bracher, Nemcova
WT 25/26	2500012		2 SWS	Seminar / 🗣	Grothe, Liu
WT 25/26	2521310	Topics in Econometrics	2 SWS	Seminar	Schienle, Krüger, Bracher, Buse, Koster, Rüter, Leimenstoll, Renner, Pohle
ST 2026	2500012	Making and Evaluating Predictions	2 SWS	Seminar	Schienle, Allen, Zhang
ST 2026	2521310	Advanced Topics in Econometrics	2 SWS	Seminar	Schienle, Krüger, Bracher, Buse, Koster, Rüter, Leimenstoll, Renner
ST 2026	2550561	Fortgeschrittene Themen zu Statistik, Datenanalyse und maschinellem Lernen (Master)	2 SWS	Seminar / 🗣	Grothe, Liu
WT 26/27	2521310	Topics in Econometrics	2 SWS	Seminar	Schienle, Krüger, Bracher, Buse, Koster, Rüter, Leimenstoll, Renner

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Alternative exam assessment (§ 4(2), 3 SPO 2015). The following aspects are included:

- Regular participation in the seminar dates
- Preparation of a seminar paper on a partial aspect of the seminar topic according to scientific methods
- Lecture on the topic of the seminar paper.

The point scheme for the assessment is determined by the lecturer of the respective course. It will be announced at the beginning of the course.

Prerequisites
None.

Recommendations
See seminar description in the course catalogue of the KIT (https://campus.kit.edu/)

Additional Information
The listed seminar titles are placeholders. Currently offered seminars of each semester will be published on the websites of the institutes and in the course catalogue of the KIT. In general, the current seminar topics of each semester are already announced at the end of the previous semester. Furthermore for some seminars there is an application required.

The available places are listed on the internet: https://portal.wiwi.kit.edu.

Workload
90 hours

Below you will find excerpts from courses related to this module component:

V | **Statistics and Epidemics** | **Seminar (S)**
25000111, WS 25/26, 2 SWS, Language: English, Open in study portal | **On-Site**

Content
Motivation
Infectious disease epidemiology gives rise to a large variety of real-time data streams. During the COVID-19 pandemic, the interpretation and statistical analysis of these data has proven crucial, but also highly challenging. In this seminar, students will get to know central concepts of infectious disease surveillance and modelling from a statistical perspective. Following an overview of various aspects in the form of blocked lectures, students will choose a more specific topic for their seminar thesis.

Learning Goals

Students develop an understanding of central modeling tasks and methods, including

- estimation of reproductive numbers

- compartment models of disease spread

- nowcasting and short-term forecasting of disease spread

- detection of outbreaks

- diagnostic testing

Moreover, they get to know various data types commonly used in the analysis of disease spread.

Logistics

The project seminar is worth 4.5 credit points (Leistungspunkte). There will be three blocked lectures (approx. 135 minutes each) in the beginning of the lecture period. For the various topics covered, subjects for seminar theses will be proposed (and students are allowed to propose their own topics). Towards the end of the semester, students present their progress on the chosen topics to the group. Grades will be based on this presentation (25%) and the final report (75%).

Organizational issues
Prerequisites

Students should have a very good working knowledge of statistics, including proficiency in a programming language for applied data analysis. The lecture VWL3 Introduction to Econometrics is a prerequisite for the project seminar. Most available software in the field is in R, but in principle Python can be used as well. Advanced knowledge of biology, medicine or epidemiology is not required.

Application Procedure
Please submit a transcript of records as well as a short letter of motivation (roughly 200 words) via WIWI-Portal: https://portal.wiwi.kit.edu/ys/8902

Application time frame: September 5th, 2025 to October, 31th, 2025.

V | **Topics in Econometrics** | **Seminar (S)**
2521310, WS 25/26, 2 SWS, Language: German, Open in study portal

Organizational issues
Wöchentliche Veranstaltung, Donnerstags 13:00-14:30 Uhr in B17 - R320

V | **Making and Evaluating Predictions** | **Seminar (S)**
2500012, SS 2026, 2 SWS, Language: German/English, Open in study portal

Content
A key goal of statistics and data science is to make predictions. Good predictions are important for decision-making, allowing us to balance risks and mitigate the impacts of potentially damaging events. In this seminar, we will study theoretical questions regarding the evaluation of predictions, such as “What does it mean for a prediction to be good?”, before introducing statistical and machine learning methods to generate both point-valued and probabilistic predictions. We will discuss applications to economics, meteorology, and energy forecasting.

Format: The course will start with a short block course at the beginning of the lecture period. Evaluation will be based on a seminar thesis, which students are required to hand in at the end of the lecture period, and a short presentation. The presentations will also take place towards the end of the lecture period.

Organizational issues
Blockveranstaltung, Termine werden bekannt gegeben

V | **Advanced Topics in Econometrics** | **Seminar (S)**
2521310, SS 2026, 2 SWS, Language: German/English, Open in study portal

Organizational issues
wöchentlich, Donnerstags 13:00-14:30, B17, R320

V **Topics in Econometrics** **Seminar (S)**
2521310, WS 26/27, 2 SWS, Language: German, Open in study portal

Organizational issues
Wöchentliche Veranstaltung, Donnerstags 13:00-14:30 Uhr in B17 - R320

T 6.673 Module component: Seminar in Statistics B (Master) [T-WIWI-103484]

Coordinators: Prof. Dr. Oliver Grothe
Prof. Dr. Melanie Schienle

Organisation: KIT Department of Business and Economics

Part of: M-WIWI-104902 - Statistics
M-WIWI-107204 - Seminar Module

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Each term	1

Courses					
WT 25/26	25000111	Statistics and Epidemics	2 SWS	Seminar / 🗣	Bracher, Nemcova
WT 25/26	2500012		2 SWS	Seminar / 🗣	Grothe, Liu
WT 25/26	2521310	Topics in Econometrics	2 SWS	Seminar	Schienle, Krüger, Bracher, Buse, Koster, Rüter, Leimenstoll, Renner, Pohle
ST 2026	2500012	Making and Evaluating Predictions	2 SWS	Seminar	Schienle, Allen, Zhang
ST 2026	2521310	Advanced Topics in Econometrics	2 SWS	Seminar	Schienle, Krüger, Bracher, Buse, Koster, Rüter, Leimenstoll, Renner
ST 2026	2550561	Fortgeschrittene Themen zu Statistik, Datenanalyse und maschinellem Lernen (Master)	2 SWS	Seminar / 🗣	Grothe, Liu
WT 26/27	2521310	Topics in Econometrics	2 SWS	Seminar	Schienle, Krüger, Bracher, Buse, Koster, Rüter, Leimenstoll, Renner

Legend: 🖥 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Alternative exam assessment (§ 4(2), 3 SPO 2015). The following aspects are included:

- Regular participation in the seminar dates
- Preparation of a seminar paper on a partial aspect of the seminar topic according to scientific methods
- Lecture on the topic of the seminar paper.

The point scheme for the assessment is determined by the lecturer of the respective course. It will be announced at the beginning of the course.

Prerequisites
None.

Recommendations
See seminar description in the course catalogue of the KIT (https://campus.kit.edu/)

Additional Information
The listed seminar titles are placeholders. Currently offered seminars of each semester will be published on the websites of the institutes and in the course catalogue of the KIT. In general, the current seminar topics of each semester are already announced at the end of the previous semester. Furthermore for some seminars there is an application required.

The available places are listed on the internet: https://portal.wiwi.kit.edu.

Workload
90 hours

Below you will find excerpts from courses related to this module component:

V **Statistics and Epidemics** **Seminar (S) On-Site**
25000111, WS 25/26, 2 SWS, Language: English, Open in study portal

Content
Motivation
Infectious disease epidemiology gives rise to a large variety of real-time data streams. During the COVID-19 pandemic, the interpretation and statistical analysis of these data has proven crucial, but also highly challenging. In this seminar, students will get to know central concepts of infectious disease surveillance and modelling from a statistical perspective. Following an overview of various aspects in the form of blocked lectures, students will choose a more specific topic for their seminar thesis.

Learning Goals

Students develop an understanding of central modeling tasks and methods, including

- estimation of reproductive numbers
- compartment models of disease spread
- nowcasting and short-term forecasting of disease spread
- detection of outbreaks
- diagnostic testing

Moreover, they get to know various data types commonly used in the analysis of disease spread.

Logistics

The project seminar is worth 4.5 credit points (Leistungspunkte). There will be three blocked lectures (approx. 135 minutes each) in the beginning of the lecture period. For the various topics covered, subjects for seminar theses will be proposed (and students are allowed to propose their own topics). Towards the end of the semester, students present their progress on the chosen topics to the group. Grades will be based on this presentation (25%) and the final report (75%).

Organizational issues
Prerequisites

Students should have a very good working knowledge of statistics, including proficiency in a programming language for applied data analysis. The lecture VWL3 Introduction to Econometrics is a prerequisite for the project seminar. Most available software in the field is in R, but in principle Python can be used as well. Advanced knowledge of biology, medicine or epidemiology is not required.

Application Procedure
Please submit a transcript of records as well as a short letter of motivation (roughly 200 words) via WIWI-Portal: https://portal.wiwi.kit.edu/ys/8902

Application time frame: September 5th, 2025 to October, 31th, 2025.

V **Topics in Econometrics** **Seminar (S)**
2521310, WS 25/26, 2 SWS, Language: German, Open in study portal

Organizational issues
Wöchentliche Veranstaltung, Donnerstags 13:00-14:30 Uhr in B17 - R320

V **Making and Evaluating Predictions** **Seminar (S)**
2500012, SS 2026, 2 SWS, Language: German/English, Open in study portal

Content
A key goal of statistics and data science is to make predictions. Good predictions are important for decision-making, allowing us to balance risks and mitigate the impacts of potentially damaging events. In this seminar, we will study theoretical questions regarding the evaluation of predictions, such as “What does it mean for a prediction to be good?”, before introducing statistical and machine learning methods to generate both point-valued and probabilistic predictions. We will discuss applications to economics, meteorology, and energy forecasting.

Format: The course will start with a short block course at the beginning of the lecture period. Evaluation will be based on a seminar thesis, which students are required to hand in at the end of the lecture period, and a short presentation. The presentations will also take place towards the end of the lecture period.

Organizational issues
Blockveranstaltung, Termine werden bekannt gegeben

V **Advanced Topics in Econometrics** **Seminar (S)**
2521310, SS 2026, 2 SWS, Language: German/English, Open in study portal

Organizational issues
wöchentlich, Donnerstags 13:00-14:30, B17, R320

V	**Topics in Econometrics** 2521310, WS 26/27, 2 SWS, Language: German, Open in study portal	**Seminar (S)**

Organizational issues
Wöchentliche Veranstaltung, Donnerstags 13:00-14:30 Uhr in B17 - R320

T 6.674 Module component: Seminar in Transportation [T-BGU-100014]

Coordinators: Prof. Dr.-Ing. Martin Kagerbauer
Prof. Dr.-Ing. Peter Vortisch

Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences

Part of: M-BGU-101064 - Fundamentals of Transportation
M-BGU-101065 - Transportation Modelling and Traffic Management
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Expansion	Version
Examination of another type	3 CP	graded	Each term	1 semesters	1

Courses					
WT 25/26	6232903	Seminar Transport Studies	2 SWS	Seminar / 🗣	Vortisch, Kagerbauer
ST 2026	6232903	Seminar on Transportation Systems	2 SWS	Seminar / 🧩	Vortisch, Kagerbauer

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
seminar paper, appr. 10 pages, and presentation, appr. 10 min.

Prerequisites
none

Recommendations
none

Additional Information
none

Workload
90 hours

T 6.675 Module component: Seminar Industrial Process and Plant Engineering [T-ETIT-113932]

Coordinators: Prof. Dr.-Ing. Mike Barth
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-106970 - Seminar Industrial Process and Plant Engineering

Type	Credits	Grading	Term offered	Version
Examination of another type	4 CP	graded	Each summer term	1

Courses					
ST 2026	2300009	Seminar Industrial Process and Plant Engineering	3 SWS	Seminar / 🗣	Barth, Jilg
ST 2026	2303650	Seminar Industrial Process and Plant Engineering	3 SWS	Seminar / 🗣	Barth, Jilg

Legend: 🖥 Online, Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The examination will be the seminar presentation at the end of the semester. The criteria are:

- Live presentation of the created CAD and simulation models
- Poster design and usage within the presentation
- Answering the questions from the examiners
- Structure of the talk

Prerequisites
none

T 6.676 Module component: Seminar Methods along the Innovation process [T-WIWI-110987]

Coordinators: Dr. Daniela Beyer
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	see Annotations	1

Assessment
Alternative exam assessment.

Recommendations
Prior attendance of the course Innovation Management [2545015] is recommended.

Additional Information
The course is no longer offered.

Workload
90 hours

T 6.677 Module component: Seminar New Power Electronic Systems and Technologies [T-ETIT-114862]

Coordinators: Prof. Dr.-Ing. Marc Hiller
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-107588 - Seminar New Power Electronic Systems and Technologies

Type	Credits	Grading	Term offered	Version
Examination of another type	4 CP	graded	Each term	1

Courses					
WT 25/26	2306317	Seminar New Power Electronic Systems and Technologies	3 SWS	Seminar / 🗣	Hiller
ST 2026	2306317	Seminar New Power Electronic Systems and Technologies	3 SWS	Seminar / 🗣	Hiller

Legend: 🖥 Online, ⚄ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The success control takes place in the form of an examination of another type. It consists of a 15-minute final presentation followed by a discussion and a 2- to 3-page written summary paper. The overall impression is evaluated.

The following aspects are assessed:

Presentation:

- Quality of the presentation (form and content)
- Presentation style (structure, style, content)
- Behavior during the discussion

Paper with a summary of the main content:

- Format, spelling, linguistic style (academic/factual)
- Content, (graphical) presentation of the researched findings
- Relevance and quality of the references used, citation style

Prerequisites
none

Additional Information
The number of participants in the seminar may be limited for capacity reasons and depends on the number of seminar topics offered.

Selection may be based on degree program, academic progress, waiting time, and topic assignment.

Presentations and summaries may be given/written in English or German.

T 6.678 Module component: Seminar on Modeling and Simulation in Transportation [T-BGU-112552]

Coordinators: Prof. Dr.-Ing. Martin Kagerbauer
Prof. Dr.-Ing. Peter Vortisch
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: M-BGU-101064 - Fundamentals of Transportation
M-BGU-101065 - Transportation Modelling and Traffic Management

Type	Credits	Grading	Term offered	Expansion	Version
Examination of another type	3 CP	graded	Each winter term	1 semesters	1

Courses					
WT 25/26	6232907	Seminar Modeling and Simulation in Transportation	2 SWS	Seminar / 🗣	Vortisch, Kagerbauer, Mitarbeiter/innen

Legend: 🖥 Online, ❂ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
work on a practical problem in the area of traffic engineering, traffic simulation or in the area of microscopic travel demand modeling:

final report, appr. 5 pages, and presentation, appr. 10 min.

Prerequisites
none

Recommendations
none

Additional Information
none

Workload
90 hours

T 6.679 Module component: Seminar on Transport Policy [T-BGU-114772]

Coordinators: Prof. Dr.-Ing. Peter Vortisch
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: M-BGU-101064 - Fundamentals of Transportation

Type	Credits	Grading	Term offered	Expansion	Version
Examination of another type	3 CP	graded	Each winter term	1 semesters	1

Courses					
WT 25/26	6232908	Seminar on Transport Policy	2 SWS	Seminar / 🗣	Vortisch

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, × Cancelled

Assessment
engagement and quality of contributions to debates in the seminar meetings;

final presentation, appr. 15 min.

Prerequisites
none

Recommendations
The seminar is aimed at students who have already acquired initial knowledge in the field of mobility and infrastructure. In addition, an interest in transport policy and debating is desirable.

Additional Information
new selection option offered as from winter term 2025/26

The seminar requires a minimum number of 3 and a maximum number of 10 participants. Please register by e-mail via the person listed as the contact person for the course. Places will be allocated according to the progress of your studies.

Workload
90 hours

Below you will find excerpts from courses related to this module component:

V Seminar on Transport Policy

6232908, WS 25/26, 2 SWS, Language: German, Open in study portal

Seminar (S)
On-Site

Content
see German version

Organizational issues
Anmeldung erforderlich - Die Anmeldung erfolgt per E-Mail an christian.klinkhardt@kit.edu bis einschließlich dem ersten Tag der Vorlesungszeit. Die Lehrveranstaltung hat eine Mindestzahl von 3 und eine Höchstzahl von 10 Teilnehmenden. Die Plätze werden unter Berücksichtigung des Studienfortschritts vergeben.

Der erste Termin findet am 11.11.2025 statt.

T 6.680 Module component: Seminar Production Technology [T-MACH-109062]

Coordinators: Prof. Dr.-Ing. Jürgen Fleischer
Prof. Dr.-Ing. Gisela Lanza
Prof. Dr.-Ing. Volker Schulze

Organisation: KIT Department of Mechanical Engineering

Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Each term	2

Courses					
ST 2026	2149665	Seminar Production Technology	1 SWS	Seminar / 🗣	Fleischer, Lanza, Schulze, Zanger

Legend: 🖥 Online, ♲ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Alternative test achievement (graded):

- written elaboration (workload of at least 80 h)
- oral presentation (approx. 30 min)

Prerequisites
none

Additional Information
The course is offered in German.

The specific topics are published on the homepage of the wbk Institute of Production Science.

Workload
90 hours

Below you will find excerpts from courses related to this module component:

V Seminar Production Technology

Seminar (S)
On-Site

2149665, SS 2026, 1 SWS, Language: German, Open in study portal

Content
In course of the seminar Production Technology current issues of the wbk main fields of research "Manufacturing and Materials Technology", "Machines, Equipment and Process Automation" as well as "Production Systems" are discussed.

The specific topics are published on the homepage of the wbk Institute of Production Science.

Learning Outcomes:

The students …

- are in a position to independently handle current, research-based tasks according to scientific criteria.
- are able to research, analyze, abstract and critically review the information.
- can draw own conclusions using their interdisciplinary knowledge from the less structured information and selectively develop current research results.
- can logically and systematically present the obtained results both orally and in written form in accordance with scientific guidelines (structuring, technical terminology, referencing). They can argue and defend the results professionally in the discussion.

Workload:

regular attendance: 10 hours
self-study: 80 hours

Organizational issues
siehe http://www.wbk.kit.edu/seminare.php

T 6.681 Module component: Seminar: Commercial and Corporate Law in the IT Industry [T-INFO-111405]

Coordinators: Dr. Georg Nolte
Organisation: KIT Department of Informatics
Part of: M-INFO-101216 - Private Business Law

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Each winter term	1

Courses					
WT 25/26	2400165	Seminar Commercial and Corporate Law in Information Technology	2 SWS	Seminar / 🖥	Nolte

Legend: 🖥 Online, Blended (On-Site/Online), On-Site, ✕ Cancelled

T 6.682 Module component: Seminar: Governance, Risk & Compliance [T-INFO-102047]

Coordinators: Prof. Dr. Thomas Dreier
Organisation: KIT Department of Informatics
Part of: M-WIWI-104903 - Law
M-WIWI-104909 - Informatics (Department of Informatics)

Type	Credits	Grading	Version
Examination of another type	3 CP	graded	1

Courses					
ST 2026	2400005	Governance, Risk & Compliance	2 SWS	Seminar / 🗣	Herzig, Siddiq

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

T 6.683 Module component: Seminar: IT- Security Law [T-INFO-111404]

Coordinators: Martin Schallbruch
Organisation: KIT Department of Informatics
Part of: M-INFO-106754 - Public Economic and Technology Law

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Each winter term	1

Courses					
WT 25/26	2424389	Seminar "IT security law"	2 SWS	Seminar / 🗣	Schallbruch

Legend: 🖥 Online, ⚃ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

T 6.684 Module component: Seminar: Legal Studies Bachelor [T-INFO-115024]

Coordinators: Professorenschaft des Fachbereichs Recht
Organisation: KIT Department of Informatics
Part of: M-WIWI-104903 - Law

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Each term	1

Courses					
WT 25/26	2400014	Current Issues in Patent Law	2 SWS	Seminar / 🗣	Melullis
WT 25/26	2400165	Seminar Commercial and Corporate Law in Information Technology	2 SWS	Seminar / 🖥	Nolte
WT 25/26	2400177	Human vs. Machine: (Un-)Biased Legal Decision-Making?	2 SWS	Seminar / 🗣	Holzhausen
WT 25/26	2400184	EU Digital Regulatory Framework	2 SWS	Seminar / 🗣	Zufall
WT 25/26	2400203	EU Fundamental Rights and EU Economic Law in Smart Cities	2 SWS	Seminar / 🗣	Kasper-Schöniger
WT 25/26	2424186	Seminar Patent Law	2 SWS	Seminar / 🗣	Dammler
WT 25/26	2424389	Seminar "IT security law"	2 SWS	Seminar / 🗣	Schallbruch
WT 25/26	2513214	Seminar Information security and Data protection (Bachelor)	2 SWS	Seminar / 🗣	Volkamer, Raabe, Schiefer, Werner
ST 2026	2400005	Governance, Risk & Compliance	2 SWS	Seminar / 🗣	Herzig, Siddiq
ST 2026	2400078	Intelligente Chatbots und Recht	2 SWS	Seminar / 🗣	Raabe
ST 2026	2400171	Regulating AI: from ethics to law	2 SWS	Seminar / 🗣	Gil Gasiola
ST 2026	2400177	Designing Data Governance of Digital Systems	2 SWS	Seminar / 🗣	Pathak
ST 2026	2400225	Rechtlicher Rahmen für Künstliche Intelligenz	2 SWS	Seminar / 🗣	Sattler
ST 2026	24820	Current Issues in Patent Law	2 SWS	Seminar / 🗣	Melullis
WT 26/27	2513214	Seminar Information security and Data protection (Bachelor)	2 SWS	Seminar / 🗣	Volkamer, Raabe, Schiefer, Werner

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Below you will find excerpts from courses related to this module component:

V Human vs. Machine: (Un-)Biased Legal Decision-Making?

Seminar (S)
On-Site

2400177, WS 25/26, 2 SWS, Language: English, Open in study portal

Content
This seminar provides an overview of the foundational principles of behavioral economics and their relevance to public law. Special attention is given to (biased) decision-making, both from a substantive and procedural perspective. Students will explore how regulation should be designed to account for cognitive biases and achieve intended policy outcomes. In addition, the seminar will address the role of automated decision-making systems - examining how such systems are influenced by, and may also reinforce or mitigate, behavioral biases.

The seminar will cover the following topics:

- Introduction to (behavioral) economic theory and its application to law
- Cognitive biases and their impact on (public) decision-making
- Different types of regulation in light of behavioral insights
- Implications for automated decision-making systems
- The role of human biases in the AI Act

Algorithmic decision-making is often treated as a black box—something that should not be permitted to make decisions in legal contexts. In response, Article 14 of the EU AI Act requires high-risk AI systems to be designed in such a way that they can be effectively overseen by natural persons. This human-in-the-loop requirement stems, among other things, from the perception that AI systems operate ambiguously and produce decisions that are not fully comprehensible.

However, treating humans in comparison as fundamentally transparent or fully rational decision-makers is equally naïve. Understanding how people actually behave—not just how they should behave—is essential for designing effective, fair, and legitimate legal systems. Traditional legal research often assumes rational actors, yet behavioral research shows that cognitive biases, heuristics, and emotional influences shape the decisions of both citizens and public officials.

This seminar addresses that gap by equipping students with behavioral insights to critically assess and improve legal and administrative. A particular focus is placed on how human biases are already embedded in the data underlying automated decision-making systems, potentially reproducing or amplifying unfairness. At the same time, students will explore how such systems can be intentionally designed to mitigate behavioral distortions and support better decision-making.

Organizational issues
<u>1. Termin:</u> 13.11.2025 9:00-15:00 Uhr (Seminarrum Nr. 313, bei uns am Institut)

<u>2. Termin:</u> 05.02.2025 9:00- 16:00 Uhr (Seminarrum Nr. 313, bei uns am Institut).

V	**EU Digital Regulatory Framework** 2400184, WS 25/26, 2 SWS, Language: English, Open in study portal	**Seminar (S)** **On-Site**

Content
This class aims to provide an overview on the legal instruments forming the EU digital regulatory framework. Following its Digital Single Market Strategy, the EU has set up a new strategic programme for a "Digital Decade". Existing regulations like the General Data Protection Regulation (GDPR), or the E-Commerce Directive, are being complemented by a variety of new instruments that aim to set binding rules on online markets, to regulate data flows in various ways, but also to pioneer a legal framework on AI. Prominent instruments include the new AI Act, the Digital Services Act (DSA) and Digital Markets Act (DMA), the Data Act, Data Governance Act, or Open Data Directive.

The class will provide an overview on the existing framework: Which regulations and directives are relevant? How do they apply and interact which each other in a broader context?

Another objective is to provide students with the ability to read these legal instruments: How to access regulatory instruments that often have more than 100 pages (without having to read every single sentence)? How to gain a comprehensive, high-level understanding of the instrument? How to identify parts relevant to a particular legal problem?

The class will start with an introduction into EU law and regulatory instruments in general. Concrete guidance on reading, analysing and working with legal instruments in English will be given. Based on these instructions, students will be assigned legal instruments to present in the final unit along with a two-pages report.

Grades will be assigned based on the quality of these presentations and the report, as well as participation in the discussion (presentation: 40 %, two-pages report: 40 %, discussion: 20 %).

Organizational issues
WS 2024/25

Hierbei handelt es sich NICHT um eine Pro-Seminar, sondern um ein Seminar (aus Rechtswissenschaften).

Anmeldungen für das Seminar bitte NUR! über das WiWi-Portal!

***Für die Prüfung bitte NUR über CAS (Campus-Portal) anmelden!**

*Erläuterung: n<u>ach</u> der für die Teilnahme am Seminar verbindlichen Teilnahme an der <u>Einführungsveranstaltung bitte Anmeldung über das Campus-System</u> (notwendig für die Erfassung der Note der Seminararbeit).

V	**EU Fundamental Rights and EU Economic Law in Smart Cities** 2400203, WS 25/26, 2 SWS, Language: English, Open in study portal	**Seminar (S)** **On-Site**

Content
This seminar is designed to instruct students in recognising legal questions as they relate to an interdisciplinary topic, classifying such questions along the hierarchy of norms, and solving them in a convincing argumentative manner. By focusing on „EU Fundamental Rights and EU Economic Law in Smart Cities“, we will discuss the influence of large cities and metropolises on their more rural periphery. Possible aspects of discussion are, inter alia,

- democratic legitimacy questions in such cases,
- elements of structural support/aid
- approaches to implement technical innovations in smart cities,
- questions relating to the development of (large) infrastructures,
- aspects of the efficient use of infrastructure, and
- environmental aspects.

Particularly in times of increasingly comprehensive European regulation, European standards and regulatory approaches will be at the heart of the discussion while not neglecting their interplay with national regulation. Learning in more depth about the interplay of European and national regulation will further accustom participating students to discuss relevant legal questions across (legal) sectors and with a broader view on regulations' effects.

Conducting the seminar in English intends to facilitate students to link their ideas and arguments to current European and international debates.

Organizational issues
This is a seminar (not a pro-seminar).

Registrations only via Wiwi-Portal.

Kick-off-meeting WS 2025/26:

Friday, 7th November 2025, 14:00 - 18:00 h (online)

(Mr. Kasper-Schöniger will send you the according link via ILIAS a few days before the kick-off date.)

Presentations:

- Friday, 6th February 2026, 09:00 - 18:00 h

and

- Saturday, 7th February 2026, 10:00 - 16:00

Place: see section "TERMINE" (below).

V	**Regulating AI: from ethics to law** 2400171, SS 2026, 2 SWS, Language: English, Open in study portal	**Seminar (S) On-Site**

Content
This seminar will provide an overview of technology regulation, with a particular focus on the regulation of AI systems. Following an introduction to the different forms of regulation, students will explore the various regulatory instruments in light of the key principles of AI, such as

- fairness / non-discrimination,
- transparency / explainability,
- non-maleficence,
- privacy / data protection,
- sustentability
- security / safety, and
- accountability.

This will enable them to discuss how these principles and rules can be applied in real-world scenarios.

Organizational issues
Please register for the seminar ONLY via the WiWi-Portal!
Please register for the exam ONLY via CAS (Campus-Portal)!*

*ExplanationAfter attending the introductory event, which is mandatory for participation in the seminar, please register via the campus system (necessary for recording the grade of the seminar paper).

Block Seminar in Summer term 2026:

- **Wednesday, 22th April 2026, 09:00 - 13:00 h;**
- **Wednesday, 22th July 2026, 09:00 - 13:00 h.**

Room: each seminar takes place in room no. 313 (building 07.08) .

V	**Designing Data Governance of Digital Systems** 2400177, SS 2026, 2 SWS, Language: English, Open in study portal	**Seminar (S) On-Site**

Content
The latest regulations in the digital sector at EU level represent a highly topical and important regulatory instrument with enormous practical relevance for students of computer science and business informatics. The seminar not only enables students to acquire important knowledge in this area, but also to apply it specifically to the governance of digital systems and to learn the practical design of digital systems against the background of legal framework conditions.

Organizational issues
***Please register for the exam ONLY via CAS (Campus-Portal)**!

***Explanation:after attending the introductory event,** which is mandatory for participation in the seminar**, please register via the campus system** (necessary for recording the grade of the seminar paper).

Due to the minimum number of text characters that can be used, please check ILIAS and the WiWi portal for details such as dates and location. Thank you.

T 6.685 Module component: Seminar: Legal Studies Master I [T-INFO-115025]

Coordinators: Professorenschaft des Fachbereichs Recht
Organisation: KIT Department of Informatics
Part of: M-WIWI-104903 - Law

Type	Credits	Grading	Version
Examination of another type	3 CP	graded	1

Courses					
WT 25/26	2400014	Current Issues in Patent Law	2 SWS	Seminar / On-Site	Melullis
WT 25/26	2400060	Data in Software-Intensive Technical Systems – Modeling – Analysis – Protection	2 SWS	Seminar / On-Site	Reussner, Raabe, Werner, Müller-Quade
WT 25/26	2400125	Security and Privacy Awareness	2 SWS	Seminar / Blended	Seidel-Saul, Volkamer, Boehm, Ballreich, Sikra, Veit
WT 25/26	2400165	Seminar Commercial and Corporate Law in Information Technology	2 SWS	Seminar / Online	Nolte
WT 25/26	2400177	Human vs. Machine: (Un-)Biased Legal Decision-Making?	2 SWS	Seminar / On-Site	Holzhausen
WT 25/26	2400184	EU Digital Regulatory Framework	2 SWS	Seminar / On-Site	Zufall
WT 25/26	2400203	EU Fundamental Rights and EU Economic Law in Smart Cities	2 SWS	Seminar / On-Site	Kasper-Schöniger
WT 25/26	2424186	Seminar Patent Law	2 SWS	Seminar / On-Site	Dammler
WT 25/26	2424389	Seminar "IT security law"	2 SWS	Seminar / On-Site	Schallbruch
ST 2026	2400005	Governance, Risk & Compliance	2 SWS	Seminar / On-Site	Herzig, Siddiq
ST 2026	2400078	Intelligente Chatbots und Recht	2 SWS	Seminar / On-Site	Raabe
ST 2026	2400171	Regulating AI: from ethics to law	2 SWS	Seminar / On-Site	Gil Gasiola
ST 2026	2400177	Designing Data Governance of Digital Systems	2 SWS	Seminar / On-Site	Pathak
ST 2026	2400207	(Generative) AI and Law	2 SWS	Seminar / Blended	Boehm, Turkovic-Popovski
ST 2026	2400225	Rechtlicher Rahmen für Künstliche Intelligenz	2 SWS	Seminar / On-Site	Sattler
ST 2026	24820	Current Issues in Patent Law	2 SWS	Seminar / On-Site	Melullis
WT 26/27	2400060	Seminar: Software Architecture, Security and Privacy	2 SWS	Seminar / On-Site	Reussner, Raabe, Werner, Müller-Quade
WT 26/27	2400125	Security and Privacy Awareness	2 SWS	Seminar / Blended	Volkamer, Boehm, Ballreich, Algedri

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Below you will find excerpts from courses related to this module component:

V Security and Privacy Awareness

2400125, WS 25/26, 2 SWS, Language: German/English, Open in study portal

Seminar (S)
Blended (On-Site/Online)

Content
Within the framework of this interdisciplinary seminar, the topics security awareness and privacy awareness are to be considered from different perspectives. It deals with legal, information technology, psychological, social as well as philosophical aspects.

Important notes:

- The seminar language is English. This means that English is the standard for the presentation and the seminar paper, and German is only permitted for the paper after consultation with the supervisor, for example, if the topic of the thesis requires it.
- The seminar is only for MASTER students (or Mastervorzug)
- The link to enrol is for every student, regardless of the study background

Topics:

The advertised topics can be found in the wiwi portal.

V **Human vs. Machine: (Un-)Biased Legal Decision-Making?** **Seminar (S) On-Site**
2400177, WS 25/26, 2 SWS, Language: English, Open in study portal

Content
This seminar provides an overview of the foundational principles of behavioral economics and their relevance to public law. Special attention is given to (biased) decision-making, both from a substantive and procedural perspective. Students will explore how regulation should be designed to account for cognitive biases and achieve intended policy outcomes. In addition, the seminar will address the role of automated decision-making systems - examining how such systems are influenced by, and may also reinforce or mitigate, behavioral biases.

The seminar will cover the following topics:

- Introduction to (behavioral) economic theory and its application to law
- Cognitive biases and their impact on (public) decision-making
- Different types of regulation in light of behavioral insights
- Implications for automated decision-making systems
- The role of human biases in the AI Act

Algorithmic decision-making is often treated as a black box—something that should not be permitted to make decisions in legal contexts. In response, Article 14 of the EU AI Act requires high-risk AI systems to be designed in such a way that they can be effectively overseen by natural persons. This human-in-the-loop requirement stems, among other things, from the perception that AI systems operate ambiguously and produce decisions that are not fully comprehensible.

However, treating humans in comparison as fundamentally transparent or fully rational decision-makers is equally naïve. Understanding how people actually behave—not just how they should behave—is essential for designing effective, fair, and legitimate legal systems. Traditional legal research often assumes rational actors, yet behavioral research shows that cognitive biases, heuristics, and emotional influences shape the decisions of both citizens and public officials.

This seminar addresses that gap by equipping students with behavioral insights to critically assess and improve legal and administrative. A particular focus is placed on how human biases are already embedded in the data underlying automated decision-making systems, potentially reproducing or amplifying unfairness. At the same time, students will explore how such systems can be intentionally designed to mitigate behavioral distortions and support better decision-making.

Organizational issues
1. Termin: 13.11.2025 9:00-15:00 Uhr (Seminarrum Nr. 313, bei uns am Institut)

2. Termin: 05.02.2025 9:00- 16:00 Uhr (Seminarrum Nr. 313, bei uns am Institut).

V **EU Digital Regulatory Framework** **Seminar (S) On-Site**
2400184, WS 25/26, 2 SWS, Language: English, Open in study portal

Content
This class aims to provide an overview on the legal instruments forming the EU digital regulatory framework. Following its Digital Single Market Strategy, the EU has set up a new strategic programme for a "Digital Decade". Existing regulations like the General Data Protection Regulation (GDPR), or the E-Commerce Directive, are being complemented by a variety of new instruments that aim to set binding rules on online markets, to regulate data flows in various ways, but also to pioneer a legal framework on AI. Prominent instruments include the new AI Act, the Digital Services Act (DSA) and Digital Markets Act (DMA), the Data Act, Data Governance Act, or Open Data Directive.

The class will provide an overview on the existing framework: Which regulations and directives are relevant? How do they apply and interact which each other in a broader context?

Another objective is to provide students with the ability to read these legal instruments: How to access regulatory instruments that often have more than 100 pages (without having to read every single sentence)? How to gain a comprehensive, high-level understanding of the instrument? How to identify parts relevant to a particular legal problem?

The class will start with an introduction into EU law and regulatory instruments in general. Concrete guidance on reading, analysing and working with legal instruments in English will be given. Based on these instructions, students will be assigned legal instruments to present in the final unit along with a two-pages report.

Grades will be assigned based on the quality of these presentations and the report, as well as participation in the discussion (presentation: 40 %, two-pages report: 40 %, discussion: 20 %).

Organizational issues
WS 2024/25

Hierbei handelt es sich NICHT um eine Pro-Seminar, sondern um ein Seminar (aus Rechtswissenschaften).

Anmeldungen für das Seminar bitte NUR! über das WiWi-Portal!

***Für die Prüfung bitte NUR über CAS (Campus-Portal) anmelden!**

*Erläuterung: nach der für die Teilnahme am Seminar verbindlichen Teilnahme an der Einführungsveranstaltung bitte Anmeldung über das Campus-System (notwendig für die Erfassung der Note der Seminararbeit).

V	**EU Fundamental Rights and EU Economic Law in Smart Cities** 2400203, WS 25/26, 2 SWS, Language: English, Open in study portal	**Seminar (S)** **On-Site**

Content
This seminar is designed to instruct students in recognising legal questions as they relate to an interdisciplinary topic, classifying such questions along the hierarchy of norms, and solving them in a convincing argumentative manner. By focusing on „EU Fundamental Rights and EU Economic Law in Smart Cities“, we will discuss the influence of large cities and metropolises on their more rural periphery. Possible aspects of discussion are, inter alia,

- democratic legitimacy questions in such cases,
- elements of structural support/aid
- approaches to implement technical innovations in smart cities,
- questions relating to the development of (large) infrastructures,
- aspects of the efficient use of infrastructure, and
- environmental aspects.

Particularly in times of increasingly comprehensive European regulation, European standards and regulatory approaches will be at the heart of the discussion while not neglecting their interplay with national regulation. Learning in more depth about the interplay of European and national regulation will further accustom participating students to discuss relevant legal questions across (legal) sectors and with a broader view on regulations' effects.

Conducting the seminar in English intends to facilitate students to link their ideas and arguments to current European and international debates.

Organizational issues
This is a seminar (not a pro-seminar).

Registrations only via Wiwi-Portal.

Kick-off-meeting WS 2025/26:

Friday, 7th November 2025, 14:00 - 18:00 h (online)

(Mr. Kasper-Schöniger will send you the according link via ILIAS a few days before the kick-off date.)

Presentations:

- Friday, 6th February 2026, 09:00 - 18:00 h

and

- Saturday, 7th February 2026, 10:00 - 16:00

Place: see section "TERMINE" (below).

V	**Regulating AI: from ethics to law** 2400171, SS 2026, 2 SWS, Language: English, Open in study portal	**Seminar (S)** **On-Site**

Content
This seminar will provide an overview of technology regulation, with a particular focus on the regulation of AI systems. Following an introduction to the different forms of regulation, students will explore the various regulatory instruments in light of the key principles of AI, such as

- fairness / non-discrimination,
- transparency / explainability,
- non-maleficence,
- privacy / data protection,
- sustentability
- security / safety, and
- accountability.

This will enable them to discuss how these principles and rules can be applied in real-world scenarios.

Organizational issues
Please register for the seminar ONLY via the WiWi-Portal!
Please register for the exam ONLY via CAS (Campus-Portal)!*

*ExplanationAfter attending the introductory event, which is mandatory for participation in the seminar, please register via the campus system (necessary for recording the grade of the seminar paper).

Block Seminar in Summer term 2026:

- **Wednesday, 22th April 2026, 09:00 - 13:00 h;**
- **Wednesday, 22th July 2026, 09:00 - 13:00 h.**

Room: each seminar takes place in room no. 313 (building 07.08) .

V	**Designing Data Governance of Digital Systems** 2400177, SS 2026, 2 SWS, Language: English, Open in study portal	**Seminar (S)** **On-Site**

Content
The latest regulations in the digital sector at EU level represent a highly topical and important regulatory instrument with enormous practical relevance for students of computer science and business informatics. The seminar not only enables students to acquire important knowledge in this area, but also to apply it specifically to the governance of digital systems and to learn the practical design of digital systems against the background of legal framework conditions.

Organizational issues
***Please register <u>for the exam ONLY via CAS</u> (Campus-Portal)**!

***Explanation:<u>after </u>attending the introductory event,** which is mandatory for participation in the seminar**, please register via the campus system** (necessary for recording the grade of the seminar paper).

Due to the minimum number of text characters that can be used, please check ILIAS and the WiWi portal for details such as dates and location. Thank you.

V	**Security and Privacy Awareness** 2400125, WS 26/27, 2 SWS, Language: German/English, Open in study portal	**Seminar (S)** **Blended (On-Site/Online)**

Content
Within the framework of this interdisciplinary seminar, the topics security awareness and privacy awareness are to be considered from different perspectives. It deals with legal, information technology, psychological, social as well as philosophical aspects.

<u>Important notes:</u>

- The seminar language is English. This means that English is the standard for the presentation and the seminar paper, and German is only permitted for the paper after consultation with the supervisor, for example, if the topic of the thesis requires it.
- The seminar is only for MASTER students (or Mastervorzug)
- The link to enrol is for every student, regardless of the study background

<u>Topics:</u>

The advertised topics can be found in the wiwi portal.

T 6.686 Module component: Seminar: Legal Studies Master II [T-INFO-115026]

Coordinators: Professorenschaft des Fachbereichs Recht
Organisation: KIT Department of Informatics
Part of: M-WIWI-104903 - Law

Type	Credits	Grading	Version
Examination of another type	3 CP	graded	1

Courses					
WT 25/26	2400014	Current Issues in Patent Law	2 SWS	Seminar / On-Site	Melullis
WT 25/26	2400060	Data in Software-Intensive Technical Systems – Modeling – Analysis – Protection	2 SWS	Seminar / On-Site	Reussner, Raabe, Werner, Müller-Quade
WT 25/26	2400125	Security and Privacy Awareness	2 SWS	Seminar / Blended	Seidel-Saul, Volkamer, Boehm, Ballreich, Sikra, Veit
WT 25/26	2400165	Seminar Commercial and Corporate Law in Information Technology	2 SWS	Seminar / Online	Nolte
WT 25/26	2400177	Human vs. Machine: (Un-)Biased Legal Decision-Making?	2 SWS	Seminar / On-Site	Holzhausen
WT 25/26	2400184	EU Digital Regulatory Framework	2 SWS	Seminar / On-Site	Zufall
WT 25/26	2400203	EU Fundamental Rights and EU Economic Law in Smart Cities	2 SWS	Seminar / On-Site	Kasper-Schöniger
WT 25/26	2424186	Seminar Patent Law	2 SWS	Seminar / On-Site	Dammler
WT 25/26	2424389	Seminar "IT security law"	2 SWS	Seminar / On-Site	Schallbruch
ST 2026	2400005	Governance, Risk & Compliance	2 SWS	Seminar / On-Site	Herzig, Siddiq
ST 2026	2400078	Intelligente Chatbots und Recht	2 SWS	Seminar / On-Site	Raabe
ST 2026	2400171	Regulating AI: from ethics to law	2 SWS	Seminar / On-Site	Gil Gasiola
ST 2026	2400177	Designing Data Governance of Digital Systems	2 SWS	Seminar / On-Site	Pathak
ST 2026	2400207	(Generative) AI and Law	2 SWS	Seminar / Blended	Boehm, Turkovic-Popovski
ST 2026	2400225	Rechtlicher Rahmen für Künstliche Intelligenz	2 SWS	Seminar / On-Site	Sattler
ST 2026	24820	Current Issues in Patent Law	2 SWS	Seminar / On-Site	Melullis
WT 26/27	2400060	Seminar: Software Architecture, Security and Privacy	2 SWS	Seminar / On-Site	Reussner, Raabe, Werner, Müller-Quade
WT 26/27	2400125	Security and Privacy Awareness	2 SWS	Seminar / Blended	Volkamer, Boehm, Ballreich, Algedri

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Below you will find excerpts from courses related to this module component:

V Security and Privacy Awareness

2400125, WS 25/26, 2 SWS, Language: German/English, Open in study portal

Seminar (S)
Blended (On-Site/Online)

Content
Within the framework of this interdisciplinary seminar, the topics security awareness and privacy awareness are to be considered from different perspectives. It deals with legal, information technology, psychological, social as well as philosophical aspects.

Important notes:

- The seminar language is English. This means that English is the standard for the presentation and the seminar paper, and German is only permitted for the paper after consultation with the supervisor, for example, if the topic of the thesis requires it.
- The seminar is only for MASTER students (or Mastervorzug)
- The link to enrol is for every student, regardless of the study background

Topics:

The advertised topics can be found in the wiwi portal.

V	**Human vs. Machine: (Un-)Biased Legal Decision-Making?** 2400177, WS 25/26, 2 SWS, Language: English, Open in study portal	**Seminar (S)** **On-Site**

Content
This seminar provides an overview of the foundational principles of behavioral economics and their relevance to public law. Special attention is given to (biased) decision-making, both from a substantive and procedural perspective. Students will explore how regulation should be designed to account for cognitive biases and achieve intended policy outcomes. In addition, the seminar will address the role of automated decision-making systems - examining how such systems are influenced by, and may also reinforce or mitigate, behavioral biases.

The seminar will cover the following topics:

- Introduction to (behavioral) economic theory and its application to law
- Cognitive biases and their impact on (public) decision-making
- Different types of regulation in light of behavioral insights
- Implications for automated decision-making systems
- The role of human biases in the AI Act

Algorithmic decision-making is often treated as a black box—something that should not be permitted to make decisions in legal contexts. In response, Article 14 of the EU AI Act requires high-risk AI systems to be designed in such a way that they can be effectively overseen by natural persons. This human-in-the-loop requirement stems, among other things, from the perception that AI systems operate ambiguously and produce decisions that are not fully comprehensible.

However, treating humans in comparison as fundamentally transparent or fully rational decision-makers is equally naïve. Understanding how people actually behave—not just how they should behave—is essential for designing effective, fair, and legitimate legal systems. Traditional legal research often assumes rational actors, yet behavioral research shows that cognitive biases, heuristics, and emotional influences shape the decisions of both citizens and public officials.

This seminar addresses that gap by equipping students with behavioral insights to critically assess and improve legal and administrative. A particular focus is placed on how human biases are already embedded in the data underlying automated decision-making systems, potentially reproducing or amplifying unfairness. At the same time, students will explore how such systems can be intentionally designed to mitigate behavioral distortions and support better decision-making.

Organizational issues
1. Termin: 13.11.2025 9:00-15:00 Uhr (Seminarrum Nr. 313, bei uns am Institut)

2. Termin: 05.02.2025 9:00- 16:00 Uhr (Seminarrum Nr. 313, bei uns am Institut).

V	**EU Digital Regulatory Framework** 2400184, WS 25/26, 2 SWS, Language: English, Open in study portal	**Seminar (S)** **On-Site**

Content
This class aims to provide an overview on the legal instruments forming the EU digital regulatory framework. Following its Digital Single Market Strategy, the EU has set up a new strategic programme for a "Digital Decade". Existing regulations like the General Data Protection Regulation (GDPR), or the E-Commerce Directive, are being complemented by a variety of new instruments that aim to set binding rules on online markets, to regulate data flows in various ways, but also to pioneer a legal framework on AI. Prominent instruments include the new AI Act, the Digital Services Act (DSA) and Digital Markets Act (DMA), the Data Act, Data Governance Act, or Open Data Directive.

The class will provide an overview on the existing framework: Which regulations and directives are relevant? How do they apply and interact which each other in a broader context?

Another objective is to provide students with the ability to read these legal instruments: How to access regulatory instruments that often have more than 100 pages (without having to read every single sentence)? How to gain a comprehensive, high-level understanding of the instrument? How to identify parts relevant to a particular legal problem?

The class will start with an introduction into EU law and regulatory instruments in general. Concrete guidance on reading, analysing and working with legal instruments in English will be given. Based on these instructions, students will be assigned legal instruments to present in the final unit along with a two-pages report.

Grades will be assigned based on the quality of these presentations and the report, as well as participation in the discussion (presentation: 40 %, two-pages report: 40 %, discussion: 20 %).

Organizational issues
WS 2024/25

Hierbei handelt es sich NICHT um eine Pro-Seminar, sondern um ein Seminar (aus Rechtswissenschaften).

Anmeldungen für das Seminar bitte NUR! über das WiWi-Portal!

***Für die Prüfung bitte NUR über CAS (Campus-Portal) anmelden!**

*Erläuterung: nach der für die Teilnahme am Seminar verbindlichen Teilnahme an der Einführungsveranstaltung bitte Anmeldung über das Campus-System (notwendig für die Erfassung der Note der Seminararbeit).

V **EU Fundamental Rights and EU Economic Law in Smart Cities** — **Seminar (S) On-Site**
2400203, WS 25/26, 2 SWS, Language: English, Open in study portal

Content
This seminar is designed to instruct students in recognising legal questions as they relate to an interdisciplinary topic, classifying such questions along the hierarchy of norms, and solving them in a convincing argumentative manner. By focusing on „EU Fundamental Rights and EU Economic Law in Smart Cities“, we will discuss the influence of large cities and metropolises on their more rural periphery. Possible aspects of discussion are, inter alia,

- democratic legitimacy questions in such cases,
- elements of structural support/aid
- approaches to implement technical innovations in smart cities,
- questions relating to the development of (large) infrastructures,
- aspects of the efficient use of infrastructure, and
- environmental aspects.

Particularly in times of increasingly comprehensive European regulation, European standards and regulatory approaches will be at the heart of the discussion while not neglecting their interplay with national regulation. Learning in more depth about the interplay of European and national regulation will further accustom participating students to discuss relevant legal questions across (legal) sectors and with a broader view on regulations’ effects.

Conducting the seminar in English intends to facilitate students to link their ideas and arguments to current European and international debates.

Organizational issues
This is a seminar (not a pro-seminar).

Registrations only via Wiwi-Portal.

Kick-off-meeting WS 2025/26:

Friday, 7th November 2025, 14:00 - 18:00 h (online)

(Mr. Kasper-Schöniger will send you the according link via ILIAS a few days before the kick-off date.)

Presentations:

- Friday, 6th February 2026, 09:00 - 18:00 h

and

- Saturday, 7th February 2026, 10:00 - 16:00

Place: see section "TERMINE" (below).

V **Regulating AI: from ethics to law**
2400171, SS 2026, 2 SWS, Language: English, Open in study portal
Seminar (S)
On-Site

Content
This seminar will provide an overview of technology regulation, with a particular focus on the regulation of AI systems. Following an introduction to the different forms of regulation, students will explore the various regulatory instruments in light of the key principles of AI, such as

- fairness / non-discrimination,
- transparency / explainability,
- non-maleficence,
- privacy / data protection,
- sustentability
- security / safety, and
- accountability.

This will enable them to discuss how these principles and rules can be applied in real-world scenarios.

Organizational issues
Please register for the seminar ONLY via the WiWi-Portal!
Please register for the exam ONLY via CAS (Campus-Portal)!*

*ExplanationAfter attending the introductory event, which is mandatory for participation in the seminar, please register via the campus system (necessary for recording the grade of the seminar paper).

Block Seminar in Summer term 2026:

- **Wednesday, 22th April 2026, 09:00 - 13:00 h;**
- **Wednesday, 22th July 2026, 09:00 - 13:00 h.**

Room: each seminar takes place in room no. 313 (building 07.08) .

V **Designing Data Governance of Digital Systems**
2400177, SS 2026, 2 SWS, Language: English, Open in study portal
Seminar (S)
On-Site

Content
The latest regulations in the digital sector at EU level represent a highly topical and important regulatory instrument with enormous practical relevance for students of computer science and business informatics. The seminar not only enables students to acquire important knowledge in this area, but also to apply it specifically to the governance of digital systems and to learn the practical design of digital systems against the background of legal framework conditions.

Organizational issues
***Please register for the exam ONLY via CAS (Campus-Portal)**!

***Explanation:after attending the introductory event,** which is mandatory for participation in the seminar**, please register via the campus system** (necessary for recording the grade of the seminar paper).

Due to the minimum number of text characters that can be used, please check ILIAS and the WiWi portal for details such as dates and location. Thank you.

V **Security and Privacy Awareness**
2400125, WS 26/27, 2 SWS, Language: German/English, Open in study portal
Seminar (S)
Blended (On-Site/Online)

Content
Within the framework of this interdisciplinary seminar, the topics security awareness and privacy awareness are to be considered from different perspectives. It deals with legal, information technology, psychological, social as well as philosophical aspects.

Important notes:

- The seminar language is English. This means that English is the standard for the presentation and the seminar paper, and German is only permitted for the paper after consultation with the supervisor, for example, if the topic of the thesis requires it.
- The seminar is only for MASTER students (or Mastervorzug)
- The link to enrol is for every student, regardless of the study background

Topics:

The advertised topics can be found in the wiwi portal.

T 6.687 Module component: Seminar: Machine Learning and Optimization in Engineering [T-MACH-115011]

Coordinators: Prof. Dr.-Ing. Anne Meyer
Organisation: KIT Department of Mechanical Engineering
Part of: M-MACH-107716 - Data Driven Engineering
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Expansion	Version
Examination of another type	4,5 CP	graded	Each term	1 semesters	1

Courses					
ST 2026	2122353	Seminar: Machine Learning and Optimization in Engineering	2 SWS	Seminar / 🗣	Meyer

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The grade is determined by an examination of another type. The following aspects are included in the assessment:

- Preparation of a seminar paper on (a specific aspect of) the seminar topic using scientific methods (about 15 till 20 pages).
- Presentation on the topic of the seminar paper (about 15 minutes).

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Seminar: Machine Learning and Optimization in Engineering

Seminar (S)
On-Site

2122353, SS 2026, 2 SWS, Language: English, Open in study portal

Content
This seminar focuses on advanced methodological aspects of machine learning and optimization in engineering applications, with an emphasis on hybrid approaches that combine data-driven learning with optimization techniques. Current research contributions are analyzed, critically assessed, and discussed, with a strong focus on modeling, algorithm design, and computational performance. **Topics may include method-oriented projects in areas such as product design, industrial process optimization, or logistics systems.**

Participants are required to independently study scientific literature, prepare a seminar paper, and present their findings in an oral presentation. Depending on the topic, a prototypical implementation and computational evaluation of the considered models or algorithms using modern software frameworks (e.g., Python-based ML libraries, optimization and simulation tools, or hybrid ML+optimization frameworks) is expected. All topics are designed to enable further development into a Master's thesis.

Learning Objectives
After successfully completing the seminar, students will be able to:

- independently evaluate hybrid ML and optimization approaches for engineering problems and implement them in prototypes.
- methodically review research literature,
- document their findings in writing, and present them in a scientific discussion.

The seminar will prepare students for writing a master's thesis in the field of Machine Learning and Optimization in Engineering.

T 6.688 Module component: Sensor Systems [T-ETIT-100709]

Coordinators: Dr. Wolfgang Menesklou
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Oral examination	3 CP	graded	Each summer term	1

T 6.689 Module component: Sensors [T-ETIT-101911]

Coordinators: TT-Prof. Dr. Johanna Schröder
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-101158 - Sensor Technology I
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Written examination	3 CP	graded	Each summer term	2

Courses					
ST 2026	2304231	Sensors	2 SWS	Lecture / 🗣	Schröder

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, × Cancelled

Assessment
Success is assessed in the form of a written examination lasting 2 hours.

Prerequisites
none

Recommendations
Basic knowledge of materials science (e.g. lecture "Passive components") is helpful.

T 6.690 Module component: Service Design Thinking [T-WIWI-102849]

Coordinators: Prof. Dr. Gerhard Satzger
Prof. Dr. Orestis Terzidis
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101503 - Service Design Thinking
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Examination of another type	9 CP	graded	Irregular	5

Courses					
WT 25/26	2595600	Service Design Thinking	2 SWS	Lecture / 🗣	Osaro, Terzidis, Feldmann

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success is assessed in the form of an alternative exam assessment which consists of a case study, workshops, and a final presentation. The weighting of these components for the grade will be announced at the beginning of the course.

Prerequisites
None

Recommendations
This course is held in English – proficiency in writing and communication is required.

Our past students recommend to take this course at the beginning of the masters program.

Additional Information
Due to practical project work as a component of the program, access is limited. The module (as well as the module component) spans two semesters. It starts in September every year and runs until end of June in the subsequent year. Entering the program is only possible at its beginning - after prior application in May/June. For more information on the application process and the program itself are provided in the module component description and the program's website (https://sdtkarlsruhe.de/). Furthermore, the lecturers provide an information event for applicants every year in May.

Below you will find excerpts from courses related to this module component:

V Service Design Thinking

2595600, WS 25/26, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content
The Service Design Thinking program is much more than a normal course. Through this program, we provide the knowledge and skills that true innovators need. In this context, we train our participants in the human-centric innovation approach “Design Thinking”. In addition, participants work in small international and interdisciplinary teams on real innovation challenges from practice.

The teams are made up of students from KIT and another university from the global SUGAR network. These include, for example, the Hasso Plattner Institute in Potsdam, Trinity College in Dublin and the University of Science and Technology of China. The program includes visits to international events of the SUGAR Network, which are usually held in places known for their high level of innovation. At these events, our participants present their (interim) results to a large audience consisting of employees from the partner companies and the universities involved.

What students will learn:

- A comprehensive understanding of the globally recognized innovation approach “Design Thinking” as introduced and promoted by the Stanford University
- Development of new, creative solutions through extensive need finding, in particular with regard to the relevant service users
- to develop prototypes of the collected ideas early and independently, to test them and improve them iteratively, thereby solving the issue defined by the partner company
- to communicate, present and network in an interdisciplinary and international environment
- to apply the learned approach in the context of a real innovation project provided by a practical partner.

Course phases (roughly 4 weeks each):

- **Kick off:**
  Learning the basic method elements by solving an exercise challenge. Participation in the Global Kick-Off of the SUGAR Network consisting of method workshops, working on team challenges, networking with other universities and forming project teams for the challenges of the practical partners.
- **Design Space Exploration**:
  Exploring the problem space by questioning the given innovation challenge from practice. Familiarization with the topic area of the respective challenge. Gathering first impressions of the requirements and needs of people related to the problem.
- **Critical Function Prototype:**
  Building an intensive understanding of the needs of the target group of the respective challenge. Deriving critical functions from the customer's perspective that could help solve the overall problem. Building prototypes for the critical functions and testing them in real customer situations.
- **Dark Horse Prototype**:
  Reversal of assumptions and experiences made so far. The goal is to develop radically new and unconventional ideas. Implementation of the ideas into simple prototypes and subsequent testing.
- **Funky Prototype**:
  Integration of the individual successfully tested functions from the critical function and dark horse phase into solution concepts. These are also tested and further developed.
- **Functional Prototype**:
  Selection of successful funky prototypes and development of these towards high-resolution prototypes. The final solution approach for the project is written down in detail and feedback is obtained.
- **Final Prototype**:
  Implementing the final prototype and presenting it to the practical partner as well as the SUGAR Network.

Organizational issues
Bei der Vorlesung handelt es sich um eine zweisemestrige Veranstaltung, die jährlich im September startet.

Literature

- Design Thinking: Das Handbuch; Falk Uebernickel, Walter Brenner, Therese Naef, Britta Pukall, Bernhard Schindlholzer
- The Design Thinking Playbook: Mindful Digital Transformation of Teams, Products, Services, Businesses and Ecosystems; Michael Lewrick, Patrick Link, Larry Leifer
- The Design Thinking Toolbox: A Guide to Mastering the Most Popular and Valuable Innovation Methods; Michael Lewrick, Patrick Link, Larry Leifer
- Frame Innovation: Create New Thinking by Design (Design Thinking, Design Theory); Kees Dorst

T 6.691 Module component: Service Operations and Cyber Security [T-WIWI-114109]

Coordinators: Esther Mohr
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-102805 - Service Operations
M-WIWI-102808 - Digital Service Systems in Industry
M-WIWI-104899 - Operations Research

Type	Credits	Grading	Term offered	Version
Examination of another type	4,5 CP	graded	Each summer term	1

Assessment
The assessment is carried out as a single examination of another type that combines two mandatory components:

1. written dissertation
2. oral final exam (duration ≈ 30 – 40 minutes)

Both components are integrated into one overall assessment, which reflects the overall impression of the combined performance.

Further details on the concrete design (e.g., grading criteria) will be announced during the lecture.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-WIWI-102872 - Challenges in Supply Chain Management must not have been started.

Recommendations
Basic knowledge as conveyed in the module "Introduction to Operations Research" is assumed.

Additional Information
The number of course participants is limited to 12 due to the collaborative work in project teams. As a result of this limitation, registration is required before the course begins. Further information can be found on the course's website.
The event takes place irregularly. The planned lectures and courses for the next three years will be announced online.

T 6.692 Module component: Services Marketing and B2B Marketing [T-WIWI-102806]

Coordinators: Dr. Sven Feurer
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	3 CP	graded	see Annotations	1

Assessment
The examination will be open to first-time writers for the last time in the 2021 summer semester.

The assessment consists of a written exam (60 minutes) (following §4(2), 1 of the examination regulation).

Prerequisites
None

Additional Information
The course "Services Marketing and B2B Marketing" will be offered for the last time in winter semester 2020/21. For further information please contact the Marketing & Sales Research Group (marketing.iism.kit.edu).

T 6.693 Module component: SIL Entrepreneurship Emphasis [T-WIWI-110287]

Coordinators: Prof. Dr. Orestis Terzidis
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-105010 - Student Innovation Lab (SIL) 1

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Each winter term	1

Courses					
WT 25/26	2500002	SIL Entrepreneurship Emphasis	4 SWS	Seminar	Terzidis

Assessment
Alternative exam assessment (§4(2), 3 SPO). The final grade is a result from both, the grade of the term paper and its presentation, as well as active participation during the seminar. In addition, smaller, ungraded tasks are provided in the course to monitor progress.

Prerequisites
None

Recommendations
None

Workload
90 hours

T 6.694 Module component: SIL Entrepreneurship Project [T-WIWI-110166]

Coordinators: Prof. Dr. Orestis Terzidis
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-105010 - Student Innovation Lab (SIL) 1

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Each winter term	1

Courses					
WT 25/26	2545082	SIL Entrepreneurship Project	4 SWS	Seminar	Terzidis

Assessment
Success is assessed in the form of an alternative exam assessment. The grade results from the evaluation of two seminar papers. Further details will be provided at the beginning of the course. The points system for the assessment of the two seminar papers is determined by the lecturer of the course. It will be announced at the beginning of the course.

Prerequisites
None

Recommendations
None

Workload
90 hours

T 6.695 Module component: SIL-LaunchLab [T-WIWI-114635]

Coordinators: Prof. Dr. Orestis Terzidis
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Examination of another type	9 CP	graded	Each summer term	1

Courses					
ST 2026	2545022	SIL - LaunchLab	6 SWS	Seminar / 🗣	Böhrer, Terzidis

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment is conducted as an alternative exam assessment. The specific form of assessment depends on the course and may include, for example, the following components:

- **Project Report or Reflection Paper**
- **Mid-term and Final Presentation**
- **Group Work with Individual Contribution**
- **Active Participation and Mandatory Attendance**

The overall grade is determined by the weighted evaluation of these individual assessment components. The detailed weighting of these components for grade calculation will be communicated to students at the beginning of the course.

Prerequisites
None

Recommendations
An interest in entrepreneurial thinking and action, teamwork, and interdisciplinary project work is recommended. Basic knowledge in business administration and/or innovation methods is helpful but not required.

Workload
270 hours

Below you will find excerpts from courses related to this module component:

V SIL - LaunchLab

2545022, SS 2026, 6 SWS, Open in study portal

Seminar (S)
On-Site

Content
Course content:

In the SIL LaunchLab, you will work in interdisciplinary teams on real innovation projects from the Student Innovation Lab (SIL) at KIT. As part of the seminar, you will take on the role of a project mentor and support an existing SIL team in further developing its start-up idea. This can include tasks such as market analysis, business model development, or pitch preparation, and will be defined individually at the start of the seminar. Through this, you gain direct insights into technology-based start-up projects at KIT and apply entrepreneurial methods to real start-up cases.

Learning Objectives:

After completing the course, participants will be able to:

- Analyze and support real start-up projects in a structured way
- Conduct market and user analyses
- Evaluate and further develop a business model using the Business Model Canvas tool
- Assist teams in preparing for investor and customer pitches
- Develop a basic understanding of technology-driven spin-offs

Seminar Information:

Participation is only possible during the summer semester. The seminar is aimed at bachelor students from the 4th semester onwards from all disciplines. Places are limited. Registration takes place via the WiWi portal and, after admission, via CAS for the corresponding examination.

T 6.696 Module component: Simulation Game in Energy Economics [T-WIWI-108016]

Coordinators: Dr. Massimo Genoese
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101451 - Energy Economics and Energy Markets
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Examination of another type	3,5 CP	graded	Each summer term	2

Courses					
ST 2026	2581025	Simulation Game in Energy Economics	3 SWS	Lecture / Practice / 🗣	Genoese, Zimmermann

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Examination as written assignment and oral presentation (§4 (2), 1 SPO).

Prerequisites
None

Recommendations
Visiting the course "Introduction to Energy Economics"

Additional Information
The number of participants is limited.
There is a registration procedure via CAS followed by a selection of the participants.

Below you will find excerpts from courses related to this module component:

V Simulation Game in Energy Economics

2581025, SS 2026, 3 SWS, Language: German, Open in study portal

Lecture / Practice (VÜ)
On-Site

Content

- Introduction
- Agents and market places in the electricity industry
- Selected planning tasks of energy service companies
- Methods of modelling in the energy sector
- Agent-based simulation: The PowerACE model
- Simulation game: Simulation in energy economics (electricity and emission trading, investment decisions)

The lecture is structured in a theoretical and a practical part. In the theoretical part, the students are taught the basics to carry out simulations themselves in the practical part which comprises amongst others the simulation of the power exchange. The participants of the simulation game take a role as a power trader in the power market. Based on various sources of information (e.g. prognosis of power prices, available power plants, fuel prices), they can launch bids in the power exchange.

Assessment: presentation and written summary

Prerequisites: Basics in Energy economics ad markets are advantageous.

Organizational issues
CIP-Pool West, Raum 102, Geb. 06.41 - siehe Institutsaushang

Literature
Weiterführende Literatur:

Möst, D. und Genoese, M. (2009): Market power in the German wholesale electricity market. The Journal of Energy Markets (47–74). Volume 2/Number 2, Summer 2009

T 6.697 Module component: Simulation of Nanoscale Systems, without Seminar [T-PHYS-102504]

Coordinators: Prof. Dr. Wolfgang Wenzel
Organisation: KIT Department of Physics
Part of: M-MACH-101294 - Nanotechnology

Type	Credits	Grading	Term offered	Version
Oral examination	6 CP	graded	Irregular	1

Prerequisites
none

6.698 Module component: Simulation with Lumped Parameters [T-MACH-113862]

Coordinators: Prof. Dr.-Ing. Marcus Geimer
Organisation: KIT Department of Mechanical Engineering
Part of: M-MACH-101265 - Vehicle Development
M-MACH-101267 - Mobile Machines

Type	Credits	Grading	Term offered	Expansion	Version
Oral examination	3 CP	graded	Each summer term	1 semesters	3

Courses					
ST 2026	2114071	Simulation with Lumped Parameters	2 SWS	Lecture / 🗣	Geimer

Legend: 🖥 Online, Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Oral examination, duration approx. 20 minutes

Prerequisites
none

Recommendations
Knowledge of mechanics and hydraulics

Qualification goals

After completing this part of the course, students will be able to evaluate how a simulation with concentrated parameters can be used sensibly and which simulation methods are suitable for a given problem. They will be able to create a model for a given problem and explain and implement algorithms for solving a model. You will acquire in-depth knowledge of how a system can be modelled and parameterized with concentrated parameters. You will be able to carry out simulation studies, evaluate simulation results and recognize and avoid errors in the simulation.

Contents:

The basics of discrete-time modeling are taught using the example of simulation with concentrated parameters. For this purpose, modeling in the disciplines of mechanics, electrics and hydraulics is shown by way of example and analogies are drawn. Furthermore, possibilities for simulation coupling of the discipline are shown. The students solve exemplary tasks with the help of simulation and briefly summarize the solutions in a report.

Recommended are:

Basic knowledge of Matlab/Simulink and hydraulics.

Knowledge of the dynamics of mechanical systems and the fundamentals of electrical engineering is assumed.

Additional Information
The course is offered in German.

Workload
90 hours

Below you will find excerpts from courses related to this module component:

V Simulation with Lumped Parameters

2114071, SS 2026, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content

Learning objectives (qualification objectives):

After completing this part of the course, students will be able to evaluate how a simulation with concentrated parameters can be used sensibly and which simulation methods are suitable for a given problem. They can create a model for a problem and can explain and implement algorithms for solving a model. You will acquire in-depth knowledge of how a system can be modelled and parameterized with concentrated parameters. You will be able to carry out simulation studies, evaluate simulation results and recognize and avoid errors in the simulation.

Contents:

The basics of discrete-time modeling are taught using the example of simulation with concentrated parameters. For this purpose, modeling in the disciplines of mechanics, electrics and hydraulics is shown by way of example and analogies are drawn. Furthermore, possibilities for simulation coupling of the disciplines are shown. The students solve exemplary tasks with the help of simulation and briefly summarize the solutions in a report.

Recommended are:

Basic knowledge of Matlab/Simulink and hydraulics.

Knowledge of the dynamics of mechanical systems and the fundamentals of electrical engineering is assumed.

T 6.699 Module component: Single-Cell Technologies [T-CIWVT-113231]

Coordinators: Prof. Dr.-Ing. Alexander Grünberger
Organisation: KIT Department of Chemical and Process Engineering
Part of: M-CIWVT-106564 - Single-Cell Technologies

Type	Credits	Grading	Version
Oral examination	4 CP	graded	1

Courses					
WT 25/26	2213030	Single-Cell Technologies	2 SWS	Lecture / 🗣	Grünberger
WT 26/27	2213030	Single-Cell Technologies	2 SWS	Lecture / 🗣	Grünberger

Legend: 🖥 Online, ✥ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The learning control is an oral examination.

Prerequisites
None

T 6.700 Module component: Single-Photon Detectors [T-ETIT-108390]

Coordinators: Dr. Konstantin Ilin
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-101971 - Single-Photon Detectors

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each winter term	1

Courses					
WT 25/26	2312680	Single-Photon Detectors	2 SWS	Lecture / 🗣	Ilin
WT 25/26	2312694	Tutorial for 2312680 Single-Photon Detectors	1 SWS	Practice / 🗣	Ilin
WT 26/27	2312680	Single-Photon Detectors	2 SWS	Lecture / 🗣	Ilin
WT 26/27	2312694	Tutorial for 2312680 Single-Photon Detectors	1 SWS	Practice / 🗣	Ilin

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success controll takes place in form of an overall oral examination lasting approx. 20 minutes.

Prerequisites
none

6.701 Module component: Site Management [T-BGU-103427]

Coordinators: Prof. Dr.-Ing. Shervin Haghsheno
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: M-BGU-101884 - Lean Management in Construction
M-BGU-101888 - Project Management in Construction
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Expansion	Version
Oral examination	1,5 CP	graded	Each summer term	1 semesters	1

Courses					
ST 2026	6241801	Site Management	1 SWS	Lecture / Practice / 🗣	N.N.

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
oral exam, appr. 15 min.

Prerequisites
none

Recommendations
none

Additional Information
none

Workload
45 hours

T 6.702 Module component: Smart Energy Infrastructure [T-WIWI-107464]

Coordinators: Dr. Armin Ardone
Dr. Dr. Andrej Marko Pustisek
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101452 - Energy Economics and Technology
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	5,5 CP	graded	Each winter term	2

Courses					
WT 25/26	2581023	(Smart) Energy Infrastructure	4 SWS	Lecture / 🗣	Ardone, Pustisek, Schuler

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment consists of a written exam (90 minutes). The exam takes place in every semester. Re-examinations are offered at every ordinary examination date.

Workload
165 hours

Below you will find excerpts from courses related to this module component:

V (Smart) Energy Infrastructure

2581023, WS 25/26, 4 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content
The lecture provides a techno-economic overview of different infrastructures of the energy system and their importance regarding the future energy system (“Energiewende”) – in particular

- for electricity:
  - the supply side (e.g. power plants)
  - the demand side (e.g. load structures of appliances, flexibilities) as well as
  - transport infrastructures (electricity grids)
- for fuel transportation:
  - pipeline infrastructures (focus on natural gas)
  - shipping of LNG
  - crude oil and oil product transportation
  - hydrogen transportation
  - comparison of potential energy carriers for global trade of renewable energy (e.g., hydrogen and its derivates, e-fuels, reactive metals)
- storage systems (e.g. batteries)

Additionally, the lecture provides a toolbox for energy system analysis such as an overview and classification of energy systems modelling approaches as well as the usage of scenario techniques for energy systems analysis.

The lecture also provides practical examples for the relevant methods presented.

T 6.703 Module component: Smart Grid Applications [T-WIWI-107504]

Coordinators: Prof. Dr. Christof Weinhardt
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101446 - Market Engineering
M-WIWI-103720 - eEnergy: Markets, Services and Systems
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	see Annotations	2

Assessment
The assessment consists of a written exam (60 min) (according to §4(2), 1 of the examination regulations). By successful completion of the exercises (§4 (2), 3 SPO 2007 respectively §4 (3) SPO 2015) a bonus can be obtained. If the grade of the written exam is at least 4.0 and at most 1.3, the bonus will improve it by one grade level (i.e. by 0.3 or 0.4).

Prerequisites
None

Recommendations
None

Additional Information
The lecture will no longer be offered from the coming winter semester 2023/24. It is only possible to take part in the main exam (first-time writer) and follow-up exam (repeater).

T 6.704 Module component: Social Choice Theory [T-WIWI-102859]

Coordinators: Prof. Dr. Clemens Puppe
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101500 - Microeconomic Theory
M-WIWI-101504 - Collective Decision Making
M-WIWI-104908 - Economics

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each summer term	3

Courses					
ST 2026	2520537	Social Choice Theory	2 SWS	Lecture / 🗣	Puppe, Kretz
ST 2026	2520539	Übung zu Social Choice Theory	1 SWS	Practice / 🗣	Puppe, Kretz

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment consists of a written exam (60 min.). The examination is offered every summer semester.

Prerequisites
None

Below you will find excerpts from courses related to this module component:

V Social Choice Theory
2520537, SS 2026, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content
How should (political) candidates be elected? What are good ways of merging individual judgments into collective judgments? Social Choice Theory is the systematic study and comparison of how groups and societies can come to collective decisions.

The course offers a rigorous and comprehensive treatment of judgment and preference aggregation as well as voting theory. It is divided into two parts. The first part deals with (general binary) aggregation theory and builds towards a general impossibility result that has the famous Arrow theorem as a corollary. The second part treats voting theory. Among other things, it includes prooving the Gibbard-Satterthwaite theorem.

Workload:
Total workload for 4.5 credit points: approx. 135 hours
Attendance: 30 hours
Self-study: 105 hours

Literature
Main texts:

- Moulin, H. 1988. *Axioms of Cooperative Decision Making*. Cambridge University Press.
- List, C. and Puppe, C. 2009. Judgement Aggregation. A survey. In: *The Handbook of rational & social choice*. P. Anand, P. Pattanaik, C. Puppe (Eds.). Oxford University Press.

Secondary texts:

- Sen, A. K. 1970. *Collective Choice and Social Welfare*. Holden-Day.
- Gaertner, W. 2009. *A Primer in Social Choice Theory*. Revised edition. Oxford University Press.
- Gaertner, W. 2001. *Domain Conditions in Social Choice Theory*. Cambridge University Press.

T 6.705 Module component: Social Science A (WiWi) [T-GEISTSOZ-109048]

Coordinators: Prof. Dr. Gerd Nollmann
Organisation: KIT Department of Humanities and Social Sciences
Part of: M-WIWI-104906 - Social Sciences

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Each winter term	1

Courses					
WT 25/26	5011011	Artificial intelligence in the research process	2 SWS	Seminar / Blended	Banisch
WT 25/26	5011014	Advanced module:Technology and Future: Theories of prospective knowledge	2 SWS	Seminar / On-Site	Lösch
ST 2026	5000048	Socio-scientific Theories of Technology Assessment	2 SWS	Proseminar / On-Site	Lösch
ST 2026	5011013	Digital democracy: worries, opportunities, and challenges	2 SWS	Seminar / Blended	Mäs
ST 2026	5011019	How to control algorithms?	2 SWS	Seminar / On-Site	Mäs

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Below you will find excerpts from courses related to this module component:

V Artificial intelligence in the research process

5011011, WS 25/26, 2 SWS, Language: German, Open in study portal

Seminar (S)
Blended (On-Site/Online)

Content

ChatGPT und andere Large Language Models (LLMs) transformieren unsere Gesellschaft auf vielen Ebenen. Auch Studium und Wissenschaft stehen vor tiefgreifenden Veränderungen. Im Seminar „Künstliche Intelligenz im Forschungsprozess" nähern wir uns diesen neuen Technologien und erproben, wie sie sinnvoll eingesetzt werden können, um aktuelle Forschungsfragen zu adressieren. Wir orientieren uns dabei an den Methoden und Fragestellungen der Computer-gestützen Sozialwissenschaft (Computational Social Science) mit besonderem Fokus auf die Extraktion komplexer Bedeutungsmuster (z.B. Meinungen, Argumente, Narrative, etc.). Das Seminar ist als Blockseminar mit zwei Blöcken konzipiert (voraussichtlich Januar and März). Gemeinsam erarbeiten wir Themen für Miniprojekte, die zwischen den beiden Blöcken von den Studierenden bearbeitet werden. Im Vorfeld wird es eine online-Sitzung geben.

Organizational issues

Diese Veranstaltung wird als Blockseminar angeboten.

NEUE TERMINE:

Di: 03.03.2026 09.00 - 17.00

Mi: 18.03.2026 09.00 - 17.00

DO: 19.03.2026 09.00 - 17.00

V Digital democracy: worries, opportunities, and challenges

5011013, SS 2026, 2 SWS, Language: German, Open in study portal

Seminar (S)
Blended (On-Site/Online)

Content

Democracy is in danger as a growing, increasingly vocal segment of the population feels unrepresented by democratic institutions. Populist movements with overtly anti-democratic goals attract strong support and are achieving electoral successes more often. We examine possible responses to these trends and the digital solutions that could help. Key questions include: Which strategies can curb the root causes of populist success? Can regulating social-media platforms mitigate the problem? How can civil society be strengthened, and which new democratic institutions can better integrate citizens into the legislative process? We discuss the potential of citizens' assemblies and other digital democracy initiatives to enhance participation. The seminar also explores how such tools can be combined with traditional mechanisms to create more inclusive decision-making. The central issue is identifying the research needed to develop, pilot, and iteratively improve new democratic models. We consider how to conduct this research effectively in a context where democratic systems are already under stress.

V **How to control algorithms?**
5011019, SS 2026, 2 SWS, Language: German, Open in study portal

Seminar (S)
On-Site

Content
It is warned that algorithms increasingly harm societies: they can amplify opinion polarization, facilitate the spread of fake news, and even make democratic elections vulnerable. In this seminar we examine these warnings and the scientific research behind them. We ask how unintended societal consequences of algorithms can be tackled: What can users do? How might governments respond? What responsibilities do companies such as Facebook and Google bear? To answer these questions, we develop agent-based models using the NetLogo software. Programming skills are helpful but not required, because NetLogo can be learned quickly.

Organizational issues
Teilnehmende halten einen Kurzvortrag und erstellen einen Seminararbeit.

T 6.706 Module component: Social Science B (WiWi) [T-GEISTSOZ-109049]

Coordinators: Prof. Dr. Gerd Nollmann
Organisation: KIT Department of Humanities and Social Sciences
Part of: M-WIWI-104906 - Social Sciences

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Each winter term	1

Courses					
WT 25/26	5011011	Artificial intelligence in the research process	2 SWS	Seminar / Blended	Banisch
WT 25/26	5011014	Advanced module:Technology and Future: Theories of prospective knowledge	2 SWS	Seminar / On-Site	Lösch
ST 2026	5000048	Socio-scientific Theories of Technology Assessment	2 SWS	Proseminar / On-Site	Lösch

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Below you will find excerpts from courses related to this module component:

V Artificial intelligence in the research process

5011011, WS 25/26, 2 SWS, Language: German, Open in study portal

Seminar (S)
Blended (On-Site/Online)

Content
ChatGPT und andere Large Language Models (LLMs) transformieren unsere Gesellschaft auf vielen Ebenen. Auch Studium und Wissenschaft stehen vor tiefgreifenden Veränderungen. Im Seminar „Künstliche Intelligenz im Forschungsprozess“ nähern wir uns diesen neuen Technologien und erproben, wie sie sinnvoll eingesetzt werden können, um aktuelle Forschungsfragen zu adressieren. Wir orientieren uns dabei an den Methoden und Fragestellungen der Computer-gestützen Sozialwissenschaft (Computational Social Science) mit besonderem Fokus auf die Extraktion komplexer Bedeutungsmuster (z.B. Meinungen, Argumente, Narrative, etc.). Das Seminar ist als Blockseminar mit zwei Blöcken konzipiert (voraussichtlich Januar and März). Gemeinsam erarbeiten wir Themen für Miniprojekte, die zwischen den beiden Blöcken von den Studierenden bearbeitet werden. Im Vorfeld wird es eine online-Sitzung geben.

Organizational issues
Diese Veranstaltung wird als Blockseminar angeboten.

NEUE TERMINE:

Di: 03.03.2026 09.00 - 17.00

Mi: 18.03.2026 09.00 - 17.00

DO: 19.03.2026 09.00 - 17.00

T 6.707 Module component: Software Architecture and Quality [T-INFO-114261]

Coordinators: Prof. Dr. Ralf Reussner
Organisation: KIT Department of Informatics
Part of: M-WIWI-104909 - Informatics (Department of Informatics)

Type	Credits	Grading	Term offered	Version
Written examination	3 CP	graded	Each winter term	3

Courses					
WT 25/26	2424667	Software Architecture and Quality	2 SWS	Lecture / 🗣	Reussner
WT 26/27	2424667	Software Architecture and Quality	2 SWS	Lecture / 🗣	Reussner

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment is carried out as a written examination (§ 4 Abs. 2 No. 1 SPO) lasting 60 minutes.

Prerequisites
This lecture and the lectures *Component-Based Software Development* and *Software Architecture* are mutually exclusive.

T 6.708 Module component: Software Engineering [T-WIWI-100809]

Coordinators: Prof. Dr. Andreas Oberweis
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104901 - Informatics (KIT-Department of Economics and Management)

Type	Credits	Grading	Term offered	Version
Written examination	4 CP	graded	Each summer term	3

Assessment
The assessment consists of an 1h written exam in the first week after lecture period.

Prerequisites
None

T 6.709 Module component: Software Quality Management [T-WIWI-102895]

Coordinators: Prof. Dr. Andreas Oberweis
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101472 - Informatics
M-WIWI-101477 - Development of Business Information Systems
M-WIWI-101628 - Emphasis in Informatics
M-WIWI-104901 - Informatics (KIT-Department of Economics and Management)

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each summer term	3

Courses					
ST 2026	2511208	Software Quality Management	2 SWS	Lecture / 🗣	Oberweis
ST 2026	2511209	Übungen zu Software-Qualitätsmanagement	1 SWS	Practice / 🗣	Kruse

Legend: 🖥 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment of this course is a written examination (60 min) according to §4(2), 1 of the examination regulation in the first week after lecture period.

Prerequisites
None

Below you will find excerpts from courses related to this module component:

V Software Quality Management

2511208, SS 2026, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content
This lecture imparts fundamentals of active software quality management (quality planning, quality testing, quality control, quality assurance) and illustrates them with concrete examples, as currently applied in industrial software development. Keywords of the lecture content are: software and software quality, process models, software process quality, ISO 9000-3, CMM(I), BOOTSTRAP, SPICE, software tests.

Learning objectives:

Students

- explain the relevant quality models,
- apply methods to evaluate the software quality and evaluate the results,
- know the mail models of sofware certification, compare and evaluate these models,
- write scientific theses in the area of software quality management and find own solutions for given problems.

Recommendations:

Programming knowledge in Java and basic knowledge of computer science are expected.

Workload:

- Lecture 30h
- Exercise 15h
- Preparation of lecture 24h
- Preparation of exercises 25h
- Exam preparation 40h
- Exam 1h

Literature

- Helmut Balzert: Lehrbuch der Software-Technik. Spektrum-Verlag 2008
- Peter Liggesmeyer: Software-Qualität, Testen, Analysieren und Verifizieren von Software. Spektrum Akademischer Verlag 2002
- Mauro Pezzè, Michal Young: Software testen und analysieren. Oldenbourg Verlag 2009

Weitere Literatur wird in der Vorlesung bekanntgegeben.

T 6.710 Module component: Solar Thermal Energy Systems [T-MACH-106493]

Coordinators: apl. Prof. Dr. Ron Dagan
Organisation: KIT Department of Mechanical Engineering

Part of: M-WIWI-104907 - Engineering Sciences
M-WIWI-107718 - Mechanical Engineering I
M-WIWI-107730 - Mechanical Engineering II

Type	Credits	Grading	Term offered	Version
Oral examination	4,5 CP	graded	Each winter term	6

Courses					
WT 25/26	2189400	Solar Thermal Energy Systems	2 SWS	Lecture / 🗣	Dagan
WT 26/27	2189400	Solar Thermal Energy Systems	2 SWS	Lecture / 🗣	Dagan

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
oral exam of about 30 minutes

Prerequisites
none

Recommendations
Literature

1. "Solar Engineering of Thermal Processes", 4th Edition, J. Duffie &W. Beckman. Published by Wiley & Sons
2. “Heat Transfer”, 10th Edition, J. P. Holman Mc. Graw Hill publisher
3. “Fundamentals of classical Thermodynamics”, G. Van Wylen & R. E. Sonntag. Published by Wiley &Sons

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Solar Thermal Energy Systems

2189400, WS 25/26, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content

The course deals with fundamental aspects of solar energy

1. Introduction to solar energy – global energy panorama

2. Solar energy resource-

Structure of the sun, Black body radiation, solar constant, solar spectral distribution

Sun-Earth geometrical relationship

3. Passive and active solar thermal applications.

4. Solar thermal systems- solar collector-types, concentrating collectors, solar towers,

Heat losses, efficiency

5. Selected topics on thermodynamics and heat transfer which are relevant for solar systems.

6. Introduction to Solar induced systems: Wind , Heat pumps, Biomass , Photovoltaic

7. Energy storage

The course deals with fundamental aspects of solar energy. Starting from a global energy panorama the course deals with the sun as a thermal energy source. In this context, basic issues such as the sun's structure, blackbody radiation and solar–earth geometrical relationship are discussed. In the next part, the lectures cover passive and active thermal applications and review various solar collector types including concentrating collectors and solar towers and the concept of solar tracking. Further, the collector design parameters determination is elaborated, leading to improved efficiency. This topic is augmented by a review of the main laws of thermodynamics and relevant heat transfer mechanisms.

The course ends with an overview on energy storage concepts which enhance practically the benefits of solar thermal energy systems.

The students get familiar with the global energy demand and the role of renewable energies learn about improved designs for using efficiently the potential of solar energy gain basic understanding of the main thermal hydraulic phenomena which support the work on future innovative applications will be able to evaluate quantitatively various aspects of the thermal solar systems.

Total 120 h, hereof 30 h contact hours and 90 h homework and self-studies

oral exam about 30 min.

Organizational issues

Die Vorlesung "Thermische Solarenergie" findet ab dem WS 2024/25 nicht mehr statt. Sie wurde zusammengelegt mit der engl. Version "Solar Thermal Energy Systems"

Literature

- "Solar Engineering of Thermal Processes "4th Edition, J. Duffie &W. Beckman. Published by Wiley & Sons.
- "Heat Transfer", 10th Edition, P. Holman Mc. Graw Hill publisher.
- "Fundamentals of classical Thermodynamics", G. Van Wylen & R. E. Sonntag. Published by Wiley & Sons

V **Solar Thermal Energy Systems**
2189400, WS 26/27, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content

The course deals with fundamental aspects of solar energy

1. Introduction to solar energy – global energy panorama

2. Solar energy resource-

Structure of the sun, Black body radiation, solar constant, solar spectral distribution

Sun-Earth geometrical relationship

3. Passive and active solar thermal applications.

4. Solar thermal systems- solar collector-types, concentrating collectors, solar towers,

Heat losses, efficiency

5. Selected topics on thermodynamics and heat transfer which are relevant for solar systems.

6. Introduction to Solar induced systems: Wind , Heat pumps, Biomass , Photovoltaic

7. Energy storage

The course deals with fundamental aspects of solar energy. Starting from a global energy panorama the course deals with the sun as a thermal energy source. In this context, basic issues such as the sun’s structure, blackbody radiation and solar–earth geometrical relationship are discussed. In the next part, the lectures cover passive and active thermal applications and review various solar collector types including concentrating collectors and solar towers and the concept of solar tracking. Further, the collector design parameters determination is elaborated, leading to improved efficiency. This topic is augmented by a review of the main laws of thermodynamics and relevant heat transfer mechanisms.

The course ends with an overview on energy storage concepts which enhance practically the benefits of solar thermal energy systems.

The students get familiar with the global energy demand and the role of renewable energies learn about improved designs for using efficiently the potential of solar energy gain basic understanding of the main thermal hydraulic phenomena which support the work on future innovative applications will be able to evaluate quantitatively various aspects of the thermal solar systems.

Total 120 h, hereof 30 h contact hours and 90 h homework and self-studies

oral exam about 30 min.

Organizational issues

Die Vorlesung "Thermische Solarenergie" findet ab dem WS 2024/25 nicht mehr statt. Sie wurde zusammengelegt mit der engl. Version "Solar Thermal Energy Systems"

Literature

- “Solar Engineering of Thermal Processes “4th Edition, J. Duffie &W. Beckman. Published by Wiley & Sons.
- “Heat Transfer”, 10th Edition, P. Holman Mc. Graw Hill publisher.
- “Fundamentals of classical Thermodynamics”, G. Van Wylen & R. E. Sonntag. Published by Wiley & Sons

T 6.711 Module component: Spatial Economics [T-WIWI-103107]

Coordinators: Prof. Dr. Ingrid Ott
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101485 - Transport Infrastructure Policy and Regional Development
M-WIWI-104908 - Economics
M-WIWI-107010 - Economics in a Connected World

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each winter term	1

Courses					
WT 25/26	2561260	Spatial Economics	2 SWS	Lecture / 🗣	Ott
WT 25/26	2561261	Exercise for Spatial Economics	1 SWS	Practice / 🗣	Ott, Mirzoyan
WT 26/27	2561260	Spatial Economics	2 SWS	Lecture / 🗣	Ott
WT 26/27	2561261	Exercise for Spatial Economics	1 SWS	Practice / 🗣	Ott, Mirzoyan

Legend: 🖥 Online, ⚄ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Depending on further pandemic developments, the examination will be offered either as an open-book examination, or as a 60-minute written examination.

Prerequisites
None

Recommendations
Basic micro- and macroeconomic knowledge is required, such as that taught in the courses "Economics I" [2600012] and "Economics II" [2600014], attendance of which is strongly recommended (but not mandatory). An interest in quantitative-mathematical modeling is also a prerequisite. Attendance of the course "Introduction to Economic Policy" [2560280] is recommended.

Below you will find excerpts from courses related to this module component:

V Spatial Economics

2561260, WS 25/26, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content
The course covers the following topics:

- Geography, trade and development
- Geography and economic theory
- Core models of economic geography and empirical evidence
- Agglomeration, home market effect, and spatial wages
- Applications and extensions

Learning objectives:

The student

- analyses how spatial distribution of economic activity is determined.
- uses quantitative methods within the context of economic models.
- has basic knowledge of formal-analytic methods.
- understands the link between economic theory and its empirical applications.
- understands to what extent concentration processes result from agglomeration and dispersion forces.
- is able to determine theory based policy recommendations.

Recommendations:

Basic knowledge of micro- and macroeconomics is assumed, as taught in the courses Economics I [2600012], and Economics II [2600014]. An interest in mathematical modeling is advantageous.

Workload:

The total workload for this course is approximately 135 hours.

- Classes: ca. 30 h
- Self-study: ca. 45 h
- Exam and exam preparation: ca. 60 h

Assessment:

The assessment consists of a written exam (60 minutes) (following §4(2), 1 of the examination regulation).

Literature
Steven Brakman, Harry Garretsen, Charles van Marrewijk (2009): The New Introduction to Geographical Economics, 2nd ed, Cambridge University Press.

Weitere Literatur wird in der Vorlesung bekanntgegeben.
(Further literature will be announced in the lecture.)

V	**Spatial Economics** 2561260, WS 26/27, 2 SWS, Language: English, Open in study portal	**Lecture (V)** **On-Site**

Content
The course covers the following topics:

- Geography, trade and development
- Geography and economic theory
- Core models of economic geography and empirical evidence
- Agglomeration, home market effect, and spatial wages
- Applications and extensions

Learning objectives:

The student

- analyses how spatial distribution of economic activity is determined.
- uses quantitative methods within the context of economic models.
- has basic knowledge of formal-analytic methods.
- understands the link between economic theory and its empirical applications.
- understands to what extent concentration processes result from agglomeration and dispersion forces.
- is able to determine theory based policy recommendations.

Recommendations:

Basic knowledge of micro- and macroeconomics is assumed, as taught in the courses Economics I [2600012], and Economics II [2600014]. An interest in mathematical modeling is advantageous.

Workload:

The total workload for this course is approximately 135 hours.

- Classes: ca. 30 h
- Self-study: ca. 45 h
- Exam and exam preparation: ca. 60 h

Assessment:

The assessment consists of a written exam (60 minutes) (following §4(2), 1 of the examination regulation).

Literature
Steven Brakman, Harry Garretsen, Charles van Marrewijk (2009): The New Introduction to Geographical Economics, 2nd ed, Cambridge University Press.

Weitere Literatur wird in der Vorlesung bekanntgegeben.
(Further literature will be announced in the lecture.)

T 6.712 Module component: Special Topics in Highway Engineering and Environmental Impact Assessment [T-BGU-101860]

Coordinators: Dr.-Ing. Matthias Zimmermann
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Expansion	Version
Oral examination	3 CP	graded	Each term	1 semesters	1

Assessment
oral exam with 15 minutes

Prerequisites
None

Recommendations
None

Additional Information
will not be offered anymore as from summer term 2026

Workload
90 hours

T 6.713 Module component: Special Topics in Information Systems [T-WIWI-113723]

Coordinators: Prof. Dr. Christof Weinhardt
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Examination of another type	4,5 CP	graded	Each term	1

Assessment
The assessment of this course is in form of a written documentation, a presentation of the outcome of the conducted practical components and an active participation in class.
Please take into account that, beside the written documentation, also a practical component (such as a survey or an implementation of an application) is part of the course. Please examine the course description for the particular tasks.

The overall grade is composed as follows:

A total of 60 points can be achieved, of which

- A maximum of 30 points for the written documentation
- A maximum of 30 points for the practical component

In order to pass the success control, at least 15 points (written documentation / practical component) must be achieved.

Prerequisites
see below

Recommendations
None

Additional Information
All the practical seminars offered at the chair of Prof. Dr. Weinhardt can be chosen in the Special Topics in Information Systems course. The current topics of the practical seminars are available at the following homepage: www.iism.kit.edu/im/lehre.

The Special Topics Information Systems is equivalent to the practical seminar, as it was only offered for the major in “Information Systems” so far. With this course students majoring in “Industrial Engineering and Management“ and “Economics Engineering” also have the chance of getting practical experience and enhance their scientific capabilities.

The Special Topics Information Systems can be chosen instead of a regular lecture (see module description). Please take into account, that this course can only be accounted once per module.

T 6.714 Module component: Special Topics in Information Systems [T-WIWI-113727]

Coordinators: Prof. Dr. Christof Weinhardt
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101411 - Information Engineering

Type	Credits	Grading	Term offered	Version
Examination of another type	4,5 CP	graded	Each term	1

Assessment
The assessment of this course is in form of a written documentation, a presentation of the outcome of the conducted practical components and an active participation in class.
Please take into account that, beside the written documentation, also a practical component (such as a survey or an implementation of an application) is part of the course. Please examine the course description for the particular tasks.

The overall grade is composed as follows:

A total of 60 points can be achieved, of which

- A maximum of 30 points for the written documentation
- A maximum of 30 points for the practical component

In order to pass the success control, at least 15 points (written documentation / practical component) must be achieved.

Prerequisites
see below

Recommendations
None

Additional Information
All the practical seminars offered at the chair of Prof. Dr. Weinhardt can be chosen in the Special Topics in Information Systems course. The current topics of the practical seminars are available at the following homepage: www.iism.kit.edu/im/lehre.

The Special Topics Information Systems is equivalent to the practical seminar, as it was only offered for the major in “Information Systems” so far. With this course students majoring in “Industrial Engineering and Management“ and “Economics Engineering” also have the chance of getting practical experience and enhance their scientific capabilities.

The Special Topics Information Systems can be chosen instead of a regular lecture (see module description). Please take into account, that this course can only be accounted once per module.

T 6.715 Module component: Special Topics in Information Systems [T-WIWI-113725]

Coordinators: Prof. Dr. Christof Weinhardt
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101506 - Service Analytics
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Examination of another type	4,5 CP	graded	Each term	1

Assessment
The assessment of this course is in form of a written documentation, a presentation of the outcome of the conducted practical components and an active participation in class.
Please take into account that, beside the written documentation, also a practical component (such as a survey or an implementation of an application) is part of the course. Please examine the course description for the particular tasks.

The overall grade is composed as follows:

A total of 60 points can be achieved, of which

- A maximum of 30 points for the written documentation
- A maximum of 30 points for the practical component

In order to pass the success control, at least 15 points (written documentation / practical component) must be achieved.

Prerequisites
see below

Recommendations
None

Additional Information
All the practical seminars offered at the chair of Prof. Dr. Weinhardt can be chosen in the Special Topics in Information Systems course. The current topics of the practical seminars are available at the following homepage: www.iism.kit.edu/im/lehre.

The Special Topics Information Systems is equivalent to the practical seminar, as it was only offered for the major in “Information Systems” so far. With this course students majoring in “Industrial Engineering and Management“ and “Economics Engineering” also have the chance of getting practical experience and enhance their scientific capabilities.

The Special Topics Information Systems can be chosen instead of a regular lecture (see module description). Please take into account, that this course can only be accounted once per module.

T 6.716 Module component: Special Topics in Information Systems [T-WIWI-113726]

Coordinators: Prof. Dr. Christof Weinhardt
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-103720 - eEnergy: Markets, Services and Systems
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Examination of another type	4,5 CP	graded	Each term	1

Assessment
The assessment of this course is in form of a written documentation, a presentation of the outcome of the conducted practical components and an active participation in class.
Please take into account that, beside the written documentation, also a practical component (such as a survey or an implementation of an application) is part of the course. Please examine the course description for the particular tasks.

The overall grade is composed as follows:

A total of 60 points can be achieved, of which

- A maximum of 30 points for the written documentation
- A maximum of 30 points for the practical component

In order to pass the success control, at least 15 points (written documentation / practical component) must be achieved.

Prerequisites
see below

Recommendations
None

Additional Information
All the practical seminars offered at the chair of Prof. Dr. Weinhardt can be chosen in the Special Topics in Information Systems course. The current topics of the practical seminars are available at the following homepage: www.iism.kit.edu/im/lehre.

The Special Topics Information Systems is equivalent to the practical seminar, as it was only offered for the major in "Information Systems" so far. With this course students majoring in "Industrial Engineering and Management" and "Economics Engineering" also have the chance of getting practical experience and enhance their scientific capabilities.

The Special Topics Information Systems can be chosen instead of a regular lecture (see module description). Please take into account, that this course can only be accounted once per module.

T 6.717 Module component: Special Topics in Information Systems [T-WIWI-109940]

Coordinators: Prof. Dr. Christof Weinhardt
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Examination of another type	4,5 CP	graded	Each term	2

Assessment
The assessment of this course is in form of a written documentation, a presentation of the outcome of the conducted practical components and an active participation in class.
Please take into account that, beside the written documentation, also a practical component (such as a survey or an implementation of an application) is part of the course. Please examine the course description for the particular tasks.

The overall grade is composed as follows:

A total of 60 points can be achieved, of which

- A maximum of 30 points for the written documentation
- A maximum of 30 points for the practical component

In order to pass the success control, at least 15 points (written documentation / practical component) must be achieved.

Prerequisites
see below

Recommendations
None

Additional Information
All the practical seminars offered at the chair of Prof. Dr. Weinhardt can be chosen in the Special Topics in Information Systems course. The current topics of the practical seminars are available at the following homepage: www.iism.kit.edu/im/lehre.

The Special Topics Information Systems is equivalent to the practical seminar, as it was only offered for the major in “Information Systems” so far. With this course students majoring in “Industrial Engineering and Management“ and “Economics Engineering” also have the chance of getting practical experience and enhance their scientific capabilities.

The Special Topics Information Systems can be chosen instead of a regular lecture (see module description). Please take into account, that this course can only be accounted once per module.

6.718 Module component: Special Topics in Information Systems [T-WIWI-113724]

Coordinators: Prof. Dr. Christof Weinhardt
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101410 - Business & Service Engineering
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Examination of another type	4,5 CP	graded	Each term	1

Assessment
The assessment of this course is in form of a written documentation, a presentation of the outcome of the conducted practical components and an active participation in class.
Please take into account that, beside the written documentation, also a practical component (such as a survey or an implementation of an application) is part of the course. Please examine the course description for the particular tasks.

The overall grade is composed as follows:

A total of 60 points can be achieved, of which

- A maximum of 30 points for the written documentation
- A maximum of 30 points for the practical component

In order to pass the success control, at least 15 points (written documentation / practical component) must be achieved.

Prerequisites
see below

Recommendations
None

Additional Information
All the practical seminars offered at the chair of Prof. Dr. Weinhardt can be chosen in the Special Topics in Information Systems course. The current topics of the practical seminars are available at the following homepage: www.iism.kit.edu/im/lehre.

The Special Topics Information Systems is equivalent to the practical seminar, as it was only offered for the major in “Information Systems” so far. With this course students majoring in “Industrial Engineering and Management“ and “Economics Engineering” also have the chance of getting practical experience and enhance their scientific capabilities.

The Special Topics Information Systems can be chosen instead of a regular lecture (see module description). Please take into account, that this course can only be accounted once per module.

T 6.719 Module component: Startup Experience [T-WIWI-111561]

Coordinators: Prof. Dr. Orestis Terzidis
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101488 - Entrepreneurship (EnTechnon)
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Examination of another type	6 CP	graded	Each term	1

Courses					
WT 25/26	2545004	Startup Experience	4 SWS	Seminar / 🗣	Weimar, Martjan, Terzidis
ST 2026	2545004	Startup Experience	4 SWS	Seminar / 🗣	Weimar, Terzidis, Martjan, Osaro

Legend: 🖥 Online, ✲ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Alternative exam assessment. Details on the design of the examination performance of other types will be announced in the course.The grade is composed of a presentation and a written paper (plus any specified documentation, e.g. work results, experience diary, reflection).

Recommendations
Lecture Entrepreneurship already completed

Additional Information
The language in the seminar is English. The seminar contents will be published on the chair homepage.

Workload
180 hours

Below you will find excerpts from courses related to this module component:

V Startup Experience
2545004, WS 25/26, 4 SWS, Language: English, Open in study portal

Seminar (S)
On-Site

Content
Content

In the Startup Experience seminar you will develop entrepreneurial competences that will enable you to build a new business. In an entrepreneurial project, you have three main objectives:

1. Identify and develop an opportunity. Who is your target customer and what problem or task does he or she have? How attractive and how big is this market?
2. How will you add value to it? How can you use specific resources, including technology, to develop a solution?
3. How can you design and set up a viable organisation? What business model do you propose to create, deliver and capture value?

Our primary focus is on digital healthcare ventures, granting you the opportunity to delve into the realm of entrepreneurship within the healthcare system. After gaining a deep understanding of healthcare needs, you will utilize creativity techniques to uncover potential business ideas that provide value for patients and doctors. Additionally, you will learn how to create viable business models, dive into health regulations, and pitch your idea to a jury.

Learning Objectives

After completing this course, the course participants will be able to:

- Work effectively in a cohesive team
- Understand the role of digital entrepreneurship in healthcare
- Apply creativity techniques to ideate
- Use utility analysis approaches to select promising solutions
- Develop a value proposition based on techniques like the value proposition canvas or the jobs-to-be-done method
- Apply advanced business modeling methods to develop a sound business concept
- Develop and deliver a concise presentation ("pitch") to communicate your project
- Gain basic knowledge of healthcare regulations and reimbursement ways

Additional information:

Alternative exam assessment. The grade consists of the presentation and the written elaboration. Potentially, a 'project diary' of the seminar progress may be part of the deliverables (depends on tutor and will be communicated at the kick-off).

For a successful course completion, we expect you to submit a Business Plan with the following features:

- Scope: 9000 words,
- Sound and clear structure,
- Expression and spelling are correct
- Complete and correct references, quotations, etc.
- Visual elements are chosen appropriately
- Documentation and traceability of data acquisition, analysis and evaluation,
- Content is developed according to the course instructions.

Furthermore, we expect you to deliver a team Pitch.

- Duration: will be communicated (typically 5-10 minutes)
- Content: Introduction/Purpose; Problem; Solution; Business Model; Prototype; Competition; Management Team; Current Status and next steps,
- Layout and form: appropriate choice,
- Appearance: appropriate amount of visual elements,
- Data: well researched and organized visually
- Story Line: is sound; clear and convincing.

Organizational issues
Registration is via the Wiwi portal.

In the seminar you will work on a project in teams of max. 5 persons. The groups are formed in the seminar.

V	**Startup Experience** 2545004, SS 2026, 4 SWS, Language: English, Open in study portal	**Seminar (S)** **On-Site**

Content
Content

In the Startup Experience seminar you will develop entrepreneurial competences that will enable you to build a new business. In an entrepreneurial project, you have three main objectives:

1. Identify and develop an opportunity. Who is your target customer and what problem or task does he or she have? How attractive and how big is this market?
2. How will you add value to it? How can you use specific resources, including technology, to develop a solution?
3. How can you design and set up a viable organisation? What business model do you propose to create, deliver and capture value?

Our primary focus is on digital healthcare ventures, granting you the opportunity to delve into the realm of entrepreneurship within the healthcare system. After gaining a deep understanding of healthcare needs, you will utilize creativity techniques to uncover potential business ideas that provide value for patients and doctors. Additionally, you will learn how to create viable business models, dive into health regulations, and pitch your idea to a jury.

Learning Objectives

After completing this course, the course participants will be able to:

- Work effectively in a cohesive team
- Understand the role of digital entrepreneurship in healthcare
- Apply creativity techniques to ideate
- Use utility analysis approaches to select promising solutions
- Develop a value proposition based on techniques like the value proposition canvas or the jobs-to-be-done method
- Apply advanced business modeling methods to develop a sound business concept
- Develop and deliver a concise presentation (“pitch”) to communicate your project
- Gain basic knowledge of healthcare regulations and reimbursement ways

Additional information:

Alternative exam assessment. The grade consists of the presentation and the written elaboration. Potentially, a ‘project diary’ of the seminar progress may be part of the deliverables (depends on tutor and will be communicated at the kick-off).

For a successful course completion, we expect you to submit a Business Plan with the following features:

- Scope: 9000 words,
- Sound and clear structure,
- Expression and spelling are correct
- Complete and correct references, quotations, etc.
- Visual elements are chosen appropriately
- Documentation and traceability of data acquisition, analysis and evaluation,
- Content is developed according to the course instructions.

Furthermore, we expect you to deliver a team Pitch.

- Duration: will be communicated (typically 5-10 minutes)
- Content: Introduction/Purpose; Problem; Solution; Business Model; Prototype; Competition; Management Team; Current Status and next steps,
- Layout and form: appropriate choice,
- Appearance: appropriate amount of visual elements,
- Data: well researched and organized visually
- Story Line: is sound; clear and convincing.

Organizational issues
Registration is via the Wiwi-Portal.

Attention: The Startup X seminar overlaps in some instances with the entrepreneurship lecture. Please be aware of this before applying for the seminar.

In the seminar you will work on a project in teams of max. 5 persons. Team applications are welcome but not a prerequisite for participation. The seminars will be held in English.

T 6.720 Module component: Statistical Modeling of Generalized Regression Models [T-WIWI-103065]

Coordinators: apl. Prof. Dr. Wolf-Dieter Heller
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101638 - Econometrics and Statistics I
M-WIWI-101639 - Econometrics and Statistics II
M-WIWI-104902 - Statistics

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each winter term	1

Courses					
WT 25/26	2521350	Statistical Modeling of Generalized Regression Models	2 SWS	Lecture	Heller
WT 26/27	2521350	Statistical Modeling of Generalized Regression Models	2 SWS	Lecture	Heller

Assessment
The assessment of this course is a written examination (60 min) according to §4(2), 1 of the examination regulation.

Prerequisites
None

Recommendations
Knowledge of the contents covered by the course "Economics III: Introduction in Econometrics" [2520016]

Below you will find excerpts from courses related to this module component:

V Statistical Modeling of Generalized Regression Models
2521350, WS 25/26, 2 SWS, Open in study portal

Lecture (V)

Content
Learning objectives:

The student has profound knowledge of generalized regression models.

Requirements:

Knowledge of the contents covered by the course *Economics III: Introduction in Econometrics" [2520016]*.

Workload:

Total workload for 4.5 CP: approx. 135 hours

Attendance: 30 hours

Preparation and follow-up: 65 hours

V Statistical Modeling of Generalized Regression Models
2521350, WS 26/27, 2 SWS, Open in study portal

Lecture (V)

Content
Learning objectives:

The student has profound knowledge of generalized regression models.

Requirements:

Knowledge of the contents covered by the course *Economics III: Introduction in Econometrics" [2520016]*.

Workload:

Total workload for 4.5 CP: approx. 135 hours

Attendance: 30 hours

Preparation and follow-up: 65 hours

T 6.721 Module component: Statistics I [T-WIWI-102737]

Coordinators: Prof. Dr. Oliver Grothe
Prof. Dr. Melanie Schienle
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104902 - Statistics

Type	Credits	Grading	Term offered	Version
Written examination	5 CP	graded	Each summer term	1

Courses					
ST 2026	2600008	Statistics I	4 SWS	Lecture / 🗣	Grothe
ST 2026	2600009	Tutorien zu Statistik I	2 SWS	Tutorial	Grothe, Becker

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success is assessed in the form of a written examination (120 minutes). The examination is offered towards the end of the lecture period or at the beginning of the lecture-free period. The repeat examination is offered in the following semester.

Bonus: It is planned that, from the summer semester 2025, a grade bonus for the Statistics I exam can be earned through successful participation in the tutorials. If the grade of the written exam is between 4.0 and 1.3, the bonus will generally improve the grade by one grade level (0.3 or 0.4). The exact criteria for awarding a bonus will be announced at the beginning of the lecture.

Prerequisites
None

Below you will find excerpts from courses related to this module component:

V Statistics I

2600008, SS 2026, 4 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content

Learning objectives:

Students understand and apply

- basic concepts of statistical data exploration as well as
- basic definitions and theorems of probability theory.

Content:

A. Descriptive Statistics: univariate und bivariate analysis

B. Probability Theory: probability space, conditional and product probabilities

C. Random variables: location and shape parameters, dependency measures, concrete distribution models

Workload:

Total workload for 5 CP: approx. 150 hours

Attendance: 60 hours

Preparation and follow-up: 90 hours

Literature
Skript: Kurzfassung Statistik I. Dieses enthält ausführliche Angaben zu weiterführender Literatur.

T 6.722 Module component: Statistics II [T-WIWI-102738]

Coordinators: Prof. Dr. Oliver Grothe
Prof. Dr. Melanie Schienle
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104902 - Statistics

Type	Credits	Grading	Term offered	Version
Written examination	5 CP	graded	Each winter term	1

Courses					
WT 25/26	2610020	Statistics II	4 SWS	Lecture / 🗣	Krüger
WT 25/26	2610021		2 SWS	Tutorial	Krüger, Becker, Koster
WT 25/26	2610022	PC-Praktikum zu Statistik II	2 SWS		Krüger, Koster

Legend: 🖥 Online, ✲ Blended (On-Site/Online), 🗣 On-Site, × Cancelled

Assessment
Success is assessed in the form of a written examination (120 minutes). The examination is offered towards the end of the lecture period or at the beginning of the lecture-free period. The repeat examination is offered in the following semester.
Bonus: It is planned that from the winter semester 2025/2026, a grade bonus for the Statistics II exam can be earned through successful participation in the tutorials. If the grade of the written examination is between 4.0 and 1.3, the bonus will generally improve the grade by one grade level (0.3 or 0.4). The exact criteria for awarding a bonus will be announced at the beginning of the lecture.

Prerequisites
None

Recommendations
It ist recommended to attend the course *Statistics I* [2600008] before the course *Statistics II* [2610020].

Below you will find excerpts from courses related to this module component:

V Statistics II

2610020, WS 25/26, 4 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content
Learning objectives:

The student

- understands and applies the basic definitions and theorems of probability theory,
- transfers these theoretical foundations to problems in parametric mathematical statistics.

Content:

D. Sampling and estimation theory: Sampling distributions, estimators, point and interval estimation
E. Test theory: General principles of hypothesis testing, specific 1- and 2-sample tests
F. Regression analysis: Simple and multiple linear regression, statistical inference

Requirements:

It is recommended to attend the course *Statistics I* [2600008] before the course *Statistics II* [2600020].

Workload:

Total workload: 150 hours (5.0 Credits).

Attendance: 60 hours

Preparation and follow-up: 90 hours

Literature
Skript: Kurzfassung Statistik II. Dieses enthält ausführliche Angaben zu weiterführender Literatur.

T 6.723 Module component: Stochastic Calculus and Finance [T-WIWI-103129]

Coordinators: Dr. Mher Safarian
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101639 - Econometrics and Statistics II
M-WIWI-104902 - Statistics

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each winter term	1

Courses					
WT 25/26	2521331	Stochastic Calculus and Finance	2 SWS	Lecture	Safarian
WT 25/26	2521332	Übungen zu Stochastic Calculus and Finance	2 SWS	Practice	Safarian
WT 26/27	2521331	Stochastic Calculus and Finance	2 SWS	Lecture	Safarian
WT 26/27	2521332	Übungen zu Stochastic Calculus and Finance	2 SWS	Practice	Safarian

Assessment
The assessment of this course consists of a written examination (§4(2), 1 SPOs, 180 min.).

Prerequisites
None

Additional Information
For more information see http://statistik.econ.kit.edu/

Below you will find excerpts from courses related to this module component:

V Stochastic Calculus and Finance

2521331, WS 25/26, 2 SWS, Language: English, Open in study portal

Lecture (V)

Content
Learning objectives:

After successful completion of the course students will be familiar with many common methods of pricing and portfolio models in finance. Emphasis we be put on both finance and the theory behind it.

Content:

The course will provide rigorous yet focused training in stochastic calculus and mathematical finance. Topics to be covered:

1. Stochastic Calculus: Stochastic Processes, Brownian Motion and Martingales, Entropy, Stopping Times, Local martingales, Doob-Meyer Decomposition, Quadratic Variation, Stochastic Integration, Ito Formula, Girsanov Theorem, Jump-diffusion Processes, Stable and Levy processes.
2. Mathematical Finance: Pricing Models, The Black-Scholes Model, State prices and Equivalent Martingale Measure, Complete Markets and Redundant Security Prices, Arbitrage Pricing with Dividends, Term-Structure Models (One Factor Models, Cox-Ingersoll-Ross Model, Affine Models), Term-Structure Derivatives and Hedging, Mortgage-Backed Securities, Derivative Assets (Forward Prices, Future Contracts, American Options, Look-back Options), Incomplete Markets, Markets with Transaction Costs, Optimal Portfolio and Consumption Choice (Stochastic Control and Merton continuous time optimization problem, CAPM), Equilibrium models, Numerical Methods.

Workload:

Total workload for 4.5 CP: approx. 135 hours
Attendance: 30 hours
Preparation and follow-up: 65 hours

Organizational issues
Blockveranstaltung, Termine werden über Ilias bekannt gegeben

Literature

- Dynamic Asset Pricing Theory, Third Edition by D. Duffie, Princeton University Press, 1996
- Stochastic Calculus for Finance II: Continuous-Time Models by S. E. Shreve, Springer, 2003
- Stochastic Finance: An Introduction in Discrete Time by H. Föllmer, A. Schied, de Gruyter, 2011
- Methods of Mathematical Finance by I. Karatzas, S. E. Shreve, Springer, 1998
- Markets with Transaction Costs by Yu. Kabanov, M. Safarian, Springer, 2010
- Introduction to Stochastic Calculus Applied to Finance by D.Lamberton, B. Lapeyre, Chapman&Hall,1996

V **Stochastic Calculus and Finance** **Lecture (V)**
2521331, WS 26/27, 2 SWS, Language: English, Open in study portal

Content
Learning objectives:

After successful completion of the course students will be familiar with many common methods of pricing and portfolio models in finance. Emphasis we be put on both finance and the theory behind it.

Content:

The course will provide rigorous yet focused training in stochastic calculus and mathematical finance. Topics to be covered:

1. Stochastic Calculus: Stochastic Processes, Brownian Motion and Martingales, Entropy, Stopping Times, Local martingales, Doob-Meyer Decomposition, Quadratic Variation, Stochastic Integration, Ito Formula, Girsanov Theorem, Jump-diffusion Processes, Stable and Levy processes.
2. Mathematical Finance: Pricing Models, The Black-Scholes Model, State prices and Equivalent Martingale Measure, Complete Markets and Redundant Security Prices, Arbitrage Pricing with Dividends, Term-Structure Models (One Factor Models, Cox-Ingersoll-Ross Model, Affine Models), Term-Structure Derivatives and Hedging, Mortgage-Backed Securities, Derivative Assets (Forward Prices, Future Contracts, American Options, Look-back Options), Incomplete Markets, Markets with Transaction Costs, Optimal Portfolio and Consumption Choice (Stochastic Control and Merton continuous time optimization problem, CAPM), Equilibrium models, Numerical Methods.

Workload:

Total workload for 4.5 CP: approx. 135 hours
Attendance: 30 hours
Preparation and follow-up: 65 hours

Organizational issues
Blockveranstaltung, Termine werden über Ilias bekannt gegeben

Literature

- Dynamic Asset Pricing Theory, Third Edition by D. Duffie, Princeton University Press, 1996
- Stochastic Calculus for Finance II: Continuous-Time Models by S. E. Shreve, Springer, 2003
- Stochastic Finance: An Introduction in Discrete Time by H. Föllmer, A. Schied, de Gruyter, 2011
- Methods of Mathematical Finance by I. Karatzas, S. E. Shreve, Springer, 1998
- Markets with Transaction Costs by Yu. Kabanov, M. Safarian, Springer, 2010
- Introduction to Stochastic Calculus Applied to Finance by D.Lamberton, B. Lapeyre, Chapman&Hall,1996

T 6.724 Module component: Strategic Decision-Making in Global Production Networks through Optimization and Simulation [T-MACH-114969]

Coordinators: Prof. Dr.-Ing. Gisela Lanza
Michael Martin

Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-105455 - Strategic Design of Modern Production Systems
M-MACH-106590 - Production Engineering
M-WIWI-104907 - Engineering Sciences
M-WIWI-107718 - Mechanical Engineering I
M-WIWI-107730 - Mechanical Engineering II

Type	Credits	Grading	Term offered	Version
Examination of another type	4,5 CP	graded	Each summer term	2

Courses					
ST 2026	2150659	Strategic Decision-Making in Global Production Networks through Optimization and Simulation	2 SWS	Lecture / 🗣	Lanza, Martin

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment of success takes the form of an examination with a different type of assessment in accordance with Section 4 (2) No. 3 of the SPO. The project work, two milestone-based presentations of the results in presentation form (each lasting approx. 20 minutes) and a final presentation (lasting approx. 30 minutes) are included in the assessment. The project work and the presentations are carried out in small groups.

Prerequisites
Depending on the degree program, one of the following partial achievements must be passed in order to register for an examination:
T-MACH-110991 - Global ProductionPartial achievement
T-MACH-105158 - Global Production and Logistics - Part 1: Global ProductionPartial
T-MACH-108848 - Global Production and Logistics - Part 1: Global Production
T-MACH-110337 - Global Production and LogisticsPartial
T-MACH-108849 - Integrated Production Planning in the Age of Industry 4.0
T-MACH-109054 - Integrated Production Planning in the Age of Industry 4.0
T-MACH-114031 - Global ProductionPartial service
T-MACH-113832 - Global ProductionPartial service
T-MACH-114800 - Global ProductionPartial service

The partial achievement T-MACH-113372 - Strategic Decision-Making in Global Production Network Design: A Seminar on Optimization and Simulation must not have been started for exam registration.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. You have to fulfill one of 7 conditions:
   1. The module component T-MACH-108849 - Integrated Production Planning in the Age of Industry 4.0 must have been passed.
   2. The module component T-MACH-109054 - Integrated Production Planning in the Age of Industry 4.0 must have been passed.
   3. The module component T-MACH-114031 - Global Production must have been passed.
   4. The module component T-MACH-114800 - Global Production must have been passed.
   5. The module component T-MACH-115031 - Global Production and Logistics must have been passed.
   6. The module component T-MACH-113565 - Global Logistics must have been passed.
   7. The module component T-MACH-115017 - Global Logistics must have been passed.
2. The following conditions have to be fulfilled:

Recommendations
Participation in the following lectures:
Introduction to Operations Research I [2550040] + II [2530043]

Additional Information
The course is offered in English.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V	**Strategic Decision-Making in Global Production Networks through Optimization and Simulation** 2150659, SS 2026, 2 SWS, Language: English, Open in study portal	**Lecture (V)** **On-Site**

Content
The lecture "Strategic Decision-Making in Global Production Networks through Optimization and Simulation" offers students a comprehensive insight into the application of quantitative models from operations research in global production networks. The course places special emphasis on practical applications and allows students to deepen their skills through a real-world use case during the semester.

The classroom sessions serve to convey important basics and to introduce and present the practice-relevant cases. In the self-study phase, the topics covered are worked on in greater depth. The curriculum covers various phases. Optimization techniques for network design are covered first, followed by simulation methods for network management. Subsequently, open questions are dealt with, e.g. from the consideration of uncertainty, sustainability aspects or the search for the overall optimum in the production network.

The students are divided into small groups to work together on the questions. The methods taught in the course are implemented in python. In order to strengthen the students' presentation skills, regular presentations of interim results are planned. The progress made is supported by feedback and interaction with an internationally operating consulting firm.

The practical orientation of the course, combined with the application of quantitative models and the use of Python, enables students to prepare holistically for complex challenges in global production.

Learning Outcomes:

The students are able to

1. **put concepts of global production into practice:**
   • Understand how global production networks can be implemented in real business scenarios.
   • Develop and implement strategies for adapting global production networks to specific business requirements.
2. **in-depth knowledge and use of optimization in global production:**
   • Develop an in-depth understanding of various optimization techniques in global production processes.
   • Apply optimization models to complex production networks and continuously improve them.
3. **approach to improving network configuration, site selection and transportation routes:**
   • Understand methods to evaluate and optimize production networks.
   • Effectively plan and improve site selection decisions and transportation routes.
4. **deepen knowledge and use of simulations in global production:**
   • Understand how simulations can be used as a tool to analyze and optimize global production processes.
   • Gain experience in the application of simulation techniques for modeling and analyzing production processes.
5. **approach to improving delivery reliability:**
   • Develop and implement strategies to improve delivery reliability.
   • Optimize processes that can affect delivery reliability.
6. **consider uncertainties, aspects of sustainability and multidimensionality:**
   • Recognize and manage uncertainties in global production environments.
   • Consider sustainability aspects and multidimensional challenges when making decisions in global production.
7. **linking results and models:**
   • Link models and analytical results to create holistic solutions to complex problems in global production.
   • Strengthen the ability to iteratively improve models based on real-world results.
8. **presentations to management:**
   • Present complex global manufacturing concepts to management in an understandable and persuasive manner.
   • Build confidence in the use of visual aids and effective communication techniques in front of management levels.

Workload:

regular attendance: ~ 30 hours
self-study: ~ 99 hours

Media:

E-learning plattform Ilias, Powerpoint, photo protocol.
The Media are provided through Ilias (https://ilias.studium.kit.edu/).

Organizational issues

Aus organisatorischen Gründen ist die Teilnehmerzahl für die Lehrveranstaltung auf 20 Studierende begrenzt. Termine und Fristen zur Veranstaltung werden über die Homepage des wbk (https://www.wbk.kit.edu/studium-und-lehre.php) bekannt gegeben.

For organizational reasons the number of students is limited to 20. Dates and deadlines for the seminar will be announced via the homepage of wbk (https://www.wbk.kit.edu/studium-und-lehre.php).

Literature

Vorlesungsskript der Lehrveranstaltungen / Lecture notes of the courses:
Abele et al. (2008): Global Production [978-3-540-71652-5]
Domschke et al. (2015): Einführung in das Operations Research [Einführung in Operations Research]
Friedli et al. (2021): Global Manufacturing Management: From Excellent Plants Toward Network Optimization [978-3-030-72739-0]

T 6.725 Module component: Strategic Management [T-WIWI-113090]

Coordinators: Prof. Dr. Hagen Lindstädt
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each summer term	2

Assessment
The assessment consists of a written exam (60 min) taking place at the beginn of the recess period (according to §4 (2), 1 of the examination regulation). The exam takes place in every semester. Re-examinations are offered at every ordinary examination date.

Prerequisites
None

Workload
135 hours

T 6.726 Module component: Strategic Transport Planning [T-BGU-103426]

Coordinators: Volker Waßmuth
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: M-BGU-101064 - Fundamentals of Transportation
M-BGU-101065 - Transportation Modelling and Traffic Management
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Expansion	Version
Written examination	3 CP	graded	Each term	1 semesters	2

Courses					
ST 2026	6232808	Strategic Traffic Planning	2 SWS	Lecture / 🗣	Waßmuth

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
written exam, 60 min.

Prerequisites
None

Recommendations
None

Additional Information
None

Workload
90 hours

T 6.727 Module component: Strategy and Management Theory: Developments and “Classics” [T-WIWI-106190]

Coordinators: Prof. Dr. Hagen Lindstädt
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-103119 - Advanced Topics in Strategy and Management
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Irregular	1

Courses					
WT 25/26	2577921	Strategy and Management Theory: Developments and "Classics" (Master)	2 SWS	Lecture / 🗣	Lindstädt
WT 26/27	2577921	Strategy and Management Theory: Developments and "Classics" (Master)	2 SWS	Lecture / 🗣	Lindstädt

Legend: 🖥 Online, Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The control of success according to § 4(2), 3 SPO takes place by writing a scientific work and a presentation of the results of the work in the context of a conclusion meeting. Details on the design of the performance review will be announced during the lecture.

Prerequisites
None

Recommendations
Basic knowledge as conveyed in the bachelor module „Strategy and Organization“ is recommended.

Additional Information
This course is admission restricted. If you were already admitted to another course in the module “Advanced Topics in Strategy and Management” the participation at this course will be guaranteed.

The course is planned to be held for the first time in the winter term 2017/18.

Workload
90 hours

Below you will find excerpts from courses related to this module component:

V Strategy and Management Theory: Developments and "Classics" (Master)
Lecture (V)
On-Site

2577921, WS 25/26, 2 SWS, Language: German, Open in study portal

Content

This course covers highly topical issues of great relevance to the management of organizations. Students will be enabled to take strategic management positions. By applying appropriate models from the fields of strategy and management - or models developed in-house - participants will learn to evaluate the strategic starting position of an organization and derive precise and well-founded recommendations for action based on this.

This course offers students the opportunity to explore current management issues and sharpen their skills in strategic analysis and evaluation. Through intensive collaboration and practical application of the knowledge learned, students are optimally prepared for the demands and challenges of modern business management.

Structure

The course begins with an overarching theme, based on which students are divided into groups of two. The core of the course consists of the preparation of a written paper as well as the presentation and discussion of the results.

Learning Objectives

Upon completion of the course, students will be able to,

- analyze complex business situations, think strategically and derive sound management decisions.
- compose clear and convincing written papers that accurately present the analyses and recommendations developed.
- present results in an engaging manner and actively participate in substantive discussions.

Recommendations:

Prior attendance of the Bachelor's module "Strategy and Organization" or another module with comparable content at another university is recommended.

Workload:

Total effort approx. 90 hours

Attendance time: 15 hours

Preparation and follow-up: 75 hours

Examination and preparation: not applicable

Success Monitoring:

The success control according to § 4(2), 3 SPO is done by writing a scientific paper and a presentation of the results of the paper in the context of a final event. Details on the design of the performance review will be announced during the lecture.

Annotation:

The course is admission restricted. In case of prior admission to another course in the module "Strategy and Management: Advanced Topics" [M-WIWI-103119], participation in this course is guaranteed. For more information on the application process, see the IBU website.

Exams are offered at least every other semester, so the entire module can be completed in two semesters.

V	**Strategy and Management Theory: Developments and "Classics" (Master)** 2577921, WS 26/27, 2 SWS, Language: German, Open in study portal	**Lecture (V)** **On-Site**

Content

This course covers highly topical issues of great relevance to the management of organizations. Students will be enabled to take strategic management positions. By applying appropriate models from the fields of strategy and management - or models developed in-house - participants will learn to evaluate the strategic starting position of an organization and derive precise and well-founded recommendations for action based on this.

This course offers students the opportunity to explore current management issues and sharpen their skills in strategic analysis and evaluation. Through intensive collaboration and practical application of the knowledge learned, students are optimally prepared for the demands and challenges of modern business management.

Structure

The course begins with an overarching theme, based on which students are divided into groups of two. The core of the course consists of the preparation of a written paper as well as the presentation and discussion of the results.

Learning Objectives

Upon completion of the course, students will be able to,

- analyze complex business situations, think strategically and derive sound management decisions.
- compose clear and convincing written papers that accurately present the analyses and recommendations developed.
- present results in an engaging manner and actively participate in substantive discussions.

Recommendations:

Prior attendance of the Bachelor's module "Strategy and Organization" or another module with comparable content at another university is recommended.

Workload:

Total effort approx. 90 hours

Attendance time: 15 hours

Preparation and follow-up: 75 hours

Examination and preparation: not applicable

Success Monitoring:

The success control according to § 4(2), 3 SPO is done by writing a scientific paper and a presentation of the results of the paper in the context of a final event. Details on the design of the performance review will be announced during the lecture.

Annotation:

The course is admission restricted. In case of prior admission to another course in the module "Strategy and Management: Advanced Topics" [M-WIWI-103119], participation in this course is guaranteed. For more information on the application process, see the IBU website.

Exams are offered at least every other semester, so the entire module can be completed in two semesters.

T 6.728 Module component: Structural and Phase Analysis [T-MACH-102170]

Coordinators: Dr.-Ing. Susanne Wagner
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101268 - Specific Topics in Materials Science
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each winter term	1

Assessment
Oral examination

Prerequisites
none

T 6.729 Module component: Student2Startup [T-WIWI-114636]

Coordinators: Prof. Dr. Orestis Terzidis
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Examination of another type	4,5 CP	graded	Each winter term	1

Courses					
WT 25/26	2500165	Student2Startup	3 SWS	Seminar /	Böhrer, Terzidis
ST 2026	2500165	Student2Startup (Edition HAFIS)	3 SWS	Seminar /	Terzidis, Böhrer

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment

The assessment is conducted as an alternative exam assessment. The specific form of assessment depends on the course and may include, for example, the following components:

- **Project Report or Reflection Paper**
- **Mid-term and Final Presentation**
- **Group Work with Individual Contribution**
- **Active Participation and Mandatory Attendance**

The overall grade is determined by the weighted evaluation of these individual assessment components. The detailed weighting of these components for grade calculation will be communicated to students at the beginning of the course.

Prerequisites

None

Recommendations

An interest in entrepreneurial thinking and action, teamwork, and interdisciplinary project work is recommended. Basic knowledge in business administration and/or innovation methods is helpful but not required.

Workload

135 hours

Below you will find excerpts from courses related to this module component:

V Student2Startup

2500165, WS 25/26, 3 SWS, Language: English, Open in study portal

Seminar (S)
Blended (On-Site/Online)

Content
Content:

This seminar offers a unique opportunity to engage in entrepreneurial tasks within real pre-seed start-up projects at KIT. The start-ups define specific challenges, and student teams will develop solutions and recommendations in response.

At the kick-off event, students will select a team challenge to work on. We will introduce a toolbox of methods and resources, after which you will work independently in teams, collaborating directly with the start-up. But you won't be on your own: each team will be supported by a mentor, and we will hold regular seminar sessions for feedback and guidance. At the end of the seminar, each team will present their results in a final pitch.

Learning Objectives:

After completing this course, the course participants will be able to

- Understand and apply fundamental concepts of entrepreneurship, such as business modelling, lean start-up methodologies, and market analysis.
- Work effectively in a team, organize the division of labor into distinct tasks, and coordinate efforts to achieve a common goal—gaining practical experience in project management.
- Recognize the specific challenges faced by start-up projects and acquire hands-on start-up experience.
- Collaborate with real start-up teams, industry experts, and potential users to develop solutions to a defined challenge.
- Present their solutions and results to start-ups and industry experts.

Exam:

Team presentation at the final event, detailed presentation appendix with background information, and active participation in all sessions

Target group:

Bachelor students

Organizational issues
Registration is via the Wiwi-Portal.

In the seminar, you will work on a project in teams of max five people. The groups are formed in the seminar.

V	**Student2Startup (Edition HAFIS)** 2500165, SS 2026, 3 SWS, Language: English, Open in study portal	**Seminar (S)** **Blended (On-Site/Online)**

Content

Content:

This seminar is a perfect complement to the *Entrepreneurship Basics* seminar. While you develop your own ideas in *Entrepreneurship Basics* within a protected environment, you will work on real entrepreneurial projects in this seminar. HAFIS teams at KIT consist of researchers who aim to maximize the impact of their research on society and the economy. This seminar offers a unique opportunity to engage with them by solving entrepreneurial and economic challenges within real KIT projects and by recommending suitable entrepreneurial activities:

The HAFIS teams define specific challenges, and student teams will develop solutions and recommendations in response. At the kick-off event, students will select a team challenge to work on. We will discuss and refine the necessary methods and activities (some of which are already covered in *Entrepreneurship Basics*), after which you will work independently in teams, collaborating directly with the HAFIS contact. However, you will not be on your own: each team will be supported by the lecturer through regular seminar sessions that provide feedback and guidance. At the end of the seminar, each team will present its results in a final pitch. To prepare for this, you will train your presentation skills through pitch training, a midterm pitch, and a dry run.

Learning Objectives:

After completing this course, the course participants will be able to

- Apply fundamental concepts of entrepreneurship, such as business modelling, lean start-up methodologies, and market analysis.
- Work effectively in a team, organize the division of labor into distinct tasks, and coordinate efforts to achieve a common goal, gaining practical experience in project management.
- Recognize the specific challenges faced by start-up projects and acquire hands-on start-up experience.
- Collaborate with real KIT projects and potential users to develop solutions to a defined challenge.
- Present solutions and recommendations.

Target group:

Bachelor students

In the seminar, you will work individually on your challenge in teams of up to five people. The groups will be formed at the seminar kick-off.

Teaching language: expected to be English.

Exam:

The team Pitch & Q&A session will be your main examination performance. Additionally, in written, you will document your Pitch and make a short team and personal reflection (till 25.07.2026).
Attendance at Kick-off and at the pitches is mandatory; one other seminar session may be missed for important reasons.

Dates and Topics:

Wednesday, 29.04.2026, 10:00 – 13:00 (K26) - Kick-off: Team Forming, All participants

Tuesday, 05.05.2026, 09:00 – 13:00: PM+Knowledge, All

Tuesday, 12.05.2026, 09:00 – 13:00: Feedback Session, 90 min, individual

Tuesday, 19.05.2026, 09:00 – 13:00: Pitch Training, All

Wednesday, 10.06.2026, 10:00 – 13:00 (K26): Midterm Pitch, All

Tuesday, 16.06.2026, 09:00 – 13:00: Feedback Session, 90 min, individual

Tuesday, 23.06.2026, 09:00 – 13:00: Feedback Session, 90 min, individual

Tuesday, 30.06.2026, 09:00 – 13:00: Feedback Session, 90 min, individual

Tuesday, 07.07.2026, 09:00 – 13:00: Dry Run, 90 min, individual

Wednesday, 15.07.2026, 9:00 – 13:00 (K26): Pitch & Q&A, All

Location (unless otherwise specified):

Bld. 06.35 Westhochschule - Bau 35 R 219, Hertzstraße 16

Module:

For wiwi students, the course is credited within the module *‘**Entrepreneurial Thinking and Action: Fundamentals of Entrepreneurship’** (M-WIWI-107451)* with 4,5 ECTS.

You can complete the module by participating in our seminar Entrepreneurship Basics (4,5 ECTS, LV 2545010).

Organizational issues
Registration is via the Wiwi-Portal.

In the seminar, you will work on a project in teams of max five people. The groups are formed in the seminar.

Attention:

The dates are the following:

- Wed, 29.04.26 Kick-Off (10:00-13:00, K26) Team Forming (all)
- Tue, 05.05.26 Session 1 (09:30-13:00) PM+Knowledge (all)
- Tue, 12.05.26 Session 2 (09:30-13:00) Feedback Session (90 minutes, individual)
- Tue, 19.05.26 Session 3 (09:30-13:00) Pitch Training (all)
- Wed, 10.06.26 Session 4 (10:00-13:00, K26) Midterm Pitch (all)
- Tue, 16.06.26 Session 5 (09:30-13:00) Feedback Session (90 minutes, individual)
- Tue, 23.06.26 Session 6 (09:30-13:00) Feedback Session (90 minutes, individual)
- Tue, 30.06.26 Session 7 (09:30-13:00) Feedback Session (90 minutes, individual)
- Tue, 07.07.26 Session 8 (09:30-13:00) Dry Run (90 minutes, individual)
- Wed, 15.07.26 Final Pitch (09:00-13:00) Pitch & Q&A (all)

T 6.730 Module component: Successful Transformation Through Innovation [T-WIWI-111823]

Coordinators: Malte Busch
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101507 - Innovation Management
M-WIWI-101507 - Innovation Management

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Each summer term	1

Courses					
ST 2026	2500018	Successful transformation through innovation	2 SWS	Seminar / 🗣	Busch

Legend: 🖥 Online, ❂ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Alternative exam assessments. The grade consists of an presentation of the results (50%) and a seminar paper (50%).

Recommendations
Prior attendance of the course Innovation Management [2545015] is recommended.

Additional Information
Teaching and learning format: Seminar

Workload
90 hours

Below you will find excerpts from courses related to this module component:

V Successful transformation through innovation
2500018, SS 2026, 2 SWS, Language: German, Open in study portal

Seminar (S)
On-Site

Content
This seminar uses strategic innovation management and concepts such as organizational ambidexterity, boundary spanning and stakeholder approaches to shed light on how companies can increase their innovative capacity through innovation. Students will use a core paper to understand the steps a company takes to become an innovative organization. The aim is to understand how medium-sized companies can develop into innovation-driven organizations in the context of organizational inertia and path dependency with the help of the aforementioned concepts. Part of the seminar will be to analyze the role of different stakeholders and how companies can become part of innovation ecosystems.

Based on the impulses and the core paper, students will apply the concepts they have learned to selected companies and present their results. These are also incorporated into academic seminar papers. As part of this work process, students deal independently with a current, research-oriented question and apply scientific criteria consistently. They research relevant information, analyze and abstract it critically and draw their own conclusions on this basis, reflecting their interdisciplinary knowledge and selectively developing current research findings.

The results obtained are systematically presented in written and oral form in compliance with academic standards - including structured presentation, correct specialist terminology and comprehensible references. In doing so, the students argue in a technically sound manner and are able to defend their positions in scientific discourse with experts. In addition, they are familiar with the DFG Code of Conduct “Guidelines for Safeguarding Good Scientific Practice” and successfully apply these guidelines in the preparation of their scientific work.

Organizational issues
Weblink: https://itm.entechnon.kit.edu/192_1281.phphttps://itm.entechnon.kit.edu/192_1673.php

T 6.731 Module component: Superconducting Magnet Technology [T-ETIT-113440]

Coordinators: Prof. Dr. Tabea Arndt
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-106684 - Superconducting Magnet Technology

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each summer term	3

Courses					
ST 2026	2312698	Superconducting Magnet Technology	3 SWS	Lecture / Practice /	Arndt

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
The examination takes place in form of an oral exam (abt. 30 minutes).

Two timeslots (weeks) for examination dates will be announced (usually near end of lecture period & end of semester).

The module grade is the grade of the oral exam.

Prerequisites
none

T 6.732 Module component: Superconducting Materials Lab [T-ETIT-114730]

Coordinators: Dr. Jens Hänisch
Prof. Dr. Bernhard Holzapfel

Organisation: KIT Department of Electrical Engineering and Information Technology

Part of: M-ETIT-105521 - Superconducting Materials

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Each summer term	1

Courses					
ST 2026	2312696	Superconducting Materials Part II	2 SWS	Practical course / 🗣	Holzapfel, Hänisch

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The success control takes place in the form of other types of examination. It consists of a written protocol (approx. 40 pages).

Prerequisites
none

Recommendations
It is recommanded to pass the lecture (part I) before the lab (part II).

T 6.733 Module component: Superconducting Materials Lecture [T-ETIT-114729]

Coordinators: Dr. Jens Hänisch
Prof. Dr. Bernhard Holzapfel
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-105521 - Superconducting Materials

Type	Credits	Grading	Term offered	Version
Oral examination	3 CP	graded	Each winter term	1

Courses					
WT 25/26	2312717	Superconducting Materials Part I	2 SWS	Lecture / 🗣	Holzapfel, Hänisch

Legend: 🖥 Online, ❂ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The Success control takes place in the form of an oral examination lasting approx. 30 minutes.

Prerequisites
none

Recommendations
It is recommanded to pass the lecture (part I) before the lab (part II).

T 6.734 Module component: Superconducting Nanowire Detectors [T-ETIT-111236]

Coordinators: Dr. Konstantin Ilin
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-105609 - Superconducting Nanowire Detectors

Type	Credits	Grading	Term offered	Expansion	Version
Oral examination	4 CP	graded	Each summer term	1 semesters	2

Courses					
ST 2026	2312671	Superconducting Nanowire Detectors	2 SWS	Lecture / 🗣	Ilin
ST 2026	2312673	Practice to 2312671 Superconducting Nanowire Detectors	1 SWS	Practice / 🗣	Ilin

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Oral Exam (approx. 20 min.)

Prerequisites
none

T 6.735 Module component: Superconductivity for Engineers [T-ETIT-111239]

Coordinators: Prof. Dr. Bernhard Holzapfel
Organisation: KIT Department of Electrical Engineering and Information Technology
Part of: M-ETIT-105611 - Superconductivity for Engineers

Type	Credits	Grading	Term offered	Expansion	Version
Written examination	6 CP	graded	Each winter term	1 semesters	4

Courses					
WT 25/26	2312708	Superconductivity for Engineers	3 SWS	Lecture / 🗣	Kempf, Holzapfel
WT 25/26	2312709	Exercise for 2312708 Superconductivity for Engineers	1 SWS	Practice / 🗣	Hänisch, Arldt
WT 26/27	2312708	Superconductivity for Engineers	3 SWS	Lecture / 🗣	Kempf, Holzapfel
WT 26/27	2312709	Exercise for 2312708 Superconductivity for Engineers	1 SWS	Practice / 🗣	Hänisch, Arldt

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment of success takes place in the form of a written examination lasting 120min.

The module grade is the grade of the written examination.

Prerequisites
none

T 6.736 Module component: Superhard Thin Film Materials [T-MACH-102103]

Coordinators: Prof. Sven Ulrich
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101268 - Specific Topics in Materials Science
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each winter term	3

Courses					
WT 25/26	2177618	Superhard Thin Film Materials	2 SWS	Lecture / 🗣	Ulrich
WT 26/27	2177618	Superhard Thin Film Materials	2 SWS	Lecture / 🗣	Ulrich

Legend: 🖥 Online, ❖ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
oral examination (ca. 30 Minuten)

Prerequisites
none

Additional Information
The course is offered in German

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Superhard Thin Film Materials

Lecture (V)
On-Site

2177618, WS 25/26, 2 SWS, Language: German, Open in study portal

Content
oral examination (about 30 min), no tools or reference materials

Teaching Content:

Introduction

Basics

Plasma diagnostics

Particle flux analysis

Sputtering and ion implantation

Computer simulations

Properties of materials, thin film deposition technology,
thin film analysis and modelling of superhard materials

Amorphous hydrogenated carbon

Diamond like carbon

Diamond

Cubic Boronnitride

Materials of the system metall-boron-carbon-nitrogen-silicon

regular attendance: 22 hours
self-study: 98 hours

Superhard materials are solids with a hardness higher than 4000 HV 0,05. The main topics of this lecture are modelling, deposition, characterization and application of superhard thin film materials.

Recommendations: none

Organizational issues
Falls die Vorlesung online stattfinden muss, bitte um Anmeldung unter sven.ulrich@kit.edu bis zum 28.10.25.

Den entsprechenden MS Teams Link erhalten Sie dann per E-Mail am 29.10.25.

Literature
G. Kienel (Herausgeber): Vakuumbeschichtung 1 - 5, VDI Verlag, Düsseldorf, 1994

Abbildungen und Tabellen werden verteilt; Copies with figures and tables will be distributed

V **Superhard Thin Film Materials**
2177618, WS 26/27, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content
oral examination (about 30 min), no tools or reference materials

Teaching Content:

Introduction

Basics

Plasma diagnostics

Particle flux analysis

Sputtering and ion implantation

Computer simulations

Properties of materials, thin film deposition technology,
thin film analysis and modelling of superhard materials

Amorphous hydrogenated carbon

Diamond like carbon

Diamond

Cubic Boronnitride

Materials of the system metall-boron-carbon-nitrogen-silicon

regular attendance: 22 hours
self-study: 98 hours

Superhard materials are solids with a hardness higher than 4000 HV 0,05. The main topics of this lecture are modelling, deposition, characterization and application of superhard thin film materials.

Recommendations: none

Organizational issues
Achtung: Die Vorlesung beginnt erst am 12.11.2026!

Falls die Vorlesung online stattfinden muss, bitte um Anmeldung unter sven.ulrich@kit.edu bis zum 28.10.26.

Den entsprechenden MS Teams Link erhalten Sie dann per E-Mail am 29.10.26.

Literature
G. Kienel (Herausgeber): Vakuumbeschichtung 1 - 5, VDI Verlag, Düsseldorf, 1994

Abbildungen und Tabellen werden verteilt; Copies with figures and tables will be distributed

T 6.737 Module component: Supervised Profile Definition [T-WIWI-114229]

Coordinators: Studiendekan des KIT-Studienganges
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-107204 - Seminar Module

Type	Credits	Grading	Version
Coursework	1 CP	pass/fail	1

Assessment
This study achievement serves the active engagement with your course of studies and the conscious selection of your modules. At the beginning of your studies, you will engage intensively with the content and requirements of the modules offered. This includes:

- **Attendance at introductory events:** Take the opportunity to get to know lecturers and to learn about the contents and objectives of the modules.
- **Use of the module handbook:** Familiarize yourself with the descriptions, learning objectives, and examination modalities of the modules.
- **Own interests:** Think about which modules best suit your interests and goals.

Then, you choose the modules relevant to you in the campus system. This module selection is reviewed by the program coordinator. Upon successful selection, the study achievement will be credited to you in the campus system with one credit point (CP).

Note: This study achievement serves not only for the crediting of credit points, but above all for your personal study planning. A careful module selection is crucial for your study success.

Prerequisites
None

Workload
30 hours

6.738 Module component: Supplement Enterprise Information Systems [T-WIWI-110346]

Coordinators: Prof. Dr. Andreas Oberweis
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101472 - Informatics
M-WIWI-101477 - Development of Business Information Systems
M-WIWI-101628 - Emphasis in Informatics
M-WIWI-104901 - Informatics (KIT-Department of Economics and Management)

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each term	1

Assessment
Success is assessed in the form of a written examination (60 minutes) or, if applicable, an oral examination (approx. 30 minutes).

Prerequisites
None

Additional Information
This course can be used in particular for the acceptance of external courses whose content is in the broader area of applied informatics, but is not equivalent to another course of this topic.

Workload
135 hours

T 6.739 Module component: Supplement Software- and Systemsengineering [T-WIWI-110372]

Coordinators: Prof. Dr. Andreas Oberweis
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101472 - Informatics
M-WIWI-101628 - Emphasis in Informatics
M-WIWI-104901 - Informatics (KIT-Department of Economics and Management)

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each term	1

Assessment
The assessment of this course is a written or (if necessary) oral examination.

Prerequisites
None

Additional Information
This course can be used in particular for the acceptance of external courses whose content is in the broader area of software and systems engineering, but cannot assigned to another course of this topic.

T 6.740 Module component: Supply Chain Management with Advanced Planning Systems [T-WIWI-102763]

Coordinators: Claus J. Bosch
Dr. Mathias Göbelt

Organisation: KIT Department of Business and Economics

Part of: M-WIWI-101412 - Industrial Production III
M-WIWI-101471 - Industrial Production II
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	3,5 CP	graded	Each summer term	1

Courses					
ST 2026	2581961	Supply Chain Management with Advanced Planning Systems	2 SWS	Lecture / 🗣	Göbelt, Bosch

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment consists of an oral (30 minutes) or written exam (60 minutes) (following §4(2) of the examination regulation). The exam takes place in every semester. Re-examinations are offered at every ordinary examination date.

Prerequisites
None

Recommendations
None

Below you will find excerpts from courses related to this module component:

V Supply Chain Management with Advanced Planning Systems
Lecture (V)
On-Site

2581961, SS 2026, 2 SWS, Language: English, Open in study portal

Content
This lecture deals with supply chain management from a practitioner's perspective with a special emphasis Advanced Planning Systems (APS) and the planning domain. The software solution SAP SCM, one of the most widely used Advanced Planning Systems, is used as an example to show functionality and application of an APS in practice.

First, the term supply chain management is defined and its scope is determined. Methods to analyze supply chains as well as indicators to measure supply chains are derived. Second, the structure of an APS (advanced planning system) is discussed in a generic way. Later in the lecture, the software solution SAP SCM is mapped to this generic structure. The individual planning tasks and software modules (demand planning, supply network planning / sales & operations planning, production planning / detailed scheduling, deployment, transportation planning, global available-to-promise) are presented by discussing the relevant business processes, providing academic background, describing typical planning processes and showing the user interface and user-related processes in the software solution. At the end of the lecture, implementation methodologies and project management approaches for SAP SCM are covered.

Contents

1. Introduction to Supply Chain Management

- Supply Chain Management Fundamentals
- Supply Chain Management Analytics

2. Structure of Advanced Planning Systems

3. SAP SCM

- Introduction / SCM Solution Map
- Demand Planning
- Supply Network Planning / Sales & Operations Planning
- Production Planning and Detailed Scheduling
- Deployment
- Transportation Planning / Global Available to Promise
- Cloud-based Supply Chain Planning

4. SAP SCM in Practice

- Project Management and Implementation
- SAP Implementation Methodology

Literature
will be announced in the course

T 6.741 Module component: Sustainability in Mobility Systems [T-BGU-111057]

Coordinators: Prof. Dr.-Ing. Martin Kagerbauer
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: M-BGU-101064 - Fundamentals of Transportation
M-BGU-101065 - Transportation Modelling and Traffic Management

Type	Credits	Grading	Term offered	Expansion	Version
Written examination	3 CP	graded	Each term	1 semesters	1

Courses					
WT 25/26	6232906	Sustainability in Mobility Systems	2 SWS	Lecture / 🗣	Kagerbauer, Plötz, Gnann

Legend: 🖥 Online, ❂ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
written exam, 60 min., computer-based

Prerequisites
none

Recommendations
none

Additional Information
none

Workload
90 hours

T 6.742 Module component: Sustainable Information Systems [T-WIWI-114995]

Coordinators: TT-Prof. Dr. Philipp Staudt
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration
M-WIWI-107649 - Sustainable Information Systems

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each summer term	1

Courses					
ST 2026	2500007	Sustainable Information Systems	3 SWS	Lecture	Staudt
ST 2026	2500039	Exercise on Sustainable Information Systems	0 SWS	Practice	Staudt

Assessment
Written examination (60 minutes). The examination is offered every semester and can be repeated at any regular examination date.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Sustainable Information Systems

2500007, SS 2026, 3 SWS, Open in study portal

Lecture (V)

Content
This lecture provides students with essential research methods for studying sustainability challenges in information systems. The course begins with an introduction to Green IS and sustainability fundamentals, establishing the theoretical foundation for understanding how information systems can contribute to environmental and social goals. Students will gain insight into current research topics and debates in the field of sustainable information systems. The core of the course focuses on key research methodologies used in MIS research, including qualitative approaches such as case study research, as well as quantitative methods including surveys, laboratory experiments, and field experiments. Each method is introduced through published research papers from the sustainability domain, allowing students to see how these approaches are applied in practice. By the end of the course, students will be equipped with the methodological toolkit needed to design and conduct rigorous research on sustainability questions in information systems contexts.

T 6.743 Module component: Sustainable Materials [T-MACH-113987]

Coordinators: Prof. Dr. Christian Greiner
Organisation: KIT Department of Mechanical Engineering

Part of: M-WIWI-107718 - Mechanical Engineering I
M-WIWI-107730 - Mechanical Engineering II

Type	Credits	Grading	Term offered	Version
Oral examination	4,5 CP	graded	Each winter term	2

Assessment
oral exam ca. 30 minutes

Prerequisites
The course T--MACH-114606 - Sustainability in Materials Science and Engineering must not have been started.

Recommendations
Basic knowledge in materials science (e.g. lecture materials science I and II)

Additional Information
The course is offered in English.

The course will be offered for the first time in the winter semester of 2026/2027!

Workload
135 hours

T 6.744 Module component: Sustainable Materials Production [T-MACH-114608]

Coordinators: Dr.-Ing. Antje Dollmann
Prof. Dr. Christian Greiner

Organisation: KIT Department of Mechanical Engineering

Part of: M-WIWI-107718 - Mechanical Engineering I
M-WIWI-107730 - Mechanical Engineering II

Type	Credits	Grading	Term offered	Version
Oral examination	4,5 CP	graded	Each winter term	2

Assessment
oral exam ca. 30 minutes

Prerequisites
The course T--MACH-114606 - Sustainability in Materials Science and Engineering must not have been started.

Recommendations
Basic knowledge in materials science (e.g. lecture materials science I and II)

Additional Information
The course is offered in English.

The course will be offered for the first time in the winter semester of 2026/2027!

Workload
135 hours

T 6.745 Module component: Sustainable Vehicle Drivetrains [T-MACH-111578]

Coordinators: Dr.-Ing. Olaf Toedter
Organisation: KIT Department of Mechanical Engineering
Part of: M-MACH-101303 - Combustion Engines II

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each winter term	1

Courses					
WT 25/26	2133132	Sustainable Vehicle Drivetrains	3 SWS	Lecture / 🗣	Toedter

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
oral exam (approx. 20 minutes)

Prerequisites
none

Additional Information
Starting in winter term 25/26, the course consists of a lecture (2h / week) and a tutorial (1 h / week).

This course is offered in German.

Workload
120 hours

T 6.746 Module component: System Integration in Micro- and Nanotechnology I [T-MACH-114552]

Coordinators: apl. Prof. Dr. Ulrich Gengenbach
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101294 - Nanotechnology
M-WIWI-107718 - Mechanical Engineering I
M-WIWI-107730 - Mechanical Engineering II

Type	Credits	Grading	Term offered	Version
Oral examination	4,5 CP	graded	Each summer term	2

Courses					
ST 2026	2106033	System Integration in Micro- and Nanotechnology I	2 SWS	Lecture / 🗣	Gengenbach

Legend: 🖥 Online, ❂ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Oral exam (Duration: 30 min)

Prerequisites
None

Additional Information
The course is offered in English.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V System Integration in Micro- and Nanotechnology I

Lecture (V)
On-Site

2106033, SS 2026, 2 SWS, Language: English, Open in study portal

Content
Content:

- Introduction to system integration (fundamentals)
- Brief introduction to MEMS processes
- Flexures
- Surfaces and plasma processes for surface treatment
- Adhesive bonding in engineering
- Mounting techniques in electronics
- Molded Interconnect devices (MID)
- Functional Printing
- Low temperature cofired ceramics in system integration

Learning objectives:
The students acquire basic knowledge of challenges and system integration technologies from mechanical engineering, precision engineering and electronics.

Literature

- A. Risse, Fertigungsverfahren der Mechatronik, Feinwerk- und Präzisionsgerätetechnik, Vieweg+Teubner Verlag, Wiesbaden, 2012
- M. Madou, Fundamentals of microfabrication and nanotechnology, CRC Press Boca Raton, 2012
- G. Habenicht, Kleben Grundlagen, Technologien, Anwendungen, Springer-Verlag Berlin Heidelberg, 2009
- J. Franke, Räumliche elektronische Baugruppen (3D-MID), Carl Hanser-Verlag München, 2013

T 6.747 Module component: System Integration in Micro- and Nanotechnology II [T-MACH-114553]

Coordinators: apl. Prof. Dr. Ulrich Gengenbach
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101294 - Nanotechnology
M-WIWI-107718 - Mechanical Engineering I
M-WIWI-107730 - Mechanical Engineering II

Type	Credits	Grading	Term offered	Version
Oral examination	4,5 CP	graded	Each winter term	2

Courses					
WT 25/26	2105040	System Integration in Micro- and Nanotechnology II	2 SWS	Lecture / 🗣	Gengenbach

Legend: 🖥 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Oral exam (Duration: 30 min)

Prerequisites
None

Additional Information
The course is offered in English.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V System Integration in Micro- and Nanotechnology II

Lecture (V)
On-Site

2105040, WS 25/26, 2 SWS, Language: English, Open in study portal

Content
Introduction to system integration (novel processes and applications)

Assembly of hybrid microsystems

Packaging processes

Applications:

- Lab-on-chip systems
- Microoptical systems
- Silicon Photonics

Novel integration processes:

- Direct Laser Writing
- Self Assembly

Learning objectives

The students acquire knowledge of novel system integration technologies and their application in microoptic and microfluidic systems.

Literature
N.-T. Nguyen, Fundamentals and Applications of Microfluidics, Artech House

G. T. Reed, Silicon Photonics: An Introduction, Wiley

T 6.748 Module component: Systematic Materials Selection [T-MACH-100531]

Coordinators: Dr.-Ing. Stefan Dietrich
Prof. Dr.-Ing. Volker Schulze

Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101268 - Specific Topics in Materials Science
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Written examination	4 CP	graded	Each summer term	6

Courses					
ST 2026	2174576	Systematic Materials Selection	3 SWS	Lecture / 🗣	Dietrich
ST 2026	2174577	Excercises in Systematic Materials Selection	1 SWS	Practice / 🗣	Dietrich

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The assessment is carried out as a written exam of 2 h.

Prerequisites
none

Recommendations
Basic knowledge in materials science, mechanics and mechanical design due to the lecture Materials Science I/II.

Additional Information
The course is offered in German.

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Systematic Materials Selection

2174576, SS 2026, 3 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content
Important aspects and criteria of materials selection are examined and guidelines for a systematic approach to materials selection are deeloped. The following topics are covered:

- Information and introduction
- Necessary basics of materials
- Selected methods / approaches of the material selection
- Examples for material indices and materials property charts
- Trade-off and shape factors
- Sandwich materials and composite materials
- High temperature alloys
- Regard of process influences
- Material selection for production lines
- Incorrect material selection and the resulting consequences
- Abstract and possibility to ask questions

learning objectives:
The students are able to select the best material for a given application. They are proficient in selecting materials on base of performance indices and materials selection charts. They can identify conflicting objectives and find sound compromises. They are aware of the potential and the limits of hybrid material concepts (composites, bimaterials, foams) and can determine whether following such a concept yields a useful benefit.
requirements:

WiIng SPO 2007 (B.Sc.)

The course Material Science I [21760] has to be completed beforehand.

WiIng (M.Sc.)

The course Material Science I [21760] has to be completed beforehand.

workload:

The workload for the lecture is 120 h per semester and consists of the presence during the lecture (30 h) as well as preparation and rework time at home (30 h) and preparation time for the oral exam (60 h).

Literature
Vorlesungsskriptum; Übungsblätter; Lehrbuch: M.F. Ashby, A. Wanner (Hrsg.), C. Fleck (Hrsg.);
Materials Selection in Mechanical Design: Das Original mit Übersetzungshilfen
Easy-Reading-Ausgabe, 3. Aufl., Spektrum Akademischer Verlag, 2006
ISBN: 3-8274-1762-7

Lecture notes; Problem sheets; Textbook: M.F. Ashby, A. Wanner (Hrsg.), C. Fleck (Hrsg.);
Materials Selection in Mechanical Design: Das Original mit Übersetzungshilfen
Easy-Reading-Ausgabe, 3. Aufl., Spektrum Akademischer Verlag, 2006
ISBN: 3-8274-1762-7

T 6.749 Module component: Systems of Remote Sensing, Prerequisite [T-BGU-101637]

Coordinators: Prof. Dr. Jan Cermak
Prof. Dr.-Ing. Stefan Hinz
Dr.-Ing. Uwe Weidner

Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences

Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Coursework	1 CP	pass/fail	Each summer term	1

Courses					
ST 2026	6020242	Remote Sensing Systems, Excercise	1 SWS	Practice / 🗣	Pauli

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Prerequisites
None

Recommendations
None

Additional Information
None

Workload
30 hours

T 6.750 Module component: Tactical and Operational Supply Chain Management [T-WIWI-102714]

Coordinators: Prof. Dr. Stefan Nickel
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104899 - Operations Research

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each summer term	3

Courses					
ST 2026	2550486	Tactical and operational SCM	3 SWS	Lecture / 🗣	Nickel
ST 2026	2550487	Übungen zu Taktisches und operatives SCM	1.5 SWS	Practice / 🗣	Pomes, Hoffmann

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Success is assessed in the form of a written examination (60 minutes). The examination is offered every semester and can be repeated at any regular examination date.

Successful participation in the online exercises is a prerequisite for admission to the written exam.

Prerequisites
Prerequisite for admission to examination is the succesful completion of the online assessments.

Recommendations
None

Additional Information
The lecture is held in every summer term. The planned lectures and courses for the next three years are announced online at https://dol.ior.kit.edu/Lehrveranstaltungen.php.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Tactical and operational SCM

2550486, SS 2026, 3 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content
The planning of material transport is an essential element of Supply Chain Management. By linking transport connections across different facilities, the material source (production plant) is connected with the material sink (customer). The general supply task can be formulated as follows (cf. Gudehus): For given material flows or shipments, choose the optimal (in terms of minimal costs) distribution and transportation chain from the set of possible logistics chains, which asserts the compliance of delivery times and further constraints. The main goal of the inventory management is the optimal determination of order quantities in terms of minimization of fixed and variable costs subject to resource constraints, supply availability and service level requirements. Similarly, the problem of lot sizing in production considers the determination of the optimal amount of products to be produced in a time slot. The course includes an introduction to basic terms and definitions of Supply Chain Management and a presentation of fundamental quantitative planning models for distribution, vehicle routing, inventory management and lot sizing. Furthermore, case
studies from practice will be discussed in detail.

Passing the online exercise is a prerequisite for admission to the exam.

Literature
Weiterführende Literatur

- Domschke: Logistik: Transporte, 5. Auflage, Oldenbourg, 2005
- Domschke: Logistik: Rundreisen und Touren, 4. Auflage, Oldenbourg, 1997
- Ghiani, Laporte, Musmanno: Introduction to Logistics Systems Planning and Control, Wiley, 2004
- Gudehus: Logistik, 3. Auflage, Springer, 2005
- Simchi-Levi, Kaminsky, Simchi-Levi: Designing and Managing the Supply Chain, 3rd edition, McGraw-Hill, 2008
- Silver, Pyke, Peterson: Inventory management and production planning and scheduling, 3rd edition, Wiley, 1998

T 6.751 Module component: Tax Law [T-INFO-111437]

Coordinators: Detlef Dietrich
Organisation: KIT Department of Informatics
Part of: M-INFO-101216 - Private Business Law
M-WIWI-104903 - Law

Type	Credits	Grading	Term offered	Version
Written examination	3 CP	graded	Each summer term	1

Courses					
ST 2026	24646	Tax Law	2 SWS	Lecture / 🖥	Dietrich

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, × Cancelled

T 6.752 Module component: Team Project Management and Technology [T-WIWI-110968]

Coordinators: Prof. Dr. Martin Klarmann
Prof. Dr. Alexander Mädche
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Expansion	Version
Examination of another type	9 CP	graded	Each term	1 semesters	1

Courses					
WT 25/26	2571176	Team Project Management and Technology	6 SWS	Project / Blended	Klarmann, Mädche
ST 2026	2571176	Teamprojekt Wirtschaft und Technologie		Project / Blended	Klarmann, Mädche

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
Alternative exam assessment. The basis for grading is the documents produced, the presentations during the course of the project, the artifact to be produced (e.g. algorithm, method, model, software, component) and the final presentation.

Workload
270 hours

T 6.753 Module component: Team Project Management and Technology (BUS/ENG) [T-WIWI-110977]

Coordinators: Prof. Dr. Martin Klarmann
Prof. Dr. Alexander Mädche
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Expansion	Version
Examination of another type	9 CP	graded	Each term	1 semesters	1

Courses					
WT 25/26	2571176	Team Project Management and Technology	6 SWS	Project / Blended	Klarmann, Mädche
ST 2026	2571176	Teamprojekt Wirtschaft und Technologie		Project / Blended	Klarmann, Mädche

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
Alternative exam assessment. The basis for grading is the documents produced, the presentations during the course of the project, the artifact to be produced (e.g. algorithm, method, model, software, component) and the final presentation.

Workload
270 hours

T 6.754 Module component: Telecommunications and Internet – Economics and Policy [T-WIWI-113147]

Coordinators: Prof. Dr. Kay Mitusch
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101406 - Network Economics
M-WIWI-101409 - Electronic Markets
M-WIWI-104908 - Economics
M-WIWI-107010 - Economics in a Connected World

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each winter term	1

Courses					
WT 25/26	2561232	Telecommunication and Internet - Economics and Policy	2 SWS	Lecture /	Mitusch
WT 25/26	2561233	Excercises to Telecommunication and Internet - Economics and Policy	1 SWS	Practice /	Mitusch

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
Students' understanding and knowledge will be assessed through either an oral or a written exam. The actual method used will be announced during the course. The course takes place every winter term, and exams are offered two times a year, in March and in September.

Recommendations
Basic knowledge of microeconomics is a precondition. Further knowledge of industrial economics or networks economics is useful, but not necessary. No prior knowledge of telecommunications or internet technologies is required.

Additional Information
Disclaimer:
German wording is sometimes provided in parallel. Some German original literature is used (especially official and legislative texts) where we will try to provide English translations in parallel.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Telecommunication and Internet - Economics and Policy

2561232, WS 25/26, 2 SWS, Language: German/English, Open in study portal

Lecture (V)
Blended (On-Site/Online)

Content

Description:

The course provides students with a comprehensive understanding of the economic principles, dynamics, and policies that govern the telecommunication and internet industries and markets. It focuses on the infrastructure of the internet, both physical and logical.

Course Objectives:

Understand the telecommunication and internet landscape: Students will be introduced to the historical development, evolution, and current state of the telecommunication and internet industries. This includes technology, industrial organization, regulation, and other policies. Students will explore the emergence of modern telecommunication networks, the birth of the internet, and key milestones that have shaped the global communication landscape.

Examine network economics: Students will explore the unique economic characteristics of telecommunications networks, including network effects, economies of scale, the implications for investment decisions and market entry barriers, and regulatory responses.

Analyse market structures and competition policies: Students will dive into the various market structures that exist within the telecommunication and internet industries, including: access to the internet by users, access to the infrastructure by firms, economic interactions between the autonomous systems (i.e. sub-networks) and other players (like internet exchange points) of the internet, implications for quality of services and network neutrality. Emphasis will be placed on competitiveness of markets, resp. market power, on the role of regulation, and how they impact market dynamics.

Investigate infrastructure investment and policy: The course will address the significant role of infrastructure investment in the telecommunication and internet sectors. Students will analyse the economic drivers behind infrastructure construction, government policies, and regulatory frameworks that influence investment decisions.

Address emerging trends: The course will address the latest trends and technologies in telecommunication and the internet, such as 5G, Internet of Things (IoT), and cloud computing, content delivery networks, and their economic implications.

Assess platform economics: The role of digital platforms in the telecommunication and internet industries will be addressed. Students will understand platform business models and the economics of multisided markets. In this context, the "hypergiants" of the internet get into the focus as well as the challenges and opportunities they present.

Teaching Methodology:

The course will adopt a combination of lectures, case studies, and guest lectures from (industry) experts. Real-world examples will be used to illustrate economic principles in action within the telecommunication and internet sectors. A few economic models will be analysed, but most of the issues will be addressed verbally.

T 6.755 Module component: Telecommunications Law [T-INFO-101309]

Organisation: KIT Department of Informatics
Part of: M-INFO-106754 - Public Economic and Technology Law
M-WIWI-104903 - Law

Type	Credits	Grading	Term offered	Version
Written examination	3 CP	graded	Each summer term	1

Courses					
ST 2026	2424632	Telekommunikationsrecht	2 SWS	Lecture / 🗣	Döveling

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

T 6.756 Module component: Tendering, Planning and Financing in Public Transport [T-BGU-101005]

Coordinators: Prof. Dr.-Ing. Peter Vortisch
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: M-BGU-101064 - Fundamentals of Transportation
M-BGU-101065 - Transportation Modelling and Traffic Management
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Expansion	Version
Oral examination	3 CP	graded	Each term	1 semesters	1

Courses					
ST 2026	6232807	Competition, Planning and Financing in Public Transport	2 SWS	Lecture / 🗣	Pischon

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, x Cancelled

Assessment
oral exam, appr. 20 min.

Prerequisites
none

Recommendations
none

Additional Information
none

Workload
90 hours

T 6.757 Module component: Time Series Analysis [T-WIWI-114325]

Coordinators: Prof. Dr. Melanie Schienle
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-107187 - Quantitative Core

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each winter term	1

Assessment
The assessment consists of a written exam (90 minutes) (following §4(2), 1 of the examination regulation).

Prerequisites
None

Recommendations
Knowledge of the contents covered by the course "Economics III: Introduction in Econometrics"[2520016]

Additional Information
The next lecture will take place in the winter semester 2022/23.

T 6.758 Module component: Tires and Wheel Development for Passenger Cars [T-MACH-102207]

Coordinators: Prof. Dr.-Ing. Günter Leister
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101265 - Vehicle Development
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each summer term	1

Courses					
ST 2026	2114845	Tires and Wheel Development for Passenger Cars	2 SWS	Lecture / 🗣	Leister

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, × Cancelled

Assessment
Oral Examination

Duration: 30 up to 40 minutes

Auxiliary means: none

Prerequisites
none

Additional Information
The course is offered in German.

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Tires and Wheel Development for Passenger Cars

2114845, SS 2026, 2 SWS, Language: German, Open in study portal

Lecture (V)
On-Site

Content

1. The role of the tires and wheels in a vehicle
2. Geometrie of Wheel and tire, Package, load capacity and endurance, Book of requirement
3. Mobility strategy, Minispare, runflat systems and repair kit.
4. Project management: Costs, weight, planning, documentation
5. Tire testing and tire properties
6. Wheel technology incuding Design and manifacturing methods, Wheeltesting
7. Tire presssure: Indirect and direct measuring systems
8. Tire testing subjective and objective

Learning Objectives:

The students are informed about the interactions of tires, wheels and chassis. They have an overview of the processes regarding the tire and wheel development. They have knowledge of the physical relationships.

Organizational issues
Vorlesung findet am Campus Ost, Geb.70.04 statt. Voraussichtliche Termine, nähere Informationen und eventuelle Terminänderungen:

siehe Institutshomepage.

Literature
Manuskript zur Vorlesung

Manuscript to the lecture

T 6.759 Module component: Topics in Experimental Economics [T-WIWI-102863]

Coordinators: Prof. Dr. Johannes Philipp Reiß
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101505 - Experimental Economics
M-WIWI-104908 - Economics

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Irregular	1

Assessment
The assessment consists of a written exam (following §4(2), 1 of the examination regulation).

Prerequisites
None

Recommendations
Basic knowledge of Experimental Economics is assumed. Therefore, it is strongly recommended to attend the course Experimental Economics beforehand.

Additional Information
The repeat examination can be taken at any later, regular examination date. The examination dates are only offered in the semester in which the course is offered and in the semester immediately following. The subject matter relates to the last course held.

Workload
135 hours

T 6.760 Module component: Topics in Stochastic Optimization [T-WIWI-112109]

Coordinators: Prof. Dr. Steffen Rebennack
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101473 - Mathematical Programming
M-WIWI-101637 - Analytics and Statistics
M-WIWI-102832 - Operations Research in Supply Chain Management
M-WIWI-103289 - Stochastic Optimization
M-WIWI-104899 - Operations Research

Type	Credits	Grading	Term offered	Version
Examination of another type	4,5 CP	graded	Each winter term	1

Assessment
Students will be given problem sets on which they work in groups. The problem sets will involve the implementation of the models presented in the course, and exploring features of these models. The groups will present their findings in front of the class. The grading will be based on the presentation.

Recommendations
A solid understanding of Stochastic Optimization and/or Optimization under Uncertainty as well as optimization in general is highly recommended, since we will heavily build upon basics of these areas.

Additional Information
Teaching and learning format: Lecture and exercise

Workload
135 hours

T 6.761 Module component: Topology Optimisation in Engineering [T-MACH-113949]

Coordinators: Dr. Yongbo Deng
Organisation: KIT Department of Mechanical Engineering
Part of: M-WIWI-107718 - Mechanical Engineering I
M-WIWI-107730 - Mechanical Engineering II

Type	Credits	Grading	Term offered	Expansion	Version
Written examination	4,5 CP	graded	Each summer term	1 semesters	2

Courses					
WT 25/26	2142830	Topology Optimisation in Engineering	2 SWS	Lecture / Practice / 🗣	Deng, Mager
ST 2026	2142830	Topology Optimisation in Engineering	2 SWS	Lecture / Practice / 🗣	Deng

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Written exam 60 minutes

Additional Information
The course is offered in English.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Topology Optimisation in Engineering

2142830, WS 25/26, 2 SWS, Language: English, Open in study portal

Lecture / Practice (VÜ)
On-Site

Content
The aim of this course is to learn the basic knowledge of topology optimization which is one of the current most powerful structural design methods. This course also provides the related implementing details and numerical skills of topology optimization.

V Topology Optimisation in Engineering

2142830, SS 2026, 2 SWS, Language: English, Open in study portal

Lecture / Practice (VÜ)
On-Site

Content
The aim of this course is to learn the basic knowledge of topology optimization which is one of the current most powerful structural design methods. This course also provides the related implementing details and numerical skills of topology optimization.

T 6.762 Module component: Trademark and Unfair Competition Law [T-INFO-101313]

Coordinators: Dr. Yvonne Matz
Organisation: KIT Department of Informatics
Part of: M-INFO-101215 - Intellectual Property Law
M-WIWI-104903 - Law

Type	Credits	Grading	Term offered	Version
Written examination	3 CP	graded	Each term	1

Courses					
WT 25/26	2424136	Trademark and Unfair Competition Law	2 SWS	Lecture / 🗣	Matz
ST 2026	24609	Trademark and Unfair Competition Law	2 SWS	Lecture / 🗣	Matz
WT 26/27	2424136	Trademark and Unfair Competition Law	2 SWS	Lecture / 🗣	Matz

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, × Cancelled

Assessment
The assessment is carried out as a written examination (§ 4 Abs. 2 No. 1 SPO) lasting 60 minutes.

Prerequisites
None.

T 6.763 Module component: Traffic Engineering [T-BGU-101798]

Coordinators: Prof. Dr.-Ing. Peter Vortisch
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: M-BGU-101065 - Transportation Modelling and Traffic Management
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Expansion	Version
Oral examination	3 CP	graded	Each term	1 semesters	1

Courses					
WT 25/26	6232703	Road Traffic Engineering	2 SWS	Lecture / Practice / 🗣	Vortisch, Mitarbeiter/innen

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
oral exam, appr. 20 min.

Prerequisites
None

Recommendations
None

Additional Information
None

Workload
90 hours

Below you will find excerpts from courses related to this module component:

V Road Traffic Engineering

6232703, WS 25/26, 2 SWS, Language: German, Open in study portal

Lecture / Practice (VÜ)
On-Site

Content
The lecture teaches basic principles and skills necessary to understand the methods and tools of traffic engineering, including theoretical background information as well as application of the relevant manuals and guidelines.

- Applications of traffic engineering: design of infrastructure and traffic control
- Description and analysis of traffic flow: Basic principles (kinematics, measurements of traffic flows, microscopic and macroscopic traffic parameters, Fundamental diagram)
- Methods in traffic engineering: travel demand structure, traffic flow characteristics, Queuing theory, Level-of-Service-concepts
- Capacity analysis for intersections with and without signalisation (entries and weaving sections, roundabouts and signal-controlled intersection),
- Backgrounds and application of the German Highway Capacity Manual
- Design of signal control (Fixed time signal controls, vehicle actuated control, „green waves“, network control, progressive signal systems) including public transport (prioritizing systems) and other transport modes (bicycles, pedestrians)
- Introduction to traffic management (for more detailed information see lecture “Transport Management and Transport Telematics [6232802])

Coordination: Baumann, Marvin

T 6.764 Module component: Traffic Flow Simulation [T-BGU-101800]

Coordinators: Prof. Dr.-Ing. Peter Vortisch
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: M-BGU-101065 - Transportation Modelling and Traffic Management
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Expansion	Version
Oral examination	3 CP	graded	Each term	1 semesters	1

Courses					
ST 2026	6232804	Traffic Simulation	2 SWS	Lecture / Practice / 🗣	Vortisch, Mitarbeiter/innen

Legend: Online, Blended (On-Site/Online), On-Site, × Cancelled

Assessment
oral exam, appr. 20 min.

Prerequisites
None

Recommendations
None

Additional Information
None

Workload
90 hours

Below you will find excerpts from courses related to this module component:

V Traffic Simulation

6232804, SS 2026, 2 SWS, Language: German, Open in study portal

Lecture / Practice (VÜ)
On-Site

Content
The lecture teaches basic principles and application of traffic flow simulation tools in traffic engineering and transport planning.

This includes application of simulation software as well as the knowledge about models and how to deal with the stochastic nature of simulation results.

The lecture teaches the application of microscopic traffic flow simulation using the simulation software PTV Vissim, combining practical and theoretical aspects. Theoretical aspects include car following models, lane changing behavior and route choice models. Calibration and validation of the models will be explained and demonstrated by practical examples. Furthermore, German and American guidelines for the application of simulation models will be discussed and background information will be given.

In addition to the lectures, students will build a microscopic traffic flow model of an intersection. The aim is to practically apply what has been learned and to deepen the modeling knowledge.

Coordination: Grau, Josephine

T 6.765 Module component: Traffic Management and Transport Telematics [T-BGU-101799]

Coordinators: Prof. Dr.-Ing. Peter Vortisch
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: M-BGU-101065 - Transportation Modelling and Traffic Management
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Expansion	Version
Oral examination	3 CP	graded	Each term	1 semesters	2

Courses					
ST 2026	6232802	Traffic Management and Telematics	2 SWS	Lecture / Practice / 🗣	Vortisch

Legend: 🖥 Online, Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
oral Exam, appr. 20. min.

Prerequisites
Exercise Transportation Data Analysis must be passed

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-BGU-113971 - Exercise Transportation Data Analysis must have been passed.

Recommendations
None

Additional Information
None

Workload
90 hours

T 6.766 Module component: Transport Economics [T-WIWI-100007]

Coordinators: Prof. Dr. Kay Mitusch
Dr. Eckhard Szimba

Organisation: KIT Department of Business and Economics

Part of: M-WIWI-101406 - Network Economics
M-WIWI-101468 - Environmental Economics
M-WIWI-101485 - Transport Infrastructure Policy and Regional Development
M-WIWI-104908 - Economics

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each summer term	1

Courses					
ST 2026	2560230	Transport Economics	2 SWS	Lecture	Mitusch, Szimba
ST 2026	2560231	Übung zu Transportökonomie	1 SWS	Practice	Krenn

Assessment
The assessment is made by a 60 minutes written examination during the semester break (according to §4(2), 1 ERSC). Examination is offered every semester and can be retried at any regular examination date.

Below you will find excerpts from courses related to this module component:

V Transport Economics

2560230, SS 2026, 2 SWS, Language: German, Open in study portal

Lecture (V)

Content
The course shall provide an overview of transport economics. It will be demonstrated, using new microeconomic models, which impacts regulation and pricing in transport have on the economic actions of individuals and logisticans and which benefits and costs apply. The following topics will be discussed:

- demand and supply in transport
- empirical analysis of transport demand
- assessment of transport infrastructure projects
- external effects in transport
- transport policy
- cost structures of transport infrastructure
- Project evaluation from the perspective of the public sector

Literature
Literatur:

Aberle, G: Transportwirtschaft: einzelwirtschaftliche und gesamtwirtschaftliche Grundlagen München; Wien: Oldenbourg, 2003.

Blauwens, G., De Baere, P. and Van der Voorde, E. (2006): Transport Economics.

Frerich, J; Müller, G: Europäische Verkehrspolitik, Landverkehrspolitik München; Wien: Oldenbourg, 2004.

Dasgupta, A, Pearce, D (1972): Cost-Benefit Analysis, MacMillan, London.

Europäische Kommission (2008): Guide to Cost Benefit Analysis of Investment Projects, online unter http://ec.europa.eu/regional_policy/sources/Ben-Akiva, M., Meerseman, H., and Van de Voorde, E. (2008): Recent developments in transport modelling: Lessons for the freight sector.

Ortúzar, J. d. D. and Willumsen, L. (1990): Modelling Transport.

T 6.767 Module component: Transportation Data Analysis [T-BGU-100010]

Coordinators: Prof. Dr.-Ing. Martin Kagerbauer
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: M-BGU-101065 - Transportation Modelling and Traffic Management
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Expansion	Version
Oral examination	3 CP	graded	Each term	1 semesters	2

Courses					
WT 25/26	6232901	Empirical Data in Transportation	2 SWS	Lecture / Practice /	Kagerbauer

Legend: Online, Blended (On-Site/Online), On-Site, Cancelled

Assessment
oral exam, appr. 20 min.

Prerequisites
The Exercise Transportation Data Analysis (T-BGU-113671) has to be passed.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-BGU-113671 - Exercise Transportation Data Analysis must have been passed.

Recommendations
None

Additional Information
None

Workload
90 hours

T 6.768 Module component: Transportation Systems [T-BGU-106610]

Coordinators: Prof. Dr.-Ing. Peter Vortisch
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Expansion	Version
Written examination	3 CP	graded	Each term	1 semesters	2

Courses					
ST 2026	6200406	Transportation Systems	2 SWS	Lecture / 🗣	Vortisch
ST 2026	6200407	Exercises to Transportation Systems	1 SWS	Practice / 🗣	Vortisch, Mitarbeiter/ innen

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Prerequisites
None

Recommendations
None

Additional Information
None

Workload
90 hours

T 6.769 Module component: Tunnel Construction and Blasting Engineering [T-BGU-101846]

Coordinators: Prof. Dr.-Ing. Shervin Haghsheno
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: M-BGU-101110 - Process Engineering in Construction
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Expansion	Version
Oral examination	3 CP	graded	Each term	1 semesters	1

Courses					
WT 25/26	6241903	Tunnel Construction and Blasting Engineering	2 SWS	Lecture / 🗣	Haghsheno, Mitarbeiter/innen

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
oral exam, appr. 20 min.

Prerequisites
none

Recommendations
none

Additional Information
none

Workload
90 hours

T 6.770 Module component: Turnkey Construction [T-BGU-111921]

Coordinators: Prof. Dr.-Ing. Shervin Haghsheno
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: M-BGU-101884 - Lean Management in Construction
M-BGU-101888 - Project Management in Construction
M-BGU-105592 - Digitalization in Facility Management

Type	Credits	Grading	Term offered	Expansion	Version
Written examination	3 CP	graded	Each term	1 semesters	1

Courses					
ST 2026	6241808	Turnkey Construction	2 SWS	Lecture / Practice / 🗣	Teizer

Legend: 🖥 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
written exam, 60 min.

Prerequisites
none

Recommendations
none

Additional Information
none

Workload
90 hours

T 6.771 Module component: Tutorial Mathematical Methods in Hydraulics [T-MACH-113913]

Coordinators: Prof. Dr.-Ing. Marcus Geimer
Organisation: KIT Department of Mechanical Engineering
Part of: M-MACH-101266 - Automotive Engineering
M-MACH-101267 - Mobile Machines

Type	Credits	Grading	Term offered	Expansion	Version
Coursework	2 CP	pass/fail	Each winter term	1 semesters	1

Assessment
Successful completion of Ilias tests. Details will be announced in the first lecture.

Prerequisites
none

Additional Information
See "Mathematical methods of hydraulics".

The course is offered in German.

Workload
60 hours

T 6.772 Module component: Tutorial Simulation with Lumped Parameters [T-MACH-113863]

Coordinators: Prof. Dr.-Ing. Marcus Geimer
Organisation: KIT Department of Mechanical Engineering
Part of: M-MACH-101265 - Vehicle Development
M-MACH-101267 - Mobile Machines

Type	Credits	Grading	Term offered	Expansion	Version
Coursework	1 CP	pass/fail	Each summer term	1 semesters	1

Courses					
ST 2026	2114072	Simulation with Lumped Parameters (Tutorial)	2 SWS	Practice / 🗣	Geimer

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Submission of a report at the end of the lecture period.

Prerequisites
none

Recommendations
Knowledge of mechanics and hydraulics

Additional Information
This tutorial is a prerequisite for the module components T-MACH-113862 - Simulation with concentrated parameters (examination) in the MSc Mechanical Engineering program (SPO 2025).

The course is offered in German.

Workload
30 hours

6.773 Module component: Tutoring: Training and Practice [T-WIWI-112967]

Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104910 - Interdisciplinary Qualifications

Type	Credits	Grading	Term offered	Version
Coursework	2 CP	pass/fail	Each term	1

Assessment

- Successful participation in the KIT-PEBA tutor training course "Start in die Lehre": 2 credit points.
- Successful participation in the tutor training course "Start in die Lehre" and supplementary tutoring activity over at least two semesters: 3 credit points.

Additional Information

The successful participation in the tutor training "Start in die Lehre" of KIT-PEBA can be credited in the seminar module WiIng/TVWL M.Sc. as interdisciplinary qualification with two or three credit points.

The online application with further information can be found at https://portal.wiwi.kit.edu/forms/form/AnerkennungTutorent%C3%A4tigkeit.

T 6.774 Module component: Ubiquitous Computing [T-INFO-114188]

Coordinators: Prof. Dr.-Ing. Michael Beigl
Organisation: KIT Department of Informatics
Part of: M-WIWI-104909 - Informatics (Department of Informatics)

Type	Credits	Grading	Term offered	Version
Oral examination	5 CP	graded	Each winter term	1

Courses					
WT 25/26	2424146	Ubiquitous Computing		Lecture / Practice	Beigl, Röddiger
WT 26/27	2424146	Ubiquitous Computing		Lecture / Practice	Beigl

Assessment
The assessment is carried out as an oral examination (§ 4 Abs. 2 Nr. 2 SPO) lasting 20 minutes.

Prerequisites
None.

Below you will find excerpts from courses related to this module component:

V Ubiquitous Computing **Lecture / Practice (VÜ)**
2424146, WS 25/26, SWS, Language: English, Open in study portal

Content
In the practical part of the lecture, students deepen their understanding of ubiquitous systems through the practical application of the knowledge base of the lecture. To this end, students design and develop their own appliance and test it. The aim is to have gone through the steps towards a prototypical and possibly marketable appliance.

Description:

The lecture provides an overview of concepts, theories, and methods of ubiquitous information technology (ubiquitous computing).

Based on the appliance concept, students then develop their own appliance in the exercise. The necessary technical and methodological basics such as hardware for ubiquitous systems, software for ubiquitous systems, principles of context recognition for ubiquitous systems, networking of ubiquitous systems, and design principles such as design thinking are discussed. Methods of design and testing for human-machine interaction and human-machine interfaces developed in ubiquitous computing are explained in detail. An introduction to the economic aspects of a ubiquitous system is also provided.

In the practical part of the lecture, students deepen their understanding of ubiquitous systems by applying the knowledge gained in the lecture. Students design and develop their own appliance and test it. The aim is to go through the steps towards a prototypical and potentially marketable appliance.

Course content:

The lecture provides an overview of concepts, theories, and methods of ubiquitous information technology (ubiquitous computing). Based on the appliance concept, students then design their own appliances in the exercise, plan the construction, and then develop them. The necessary technical and methodological basics such as hardware for ubiquitous systems, software for ubiquitous systems, principles of context recognition for ubiquitous systems, networking of ubiquitous systems, and design of ubiquitous systems (e.g., design thinking) and topics related to the development of miniature electronics and 3D printing are discussed. Methods of design and testing for human-machine interaction and human-machine interfaces developed in ubiquitous computing are also explained. There is also an introduction to the economic aspects of a ubiquitous system.

In the practical part of the lecture, understanding of ubiquitous systems is deepened through practical application of the knowledge base of the lecture. Students design and develop their own devices, including electronics and programming, and test them. The goal is to go through the steps toward a prototypical and potentially marketable appliance.

Workload:

The total workload for this learning unit is approximately 150 hours (5.0 credits).

Activity

Workload

Attendance time: Attendance at lectures

15 x 90 min

22 h 30 min

Attendance time: Attendance at exercises

15 x 45 min

11 h 15 min

Preparation/follow-up for lectures and exercises

15 x 90 min

22 h 30 min

Develop a self-developed concept for an information appliance.

33 hours 45 min

Go through the slide set twice.

2 x 12 hours

24 hours

Prepare for the exam.

36 hours

TOTAL

150 hours

Workload for the learning unit “Ubiquitous Information Technologies”

Learning objectives:

The aim of the lecture is to impart knowledge about the fundamentals and advanced methods and techniques of ubiquitous computing. After completing the lecture, students will be able to

- Reproduce and discuss the knowledge they have acquired about existing ubiquitous computing systems.
- Evaluate general knowledge about ubiquitous systems and apply statements and principles to special cases.
- evaluate and assess different methods for design processes and user studies, and select suitable methods for developing new solutions.
- invent, plan, design, build (electronics, software, 3D printing) and evaluate new ubiquitous systems for use in everyday or industrial process environments, and assess the costs and technical implications.

Organizational issues
Mündliche Prüfung nach Vereinbarung.

Die Erfolgskontrolle erfolgt in Form einer mündlichen Prüfung im Umfang von 20 min. nach § 4 Abs. 2 Nr. 2 SPO.

Vorlesung: Dienstags, 11:30 bis 13:00 Uhr, Geb. 50.34, Raum -102

Übung: Mittwochs, 08:00 bis 09:30 Uhr, Geb. 07.07, Raum 222

Oral exam by arrangement.

Performance is assessed in the form of a 20-minute oral exam in accordance with § 4 (2) No. 2 SPO.

Lecture: Tuesdays, 11:30 a.m. to 1:00 p.m., Building 50.34, Room -102
Exercise: Wednesdays, 8:00 a.m. to 9:30 a.m., Building 07.07, Room 222

V **Ubiquitous Computing** **Lecture / Practice (VÜ)**
2424146, WS 26/27, SWS, Language: English, Open in study portal

Content
In the practical part of the lecture, students deepen their understanding of ubiquitous systems through the practical application of the knowledge base of the lecture. To this end, students design and develop their own appliance and test it. The aim is to have gone through the steps towards a prototypical and possibly marketable appliance.

Description:

The lecture provides an overview of concepts, theories, and methods of ubiquitous information technology (ubiquitous computing).

Based on the appliance concept, students then develop their own appliance in the exercise. The necessary technical and methodological basics such as hardware for ubiquitous systems, software for ubiquitous systems, principles of context recognition for ubiquitous systems, networking of ubiquitous systems, and design principles such as design thinking are discussed. Methods of design and testing for human-machine interaction and human-machine interfaces developed in ubiquitous computing are explained in detail. An introduction to the economic aspects of a ubiquitous system is also provided.

In the practical part of the lecture, students deepen their understanding of ubiquitous systems by applying the knowledge gained in the lecture. Students design and develop their own appliance and test it. The aim is to go through the steps towards a prototypical and potentially marketable appliance.

Course content:

The lecture provides an overview of concepts, theories, and methods of ubiquitous information technology (ubiquitous computing). Based on the appliance concept, students then design their own appliances in the exercise, plan the construction, and then develop them. The necessary technical and methodological basics such as hardware for ubiquitous systems, software for ubiquitous systems, principles of context recognition for ubiquitous systems, networking of ubiquitous systems, and design of ubiquitous systems (e.g., design thinking) and topics related to the development of miniature electronics and 3D printing are discussed. Methods of design and testing for human-machine interaction and human-machine interfaces developed in ubiquitous computing are also explained. There is also an introduction to the economic aspects of a ubiquitous system.

In the practical part of the lecture, understanding of ubiquitous systems is deepened through practical application of the knowledge base of the lecture. Students design and develop their own devices, including electronics and programming, and test them. The goal is to go through the steps toward a prototypical and potentially marketable appliance.

Workload:

The total workload for this learning unit is approximately 150 hours (5.0 credits).

Activity

Workload

Attendance time: Attendance at lectures

15 x 90 min

22 h 30 min

Attendance time: Attendance at exercises

15 x 45 min

11 h 15 min

Preparation/follow-up for lectures and exercises

15 x 90 min

22 h 30 min

Develop a self-developed concept for an information appliance.

33 hours 45 min

Go through the slide set twice.

2 x 12 hours

24 hours

Prepare for the exam.

36 hours

TOTAL

150 hours

Workload for the learning unit “Ubiquitous Information Technologies”

Learning objectives:

The aim of the lecture is to impart knowledge about the fundamentals and advanced methods and techniques of ubiquitous computing. After completing the lecture, students will be able to

- Reproduce and discuss the knowledge they have acquired about existing ubiquitous computing systems.
- Evaluate general knowledge about ubiquitous systems and apply statements and principles to special cases.
- evaluate and assess different methods for design processes and user studies, and select suitable methods for developing new solutions.
- invent, plan, design, build (electronics, software, 3D printing) and evaluate new ubiquitous systems for use in everyday or industrial process environments, and assess the costs and technical implications.

Organizational issues

Mündliche Prüfung nach Vereinbarung.

Die Erfolgskontrolle erfolgt in Form einer mündlichen Prüfung im Umfang von 20 min. nach § 4 Abs. 2 Nr. 2 SPO.

Vorlesung: Dienstags, 11:30 bis 13:00 Uhr, Geb. 50.34, Raum -102

Übung: Mittwochs, 08:00 bis 09:30 Uhr, Geb. 07.07, Raum 222

Oral exam by arrangement.

Performance is assessed in the form of a 20-minute oral exam in accordance with § 4 (2) No. 2 SPO.

Lecture: Tuesdays, 11:30 a.m. to 1:00 p.m., Building 50.34, Room -102
Exercise: Wednesdays, 8:00 a.m. to 9:30 a.m., Building 07.07, Room 222

T 6.775 Module component: Upgrading of Existing Buildings [T-BGU-111218]

Coordinators: Prof. Dr.-Ing. Kunibert Lennerts
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: M-BGU-106453 - Facility Management in Hospitals for Industrial Engineering

Type	Credits	Grading	Term offered	Expansion	Version
Written examination	3 CP	graded	Each term	1 semesters	1

Courses					
WT 25/26	6240901	Upgrading of Existing Buildings	3 SWS	Lecture / Practice / 🗣	Lennerts, Schneider

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
written exam, 60 min.

Prerequisites
none

Recommendations
none

Additional Information
none

Workload
90 hours

T 6.776 Module component: Urban Water Technologies [T-BGU-112365]

Coordinators: Dr.-Ing. Mohammad Ebrahim Azari Najaf Abad
apl. Prof. Dr. Stephan Fuchs
Organisation: KIT Department of Civil Engineering, Geo and Environmental Sciences
Part of: M-BGU-104448 - Urban Water Technologies

Type	Credits	Grading	Term offered	Expansion	Version
Oral examination	9 CP	graded	Each term	1 semesters	1

Courses					
WT 25/26	6223701	Urban Water Infrastructure and Management	4 SWS	Lecture / Practice / 🗣	Fuchs
WT 25/26	6223801	Wastewater Treatment Technologies	4 SWS	Lecture / Practice / 🗣	Fuchs, Azari Najaf Abad
WT 26/27	6223701	Urban Water Infrastructure and Management	4 SWS	Lecture / Practice / 🗣	Fuchs

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
oral exam, appr. 30 min.

Prerequisites
none

Recommendations
none

Additional Information
none

Workload
270 hours

T 6.777 Module component: Validation of Technical Systems [T-MACH-113982]

Coordinators: Prof. Dr.-Ing. Tobias Düser
Organisation: KIT Department of Mechanical Engineering

Part of: M-WIWI-107718 - Mechanical Engineering I
M-WIWI-107730 - Mechanical Engineering II

Type	Credits	Grading	Term offered	Expansion	Version
Written examination	4,5 CP	graded	Each summer term	1 semesters	5

Courses					
ST 2026	2146230	Validation of technical Systems	2 SWS	Lecture / 🗣	Düser

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Written exam with a duration of 90 min

Recommendations
The course is offered in English.

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V Validation of technical Systems

2146230, SS 2026, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content

- Discussion and analysis of various validation environments from technical areas such as automotive, medical technology, device technology (focus on automotive)
- Teaching methodological aspects of how validation of complex cyber-physical systems is planned and operationalized
- Learning content on power test benches with their mechanical and electrical design, as well as measurement and control technology, actuators and modeling
- Understanding the use of simulations, their scaling and connection to the real system
- Application of theoretical knowledge in the context of a leading example in the field of automated driving
- Outlook on the role of large language models and gamification in validation

T 6.778 Module component: Valuation [T-WIWI-102621]

Coordinators: Prof. Dr. Martin Ruckes
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101480 - Finance 3
M-WIWI-101482 - Finance 1
M-WIWI-101483 - Finance 2
M-WIWI-101510 - Cross-Functional Management Accounting
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Each winter term	1

Courses					
WT 25/26	2530212	Valuation	2 SWS	Lecture / 🗣	Ruckes
WT 25/26	2530213	Übungen zu Valuation	1 SWS	Practice / 🗣	Ruckes, Luedecke
WT 26/27	2530212	Valuation	2 SWS	Lecture / 🗣	Ruckes, Kohl, Sarac
WT 26/27	2530213	Übungen zu Valuation	1 SWS	Practice / 🗣	Ruckes, Kohl, Sarac

Legend: 🖥 Online, 🔀 Blended (On-Site/Online), 🗣 On-Site, × Cancelled

Assessment
See German version.

Prerequisites
None

Recommendations
None

Below you will find excerpts from courses related to this module component:

V Valuation
2530212, WS 25/26, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Literature
Weiterführende Literatur

Titman/Martin (2013): *Valuation - The Art and Science of Corporate Investment Decisions*, 2nd. ed. Pearson International.

V Valuation
2530212, WS 26/27, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Literature
Weiterführende Literatur

Titman/Martin (2013): *Valuation - The Art and Science of Corporate Investment Decisions*, 2nd. ed. Pearson International.

T 6.779 Module component: Vehicle Drive Technology [T-MACH-113997]

Coordinators: Prof. Dr.-Ing. Marcus Geimer
Organisation: KIT Department of Mechanical Engineering
Part of: M-MACH-101267 - Mobile Machines

Type	Credits	Grading	Term offered	Expansion	Version
Oral examination	4 CP	graded	Each winter term	1 semesters	1

Courses					
WT 25/26	2113091	Machine Drive Technology	2 SWS	Lecture / 🗣	Geimer
WT 25/26	2113095	Exercise Machine Drive Technology	2 SWS	Practice / 🗣	Geimer

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Oral examination, duration approx. 20 minutes

Prerequisites
none

Additional Information
The course is offered in German

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Machine Drive Technology

2113091, WS 25/26, 2 SWS, Open in study portal

Lecture (V)
On-Site

Content
This lecture presents and discusses the possible variations of the traction drive trains of mobile machinery. The focus of the lecture is as follows:
-Mechanical transmissions
-Hydrodynamic converters
-Hydrostatic drives
-Power-split transmissions
-Electric drives
-Hybrid drives
-axles
-Terra mechanics (wheel-ground effects)

Organizational issues
Findet am Campus Ost, Geb. 70.04, Raum 219 statt.

T 6.780 Module component: Vehicle Systems for Urban Mobility [T-MACH-113069]

Coordinators: Prof. Dr.-Ing. Martin Cichon
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-106496 - Modern Mobility on Rails and Roads

Type	Credits	Grading	Version
Oral examination	4 CP	graded	3

Courses					
WT 25/26	2115922	Vehicle Systems for Urban Mobility	2 SWS	Lecture / 🗣	Ziesel, Cichon
ST 2026	2115922	Vehicle Systems for Urban Mobility	2 SWS	Lecture / 🗣	Ziesel, Cichon
WT 26/27	2115922	Vehicle Systems for Urban Mobility	2 SWS	Lecture / 🗣	Ziesel

Legend: 🖥 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Oral examination
Duration: approx. 20 minutes
No tools or reference material may be used during the exam.

Additional Information
The course is offered in German.

Workload
120 hours

T 6.781 Module component: Warehousing and Distribution Systems [T-MACH-105174]

Coordinators: Prof. Dr.-Ing. Kai Furmans
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101278 - Material Flow in Networked Logistic Systems
M-MACH-104888 - Advanced Module Logistics
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Written examination	4 CP	graded	Each summer term	3

Courses					
ST 2026	2118097	Warehousing and distribution systems	2 SWS	Lecture / 🗣	Furmans

Legend: 📱 Online, ❖ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The success control takes place in form of a written examination (60 min) during the semester break (according to §4(2), 1 SPO). If the number of participants is low, an oral examination (according to §4 (2), 2 SPO) may also be offered.

Prerequisites
none

Additional Information
The course is offered in German

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Warehousing and distribution systems

Lecture (V)
On-Site

2118097, SS 2026, 2 SWS, Language: German, Open in study portal

Organizational issues
Vorlesung:

Die Vorlesung wird in diesem Semester als **Blockveranstaltung** angeboten.

Die Veranstaltungstermine sind:

- Di., 19. Mai 2026
- Mi., 20. Mai 2026
- Fr., 22. Mai 2026

Die Vorlesung findet jeweils von 08:00 Uhr bis 15:30 Uhr im **Selmayr-HS (Geb. 50.38)** statt. Bitte beachten Sie für mögliche kurzfristige Raumänderungen die Informationen im ILIAS-Kurs.

Die Vorlesung wird von einem externen Dozenten aus der Logistikberatung gehalten.

Während der Vorlesung werden u.a. als Teil der Klausurvorbereitung einzelne Übungsaufgaben bereitgestellt und besprochen.

Fokusthemen:

- Grundlagen von Lagerplanung und Intralogistik-Technik
- Je eine Vorlesungseinheit für alle Bereiche eines Lagers
- Praxisbeispiele und Fallstudien
- Übersicht über Logistiknetzwerke und Lean Logistics

Klausur:

- Informationen zur Klausur werden zeitnah über den Ilias-Kurs bekanntgegeben.

Vorlesungsbegleitende Unterlagen:

- Die vorlesungsbegleitenden Unterlagen finden Sie ebenfalls im Ilias-Kurs.

Ansprechpartner:

Keno Büscher

Literature
ARNOLD, Dieter, FURMANS, Kai (2005)
Materialfluss in Logistiksystemen, 5. Auflage, Berlin: Springer-Verlag

ARNOLD, Dieter (Hrsg.) et al. (2008)
Handbuch Logistik, 3. Auflage, Berlin: Springer-Verlag

BARTHOLDI III, John J., HACKMAN, Steven T. (2008)
Warehouse Science

GUDEHUS, Timm (2005)
Logistik, 3. Auflage, Berlin: Springer-Verlag

FRAZELLE, Edward (2002)
World-class warehousing and material handling, McGraw-Hill

MARTIN, Heinrich (1999)
Praxiswissen Materialflußplanung: Transport, Hanshaben, Lagern, Kommissionieren, Braunschweig, Wiesbaden: Vieweg

WISSER, Jens (2009)
Der Prozess Lagern und Kommissionieren im Rahmen des Distribution Center Reference Model (DCRM); Karlsruhe : Universitätsverlag

Eine ausführliche Übersicht wissenschaftlicher Paper findet sich bei:

ROODBERGEN, Kees Jan (2007)
Warehouse Literature

T 6.782 Module component: Water Technology [T-CIWVT-106802]

Coordinators: Prof. Dr. Harald Horn
Organisation: KIT Department of Chemical and Process Engineering
Part of: M-CIWVT-101121 - Water Chemistry and Water Technology I
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Oral examination	6 CP	graded	Each winter term	1

Courses					
WT 25/26	2233030	Water Technology	2 SWS	Lecture / 🗣	Horn
WT 25/26	2233031	Exercises to Water Technology	1 SWS	Practice / 🗣	Horn, und Mitarbeitende

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

6.783 Module component: Web App Programming for Finance [T-WIWI-110933]

Coordinators: Prof. Dr. Julian Thimme
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-101480 - Finance 3

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	Once	1

Assessment
Non exam assessment according to § 4 paragraph 3 of the examination regulation. (Anmerkung: gilt nur für SPO 2015). The grade is made up as follows: 50% result of the project (R-code), 50% presentation of the project.

Prerequisites
None

Recommendations
The content of the bachelor course Investments is assumed to be known and necessary to follow the course.

Workload
135 hours

T 6.784 Module component: Web Applications and Service-Oriented Architectures (II) [T-INFO-101271]

Coordinators: Prof. Dr. Sebastian Abeck
Organisation: KIT Department of Informatics
Part of: M-WIWI-104909 - Informatics (Department of Informatics)

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each summer term	1

Courses					
ST 2026	24677	Web Applications and Service oriented Architectures (II)	2 SWS	Lecture / 🗣	Abeck, Schneider, Throner

Legend: 🖥 Online, ❂ Blended (On-Site/Online), 🗣 On-Site, x Cancelled

6.785 Module component: Welding Technology [T-MACH-105170]

Coordinators: Dr.-Ing. Majid Farajian
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101268 - Specific Topics in Materials Science
M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Oral examination	4 CP	graded	Each winter term	1

Courses					
WT 25/26	2173571	Welding Technology	2 SWS	Block / 🗣	Farajian
WT 26/27	2173571	Welding Technology	2 SWS	Block / 🗣	Farajian

Legend: 🖥 Online, Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Oral exam, about 20 minutes

Prerequisites
none

Recommendations
Basics of material science (iron- and non-iron alloys), materials, processes and production, design.

All the relevant books of the German Welding Institute (DVS: Deutscher Verband für Schweißen und verwandte Verfahren) in the field of welding and joining is recommended.

Additional Information
The course is offered in German.

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Welding Technology

2173571, WS 25/26, 2 SWS, Language: German, Open in study portal

Block (B)
On-Site

Content
definition, application and differentiation: welding,

welding processes, alternative connecting technologies.

history of welding technology

sources of energy for welding processes

Survey: Fusion welding,

pressure welding.

weld seam preparation/design

welding positions

weldability

gas welding, thermal cutting, manual metal-arc welding

submerged arc welding

gas-shielded metal-arc welding, friction stir welding, laser beam and electron beam welding, other fusion and pressure welding processes

static and cyclic behavior of welded joints,

fatigue life improvement techniques

learning objectives:

The students have knowledge and understanding of the most important welding processes and its industrial application.

They are able to recognize, understand and handle problems occurring during the application of different welding processes relating to design, material and production.

They know the classification and the importance of welding technology within the scope of connecting processes (advantages/ disadvantages, alternatives).

The students will understand the influence of weld quality on the performance and behavior of welded joints under static and cyclic load.

How the fatigue life of welded joints could be increased, will be part of the course.

requirements:
basics of material science (iron- and non-iron alloys), of electrical engineering, of production processes.

workload:

The workload for the lecture Welding Technology is 120 h per semester and consists of the presence during the lecture (18 h) as well as preparation and rework time at home (102 h).

exam:

oral, ca. 20 minutes, no auxiliary material

Organizational issues
Die Blockveranstaltung findet am 29.01.26, 30.01.2026, 05.02.2026, 06.02.2026, 13.02.2026 jeweils von 09:00 bis 15:00 Uhr in Gebäude 10.91 Raum 380 statt. Anmeldungen erfolgen über den Beitritt zum ILIAS-Kurs. Bei Fragen wenden Sie sich gerne an majid.farajian@kit.edu

Literature
Für ergänzende, vertiefende Studien gibt das

Handbuch der Schweißtechnik von J. Ruge, Springer Verlag Berlin, mit seinen vier Bänden

Band I: Werkstoffe

Band II: Verfahren und Fertigung

Band III: Konstruktive Gestaltung der Bauteile

Band IV: Berechnung der Verbindungen

einen umfassenden Überblick. Der Stoff der Vorlesung Schweißtechnik findet sich in den Bänden I und II. Einen kompakten Einblick in die Lichtbogenschweißverfahren bietet das Bändchen

Nies: Lichtbogenschweißtechnik, Bibliothek der Technik Band 57, Verlag moderne Industrie AG und Co., Landsberg / Lech

Im Übrigen sei auf die zahlreichen Fachbücher des DVS Verlages, Düsseldorf, zu allen Einzelgebieten der Fügetechnik verwiesen.

V **Welding Technology** — Block (B) On-Site
2173571, WS 26/27, 2 SWS, Language: German, Open in study portal

Content
definition, application and differentiation: welding,

welding processes, alternative connecting technologies.

history of welding technology

sources of energy for welding processes

Survey: Fusion welding,

pressure welding.

weld seam preparation/design

welding positions

weldability

gas welding, thermal cutting, manual metal-arc welding

submerged arc welding

gas-shielded metal-arc welding, friction stir welding, laser beam and electron beam welding, other fusion and pressure welding processes

static and cyclic behavior of welded joints,

fatigue life improvement techniques

learning objectives:

The students have knowledge and understanding of the most important welding processes and its industrial application.

They are able to recognize, understand and handle problems occurring during the application of different welding processes relating to design, material and production.

They know the classification and the importance of welding technology within the scope of connecting processes (advantages/ disadvantages, alternatives).

The students will understand the influence of weld quality on the performance and behavior of welded joints under static and cyclic load.

How the fatigue life of welded joints could be increased, will be part of the course.

requirements:
basics of material science (iron- and non-iron alloys), of electrical engineering, of production processes.

workload:

The workload for the lecture Welding Technology is 120 h per semester and consists of the presence during the lecture (18 h) as well as preparation and rework time at home (102 h).

exam:

oral, ca. 20 minutes, no auxiliary material

Organizational issues
Die Blockveranstaltung findet am 29.01.26, 30.01.2026, 05.02.2026, 06.02.2026, 13.02.2026 jeweils von 09:00 bis 15:00 Uhr in Gebäude 10.91 Raum 380 statt. Anmeldungen erfolgen über den Beitritt zum ILIAS-Kurs. Bei Fragen wenden Sie sich gerne an majid.farajian@kit.edu

Literature
Für ergänzende, vertiefende Studien gibt das

Handbuch der Schweißtechnik von J. Ruge, Springer Verlag Berlin, mit seinen vier Bänden

Band I: Werkstoffe

Band II: Verfahren und Fertigung

Band III: Konstruktive Gestaltung der Bauteile

Band IV: Berechnung der Verbindungen

einen umfassenden Überblick. Der Stoff der Vorlesung Schweißtechnik findet sich in den Bänden I und II. Einen kompakten Einblick in die Lichtbogenschweißverfahren bietet das Bändchen

Nies: Lichtbogenschweißtechnik, Bibliothek der Technik Band 57, Verlag moderne Industrie AG und Co., Landsberg / Lech

Im Übrigen sei auf die zahlreichen Fachbücher des DVS Verlages, Düsseldorf, zu allen Einzelgebieten der Fügetechnik verwiesen.

T 6.786 Module component: Welfare Economics [T-WIWI-102610]

Coordinators: Prof. Dr. Clemens Puppe
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-104908 - Economics

Type	Credits	Grading	Term offered	Version
Written examination	4,5 CP	graded	see Annotations	3

Assessment
The assessment consists of a written exam (60 min.).

Prerequisites
The course *Economics I: Microeconomics* [2610012] has to be completed beforehand.

Modeled Prerequisites
The following conditions have to be fulfilled:

1. The module component T-WIWI-102708 - Economics I: Microeconomics must have been passed.

Recommendations
None

Additional Information
The course only takes place every second summer semester, the next course is planned for summer semester 2027. The last iteration of this course is expected to take place in the summer semester of 2029. The last exam will therefore be offered in the summer semester of 2030.

T 6.787 Module component: Windpower [T-MACH-105234]

Coordinators: Norbert Lewald
Organisation: KIT Department of Mechanical Engineering
Institute of Thermal Turbomachinery
Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Written examination	4 CP	graded	Each winter term	3

Courses					
WT 25/26	2157381	Windpower	2 SWS	Lecture / 🗣	Lewald
WT 26/27	2157381	Windpower	2 SWS	Lecture / 🗣	Lewald

Legend: 🖥 Online, 🧩 Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
Written exam.

Duration: 80 min

Prerequisites
none

Additional Information
The course is offered in German.

Workload
120 hours

Below you will find excerpts from courses related to this module component:

V Windpower — **Lecture (V) On-Site**
2157381, WS 25/26, 2 SWS, Language: German, Open in study portal

V Windpower — **Lecture (V) On-Site**
2157381, WS 26/27, 2 SWS, Language: German, Open in study portal

T 6.788 Module component: Workshop Business Wargaming – Analyzing Strategic Interactions [T-WIWI-106189]

Coordinators: Prof. Dr. Hagen Lindstädt
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-103119 - Advanced Topics in Strategy and Management
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Irregular	1

Courses					
ST 2026	2577922	Business Wargaming Workshop - Analysis of Strategic Interactions (Master)	2 SWS	Lecture / 🗣	Lindstädt

Legend: 📱 Online, ⚙ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
In this course, real conflict situations are simulated and analyzed using various methods from business wargaming. Details on the design of the performance review will be announced during the lecture.

Prerequisites
None

Recommendations
Basic knowledge as conveyed in the bachelor module „Strategy and Organization“ is recommended.

Additional Information
This course is admission restricted. If you were already admitted to another course in the module “Advanced Topics in Strategy and Management” the participation at this course will be guaranteed.

The course is planned to be held for the first time in the summer term 2018.

Workload
90 hours

Below you will find excerpts from courses related to this module component:

V Business Wargaming Workshop - Analysis of Strategic Interactions (Master)
Lecture (V)
On-Site

2577922, SS 2026, 2 SWS, Language: English, Open in study portal

Content

This course enables the simulation of strategic conflicts in which the participants assume the roles of selected actors. With the help of specially programmed wargaming software, strategic conflicts are simulated interactively and then reflected upon and discussed.

The course focuses on the simulation and analysis of real conflict situations with strategic interaction. Students gain a better understanding of the structural characteristics of strategic conflicts in the fields of economics and politics as well as the ability to derive their own strategies for action.

Through a combination of group work, simulation, and reflection, the seminar provides a learning experience that both strengthens team skills and develops analytical skills in strategic conflict. Join this seminar to gain sound insights into conflict dynamics and develop effective action strategies for complex situations.

Learning Objectives

Upon completion of the course, students will be able to,

- learn the basic methodologies, features and benefits of business wargaming
- improve their understanding of conflict dynamics by reflecting on strategic conflicts
- Strengthen analytical skills by processing a variety of courses of action and deriving strategies for action

Recommendations:

Prior attendance of the Bachelor's module "Strategy and Organization" or another module with comparable content at another university is recommended.

Workload:

- Total workload: approx. 90 hours
- Attendance time: 15 hours
- Preparation and follow-up: 75 hours
- Examination and preparation: not applicable

Competence Certificate:

In this course, real conflict situations are simulated and analyzed with the help of various methods from business wargaming. Details on the design of the performance review will be announced during the lecture.

Note:

The course is admission restricted. In case of prior admission to another course in the module "Strategy and Management: Advanced Topics" [M-WIWI-103119], participation in this course is guaranteed. For more information on the application process, see the IBU website.

Exams are offered at least every other semester, so the entire module can be completed in two semesters.

Organizational issues

IBU-Seminarraum, Geb. 05.20, Raum 2A-12.1

T 6.789 Module component: Workshop Current Topics in Strategy and Management [T-WIWI-106188]

Coordinators: Prof. Dr. Hagen Lindstädt
Organisation: KIT Department of Business and Economics
Part of: M-WIWI-103119 - Advanced Topics in Strategy and Management
M-WIWI-104900 - Business Administration

Type	Credits	Grading	Term offered	Version
Examination of another type	3 CP	graded	Irregular	1

Courses					
WT 25/26	2577923	Workshop aktuelle Themen Strategie und Management (Master)	2 SWS	Lecture / 🗣	Lindstädt
WT 26/27	2500006	Workshop aktuelle Themen Strategie und Management (Master)	2 SWS	Lecture / 🗣	Lindstädt

Legend: 🖥 Online, Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
The evaluation of the performance takes place through the active participation in the discussion rounds; an appropriate preparation is expressed here and a clear understanding of the topic and framework becomes recognizable. Further details on the design of the performance review will be announced during the lecture.

Prerequisites
None

Recommendations
Basic knowledge as conveyed in the bachelor module „Strategy and Organization“ is recommended.

Additional Information
This course is admission restricted. If you were already admitted to another course in the module “Advanced Topics in Strategy and Management” the participation at this course will be guaranteed.

The course is planned to be held for the first time in the winter term 2017/18.

Workload
90 hours

Below you will find excerpts from courses related to this module component:

V Workshop aktuelle Themen Strategie und Management (Master)

Lecture (V)
On-Site

2577923, WS 25/26, 2 SWS, Language: German, Open in study portal

Content
Aspects of strategic management can be found in a variety of daily events. In this course, current strategic and industrial policy issues are discussed and the exchange of ideas on current management topics is promoted.

For this purpose, practice-relevant case studies and dedicated questions are communicated to the students in advance so that they can prepare themselves individually for the discussion. The chair team actively moderates the discussion and creates typical discussion situations such as pro/con discussions and conflicting interests of different groups in order to bring opposing opinions into an exchange and to promote the power of argumentation. In this way, the discussion not only imparts knowledge about the content, but also strengthens the participants' skills by simulating real discussion situations in a management team.

In addition, company representatives and managers participate in individual case studies to strengthen the context of the content and experience the daily dynamics of discussion in strategic business areas.

Learning Objectives:

Students will

- are able to evaluate strategic decisions using appropriate models of strategic business management,
- are able to present and critically evaluate theoretical approaches and models in the field of strategic business management and illustrate them using practical examples, and
- have the ability to present their position convincingly through a reasoned argumentation in structured discussions.

Recommendations:

Previous attendance of the Bachelor's module "Strategy and Organization" or another module with comparable content at another university is recommended.

Workload:

Total effort approx. 90 hours

Attendance time: 15 hours

Preparation and follow-up: 75 hours

Examination and preparation: not applicable

Competence Certificate:

Performance will be assessed through active discussion participation in the discussion rounds; here, adequate preparation will be expressed and a clear understanding of the topic and framework will be evident. Further details on the design of the performance assessment will be announced during the lecture.

Annotation:

This course is admission restricted. In case of prior admission to another course in the module "Strategy and Management: Advanced Topics"[M-WIWI-103119], participation in this course is guaranteed. For more information on the application process, see the IBU website.

Exams are offered at least every other semester so that the entire module can be completed in two semesters.

V	**Workshop aktuelle Themen Strategie und Management (Master)** 2500006, WS 26/27, 2 SWS, Language: German, Open in study portal	**Lecture (V)** **On-Site**

Content
Aspects of strategic management can be found in a variety of daily events. In this course, current strategic and industrial policy issues are discussed and the exchange of ideas on current management topics is promoted.

For this purpose, practice-relevant case studies and dedicated questions are communicated to the students in advance so that they can prepare themselves individually for the discussion. The chair team actively moderates the discussion and creates typical discussion situations such as pro/con discussions and conflicting interests of different groups in order to bring opposing opinions into an exchange and to promote the power of argumentation. In this way, the discussion not only imparts knowledge about the content, but also strengthens the participants' skills by simulating real discussion situations in a management team.

In addition, company representatives and managers participate in individual case studies to strengthen the context of the content and experience the daily dynamics of discussion in strategic business areas.

Learning Objectives:

Students will

- are able to evaluate strategic decisions using appropriate models of strategic business management,
- are able to present and critically evaluate theoretical approaches and models in the field of strategic business management and illustrate them using practical examples, and
- have the ability to present their position convincingly through a reasoned argumentation in structured discussions.

Recommendations:

Previous attendance of the Bachelor's module "Strategy and Organization" or another module with comparable content at another university is recommended.

Workload:

Total effort approx. 90 hours

Attendance time: 15 hours

Preparation and follow-up: 75 hours

Examination and preparation: not applicable

Competence Certificate:

Performance will be assessed through active discussion participation in the discussion rounds; here, adequate preparation will be expressed and a clear understanding of the topic and framework will be evident. Further details on the design of the performance assessment will be announced during the lecture.

Note:

This course is admission restricted. In case of prior admission to another course in the module "Strategy and Management: Advanced Topics"[M-WIWI-103119], participation in this course is guaranteed. For more information on the application process, see the IBU website.

Exams are offered at least every other semester so that the entire module can be completed in two semesters.

T 6.790 Module component: Workshop Mechatronical Systems and Products [T-MACH-112648]

Coordinators: Prof. Dr.-Ing. Sören Hohmann
Prof. Dr.-Ing. Sven Matthiesen

Organisation: KIT Department of Mechanical Engineering

Part of: M-WIWI-104907 - Engineering Sciences

Type	Credits	Grading	Term offered	Version
Coursework	5 CP	pass/fail	Each winter term	2

Courses					
WT 25/26	2145162	Workshop Mechatronical Systems and Products	3 SWS	Practical course / 🗣	Matthiesen, Hohmann, Teltschik
WT 26/27	2145162	Workshop Mechatronical Systems and Products	3 SWS	Practical course / 🗣	Matthiesen, Hohmann, Teltschik

Legend: 🖥 Online, ✲ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment

The assessment is carried out in the form of a course achievement consisting of a semester-long workshop.
The partial achievement is not graded. It is considered passed upon successful evaluation of the course achievement.

Throughout the semester, accompanying the workshop, submission tasks are required at given milestones. These tasks assess the application of knowledge developed within the module. Examples of such submissions include CAD designs, control software, and reflection reports, which are specified in a workshop assignment at the beginning of the semester. The milestones are announced in a calendar at the start of the semester and are made available to students via ILIAS.

Prerequisites

None

Knowledge of the contents of the lecture Mechatronische Systeme und Produkte [T-MACH-105574] will be assumed to be known.

Additional Information

All relevant materials (script, exercise sheets, etc.) for the course can be accessed via the eLearning platform ILIAS.

To participate in the course, please complete the survey, registration and group allocation in ILIAS before the start of the semester.

The course is offered in German.

Workload

150 hours

T 6.791 Module component: X-ray Optics [T-MACH-109122]

Coordinators: Dr. Arndt Last
Organisation: KIT Department of Mechanical Engineering

Part of: M-MACH-101291 - Microfabrication
M-MACH-101292 - Microoptics
M-WIWI-104907 - Engineering Sciences
M-WIWI-107718 - Mechanical Engineering I
M-WIWI-107730 - Mechanical Engineering II

Type	Credits	Grading	Term offered	Version
Oral examination	4,5 CP	graded	Each term	2

Courses					
WT 25/26	2141007	X-ray Optics	2 SWS	Lecture / 🗣	Last
ST 2026	2141007	X-ray optics	2 SWS	Lecture / 🗣	Last

Legend: 🖥 Online, ❂ Blended (On-Site/Online), 🗣 On-Site, ✕ Cancelled

Assessment
oral exam (about 20 min)

Prerequisites
none

Workload
135 hours

Below you will find excerpts from courses related to this module component:

V X-ray Optics

2141007, WS 25/26, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content
Introduction: X-ray optics and applications
X-ray generation: sources
Propagation of X-rays in matter
Principles of optics and imaging
Types of X-ray optics: reflecting, refracting, diffracting, absorbing
Characteristics of X-ray optics
Methods to simulate X-ray optics (ray tracing, wave propagation)
Manufacturing of X-ray optics
Characterization of X-ray optics
X-ray detection
Applications: Medical, material sciences…
X-ray analytical methods

Organizational issues
Termin und Ort nach Absprache mit den Angemeldeten

Literature
M. Born und E. Wolf
Principles of Optics, 7th (expanded) edition
Cambridge University Press, 2010

A. Erko, M. Idir, T. Krist und A. G. Michette
Modern Developments in X-Ray and Neutron Optics
Springer Series in Optical Sciences, Vol. 137
Springer-Verlag Berlin Heidelberg, 2008

D. Attwood
Soft X-Rays and Extreme Ultraviolet Radiation: Principles and Applications
Cambridge University Press, 1999

V

X-ray optics

2141007, SS 2026, 2 SWS, Language: English, Open in study portal

Lecture (V)
On-Site

Content
Introduction: X-ray optics and applications
X-ray generation: sources
Propagation of X-rays in matter
Principles of optics and imaging
Types of X-ray optics: reflecting, refracting, diffracting, absorbing
Characteristics of X-ray optics
Methods to simulate X-ray optics (ray tracing, wave propagation)
Manufacturing of X-ray optics
Characterization of X-ray optics
X-ray detection
Applications: Medical, material sciences…
X-ray analytical methods

Organizational issues
Viertägiger Blockkurs im Juni oder Juli 2026. Interessenten melden sich bitte zur Terminabsprache bis zum 30.5.2026 bei arndt.last@kit.edu